

A Guide to IMF Stress Testing

Methods and Models



Editor
Li Lian Ong

I N T E R N A T I O N A L M O N E T A R Y F U N D

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For

Mark W. Swinburne
June 17, 1955–September 3, 2009

Assistant Director
Monetary and Capital Markets Department
International Monetary Fund

Manager, mentor, friend

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Stress Tester 3.0

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Excel Spreadsheet Macro for the Next-Generation Solvency Stress Test

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Excel Spreadsheet Macro for the Market and Funding Liquidity Stress Test

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Excel Spreadsheet Macro for Forward-Looking Macroprudential Stress Test

Foreword

The global financial crisis has placed a spotlight on the stress testing of financial systems. Although weaknesses in stress tests were exposed by the crisis, the recent experience of several countries has conversely provided a stark illustration of their potential benefit in examining the resilience of bank balance sheets when performed credibly and transparently. Nonetheless, the large menu of stress testing approaches, methods, and models raises questions about their appropriate application under different situations and, consequently, the comparability and reliability of the associated analyses.

The International Monetary Fund (IMF) has had a long and unique involvement in the stress testing of financial systems. Since the introduction of the Financial Sector Assessment Program (FSAP) more than a decade ago, IMF staff has conducted stress tests of banking sectors in over 120 countries, typically in close collaboration with country authorities. Stress testing is also playing an increasingly important role in the IMF's multilateral surveillance, through the analysis in our *Global Financial Stability Report*. Separately, member countries are increasingly requesting IMF technical assistance in stress testing as they develop their own expertise in this area. As a result, our staff has amassed a wealth of hands-on experience with stress testing techniques and their practical application.

This book represents a compendium of stress testing methods, models, and tools developed or adapted by IMF staff over the years. Almost all the methods and models that are included in this volume have, at one time or another, been applied in our surveillance of, or our technical assistance to, member countries. To guide users, each chapter offers a summary describing the application of a method or model, its strengths and weaknesses, and the data requirements. Where available, the stress testing tools or program codes are also provided for wider public use.

Although I trust that this volume will provide a valuable resource for policymakers, supervisors, academics, and private sector participants alike, caveats still apply. The crisis has underscored that stress tests, irrespective of their level of sophistication, are not fail-safe, stand-alone diagnostic tools. Assessments of the soundness of any financial system cannot and should not be based solely on a "model" and must be complemented by other quantitative analyses, qualitative information, and, most important, expert judgment. Especially in light of evolving market practices, risks, and regulatory requirements, stress testing will necessarily continue to be art rather than science.

IMF staff is continually working to strengthen the analytical underpinnings of its stress testing, in ways that will help bolster its consistency and comparability and hence its credibility. Key areas of focus include extending the analysis to better cover nonbank financial institutions and infrastructures; to take account of spillovers between institutions and across borders; to consider the interaction between liquidity and solvency risks; and to address data gaps. In addition, IMF staff is developing the policy-related aspects of stress testing, namely, "best practice" principles, concepts, and frameworks, to complement and strengthen the application of the models. These efforts represent a challenging and exciting part of the IMF's broader support of global efforts to improve financial surveillance and promote sound macroprudential frameworks.

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Abbreviations

ABO	accrued benefit obligation
ABS	asset-backed security
AIB	Allied Irish Banks PLC
AIG	American International Group
AIRB	Advanced Internal Ratings Based
Anglo IB	Anglo Irish Bank Corp. PLC
ARCH	autoregressive conditional heteroskedasticity
ASF	available stable funding
BBVA	Banco Bilbao Vizcaya Argentaria
BCBS	Basel Committee on Banking Supervision
BCCH	Central Bank of Chile/Banco Central de Chile
BCP	Basel Core Principles for Banking Supervision
BHC	bank holding company
BIS	Bank for International Settlements
BoE	Bank of England
BoI	Bank of Ireland
BSI	Banking Stability Index
BSM	banking stability measure
BSMD	Banking System's (portfolio) Multivariate Density
BSoM	Black-Scholes-Merton
BU	bottom-up
CAPM	capital asset pricing model
CAR	capital adequacy ratio (regulatory capital to risk-weighted assets)
CCA	contingent claims analysis
CCP	Copula Choice Problem
CDO	collateralized debt obligation
CDS	credit default swap
CEBS	Committee of European Banking Supervisors
CEDF	cumulative expected default frequency
CESE	Central, Eastern, and Southern Europe
C&I	commercial and industrial
CIMDO	Consistent Information Multivariate Density Optimizing
CN	capital need
CNB	Croatian National Bank
CoPoD	Conditional Probability of Default
CPI	consumer price index
CRE	commercial real estate
CRI	credit risk indicator

CRT	credit risk transfer
CSFP	Credit Suisse Financial Products
DAX	Deutscher Aktien IndeX
DB	defined benefit
DCC	dynamic conditional correlation
DD	distance to default
DiDe	Distress Dependence Matrix
DNB	De Nederlandsche Bank
DSI	debt service-to-income ratio
DTA	deferred tax assets
DtD	distance to distress
EAD	exposure at default
EC	economic capital
ECB	European Central Bank
EDF	expected default frequency
EL	expected loss
EMBI+	Emerging Market Bond Index
EMBIG	Emerging Market Bond Index Global
ES	expected shortfall
EU	European Union
EVT	extreme value theory
FFT	fast Fourier transform
FIRB	Foundation Internal Ratings Based
FME	Financial Supervisory Authority/Fjármálaeftirlitsins
FMI	financial market infrastructure
FSAP	Financial Sector Assessment Program
FSB	Financial Stability Board
FSC	Financial Services Center
FSI	financial soundness indicator
FSR	Financial Stability Report
FVCDS	fair value CDS
FVOAS	fair value option adjusted spread
FX	foreign exchange
GARCH	generalized autoregressive conditional heteroskedasticity
GEV	generalized extreme value
GFSR	<i>Global Financial Stability Report</i>
GMM	Generalized Method of Moments
GOB	Government of Brazil
HBOS	Halifax Bank of Scotland
HHI	Herfindahl-Hirschman Index
HSBC	Hongkong and Shanghai Banking Corporation
IBB	immediate borrower basis
ICR	interest coverage ratio
IFRS	International Financial Reporting Standards
IFS	International Financial Statistics

IMACEC	Indicador Mensual de Actividad Económica
IPSA	Indice de Precios Selectivo de Acciones
IRB	Internal Ratings Based
IRF	impulse response function
ISEQ	Irish Stock Exchange Overall Index
IT	information technology
JPoD	joint probability of distress
KMV	Kealhofer, McQuown, and Vasicek (a part of Moody's Analytics)
LCR	liquidity coverage ratio
LGD	loss given default
LHS	left-hand side
LIBOR	London Interbank Offered Rate
LLR	loan loss reserve
LS	least squares
LTCM	Long Term Capital Management
LTV	loan-to-value
MCAR	market-implied capital adequacy ratio
MES	marginal expected shortfall
MfRisk	Macro-Financial Risk
MGF	moment generating function
MIDP	market implied default probabilities
MKMV	Moody's KMV
ML	maximum likelihood
MPS	macroprudential policy and surveillance
MSCI	Morgan Stanley Capital International
MSE	mean squared error
MXED	minimum cross-entropy distribution
NASDAQ	National Association of Securities Dealers Automated Quotations
NBB	National Bank of Bankistan
NBFI	nonbank financial institution
NFI	net foreign investment
NPL	nonperforming loan
NSFR	net stable funding ratio
OBS	off-balance-sheet
OeNB	Austrian National Bank/Oesterreichische Nationalbank
OIS	overnight indexed swap
OLS	ordinary least squares
OOM	out-of-the-money
PAO	probability that at least one bank becomes distressed
PBO	projected benefit obligation
PBOcd	projected benefit obligation constant dollar
PCA	principal component analysis
PD	probability of default
PGF	probability generating function
PIT	Probability Integral Transformation

PLD	profit and loss distribution
PMD	portfolio multivariate distribution
PoD	probability of distress
QIRB	quasi-Internal Ratings Based
QIS	Quantitative Impact Study
RAMSI	Risk Assessment Model for Systemic Institutions
RHS	right-hand side
RBO	retirement benefit obligation
RBS	Royal Bank of Scotland
RNDP	risk-neutral default probability
RNS	risk-neutral credit spread
ROA	return on assets
RRE	residential real estate
RSF	required stable funding
RWA	risk-weighted assets
SAS	stand-alone subsidiarization
SBIF	Banking Supervisory Agency/Superintendencia de Bancos e Instituciones Financieras
SCAP	Supervisory Capital Assessment Program
SELIC	Sistema Especial de Liquidação e Custodia
SLOOS	Senior Loan Officer Opinion Survey of Bank Lending Practices
SME	small- and medium-sized enterprise
S&P	Standard and Poor's
SPD	state-price density
SRL	Systemic Risk-Adjusted Liquidity
SRM	Systemic Risk Monitor
StA	Standardized Approach
SWF	sovereign wealth fund
TARP	Troubled Asset Relief Program
TBTF	too-big-to-fail
TCTF	too-connected-to-fail
TD	top-down
TMTF	too-many-to-fail
TTC	through-the-cycle
UL	unexpected loss
URB	ultimate risk basis
VaR	value at risk
VAR	vector autoregression
VDAX	implied volatility of the Deutscher Aktien Index (DAX)
VIX	Chicago Board Options Exchange Market Volatility Index (implied volatility of the S&P 500 index)
WaMu	Washington Mutual
WEO	<i>World Economic Outlook</i>

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Disclaimer

The views expressed in this volume are those of the authors and do not necessarily represent those of their respective institutions.

Stress Testing at the International Monetary Fund: Methods and Models

LI LIAN ONG • MARTIN ČIHÁK

“Down one road lies disaster, down the other utter catastrophe. Let us hope we have the wisdom to choose wisely.”
—Woody Allen

Stress testing is a “what if” exercise. It measures the sensitivity of a portfolio, an institution, or a financial system to *exceptional but plausible shocks*. The answer involves identifying relevant risk drivers; selecting the appropriate method or model; using that particular method or model to calculate the effects of large shocks; and interpreting the results correctly. A number of studies provide a general introduction to stress testing, discussing its nature and purpose (e.g., Blaschke and others, 2001; Jones, Hilbers, and Slack, 2004; and Čihák, 2007). Stress tests are also being designed from another angle, which is to ask the question: What would it take to “break” a financial institution or a financial system? (e.g., Financial Services Authority, 2009).

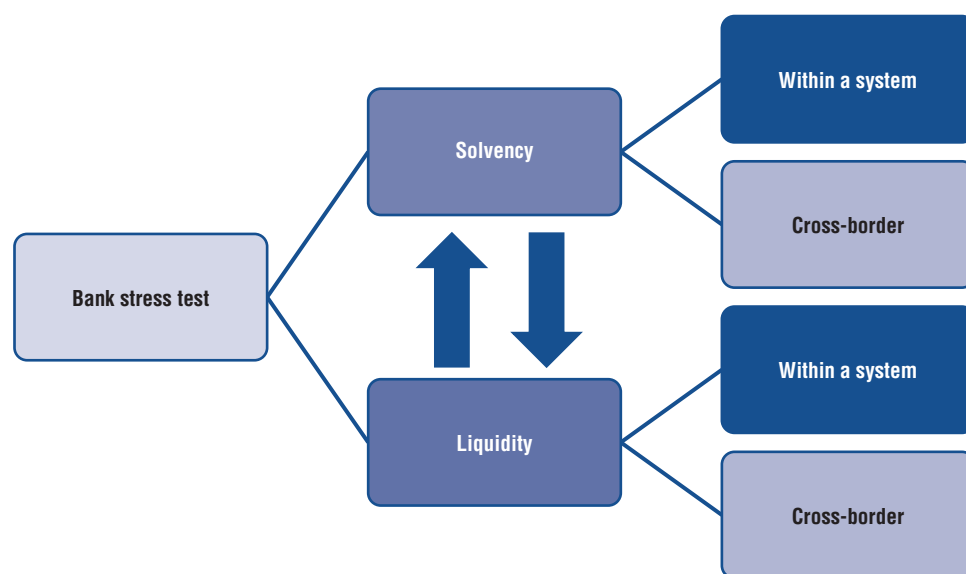
From a technical perspective, stress testing has become more complex and sophisticated over time. A wide range of methods and statistical and mathematical models developed by academics and practitioners are now available for estimating the impact of various financial or economic shocks on financial systems. For potential users, the wealth of available techniques can be confusing—their relevance and applicability under different conditions and situations may not be always clear, and it may not be obvious how they supplement or complement each other.

Over the years, staff at the IMF also has developed a suite of stress testing methods and models, and has adapted existing ones, for use in their financial surveillance work. Indeed, this area of macroprudential risk analysis has become a central aspect of IMF staff’s assessment of individual financial systems and of the international financial system itself. It is a key component of the Financial Sector Assessment Program (FSAP) and has become an important part of the conjunctural and structural analyses in the *Global Financial Stability Report*

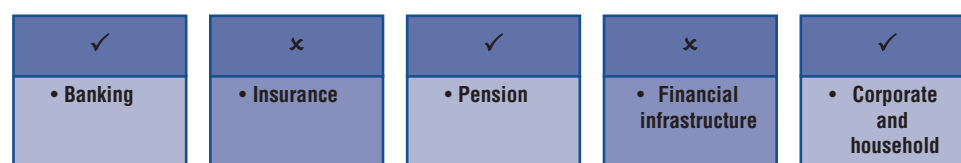
(GFSR). Stress testing is also being undertaken increasingly in Article IV and crisis program work. Correspondingly, the demand by IMF member countries for technical assistance from IMF staff on stress testing has risen as well, as country authorities seek to develop and enhance their own capacity in this area.

The global financial crisis injected a dose of caution into the enthusiasm surrounding the usefulness of stress tests. It raised questions about the credibility of the exercises conducted in the run-up to the crisis, many of which were unable to adequately capture the relevant risks and exposures and hence did not provide sufficient early warning of potential vulnerabilities. Critics attributed the failures to poor data quality, weaknesses in scenario design, inadequate methods and models, or their incorrect application. At the IMF, lessons learned from the crisis have spurred staff to improve the robustness and versatility of stress tests. One of the main areas is improving the design and application of stress testing methods and models. IMF staff has developed new models and are adapting or calibrating existing ones to better capture the risks (including those that manifested during the crisis) and are working to ensure their consistency and appropriate use in different settings.

This volume puts together, for the first time, the applied stress testing methods and models built or adapted by IMF staff, some in collaboration with external colleagues, before and during the crisis. Most chapters have previously been released as IMF working papers, while some have been published in refereed journal articles. Given the very technical nature of the material presented, this book is not for the faint-hearted. But for those who are interested in understanding staff’s stress testing methods and models, this book provides essential insight into the strengths and shortcomings of each technique and details the data required for implementation.



Source: Authors.

Figure 1.1 Stress Test Methods and Models by IMF Staff: Banking Sector

Source: Authors.

Figure 1.2 Stress Test Methods and Models by IMF Staff: Sector Coverage

In each case, related work by academics and other practitioners also are reviewed and put into context. This book aims to be innovative in several ways, by

- offering a suite of stress testing methods and models that can be
 - applied to the gamut of financial systems across the entire spectrum of development—from the most basic banking sectors, where data may be limited or of poor quality, to the most sophisticated, with a wealth of accounting and market data;
 - used for surveillance or supervisory purposes;
 - applied to individual banks or the financial system as a whole.
- categorizing the methods and models into various approaches and subapproaches and providing summary guidance to users at the beginning of each chapter on their appropriate application, the data requirements, as well as their strengths and weaknesses; and
- making available the accompanying tools or programming codes, where possible.

Stress tests can be conducted on the various sectors of the financial system, as well as on corporate and household balance sheets. At the IMF, work on stress tests of the banking

sector is clearly the most advanced, reflecting the systemic importance of banks in practically all member countries. Within the banking sphere, stress testing for solvency risk has been the main focus, although work on liquidity risk has also come to the fore in the wake of the global financial crisis. IMF staff now has the tools to separately carry out solvency and liquidity stress tests within banking systems and across borders (Figure 1.1), and work is progressing on models to explicitly link the two risks (Part I.A). However, staff's work on the nonbank sector remains nascent: techniques have been developed to “stress test” the pension sector (Part I.A) and to determine the impact of the corporate and household sector on the financial positions on banks (Part II.C), but little has been done to date on stress tests for the insurance and infrastructure sectors (Figure 1.2).

APPROACHES, METHODS, AND MODELS

No single stress testing method or model is perfectly suited for all financial systems, and an important challenge for IMF staff is to ensure that the appropriate stress test is applied on

each occasion. If the stress test is to be informative, it is critical that the method adequately captures the important risk drivers amid the complexity, uniqueness, and idiosyncrasies of a particular system. It is also crucial to be able to reconcile the differences across different stress testing methods and their implications for the results. As a general rule:

- Stress tests of simple financial systems dominated by domestic banks offering “plain vanilla” products normally require less sophisticated models and are less resource intensive. In contrast, stress tests of more complex systems typically apply more advanced stress testing methods to capture the gamut of risks and require more resources to implement.

- The more sophisticated the model, the greater the estimation uncertainty, an issue to be taken into account when drawing policy conclusions. The dilemma is that simpler methods might be inadequate for highly interconnected and complex banking sectors with large credit and market risk exposures.

This book categorizes stress testing methods and models into three main approaches, namely, the accounting-based, the market price-based, and the macro-financial approach. The first two are presented and compared in IMF (2012a) across several dimensions (Table 1.1). This book builds on that comparison and adds the third approach, which has attracted much attention in recent public supervisory stress tests for

TABLE 1.1

Comparing the Accounting-Based and Market Price-Based Approaches

	Accounting Based	Market Price Based
Primary input data	Accounting data (balance sheet, profit and loss account, matrix of interbank exposures).	Financial market data (equity prices, bond yields, CDS spreads, or equity option-based probability of distress of banks).
Secondary input data	Probability of distress (and loss-given-default) or NPL ratios/loan classification of borrowers (for credit risk); market data (equity prices, exchange rates, interest rates, price volatility, term premium) to calibrate shocks.	Balance sheet data (combination of equity prices and accounting data obtain key input variable, such as the expected default frequency by Moody's).
Type of test	Solvency, liquidity, and network analyses.	To date, largely focused on solvency and its interdependence among key financial institutions. Initial attempts at testing for liquidity stress.
Frequency	Varies depending on the reporting cycle (quarterly, semiannual, annual).	Daily or lower frequency.
Application	Most banks or financial systems (including emerging markets and low-income countries) as long as financial reporting or supervisory data exist and are available.	Limited to market data—rich countries and institutions that are quoted on the market (it generally cannot cover mutuals, privately held, or government-owned companies). Stand-alone analysis for subsidiaries may be difficult.
Link(s) to macro scenarios	Possible, by estimating additional macro-financial model(s), linking macro scenario variables and risk factors (PDs of borrowers, NPL ratios, etc.).	Possible, by estimating additional macro-financial model(s), linking macro scenario variables and risk factors (PDs of banks, volatility, or leverage of banks, etc.).
Estimation of systemic effects	By considering common macro shocks across banks (e.g., GDP, inflation); and incorporating network effects (interbank exposures).	By considering common macro shocks across banks and incorporating interdependence (portfolio) effects among banks, which may be estimated using asset prices.
Output	Various capital ratios. Liquidity ratios. Capital shortfalls. The number/share of banks breaching minimum requirements.	Expected losses. Unexpected or tail losses. Contingent liabilities for the government. Probabilities of spillover among banks.
Strength	Pinpoints the type of risk that creates the vulnerability (e.g., credit losses from housing loans, market valuation losses from exposures to sovereigns, losses from currency mismatches). Possible to adjust for supervisory weakness (e.g., underprovisioning, forbearance).	Less data-intensive than the accounting-based approach. Focuses on systemic risks/losses and tail events. Incorporates risk factors priced by the market.
Weakness	Data intensive (especially for network analysis). Quality of the analysis depends on the granularity and availability of the data.	The causes of different risks are difficult to disentangle (“black box”). Estimated vulnerability measures may be very volatile during periods when markets are under significant stress, and links with balance sheet fundamentals may be obscured.

Source: IMF (2012a).

Note: CDS = credit default swap; NPL = nonperforming loan; PD = probability of default.

crisis management purposes in Europe and the United States. Most of the included methodologies are “bread and butter” stress tests. They estimate the capital needs or “hole” in a bank or banking system following the imposition of an adverse shock. Less conventional methods also are presented to demonstrate the variety of techniques that have been developed by IMF staff. These comprise methods that identify interconnectedness, spillover or systemic risks under stress, and feedback between the macroeconomy and financial sector and also include innovative adaptations of existing theory and models to stress testing.

The accounting-based approach

The accounting-based approach uses accounting data from the financial statements of individual institutions or systems. It is also popularly known as the *balance sheet–based* approach, although that is a shorthand expression given that the approach requires not only data from standard balance sheets but also profit and loss statements, off-balance-sheet items, and additional information available in the financial statements. As noted in Schmieder and Schumacher’s introduction in Chapter 2, it is a “natural” approach to stress testing in that the information is usually readily and publicly available and allows a bottom-up analysis of individual institutions. At the IMF, balance sheet–based methods were introduced in FSAPs. They remain the cornerstone among stress tests and have been calibrated and enhanced to adapt to the expanding variety of financial systems covered and evolving regulatory regimes.

This book discusses the *network analysis* approach as a subcategory of the accounting-based approach, separate from the other balance sheet–based techniques. Stress tests using network analysis are a relatively recent development and took center stage during the global financial crisis. At the IMF, network analysis was first applied by staff in the GFSR (IMF, 2009) and has since proliferated into other areas of staff’s multilateral surveillance as well as bilateral country work. In Chapter 13 Espinosa-Vega and Solé discuss the conceptual underpinnings and its growing application in the international context as policymakers and analysts seek to better understand the impact of spillover shocks in an increasingly interconnected international financial system.

The market price–based approach

The market price–based approach largely uses market prices of various financial instruments. Although more timely from a data perspective, it is typically applied to supplement the accounting-based approach in the IMF’s surveillance work, given the relatively nascent nature of the modeling for stress testing purposes. This book lists three subcategories:

- The first relies on the use of *equity indicators* as an alternative to accounting data, as explained by Chan-Lau in Chapter 16. Equity indicators may be used to estimate default risk of individual financial and nonfinan-

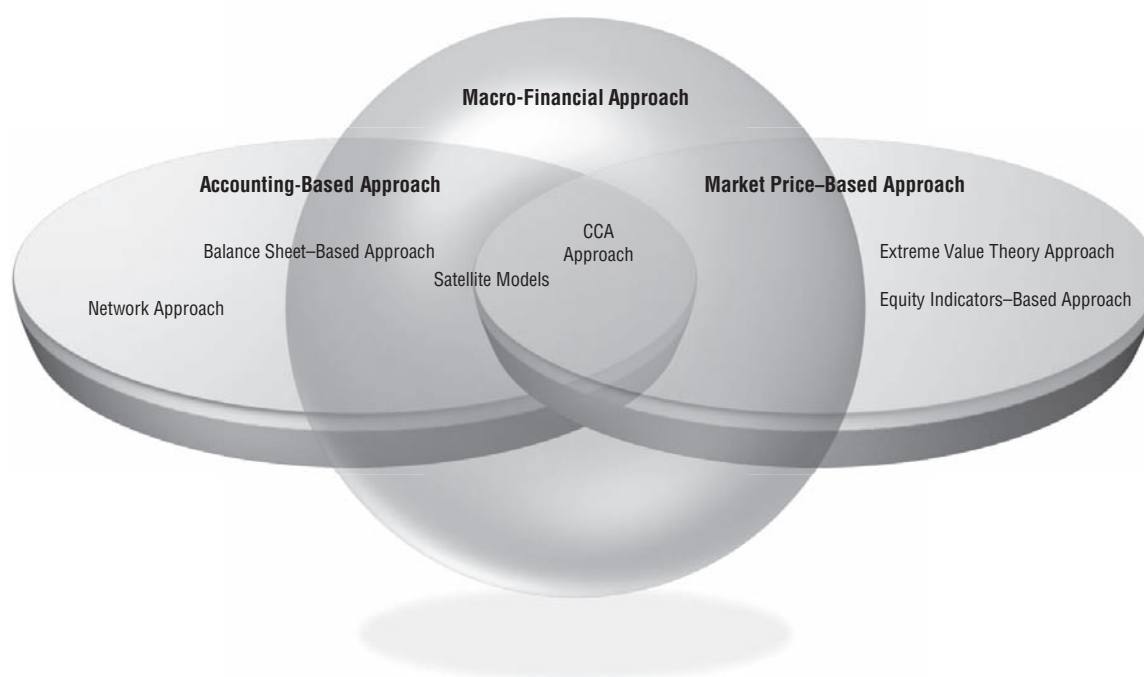
cial institutions and, consequently, to assess losses under different scenarios or to evaluate the level of systemic risk under stressed scenarios. The findings could then be applied in policy decisions aimed at reducing systemic risk—for example, levying capital charges on highly connected institutions.

- The second is the *extreme value theory* (EVT) approach, presented by Mitra in Chapter 19. This method is used to identify all extreme events (tail risks) that could have a severe impact on the soundness of financial institutions or systems and then estimates distress dependence using a logit model. Unlike the standard stress test, the EVT method does not estimate capital needs; rather, it shows the potential spillover countries or financial institutions following a shock to a particular country or institution.
- The third is based on the *contingent claims analysis* (CCA) framework, which is discussed by Gray and others in Chapter 22. This methodology applies Black and Scholes’s (1973) as well as Merton’s (1973) model to stress testing. It enables the estimation of the relationship of macroeconomic factors (including sovereign risk) to the time pattern of bank assets or credit risk indicators, which is then integrated with stress scenarios to project the risks to the banking sector. Systemic tail risk in the financial system also can be analyzed by considering the dependence between CCA risk indicators for multiple financial institutions.

The macro-financial approach

For completeness, this book introduces a third category of stress tests—the *macro-financial* approach, which focuses on linkages between the financial and the nonfinancial sectors of the economy. Arguably, the macro-financial approach might be considered a separate dimension of the other two approaches, as it can be implemented with both accounting and market price data by estimating additional macro-financial linkages models (“satellite models”) that directly connect macroeconomic assumptions and risk parameters (Figure 1.3). However, although satellite models are typically part of the methodology applied in the other two approaches, they represent the main technique in some stress tests, hence the separate category.

The macro-financial approach gained prominence during the global financial crisis. It was used, for example, by the IMF to assess global capital shortfalls, accounting for the complex dynamics of marked-to-market repricing of structured products (IMF, 2008, 2009). Separately, country authorities and third-party consultants used macro-financial models (linking bank financial statements to macroeconomic factors) to analyze banking sector vulnerabilities in a forward-looking manner, as illustrated by the high-profile exercises conducted in the United States, Ireland, and Spain. In Chapter 28, Maechler discusses an eclectic selection of methods developed by IMF staff. These include mainstream econometric credit risk models;



Source: Authors.
Note: CCA = contingent claims analysis.

Figure 1.3 Stress Testing Models Developed by IMF Staff

copula-based models; a model that replicates the one applied in the U.S. Supervisory Capital Assessment Program; and a model that attempts to capture feedback effects between banks and the real economy.

OPERATIONAL CONSIDERATIONS

The approaches that fall into the three broad categories are not completely exclusive in terms of their methodologies. Some straddle two or even all three approaches and may be further grouped according to their methodologies, as shown in Figure 1.3. Correspondingly, the data requirements vary with the approaches and methods and may include public or supervisory information and accounting data or market prices. The method selected also determines the nature of the shock applied in the stress test and hence the complexity of the task (Table 1.2).

Importantly, almost all the techniques included in this volume have been operationalized and implemented. They have been used by staff in one or more core areas of IMF work, namely, bilateral surveillance (e.g., FSAP, Article IV), multi-lateral surveillance (e.g., GFSR, Spillover Report), and technical assistance (Table 1.3). Some have also been applied to internal IMF analyses, such as the Early Warning Exercise, whereas the rest were designed to highlight specific concepts relating to ongoing work in a particular area or for further work. Where possible, the tools to implement the methods or models presented here are either provided with this book or will be made available to readers upon request to the authors.

LOOKING AHEAD

IMF staff is pushing the work on stress testing forward on several fronts. The goal is to create a comprehensive suite of models and to ensure that their application adequately captures the relevant risks to domestic and international financial systems, in a consistent and comparable manner. To this end, IMF staff is focusing on four key areas (Figure 1.4).

Methodology

There is still much room for improvement in stress test modeling. IMF staff is continuing to develop, adapt, and calibrate existing methods and models as new ideas, information, or techniques come to light. In addition, work is being expanded to try and capture feedback loops between macroeconomic and banking system shocks—one of the glaring gaps in the literature as revealed by developments since the onset of the global financial crisis (Alfaro and Drehmann, 2009).

Nonbank financial institutions and financial market infrastructures (FMIs)

The need to stress test the vulnerabilities of the nonbank financial sector has become clear during the global financial crisis. At the IMF, work on and understanding of related issues has advanced, but much more remains to be done compared with the banking sector. For example:

TABLE 1.2

Scorecard: IMF Staff Stress Test Methods and Models

Details	Data Complexity						
	Basic Data	More Sophisticated Data					
<i>Elements</i>	Accounting based	Accounting based	Accounting based	Market price based	Market price based	Accounting and market	Market price based
Approach(es)		incorporating macro-financial models				price based incorporating macro-financial models	incorporating macro-financial models
Method	Balance sheet–based approach	Balance sheet–based approach (including with satellite models)	Network analysis approach	Equity indicator approach	Extreme value theory approach	Contingent claims analysis approach	Distress dependence framework
Design of shock	Sensitivity analysis	Macro scenarios	Sensitivity analysis	Sensitivity analysis	Sensitivity analysis	Macro scenarios	Macro scenarios
Examples	Ong, Maino, and Duma (Chapter 4)	Schmieder, Puhr, and Hasan (Chapter 5)	Espinosa-Vega and Solé (Chapter 14)	Chan-Lau (Chapter 17)	Duggar and Mitra (Chapter 20)	Jobst and Gray (Chapter 26)	Segoviano and Goodhart (Chapter 32)
<i>Data type</i>							
Accounting							
Public	✓	✓	✓		✓	✓	
Supervisory	✓	✓	✓				
Interbank			✓				
Market				✓	✓	✓	✓
Macroeconomic	✓	✓				✓	✓

Source: Authors.

TABLE 1.3

Book Structure, Stress Test Applications, and Tool Availability						
Chapter	Author(s)	Application of Method or Model at the IMF				Tool ¹
		Surveillance		Technical Assistance	Other	
		Bilateral	Multilateral			
		Article IV	FSAP	GFSR	Spillover Report	
1	Ong and Čihák (Book introduction)					
The Accounting-Based Approach						
<i>The Balance Sheet–Based Approach</i>						
2	Schmieder and Schumacher (Introduction)					
3	Čihák	✓	✓			Available with book.
4	Ong, Maino, and Duma		✓			Available with book.
5	Schmieder, Puhr, and Hasan		✓			Available with book.
6	Ong and Čihák					✓ ² Available with book.
7	Schmieder, Hesse, Neudorfer, Puhr, and Schmitz		✓			✓ ³ Available with book.
8	Barnhill and Souto					✓ ⁴ Third-party copyright.
9	Barnhill and Schumacher		✓	✓		Not yet available.
10	Avesani, Liu, Mirestean, and Salvati		✓		✓	Available with book.
11	Cerutti, Ilyina, Makarova, and Schmieder		✓			Std. econometrics pkg.
12	Impavido		✓		✓	Available with book.
<i>The Network Analysis Approach</i>						
13	Espinosa-Vega and Solé (Introduction)					
14	Espinosa-Vega and Solé		✓	✓	✓	Available with book.
15	Chan-Lau	✓			✓	Available upon request.
The Market Price–Based Approach						
<i>The Equity Indicators–Based Approach</i>						
16	Chan-Lau (Introduction)					
17	Chan-Lau	✓		✓	✓	Std. econometrics pkg.
18	Chan-Lau	✓		✓	✓	Std. econometrics pkg.
<i>The Extreme Value Theory Approach</i>						
19	Mitra (Introduction)					
20	Duggar and Mitra		✓			Available with book.
21	Chan-Lau, Čihák, Mitra, and Ong	✓			✓	Available with book.
<i>The Contingent Claims Analysis Approach</i>						
22	Gray, Jobst, Lim, and Xiao (Introduction)					
23	Ruiz-Arranz		✓	✓ ⁵		Available upon request.
24	Gapen, Gray, Lim, and Xiao	✓			✓	Available with book.
25	Gray and Walsh	✓				Std. econometrics pkg.
26	Jobst and Gray		✓	✓	✓	Available upon request.
27	Jobst			✓		Not yet available.
The Macro-Financial Approach						
28	Maechler (Introduction)					
29	Vazquez, Tabak, and Souto				✓	Available upon request.
30	Wezel, Canta, and Luy				✓	Std. econometrics pkg.
31	Segoviano and Padilla		✓		✓	Available upon request.
32	Segoviano and Goodhart		✓	✓	✓	Available upon request.
33	Keim and Maechler		✓			Std. econometrics pkg.
34	Tieman and Maechler					✓ ⁶ Macro avail. with book. Std. econometrics pkg.

Source: Authors.

Note: FSAP = Financial Sector Assessment Program; GFSR = *Global Financial Stability Report*.

1. The available tools are typically Excel-based or program codes.

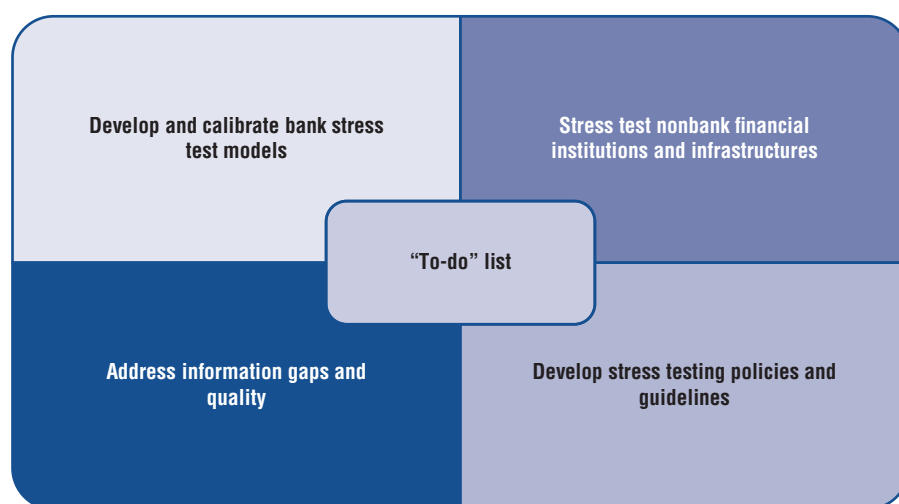
2. Back-testing liquidity risk in stress tests.

3. Also used in the IMF's early-warning exercise.

4. Bilateral work with central bank.

5. Debt-at-risk method.

6. Demonstration of feedback loops.



Source: Authors.

Figure 1.4 Stress Testing by IMF Staff: Areas for Further Development

- Since 2003, FSAP stress tests on the insurance sector have been conducted by IMF staff in fewer than 15 countries and on the pension funds sector in two countries, compared with more than 50 on the banking sector since the onset of the global financial crisis in 2008 alone.
- IMF staff has contributed to the specifications of stress testing requirements of FMIs in the Committee on Payment and Settlement Systems-International Organization of Securities Commissions standards and assessed their implementation during FSAPs but have thus far not conducted stress tests on FMIs.

Policies

In addition to the technical work on modeling, IMF staff is also turning their focus toward developing stress testing policies and improving their implementation. The large menu of choices in terms of stress testing approaches, methods, scenarios, and underlying assumptions applied in staff's analyses—all within changing stability environments and regulatory regimes—has given rise to questions about the effectiveness of such exercises, the interpretation of the results, and their comparability across countries. In this context, IMF staff is making every attempt to improve the usefulness and credibility of stress tests and to ensure a modicum of uniformity for comparison purposes, both within a financial system and, at the very least, across “peer” countries. Recent efforts include

- developing “best practice” principles (IMF, 2012a);
- presenting guidelines within defined stress testing frameworks derived from IMF staff's own experiences (Jobst, Ong, and Schmieder, 2013; Jobst and Hesse, forthcoming);
- identifying rules of thumb for key dimensions of bank solvency (credit losses, pre-impairment income

and credit growth during crises) to support the simulation of capital ratios under stress (Hardy and Schmieder, 2013).

IMF staff has also analyzed the application of stress tests for crisis management (macroprudential) purposes during the global financial crisis. Recent developments have highlighted additional concepts, issues, and nuances that need to be taken into account in the design of such exercises to ensure their effectiveness (Ong and Pazarbasioglu, 2014).

Information gaps and quality

The global financial crisis revealed the costliness of the lack of comprehensive, timely, and accurate information for surveillance and crisis management. IMF staff continues to be involved in international efforts toward addressing these shortcomings:

- The G20 has established the Data Gaps Project, in which the IMF is a key participant, to improve both the quality and quantity of economic and financial data available for policy analysis (see IMF/Financial Stability Board, 2009). From a stress testing perspective, data problems were manifest for IMF staff in terms of the reliability of the FSAP results when the asset quality on banks' books came into question during the crisis. Consequently, some of staff's analyses had to be qualified to ensure transparency (e.g., IMF, 2011, 2012b).
- Similar concerns in crisis stress tests in Europe have led authorities to undertake asset quality reviews of banks' portfolios in order to regain market confidence (e.g., Ireland, Spain, and the 2014 European Union stress testing exercise). IMF staff has been and remains involved in ongoing discussions on this topic.

In short, much has been achieved at the IMF in stress testing, but continuing improvements are necessary in several

areas to enhance the credibility of the institution's work going forward. IMF staff engages widely with country authorities, counterpart agencies, and the private sector on stress testing, and these partnerships and interactions are set to grow further. This book represents a contribution by IMF staff toward improving the understanding of users of the myriad of methods and models that IMF staff has developed or adapted, which are being applied in the day-to-day work of the IMF.

REFERENCES

- Alfaro, Rodrigo, and Mathias Drehmann, 2009, "Macro Stress Tests and Crises: What Can We Learn," *BIS Quarterly Review* (December), pp. 29–41 (Basel: Bank for International Settlements). Available via the Internet: http://www.bis.org/publ/qtrpdf/r_qt0912.htm
- Black, Fischer, and Myron S. Scholes, 1973, "The Pricing of Options and Corporate Liabilities," *Journal of Political Economy*, Vol. 81, No. 3, pp. 637–54.
- Blaschke, Winfrid, Matthew T. Jones, Giovanni Majnoni, and Soledad Martinez Peria, 2001, "Stress Testing of Financial Systems: An Overview of Issues, Methodologies and FSAP Experiences," IMF Working Paper 01/88 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.cfm?sk=15166.0>
- Čihák, Martin, 2007, "Introduction to Applied Stress Testing," IMF Working Paper 07/59 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=20222>
- Financial Services Authority, 2009, "Stress and Scenario Testing: Feedback on CP08/24 and Final Rules," Policy Statement 09/20 (London, December). Available via the Internet: www.fsa.gov.uk/pubs/policy/ps09_20.pdf
- Hardy, Daniel C., and Christian Schmieder, 2013, "Rules of Thumb for Bank Solvency Stress Testing," IMF Working Paper (Washington: International Monetary Fund).
- International Monetary Fund, 2008, *Global Financial Stability Report*, World Economic and Financial Surveys, Chapter 1 (Washington, April). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2008/01/index.htm>
- , 2009, *Global Financial Stability Report*, World Economic and Financial Surveys, Chapter 1 (Washington, April). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2009/01/index.htm>
- , 2011, "United Kingdom: Financial System Stability Assessment," IMF Country Report 11/222 (Washington, July). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=25111>
- , 2012a, "Macrofinancial Stress Testing: Principles and Practices," IMF Policy Paper (Washington, August 22). Available via the Internet: <http://www.imf.org/external/pp/longres.aspx?id=4702>
- , 2012b, "Spain: Financial System Stability Assessment," IMF Country Report 12/137 (Washington, June). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=25977>
- International Monetary Fund/Financial Stability Board, 2009, "The Financial Crisis and Information Gaps: Report to the G20 Finance Ministers and Central Bank Governors" (Washington and Basel, October). Available via the Internet: <http://www.imf.org/external/np/g20/pdf/102909.pdf>
- Jobst, Andreas A., and Heiko Hesse, forthcoming, "A Framework for Macroprudential Bank Liquidity Stress Testing at the IMF: Concepts and Applications," IMF Working Paper (Washington: International Monetary Fund).
- Jobst, Andreas A., Li Lian Ong, and Christian Schmieder, 2013, "A Framework for Macroprudential Bank Solvency Stress Testing: Application to S-25 and other G-20 Country FSAPs," IMF Working Paper 13/68 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=40390>
- Jones, Matthew T., Paul Hilbers, and Graham Slack, 2004, "Stress Testing Financial Systems: What to Do When the Governor Calls," IMF Working Paper 04/127 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=17517>
- Merton, Robert C., 1973, "Theory of Rational Option Pricing," *Bell Journal of Economics and Management Science*, Vol. 4, No. 1, pp. 141–83.
- Ong, Li Lian, and Ceyla Pazarbasioglu, 2014, "Credibility and Crisis Stress Testing," *International Journal of Financial Studies*, Vol. 2, No. 1, pp. 15–81.

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PART I

The Accounting-Based Approach

SECTION A

The Balance Sheet–Based Approach

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Introduction to the Balance Sheet–Based Approach to Stress Testing

CHRISTIAN SCHMIEDER • LILIANA SCHUMACHER

The balance sheet approach is the “natural” approach to stress testing banks and other financial institutions. The reason is twofold: first, balance sheet information is publicly available for financial institutions in a standardized format,¹ which facilitates peer comparison and systemwide stress tests; second, the approach allows a bottom-up view on the vulnerabilities of institutions and hence the identification of important risk drivers. These two dimensions are key advantages of the approach compared with market-based models, which are based on markets’ perceptions of default probabilities and other risk drivers embedded in market prices (such as stock and bond prices and credit default swap spreads), which are not necessarily available for all banks, even in developed markets.

The balance sheet approach to stress testing has, in part, been spurred by the Financial Sector Assessment Program (FSAP), conducted by the IMF and the World Bank since the late 1990s. This method, introduced by IMF staff in the earlier years of the FSAP, represents the “first generation” of (balance sheet–based) stress tests and is documented by Čihák in Chapter 3. The contribution was not only to allow for stress tests covering the key risks faced by banks (credit, market, liquidity, and contagion risk) but to do so in a meaningful yet practical, transparent, and flexible way to account for differences in financial systems and their vulnerabilities. Work outside the IMF in this area has also been abundant in recent

years, as documented in central banks’ Financial Stability Reports. By now, many central banks and/or regulatory agencies run stress tests on a regular basis, and some publish the results of the tests on their respective Web sites.

At present, the key balance sheet frameworks used at the IMF are the Excel-based balance sheet approaches introduced in Chapter 3 and subsequent calibrations and enhancements. The choices depend on the complexity of the banking system being assessed, the data at hand, and the experience of the stress tester. More integrated stress testing frameworks (such as the ones used by the Bank of England and the Austrian Central Bank) are more complex and are perceived to be “black boxes” to some extent, although the recent prominent crisis stress tests conducted by the U.S. and European authorities have leaned in the direction of the simpler balance sheet–based approach.

Despite its popularity and many advantages, the balance sheet approach also faces its own challenges. These include the ability to adequately capture the risks in a comprehensive, meaningful, timely, and preferably forward-looking manner; as well as systemic aspects, such as the risk-transmission process among banks, financial institutions, and other market participants that can create or help propagate systemic risks (e.g., sovereigns). Extensions of the basic balance sheet described in Chapter 3 and the related approaches presented in this volume seek to tackle these two dimensions as further outlined in the following.

DATA CONSIDERATIONS

The “quality” of a balance sheet–based stress test depends on the scope and quality of available data and the selection of an

¹ This assumes that banks use the International Financial Reporting Standards rather than national Generally Accepted Accounting Principles. However, even in the latter case, it is usually relatively straightforward to run countrywide stress tests and, with expert judgment, to make the necessary cross-country comparisons.

adequate technique. Timely and high-quality data are prerequisites for a meaningful balance sheet–based stress test but will likely remain a major limitation for some time, for the following reasons: data are very often limited in scope (e.g., typically on new products), and this can be crucial (as the crisis has shown); data quality is also often limited (such as in the case of missing values and short time series).²

The practical implication of this is that the information used to conduct balance sheet–based stress tests ideally should be more granular than the “pure” financial statements information (i.e., assets, liabilities, and items from the profit and loss account). Additional information may be available in banks’ regulatory Pillar 3 reports and from publications by regulatory authorities.³ Valuable data provided by authorities and banks may also be found on the Web pages of supervisory authorities (e.g., most notably the European Banking Authority with the 2011 EU bank stress tests, when more than 3,000 bank variables were published).⁴ The most detailed information on banks is available to supervisors (e.g., in the credit register and other supervisory data) and to the banks themselves, but these sources of information are confidential and not available to the public.

Several chapters in this book address the data issues:

- Ong, Maino, and Duma (Chapter 4) demonstrate how solvency stress tests could be conducted in countries with weak data. Separately, Schmieder, Puhr, and Hasan (Chapter 5) provide examples of how to integrate information from other sources into balance sheet data, in order to make the assessment more meaningful and forward-looking. Their framework also accounts for data limitations and allows for risk-sensitive solvency and liquidity stress tests even in cases where only limited data are available.
- Ong and Čihák (Chapter 6) apply a “traditional” liquidity test to Iceland and document the limitations of solvency stress tests in identifying the buildup of risks in the lead-up to the crisis. This contribution remains one of the very few studies “validating” stress tests, an area to be explored further in order to improve stress testing framework on the one hand and

also to be reminded of their limitations on the other. Separately, the work by Schmieder and others (Chapter 7) represents the balance sheet–based liquidity risk compendium to their solvency stress test in Chapter 5 and allows stress tests to be carried out even with limited information on liquidity.

THE CRISIS AND REGULATORY REFORMS

In recent years, various contributions have been added to the domain of “balance sheet–based” approaches. The aims include (1) accounting for the lessons learned from the crisis (e.g., that the quality of capital matters); (2) making use of recent risk management techniques; and (3) making use of more granular information once it becomes available. A catalyst in this context was the introduction of Basel II, which lifted bankwide stress tests (beyond market risk) “formally” into the regulatory framework. The other driver was, naturally, the financial crisis, whereby the role of stress tests moved beyond “what-if” computations (which more often than not omitted policy actions) to that of crisis management tools used with the aim of regaining the confidence of markets by removing asymmetric information on the value of banks’ assets and liabilities between bank managers and market participants, as in the case of the U.S. and EU stress tests. These issues are explicitly targeted in the frameworks presented by Schmieder and his respective coauthors in Chapters 5 and 7:

- Chapter 5 introduces IMF staff’s “next-generation” balance sheet–based framework for assessing solvency risks, which is centered on Basel II/III developments. The framework allows the user to run multiyear tests for hundreds of banks and is more risk sensitive than previous frameworks (by adjusting risk-weighted assets for increasing portfolio risk) and more comprehensive.
- Chapter 7 presents the equivalent of the next-generation balance sheet–based stress tests for liquidity risk. As in the solvency case, the purpose of the framework is to allow for more comprehensive, more risk-sensitive tests while keeping it transparent and relatively simple. In addition to “traditional” bank-run-type liquidity tests, the framework enables the assessment of maturity mismatches, including full-fledged cash flow tests used by banks and regulators. Likewise, the framework allows an analysis of the links between solvency and liquidity risks and the computation of proxies for the Basel III liquidity ratios, in all cases provided that data are available.

PORTFOLIO RISKS AND SYSTEMIC ASPECTS

Given their (static) nature, traditional balance sheet approaches need to be complemented with satellite models.

² See Hardy and Schmieder (2013).

³ Pillar 3 of the Basel framework seeks to facilitate market discipline. Banks’ Pillar 3 reports are particularly useful for balance sheet–based stress tests. They contain information on banks’ capital adequacy and comprehensive information on specific risk types—namely, (counterparty) credit risk (including structured credit), market risk, operational risk, and liquidity risk—and their relevance, some of which is standardized as foreseen by Basel II regulation. A cornerstone of this reporting is to provide information to market participants and thus reduce information asymmetries. An illustrative example is the Pillar 3 information provided by Deutsche Bank; the bank has published Pillar 3 reports since 2008 (http://www.db.com/ir/en/content/reports_2010.htm). Further information can be found in Basel Committee on Banking Supervision (2006, pp. 226–42) and the pertinent EU regulation (DIRECTIVE 2006/48/EC). It is anticipated that Pillar 3 disclosure requirements will be amended (beyond disclosure on remuneration). U.S. banks do not (yet) publish Pillar 3 reports.

⁴ See <http://stress-test.eba.europa.eu/> for further information.

Satellite models explore statistical regularities between macroeconomic variables and the quality of bank assets or income accounts. In this way, balance sheet items, including capital, can be projected under stress scenarios. An alternative approach is taken by portfolio models that use valuation techniques to project balance sheet items under stress scenarios (defined by the properties of the multivariate distribution of price changes).

The focus of these approaches is the simulation of the stressed macroeconomic environment as a set of correlated changes in prices that are used to reprice all balance and off-balance-sheet transactions. Because these models target valuation, which is the core source of most changes in bank capital, they are good at integrating in a straightforward manner different kinds of risks, such as market and credit risks.

- In Chapter 8, Barnhill and Souto extend the basic portfolio approach presented in Barnhill, Papapanagiotou, and Schumacher (2002) to a set of institutions operating in the same financial environment. For illustration, they apply the approach to a set of Brazilian banks to assess systemic risks coming from different channels of interconnection, such as a common exposure to changes in prices and claims in the inter-bank market.
- Using the same portfolio approach and following Aikman and others (2009), Barnhill and Schumacher (Chapter 9) integrate systemic solvency and liquidity risks, including their simultaneous feedback, by simulating the market response to deteriorating bank asset quality (funding liquidity) as well as the impact of asset sales (to address liquidity demands) on asset values (market liquidity). The framework is applied to a set of stylized U.S. banks using publicly available information.
- Separately, Avezani and others illustrate an alternative portfolio approach using an actuarial model in Chapter 10. This application of the CreditRisk+ model is based on the framework developed by Credit Suisse Financial Products.

Prior to the crisis, the stress tests developed by IMF staff to date had largely focused on shocks to and within a particular financial system. However, the global financial crisis underscored the importance of anticipating cross-border shocks in an increasingly interconnected world. Among those who tried to address this gap, Cerutti and others (Chapter 11) com-

bine satellite modeling with balance sheet-based analysis to estimate the impact of potential ring-fencing on international banks (see also IMF staff work in the areas of network analysis and extreme value theory later in the book).

CONCLUSION

Notwithstanding the recent welcome developments in the balance sheet approach to stress testing, challenges remain. Frameworks need to be made gradually (even) more comprehensive and risk sensitive (e.g., by accounting for the evolution of business and/or risk-management techniques), while keeping them tractable. The modeling of macro-financial linkages and interconnectedness needs to be extended to capture all channels of risk transmission as well as the links between solvency, liquidity, and contagion risks. Most important, data need to be made more available to ensure reliability and to allow for up-to-date, preferably forward-looking, analysis. Finally, stress tests need to integrate the universe of financial institutions operating in a common financial environment. An example of the extension of stress testing techniques to non-banks is provided by Impavido in Chapter 12, representing a nascent area of work at the IMF.

REFERENCES

- Aikman, David, Piergiorgio Alessandri, Bruno Eklund, Prasanna Gai, Sujit Kapadia, Elizabeth Martin, Nada Mora, Gabriel Sterne, and Matthew Willison, 2009, "Funding Liquidity Risk in a Quantitative Model of Systemic Stability," Working Paper No. 372 (London: Bank of England). Available via the Internet: <http://www.bankofengland.co.uk/publications/Pages/workingpapers/2009/wp372.aspx>
- Barnhill, Theodore, Panagiotis Papapanagiotou, and Liliana Schumacher, 2002, "Measuring Integrated Market and Credit Risk in Bank Portfolios: An Application to a Set of Hypothetical Banks Operating in South Africa," *Financial Markets, Institutions and Instruments*, Vol. 11, No. 5, pp. 401–43.
- Basel Committee on Banking Supervision, 2006, "International Convergence of Capital Measurement and Capital Standards: A Revised Framework—Comprehensive Version" (Basel, June). Available via the Internet: <http://www.bis.org/publ/bcbs128.htm>
- Hardy, Daniel C., and Christian Schmieder, 2013, "Rules of Thumb for Bank Solvency Stress Testing," IMF Working Paper 13/232 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=41047.0>

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Stress Tester: A Toolkit for Bank-by-Bank Analysis with Accounting Data

MARTIN ČIHÁK

Stress testing is a useful and increasingly popular, yet sometimes misunderstood, method of analyzing the resilience of financial systems to adverse events. This chapter aims to help demystify stress tests and illustrate their strengths and weaknesses. Using an Excel-based exercise with institution-by-institution data, readers are walked through stress testing for credit risk, interest rate and exchange rate risks, liquidity risk, and contagion risk and are guided in the design of stress testing scenarios. The chapter also describes the links between stress testing and other analytical tools, such as financial soundness indicators and supervisory early-warning systems. Furthermore, it includes surveys of stress testing practices in central banks and the IMF.

METHOD SUMMARY

Overview	The Stress Tester 3.0 is a basic yet flexible model for implementing bank-by-bank stress tests for a set of risks. It covers basic solvency and liquidity risks, contagion, reverse stress testing, and links between stress tests and early-warning systems.
Application	The method can be applied to a wide variety of banking systems but is particularly appropriate for small, noncomplex banking systems.
Nature of approach	Basic balance sheet shocks, combined with basic contagion analysis.
Data requirements	Accounting information on capital, loans, and risk-weighted assets. Supervisory data on classified loans and provisions.
Strengths	Easy to use, flexible (institutions, shocks, assumptions can be easily added), transparent (all assumptions are in one sheet), and covers a relatively wide range of risks.
Weaknesses	Numerous assumptions are required. The mechanism for generating the shocks (e.g., historical scenario and macroeconomic model) is left unspecified. Reliability and comparability of results depend on quality of input data (which includes, e.g., accounting data on nonperforming loans and provisions).
Tool	The Excel spreadsheet macro (Stress Tester 3.0) is available in the toolkit, which is on the companion CD and at www.elibrary.imf.org/stress-test-toolkit .

The subject of this chapter and the accompanying Excel file are system-oriented stress tests carried out on bank-by-bank data.¹ The chapter and the accompanying file aim to illustrate

strengths and weaknesses of stress testing, using concrete examples. Readers will familiarize themselves with how common types of stress tests can be implemented in practice, in

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¹ We concentrate on banks, even though the impact of risks in nonbank financial institutions is also discussed.

a small and noncomplex banking system. They should gain an understanding of how the various potential shocks can be fitted together and how stress testing complements other analytical tools, such as financial soundness indicators (FSIs) and supervisory early-warning systems. They should also learn how to interpret the results of stress tests.

As the title indicates, this chapter is about *applying* stress tests to actual data. It devotes relatively little space to general discussions of what stress testing is. There is already a plethora of studies that provide a general introduction to stress testing, discussing its nature and purpose. For example, Blaschke and others (2001), Jones, Hilbers, and Slack (2004), Čihák (2004, 2005), and IMF and World Bank (2005) provide a general introduction to stress testing. In contrast, relatively little is available in terms of practical technical guidance on how to actually implement stress tests for financial systems, using concrete data as examples. This study and the accompanying Excel file are an attempt to fill that gap.

As the title also suggests, this is an introduction, not a comprehensive “stress testing cookbook.” The chapter covers basic versions of the most common stress tests. Depending on the sophistication of the financial system and the type of its exposures, it may be necessary to elaborate on these basic versions of tests (e.g., by estimating econometrically some relationships that are only assumed in this file) and perhaps include also tests for other risks (e.g., asset price risks or commodity risks) if financial institutions have material exposures to those risks. The accompanying file can be developed in a modular fashion, with additional modules capturing additional risks or elaborating on the existing ones. The final part of this chapter provides an overview of the main extensions that could be considered.

One of the key messages of this chapter is that assumptions matter and that they particularly matter in stress testing. The chapter calls for transparency in presenting stress testing assumptions and for assessing robustness of results to the assumptions. **To highlight the importance of assumptions in stress testing, this chapter uses boldface for references to assumptions used in the accompanying Excel file.** To achieve transparency in the Excel file itself, assumptions are highlighted (in blue and green) and grouped in one worksheet.

The remainder of this chapter is structured as follows. Section 1 provides an overview of the general issues one needs to address when carrying out stress tests and describes the design of the accompanying Excel file. It also introduces the general setting of the fictional economy described by the Excel file. Section 2 discusses the input data. Sections 3–7 discuss the stress tests for the individual risk factors, namely, credit risk (Section 3), interest rate risk (Section 4), foreign exchange risk (Section 5), interbank contagion risk (Section 6), and liquidity risk (Section 7). Section 8 shows how to create scenarios from the individual risk factors, how to present the results, and how to link the results to supervisory early-warning systems. Section 9 concludes and discusses possible extensions.

The accompanying Excel file, “Stress Tester 3.0.xls,” constitutes an essential part of this chapter. For each of the concepts

introduced in the subsequent sections, we include specific references to the file. To distinguish references to tables in the Excel file from those in this chapter, the Excel table names start with a capital letter A–H (denoting the order of the spreadsheet) followed by a number (denoting the order of the table within the spreadsheet). For instance, Table A2 denotes the second table in the first Excel spreadsheet.

1. OVERVIEW OF THE FILE AND OF THE STRESS TESTING PROCESS

Stress testing can be thought of as a process that includes (1) identifying specific vulnerabilities or areas of concern; (2) constructing a scenario; (3) mapping the outputs of the scenario into a form that is usable for an analysis of financial institutions’ balance sheets and income statements; (4) performing the numerical analysis; (5) considering any second-round effects; and (6) summarizing and interpreting the results (Jones, Hilbers, and Slack, 2004; IMF and World Bank, 2005). The aim of this exercise is to illustrate the stages of this process. It will also illustrate that these stages are not necessarily sequential, as some modification or review of each component of the process may be needed as work progresses.

The stress testing exercise is performed on the banking system in a fictional country named Bankistan. Given that confidentiality restrictions do not allow the IMF to pass on individual bank data, we refer to a fictional country rather than an actual one. The exercise is modeled on stress tests conducted in a number of Financial Sector Assessment Program (FSAP) missions, and the input data were created to make the exercise realistic. However, compared with typical FSAP stress tests, the exercise was simplified significantly to make it suitable for a short workshop that would give an overview of the FSAP stress tests (see IMF and World Bank, 2003). It is a version that could be used for a noncomplex banking system. For larger systems characterized by complex financial institutions and markets, more elaborate tests may be necessary.

To understand how to design stress testing shocks and scenarios, it is important to have a good understanding of the structure of the financial system and the overall environment in which the system operates. Box 3.1 therefore provides short briefing information on the macroeconomic and macroprudential situation in Bankistan.

A. How to operate Stress Tester 3.0: A quick guide to the accompanying file

This section introduces the accompanying Excel file, “Stress Tester 3.0.xls.” Working with the file requires relatively little prior knowledge. Users should be proficient in operating standard Excel files. Knowledge of intermediate macroeconomics is useful for understanding the linkages between the financial sector and the broader macroeconomic framework.

The file can be viewed as a module belonging to a broader stress testing framework (Figure 3.1). Such a framework would

Box 3.1. Background Information on Bankistan's Economy and Banking Sector

The economic environment in which banks are operating in Bankistan is challenging, with increasing macroeconomic imbalances, inappropriate macroeconomic policies, and deep uncertainty fueled by political tensions. Real activity is sharply contracting, and inflation has almost doubled to 65 percent.

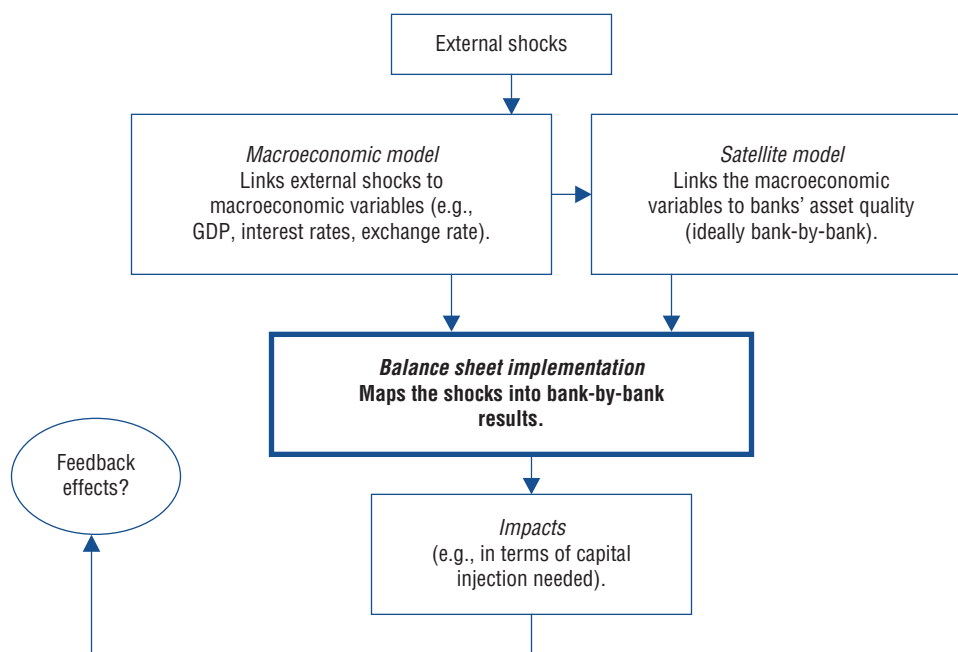
Unsustainable fiscal imbalances and loose monetary conditions were key to the deteriorating situation in Bankistan. The government deficit more than doubled in 2005, and a sharp increase in central bank financing of the government has significantly accelerated money growth.

The policy response to the deteriorating situation has been inappropriate. Expansionary monetary policy measures (e.g., a lowering of reserve requirements) have induced a further easing of liquidity conditions. The ensuing excess liquidity induced a drop in treasury bill rates from 60 percent to below 15 percent. This means that together with an inflation rate of 65 percent, real interest rates are sharply negative. (Note: This is used for assessing interest rate risk.)

The official exchange rate of the Bankistan currency, Bankistan dollar (B\$), is fixed at 55 B\$/US\$. However, the black market exchange rate has depreciated in recent months from about 60 B\$/US\$ to about 85 B\$/US\$. (Note: This is important information for assessing foreign exchange risk.)

The deteriorating macroeconomic environment has put considerable strain on the financial condition of the banking system. Even though the system has proved so far to be remarkably resilient, some banks have been weakened considerably and are prone to further deterioration in light of the significant risks. Reported high-capital adequacy ratios were found to be overstated because of insufficient provisioning. (Note: This information is used for the assessment of asset quality.) In addition, asset quality has deteriorated. The ratio of gross nonperforming loans (NPLs) to total loans has increased from 15 percent at end-2009 to 20 percent at end-2010. (Note: This information will be used for assessing credit risk.)

The banking system of Bankistan consists of 12 banks. Three of them are state owned (with code names SB1 to SB3), 5 are domestic privately owned banks (DB1 to DB5), and 4 are foreign owned (FB1 to FB4). The banking system, and particularly the state-owned banks, have been plagued by a large stock of NPLs and weak provisioning practices. Data on the structure and performance of the 12 banks are provided in the "Data" sheet of the accompanying Excel file. An assessment of Bankistan's compliance with the Basel Core Principles for Banking Supervision (BCP) suggests that even though existing loan classification and provisioning rules in Bankistan are broadly adequate, they are not well implemented in practice, and banks are underprovisioned.



Source: Author.

Figure 3.1 Stress Testing Framework

typically include a model characterizing linkages among key macroeconomic variables, such as GDP, interest rates, the exchange rate, and other variables. Medium-scale macroeconomic models (e.g., those used by a central bank for

macroeconomic forecasts) including dozens of estimated or calibrated relationships are often used for this purpose (if such models are not available, vector autoregression or vector error-correction models can be estimated). Given that such

TABLE 3.1

Stress Tester 3.0: Description of the Worksheets

Worksheet	Description
Read Me	Basic information, acknowledgments, and explanation of the workbook.
Data	Six tables. Input data as compiled by the National Bank of Bankistan (NBB). The data were collected in March 2011 and generally relate to end-December 2010, unless noted otherwise. Table A1 contains basic financial statement data. Table A2 contains other relevant data, including more detailed breakdowns, results of bottom-up stress tests reported by individual banks, and GDP number (for calculating ratios). The next two tables include key ratios based on the input data. Specifically, Table A3 contains the financial soundness indicators, and Table A4 characterizes the structure of the banking sector. The following two tables show how the financial soundness indicators can be combined into institution-by-institution rankings, using a simple early warning system calibrated by the NBB (see the Assumptions sheet). Specifically, Table A5 provides the rankings, and Table A6 converts them into probabilities of default.
Assumptions	One table. Table B puts together all the assumptions. This worksheet also contains several charts allowing the user to see how changes in the assumptions affect the results.
Credit Risk	Two tables. Table C1 summarizes the reported data on asset quality. Table C2 shows the credit risk stress test. It consists of four components: (i) a correction for underprovisioning of NPLs; (ii) an aggregate NPL shock; (iii) a sectoral shock, allowing different shocks to different sectors; and (iv) a shock for credit concentration risk (large exposures).
Interest Risk	Two tables. Table D1 sorts assets and liabilities into three time-to-repricing buckets, using the input data provided by the NBB. Table D2 shows the corresponding interest rate stress test. The test itself consists of two components: (i) flow impact from a gap between interest-sensitive assets and liabilities; and (ii) stock impact resulting from the repricing of bonds.
FX Risk	Two tables. Table E1 contains information on the foreign exchange exposure of the banks and the direct exchange rate risk shock. Table E2 shows a basic calculation of the indirect foreign exchange shock (using FX loans to approximate impact on credit quality).
Interbank	Three tables. Table F1 is a matrix of net interbank exposures. Table F2 uses the interbank exposure data to show “pure” interbank contagion, i.e., to illustrate what happens to the other banks when one bank fails to repay its obligations in the interbank market. Table F3 shows a “macro” contagion exercise, in which banks’ failures to repay obligations in the interbank market are not assumed, but rather a result of the “macro” shocks modeled in the sheet “Scenarios.”

models generally do not include financial sector variables, the stress testing framework can also include a “satellite model” that maps (a subset of) the macroeconomic variables into financial sector variables, in particular asset quality. Such a satellite model can be built on data on individual banks over a period of time: using panel data techniques, asset quality in individual banks can be explained as a function of individual bank variables and system-level variables. Together with the macroeconomic model, the satellite model can be used to map assumed external shocks (e.g., a slowdown in world GDP) into bank-by-bank asset quality shocks.

We focus on calculating the bank-by-bank impacts resulting from external shocks and expressing the impacts in terms of variables such as capital adequacy or capital injection as a percentage of GDP. We spend relatively little time discussing the broader macroeconomic framework or possible feedback effects to the broader economy. The cells that contain sizes of shocks and numerical assumptions in this file can be thought of as interfaces between this module and the other modules of the stress testing framework. The cells that contain results (e.g., capital injections as a percentage of GDP) can be viewed as interfaces to a module analyzing the feedback effects.

The modular design has the advantage that as a user becomes more experienced in stress testing or as more data become available for analysis, additional modules can be added or developed. Incorporating, for example, the underlying econometric calculations in Excel would make the resulting file too big and unwieldy (it also may not be possible because the econometric tools in Excel are more limited than in other packages).

All data in the file relate to end-2010, unless indicated otherwise, and are expressed in millions of Bankistan dollars (B\$), except for ratios (shown in percent).

The file contains the following eleven worksheets: Read Me, Data, Assumptions, Credit Risk, Interest Risk, Foreign Exchange (FX) Risk, Interbank, Liquidity, Scenarios, Reverse, and Bottom Up. Table 3.1 contains a description of the individual worksheets.

To differentiate the various types of cells (input data, numerical assumptions, and formulas), different colors are used in the file. The following color coding is used:

- *Yellow* denotes data reported by the National Bank of Bankistan (NBB). The yellow cells are found only in the Data worksheet. When a new set of data arrives from the NBB, the contents of the yellow cells should be replaced with the new data, and all the results are recalculated automatically.
- *Blue* denotes the assumed sizes of the shocks to risk factors, for example, an increase in interest rates. The blue cells are found only in the Assumptions worksheet. The users can change the values of these factors in the Assumptions worksheet and observe the impact of these changes on the stress test results.
- *Green* denotes numerical assumptions (parameters) of the stress test. Like the blue cells, the green cells are found only in the Assumptions worksheet. As with the blue cells, the users can change the values of these factors in the Assumptions worksheet and observe the impact of these changes on the stress test results.

- *No background.* Cells that have no background and generally normal black font contain formulas linked to the yellow, green, and blue cells. If the values of the blue or green cells are changed (or if new input data are entered in the yellow cells), the results of the stress tests are recalculated automatically.
- *Yellow stripes* indicate consistency checks. These cells contain sums or other functions of the input data. Those would normally also come from the authorities as hard numbers but are calculated in this file to avoid inconsistencies.
- *Green/white stripes* denote numerical assumptions imported from the Assumptions sheet. Under normal circumstances, it is expected that the user will leave these cells (each of which contains a link to the Assumptions sheet) unchanged and will carry out the changes in the corresponding green cells in the Assumptions sheet. However, if users want to see the impact of changes in an assumption directly in the corresponding sheet (e.g., for credit risk assumptions in the Credit Risk worksheet), they can do so by changing the value of the green/white cell rather than going back to the Assumptions sheet. Of course, users need to be aware that if they save these changes, it will result in overwriting the original links in the file and some of the links between the Assumptions sheet and the results in the Scenarios sheet may be broken. However, if they do not save those changes and afterward return to the original template, they will be able to use the file again in its original form.
- *Blue/white stripes* denote numerical assumptions imported from the Assumptions sheet. Like the green/white stripe cells, these cells allow the user to change the values of assumed shocks directly in the individual worksheets without going back to the Assumptions sheet. However, these changes should be used with caution to avoid breaking the links in the file; changes in the shock sizes should primarily be done in the Assumptions sheet.

In addition to explanations in the Read Me sheet, many of the cells in the stress testing file contain comments explaining the calculations carried out in these cells.

All assumptions and shock parameters are in the Assumptions sheet. This is the sheet that a regular user would use the most, changing the assumptions (in green) and shock sizes (in blue) and observing the results. Given that a summary presentation of the stress test results is provided in the Assumptions sheet in charts, the user can change the assumptions and directly see the impacts in a graphical form. If the user wants to examine the overall stress test results in a tabular form, the results are available in the Scenarios and other sheets.

Expert users who have become familiar with the file are invited to suggest improvements in the file or to develop the file further themselves. Such developments can include new types of risks, making the modeling of the existing risks more

realistic or including more institutions in the system.² Not all the developments need to take place in the same file: users can think about some of the blue or green cells as interfaces between this tool (module) and other tools (modules), such as macroeconomic models that provide scenarios. Those can provide inputs that feed into this stress testing tool. The main advantage of Excel-based tools, such as this one, is the relative ease with which they can be adapted and extended. For longer-term usage, it may be useful to develop the file into a program, for example, in Microsoft Access. This may reduce the flexibility for regular users, but it may, among other things, allow development of the file from the current one-period snapshot to a multiperiod framework.

B. Top-down or bottom-up?

There are two main approaches to translating macroeconomic shocks and scenarios into financial sector variables: the “bottom-up” approach, where the impact is estimated using data on individual portfolios, and the “top-down” approach, where the impact is estimated using aggregated data.³ Among central banks’ Financial Stability Reports (FSRs), reports by the Bank of England and Norges Bank can be used as examples of FSRs that rely more on top-down approaches to stress testing, whereas reports by the Austrian National Bank and Czech National Bank are examples of stress tests using more bottom-up approaches, even though in all these cases, the reports in fact combine elements from both approaches.

The disadvantage of a top-down approach is that applying the tests only to aggregated data could overlook the concentration of exposures at the level of individual institutions and linkages among the institutions. This approach may therefore overlook the risk that failures in a few weak institutions can spread to the rest of the system. The bottom-up approach should be able to capture the concentration of risks and contagion and therefore should lead generally to more precise results, but it may be hampered by insufficient data and by calculation complexities. Having detailed information on exposures of individual banks to individual borrowers should in principle lead to more accurate results than using more aggregated data, but, especially for large and complex financial systems, it may lead to insurmountable computational problems. Most macroprudential stress tests therefore try to combine the advantages and minimize the disadvantages of the bottom-up and top-down approaches.

This chapter and the accompanying Excel file focus on the bottom-up implementation of stress tests in a relatively small, noncomplex banking system. The spreadsheet illustrates why

² Including more institutions requires adding columns (and some rows in the “Interbank” worksheet), copying the relevant links in other worksheets, and checking summation formulas for the peer groups and the system.

³ For a longer discussion of this distinction, see, e.g., Čihák (2004), Jones, Hilbers, and Slack (2004), and IMF and World Bank (2005).

using institution-by-institution data is important: a relatively minor change in the distribution of risks among banks can result in substantial changes in the overall impacts. The spreadsheet also illustrates how the bottom-up approach can be complemented by a top-down approach. For example, a model estimated on aggregate data can be used to identify how a combination of shocks to macroeconomic variables can translate into an increase in nonperforming loans. Stress Tester 3.0 can then be used to calculate how this aggregate impact influences individual banks and the system as a whole.

Although much of the Excel file illustrates a “centralized” approach to bank-by-bank stress testing, where all the calculations are done in one center (e.g., at the central bank or a supervisory agency or by an IMF expert), the Bottom Up worksheet illustrates an alternative, more “decentralized” approach. The approach often used in advanced-economy FSAPs is to involve banks themselves in carrying out the stress testing calculations. The advantage of the “decentralized” approach is that it can provide a richer, more detailed modeling, using a wider set of data and leveraging on the expertise and calculation capacity of banks’ risk management.⁴ The disadvantage is that the “decentralized” calculations may not sufficiently reflect contagion effects among banks, so just adding up the results for individual banks may not be a good proxy for the systemic impacts. Also, if the calculations are complex and done by many institutions, it may be a major challenge to ensure that all banks implement the assumed shocks or scenarios in a consistent fashion. In comparison, the “centralized” stress tests discussed in this chapter are less refined (for reasons of computational complexity and data availability), but they (1) are more focused on linkages to macroeconomic factors; (2) can better integrate credit and market risks; (3) are implemented consistently across institutions; (4) can better analyze correlation across institutions (interportfolio correlation); and (5) can analyze network effects (contagion). Therefore, even if the “decentralized” calculations are carried out, it is important to complement them with the “centralized” calculations along the lines discussed in this chapter.

C. Presenting stress test results: What variables can be stressed?

For a variable to be used to measure the impacts of the stress tests, it should have two key properties: (1) it should be possible to interpret the variable as a measure of financial soundness of the system in question; and (2) it can be credibly linked to the risk factors. Of the various variables that have been used in the literature so far, each has advantages and disadvantages, which should be clear to the reader and user. Here is a list of the commonly used variables:

- *Capital.* Using capital as a measure of impact has a clear motivation. If a risk has a material impact on solvency, it has an impact on capital. Also, commercial banks’ capital is part of the Other Items Net in monetary surveys, so expressing impacts in terms of capital could be used to directly link stress tests to other parts of the financial programming framework (even though it is only a small part of the possible feedback effects from the financial sector to the macroeconomy, as discussed in Section 8.C). The disadvantage of using impact in terms of capital is that it is just a number in B\$ that needs to be compared with something else to give the reader an idea about the impact on soundness (e.g., dividing it by risk-weighted assets) and on the macroeconomic framework (e.g., dividing it by GDP). Nonetheless, it is a key measure, and the accompanying Excel file illustrates the presentation of stress testing impact in terms of capital.
- *Capitalization.* The advantage of capitalization measures (capital or equity to assets or capital to risk-weighted assets) is that capital adequacy is a commonly recognized soundness indicator. Compared with capital, this measure is scaled, so it allows comparison among institutions of different size. For this reason, the accompanying Excel file uses capitalization as a key indicator of impact. The disadvantage of this measure is that a change in capitalization does not by itself indicate macroeconomic relevance of the calculated impacts. It therefore needs to be accompanied by other measures.
- *Capital injection needed* (e.g., as a percentage of GDP). This indicator provides a direct link to the macroeconomy. It provides an upper bound on the potential fiscal costs of bank failures associated with the assumed stressful scenario. The accompanying Excel file illustrates the use of capital injection.
- *Profits.* In a normal, nonstressful situation (“baseline scenario”), banks would typically create profits. When carrying out stress tests, it is important to bear in mind that we are evaluating impacts against such a baseline, as banks would normally use profits as the first line of defense before dipping into capital. Expressing shocks only in terms of capital may result in overestimating the actual impacts if banks were profitable in the baseline. The accompanying Excel file allows profits to be taken into account. More specifically, it indicates the “profit buffer” that banks would have available in the baseline. The profit buffer is based on the average annual profits over the last 10 years, but the file also allows the user to run a separate “test” for the impact of an autonomous shock affecting profits (one can think about autonomous shocks to net interest income, e.g., related to an increase in competition from abroad; alternatively, this can be viewed as a measure of the risks not reflected in the other shocks specified in the stress testing scenario). However, to

⁴ Čihák and Heřmánek (2005) provide a more detailed discussion of such “decentralized” stress tests. Hoggarth, Logan, and Zicchino (2005) discuss how such stress tests were carried out in the case of the United Kingdom.

reflect the views of some observers that it is better to be prudent and disregard profits, it shows them as a separate item rather than directly deducting the impacts from profits.

- *Profitability* (return on equity, assets, or risk-weighted assets). Compared with profits, these measures are scaled by bank size, thus allowing comparison among banks of different size. The accompanying Excel file shows the profit buffers as ratios to the risk-weighted assets, to make them comparable with the impact shown in terms of risk-weighted assets.
- *Net interest income and other components of profits*. Sometimes it can be useful to stress test separately individual components of profits. For example, net interest income is likely to have a more direct relationship to interest rates and may therefore be more amenable to econometric analysis. However, such an approach provides only a partial picture of the economic value of a bank and its resilience to adverse events.
- *z-scores*. The *z*-score has become a popular measure of bank soundness (e.g., Boyd and Runkle, 1993; or Hesse and Čihák, 2007). Its popularity stems from the fact that it is related to the probability of a bank's insolvency, that is, the probability that the value of its assets becomes lower than the value of the debt. The *z*-score can be summarized as $z = (k + \mu) / \sigma$, where *k* is equity capital as percentage of assets, μ is average aftertax return as percentage on assets, and σ is standard deviation of the aftertax return on assets, as a proxy for return volatility. The *z*-score measures the number of standard deviations a return realization has to fall in order to deplete equity, assuming normality of returns. A higher *z*-score therefore implies a lower probability of insolvency risk. The accompanying Excel file illustrates this presentation.⁵
- *Loan losses*. Stress tests presented by staff from Norges Bank (e.g., Evjen and others, 2005) and the Bank of England (e.g., Bunn, Cunningham, and Drehmann, 2005) are just two examples of presenting stress testing results in terms of loan losses. Although this approach has its advantages (in particular, it is easier to implement in top-down calculations than measures calculating losses to capital), its drawback is that it does not take into account banks' buffers (profits and capital) against those losses. It may underestimate the overall impact if losses are concentrated in weak institutions.
- *Liquidity indicators*. For liquidity stress tests, the impacts have to be measured differently than for solvency

tests, namely, in terms of liquidity indicators. The accompanying Excel file illustrates this presentation.

- *Ratings and probabilities of default (PDs)*. Ratings and PDs provide a useful way of combining solvency and liquidity risks. By definition, ratings try to combine various solvency and liquidity risks into a single measure. We can use the system designed for ratings and see how changes in the various variables translate into changes in ratings. If we have a model linking ratings and probabilities of default, we can also calculate how a stressful scenario influences PDs. The accompanying Excel file illustrates this presentation.

This list is far from complete. It is possible to calculate and present stress test impacts in terms of other variables that capture soundness of financial institutions and can be credibly linked to the development of risk factors. For example, instead of the accounting-based data discussed earlier, it is possible to present the impact of a stressful scenario in terms of market-based indicators of financial sector soundness, such as relative prices of securities issued by financial securities, the distance to default for banks' stocks,⁶ or credit default swap premia (for a review, see, e.g., Čihák, 2007b). One of the advantages of the market-based indicators is that they are usually available on a much more frequent basis than accounting data. However, one of their major disadvantages is the absence in many countries of sufficiently deep markets from which such indicators can be derived (e.g., bank stocks are not traded, or the market for such stocks is illiquid). There have been some attempts to link market-based indicators, such as distance to default, to macroeconomic variables (see, e.g., IMF, 2005), but work on using such relationships for stress tests has been limited so far.

D. How are the results presented in Stress Tester 3.0?

Each stress test conducted in this exercise aims to address two main questions: (1) Which banks could withstand the assumed shocks and which ones would fail? (2) What are the associated potential costs for the government given the failure of banks in times of stress?

A commonly used approach to assessing question (1), as mentioned in the previous section, is to look at banks' capital adequacy ratio (CAR). According to the original Basel Agreement, a bank has to hold a minimum CAR, defined as total regulatory capital to risk-weighted assets (RWA), of 8 percent. In our example, given that Bankistan as an emerging market country faces more risks than an industrial country, it is appropriate to require from banks a higher CAR. On the basis of these considerations, Bankistan's supervisors use a minimum CAR of 10 percent. Whenever the CAR of a bank in Bankistan falls below 10 percent, its owners are obliged to inject capital

⁵ The file also shows the *z*-scores for the peer groups of banks, using a similar methodology as used by some authors to translate bank-by-bank distance to default measures to a "portfolio distance to default" (e.g., IMF, 2005a). These calculations need to be treated with a degree of caution, given that they overlook issues related to contagion (see, e.g., Čihák, 2007b).

⁶ Distance to default is in effect an implementation of the *z*-score for banks with stocks listed in liquid equity markets. Distance to default uses stock price data to estimate the volatility in the economic capital of the bank (see, e.g., Danmarks Nationalbank, 2004).

in order to stay in business.⁷ If they fail to do so, the bank will be closed and its banking license withdrawn. A CAR below 0 means that a bank has negative capital and is insolvent.⁸

Stress Tester 3.0 allows profits to be taken into account, which—as explained earlier—is important because shocks always take place over time and therefore need to be compared with a “baseline scenario.” The Scenarios sheet indicates the profit buffer that banks would have available in the baseline scenario. We refer to annual profits, which is consistent with the fact that we evaluate the shocks in a horizon of one year (see the interest rate shock). The value of the profit buffer is based on the average annual profits over the last 10 years, but the file also allows the user to run a separate “test” for the impact of an autonomous shock affecting profits or net interest income. Take, for instance, autonomous shocks to net interest income, for example, related to an increase in competition from abroad; alternatively, this can be viewed as a measure of the risks not reflected in the other shocks specified in the stress testing scenario. However, to reflect the views of some papers that it is more prudent to disregard profits and measure shocks directly against capital (e.g., Blaschke and others, 2001), the file shows the profit buffers as a separate item rather than directly deducting the impacts from profits. For some banks, the profit “buffer” is nonexistent or negative (they have been creating losses).

An assessment of question (2) requires consideration of the following question: if bank owners fail to inject new capital, how much capital would the government need to inject in order to bring the CAR up to 10 percent again? For state-owned banks, it is obvious that the government would have to inject capital to keep banks operating. For private-owned banks, asking this question assumes that the government has an implicit or explicit guarantee for the banking sector, which may or may not be the case. If it is, an answer to this question can be found from the following accounting relationship:

$$\frac{C+I}{RWA+qI} = \rho,$$

where C is the bank’s existing total regulatory capital, RWA are its existing risk-weighted assets, I is the capital injection, q is the percentage of the capital injection that is immediately used to increase risk-weighted assets, and ρ is the regulatory minimum CAR ($\rho = 10$ percent in the case of Bankistan).

From the above equation, we can express the necessary capital injection as follows:

$$I = \frac{\rho RWA - C}{1 - q\rho} \quad \text{if } C < \rho RWA;$$

$$= 0 \quad \text{otherwise}$$

⁷ The 10 percent minimum capital adequacy requirement is just an **assumption** in Stress Tester 3.0, contained in the B71 cell of the Assumptions worksheet. The reader can change it, for example, to 8 percent and see in the relevant chart in the Assumptions worksheet how the capital injection (in percentage of GDP) declines.

⁸ Total capital may differ in general from regulatory capital; in this workbook, we use for simplicity the same numbers, but the file is set up in a way that allows for differences between equity and regulatory capital.

If $q = 0$, that is, the capital injection is not used for an increase in RWA (at least immediately), and if we substitute for ρ the 10 percent value used in Bankistan, we can calculate the capital injection as $I = 0.1 \cdot RWA - C$. The values of the parameters ρ and q are assumed. The stress test file enables us to change the values of the two parameters in the appropriate green cells (B71 and B72 in the worksheet Assumptions). We can see that if ρ is lower than 10 percent, the necessary capital injection is lower, and vice versa. If RWA increase as a result of the increase in capital (i.e., if $q > 0$), the necessary capital injection is higher (but the impact of changes in q is generally rather small).

2. UNDERSTANDING AND ANALYZING THE INPUT DATA

The Data sheet summarizes the input data (Tables A1 and A2), as reported by the NBB. It also shows key ratios (Tables A3 and A4) and illustrates how these ratios can be used in an off-site supervisory assessment system (Tables A5 and A6).

A. Coverage of stress tests

In the Excel file, the stress tests cover all 12 commercial banks in the country. An overview of FSAP stress tests suggests that only a minority of FSAPs have covered all banks in a country. Most FSAPs have covered only a subsample of large banks, accounting for a substantial majority (generally 70–80 percent) of the banking system’s total assets. Including all banks rather than a subsample has the obvious advantage of being more comprehensive. For this reason, this approach is also often favored by supervisors, who are expected to supervise all institutions, not only the larger ones. However, for someone interested primarily in macroprudential issues (e.g., a central bank not involved in microprudential supervision), it may be sufficient to include only the systemically important institutions (that is why some authors, such as Jones, Hilbers, and Slack [2004] refer to this type of stress testing as “system-oriented stress testing” rather than “systemwide stress testing”). Excluding the other institutions may be practical for reasons of computational complexity.⁹

The Stress Tester illustrates a basic peer group analysis. To do that, the 12 banks in Bankistan are grouped into three peer groups according to their ownership: state-owned banks (SB1 to SB3), domestic privately owned banks (DB1 to DB5), and foreign-owned banks (FB1 to FB4). This is just one possible grouping. Depending on the analytical purpose, the

⁹ A practical complication with the system-oriented stress tests is the fact that the systemic relevance of an institution is established only after, not before, the stress tests. As a practical shortcut, many FSAPs and other authors use a measure of size (e.g., total assets) as a first approximation for systemic importance. It is a good proxy in most cases, but in some it can miss institutions that are small but have large potential for impact on other institutions, for example, through exposures in the interbank market. In Section VI, we provide a measure of systemic importance that takes the contagion effects into account.

banks in Bankistan can be grouped into other groups, for example, by their size (e.g., large, medium, and small) or financial performance (e.g., strong and weak).

In the Bankistan example, foreign banks are present only through locally incorporated subsidiaries. In practice, foreign banks may also be present through branches. The difference from a stress testing perspective is that branches typically do not have their own capital against which the impact of shocks could be shown (and comparing the impacts with the parent institution's capital would overlook risks faced by the same institution in other countries). However, if there are separately available data on assets, liabilities, incomes, expenses, and other input data indicated in the Data worksheet, most of the standard tests can be performed on the branches of foreign banks as well. The main practical difference, then, is that the impacts have to be expressed in terms of a different variable than capital or capital adequacy (e.g., in terms of profits).

Mirroring the practice of FSAPs and central banks' FSRs, this chapter focuses on stress tests for banks and banking systems. In most countries, banks tend to dominate the financial system and are key to assessing systemic risk. Some FSAPs and FSRs contained explicit stress tests of insurance companies and pension funds. Stress testing pension funds and insurance companies themselves can be a rather complex task. Some of the insurance sector risks, such as market risk, liquidity risk, or credit risk, can be modeled similarly to the banking stress tests. Modeling of other risks, especially some of those stemming from the liabilities side (e.g., catastrophe risk), is beyond the scope of this study.

The impact of failures in nonbank financial institutions can be assessed as part of the credit risk. The accompanying Excel file allows for two ways of incorporating such an analysis. First, it can be incorporated as part of the sectoral credit risk, with nonbank financial institutions as one of the sectors.

We can then run a basic stress test for what would happen if a certain percentage of loans to the nonbank financial sector became nonperforming. Second, it can be incorporated as part of the large exposures tests. If we have data on the largest exposures of banks to nonbank financial institutions, we can run a test on what would happen to banks' solvency should their largest counterparties in the nonbank financial sector fail.

B. Balance sheets, income statements, and other input data

Data availability is a key determinant of the quality of a stress test. The purpose of Tables A1 and A2 in the Data worksheet is to illustrate what data are typically needed for carrying out a set of stress tests. These are not the minimum requirements—it is possible to do very rudimentary stress tests with even less data, as discussed in Box 3.2. However, these are the types of data that one usually looks for when doing a basic stress test.

Throughout the sheet, as well as the file, the first data column shows aggregated data for the whole banking system, the next 3 columns show the aggregated data for the three peer groups of banks (state banks, private domestic banks, and foreign banks), and the remaining 12 columns show the data for the individual banks.

The input data presented in the top part of the Data worksheet (Table A1) form a set of balance sheets and income statements. To keep the exercise straightforward and transparent, the aggregated data and the peer group data can be calculated as sums of the bank-by-bank data. This means that we disregard interbank exposures for the time being. This assumption is relaxed later on, in the analysis of interbank contagion risk (Section 6).

Table A2 in the Data worksheet lists—again for the system in aggregate, for the peer groups, and for individual

Box 3.2. Stress Testing When Some Input Data Are Unavailable

It may happen that some of the input data in Stress Tester 3.0—such as those on nonperforming loans, net open positions, and time to repricing—are not available, for example, owing to weak reporting systems or legal restrictions on data sharing. A rudimentary version of the stress tests can still be performed with the basic financial statements, provided that they are available for a number of periods. Those tests can be based on the observed relationships among the risk factors, the various items of the income statement, and the balance sheet. For example, even when no data are available on repricing buckets of assets and liabilities and bond portfolios in banks, a rudimentary stress test for interest rate risk could be based on the net interest income on banks' income statements. In particular, past data on individual banks' net interest income over time can be regressed on interest rates and other potential variables to estimate how banks' net interest income responded to changes in interest rates, and the estimated slope coefficient(s) can be used to translate a change in interest rates into the impact in terms of profits (and potentially capital). Similarly, provisions for loan losses from the income statement can be regressed on the risk factors and other explanatory variables to analyze the impact on banks' profitability. If longtime series are not available to carry out such regressions, the slope coefficients can be calibrated based on expert information or experience from other countries.

Even if the individual items of the financial statements are not available, "reduced-form" stress tests can still be carried out if there are reliable time series data. For example, one needs just capital, asset, and return data on individual banks over time to calculate the z-score, as a proxy for individual bank soundness, discussed in Section 2.C. The z-scores for individual banks can be regressed on a range of macroeconomic variables (e.g., real GDP growth rate, interest rate, and exchange rate) and bank-level variables (e.g., asset size or loan-to-asset ratio). The slope coefficients from this regression can then be used to map a macroeconomic scenario into the z-scores, to approximate the impact of macroeconomic stress on individual bank soundness. (A similar approach can also be used with distance to default data or other market-based indicators of individual institution soundness.) The main challenge in this type of approach is how to aggregate the bank-by-bank soundness data into a systemwide indicator (an issue that is discussed in more detail in Čihák, 2007b).

banks—other key input data that are used for the stress test calculations. Such data are on the following: (1) regulatory capital and risk-weighted assets (to be able to express the stress test impacts in terms of capital adequacy); (2) asset quality and structure of lending by sector and by size of borrower; (3) provisioning and collateral (for the credit risk calculation); (4) structure of assets, liabilities, and off-balance-sheet items by time to repricing; (5) the structure of the bond portfolio (for the interest rate risk calculation); (6) net open positions in foreign exchange and lending in foreign currency (for the foreign exchange solvency risk calculation); (7) average profits and standard deviation of profits over time (to have a measure of “baseline” profitability); (8) liquidity structure of assets and liabilities (for liquidity risk calculation); and (9) bank-to-bank uncollateralized exposures, presented in matrix form (for the interbank solvency contagion risk).

Most of the data in Table A2 usually are available to bank supervisors through standard regulatory returns. However, several issues need to be mentioned:

- Stress tests analyze the economic position (*net worth of banks*). In principle, this should be aligned with the reported data on capital, but in practice, there may be important differences between the calculated economic net worth of a bank and the reported regulatory data on capital. This may be the case, for example, when some assets are overvalued in banks’ balance sheets or when regulators accept as capital some liabilities that, in fact, are not capital (e.g., some long-term loans). The person carrying out stress tests should first try to adjust the input data for such biases. In the file, we show one example of such adjustments, namely, when banks underprovision their nonperforming loans. Another example (not shown in the file) might be when banks do not mark to market some bonds that they are holding in their portfolios.
- The input data should reflect not only assets and liabilities but also *off-balance-sheet* positions. For example, the net open positions in foreign currency should reflect the delta equivalents of foreign exchange options.
- Data on *bank-to-bank exposures* may be difficult to collect in many countries. In those cases, approximate calculations using less data (e.g., only data on each bank’s exposure to the rest of the system as a whole) can be used to at least broadly assess the associated risks, even though such methods may result in overlooking some exposures in the system.
- For interest rate risk calculations, data on *time to repricing* are crucial. For example, from an interest rate risk perspective, a 20-year mortgage loan with an interest rate that can change every 6 months should be treated the same as a 6-month fixed-rate loan, not as a 20-year loan. However, getting data on time to repricing may be difficult to obtain in some cases. In many countries, banks report instead a breakdown of assets by maturity or by residual maturity. Although data on maturity or residual maturity are important

for analyzing liquidity, using them as proxy for time to repricing may lead to misleading results (typically overstating the interest rate risk).

Additional data may be needed for stress tests for other risks not covered in this file. For example, to carry out stress tests for equity price risk and commodity price risk, data on net open positions in equities and in commodities would be needed (the mechanics of the test would be similar to the direct foreign exchange solvency test). Also, breakdowns of assets and liabilities by residual maturity/time to repricing *and* currency would be needed to perform stress tests separately for foreign currency.

C. Indicators of financial sector soundness and structure

Table A3 contains a set of core FSIs and other important ratios characterizing Bankistan’s banking sector and its components. These ratios can be used to provide a summary picture of the soundness of the financial sector and its components (peer groups and individual banks) in Bankistan. For more on the definitions and compilation of FSIs, see IMF (2004).

Table A3 also includes, at the bottom, individual banks’ *z*-scores. The *z*-score is defined as $z = (k + \mu) / \sigma$, where *k* is equity capital as a percentage of assets, μ is average aftertax return as percentage on assets, and σ is standard deviation of the aftertax return on assets, as a proxy for return volatility. As mentioned earlier, *z*-scores have become popular as measures of bank soundness, because they are directly linked to the probability of a bank’s insolvency. The table also shows the *z*-scores for the peer groups of banks (for a discussion on the calculation of *z*-scores for groups of banks, see, e.g., Čihák, 2007b).

Table A4 characterizes the structure of the banking sector in Bankistan, showing the shares of the peer groups and the individual banks in total assets, loans, deposits, and capital. It shows that the share of foreign-owned banks is relatively high—about 55 percent in terms of assets and 83 percent in terms of capital (reflecting their higher capitalization). It also shows the total assets as a ratio to GDP, to indicate the size of the banking system relative to the Bankistan economy. The GDP figure is used again later, to put into macroeconomic perspective the capital injection needed to get all banks to comply with the minimum capital adequacy requirement.

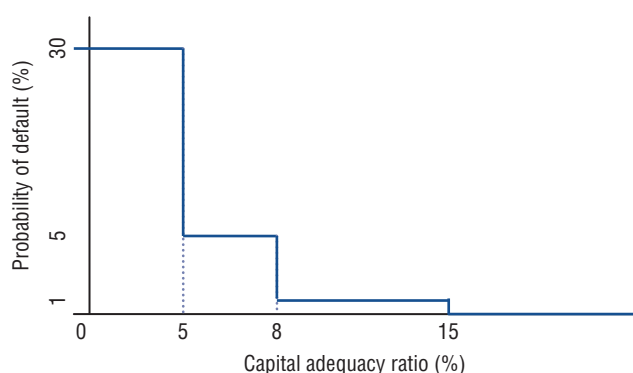
D. Ratings and probabilities of default

Tables A5 and A6 show how banking sector ratios can be combined into institution-by-institution rankings, using a supervisory early-warning system.¹⁰ Such systems (e.g., Sahajwala and Van den Bergh, 2000) are very common in supervisory

¹⁰ The early-warning system does not have to comprise only ratios. Inclusion of variables in the early-warning system should reflect their power to identify weak banks. One variable that in some cases has good discriminatory power is deposit rates: high deposit rates in a bank can indicate that it has difficulty retaining depositors, a potential sign of problems (Kraft and Galac, 2007).

agencies, and they are used typically for assessing soundness of banks in “baseline” conditions; we will show in this exercise that they can be also used for assessing soundness in stressful conditions. One caveat needs to be borne in mind, however, namely, that these early-warning systems typically treat each bank separately and do not look at contagion among banks—an issue that will be analyzed as part of the stress test exercise presented here.

Table A5 provides the rankings, based on the off-site supervisory ranking system of the NBB, characterized in rows 3–22 of the Assumptions sheet. The system has three thresholds (columns B, C, and D in the Assumptions sheet) for each indicator, determining a numerical ranking for each of the indicators (with 1 indicating the best ranking and 4 the worst ranking). The rankings for the individual variables are weighted (using weights established by the NBB and provided



Source: Author, based on the default settings in the Assumptions sheet.

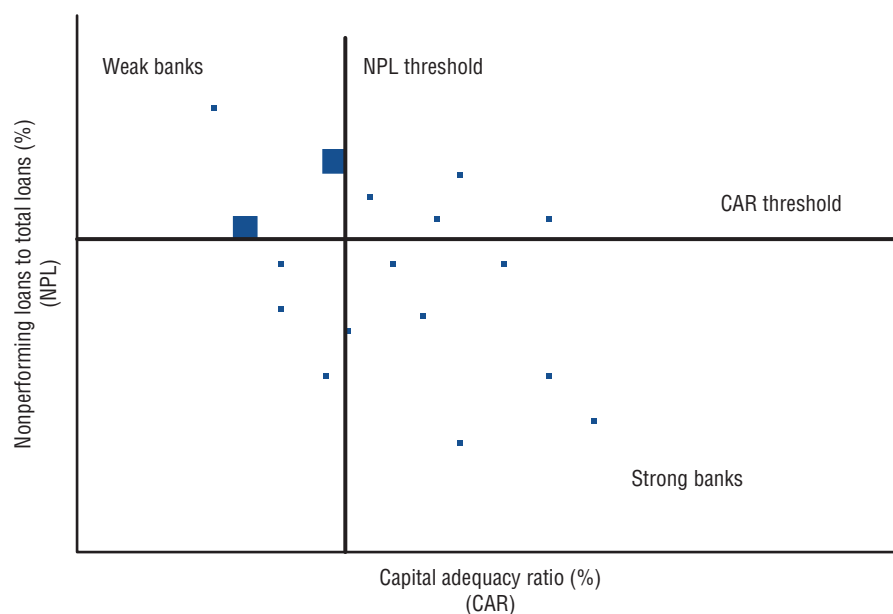
Figure 3.2 Step Function (Example)

in column E of the Assumptions worksheet) to derive an overall ranking for a bank. Similar off-site ranking systems are used by supervisors to identify banks that deserve increased attention (Sahajwala and Van den Bergh, 2000).

Underlying each ranking system is a more or less explicit link to probability of default (or probability of technical insolvency, that is, probability that the capital adequacy ratio declines below the regulatory minimum). Table A6 illustrates this by converting the rankings into probabilities of default, using a “step function” illustrated in Figure 3.2 and described in the case of Bankistan rows 6 and 22 of the Assumptions worksheet: according to NBB’s estimates, a bank with a rating of 1 has a 0.1 percent probability of default in a given year; a bank rated 2 fails with a 1 percent probability; a bank rated 3 has a 5 percent probability of default, and a bank rated 4 has a 30 percent probability of default in the coming year.

How are the parameters of such step functions derived? In some cases, they are based on expert estimates. In other cases, central banks or supervisory agencies attempt to “back-test” such systems to see whether they actually identify institutions that fail (or institutions that need interventions). When the step function is reestimated, the new parameters should be entered in row 22.

Figure 3.3 provides an illustration of such “back-testing” in the case of two variables (capital adequacy and gross nonperforming loans to total loans) and one threshold per variable. The dots represent observations of banks, and the two bigger boxes indicate two banks that have actually failed. The supervisory ranking system attempts to single out the failed banks (i.e., the bigger boxes), minimizing the signal-to-noise ratio for the estimate. If we want to capture all failed banks in Figure 3.3 (i.e., eliminate Type I errors), the early-warning



Source: Author.

Figure 3.3 Back-Testing a Supervisory Early-Warning System (example)

Box 3.3. Linking Credit Risk and Macroeconomic Models

A number of papers have attempted to link credit risk to macroeconomic variables using econometric models. For example, Pesola (2005) presents an econometric study of macroeconomic determinants of credit risk and other sources of banking fragility and distress in the Nordic countries, Belgium, Germany, Greece, Spain, and the United Kingdom from the early 1980s to 2002. An even broader cross-country analysis is presented in IMF (2003). For Austria, Boss (2002) and Boss and others (2004) provide estimates of the relationship between macroeconomic variables and credit risk. For Finland, Virolainen (2004) develops a macroeconomic credit risk model, estimating the probability of default in various industries as a function of a range of macroeconomic variables. For Norway, the Norges Bank has single-equation models for household debt and house prices and a model of corporate bankruptcies based on annual accounts for all Norwegian enterprises (Eklund, Larsen, and Berghardsen, 2003). For Hong Kong SAR, several studies are available on the topic, employing both single-equation aggregate estimates and panels using bank-by-bank data (Shu, 2002; Peng and others, 2003; and Gerlach, Peng, and Shu, 2004). For the Czech Republic, Babouček and Jančar (2005) estimate a vector autoregression model with nonperforming loans and a set of macroeconomic variables.

Similar models are also common in FSAP missions. For example, the technical note from the Spain FSAP includes an estimate of a regression explaining nonperforming loans on an aggregate level with financial sector indicators and a set of macroeconomic indicators (IMF, 2006).

Several issues need to be considered when interpreting the macroeconomic models of credit risk. In particular, the literature is dominated by linear statistical models. The linear approximation may be reasonable when shocks are small, but nonlinearities are likely to be important for large shocks: doubling the size of the shock may more than double its impact. Indeed, microlevel credit risk models often find a nonlinear relationship between the scale of shocks and the likelihood of default; for macroeconomic shocks, Drehmann (2005) also reports a nonlinear link to credit risk. Moreover, the models are subject to the Lucas critique (Lucas, 1976), because their parameters or functional forms may become unstable, especially if exposed to a major stress. As an extreme example, when considering a scenario that involves depegging in a country with a currency board regime, models estimated on past data are likely to say very little about the impact of the exchange rate change on credit risk. In such a situation, other approaches, such as calibration that uses parameters based on experience from other countries, may be more appropriate.

system characterized by the CAR threshold and the NPL threshold shown in the figure allows a decrease in the percentage of banks misclassified as failures (i.e., Type II errors) from 88 percent (15/17, if we do not have any prior information) to 33 percent (1/3, for the “northwest” subset identified by the two thresholds). That is a major improvement in forecast precision.

3. CREDIT RISK

Lending is the core of the traditional banking business. In most banking systems, credit risk is the key type of risk. At the same time, it is the type of risk where existing models are most in need of strengthening.

There are three basic groups of approaches to modeling credit risk as part of stress tests. First, there are mechanical approaches (typically used if there are insufficient data or if shocks are different from past ones). Second, there are approaches based on loan performance data (e.g., probabilities of default, losses given default, nonperforming loans, and provisions) and regressions (e.g., single equation, structural, and vector auto regression). Third, there are approaches based on corporate sector data (e.g., leverage or interest coverage) and possibly on household sector data (even though such data are typically much more difficult to collect than corporate sector data).

The exposition in this section and in the accompanying file starts with the basic mechanical approaches. We then present how these can be extended into more realistic approaches. In particular, Box 3.3 discusses the links between credit risk and macroeconomic risk.

The Credit Risk worksheet contains calculations relating to the credit risk of banks, that is, the risk that banks' borrowers will default on their contractual obligations. Table C1 in the worksheet Credit Risk summarizes the reported data for asset quality. Table C2 is used for the credit risk stress tests. It includes four different types of credit shocks, labeled 1, 2, 3, and 4.

A. Credit Shock 1 (“adjustment for underprovisioning”)

The purpose of the first part of the credit risk calculation is to make the point that stress testing should focus on the underlying economic value (net worth) of the bank. The economic value may differ in general from the bank's reported regulatory capital. For example, as part of reported capital, some banks may include items that, in fact, are not capital and should be treated rather as something else (e.g., long-term loan). Or they can overstate some assets, resulting in overstated capital. Because all the stress testing calculations relate to the economic value of the bank, the stress testing analyst needs to first adjust the reported data to get a better picture of the starting (“baseline”) economic situation of the bank. Credit Shock 1 shows an example of such an adjustment.¹¹

¹¹ In a strict sense, therefore, this is just a starting point adjustment, not a part of stress tests themselves. In a broader sense, however, it is a “stress test” that shows how the reported data would change if the reporting system changed to one that reflected more closely the economic value of a bank. Of course, if the reporting system already reflects the economic value, there is no adjustment, and one can proceed directly to Credit Shock 2 and beyond.

In Credit Shock 1, we assess what would happen if banks corrected their currently insufficient provisioning to fully meet the existing provisioning requirements. As mentioned earlier, a BCP assessment suggested that the current loan classification and provisioning standards in Bankistan are broadly adequate but not well enforced. The regulations prescribe the following loan provisioning rules: 1 percent general provision for pass loans, 3 percent general provision for special mention loans, 20 percent specific provision for substandard loans, 50 percent specific provision for doubtful loans, and 100 percent for loss loans. Reflecting the fact that it is very difficult to foreclose collateral in Bankistan, the test **assumes** that the actual value of the collateral is only 25 percent of the reported values (i.e., we assume a 75 percent “haircut” on the value of the collateral).

B. Credit Shock 2 (“increase in NPLs”)

If Credit Shock 1 is a “starting point adjustment,” then Credit Shock 2 can be seen as the first “real” stress test. It models a general decline in asset quality, affecting all banks proportionately. It is **assumed** that NPLs increase by a certain percentage, the default values being 25 percent of the existing stock of NPLs. This means that a bank would have to undertake additional provisioning by 25 percent for each of the three groups that constitute NPLs (substandard, doubtful, and loss loans). The increased provisioning requirements will reduce the value of the RWA as well as the capital. As regards the impact on the RWA, a common **assumption** is that the full increase in NPLs is subtracted from RWA. However, the impact on RWA may be smaller if the affected assets have a weight of less than one in the RWA. Typically, precise information of the distribution of NPLs across risk categories is not available. Nonetheless, a green cell (B40) in the Assumptions worksheet allows the user to change the **assumed** weight from 100 percent to a smaller number (say, 80 percent) and observe the impact of this assumption on the results.

The initial **assumption** in Credit Shock 2 is that the increase in NPLs in individual banks is proportional to the existing NPLs in these banks. In other words, banks that had more NPLs in the past are assumed to have more new NPLs as a result of the shock. This is the most straightforward calculation, but there are alternative approaches. For example, the new NPLs can be proportional to the overall stock of loans or to the stock of new performing loans. The green cells in this part of the worksheet allow the relaxation of this **assumption** and choose the weight of the existing NPLs and performing loans in determining the bank-by-bank increases in NPLs.

Whether to use existing NPLs or existing performing loans as a basis for assessing future credit risk is an open question. It is an empirical question that—if there are sufficient empirical data—can and should be decided empirically, by testing for the factors explaining bank-by-bank changes in NPLs (see Box 3.3 for a discussion of this issue). In the absence of reliable or sufficiently detailed empirical data, however, it is

often necessary to resort to simplifying assumptions. As a rule of thumb, using the existing NPLs is correct if the existing bad loans are a good proxy for the quality of a bank’s risk management and therefore of the risk faced by the bank going forward. Conversely, using the existing performing loans can be justified by the fact that performing loans are those loans that may “go bad,” that is, be shifted into the NPL category, and therefore indicate the potential for credit risk. Using performing loans as a basis may be warranted if there has been a structural change in the economy (and therefore the past NPL ratios are of limited guidance in assessing future credit risks). For example, in a number of emerging markets in Central and Eastern Europe in the early 2000s, there was a marked shift from corporate lending to household lending, with very different qualities and parameters. Using past NPLs as a basis for the bank-by-bank increases in such a situation might lead to misleading results.

The importance of the bank-by-bank distribution of NPL increases can be illustrated in the Credit Risk sheet. For example, in the default settings, NPLs increase by 25 percent in each bank (i.e., cell B46 in the Credit Risk sheet has a value of 25, and cells B48 and B49 have the values of 1 and 0, respectively).¹² The total volume of additional NPLs in the system (cell B50) is B\$2,206 million and the state-owned banks’ postshock capital is B\$–437 million (cell C53). If we instead make the increase in NPLs proportional to the existing performing loans (i.e., we make cells B48 and B49 equal 0 and 1, respectively) and make the overall volume of additional NPLs the same (which we can achieve by using the Tools/Goal Seek function, setting the value of B50 back to B\$2,206 million by changing the values of B46), the increase in NPLs has to be about 4.3 percent of performing loans, and the state-owned banks’ postshock capital is B\$–296 billion (cell C53), that is, smaller than if the increase in NPLs is proportional to the stock of NPLs. So, if we are concerned about the fiscal costs of the state-owned banks, the distribution of credit risks and buffers is important.

C. Credit Shock 3 (“sectoral shocks”)

As an illustration of how bank-by-bank credit risk can be modeled in a more realistic fashion, the stress testing file also includes sectoral shocks (see the bottom part of Table C2).

This exercise allows the user to select different shocks to economic sectors and observe how each bank would be affected, depending on the relative sizes of the banks’ credit exposures to these sectors. The case shown in the file as a starting example models a “terrorist attack” scenario, increasing credit risk in the tourism and trade sectors. The calibration of the sectoral shocks can be based on a historical scenario (e.g., a concrete example of a terrorist attack in Bankistan or in a neighboring country) or on empirical models explaining, based on past data, default rates in different sectors as a function of

¹² These values originate in cells B35, B37, and B38 of the Assumptions worksheet.

macroeconomic and other explanatory variables. The B43–B49 cells in the Assumptions worksheet can then be seen as an interface between the econometric model (or historical scenario) and the balance sheet implementation calculated in Stress Tester 3.0.

The definitions of sectors may be adapted depending on the country and the analyzed topic. For example, the “other” sector may be broken down into a number of subsectors. Alternatively, instead of using the main economic activity of the counterparty as the defining feature, sectors can be defined by the nature of the counterparty. For example, the “sectors” can be households (residents/nonresidents), nonfinancial enterprises (residents/nonresidents), nonbank financial institutions, and government (domestic/foreign).

The increase in NPLs is assumed to be proportional to the particular bank’s credit exposure to a sector, approximated by the bank’s total loans to that sector. Along the lines of the discussion in the previous subsection, it is possible to envisage that the increases reflect the bank’s existing NPLs or total loans to each sector and relax this assumption accordingly.

D. Credit Shock 4 (“concentration risk”)

Stress Tester 3.0 incorporates a block at the end of Table C2 that allows testing for the failure of the largest counterparties of individual banks. The user can change the **assumed** number of failures per institution and the **assumed** provisioning rate for those failures. If supervisors have data on banks’ exposures to nonbank financial institutions (NBFIs), this type of test can also be used to model the credit impact on banks of failures of the largest NBFIs.

4. INTEREST RATE RISK

The interest rate shock in the Interest Risk worksheet tests for direct interest rate risk. Direct interest rate risk is the risk incurred by a financial institution when the interest rate sensitivities of its assets and liabilities are mismatched. In addition, the financial institution is also exposed to indirect interest rate risk, resulting from the impact of interest rate changes on borrowers’ creditworthiness and ability to repay. The indirect interest rate risk is a part of credit risk. We discuss how to account for the interest rate–related credit risk in the section devoted to designing scenarios involving several shocks (Section 8).

A. Direct interest rate risk

The direct interest rate risk calculation in Stress Tester 3.0 consists of two parts, reflecting, respectively, flow and stock impacts of interest rate changes. The upper part, in Table D1, works with the repricing gap information from the Data worksheet. It calculates the changes in interest income and interest expenses resulting from the “gap” between the flow of interest on the holdings of assets and liabilities in each bucket. The “gap” in each time band or time-to-repricing bucket shows

how net interest income will be affected by a given change in interest rates. It sorts assets and liabilities into three time-to-repricing buckets (due in less than 3 months, due in 3 to 6 months, due in 6 to 12 months).¹³

The bottom part of the calculation, in Table D2, shows the impact of interest rate changes on the value of bonds held by the commercial banks. The calculations assume that the bonds are “marked-to-market,” that is, changes in their market value have a direct impact on the capitalization of the banks. The impact of an interest rate change on the market value is approximated using the duration of the bonds held by the banks. The data on duration are given to us by the NBB, which calculated it using more detailed data on the parameters of the bonds and the structure of their holdings by banks—see the formula for duration in the *Financial Soundness Indicators Compilation Guide* (IMF, 2004).

The direct impact of higher nominal interest rates on capital and capital adequacy is typically negative, resulting from the fact that financial institutions operate with a duration gap between their assets and liabilities. Duration of assets (liabilities), D_A (D_L), is defined as the weighted average, term-to-maturity of an asset’s (liability’s) cash flow, the weights being the present value of each future cash flow as a percentage of the asset’s (liability’s) full price.¹⁴ Duration approximates the elasticity of the market values of assets and liabilities to the respective rates of return:

$$\frac{\Delta A(r_A)}{A(r_A)} \equiv \frac{-D_A \Delta r_A}{(1+r_A)}, \quad \frac{\Delta L(r_L)}{L(r_L)} \equiv \frac{-D_L \Delta r_L}{(1+r_L)}, \quad (3.1)$$

where $A(r_A)$ and $L(r_L)$ are market values of assets and liabilities of the financial system, and r_A and r_L are annual interest rates of assets and liabilities.¹⁵ Differentiating the capital adequacy ratio with respect to the interest rate on assets and substituting from equation (3.1), we obtain

$$\begin{aligned} \frac{\Delta[C(r_A, r_L)/A_{RW}(r_A)]}{\Delta r_A} &\equiv -\frac{(L/A_{RW})}{1+r_A} \left(D_A - D_L \frac{1+r_A}{1+r_L} \frac{\Delta r_L}{\Delta r_A} \right) \\ &\times \frac{1 - \frac{\Delta A_{RW}}{A_{RW}} \frac{C}{\Delta C}}{1 - \frac{\Delta A}{A} \frac{C}{\Delta C}}. \end{aligned} \quad (3.2)$$

¹³ This is a crude breakdown; for more precise calculations, more maturity brackets are used, especially for the very short periods but also for longer periods.

¹⁴ See the *Financial Soundness Indicators Compilation Guide* (IMF, 2004, paragraph 3.52) for a formula. In practice, the calculation of the duration of total assets and total liabilities of a financial system is a difficult computational task, and various simplifications are used (e.g., duration is computed for groups of assets and liabilities with common features and aggregated across such groups; or duration is replaced by residual maturity and time to repricing).

¹⁵ See, for example, Bierwag (1987). The results are first-order approximations; for large changes in interest rates, second derivative terms need to be included to account for convexity of portfolios. Alternatively, the elasticity of bond prices to interest rate changes can be empirically estimated using past data.

Assuming that the risk-weighted assets move proportionately to total assets, that is, $\Delta A_{RW}/A_{RW} = \Delta A/A$, equation (3.2) can be simplified into

$$\frac{\Delta[C(r_A, r_L)/A_{RW}(r_A)]}{\Delta r_A} \equiv -\frac{(L/A_{RW})}{1+r_A} GAP_D, \quad (3.3)$$

where GAP_D is the duration gap, defined as¹⁶

$$GAP_D = D_A - D_L \frac{1+r_A}{1+r_L} \frac{\Delta r_L}{\Delta r_A}. \quad (3.4)$$

Most financial institutions, and banks in particular, operate by transforming short-term, low-interest-rate liabilities into long-term, higher-interest-rate assets. This means that $D_A > D_L$, $r_A > r_L$, and $GAP_D > 0$. Thus, an increase in interest rates has a negative impact on the institutions' net worth and capitalization, leading to increased financial sector vulnerability.

In the Bankistan example illustrated in Stress Tester 3.0, only two bonds are available to banks. Both of them are government-issued bonds. The Assumptions worksheet shows the key parameters of these bonds (in rows 57–59) and calculates their duration, using the Duration function in Excel (see cells H58 and H59).¹⁷

B. Indirect interest rate risk

The indirect effects, related to the interest-risk nexus, work in the same direction. An increase in nominal interest rates—to the extent that it increases real interest rates and makes it more difficult for borrowers to repay their debts and to obtain new credit—is likely to have a negative effect on the credit risk of the financial institutions' borrowers. Other things being equal, higher risk eventually translates into higher losses and a decline in the financial institutions' net worth. The exact impact depends on factors such as the borrowers' earnings in relation to interest and principal expenses, loan loss provisions, and the degree of collateralization of the loans. Country case studies find a positive relationship between higher interest rates and nonperforming loans or loan losses.

The basic calculation does not cover the impact of nominal interest rate changes on real interest rates and thereby on borrowers' creditworthiness and ability to repay. The impact size depends mostly on the corporate sector's leverage and exposure to the real estate market (which is also likely to be influenced by changes in interest rates). In order to assess this type of risk, one would usually need to estimate the impact of changes in interest rates on nonperforming loans, using a regression model. In our example, the joint impact of changes in interest rates and in credit quality can be simulated using the worksheet Scenarios.

¹⁶ If the interest rates for assets and liabilities move simultaneously, the duration gap can be approximated as a difference of the two durations, $D_A - D_L$.

¹⁷ Excel also has a Price function, which can also be used to implement this calculation.

5. FOREIGN EXCHANGE RISK

The foreign exchange risk is the risk that exchange rate changes affect the local currency value of financial institutions' assets, liabilities, and off-balance-sheet items. The foreign exchange risk is composed of three types: the direct solvency risk (resulting from banks' net open positions in foreign currency and those in local currency that are indexed to exchange rates); the indirect solvency risk (resulting from the impact of foreign exchange positions taken by borrowers on their creditworthiness and ability to repay and thereby on financial institutions); and the foreign exchange liquidity risk (resulting from liquidity mismatches in foreign currency). In this section, we will focus on the direct solvency risk, and we will discuss implementation of the indirect solvency risk; we relegate the foreign exchange liquidity risk to the section dealing with the liquidity stress test.¹⁸

The foreign exchange shock in the FX Risk worksheet is composed of two parts. The first part, shown in Table E1, tests direct foreign exchange rate risk. The second part, shown in Table E2, shows the impact of the change in the nominal exchange rate on banks through changes in the credit risk.

A. Direct foreign exchange risk

Table E1 in the FX Risk worksheet tests direct foreign exchange rate risk based on the net open position in foreign exchange at end-2010. This figure is calculated by the NBB, using the methodology described in the *Financial Soundness Indicators Compilation Guide* (IMF, 2004), and is copied to Table E1 using a link to the Data worksheet.

The *direct exchange rate risk* can be assessed using the net open position in foreign exchange, one of the "core FSIs," defined in IMF (2004). The direct exchange rate risk is arguably the easiest part of stress tests to implement. To illustrate this test, let F denote the net open position in foreign exchange, C the capital, A_{RW} the risk-weighted assets (all in domestic currency units), and e the exchange rate in units of foreign currency per unit of domestic currency. A depreciation (decline) in the exchange rate leads to a proportional decline in the domestic currency value of the net open position, that is, $\Delta e/e = \Delta F/F$ (for $F \neq 0$). Let us **assume** that this translates directly into a decline in capital, that is, $\Delta C/\Delta F = 1$.¹⁹ The impact of the exchange rate shock on the ratio of capital to risk-weighted assets would then be

$$\begin{aligned} \frac{\Delta[C(e)/A_{RW}(e)]}{\Delta e} &\equiv \frac{\frac{F}{e} A_{RW} - C}{A_{RW}^2} \frac{\Delta A_{RW}}{\Delta C} \frac{F}{e} \\ &\equiv \frac{1}{e} \frac{F}{C} \frac{C}{A_{RW}} \left(1 - \frac{\Delta A_{RW}}{\Delta C} \frac{C}{A_{RW}} \right), \quad (3.5) \end{aligned}$$

¹⁸ The direct and indirect solvency risks in foreign exchange are often also referred to as "exchange rate risks."

¹⁹ An alternative, and arguably more realistic, approach is to deduct the impact first from profits (if any) and then from capital. See Section 1 for a more detailed discussion of this issue.

which uses the fact that $\Delta C/\Delta e = \Delta F/\Delta e = F/e$. The symbol “ $\cong$ ” means that the equation is only approximate for larger than infinitesimal changes. Equation (3.1) can be rewritten as

$$\Delta[C(e)/A_{RW}(e)] \cong \frac{\Delta e}{e} \frac{F}{C} \frac{C}{A_{RW}} \left(1 - \frac{\Delta A_{RW}}{\Delta C} \frac{C}{A_{RW}} \right). \quad (3.6)$$

The term $\Delta A_{RW}/\Delta C$ can have values from 0 to 1, reflecting the degree of co-movement of capital and the risk-weighted assets.²⁰ In the special case of $\Delta A_{RW}/\Delta C = 0$, that is, if the risk-weighted assets do not change, the change in the capital adequacy ratio equals the exchange rate shock times the exposure, measured as a product of the net open position to capital (F/C) and capital adequacy (C/A_{RW}), both of which are “core FSIs” as defined by IMF (2004). This is sometimes used as a shorthand calculation of the direct exchange rate stress test. It should be noted that equation (3.6) holds only as a linear approximation, which works well in nonsophisticated financial systems. However, if financial institutions have large positions in foreign exchange options, the relationship between the exchange rate change and the impact on capital can become highly nonlinear. In such cases, stress tests based on detailed decomposition of financial institutions’ open positions are a superior analytical tool.²¹

In Stress Tester 3.0, a depreciation of the Bankistan dollar against the U.S. dollar from the official exchange rate of 55 B\$/US\$ to the currently prevailing parallel market rate of 85 B\$/US\$ is **assumed** (at constant cross-exchange rates to the U.S. dollar). Information on the banks’ foreign exchange exposure is provided in Table E1. A depreciation will benefit banks that have a long (positive) open position in foreign currency and hurt banks that have a short (negative) position in foreign currency.

Only a very limited number of banks have short positions; therefore, the direct depreciation effects are very small—some banks would even gain from a depreciation. Given that most central banks impose limits on foreign exchange positions to capital, this result is not unusual. For most banking systems, the direct foreign exchange solvency risk is rather small. Banks’ net open positions in foreign exchange are typically under close scrutiny from banks’ risk managers and supervisors. Banks in some countries have explicit limits on these positions as a percentage of the bank’s capital (the ceilings typically being in the range of 10–20 percent of capital). In other countries, this is addressed by including the net open positions in the capital adequacy calculation. In general, the open positions tend to be rather small, and consequently the direct impact of an exchange rate depreciation (or appreciation) tends to be rather small.

B. Indirect foreign exchange risk

Besides direct depreciation effects, a change in the exchange rate would also influence the creditworthiness and ability to

repay of the corporate sector. A change in the exchange rate influences the corporate sector in two main ways: first, it changes its competitiveness relative to the foreign corporate sector; second, it influences the corporate balance sheets directly via firms’ net open positions in foreign currencies (for instance, companies can borrow massively in foreign currencies).

The indirect foreign exchange risk seems to be very important. FSAP missions generally have not been able to collect comprehensive data on the corporate sector’s foreign exchange exposure, but those FSAP missions that analyzed the corporate sector in detail generally found that the banking sector’s indirect exchange rate risk was more important than its direct one. The indirect foreign exchange rate risk appears to be particularly substantial in countries with closely managed exchange rate pegs.

To illustrate the significance of the indirect risk in overall banking sector risk, let us denote the corporate sector’s debt, equity, and open foreign exchange position as $D_c(e)$, $E_c(e)$, and $F_c(e)$, respectively.²² Let us **assume** that, similar to the case of banks’ net open position, a percentage change in the exchange rate will translate into the same percentage change in the domestic currency value of the net open position, which will in turn lead to an equivalent change in the corporate sector’s equity, that is, $\Delta E_c/\Delta e = \Delta F_c/\Delta e = F/e$. The impact of the exchange rate on the corporate leverage (D_c/E_c) is then given by

$$\frac{\Delta[D_c(e)/E_c(e)]}{\Delta e} \cong \frac{\frac{\Delta D_c}{D_c} \frac{F_c}{E_c} E_c - D_c \frac{F_c}{E_c}}{E_c^2} \cong \frac{1}{e} \frac{F_c}{E_c} \left(\frac{D_c}{E_c} - \frac{\Delta D_c}{\Delta E_c} \right). \quad (3.7)$$

Thus, if the corporate sector is short in foreign exchange, a depreciation (decline) in the exchange rate would lead to an increase in its leverage. Corporate leverage typically is positively correlated with the share of banks’ nonperforming loans in total loans (denoted as NPL/TL), that is, $\Delta(NPL/TL)/\Delta(D_c/E_c) = a > 0$.²³ The impact of a change in the exchange rate on the NPL/TL ratio can then be expressed as

$$\begin{aligned} \Delta(NPL/TL) &\cong a \Delta[D_c(e)/E_c(e)] \\ &\cong - \frac{\Delta e}{e} \frac{F_c}{E_c} a \left(\frac{D_c}{E_c} - \frac{\Delta D_c}{\Delta E_c} \right). \end{aligned} \quad (3.8)$$

In the special case when $\Delta D_c/\Delta E_c = 0$, the change in the NPL/TL ratio would equal the exchange rate change times the respective FSI (the net open position), times the parameter a , which can be estimated empirically, as shown, for example, in IMF (2003) or Boss and others (2004). To find the impact on capital adequacy, we can **assume**, as has been

²⁰ Empirically, $\Delta A_{RW}/\Delta C$ could be estimated by a regression.

²¹ As a general point, stress tests should include all relevant off-balance-sheet items.

²² Given the practical difficulties involved in obtaining empirical data on open positions in the household sector, we refer here for simplicity only to the corporate sector, even though the theoretical analysis would be essentially the same even if we included the household sector.

²³ IMF (2003) shows that for a panel of 47 countries, a 10 percentage point rise in the corporate leverage was associated with a 1.1 percentage point rise in the ratio of NPLs to total loans after a one-year lag.

done in some FSAP missions, that the credit shock moves some of the previously performing loans into the nonperforming category. By differentiating C/A_{RW} with respect to NPL/TL , and substituting for NPL/TL from equation (3.8), we obtain

$$\Delta(C/A_{RW}) \cong \frac{\Delta e}{e} \frac{TL}{A_{RW}} \left(1 - \frac{C}{A_{RW}} \frac{\Delta A_{RW}}{\Delta C} \right) \times \pi \frac{F_c}{E_c} a \left(\frac{D_c}{E_c} - \frac{\Delta D_c}{\Delta E_c} \right), \quad (3.9)$$

where we **assume** (as several FSAP missions have done) that provisions are expressed as a fixed percentage (π) of nonperforming loans and that they are deducted directly from capital.

The incorporation of the indirect effect makes the analysis of foreign exchange rate risk more complex and dependent on additional assumptions or regression analysis. One of the reasons adding to the complexity of the indirect exchange rate stress test is the fact that it includes the effects on stocks as well as on flows. The calculation of the indirect effect as per equation (3.9) would need to reflect the impact of exchange rate changes on the net present value of the corporate sector, which means to take into account changes in the net present value of future earnings. For example, in export-oriented companies, a depreciation could be expected generally to increase their future earnings. In terms of the net present value, the effect would be essentially equivalent to the impact of a long position in foreign currency. However, it may be more practical to calculate the impact on flows, by estimating the elasticity of earnings to interest and principal expenses (an encouraged FSI) with respect to the exchange rate and then to estimate the relationship between this FSI and the NPL/TL ratio. Alternatively, it would be useful to compile an indicator measuring the corporate sector's flow exposure, for example, a ratio of foreign exchange earnings to total earnings or (ideally) a ratio of earnings in foreign exchange to interest and principal expenses in foreign exchange.

Table E2 in the FX Risk worksheet shows the impact of the change in the nominal exchange rate on banks through changes in the credit risk. The impact is approximated here by **assuming** that the change in the NPLs is proportional to the volume of foreign exchange loans in a bank. The idea behind this assumption is that a depreciation would increase the domestic currency value of these loans, which would make it more difficult for the borrowers to repay—for example, because some of the loans are extended to borrowers with limited access to foreign currency.

6. INTERBANK (SOLVENCY) CONTAGION RISK

We have so far **assumed** that there is no contagion among banks in the event of a failure. This assumption is relaxed in

the Interbank worksheet, which presents a basic calculation of the interbank contagion risk.²⁴

In this section, we focus on contagion through insolvencies. There is also, potentially very important, the risk of liquidity contagion through bank runs triggered by a run on another bank. A basic model of liquidity contagion is shown as part of the liquidity risk (Section 7). The principle of having a matrix of bank-to-bank “exposures” is the same in all the contagion tests, but the specification of the matrix is different for a liquidity test.

We focus here on contagion within the (domestic) banking system. The Bankistan banking system includes foreign-owned banks, but we do not study cross-border exposures and cross-border contagion. In principle, the same framework would have to be applied, but the definition of “system” would have to be wider to include the foreign banks. For simplicity (and because obtaining good bank-by-bank data on cross-border exposures is not trivial in practice), we focus here on interbank exposures in the domestic market. We have to be at least aware, however, that in addition to the domestic exposures, there may also be important cross-border exposures and the related risk of contagion.

The upper panel in the Interbank worksheet (Table F1) derives a matrix of interbank exposures for the banks in Bankistan. The first panel shows the matrix of net interbank credits, in which each cell shows, for the bank in the column (i.e., for the bank listed in the same column in row 4), its net credit to the bank in the row (i.e., the bank listed in the same row in column A). For example, the value of 70 in cell I11 means that the bank in column I (i.e., DB1) has an outstanding net credit to the bank in row 11 (i.e., SB2) in the amount of B\$70 million. A corresponding entry of –70 in cell G13 confirms this by showing that the bank in column G (i.e., SB2) has an outstanding negative net credit of B\$70 million to the bank in row 13 (i.e., DB1), that is, it has a net borrowing of the same amount. The matrix is obtained by netting out the gross figures in the interbank lending table reported in the Data worksheet, that is, reported by the NBB. In this matrix, positive figures mean that the bank in the column is a net creditor of the bank in the row, whereas negative figures mean that the bank in the column is a net borrower of the bank in the row. The diagonal cells are left empty, because the focus of this exercise is on exposures between each bank and the other banks.

To facilitate further calculation, the matrix of net interbank credit is converted into the matrix of net interbank exposures. This is done by “stripping” the matrix of net creditors by focusing on the positive numbers (because those are the banks exposed to interbank credit risk). The other cells in the

²⁴ In a real stress testing exercise, we would have to start from the very beginning with balance sheets that include the interbank exposures. This would make the aggregation of the bank-by-bank data more cumbersome (we would have to net out the bank-to-bank exposures). For simplicity, this exercise starts without the interbank exposures, which are added only later.

table are left empty. For example, the value of 70 in cell I26 means that the bank in column I (i.e., DB1) has an outstanding net exposure to the bank in row 26 (i.e., SB2) in the amount of B\$70 million. The corresponding cell, G28, is empty: the bank in column G (i.e., SB2) has no net exposure to the bank in row 28 (i.e., DB1), because SB2 is a net borrower with respect to DB1 and therefore has no direct credit exposure to DB1.

A. “Pure” interbank contagion

The middle panel in the Interbank worksheet (Table F2) provides a calculation of the “pure” interbank contagion exercise. The exercise shows what would happen with the capital of the bank in the column if the bank in the row failed and defaulted on all its interbank borrowing. This is actually a series of 12 separate stress tests, one in each row, showing for each bank what would be the direct impact of its failure on the capital of each of the other banks. The stress test is run in several iterations, as the contagion-induced failures (“first iteration”) can induce failures in other banks (“second iteration”), which can lead to further failures (“third iteration”), and so on.

The first part of Table F2 shows, in each row, the postshock capital of the individual banks in the column after the **assumed** failure of the bank in the row. For example, row 49 shows in columns F to Q what would be the postshock capital of banks SB1 to FB4 if bank SB3 failed. The results in row 49 suggest that none of the banks would have a negative postshock capital as a direct result of SB3’s failure.

The next part of Table F2 shows which banks fail as a result of the first iteration (“1” denotes a newly failed bank; all others have zeros). The table shows that two banks would fail as a result of failures in other banks, namely, that DB1 can fail as a result of failures in SB2 or FB3 (see cells I62 and I71) and DB2 can fail as a result of failures in SB1, SB2, FB1, and FB3 (see cells J61, J62, J69, and J71).

We are **assuming** for simplicity that if a bank’s capital stays positive after an iteration, the bank does not fail and remains able to repay all its interbank obligations; if its capital becomes negative, it fails and does not repay its obligations. The calculation can be made more realistic by estimating a more complex mapping between the capital adequacy ratio and the bank’s probability of failure (rows 6 and 22 in the Assumptions worksheet contain an example of such a mapping). Such mapping is likely to indicate a wider scope for interbank contagion than the introductory calculation presented here; however, the calculations would be based on similar techniques.

To keep the calculations straightforward, we **assume** here that the impact of shocks is deducted directly from capital. It is not complicated, however, to extend the file to take into account banks’ profits.

The second iteration needs to be calculated only for the failures in SB1, SB2, FB1, and FB3, because these are the only banks whose failures lead to failures in other banks (namely, DB1 or DB2). The failures in DB1 and DB2 could lead, in the second iteration, to failures in other banks if the other banks

had substantial net credit outstanding with respect to DB1 or DB2. However, as illustrated in rows 73–113, this is not the case: the banks that are creditors to SB3 and DB1 have exposures to SB3 and DB1 that are well below their capital.

Rows 73–85 start the second iteration by inverting the table of the failed banks from the first iteration. This is to illustrate that in the second iteration, the banks that were exposed to contagion in the first iteration become the source of further contagion.²⁵ Rows 88–99 show the capital after the second iteration, which generally equals the capital after the first iteration, but for failures in SB1, SB2, FB1, and FB3 (i.e., in rows 88, 89, 96, and 98), the second iteration capital of the bank in the column is lowered by the amount of exposure of this bank to the bank that failed in the first round. This is highlighted in rows 101–112, which show the change in capital between the first and second iterations: capital is lower for stress tests in rows 101, 102, 109, and 111, because the corresponding banks (SB1, SB2, FB1, and FB3) cause additional bank failures in the second iteration.

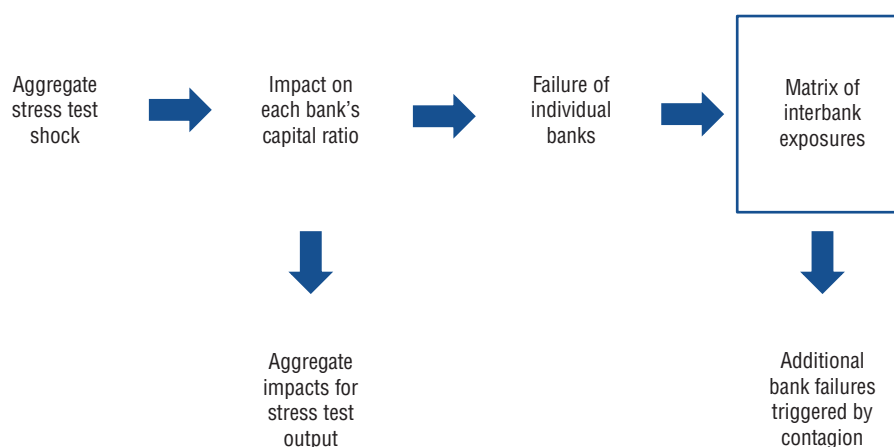
Rows 115–126 show that the third iteration is not needed in this case, as none of the banks are affected enough to fail in the second round. If such failure occurred, the third round (and any subsequent round) could be implemented in the same way as the second round.

To present the results in terms of capital adequacy, the worksheet uses an **assumption** on the risk weight of the interbank loans that become unpaid. Reflecting the typical Basel weight for interbank loans, this assumption is set at 20 percent but can be changed by changing the value of the B77 cell in the Assumptions worksheet.

The “pure” interbank contagion test could be interpreted as a measure of systemic importance of individual banks: the bigger the decline in the system’s capital (or capital adequacy ratio), the more systemically important is the bank whose default is assumed. In our example, FB2 is the systemically most important bank, using this criterion, because assuming its failure yields the lowest postshock systemic capital (B\$3,570 million in the cell B138) and postcontagion CAR (9.7 percent in the cell B152). This test (or rather set of tests) is useful, but it does not take into account the different likelihood of failures in different banks, an issue that is addressed by the “macro” interbank contagion test.²⁶

²⁵ The calculation can be implemented without the inversion, but inverting makes calculations easier, because one can use the SUMPRODUCT function. Also, the inversion highlights the nature of the calculation: in the second iteration, we look at the failed creditors from the previous iteration and analyze who are their creditors. Given that the exposure table has creditors in columns and borrowers in rows, going from the first iteration to the second one requires inverting either the matrix of exposures or the table with results from the first iteration. In this implementation, we chose the latter approach.

²⁶ “Pure” interbank contagion tests are more common in the literature. For example, Sveriges Riksbank presents its results regularly in its Financial Stability Report (for the methodology, see Blåvarg and Niemander 2002). “Macro” interbank contagion tests are less common but are presented for example by Elsinger, Lehar, and Summer (2003) for Austria and by Čihák, Heřmánek, and Hlaváček (2007) for the Czech Republic.



Source: Author.

Figure 3.4 “Macro” Interbank Contagion

B. “Macro” interbank contagion

The lower table in the Interbank worksheet (Table F3) shows a “macro” interbank contagion test. In this case, we model the case when bank failures are triggered by macroeconomic developments, in particular by the scenario that is already modeled in the Scenario sheet (Figure 3.4). The starting point for the “macro” interbank contagion is therefore the postshock values of capital and risk-weighted assets for each bank from the Scenarios sheet.²⁷ For the failed banks (SB2 and SB3 for the default settings in the spreadsheet), we run an interbank contagion exercise using the matrix of net interbank exposures. Then we search for banks that fail in this first iteration. If there were no new failures, the contagion exercise would stop here. In our example, however, there are three new failures: SB1, DB1, and DB2. We therefore run a second iteration, looking at the impact of these additional failures on other banks. We find that they lead to one additional failure, namely, in DB4. If this failure led to other new failures, we would need to run a fourth iteration. However, in this particular case, the contagion-induced failure of DB4 does not lead to other failures, and the process stops at the third iteration.

What is the key difference between the “pure” and the “macro” contagion tests? The “pure” contagion test **assumes** that a failure occurs in a single bank, for example, for some internal reason (e.g., because of a large fraud in the bank); it does not distinguish the relative likelihood of the failure of various banks. This is what the “macro” contagion test does. It analyzes situations when all banks are weakened at the same time by a common external (typically macroeconomic) shock, which affects each bank differently depending on its exposures to the various risk factors and makes some of the

banks (perhaps more than one) fail. For the default settings for Bankistan, for example, the first iteration of the “macro” contagion test involves a simultaneous failure of two banks (SB2 and SB3), resulting in a second-iteration failure of three other banks (SB1, DB1, and DB2), leading to a third-iteration failure of DB4. The process stops at the third iteration, which does not lead to additional failures.

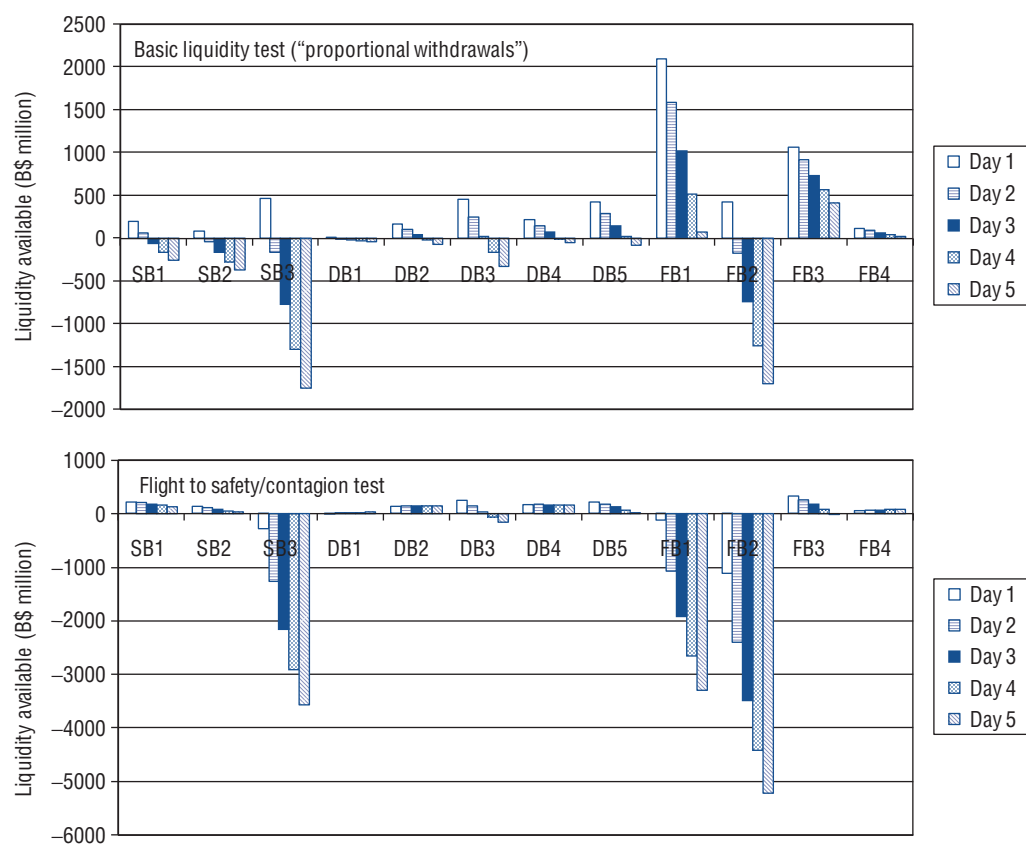
7. LIQUIDITY TESTS AND LIQUIDITY CONTAGION

Testing for liquidity risks is less common than testing for risks to solvency in central banks’ stability reports and in IMF work. This mostly reflects the fact that modeling liquidity risks is more complex. First, to properly model liquidity fluctuations in banks, one needs to have very detailed, high-frequency data, such that are typically used by commercial banks themselves in their liquidity management models. Second, to model the impact of large liquidity shocks, one needs to consider the broader liquidity management framework, in particular the lender-of-last-resort function of many central banks.

At the same time, testing for liquidity risks is important. In the last two decades, much of the attention in risk management and prudential supervision was on capital, partly in relation to the efforts to standardize capital adequacy requirements across countries. In the process, relatively less attention has been paid to cash flows and analysis of liquidity (e.g., Goodhart, 2006). Analyzing the response of liquidity to stress is an important undertaking, because liquidity is how a stressful situation often manifests itself in the short run.

The presentation of the stress test impact is different from the solvency tests discussed so far. The impact is shown for each bank in terms of the number of days it would be able to survive a liquidity drain *without resorting to liquidity from outside*

²⁷ The calculations in the Scenarios sheet disregarded these contagion effects, so the calculation in Table F3 of the Interbank sheet can be seen as an extension of the calculation in the Scenarios sheet.



Source: Author's calculations, using Stress Tester 3.0.

Figure 3.5 Results of Liquidity Stress Tests

(i.e., from other banks or the central bank).²⁸ This is a relatively narrow approach to liquidity stress testing, but it is one that allows for an introductory exposition without going into details.

The Stress Tester 3.0 file contains two basic examples of liquidity tests. The Liquidity worksheet contains these two examples in two tables. Figure 3.5 shows the results of these two tests for the default values contained in the spreadsheet.

Table G1 models a liquidity drain that affects all banks in the system proportionally to their volumes of demand and time deposits. The worksheet allows the user to change **assumptions** on the percentage of demand deposits and time deposits that get withdrawn each day and on the percentage of liquid assets and other assets that banks can convert to cash each day.²⁹

Table G2 models "liquidity contagion," where the liquidity drain starts in the smallest or weakest banks and proceeds to the larger or stronger banks. The test allows for **three possible measures** of "bank safety": (1) total assets; (2) total assets, with a premium for state ownership; and (3) preshock rating. In the first case, depositors perceive bank safety as linked to the size of the bank, approximated by total assets. In the second case, they also perceive state-owned banks as being safer than privately owned banks (because of an explicit or implicit government guarantee in the former case). In the third case, depositors' perceptions of bank safety are correlated with the banks' recent financial performance.³⁰ Table G2 also combines the liquidity impact of government default with a bank run. It allows users to change the **assumption** on the percentage of the government bonds that are in default.

8. SCENARIOS

The Scenarios worksheet illustrates, in Tables H1–H4, how shocks to the various risk factors can be combined into a

²⁸ As a rule of thumb, supervisors often see five days as an important threshold for a bank's ability to withstand a liquidity run. After five days or less, banks are likely to close for a weekend or a holiday, providing some "breathing time" for bank management and supervisors to regroup, assess the situation, and decide on measures and public announcements to make. This rule of thumb has been partly diluted with the growth of direct banking.

²⁹ Another useful distinction in many cases is between domestic and foreign exchange deposits (and assets). It is possible to extend the file to assume different withdrawal rates by currency of denomination.

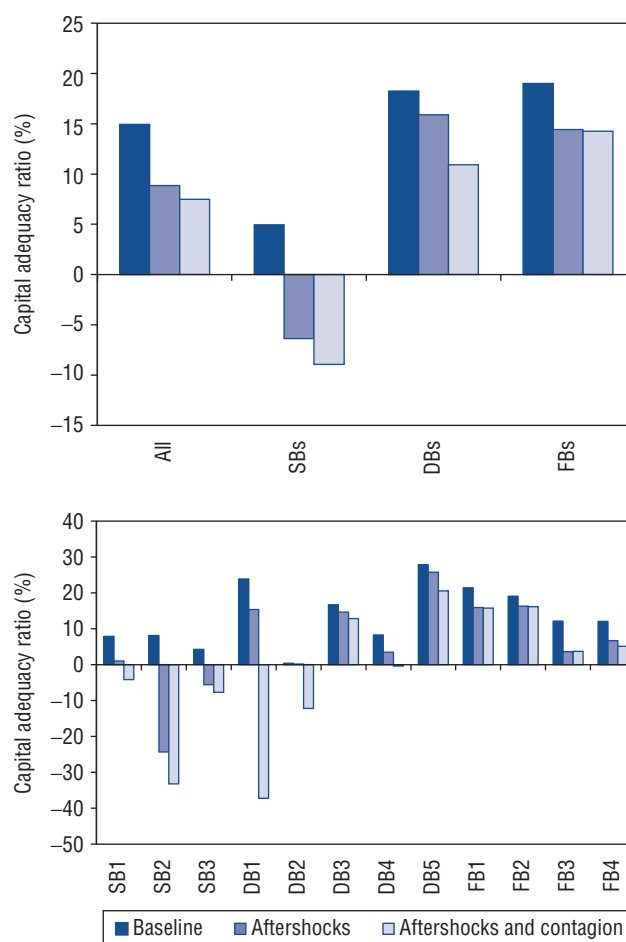
³⁰ Liquidity contagion can also be thought of in terms of a similar "exposure" matrix as used in the solvency contagion, except that instead of "net uncollateralized interbank exposures," the value of the cell in the matrix for liquidity contagion would be the difference between bank i 's and bank j 's measure of "bank safety."

single scenario. The main reason for using scenarios rather than single-factor shocks is that in the macroeconomic context, changes in several risk factors are typically interrelated. For example, a large increase in nominal interest rates can lead to an increase in real interest rates, which can (perhaps with a lag) contribute to an increase in NPLs. If this is the case, banks will be hit not only by the direct impact of the nominal increase in interest rates but also by the indirect impact through credit risk.

The purpose of this worksheet is for the users to see how different combinations of the shocks affect the capital adequacy of the system. For simplicity of presentation, the worksheet contains only formulas linked to the other worksheets, namely, Credit Risk for credit risk, Interest Risk for the interest rate risk, FX Risk for the direct and indirect foreign exchange solvency risk, Interbank for interbank contagion, and Liquidity for liquidity risk. When a number of different specifications of a stress test are available, Stress Tester 3.0 allows the user to specify which of the alternative specifications is being considered in the overall scenario. These choices are **assumptions** that can be made in cells B69–B75 of the Assumptions worksheet (a summary of the chosen scenario is then shown on top of Table H in the Scenarios worksheet).

The Scenarios worksheet adds up the impacts of the selected shocks to arrive at an aggregate impact. Two issues are important in considering whether the impacts can be added up. First, the calculations need to take into account concentration of risks in institutions. Simply adding up aggregate losses caused by individual shocks could overlook situations when risks are concentrated in an institution or a group of institutions. Stress Tester 3.0 addresses this issue by calculating the impacts bank by bank. This allows us to see whether some banks are hit by the selected combination of shocks much harder than others (indeed, they are). Second, it is not trivial to combine solvency and liquidity risks. Stress Tester 3.0 illustrates how this can be done. It uses the NBB's supervisory early-warning system to combine the changes in solvency and liquidity (and other measures) to identify the change in the supervisory rating and the implied change in probability of default. There are some limitations to this approach (in particular, the PDs cannot be easily aggregated for the system as a whole), but it provides a useful illustration.

As a basic presentational approach, the Scenarios worksheet shows the impacts of the scenarios in terms of the capital adequacy and decomposes the overall impact into the individual risk factors (in percentage points of the CAR ratio). The charts in the Assumptions worksheet allow the user to immediately and graphically see the results of the various assumptions on the outcome in terms of changes in capital adequacy ratios. For illustration, we reproduce these charts here as Figure 3.6. The values shown in these charts represent the default sizes of shocks and starting assumptions in the accompanying file. Dark blue indicates the baseline (pre-shock) values of the capital adequacy ratio, medium blue indicates the corresponding postshock values, and light blue



Source: Stress Tester 3.0.xls, Assumptions worksheet, based on default values of shocks and assumptions.

Figure 3.6 Impact of Stress on Capital Adequacy Ratios

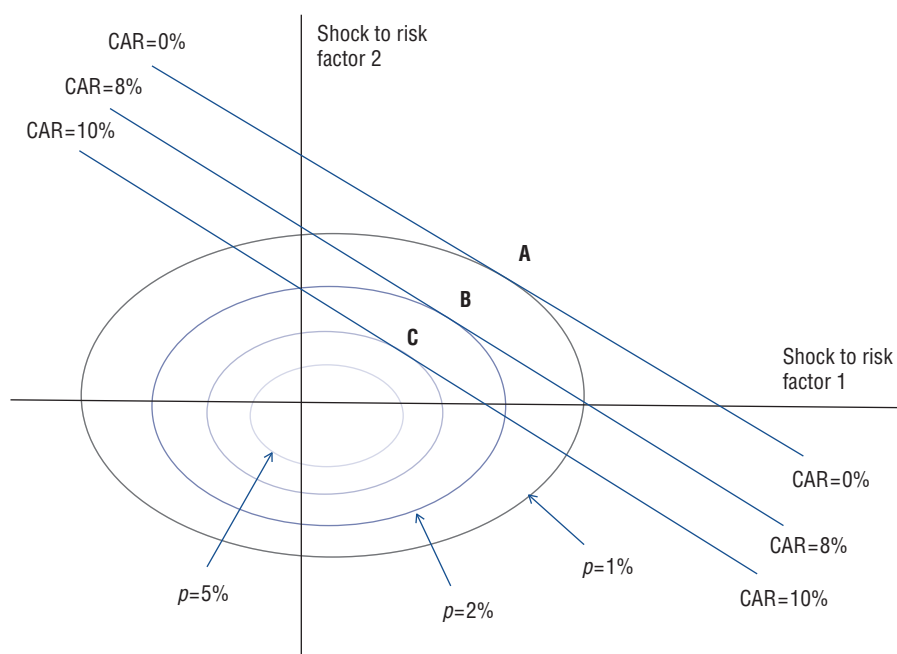
indicates values after the shocks and after taking into account the subsequent contagion among banks. The preshock (baseline) CAR values are taken from the Data worksheet, whereas the values corresponding to the stressful scenario are taken from the Scenarios worksheet.

The worksheet also compares the overall impact with the banks' profits. Although we have so far **assumed** that the impacts are deducted directly from capital, in reality banks could use profits as their first line of defense. Table H in the Scenarios worksheet shows for comparison, for each bank, what its profits were in the past. It also allows the user to **assume** (in the appropriate green cell) an autonomous shock to net interest income.

A. Designing consistent scenarios

How can one design consistent scenarios?³¹ In general, there are two ways of asking questions about exposures in the financial system. The first way is to ask, for a given level of

³¹ For more details, see Čihák (2004, 2005).



Source: Author.

Figure 3.7 “Worst Case Approach” versus “Threshold Approach”

plausibility, which scenario has the worst impact on the system (“the worst case approach”). The second way is to ask, for a given impact on the system, what is the most plausible combination of shocks that would need to occur to have that impact (“threshold approach”). In Stress Tester, the “threshold approach,” also called “reverse stress testing,” is illustrated in the Reverse worksheet.

Figure 3.7 shows the process of scenario selection under the worst case approach and the threshold approach, for a simplified case when there are only two risk factors (e.g., changes in interest rates and exchange rates). Each ellipse depicts the set of combinations of the two risk factors with the same probability of occurrence. The shape of the ellipse represents the correlation between the two factors, and its size represents the level of plausibility (the larger the ellipse, the smaller the plausibility). The diagonal lines depict combinations of the risk factors leading to the same overall impact, measured here by a change in the system’s CAR. The impact increases with the size of the shocks to the risk factors, so the CAR decreases in the northeast direction. The diagonal lines do not have to be straight; they are depicted here as such only for simplicity. Figure 3.7 illustrates that the worst case approach and the threshold approach are two essentially equivalent ways of analyzing the same problem.³²

The worst case approach starts with selecting a level of plausibility (e.g., 1 percent) and searching for the combination of

shocks with this level of plausibility that have the worst impact on the portfolio. This means searching for the point on the largest ellipse that lies as far northeast as possible. In Figure 3.7, this is point A.

The threshold approach starts with selecting the threshold, that is, the diagonal line; it then searches for the most plausible (i.e., smallest) shocks reaching this threshold. This is straightforward if there is only one risk factor; if there are two risk factors, one needs to take into account the correlation between the risk factors. For the specific correlation pattern in Figure 3.7, selecting a threshold of zero capital adequacy would lead again to the combination of shocks corresponding to point A.

Establishing the plausibility level of a scenario can be difficult in practice, given that the scenario should be a low-probability “tail” event. For risk factors with good time series of historical data (in particular, for market risks), the natural starting point is to base the scenarios on the past volatility and covariance patterns. Calibrating the shocks is particularly straightforward for single-factor stress tests: an exchange rate shock can be based on 3 standard deviations of past exchange rate changes (corresponding roughly to a 1 percent confidence level). With multiple risk factors, one needs also to look at the covariance statistics of the variables or use stochastic simulations based on macroeconomic models. Such calculations are subject to a number of caveats. In particular, models can break down for large shocks. Nonetheless, the models, if used cautiously, can help to find a first-cut approximation of stress test scenarios (see Box 3.4 for an additional discussion of choosing the “right” scenario).

³² To some readers, these two approaches may resemble the dual tasks of microeconomics.

Box 3.4. Picking the “Right” Scenario

In discussions on designing stress tests, too much is often made of establishing the “right” scenario. Of course, it is important, at least in theory, for scenarios to be internally consistent, as highlighted, for example, by Jones, Hilbers, and Slack (2004) or by Figure 3.7 in this chapter. In practice, assessing such consistency is tricky, because the scenarios are also supposed to be exceptional (but plausible).

How to address this challenge? One approach is to choose a concrete extreme historical scenario (e.g., the East Asian crisis of 1997) and calculate what would be the impact of repeating such a scenario (or an adaptation of such scenario) in the present situation of the banking system. The main advantage is that historical scenarios are easy to communicate and to implement. Also, they are plausible, because such a situation actually happened. Their main disadvantage is that past crises may not be good models for future crises. Also, the probability level of the past historical scenario may be unclear.

Another approach, used by some FSAPs and central bank FSRs, is to use an existing macroeconomic model (e.g., a model used by the central bank for macroeconomic forecasts and policy analysis) as a basis for stochastic simulations showing the distribution of the key risk factors in the case of shocks to the model’s exogenous variables. The challenge of this approach often arises from the fact that such macroeconomic models typically do not include a measure of credit risk (e.g., nonperforming loans to total loans or another measure of asset quality). Therefore, this approach usually involves estimating a “satellite” model that links a measure of credit risk to the variables from the macroeconomic model. Unlike the macroeconomic model, the satellite model can be estimated (and generally should be, if adequate data are available) on individual bank (and even individual borrower) data. The estimates from the satellite model can then be used for the balance sheet implementation.

Yet another, and perhaps more direct, approach can be to plot the existing observations of the various risk factors (in a similar fashion as shown, for two risk factors, in Figure 3.7) against a measure of soundness (e.g., capital adequacy ratio) and use this to identify the most stressful combinations of risk factors (in terms of Figure 3.7, this would mean identifying the points lying the most toward the northeast).

In sum, there is a range of methods, each with its advantages and disadvantages. Picking a scenario that is stressful and tells an interesting and consistent story is important. However, in most cases, identifying “the right scenario” is close to impossible. Much more important than fine-tuning scenarios is (1) being transparent about the underlying assumptions of the scenarios; (2) being transparent about the sensitivity of the results to those assumptions; and (3) showing how results with the same assumptions change over time. Showing results over time allows making judgments about the developments in the overall pool of risks and in the structure of risks faced by a financial system.

B. Linking stress tests to rankings and probabilities of default

The Scenarios sheet also illustrates the links between the stress tests and the supervisory early-warning system. It follows the same approach as the illustration shown in the Data worksheet, but it is derived from the bank-by-bank data after the shocks rather than those before the shocks. Specifically, Table H2 provides the postshock FSIs and other ratios for individual banks. These can be compared with the corresponding preshock ratios, shown in Table A3 (in the Data worksheet).

Table H3 converts these ratios into postshock ratings of the individual banks, using the same “step functions” that were applied in Table A5 to the preshock ratios. In both cases, the “step function” reflects the off-site supervisory assessment model and is specified in the Assumptions worksheet (rows 3–22). Table H3 also provides averages (weighted by banks’ total assets) for the three peer groups and for the banking system as a whole.

Table H4 converts the postshock ratings into postshock probabilities of default for each bank. These can be compared with the preshock probabilities of default, shown in Table A6 in the Data worksheet.

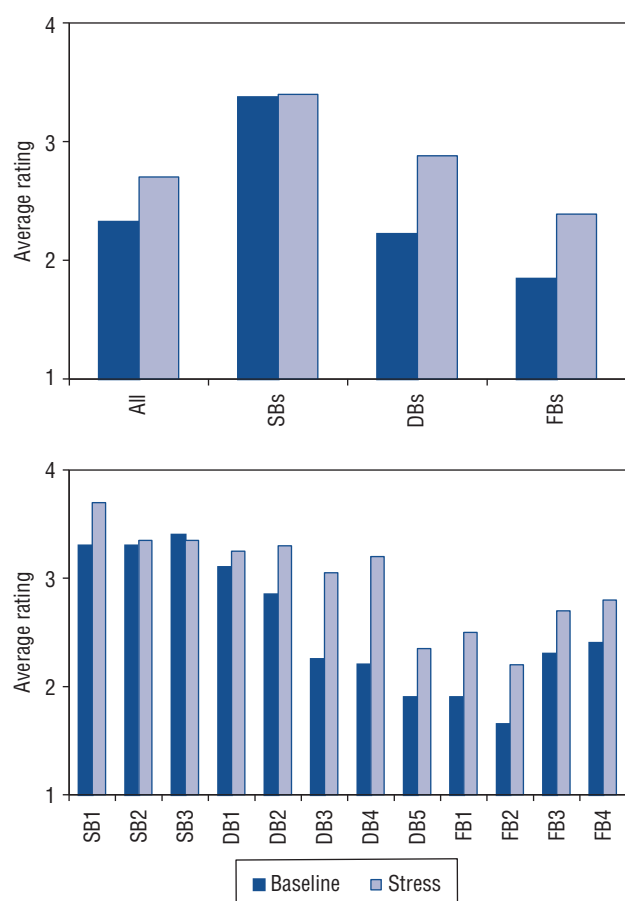
The charts in the Assumptions worksheet, reproduced in this chapter as Figure 3.8 and Figure 3.9, illustrate such comparisons. Dark blue indicates the baseline values of the indicators (ratings in Figure 3.8 and probabilities of default in Figure 3.9),

and light blue indicates the corresponding values in situations of stress. The values of the baseline are taken from the Data worksheet, whereas the values corresponding to the stressful scenario are taken from the Scenarios worksheet. The values shown in the charts reproduced here reflect the default sizes of shocks and starting assumptions in the accompanying file.

Figure 3.10 shows another form of presentation of stress testing results. It captures the impact of stress on the banks’ *z*-scores. As indicated earlier, the *z*-score has become a popular measure of bank soundness, because it is directly related to the probability of a bank’s insolvency. Figure 3.10 illustrates how the banks’ *z*-scores decline as a result of the assumed impacts, generally mirroring the decline in banks’ probabilities of default, shown in Figure 3.9 (one needs to bear in mind that higher *z*-scores correspond to lower probabilities of default). If the user changes the key sizes of shocks and assumptions in the accompanying Excel file, the charts in the file will change automatically.

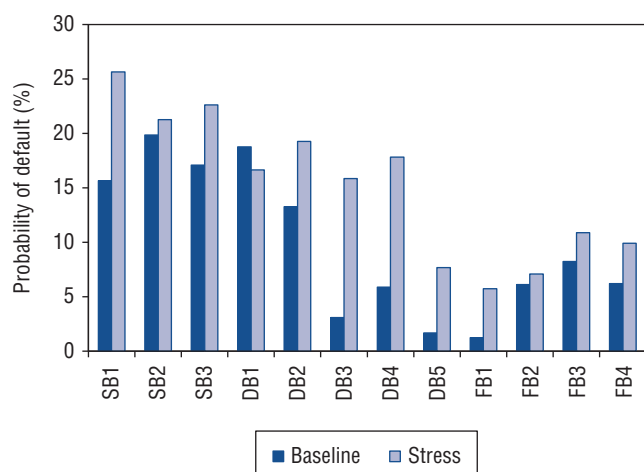
C. Modeling the feedback effects

The stress test calculations presented here focus on the impacts of shocks arising from the macroeconomic environment and affecting the financial sector. From a macroeconomic perspective, an important issue is whether the shocks in the financial sector can have feedback effects affecting the macroeconomic environment.



Source: Stress Tester 3.0.xls, Assumptions worksheet, based on default values of shocks and assumptions.

Figure 3.8 Impact of Stress on Supervisory Ratings



Source: Stress Tester 3.0.xls, Assumptions worksheet, based on default values of shocks and assumptions.

Figure 3.9 Impact of Stress on Banks' Probabilities of Default

The practical problem in modeling the feedback effects is that there are too many. One direct effect that can be incorporated easily into the financial programming framework employed by the IMF is an impact on capital on "Other Items Net" in the monetary survey and thereby on other macroeconomic variables.

However, this is just one of many potential impacts. In some cases, the effects depend on the behavior of the institutions in situations of stress. For example, if banks attempt to sell off certain types of assets (e.g., real estate) in situations of stress, they may bring down the asset prices, with repercussion effects for other sectors (e.g., household consumption). Also, bank failures triggered by stress may result in a credit crunch.

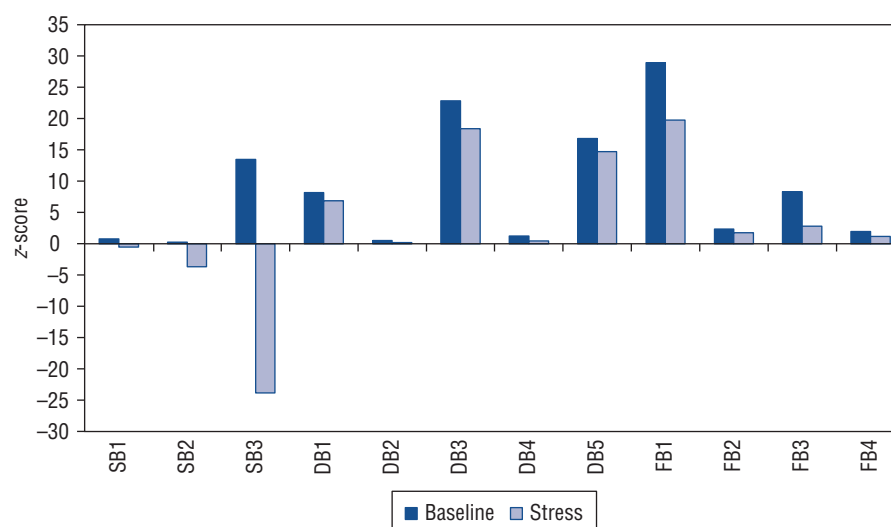
In the accompanying Excel example, we approximate the potential macroeconomic impacts by the capital injection needed to bring all banks to the minimum required capital adequacy ratio. This indicator does not capture all the potential macroeconomic effects, but it is a useful broad indicator of potential fiscal costs associated with averting failures in the banking system. It is an upper-bound estimate of such costs. The public sector will most likely inject capital into state-owned banks, but it is less clear whether (and if so, how much) it will inject into the other banks, which are privately owned (it is generally more likely to do so for larger banks that are considered "too big to fail"). For this reason, the charts and results in the file show a breakdown of the necessary capital injection by ownership. The charts are included in the Assumptions worksheet and are reproduced here as Figure 3.11. The values shown in the figure reflect the default sizes of shocks and default assumptions in the accompanying file.

9. CONCLUSION

It is difficult to really understand stress testing without going through actual stress testing calculations. This chapter enables readers to do just that: working with the accompanying Excel file, they can change the assumptions and observe changes in results.

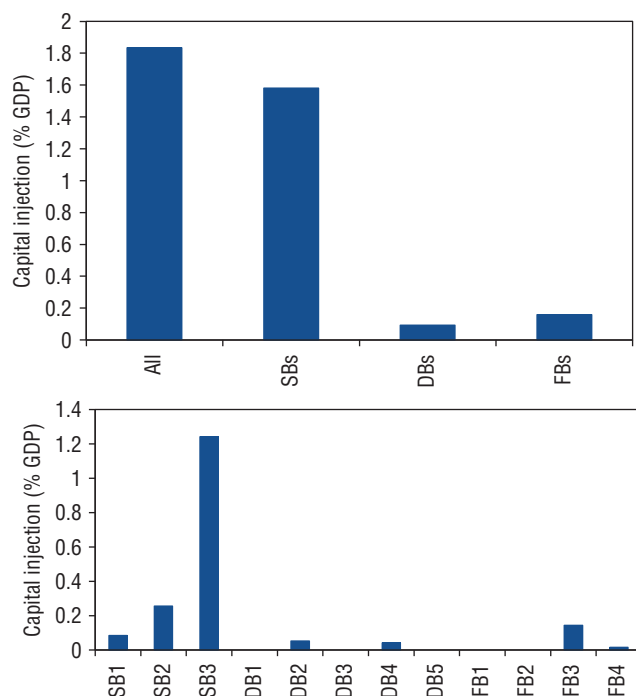
Stress tests are complementary to other tools for financial stability analysis, and the exercise illustrates this complementarity. In particular, it illustrates that stress tests are complementary to the FSIs, which allow for "benchmarking," that is, for a baseline assessment of the financial system under no stress. FSIs can also be used to describe the impact of stress on a system. Stress tests are also complementary to the supervisory early-warning system, an example of which was shown as well. The early-warning system traditionally is used to calculate ratings and probabilities of default in the "baseline" scenario, but as shown in Stress Tester 3.0, it can also be used to produce ratings and probabilities of default in a stressful scenario. Stress tests are complementary also to other tools, such as assessments of compliance with regulatory standards and codes, and assessment of the broader financial stability framework.

The exercise highlights several challenges. The basic challenge is that stress testing is *data intensive*. It deals with low-



Source: Stress Tester 3.0.xls, Assumptions worksheet, based on default values of shocks and assumptions.

Figure 3.10 Impact of Stress on Banks' z-Scores



Source: Stress Tester 3.0.xls, Assumptions worksheet, based on default values of shocks and assumptions.

Figure 3.11 Capital Injections Needed to Bring Banks to Minimum Capital Adequacy

probability events, implying that there will always be a lack of data and a need for simplifying assumptions. The main thing an analyst should do is to be transparent about the nature of the assumptions. To make things more complicated, *nonlinearities* are likely to kick in for the large shocks that are being contemplated in stress tests. Also, macroeconomic and other *models tend to break down* in crises. Past crises may not be a

good guide for the future. For example, a change in borrower characteristics may leave credit more vulnerable to interest rate risk. Another challenge is that the impact of shocks is *distributed over time*. It takes time for asset quality to deteriorate and for that deterioration to have an impact. A crisis evolves over a period of time, sometimes several years. Modeling stressful scenarios therefore has to take the time dimension into account, and it is important to clarify what the benchmark scenario is against which the stressful one is being compared. Finally, *mitigating measures* can be taken by participants and authorities, especially when looking at longer time periods. These measures and the feedback effects play an important role over time, and they make the stress testing calculations more complex.

Two main steps can be taken to address these challenges in stress tests. The first is to *keep assumptions transparent and be clear on the sensitivity of the results to the assumptions*. The accompanying stress testing file is trying to do that. All assumptions are brought into one place and highlighted. Users can experiment with the assumptions to see the impact of the changes on the results. The second step is to *present stress test results over time*. Presenting results over time helps to say whether the overall pool of risks has changed or whether the structure of risks has changed. Most financial stability reports still do not provide results over time, which makes interpretation of the presented results more difficult for readers (Čihák, 2006). The Stress Tester 3.0 exercise is based on a one-period snapshot, but the idea behind the exercise is that it is run repeatedly, replacing the input data (in yellow) by data from other periods.

The idea behind the accompanying file is that it can be developed in a modular fashion, with additional modules capturing additional risks or elaborating on the existing ones. One can think about Stress Tester 3.0 as a module in a broader stress testing tool kit and about the cells with assumptions as

interfaces between this module and other modules. The following extensions are particularly worth considering:

Credit risk—macro nexus. Credit risk is the main source of risk in most financial systems. At the same time, it is the part where this exercise is perhaps the most simplified. Ideally, we would need to have more detailed data on loan exposures and loan performance by economic sector and also data on the financial soundness of the corporate and household sectors. Time series of historical data ideally should be used to establish linkages between macroeconomic variables and loan performance (or, in the absence of reliable time series, estimates from other countries could be used). The calculations presented in this exercise try to convey the gist of the credit risk stress test, without going into technical details.

Credit VaR models. Developed country commercial banks and some banks in medium-income countries are using value-at-risk (VaR) models as a basis for stress testing for credit risk. There is a range of these models, all of which share the purpose of determining the probability distribution of the losses on a portfolio of loans and other debt instruments. Being able to compute the loss distribution of a portfolio allows the determination of the economic capital required by credit operations. Implementing VaR tests for credit risk on the macroprudential level is more complex than for market risk VaR models, because (1) fewer banks use them; and (2) the risk factors and their parametrization are likely to differ across banks, making comparability and aggregation more difficult. Nonetheless, implementing credit risk models on a macroprudential level is possible and can provide a useful benchmark for credit risk evaluation, as illustrated in Avesani and others (2006) on the CreditRisk+ model. Indeed, some recent FSAPs have made use of these approaches.

Stress testing based on factor models. This includes (1) portfolio risk management models using structural credit risk models of obligors' assets and risky debt based on domestic and international factor models, such as the portfolio manager models, CreditMetrics, and the default and conditional probability of default models (e.g., Segoviano and Padilla, 2006); and (2) Merton-type structural models of banks, which calibrate risk-adjusted balance sheets and implied assets of banks, which are then linked to domestic and international factor models. Stress testing can then analyze how changes in key international and domestic factors drive individual bank risk and systemic risk).

Other risk factors. Depending on the sophistication of the financial system and the type of its exposures, it may be necessary to perform stress tests for other risk factors. These might include asset price shocks (including, e.g., shocks to real estate prices) and shocks to commodity prices (especially in developing economies with significant exposures to commodities).

REFERENCES

- Avesani, Renzo G., Kexue Liu, Alin Mirestean, and Jean Salvati, 2006, "Review and Implementation of Credit Risk Models of the Financial Sector Assessment Program," IMF Working Paper 06/134 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=19111.0>
- Babouček, Ivan, and Martin Jančar, 2005, "Effects of Macroeconomic Shocks to the Quality of the Aggregate Loan Portfolio," Czech National Bank Working Paper 1/2005 (Prague: Czech National Bank). Available via the Internet: http://www.cnb.cz/www.cnb.cz/en/research/cnb_wp/download/cnbwp_2005_01.pdf
- Bierwag, Gerald O., 1987, *Duration Analysis: Managing Interest Rate Risk* (Cambridge, MA: Ballinger).
- Blaschke, Winfrid, Matthew T. Jones, Giovanni Majnoni, and Soledad Martinez Peria, 2001, "Stress Testing of Financial Systems: An Overview of Issues, Methodologies, and FSAP Experiences," IMF Working Paper 01/88 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/ft/wp/2004/wp04127.pdf>
- Blåvarg, M., and P. Niemand, 2002, "Inter-bank Exposures and Systemic Risk," *Sveriges Riksbank Economic Review* 2/2002 (Sweden: Sveriges Riskbank).
- Boss, Michael, 2002, "A Macroeconomic Credit Risk Model for Stress Testing the Austrian Credit Portfolio," Financial Stability Report No. 4 (Vienna: Oesterreichische Nationalbank). Available via the Internet: http://www.oenb.at/en/presse_pub/period_pub/finanzmarkt/finanzmarktstabilita/financial_stability_report_4.jsp
- Boss, Michael, Gerald Krenn, Markus Schwaiger, and Wolfgang Wegschaidner, 2004, "Stress Testing the Austrian Banking System," *Österreichisches Bankarchiv* 11/04, pp. 841–52 (Vienna: Oesterreichische Nationalbank). Available via the Internet: http://www.oenb.at/de/img/artikel_9_841852_boss_et_al_11_tcm14-27296.pdf
- Boyd, John H., and David E. Runkle, 1993, "Size and Performance of Banking Firms," *Journal of Monetary Economics*, Vol. 31, No. 3, pp. 47–67.
- Bunn, Philip, Alastair Cunningham, and Mathias Drehmann, 2005, "Stress Testing as a Tool for Assessing Systemic Risks," *Financial Stability Review*, June, pp. 116–26 (London: Bank of England). Available via the Internet: <http://www.bankofengland.co.uk/publications/Pages/fsr/2005/fsr18.aspx>
- Čihák, Martin, 2004, "Stress Testing: A Review of Key Concepts," CNB Research Policy Note 2/2004 (Prague: Czech National Bank). Available via the Internet: http://www.cnb.cz/en/pdf/IRPN_2_2004.pdf
- , 2005, "Stress Testing of Banking Systems," *Finance a úvěr/Czech Journal of Economics and Finance*, Vol. 55, No. 9–10, pp. 418–40.
- , 2006, "How Do Central Banks Write on Financial Stability?" IMF Working Paper 06/163 (Washington: International Monetary Fund). Available via the Internet: www.imf.org/external/pubs/ft/wp/2006/wp06163.pdf
- , 2007a, "Introduction to Applied Stress Testing," IMF Working Paper 07/59 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/ft/wp/2007/wp0759.pdf>
- , 2007b, "Systemic Loss: A Measure of Financial Stability," *Finance a úvěr/Czech Journal of Economics and Finance*, Vol. 57, No. 1–2, pp. 5–26. Available via the Internet: http://journal.fsv.cuni.cz/storage/1073_fau_1_2_07_00000000005.pdf

- Čihák, Martin, and Jaroslav Heřmánek, 2005, "Stress Testing the Czech Banking System: Where Are We? Where Are We Going?" CNB Research Policy Note 2/2005. Available via the Internet: www.cnb.cz/en/research/research_publications/irpn/download/irpn_2_2005.pdf
- Čihák, Martin, and Michal Hlaváček, 2007, "New Approaches to Stress Testing the Czech Banking Sector," *Finance a úvěr/Czech Journal of Economics and Finance*, Vol. 57, No. 1–2, pp. 41–59. Available via the Internet: http://journal.fsv.cuni.cz/storage/6_fau_1_2_07_00000000041.pdf
- Danmarks Nationalbank, 2004, "Market-Based Risk Measures for Banks," *Financial Stability* (Copenhagen, July). Available via the Internet: https://www.nationalbanken.dk/DNUK/FinanceStab.nsf/side/Publications_about_financial_stability!OpenDocument
- Drehmann, Mathias, 2005, "A Market Based Macro Stress Test for the Corporate Credit Exposures of U.K. Banks" (London: Bank of England). Available via the Internet: <http://www.bis.org/bcbs/events/rtf05Drehmann.pdf>
- Eklund, Trond, Kai Larsen, and Eivind Berhardsen, 2003, "Model for Analysing Credit Risk in the Enterprise Sector," Economic Bulletin Q3 01 (Oslo: Norges Bank). Available via the Internet: http://www.norges-bank.no/english/publications/economic_bulletin/2001-03/eklund-larsen.pdf
- Elsinger, Helmut, Alfred Lehar, and Martin Summer, 2003, "Risk Assessment for Banking Systems," 14th Annual Utah Winter Finance Conference Paper; EFA 2003 Annual Conference Paper No. 437.
- Evjen, Snorre, Arild J. Lund, Kjersti Haare Morka, Kjell B. Nordal, and Ingvild Svendsen, 2005, "Monetary and Financial Stability in Norway: What Can We Learn from Macroeconomic Stress Tests?" in *Investigating the Relationship between the Financial and Real Economy*, BIS Papers, No. 22, pp. 409–30. Available via the Internet: <http://www.bis.org/publ/bppdf/bispap22u.pdf>
- Gerlach, Stefan, Wensheng Peng, and Chang Shu, 2004, "Macroeconomic Conditions and Banking Performance in Hong Kong: A Panel Data Study," Hong Kong Monetary Authority Research Memorandum (Hong Kong SAR, April). Available via the Internet: http://www.info.gov.hk/hkma/eng/research/RM_macro_and_banking.pdf
- Goodhart, Charles, 2006, "A Framework for Assessing Financial Stability?" *Journal of Banking and Finance*, Vol. 30, No. 12, pp. 3415–22.
- Hesse, Heiko, and Martin Čihák, 2007, "Cooperative Banks and Financial Stability," IMF Working Paper No. 07/02 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/ft/wp/2007/wp0702.pdf>
- Hoggarth, Glenn, Andrew Logan, and Lea Zicchino, 2005, "Macro Stress Tests of U.K. Banks," in *Investigating the Relationship between the Financial and Real Economy*, BIS Papers, No. 22, pp. 392–408. Available via the Internet: <http://www.bis.org/publ/bppdf/bispap22.htm>
- International Monetary Fund, 2003, "Financial Soundness Indicators—Background Paper," IMF Policy Paper (Washington, May 14). Available via the Internet: <http://www.imf.org/external/np/sta/fsi/eng/2003/051403bp.pdf>
- , 2004, *Financial Soundness Indicators Compilation Guide* (Washington, July). Available via the Internet: <http://www.imf.org/external/np/sta/fsi/eng/guide/index.htm>
- , 2005a, "European Financial Integration, Stability and Supervision," IMF Country Report 05/266, pp. 113–46 (Washington, August). Available via the Internet: <http://www.imf.org/external/np/sta/fsi/eng/guide/index.htm>
- , 2005b, *Financial Sector Assessment: A Handbook* (Washington, September). Available via the Internet: <http://www.imf.org/external/pubs/ft/fsa/eng/index.htm>
- , 2006, "Spain: FSAP: Technical Note—Stress Testing Methodology and Results," IMF Country Report 06/216 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/ft/scr/2006/cr06216.pdf>
- International Monetary Fund and the World Bank, 2003, "Analytical Tools of the FSAP," IMF and World Bank Policy Paper (Washington, February). Available via the Internet: <http://www.imf.org/external/np/fsap/2003/022403a.pdf>
- Jones, Matthew T., Paul Hilbers, and Graham Slack, 2004, "Stress Testing Financial Systems: What to Do When the Governor Calls," IMF Working Paper 04/127 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/ft/wp/2004/wp04127.pdf>
- Kraft, Evan, and Tomislav Galac, 2007, "Deposit Interest Rates, Asset Risk and Bank Failure in Croatia," *Journal of Financial Stability*, Vol. 2, No. 4, pp. 337–55.
- Lucas, Robert E., 1976, "Econometric Policy Evaluation: A Critique," *Carnegie-Rochester Conference Series on Public Policy*, Vol. 1, No. 1, pp. 19–46.
- Peng, Wensheng, Kitty Lai, Frank Leung, and Chang Shu, 2003, "The Impact of Interest Rate Shocks on the Performance of the Banking Sector," Hong Kong Monetary Authority Research Memorandum (Hong Kong SAR, May). Available via the Internet: <http://www.info.gov.hk/hkma/eng/research/RM07-2003.pdf>
- Pesola, Jarmo, 2005, "Banking Fragility and Distress: An Econometric Study of Macroeconomic Determinants," Research Discussion Papers No. 13 (Helsinki: Bank of Finland). Available via the Internet: http://www.bof.fi/eng/6_julkaisut/6.1_SPn_julkaisut/6.1.5_Keskustelualoitteita/0513netti.pdf
- Sahajwala, Ranjana, and Paul Van den Bergh, 2000, "Supervisory Risk Assessment and Early Warning Systems," BCBS Working Paper No. 4 (Basel: Basel Committee on Banking Supervision). Available via the Internet: http://www.bis.org/publ/bcbs_wp4.pdf
- Segoviano, Miguel A., and Pablo Padilla, 2006, "Portfolio Credit Risk and Macroeconomic Shocks: Applications to Stress Testing under Data-Restricted Environments," IMF Working Paper 06/283 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/ft/wp/2006/wp06283.pdf>
- Shu, Chang, 2002, "The Impact of Macroeconomic Environment on the Asset Quality of Hong Kong's Banking Sector," Hong Kong Monetary Authority Research Memorandum (Hong Kong SAR, December). Available via the Internet: <http://www.info.gov.hk/hkma/eng/research/index.htm>
- Virolainen, Kimmo, 2004, "Macro Stress Testing with a Macroeconomic Credit Risk Model for Finland," Discussion Paper 18/2004 (Helsinki: Bank of Finland). Available via the Internet: <http://www.suomenpankki.fi/en/julkaisut/tutkimukset/keskustelualoitteet/Pages/default.aspx?year=2004>

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Into the Great Unknown: Stress Testing with Weak Data

LI LIAN ONG • RODOLFO MAINO • NOMBULELO DUMA

Stress testing has become the risk management tool du jour in the wake of the global financial crisis. In countries where the information reported by financial institutions is considered to be of sufficiently good quality, and supervisory and regulatory standards are high, stress tests can be of significant value. In contrast, the proliferation of stress testing in underdeveloped financial systems with weak oversight regimes is fraught with uncertainties, as it is unclear what the results actually represent and how they could be usefully applied. In this chapter, problems associated with stress tests using weak data are examined. We offer a potentially more useful alternative, the “breaking point” method, which also requires close coordination with on-site supervision and is complemented by other supervisory tools and qualitative information.

METHOD SUMMARY

Overview	The Breaking Point method (for solvency risk) is essentially a reverse stress test for determining the level of nonperforming loans (NPLs) required to “break the bank.”
Application	The method is appropriate in situations where data are limited and/or of poor quality.
Nature of approach	Basic balance sheet shocks.
Data requirements	Accounting information on capital, loans, and risk-weighted assets. Supervisory data on classified loans and provisions.
Strengths	The method does not require an estimation of the size of shocks in an environment of poor quality or insufficient data.
Weaknesses	• Assumptions are required on the amounts of transitioning NPLs, from one category to another. • Assumptions on the size of shocks are ad hoc and not based on satellite models; the method is essentially useful for sensitivity analysis. • Poor quality data could render the results meaningless; the results must be supplemented by on-site examination findings.
Tool	The Excel spreadsheet macro is available in the toolkit, which is on the companion CD and at www.elibrary.imf.org/stress-test-toolkit . Contact author: L. L. Ong.

The global financial crisis has placed the topic of stress testing firmly in the spotlight.¹ Issues such as the methodologies and assumptions applied in stress tests and the availability (and quality) of data used in those tests have come under close scrutiny, amid heated debate about transparency and the desirability of making stress test results public. The ongoing discussion is clearly very relevant for countries with more advanced financial systems—and oversight practices—

where well-designed stress tests could add significant value to risk management and contingency planning by both the authorities and individual financial institutions. Robust stress tests also are useful for the financial surveillance work done by international financial institutions, including the IMF.

The use of stress tests as an off-site supervision tool has also gained momentum in lower-income countries with typically

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¹ See Basel Committee on Banking Supervision (2008a, 2008b, 2008c, 2009a, 2009b).

TABLE 4.1

Baseline. Selected Bank Balance Sheet Items for Country X, at End-2009
(in millions of domestic currency units unless stated otherwise)

Row Number and Formula	Item	Provisioning Rate	(1) All Banks	(2) Bank 1	(3) Bank 2	(4) Bank 3	(5) Bank 4	(6) Bank 5
(1)	Capital		530.0	30.0	160.0	220.0	80.0	40.0
(2)	RWA		3,520.0	170.0	1,100.0	1,400.0	550.0	300.0
(3) = (1)/(2)*100	CAR (in percent)		15.1	17.6	14.5	15.7	14.5	13.3
(4) = (5) + (8)	Total loans		1,448.0	71.0	385.0	615.0	287.0	90.0
(5) = (6) + (7)	Performing loans		1,375.0	65.0	365.0	590.0	275.0	80.0
(6)	Normal and pass loans		1,250.0	55.0	330.0	530.0	260.0	75.0
(7)	Special mention loans		125.0	10.0	35.0	60.0	15.0	5.0
(8) = (9) + (10) + (11)	NPLs		73.0	6.0	20.0	25.0	12.0	10.0
(9)	Substandard loans		26.0	3.0	10.0	5.0	5.0	3.0
(10)	Doubtful loans		20.0	2.0	5.0	10.0	2.0	1.0
(11)	Loss loans		27.0	1.0	5.0	10.0	5.0	6.0
(12) = (8)/(4)*100	NPL ratio (in percent)		5.0	8.5	5.2	4.1	4.2	11.1
(13)	Total provisions currently held ¹		58.5	3.5	13.9	23.1	10.1	8.0
(14) = (15) + (17)	Total provisions that should be held		58.5	3.5	13.9	23.1	10.1	8.0
(15) = (16)	General provision		12.5	0.6	3.3	5.3	2.6	0.8
(16) = (6)*Rate	against normal and pass loans	0.01	12.5	0.6	3.3	5.3	2.6	0.8
(17) = (18) + (19) + (20) + (21)	Specific provision		46.0	2.9	10.6	17.8	7.5	7.3
(18) = (7)*Rate	against special mention loans	0.03	3.8	0.3	1.1	1.8	0.5	0.2
(19) = (9)*Rate	against substandard loans	0.20	5.2	0.6	2.0	1.0	1.0	0.6
(20) = (10)*Rate	against doubtful loans	0.50	10.0	1.0	2.5	5.0	1.0	0.5
(21) = (11)*Rate	against loss loans	1.00	27.0	1.0	5.0	10.0	5.0	6.0
(22) = (13) – (14)	Under-/over-provisioning		0.0	0.0	0.0	0.0	0.0	0.0

Source: Authors.

Note: CAR = capital adequacy ratio; NPL = nonperforming loan; RWA = risk-weighted assets.

1. It is assumed that NPLs are fully provisioned for initially.

underdeveloped financial systems.² However, in many of these countries, supervisory capacity is low, human resources are limited, the regulatory framework remains largely inadequate, and the track record on implementation and enforcement tends to be weak. As a result, the necessary data required for stress testing are usually of poor quality, that is, insufficient, incomplete, or inaccurate. In such instances, the desire to keep up with international developments by running stress tests on the respective financial systems is potentially fraught with problems. Indeed, it could do more harm than good if the flawed findings cause undue consternation or lead to inappropriate decisions and actions.

This chapter will briefly discuss the problems associated with stress tests that are usually applied to underdeveloped banking systems and then propose a more useful alternative. We focus our analysis on the stress testing of credit risk in the banking system and its impact on solvency, which tends to be

the key concern in many of these countries. We demonstrate how a modified version of Čihák's (2007) credit risk stress test could be used to complement other supervisory actions, including on-site examinations by supervisors. Consequently, we suggest that any stress testing performed in underdeveloped banking systems would have to be closely coordinated with on-site supervision and complemented by other supervisory tools and information to be useful in any way, because stand-alone stress test results would likely be meaningless.

This chapter is structured as follows. The data are briefly described in Section 1. In Section 2, we discuss the problems associated with the simple stress test model that is commonly used for countries with more basic financial systems, where data quality may be questionable. Section 3 proposes an alternative method for stress testing, which is less dependent on data quality and the highly subjective assumptions of stress testers, to complement on-site supervision. Our concluding thoughts on the topic are presented in Section 4.

1. THE DATA

We use hypothetical numerical examples to illustrate the issues raised in this chapter. The data are presented as follows:

- An assumed set of capitalization and credit data for the banking system of Country X, which comprises five individual banks, is used as the baseline (Table 4.1).

² In these countries, top-down stress tests—where the impact of macroeconomic shocks and scenarios on financial sector variables is estimated by using aggregate data—usually are introduced by counterparts from more developed financial systems providing technical assistance or by international financial institutions, such as the IMF, during their surveillance or technical assistance missions. Supervisors may then adjust the stress test models over time to suit developments in their own financial systems. Banks operating in these countries typically do not run bottom-up stress tests, either because they are not required to or because they lack capacity.

TABLE 4.2

Loan Classifications and Provisioning Requirements for Country X		
Classification	Definition	Provisioning Requirement (in percent of outstanding amount)
Performing loans		
Normal and pass loans	Assets in this category are performing in accordance with contractual terms and are expected to continue doing so.	1
Special mention loans	Any loan that is past due 30 days or more but less than 90 days.	3
NPLs		
Substandard loans	Any loan that is past due 90 days or more but less than 180 days.	20
Doubtful loans	Any loan that is past due 180 days or more but less than 360 days.	50
Loss loans	Any loan that is past due 360 days or more.	100

Source: Authors.

Note: NPL = nonperforming loan.

- The local definitions of loan classifications and their corresponding provisioning requirements are also assumed (Table 4.2).
- The required capital adequacy ratio (CAR) for banks in Country X is assumed to be 12 percent, below which banks would be required to recapitalize.

Several key assumptions are also made with regard to the calculation of the CAR:

- Risk-weighted assets (RWA) are assumed to remain the same postshock, which would translate to a more conservative result.³
- Profits are assumed to be zero for the period of the shock, so the full impact is reflected in capital.
- Where loans by classification are available, they are assumed to have been fully provisioned for prior to the shock; where less granular information is available, loans may be underprovisioned for.
- In all cases, provisions are topped up postshock to ensure that loans are again fully provisioned for.

2. WEAKNESSES IN THE “AD HOC SHOCK” METHOD

Ideally, reliable macroeconomic and financial data would be available for modeling the impact of external shocks on banks’ balance sheets in stress tests. Specifically, econometric models would be used to quantify the historical relationship between shocks to selected macroeconomic variables and nonperforming loans (NPLs).⁴ A variety of macro-scenarios would be applied, and their effect on NPLs and consequently on loan-loss provisions and capitalization would be estimated.

In the absence of such data, stress testers have to subjectively make assumptions about the size of shocks to banks’ loan

portfolios and possibly take other “shortcuts” in designing the top-down stress tests.⁵ The possible scenarios are as follows:

1. *Shocks to aggregate NPLs versus to loans by classification.* The shocks are applied directly to the aggregate NPLs of the banking system as a whole when more granular data on loan classifications are not available. Provisions may be estimated by using one of two methods, depending on data availability:
 - a. Assume provisions of 100 percent of NPLs when calculating the impact of the shocks on capital adequacy, in the absence of more granular information on provisioning requirements; or
 - b. Assume provisions using the average of performing and nonperforming provisioning rates, where information on provisioning requirements is available.
2. *Shocks to the banking system versus to individual banks.* When individual bank data are either not provided or incomplete, “back-of-the-envelope” stress tests are performed on the system as a whole with some form of aggregated data.
3. *Shocks of ever larger magnitudes to NPLs.* Credit shocks of increasingly larger magnitudes are applied to estimate the impact on capitalization. Such shocks are usually multiples of existing NPLs or involve increasingly larger proportions of performing loans becoming NPLs.

For the purposes of this study, ad hoc shocks, as defined in Table 4.3, are applied to each scenario, and the results are analyzed.

A. Analysis

The first question that should be asked prior to performing any stress test is whether the reported data are reliable, so as to determine the possible size of the shocks. If the information

³ Under the standardized approach, RWA would decline when problem loans are written off.

⁴ See Cihák (2007) for a discussion of the research using such models.

⁵ Our references to stress testers in this chapter apply to supervisors who perform stress testing and third parties, such as IMF staff, who perform stress testing as part of their surveillance work.

TABLE 4.3

Stress Test Assumptions: Ad Hoc Shocks to Asset Quality

Scenario	Type of Shock
1.	Shock to aggregate NPLs vs. to loans by classification: Aggregate NPLs increase by 400 percent with provisions assumed at 100 percent vs. each NPL classification increases by 400 percent with performing loans representing the remainder, distributed proportionally and with graduating provisions.
2.	Shock to the banking system vs. to individual banks: Each NPL classification increases by 400 percent with performing loans representing the remainder, distributed proportionally and with graduating provisions.
3.	Shocks of ever larger magnitudes to NPLs: Shock to NPLs NPLs increase by 100 percent. NPLs increase by 200 percent. NPLs increase by 400 percent. Shock to performing loans 10 percent of performing loans become NPLs. 20 percent of performing loans become NPLs. 40 percent of performing loans become NPLs.

Source: Authors.

Note: NPL = nonperforming loan.

is of sufficiently good quality, then shocks over the *short term* to banks' balance sheets should realistically be constrained by the amount of loans in each classification. By definition, past due loans typically migrate from one classification down to another over time, which means that the impact of any shock to credit quality—no matter how severe—should be limited to the outstanding amount in any one loan classification. The constraint of maximum possible loans migrating down classifications is sometimes overlooked in stress tests.

The maximum amount by which a loan category can increase in the *short term* following any shock should be equivalent to the balance in the category above it. As we show in Table 4.4, the worst possible shock to both performing and nonperforming loans (rows 6, 7, 9, and 10) should result in the total amount in each category moving down by one step (rows 25, 27, 28, and 29). With the exception of doubtful loans, other categories of loans do not become loss loans—requiring 100 percent provisioning—straightaway. The resulting impact on capitalization from the required increase in provisions appears relatively modest (row 41) at between 1.3 and 2.8 percentage points. Indeed, none of the banks in the example would be required to recapitalize following the shock. Therefore, to the extent that shocks applied to asset quality in each loan classification exceed the maximum possible amount as described above, the stress test must be assuming actual shocks plus some underreporting of NPLs or that the stress test horizon is over the medium to long term.

Another caveat is that loan books may be very different across banks and are thus affected differently when a shock occurs. For instance, a bank with a loan book that consists

predominantly of speculative commercial property is likely to be harder hit than one that has focused its lending on residential mortgages. Thus, the application of shocks of uniform magnitudes across banks may be highly unrealistic and likely uninformative. However, granular data typically are unavailable in countries where data reporting and collection are weak and incomplete. In this case, the design of the stress tests may have to take into account more qualitative information, such as anecdotal evidence about the composition of individual banks' loan books.

Scenario 1a: Shock to Aggregate NPLs Using a 100 Percent Provisioning Rate versus to Loans by Classification (Table 4.5)

Where the stress tester may not have more granular and accurate data on classified loans to work with, the tendency is to shock aggregate NPLs. The situation could occur if, say, banks do not adhere to reporting requirements for loan classifications or if supervisors do not make the data available to third-party stress testers. We demonstrate the inaccuracies in the results that could arise from shocking aggregate NPLs:

- In the absence of more detailed information on classified loans, simplistic assumptions are sometimes made. For instance, the stress tester may assume provisions at 100 percent of the additional NPLs (i.e., that all are loss loans) in order to calculate the additional provisions required (column 1, row 14) and ultimately the impact on capital. In our example, such an assumption, with a shock amounting to a 400 percent increase in NPLs, would result in systemwide CAR falling by 7 percentage points (column 1, row 17).
- There is significant downward bias when the aggregate NPL amount is used, compared with the alternate scenario, where more detailed data on classified loans data are available (column 2, rows 16, 18–21). In the latter situation, calculations of graduated provisions would be possible (column 2, rows 32, 34–37), resulting in a more moderate decline in CAR of 4.7 percentage points (column 2, row 41) following a 400 percent increase in NPLs. In other words, the estimated impact would be around two-thirds of that from using aggregate NPLs.

Scenario 1b: Shock to Aggregate NPLs Using an Average Provisioning Rate versus to Loans by Classification (Table 4.6)

A more accurate method for refining the preceding calculations may be to use the average rates for performing and nonperforming loans to determine provisions. In this example, we would use the average rate of 0.02 (arithmetic average of 0.01 and 0.03) to determine the required provisions for performing loans (column 1, row 10) and the average rate of 0.567 (arithmetic average of 0.2, 0.5, and 1.0) for calculating the required provisions for NPLs (column 1, rows 11 and 17):

TABLE 4.4

Ad Hoc Shock Stress Test: Maximum Possible Migration Down Classifications
 (in millions of domestic currency units unless stated otherwise)

		(1)	(2)	(3)	(4)	(5)	(6)	
Row Number and Formula	Item	Provisioning Rate	All Banks	Bank 1	Bank 2	Bank 3	Bank 4	Bank 5
	Preshock							
(1)	Capital		530.0	30.0	160.0	220.0	80.0	40.0
(2)	RWA		3,520.0	170.0	1,100.0	1,400.0	550.0	300.0
(3) = (1)/(2)*100	CAR (in percent)		15.1	17.6	14.5	15.7	14.5	13.3
(4) = (5) + (8)	Total loans		1,448.0	71.0	385.0	615.0	287.0	90.0
(5) = (6) + (7)	Performing loans		1,375.0	65.0	365.0	590.0	275.0	80.0
(6)	Normal and pass loans		1,250.0	55.0	330.0	530.0	260.0	75.0
(7)	Special mention loans		125.0	10.0	35.0	60.0	15.0	5.0
(8) = (9) + (10) + (11)	NPLs		73.0	6.0	20.0	25.0	12.0	10.0
(9)	Substandard loans		26.0	3.0	10.0	5.0	5.0	3.0
(10)	Doubtful loans		20.0	2.0	5.0	10.0	2.0	1.0
(11)	Loss loans		27.0	1.0	5.0	10.0	5.0	6.0
(12) = (8)/(4)	NPL ratio (in percent)		5.0	8.5	5.2	4.1	4.2	11.1
(13)	Total provisions currently held ¹		58.5	3.5	13.9	23.1	10.1	8.0
(14) = (15) + (17)	Total provisions that should be held		58.5	3.5	13.9	23.1	10.1	8.0
(15) = (16)	General provision		12.5	0.6	3.3	5.3	2.6	0.8
(16) = (6)*Rate	against normal and pass loans	0.01	12.5	0.6	3.3	5.3	2.6	0.8
(17) = (18) + (19) + (20) + (21)	Specific provision		46.0	2.9	10.6	17.8	7.5	7.3
(18) = (7)*Rate	against special mention loans	0.03	3.8	0.3	1.1	1.8	0.5	0.2
(19) = (9)*Rate	against substandard loans	0.20	5.2	0.6	2.0	1.0	1.0	0.6
(20) = (10)*Rate	against doubtful loans	0.50	10.0	1.0	2.5	5.0	1.0	0.5
(21) = (11)*Rate	against loss loans	1.00	27.0	1.0	5.0	10.0	5.0	6.0
(22) = (13) – (14)	Under/overprovisioning		0.0	0.0	0.0	0.0	0.0	0.0
	Postshock							
	Shock: All loans migrate from one classification down to the next							
= (4)	Total loans		1,448.0	71.0	385.0	615.0	287.0	90.0
(23) = (4) – (26)	Performing loans		1,250.0	55.0	330.0	530.0	260.0	75.0
(24) = 0	Normal and pass loans		0.0	0.0	0.0	0.0	0.0	0.0
(25) = (6)	Special mention loans		1,250.0	55.0	330.0	530.0	260.0	75.0
(26) = (27) + (28) + (29)	NPLs ²		198.0	16.0	55.0	85.0	27.0	15.0
(27) = (7)	Substandard loans		125.0	10.0	35.0	60.0	15.0	5.0
(28) = (9)	Doubtful loans		26.0	3.0	10.0	5.0	5.0	3.0
(29) = (11) + (10)	Loss loans		47.0	3.0	10.0	20.0	7.0	7.0
= (13)	Total provisions currently held		58.5	3.5	13.9	23.1	10.1	8.0
(30) = (31) + (33)	Total provisions that should be held		122.5	8.2	31.9	50.4	20.3	11.8
(31) = (32)	General provision		0.0	0.0	0.0	0.0	0.0	0.0
(32) = (24)*Rate	against normal and pass loans	0.01	0.0	0.0	0.0	0.0	0.0	0.0
(33) = (34) + (35) + (36) + (37)	Specific provision		122.5	8.2	31.9	50.4	20.3	11.8
(34) = (25)*Rate	against special mention loans	0.03	37.5	1.7	9.9	15.9	7.8	2.3
(35) = (9)*Rate	against substandard loans	0.20	25.0	2.0	7.0	12.0	3.0	1.0
(36) = (10)*Rate	against doubtful loans	0.50	13.0	1.5	5.0	2.5	2.5	1.5
(37) = (11)*Rate	against loss loans	1.00	47.0	3.0	10.0	20.0	7.0	7.0
(38) = (13) – (30)	Under-/over-provisioning		–64.1	–4.7	–18.1	–27.3	–10.3	–3.8
	Assuming full provisioning after shock							
(39) = (1) + (38)	New capital		466.0	25.3	142.0	192.7	69.8	36.3
(40) = (39)/(2)	New CAR (in percent)³		13.2	14.9	12.9	13.8	12.7	12.1
(41) = (40) – (3)	Impact on CAR (in percent)		–1.8	–2.8	–1.6	–2.0	–1.9	–1.3

Source: Authors.

Note: CAR = capital adequacy ratio; NPL = nonperforming loan; RWA = risk-weighted assets.

1. It is assumed that NPLs are fully provisioned for initially.

2. It is assumed that NPLs increase proportionately across all categories.

3. It is assumed that RWA remains the same.

TABLE 4.5

Scenario 1a: Ad Hoc Shock to Aggregate NPLs Using a 100 Percent Provisioning Rate vs. to Loans by Classification
(in millions of domestic currency units unless stated otherwise)

Item	(1)			(2)		
	Shock to Aggregate NPLs			Shock to NPLs by Classification		
	Row Number and Formula	Provisioning Rate	All Banks	Row Number and Formula	Provisioning Rate	All Banks
Preshock						
Capital	(1)		530.0	(1)		530.0
RWA	(2)		3,520.0	(2)		3,520.0
CAR (in percent)	(3) = (1)/(2)*100		15.1	(3) = (1)/(2)*100		15.1
Total loans	(4) = (5) + (6)		1,448.0	(4) = (5) + (8)		1,448.0
Performing loans	(5)		1,375.0	(5) = (6) + (7)		1,375.0
<i>Normal and pass loans</i>				(6)		1,250.0
<i>Special mention loans</i>				(7)		125.0
NPLs	(6)		73.0	(8) = (9) + (10) + (11)		73.0
<i>Substandard loans</i>				(9)		26.0
<i>Doubtful loans</i>				(10)		20.0
<i>Loss loans</i>				(11)		27.0
NPL ratio (in percent)	(7) = (6)/(4)*100		5.0	(12)		5.0
Total provisions currently held¹	(8)		58.5	(13)		58.5
Total provisions that should be held	(9) = (6)*Rate	1.00	73.0	(14) = (15) + (17)		58.5
General provision				(15)		12.5
<i>against normal and pass loans</i>				(16) = (6)*Rate	0.01	12.5
Specific provision				(17) = (18) + (19) + (20) + (21)		46.0
<i>against special mention loans</i>				(18) = (7)*Rate	0.03	3.8
<i>against substandard loans</i>				(19) = (9)*Rate	0.20	5.2
<i>against doubtful loans</i>				(20) = (10)*Rate	0.50	10.0
<i>against loss loans</i>				(21) = (11)*Rate	1.00	27.0
Under-/over-provisioning	(10) = (8) – (9)		–14.6	(22) = (13) – (14)		0.0
Postshock						
Shock: NPLs increase by “x” percent			400.0			400.0
Total loans	= (4)		1,448.0	= (4)		1,448.0
Performing loans	(11) = (4) – (12)		1,083.0	(23) = (4) – (26)		1,083.0
<i>Normal and pass loans</i>				(24) = (23)*(6)/(5)		984.5
<i>Special mention loans</i>				(25) = (23)*(7)/(5)		98.5
NPLs²	(12) = (6) + (1 + Shock/100)		365.0	(26) = (27) + (28) + (29)		365.0
<i>Substandard loans</i>				(27) = (9)*(1 + Shock/100)		130.0
<i>Doubtful loans</i>				(28) = (10)*(1 + Shock/100)		100.0
<i>Loss loans</i>				(29) = (11)*(1 + Shock/100)		135.0
Total provisions currently held	= (8)		58.5	= (13)		58.5
Total provisions that should be held	(13) = {(12) – (8)}*Rate	1.00	306.6	(30) = (31) + (33)		223.8
General provision				(31) = (32)		9.8
<i>against normal and pass loans</i>				(32) = (24)*Rate	0.01	9.8
Specific provision				(33) = (34) + (35) + (36) + (37)		214.0
<i>against special mention loans</i>				(34) = (25)*Rate	0.03	3.0
<i>against substandard loans</i>				(35) = (9)*Rate	0.20	26.0
<i>against doubtful loans</i>				(36) = (10)*Rate	0.50	50.0
<i>against loss loans</i>				(37) = (11)*Rate	1.00	135.0
Under-/over-provisioning	(14) = (8) – (13)		–248.1	(38) = (13) – (30)		–165.3
Assuming full provisioning after shock						
New capital	(15) = (1) + (14)		281.9	(39) = (1) + (38)		364.7
New CAR (in percent)³	(16) = (15)/(2)*100		8.0	(40) = (39)/(2)*100		10.4
Impact on CAR (in percent)	(17) = (15) – (3)		–7.0	(41) = (40) – (3)		–4.7

Source: Authors.

Note: CAR = capital adequacy ratio; NPL = nonperforming loan; RWA = risk-weighted assets.

1. It is assumed that NPLs are underprovisioned for initially in the case of aggregate NPLs because 100 percent provisioning is assumed; NPLs are fully provisioned for initially in the case where NPL data by classification are available.

2. Where data on classified loans are available, it is assumed that NPLs increase by “x” percent across classifications; balance of performing loans are distributed proportionately across classifications.

3. It is assumed that RWA remains the same.

TABLE 4.6

Scenario 1b: Ad Hoc Shock to Aggregate NPLs Using an Average Provisioning Rate vs. to Loans by Classification
(in millions of domestic currency units unless stated otherwise)

Item	(1)			(2)		
	Shock to Aggregate NPLs			Shock to NPLs by Classification		
	Row Number and Formula	Provisioning Rate	All Banks	Row Number and Formula	Provisioning Rate	All Banks
Preshock						
Capital	(1)		530.0	(1)		530.0
RWA	(2)		3,520.0	(2)		3,520.0
CAR (in percent)	(3) = (1)/(2)*100		15.1	(3) = (1)/(2)*100		15.1
Total loans	(4) = (5) + (6)		1,448.0	(4) = (5) + (8)		1,448.0
Performing loans	(5)		1,375.0	(5) = (6) + (7)		1,375.0
<i>Normal and pass loans</i>				(6)		1,250.0
<i>Special mention loans</i>				(7)		125.0
NPLs	(6)		73.0	(8) = (9) + (10) + (11)		73.0
<i>Substandard loans</i>				(9)		26.0
<i>Doubtful loans</i>				(10)		20.0
<i>Loss loans</i>				(11)		27.0
NPL ratio (in percent)	(7) = (6)/(4)*100		5.0	(12)		5.0
Total provisions currently held ¹	(8)		58.5	(13)		58.5
Total provisions that should be held	(9) = (10) + (11)		68.9	(14) = (15) + (17)		58.5
General provision <i>against normal and pass loans</i>				(15) = (16)		12.5
Specific provision <i>against special mention loans</i>	(10) = (5)*Avg PL rate	0.020	27.5	(16) = (6)*Rate	0.01	12.5
				(17) = (18) + (19) + (20) + (21)		46.0
				(18) = (7)*Rate	0.03	3.8
<i>against substandard loans</i>				(19) = (9)*Rate	0.20	5.2
<i>against doubtful loans</i>	(11) = (6)*Avg NPL rate	0.567	41.4	(20) = (10)*Rate	0.50	10.0
<i>against loss loans</i>				(21) = (11)*Rate	1.00	27.0
Under-/over-provisioning	(12) = (8) – (9)		–10.4	(22) = (13) – (14)		0.0
Postshock						
Shock: NPLs increase by “x” percent			400.0			400.0
Total loans	= (4)		1,448.0	= (4)		1,448.0
Performing loans	(13) = (4) – (14)		1,083.0	(23) = (4) – (26)		1,083.0
<i>Normal and pass loans</i>				(24) = (23)*(6)/(5)		984.5
<i>Special mention loans</i>				(25) = (23)*(7)/(5)		98.5
NPLs ²	(14) = (6)*(1 + Shock/100)		365.0	(26) = (27) + (28) + (29)		365.0
<i>Substandard loans</i>				(27) = (9)*(1 + Shock/100)		130.0
<i>Doubtful loans</i>				(28) = (10)*(1 + Shock/100)		100.0
<i>Loss loans</i>				(29) = (11)*(1 + Shock/100)		135.0
Total provisions currently held	= (8)		58.5	= (13)		58.5
Total provisions that should be held	(15) = (16) + (17)		228.5	(30) = (31) + (33)		223.8
General provision <i>against normal and pass loans</i>				(31) = (32)		9.8
Specific provision <i>against special mention loans</i>	(16) = (13)*Avg PL rate	0.020	21.7	(32) = (24)*Rate	0.01	9.8
				(33) = (34) + (35) + (36) + (37)		214.0
				(34) = (25)*Rate	0.03	3.0
<i>against substandard loans</i>				(35) = (9)*Rate	0.20	26.0
<i>against doubtful loans</i>	(17) = (14)*Avg NPL rate	0.567	206.8	(36) = (10)*Rate	0.50	50.0
<i>against loss loans</i>				(37) = (11)*Rate	1.00	135.0
Under-/over-provisioning	(18) = (8) – (15)		–170.0	(38) = (13) – (30)		–165.3
Assuming full provisioning after shock						
New capital	(19) = (1) + (18)		360.0	(39) = (1) + (38)		364.7
New CAR (in percent)³	(20) = (19)/(2)*100		10.2	(40) = (39)/(2)*100		10.4
Impact on CAR (in percent)	(21) = (20) – (3)		–4.8	(41) = (40) – (3)		–4.7

Source: Authors.

Note: CAR = capital adequacy ratio; NPL = nonperforming loan; RWA = risk-weighted assets.

1. It is assumed that NPLs are underprovisioned for initially in the case of aggregate NPLs because an average of the provisioning rates is assumed; NPLs are fully provisioned for initially in the case where NPL data by classification are available.

2. Where data on classified loans are available, it is assumed that NPLs increase by “x” percent across classifications; balance of performing loans are distributed proportionately across classifications.

3. It is assumed that RWA remains the same.

- The result is almost identical to that from using more detailed classifications, with CAR falling by 4.8 percentage points (column 1, row 21) compared with 4.7 percentage points (column 2, row 41).
- More generally, however, the similarity of the impact between the two methods would depend on the distribution across NPL classifications.

Scenario 2: Shock to the Banking System versus to Individual Banks (Table 4.7)

Another stumbling block in stress testing may be the lack of availability of information on individual banks. Sometimes, supervisors may not be inclined to share the bank-by-bank information with third-party stress testers, usually for confidentiality reasons. In such instances, the stress tester would use data for the aggregate banking system, in which case the stress test findings would need to be interpreted with caution:

- The information derived from shocks to the aggregate system could mask problems among individual banks. In our example, a shock representing a 400 percent increase in NPLs across the board would result in the system's CAR declining by 4.7 percentage points to 10.4 percent (column 1, rows 41 and 40, respectively), that is, 1.6 percentage points below the required minimum of 12 percent.
- Closer examination of the impact on individual banks shows significantly varied outcomes. The CAR of Bank 5 has declined by a massive 9.3 percentage points to 4 percent (column 6, rows 41 and 40, respectively), while the capitalization of Bank 2 has fallen by 3.4 percentage points to 11.2 percent (column 3, rows 41 and 40, respectively), not far below the required minimum 12 percent. Thus, focusing on the aggregate outcome alone could obscure the possibility that a particular institution may be very vulnerable with potentially systemic consequences.

Scenario 3: Shocks of Increasingly Larger Magnitudes to NPLs (Table 4.8)

In the absence of good-quality and sufficient historical data to model the relationship between macroeconomic developments and credit risk, the *size* of shocks applied in stress tests often lacks foundation or justification. Although historical experience could be used as a guide, many nascent banking sectors may not have experienced a complete business cycle; shocks also tend to be different from one crisis to the next. Thus, stress testers typically would apply increasingly larger shocks to estimate their impact on capital and then conclude that the banks or banking system may be vulnerable. Such stress tests seem to overlook the obvious algebraic relationship between NPLs and CARs, that is, the larger the increase in NPLs, the greater the decrease in CARs.

Using a set of increasingly larger shocks to NPLs (row 11), we demonstrate their impact on banking system and indi-

vidual bank CARs.⁶ As expected, the impact on individual banks' CARs increases as NPLs rise from 100 to 400 percent and as the amount of performing loans becoming NPLs increases from 10 to 40 percent (columns 1–6, row 14). Put another way, the CARs deteriorate as a matter of course when the shocks increase in magnitude, all to the point of falling below the required capitalization levels and, in some cases, significantly so.

The key weakness to the increasingly larger ad hoc shocks approach lies in the relevance of the results. It would be impossible to infer that the banks and banking system are significantly vulnerable in cases where the shocks translate to significant undercapitalization, as there would be little empirical evidence to support the plausible occurrence of tail shocks of such large magnitudes. As a result, it could be very difficult for the stress tester to make constructive recommendations on actions to be taken in response to the findings.

B. Summary of findings

Clearly, simple stress tests using ad hoc and extreme shocks are flawed, which begs the question of how useful they may be for risk management and contingency planning purposes. Specific caveats are as follows:

- *Any assumption of 100 percent provisioning following a shock would significantly overstate the amount of additional provisions required and thus underestimate the resulting capitalization.* NPLs are classified according to the lateness in debt service, and different provisioning rates apply across the different loan classifications. When a shock occurs, the quality of loans typically moves from one classification down to the next over the short term, with a graduating rise in the provisioning rate, rather than becoming loss loans—with a 100 percent provisioning requirement—straightaway.
- *The soundness of individual banks' balance sheets varies considerably across a particular financial system, and uniform shocks to aggregate banking system data would likely yield less than useful results.* In many financial systems, the quality of banks ranges from those that are well capitalized and well managed with conservative business models and sound risk management systems, to those that are weak, risk seeking, and profligate. Thus, the application of a particular shock to the aggregate system runs the risk that supervisors may base their contingency planning decisions on potentially meaningless information from the stress tests.
- *Applied shocks, if sufficiently large, would break any bank or banking system in the world, which suggests that such results may not be instructive.* The laws of algebra should show that the larger the ad hoc shock to NPLs, the greater its flow-through impact on capitalization, to the point where the banks appear severely undercapitalized. However, it is unclear what should be inferred

⁶ The shocks are described in Table 4.3.

TABLE 4.7

Scenario 2: Ad Hoc Shock to the Banking System vs. to Individual Banks (in millions of domestic currency units unless stated otherwise)								
			(1)	(2)	(3)	(4)	(5)	(6)
Row Number and Formula	Item	Provisioning Rate	All Banks	Bank 1	Bank 2	Bank 3	Bank 4	Bank 5
	Preshock							
(1)	Capital		530.0	30.0	160.0	220.0	80.0	40.0
(2)	RWA		3,520.0	170.0	1,100.0	1,400.0	550.0	300.0
(3) = (1)/(2)*100	CAR (in percent)		15.1	17.6	14.5	15.7	14.5	13.3
(4) = (5) + (8)	Total loans		1,448.0	71.0	385.0	615.0	287.0	90.0
(5) = (6) + (7)	Performing loans		1,375.0	65.0	365.0	590.0	275.0	80.0
(6)	Normal and pass loans		1,250.0	55.0	330.0	530.0	260.0	75.0
(7)	Special mention loans		125.0	10.0	35.0	60.0	15.0	5.0
(8) = (9) + (10) + (11)	NPLs		73.0	6.0	20.0	25.0	12.0	10.0
(9)	Substandard loans		26.0	3.0	10.0	5.0	5.0	3.0
(10)	Doubtful loans		20.0	2.0	5.0	10.0	2.0	1.0
(11)	Loss loans		27.0	1.0	5.0	10.0	5.0	6.0
(12) = (8)/(4)*100	NPL ratio (in percent)		5.0	8.5	5.2	4.1	4.2	11.1
(13)	Total provisions currently held ¹		58.5	3.5	13.9	23.1	10.1	8.0
(14) = (15) + (17)	Total provisions that <i>should</i> be held		58.5	3.5	13.9	23.1	10.1	8.0
(15) = (16)	General provision		12.5	0.6	3.3	5.3	2.6	0.8
(16) = (6)*Rate	against normal and pass loans	0.01	12.5	0.6	3.3	5.3	2.6	0.8
(17) = (18) + (19) + (20) + (21)	Specific provision		46.0	2.9	10.6	17.8	7.5	7.3
(18) = (7)*Rate	against special mention loans	0.03	3.8	0.3	1.1	1.8	0.5	0.2
(19) = (9)*Rate	against substandard loans	0.20	5.2	0.6	2.0	1.0	1.0	0.6
(20) = (10)*Rate	against doubtful loans	0.50	10.0	1.0	2.5	5.0	1.0	0.5
(21) = (11)*Rate	against loss loans	1.00	27.0	1.0	5.0	10.0	5.0	6.0
(22) = (13) – (14)	Under/overprovisioning		0.0	0.0	0.0	0.0	0.0	0.0
	Postshock							
	Shock: NPLs increase by 400 percent across loan classifications; balance of performing loans are distributed proportionately		400.0	400.0	400.0	400.0	400.0	400.0
= (4)	Total loans		1,448.0	71.0	385.0	615.0	287.0	90.0
(23) = (4) – (26)	Performing loans		1,083.0	41.0	285.0	490.0	227.0	40.0
(24)	Normal and pass loans		984.7	34.7	257.7	440.2	214.6	37.5
(25) = (7)*(22)/100	Special mention loans		98.3	6.3	27.3	49.8	12.4	2.5
(26) = (27) + (28) + (29)	NPLs ²		365.0	30.0	100.0	125.0	60.0	50.0
(27) = (9)*(1 + Shock/100)	Substandard loans		130.0	15.0	50.0	25.0	25.0	15.0
(28) = (10)*(1 + Shock/100)	Doubtful loans		100.0	10.0	25.0	50.0	10.0	5.0
(29) = (11)*(1 + Shock/100)	Loss loans		135.0	5.0	25.0	50.0	25.0	30.0
= (13)	Total provisions currently held		58.5	3.5	13.9	23.1	10.1	8.0
(30) = (31) + (33)	Total provisions that <i>should</i> be held		223.8	13.5	50.9	85.9	37.5	36.0
(31) = (32)	General provision		9.8	0.3	2.6	4.4	2.1	0.4
(32) = (24)*Rate	against normal and pass loans	0.01	9.8	0.3	2.6	4.4	2.1	0.4
(33) = (34) + (35) + (36) + (37)	Specific provision		214.0	13.2	48.3	81.5	35.4	35.6
(34) = (25)*Rate	against special mention loans	0.03	3.0	0.2	0.8	1.5	0.4	0.1
(35) = (9)*Rate	against substandard loans	0.20	26.0	3.0	10.0	5.0	5.0	3.0
(36) = (10)*Rate	against doubtful loans	0.50	50.0	5.0	12.5	25.0	5.0	2.5
(37) = (11)*Rate	against loss loans	1.00	135.0	5.0	25.0	50.0	50.0	30.0
(38) = (13) – (30)	Under/overprovisioning		–165.3	–10.1	–37.0	–62.8	–27.5	–28.0
	Assuming full provisioning after shock							
(39) = (1) + (38)	New capital		364.7	19.9	123.0	157.2	52.5	12.1
(40) = (39)/(2)*100	New CAR (in percent)³		10.4	11.7	11.2	11.2	9.6	4.0
(41) = (40) – (3)	Impact on CAR (in percent)		–4.7	–5.9	–3.4	–4.5	–5.0	–9.3

Source: Authors.

Note: CAR = capital adequacy ratio; NPL = nonperforming loan; RWA = risk-weighted assets.

1. It is assumed that NPLs are fully provisioned for initially.

2. It is assumed that NPLs increase proportionately across all categories.

3. It is assumed that RWA remains the same.

TABLE 4.8

Scenario 3: Ad Hoc Shocks to NPLs of Increasingly Larger Magnitudes
(in millions of domestic currency units unless stated otherwise)

		(1)	(2)	(3)	(4)	(5)	(6)	
Row Number and Formula	Item	Shock	All Banks	Bank 1	Bank 2	Bank 3	Bank 4	Bank 5
Preshock								
(1)	Capital		530.0	30.0	160.0	220.0	80.0	40.0
(2)	RWA		3,520.0	170.0	1,100.0	1,400.0	550.0	300.0
(3) = (1)/(2)*100	CAR (in percent)		15.1	17.6	14.5	15.7	14.5	13.3
(4)	Total loans		1,448.0	71.0	385.0	615.0	287.0	90.0
(5)	Performing loans		1,375.0	65.0	365.0	590.0	275.0	80.0
(6)	NPLs		73.0	6.0	20.0	25.0	12.0	10.0
(7) = (6)/(4)*100	NPL ratio (in percent)		5.0	8.5	5.2	4.1	4.2	11.1
(8)	Total provisions currently held ¹		58.5	3.5	13.9	23.1	10.1	8.0
(9)	Total provisions that should be held		58.5	3.5	13.9	23.1	10.1	8.0
(10) = (8) – (9)	Under-/over-provisioning		0.0	0.0	0.0	0.0	0.0	0.0
Postshock								
= (4)	Total loans		1,448.0	71.0	385.0	615.0	287.0	90.0
(11)	Total NPLs							
= (6)*(1 + Shock)	NPLs increase by 100 percent	1.0	146.0	12.0	40.0	50.0	24.0	20.0
= (6)*(1 + Shock)	NPLs increase by 200 percent	2.0	219.0	18.0	60.0	75.0	36.0	30.0
= (6)*(1 + Shock)	NPLs increase by 400 percent	4.0	365.0	30.0	100.0	125.0	60.0	50.0
= (6) + {(5)*Shock}	10 percent of performing loans become NPLs	0.1	210.5	12.5	56.5	84.0	39.5	18.0
= (6) + {(5)*Shock}	20 percent of performing loans become NPLs	0.2	348.0	19.0	93.0	143.0	67.0	26.0
= (6) + {(5)*Shock}	40 percent of performing loans become NPLs	0.4	623.0	32.0	166.0	261.0	122.0	42.0
= (8)	Total provisions currently held		58.5	3.5	13.9	23.1	10.1	8.0
(12) = (11) – (8)	Under-/over-provisioning							
	NPLs increase by 100 percent		–42.2	–2.6	–9.5	–15.5	–7.2	–7.5
	NPLs increase by 200 percent		–85.9	–5.3	–19.4	–32.0	–14.4	–14.8
	NPLs increase by 400 percent		–173.3	–10.8	–39.2	–65.1	–28.9	–29.4
	10 percent of performing loans become NPLs		–81.0	–2.8	–17.7	–38.0	–16.5	–6.0
	20 percent of performing loans become NPLs		–163.5	–5.8	–35.8	–77.0	–33.1	–11.8
	40 percent of performing loans become NPLs		–328.5	–11.7	–72.0	–155.0	–66.3	–23.5
Assuming full provisioning postshock								
(13) = (1)–(12)	New capital							
	NPLs increase by 100 percent		487.8	27.4	150.5	204.5	72.8	32.5
	NPLs increase by 200 percent		444.1	24.7	140.6	188.0	65.6	25.2
	NPLs increase by 400 percent		356.7	19.2	120.8	154.9	51.1	10.6
	10 percent of performing loans become NPLs		449.0	27.2	142.3	182.1	63.5	34.0
	20 percent of performing loans become NPLs		366.5	24.2	124.2	143.1	46.9	28.2
	40 percent of performing loans become NPLs		201.5	18.3	88.0	65.1	13.7	16.5
(14) = (13)/(2)*100	New CAR (in percent)²							
	NPLs increase by 100 percent		13.9	16.1	13.7	14.6	13.2	10.8
	NPLs increase by 200 percent		12.6	14.5	12.8	13.4	11.9	8.4
	NPLs increase by 400 percent		10.1	11.3	11.0	11.1	9.3	3.5
	10 percent of performing loans become NPLs		12.8	16.0	12.9	13.0	11.5	11.3
	20 percent of performing loans become NPLs		10.4	14.2	11.3	10.2	8.5	9.4
	40 percent of performing loans become NPLs		5.7	10.8	8.0	4.6	2.5	5.5
(15) = (14) – (3)	Impact on CAR (in percent)							
	NPLs increase by 100 percent		–1.2	–1.5	–0.9	–1.1	–1.3	–2.5
	NPLs increase by 200 percent		–2.4	–3.1	–1.8	–2.3	–2.6	–4.9
	NPLs increase by 400 percent		–4.9	–6.4	–3.6	–4.6	–5.3	–9.8
	10 percent of performing loans become NPLs		–2.3	–1.7	–1.6	–2.7	–3.0	–2.0
	20 percent of performing loans become NPLs		–4.6	–3.4	–3.3	–5.5	–6.0	–3.9
	40 percent of performing loans become NPLs		–9.3	–6.9	–6.5	–11.1	–12.1	–7.8

Source: Authors.

Note: CAR = capital adequacy ratio; NPL = nonperforming loan; RWA = risk-weighted assets.

1. It is assumed that loans are fully provisioned for initially.

2. It is assumed that RWA remains the same.

from such stress tests and how the findings could be applied usefully, given that the quality of the raw data usually precludes any ability to quantify or justify such shocks.

3. A PROPOSED COMPROMISE: THE “BREAKING POINT” METHOD

We subsequently propose a possibly more useful method for determining banks’ CAR, which could be more informative for supervisors in situations where the reporting of NPLs by banks is unreliable. The “breaking point” method is essentially a “stressing until it breaks” exercise, which is also known as reverse stress testing. This particular method of analysis is intuitively appealing in that it (1) does not depend heavily on the quality of reported data;⁷ and (2) does not require any assumption with regard to the size of the overall NPL shock(s). It estimates the amounts of classified loans that would reduce a bank’s CAR to the “breaking point”—in our example, 12 percent—below which recapitalization would be necessary. We contemplate two situations: (1) only aggregate NPL data are available; and (2) granular data on loan classifications and provisioning are available.

It should be emphasized that even the breaking point method cannot be used as a stand-alone stress test. Rather, the findings from applying this particular method of stress testing would need to be complemented by information gathered from bank examinations performed by on-site supervisors. Thus, in situations where the reported data are of poor quality, the off- and on-site teams would have to collaborate even more closely, and any stress test result can provide only guidance to the latter on which bank(s) may be undercapitalized, given their findings during on-site inspections.

Scenario A: Shock to Aggregate NPLs (Table 4.9)

Application of the breaking point method is very straightforward when only aggregate data for the banking system are available. We estimate algebraically the aggregate NPL ratio for the banking system that would bring the CAR down to 12 percent. Naturally, the calculation of provisions that should be held in the banking system following the shock would play an important role in determining postshock capital. As in Scenario 1b earlier, we use the average rates for performing and nonperforming loans to calculate the new provision amount required. The results show that

- The breaking point NPL ratio for the banking system as a whole is around 17.5 percent (row 13), compared with the current 5 percent (row 7).

⁷ It should be noted that the existing estimates of CARs of banks may also be unreliable. For instance, banks may not have adhered to regulatory requirements in determining risk-weighted assets or in their calculation of regulatory capital.

- NPLs would have to increase by almost 250 percent from current levels (row 17).

The *aggregate* breaking point information is not useful as a complementary on-site supervision tool. On-site examiners would not be able to compare their findings on individual banks with the aggregate figure to arrive at any relevant conclusion. As we show in the next scenario, the breaking point is likely to vary widely across individual banks, depending on the state of their balance sheets.

Scenario B: Shock to Loans by Classification (Table 4.10)

Where classified loans data are available, the breaking point method would require some basic assumptions with regard to the magnitude of shocks to each category of loans in order to estimate the breaking point NPL ratio. In this case, we assume that

1. A certain percentage of existing performing loans become NPLs for each of the five banks in our sample (columns 2–6, row 24).
2. The new NPL amounts for each bank (columns 2–6, rows 25–27) remain in the same proportions as the preshock balances (columns 2–6, rows 9–11).
3. The balance (columns 2–6, row 28), which represents performing loans, also remains in the same proportions for each bank (columns 2–6, rows 29 and 30) as the preshock amounts (columns 2–6, rows 6 and 7).

The percentage that reduces the CAR to the 12 percent threshold is the “breaking point” for the individual banks and the banking system (columns 1–6, row 23). From our example, off-site supervisors should observe the following:

- *The breaking point NPL ratio for the aggregate banking system should serve only as a guide to the overall health of banks.* As discussed in the previous section and in Scenario A, the aggregate numbers mask developments at individual institutions.
- *The resilience of individual banks in the system to shocks varies widely.* Breaking point NPL ratios for the banks range from around 12.7 percent for Bank 4 (column 5, row 23) to 40.5 percent for Bank 1 (column 2, row 23) and 18.5 percent for the banking system as a whole (column 1, row 23).
- *The relative size and composition of classified loans affect bank soundness.* For example, Bank 2 and Bank 4 each has a preshock CAR of 14.5 percent (columns 3 and 5, row 3), and both are fully provisioned (columns 3 and 5, rows 13 and 14), yet Bank 2 would be able to absorb any credit shock up to an NPL ratio of 20.7 percent (column 3, row 23), whereas Bank 4—which has a larger credit exposure relative to capital—would be at the CAR threshold when its NPL ratio reaches 12.7 percent (column 5, row 23).

TABLE 4.9

Scenario A: Breaking Point Shock to Aggregate NPLs (in millions of domestic currency units unless stated otherwise)			
Item	Row Number and Formula	Provisioning Rate	All Banks
Preshock			
Capital	(1)		530.0
RWA	(2)		3,520.0
CAR (in percent)	(3) = (1)/(2)*100		15.1
Total loans	(4) = (5) + (6)		1,448.0
Performing loans	(5)		1,375.0
NPLs	(6)		73.0
NPL ratio (in percent)	(7) = (6)/(4)*100		5.0
Total provisions currently held ¹	(8)		58.5
Total provisions that <i>should</i> be held	(9) = (10) + (11)		68.9
General provision against normal and pass loans	(10) = (5)*Avg PL rate	0.020	27.5
Specific provision against special mention loans			
against substandard loans	(11) = (6)*Avg NPL rate	0.567	41.4
against doubtful loans			
against loss loans			
Under-/over-provisioning	(12) = (8) – (9)		–10.4
Postshock			
Shock: NPL ratio (in percent)	(13)		17.5
Total loans	= (4)		1,448.0
Total NPLs ²	(14) = (13)/100*(4)		253.4
Balance of loans that are performing	(15) = (4) – (14)		1,194.6
Increase in NPLs	(16) = (14) – (6)		180.4
Rate of increase in NPLs (in percent)	(17) = {(14)/(6) – 1}*100		247.1
Total provisions currently held	= (8)		58.5
Total provisions that <i>should</i> be held	(18) = (19) + (20)		167.5
General provision against normal and pass loans	(19) = (15)*Avg PL rate	0.020	23.9
Specific provision against special mention loans			
against substandard loans	(20) = (6)*Avg NPL rate	0.567	143.6
against doubtful loans			
against loss loans			
Under-/over-provisioning	(21) = (8) – (18)		–109.0
Assuming full provisioning after shock			
New capital amount required	(22) = (1) + (21)		421.0
New CAR (in percent)²	(23) = (22)/(2)*100		12.0

Source: Authors.

Note: CAR = capital adequacy ratio; NPL = nonperforming loan; RWA = risk-weighted assets.

1. It is assumed that loans are fully provisioned for initially.

2. It is assumed that RWA remains the same.

4. CONCLUSION

Stress testing is increasingly seen as a key supervision, risk management, and surveillance tool in the wake of the global financial crisis. Indeed, stress testing almost seems obligatory these days, irrespective of the level of development of the financial system. Even in countries with largely underdeveloped financial systems—where the accounting and auditing frame-

work is weak, the reporting infrastructure is inadequate, and oversight, implementation, and enforcement of laws and regulations are poor—stress testing has become de rigueur.

Stress testing with insufficient, incomplete, and unreliable data yields results that may have little application. As we demonstrate throughout this chapter, the outcomes and interpretations of stress tests may be very different depending on the granularity of the data available, either in terms of

TABLE 4.10

Scenario B: Breaking Point Shock to Loans by Classification
(in millions of domestic currency units unless stated otherwise)

			(1)	(2)	(3)	(4)	(5)	(6)
Row Number and Formula	Item	Provisioning Rate	All Banks	Bank 1	Bank 2	Bank 3	Bank 4	Bank 5
Pres shock								
(1)	Capital		530.0	30.0	160.0	220.0	80.0	40.0
(2)	RWA		3,520.0	170.0	1,100.0	1,400.0	550.0	300.0
(3) = (1)/(2)*100	CAR (in percent)		15.1	17.6	14.5	15.7	14.5	13.3
(4) = (5) + (6)	Total loans		1,448.0	71.0	385.0	615.0	287.0	90.0
(5) = (6) + (7)	Performing loans		1,375.0	65.0	365.0	590.0	275.0	80.0
(6)	Normal and pass loans		1,250.0	55.0	330.0	530.0	260.0	75.0
(7)	Special mention loans		125.0	10.0	35.0	60.0	15.0	5.0
(8) = (9) + (10) + (11)	NPLs		73.0	6.0	20.0	25.0	12.0	10.0
(9)	Substandard loans		26.0	3.0	10.0	5.0	5.0	3.0
(10)	Doubtful loans		20.0	2.0	5.0	10.0	2.0	1.0
(11)	Loss loans		27.0	1.0	5.0	10.0	5.0	6.0
(12) = (8)/(4)*100	NPL ratio (in percent)		5.0	8.5	5.2	4.1	4.2	11.1
(13) = (14)	Total provisions currently held ¹		58.5	3.5	13.9	23.1	10.1	8.0
(14) = (15) + (17)	Total provisions that <i>should</i> be held		58.5	3.5	13.9	23.1	10.1	8.0
(15) = (16)	General provision		12.5	0.6	3.3	5.3	2.6	0.8
(16) = (6)*Rate	against normal and pass loans	0.01	12.5	0.6	3.3	5.3	2.6	0.8
(17) = (18) + (19) + (20) + (21)	Specific provision		46.0	2.9	10.6	17.8	7.5	7.3
(18) = (7)*Rate	against special mention loans	0.03	3.8	0.3	1.1	1.8	0.5	0.2
(19) = (9)*Rate	against substandard loans	0.20	5.2	0.6	2.0	1.0	1.0	0.6
(20) = (10)*Rate	against doubtful loans	0.50	10.0	1.0	2.5	5.0	1.0	0.5
(21) = (11)*Rate	against loss loans	1.00	27.0	1.0	5.0	10.0	5.0	6.0
(22) = (13) – (14)	Under-/over-provisioning		0.0	0.0	0.0	0.0	0.0	0.0
Post shock								
(23)	Shock: NPL ratio (in percent)		18.5	40.5	20.7	17.4	12.7	17.5
= (4)	Total loans		1,448.0	71.0	385.0	615.0	287.0	90.0
(24) = (23)/100*(4)	Total NPLs ²		267.7	28.8	79.7	107.0	36.4	15.8
(25) = (9)/(8)*(24)	Substandard loans		95.5	14.4	39.8	21.4	15.2	4.7
(26) = (10)/(8)*(24)	Doubtful loans		80.0	9.6	19.9	42.8	6.1	1.6
(27) = (11)/(8)*(24)	Loss loans		92.2	4.8	19.9	42.8	15.2	9.5
(28) = (4) – (24)	Balance of loans that are performing ³		1,180.3	42.2	305.3	508.0	250.6	74.3
(29) = (6)/(5)*(28)	Normal and pass loans		1,074.6	35.7	276.0	456.3	236.9	69.6
(30) = (7)/(5)*(28)	Special mention loans		105.7	6.5	29.3	51.7	13.7	4.6
(31) = (24) – (8)	Increase in NPLs		194.7	22.8	59.7	82.0	24.4	5.8
(32) = {(24)/(8) – 1}*100	Rate of increase in NPLs (in percent)		266.7	379.3	298.5	328.0	203.7	57.5
= (13)	Total provisions currently held		58.5	3.5	13.9	23.1	10.1	8.0
(33) = (34) + (36)	Total provisions that <i>should</i> be held		165.2	13.0	41.5	74.6	24.0	12.0
(34) = (35)	General provision		10.7	0.4	2.8	4.6	2.4	0.7
(35) = (29)*Rate	against normal and pass loans	0.01	10.7	0.4	2.8	4.6	2.4	0.7
(36) = (37) + (38) + (39) + (40)	Specific provision		154.4	12.7	38.7	70.0	21.7	11.3
(37) = (30)*Rate	against special mention loans	0.03	3.2	0.2	0.9	1.5	0.4	0.1
(38) = (25)*Rate	against substandard loans	0.20	19.1	2.9	8.0	4.3	3.0	0.9
(39) = (26)*Rate	against doubtful loans	0.50	40.0	4.8	10.0	21.4	3.0	0.8
(40) = (27)*Rate	against loss loans	1.00	92.2	4.8	19.9	42.8	15.2	9.5
(41) = (33) – (34)	Under-/over-provisioning		–106.7	–9.6	–27.6	–51.5	–14.0	–4.0
Assuming full provisioning after shock								
(42) = (1) + (41)	New capital amount required		423.3	20.4	132.4	168.5	66.0	36.0
(43) = (42)/(2)*100	New CAR (in percent)⁴		12.0	12.0	12.0	12.0	12.0	12.0

Source: Authors.

Note: CAR = capital adequacy ratio; NPL = nonperforming loan; RWA = risk-weighted assets.

1. It is assumed that loans are fully provisioned for initially.

2. It is assumed that NPLs increase proportionately across all categories.

3. It is assumed that the balance of loans, which are performing, are distributed in the same proportion as previously.

4. It is assumed that RWA remains the same.

line items, or by individual institutions. The availability of data could influence the assumptions and models used to estimate the size of the shocks. Although such stress tests may be presented as extreme, tail-shock scenarios, it is unclear what actions supervisors should take in response to the findings, in the absence of any evidence that the applied shocks represent extreme but plausible macro-financial relationships.

It should also be emphasized that any stress testing exercise needs to be comprehensive. The “breaking point” method presented in this chapter, together with other qualitative and quantitative bank supervision information, could be a useful “back-of-the-envelope” tool for both off-site and on-site supervisors in monitoring the stress points for banks. Further, testing for risks that affect bank solvency alone (e.g., through credit and market risks) is inadequate insofar as they reveal only part of the picture. They should be complemented by other types of stress tests, for example, for liquidity risk, to provide a more comprehensive picture of existing vulnerabilities (see Ong and Čihák, 2010). Again, however, data for such stress tests may be unavailable or of poor quality. In such cases, *not* performing any stress test on a particular financial system may sometimes be a more prudent course of action.

REFERENCES

- Basel Committee on Banking Supervision, 2008a, *Cross-Sectoral Review of Group-Wide Identification and Management of Risk Concentrations*, The Joint Forum (Basel, April). Available via the Internet: <http://www.bis.org/publ/joint19.htm>
- , 2008b, *Liquidity Risk: Management and Supervisory Challenges* (Basel, February). Available via the Internet: <http://www.bis.org/publ/bcbs136.htm>
- , 2008c, *Principles for Sound Liquidity Risk Management and Supervision* (Basel, September). Available via the Internet: <http://www.bis.org/publ/bcbs144.htm>
- , 2009a, “International Framework for Liquidity Risk Measurement, Standards and Monitoring,” Consultative Document (Basel, December). Available via the Internet: <http://www.bis.org/publ/bcbs165.htm>
- , 2009b, *Principles for Sound Stress Testing Practices and Supervision* (Basel, May). Available via the Internet: <http://www.bis.org/publ/bcbs155.htm>
- Čihák, Martin, 2007, “Introduction to Applied Stress Testing,” IMF Working Paper 07/59 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=20222>
- Ong, Li Lian, and Martin Čihák, 2010, “Of Runes and Sagas: Perspectives on Liquidity Stress Testing Using an Iceland Example,” IMF Working Paper 10/156 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=24019>
- Ong, Li Lian, Rodolfo Maino, and Nombulelo Duma, 2010, “Into the Great Unknown: Stress Testing with Weak Data,” IMF Working Paper 10/282 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=24488>

Next-Generation Applied Solvency Stress Testing

CHRISTIAN SCHMIEDER • CLAUS PUHR • MAHER HASAN

This chapter provides an overview of the building blocks of a new solvency stress testing tool. The framework enriches solvency stress tests by enhancing their risk sensitivity while keeping them flexible, transparent, and user-friendly. The framework is Excel-based, and explicitly allows running multiperiod scenarios (up to 5 years), providing both regulatory and economic capitalization ratios under stress. The framework can be applied to single banks with a “simple” risk structure and to systemically important financial institutions and can be used for financial stability analysis that includes hundreds (and even thousands) of banks.

METHOD SUMMARY

Overview	• The framework seeks to enrich solvency stress tests in terms of their risk sensitivity while keeping them flexible, transparent, and user-friendly. • It generates both regulatory and economic capitalization ratios under stress.
Application	The framework enables the conduct of multiperiod solvency stress tests for banks (and, in principle, also other financial institutions). Its main contributions include 1. increasing the risk sensitivity of stress tests by capturing changes in risk-weighted assets under stress, including for non-Internal Ratings Based (IRB) banks (through a quasi-IRB approach); 2. providing stress testers with a comprehensive platform for using satellite models for defining various assumptions and scenarios; 3. allowing stress testers to run multiyear scenarios (up to 5 years) for hundreds of banks, depending on the availability of data.
Nature of approach	Balance sheet data based.
Data requirements	Accounting information and supervisory data, including comprehensive information on banks’ assets, income components, and capitalization ratios, as well as banks’ off-balance-sheet positions. Satellite models are also needed; the scenarios are expert based.
Strengths	The method caters for the need to run more risk-sensitive, granular, and comprehensive solvency stress tests and takes into account the fact that meaningful data to run IRB-type stress tests remain limited at this stage.
Weaknesses	• The scenario design requires expert judgment or reliance on rules of thumb (some gathered from experiences with other countries, which may not be applicable to the country being analyzed). • The framework does not explicitly capture dependencies across banks (as portfolio models may do).
Tool	The Excel spreadsheet macro is available in the toolkit, which is on the companion CD and at www.elibrary.imf.org/stress-test-toolkit. Contact author: C. Schmieder.

This chapter is an abridged version of IMF Working Paper 11/83 (Schmieder, Puhr, and Hasan, 2011). The working paper benefited from comments from Martin Čihák, Andreas Jobst, Liliana Schumacher, Hiroko Oura, Elena Loukoianova, Joseph Crowley, Aidyn Bibolov, Jay Surti, Heiko Hesse, and Torsten Wezel; as well as from the participants of the Monetary and Capital Markets Department and European Department seminar.

The recent past has clearly revealed the importance of stress tests not only as a risk management tool and key component of financial stability analysis but also as a crisis management tool. This last role of stress tests became evident in the U.S. Supervisory Capital Assessment Program¹ (SCAP) as well as in European stress tests that were used to decide the required level of capital backstops. In addition, emerging macroprudential tools such as capital conservation buffers, countercyclical capital conservation buffers, systemic risk charges, and Pillar 2 capital charges (and even Pillar 1 charges for market risk) can be and are determined on the basis of stress tests. Stress tests also are used for internal bank capital allocation and risk management purposes, such as limit setting.

In developing the new framework, we take into account three key dimensions. First, a stress test framework needs to allow its users to conveniently run (a series of) severe yet plausible scenarios to assess the sensitivity of the stress to the underlying *assumptions*. Second, a useful stress test framework needs to be *risk sensitive*. This requires that changes to risk parameters be based on economic measures of solvency, in addition to statutory regulatory ones. Last, the effectiveness of stress testing depends on the ease with which the output can be communicated to decision makers (e.g., policymakers, senior bank managers) and market participants. Although these concepts seem straightforward, they might pose challenges in practice as higher risk-sensitivity exercises are expected typically to require the use of technically complex, “sophisticated” frameworks.

We develop a “next-generation”² Excel-based tool, designed to be *transparent*, *flexible*, and *user-friendly*. The last two features make the tool accessible to banks, regulators, and rating agencies in advanced, emerging, and developing economies. The tool is particularly geared toward evolving risk management practices, spurred by regulatory changes (Basel II/III). The key conceptual contribution of this framework is that it allows for an assessment of economic solvency under stress. This is done by adjusting the denominator of capitalization, risk-weighted assets (RWA), for potential unexpected (worst case) losses, not only in terms of volume (as in the past) but also for changes in the risk profile of a bank’s business. The rationale for that is to make potential losses visible well before they materialize in terms of losses in the numerator of capitalization ratios (in terms of a reduction in capital through losses). Although economic solvency tests (ideally) require various inputs, the framework has been designed to make economic solvency tests possible also for banks that are currently under Basel I or the Standardized Approach (StA) through a quasi-Internal Ratings Based (QIRB) approach. With various banks moving to the Foundation Internal Rat-

ings Based (FIRB) and Advanced Internal Ratings Based (AIRB) approach during the next years, the tool caters to the needs across various financial systems.

With these overarching objectives, our framework can be contrasted with other comprehensive stress testing frameworks that have been developed during the last decade. Two notable integrated ones come to mind: the Risk Assessment Model for Systemic Institutions (RAMSI) by the Bank of England (Aikman and others, 2009) and the Systemic Risk Monitor (SRM) by the Austrian Central Bank (Boss and others, 2006). Other institutions also have recently developed or upgraded their (already sophisticated) stress testing frameworks, such as in Brazil, Canada, Chile, the Czech Republic, France, Germany, Italy, Japan, the Netherlands, Norway, Spain, Sweden, Switzerland, and the United States as well as the European Central Bank (ECB).³ Most of these approaches were discussed by Foglia (2009), and details are presented in the recent Financial Stability Reports of the respective countries and/or institutions.

Our stress testing framework is based on three dimensions: *risk sensitivity*, *scope*, and *ease of use* (Figure 5.1). Risk sensitivity is key but should not come at the expense of it being analyzed through a “black box.” The scope should be wide, allowing for extensions when need arises. The ease of use is very important not only to make the output of a tool readily understandable and credible but also to focus on the stress tests as such rather than struggling with details on techniques.

The most important difference between our framework and the aforementioned integrated stress testing tools lies in the scope. Although the next-generation framework allows the use of satellite models (i.e., econometric models) to establish macro-financial linkages over the stress horizon, these models have to be estimated outside the framework (e.g., by means of an econometric software). In that sense, other stress testing tools, including those focusing on single risks (such as credit portfolio models, usually run based on market data), are more sophisticated than the next-generation framework in its standard form (i.e., without additional input).

In order to (at least partially) close this gap, certain considerations have been incorporated:

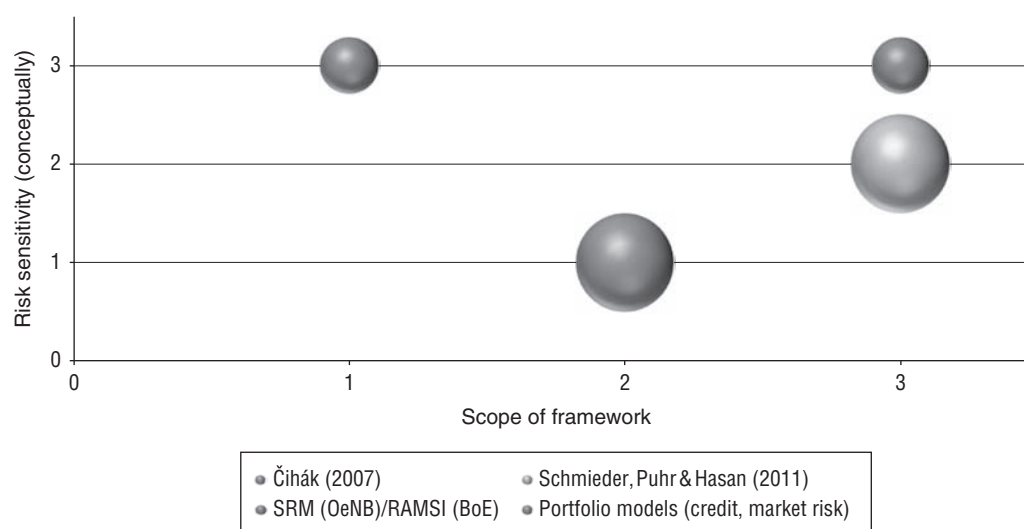
- Work by Hardy and Schmieder (2013) in establishing rules of thumb for credit risk and income under different levels of stress has been built into the tool. These rules of thumb allow credit losses (default and recovery rates), correlations, and income to be linked to macroeconomic conditions. Hence, macro stress testing is possible without calibrating satellite models—a key feature of the tool.⁴
- Likewise, the framework is based on a set of reduced-form models to run stress tests based on quasi-portfolio credit risk models—the other key instrument of the framework. The framework has been used recently

¹ The SCAP covered the 19 largest U.S. bank holding companies (BHCs), accounting for two-thirds of aggregate assets of the banking system. The SCAP assessed the capital positions of these firms against a baseline macroeconomic scenario, based on market and consensus forecasts, and an adverse scenario defined by the authorities. The results for each institution were published in May 2009.

² The “next-generation” framework presented in this chapter extends the work on applied (macro) stress testing by Čihák (2007).

³ This list is not exhaustive.

⁴ As a caveat, this simplified approach has to be carefully applied to avoid misleading results under specific circumstances and/or for specific banks.



Source: Authors.

Notes: BoE = Bank of England; OeNB = Austrian National Bank/Oesterreichische Nationalbank; RAMSI = Risk Assessment Model for Systemic Institutions; SRM = Systemic Risk Monitor.

Figure 5.1 Comparison of Key Stress Testing Frameworks

for various types of risk analysis and stress tests carried out by the IMF and the ECB.

The rest of the chapter is organized as follows: Section 1 provides an overview on the methodological framework, including the concepts on the one hand and the technical solution on the other. Section 2 concludes the chapter.

1. METHODOLOGY

A. Concept

The methodology underlying the framework is a set of reduced-form models that enable quasi-portfolio model calculations. They are discussed below.

Stress Test Metric

The major output of the tool is banks' capitalization levels under different scenarios. It allows stress testers to determine whether banks (1) are resilient enough to stay above the regulatory minima; (2) are resilient enough to meet market expectations (i.e., certain hurdle rates that are considered best practice by market participants);⁵ or (3) are sufficient to safeguard any particular bank from an additional idiosyncratic shock (in case of a common macroeconomic scenario). Moreover, the stress test can also help in determining banks' potential capital needs in case either of these thresholds is not met.

Capitalization under stress is measured as follows:

$$\text{Capitalization } (t+1) = [\text{Capital } (t) + \text{Net income } (t+1)] / \text{RWA } (t+1).$$

⁵ In recent months, Core Tier 1 ratios of at least 10 percent became more common, whereas 4–8 percent was a more common benchmark in the past.

If net income becomes negative, capital will be reduced; otherwise, capital increases subject to taxation and the earnings retention rate. The latter is common under baseline scenarios; otherwise, banks would be in a difficult position. Besides preshock capitalization (which is given as an input), the two main drivers of solvency risk for banks are credit losses and pre-impairment income.

When using a Basel I/II StA type definition of RWA for credit risk, RWA under stress evolves conditional to portfolio volume (i.e., credit growth and credit losses), except for some positions subject to risk weighting (such as sovereign bonds). For IRB banks (and when using economic capitalization measures), RWA is also adjusted for risk to reflect the change in the risk profile of banks' business and thereby the increase or decrease of potential unexpected (worst case) losses. In the latter case, RWA allows banks' portfolio quality to be monitored and thereby help to identify the buildup of risk early on—subject to the availability of timely and precise risk information. The framework also allows for an adjustment of RWA for market risk, operational risk, and other Pillar 1 and Pillar 2 risks under stress.

Income

Income is banks' first line of defense against unforeseen losses but also, at the same time, supports asset growth and represents a buffer for absorbing any impact from regulatory changes such as Basel III. Therefore, income should be a major element in any stress test framework, even more so for multi-period stress testing exercises. However, the modeling of income remains far less prominent (and developed) relative to loss modeling. The simulation of pre-impairment income (and/or of its specific components) should be guided by satellite

TABLE 5.1

Income under Stress	
Income Element	Description
Net operating income including impairments	By default, nonrecurring income is not considered; hence, net operating income is modeled as the sum of net interest income, net fee and commission income, and other operating income (including expenses). All three components can be adjusted separately if deemed appropriate.
• Add: Change in net operating income vs. reporting year	Stress testers can define changes in net interest income, both expert based and model based; in addition to that, foregone interest owing to losses and additional interest income from credit growth are taken into account. Moreover, changes in net fee and commission income and other operating income (including expenses) are accounted for (again, where available, the application of satellite models is possible).
• Less: Impairments for credit losses exceeding those in reporting year	Outcome of stress test of credit risk.
• Add: Changes in trading and investment income including marked-to-market gains/losses (interest rate shock in the banking book, foreign exchange [FX] rate shock)	The evolution of the trading income can be based on expert judgment by means of satellite models; in addition to that, marked-to-market gains/losses can be simulated through shocks affecting interest rates in the banking book and/or FX rates provided that bank-specific data are available.
• Add: Change in other income	Other sources of income are foreseen to be simulated based on expert judgment.
Equal to: Net income (scenario, year <i>t</i>)	The sum of all the above components.

Source: Authors.

models. In the published version of the tool, postshock income is determined as shown in Table 5.1.

The reported operating income net of impairments serves as a benchmark for the income of the following years.⁶ The guiding idea is that nonrecurring income as well as other sources of income (not part of operating income) is not taken into account, as these elements are not an integral part of the (medium-term) earning capacity of banks.⁷ The calculation of income strikes a balance between setting straightforward assumptions and more sophisticated modeling. If the resulting net income after stress is positive, then the portion foreseen to be retained (after tax) based on the stress test assumptions will be added to the capital; otherwise, losses will be deducted from capital. Rather than sticking to a general assumption about retained income, rules depending on the postshock capitalization of each specific bank could be referred to (e.g., in line with Basel III maximum pay-out rules or based on empirical evidence, as discussed in Hardy and Schmieder, 2013), but such rules are not part of the standard version of the tool. Further information on the subcomponent of income can be found in Schmieder, Pühr, and Hasan (2011).

Credit Risk

The treatment of credit risk is the key innovation of the framework, which is based on a Basel II/III type notion of credit risk. The simulation of credit risk under stress is foreseen to be based on the credit risk parameters used for the computation

of IRB capital charges, namely, (1) probabilities of default (PDs) and losses-given-default (LGDs); or (2) credit losses (such as impairments) as well as exposures at default and asset correlations. The tool offers a conceptual framework for determining credit losses under stress on the one hand (which informs the numerator of capital adequacy) and RWA for credit risk under stress on the other (the denominator of key capitalization ratios). It is worth highlighting that credit risk analyses (and stress tests more generally) are assumed to be carried out for all assets subject to default risk, that is, including counterparty credit risk and off-balance-sheet items. The market risk of the liquid assets is simulated separately through income.

The Relationship between PDs and LGDs. It was discovered a while ago that there is a positive correlation between the PDs and LGDs of bonds. This implies that, in times of stress (i.e., when PDs are higher), LGDs are higher than in “normal” times. With more data becoming available in recent years, this observation has also been confirmed for loans. In order to provide stress testers with the possibility of linking stressed LGDs to stressed PDs (and thus avoiding the need to make a separate assumption about the development of LGDs under stress), we combine evidence determined by Moody’s (2009) with an approximation formula proposed by the Board of Governors of the Federal Reserve System (Fed, 2006) to determine downturn LGDs, that is, LGDs under stress conditions.⁸ Using the formula in Fed (2006):

$$\text{Downturn LGD} = 0.08 + 0.92 * \text{Long-term average LGD}.$$

⁶ If a stress test is to be based on a different initial value (e.g., net interest income as the average ratio of net interest income over total loans to customers over the last *x* number of years), minor adaptations in the Excel links of the framework would be necessary.

⁷ Stress testers should alter these assumptions as appropriate under specific conditions.

⁸ Downturn LGDs are defined in paragraph 468 of the Basel II framework (Basel Committee on Banking Supervision, 2006).

A nonlinear formula is derived, accounting for the finding that the relationship is nonlinear (Standard & Poor's, 2010):^{9,10}

$$LGD \text{ (under stress)} = 0.4022 + 2.1535 * PD. \quad (5.1)$$

The intercept of equation (5.1) can be modified to match the actual levels of LGDs in specific countries. We propose using the World Bank's *Doing Business* data to do so, drawing on work by Djankov, McLiesh, and Schleifer (2007), for countries for which no specific studies have been done (unlike the United States, the United Kingdom, Germany, and France, for example). The LGDs published by the World Bank can serve as a rough (and rather benign, in some cases) proxy for corporate LGDs. To arrive at a bank-level or country-level LGD, a weighted average is computed, to account for the portion of retail and other types of credit, and expert judgment is applied. The studies by Schmieder and Schmieder (2011) and Hardy and Schmieder (2013) on the implications of the legal framework could also be used to guide the calibration of LGDs.

RWA for Credit Risk. A portfolio credit risk model that captures credit correlations is needed to compute RWA for credit risk in economic terms. This framework uses the one-factor model underlying the IRB approach to determine changes in RWA conditional on changes in credit risk parameters (PDs, correlations, name concentration). LGDs exhibit a linear relationship with RWA, so no model is necessary, although it underscores the importance of LGDs. An illustrative example is provided in Schmieder, Pühr, and Hasan (2011).

RWA Sensitivity of PDs. The stress test framework uses the Basel II IRB formula to translate increases in PDs into stressed RWA. If one keeps LGDs and correlations constant,¹¹ one can simulate the marginal effect of an increase in PDs on RWA. The RWA elasticity of PDs is higher the lower the pre-stress PDs, and the elasticity decreases when PDs increase.¹² For low levels of pre-stress PDs, the RWA elasticity of PDs is 0.6, that is, an increase in PDs by 1 percent yields an increase in RWA by 0.6 percent. For higher PDs, the elasticity goes down: for PDs of 5 percent, the effect is about 0.35 and for PDs of 10 percent about 0.2. The nonlinear effect is captured by means of a polynomial fit function. Further details are available in Schmieder, Pühr, and Hasan (2011).

RWA Sensitivity of Asset Correlations. For correlations, the elasticity depends on the characteristics of the counterpart,

that is, their relationship with macroeconomic conditions. In simplified terms, one can refer to asset classes, such as (large) corporates, small- and medium-sized enterprises (SMEs), and retail customers. Corporates are expected to be most correlated with the cycle (on average), whereas idiosyncratic performance plays a more important role in debt service by SMEs and even more so for retail counterparts. The Basel II IRB framework takes a shortcut by linking correlations to PDs, using the empirical relationship that larger firms exhibit better ratings, that is, lower PDs. For the corporate IRB formula, the effect is more than linear (i.e., the PD elasticity is above unity) for PDs lower than 2 percent (see Schmieder, Pühr, and Hasan, 2011, for details).

We compared the IRB-based results with empirical evidence provided by Mager and Schmieder (2009) as a robustness check.¹³ On the basis of stress tests of realistic German credit portfolios, we estimate the RWA elasticity of asset correlations for small banks at 0.45; for medium-sized banks, the elasticity is 0.7; and for large German banks, elasticity is 1.25, all of which confirm the above findings. As a default, the framework assumes a linear relationship between asset correlations and RWA, but stress testers can modify this assumption. The findings of Hardy and Schmieder (2013) could also be used as a reference point to determine the level of asset correlations under stress.

In sum, the framework performs similar functions to those of an economic capital model (such as CreditMetrics, CreditRisk+) but with simple means. Combined with its ability to include capital charges for name concentration, as outlined below, the framework comes close to a full-fledged portfolio model.

Name Concentration and RWA. Basel II IRB minimum capital requirements do not account for name concentration, as the underlying (one-factor portfolio) model assumes that banks' credit portfolios are perfectly granular. Although this assumption keeps the underlying model relatively simple, capital requirements may be underestimated. In order to avoid such underestimation, which is most likely for small banks, concentration risk is subject to Pillar 2, that is, supervisory scrutiny. The study by Gordy and Lütkebohmert (2007) offers a framework for estimating name concentration. The outcome of a numerical example provided by the authors was used to determine an approximation formula that translates name concentration into additional RWA (in percent). This approximation depends on the actual level of name concentration as measured in terms of the Herfindahl-Hirschman Index (HHI)¹⁴ and the aggregate bank-level PD:

$$\Delta RWA = 100 * (0.02 + 12.599 * HHI) * (1 + (PD/0.4\% - 1) * 0.1).$$

⁹ Standard and Poor's does not reveal the equation for its logarithmic approximation.

¹⁰ LGD is capped at 100 percent.

¹¹ When the Basel II formula is applied without adjustments, correlations would go down when PDs are increased, which would be inconsistent from a risk perspective.

¹² The reason is that for the highest-rated borrowers, default is very unlikely, unlike for their lower-rated counterparts. In the latter case, default is "expected" (i.e., there would be little surprise), whereas in the former case, a downgrade would make default increasingly more likely. That said, it remains relatively unlikely for highly rated counterparts, that is, not expected but rather unexpected.

¹³ This is also to account for the fact that the correlations in the IRB framework are modeled conditional on the PD.

¹⁴ The HHI is the sum of the squared exposure portions. Gordy and Lütkebohmert (2007) refer to an HHI calculated based on exposures to groups/borrower units.

Schmieder, Puhr, and Hasan (2011) provide an illustrative example.

Translation of RWA Based on the Standardized Approach into QIRB RWA. The use of a risk-sensitive measure of bank capitalization is essential for stress testing, as the evolution of risk and also the level of risk assumed before stress could otherwise be misleading. The framework using QIRB RWA to translate risk-invariant RWA into risk-sensitive measures allows banks that have not moved to the IRB approach to run meaningful stress tests. The idea is to

1. rescale the RWA of banks by means of an approximation (QIRB); and then
2. run risk-sensitive tests and simulate scenarios based on the IRB approach. In case stress testers do not feel comfortable with rescaling RWA, the reported RWA can be used as a starting point for risk-sensitive tests.¹⁵

A key precondition for calculating QIRB RWA is to use meaningful credit risk parameters. A second-best solution is needed for non-IRB banks. For many banks, nonperforming loans (NPLs) are available, usually including time series. However, NPLs are often stock figures, and definitions vary widely across countries, so a feasible solution has to be found to determine PD-like numbers (see Box 1 in Schmieder, Puhr, and Hasan, 2011). As discussed earlier, LGDs are often available at a country level only, which appears to be a good proxy for corporate exposure. For retail exposure, an alternative proxy has to be found, which may also apply to other asset classes. Aside from NPLs, data on (specific) provisions and write-offs also could be used to determine implied PDs or to directly measure credit losses, wherein PDs and LGDs do not have to be determined separately.¹⁶

An illustrative example for a hypothetical bank is shown below. Under the StA, the capital requirements are assumed to be at 8 percent of total exposure and do not fluctuate over time.¹⁷ For the IRB approach, the capital requirements have been determined as a fraction of exposures by using default rates observed by Moody's during the last decade (for the universe of the firms rated by them) and an LGD of 40 percent, which is a realistic through-the-cycle (TTC) benchmark for advanced countries.¹⁸ In the example, the capital requirements under the IRB approach fluctuate significantly over time and

are lower in most years (e.g., through 2003 to 2008).¹⁹ The scaling factor compares the relative level of capital requirements between the StA and the IRB approaches over time (Figure 5.2). The scaling factor thus adjusts the level of the StA RWA to the level of the IRB capital requirements. Given the high level of LGDs for most emerging markets and low-income countries, the scaling factor would be above unity except for very benign years with very low PDs, indicating that economic risk is often underestimated by the StA capital ratio, which might provide a false sense of security (see also Hardy and Schmieder, 2013).

Basel III

Our tool allows a general assessment of the potential impact over time from the Basel III phase-in, informed by the aggregate outcome of the Quantitative Impact Study (QIS) 6 (Basel Committee on Banking Supervision, 2010b). More specific assumptions can be defined provided that bank-specific data are available. The simulation of the effect of Basel III rules on bank solvency includes three key elements (Basel Committee on Banking Supervision, 2010a):

1. an increase in RWA in 2011;
2. the phase-out of eligible capital beginning in 2013 (Total, Tier 1) and 2014 (Common Equity/Core Tier 1), respectively; and
3. changes in minimum capital ratios over time.²⁰

Changes in the first two elements are simulated based on the outcome of the QIS 6 and are applied to banks according to their size (banks with equity of less than \$3 billion are classified as Group 2 banks). For the increase in RWA, stress testers can define a portion of behavioral adjustment. If one assumes that there is a behavioral adjustment of 50 percent, for example, then banks are assumed to mitigate 50 percent of the expected increase of RWA, for example, through a change in their asset composition. Basel III can also be applied in terms of the profit retention rate. This can be done by either using uniform payout ratios (from the drop-down menu) or by defining bank-specific behavior (see Hardy and Schmieder, 2013, for a discussion on typical bank behavior).

B. Technical overview

Architecture

The stress test framework is built on a modular kernel, which conveniently allows extensions and refinements (Figure 5.3). As a guide, the upper left-hand side contains the external

¹⁵ Although this is less than ideal from a risk perspective, it is a step forward compared with traditional tests based on Basel I type capitalization measures.

¹⁶ With the advent of Basel II, provisions are meant to reflect expected losses, which is a required input.

¹⁷ There will be some changes in external ratings and in the value of collateral, but for most banks, they will be limited.

¹⁸ For emerging markets and low-income countries, LGDs are typically higher, at 60–80 percent, which has a significant effect on the scaling factor. The reasons are manifold, but legislation is a key factor (which would be important for the duration of the workout process, for example). Further elaboration on this subject is provided in Schmieder and Schmieder (2011).

¹⁹ The reason is that the Basel II framework has been calibrated with a view that IRB capital charges are, on average, lower than the ones under StA in the primary recipient countries (i.e., not the ones that voluntarily adopt Basel II), reflecting economic reasons on the one hand and providing banks with incentives to move to the IRB on the other.

²⁰ The leverage ratio has not been added because of the late phase-in but will be part of future releases.

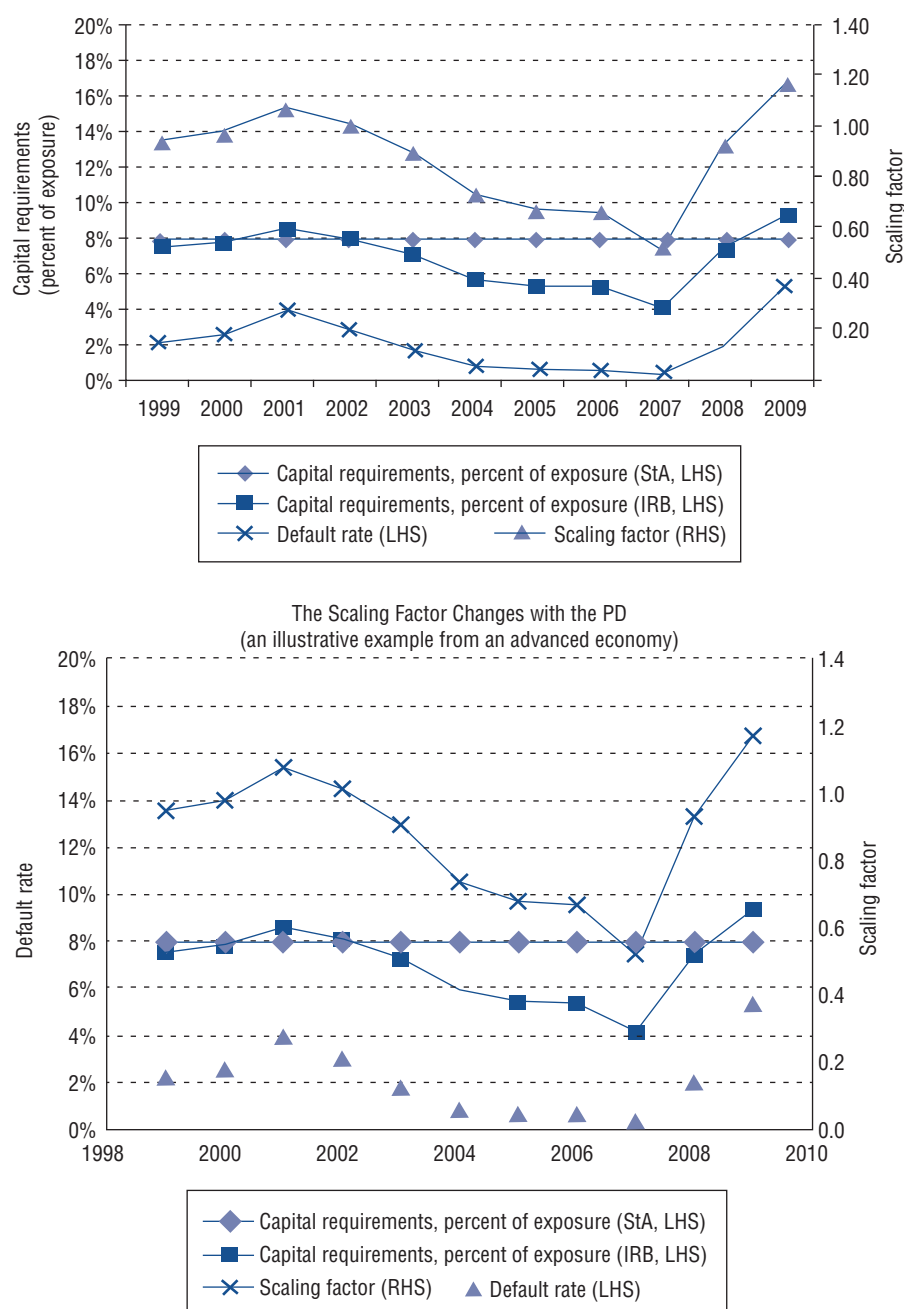


Figure 5.2 Illustrative Example for the Scaling Factor (Advanced Economy)

parameterization and models, the lower right-hand side the input data. Once these are settled, the main sequence of the framework's mechanics goes from top-right to bottom-left (i.e., define assumptions of the stress test, calculate impact on the solvency, calculate the impact of name concentration, aggregate, and finally summarize results). The solvency tool is designed to be easy to use, both to simplify stress tests for the growing community of stress testers (with heterogeneous

needs) and to provide interested stakeholders with an opportunity to smoothly access the field:

1. It is Excel-based;
2. users are systematically guided through the sheet (based on a user-friendly layout, documentation, and help menus);
3. the layout makes it easy to set assumptions, including without the use of satellite models (or using

predefined, simple rules, per Hardy and Schmieder, 2013);

4. drop-down lists allow the user to switch between different settings (for example scenarios); and
5. the framework has been tested and improved in various contexts.

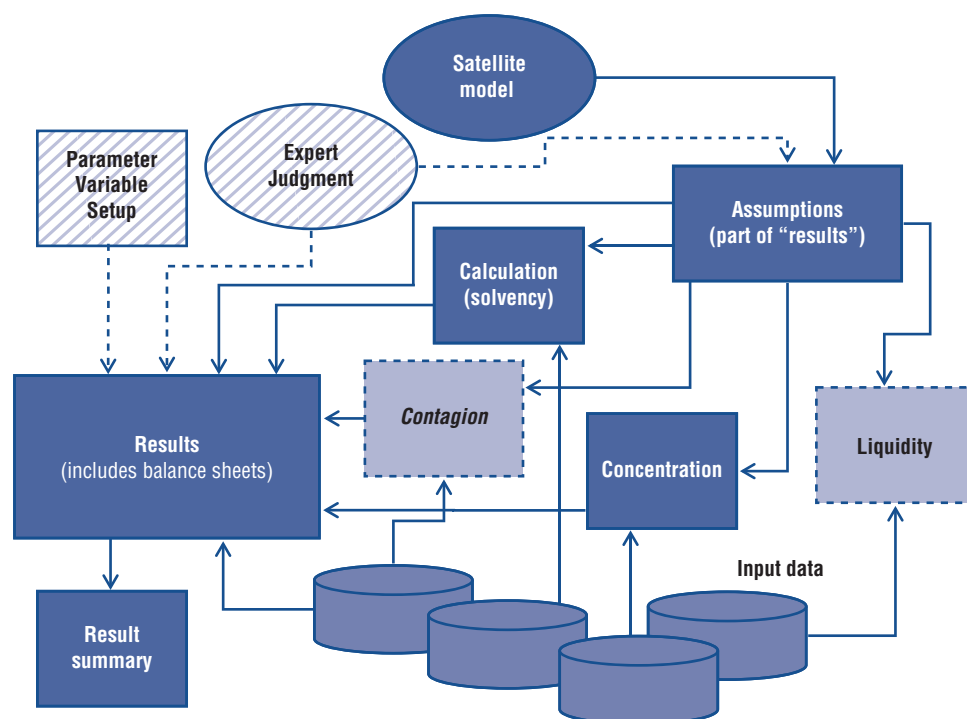
How Does the Framework Work? Running a stress test requires several steps (Figure 5.4): defining the scenario, configuring the framework, and entering the input data (see Step 1); linking the (macroeconomic) scenario to financial risks (Step 2); and executing the stress test (Step 3).

- **Step 1: The (macro) scenario definition.** Outside the tool, the stress tester has to decide on the shock and, in the case of a macro stress test, a macroeconomic scenario (either with the help of a model or using expert judgment). Within the tool, the user has to first implement a straightforward parameterization that alters the template according to the number of banks, the granularity of the (credit) portfolio, and so forth. Last but not least, bank data have to be input, which are limited to about 30 variables in the minimum setup.
- **Step 2: From (macro) scenarios to micro impact.** In the simplest case, stress testers run sensitivity tests for specific risk types, such as credit risk, market risk,

operational risk, or concentration risk. In the more demanding case, including when an assessment is linked to a macroeconomic scenario (referred to as macro stress test), multiple risk factors are accounted for at the same time. In case of a macro stress test, the test is performed by linking macroeconomic risk factors to financial risks (i.e., banks' asset quality) by means of so-called satellite models (i.e., econometric models).

- **Step 3: The execution of the stress test.** The actual run of the test happens “on the fly”—that is, once the specific setting has been chosen and the satellite models calibrated, the final outcome in terms of banks' balance sheets and capitalization is generated immediately. The solvency tests reveal bank-by-bank solvency under stress and the aggregate figures for the system (capitalization, number of banks failing the tests, capital needs, and so forth) as well as various financial soundness indicators (FSIs), including the risk contributions of various elements (credit losses, trading and investment losses, profit serving as a first buffer against losses) to stress.

The Execution of the Stress Tests. A key advantage of the framework is that the actual run of the tests takes place immediately. Once the setting has been specified, the outcome

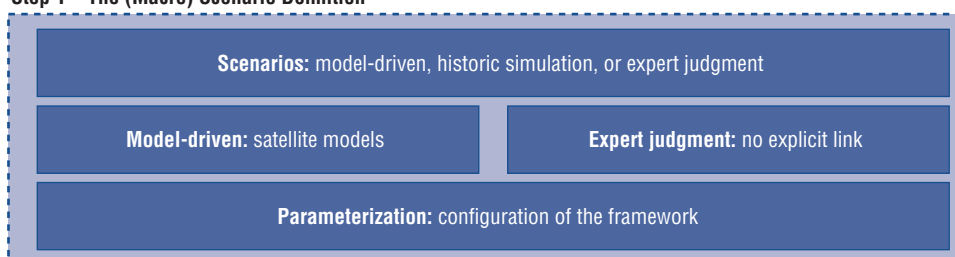
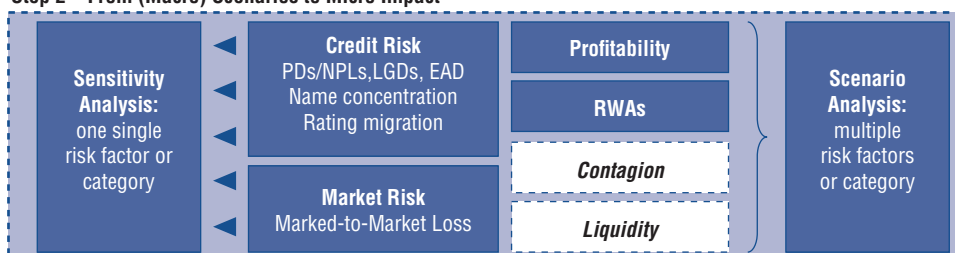
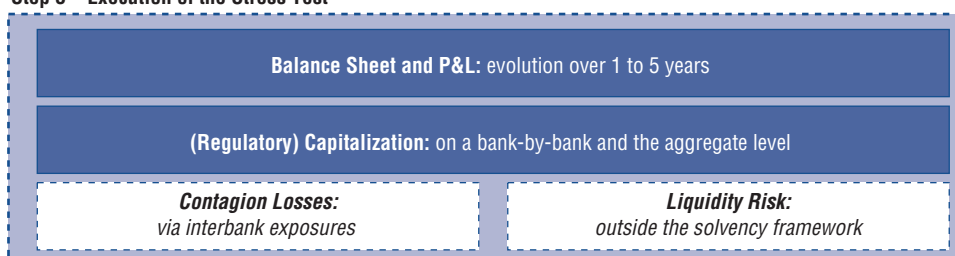


Source: Authors.

Note: Dashed boxes refer to modules made available separately.

Figure 5.3 The Modular Design of the Stress Testing Framework²¹

²¹ As the framework is based on an Excel template, the modules are characterized by different tabs. At the heart of the framework are the result sheets, which bring various sources of data, assumptions, and parameters together and summarize the outcome of the stress tests, both on the bank level and on the system level.

Step 1 – The (Macro) Scenario Definition**Step 2 – From (Macro) Scenarios to Micro Impact****Step 3 – Execution of the Stress Test**

Source: Authors.

Note: The elements in italics will be available through separate pieces of work.

EAD = exposure at default; LGDs = losses given default; NPL = nonperforming loan; PDs = probabilities of default; P&L = profit and loss; RWAs = risk-weighted assets.

Figure 5.4 Stress Testing Framework—Conceptual Overview

of the tests is generated under the respective results tabs. The mechanics of the spreadsheet are summarized step by step under the roadmap tab. The results sheets not only define the methods but also display the results of the tests at both the level of the financial system (or the overall sample of banks) and on an institution-specific level.

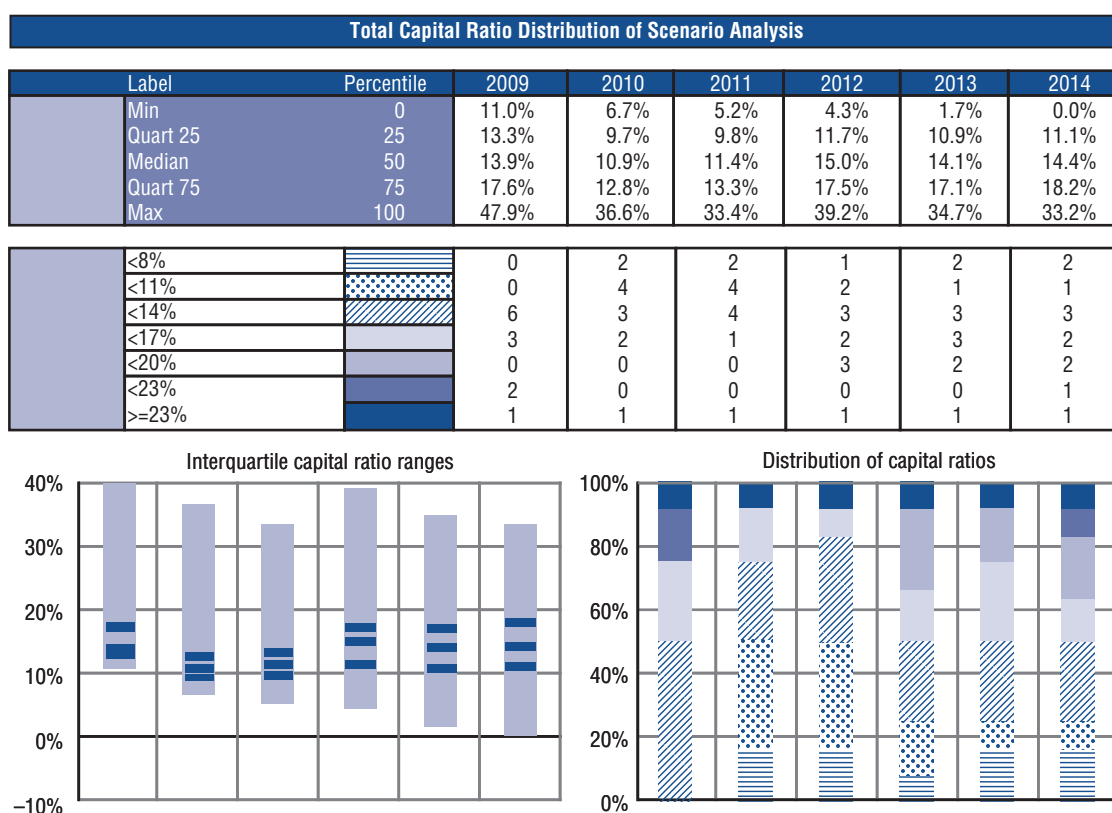
The dispersion of the bank-specific outcomes of a macro stress test for 12 international banks through 2014 is shown in Figure 5.5. The output shows the evolution of different quantiles of the total regulatory capital ratios (the setting can easily be changed to Tier 1 and Core Tier 1 ratios) over time and the number of banks in different capital buffer ranges. The outcome for the system (which is not shown here) includes the number of banks failing the tests, capitalization needs in absolute and relative terms, as well as the contribution of different risk drivers to stress. An illustrative numerical example is available in Schmieder, Puh, and Hasan (2011).

2. CONCLUSION

To date, the methods used to carry out stress tests tend either to be too simplified to identify important risks or are black

boxes, wherein the economics of the tests becomes less important than the technique itself. Our framework seeks to close this gap, using conceptual elements from more sophisticated tools but making them accessible in a convenient and flexible manner. The framework is designed for banks and could, in principle, also be used for other financial institutions provided that adjustments are made to account for the pertinent differences in the types of business in general and the associated vulnerabilities in particular.

This stress testing framework contributes to the growing body of applied stress testing work, by providing a tool that satisfies three main objectives. It (1) facilitates the design and running of a series of meaningful scenarios to derive an overview of key risk drivers and their sensitivities; (2) allows the user to run risk-sensitive tests based not only on statutory regulatory rules but also on economic measures of solvency; and (3) provides ease of use for the stress tester and facilitates the communication of the result to decision makers. In the last context, the framework is flexible to amendments, as it is Excel based and is thus a versatile tool for stress testers. An example of an amendment to the tool is the ability to simulate the impact of an increase in funding costs.



Source: Authors.

Figure 5.5 Screenshot of Bank-Specific Results

The stress test framework presented in this chapter represents a work in progress. The tool will continue to be calibrated to account for ongoing regulatory changes and the evolution of best practice. At the same time, it is important not to lose sight of the fact that the primary responsibility of the stress tester is to run meaningful stress tests. For instance, it is up to the stress tester to decide how financial risks of specific banks link to macroeconomic stresses but to also make sure that the scenarios are plausible in any given situation. Further, an important requirement for running meaningful stress tests lies in the quality of input data.

REFERENCES

- Aikman, David, Piergiorgio Alessandri, Bruno Eklund, Prasanna Gai, Sujit Kapadia, Elizabeth Martin, Nada Mora, Gabriel Sterne, and Matthew Willison, 2009, "Funding Liquidity Risk in a Quantitative Model of Systemic Stability," Working Paper No. 372 (London: Bank of England). Available via the Internet: <http://www.bankofengland.co.uk/publications/Pages/workingpapers/2009/wp372.aspx>
- Basel Committee on Banking Supervision, 2006, "Basel II: International Convergence of Capital Measurement and Capital Standards: A Revised Framework—Comprehensive Version" (Basel, June). Available via the Internet: <http://www.bis.org/publ/bcbs128.htm>
- , 2010a, "Basel III: A Global Regulatory Framework for More Resilient Banks and Banking System" (Basel, December). Available via the Internet: http://www.bis.org/publ/bcbs189_dec2010.htm
- , 2010b, "Results of the Comprehensive Quantitative Impact Study" (Basel, December). Available via the Internet: <http://www.bis.org/publ/bcbs186.htm>
- Board of Governors of the Federal Reserve System, 2006, "Draft Federal Register Notice," March 22, p. 119 (Washington). Available via the Internet: http://www.federalreserve.gov/generalinfo/Basel2/DraftNPR/npr_memo
- Boss, Michael, Gerald Krenn, Claus Puhr, and Martin Summer, 2006, "Systematic Risk Monitor: A Model for Systematic Risk Analysis and Stress Testing of Banking Systems," Financial Stability Report, No. 11, June, pp. 83–95 (Vienna: Österreichische Nationalbank). Available via the Internet: http://www.oenb.at/en/presse/pub/period_pub/finanzmarkt/finanzmarktstabilitaet/financial_stability_report_11.jsp#tcm:16-173405
- Čihák, Martin, 2007, "Introduction to Applied Stress Testing," IMF Working Paper 07/59 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=20222>
- Djankov, Simeon, Caralee McLiesh, and Andrei Shleifer, 2007, "Private Credit in 129 Countries," *Journal of Financial Economics*, Vol. 84, No. 2, pp. 299–329.
- Foglia, Antonella, 2009, "Stress Testing Credit Risk: A Survey of Authorities' Approaches," *International Journal of Central Banking*, Vol. 5, No. 3, pp. 9–45.

- Gordy, Michael, and Eva Lütkebohmert, 2007, “Granularity Adjustment for Basel II,” Discussion Paper, Series 2 (Banking and Financial Studies), No. 01/2007 (Frankfurt: Deutsche Bundesbank). Available via the Internet: http://www.bundesbank.de/Navigation/EN/Publications/Discussion_papers/discussion_papers.html?gtp=93634_list%253D3
- Hardy, Daniel, and Christian Schmieder, 2013, “Rules of Thumb for Bank Solvency Stress Testing,” IMF Working Paper 13/232 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=41047.0>
- Mager, Ferdinand, and Christian Schmieder, 2009, “Stress Testing German Credit Portfolios,” *Journal of Risk Model Validation*, Vol. 3, No. 3, pp. 61–77.
- Moody’s, 2009, “Corporate Default and Recovery Rates, 1920–2008,” Special Comment, February.
- Schmieder, Christian, Claus Pühr, and Maher Hasan, 2011, “Next Generation Balance Sheet Stress Testing,” IMF Working Paper 11/83 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=24798>
- Schmieder, Christian, and S. Philipp Schmieder, 2011, “Impact of Legislation on Credit Risk—Comparative Evidence from the United States, the United Kingdom, and Germany,” IMF Working Paper 11/55 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=24703>
- Standard & Poor’s, 2010, “Default, Transition, and Recovery: 2009 Annual Global Corporate Default Study and Rating Transitions,” March 17. Available via the Internet: <http://www.standardandpoors.com/ratings/articles/en/us/?assetID=1245207201119>

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Of Runes and Sagas: Perspectives on Liquidity Stress Testing Using an Iceland Example

LI LIAN ONG • MARTIN ČIHÁK

The global financial crisis revealed weaknesses in the stress testing exercises performed on financial institutions and systems around the world. These failures were most evident in the area of liquidity risk, where now obvious vulnerabilities were left largely undetected, with stress tests having largely focused on solvency risk. This study uses publicly available data from a now defunct bank in Iceland, where liquidity shocks were immense, to demonstrate how a combination of stress tests of the various risks would have provided a clearer picture of existing vulnerabilities. We show that, ultimately, stress test models do not necessarily need to be complex or overly sophisticated. Basic stress tests, using appropriate assumptions and shocks, could reveal key areas of vulnerability to inform contingency planning.

METHOD SUMMARY

Overview	The method applies both market liquidity and funding liquidity shocks to estimate a bank's potential short-term liquidity gaps.
Application	The method may be used to determine a bank's overall vulnerability to specific liquidity shocks.
Nature of approach	Basic on- and off-balance-sheet shocks.
Data requirements	Granular short-term (on- and off-balance-sheet) assets and liabilities, including by currency breakdown and preferably by short-term maturity buckets.
Strengths	• The method is simple to implement using publicly available data. • It allows the calibration of shocks to individual asset and liability items.
Weaknesses	• The usefulness of the exercise depends on the granularity of data. • The potential lack of timeliness of the reported data may not yield useful information in a fast-changing environment.
Tool	The Excel spreadsheet macro is available in the toolkit, which is on the companion CD and at www.elibrary.imf.org/stress-test-toolkit . Contact author: L. L. Ong.

The global financial crisis revealed weaknesses in the stress testing exercises performed on financial institutions and systems by country authorities, as well as by international financial institutions, including the IMF. Postmortems on risk management in general, and stress testing in particular, have

pointed to weaknesses in the design of these tests and have led to intensified focus on strengthening the analysis in this area.¹ The “failure to stress” was particularly evident in the area of liquidity risk, where now obvious vulnerabilities across banks and banking systems were left largely undetected.² Stress

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¹ See, for instance, Basel Committee on Banking Supervision (BCBS, 2008c, 2009a, 2009b).

² The BCBS had highlighted the issue of liquidity risk shortly before the problem reached its nadir during the latter part of 2008 (BCBS, 2008a, 2008b).

testing work by country authorities, as well as that done by the IMF, typically would highlight many of the key risks, but specific vulnerabilities were not always adequately singled out, and the potential magnitudes of the identified problems were not fully quantified.

In the lead-up to the global financial crisis, stress tests had shown that banking systems in many countries could comfortably absorb significant shocks. More often than not, the stress tests did not provide indications that banking systems or individual banks were on the brink of significant stress. A relevant example is Iceland, where the banking sector imploded in the space of 10 days in the fall of 2008, and the government took over the domestic operations of the country's three largest banks. Stress test results published by the country's financial supervisor, the Financial Supervisory Authority, or Fjármálaeftirlitsins (FME), as late as mid-2008 showed that the three largest banks could comfortably weather large, concurrent shocks to financial markets and loan quality.³ Separately, stress tests of credit risk, performed by the Seðlabanki, the country's central bank, in 2008 suggested that Icelandic banks' capital positions were sufficiently strong to meet increases in unexpected losses but that they would need to bolster their capital under certain scenarios.

The objective of this chapter is to demonstrate why stress tests focusing largely on solvency may have failed to trigger early warnings of the problems that manifested in banks in general during the global financial crisis. We apply basic stress testing techniques, using publicly available data from a now defunct Icelandic bank (to avoid implicating any existing and viable institution). The aim is not to criticize the quality stress tests that were done in Iceland *per se*—there were no “international best practice” standards available—as the weaknesses have also been manifest in the stress tests performed in many countries around the world, including work done at the IMF. Rather, the intention is to show, more broadly, how inadequate focus on certain risks—notably, liquidity risk—and the omission of crucial assumptions in designing scenarios could reduce the usefulness of stress tests and potentially lead to a false sense of security.⁴

Our findings reveal several key weaknesses in the design of stress tests in general and highlight areas for improvement. Specifically, the typical emphasis of stress testing on solvency needs to be expanded to give due attention to liquidity risk as well. Off-balance-sheet information could prove very useful for calibrating stress tests; and it is also crucial that stress tests consider “unthinkable” scenarios, however unpalatable or impolitic they may be. Importantly, we show that stress tests need not be overly complex, nor do stress testing models need

to be too sophisticated to adequately capture risks. Stress tests—viewed holistically—may be useful in highlighting key areas of risk, to inform contingency planning.

The chapter is structured as follows. Section 1 presents the Iceland case study to demonstrate how stress tests could be designed to appropriately capture pertinent risks, using data from one of the country's largest banks at the time of the crisis in order to highlight the areas for improvement in stress testing exercises in general. Section 2 concludes with recommendations of how stress tests of banks generally could be enhanced to more usefully provide early indications of potential problems.

1. THE ICELAND CASE STUDY

A. Background on the banking crisis

The size of Iceland's banking sector was about nine times the country's GDP at the end of 2007, funded largely by external debt. The banking system was dominated by three large commercial banks, Kaupthing Bank hf. (“Kaupthing”), Landsbanki Íslands hf. (“Landsbanki”), and Glitnir banki hf. (“Glitnir”). The banks had relied heavily on market funding for their operations and had previously been criticized for a lack of diversification in their funding profile, in particular, for the low proportion of deposits in their funding.⁵ As a result, these banks intensified their focus on gathering deposits and successfully so. At the end of 2007, some 40 percent of their funding was in the form of deposits, up from 28 percent in 2006, with more than two-thirds sourced from nonresidents.⁶

Iceland's banking sector collapsed in early October 2008, following severe liquidity problems at the banks. On September 29, 2008, the Prime Minister announced that an agreement had been reached between the government and the largest owners of Glitnir, the country's third largest bank, whereby the government would contribute new share capital and take up a 75 percent stake in the bank. A week later, on October 6, Iceland's parliament, the Althing, passed emergency legislation enabling the government to intervene extensively in Iceland's financial system. On October 7, the FME put Landsbanki into receivership; Glitnir and Kaupthing followed on October 8 and 9, respectively. By that stage, the

³ See <http://www.fme.is/?PageID=168>.

⁴ More broadly based stability assessments performed by IMF staff had warned of the risks but were not specific enough on the source and potential impact of the shocks. See Mitra (2006), Ong (2007), and IMF (2008) for an assessment of developments in Iceland's financial sector and risks to the outlook.

⁵ See Seðlabanki (2008).

⁶ Almost 80 percent of deposits were in foreign currency, including 45 percent in pound sterling; some 56 percent of deposits were time deposits, and 86 percent of those time deposits had a maturity of three months or less. Landsbanki introduced the Icesave retail deposit product in the United Kingdom in October 2006, which became by far its most important source of deposit funding. Icesave was introduced in continental Europe on May 30, 2008, with its launch in the Netherlands. Separately, Kaupthing introduced Kaupthing Edge, an online retail savings account, to eleven countries in Europe. The purpose was to diversify the liabilities on Kaupthing's balance sheet by currency, customer type, and country. Kaupthing Edge was first launched in Finland, in October 2007.

three banks combined had amassed debt of an estimated \$61 billion—about 12 times the size of Iceland’s economy—and were unable to secure short-term funding to continue servicing their obligations.⁷

The nadir of the crisis, during the first week of October 2008, was marked by severe shocks to Iceland’s financial system. The value of the króna plummeted, and on October 7, 2008, the Seðlabanki attempted to peg the króna—trading at 172 against the euro at the time—at 131 against the euro; the peg was abandoned the next day. Trade in the currency was subsequently suspended, and payments in and out of Iceland effectively came to a standstill. A number of private interbank credit facilities to Icelandic banks were shut down, and banks were unable to finance their debts through short-term borrowing. In an attempt to alleviate depositor concerns, the government offered an unlimited guarantee to all depositors in banks and branches in Iceland. By that stage, however, deposit runs on the overseas branches of Icelandic banks had already started.

Could the extent of the devastating liquidity crisis that struck Iceland’s banking sector have been estimated? The Icelandic authorities had placed significant importance on their stress testing of the banking system and had worked to enhance their modeling of the risks:

- The FME had performed stress tests of the country’s individual banks on a quarterly basis. Indeed, the FME was one of the most transparent of supervisors in that the shock parameters of its stress tests and the results for individual banks were made available on its Web site. The stress scenarios assumed by the FME also appeared adequately conservative, incorporating significant shock parameters and scenarios. The acknowledged weakness in the stress test was that it did not incorporate any second-round effects from shocks to the banking sector.⁸ In addition, an IMF report recommended that the FME stress tests be expanded to evaluate significant tail events (IMF, 2008).
- Separately, the Seðlabanki had estimated banks’ resilience against loan losses and shocks to their portfolios.⁹ The distribution of loan losses was divided into expected and unexpected losses—the former is covered through provisioning, while the latter is covered with capital, up to specified tolerance levels. Expected and unexpected losses were estimated based on specified shocks.

In the years leading up to the crisis, the stress tests by the FME and the Seðlabanki had focused largely on solvency risk and, in hindsight, did not adequately capture the potential tail risks of a liquidity shock. The Seðlabanki’s rules on the liquidity ratio for credit institutions subject to minimum

reserve requirements represented a liquidity stress test of sorts.¹⁰ However, as we demonstrate, the appropriateness of the weightings applied to the individual items may be questionable during crisis periods, when extreme shocks occur. Indeed, the Seðlabanki had more formally introduced liquidity stress tests of individual banks in late 2007, using severe shock scenarios, as global financial conditions continued to deteriorate, and had identified vulnerabilities. As it turned out, it was too late to adequately address the problems at the banks by that stage.¹¹

B. Example: Landsbanki Íslands hf.

In this section, we analyze the design of the stress tests that were performed on the Icelandic banking sector by replicating the actual shocks to one of the country’s largest banks.¹² Specifically, we use end-2007 data published by the then-second largest bank, Landsbanki—which had operations encompassing Iceland, Continental Europe, the Nordic countries, the United Kingdom, Ireland, and North America—to re-create the liquidity situation at the bank. Detailed financial information is available in Landsbanki’s 2007 Annual Report, in the bank’s *Notes to the Consolidated Financial Statements* (Landsbanki Íslands hf., 2008).

On the surface at least, Landsbanki appeared to be in a sound, solvent position at the end of 2007. The bank’s capital adequacy ratio (CAR) at the time was 11.7 percent, and its Tier 1 ratio was 10.1 percent. When the FME’s stress test shocks were applied, Landsbanki’s CAR fell to 10.5 percent—still above the regulatory minimum 8 percent—and its Tier 1 ratio declined to 8.8 percent (Landsbanki Íslands hf., 2008). Separately, the Seðlabanki’s credit risk stress test results suggested that individual banks, including Landsbanki, could withstand an expected increase in defaults in 2008 (Seðlabanki, 2008).

Landsbanki also seemed to have sufficient liquid assets to cover its short-term liabilities. At the end of 2007, the parent

⁷ See Seðlabanki (2009) for a detailed discussion of the financial crisis in Iceland.

⁸ See Ong (2007).

⁹ See Seðlabanki (2007, 2008).

¹⁰ Rule No. 317/2006 on the Liquidity Ratio for Credit Institutions governs the ratio of weighted liquid assets and liabilities, whereby assets and liabilities are weighted using specific coefficients reflecting the probability of recovery of particular short-term assets, and the probability of that particular short-term liabilities must be repaid. The liquidity ratio is required to be calculated on the basis of data at the end of each month and presented in a separate report to the central bank.

¹¹ The Seðlabanki’s stress tests did not apply set stress scenarios across the banking sector. Rather, shocks were varied according to the different risk factors faced by different banks and the situation developing in financial markets. They included closed access to credit lines, a significant drop in securities prices, a depreciation in the króna, and increased defaults in the bank’s loan portfolios; banks were also stressed to determine how much of their deposits could reasonably be paid out. The results of the stress tests were never made public. Subsequently, the Seðlabanki, in cooperation with the FME, gathered additional liquidity information from the banks, with greater frequency, as input into liquidity stress tests.

¹² Comparisons with stress tests performed by the FME and Seðlabanki focus on publicly available information; the authorities’ specific liquidity stress tests on individual banks are not analyzed.

TABLE 6.1

Landsbanki: Net Assets by Maturity, as of December 31, 2007
(in millions of króna)

	Up to 3 months	3–12 months	1–5 years	Over 5 years	Total
Total assets	1,321,104	337,526	945,953	706,549	3,311,132
Total liabilities	1,644,157	261,887	730,413	425,452	3,061,909
Net assets	–323,053	75,639	215,540	281,097	249,223
Cumulative net assets	–323,053	–247,414	–31,874	249,223	

Sources: Landsbanki, *Annual Report, 2007*; and authors' calculations.

bank's liquidity ratio, calculated in accordance with the central bank's rule No. 317/2006, was 2.23—well above the required ratio of higher than one for the next three-month period—even though the bank's nonderivative cash flow obligations at end-2007 revealed that its short-term assets (less than one year) were lower than its short-term liabilities (Table 6.1). Under normal conditions, Landsbanki's total assets, held to maturity, should have been sufficient to cover its total obligations.

Liquidity Position

To assess Landsbanki's liquidity position in greater depth, we first reconstruct the bank's overall liquidity position using data from the *Notes to the Consolidated Financial Statements* (Table 6.2).¹³ Specifically, we

- combine the information presented on (1) liquidity risk; (2) derivative cash flows; and (3) currency risk, to derive the balances of the individual items by currency;¹⁴
- limit our focus to the bank's short-term position, of one year or less, to highlight the key liquidity risk items; and
- consider the more liquid short-term assets, that is, those that are ostensibly the easiest to liquidate in order to obtain funds quickly. These items are in row 1 and rows 4–7 of Table 6.2.¹⁵

The short-term liquidity position at Landsbanki, at end-2007, is as follows:

- The total short-term assets amount to 1.70 trillion króna; the total liquid short-term assets are 728 billion króna.

- The total short-term liability amounts to 1.95 trillion króna.

Under normal conditions, a negative net asset position does not necessarily reflect a liquidity crisis. Some short-term assets and liabilities are usually rolled over when they mature. One such item is deposits, which are typically not required to be paid back in their entirety in a short period of time and therefore are usually considered a stable source of funding. Indeed, Landsbanki (2008, p. 25) had noted that “Deposits, particularly retail deposits, have historically provided the bank with more stable funding than capital markets.” In the case of Landsbanki, deposits from customers (Table 6.2, row 13) alone appear more than sufficient to cover any liquidity shortfall (Table 6.2, rows 23 and 24). Notably, customer deposits in the United Kingdom and Iceland represented the main sources of funding for the bank.

Stress Tests of Liquidity Risk

In a severe financial crisis, the probability of extreme and unlikely events occurring increases. In Iceland's case, it meant that the weightings assigned to liquid assets and liabilities per the Seðlabanki's requirements were unlikely to hold. During periods of extreme stress, weightings on liquid claims are likely to fall on the back of any sharp drop in the market value of securities portfolios; in addition, repayment of maturing loans and advances may fall significantly as asset quality deteriorates. Conversely, the weightings placed on liquid liabilities are likely to rise significantly as depositors pull out their funds and lenders refuse to roll over their financing.

Buiter and Sibert (2008a, 2008b) observe in their assessment of the Icelandic banking sector that even fundamentally solvent banking systems can be brought down by a severe liquidity crunch. The authors posit that liquidity crises could either be the result of conventional bank runs by depositors and other creditors (“funding liquidity crises”) or arise through illiquidity in the markets for its assets (“market liquidity crises”). More specifically, Buiter and Siebert (2008b, p. 3) note that “There is no such thing as a safe deposit-taking bank on its own, even if its assets are of good quality and it has enough liquid assets to cope with normal variations in the net flow of deposits and other short-term liabilities.” They argue that

¹³ See the Appendix.

¹⁴ The individual country amounts for each asset and liability item as a percentage of the total are used to prorate the individual country amounts across each maturity group. The total for “other” is assumed to be held in equal proportions in Denmark, Norway, and Sweden.

¹⁵ We define cash, bonds, and equities, as well as hedged securities and derivatives held for trading, as being very liquid assets, assuming a market exists at the time of sale. Derivatives held for hedging are likely to be more difficult to unwind given the potential implications for capital adequacy, whereas loans and advances may be more difficult to claim from debtors, who may default.

TABLE 6.2

Landsbanki: Initial 12-Month Liquidity Position as of December 31, 2007 (in millions of ISK)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Short-Term Assets and Liabilities	ISK	EUR	USD	GBP	CHF	DKK	NOK	SEK	Total
Short-term assets									
(1) Cash and cash balances with Central Bank	69,285	8,510	0	1,145	0	873	873	873	81,559
(2) Loans and advances to financial institutions	22,066	58,378	14,935	57,015	1,184	2,281	2,281	2,281	160,423
(3) Loans and advances to customers	212,113	170,879	70,921	120,789	79,598	33,434	33,434	33,434	754,600
(4) Bonds	116,166	170,072	59,917	11,730	823	1,303	1,303	1,303	362,617
(5) Equities	19,656	17,802	1,199	6,421	7	6,441	6,441	6,441	64,407
(6) Hedged securities	92,931	40,169	23,157	11,440	399	2,695	2,695	2,695	176,181
(7) Derivatives held for trading	15,430	11,696	7,934	2,149	505	1,721	1,721	1,721	42,876
(8) Derivatives held for hedging	131	81	1,193	36	1	5	5	5	1,457
(9) Unsettled securities trading	3,724	26,270	799	27,982	0	23	23	23	58,845
(10) Total short-term assets	551,503	503,857	180,055	238,706	82,517	48,776	48,776	48,776	1,702,965
(11) = (1) + (4) + (5) + (6) + (7) Total liquid short-term assets	313,468	248,249	92,207	32,885	1,734	13,032	13,032	13,032	727,640
Short-term liabilities									
(12) Deposits from financial institutions	19,202	91,010	20,410	25,130	6,010	58,923	58,923	58,923	338,530
(13) Deposits from customers	291,329	186,174	19,851	870,148	1,324	3,154	3,154	3,154	1,378,288
(14) Borrowings	14,731	68,347	23,881	915	2,589	1,490	1,490	1,490	114,934
(15) Financial liabilities designated at fair value	0	1,797	43	0	0	359	359	359	2,918
(16) Subordinated loans	453	5,128	1,955	0	0	62	62	62	7,721
(17) Trading liabilities	843	2,387	1,414	2,110	67	95	95	95	7,105
(18) Derivatives held for hedging	0	765	0	0	12	5	5	5	792
(19) Tax liabilities	5,255	0	0	0	0	0	0	0	5,255
(20) Unsettled securities trading	0	22,122	443	23,516	2,285	11	11	11	48,399
(21) Derivatives held for trading	20,844	12,803	153	12,850	750	14	14	14	47,443
(22) Total short-term liabilities	352,657	390,534	68,149	934,669	13,036	64,113	64,113	64,113	1,951,385

Sources: Authors; and Landsbanki, *Annual Report 2007*.

Note:

(1) Foreign holdings of cash in foreign currency are adjusted for exchange rate changes.

(2) Loans and advances are not adjusted for changes in the value of NPLs as they are net of provisions; foreign currency amounts are also adjusted for exchange rate changes.

(3) Loans and advances are not adjusted for changes in the value of NPLs as they are net of provisions; foreign currency amounts are also adjusted for exchange rate changes.

(4) Bond holdings are adjusted for price changes; foreign currency holdings are also adjusted for exchange rate changes.

(5) Equity holdings are adjusted price changes; foreign currency holdings are also adjusted for exchange rate changes.

(6) It is assumed that hedged securities do not change in value as any changes would have been offset by the hedge.

(7) Given the difficulty in determining the unrealized gains or losses on derivatives agreements, derivatives price changes are used instead, and are assumed to be a weighted average of bond and equity price changes; foreign currency values are adjusted for exchange rate changes.

(8) It is assumed that derivatives held for hedging do not change in value as any changes would have been offset by underlying asset.

(9) Foreign currency amounts of unsettled securities trading are adjusted for exchange rate changes.

(12) Deposits in foreign currency are adjusted for exchange rate changes.

(13) Deposits in foreign currency are adjusted for exchange rate changes.

(14) Borrowings in foreign currency are adjusted for exchange rate changes.

(15) Financial liabilities in foreign currency are adjusted for exchange rate changes.

(16) Subordinated loans in foreign currency are adjusted for exchange rate changes.

(17) Trading liabilities in foreign currency are adjusted for exchange rate changes.

(18) It is assumed that derivatives held for hedging do not change in value as any changes would have been offset by underlying liability.

(19) Tax liabilities in foreign currency are adjusted for exchange rate changes.

(20) Foreign currency amounts of unsettled securities trading are adjusted for exchange rate changes.

(21) Given the difficulty in determining the unrealized gains or losses on derivatives agreements, derivatives price changes are used instead, and are assumed to be a weighted average of bond and equity price changes; foreign currency values are adjusted for exchange rate changes.

TABLE 6.3

Iceland: Stress Test Shock Scenarios (in percent unless stated otherwise)

Scenario	FME ¹	FME and Seðlabanki ¹	Actual: Between December 31, 2007, and October 7, 2008
Change in domestic stock prices	–35	–35	–52
Change in foreign stock prices	–25	–25	Between –24 and –44
Change in value of NPLs ²	–20	–20	–100*
Change in bond prices ³	–7	–50	Up to +700 basis points in yields
Change in krona levels	–20	–20	Between –35 and –55
Liquid claims⁴			
Repayment of claims against domestic credit undertakings	—	90	100 less NPLs
Repayment of claims against foreign credit undertakings	—	100	100 less NPLs
Repayments of claims against other parties	—	80	100 less NPLs
Positive market value of derivative agreements	—	90	—
Liquid liabilities			
Withdrawal of approved loan commitments	100	–80	–100
Withdrawal of time deposits in krona ⁵	—	–5	–43
Withdrawal of other deposits	—	–10	–43
Borrowings	100	–100	–100

Sources: Bloomberg; FME; Seðlabanki; and various media reports.

Note: FME = Financial Supervisory Authority/Fjármálaeftirlitsins; NPL = nonperforming loan.

1. The FME stress test does not assume shocks to specific liability items; for demonstration purposes, we assume that all short-term liabilities are redeemed except for deposits from customers which remain stable.

2. Where FME and Seðlabanki shocks overlap, the larger shock is applied (e.g., the loss of value in bonds and stocks).

3. Given the illiquidity of Icelandic bonds at the time of the crisis—partly as a result of the suspension in krona trading—we use the change in credit default swap spreads to price Icelandic bonds during that period. In calculating the change in bond prices under the actual scenario, we make the simple assumption that the bonds which have maturities of up to 12 months each have a duration of one year.

4. The NPL ratio as at end-2007 was 0.9 percent. The ratio at around the time of the banking crisis was estimated at about 3 percent although it was likely to have been higher; we assume a 50 percent loss in NPLs. Under the actual shock scenario, it is assumed that the repayment of all claims by debtors are for the full amount less the reported NPLs, as there is no information of debtors being unable to repay the bank at the time of the shock.

5. It is assumed that 43 percent of all deposits are withdrawn under the “actual” scenario—the same proportion as those calculated for the United Kingdom—before the accounts were frozen.

any highly leveraged institution with assets that are mostly long term and illiquid and liabilities that are mostly short term could suffer “catastrophic” liquidity shortages.

We consider such possibilities by applying different sets of shocks to the Landsbanki liquidity data constructed in Table 6.2. The different sets of shocks are summarized in Table 6.3:

- The first set of shocks (the first column) comprises the shocks used in the FME’s stress tests. Although the FME shocks do not include shocks to specific liability items, we assume that all short-term liabilities are redeemed except for deposits from customers, which remain stable, for demonstration purposes.
- The second set of shocks comprises a combination of the FME shocks with the Seðlabanki’s liquidity shocks per rule No. 317/2006.
- The third set of shocks is a replication of the actual shocks that hit Icelandic markets and banks between end-2007 and October 2008.

The differences among these three scenarios lie in the types of shocks that are applied and the magnitudes of the common shocks. A key assumption is that the bank’s financial position as of December 31, 2007, remains static. In our presentation of the results, short-term liabilities that are assumed to be redeemed are recorded as outflows, whereas short-term assets available for liquidation are presented as inflows.

Stress Test with FME Shocks. Under the scenario where the FME shocks are applied, the impact is largely on the asset side of the balance sheet, given the focus on solvency risk (Table 6.4). Specifically, the items in rows 2–5 are shocked. The exchange rate shock would have an impact on both foreign currency assets and liabilities (all items in columns 2–8, which are reported in króna). The stress test results suggest that

- Landsbanki’s deposit funding from customers clearly drives its liquidity position:

TABLE 6.4

Landsbanki: 12-Month Liquidity Situation as of December 31, 2007, after FME Shocks Are Applied (in millions of ISK)									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Short-Term Assets and Liabilities	ISK	EUR	USD	GBP	CHF	DKK	NOK	SEK	Total
Short-term assets available for liquidation									
(1) Cash and cash balances with Central Bank	69,285	10,212	0	1,374	0	1,048	1,048	1,048	84,014
(2) Loans and advances to financial institutions	22,106	70,054	17,922	68,418	1,421	2,738	2,738	2,738	1,88,134
(3) Loans and advances to customers	212,495	205,055	85,105	144,946	95,518	40,120	40,120	40,120	8,63,479
(4) Bonds	108,034	189,800	66,867	13,091	918	1,454	1,454	1,454	3,83,074
(5) Equities	12,776	16,022	1,079	5,779	6	5,797	5,797	5,797	53,052
(6) Hedged securities	92,931	40,169	23,157	11,440	399	2,695	2,695	2,695	1,76,181
(7) Derivatives held for trading	12,190	11,790	7,997	2,166	509	1,734	1,734	1,734	39,855
(8) Derivatives held for hedging	131	81	1,193	36	1	5	5	5	1,457
(9) Unsettled securities trading	3,724	31,524	959	33,578	0	28	28	28	69,869
(10) Total short-term assets available for liquidation	533,673	574,706	204,279	280,828	98,772	55,619	55,619	55,619	18,59,115
(11) Total liquid short-term assets available for liquidation	295,217	267,993	99,101	33,850	1,833	12,728	12,728	12,728	7,36,176
Less: Short-term liabilities that are redeemed									
(12) Deposits from financial institutions (assuming 100 percent redemption by other credit institutions)	7,713	43,870	9,838	12,113	2,897	28,402	28,402	28,402	1,61,638
(13) Deposits from customers (assuming 100 percent redemption)	291,329	223,409	23,821	1,044,178	1,588	3,785	3,785	3,785	15,95,680
(14) Borrowings	14,731	82,016	28,657	1,099	3,107	1,788	1,788	1,788	1,34,975
(15) Financial liabilities designated at fair value	0	2,156	51	0	0	431	431	431	3,502
(16) Subordinated loans	453	6,153	2,346	0	0	74	74	74	9,175
(17) Trading liabilities	843	2,864	1,697	2,532	80	114	114	114	8,357
(18) Derivatives held for hedging	0	765	0	0	12	5	5	5	792
(19) Tax liabilities	5,255	0	0	0	0	0	0	0	5,255
(20) Unsettled securities trading	0	26,546	532	28,219	2,742	13	13	13	58,079
(21) Derivatives held for trading	16,467	12,906	154	12,953	756	14	14	14	43,279
(22) Total short-term liabilities that are redeemed	336,791	400,687	67,095	1,101,094	11,182	34,627	34,627	34,627	20,20,730
(23) = (10) – (22) Total short-term assets available less total short-term liabilities redeemed (assuming full redemption of deposits)	196,881	174,019	137,184	–820,265	87,591	20,992	20,992	20,992	–1,61,615
(24) = (11) – (22) Total liquid short-term assets available less total short-term liabilities redeemed (assuming full redemption of deposits)	–41,575	–132,694	32,005	–1,067,244	–9,349	–21,899	–21,899	–21,899	–12,84,555
(25) = (23) + (13) Total short-term assets available less total short-term liabilities redeemed (assuming stability of deposits)	488,210	397,429	161,005	223,912	89,179	24,776	24,776	24,776	14,34,065
(26) = (24) + (13) Total liquid short-term assets available less total short-term liabilities redeemed (assuming stability of deposits)	249,754	90,715	55,827	–23,066	–7,761	–18,115	–18,115	–18,115	3,11,125

(continued)

TABLE 6.4 (continued)

Sources: Authors; and Landsbanki, *Annual Report 2007*.

Note: FME = Financial Supervisory Authority/Fjármálaeftirlitsins.

- (1) Foreign holdings of cash in foreign currency are adjusted for exchange rate changes.
- (2) Loans and advances are not adjusted for changes in the value of NPLs as they are net of provisions; foreign currency amounts are also adjusted for exchange rate changes.
- (3) Loans and advances are not adjusted for changes in the value of NPLs as they are net of provisions; foreign currency amounts are also adjusted for exchange rate changes.
- (4) Bond holdings are adjusted for price changes; foreign currency holdings are also adjusted for exchange rate changes.
- (5) Equity holdings are adjusted price changes; foreign currency holdings are also adjusted for exchange rate changes.
- (6) It is assumed that hedged securities do not change in value as any changes would have been offset by the hedge.
- (7) Given the difficulty in determining the unrealized gains or losses on derivatives agreements, derivatives price changes are used instead, and are assumed to be a weighted average of bond and equity price changes; foreign currency values are adjusted for exchange rate changes.
- (8) It is assumed that derivatives held for hedging do not change in value as any changes would have been offset by underlying asset.
- (9) Foreign currency amounts of unsettled securities trading are adjusted for exchange rate changes.
- (12) Deposits are apportioned according to their sources; 100 percent of the amount owing to other credit institutions are assumed to be redeemed, the central bank does not redeem its deposits; those denominated in foreign currency are adjusted for exchange rate changes.
- (13) Deposits are assumed to remain stable.
- (14) All borrowings are assumed to be redeemed; balances in foreign currency are adjusted for exchange rate changes.
- (15) All financial liabilities are assumed to be redeemed; balances in foreign currency are adjusted for exchange rate changes.
- (16) All trading liabilities are assumed to be redeemed; balances in foreign currency are adjusted for exchange rate changes.
- (17) All subordinated loans are assumed to be redeemed; balances in foreign currency are adjusted for exchange rate changes.
- (18) It is assumed that derivatives held for hedging do not change in value as any changes would have been offset by changes in the values of the underlying liabilities.
- (19) Tax liabilities in foreign currency are adjusted for exchange rate changes.
- (20) Foreign currency amounts of unsettled securities trading are adjusted for exchange rate changes.
- (21) Given the difficulty in determining the unrealized gains or losses on derivatives agreements, derivatives price changes are used instead, and are assumed to be a weighted average of bond and equity price changes; foreign currency values are adjusted for exchange rate changes.

- If all deposits from customers were to be redeemed, the net short-term asset balance would show a deficit of 162 billion króna, widening to a deficit of 1.28 trillion króna if only liquid short-term assets are taken into account (rows 23 and 24, column 9).
- Assuming the stability of deposits from customers, of about 1.60 trillion króna, the overall liquidity position shows a positive net balance of 1.43 trillion króna if the redemption of liabilities is covered by all short-term assets (row 25, column 9); the balance falls to 311 billion króna if only liquid short-term assets are realized to cover redemptions (row 26, column 9).
- Naturally, the liquidity position would improve if it is assumed that some creditors providing market funding would continue to roll over their short-term loans (rows 12 and 14).
- The bank's liquidity position differs across currencies following the shock:
 - If all deposits were redeemed, net short-term assets would be positive for the króna, euro, U.S. dollar, and Swiss franc positions (row 23, columns 1–3 and 5), but only the U.S. dollar operations would show a surplus if liquid short-term assets are recoverable (row 24, column 3).
 - Assuming stable deposits, the net short-term asset balances for the króna, euro, and U.S. dollar show surpluses when either the short-term assets or the liquid short-term assets are taken into account (rows 23 and 24, columns 1–3); liquidity positions in the other currencies are largely negative (rows 23 and 24, columns 4–8), suggesting that the bank's operations in the former set of currencies may have to cover the liquidity needs in the other currencies.
- The bank's liquidity position in pound sterling is notable in that short-term deposits from customers in this currency are the largest by far, accounting for 65 percent of total short-term deposits (row 13, column 4). Any run on those accounts alone would have caused significant funding problems for the bank.

Stress Test Combining FME and Seðlabanki Liquidity Shocks. Under the scenario where the combined FME and Seðlabanki liquidity shocks are applied, both the asset and liability sides of Landsbanki's balance sheet are affected (Table 6.5):

- This time, the value of the bank's obligations is also directly shocked using the magnitudes described in the Seðlabanki rule No. 317/2006 on the Liquidity Ratio for Credit Institutions.¹⁶ In sum, the items in

rows 2–5 are directly shocked, as are those in rows 12–14 and 21. Where FME and Seðlabanki shocks overlap, the larger shock is applied.

- The remaining liabilities, which are relatively small, are assumed to be redeemed in full.
- The exchange rate shock would affect both foreign currency assets and liabilities (all items in columns 2–8, which are reported in króna).

The stress test results suggest that

- There would be sufficient short-term assets to cover the outflows.
 - Short-term assets available for liquidation less the redemptions on the short-term liabilities show an aggregate surplus of 960 billion króna (row 23, column 9).
 - The short-term assets in each currency would also be adequate to cover the corresponding short-term liabilities (row 23).
- Liquid short-term assets, in aggregate, would be sufficient to repay short-term liabilities (row 24, column 9).
 - Liquid short-term assets less redemptions of short-term liabilities result in an aggregate surplus of 12 billion króna (row 24, column 9).
 - On a currency-by-currency basis, however, the surpluses in króna and U.S. dollars (row 24, columns 1 and 3) may be required to cover the deficits in the other currencies.

Stress Test with Actual Shocks.

Liquidity Situation. Under the *scenario* where the actual shocks are replicated, the impact on the liability side of the balance sheet is especially marked (Table 6.6). As before, the items in rows 2–5 are shocked, and the items in columns 2–8 are adjusted by the actual changes in the króna exchange rate. In addition, the funding (liability) side of the balance sheet is significantly affected, with counterparties demanding repayment as a result of Landsbanki's sharply rising counterparty risk:

- Some 43 percent of deposits in the United Kingdom, in Landsbanki's Icesave accounts, were withdrawn during the crisis. In the absence of information, we conservatively assume runs of a similar magnitude on the rest of the deposits at Landsbanki.¹⁷
- Short-term market funding, through deposits from other credit institutions, and borrowings through securities issuances and syndicated loans, became inaccessible as lenders refused to roll over these loans or to make new ones, and the bank was required to repay maturing liabilities (rows 12 and 14).

As actually happened, the impact of the actual shocks to Iceland's financial system on Landsbanki's overall liquidity situation was quite severe. At the point where deposit withdrawals at branches abroad were suspended by Landsbanki,

¹⁶ The Seðlabanki rule No. 317/2006 assigns a weighting of 5 percent to time deposits and 10 percent to other deposits in its calculation of liquid liabilities. In other words, it assumes that only 5 percent of time deposits and 10 percent of other deposits would be withdrawn. In the months leading up to the crisis, the Seðlabanki had applied greater shocks to banks' deposits, in stress tests of liquidity risk (see earlier discussion).

¹⁷ The extent of the runs was likely to have been smaller as the outflow of deposits from banks in Iceland subsided following the government's announcement of an unlimited guarantee.

TABLE 6.5

Landsbanki: 12-Month Liquidity Situation as of December 31, 2007, after a Combination of FME and Seðlabanki Shocks Are Applied (in millions of ISK)									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Short-Term Assets and Liabilities	ISK	EUR	USD	GBP	CHF	DKK	NOK	SEK	Total
Short-term assets available for liquidation									
(1) Cash and cash balances with Central Bank	69,285	10,212	0	1,374	0	1,048	1,048	1,048	84,014
(2) Loans and advances to financial institutions	19,860	70,054	17,922	68,418	1,421	2,738	2,738	2,738	185,888
(3) Loans and advances to customers	169,691	164,044	68,084	115,957	76,414	32,096	32,096	32,096	690,478
(4) Bonds	58,083	102,043	35,950	7,038	494	782	782	782	205,954
(5) Equities	7,371	17,090	1,151	6,164	7	6,183	6,183	6,183	50,332
(6) Hedged securities	92,931	40,169	23,157	11,440	399	2,695	2,695	2,695	176,181
(7) Derivatives held for trading	6,751	6,140	4,165	1,128	265	903	903	903	21,160
(8) Derivatives held for hedging	131	81	1,193	36	1	5	5	5	1,457
(9) Unsettled securities trading	3,724	31,524	959	33,578	0	28	28	28	69,869
(10) Total short-term assets available for liquidation	427,826	441,357	152,581	245,134	79,001	46,478	46,478	46,478	1,485,332
(11) Total liquid short-term assets available for liquidation	234,421	175,655	64,423	27,144	1,165	11,611	11,611	11,611	537,640
Less: Short-term liabilities that are redeemed									
(12) Deposits from financial institutions	6,171	35,096	7,871	9,691	2,318	22,722	22,722	22,722	129,311
(13) Deposits from customers	22,420	22,341	2,382	104,418	159	378	378	378	152,855
(14) Borrowings	14,731	82,016	28,657	1,099	3,107	1,788	1,788	1,788	134,975
(15) Financial liabilities designated at fair value	0	2,156	51	0	0	431	431	431	3,502
(16) Subordinated loans	453	6,153	2,346	0	0	74	74	74	9,175
(17) Trading liabilities	843	2,864	1,697	2,532	80	114	114	114	8,357
(18) Derivatives held for hedging	0	765	0	0	12	5	5	5	792
(19) Tax liabilities	5,255	0	0	0	0	0	0	0	5,255
(20) Unsettled securities trading	0	26,546	532	28,219	2,742	13	13	13	58,079
(21) Derivatives held for trading	9,119	6,722	80	6,746	394	8	8	8	23,084
(22) Total short-term liabilities that are redeemed	58,992	184,660	43,615	152,705	8,811	25,534	25,534	25,534	525,384
(23) = (10) – (22) Total short-term assets available less total short-term liabilities redeemed	368,833	256,696	108,966	92,429	70,190	20,944	20,944	20,944	959,948
(24) = (11) – (22) Total liquid short-term assets available less total short-term liabilities redeemed	175,428	–9,006	20,809	–125,560	–7,646	–13,923	–13,923	–13,923	12,256

Sources: Authors; Landsbanki, *Annual Report 2007*; and Seðlabanki.

Note: FME = Financial Supervisory Authority/Fjármálaeftirlitsins.

TABLE 6.6

Landsbanki: 12-Month Liquidity Situation as of December 31, 2007, after Actual Shocks and Freezing of Deposit Withdrawals Are Applied (in millions of ISK)									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Short-Term Assets and Liabilities	ISK	EUR	USD	GBP	CHF	DKK	NOK	SEK	Total
Short-term assets available for liquidation									
(1) Cash and cash balances with Central Bank	69,285	11,077	0	1,443	0	1,136	1,099	1,116	81,559
(2) Loans and advances to financial institutions	21,404	73,710	19,552	69,697	1,547	2,879	1,640	2,830	193,258
(3) Loans and advances to customers	205,750	215,758	92,845	147,654	103,932	42,197	24,031	41,474	873,641
(4) Bonds	108,826	222,884	82,090	14,894	1,087	1,706	1,648	1,673	434,807
(5) Equities	9,469	15,298	1,098	5,771	7	5,940	4,568	5,386	47,538
(6) Hedged securities	92,931	40,169	23,157	11,440	399	2,695	2,695	2,695	176,181
(7) Derivatives held for trading	10,944	12,689	9,067	2,330	591	1,920	1,698	1,824	41,064
(8) Derivatives held for hedging	131	81	1,193	36	1	5	5	5	1,457
(9) Unsettled securities trading	3,724	34,195	1,078	35,264	0	30	29	30	74,351
(10) Total short-term assets available for liquidation	522,465	625,861	230,080	288,528	107,564	58,508	37,415	57,034	1,923,857
(11) Total liquid short-term assets available for liquidation	291,456	302,117	115,412	35,878	2,084	13,396	11,709	12,696	781,150
Less: Short-term liabilities that are redeemed									
(12) Deposits from financial institutions	7,713	47,587	11,065	12,721	24,977	30,797	29,799	30,269	194,927
(13) Deposits from customers	124,345	103,435	11,435	426,946	760	1,752	1,695	1,722	672,089
(14) Borrowings	14,731	88,966	32,230	1,154	3,485	1,939	1,876	1,906	146,287
(15) Financial liabilities designated at fair value	0	2,339	58	0	0	468	453	460	3,777
(16) Subordinated loans	453	6,675	2,638	0	0	80	78	79	10,003
(17) Trading liabilities	843	3,107	1,908	2,659	90	123	119	121	8,971
(18) Derivatives held for hedging	0	765	0	0	12	5	5	5	792
(19) Tax liabilities	5,255	0	0	0	0	0	0	0	5,255
(20) Unsettled securities trading	0	28,796	598	29,635	3,076	14	14	14	62,147
(21) Derivatives held for trading	14,784	13,891	175	13,933	878	16	14	14	43,705
(22) Total short-term liabilities that are redeemed	168,124	295,561	60,106	487,048	33,277	35,194	34,052	34,589	1,147,953
(23) = (10) – (22) Total short-term assets available less total short-term liabilities redeemed	354,340	330,300	169,974	–198,521	74,287	23,314	3,363	22,445	775,904
(24) = (11) – (22) Total liquid short-term assets available less total short-term liabilities redeemed	123,331	6,555	55,306	–451,170	–31,193	–21,798	–22,343	–21,894	–366,803

(continued)

TABLE 6.6 (continued)

Sources: Authors; Landsbanki, *Annual Report 2007*; and various media reports.

Note:

- (1) Foreign holdings of cash in foreign currency are adjusted for exchange rate changes.
- (2) Loans and advances are adjusted for NPLs; foreign currency amounts are also adjusted for exchange rate changes.
- (3) Loans and advances are adjusted for NPLs; foreign currency amounts are also adjusted for exchange rate changes.
- (4) Bond holdings are adjusted for actual price changes; foreign currency holdings are also adjusted for exchange rate changes.
- (5) Equity holdings are adjusted for actual price changes; foreign currency holdings are also adjusted for exchange rate changes.
- (6) It is assumed that hedged securities do not change in value as any changes would have been offset by the hedge.
- (7) Given the difficulty in determining the unrealized gains or losses on derivatives agreements, derivatives price changes are used instead, and are assumed to be a weighted average of bond and equity price changes; foreign currency values are adjusted for exchange rate changes.
- (8) It is assumed that derivatives held for hedging do not change in value as any changes would have been offset by changes in the values of the underlying assets.
- (9) Foreign currency amounts of unsettled securities trading are adjusted for exchange rate changes.
- (12) Deposits are apportioned according to their sources; all amounts owing to other credit institutions are assumed to be redeemed, the central bank does not redeem its deposits; those denominated in foreign currency are adjusted for exchange rate changes.
- (13) It is assumed that 43 percent of all deposits are withdrawn, in the same proportion as those in the United Kingdom, before the accounts were frozen; the amounts denominated in foreign currency are adjusted for exchange rate changes.
- (14) All financial liabilities are assumed to be redeemed; balances in foreign currency are adjusted for exchange rate changes.
- (15) All borrowings are assumed to be redeemed; balances in foreign currency are adjusted for exchange rate changes.
- (16) All trading liabilities are assumed to be redeemed; balances in foreign currency are adjusted for exchange rate changes.
- (17) All subordinated loans are assumed to be redeemed; balances in foreign currency are adjusted for exchange rate changes.
- (18) It is assumed that derivatives held for hedging do not change in value as any changes would have been offset by changes in the values of the underlying liabilities.
- (19) Tax liabilities in foreign currency are adjusted for exchange rate changes.
- (20) Foreign currency amounts of unsettled securities trading are adjusted for exchange rate changes.
- (21) Given the difficulty in determining the unrealized gains or losses on derivatives agreements, derivatives price changes are used instead, and are assumed to be a weighted average of bond and equity price changes; foreign currency values are adjusted for exchange rate changes.

- The bank's net short-term liquidity position remained in surplus of 776 billion króna (row 23, column 9).
- However, its liquid short-term assets would have been insufficient to cover demands for the repayment of its short-term liabilities, by 367 trillion króna (row 24, column 9).
- In addition, the shutdown of the payment system would have significantly affected its ability to sell off its traditionally more liquid assets.
- On a by-currency basis:
 - Landsbanki would have had sufficient assets to cover its króna, euro, and U.S. dollar liabilities (rows 23 and 24, columns 1–3) at the point when repayments were halted.
 - The largest deficits were in the bank's pound sterling operations—its high reliance on retail deposit funding from its United Kingdom Icesave accounts meant that the run on deposits significantly affected its liquidity position (rows 23 and 24, column 4).
 - Furthermore, the shutdown in the international payment system would have prevented any transfer of surpluses by the bank to meet its other foreign currency liabilities, even if it had wanted to do so.

Funding Gap. The key question, then, is to what extent Landsbanki's estimated funding gap would have been at the time of the actual shocks. If we assume that the bank would have been able to realize its short-term assets in the market, and that all its short-term creditors would have demanded repayment of their maturing claims from the bank (Table 6.7), we find the following:

- Short-term assets less short-term liabilities would have shown a funding gap of 231 billion króna (row 23, column 9); liquid short-term assets less short-term liabilities would have resulted in a funding shortfall of about 1.37 trillion króna (row 24, column 9).
- The funding gap in pound sterling would have been the most severe, at between 868 billion króna, if all short-term assets could be realized, and 1.12 trillion króna, if only liquid short-term assets could be realized (rows 23 and 24, column 4). Indeed, the surplus net assets in all other currencies would have been insufficient to cover the funding gap in pound sterling (row 23).

If the bank would not have been able to sell its securities assets, such as bonds and equities, as a result of the closure of the payment system, then the funding gap would have widened considerably.

Summary of Results. Our exercise illustrates that Landsbanki's liquidity position worsens when more realistic shocks are introduced to include the funding (liability) side of the balance sheet (Table 6.8). In particular, we find that

- (1) The short-term balance sheet of the bank is more than 1.5 times the country's GDP.

- (2) Under the FME shocks, the bank would still be able to comfortably meet its short-term obligations with its short-term assets, if deposits remain stable.
- (3) The bank's liquidity position worsens somewhat when the Seðlabanki liquidity shocks are added, although it would still be able to cover its obligations.
- (4) The introduction of the actual shocks to the bank's deposit funding further reduces the availability of liquidity. At the point where the withdrawal of deposits is frozen, the bank would not have been able to cover a large portion of its short-term liabilities with its liquid assets alone.
- (5) With the actual shocks, Landsbanki's funding gap amounts to an estimated 18 percent of Iceland's 2007 GDP, assuming all short-term assets are recoverable. The gap would have widened to more than 100 percent of GDP if liquid short-term assets only are taken into account; the funding gap would have worsened if liquid assets such as bonds and equities could not have been sold down following the closure of the international payment system.

2. LESSONS LEARNED

The global financial crisis provided an opportunity to assess stress testing efforts by country authorities and others, including international financial institutions. Many stress tests have failed to capture the key shocks that subsequently materialized during the crisis, as a result of the emphasis on solvency risk, instead of taking a more comprehensive view, which includes a perspective on the liquidity situation. Using the example of a particular (failed) bank in Iceland, we demonstrate the basic weaknesses of stress tests that have been applied in many countries and provide suggestions for improvement.

In the case of Iceland, it has been posited that the problems that beset the country's banking sector were largely attributable to its unviable banking model (e.g., Buiter and Sibert, 2008b; Danielsson, 2008; and Jackson, 2008). Iceland is a small, open economy, with its own currency and an internationally integrated financial sector that was very large relative to its GDP and the government's fiscal capacity. Thus, the central bank could not act as an effective foreign currency lender of last resort, nor did the government have the capacity to provide credible guarantees, either explicit or implicit, to the banks to back up their assets. In other words, the banks were "too big to save." It has also been contended that the Icelandic banks might not have been brought down by illiquidity alone, if Iceland had been a full member of the euro area, with the full liquidity facilities of the euro system at its disposal.

Our observations on the design of stress tests are broadly applicable, notwithstanding the relatively unique situation of Iceland's banking sector. Many countries around the globe are not members of a common currency area, and their governments may still need to provide backing to the banking system in the event of a crisis. Even if those governments

TABLE 6.7

Landsbanki: Estimated Short-Term Funding Gap as of December 31, 2007, after Actual Shocks Are Applied (in millions of ISK)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Short-Term Assets and Liabilities	ISK	EUR	USD	GBP	CHF	DKK	NOK	SEK	Total
Short-term assets									
(1) Cash and cash balances with Central Bank	69,285	11,077	0	1,443	0	1,136	1,099	1,116	85,157
(2) Loans and advances to financial institutions	21,404	73,710	19,552	69,697	1,547	2,879	1,640	2,830	193,258
(3) Loans and advances to customers	205,750	215,758	92,845	147,654	103,932	42,197	24,031	41,474	873,641
(4) Bonds	108,826	222,884	82,090	14,894	1,087	1,706	1,648	1,673	434,807
(5) Equities	9,469	15,298	1,098	5,771	7	5,940	4,568	5,386	47,538
(6) Hedged securities	92,931	40,169	23,157	11,440	399	2,695	2,695	2,695	176,181
(7) Derivatives held for trading	10,944	12,689	9,067	2,330	591	1,920	1,698	1,721	40,960
(8) Derivatives held for hedging	131	81	1,193	36	1	5	5	5	1,457
(9) Unsettled securities trading	3,724	34,195	1,078	35,264	0	30	29	30	74,351
(10) Total short-term assets	522,465	625,861	230,080	288,528	107,564	58,508	37,415	56,931	1,927,351
(11) Total liquid short-term assets	291,456	302,117	115,412	35,878	2,084	13,396	11,709	12,592	784,644
Short-term liabilities									
(12) Deposits from financial institutions	19,202	47,587	11,065	12,721	24,977	30,797	29,799	30,269	206,415
(13) Deposits from customers	291,329	242,340	26,791	1,096,582	1,782	4,104	3,971	4,033	1,670,933
(14) Borrowings	14,731	88,966	32,230	1,154	3,485	1,939	1,876	1,906	146,287
(15) Financial liabilities designated at fair value	0	2,339	58	0	0	468	453	460	3,777
(16) Subordinated loans	453	6,675	2,638	0	0	80	78	79	10,003
(17) Trading liabilities	843	3,107	1,908	2,659	90	123	119	121	8,971
(18) Derivatives held for hedging	0	996	0	0	16	6	6	6	1,031
(19) Tax liabilities	5,255	0	0	0	0	0	0	0	5,255
(20) Unsettled securities trading	0	28,796	598	29,635	3,076	14	14	14	62,147
(21) Derivatives held for trading	14,784	13,891	175	13,933	878	16	14	14	43,705
(22) Total short-term liabilities	346,598	434,697	75,463	1,156,684	34,303	37,548	36,329	36,903	2,158,524
(23) = (10) – (22) Total short-term assets less total short-term liabilities	175,867	191,164	154,618	–868,157	73,261	20,960	1,085	20,028	–231,173
(24) = (11) – (22) Total liquid short-term assets less total liabilities	–55,142	–132,581	39,950	–1,120,806	–32,218	–24,152	–24,620	–24,311	–1,373,881

Sources: Authors; Landsbanki, *Annual Report 2007*; and various financial press.

Note:

- (1) Foreign holdings of cash in foreign currency are adjusted for exchange rate changes.
- (2) Loans and advances are adjusted for NPLs; foreign currency amounts are also adjusted for exchange rate changes.
- (3) Loans and advances are adjusted for NPLs; foreign currency amounts are also adjusted for exchange rate changes.
- (4) Bond holdings are adjusted for actual price changes; foreign currency holdings are also adjusted for exchange rate changes.
- (5) Equity holdings are adjusted for actual price changes; foreign currency holdings are also adjusted for exchange rate changes.
- (6) It is assumed that hedged securities do not change in value as any changes would have been offset by the hedge.
- (7) Given the difficulty in determining the unrealized gains or losses on derivatives agreements, derivatives price changes are used instead, and are assumed to be a weighted average of bond and equity price changes; foreign currency values are adjusted for exchange rate changes.
- (8) It is assumed that derivatives held for hedging do not change in value as any changes would have been offset by changes in the values of the underlying assets.
- (9) Foreign currency amounts of unsettled securities trading are adjusted for exchange rate changes.
- (12) Deposits are apportioned according to their sources; all amounts owing to other credit institutions are assumed to be redeemed, the central bank does not redeem its deposits; those denominated in foreign currency are adjusted for exchange rate changes.
- (13) It is assumed that all deposits are withdrawn; the amounts denominated in foreign currency are adjusted for exchange rate changes.
- (14) All financial liabilities are assumed to be redeemed; balances in foreign currency are adjusted for exchange rate changes.
- (15) All borrowings are assumed to be redeemed; balances in foreign currency are adjusted for exchange rate changes.
- (16) All trading liabilities are assumed to be redeemed; balances in foreign currency are adjusted for exchange rate changes.
- (17) All subordinated loans are assumed to be redeemed; balances in foreign currency are adjusted for exchange rate changes.
- (18) It is assumed that derivatives held for hedging do not change in value as any changes would have been offset by changes in the values of the underlying liabilities.
- (19) Tax liabilities in foreign currency are adjusted for exchange rate changes.
- (20) Foreign currency amounts of unsettled securities trading are adjusted for exchange rate changes.
- (21) Given the difficulty in determining the unrealized gains or losses on derivatives agreements, derivatives price changes are used instead, and are assumed to be a weighted average of bond and equity price changes; foreign currency values are adjusted for exchange rate changes.

TABLE 6.8

Landsbanki: Summary of Stress Test Results

Shock	Outcome
Iceland 4-quarter GDP to 30 September 2008 (in millions of krona)	1,301,409
(1) Initial liquidity position, assuming no crisis	
Total short-term assets	
In millions of krona	1,702,965
As a percentage of GDP	131
Total liquid short-term assets	
In millions of krona	727,640
As a percentage of GDP	56
Total short-term liabilities	
In millions of krona	1,951,385
As a percentage of GDP	150
(2) FME shocks, assuming stability of deposits	
Total short-term assets available less total short-term liabilities redeemed	
In millions of krona	1,434,065
As a percentage of GDP	110
Total liquid short-term assets available less total short-term liabilities redeemed	
In millions of krona	311,125
As a percentage of GDP	24
(3) Combined FME and Seðlabanki liquidity shocks	
Total short-term assets available less total short-term liabilities redeemed	
In millions of krona	959,948
As a percentage of GDP	74
Total liquid short-term assets available less total short-term liabilities redeemed	
In millions of krona	12,256
As a percentage of GDP	1
(4) Actual shocks, with deposit freeze	
Total short-term assets available less total short-term liabilities redeemed	
In millions of krona	775,904
As a percentage of GDP	60
Total liquid short-term assets available less total short-term liabilities redeemed	
In millions of krona	–366,803
As a percentage of GDP	–28
(5) Funding gap, following actual shocks	
Total short-term assets less total short-term liabilities	
In millions of krona	–231,173
As a percentage of GDP	–18
Total liquid short-term assets less total liabilities	
In millions of krona	–1,373,881
As a percentage of GDP	–106

Sources: Authors; Tables 6.2, 6.4–7; and International Financial Statistics, IMF.

have the capacity to do so, liquidity crises in the banking sector could still cause significant stress for the stability of a financial system while the problem is being addressed by the authorities (e.g., the Northern Rock incident in the United Kingdom). Thus, stress tests that are able to capture key risks early on could contribute toward the design of or improvements in crisis mitigation and resolution frameworks.

The stress testing exercise presented in this chapter highlights several salient points about the design of stress tests in general. These are as follows:

- Stress testing for solvency risk alone is insufficient. Even solvent financial institutions, when faced with severe liquidity shortages, could rapidly become insolvent as demonstrated by the recent global financial crisis. As noted by the Basel Committee on Banking Supervision (BCBS, 2008c, p. 4), “Even though strong capital positions reduce the likelihood of liquidity pressure, apparently solvent banks can suffer liquidity problems.” Conversely, financial institutions that may be insufficiently capitalized would be able to “survive” for some time if they have sufficient liquidity to service their short-term debts.¹⁸
- There is a need to look beyond the consolidated numbers in profit and loss statements and balance sheets. Ideally, one would need to have access to supervisory information with detailed breakdowns of the various off-balance-sheet positions, identifying more precisely their impact on liquidity. However, the analysis in this study shows that the publicly available *Notes to the Consolidated Accounts* already provide a substantial amount of detail (e.g., claims and obligations by currencies; spot and derivatives positions; maturities), as well as information on off-balance-sheet items, which enable closer focus on the key risk areas.
- Stress tests should consider “unthinkable” developments. BCBS (2008c) observes that, regardless of how strong a bank’s current liquidity situation may seem, it should consider the potential impact of severe stress scenarios and not discount those severe scenarios as “implausible.” They may need to take into account sensitive issues, such as a sustained run on deposits (typically considered a stable source of funding), a refusal by lenders in wholesale capital markets to roll over their loans, or the possibility of a shutdown in the payment and settlement infrastructure, paralyzing the transfer of funds to meet maturing obligations and the trading of securities. Indeed, performing such exercises on a routine basis could make them more acceptable.
- A useful way of performing and presenting extreme shocks is to “stress test till it breaks.” This means that, instead of focusing on identifying an “extreme but plausible” scenario, the focus would be on identifying a set of scenarios under which the system reaches a predefined threshold, such as a low capitalization ratio or a low level of liquidity (Čihák, 2007). If such tests are performed regularly, their results can be used

¹⁸ In the case of Landsbanki, the short-term funding problems eventually revealed the bank to be badly insolvent as asset quality deteriorated sharply and quickly amid the unprecedented magnitude of the financial crisis, exacerbated by the bank’s weak lending practices.

for discussing how vulnerabilities in the system evolve over time.

Within the specific framework of a liquidity stress test, there is a need to focus on key vulnerabilities in terms of asset recovery and funding sources. For instance, tail risks should be taken into account when shocking key liquid short-term assets that may have to be realized quickly (e.g., the impact from a closure of the payment system on securities trading) or when shocking the main funding sources on which significant reliance is placed (e.g., withdrawals of deposits or wholesale funding). For the items described, shocks could be focused on particularly risky items or currencies, where such breakdown is available. Ultimately, stress test models do not need to be overly complex. Basic stress tests, using appropriate assumptions and shocks, could reveal important vulnerabilities well ahead of time.

3. CONCLUSION

Clearly, our findings in this study have benefited from hindsight. Thinking the unthinkable or unpalatable is easier said than done, *ex ante*. For instance, few banks can survive a sustained deposit run, which could make the specification of that particular stress difficult. That said, well-designed stress tests—ones that assess risks holistically, rather than each one in isolation—could help identify key areas where risks may be concentrated.

Once identified, supervisory authorities and banks themselves could take some actions, *ex ante*, to mitigate potential risks. For instance, BCBS (2008b) notes that in the lead-up to the global financial crisis, many banks had failed to take account of a number of basic principles of liquidity risk management. In particular, many of the most exposed banks did not have an adequate framework that satisfactorily accounted for the liquidity risks posed by individual products and business lines; many had not considered the amount of liquidity they might need to satisfy contingent obligations. Numerous banks also viewed severe and prolonged liquidity disruptions as implausible and did not conduct stress tests that factored in the possibility of marketwide strain or the severity or duration of the disruptions. And finally, contingency funding plans were not always properly linked to stress test results and sometimes failed to take account of the potential closure of some funding sources.

Stress testing and contingency planning should be closely intertwined. The results of stress tests should also play a key role in shaping the contingency planning by banks and the authorities and in determining the appropriate strategies and tactics to deal with events of stress or, ideally, to forestall significant problems from developing. In the Iceland example presented here, the banks were simply too big to save; appropriate prudential measures should have been implemented and risk management strategies should have been strictly enforced while those banks were expanding rapidly.

REFERENCES

- Basel Committee on Banking Supervision, 2008a, *Cross-Sectoral Review of Group-Wide Identification and Management of Risk Concentrations*, The Joint Forum (Basel, April). Available via the Internet: <http://www.bis.org/publ/joint19.htm>
- , 2008b, *Liquidity Risk: Management and Supervisory Challenges* (Basel, February). Available via the Internet: <http://www.bis.org/publ/bcb136.htm>
- , 2008c, *Principles for Sound Liquidity Risk Management and Supervision* (Basel, September). Available via the Internet: <http://www.bis.org/publ/bcb144.htm>
- , 2009a, “International Framework for Liquidity Risk Measurement, Standards and Monitoring,” Consultative Document (Basel, December). Available via the Internet: <http://www.bis.org/publ/bcb165.htm>
- , 2009b, *Principles for Sound Stress Testing Practices and Supervision* (Basel, May). Available via the Internet: <http://www.bis.org/publ/bcb155.htm>
- Buiter, Willem, and Anne Sibert, 2008a, “The Collapse of Iceland’s Banks: The Predictable End of a Non-Viable Business Model,” VoxEU.org, October 30.
- , 2008b, “The Icelandic Banking Crisis and What to Do about It: The Lender of Last Resort Theory of Optimal Currency Areas,” CEPR Policy Insight No. 26 (Washington: Center for Economic Policy Research).
- Čihák, Martin, 2007, “Introduction to Applied Stress Testing,” IMF Working Paper 07/59 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=20222>
- Danielsson, Jon, 2008, “The First Casualty of the Crisis: Iceland,” VoxEU.org, November 12.
- International Monetary Fund, 2008, “Iceland: Financial System Stability Assessment—Update,” IMF Country Report 08/368 (Washington, December). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=22529>
- Jackson, James K., 2008, “Iceland’s Financial Crisis,” CRS Report for Congress (Washington: Library of Congress).
- Landsbanki Íslands hf., 2008, *Annual Report 2007* (Reykjavik, January).
- Mitra, Srobona, 2006, “Risks and Vulnerabilities in Icelandic Banks,” IMF Country Report 06/297, pp. 11–44 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=19528>
- Ong, Li Lian, 2007, “Iceland: Financial Sector Developments and Risks to the Outlook,” IMF Country Report 07/296, pp. 19–37 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/ft/scr/2007/cr07296.pdf>
- Ong, Li Lian, and Martin Čihák, 2010, “Of Runes and Sagas: Perspectives on Liquidity Stress Testing Using an Iceland Example,” IMF Working Paper 10/156 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=21292>
- Seðlabanki, 2007, *Financial Stability* (Reykjavik, April). Available via the Internet: <http://www.cb.is/default.aspx?pageid=3364d871-979d-4bd5-ad6c-2888afa54fe1>
- , 2008, *Financial Stability* (Reykjavik, May). Available via the Internet: <http://www.cb.is/default.aspx?pageid=3364d871-979d-4bd5-ad6c-2888afa54fe1>
- , 2009, *Financial Stability* (Reykjavik, October). Available via the Internet: <http://www.cb.is/default.aspx?pageid=3364d871-979d-4bd5-ad6c-2888afa54fe1>

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Appendix

Glossary

TABLE 6.9

Landsbanki: Composition of Balance Sheet Items

Item	Composition
Assets	
Cash and cash balances with Central Bank	N/A
Loans and advances to financial institutions	N/A
Loans and advances to customers	Customers include public entities, corporates and individuals; net of provisions for credit losses.
Bonds	Domestic and foreign, listed and unlisted.
Equities	Domestic and foreign, listed and unlisted.
Hedged securities	Bonds and equities.
Derivatives held for trading	Foreign exchange, interest rate, equity, and commodity derivatives.
Derivatives held for hedging	Interest rate and cross-currency swaps.
Unsettled securities trading	N/A
Liabilities	
Deposits from financial institutions	Loans from Central Bank and repurchase agreements; loans from other credit institutions.
Deposits from customers	Demand and time deposits.
Borrowings	Securities issued (medium-term notes, commercial paper, other); syndicated loans and other borrowings.
Financial liabilities designated at fair value	Under unit-linked investment contracts; with matched interest rate swaps; with embedded derivatives.
Subordinated loans	Tier I (noninnovative and innovative hybrid capital); Tier II
Trading liabilities	N/A
Derivatives held for hedging	Interest rate and cross-currency swaps.
Tax liabilities	N/A
Unsettled securities trading	N/A
Derivatives held for trading	Foreign exchange, interest rate, equity, and credit derivatives.

Source: Landsbanki, *Annual Report 2007*.

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Next-Generation Systemwide Liquidity Stress Testing

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This article presents a framework for stress testing and monitoring banks' liquidity positions from various perspectives. The associated Excel-based tool can be used to run systemwide, balance sheet data-based liquidity stress tests and is part of the new-generation stress testing framework introduced in Schmieder, Pühr, and Hasan (2011). The liquidity framework includes three elements: (1) a module to simulate the impact of bank-run-type scenarios; (2) a module to assess risks arising from maturity transformation and rollover risks (it enables simplified tests for rollover risk or a full-fledged cash flow-based approach to simulating liquidity shortfalls in different maturity buckets; and (3) a framework for linking liquidity and solvency risks. Our framework can also be used to simulate how banks would cope with upcoming regulatory changes (Basel III) and takes into account the fact that data availability differs widely.

METHOD SUMMARY

Overview	- The method provides a framework for stress testing and monitoring banks' liquidity positions from various perspectives. - It also simulates how banks would cope with upcoming regulatory changes (e.g., Basel III).
Application	The tool enables the running of systemwide liquidity stress tests for banks (and, in principle, other financial institutions) from complementary perspectives: 1. to simulate bank-run-type tests; 2. to assess maturity mismatch, accounting for both contractual and behavioral cash flows; and 3. to establish links between liquidity and solvency. The tool also enables computation (of proxies) for the Basel III ratios.
Nature of approach	Balance sheet data-based.
Data requirements	- Accounting information on assets and liabilities, including maturity mismatch and, if available, detailed contractual and behavioral cash flow data as well as banks' off-balance-sheet positions. - To establish a (dynamic) link between liquidity and solvency risk, capitalization ratios as reported and under stress are needed, as well as the relationship between capitalization and funding costs.
Strengths	The method allows more granular and comprehensive liquidity stress tests and accounts for the fact that data availability differs widely across financial systems.
Weaknesses	- The quality of the results depends on the data available for running the tests. - Strong assumptions are required on liquidity requirements associated with contingent liabilities, such as derivatives (which is a common challenge). - The framework needs to evolve, provided that more data become available, to implicitly (or more explicitly) capture the dynamic nature of liquidity risks.
Tool	The Excel spreadsheet macro is available in the toolkit, which is on the companion CD and at www.elibrary.imf.org/stress-test-toolkit. Contact author: C. Schmieder.

This chapter is an abridged version of IMF Working Paper 11/203 (Schmieder and others, 2011). The working paper benefited from comments by Martin Čihák, Daniel Hardy, Maher Hasan, Torsten Wezel, Nico Valckx, Martha Ruiz-Arranz, Fabian Valencia, and Minsuk Kim as well as from seminar participants at the IMF and the European Central Bank, participants at an IMF-Oesterreichische Nationalbank stress testing workshop in Vienna, and numerous central bankers and bank supervisors during IMF Financial Sector Assessment Program and technical assistance missions.

Traditionally speaking, liquidity risk has been at least as important for banks as solvency risk. It was only with the introduction of Basel I in 1988, when liquidity risk was omitted from the framework, that solvency risk gained greater prominence and the erosion of capitalization came to a halt (Goodhart, 2008). However, the global financial crisis has shown very clearly that neglecting liquidity risk comes at a substantial price. Liquidity risk at banks is triggered by maturity transformation (long-term maturity profile on the assets side versus short-term maturities on the liabilities side). Over the last decade, the large banks have become increasingly reliant on short-term wholesale funding (especially in interbank markets) to finance their rapid asset growth. At the same time, the supply of funding from nondeposit sources such as money market mutual funds has soared. As the global financial crisis unfolded, interbank markets froze as counterparty risk rose, resulting in funding liquidity challenges for many banks.¹

The liquidity stress testing framework presented in this chapter was developed for application in the IMF's Financial Sector Assessment Programs (FSAPs).² It extends the seminal work of Čihák (2007) and that of the Austrian National Bank/*Oesterreichische Nationalbank* (OeNB). Four key aspects were taken into account in the development of the framework, namely that

1. the availability of data varies widely across countries;
2. liquidity risk has several dimensions, and assessing banks' resilience vis-à-vis funding risks requires multidimensional analysis;
3. designing liquidity scenarios is more challenging than for solvency risks, mainly because liquidity crises are relatively rare and can originate from many different sources; and
4. there is a close link between solvency and liquidity risks.

The resulting framework is an Excel-based, easy-to-use balance sheet-type liquidity stress testing tool that allows the user to run bottom-up tests for hundreds (and even thousands) of banks.³ It has the following features:

- First, the tool can be used in circumstances where data are very limited to run some basic liquidity tests. Likewise, a cash flow based module allows the user to run detailed liquidity analysis like those carried out by (large) banks for their internal purposes.
- Second, the framework includes three broad dimensions (based on four modules) that allow for complementary views on liquidity risks, including the link to solvency risks.
- Third, we provide benchmark scenarios based on historical evidence on the one hand and common scenarios

used by FSAP missions on the other, and these also are informed by Basel III.

- Fourth, the framework allows an assessment of the link between liquidity and solvency, although additional effort is needed in this context, including work that captures the dynamic aspects of this relationship and spillover effects.⁴

Overall, the framework is aimed at giving users the ability to run a meaningful systemwide liquidity stress test.

The framework consists of three types of stress testing concepts:

1. *Stress testing funding liquidity based on implied cash flow-type methods*, which comprises (a) simulating bank-run-type scenarios while accounting for fire sales of liquid assets; (b) liquidity gap analysis (assessing different maturity buckets); and (c) calculating Basel III liquidity ratios.
2. *Cash flow-based liquidity tests*, which provides an intuitive view of each bank's liquidity risk-bearing capacity in the form of the cumulated counterbalancing capacity at the end of each maturity bucket.
3. *Tests linking solvency and liquidity risk* through (a) an increase in funding costs; (b) the closure of funding markets; and (c) potential vulnerabilities stemming from funding concentration.

The chapter is organized as follows: Section 1 first provides some generic considerations on concepts and methods to assess liquidity risks. Section 2 presents the newly developed methodological framework. Section 3 is devoted to designing "extreme yet plausible" scenarios, complemented by an illustrative study in Schmieder, Hesse, and others (2011). Section IV concludes the chapter.

1. REVIEW OF GENERAL CONCEPTS TO ASSESS LIQUIDITY RISKS

A. General considerations and motivation

Liquidity stress tests are less developed compared with solvency stress tests (particularly market risk). This development can be attributed to several factors: (1) liquidity risk management was considered to be "less of an issue" until the current crisis and hence has a shorter history of making use of information technology (IT) systems, which would have facilitated its development; (2) liquidity crises are very *low-frequency-high-impact* events, which greatly reduces the opportunity to draw on historical experience to calibrate models;⁵ and (3) all liquidity crises are somehow different, at least if one seeks to analyze the relationship between the

¹ See Appendix I of Schmieder and others (2011) for the evolution of liquidity evaporation during the crisis.

² Examples include Germany, Turkey, Chile, and India.

³ In principle, the tool could also be used to assess nonbank financial institutions but potentially would need some adjustment.

⁴ See Barnhill and Schumacher (2011) and IMF (2011).

⁵ In some cases, central bank support via liquidity provisions has masked the extent of an explicit liquidity squeeze, with many banks hoarding liquidity and banks faced with funding liquidity challenges merely substituting their loss of market wholesale and/or retail funding with central bank funding.

sources and the resulting liquidity shortfalls, reducing the usefulness of “standard” stress assumptions.

Liquidity crises pose significant challenges in that they typically occur very suddenly and spread very quickly, giving banks very little time to react. Idiosyncratic liquidity crises can be triggered by various events, most notably solvency problems, but also political instability and fraud. Contagion can escalate idiosyncratic shocks into marketwide shocks, as seen during the recent crisis period. This dynamic warrants requiring liquidity buffers to be based on highly conservative principles—an important consideration in the design of stress scenarios (Section 3). A key principle applied by the U.S. authorities during the height of the financial crisis in late 2008 was to ensure that ailing (investment) banks made it through a business week, so that a viable solution could be found during the weekend if necessary. With the regulatory framework to be established by Basel III, the aim is to ensure banks’ resilience against a “significant stress scenario lasting 30 days” (Basel Committee on Banking Supervision [BCBS], 2010), that is, banks should be able to survive a month under (medium to severe) stress.

Liquidity crises may be sharp and short or more drawn out, each requiring different mitigating actions and considerations:

- The obvious solution to counterbalancing bank-run-type outflows is to liquidate assets through fire sales. The dilemma for banks lies in the cost of holding high-quality liquid assets, particularly cash and “prime” government bonds. More illiquid securities are less costly (i.e., they qualify as some substitute for traditional bank business) but subject to higher haircuts, at best, or cannot be sold at all (i.e., become illiquid) during periods of market stress and/or may no longer qualify as eligible collateral (to ensure secured funding).
- In the case of longer-lasting liquidity disturbances, the maturity profile of assets and liabilities plays an important role, as maturing assets can then be used to deleverage in an “orderly” manner, provided that (at least partly) maturing (longer-term) debt can be rolled over. In fact, the analysis of rollover risk has become an important aspect of liquidity risk analyses as many large banks are facing a “wall of funding” in the coming years.⁶

Central bank funding plays a natural counterbalancing role. In a severe crisis, central banks can act as a lender of last resort. For instance, a number of central banks entered into currency swap agreements with the Board of Governors of the Federal Reserve during the financial crisis so that they could supply their domestic banks with much needed U.S. dollar funding.⁷ Parent banks can also step in to increase or main-

tain credit lines to subsidiaries if a subsidiary or branch loses access to funding sources. The Committee of European Banking Supervisors (CEBS) found that the majority of instances in which parent institutions did not provide additional liquidity for subsidiaries were attributable to idiosyncratic liquidity shocks hitting the parent as a consequence of severe (perceived) solvency problems of the banking group (CEBS, 2009).

B. Methodological aspects

Overview of Recent Methods to Assess Liquidity Risks

Methods to assess liquidity risk run from use of simple indicators to very sophisticated, complex techniques. A natural starting point for assessing liquidity risk is through financial soundness indicators, which provide relevant information on the liquidity position of banks, both vis-à-vis peers (banks and/or countries) and over time. Liquidity stress tests are more forward-looking and several methods have been developed by central banks and supervisory authorities:

- The OeNB is one of the early users of cash flow-based (top-down and bottom-up) liquidity stress tests (Boss and others, 2008);⁸ more recently, the work by Schmitz (2010) has had an important influence on the European approach (see European Central Bank [ECB], 2008a).
- Another liquidity stress testing approach was developed by Wong and Hui (2009) at the Hong Kong Monetary Authority. In their framework, banks’ liquidity risk arises from the impact of marked-to-market losses on banks’ solvency, leading to deposit outflows and evaporating asset fire sales and, consequently, sharply rising contingent liquidity risk.
- The Dutch Central Bank developed a stress testing model for both market and funding liquidity risk, which incorporates feedback effects as well as a Monte Carlo approach that simulates the impact on liquidity risk (van den End, 2008).
- The Bank of England attempted to link funding liquidity risk with solvency risk in the Risk Assessment Model for Systemic Institutions (RAMSI; Aikman and others, 2009). The framework simulates banks’ liquidity positions conditional on their capitalization under stress and incorporates other relevant dimensions, such as a decrease in confidence among market participants under stress.

At the IMF, liquidity stress tests have been developed to support staff’s bilateral and multilateral surveillance work. Applied stress tests in the context of the FSAP were introduced in Čihák (2007); bank balance sheet data were used to implement bank-run-type stress tests on a bank-by-bank basis. More recently, the *Global Financial Stability Report*

⁶ Banks’ debt maturity profiles are monitored in the IMF’s *Global Financial Stability Report*, for example.

⁷ Recently released figures by U.S. authorities on U.S. dollar liquidity support granted to large banks (including major European banks) at the onset of the crisis are also a telling example in this context.

⁸ In the liquidity stress tests conducted for the IMF FSAP in 2007, highly adverse scenarios were adopted to test the resilience of Austrian banks. See Section III.C for further information.

focused on analyzing systemic liquidity, using a Merton-type approach on market data and balance sheet information to estimate liquidity risk at individual banks and to calculate the joint probability of all institutions experiencing a systemic liquidity event (IMF, 2011).⁹ Barnhill and Schumacher (2011) simulated the impact of an asset price shock on banks' solvency position, resulting in a withdrawal of bank liabilities; they subsequently tested whether banks could counterbalance these outflows with fire sales.

On the regulatory side, substantial microprudential efforts have been made to contain liquidity risk at the bank level. In 2008, the BCBS published guiding principles for sound liquidity risk management (BCBS, 2008); an overhaul of the regulatory framework followed in December 2010 (BCBS, 2010), when the Committee introduced two measures to contain short-term vulnerabilities and excessive maturity mismatches. The minimum liquidity standard incorporates (1) a 30-day liquidity coverage ratio (LCR), essentially a pre-specified substantial bank-run-type stress test lasting a month that banks have to pass in order to be considered safe in the short term; and (2) a longer-term structural liquidity ratio, the net stable funding ratio (NSFR), which aims to limit maturity mismatches, with a 12-month horizon. Both ratios are subject to a transition phase, during which they will be calibrated, and are scheduled to be fully implemented by 2015 and 2018, respectively.

In addition, several macroprudential approaches to manage systemic liquidity risk have been brought forward during the last two years. Some have been used, at least partially, in some emerging-market countries for many years. These approaches provide incentives to limit systemic liquidity risks, including through levies, capital charges, and the introduction of minimum liquidity ratios and haircuts. However, implementation seems unlikely at this stage, mainly owing to the complexity of measuring systemic risk (see IMF, 2011).

Finally, stress tests in the banking industry are centered on maturity mismatch approaches, sometimes complemented by stochastic value-at-risk (VaR) components for those funding sources for which sufficient history on high-frequency data is available. The ultimate goal of liquidity tests is to determine a bank's risk tolerance for liquidity risk, that is, the maximum level of risk that the bank is willing to accept under stress conditions. Most large European banks compute their maximum risk tolerance (ECB, 2008a). The stochastic approach aims at determining Liquidity at Risk¹⁰ (maximum liquidity gap within a certain time horizon and for a given confidence level) or Liquidity VaR (maximum cost of liquidity under certain assumptions). Although instructive under business-as-usual and mild stress scenarios, these models face limitations in

stress testing more severe liquidity shocks. Given that liquidity risk is a low-frequency, high-impact risk, historical volatilities and correlations tend to underestimate funding risk under severe stress, which is highly nonlinear (ECB, 2008a).

Top-down or Bottom-up?

The most intuitive way to stress test liquidity risk is to use cash flow-level data, usually available only within banks. Provided that the cash flow structure and the maturity of all cash flows are monitored through IT systems, the challenge is how to deal with

1. the volatility of funding, that is, cash flows with non-predefined cash flow structures, such as contingent liabilities (e.g., credit lines) on the asset side and demand deposits or short-term interbank market access on the liability side; as well as
2. the strategy of managing maturity mismatch.

For systemwide liquidity stress tests, the subject matter of our framework, there are two ways to stress test liquidity risk:

1. Define common scenarios that are run by banks themselves, so-called bottom-up (BU) tests, making use of granular data; or
2. Collect data by broader liability and asset types, currency, and maturity, and apply scenarios accordingly in a top-down (TD) fashion (i.e., run by authorities).

Table 7.1 summarizes the relative strengths and weaknesses of the two approaches for liquidity risk, omitting the hybrid case (TD, run by banks).

Outcome of Liquidity Stress Tests

The outcome of TD liquidity tests is threefold:

1. They show the counterbalancing ability of banks (and their specific limits in case of reverse stress tests)¹¹ to remain liquid.
2. They allow a peer comparison, that is, the relative performance of banks under liquidity stress.
3. They can provide a link between the joint resistance to liquidity and solvency risks if the feedback between solvency and liquidity risks is modeled (in the TD stress testing framework).

C. Framework of next-generation liquidity stress tests

Our framework originates in the balance sheet-based liquidity stress tests of Čihák (2007). We take into account (1) lessons learned from the crisis on the one hand and (2) the evolution of conceptual and regulatory initiatives on the other. The framework is part of a larger project on next-generation balance sheet stress testing at the IMF, following

⁹ The framework can also be used to compute each institution's contribution to systemic risk and systemic risk shortfalls, respectively, which could trigger an insurance premium.

¹⁰ Liquidity at Risk denotes computing a 99.9 percent event based on the cumulative probability distribution (as is done for market risk and credit risk).

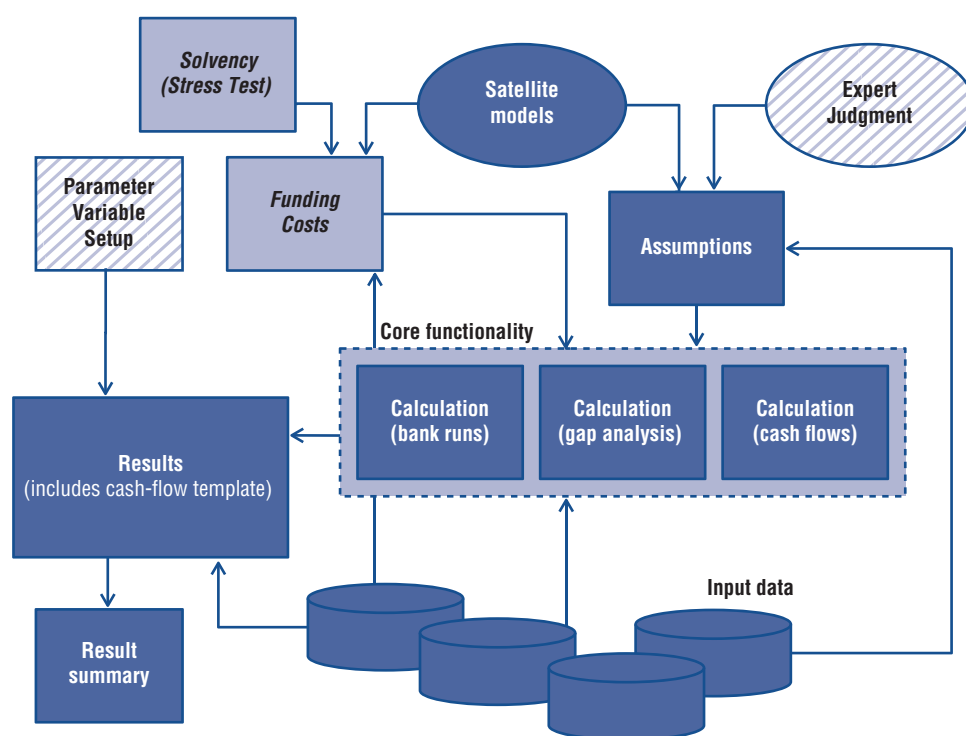
¹¹ Reverse stress tests seek to identify maximum stress resistance of banks or the banking system by increasing the risk factors (e.g., haircuts, run-off rates) until a predefined threshold (e.g., positive counterbalancing capacity) is reached.

TABLE 7.1

Comparison of Pros and Cons of Balance Sheet–Type TD and BU Liquidity Stress Tests		
Type of Test	Pros	Cons
BU tests (run by banks)	- Cash flow–level data. - Use of models developed by banks. - The effects of liquidity shocks on the profit and loss account and the cost of funding shocks can be incorporated more easily.	Less consistent than TD.
TD tests (run by authorities)	- Consistent approach. - Authorities have the flexibility to run various scenarios. - Transparency of situation to authorities.	- Less detailed data. - Bank-specific situation less recognized. - Data are rapidly outdated, which can be prevented by a high but burdensome frequency of reporting.

Source: Authors.

Note: BU = bottom-up; TD = top-down.



Source: Authors

Figure 7.1 Overview on Liquidity Risk Framework

Schmieder, Puhr, and Hasan (2011). Our tool provides extensions in five dimensions (Figure 7.1):

- A more granular balance sheet structure that can be exploited.
- Maturity mismatches that are explicitly taken into account through separate tests.
- A framework that allows computation of (simplified) Basel III liquidity ratios, both the LCR and NSFR.
- A full-fledged cash flow test that shows detailed information on banks' vulnerabilities in different maturity buckets (subject to availability of granular information).
- A framework linking liquidity and solvency risks that considers liquidity from complementary angles and

allows an assessment of the impact of changes in funding costs and a (partial) closure of funding sources on solvency and liquidity, as well as funding concentration risks.

Given the lack of empirical evidence, as argued previously, the calculation of satellite models (i.e., econometric models) that link the outflow of deposits to macroeconomic conditions is not (yet) feasible. Such models can be used to determine the haircuts for assets under stress (i.e., market liquidity risk). In addition, satellite models can be used to link banks' solvency under stress (e.g., capital ratios or banks' default probabilities) to funding costs. Accordingly, a multiperiod solvency test could link the deterioration in liquidity conditions to the evolution of bank solvency and vice versa.

TABLE 7.2

Overview on the Main Elements of Three Liquidity Tests

Type of Test	Description
Bank-run-type tests	Simulates banks' counterbalancing capacity in case of bank-run-type scenarios for a period of (1) 5 and (2) 30 consecutive periods (e.g., days). Allows for a high level of granularity with respect to fire sales (defining liquid assets, haircuts, and so forth). (A proxy for) the LCR assesses banks' counterbalancing strategy to a prespecified bank-run-type scenario lasting a month.
Maturity mismatch/rollover risk tests	The liquidity gap simulation matches liability and asset maturities and identifies liquidity gaps at each maturity and under different scenarios, taking into account rollover risk. The (proxied) NSFR assesses the stability of banks' funding sources in more structural terms, with a focus on the next 12 months. Cash flow-based liquidity tests allow for an intuitive view on each bank's liquidity risk-bearing capacity in the form of the cumulative counterbalancing capacity at the end of each maturity bucket.
Integrated liquidity and solvency test	Simulates the impact of changes in solvency, rating downgrades, and concentration risk on funding costs (the first two modules require input from a solvency test and funding cost model, respectively).

Source: Authors.

Note: LCR = liquidity coverage ratio; NSFR = net stable funding ratio.

Our framework consists of three liquidity stress test modules (Table 7.2). They assess (1) the capacity of a bank to withstand a bank run (*short-term counterbalancing capacity*); (2) the extent and capacity of a bank to deal with maturity mismatches; and (3) potential threats to liquidity arising from solvency risks. A case study simulating banks' short-term counterbalancing capacity is presented in Schmieder, Hesse, and others (2011), along with more detailed information on each of the stress tests.

2. DESIGN OF STRESS SCENARIOS

A. General considerations

In line with the overarching principles for sound stress testing, scenarios should be "extreme yet plausible." This is even more important for liquidity risk than it is for solvency risk as only solid liquidity buffers can ultimately safeguard banks, unless there is a major systemic event when even those would no longer suffice. Given that liquidity crises are infrequent (more so than solvency crises), evidence is scarce, and, moreover, stress levels vary widely. However, conditions tend to be very unfavorable once stress occurs as it tends to be highly nonlinear.

Our aim is to present some crude benchmark scenarios by introducing a new unit: "Lehman." This unit is meant to be a benchmark for funding conditions faced by banks that were particularly badly hit in the month after the Lehman collapse.¹² In Schmieder, Hesse, and others (2011), a case study is applied to stylized but representative banks in advanced economies, emerging markets, and low-income countries.¹³ The simulated stress levels were equal to half Lehman, Lehman, two times Lehman, and three times Lehman. In the following, evidence, regulatory parameters, and scenarios used by authorities and banks are presented.

The classic alternative to point estimate-based scenarios is to "stress it until it breaks." Ong and Čihák (2010) also refer to *reverse stress tests*, where tests are used to determine a set of scenarios that would cause an increasing part of the system (or specific banks) to run short on liquidity. Reverse stress tests and tests simulating "extreme yet plausible" scenarios complement each other, and thus there is a good reason to run both, especially for liquidity risk.¹⁴ The advantage of running a series of stress tests is to gain a meaningful view on the sensitivities of banks (and systems) to specific stress drivers (e.g., outflow of funding, drain of liquid assets, contingent liabilities, margin calls)—key preconditions for robust stress test results (taking into account the Black Swan theory).

In general, three basic types of inputs can be useful in designing extreme-but-plausible scenarios: (1) past experience; (2) expert judgment; or (3) an individual, reverse test-type assessment of the limit for each bank. Scenarios should take into account both marketwide shocks (a worsening of market conditions and investor confidence) that affect all banks in the system as well as idiosyncratic shocks, for example, resulting from deterioration in the solvency of individual banks. Given that market confidence in individual banks is more fragile under marketwide stress, combined scenarios should complement pure marketwide shock and idiosyncratic scenarios (e.g., the LCR is modeled around such a combined scenario).

If possible, scenarios should be accompanied by a consistent "story line" that underpins the assumptions on all relevant elements. These include

1. runoff rates for funding;
2. haircuts for assets sold at fire sale prices and draw-down of contingent liabilities;

¹² This benchmark scenario remains hypothetical and is geared toward large banks in the Organization for Economic Cooperation and Development countries. For smaller banks, the benchmark could be different, with (even) higher runoff rates for customer deposits in case of a crisis. Expert judgment is needed to design the most plausible scenario for the situation at hand.

¹³ The stylized banks were constructed using the universe of Bankscope data, which remains limited for low-income countries and (although less so) also for emerging-market countries.

¹⁴ The challenge is how to deal with the outcome of reverse stress tests in the context of authorities' stress tests. Given the sensitivity of liquidity risk, an appropriate way to disseminate the results needs to be found.

- the impact of banks' rating downgrades, that is, a deterioration in bank solvency, resulting in margin calls (i.e., higher levels of encumbered assets), an increase in funding costs, and, ultimately, a gradual closure of funding sources.

For the analysis of maturity mismatches, additional parameters (e.g., rollover rates) need to be modeled in a consistent manner. In the case of retail deposits, the following could be guiding questions for the design of scenarios and the development of story lines:

- Which retail deposits are the most vulnerable (e.g., foreign currency denominated deposits, deposits held by foreigners abroad, demand deposits in case of an increase in policy rates from very low levels) and would go first?
- Would depositors hoard cash or shift deposits outside the national banking system in the event of a crisis?
- Under what conditions would flight to quality initiate deposit inflows at a subset of banks in the system and an outflow at others?

B. Runoff rates for different funding sources

The financial crisis provided ample evidence of solvency and liquidity crises at banks. From a liquidity perspective, the most prominent victims were probably the U.S. investment banks, which suffered from a drying up of interbank markets combined with solvency concerns, which pushed them to seek much-needed capital (e.g., from sovereign wealth funds). Others became victims of their rapid and aggressive growth strategy and heavy reliance on wholesale funding, such as Northern Rock in the United Kingdom, which experienced a textbook retail bank run following a quiet wholesale run. A third group of banks were those domiciled in countries experiencing major recessions and/or banking crises, such as in the Baltic countries and Kazakhstan. More recently, banks in peripheral European countries became highly dependent on ECB funding after they were shut out of the interbank market and debt capital markets and experienced protracted outflows of funds, including retail deposits, in some cases.

The two prudential Basel III ratios provide benchmark parameters for runoff rates of funding sources: the LCR for a period of 1 month and the NSFR for 12 months (BCBS, 2010). For retail deposits, the LCR foresees minimum outflow ratios (runoff rates) of 5 percent for stable funding provided by retail customers and small- and medium-sized enterprises and 10 percent for less stable funding.¹⁵ For the NSFR, the level is twice as high (10 and 20 percent, respectively). Secured wholesale funding is subject to withdrawal between 0 and 25 percent, provided that it is secured with higher-quality collateral, whereas unsecured wholesale funding is associated with runoff rates of at least 50 percent (for

nonfinancial corporates), most of it 100 percent (especially for financial institutions). European banks¹⁶ use similar parameters for their internal stress tests, with retail deposit runoff rates mostly at 10 percent (up to 30 percent) and wholesale runoff rates ranging from 0 to 100 percent (100 percent is assumed by one-fifth of the banks in the survey; ECB, 2008a).

C. Asset side of the balance sheet: Fire sales and rollover

The counterbalancing ability of banks depends on their ability to generate cash inflows from liquid assets. It includes three elements:

- defining which asset types remain liquid;
- defining the market liquidity for those assets, that is, the loss in value (haircut) banks have to accept to sell the asset;
- defining the portion of liquid assets that remains unencumbered (i.e., available for fire sales or to obtain secured funding). Given recent events and the increased importance of secured funding (e.g., repos and covered bonds), it is crucial to have information on the level of unencumbered liquid assets on the one hand and to make assumptions about their availability under stress on the other, to account for potential margin calls.¹⁷

Haircuts should differentiate among asset categories, accounting for the level of stress simulated as well as the "quality" of assets (e.g., in case of debt securities, the type and rating of the counterparty). In any case, double counting should be avoided—(unencumbered) liquid assets can be used only to either generate cash or maintain the level of funding (as a substitute of encumbered assets used as collateral that have lost in value). Potential haircuts to be modeled comprise

- haircuts for (unencumbered) liquid assets;
- haircuts for encumbered liquid assets (i.e., collateral/margin calls); and
- add-ons (positive haircut) for contingent liabilities.

Deriving model-based haircuts requires a substantial commitment of time and resources but comes with the advantage of developing expert knowledge on the value of assets under stress.¹⁸ Alternatively, stress testers can use supervisory haircuts under the (comprehensive) Standardized Approach for solvency purposes (BCBS, 2006, paragraph 147f.). These haircuts constitute a proxy for the 99th confidence interval for different holding periods. The Basel III framework distinguishes between two levels of high-quality liquid assets (so-called flight-to-quality assets) and refers to factors that can be used to define whether funding remains liquid (BCBS,

¹⁵ See BCBS (2010) for further information.

¹⁶ The survey is based on responses by 30 European banks in 2008.

¹⁷ See BCBS (2010, paragraph 27).

¹⁸ Calculating the volatility of market prices of assets allows assigning probabilities for the occurrence of scenarios. A useful guideline on how to do so is provided in BCBS (2006, paragraph 156ff.).

TABLE 7.3

Supervisory Haircuts Based on Solvency Regime and Basel III Liquidity Regime							
Haircut/Weight for ...	Basel II (2006, 2010)					Basel III (LCR)	Basel III (NSFR)
(1) Unencumbered liquid assets	Cash	0				0	0, includes also short-term securities (less than 1 year)
	Issue rating for debt security	Residual maturity	Sovereign	Other issuers	Securitization		
	AAA to AA–/A-1	< 1 year	0.5	1	2	Sovereign (RW 0%): 0; Corporate: 15	Sovereign-type: 5; Corporate (< 1 year): 20
		1 to 5 years	2	4	8		
		> 5 years	4	8	16		
	A+ to BBB– and unrated	< 1 year	1	2	4	Sovereign (RW 20%, Rating A+ to A–): 15	
		1 to 5 years	3	6	12		
		> 5 years	6	12	24		
	BB+ to BB–	All	15	Not eligible			
	Equity (main index) and gold	15				100	50
Other equity	25						
Mutual funds (max of allowed asset mix)	Up to 25 (highest haircut applicable to any security in the fund)						
(2) Encumbered liquid assets (collateral)	N/A	N/A				Haircut on collateral for potential margin calls (3-notch downgrade)	N/A
(3) Add-ons (contingent liabilities)	N/A	N/A				Lines – Retail: 5; corporate, credit lines: 10; corporate, liquidity lines: 100; other: 100	Undrawn credit and liquidity lines: 5

Sources: Basel Committee on Banking Supervision (2006, 2010).

Note: LCR = liquidity coverage ratio; RW = ?

2010, paragraph 22f.). Table 7.3 provides an overview of supervisory haircuts based on the Basel II solvency framework and the haircuts to be used for the Basel III liquidity tests. It is important to recognize that the purpose of the parameters is different—for solvency purposes, maturity is linked to the assets, while the maturity for liquidity purposes depends on the reference ratio (and the corresponding time frame). One can see, for example, that the LCR assumes substantial stress, with equities becoming illiquid. A more granular classification of marketable assets is the one by the ECB, which distinguishes among five categories (ECB, 2008b).

Reputational considerations, which featured prominently in the crisis, need to be built into the scenario assumptions. In particular, stress testers need to take into account contingent liabilities such as committed credit/liquidity lines to customers, sponsorships of Special Purpose Vehicles, and outflows related to derivatives (i.e., margin calls). This risk is particularly high under marketwide funding market dislocations. The LCR provides a valuable benchmark for contingent liabilities, including for derivatives.

As a general rule, stress tests should focus on the ability of banks to weather severe but plausible liquidity shocks as go-

ing concerns. That implies that the bank is able to maintain its franchise value. To do so, it needs to keep generating new business (i.e., roll over maturing assets) and honor its commitments, which is the underlying assumption under Basel III and also applied by banks (see Deutsche Bank, 2010, for example).

D. Link between liquidity and solvency

There have been some recent attempts to link solvency and liquidity risk. In particular, the link applies to (1) funding costs and (2) the closure of funding markets once solvency conditions deteriorate further. The link between solvency and funding costs consists of two dimensions: (1) an increase in the price to be paid for funding (e.g., wholesale funding is particularly sensitive to changes in solvency conditions, but the recent competition for retail deposits is an indication that retail deposits may also be more price sensitive going forward); and (2) an increase in collateral needs for secured funding sources (margin calls).

The price of funding can be derived on the basis of empirical evidence. One way is to use econometric models to deter-

TABLE 7.4

Benchmark Scenarios				
Scenario	Moderate Stress Scenario	Lehman-Type Stress Scenario (Medium)	Severe Stress Scenario	Very Severe Stress Scenario
Severity (x times Lehman ¹)	0.5	1	2	3
Liquidity outflows				
Customer deposits				
Customer deposits (Term)	2.5 percent	5 percent	10 percent	20 percent
Customer deposits (Demand)	5 percent	10 percent	20 percent	40 percent
Wholesale funding				
Short-term (secured)	5 percent	10 percent	20 percent	40 percent
Short-term (unsecured)	15 percent	30 percent	60 percent	100 percent
Contingent liabilities	0 percent need funding	5 percent need funding	10 percent need funding	20 percent need funding
Liquidity inflows				
Haircut for cash	0 percent	0 percent	0 percent	0 percent
Haircut for government securities ²	1 percent	2 percent	5 percent	10 percent
Haircut for trading assets ³	3 percent	6 percent	30 percent	100 percent
Haircut for other securities	10 percent	30 percent	75 percent	100 percent
Percent of liquid assets encumbered ⁴	10 percent (or actual figure)	20 percent (or actual figure plus 10 ppt)	30 percent (or actual figures plus 20 ppt)	40 percent (or actual figures plus 30 ppt)

Source: Authors.

1. The Lehman-type scenario would correspond to a scenario encountered by banks that were hit severely during the 30-day period after the Lehman collapse, i.e., a stress situation within a stress period rather than an average. The scenario has been put together based on expert judgment, using evidence as available.

2. The haircut highly depends on the specific features of the government debt held (rating, maturity, market depth) and can be higher or lower. The figures displayed herein are meant for high-quality investment-grade bonds, taking into account recent market conditions. The same applies for the remainder of the liquid assets. For the securities in the trading book, it is assumed that they are liquidated earlier, resulting in lower haircuts. Benchmark haircuts (independent of in which book securities are held) for equities are 5 percent (moderate), 10 percent (Lehman), 25 percent (severe), and 100 (very severe). For bonds, the corresponding haircuts are 5 percent, 10 percent, 15 percent (but high-quality bonds remain liquid—see LCR definition), and 100 percent, respectively.

3. A haircut of 100 percent means that the asset is illiquid, i.e., the market has closed.

4. The figures account for a downgrade of the bank, which triggers margin calls, and higher collateral requirements more generally. Please note that the unencumbered portion applies to a gradually narrower definition of liquid assets.

mine the increase in funding costs (i.e., interest expense) on the liability side, while also accounting for the effect on earnings on the asset side (interest income). As an illustrative example, our tool uses the estimated nonlinear relationship between solvency (measured as economic capitalization) and funding costs for Germany, as presented in Schmieder, Hesse, and others (2011). Stress testers can make assumptions with respect to the portion of the increase in funding costs that can be passed on to customers, thereby reducing the severity of the scenario. The model should be recalibrated on the basis of country-specific circumstances, taking account of the portion of wholesale funding, the level of competition, and other relevant factors. Once market conditions deteriorate further, driven by general market conditions and/or idiosyncratic strains at single banks, funding markets will close.¹⁹ Our tool provides a template to simulate different scenarios with respect to funding costs on the one hand and the (partial) closure of funding sources on the other. It is essential

to ensure that the calibration is adequate for the banks and/or system at hand, which remains at the discretion of the stress tester.

E. Liquidity stress test benchmark scenarios

In Table 7.4, we define *severe* benchmark scenarios that stress testers could refer to in order to simulate moderate, medium, severe, and very severe stress. The level of severity of stress is benchmarked to Lehman. Although the Lehman calibration is not to be understood as being scientific, it is meant to represent the situation of banks hit hard during the first month after the Lehman collapse, and, very important, it is intuitive. Accordingly, the moderate scenario is half of Lehman crisis conditions, whereas medium, severe, and very severe are one, two, and three times that of Lehman, respectively. As a caveat, it should be highlighted that the level of stress to be considered medium, for example, depends on the specific circumstances.²⁰ We recommend running a series of stresses to assess the sensitivity of

¹⁹ The RAMSI model attempts to model the deterioration in various factors and to link it to the closure of funding markets in the United Kingdom (see Aikman and others, 2009).

²⁰ “Moderate” is already a substantial stress event in terms of overall stress, but in some countries this could be rather “mild.”

systems, including reverse stress to test the threshold that systems can withstand, and to alter the runoff rates as applicable (e.g., by using higher runoff rates for retail deposits).

3. CONCLUSION

We have argued that liquidity risk has—unjustifiably—flown under the regulatory radar with the advent of the Basel I framework and its focus on bank capitalization. However, the fact that liquidity risk turned out to be one of the key threats to financial stability during the recent financial crisis has led to a reevaluation, with an emerging focus on liquidity in industry as well as regulatory circles, not least because of the ongoing challenges in this area. This study and its associated tool aim to provide stress testers with a flexible and easy-to-use platform to assess the liquidity situation of banks from different perspectives. The predefined tests can easily be adapted to bank-specific situations and/or specificities of banking systems (or an international sample of banks). A key objective is to strike a balance between data requirements and stress test sophistication, allowing for tests with parsimonious data on the one hand and more complex/demanding tests on the other.

Although the obvious way to stress test liquidity risk is the use of cash flow data, these are often not available (yet) at regulatory or supervisory institutions. One of the main contributions of this study consists of providing input templates for cash flow–based tests that could also serve as a first step toward full-fledged cash flow analysis based on regular data collection from banks by supervisors or regulators. Once available, the cash flow module allows the simulation of detailed funding structures of single banks, which would enable the stress tester to draw some broad conclusions for the system and potential contagion effects. In the latter case, liquidity monitoring (rather than stress testing) could be another focus. Moreover, our tool enables peer comparisons that should always play an important role for liquidity stress tests as they can readily reveal vulnerabilities.

Finally, our study contributes to existing work on liquidity by explicitly modeling the link to solvency stress (tests). Although this should not be misinterpreted as the ultimate solution to this highly complex problem, the inclusion of a module in the tool to analyze this link in an easy-to-use fashion should facilitate the practitioner's work.

REFERENCES

- Aikman, David, Piergiorgio Alessandri, Bruno Eklund, Prasanna Gai, Sujit Kapadia, Elizabeth Martin, Nada Mora, Gabriel Sterne, and Matthew Willison, 2009, "Funding Liquidity Risk in a Quantitative Model of Systemic Stability," Working Paper No. 372 (London: Bank of England). Available via the Internet: <http://www.bankofengland.co.uk/publications/Pages/workingpapers/2009/wp372.aspx>
- Barnhill, Theodore, and Liliana Schumacher, 2011, "Modeling Correlated Systemic Liquidity and Solvency Risks in a Financial Environment with Incomplete Information," IMF Working Paper 11/263 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=25356>
- Basel Committee on Banking Supervision, 2006, "Basel II: International Convergence of Capital Measurement and Capital Standards: A Revised Framework—Comprehensive Version" (Basel, June). Available via the Internet: <http://www.bis.org/publ/bcbs128.htm>
- , 2008, "Principles for Sound Liquidity Risk Management and Supervision" (Basel, September). Available via the Internet: <http://www.bis.org/publ/bcbs144.htm>
- , 2010, "Basel III: International Framework for Liquidity Risk Measurement, Standards and Monitoring" (Basel, December). Available via the Internet: <http://www.bis.org/publ/bcbs188.pdf>
- Boss, Michael, Gerhard Fenz, Gerald Krenn, Johannes Pann, Claus Pühr, Thomas Scheiber, Stefan W. Schmitz, Martin Schneider, and Eva Ubl, 2008, "Stress Tests for the Austrian FSAP Update 2007: Methodology, Scenarios and Results," *Financial Stability Report*, No. 15, June, pp. 68–92 (Vienna: Österreichische Nationalbank). Available via the Internet: http://www.oenb.at/en/presse_pub/period_pub/finanzmarkt/finanzmarktstabilita/financial_stability_report_15.jsp#tcm:16-173975
- Čihák, Martin, 2007, "Introduction to Applied Stress Testing," IMF Working Paper 07/59 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=20222>
- Committee of European Banking Supervisors, 2009, "Report on Supervisors' Lessons Learnt—A Liquidity Risk Perspective" (unpublished; London).
- Deutsche Bank, 2010, "Basel II Pillar 3 Report 2010," Frankfurt. Available via the Internet: http://www.db.com/ir/en/download/Basel_II_Pillar_3_Report_2010.pdf
- European Central Bank, 2008a, "EU Banks Liquidity Stress Tests and Contingency Funding Plans" (Frankfurt, November). Available via the Internet: http://www.ecb.int/pub/pdf/other/eubank_sliquiditystresstesting200811en.pdf
- , 2008b, "The Implementation of Monetary Policy in the Euro Area," General Documentation of Eurosystem Monetary Policy Instruments (Frankfurt, November). Available via the Internet: <http://www.ecb.europa.eu/pub/pdf/other/gendoc2008en.pdf>
- Goodhart, Charles, 2008, "Liquidity Risk Management," *Financial Stability Review—Special Issue on Liquidity*, No. 11, February, pp. 39–44 (Paris: Banque de France). Available via the Internet: <http://www.banque-france.fr/en/publications/financial-stability-review.html>
- International Monetary Fund, 2011, *Global Financial Stability Report*, World Economic and Financial Surveys, Chapter 2 (Washington, April). Available via the Internet: <http://www.imf.org/external/pubs/ft/gfsr/2011/01/index.htm>
- Ong, Li Lian, and Martin Čihák, 2010, "Of Runes and Sagas: Perspectives on Liquidity Stress Testing Using an Iceland Example," IMF Working Paper 10/156 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=24019>
- Schmieder, Christian, Heiko Hesse, Benjamin Neudorfer, Claus Pühr, and Stefan W. Schmitz, 2011, "Next Generation System-Wide Liquidity Stress Testing," IMF Working Paper 11/203 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.cfm?sk=25509.0>

- Schmieder, Christian, Claus Puhr, and Maher Hasan, 2011, "Next Generation Balance Sheet Stress Testing," IMF Working Paper 11/83 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=24798>
- Schmitz, Stefan W., 2010, "Liquidity Regulation and Stress Tests," Presentation at RiskMinds Conference (Geneva, December 6–10).
- van den End, Jan Willem, 2008, "Liquidity Stress Tester: A Macro Model for Stress-Testing Banks' Liquidity Risk," DNB Working Paper No. 175 (Amsterdam: De Nederlandsche Bank). Available via the Internet: <http://www.dnb.nl/en/publications/dnb-publications/dnb-working-papers-series/dnb-working-papers/auto175528.jsp>
- Wong, Eric, and Cho-Hoi Hui, 2009, "A Liquidity Risk Stress-Testing Framework with Interaction between Market and Credit Risks," HKMA Working Paper 06/2009 (Hong Kong SAR: Hong Kong Monetary Authority). Available via the Internet: <http://www.hkma.gov.hk/eng/publications-and-research/research/working-papers/2009>

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Systemic Bank Risk in Brazil: A Comprehensive Simulation of Correlated Market, Credit, Sovereign, and Interbank Risks

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In this study, we present a comprehensive forward-looking portfolio simulation methodology for assessing the correlated impacts of market risk, private sector and sovereign credit risk, and interbank default risk. In order to produce better integrated risk assessment for banks and systemic risk assessments for financial systems, we argue that reasonably detailed modeling of bank asset and liability structures, loan portfolio credit quality, and loan concentrations by sector, region, and type, as well as a number of financial and economic environment risk drivers, is required. Sovereign and interbank default risks are increasingly important in the current economic environment, and their inclusion is an important model extension. This extended model is demonstrated through an application to both individual Brazilian banks (i.e., 28 of the largest banks) and groups of banks (i.e., the Brazilian banking system) as of December 2004. Our results also show that a commonly used approach of aggregating all banks into one single bank for purposes of undertaking a systemic banking system risk assessment results in a misestimation of both the probability and cost of systemic banking system failures. Our analysis also indicates that, in the event of a sovereign default, the Government of Brazil would face constrained debt management alternatives. We conclude that such forward-looking risk assessment methodologies for assessing multiple correlated risks, combined with the targeted collection of specific types of data on bank portfolios, have the potential to better quantify overall bank and banking system risk levels, which can assist bank management, bank regulators, sovereigns, rating agencies, and investors to make better informed and proactive risk management and investment decisions.

METHOD SUMMARY

Overview	Monte Carlo simulations are conducted over a selected time period, with each bank balance sheet account updated accordingly for correlated underlying stochastic processes governing interest rates, equity market indices, and exchange rates.
Application	The method enables a forward-looking risk assessment of the impact of multiple correlated risks.
Nature of approach	Simulation based.
Data requirements	Detailed data on banks' balance sheets and portfolios, including transition probabilities from one credit risk category to another.
Strengths	The method enables the assessment of various correlated risks (market risk and private sector credit risk, including sovereign and interbank default risks) to produce an overall bank risk assessment.
Weaknesses	• The method is based on normally distributed stochastic shocks. It is particularly important to properly account for fat tail and skewness in the distribution during crisis periods. An attempt to include stochastic volatilities improved simulation results somewhat. Still more research is warranted to further refine simulated distributions. • Market and credit risk are integrated through a capital asset pricing model (CAPM) formulation and a distributional analysis of the loans-to-value ratio. In addition to the limitations of the CAPM model that are well documented in the literature,

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this formulation also requires market data to estimate the asset's beta, which can prove to be difficult in countries where equity markets are thin and sensitive to the time period used.

- This method is very data intensive, and simulations may take a significant amount of time to run.

Tool

ValueCalc Global Portfolio and Credit Risk software, copyright FinSoft, Inc. (1996–2005).

Bear Stearns, Lehman Brothers, American International Group (AIG), Fannie Mae, Freddie Mac, Wachovia, Citigroup, Washington Mutual, and a substantial number of other global financial institutions experienced correlated financial problems during the 2007–08 period. The inability to anticipate these problems and adopt proactive measures to minimize their impact on the global financial system and economy has imposed trillions of dollars of losses on investors and taxpayers. The need for improvements and innovation in risk modeling is apparent.

Financial institutions hold portfolios of debt, equity, and derivative securities that face a variety of correlated risks, including (1) asset and liability structure and loan portfolio concentration risk;¹ (2) credit risk on business, consumer, sovereign, and interbank loans and other transactions; (3) interest rate risk; (4) interest rate spread risk; (5) foreign exchange risk; and (6) equity price. The current practice is to undertake market and credit risk assessments separately (e.g., Basel Committee on Banking Supervision, 1988, 1996, 2001). Combining such separate risk measures into one overall portfolio risk measure is not easily accomplished (see, e.g., Jarrow and Turnbull, 2000; and Barnhill and Maxwell, 2002). The absence of reliable overall portfolio risk measures creates problems for determining capital adequacy requirements, capital-at-risk measures, hedging strategies, and so forth.

At any future time, individuals, businesses, banks, groups of banks, and sovereign will all face the same financial and economic environment (for better or worse). There is every reason to expect that the credit quality (default probability) for each of the above entities is nonzero, stochastic, and correlated with future financial and economic environment conditions (e.g., default rates will go up during periods of economic stress). There is also reason to believe that bank market risk is driven systematically by the same future financial and economic conditions. Barnhill, Papapanagiotou, and Schumacher (2003) and Barnhill, Papapanagiotou, and Souto (2004) have shown that banks with high credit risk and concentrated portfolios have a high risk of failure during periods of financial stress.

We present and apply a methodology² in this chapter that is capable of capturing all the correlated risks mentioned

here in a comprehensive way. Banks' balance sheets and portfolios are modeled with a substantial amount of details that we have not seen in any other methodology, to the best of our knowledge. For example, the loans portfolio of a bank is considered to be distributed across different regions for consumer loans and industry sectors for business loans. The initial distribution of the credit quality of the loan portfolio is assumed to be that reported by the bank. Systematic changes in the credit quality of these business and consumer loans are driven by stochastic economic factors relevant for their sector and region, which are then modeled as correlated stochastic variables. Borrower-specific risk factors are also modeled as independent random events. Loans are priced using different spreads, consistent with their simulated loan-to-value ratios and credit quality. We will discuss the features of the loan credit risk assessment methodology with more detail later.³

In this chapter, we further extend this methodology for assessing the correlated impacts of market risk and private sector credit risk to include sovereign and interbank default risk and produce one overall bank risk assessment. In the current economic environment, we believe that sovereign risk is a significant factor that should be considered in bank and systemic banking system risk assessments. In a further significant innovation, financial and economic environment variables are modeled with stochastic volatilities and correlations. This framework produces simulated distributions of future bank financial performance measures (returns, capital ratios, etc.) that reflect the collective impact of all of the modeled correlated variables. The simulated distributions allow the estimation of the probability and cost of various potential future events, such as bank failures or multiple bank failures, which reflect all of the modeled correlated risks. By modeling multiple banks simultaneously, we also provide a methodology for assessing the probability of multiple bank failures occurring at the same time period.

We incorporate systemic risk through different channels. First, banks are directly interdependent through a nexus of financial interbank contracts. One insolvent bank might become unable to honor its contracts, provoking financial distress in its counterparts (e.g., Rochet and Tirole, 1986; and Elsinger, Lehar, and Summer, 2003). When simulating the banks simultaneously, we assumed that if one bank

¹ For example, in previous years, there were a number of correlated failures of "agricultural" and "energy" banks in the United States. In the 2007–08 period, it was notable the number of correlated failures of institutions with concentrated exposures to real estate mortgages and mortgage-backed securities.

² Based on the work of Barnhill and Maxwell (2002).

³ The ValueCalc Global Portfolio and Credit Risk software, copyright FinSoft, Inc. (1996–2005), was used to undertake the risk analyses reported in this study.

fails, it will default on its interbank obligations, with some recovery rate. Second, banks' assets have some degree of correlation, as banks might invest in the same industries or geographical regions. If a shock affects one or more particular industry (or geographical region), then banks with exposure to that industry (or geographical region) will be systematically affected (e.g., Lehar, 2003; Acharya, 2009). By modeling the banks' portfolios as distributed across geographic regions and industry sectors, we can capture underlying correlations among portfolios of banks that lend to similar regions or industries. Third, a downward business cycle may cause companies' distress, rendering many loans delinquent and causing banks to further reduce business lending. This can deepen the business cycle, worsening the financial crises and affecting more banks (e.g., Gorton, 1988). We capture this effect, to some extent, through the simulation of the macroeconomic environment under which banks operate. In the course of a Monte Carlo simulation, it is possible to obtain a combination of macroeconomic factors that would be consistent with a downward business cycle. Fourth, correlated interest rate, exchange rate, and equity price risk (i.e., market risk) also may have an impact on multiple banks simultaneously.⁴ We explicitly incorporate the correlation between these variables in the simulation. Fifth, various factors including bank lending policies, the efficiency of the legal system in resolving loan defaults, and the aggregate level of defaults can have a substantial impact on bank risk through variations in the recovery rates on defaulted loans. We model recovery rates as distributed according to a beta distribution. Finally, sovereign defaults may simultaneously impose direct losses on banks through a reduction in the market value of their government securities and indirect losses brought about by economic and contract disruptions that incrementally increase the default rates on private sector loans. Although the characterization of a sovereign default event may be a difficult task, we assume that the government may impose some losses on the market value of its securities, with significant impact on banks' assets. This channel is of particular importance when analyzing Brazilian banks, as they hold a massive amount of government securities in their portfolios, sometimes above 80 percent of their total assets.

In this study, we model government default in a relatively simplistic way, as a very large "corporate borrower" whose known default probability is systematically related to returns on the Brazilian equity market index plus an idiosyncratic component. The impact of interbank exposure is modeled in a second step after the Monte Carlo simulations are done and the initial bank risk assessments completed.⁵ It is important

to keep in mind the significance of modeling all of the correlated risks above. In particular during times of economic stress, it is likely that default losses on private sector loans will increase, market volatility and risk will also likely increase, and so will the risk of sovereign default. Thus, should a sovereign default occur, it will likely be at a time when many banks are already being adversely affected by other risk factors. This is just the time when the failure of several banks could, through interbank credit defaults, precipitate a number of additional bank defaults and a systemic banking crisis.

We simulate bank risk at different levels: (1) for individual banks; (2) for individual aggregated groups of banks; and (3) simultaneously for three aggregated groups of banks. For our individual bank risk assessments, we simulate 28 of the largest Brazilian banks for two different cases: (1) the Government of Brazil (GOB) is assumed to never default; and (2) GOB can default on its debt with a 4.5 percent probability, which is consistent with the average default probability of countries rated by Fitch in the same grade as Brazil (B rating), as of December 2004.⁶

Our analysis, as of December 2004, finds that most Brazilian banks have a low risk of failure so long as the government does not default.⁷ However, in the event of a sovereign default, a number of Brazilian banks face potential solvency problems and could, if customers lose confidence, face liquidity problems as well. This highlights the danger of concentrated lending to an entity that has a nonzero probability of default.

When assessing the systemic risk of the Brazilian financial system, we use different approaches, with the intention of illustrating the importance of modeling the banks individually and simultaneously, as operating under the same macroeconomic environment. We first consider a single bank that is a combination of all 28 banks (these banks represent approximately 75 percent of the banking system total assets and are the banks for which we managed to gather the best available data to carry out this exercise). Subsequently, we simulate three aggregate banks simultaneously and include the risk component associated with interbank default (interbank propagation channel). Our results show that aggregating the banks into one single bank underestimates the cost associated with a systemic risk crisis, when compared with the three-bank simultaneous simulation case. Our analysis also highlights the danger of modeling the financial system as one single financial institution and not accounting for the differential risk characteristics of various banks and for the interbank risk channel, through which a systemic crisis may propagate.

⁴ Several authors have stressed the importance of macroeconomic factors, such as cyclical GDP downturns, interest rate increases, or exchange rate devaluations, on banks' performance (e.g., Gorton, 1988; and Lindgren, Garcia, and Saal, 1996; among others).

⁵ Each simulation run produces a random path over a certain time period (e.g., one year). To minimize computational effort and time, balance sheet accounts are recalculated only at the end of the time period.

⁶ According to data provided by the Brazilian Treasury Department, the spread on GOB U.S. dollar-denominated external borrowings averaged over 5 percent during 2004. This is generally consistent with an expected default rate of 4.5 percent.

⁷ We note that during the very stressful times of 2007–08, none of the banks that we model in this study failed, consistent with our results.

The remainder of this chapter proceeds as follows: Section 1 provides an abbreviated description of the portfolio simulation model we use to undertake the various integrated risk assessments.⁸ In Section 2, we present and discuss the risk assessments for Brazilian banks and the Brazilian banking system. Concluding remarks and final comments are given in Section 3.

1. AN INTEGRATED MODEL OF CORRELATED MARKET, CREDIT, SOVEREIGN, AND INTERBANK RISK IN AN ENVIRONMENT WITH STOCHASTIC VOLATILITIES AND CORRELATIONS

An integrated analysis of correlated market risk and private sector, sovereign, and interbank default risk on both individual banks (i.e., 28 of the largest Brazilian banks) and groups of banks (i.e., the Brazilian banking system) requires a powerful and flexible forward-looking risk assessment methodology.⁹

We propose that it is possible to identify a set of correlated financial and economic variables, for example, sector equity returns and regional unemployment rates,¹⁰ which *systematically* affect the credit quality and produce correlated default rates among various groups of borrowers (e.g., businesses in various sectors and individuals in various regions). We also model many uncorrelated borrower-specific risk factors. We find such detailed individual borrower and portfolio modeling necessary to assess the risk level of portfolios of loans with varying credit quality distributed across various sectors and regions. In particular, it allows for explicit modeling of loan portfolio concentration and diversification impacts on bank risk levels.

We further propose that the identified set of correlated financial and economic variables can include ones that have *systematic* impacts on the market values of securities, loans, and other assets. Examples of such variables are interest rates, interest rate spreads, foreign exchange (FX) rates, equity index returns, and commodity prices. Also, a significant innovation in this chapter is that financial and economic environment variables are modeled with stochastic volatilities and correlations.

The structure of the systemic bank risk simulation model is as follows:

- Simulate the future financial environment (e.g., monthly steps for one year) as a set of correlated stochastic variables that systematically affect each bank's market risk (interest rates, interest rate spreads, FX rates) and credit risk (sector equity returns, regional real estate prices, regional unemployment rates, etc.).
- Simulate the correlated evolution of the credit rating (default) for each borrower in each bank's loan portfolio as a function of the simulated financial environment.
- Simulate the correlated evolution of the credit rating (default) for the Government of Brazil as a function of the simulated financial environment.
- Revalue each asset and liability in each bank's balance sheet as a function of the simulated financial environment and credit ratings. Note that for each run of the simulation, the simulated value of each bank's assets and liabilities reflects correlated market and credit risk.
- Estimate each bank's pretax income, operating costs, taxes, and aftertax income over the simulation period.
- Calculate financial performance measures for each bank (e.g., economic capital ratio, rate of return on assets) under the simulated conditions. Estimate whether the bank will fail.
- Estimate each bank's potential losses on defaulted interbank loans.
- Recalculate financial performance measures for each bank (e.g., economic capital ratio, rate of return on assets) under the simulated conditions. Reestimate whether each bank will fail.
- Repeat the simulation a large number of times.
- Analyze the distributions of simulated bank financial performance measures (e.g., economic capital ratio) to determine individual bank and banking system systemic risk levels.¹¹

⁸ For a more detailed discussion of the simulation methodologies used in this study see Barnhill and Souto (2008b).

⁹ See Table 8.1 for a description of the data used in this study along with respective sources.

¹⁰ In circumstances where banks hold significant mortgage loans, it is useful to model regional real estate prices as a systematic credit risk driver (see Barnhill, Papapanagiotou, and Schumacher, 2003; and Barnhill, Papapanagiotou, and Souto, 2004) that produces correlated changes in the loan-to-value ratio and defaults on mortgage loans in various regions. Given that most Brazilian banks hold very few mortgage loans, we did not model Brazilian real estate prices.

¹¹ In the present study, the composition of the bank asset and liability portfolio is held constant over the risk horizon (e.g., one year) and is repriced after each run of the simulation using the simulated financial and economic variables and credit qualities of the borrowers. Simulations were run 2,000 times using monthly (i.e., 12) time steps. To infer the *potential* size of Monte Carlo simulation errors, we simulated some of the banks multiple times. The difference between the maximum and minimum simulated capital ratios over the various simulation run depends on the confidence level considered. At the extremities (e.g., 99th percentile), the variation in simulated capital ratios is typically on the order of 0.1 to 0.2 percent. At the 50th percentile, the variation in simulated capital ratios is on the order of 0.02 to 0.03 percent. The critical capital ratio at which banks are assumed to fail is 3 percent. Given the relatively small size of the above simulation errors, we believe that 2,000 runs are adequate for the purposes of this study.

TABLE 8.1

Data Description and Sources		
Analysis	Data	Data Sources
Estimation of initial volatilities and correlations for financial and economic environment simulation	Brazilian short-term domestic nominal interest rate	Central Bank of Brazil (CBB) Web site
	U.S. nominal interest rate	FED St. Louis Web site
	Brazilian FX rate	CBB Web site
	Gold	CBB Web site
	Oil	International Monetary Fund Commodity Prices Database
	Brazilian CPI level	CBB Web site
	Ibovespa (the broad Brazilian equity market index)	DataStream
	Ten business sector equity indices (basic industry, construction material, chemicals, cyclical services, food production, food retail, forest and paper, telecommunications, and utilities)	DataStream
	Unemployment rates for five big cities in Brazil (Recife, Belo Horizonte, Salvador, Rio de Janeiro and São Paulo, and Porto Alegre)	Brazilian Institute of Geography and Statistics Web site
	Bank-by-bank net interest margin	Bankscope
Estimation of bank-by-bank interest rate spreads by credit rating category	Default rates by credit rating (for 2 of the largest Brazilian banks) ¹	Barnhill, Souto, and Tabak (2006)
	Bank-by-bank average interest rate spread for business and customers' loans	CBB Web site
Estimation of betas and firm-specific risk	Stock prices for 543 Brazilian companies listed in the Brazilian Stock Exchange	DataStream
	Price indices for 9 business sector equity indices (basic industry, construction material, chemicals, cyclical services, food production, food retail, forest and paper, telecommunications, and utilities)	DataStream
Estimation of debt ratio ranges for businesses in various credit rating categories	Debt-to-value ratios for 543 companies listed in the Brazilian Stock Exchange	DataStream
Borrowers' credit ratings ¹	Bank loan ratings for 543 Brazilian companies listed in the Brazilian Stock Exchange, as assigned by Brazilian banks	Barnhill, Souto, and Tabak (2006)
Bank-by-bank balance sheet data	Bank-by-bank balance sheet data for 28 of the largest Brazilian banks	CBB Web site and Bankscope
Bank-by-bank balance sheet maturity structure	Bank-by-bank asset and liability maturity structures for 28 of the largest Brazilian banks	CBB Web site
Bank-by-bank portfolio composition	Bank-by-bank business loan portfolio distribution by 9 business sector equity indices (basic industry, construction material, chemicals, cyclical services, food production, food retail, forest and paper, telecommunications, and utilities) and customers' loan portfolio distribution by five geographical regions (South, Southeast, Middle-West, North, and Northeast)	CBB Web site
Bank-by-bank distributions of loan credit ratings	Bank-by-bank business loan portfolio distribution by credit quality	CBB
Bank-by-bank credit rating	Bank-by-bank credit rating (for the ones among the 28 of the largest Brazilian banks for which this information was available)	Standard and Poor's, Moody's, and Fitch Web sites

Source: Authors.

Note: CPI = Consumer price index; FX = foreign exchange.

1. This information is protected by Brazilian law and cannot be disclosed.

Data for financial and economic variables were available monthly for the period 1999–2004.

When modeling the macro-financial environment under which banks operate, we use the Hull and White extended Vasicek model (Hull and White, 1990, 1993, 1994) to model stochastic risk-free (e.g., U.S. Treasury) interest rates. Once the risk-free term structure has been estimated, then the AA term structure is modeled as a sto-

chastic lognormal spread over the risk-free interest rate, the A term structure is modeled as a stochastic spread over AA, and so on. The mean value of these simulated credit spreads is set approximately equal to the forward rates implied by the initial term structures for various credit qualities (e.g., AA). This procedure ensures that all simulated

credit spreads¹² are always positive and that the simulated risky term structures are approximately arbitrage free.¹³

The model utilized to simulate the value of the equity market indices and FX rate (S) assumes that (S) follows a geometric Brownian motion where the expected growth rate (m) and volatility (σ) are constant (Hull, 2008).

In the current portfolio risk assessment model, the equity indices and FX rate returns are simulated as stochastic variables correlated with the future risk-free interest rate and interest rate spreads. Error terms are drawn from univariate normal distributions and decomposed using the Cholesky factorization. In this particular study, a total of 22 independent random variables (see Table 8.1 for a list of the variables used) are used to describe the macro-financial environment underlying banks' operations.

Variances and covariances are updated for each simulation time step via the following model:

$$\sigma_{ji,t} = \lambda \sigma_{ji,t-1} + (1 - \lambda) \sigma_{ji,t-1} v_t, \quad (8.1)$$

where $v_t \sim N(0, 1)$, $\sigma_{ji,t}$ is the covariance between variables i and j , and λ is the decay factor.¹⁴ Initial variances and covariances at the start of each simulation run are estimated as the average realized variances and covariances over the entire January 2003 to December 2004 period. Similar to modeling stochastic returns, volatility shocks also are decomposed via Cholesky factorization and thus are affected by the correlation structure of the state variables.

Barnhill and Souto (2008a) show that the decay factor (λ) can be changed appropriately, so as to increase the probability mass of index returns in the tails of the simulated distributions. They further show that simulated volatilities and correlations match reasonably well the evolution of historical values. This model is implemented to simulate the credit transition matrix for Brazilian bank loans, and it is shown that this methodology has the potential to improve simulated transition probabilities as compared with the constant volatility case. In particular, it increases the simulated probability of default in lower credit risk categories to values closer to observed historical levels.

¹² Considering the large interest rate spreads charged by the banks and the fact that this information was not available to us on a bank-by-bank basis, different credit spreads were estimated for each bank so as to match the historical net interest income of each bank and produce average spreads that are consistent with the reported bank-by-bank interest rate spread (Central Bank of Brazil Web site) and assuming that the credit spread profile across the different credit risk categories was proportional to the expected default rate for each category.

¹³ The use of an arbitrage-free interest rate model such as the Hull and White extended Vasicek model (Hull and White, 1990, 1993, 1994) allows the estimation of an entire term structure of interest rates at the end of each simulation run that is needed to value financial instruments. The volatilities and correlations between the various financial and economic environment variables for this study were estimated from historical Brazilian data.

¹⁴ The optimal decay factor was estimated by minimizing the difference between historical and simulated volatility in the expression $(\sigma_{i,t}^2 - \sigma_{i,t-1}^2) + (\sigma_{i,t}^2 - \sigma_{i,t-6}^2) + (\sigma_{i,t}^2 - \sigma_{i,t-12}^2)$. For more details, please see Barnhill and Souto (2008a).

A. Modeling banks' assets, liabilities, and income

Banks' Balance Sheets

Banks' balance sheets are modeled with some degree of detail. On the liabilities side, we have (1) domestic funding, which includes interbank, demand, savings, and fixed deposits, NCDs, and repos; (2) foreign funding (in foreign currency); (3) debt; (4) non-interest-bearing liabilities; (5) capital and reserves; and (6) equity. Domestic and foreign funding and debt can be broken down, in the simulation framework, in up to three different maturities¹⁵ and then properly linked to different correlated stochastic interest rates (domestic or foreign) and foreign exchange rates. As we shall see, some asset accounts also can be modeled through multiple maturities, and this structure allows us to incorporate asset and liability maturity and currency mismatches (i.e., market risk) as a component of banks' integrated risk assessments.

On the asset side, the main accounts are (1) money (cash and gold reserves); (2) GOB securities; (3) business loans; (4) consumer loans; (5) foreign loans; (6) equity investments; (7) real estate investments; and (8) non-interest-earning assets. Money and non-interest-earning assets are not updated in the course of the simulations (we thus assume that the bank does not change the amount of money and non-interest-earning assets it holds, over the course of the simulation). Because they are usually the main risk elements in any bank's portfolio, loans are modeled in more detail. Absent other information, loans (which usually comprise the biggest fraction of most banks' portfolios) typically are modeled as bonds. We allocate business and consumer loans separately across eight credit risk categories defined by Banco Central do Brasil (AA,..., H). In the case of business loans, we also distribute them across 9 business sectors, whereas loans to individuals are distributed across six regions.¹⁶ This breakdown is important for capturing the portfolio diversification or concentration aspects of the risk analysis as well as the integrating of market and credit risk. In total, we use approximately 450 stylized securities to model each bank. Although a larger number of currencies and securities easily could have been modeled, we believe that this number is adequate to capture statistically much of the impact of bank asset and liability portfolio maturity, currency, credit quality, and sector and region concentration characteristics.

¹⁵ On the basis of the data we obtained, we categorized the liabilities into the following three maturity groups: <1 year, between 1 and 3 years, and >3 years. The simulation methodology, however, is flexible enough to handle multiple maturities for both liabilities and assets as well as multiple currency denominations.

¹⁶ The allocation of the 400 stylized business and consumer loans across sectors and regions reflects the amount of the bank's lending in the sectors and regions as well as the credit quality of the loans. Sectors and regions with larger amounts of higher credit risk loans (which default at a higher rate) are allocated a larger portion of the available loans. We find that this approach produces more consistent bank risk assessments.

TABLE 8.2
Brazilian Banks Balance Sheets—Main Statistics

Balance Sheet Items	Min.	Median	SD	Max.
Capital and liabilities				
Public funding				
Domestic funding	26.0%	59.8%	17.9%	85.2%
Foreign funding	0.9%	7.3%	12.7%	54.5%
Capital and other liabilities	0.0%	0.0%	0.0%	0.0%
Non-interest-bearing	3.9%	12.2%	12.4%	50.2%
Equity and reserves less impairments	5.0%	10.1%	7.4%	38.4%
Debt	0.0%	0.0%	1.3%	4.0%
Total	100.0%	100.0%	0.0%	100.0%
Assets				
Money	0.0%	1.3%	2.1%	9.4%
Gold	0.0%	0.0%	0.0%	0.0%
Domestic risk-free loans	25.0%	54.2%	18.3%	90.5%
Domestic business loans	1.0%	21.2%	13.9%	55.6%
Domestic individual loans	0.0%	3.5%	11.3%	57.0%
Foreign loans	0.0%	0.0%	0.0%	0.0%
Equity investments	0.0%	0.4%	0.8%	3.8%
Real estate investments	0.0%	1.1%	1.2%	3.6%
Other assets	2.5%	10.6%	9.1%	44.4%
Total	100.0%	100.0%	0.0%	100.0%
Net interest margin	0.027	0.075	0.042	0.203
Net noninterest income/total assets	−0.080	−0.032	0.031	0.040

Sources: Bankscope and Central Bank of Brazil Web site.

Note: This table provides statistics on the stylized balance sheets used in the simulation to describe the main assets and liabilities accounts for Brazilian banks, as of December 2004.

Unlike state variable returns that are estimated for each time step in each simulation path, all balance sheet accounts are recalculated only at the end of each simulation period.¹⁷ In Table 8.2, we present some descriptive statistics on the stylized balance sheet for 28 Brazilian banks. These data were obtained from Bankscope and the Web site of the Brazilian central bank.¹⁸

Banks' Operating Expenses, Taxes, and Income

In addition to the bank's assets and liabilities, we also model net noninterest income. Net noninterest income is defined as fee income plus other noninterest income minus operating expenses. We use historical information to estimate the ratio of noninterest income to total assets, and at the end of each simulation period, we calculate how much the bank will have spent on operations. This amount is then deducted from the bank's simulated net interest income plus (minus) capital gains (losses) to estimate pretax income. We also estimate taxes paid by banks as an assumed percentage of taxable income over the simulation period. Aftertax income is estimated as pretax income less estimated taxes.

¹⁷ We also specify the time horizon and the time steps over which price paths and balance sheet accounts will be simulated and the number of simulation runs.

¹⁸ It is possible to model a great variety of other financial instruments (e.g., bonds, zero coupon bonds, variable rate loans, forward contracts, swaps, options) within our simulation framework.

Banks in Brazil operate with very high net interest margins. During the period under analysis, information from Banco Central do Brasil indicates that the spreads between the average yield on bank business loans (consumer loans) and government securities were approximately 20 percent (40 percent). Considering all loans (government, business, and consumer) information from Bankscope (see Table 8.2), the median net interest margin for the 28 Brazilian banks analyzed in this study was 7.5 percent.

Brazilian banks also have high operating expense ratios. Information from Bankscope indicates that the median ratio of net noninterest income to total assets for the 28 Brazilian banks analyzed in this study was −3.2 percent.

B. Modeling bank loan portfolio credit risk

The methodology used to model business loan credit transition probabilities and defaults starts with an extensive empirical analysis of all publicly traded companies in Brazil and all bank loans made to those firms by all Brazilian banks.¹⁹ The purpose of this analysis is to identify typical debt-to-value ratios, beta coefficients, and firm-specific equity return volatilities for Brazilian firms with various bank-assigned credit ratings. The results are found in Table 8.3.

¹⁹ For a more detailed discussion of the credit risk modeling methodologies used in this chapter, see Barnhill and Souto (2008b).

TABLE 8.3

Debt Ratios, Betas, and Unsystematic Risk Used in the Portfolio Simulation Risk Assessments¹

	AA	A	B	C	D	E	F	Default
Debt-to-value ratios								
Lower bound	0.270	0.457	0.660	0.745	0.785	0.798	0.802	0.960
Target	0.405	0.620	0.810	0.838	0.890	0.902	0.894	0.960
Upper bound	0.516	0.706	0.865	0.917	0.922	0.935	0.940	0.960
Beta	0.67	0.85	1.00	1.10	1.20	1.30	1.36	—
Unsystematic risk	0.38	0.55	0.69	0.71	0.77	0.78	0.72	—

Source: Authors.

Note: This table provides results for distributional analysis of equity return risk characteristics, debt-to-value ratios, and internal bank credit ratings for all publicly traded Brazilian companies.

1. See Barnhill, Souto, and Tabak (2003).

As expected, typical debt ratios increase as company bank loan ratings decline.²⁰ For simulation runs reported later in this study, we assume that debt ratios start at the midpoint between the maximum and minimum values for the assumed initial credit rating category. Credit ratings are generally assumed to change when simulated debt ratios cross the min-max boundaries set for the initial credit rating.²¹ We use this information to develop stylized firms for each credit rating (i.e., target debt-to-value ratio and assumed beta and firm-specific risk factors). For each run of the simulation, the return on each borrowing firm's equity is estimated as a function of the simulated return on an equity market index plus a firm-specific random change. This simulated equity return allows the estimation of a new debt-to-value ratio and credit rating (including default²²) for each firm in the bank's

loan portfolio.²³ Using a metric proposed by Jafry and Schuermann (2004), Barnhill and Souto (2008a) report simulated Brazilian bank loan credit transition probability matrices that are very similar to historical Brazilian credit transition matrices.

After simulating the loans' future credit rating, we calculated their values using the simulated term structure of interest rates appropriate for each risk class. If the loan is simulated to default, the recovery rate (representing the present value of all future recoveries) on the loan is set at either the higher (45 percent) or lower (15 percent) level discussed earlier. If the loan is denominated in a foreign currency, then its numeraire currency value is calculated by multiplying the simulated loan value by the simulated foreign exchange rate that by construction is also a correlated stochastic variable.

C. Modeling bank capital ratios

The previous analysis allows the market value of the bank's assets, liabilities, equity, and capital ratio to be calculated for each simulation run:

$$MVE_t = \sum_{i=1}^n A_{i,t} - \sum_{i=1}^n L_{i,t}, \quad (8.2)$$

where MVE_t is the simulated market value of the bank's equity at time t ; $A_{i,t}$ is the simulated market value of the i th asset at time t , which represents the simulated financial environment variables (e.g., interest rates, exchange rates, equity prices); and, where appropriate, the simulated credit rating of the borrower, $L_{i,t}$, is the simulated market value of the i th

²⁰ We also find that typical betas and firm-specific risk levels increase as a firm's credit quality decreases. The increasing betas are due in large part to the typically increasing debt-to-value ratios for lower credit-quality companies, which increase the systematic volatility of equity returns. Higher firm-specific risk levels are consistent with firms having a lower credit rating and higher default risk.

²¹ Once outside of the assumed debt-to-value range for the initial credit rating, subsequent upgrades (downgrades) occur when the simulated debt-to-value ratio goes below (above) the minimum (maximum) debt-to-value ratios for the sequential rating categories.

²² The recovery rate on defaulted business and consumer loans ("private sector loans") is an important risk factor. In the case of Brazil, historically the legal system was viewed as taking an exceptionally long time to resolve loan defaults and bankruptcies, and the recovery rate on defaulted loans was low (e.g., 15 percent). In June 2005, a decade-long effort to modernize Brazil's bankruptcy laws came into effect. This change is projected to improve the recovery prospects for defaulted Brazilian bank loans (see Eccles, 2005). Thus, for the current study, when we undertake forward-looking risk assessments for Brazilian banks and the banking system, our base case assumption is that the recovery rate on defaulted private sector loans is drawn from a beta distribution with a mean value of 0.45 and a standard deviation of 0.25. However, we will also report simulation results based on an assumed recovery rate of 15 percent to illustrate the potential impacts of a lower recovery rate on systemic banking system risk levels. Although we have not done so in this study, the methodology could be modified to dynamically adjust the assumed recovery rate on defaulted loans to reflect a variety of variables, including end-of-period borrower debt ratios or loan-to-value ratios and banking system total loan default levels.

²³ We were unable to obtain the data required to develop a separate credit risk model for consumer loans. Thus, in this study, we model consumer loans as business loans. For this purpose, we model a systematic risk factor for each region that has the return and volatility characteristics of the Ibovespa stock index but that has correlations with other financial and economic environment variables equal to those of the regional unemployment rates. This model allows us to generate credit transition probabilities and default rates similar to those reported by Brazilian banks and also account for some of the diversification benefits of lending to different types of borrowers (i.e., businesses in various sectors and consumers in various regions).

liability at time t , which represents the simulated financial environment variables (e.g., interest rates, exchange rates). The bank's asset level is also adjusted to reflect bank net interest income, operating costs, and taxes over the simulation period.

The final outcome of the model after many simulation runs is an estimated distribution of the bank's capital-to-asset ratio, characterized by a mean, a standard deviation, a maximum, and a minimum value, as well as a value-at-risk measure indicating how frequently the bank's capital-to-asset ratio falls below a certain threshold (e.g., 3 percent), which is used to estimate the bank's default probability. Declines in the capital ratio (i.e., potential losses) under each simulation run are estimated as the difference between the initial bank capital and the simulated capital ratio.²⁴

$$Capital_Ratio_t = MVE_t / \sum_{i=1}^n A_{i,t}, \quad (8.3)$$

where $Capital_Ratio_t$ is the simulated bank capital ratio at time t .

D. Modeling sovereign risk

The Government of Brazil is modeled as a large business borrower with an assumed debt-to-value ratio and an assumed market value of equity.²⁵ The government's market value of equity is assumed to be systematically related to returns on the Brazilian equity market index and to also have an idiosyncratic component. On each run of the simulation, we estimate a new market value of equity and debt-to-value ratio for the GOB, and if this value exceeds some critical level, then the GOB is assumed to default. The initial debt-to-value ratio, beta, and idiosyncratic risk components are selected to produce a targeted average sovereign default rate of 4.5 percent. The simulated sovereign defaults thus are systematically related to returns on the Brazilian equity market (and other correlated state variables), which also are systematically related to simulated defaults on private sector loans. Through this mechanism, we end up modeling correlated sovereign and private sector loan defaults,²⁶ which also are correlated with simulated financial and economic variables (e.g., equity returns, interest rates, foreign exchange rates).

E. Modeling interbank default risk

The potential impact of interbank defaults is modeled in a second step after the initial Monte Carlo simulation bank risk assessments are completed.²⁷ For this purpose, we first aggregate the 28 banks into three groups according to their individual risk characteristics and then simulate the three groups simultaneously,²⁸ under the same financial/macroscopic scenarios. Given that we do not have precise information on interbank borrower or lender identities (we have info only on aggregate interbank lending for each bank), we assume interbank lending from one bank to be proportional to the other three aggregate banks' total assets. We then assume, in a second lance, that if one bank falls below a 3 percent capital ratio,²⁹ it becomes incapable of honoring its interbank obligations and defaults on them. Only 50 percent of these interbank obligations are assumed to be recovered, with subsequent impact on other banks' capital ratios. Eventually, one bank's failure can induce other banks to become insolvent as well, depending on the size of interbank exposure, combined with other factors, as mentioned earlier. It is important to keep in mind the significance of modeling all of the correlated risks above. In particular, during times of economic stress, it is likely that default losses on private sector loans will increase, market volatility and risk also will likely increase, and so will the risk of sovereign default. Thus, should a sovereign default occur, it will likely be at a time when many banks are already being adversely affected by other risk factors. This is just the time when the failure of several banks could, through interbank credit defaults, precipitate a number of additional bank defaults and a systemic banking crisis.

F. Modeling systemic bank risk

Finally, we assess the systemic risk of the Brazilian financial system, in different ways. We first consider a single bank that is a combination of all 28 banks,³⁰ and then we simulate the three aggregate banks simultaneously and include potential correlated interbank default losses (interbank propagation channel). Our results show that aggregating the banks into

²⁴ It is important to note that we are modeling banks' economic equity capital ratio.

²⁵ As of December 31, 2008, Standard and Poor's (S&P) rated 123 sovereigns. Over the previous 12 months, S&P had downgraded 29 sovereigns and upgraded 11 additional ones. At that same date, S&P indicated 2 positive outlooks for upgrades and 26 negative outlooks for downgrades. "These negative trends stem largely from difficulties adjusting to mounting economic pressures" (Cavanaugh, 2009). The recent crisis has evidenced how sovereign risk has become an important risk consideration for banks.

²⁶ We chose this methodology for modeling the sovereign risk because of simplicity and to save computational time. For more elaborate alternatives to this method of modeling sovereign default risk, which also could be used in this type of systemic bank risk assessment model, see Barnhill and Kopits (2004) and Barnhill (2006).

²⁷ Each simulation run produces a random path over a certain time period (e.g., one year). To minimize computational effort and time, balance sheet accounts are recalculated only at the end of the time period.

²⁸ The current version of the ValueCalc programs allows for simulating only three banks simultaneously (for computer memory reasons), although it has the potential to simulate any number of banks simultaneously.

²⁹ Discussions with bank regulators in several countries indicate that at some low but positive capital ratio, banks start having liquidity and other problems that make it difficult for them to survive. For this study, we assume that banks are likely to fail if their capital ratio (total asset minus total liabilities, over total assets) falls below 3 percent. Our model is, of course, flexible to accommodate other assumptions.

³⁰ Some international institutions and many central banks create a signage aggregate bank for conducting vulnerability analysis of a country's banking sector. We believe that this approach has serious limitations in that it masks the fact that many individual banks may be much more risky than the "average aggregate bank."

one single bank underestimates the cost associated with a systemic risk crisis, when compared with the three-bank simultaneous simulation case. The analysis highlights the danger of modeling the financial system as one single financial institution and not accounting for the differential risk characteristics of various banks and for the interbank channel, through which a systemic crisis may propagate.

2. SIMULATION RESULTS

Our risk assessments relate to three sets of Monte Carlo simulations:³¹ (1) individual banks, with no government default; (2) individual banks, with government default; and (3) systemic banking system risk. In the first and second sets of simulations, we wish to compare individual bank risk with and without sovereign default risk. We also compare the risk assessment produced by our portfolio simulations with the bank credit ratings provided by Moody's and Standard & Poor's. Using the risk assessments from the second set of simulations, we group banks into three risk categories and construct three aggregate banks. In the third set of simulations, we undertake a risk analysis for a single aggregate bank and subsequently a simultaneous risk analysis for three aggregate banks with different risk levels to assess systemic banking system risk (i.e., the likelihood of multiple banks defaulting), reflecting market, credit, sovereign, and interbank risk. These correlated market and credit risk assessments are based on analyses of the output from 2,000 simulation runs.

A. Individual banks, no government default

The first set of our simulations comprises all 28 banks, simulated individually, assuming that the GOB will never default on its domestic debt over the simulation horizon (one year).³² Simulation results for selected banks from this group are presented in Table 8.4, and a few comments are in order. First, reported value-at-risk (VaR) levels indicate the percentage of time that simulated bank capital ratios fall above a certain threshold. For example, Bank 20 VaR at the 99 percent confidence level is 0.147, indicating that 1 percent of the time, the simulated capital ratio for Bank 20 fell below 0.147. Second, with a few exceptions, Brazilian banks are well capitalized. For example, Bank 12 has a 0.252 capital ratio, and Bank 6 has a 0.382 capital ratio. Banks with capital ratios below 0.07 are Bank 5 (0.059), Bank 14 (0.062), Bank 16 (0.065), Bank 21 (0.067), and Bank 23 (0.063). Third, in general, *if* we do not factor government default

into the simulations, Brazilian banks are profitable and have an increasing capital ratio on average over the one-year simulation period. None of the banks produces simulated capital ratios that would indicate significant default risk problems over a one-year time step. Even the minimum simulated capital ratios are well above the 3 percent level that we have set as the critical bank capital level at which banks begin to default, although in several cases the simulated capital ratios are below a 0.08 capital ratio on an economic capital basis (e.g., Bank 5 with minimum capital ratio of 0.004, Bank 7 with 0.040, and Bank 21 with 0.046, among others). Finally, simulated capital ratios have a small standard deviation for all banks, ranging from 0.001 (Banks 14, 19, and 21) to 0.008 (Bank 5 and Bank 24). Given the substantial amount of government securities held by these banks, this result is not surprising at all. Bank 9 holds a more modest fraction of government securities. However, the credit quality of its portfolio is fairly high, with a corresponding small default rate.

B. Individual banks, government default

Given that Brazilian banks hold a significant amount of GOB debt, it is very important to assess the impact of correlated sovereign risk on banks' default probabilities.³³ For this exercise, we propose to model government default in a relatively simplistic way, which we claim can still provide reasonable insights on banks' exposure to sovereign risk. In particular, in the current study we model the GOB as if it were a large corporate borrower, subject to systematic and idiosyncratic risk factors. When determining these parameters, we need to be able to reproduce reasonably closely the rate at which a sovereign country like Brazil is expected to default, given its current credit rating. We also wish to capture appropriately the correlations between market risk, private sector loan defaults, and sovereign defaults.

The definition of government default itself is neither simple nor consensual. Fitch defines default as a failure to make timely payment of principal and/or interest on either (1) rated foreign currency debt or (2) other material foreign currency debt obligations, such as Paris or London Club liabilities. For the present study, using the model discussed earlier, we chose to produce GOB average default rates that are comparable to the average default rates of countries' sovereign debt rated at the same risk level as Brazil (B). We assume the GOB to have a default probability of 4.5 percent on foreign currency loans, consistent with Klaar, Rawkins, and Riley (2004). An important issue in the event of a sovereign default on its foreign debt is the government's willingness or ability to honor its domestic debt obligations. Although the government may want to expand its monetary base in order to repay domestic

³¹ Barnhill and Souto (2008b) report a fourth set of simulations that demonstrate that the means and standard deviations of the simulated rates of return on average bank assets and equity are unbiased predictors of historical mean returns and standard deviations of returns.

³² In fact, as we shall see in the next section, countries at the same sovereign rating as Brazil (B) have a one-year default rate of 4.5 percent on average in the 1994–2004 period, according to Klaar, Rawkins, and Riley (2004).

³³ This study was conducted when Brazil was rated as B. Currently Brazil is being rated as BBB– by most rating agencies. With this higher credit rating (and consequently lower expected sovereign default probability), if this study were to be repeated today, then our bank and systemic banking system risk estimates would be much lower.

TABLE 8.4

Simulated Capital Ratios: Individual Banks, No Government Default

Bank	Initial	Mean	SD	Maximum	Minimum	VaR Levels		
						99%	95%	90%
Bank 1	0.189	0.201	0.004	0.214	0.183	0.191	0.195	0.197
Bank 2	0.141	0.144	0.005	0.162	0.124	0.131	0.136	0.138
Bank 3	0.083	0.089	0.004	0.097	0.058	0.077	0.082	0.084
Bank 4	0.149	0.161	0.005	0.173	0.136	0.145	0.152	0.155
Bank 5	0.059	0.068	0.008	0.088	0.004	0.047	0.054	0.058
Bank 6	0.382	0.371	0.003	0.380	0.362	0.365	0.367	0.368
Bank 7	0.073	0.072	0.004	0.091	0.040	0.059	0.065	0.067
Bank 8	0.096	0.111	0.004	0.126	0.093	0.100	0.104	0.106
Bank 9	0.184	0.224	0.005	0.246	0.204	0.211	0.216	0.218
Bank 10	0.091	0.096	0.007	0.123	0.059	0.078	0.084	0.087
Bank 11	0.091	0.107	0.003	0.119	0.097	0.100	0.102	0.104
Bank 12	0.252	0.248	0.004	0.260	0.231	0.238	0.241	0.242
Bank 13	0.132	0.149	0.007	0.188	0.127	0.134	0.138	0.140
Bank 14	0.062	0.073	0.001	0.080	0.068	0.070	0.071	0.071
Bank 15	0.103	0.150	0.004	0.168	0.136	0.143	0.145	0.146
Bank 16	0.065	0.069	0.002	0.078	0.060	0.063	0.065	0.066
Bank 17	0.094	0.106	0.006	0.134	0.085	0.090	0.096	0.098
Bank 18	0.200	0.234	0.005	0.247	0.213	0.222	0.226	0.229
Bank 19	0.201	0.200	0.001	0.206	0.196	0.197	0.198	0.199
Bank 20	0.174	0.163	0.007	0.190	0.125	0.147	0.152	0.155
Bank 21	0.067	0.066	0.004	0.074	0.046	0.055	0.059	0.061
Bank 22	0.073	0.078	0.001	0.080	0.075	0.076	0.077	0.077
Bank 23	0.063	0.121	0.005	0.152	0.098	0.107	0.113	0.115
Bank 24	0.139	0.156	0.008	0.186	0.122	0.136	0.143	0.146
Bank 25	0.092	0.101	0.003	0.113	0.089	0.093	0.096	0.097
Bank 26	0.161	0.167	0.003	0.181	0.156	0.160	0.162	0.163
Bank 27	0.206	0.207	0.004	0.225	0.189	0.196	0.199	0.201
Bank 28	0.090	0.102	0.002	0.112	0.094	0.097	0.099	0.100

Source: Authors.

Note: VaR=value at risk. The table uses 2,000 simulated capital ratios and assumes that the Government of Brazil does not default on its domestic debt. Reported VaR capital ratio levels indicate the percentage of time that simulated values fall below a certain threshold. The assumed recovery rate on defaulted private sector loans is 45 percent in these simulations.

loans, this practice has well-known nocive effects on the economy, which might eventually spill over into the banking system. It is also possible that the government may force debt holders to change their contracts for something that has a lower market value than the original ones. In any instance, even if the government does not fully default on its domestic debt, the banks ultimately may incur market value losses. To be able to model this explicitly is not an easy task. No one knows for sure which set of actions will be taken by the government during such events. There are innumerable possible outcomes for banks, each of which will affect banks' portfolios differently. Given this limitation, we propose to construct a matrix of some potential government default implications that will provoke additional losses on banks' portfolios, through two different channels. The first channel incorporates losses directly on the government securities, by assuming that banks may lose 0 percent, 10 percent, or 25 percent of the market value of their government securities. Such losses may result from increases in required market interest rates and interest rate spreads, because the government may defer payment of the debt or may force banks to change the terms of the debt in ways that reduce its value.

In addition, government default events usually are associated with major disruptions in the whole economy, affecting all sectors, and banks' borrowers may become incapable of repaying some of their debts. We conjecture that these events will affect firms with different creditworthiness differently. That is, we assume that firms with higher credit quality are better prepared to handle these crisis events. The way we capture the differential impact of a sovereign default is by imposing an increase on the default rate on private sector loans in different credit categories. We assume three different scenarios: (1) businesses and individuals have a zero increment to their default rates; (2) businesses and individuals in each credit risk category have an increase in their default rates equal to the average default of that credit risk category; and (3) businesses and individuals in each credit risk category have an increase in their default rates equal to two times the average default of that credit risk category.

The combination of all these possible outcomes leads to nine potential government and private sector incremental loan loss scenarios banks may face in the event of a government default. Although these scenarios are far from exhausting the innumerable possibilities, they do provide a reasonable

TABLE 8.5

Simulated Capital Ratios: Individual Banks, No Government Default

Bank	Losses on Government Loans and Incremental Defaults on Business and Consumer Loans								
	0%	0% + 1 times the average historical default rates	0% + 2 times the average historical default rates	10%	10% + 1 times the average historical default rates	10% + 2 times the average historical default rates	25%	25% + 1 times the average historical default rates	25% + 2 times the average historical default rates
Bank 1	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 2	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 3	0.000	0.000	0.000	0.001	0.001	0.001	0.046	0.046	0.046
Bank 4	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.002
Bank 5	0.000	0.000	0.001	0.038	0.045	0.048	0.048	0.048	0.048
Bank 6	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 7	0.000	0.000	0.000	0.047	0.047	0.047	0.047	0.047	0.047
Bank 8	0.000	0.000	0.000	0.000	0.000	0.000	0.034	0.038	0.041
Bank 9	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 10	0.000	0.000	0.000	0.005	0.007	0.007	0.060	0.060	0.060
Bank 11	0.000	0.000	0.000	0.000	0.000	0.000	0.045	0.045	0.045
Bank 12	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 13	0.000	0.000	0.000	0.000	0.000	0.000	0.003	0.004	0.005
Bank 14	0.000	0.000	0.000	0.050	0.050	0.050	0.050	0.050	0.050
Bank 15	0.000	0.000	0.000	0.000	0.000	0.000	0.045	0.045	0.045
Bank 16	0.000	0.000	0.000	0.041	0.041	0.041	0.041	0.041	0.041
Bank 17	0.000	0.000	0.000	0.000	0.000	0.000	0.034	0.035	0.037
Bank 18	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 19	0.000	0.000	0.000	0.000	0.000	0.000	0.049	0.049	0.049
Bank 20	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 21	0.000	0.000	0.000	0.050	0.050	0.050	0.050	0.050	0.050
Bank 22	0.000	0.000	0.000	0.050	0.050	0.050	0.050	0.050	0.050
Bank 23	0.000	0.000	0.000	0.000	0.000	0.000	0.051	0.051	0.051
Bank 24	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 25	0.000	0.000	0.000	0.000	0.000	0.000	0.048	0.048	0.048
Bank 26	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.001
Bank 27	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 28	0.000	0.000	0.000	0.000	0.000	0.000	0.046	0.046	0.046

Source: Authors.

Note: This table provides statistics and VaR values at different percentage levels, using 2,000 simulated capital ratios, for each of the 28 banks in our Simulation sample, assuming that the GOB does not default on its domestic debt. Reported VaR capital ratio levels indicate the percentage of time that simulated values fall below a certain threshold. For example, Bank 1 VaR at 99 percent level is 0.191, indicating that 1 percent of the time simulated capital ratios for Bank 1 fell below 0.191. The assumed recovery rate on defaulted private sector loans is 45 percent in these simulations.

range for incremental bank loan losses. As indicated earlier, in cases where we assess systemic banking system risk, we will evaluate these nine potential loan loss scenarios for two assumed private sector defaulted loan recovery rates (15 percent and 45 percent).

In Tables 8.5 to 8.7, we report the average simulated bank default rate, the associated average cost to bring banks' capital ratio back to a 0.08 level (estimated only for the times when the bank defaults), and the 99 percent VaR capital level (banks have 1 percent probability of having their capital ratios fall below this level). Given that banks hold a significant amount of government securities, bank default rates start to appear significant only when there is some degree of losses on government securities. For example, if banks lose an average of 10 percent in the market value of their government debt, then 6 banks (e.g., Bank 5 and Bank 16) will have a default rate of around 4 percent to 5 percent. If banks lose 25 percent of the market value of their government securities, then 15 banks (e.g., Bank 3, Bank 8, Bank 10) will

have a default rate of around 4 percent to 6 percent. In either of these cases (i.e., 10 percent or 25 percent loss rates), the bank failures would result from a sovereign default and thus would be highly correlated and could precipitate a systemic banking system problem. Interestingly, some banks will not default at all (e.g., Bank 1 and Bank 2), even in the worst case scenario that we analyze in this chapter (losing 25 percent of the market value of government securities and suffering incremental defaults on business and individual loans equal to twice their average historical default rates). Either these banks are highly capitalized, and thus capable of absorbing larger losses on their government securities, or they have a balance between government, business, and consumer loans that allow them to diversify the risk of a potential government default, along with a high net interest margin stemming from large interest rate spreads on business and consumer loans. The average cost (as percentage of total assets) to bring the banks back to a 0.08 capital level whenever they default is usually high. For example, if the GOB de-

TABLE 8.6

Simulated Average “Bailout” Cost: Individual Banks, Government Default

Losses on Government Loans and Incremental Defaults on Business and Consumer Loans									
Bank	0%	0% + 1 times the average historical default rates	0% + 2 times the average historical default rates	10%	10% + 1 times the average historical default rates	10% + 2 times the average historical default rates	25%	25% + 1 times the average historical default rates	25% + 2 times the average historical default rates
Bank 1	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 2	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 3	0.000	0.000	0.000	0.051	0.054	0.056	0.102	0.105	0.108
Bank 4	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.052
Bank 5	0.000	0.000	0.053	0.064	0.068	0.074	0.163	0.170	0.178
Bank 6	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 7	0.000	0.000	0.000	0.077	0.078	0.080	0.197	0.198	0.200
Bank 8	0.000	0.000	0.000	0.000	0.000	0.000	0.059	0.060	0.062
Bank 9	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 10	0.000	0.000	0.000	0.059	0.059	0.061	0.150	0.153	0.156
Bank 11	0.000	0.000	0.000	0.000	0.000	0.000	0.073	0.076	0.076
Bank 12	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 13	0.000	0.000	0.000	0.000	0.000	0.000	0.055	0.055	0.055
Bank 14	0.000	0.000	0.000	0.061	0.061	0.061	0.173	0.173	0.173
Bank 15	0.000	0.000	0.000	0.000	0.000	0.000	0.158	0.158	0.158
Bank 16	0.000	0.000	0.000	0.084	0.085	0.086	0.233	0.234	0.234
Bank 17	0.000	0.000	0.000	0.000	0.000	0.000	0.060	0.062	0.062
Bank 18	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 19	0.000	0.000	0.000	0.000	0.000	0.000	0.083	0.083	0.084
Bank 20	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 21	0.000	0.000	0.000	0.088	0.091	0.094	0.239	0.243	0.248
Bank 22	0.000	0.000	0.000	0.095	0.095	0.095	0.278	0.278	0.278
Bank 23	0.000	0.000	0.000	0.000	0.000	0.000	0.085	0.086	0.087
Bank 24	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 25	0.000	0.000	0.000	0.000	0.000	0.000	0.065	0.068	0.070
Bank 26	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.050
Bank 27	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bank 28	0.000	0.000	0.000	0.000	0.000	0.000	0.168	0.168	0.168

Source: Authors.

Note: Average “bailout” cost is the average capital (as percentage of total assets) necessary to bring banks’ capital ratio back to the 0.08 level, whenever they fall below 0.03 (assumed default). We assume various different scenarios. On the business and individuals loans, we assume that (1) businesses and individuals have a zero increment to their default rates; (2) businesses and individuals in each credit risk category have an increase in their default rates equal to the average default of that credit risk category; and (3) businesses and individuals in each credit risk category have an increase in their default rates equal to two times the average default of that credit risk category. The second channel incorporates losses directly on the government securities, by assuming that banks may lose 0 percent, 10 percent, or 25 percent of the market value of their government securities for a variety of reasons. The assumed recovery rate on defaulted private sector loans is 45 percent in these simulations.

faults and the banks lose 25 percent of the market value of their government securities, this “bailout” cost ranges from 0.055 of total assets (Bank 13) to 0.278 (Bank 22). Again, the large amount of government securities held by Brazilian banks make them quite vulnerable in the event of a sovereign default resulting in very high potential bank bailout costs. The simulated 99 percent VaR capital ratios just reinforce the size of the losses these banks could face in the event of government default. If banks lose 25 percent of the market value of government securities, capital ratios may range from 0.248 (Bank 6) to –0.199 (Bank 22).

While having the previously noted limitations, this approach to modeling correlated market, credit, and sovereign risk highlights the danger of the exposure of many Brazilian banks to very high levels of GOB securities. It is true that Brazil has been implementing more responsible fiscal and monetary policies, controlling inflation, and obtaining im-

portant positive balance in exports and imports, among other positive indicators. However, Brazil is still an emerging economy, vulnerable to flow of capitals, with a huge stock of debt.³⁴ Thus, government securities are not free of risk, and the correlated risk of government default should be accounted for.

C. Rating Brazilian banks

We will now compare the simulated results obtained in the previous section with rating agency ratings of Brazilian banks’ short-term debt instruments. We will focus our attention on a single output derived from the simulations, including the risk of a sovereign default. In particular, we will look at the 99 percent confidence level capital ratio (99 percent VaR level)

³⁴ Total debt to GDP is around 55 percent, as of December 2004 (Government Financial Statistics, IMF).

TABLE 8.7

Simulated 99 Percent VaR Capital Ratio: Individual Banks, Government Default

Losses on Government Loans and Incremental Defaults on Business and Consumer Loans									
Bank	0%	0% + 1 times the average historical default rates	0% + 2 times the average historical default rates	10%	10% + 1 times the average historical default rates	10% + 2 times the average historical default rates	25%	25% + 1 times the average historical default rates	25% + 2 times the average historical default rates
Bank 1	0.189	0.189	0.189	0.175	0.174	0.173	0.126	0.124	0.122
Bank 2	0.129	0.129	0.129	0.115	0.114	0.113	0.051	0.050	0.050
Bank 3	0.085	0.085	0.084	0.051	0.048	0.045	-0.026	-0.029	-0.032
Bank 4	0.152	0.151	0.151	0.128	0.125	0.122	0.058	0.055	0.052
Bank 5	0.058	0.056	0.053	0.011	0.005	-0.002	-0.092	-0.100	-0.108
Bank 6	0.365	0.364	0.364	0.335	0.334	0.333	0.248	0.247	0.245
Bank 7	0.065	0.064	0.063	-0.001	-0.002	-0.003	-0.121	-0.123	-0.124
Bank 8	0.103	0.103	0.102	0.084	0.083	0.080	0.018	0.016	0.013
Bank 9	0.213	0.213	0.213	0.210	0.208	0.206	0.190	0.188	0.186
Bank 10	0.077	0.077	0.077	0.040	0.038	0.035	-0.081	-0.085	-0.088
Bank 11	0.102	0.102	0.102	0.077	0.077	0.077	0.003	0.003	0.003
Bank 12	0.238	0.238	0.238	0.213	0.211	0.210	0.134	0.131	0.129
Bank 13	0.132	0.132	0.132	0.108	0.107	0.106	0.038	0.036	0.035
Bank 14	0.080	0.080	0.080	0.018	0.018	0.018	-0.094	-0.094	-0.094
Bank 15	0.147	0.147	0.147	0.086	0.086	0.086	-0.080	-0.080	-0.080
Bank 16	0.067	0.067	0.067	-0.006	-0.007	-0.008	-0.156	-0.157	-0.158
Bank 17	0.094	0.094	0.094	0.081	0.080	0.079	0.017	0.015	0.014
Bank 18	0.225	0.225	0.225	0.208	0.207	0.206	0.137	0.134	0.132
Bank 19	0.196	0.196	0.196	0.135	0.135	0.134	-0.005	-0.006	-0.006
Bank 20	0.155	0.154	0.154	0.139	0.137	0.134	0.087	0.084	0.081
Bank 21	0.061	0.061	0.060	-0.012	-0.015	-0.019	-0.164	-0.169	-0.173
Bank 22	0.076	0.076	0.076	-0.016	-0.016	-0.016	-0.199	-0.199	-0.199
Bank 23	0.110	0.110	0.110	0.080	0.079	0.079	-0.011	-0.012	-0.013
Bank 24	0.140	0.139	0.139	0.136	0.134	0.133	0.113	0.110	0.107
Bank 25	0.095	0.095	0.095	0.073	0.071	0.069	0.010	0.007	0.005
Bank 26	0.161	0.161	0.161	0.132	0.131	0.130	0.046	0.045	0.043
Bank 27	0.196	0.196	0.196	0.177	0.175	0.173	0.112	0.110	0.107
Bank 28	0.092	0.092	0.092	0.040	0.040	0.040	-0.095	-0.095	-0.095

Source: Authors.

Note: The 99 percent value-at-risk (VaR) simulated capital ratio is the threshold below which banks' capital ratio will fall 1 percent of the time. We assume various different scenarios. On the business and individuals loans, we assume that (1) businesses and individuals have a zero increment to their default rates; (2) businesses and individuals in each credit risk category have an increase in their default rates equal to the average default of that credit risk category; and (3) businesses and individuals in each credit risk category have an increase in their default rates equal to two times the average default of that credit risk category. The second channel incorporates losses directly on the government securities, by assuming that banks may lose 0 percent, 10 percent, or 25 percent of the market value of their government securities for a variety of reasons. The assumed recovery rate on defaulted private sector loans is 45 percent in these simulations.

for the scenario where in the event of a sovereign default banks lose 10 percent of the market value of their government securities, experience an incremental increase in defaults on their private sector loans equal to twice the average default rate, and have a 45 percent average recovery rate on defaulted private sector loans. This measure embeds both the default probability and the size of associated monetary loss and is consistent with what the rating agencies also utilize for rating banks, businesses, and sovereigns. For this purpose, we will divide our sample of banks into three groups, according to their 99 percent VaR capital level. Group 1 will be composed of banks with 99 percent VaR capital ratios less than 0.07, Group 2 with banks with 99 percent VaR capital ratios between 0.07 and 0.13, and Group 3 with 99 percent VaR capital ratios above 0.13. Given that the 99 percent VaR sets the threshold below which banks' capital ratio will fall 1 percent of the time, the lower the 99 percent VaR level, the

closer the bank is to the assumed critical 0.03 capital ratio level and the higher the probability of default (thus the riskier the bank).

Our ratings are generally consistent with Moody's and Standard & Poor's. Out of 16 banks also rated by Moody's, we had a similar rating on 12 banks. Out of 10 banks also rated by Standard & Poor's, we had a similar rating on 7 banks. However, out of 18 banks also rated by Fitch, we had a similar rating on only 9 banks.

D. Systemic banking system risk

For assessing systemic banking system risk, we will consider two cases. First, we aggregate the 28 banks into one single bank. Second, we aggregate banks according to their risk rating, in the three groups described earlier. Aggregated balance sheet accounts were obtained by simple addition from all banks in

TABLE 8.8

Simulated Default Probabilities, Bailout Costs, and 99 Percent Capital Ratios: Government Default Risk Considered, All 28 Banks Modeled as a Single Bank

Losses on Government Loans and Incremental Defaults on Business and Consumer Loans									
		0% + 1 times the average historical default rates	0% + 2 times the average historical default rates	10%	10% + 1 times the average historical default rates	10% + 2 times the average historical default rates	25%	25% + 1 times the average historical default rates	25% + 2 times the average historical default rates
All 28 Banks	0%			10%			25%		
45 percent recovery rate on business and customers' loans									
Default probabilities	0.000	0.000	0.000	0.000	0.000	0.000	0.017	0.022	0.030
Bailout cost	0.000	0.000	0.000	0.000	0.000	0.000	0.055	0.057	0.057
99% VaR level	0.143	0.143	0.143	0.110	0.108	0.105	0.027	0.024	0.021
15 percent recovery rate on business and customers' loans									
Default probabilities	0.000	0.000	0.000	0.000	0.000	0.000	0.025	0.033	0.036
Bailout cost	0.000	0.000	0.000	0.000	0.000	0.000	0.058	0.059	0.061
99% VaR level	0.142	0.141	0.141	0.106	0.103	0.101	0.022	0.019	0.016

Source: Authors.

Note: This table presents simulated default probabilities on groups of banks; average "bailout cost as the average capital (as percentage of total assets) necessary to bring banks' capital ratio back to 0.08 level, whenever they fall below 0.03 (assumed default); and the 99 percent value-at-risk (VaR) simulated capital ratio as the threshold below which banks' capital ratio will fall 1 percent of the time. We assume various different scenarios. On the business and individuals loans, we assume that (1) businesses and individuals have a zero increment to their default rates; (2) businesses and individuals in each credit risk category have an increase in their default rates equal to the average default of that credit risk category; and (3) businesses and individuals in each credit risk category have an increase in their default rates equal to two times the average default of that credit risk category. The second channel incorporates losses directly on the government loans, by assuming that banks may lose 0 percent, 10 percent, or 25 percent of the market value of their government loans for a variety of reasons.

each group, whereas loan credit quality distributions and assets and liability maturity structures were obtained by a weighted average (size of each category relative to total assets, in each bank).

One Single Aggregate Bank

The simulation results for a single aggregate bank are consistent with what we have obtained for individual banks. When we model correlated market and private sector loan credit risk but do not consider government default, simulated capital ratios are comfortably above an assumed 0.08 target capital level, with small standard deviation (0.003). This result is intuitive given the substantial amount of government securities that are collectively held by the 28 banks in our simulation sample. Because the single aggregate bank has a 0.154 initial capital ratio, it does not face solvency problems when a sovereign default imposes market value losses on government securities of 10 percent. However, Table 8.8 indicates that at a 25 percent loss rate on government securities, the default rate on the single aggregate bank may reach 3 percent (3.6 percent) when the assumed average recovery rate on private sector loans is 45 percent (15 percent). The associated cost to bring the single bank's capital ratio back to a 0.08 level averages 5.7 percent (6.1 percent) of the bank's total assets. The 99 percent VaR capital ratio also deteriorates substantially. Under the no-government-default assumption, it is 0.147, while it drops to 0.021 (0.016) under the above sovereign default scenario when the average recovery rates on defaulted private sector loans are 45 percent (15 percent).

Although used frequently, modeling systemic risk via a single aggregate bank has significant limitations. First, it masks the fact that under certain conditions (e.g., 10 percent loss rate on government securities), a number of individual banks could fail simultaneously.³⁵ Second, this approach does not account for interbank exposures that can trigger sequential failures. Still, the simulated results do show the importance of modeling correlated sovereign risk not only at an individual bank level but also, and perhaps most important, at a systemic level.

Multiple Aggregate Banks

So far we have simulated the banks individually and taken account of correlated market, credit, and sovereign risk. However, we have not taken into account one important channel for propagating a systemic crisis, through interbank credit exposures. For this purpose, we will simulate the three aggregated banks (Groups 1, 2, and 3) simultaneously, under the same financial and economic environment.³⁶ Then we will examine the simulated capital ratios for all three banks to determine whether one or more banks fail. In that event, we will then calculate the default impacts on other banks as the failed banks become incapable of repaying their interbank debts. This exercise will be explained in more detail in

³⁵ Also, we are simulating only 28 Brazilian banks (70 percent of financial system total assets). Including more banks would certainly push the cost for bailing the financial system out to a higher level.

³⁶ The choice of using three banks as opposed to all 28 banks was made primarily because of memory limitations in the ValueCalc software.

TABLE 8.9

Simulated Default Probabilities, Average “Bailout” Cost, and 99 Percent Capital Ratios: Individual Aggregate Banks, Government Default

Aggregate Banks	Losses on Government Loans and Incremental Defaults on Business and Consumer Loans								
	0%	0% + 1 times the average historical default rates	0% + 2 times the average historical default rates	10%	10% + 1 times the average historical default rates	10% + 2 times the average historical default rates	25%	25% + 1 times the average historical default rates	25% + 2 times the average historical default rates
Recovery rate on defaulted private sector loans = 45%									
Default probabilities									
Group 1	0.000	0.000	0.000	0.001	0.001	0.001	0.048	0.048	0.048
Group 2	0.000	0.000	0.000	0.000	0.000	0.000	0.001	0.004	0.018
Group 3	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bailout cost									
Group 1	0.000	0.000	0.000	0.054	0.055	0.055	0.122	0.123	0.124
Group 2	0.000	0.000	0.000	0.000	0.000	0.000	0.051	0.054	0.057
Group 3	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
99% VaR level									
Group 1	0.087	0.087	0.086	0.046	0.046	0.045	−0.048	−0.049	−0.049
Group 2	0.132	0.130	0.126	0.109	0.103	0.096	0.045	0.035	0.025
Group 3	0.211	0.211	0.211	0.186	0.184	0.183	0.112	0.110	0.108
Recovery rate on defaulted private sector loans = 15%									
Default probabilities									
Group 1	0.000	0.000	0.000	0.002	0.002	0.002	0.048	0.048	0.048
Group 2	0.000	0.000	0.000	0.000	0.000	0.000	0.007	0.017	0.034
Group 3	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Bailout cost									
Group 1	0.000	0.000	0.000	0.058	0.057	0.057	0.128	0.129	0.130
Group 2	0.000	0.000	0.000	0.000	0.000	0.000	0.057	0.060	0.063
Group 3	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
99% VaR level									
Group 1	0.081	0.081	0.081	0.040	0.039	0.038	−0.055	−0.056	−0.057
Group 2	0.125	0.123	0.119	0.103	0.095	0.086	0.034	0.024	0.013
Group 3	0.208	0.208	0.207	0.182	0.180	0.178	0.106	0.105	0.103

Source: Authors.

Note: This table presents simulated default probabilities on groups of banks; average “bailout” cost as the average capital (as percentage of total assets) necessary to bring banks’ capital ratio back to 0.08 level, whenever they fall below 0.03 (assumed default); and the 99 percent value-at-risk (VaR) simulated capital ratio as the threshold below which banks’ capital ratio will fall 1 percent of the time. We assume various different scenarios. On the business and individuals loans, we assume that (1) businesses and individuals have a zero increment to their default rates; (2) businesses and individuals in each credit risk category have an increase in their default rates equal to the average default of that credit risk category; and (3) businesses and individuals in each credit risk category have an increase in their default rates equal to two times the average default of that credit risk category. The second channel incorporates losses directly on the government securities, by assuming that banks may lose 0 percent, 10 percent, or 25 percent of the market value of their government securities for a variety of reasons.

the subsection below on *simultaneous aggregate banks*. Before we get there, however, it is useful to describe the groups separately and to simulate them individually, to have a more precise understanding of their default risk.

Individual Aggregate Banks. Simulated capital ratios, without sovereign risk, show that all three groups of banks are profitable and on average have increasing capital ratios.³⁷ When sovereign risk is considered (Table 8.9), then only Group 3 survives all our simulated scenarios (same as for the individual banks). Group 1 has a small nonzero default prob-

ability at a 10 percent loss rate on government securities. Group 2 also has a nonzero default rate when the assumed losses in the market value of government debt increases from 10 percent to 25 percent. Under the scenario of 25 percent average losses on government securities and a 45 percent average recovery rate on defaulted private sector loans, Group 1 defaults at an average rate of 0.048, and Group 2 defaults at rates in the range of 0.1 percent to 1.8 percent. The average cost of “bailing out” these banks is large: for Group 1, it surpasses 12 percent of total assets required to bring its capital ratio back to a 0.08 level. For Group 2, the average bailout cost is around 5 to 6 percent of its total assets. Under the assumption of a lower 15 percent recovery rate on defaulted

³⁷ For more detail on these simulations, see Barnhill and Souto (2008b).

private sector loans, Group 1's and Group 2's default probabilities and bailout costs increase.

Another important risk measure, the 99 percent VaR capital level, provides an even more distinct picture. All banks have their 99 percent VaR capital level deteriorate when a sovereign default results in market value losses on government loan. If such losses reach 25 percent of market value, then the 99 percent VaR capital levels for Groups 1, 2, and 3 fall to the range of -0.049 (-0.056), 0.035 (0.024), and 0.11 (0.105), respectively, for the 45 percent (15 percent) assumed recovery rates on defaulted private sector loans.

Simultaneous Aggregate Banks. For assessing systemic risk, we simulate correlated market, credit, and sovereign risk for the three groups of aggregate banks simultaneously, under the same financial and economic environment. Then, in a second analytical step, we introduce the interbank propagation channel by adjusting each group's simulated capital ratio whenever one of the other two groups' simulated capital ratios fall below 0.03, using the information on interbank lending and assuming that the groups borrow money from others proportionately to their total assets. For example, Group 1 lends 3.6 percent of its total assets or equivalently R\$5,503.60 million. Given that Group 2 has total assets of R\$551,250.10 million, and the sum of Group 2 and Group 3 total assets is R\$878,571.10 million, we assume that $(R\$551,250.10/R\$878,571.10 =) 0.63$ of Group 1's total interbank lending is lent to Group 2. The remaining $(1 - 0.63 =) 0.37$ is lent to Group 3.

For estimating the impact of interbank lending, when banks default, we assume interbank lending to be risk free in the first step of the simulation. Subsequently, if a bank fails, then we assume that the other banks will recover 50 percent of the value of the defaulted interbank loans. Obviously, higher recovery rates would certainly diminish the impact of interbank default. We then recalculate simulated total assets and simulated capital ratios, after deducting the defaulted amounts that are not recovered. With new simulated capital ratios, we replicate the same VaR analysis and also estimate default rates and the monetary cost to bring each group's capital ratios back to a 0.08 capital level. We also estimate the default rates and monetary cost when two banks and three banks default simultaneously. These results are presented in Tables 8.10 and 8.11, which assume a 45 percent and 15 percent recovery rate on defaulted private sector loans, respectively. A few comments are in order.

First, in Table 8.10, as expected, the interbank propagation channel moves from the riskier groups to the less risky groups. Second, because Group 3 is very well capitalized and has the highest credit profile, it defaults only when incurring a much bigger loss on the government securities. Group 3 starts having solvency problems only when a sovereign default imposes a high average loss of 40 percent or more on the market value of government securities. This reveals another novice facet of banks holding exceedingly large

TABLE 8.10

Systemic Risk: Simultaneously Simulated Aggregate Banks' Recovery Rate on Defaulted Private Sector Loans = 45 Percent

Losses on Government Loans	Incremental Defaults on Business and Consumers' Loans		
	+1 times the average historical default rates	+2 times the average historical default rates	
Probability of Groups 1 and 2 defaulting at the same time (and associated cost given default as a proportion of total assets, in parentheses), to bring both banks' capital ratios to 0.08			
00%	0.000	0.000	0.000
10%	0.000	0.000	0.000
25%	0.016 (0.109)	0.021 (0.110)	0.030 (0.111)
40%	0.048 (0.239)	0.048 (0.251)	0.048 (0.258)
50%	0.048 (0.358)	0.048 (0.362)	0.048 (0.366)
Probability of all groups defaulting at the same time (and associated cost given default as a proportion of total assets, in parentheses), to bring all banks' capital ratios to 0.08			
0%	0.000	0.000	0.000
10%	0.000	0.000	0.000
25%	0.000	0.000	0.000
40%	0.048 (0.168)	0.048 (0.177)	0.048 (0.182)
50%	0.048 (0.271)	0.048 (0.275)	0.048 (0.279)

Source: Authors.

Note: This table presents simultaneously simulated default probabilities on groups of banks; average "bailout" cost as the average capital (as percentage of total assets) necessary to bring banks' capital ratio back to 0.08 level, whenever they fall below 0.03 (assumed default); and the 99 percent value-at-risk simulated capital ratio as the threshold below which banks' capital ratio will fall 1 percent of the time.

concentrations of government securities. In the event of a sovereign default, the government has constrained debt management alternatives. Should the government take actions that result in a heavy loss on the market value of government securities, then it may trigger a systemic risk default in the financial system. In the particular case of GOB, our analysis suggests that losses on government securities of 10 percent could create significant solvency problems for 5 or 6 out of 28 banks analyzed. Higher loss rates would, of course, create even larger systemic banking system risks. Given that the amount of GOB securities in Brazilian balance sheets is much larger than the amount of business and consumer loans, the impact of incremental defaults on businesses and households has much smaller effects on banks' default rates and monetary losses. Third, when we integrate the interbank risk into our framework, systemic default rates (i.e., risk of multiple bank failures in the same time period) are smaller compared with the individual banks' default rates. For example, at a 10 percent loss rate on government securities, there is a zero probability of both Group 1 and Group 2 defaulting

TABLE 8.11

Systemic Risk: Simultaneously Simulated Aggregate Banks' Recovery Rate on Defaulted Private Sector Loans = 15 Percent

Incremental Defaults on Business and Consumers' Loans			
Losses on Government Loans	+1 times the average historical default rates	+2 times the average historical default rates	
Probability of Groups 1 and 2 defaulting at the same time (and associated cost given default as a proportion of total assets, in parentheses), to bring both banks' capital ratios to 0.08			
0%	0.000	0.000	0.000
10%	0.000	0.000	0.000
25%	0.034 (0.114)	0.036 (0.117)	0.039 (0.119)
40%	0.048 (0.259)	0.048 (0.267)	0.048 (0.274)
50%	0.048 (0.369)	0.048 (0.373)	0.048 (0.378)
Probability of all groups defaulting at the same time (and associated cost given default as a proportion of total assets, in parentheses), to bring all banks' capital ratios to 0.08.			
0%	0.000	0.000	0.000
10%	0.000	0.000	0.000
25%	0.000	0.000	0.000
40%	0.048 (0.183)	0.048 (0.189)	0.048 (0.195)
50%	0.048 (0.280)	0.048 (0.284)	0.048 (0.289)

Source: Authors.

Note: This table presents simultaneously simulated default probabilities on groups of banks; average "bailout" cost as the average capital (as percentage of total assets) necessary to bring banks' capital ratio back to 0.08 level, whenever they fall below 0.03 (assumed default); and the 99 percent value-at-risk simulated capital ratio as the threshold below which banks' capital ratio will fall 1 percent of the time.

simultaneously. At a 25 percent loss rate on government securities, the probability of both Group 1 and Group 2 defaulting at the same time reaches 3.0 percent (3.9 percent) for average recovery rates on private sector loans of 45 percent (15 percent). However, when the systemic event takes place, it is a lot more costly (as a percentage of the assets of the defaulting banks) than previously assessed when simulating all 28 banks as a single bank. We have estimated the average cost to "bail out" the single bank to be 5.5 percent of its total capital under the scenario of a 25 percent average loss on government securities. Under the same scenario, the two riskier groups default at a rate in the range of 1.6 percent to 3.9 percent, with the average cost to bring their capital level back to the 0.08 level equal to over 10 percent of their total assets. A comparison of Tables 8.9 and 8.10, which assume the higher (45 percent) and lower (15 percent) recovery rates on defaulted private sector loans, illustrates the potential significance of this variable on systemic risk levels.

3. CONCLUSION

We have simulated 28 individual Brazilian banks considering two different scenarios. In one scenario, we access correlated market and business loan credit risk but assume that the Brazilian government will never default on its debt obligations. With few exceptions, Brazilian banks perform very well in these simulations. In a second set of simulations, we model correlated market, credit, and sovereign risk. At a 10 percent loss rate on government securities in the event of a sovereign default, 6 of the 28 banks could fail. At a 25 percent loss rate on government securities, over half of the 28 banks could fail. Banks with more diversified portfolios of government, business, and consumer loans are found to be more profitable on average and to also perform better in the case of a potential government default. These results illustrate the well-known risk of concentrated lending to an entity with a nonzero default rate (e.g., GOB). It also reveals another nocive facet of banks holding exceedingly large concentrations of risky government securities. In the event of a sovereign default, the government has constrained debt management alternatives. Should the government take actions that significantly reduce the market value of government securities, it could trigger a systemic banking system failure. This suggests that the development of global markets for sovereign debt denominated in local currency could have the dual benefits of allowing sovereigns to diversify their borrowing sources and banks to diversify their loan portfolios and default risk.

Our simulations also provide a way of grouping the banks, based on their creditworthiness, under an integrated risk framework. We find that our bank risk categorization is generally consistent with Moody's and Standard & Poor's ratings but less so when compared with Fitch ratings.

We utilize two different approaches to measure systemic risk. First, we aggregate all the banks in one single bank and simulate it individually, with the possibility of government default. Results for this approach show that the financial system might be dragged down by a government default only when the average loss on government securities is 25 percent or higher and that the cost to bail out the system (to bring the average capital ratio to a 0.08 level) is about 5.7 percent of the total assets, under this scenario. In the second approach, we simulate three aggregated groups of banks simultaneously, under the same financial and economic environment. Our results show that if a bank has heavy interbank credit exposure as compared with its initial capital, then correlated interbank default losses may become "the straw that breaks the camel's back." We are also able to estimate the very substantial cost of recapitalizing the banking system in the event of a systemic banking system failure. In the particular case of GOB, in the event of government default, losses of 10 percent or higher on government securities could have a significant systemic impact on the banking system. We also find that in extreme events (sovereign defaults), should loan recovery rates on private sector loans be system-

atically depressed, then banking system systemic risk also increases significantly.

Modeling correlated market, credit, sovereign, and inter-bank risk is a challenging task. All methodologies which might be used have limitations. In spite of the limitations we have identified, we believe that the portfolio simulation methodology we present in this chapter is able to deal with all of these correlated risks and has the potential to provide important insights into individual bank and systemic banking system risk levels. To the best of our knowledge, no one else has put forward a systematic methodology for assessing *correlated* market and credit (sovereign, private sector, and interbank) default risk for a financial system. Provided that the required input information is available, we believe that the proposed models can provide useful risk assessment and management information to bank management, bank regulators, sovereigns, rating agencies, and investors.

REFERENCES

- Acharya, Viral V., 2009, "A Theory of Systemic Risk and Design of Prudential Bank Regulation," *Journal of Financial Stability*, Vol. 5, No. 3, pp. 224–55.
- Barnhill, Theodore, 2006, "Modeling and Managing Sovereign Risk," Working Paper (Washington: George Washington University Global and Entrepreneurial Finance Research Institute).
- Barnhill, Theodore, and George Kopits, 2004, "Assessing Fiscal Sustainability under Uncertainty," *Journal of Risk*, Vol. 6, No. 4, pp. 31–53.
- Barnhill, Theodore, and William F. Maxwell, 2002, "Modeling Correlated Interest Rate, Exchange Rate, and Credit Risk in Fixed Income Portfolios," *Journal of Banking and Finance*, Vol. 26, Nos. 2–3, pp. 347–74.
- Barnhill, Theodore, Panagiotis Papapanagiotou, and Liliana Schumacher, 2003, "Measuring Integrated Market and Credit Risk in Bank Portfolios: An Application to a Set of Hypothetical Banks Operating in South Africa," *Financial Markets, Institutions, and Instruments*, Vol. 11, No. 5, pp. 401–43.
- Barnhill, Theodore, Panagiotis Papapanagiotou, and Marcos R. Souto, 2004, "Preemptive Strategies for the Assessment and Management of Financial System Risk Levels: An Application to Japan with Implications to Emerging Economies," *Review of Pacific-Basin Financial Markets and Policies*, Vol. 7, No. 1, pp. 1–42.
- Barnhill, Theodore, and Marcos R. Souto, 2008a, "Stochastic Volatilities and Correlations, Extreme Values and Modeling the Financial and Economic Environment under Which Brazilian Banks Operate," IMF Working Paper 07/290 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=21502>
- , 2008b, "Systemic Bank Risk in Brazil: An Assessment of Correlated Market, Credit, Sovereign, and Inter-Bank Risk in an Environment with Stochastic Volatilities and Correlations," Working Paper (Washington: George Washington University Global and Entrepreneurial Finance Research Institute).
- , 2009, "Systemic Bank Risk in Brazil: A Comprehensive Simulation of Correlated Market, Credit, Sovereign and Inter-Bank Risks," *Financial Markets, Institutions, and Instruments*, Vol. 18, No. 4, pp. 243–83.
- Barnhill, Theodore, Marcos R. Souto, and Benjamin Tabak, 2003, "Modeling Business Loan Credit Risk in Brazil," *Financial Stability Report*, Vol. 2, No. 1, pp. 159–74. Available via the Internet: <http://www.bcb.gov.br/?FSR200305>
- , 2006, "An Analysis of Off-Site Supervision of Banks' Profitability, Risk and Capital Adequacy: A Portfolio Simulation Approach Applied to Brazilian Banks," *Financial Stability Report*, Vol. 5, No. 1, pp. 209–36 (Brasilia: Banco Central do Brasil). Available via the Internet: <http://www.bcb.gov.br/?FSR200605>
- Basel Committee on Banking Supervision, 1988, "International Convergence of Capital Measurement and Capital Standards" (Basel, July). Available via the Internet: <http://www.bis.org/publ/bcb04a.htm>
- , 1996, "Amendment to the Capital Accord to Incorporate Market Risks" (Basel, January). Available via the Internet: <http://www.bis.org/publ/bcb24.pdf>
- , 2001, "The New Basel Capital Accord," Second Consultative Paper (Basel, January). Available via the Internet: <http://www.bis.org/bcb/bcb24.htm>
- Cavanaugh, Marie, 2009, "Sovereign Rating and Country T&C Assessment Histories," Ratings Direct (New York: Standard and Poor's).
- Eccles, Peter, 2005, "New Brazil Bankruptcy Law Likely to Improve Recovery Prospects for Creditors, but Challenges Remain," S&P Structured Finance (New York: Standard and Poor's).
- Elsinger, Helmut, Alfred Lehar, and Martin Summer, 2003, "Risk Assessment for Banking Systems," Working Paper (Vienna: University of Vienna).
- FinSoft, Inc., 1996–2005, "ValueCalc 3.0: Global Portfolio Risk Management Software" (Annandale, Virginia: FinSoft, Inc.).
- Gorton, Gary, 1988, "Banking Panics and Business Cycles," *Oxford Economic Papers*, Vol. 40, No. 4, pp. 751–81.
- Hull, John C., 2008, *Options, Futures, and Other Derivatives* (Upper Saddle River, New Jersey: Prentice-Hall, 7th ed.).
- Hull, John C., and Alan White, 1990, "Pricing Interest-Rate Derivative Securities," *Review of Financial Studies*, Vol. 3, No. 4, 573–92.
- , 1993, "One-Factor Interest-Rate Models and the Valuation of Interest-Rate Derivative Securities," *Journal of Financial and Quantitative Analysis*, Vol. 28, No. 2, pp. 235–54.
- , 1994, "Numerical Procedures for Implementing Term Structure Models: Single-Factor Models," *Journal of Derivatives*, Vol. 2, No. 1, pp. 7–16.
- Jafry, Yusuf, and Til Schuermann, 2004, "Measurement, Estimation and Comparison of Credit Migration Matrices," *Journal of Banking and Finance*, Vol. 28, No. 11, pp. 2603–39.
- Jarrow, Robert A., and Stuart M. Turnbull, 2000, "The Intersection of Market and Credit Risk," *Journal of Banking and Finance*, Vol. 24, Nos. 1–2, 271–99.
- Klaar, Andres, Paul Rawkins, and David Riley, 2004, "Fitch Sovereign Rating Transition and Default Study," Fitch Ratings Special Report (New York: Fitch Ratings).
- Lehar, Alfred, 2003, "Measuring Systemic Risk: A Risk Management Approach," Working Paper (Vancouver: University of British Columbia).
- Lindgren, Carl-Johan, Gillian Garcia, and Matthew I. Saal, 1996, *Bank Soundness and Macroeconomic Policy* (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.cfm?sk=1538.0>
- Rochet, Jean-Charles, and Jean Tirole, 1986, "Inter-Bank Lending and Systemic Risk," *Journal of Money, Credit and Banking*, Vol. 28, No. 4, 733–62.

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Modeling Correlated Systemic Bank Liquidity Risks

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This chapter proposes and demonstrates a methodology for modeling correlated systemic solvency and liquidity risks for a banking system. Using a forward-looking simulation of many risk factors applied to detailed balance sheets for a 10-bank stylized U.S. banking system, we analyze correlated market and credit risks and estimate the probability that multiple banks will fail or experience liquidity runs simultaneously. Significant systemic risk factors are shown to include financial and economic environment regime shifts to stressful conditions, poor initial loan-credit quality, loan portfolio sector and regional concentrations, bank creditors' sensitivity to and uncertainties regarding solvency risk, and inadequate capital. Systemic banking system solvency risk is driven by the correlated defaults of many borrowers, other market risks, and interbank defaults. Liquidity runs are modeled as a response to elevated solvency risk and uncertainties and are shown to increase correlated bank failures. Potential bank funding outflows and contractions in lending with significant real economic impacts are estimated. Increases in equity capital levels needed to reduce bank solvency and liquidity risk levels to a target confidence level also are estimated to range from 3 percent to 20 percent of assets. For a future environment that replicates the 1987–2006 volatilities and correlations, we find only a small risk of U.S. bank failures focused on thinly capitalized and regionally concentrated smaller banks. For the 2007–10 financial environment calibration, we find substantially elevated solvency and liquidity risks for all banks and the banking system.

METHOD SUMMARY

Overview	The method provides a valuation of bank on- and off-balance-sheet positions under stress.
Application	The method is particularly useful when a creditor perspective is necessary—a creditor does not care about accounting conventions or about regulatory compliance; the creditor cares about value.
Nature of approach	Monte Carlo simulation; valuation of bank transactions.
Data requirements	Accounting information on capital, loans, and risk-weighted assets. Supervisory data on classified loans and provisions. Information about leverage of corporates and households. Prices that characterize the economic environment (such as interest rates, exchange rates, real estate prices).
Strengths	The model captures the correlated nature of main risks, such as credit and market risks, overall solvency and liquidity risks, and risks in the interbank market.
Weaknesses	• The model assumes that variables are represented as an n-variate normal distribution and that they have constant trends, volatilities, and correlations. (However, the model could be programmed to allow for stochastic volatilities and correlations.) • Data intensive. • For a large number of institutions modeled simultaneously, simulations may take a significant amount of time.
Tool	The Excel spreadsheet macro is not yet available.

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A systemic liquidity shock is an aggregate shortage of liquidity, that is, a situation in which many institutions face liquidity shortages simultaneously, as opposed to one institution suffering a liquidity shortage.¹ Systemic liquidity risk is the probability that this situation takes place. A liquidity shortage can manifest as an inability for institutions to roll over funding (funding liquidity risk), the inability to trade assets at normal bid or ask spreads (market liquidity risk), or, very frequently, both.

The stress test approach developed in this chapter takes the view that absent an aggregate preference shock (i.e., a sudden shift of preferences in favor of higher present consumption) or infrastructure malfunctioning, a systemic liquidity shock is more likely to happen in the presence of a shock to fundamentals that depresses asset values and makes the market reluctant to fund these (suddenly) lower-quality assets or the institutions that hold them.² In the presence of incomplete and asymmetric information on the values of assets and the financial condition of banks, this reluctance can also be extended to good assets and solvent institutions.

A systemic liquidity shock is closely associated with the notion of bank panics. Traditionally, there have been two leading alternative views to explain the triggers of panics in the more traditional setting of depositors' behavior: the random withdrawals theory and the information-based theory. The random withdrawal approach, as developed in Diamond and Dybvig (1983), postulates that a panic is the realization of a bad equilibrium that is due to the fulfillment of depositors' self-expectations concerning the behavior of other depositors (a pure liquidity shock). Conversely, the information-based approach, as reflected in Allen and Gale (2000), claims that a panic is an episode of market discipline during which depositors attempt to sort among ex ante "good" (solvent) and ex ante "bad" (insolvent) banks in a world of asymmetric information regarding bank asset values. In this context, bank panics can be a normal outcome of business cycles: an economic downturn will reduce the value of bank assets, raising the possibility that banks cannot meet their commitments. Gorton (1988) undertook an empirical study to differentiate between the "sunspot" view and the business-cycle view of banking panics. He found evidence consistent with the view that banking panics are related to the business cycle.

There is also consensus that the global financial crisis has not been a pure liquidity shock but was triggered instead by concerns about the value of bank assets—subprime mortgages and structured products affected by the fall in house prices. Gorton and Metrick (2009) characterized the global crisis as a systemwide "run" in the securitized banking system—more precisely a "run on the repo market"—similar to the banking panics of the 19th century. Both episodes, in

their view, were triggered by insolvency problems. They find that during 2007–08, changes in the London Interbank Offered Rate–Overnight Indexed Swap (LIBOR–OIS) spread, a proxy for counterparty risk in the (interbank) repo market, was strongly correlated with changes in credit spreads and repo rates for securitized bonds. These changes implied higher uncertainty about bank solvency and lower values for repo collateral. They conclude that the market slowly became aware of the risks associated with the subprime market, which then led to doubts about repo collateral and bank solvency. At some point—August 2007—a critical mass of such fears led to the first run on repo, with lenders no longer willing to provide short-term finance at historical spreads and haircuts.

Afonso, Kovner, and Schoar (2010) examined the connections between solvency and liquidity over the global crisis. They tested two hypotheses by which shocks to individual banks can lead to marketwide reductions in liquidity: (1) an increase in counterparty risk, leading to a drying up in liquidity; and (2) liquidity hoarding, that is, banks not willing to lend even to high-quality counterparties in order to keep liquidity for precautionary reasons. Their findings suggest that concerns about counterparty risks played a much larger role than liquidity hoarding. Moreover, in the days after Lehman's bankruptcy, loan amounts and spreads became more sensitive to borrowers' characteristics: they observe that large borrowers accessed the federal funds market less after Lehman's bankruptcy and from fewer counterparties. Furthermore, it was the worst performing large banks (the "bad" banks) that accessed the market least. They do not observe the complete cessation of lending predicted by some theoretical models that focus on liquidity hoarding.

The IMF's October 2008 *Global Financial Stability Report* showed that systemic (joint) default risk has been the dominant factor in the explanation of the spreads and that systemic (joint) default risk has influenced the spread since July 2007 (IMF, 2008). It also showed that the repo spread began to present signs of stress in 2005 when the U.S. housing market began its downturn. It then concluded that broadening access to emergency liquidity alone would not resolve bank funding stresses until broader policy measures, including those aimed at the underlying counterparty credit concerns, were implemented.

Consistently, we propose a stress test of systemic liquidity in which systemic liquidity shocks are modeled as a reaction to shocks to asset values resulting from borrower defaults and other factors. In our approach, a liquidity shock (or a "run") is an extreme episode of "market discipline" by which those providing funding (depositors, wholesale investors, other banks, etc.) attempt to sort among ex ante "good" (solvent) and ex ante "bad" (insolvent) users of funds in a world of asymmetric information regarding asset values. Although the exact timing of a systemic liquidity shock is difficult to forecast, we postulate that they are highly correlated with solvency concerns and contractions in bank lending. Our approach is also consistent with the stress testing literature in which liquidity withdrawals are linked to banks' solvency risk (Table 9.1).

¹ The risk assessments reported in this analysis were undertaken with the ValueCalc Banking System Risk Modeling Software, copyright FinSoft, Inc.

² These triggers explain a large portion of the liquidity shortages experienced during the global crisis and were discussed in IMF (2010; e.g., need for centralized repo counterparties, better recording of OTC transactions in repositories).

TABLE 9.1

Selected Liquidity Stress Testing Frameworks

Framework	Bank of England	De Nederlandsche Bank	Hong Kong Monetary Authority	Proposed Stress Testing Framework
Data	Bank-by-bank financial reporting	Bank-by-bank financial reporting	Bank-by-bank financial reporting	Bank-by-bank financial reporting
Origin of liquidity shocks	Funding liquidity shock (cost and access) upon downgrade from solvency shocks (credit and market losses in macro stress testing).	Valuation losses and/or funding withdrawal to selected liquidity items.	Deposits are withdrawn in line with stressed probability of default (PD) (owing to a loss from asset price declines) of the bank.	Asset price shocks. Bank liabilities are withdrawn following stressed PD of the bank.
Feedback, spillover, amplification effects	Linear, normal time linkages. Nonlinear effects using subjective but simple scoring system. Second-round effects through impact on asset price upon bank deleveraging and network effects.	Nonlinear effects as banks take deleveraging actions for larger shocks, and they feed back to asset valuation and funding availability (second-round effects).	Deleveraging to restore lost funding is costly owing to distress in asset markets. Interbank contagion (network effects).	Banks attempt to restore net cash flow by selling assets, which affect market liquidity of the assets, further tightening funding liquidity (through higher haircuts).
Measurement of stress	Various standard metrics (solvency ratio, liquidity ratio, asset value, credit losses, ratings, profit, etc.).	Distribution of liquidity buffer across banks and across severity of shocks.	Probability of cash shortage and default; expected first cash shortage time; expected default time.	Solvency ratio; distributions of net cash flows and equity; joint probability of multiple institutions suffering from simultaneous cash shortfalls.
Origin of “systemic liquidity” characteristics	Initial macroeconomic shocks and various second-round effects.	From second-round effects.	From initial aggregate shock on asset prices, network effects.	Initial aggregate shock on asset prices and various second-round effects.
Pros	Nonlinear liquidity shocks and various second-round effects.	Nonlinear second-round effects.	Interaction among credit and funding and market liquidity risks.	Nonlinear second-round effects, assess joint probability of liquidity distress, and contribution of individual bank.
Cons	Includes subjective components to model nonlinearity.	Bank behavioral assumption and feedback effect formulated without strong micro-foundation.	No feedback effects from distress on banks to asset prices.	Bank behavioral assumption and feedback effect formulated without strong micro-foundation.

Source: IMF (2011).

Note: Bank of England reflects the stress testing framework proposed by Aikman and others (2009); De Nederlandsche Bank reflects the stress testing framework proposed by van den End (2008); and the Hong Kong Monetary Authority reflects the stress testing framework proposed by Wong and Hui (2009).

This chapter also highlights the importance of relating the policy response to the diagnosis of the shock. Systemic liquidity shocks may all look similar, despite their origin (aggregate liquidity-preference-shock infrastructure malfunctioning or solvency concerns). However, the different origin is important to inform the policy response, both preemptively, to minimize the probability of a systemic liquidity crisis, and to manage the crisis, once it happens.³ For our application to the U.S. banks, we develop a capital surcharge aimed at minimizing the probability that any given bank would experience a destabilizing run. For crisis management, we propose recapitalizing or closing insolvent banks and disclosing enough information to eliminate uncertainties about bank solvency. Liquidity injections by a central bank—that can solve the problem in the case of a change in intertemporal preferences for consumption—would likely not be effective if there is reluctance to provide funding for suddenly poor-quality assets.⁴ Balance between supply and demand of liquidity would be achieved only by deleveraging, restoring asset quality and confidence, and providing enough information to avoid contagion problems for solvent institutions.

1. MODELING STEPS AND DATA REQUIREMENTS

A. Modeling steps

Our approach starts with a detailed solvency stress test for multiple banks and then adds, as an innovation, a systemic liquidity component. It can be used to measure correlated systemic solvency and liquidity risks, assess a bank's vulnerability to a liquidity shortfall, and develop a capital surcharge aimed at minimizing the probability that any given bank would experience a destabilizing run.

We model three channels for a systemic liquidity event:

- a stressed macro and financial environment leading to a reduction in funding from the unsecured funding markets that is due to a heightened perception of counterparty and default risks;
- a fire sale of assets as stressed banks seek to meet their cash flow obligations. Lower-asset prices affect asset valuations and margin requirements for all banks in the system, and these in turn affect funding costs, profitability, and generate systemic solvency concerns; and
- lower funding liquidity because increased uncertainty over counterparty risk and lower asset valuations induce banks and investors to hoard liquidity, leading to systemic liquidity shortfalls.

³ An interesting reflection about the policy response to the global crisis and the importance of relating the diagnosis to the policy response can be found in Taylor (2008).

⁴ To the extent that a central bank provides liquidity in exchange for a trouble asset, liquidity injections can work because it would allow the bank to deleverage—by getting rid of the trouble asset.

The approach proceeds in four stages as illustrated in Figure 9.1: (1) modeling the financial and economic environment; (2) credit risk modeling; (3) systemic solvency risk modeling; and (4) correlated systemic liquidity risk modeling. First, thousands of Monte Carlo simulations are used to simulate correlated changes in many asset prices (foreign exchange rates, interest rates, real estate prices, and market equity indexes), as well as macroeconomic factors that drive bank clients' defaults (equity values, bank clients' leverage) between the current time (T_0) and a future time (T_1). These simulated prices and macroeconomic factors are used to re-value banks' balance sheets.

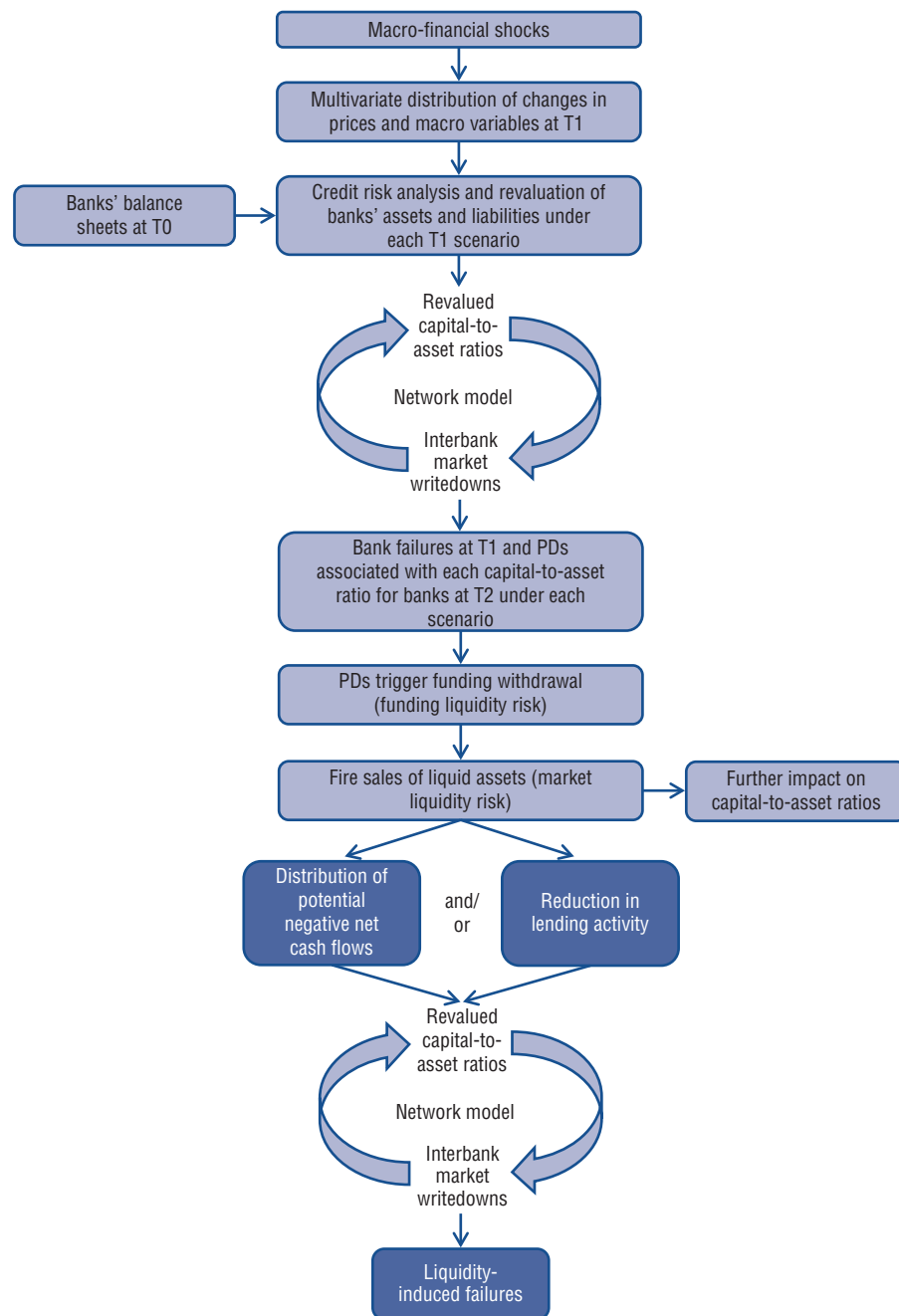
A large shock to these prices and macroeconomic factors affect the quality of bank assets directly (higher credit and market risk) and also indirectly through a network of interbank claims. Our model estimates the economic value of banks' capital, economic capital-to-asset ratios, and bank solvency defaults at T_1 and the probability of future solvency defaults at T_2 (as measured at T_1).

In simulations with higher bank probabilities of default, bank creditors react by showing reluctance to fund bank assets. Confronted with increasing difficulties to roll over their liabilities, banks need to fire sell assets at distressed prices. This in turn aggravates their economic capital-to-asset ratios. At the end of the simulation, the model generates a final distribution of economic capital to asset values as well as a distribution of cash flows. Banks are modeled as failing when their capital-to-asset ratios reach a critical threshold value (2 percent) or in the presence of liquidity shortages. Periods with multiple bank failures are likely to also have multiple banks in weakened financial conditions. This is just the time when losses on interbank credit defaults can lead to correlated banking system solvency and liquidity crises.

B. Data requirements

The stress test approach has the following intensive data requirements. In some cases, it may be possible to substitute expert opinion for data that may not be available.

- Time series related to the financial and economic environment in which banks operate. These series need to be of sufficient length to allow trends, volatilities, and correlations to be estimated during both "normal" and "stress" periods. The following data are of interest:
 - short-term domestic and foreign interest rates and their term structures
 - interest rate spreads for loans of various credit qualities (securities)
 - foreign exchange rates (as relevant)
 - economic indicators (GDP, consumer price index, unemployment, and so on)
 - commodity prices (oil, gold, and so on)
 - sector equity indices
 - regional real estate prices



Source: IMF (2011).
 Note: PDs=probabilities of default.

Figure 9.1 Modeling Steps

- Information on banks' assets, liabilities, and, ideally, off-balance-sheet transactions, including hedges, such as
 - various categories of loans, including information about their credit quality, maturity structure, and currencies of denomination
 - currency and maturity structure of the other assets and liabilities
 - capital as well as operating expenses and tax rates
 - clients' leverage ratios and recovery rates, to be able to calibrate credit risk models
 - interbank exposures, including bilateral credit exposures among the various banks
- Information to enable calibration of behavioral relationships, such as

- between banks' default probabilities and funding reduction because of bank creditors' concerns about solvency
- between asset fire sales and asset values (including haircuts), which in turn affect liquidity and solvency ratios.

2. MODEL CALIBRATION TO THE U.S. FINANCIAL ENVIRONMENT AND THE U.S. BANKING SYSTEM

For the characterization of the U.S. financial and economic environment, we use a set of variables, including interest rates, interest rate spreads, foreign exchange rates, U.S. economic indicators, global equity indices, 14 Standard and Poor's (S&P) sector equity returns, and 20 Case-Shiller regional real estate price returns. The Financial and Economic Environment Model is calibrated for two different "regimes." The first calibration is based on data from the 20-year period 1987–2006. The second calibration is based on data from the period 2007–10.

Using Call Report Data,⁵ we constructed 10 stylized U.S. banks in four categories: two megabanks that aggregate the assets and liabilities of two groups of banks, with higher and lower equity capital-to-asset ratios and assets above \$500 billion; three large banks that aggregate the assets and liabilities of groups of banks with assets between \$100 billion and \$500 billion; three medium-sized banks that aggregate groups of banks with assets between \$10 billion and \$100 billion; and two small banks that aggregate groups of banks with assets below \$10 billion. The various banks are sized so that they have an appropriate weighting relative to the overall U.S. banking system. For example, the megabanks have approximately 62 percent of the total assets for the model banking system. A larger or smaller number of banks could be modeled.

Sector and regional concentrations of bank loan portfolios are also a significant risk factor. The smaller banks are modeled as making mortgage loans in one or two states (e.g., California, or Florida and Georgia) and three sectors of the economy (industrial, retail, and services). Medium-sized banks are modeled as lending in larger regions (West Coast, Mid-America, or East Coast) and four sectors of the economy. Large and megabanks are modeled as lending nationally in 20 regions and 14 sectors of the economy.

Systemic solvency risk in our model depends on bank exposure to borrower creditworthiness (credit risk), including credit concentration, as well as correlated market prices (market risk). The risk assessment horizon was set at one year. The model is flexible to accommodate other risk modeling time steps.

⁵ We assumed that the reported balance sheets represent reasonable estimates of current market values for bank assets and liabilities.

3. CREDIT RISK MODELING

Business and mortgage loan credit risk assessments are based on simulations of business debt-to-value ratios and property loan-to-value ratios using a contingent claims type model.⁶ The future values of companies are systematically related to simulated sector-equity returns plus a company-specific random return. When business debt-to-value ratios cross critical boundaries, the credit rating of the loans is assumed to change. At identified high debt-to-value ratios, the loans are assumed to default.⁷ In this study, we use the U.S. business credit risk model estimated by Barnhill and Maxwell (2002). Correlated variations in recovery rates on business loans are also an important systematic risk factor. In our analysis, recovery rates on business loans are modeled as increasing (decreasing) as stock market returns increase (decrease).⁸

A. Corporate loan portfolio modeling

Given that we did not have information on the credit quality of corporate borrowers for each bank, we assumed that initially, the set of business loans in U.S. bank portfolios have the same credit quality distribution for all banks and that this distribution is the one described in the *Shared National Credits Review* issued annually by the Board of Governors of the Federal Reserve System, Federal Deposit Insurance Corporation (FDIC), Office of the Comptroller of the Currency, and Office of Thrift Supervision.⁹ Foreign corporate loans are modeled following the same credit risk analysis procedures used for domestic loans. However, we do account for foreign exchange rate risk. For those assets and liabilities where credit risk is not modeled, valuation is based on a present value approach where the cash flows are discounted using the simulated interest rates of a selected term structure and the simulated values for the correlated exchange rates, in the case of securities denominated in foreign currency.

B. Individual loan portfolio modeling

Given the lack of an alternative model, loans to individuals were modeled entirely as a portfolio of mortgage loans. This approach has obvious limitations but does capture any correlations among the default rates on other loans to individual and mortgage loans resulting from unemployment rates, low property prices, and so on. The initial loan-to-value ratios for mortgage loans were estimated from data given in Fannie Mae's and Freddie Mac's annual report plus esti-

⁶ See Black and Scholes (1973) and Merton (1973).

⁷ For a more detailed discussion, see Barnhill and Maxwell (2002), Barnhill, Papapanagiotou, and Schumacher (2002), and Barnhill and Souto (2009).

⁸ See Cantor and Varma (2005).

⁹ In the U.S. bank calibration of our model, loans classified as "substandard," "doubtful," and "loss" were modeled as having a credit risk similar to B, C, and D rated bonds. The balance of the business loan portfolio was divided evenly between the A, BBB, and BB rating categories.

mates of the likely distributions of mortgage portfolio loan-to-value (LTV) ratios based on assumed initial LTV ratios and trends in national real estate prices.

In our model, the future values of properties are systematically related to regional real estate returns plus a property-specific random term. The probability of a mortgage loan defaulting is modeled as being related to its LTV ratio.¹⁰ For LTV ratios between 1.2 and 1.4, the default rate is set at 20 percent. For LTV ratios between 1.4 and 1.6, the default rate is set at 40 percent. For LTV ratios between 1.6 and 1.8, the default rate is set at 60 percent. For LTV ratios over 1.8, the default rate is set at 80 percent. Recovery rates on mortgage loans are correlated with real estate prices and are assumed to be the LTV ratio less a 30 percent liquidation cost.

Correlated changes in the values of real estate assets by region and business assets by sector are driven by the correlated returns on regional real estate indices and sector equity indices in the financial and economic environment. Correlated default rates on mortgage loans in various regions and business loans in various sectors are driven by the assumed initial LTV ratios and correlated changes in the values of the real estate and business assets securing the bank loans. Such correlated defaults on bank loan portfolios produce correlated banking-system systemic solvency and liquidity risks.

4. LOAN PORTFOLIO CONCENTRATION MODELING

The concentration of bank loans in various sectors (e.g., energy), regions (e.g., Florida), and security types (e.g., mortgage loans) are particularly significant bank risk factors that are often not modeled adequately. To account for loan portfolio concentration risk, we model the correlated market and credit risk on 200 business loans distributed across up to 20 sectors of an economy and 200 mortgage loans distributed across up to 20 regions of a country. We find this to be an adequate number of loans, sectors, and regions to statistically distinguish between more concentrated and more diversified portfolios. More sectors, regions, and loans could be modeled.

We also model correlated market risk for approximately 100 other bank assets and liabilities.

5. SYSTEMIC SOLVENCY RISK MODELING

One of the outcomes of the risk assessments of the financial and economic environment and bank portfolios after many simulation runs is joint distributions of each of the 10 banks' market value of equity capital at T1.

$$MVE_t = \sum_{i=1}^n A_{i,t} - \sum_{i=1}^n L_{i,t}$$

where MVE_t is the simulated market value of the bank's equity at time t ; $A_{i,t}$ is the simulated market value of the i th asset at time t , which reflects the simulated financial environment variables (e.g., interest rates, exchange rates, equity prices, real estate prices) and, where appropriate, the simulated credit rating of the borrower; and $L_{i,t}$ is the simulated market value of the i th liability at time t , which reflects the simulated financial environment variables (e.g., interest rates, exchange rates). The bank's asset and liability levels are also adjusted to reflect bank net interest income, fee income plus other income less operating expenses, and taxes over the simulation period.

After many simulation runs, joint distributions of the various banks' capital-to-asset ratios are estimated and used to assess bank defaults and systemic banking system solvency risks at T1.

$$Capital_Ratio_t = MVE_t / \sum_{i=1}^n A_{i,t}$$

For each run of the simulation, the simulated capital ratio is also used to estimate each bank's correlated probability of defaulting at T2. These future default probabilities are derived under the assumption that the distribution of changes in capital ratios between T1 and T2 is identical to the distribution of changes in capital ratios between T0 and T1.

During times of economic stress, it is likely that default losses on loans will increase, and many banks will either fail or be weakened significantly, particularly if they have similar asset and liability structures. This is just the time when the failure of several banks could, through interbank credit defaults, precipitate a number of simultaneous bank failures. Interbank credit risk is modeled using a network methodology. Given that we do not have precise information on interbank borrowers' and lenders' identities, we assumed that the amount of interbank loans made between each bank is proportional to their total interbank borrowing and lending.

In the current study, and consistent with current U.S. regulations, we model a bank as failing when its ratio of equity capital-to-assets falls below 2 percent.¹¹ In this case, the bank becomes incapable of honoring its interbank obligations and defaults on them.¹² The recovery rate on defaulted interbank obligations is assumed to be 40 percent. Such losses could affect counterparty banks' capital ratios and potentially lead to additional bank failures. A network methodology is applied repeatedly until no additional banks fail, after which the probability of multiple simultaneous bank failures (i.e., systemic solvency risk) can be computed. The outcome of this step is again equity-to-asset ratios and bank

¹⁰ See Bhutta, Dokko, and Shan (2010).

¹¹ The FDIC Improvement Act of 1991's Prompt Corrective Action provision states that a bank should be closed when its tangible capitalization reaches 2 percent. The trigger point for bank failure could be set in the stress test framework model at any relevant regulatory level, including the new leverage ratio as proposed under Basel III.

¹² In the current study, precise information on interbank borrowers' and lenders' identities are unavailable; hence, the amount of interbank loans made between each bank is assumed to be proportional to their total interbank borrowing and lending.

failures for T1—that includes losses due to defaults on interbank claims—and a probability of default for each bank at T2 (estimated as discussed in the previous paragraph) for each run of the simulation.

6. MODELING CORRELATED SYSTEMIC LIQUIDITY RISK

A primary contribution of the model to stress testing is the addition of correlated liquidity runs on banks, driven by heightened risks or uncertainties, regarding future bank solvency.¹³ Changes in bank liabilities observed over the period from 2007 to the first quarter of 2010 were used to develop an estimated relationship between a bank's probability of default and the rate of withdrawal of total liabilities over the period T1 to T2.

Bank liquidity outflows are estimated under two cases. In Case 1, total liability withdrawal rates match those experienced by bank holding companies (BHCs) with elevated default probabilities during the 2007–10 period.¹⁴ In Case 2, higher total liability withdrawal rates match those experienced by investment banks; because investment banks have a very low level of insured deposits, this case provides a way to calibrate a more stressed scenario where funding sources may dry up very quickly. In Case 2, for lower default probabilities we modeled potential reductions in specific liability accounts (e.g., demand deposits, time deposits, jumbo time deposits, federal funds, repos), which reflect actual liability structures of the stylized BHCs.

Because of incomplete information on particular banks, we assume that bank creditors are also aware of and react to developments in the overall banking system. We thus model systemwide weighted average banking system default proba-

TABLE 9.2

Withdrawal Rate Assumptions for Decline in Total Liabilities (in percent)

Default Probability	Withdrawal Rate	
	Case 1	Case 2
10–20	5	7–10
20–35	10	14–21
>35	25	42

Sources: Authors; and SNL Financial.

bilities and assume that they have some impact on liquidity runs. In particular, liquidity runs for a particular bank are modeled as being driven by the probability of failure for that bank at T2 plus a factor equal to 10 percent of the systemwide weighted average default probability (i.e., the adjusted probability of failure). Table 9.2 summarizes assumptions on total liability withdrawal rates associated with different default probability ranges for each case.

Banks that face a liquidity run are assumed to follow one of two strategies. In the first strategy, banks stop lending in the interbank and repo markets, liquidate interest-bearing bank deposits, sell government securities, and sell other securities. If these steps do not produce adequate liquidity, they ultimately default on their obligations.¹⁵ In the second strategy, banks sell their liquid securities and reduce their loan portfolios in proportions similar to that observed in U.S. BHCs having elevated failure probabilities.¹⁶

Banks pay a high cost when they are forced to sell assets during periods of extreme financial market stress. We model bank losses resulting from the fire sale of assets. This cost is given by a selling price with an embedded high liquidity premium and consequently well below its fundamental price. In this way, the model captures the interaction between funding and market liquidity and the second-round feedback between solvency and liquidity risks. Developments in bid-ask spreads in several securities markets during the 2000–2009 period were used as a proxy for fire sale prices. At the peak of the crisis (September 2008), the size of the bid-ask spread was in the 5–10 percent range across different asset qualities, suggesting a discount factor of 3–5 percent to represent the loss suffered by the bank under distress when forced to liquidate assets. These values are in line with Duffie, Gârleanu, and Pedersen (2006), Coval and Stafford (2007), and Aikman and others (2009).

7. RESULTS

In the 2007–10:Q1 financial environment under Case 1 (BHC withdrawal rate), the probability that about 3 out of

¹³ This approach is in line with current approaches attempting to model bank creditors' behavior in the presence of asymmetric information. For example, for the purpose of calibrating the closure of funding markets in the Risk Assessment Model of Systemic Institutions (RAMSI), the Bank of England uses an index of default risk called the "danger zone." This index is an equally weighted average of three factors: factors related to solvency, factors related to banks' liquidity position, and factors related to confidence. Based on case studies, it is assumed that when the index approaches 35, banks are assumed to default and short-term unsecured markets close to them. Wong and Hui (2009) estimated the relation between bank default risk and the outflow rate of interbank deposits over the Bear Stearns debacle. They found that interbank deposits started to be withdrawn when Bear Stearns's default probability reached 8 percent. At 69 percent or higher, all interbank deposits were not renewed at maturity. Our approach is closer to the "danger zone" used by RAMSI but it does not include banks' liquidity position as one of the factors. This exclusion is in line with the assumption made in our model that all banks' projected cash flows are zero, in the absence of a systemic liquidity shock. This is also in line with the empirical evidence that liquidity mismatches per se do not tend to cause systemic liquidity shocks, unless there are also concerns about bank solvency.

¹⁴ During the crisis, some BHCs were able to increase their access to insured liabilities by converting large uninsured deposits into smaller insured deposits. These assumptions are based on the analysis of changes in total liabilities for a group of about 700 insured BHCs relative to their estimated probability of default.

¹⁵ This case may be illustrative of a very rapidly developing liquidity crisis where banks have little opportunity to adjust their loan portfolios.

¹⁶ This case may be illustrative of banks that face liquidity outflows over time and have an opportunity to adjust their entire asset and liability structure.

10 banks will simultaneously find themselves unable to make payments (i.e., have a negative cash flow) is 3.8 percent (Table 9.3). That is, the risk of a systemic liquidity shock for this hypothetical U.S. banking system as of June 2010 would be low. In this example, the smaller banks are more affected than the larger ones because of their higher credit risk concentration and exposure to the macro risk factors that triggered the recent crisis. In addition, although banking failures occurred among smaller banks, their liquidity shortages did not lead to a systemic liquidity crisis. In the 2007–10:Q1 financial environment under Case 2 (investment bank withdrawal rate), the probability that 3 out of 10 banks suffer a liquidity shortage increases to 12.7 percent.

Such potential liquidity shortages can create pressures for substantial reductions in bank lending and thus affect the real economy. Indeed, both liquidity shortages and reductions in bank lending were observed during the global crisis. In Case 1, if the stylized banks facing liquidity runs reduce both securities and loan portfolios, the impact on total loans would be small (Figure 9.2, left panel, vertical axis). In Case 2, by contrast, a potential liquidity run could lead to a significant reduction in total loans, of up to 43 percent, although with a low probability of less than 1 percent attached to this event (Figure 9.2, right panel, horizontal axis).

TABLE 9.3
Probability of Banks Ending the Simulation with a Liquidity Shortage

Number of banks	Probability	
	Case 1	Case 2
1	98.49	98.49
2	20.28	23.68
3	3.77	12.74
4	1.60	4.25
5	1.13	1.98
6	0.75	1.51
7	0.09	1.13
8	0.00	1.04
9	0.00	1.04
10	0.00	0.09

Sources: Authors; and SNL Financial.

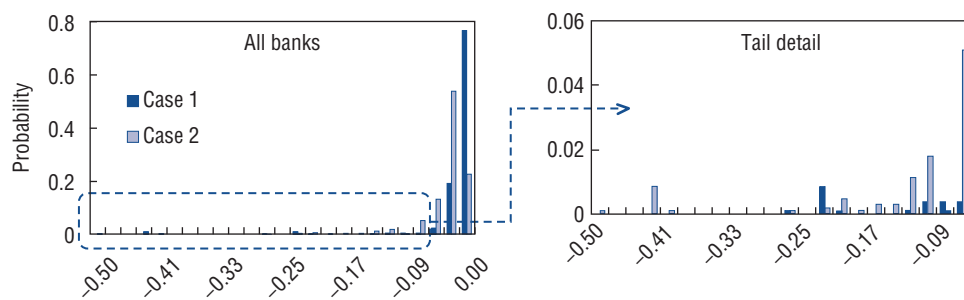
These stress test results generally show that the ability of banks to weather a financial and economic shock, and its impact on solvency and liquidity depends on a number of factors, including (1) the size of the shock; (2) the adequacy of capital; (3) the availability of liquid assets; and (4) the exposure to short-term wholesale liabilities (in this model, interbank exposures). In this framework, if institutions were sufficiently capitalized and, hence, able to sell liquid assets and deleverage in an orderly manner, then there would be no systemic liquidity shock.

The methodology can be used to estimate an additional required capital surcharge or buffer to reduce the risk of future liquidity runs by lowering bank default risk. Using the distribution of capital ratio changes, the methodology can estimate the additional capital buffer required to reduce the probability of a bank experiencing a liquidity run over the next year to below 10 percent at a 99 percent confidence level (see Table 9.4). Of the 10 stylized banks, the small banks need to add the most capital because of their undiversified asset exposures to regional real estate sectors, where credit losses have been the highest.

8. CONCLUSION

Systemic banking system risks were poorly understood and managed prior to the current crises. We propose and demonstrate a forward-looking simulation methodology applied to the financial and economic environment and detailed balance sheets for a 10-bank model banking system. The model estimates the magnitude and probability of correlated solvency and liquidity risks that affect bank borrowers, banks, the banking system, and the real economy.

In our model, banks fail from a solvency perspective when their simulated capital ratios fall below some critical level (e.g., 2 percent). Banks experience liquidity problems when their risk of future insolvency or the banking system's overall risk of insolvency rises to an unacceptable level (e.g., 10 percent). Correlated systemic risks materialize when multiple banks become insolvent or face liquidity risks simultaneously. Systemic risks are driven in part by large adverse regime shifts in the financial and economic environment that



Sources: Authors; and SNL Financial.

Note: The right panel shows the results of the left panel with a finer set of x-axis losses of loans and lower probability (y-axis).

Figure 9.2 Total Loan Reductions

TABLE 9.4

Additional Equity Capital Required at T0 for Banks to Have a 99 Percent Confidence Level That at T1 They Would Have Less Than a 10 Percent Probability of Failing by T2

	Capital Ratio Bank 1	Capital Ratio Bank 2	Capital Ratio Bank 3	Capital Ratio Bank 4	Capital Ratio Bank 5	Capital Ratio Bank 6	Capital Ratio Bank 7	Capital Ratio Bank 8	Capital Ratio Bank 9	Capital Ratio Bank 10
Initial capital ratios	0.104	0.057	0.124	0.104	0.08	0.134	0.124	0.095	0.101	0.088
Approximate additional equity capital required at T0 to have 1% probability of a 10% probability of failure at T1	0.111	0.216	0.045	0.056	0.123	0.031	-0.011	0.049	0.046	0.026

Source: ValueCalc estimates.

catch many entities by surprise. Such economic shocks drive down the value of large asset classes (e.g., real estate and businesses), resulting in dramatically higher default rates and loss rates on loans for many banks simultaneously. Banks experiencing funding outflows are likely to contract lending with significant potential impacts on the real economy.

In our model, bank solvency and liquidity risks also are driven by asset and liability structures, loan credit quality, sector and regional loan concentrations, and equity capital levels, all of which can be changed by bank managers and affected by bank regulators. Such potential systemic risks may materialize only in the context of very adverse financial and economic regime shifts that may occur only infrequently (but have extreme impacts).

Through the interbank lending market, bank failures may impose additional losses on otherwise solvent banks that cause them to also fail or increase their probability of failing, thus further increasing systemic risk levels. We find high correlations between insolvencies in megasized banks and incremental insolvencies of other banks throughout the banking system, which increases systemic risk.

Insufficient liquid assets that prevent an otherwise orderly “asset shrinking” to offset reductions in funding may also be a significant risk factor. The forced sale of assets at “fire sale” prices may also affect banks’ correlated solvency and liquidity risks.

The model was calibrated with publicly available data on the U.S. financial and economic environment and the U.S. banking system. For the 1987–2006 financial environment calibration, we find only a small risk of bank failures focused on thinly capitalized and regionally concentrated smaller banks. We find no likelihood of systemic solvency or systemic liquidity risks. For the 2007–10 financial environment calibration, we find substantially elevated solvency and liquidity risks for all banks and the banking system. When potential interbank default losses and liquidity runs are modeled, 7 of the 10 banks (including one megabank) fail at the same time with a 1 percent probability.

Within our model, we can estimate the current bank equity capital levels that are needed to reduce bank solvency and liquidity risk levels to an agreed target at a given confidence level. We conclude that substantial additional equity capital

(e.g., 3 percent to 20 percent of assets) is needed to minimize potential solvency and liquidity risks. We can also assess the bank and systemic banking system impacts of changes in the other identified risk variables.

Important areas for future research on correlated solvency and liquidity risk include assessing (1) the relationship between systemwide stress levels and liquidity risk for individual banks; (2) correlated changes in all liability accounts for banks with elevated solvency risk; (3) how volatility in bank loan collateral values increases bank solvency and liquidity risk; (4) correlations between the volume of repossessed collateral (e.g., real estate), subsequent price declines for that asset type, and subsequent default rates on related bank loans; and (5) correlated sovereign risk. Modeling potential economic regime shifts is also an exceptionally important risk assessment topic.

REFERENCES

- Afonso, Gara, Anna Kovner, and Antoinette Schoar, 2010, “Stressed, Not Frozen: The Federal Funds Market in the Financial Crisis,” Staff Report No. 437 (New York: Federal Reserve Bank of New York). Available via the Internet: http://www.newyorkfed.org/research/staff_reports/sr437.html
- Aikman, David, Piergiorgio Alessandri, Bruno Eklund, Prasanna Gai, Sujit Kapadia, Elizabeth Martin, Nada Mora, Gabriel Sterne, and Matthew Willison, 2009, “Funding Liquidity Risk in a Quantitative Model of Systemic Stability,” Working Paper No. 372 (London: Bank of England). Available via the Internet: <http://www.bankofengland.co.uk/publications/Pages/workingpapers/2009/wp372.aspx>
- Allen, Franklin, and Douglas Gale, 2000, “Financial Contagion,” *Journal of Political Economy*, Vol. 108, No. 1, pp. 1–33.
- Barnhill, Theodore, and William Maxwell, 2002, “Modeling Correlated Interest Rate, Exchange Rate, and Credit Risk in Fixed Income Portfolios,” *Journal of Banking and Finance*, Vol. 26, Nos. 2–3, pp. 347–74.
- Barnhill, Theodore, Panagiotis Papapanagiotou, and Liliana Schumacher, 2002, “Measuring Integrated Market and Credit Risk in Bank Portfolios: An Application to a Set of Hypothetical Banks Operating in South Africa,” *Financial Markets, Institutions, and Instruments*, Vol. 11, No. 5, pp. 401–43.
- Barnhill, Theodore, and Liliana Schumacher, 2011, “Modeling Correlated Systemic Bank Liquidity Risks,” IMF Working Paper 11/263 (Washington: International Monetary Fund). Available

- via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=25356>
- Barnhill, Theodore, and Marcos R. Souto, 2009, "Systemic Bank Risk in Brazil: A Comprehensive Simulation of Correlated Market, Credit, Sovereign and Inter-Bank Risks," *Financial Markets, Institutions and Instruments*, Vol. 18, No. 4, pp. 243–83.
- Bhutta, Neil, Jane Dokko, and Hui Shan, 2010, "Consumer Ruthlessness and Strategic Default during the 2007–2009 Housing Bust," Board of Governors of the Federal Reserve (Washington, July). Available via the Internet: http://www.fdic.gov/bank/analytical/cfr/mortgage_future.../Dokko.pdf
- Black, Fischer, and Myron Scholes, 1973, "The Pricing of Options and Corporate Liabilities," *Journal of Political Economy*, Vol. 81, No. 3, pp. 637–54.
- Cantor, Richard Martin, and Praveen Varma, 2005, "Determinants of Recovery Rates on Defaulted Bonds and Loans for North American Corporate Issuers: 1983–2003," *Journal of Fixed Income*, Vol. 14, No. 4, pp. 29–44.
- Coval, Joshua, and Erik Stafford, 2007, "Asset Fire Sales (and Purchases) in Equity Markets," *Journal of Financial Economics*, Vol. 86, No. 2, pp. 479–512.
- Diamond, Douglas, and Philip Dybvig, 1983, "Bank Runs, Deposit Insurance, and Liquidity," *Journal of Political Economy*, Vol. 91, No. 3, pp. 401–19.
- Duffie, Darrell, Nicolae Gârleanu, and Lasse Pedersen, 2006, "Valuation in Over-the-Counter Markets," NBER Working Paper No. 12020 (Cambridge, Massachusetts: National Bureau of Economic Research). Available via the Internet: <http://www.nber.org/papers/w12020>
- Gorton, Gary, 1988, "Banking Panics and Business Cycles," *Oxford Economic Papers*, Vol. 40, No. 4, pp. 751–81.
- Gorton, Gary, and Andrew Metrick, 2009, "Securitized Banking and the Run on the Repo," NBER Working Paper No. 15223 (Cambridge, Massachusetts: National Bureau of Economic Research). Available via the Internet: <http://www.nber.org/papers/w15223>
- International Monetary Fund, 2008, *Global Financial Stability Report*, World Economic and Financial Surveys (Washington, October). Available via the Internet: <http://www.imf.org/external/pubs/ft/gfsr/2008/02/index.htm>
- , 2010, *Global Financial Stability Report*, World Economic and Financial Surveys (Washington, October). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2010/02/index.htm>
- , 2011, *Global Financial Stability Report*, World Economic and Financial Surveys (Washington, September). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2011/02/index.htm>
- Merton, Robert C., 1973, "Theory of Rational Option Pricing," *Bell Journal of Economics and Management Science*, Vol. 4, No. 1, pp. 141–83.
- Taylor, John B., 2008, "The Financial Crisis and the Policy Responses: An Empirical Analysis of What Went Wrong," in *A Festschrift in Honour of David Dodge's Contributions to Canadian Public Policy*, November, pp. 1–18 (Ottawa: Bank of Canada). Available via the Internet: <http://www.bankofcanada.ca/publications-research/research/conferences-seminars-and-workshops/festschrift-honour-david-dodge-contributions-canadian-public-policy>
- van den End, Jan Willem, 2008, "Liquidity Stress-Tester: A Macro Model for Stress-Testing Banks' Liquidity Risk," DNB Working Paper No. 175 (Amsterdam: De Nederlandsche Bank). Available via the Internet: <http://www.dnb.nl/en/publications/dnb-publications/dnb-working-papers-series/dnb-working-papers/auto175528.jsp>
- Wong, Eric, and Cho-Hoi Hui, 2009, "A Liquidity Risk Stress-Testing Framework with Interaction between Market and Credit Risks," HKMA Working Paper 06/2009, March (Hong Kong SAR: Hong Kong Monetary Authority). Available via the Internet: <http://www.hkma.gov.hk/eng/publications-and-research/research/working-papers/2009>

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Review and Implementation of Credit Risk Models

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The chapter presents the basic CreditRisk+ model along with extensions suggested in the existing literature and proposes some modifications. The purpose of these models is to determine the probability distribution of the losses on a portfolio of loans and other debt instruments so that they could be used for stress testing in the IMF's Financial Sector Assessment Program, as a benchmark for credit risk evaluations. First, we present the setting and basic definitions common to all the model specifications used in this chapter. Then, we proceed from the simplest model based on Bernoulli-distributed default events and known default probabilities to the full-fledged CreditRisk+ implementation. The latter is based on the Poisson approximation and uncertain default probabilities determined by mutually independent risk factors. As an extension, we present a CreditRisk+ specification with correlated risk factors as in Giese (2003). Finally, we illustrate the characteristics and the results obtained from the different models using a specific portfolio of obligors.

METHOD SUMMARY

Overview	The CreditRisk+ model along with extensions suggested in the existing literature can be used to determine the probability distribution of the losses on a portfolio of loans and other debt instruments.
Application	The ability to compute the loss distribution of a portfolio enables the determination of the credit value at risk and, therefore, the economic capital required by credit operations.
Nature of approach	Actuarial approach.
Data requirements	Individual exposures, the mean individual default probabilities, the standard deviations of the individual default probabilities, and the matrix of weights representing the exposure of each obligor to the risk factors.
Strengths	Under certain assumptions, the model provides an analytical solution for determining the loss distribution.
Weaknesses	• The default probabilities are assumed to be given by a linear multifactor model. • Unlike a probit or logit model, this model cannot restrict default probabilities from falling outside the [0,1] interval. • The analytical approximation may be imprecise if the probabilities of default are too high, that is, if they exceed single digits.
Tool	The Excel add-in is available in the toolkit, which is on the companion CD and at www.elibrary.imf.org/stress-test-toolkit . Contact author: A. Mirestean.

Over the last 15 years, we have witnessed major advances in the field of modeling credit risk. There are now three main approaches to quantitative credit risk modeling: the “Merton-style” approach, the purely empirical econometric approach, and the actuarial approach.¹ Each of these approaches has, in turn, produced several models that are widely used by financial institutions around the world. All these models share a common purpose: to determine the probability distribution

of the losses on a portfolio of loans and other debt instruments. Being able to compute the loss distribution of a portfolio is critical, because it allows the estimation of the credit value at risk (VaR) and, therefore, the economic capital required by credit operations.

In this chapter, we present the theoretical background that underpins one of the most frequently used models for loss distribution determination: CreditRisk+. This model, originally

This chapter is an abridged version of IMF Working Paper 06/134 (Avesani and others, 2006).

¹ See Koyluoglu and Hickman (1998) and Crouhy, Galai, and Mark (2000) for more details.

developed by Credit Suisse Financial Products (CSFP), is based on the actuarial approach and has quickly become one of the financial industry benchmarks in the field of credit risk modeling. Its popularity spilled over into the regulatory and supervisory community, prompting some supervisory authorities to start using it in their monitoring activities.² There are several reasons why this model has become so popular:

1. It requires a limited amount of input data and assumptions.
2. It uses as basic input the same data as required by the Basel II Internal Ratings Based (IRB) approach.
3. It provides an analytical solution for determining the loss distribution.
4. It brings out one of the most important credit risk drivers—concentration.

We illustrate our implementation of the model and suggest that it could be used as a toolbox in the different surveillance activities of the IMF. We also analyze the problems that may arise by directly applying CreditRisk+, in its original formulation, to certain portfolio compositions. We subsequently propose some solutions.

The chapter is organized as follows. Initially, we present the setting and basic definitions common to all the model specifications used in this chapter. Then, we gradually proceed from the simplest model based on Bernoulli-distributed default events and default probabilities known with certainty to the full-fledged version of CreditRisk+. The latter is based on the Poisson approximation and uncertain default probabilities determined by mutually independent risk factors. We then go beyond CreditRisk+ by presenting a specification that allows for correlation among risk factors, as in Giese (2003). We also apply the implemented models to a specific portfolio of exposures to illustrate their characteristics and discuss in detail the results. Finally, we present some conclusions.

1. THE BASIC MODEL SETTING

In this section, we present the setting and basic definitions common to all the model specifications used in the chapter. We consider a portfolio of N obligors indexed by $n = 1, \dots, N$. Obligor n constitutes an exposure E_n . The probability of default of obligor n over the period considered is p_n .

A. Default events

The default of obligor n can be represented by a Bernoulli random variable D_n such that the probability of default over a given period of time is $P(D_n = 1) = p_n$, while the probability of survival over the same span is $P(D_n = 0) = 1 - p_n$. Then the Probability Generating Function (PGF) of D_n is given by the following equation:³

$$G_{D,n}(z) = \sum_{x=0}^{\infty} P(D_n = x) \cdot z^x.$$

Because D_n can take only two values (0 and 1), $G_{D,n}(z)$ can be rewritten as follows:

$$G_{D,n}(z) = (1 - p_n) \cdot z^0 + p_n \cdot z^1 = (1 - p_n) + p_n \cdot z.$$

B. Losses

The loss on obligor n can be represented by the random variable $L_n = D_n \cdot E_n$. The probability distribution of L_n is given by $P(L_n = E_n) = p_n$ and $P(L_n = 0) = 1 - p_n$. The total loss on the portfolio is represented by the random variable L :

$$L = \sum_{n=1}^N L_n = \sum_{n=1}^N D_n \cdot E_n.$$

The objective is to determine the probability distribution of L under various sets of assumptions. Knowing the probability distribution of L will allow us to compute the VaR and other risk measures for the portfolio.

C. Normalized exposures

When implementing the model, it has become common practice to normalize and round the individual exposures and then group them in exposure bands. The process of normalization and rounding limits the number of possible values for L and hence reduces the time required to compute the probability distribution of L . When the normalization factor is small relative to the total portfolio exposure, the rounding error is negligible. Let F be the normalization factor.⁴ The rounded normalized exposure of obligor n is denoted v_n and is defined by the following:

$$v_n = \text{ceil}(E_n / F). \quad (10.1)$$

D. Normalized losses

The normalized loss on obligor n is denoted by λ_n and is defined by the following:

$$\lambda_n = D_n \cdot v_n$$

where D_n is the default and v_n is the normalized exposure for obligor n . Hence, λ_n is a random variable that takes value v_n with probability p_n , and value 0 with probability $1 - p_n$. The total normalized loss on the portfolio is represented by the random variable λ :

$$\lambda = \sum_{n=1}^N \lambda_n.$$

Finding the probability distribution of λ is equivalent to finding the probability distribution of L .

² See Austrian Financial Market Authority and Oesterreichische Nationalbank (2004).

³ Probability concepts are reviewed in the Appendix.

⁴ In a portfolio with high variation of the exposure levels, the smallest exposure could be chosen as the normalization factor.

E. Exposure bands

The use of exposure bands has become a common technique used in the literature to simplify the computational process. Once the individual exposures have been normalized and rounded, as shown in equation (10.1), common exposure bands can be defined in the following manner. The total number of exposure bands, J , is given by the highest value of the normalized individual exposures, $J = \max\{v_n\}_{n=1}^N$. Let j represent the index for the exposure bands. Then, the common exposure in band j is $v_j = j$. The distribution of obligors among exposure bands is set such that each obligor n is assigned to band j if $v_n = v_j = j$. Then, the expected number of defaults in band j , μ_j , is given by

$$\mu_j = \sum_{n, v_n=j} p_n.$$

Consequently, the total expected number of defaults in the portfolio, μ , is given by

$$\mu = \sum_{j=1}^J \mu_j.$$

2. A SIMPLE MODEL WITH NONRANDOM DEFAULT PROBABILITIES

In this section, we present a simple model with Bernoulli-distributed default events and nonrandom default probabilities. The advantage of this model is that it relies on the smallest set of assumptions. As a result, the loss distribution in this model can be efficiently computed as a simple convolution, without making any approximation for the distribution of default events. This approach is particularly appropriate when default probabilities are high and when there is little uncertainty concerning the values of these probabilities.

The key assumptions of the model are that individual default probabilities over the period considered are known with certainty and that default events are independent among obligors. The objective is to determine the probability distribution of the total normalized loss, λ , or, equivalently, the PGF of λ . Given the assumption of independence among obligors, the PGF of the total normalized loss, λ , can easily be computed from the PGF of the individual normalized loss, λ_n . As λ_n can take only the values 0 and v_n , the PGF of λ_n is given by the following:

$$G_{\lambda,n}(z) = P(\lambda = 0) \cdot z^0 + P(\lambda = v_n) \cdot z^{v_n} = (1 - p_n) + p_n \cdot z^{v_n}.$$

Taking in account that λ is the sum of λ_n over n and because default events are independent among obligors, the PGF of λ is simply given by

$$G_\lambda(z) = \prod_{n=1}^N G_{\lambda,n}(z) = \prod_{n=1}^N [(1 - p_n) + p_n \cdot z^{v_n}].$$

This product of the individual loss PGF is a simple convolution that can be computed using the fast Fourier transform (FFT). From the convolution theorem,

$$G_\lambda(z) = IFFT \left(\prod_{n=1}^N \{FFT[G_{\lambda,n}(z)]\} \right) \equiv G_{\lambda,B}(z),$$

where $IFFT$ is the inverse fast Fourier transform. This algorithm can be efficiently implemented as long as the portfolio does not contain more than a few thousand obligors.⁵ This is about as far as the model can be analytically developed under the assumption that the probability distributions of default events are Bernoulli distributions. In order to find a closed-form solution for the PGF of λ , when default events are represented by Bernoulli random variables, it is also necessary to assume that default events are independent among obligors and that default probabilities are known with certainty (nonrandom). Finding an analytical solution when probabilities are random and default events are no longer independent implies using an approximation for the distribution of the default events. This is precisely the path taken by the CreditRisk+ model when deriving closed-form expressions for the PGF of λ .

3. THE POISSON APPROXIMATION

In this section, we describe in detail one of the essential assumptions of the CreditRisk+ model: the individual probabilities of default are assumed to be sufficiently small for the compound Bernoulli distribution of the default events to be approximated by a Poisson distribution. This assumption makes it possible to obtain an analytical solution even when the default probabilities are not known with certainty.

Under the assumption that default events follow Bernoulli distribution, the PGF of obligor n 's default is

$$G_{D,n}(z) = (1 - p_n) + p_n \cdot z = 1 + p_n(z - 1).$$

Equivalently:

$$G_{D,n}(z) = \exp[\ln(G_{D,n}(z))] = \exp[\ln(1 + p_n(z - 1))]. \quad (10.2)$$

If we assume that p_n is very small, then $p_n(z - 1)$ is also very small under the assumption that $|z| \leq 1$. Defining $w = p_n(z - 1)$, we can perform a Taylor expansion of $\ln(1 + w)$ in the vicinity of $w = 0$:

$$\ln(1 + w) = w - \frac{w^2}{2} + \frac{w^3}{3} \dots$$

Neglecting the terms of order 2 and above and going back to the original notation yields the following:

$$\ln[1 + p_n(z - 1)] \approx p_n(z - 1). \quad (10.3)$$

The assumption that justifies neglecting the terms of order 2 and above is that p_n is "small": the smaller p_n , the smaller the (absolute) difference between $\ln[1 + p_n(z - 1)]$ and $p_n(z - 1)$.

Going back to equation (10.2) and using the approximation from equation (10.3), we have

$$G_{D,n}(z) \approx \exp[p_n(z - 1)].$$

⁵ See Section 9 for more details.

Performing again a Taylor expansion of $\exp[p_n(z-1)]$ in the vicinity of $p_n=0$ finally yields the following:

$$G_{D,n}(z) \approx \exp[p_n(z-1)] = \exp(-p_n) \sum_{x=0}^{\infty} \frac{p_n^x}{x!} z^x. \quad (10.4)$$

The last member of equation (10.4) is the PGF of a Poisson distribution with intensity p_n . Therefore, for small values of p_n , the Bernoulli distribution of D_n can be approximated by a Poisson distribution with intensity p_n .

Using the Poisson approximation, the probability distribution of D_n is then given by

$$P(D_n = x) = \exp(-p_n) \frac{p_n^x}{x!}.$$

A new expression for the PGF of the individual normalized loss λ_n can also be derived using the Poisson approximation in equation (10.4). The PGF of λ_n is defined by

$$G_{\lambda,n}(z) = E(z^{\lambda_n}) = E(z^{D_n v_n}), \quad (10.5)$$

where E is the expectation operator. Given that v_n is not random, equation (10.5) can be written as follows:

$$\begin{aligned} G_{\lambda,n}(z) &= \sum_{x=0}^{\infty} P(D_n = x) \cdot z^{D_n v_n} \\ &= \sum_{x=0}^{\infty} \exp(-p_n) \cdot \frac{p_n^x}{x!} \cdot (z^{v_n})^x. \end{aligned} \quad (10.6)$$

Given that D_n takes only two values (0 and 1), we can express the PGF of λ_n as

$$G_{\lambda,n}(z) = (1 + p_n z^{v_n}) \exp(-p_n). \quad (10.7)$$

The second term of the right side in equation (10.7) is the first-order Taylor expansion of $\exp(p_n z^{v_n})$, that is, $\exp(p_n z^{v_n}) \approx 1 + p_n z^{v_n}$. Therefore, for small values of probabilities p_n , the PGF of the individual normalized loss, λ_n , can be approximated by

$$G_{\lambda,n}(z) = \exp[p_n(z^{v_n} - 1)]. \quad (10.8)$$

4. THE MODEL WITH KNOWN PROBABILITIES REVISITED

In this section, we apply the Poisson approximation to the basic model with known default probabilities, as presented in Section 2. Hence, we derive a new expression for the PGF of the total normalized loss on the portfolio, $G_{\lambda}(z)$, based on the expression for $G_{\lambda,n}(z)$ from equation (10.8). Individual default probabilities are still assumed to be known with certainty (nonrandom), and default events are still assumed to be independent among obligors.

Given that individual losses are mutually independent, $G_{\lambda}(z)$ is simply the product of the individual loss PGF, as in Section 2:

$$G_{\lambda}(z) = \prod_{n=1}^N G_{\lambda,n}(z). \quad (10.9)$$

Replacing $G_{\lambda,n}(z)$ in equation (10.9) with the expression from equation (10.8), we have

$$\begin{aligned} G_{\lambda}(z) &= \prod_{n=1}^N \exp[p_n(z^{v_n} - 1)] \\ &= \exp \left[\sum_{n=1}^N p_n(z^{v_n} - 1) \right]. \end{aligned}$$

Using the definition of the exposure bands, as presented in Section 1, $G_{\lambda}(z)$ can be written as follows:

$$G_{\lambda}(z) = \exp \left[\sum_{j=1}^J \sum_{n: v_n = v_j} p_n(z^{v_j} - 1) \right].$$

This expression can be finally simplified using the definitions of the expected number of defaults in band j , μ_j , and the expected total number of defaults in the portfolio, μ :

$$\begin{aligned} G_{\lambda}(z) &= \exp \left[\sum_{j=1}^J \mu_j (z^{v_j} - 1) \right] \\ &= \exp \left[\mu \left(\sum_{j=1}^J \frac{\mu_j}{\mu} z^{v_j} - 1 \right) \right] \\ &\equiv G_{\lambda, \text{FIXED}}(z). \end{aligned} \quad (10.10)$$

Equation (10.10) defines the probability distribution of the total loss on the portfolio when default events are mutually independent, default probabilities are known with certainty, and Poisson distributions are used to approximate the distributions of individual default events.

5. RANDOM DEFAULT PROBABILITIES

In this section, we present the full-fledged version of Credit-Risk+. We do so by removing the assumptions used so far—that is, that the default probabilities are known with certainty and that default events are unconditionally mutually independent. Instead, we assume that individual default probabilities are random and are influenced by a common set of Gamma-distributed systematic risk factors. Consequently, the default events are assumed to be mutually independent only *conditional* on the realizations of the risk factors. Under these assumptions, the use of the Poisson approximation still allows us to obtain an analytical solution for the loss distribution.

Let us assume that there are K risk factors indexed by k . Each factor k is associated with a “sector” (an industry or a geographic zone, for example) and is represented by a random variable γ_k . The probability distribution of γ_k is assumed to be a Gamma distribution with shape parameter $\alpha_k = 1/\sigma_k^2$ and scale parameter $\beta_k = \sigma_k^2$. Therefore, the mean and the variance are given by the following:

$$\begin{aligned} E(\gamma_k) &= \alpha_k \cdot \beta_k = 1 \\ \text{Var}(\gamma_k) &= \alpha_k \cdot \beta_k^2 = \sigma_k^2. \end{aligned}$$

The moment generating function (MGF) of γ_k is the function defined by

$$M_{\gamma,k}(z) = E[\exp(\gamma_k \cdot z)] = (1 - \beta_k \cdot z)^{-\alpha_k} = (1 - \sigma_k^2 \cdot z)^{-1/\sigma_k^2}.$$

The random variables γ_k are assumed to be mutually independent. In addition, the default probability of obligor n is assumed to be given by the following model:

$$p_n = \bar{p}_n \left(\sum_{k=1}^K \gamma_k \omega_{k,n} \right), \quad (10.11)$$

where $\bar{p}_n$ is the average default probability of obligor n , and $\omega_{k,n}$ is the share of obligor n 's exposure in sector k (or the share of obligor n 's debt that is exposed to factor k). According to this model, the default probability of obligor n is a random variable with mean $\bar{p}_n$.

It is important to note that exposure to common risk factors introduces unconditional correlation between individual default events. Consequently, default events are no longer unconditionally mutually independent.⁶ However, individual default events—and therefore individual losses—are assumed to be mutually independent conditional on the set of factors $\gamma = (\gamma_1, \dots, \gamma_k, \dots, \gamma_K)$.

As in Sections 3 and 4, D_n is assumed to follow a Poisson distribution with intensity p_n , and the PGF of the individual normalized loss λ_n is assumed to be

$$G_{\lambda,n}(z) = \exp[p_n(z^{v_n} - 1)].$$

Using the multifactor model for p_n defined in equation (10.11), $G_{\lambda,n}(z)$ can be written as follows:

$$\begin{aligned} G_{\lambda,n}(z) &= \exp \left[\bar{p}_n \left(\sum_{k=1}^K \gamma_k \omega_{k,n} \right) (z^{v_n} - 1) \right] \\ &= \prod_{k=1}^K \exp \left[\bar{p}_n \gamma_k \omega_{k,n} (z^{v_n} - 1) \right]. \end{aligned}$$

Let $G_\lambda(z|\gamma)$ be the PGF of the total normalized loss, conditional on γ . Given that individual losses are mutually independent conditional on γ , $G_\lambda(z|\gamma)$ is simply the product of the individual loss PGF conditional on γ :

$$\begin{aligned} G_\lambda(z|\gamma) &= \prod_{n=1}^N G_{\lambda,n}(z|\gamma) \\ &= \prod_{n=1}^N \prod_{k=1}^K \exp \left[\bar{p}_n \gamma_k \omega_{k,n} (z^{v_n} - 1) \right] \\ &= \exp \left[\sum_{n=1}^N \sum_{k=1}^K \bar{p}_n \gamma_k \omega_{k,n} (z^{v_n} - 1) \right]. \end{aligned}$$

If we define $P_k(z)$ as

$$P_k(z) = \sum_{n=1}^N \bar{p}_n \omega_{k,n} (z^{v_n} - 1), \quad (10.12)$$

then $G_\lambda(z|\gamma)$ can be written as

$$G_\lambda(z|\gamma) = \exp \left[\sum_{k=1}^K \gamma_k \cdot P_k(z) \right].$$

Let E_γ denote the expectation operator under the probability distribution of γ , and let E_{γ_k} denote the expectation operator under the probability distribution of γ_k . The unconditional

PGF of the total normalized loss, denoted by $G_\lambda(z)$, is the expectation of $G_\lambda(z|\gamma)$ under γ probability distribution:

$$\begin{aligned} G_\lambda(z) &= E_\gamma [G_\lambda(z|\gamma)] \\ &= E_\gamma \left\{ \exp \left[\sum_{k=1}^K \gamma_k \cdot P_k(z) \right] \right\}. \end{aligned}$$

Define $P(z) = [P_1(z), \dots, P_k(z), \dots, P_K(z)]$. Using the definition of the joint MGF of γ , M_γ , we can write $G_\lambda(z)$ as follows:

$$G_\lambda(z) = E_\gamma \{ \exp [P(z) \cdot \gamma'] \} = M_\gamma [P(z)]. \quad (10.13)$$

It is important to note that this equation does not rely on the fact that γ_k is mutually independent.

Let us consider now a vector $\zeta = (\zeta_1, \dots, \zeta_k, \dots, \zeta_K)$ of auxiliary variables such that $0 \leq \zeta_k < 1$ for all $k = 1, \dots, K$. The joint MGF of the vector γ is given by

$$\begin{aligned} M_\gamma(\zeta) &= E_\gamma [\exp(\zeta \cdot \gamma')] = E_\gamma \left[\exp \left(\sum_{k=1}^K \zeta_k \cdot \gamma_k \right) \right] \\ &= E_\gamma \left[\prod_{k=1}^K \exp(\zeta_k \cdot \gamma_k) \right]. \end{aligned} \quad (10.14)$$

Recall that the MGF of γ_k is defined by

$$M_{\gamma,k}(\zeta_k) = E_{\gamma,k} [\exp(\gamma_k \cdot \zeta_k)] = (1 - \sigma_k^2 \cdot \zeta_k)^{-1/\sigma_k^2}.$$

Given that variables γ_k are mutually independent, $M_\gamma(\zeta)$ can be rewritten as follows:

$$\begin{aligned} M_\gamma(\zeta) &= \prod_{k=1}^K E_{\gamma,k} [\exp(\zeta_k \cdot \gamma_k)] \\ &= \prod_{k=1}^K M_{\gamma,k}(\zeta_k). \end{aligned}$$

Therefore, setting $\zeta_k = P_k(z)$, $G_\lambda(z)$ is given by

$$\begin{aligned} G_\lambda(z) &= \prod_{k=1}^K M_{\gamma,k}[P_k(z)] \\ &= \prod_{k=1}^K [1 - \sigma_k^2 \cdot P_k(z)]^{-1/\sigma_k^2}. \end{aligned} \quad (10.15)$$

Equation (10.15) can be further transformed as follows:

$$\begin{aligned} G_\lambda(z) &= \exp [\ln(G_\lambda(z))] \\ &= \exp \left\{ - \sum_{k=1}^K \frac{1}{\sigma_k^2} \ln [1 - \sigma_k^2 \cdot P_k(z)] \right\} \equiv G_{\lambda,CR+}(z). \end{aligned}$$

Hence, we obtain the PGF of the total normalized loss on the portfolio for the CreditRisk+ model, $G_{\lambda,CR+}(z)$. It is worth noting that $G_{\lambda,CR+}(z)$ is obtained under the following assumptions: default probabilities are random; the factors that determine default probabilities are Gamma distributed and mutually independent; and default events are mutually independent conditional on these factors.

6. LATENT FACTORS

In CreditRisk+, the factors γ_k are treated as latent variables. That is to say, the factors that influence the default probabilities are assumed to be unobservable. It is further assumed that

⁶ An approximate method for computing the correlation matrix of individual default events is presented in CSFP (1997).

the means and standard deviations of the default probabilities, as well as their sensitivities to the latent factors, are known or can be estimated without using observations of the factors.

The latent factors approach is required by the linear multifactor model for the default probabilities (equation (10.11)). This model has the advantage of enabling the derivation of an analytical solution for the loss distribution. However, unlike a probit or logit model, this model has little empirical relevance, because it allows default probabilities to exceed one (albeit with very low probability). If equation (10.11) were to be estimated using empirical observations of the factors γ_k , it is likely that the default probabilities implied by the econometric model would often fall outside the interval $[0,1]$. To avoid having to deal with unreasonable values for the default probabilities, instead of estimating equation (10.11) using observations of the factors γ_k , these factors are treated as unobserved latent variables.

In CreditRisk+, the means of the latent factors can be normalized to one without loss of generality. As a result, the latent factors play a role only via their standard deviations after normalization (σ_k , $k=1, \dots, K$), which can be estimated from the means, standard deviations, and sensitivities of the default probabilities. Therefore, when CreditRisk+ is implemented in practice, the inputs of the model are the mean individual default probabilities, denoted by $\bar{p}_n$, $n=1, \dots, N$; the standard deviations of the individual default probabilities, denoted by σ_n , $n=1, \dots, N$; and the matrix of weights $\{\omega_{k,n}\}$ for $k=1 \dots K$, $n=1 \dots N$, representing the exposure of each obligor n to each factor k . Then the σ_k are estimated from the σ_n .

Let us assume that some of the obligors in the portfolio are exposed to a single factor. For example, consider an obligor n who is exposed only to factor κ :

$$\begin{aligned}\omega_{\kappa,n} &= 1, \quad k = \kappa \\ \omega_{k,n} &= 0, \quad k \neq \kappa.\end{aligned}$$

The default probability of this obligor is given by

$$p_n = \bar{p}_n \cdot \gamma_\kappa.$$

Therefore,

$$\sigma_n = \bar{p}_n \cdot \sigma_\kappa.$$

Note that this equation implies that the ratio $\sigma_n/\bar{p}_n$ is the same for all obligors such that $\omega_{\kappa,n}=1$. Nevertheless, because this relationship holds for any obligor n such that $\omega_\kappa^n=1$, σ_κ can be computed from σ_n as follows:

$$\sigma_\kappa = \frac{1}{N_\kappa} \sum_{n, \omega_\kappa^n=1} \frac{\sigma_n}{\bar{p}_n},$$

where N_κ is the number of obligors such that $\omega_{\kappa,n}=1$. In the case where all obligors are exposed to more than one factor, the original CreditRisk+ implementation uses weighted averages of σ_n to estimate σ_k :

$$\begin{aligned}\sigma_k &= \frac{1}{\mu_k} \sum_{n=1}^N \omega_{k,n} \cdot \sigma_n = \frac{1}{\mu_k} \sum_{n=1}^N \omega_{k,n} \cdot \bar{p}_n \cdot \frac{\sigma_n}{\bar{p}_n} \\ \text{with } \mu_k &= \sum_{n=1}^N \omega_{k,n} \cdot \bar{p}_n.\end{aligned}\quad (10.16)$$

This expression is actually not consistent with the linear multifactor model for individual default probabilities used in CreditRisk+, and it results in an underestimation of σ_k . However, in practice, it gives reasonably accurate results, as will be shown in Section 10.

An alternative is to use least squares to estimate σ_k from σ_n . According to the multifactor model for the default probabilities, σ_k^2 is the solution of the following linear system:

$$\sigma_n^2 = \bar{p}_n^2 \left(\sum_{k=1}^K \sigma_k^2 \cdot \omega_{k,n} \right), \quad n = 1, \dots, N.$$

If $N \geq K$, then a solution to this system can be found using constrained least squares. However, there is no way to guarantee that σ_k^2 , $k=1 \dots K$ will all be strictly positive. The least-squares method and the weighted average method are compared in Section 10.

7. EXTENSION OF CREDITRISK+ WITH CORRELATED FACTORS

One of the limitations of CreditRisk+ is the assumption that the factors determining the default probabilities are mutually independent. In this section, following the approach developed by Giese (2003), we integrate the factors correlation in the model. One of the main limitations of CreditRisk+ is the assumption that the factors that determine default probabilities are mutually independent. Giese (2003) has derived an extension of CreditRisk+ that allows for some form of correlation between the factors.

In CreditRisk+, the k th factor γ_k follows a Gamma distribution with nonrandom shape and scale parameters (α_k and β_k , respectively). In other words, CreditRisk+ assumes that the means and variances of γ_k are nonrandom. Giese (2003) introduces an additional factor, which will be denoted by Γ , that affects the distributions of all γ_k , thereby introducing some correlation between the factors. Γ is assumed to follow a Gamma distribution with shape parameter $\alpha=1/\sigma^2$ and scale parameter $\beta=\sigma^2$. Consequently, the mean and variance are given by

$$\begin{aligned}E(\Gamma) &= \alpha \cdot \beta = 1 \\ \text{Var}(\Gamma) &= \alpha \cdot \beta^2 = \sigma^2.\end{aligned}$$

Now variables γ_k are assumed to be mutually independent *conditional* on Γ .

The probability distribution of γ_k is still a Gamma distribution with shape parameter α_k and scale parameter β_k . However, while β_k is still constant, α_k is now a function of Γ and hence a random variable:

$$\alpha_k = \Gamma \cdot \bar{\alpha}_k,$$

where $\bar{\alpha}_k$ is a constant.

Let E_Γ denote the expectation under the Γ distribution, and let E_{γ_k} denote the expectation under the γ_k distribution conditional on Γ . Similarly, let Var_Γ denote the variance under the Γ distribution, and let Var_{γ_k} denote the variance under the γ_k distribution conditional on Γ .

The expectation of γ_k conditional on Γ is given by

$$E_{\gamma,k}(\gamma_k | \Gamma) = \alpha_k \cdot \beta_k = \Gamma \cdot \bar{\alpha}_k \cdot \beta_k.$$

The unconditional expectation of γ_k is given by

$$E(\gamma_k) = E_{\Gamma}[E_{\gamma,k}(\gamma_k | \Gamma)] = E_{\Gamma}(\alpha_k \cdot \beta_k) = E_{\Gamma}(\Gamma \cdot \bar{\alpha}_k \cdot \beta_k).$$

Given that $\bar{\alpha}_k$ and β_k are not random, $E(\gamma_k)$ can be rewritten as follows:

$$E(\gamma_k) = \bar{\alpha}_k \cdot \beta_k \cdot E_{\Gamma}(\Gamma) = \bar{\alpha}_k \cdot \beta_k.$$

The unconditional expectation of γ_k is assumed, without loss of generality, to be equal to 1, so that $\beta_k = 1/\bar{\alpha}_k$. The variance of γ_k conditional on Γ is given by

$$\sigma_k^2 = \text{Var}_{\gamma,k}(\gamma_k | \Gamma) = \alpha_k \cdot \beta_k^2 = \Gamma \cdot \bar{\alpha}_k \cdot \beta_k^2 = \Gamma / \bar{\alpha}_k.$$

The unconditional variance of γ_k is given by

$$\begin{aligned} \text{Var}(\gamma_k) &= E_{\Gamma}[\text{Var}_{\gamma,k}(\gamma_k | \Gamma)] + \text{Var}_{\Gamma}[E_{\gamma,k}(\gamma_k | \Gamma)] \\ &= E_{\Gamma}\left(\frac{1}{\bar{\alpha}_k} \cdot \Gamma\right) + \text{Var}_{\Gamma}(\Gamma \cdot \bar{\alpha}_k \cdot \beta_k) \\ &= \beta_k \cdot E_{\Gamma}(\Gamma) + \bar{\alpha}_k \cdot \beta_k \cdot \text{Var}_{\Gamma}(\Gamma) = \beta_k + \sigma^2. \end{aligned}$$

The unconditional covariance between any two factors γ_k and γ_l is given by

$$\begin{aligned} \text{Cov}(\gamma_k, \gamma_l) &= E(\gamma_k \cdot \gamma_l) - E(\gamma_k) \cdot E(\gamma_l) \\ &= E_{\Gamma}[E_{\gamma,k,l}(\gamma_k \cdot \gamma_l | \Gamma)] - E(\gamma_k) \cdot E(\gamma_l) \end{aligned}$$

Given that γ_k and γ_l are independent conditional on Γ , this expression becomes

$$\begin{aligned} \text{Cov}(\gamma_k, \gamma_l) &= E_{\Gamma}[E_{\gamma,k}(\gamma_k | \Gamma) \cdot E_{\gamma,l}(\gamma_l | \Gamma)] - E(\gamma_k) \cdot E(\gamma_l) \\ &= \bar{\alpha}_k \cdot \beta_k \cdot \bar{\alpha}_l \cdot \beta_l \cdot [E(\Gamma^2) - 1] \\ &= \bar{\alpha}_k \cdot \beta_k \cdot \bar{\alpha}_l \cdot \beta_l \cdot [\text{Var}(\Gamma) + E(\Gamma)^2 - 1]. \end{aligned}$$

Given that $\bar{\alpha}_k \cdot \beta_k = \bar{\alpha}_l \cdot \beta_l = 1$ and $E(\Gamma)^2 = 1$, the unconditional covariance between γ_k and γ_l is simply represented as

$$\text{Cov}(\gamma_k, \gamma_l) = \sigma^2 > 0.$$

Consider a vector $Z = (z_1, \dots, z_k, \dots, z_K)$ of auxiliary variables such that $0 \leq z_k < 1$ for $k = 1, \dots, K$. The joint MGF of γ is the function M_{γ} defined by

$$M_{\gamma}(Z) = E_{\Gamma}\left[\exp\left(\sum_{k=1}^K \gamma_k z_k\right)\right].$$

Using the properties of conditional expectations, $M_{\gamma}(Z)$ can be rewritten as follows:

$$M_{\gamma}(Z) = E_{\gamma}\left\{E_{\gamma}\left[\exp\left(\sum_{k=1}^K \gamma_k z_k\right) \middle| \Gamma\right]\right\},$$

where E_{γ} is the expectation under the γ joint distribution.

Given that γ_k is mutually independent conditional on Γ , $M_{\gamma}(Z)$ can be further rewritten as follows:

$$\begin{aligned} M_{\gamma}(Z) &= E_{\Gamma}\left[\prod_{k=1}^K E_{\gamma,k}[\exp(\gamma_k z_k) | \Gamma]\right] \\ &= E_{\Gamma}\left[\prod_{k=1}^K M_{\gamma,k}(z_k | \Gamma)\right], \end{aligned}$$

where $M_{\gamma,k}(\cdot | \Gamma)$ is the MGF of γ_k conditional on Γ :

$$M_{\gamma,k}(z | \Gamma) = E_{\gamma,k}[\exp(\gamma_k \cdot z) | \Gamma] = (1 - \beta_k \cdot z)^{-\alpha_k} = (1 - \beta_k \cdot z)^{-\Gamma \bar{\alpha}_k}.$$

$M_{\Gamma}(Z)$ can now be rewritten as follows:

$$\begin{aligned} M_{\Gamma}(Z) &= E_{\Gamma}\left\{\exp\left(\log\left[\prod_{k=1}^K M_{\gamma,k}(z_k | \Gamma)\right]\right)\right\} \\ &= E_{\Gamma}\left\{\exp\left(\sum_{k=1}^K \log[M_{\gamma,k}(z_k | \Gamma)]\right)\right\} \\ &= E_{\Gamma}\left\{\exp\left[-\sum_{k=1}^K \Gamma \cdot \bar{\alpha}_k \cdot \ln(1 - \beta_k \cdot z_k)\right]\right\} \\ &= E_{\Gamma}\left\{\exp\left[-\Gamma \cdot \sum_{k=1}^K \frac{1}{\beta_k} \cdot \ln(1 - \beta_k \cdot z_k)\right]\right\}. \end{aligned}$$

Define a new auxiliary variable t :

$$t = -\sum_{k=1}^K \frac{1}{\beta_k} \cdot \ln(1 - \beta_k \cdot z_k).$$

Using t , $M_{\gamma}(Z)$ can be rewritten as the MGF of Γ , denoted by M_{Γ} :

$$M_{\gamma}(Z) = E_{\Gamma}[\exp(\Gamma \cdot t)] = M_{\Gamma}(t) = -\frac{1}{\sigma^2} \ln(1 - \sigma^2 \cdot t).$$

Replacing t with its expression gives us the final expression for $M_{\gamma}(Z)$:

$$M_{\gamma}(Z) = \exp\left\{-\frac{1}{\sigma^2} \ln\left[1 + \sigma^2 \sum_{k=1}^K \frac{1}{\beta_k} \ln(1 - \beta_k \cdot z_k)\right]\right\}.$$

Recall equation (10.13) established in Section 5:

$$G_{\lambda}(z) = M_{\gamma}[P(z)].$$

This equation is still valid in the context of this section, and, when M_{γ} is replaced with its new expression, it gives us the PGF of the total normalized loss in the model with correlated sectors:

$$\begin{aligned} G_{\lambda}(z) &= \exp\left\{-\frac{1}{\sigma^2} \ln\left[1 + \sigma^2 \sum_{k=1}^K \frac{1}{\beta_k} \ln(1 - \beta_k \cdot P_k(z))\right]\right\} \\ &\equiv G_{\lambda, \text{CORR}}(z), \end{aligned}$$

with

$$P_k(z) = \sum_{n=1}^N \bar{p}_n \omega_k^n (z^{v_n} - 1).$$

8. MODEL SUMMARY

In this section, we provide a summary presentation for the loss distributions of the various models described so far. The following equations show the four expressions for the PGF of the normalized loss on the portfolio. Each expression corresponds to the PGF of a particular model:

$$G_{\lambda, B}(z) = \text{IFFT}\left(\prod_{n=1}^N \{\text{FFT}[G_{\lambda, n}(z)]\}\right)$$

$$G_{\lambda, \text{FIXED}}(z) = \exp\left[\mu \left(\sum_{j=1}^J \frac{\mu_j}{\mu} z^{v_j} - 1\right)\right]$$

$$G_{\lambda, CR+}(z) = \exp \left\{ - \sum_{k=1}^K \frac{1}{\sigma_k^2} \ln[1 - \sigma_k^2 \cdot P_k(z)] \right\}$$

$$G_{\lambda, CORR}(z) = \exp \left\{ - \frac{1}{\sigma^2} \ln \left[1 + \sigma^2 \sum_{k=1}^K \frac{1}{\beta_k} \ln(1 - \beta_k \cdot P_k(z)) \right] \right\}$$

- $G_{\lambda, B}(z)$ is the PGF of the normalized loss on the portfolio in the “basic” model with Bernoulli default events and nonrandom default probabilities;
- $G_{\lambda, FIXED}(z)$ is the PGF of the normalized loss on the portfolio in the model with fixed default probabilities as approximated by a Poisson distribution;
- $G_{\lambda, CR+}(z)$ is the PGF of the normalized loss on the portfolio in the model with random probabilities and mutually independent factors; this is the CreditRisk+ model;
- $G_{\lambda, CORR}(z)$ is the PGF of the normalized loss on the portfolio in the model with random probabilities and correlated factors; this is an extension of the CreditRisk+ model.

The basic model with Bernoulli default events and nonrandom default probabilities is the only one that does not rely on the Poisson approximation. Both $G_{\lambda, B}(z)$ and $G_{\lambda, FIXED}(z)$ were derived under the assumption that default probabilities are nonrandom. The Poisson approximation is the only source of discrepancies between the resulting loss distributions in these two models. Therefore, the accuracy of that approximation can be evaluated by comparing $G_{\lambda, FIXED}(z)$ and $G_{\lambda, B}(z)$.

9. NUMERICAL IMPLEMENTATION

In this section, we discuss numerical issues that arise in the implementation of the models described above, and then we present two algorithms that address these problems. As explained in Section 2, the PGF of the portfolio loss in the basic model with Bernoulli default events and fixed probabilities is a simple convolution of the individual loss PGFs. The models based on the Poisson approximation (including CreditRisk+) are implemented using more sophisticated algorithms.

The original algorithm proposed by CSFP (1997) to implement the CreditRisk+ model is based on a recursive formula known as the Panjer recursion.⁷ In the context of credit risk models, the usefulness of the Panjer recursion is limited by two numerical issues. Both issues arise from the fact that a computer does not have infinite precision.

- First, the Panjer recursion cannot deal with arbitrarily large numbers of obligors: as the expected number of defaults in the portfolio increases, the computation of the first term of the recursion becomes increasingly imprecise; above a certain value for the expected number of defaults in the portfolio, the value found for the first term of the recursion becomes meaningless.

- Second, the Panjer recursion is numerically unstable in the sense that numerical errors accumulate as more terms in the recursion are computed. This can result in significant errors in the upper tail of the loss distribution and hence in the computation of the portfolio's VaR.

This section briefly presents two alternative algorithms that can be used to implement CreditRisk+ and the other models based on the Poisson approximation.⁸

- The first algorithm is an alternative recursive scheme introduced by Giese (2003). Haaf, Reiss, and Schoenmakers (2003) have shown that this algorithm is numerically stable, in the sense that precision errors are not propagated and amplified by the recursive formulas. This algorithm can provide a unified implementation of all versions of the model. However, like the Panjer recursion, this algorithm can fail for very large numbers of obligors.
- The second algorithm is attributable to Melchiori (2004). It is based on the fast Fourier transform (FFT) and can deal with very large numbers of obligors. It can easily be applied to the model with nonrandom default probabilities and to the model with random probabilities and uncorrelated factors.

A. Alternative recursive scheme

Recall the definition of the PGF of the total normalized loss on the portfolio:

$$G_{\lambda}(z) = \sum_{x=0}^{\infty} P(\lambda = x) \cdot z^x.$$

$G_{\lambda}(z)$ is a polynomial function of the auxiliary variable z . The coefficients of this polynomial are the probabilities associated with all the possible values for λ . One way to implement a particular version of the model is to derive a polynomial representation for the corresponding version of $G_{\lambda}(z)$ and to compute the coefficients of that polynomial.

Note that all three versions of $G_{\lambda}(z)$ derived using the Poisson approximation involve exponential and/or logarithmic transformations of polynomials in z .⁹ The first step in the computation of $G_{\lambda, FIXED}(z)$ is the computation of the coefficients of $G_I(z)$. Similarly, the first step in the computation of $G_{\lambda, CR+}(z)$ and $G_{\lambda, CORR}(z)$ is the computation of the coefficients of $P_k(z)$, for $k=1, \dots, K$. Once these coefficients have been determined, exponential and logarithmic transformations of $G_I(z)$ and/or $P_k(z)$ must be computed.

Exponential and logarithmic transformations of polynomials can be computed using recursive formulas derived from

⁷ See CSFP (1997).

⁸ Gordy (2002) presents a third algorithm based on saddlepoint approximation of the loss distribution. That algorithm computes only the cumulative distribution function of the loss distribution—not the probability distribution function. Furthermore, it is much more accurate for large portfolios than for small ones.

⁹ See Section 8.

the power series representations of the exponential and logarithm functions.

Consider two polynomials of degree $x_{\max}$ in z , $Q(z)$ and $R(z)$:

$$Q(z) = \sum_{x=0}^{x_{\max}} q_x \cdot z^x$$

$$R(z) = \sum_{x=0}^{x_{\max}} r_x \cdot z^x.$$

If $R(z) = \exp[Q(z)]$, the coefficients of $R(z)$ can be computed using the following recursive formula:

$$r_0 = \exp(q_0)$$

$$r_m = \sum_{s=1}^m \frac{s}{m} \cdot q_s \cdot r_{m-s}.$$

If $R(z) = \ln[Q(z)]$, the coefficients of $R(z)$ can be computed using the following recursive formula:

$$r_0 = \ln(q_0)$$

$$r_m = \frac{1}{q_0} \left[q_m - \sum_{s=1}^{m-1} \frac{s}{m} \cdot q_s \cdot r_{m-s} \right].$$

These recursive formulas can be used to compute the coefficients of $G_{\lambda, \text{FIXED}}(z)$, $G_{\lambda, \text{CR+}}(z)$, and $G_{\lambda, \text{CORR}}(z)$.

This algorithm produces accurate results for portfolios such that

$$\sum_{n=1}^N \omega_k^n \cdot \bar{p}_n < \mu_k^{\max} \approx 750 \quad k = 1, \dots, K. \quad (10.17)$$

For example, if $K=5$, $\bar{p}_n = 0.05$ for all obligors, and $\omega_{n,k} = 0.2$ for all obligors and all sectors, then the maximum number of obligors in the portfolio is approximately 75,000.

B. Fast Fourier transform–based algorithm

To describe the FFT-based algorithm proposed by Melchiori (2004), we will use the PGF of the portfolio loss for the model with Poisson default events and nonrandom default probabilities:

$$G_{\lambda, \text{FIXED}}(z) = \exp \left[\mu \left(\sum_{j=1}^J \frac{\mu_j}{\mu} z^{v_j} - 1 \right) \right].$$

Define the function $G_l(z)$ as follows:

$$G_l(z) = \sum_{j=1}^J \frac{\mu_j}{\mu} z^{v_j}.$$

Note that $G_l(z)$ can be interpreted as the PGF of a random variable l such that

$$P(l = v_j) = \frac{\mu_j}{\mu} \equiv \pi_j.$$

Using the definition of $G_l(z)$, $G_{\lambda, \text{FIXED}}(z)$ can be rewritten as follows:

$$G_{\lambda, \text{FIXED}}(z) = \exp \{ \mu [G_l(z) - 1] \}. \quad (10.18)$$

The characteristic function of a random variable X is defined as

$$\Phi_X(z) = E_X[\exp(i \cdot X \cdot z)] = G_X[\exp(i \cdot z)], \quad 0 \leq z < 1,$$

where E_X is the expectation under the X probability distribution, and G_X is the PGF of X .

Using equation (10.18) and this definition, we obtain the following expression for the characteristic function of the normalized portfolio loss:

$$\begin{aligned} \Phi_{\lambda, \text{FIXED}}(Z) &= G_{\lambda, \text{FIXED}}[\exp(i \cdot z)] \\ &= \exp \{ \mu [G_l(\exp(i \cdot z)) - 1] \} \\ &= \exp \{ \mu [\Phi_l(z) - 1] \}, \end{aligned} \quad (10.19)$$

where $\Phi_l(z)$ is the characteristic function of l .

The characteristic function of a random variable can also be defined as the Fourier transform of its density. This fact and equation (10.19) suggest a simple and efficient algorithm for computing the portfolio loss distribution.

Let Π denote the vector of probabilities representing the distribution of λ , and let $\pi = (\pi_1, \dots, \pi_J, \dots, \pi_J)$ denote the vector of probabilities representing the distribution of l . Π can be computed as follows:

$$\Pi = \text{IFFT}[\exp \{ \mu [\text{FFT}(\pi) - 1] \}],$$

where FFT is the fast Fourier transform, and IFFT is the inverse fast Fourier transform. This algorithm can easily be extended to the model with random default probabilities and uncorrelated factors. It has been successfully applied to a portfolio containing 679,000 obligors.

10. NUMERICAL EXAMPLES USING THE CREDIT RISK TOOLBOX

All the models discussed in this chapter may be implemented in MATLAB. In this section, we are going to present a very short description of the capabilities of the toolbox, and then we will offer some numerical examples. To illustrate the models presented in this chapter, we use the same portfolio as in the CreditRisk+ demonstration spreadsheet from CSFP. This portfolio is presented in Table 10.1:

- In order to run the basic model with fixed probabilities, whether the distribution of the default events is assumed to be Bernoulli or Poisson, one needs to have a set of individual exposures and a set of individual default probabilities.¹⁰ This corresponds to the first two columns in Table 10.1. If data are available only in aggregate form for different classes of obligors, the programs can still be used by making additional assumptions. For example, obligors can be classified according

¹⁰ There are three main approaches to estimating the probabilities of default. One approach is to use historical frequencies of default events for different classes of obligors in order to infer the probability of default for each class. Alternatively, econometric models such as logit and probit could be used to estimate probabilities of default. Finally, when available, one could use probabilities implied by market rates and spreads.

TABLE 10.1

Credit Suisse Financial Products Reference Portfolio

Obligor Exposure	Mean Default Rate	Standard Deviation of Default Rate	Factor Weights				
			Sector 1	Sector 2	Sector3	Sector4	Total
358,475	30.00%	15.00%	50%	30%	10%	10%	100%
1,089,819	30.00%	15.00%	25%	25%	25%	25%	100%
1,799,710	10.00%	5.00%	25%	25%	20%	30%	100%
1,933,116	15.00%	7.50%	75%	5%	10%	10%	100%
2,317,327	15.00%	7.50%	50%	10%	10%	30%	100%
2,410,929	15.00%	7.50%	50%	20%	10%	20%	100%
2,652,184	30.00%	15.00%	25%	10%	10%	55%	100%
2,957,685	15.00%	7.50%	25%	25%	20%	30%	100%
3,137,989	5.00%	2.50%	25%	25%	25%	25%	100%
3,204,044	5.00%	2.50%	75%	10%	5%	10%	100%
4,727,724	1.50%	0.75%	50%	10%	10%	30%	100%
4,830,517	5.00%	2.50%	50%	20%	10%	20%	100%
4,912,097	5.00%	2.50%	25%	25%	25%	25%	100%
4,928,989	30.00%	15.00%	25%	10%	10%	55%	100%
5,042,312	10.00%	5.00%	25%	25%	30%	20%	100%
5,320,364	7.50%	3.75%	75%	10%	5%	10%	100%
5,435,457	5.00%	2.50%	50%	20%	10%	20%	100%
5,517,586	3.00%	1.50%	50%	10%	10%	30%	100%
5,764,596	7.50%	3.75%	25%	25%	20%	30%	100%
5,847,845	3.00%	1.50%	25%	10%	10%	55%	100%
6,466,533	30.00%	15.00%	25%	25%	20%	30%	100%
6,480,322	30.00%	15.00%	75%	10%	5%	10%	100%
7,727,651	1.60%	0.80%	25%	25%	20%	30%	100%
15,410,906	10.00%	5.00%	50%	20%	10%	20%	100%
20,238,895	7.50%	3.75%	75%	10%	10%	5%	100%

Source: CSFP (1997).

to exposure, rating, or type (corporate vs. individual, for instance). When the average exposure and the average default probability are the only data available in each class, then the program assumes that all obligors in a given class have the same exposure and the same default probability.

- Computation time for the model with Bernoulli default events and fixed probabilities is determined by three variables: the number of obligors, the total normalized exposure, and the granularity of the exposure bands. The last two factors can be adjusted to reduce computation time, at the expense of accuracy. Generally speaking, the convolution can be efficiently computed in MATLAB as long as the number of obligors does not exceed a few thousand.
- This model has the advantage of being the most accurate when the default probabilities can be treated as nonrandom. As shown later on in this section, this is an important consideration when the default probabilities are relatively high. The model of Section 4 with fixed probabilities and Poisson default events has been implemented using both algorithms presented in Section 9.
- When running the CreditRisk+ model as described in Sections 5 and 6, one needs to have available the mean individual default probabilities, the standard deviations of the individual default probabilities, and

the matrix of weights, representing the exposure of each obligor n to each factor k . As mentioned above, if individual data are not available, the programs can use aggregate data for different classes of obligors by making additional assumptions. CreditRisk+ has also been implemented using both algorithms presented in Section 9.

- When running the CreditRisk+ model with its extension as described in Section 7, in addition to the data required to implement CreditRisk+, this model requires a value for the intersector covariance. Thus, one needs to have available the mean individual default probabilities, the standard deviations of the individual default probabilities, the matrix of weights, and a value for the intersector covariance. As mentioned above, if individual data are not available, the programs can use aggregate data for different classes of obligors by making additional assumptions. The model with correlated factors of Section 7 has been implemented using the numerically stable recursive algorithm. Consequently, it presents a limitation on the total number of obligors it can handle.¹¹

The exposures in Table 10.1 correspond to E_n in this study; the mean default rates correspond to p_n in the case with non-random default probabilities and to $\bar{p}_n$ in the case with random default probabilities; the standard deviations corre-

¹¹ See Section 9 for more details.

TABLE 10.2

Summary of the Assumptions Used in the Different Models

Model	Default Events	Default Probabilities	Correlated Factors	Latent Factors	σ_k^2 Estimation Method
1	Bernoulli	Fixed	N/A	N/A	N/A
2	Poisson	Fixed	N/A	N/A	N/A
3	Poisson	Random	No	Yes	Averages
4	Poisson	Random	No	Yes	LS
5	Poisson	Random	Yes	Yes	Averages

Source: Authors.

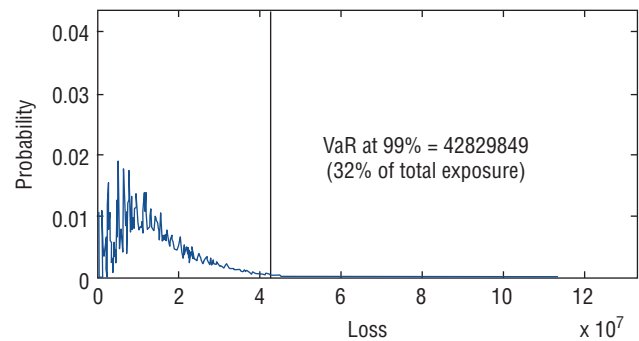
Note: LS=least squares.

spond to the values of σ_n in the models with random probabilities. The toolbox treats the factors as latent variables, and the values for σ_k are estimated from the values of σ_n . Two estimation methods are compared: least squares (LS) on the one hand and the weighted averages used by CSFP on the other hand. The standard deviations and the sector weights are required only to implement the models with random default probabilities; the models with nonrandom default probabilities use only the exposures and the mean default probabilities. Also, in general, individual exposures and default probabilities are not required to run the model; aggregate data per class of obligor can be used instead by making additional assumptions.

The loss distribution for the CSFP sample portfolio is computed using five models. The assumptions underlying these models are summarized in Table 10.2. Model 3 is CSFP's CreditRisk+. By default, the CSFP implementation treats Sector 1 as a special sector representing diversifiable idiosyncratic risk. This cannot be done in the model with correlated factors. Therefore, all the models discussed in this section treat Sector 1 as a regular sector (which is also an option in CSFP's CreditRisk+). The portfolio loss distribution is also computed by Monte Carlo simulation under the following assumptions: Bernoulli default events; random default probabilities; and independent Gamma-distributed factors, with σ_k^2 estimated by LS. This loss distribution computed by Monte Carlo simulation constitutes a benchmark. It can be used to assess the impact of the Poisson approximation and of the σ_k^2 estimation method on the models' accuracy.

A. Effects of Poisson approximation: nonrandom default probabilities

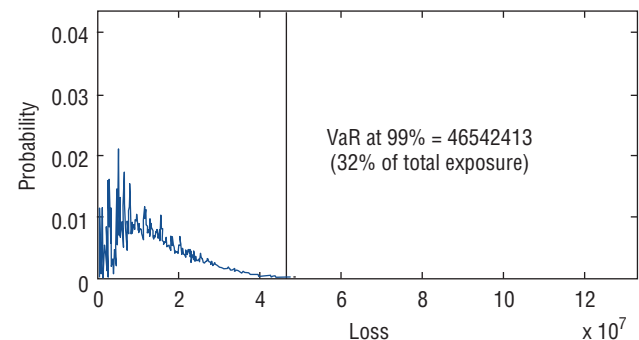
To assess the impact of the Poisson approximation, we first compare Models 1 and 2. The only difference between these two models is the distribution of default events: Bernoulli for Model 1, Poisson for Model 2. Figures 10.1 and 10.2 present the portfolio's loss distributions for these two models, respectively. They also show the VaR at the 99 percent level for each model. The VaR is 8.67 percent larger in the model with Poisson defaults than in the model with Bernoulli defaults. This is the error introduced by the Poisson approximation for this particular portfolio with 25 obligors and with default probabilities ranging from 3 percent to 30 percent.



Source: Authors.

Note: VaR=value at risk.

Figure 10.1 Model 1: Fixed Probabilities, Bernoulli Defaults



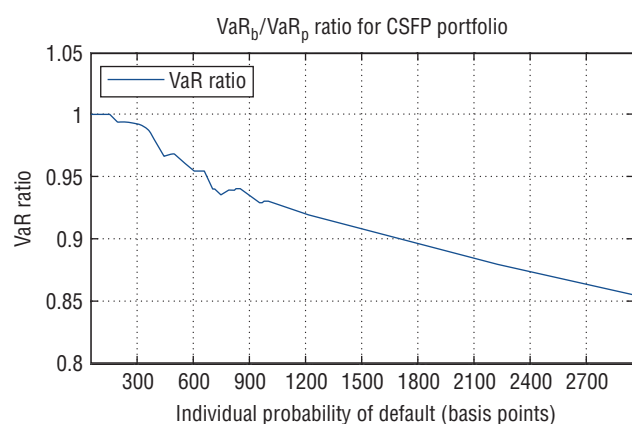
Source: Authors.

Note: VaR=value at risk.

Figure 10.2 Model 2: Fixed Probabilities, Poisson Defaults

The Poisson approximation generally results in an overestimation of the VaR. This result is observed for portfolios with very different structures.

An additional numerical experiment was performed to illustrate the relationship between the magnitude of the default probabilities and the error due to the Poisson approximation. This experiment uses the same exposures as in the CSFP portfolio, but it assumes that all obligors have the same default probability. The ratio between the VaR_b when the defaults are assumed to follow a Bernoulli distribution and the VaR_p when the defaults are assumed to follow a Poisson distribution, (VaR_b/VaR_p) , is computed for multiple values of



Source: Authors.

Note: CSFP = Credit Suisse Financial Products; VaR = value at risk.

Figure 10.3 Ratio of Bernoulli VaR to Poisson VaR for CSFP Portfolio

the common default probability, ranging from 1 percent to 30 percent. The results of this experiment are presented in Figure 10.3. One can see that VaR_b/VaR_p decreases steadily from 1 to 0.86 as the common default probability increases from 1 percent to 30 percent. In other words, as could be expected from the derivation of the Poisson approximation, the error due to this approximation increases with the magnitude of the default probability.

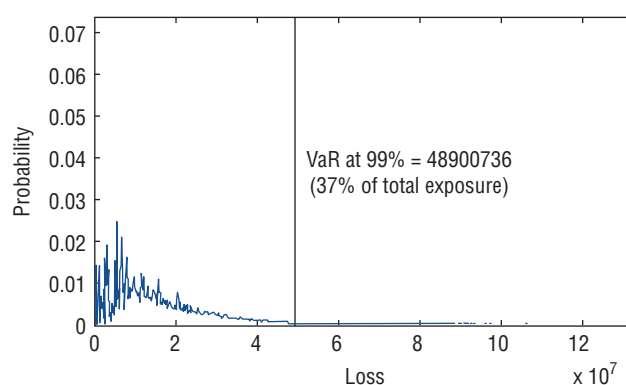
B. Random default probabilities, uncorrelated factors

This section compares the loss distributions for Models 3 and 4 to the loss distribution computed by Monte Carlo simulation. Models 3 and 4 differ only by the method used to estimate σ_k : Model 3 uses the CSFP weighted average; Model 4 uses LS. With CSFP's portfolio, σ_k^2 estimated by least squares is strictly positive. Both models assume Poisson default events and random default probabilities driven by Gamma-distributed factors.

Monte Carlo simulation is used to estimate the loss distribution of a model with random default probabilities driven by Gamma-distributed factors but with Bernoulli default events. Monte Carlo simulation is required because there is no analytical solution for the loss distribution when Bernoulli defaults are combined with random default probabilities. To perform the Monte Carlo simulation, σ_k is estimated from σ_n using least squares. Five thousand random draws of γ_k are then performed to determine p_n . For each set of γ_k , 5,000 random draws of D_n (the Bernoulli random variables representing default events) are then generated. Therefore, overall, 25 million random combinations are used for the Monte Carlo simulation. The loss distribution computed by Monte Carlo simulation is presented in Figure 10.4.

The loss distributions for Models 3 and 4 are presented in Figures 10.5 and 10.6, respectively:

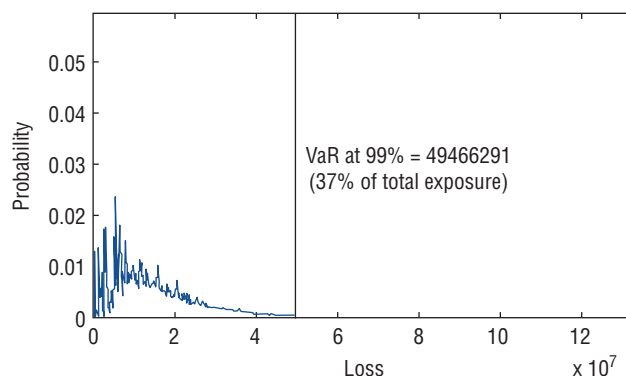
- The VaR for Model 4 is 9 percent higher than the VaR computed by Monte Carlo simulation. This is



Source: Authors.

Note: VaR = value at risk.

Figure 10.4 Bernoulli Defaults, Random Probabilities, Monte Carlo Simulation



Source: Authors.

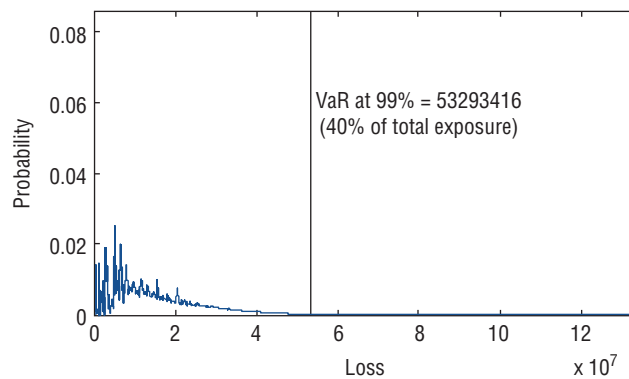
Note: VaR = value at risk.

Figure 10.5 Model 3: Poisson Defaults, Uncorrelated Factors, Weighted Average

more evidence of the fact that the Poisson estimation results in an overestimation of the VaR: the only difference between Model 3 and the model used for the Monte Carlo simulations is the distribution of default events.

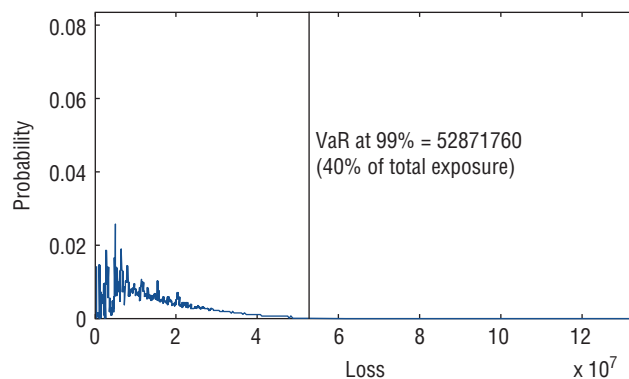
- The VaR in Model 3 (which does not use a rigorous method to estimate σ_n) is only 1 percent higher than the VaR computed by Monte Carlo simulation. This result is actually not surprising. Using weighted averages of σ_n to estimate σ_k leads to an underestimation of σ_k and hence to an underestimation of the VaR. However, this underestimation partially offsets the overestimation resulting from the Poisson approximation.

Overall, using the simple method suggested by CSFP to estimate the σ_k gives a value for the VaR that is very close to the value computed by Monte Carlo simulation. Note that the VaR is higher in all the models with random default probabilities than in the models with nonrandom probabilities. This reflects the additional risk resulting from the uncertainty concerning the default probabilities.



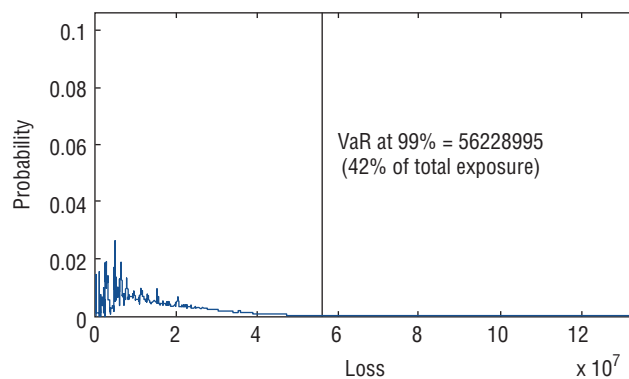
Source: Authors.
Note: VaR = value at risk.

Figure 10.6 Model 4: Poisson Defaults, Uncorrelated Factors, Least Squares



Source: Authors.
Note: VaR = value at risk.

Figure 10.7 Model 5: Correlated Sectors, Intersector Covariance = 0.1



Source: Authors.
Note: VaR = value at risk.

Figure 10.8 Model 5: Correlated Sectors, Intersector Covariance = 0.2

C. Random default probabilities, correlated factors

In this section, we present the portfolio loss distribution computed using Model 5, that is, using the model with random default probabilities and correlated factors described in Section 7. The only difference between Model 5 and Model 3 is the presence of the common factor Γ . This common factor affects the distributions of γ_k . In particular, it introduces some correlation between these factors. The portfolio loss distribution was computed for two different values of the variance of Γ (which is also the covariance between any two factors): 0.1 and 0.2. The results of these numerical experiments are presented in Figures 10.7 and 10.8, respectively. Not surprisingly, the VaR is higher in Model 5 than in Model 3, and it increases with the variance of Γ . This reflects the additional risk of incurring a large loss resulting from the positive intersector correlation, as well as the increased uncertainty concerning the default probabilities.

11. CONCLUSION

Each of the models presented here has specific features that make them useful in particular situations. The basic model with known probabilities and Bernoulli-distributed default events is useful when there is little uncertainty concerning default probabilities, when default probabilities are relatively high, and when the portfolio does not contain more than a few thousand obligors. At the cost of some approximations, CreditRisk+ and its extensions provide quasi-instantaneous solutions—even for very large portfolios—when default probabilities are influenced by a number of random latent factors. The alternative to using these models is to perform time-consuming Monte Carlo simulations. For our sample portfolio, the results of CreditRisk+ are very close to those of Monte Carlo simulations, even though this portfolio contains only 25 obligors with default probabilities as high as 30 percent.

This chapter provides a toolbox that has been used in the IMF's Financial Sector Assessment Program exercises to estimate credit risk parameters, including expected losses and credit VaR. The latter is the fundamental risk measure used to determine the economic capital required by a certain portfolio. This measure plays an important role in the IMF's bilateral financial surveillance of member countries—the need to understand how the gap between regulatory and economic capital could be bridged continues to be an important aspect of IMF staff's analysis. The instruments presented in this chapter add a rigorous quantitative dimension to this process.

REFERENCES

- Austrian Financial Market Authority and Oesterreichische Nationalbank, 2004, "New Quantitative Models of Banking Supervision" (Vienna, July).
- Avesani, Renzo G., Kexue Liu, Alin Mirestean, and Jean Salvati, 2006, "Review and Implementation of Credit Risk Models in the Financial Sector Assessment Program," IMF Working Paper

- 06/134 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=19111>
- Credit Suisse Financial Products, 1997, "CreditRisk+: A Credit Risk Management Framework," Technical Report (London: Credit Suisse First Boston). Available via the Internet: <http://www.csfb.com/institutional/research/assets/creditrisk.pdf>
- Crouhy, Michel, Dan Galai, and Robert Mark, 2000, "A Comparative Analysis of Current Credit Risk Models," *Journal of Banking and Finance*, Vol. 24, Nos. 1–2, pp. 59–117.
- Giese, Gotz, 2003, "Enhancing CreditRisk+," *Risk*, Vol. 16, No. 4, pp. 73–77.
- Gordy, Michael B., 2002, "Saddlepoint Approximation of Credit Risk," *Journal of Banking and Finance*, Vol. 26, No. 6, pp. 1335–53.
- Haaf, Hermann, Oliver Reiss, and John Schoenmakers, 2003, "Numerically Stable Computation of CreditRisk+," *Weierstrass-Institut für Angewandte Analysis und Stochastik*, Vol. 846, Berlin.
- Koyluoglu, H. Ugur, and Andrew Hickman, 1998, "A Generalized Framework for Credit Risk Portfolio Models," *DefaultRisk.com*, September 14. Available via the Internet: http://www.defaultrisk.com/pp_model_17.htm
- Melchiori, Mario R., 2004, "CreditRisk+ by FFT," Working Paper (Santa Fe: Universidad Nacional del Litoral).

Appendix

Probability and Moment Generating Functions

Consider a discrete random variable X that can take nonnegative integer values. The probability generating function (PGF) of X is the function G_X defined by

$$G_X(z) = E(z^X) = \sum_{x=0}^{\infty} P(X = x) \cdot z^x,$$

with $0 \leq z < 1$ if z is real and $|z| < 1$ if z is complex. $G_X(z)$ is simply a polynomial whose coefficients are given by the probability distribution of X . A PGF uniquely identifies a probability distribution: if two probability distributions have the same PGF, then they are identical.

Consider a second discrete random variable Y that can take nonnegative integer values. If X and Y are independent, then the PGF of $X+Y$ is given by

$$G_{X+Y}(z) = G_X(z) \cdot G_Y(z).$$

The moment generating function (MGF) of X is the function defined by

$$M_X(z) = E(e^{zX}) = \sum_{x=0}^{\infty} P(X = x) \cdot e^{zx}.$$

As its name indicates, the MGF of X can be used to compute the moments of X . The m th moment of X about the origin is given by the m th derivative of M_X valued at 0. This implies in particular that $E(X) = M'_X(0)$ and $Var(x) = M''_X(0) - [M'_X(0)]^2$.

The joint MGF of two random variables X and Y is defined as

$$M_{X,Y}(z_1, z_2) = E(e^{z_1 \cdot X + z_2 \cdot Y}),$$

where z_1 and z_2 are two auxiliary variables with the same properties as z .

If X and Y are two independent random variables, then the MGF of $X+Y$ is given by:

$$M_{X+Y}(z) = M_X(z) \cdot M_Y(z),$$

and the joint MGF of X and Y becomes

$$M_{X,Y}(z_1, z_2) = E(e^{z_1 \cdot X}) \cdot E(e^{z_2 \cdot Y}) = M_X(z_1) \cdot M_Y(z_2).$$

If $z_1 = z_2$, then $M_{X,Y}(z_1, z_2) = M_{X+Y}(z)$.

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Bankers without Borders?

Implications of Ring-Fencing for European Cross-Border Banks

EUGENIO CERUTTI • ANNA ILYINA • YULIA MAKAROVA •
CHRISTIAN SCHMIEDER

This chapter presents a stylized analysis of the effects of ring-fencing (i.e., different restrictions on cross-border transfers of excess profits and/or capital between a parent bank and its subsidiaries located in different jurisdictions) on cross-border banks. Using a sample of 25 large European banking groups with subsidiaries in Central, Eastern, and Southern Europe (CESE), we analyze the impact of a CESE credit shock on the capital buffers needed by the sample banking groups under different forms of ring-fencing. Our simulations show that under stricter forms of ring-fencing, sample banking groups have substantially larger needs for capital buffers at the parent and/or subsidiary level than under less strict (or in the absence of any) ring-fencing.

METHOD SUMMARY

Overview	The method is an extension of the standard balance sheet stress testing framework that allows explicit modeling of risks associated with different types of constraints on intragroup cross-border transfers of liquidity/capital.
Application	This framework may be used for estimating the liquidity/capital shortfall of a banking group that is restricted in its ability to reallocate excess liquidity/capital across jurisdictions following a shock that affects some of the affiliates within a group.
Nature of approach	Econometric and balance sheet–based analyses. Credit shock is modeled as deterioration in macroeconomic conditions leading to an increase in nonperforming loans and a decrease in the return on assets of banks.
Data requirements	Accounting information and data on both the banking group and each subsidiary of the group to be analyzed. Economic and market data (GDP, interest rates).
Strengths	The framework enables the inclusion of ring-fencing scenarios (i.e., different types of restrictions on cross-border transfers of excess profits and/or capital between a parent bank and its subsidiaries located in different jurisdictions) when stress testing cross-border banks.
Weaknesses	• The framework does not explicitly capture dependencies across banks and banking systems (as portfolio models may do). • This method is data intensive, as it requires up-to-date information on the geographical structure of a banking group, as well as detailed balance sheet data for the group (consolidated), parent bank, and all (or at least all the major) foreign banking subsidiaries.
Tool	Standard econometrics package.

This chapter was previously published as IMF Working Paper 10/247 (Cerutti and others, 2010). It has benefited from comments by Bas Bakker, Jan Brockmeijer, Miroslav Kollar, Inci Otter-Robe, Daniel Hardy, Jérôme Vandenbussche, and Rachel van Elkan, as well as by the participants of a Monetary and Capital Markets Department and European Department seminar.

The concept of centralized capital and liquidity management by internationally active banks was challenged by the recent crisis, sparking a debate about the desirable organizational and regulatory arrangements for cross-border banking groups. This chapter focuses on the costs for these banking groups that are associated with different restrictions on intragroup cross-border transfers imposed by the host or home country regulators (henceforth referred to as “ring-fencing”). More specifically, it provides a stylized analysis of how much additional capital might be needed if the banking groups are restricted, to different degrees, in their ability to reallocate funds across jurisdictions following a credit shock affecting their lending activities in a given region. The study does not estimate the group-level potential recapitalization needs under an extreme scenario (which is typically done in a stress test) but rather considers the implications of adverse economic conditions for cross-border banking groups under different forms of ring-fencing. The analysis is based on bank-level data for European banking groups and their subsidiaries in Central, Eastern, and Southern Europe (CESE).¹

At present, a number of European countries have legal restrictions on intragroup cross-border asset transfers. These limits are aimed at preventing undue influence by a foreign parent on its subsidiaries (e.g., in the form of disproportionate transfers of assets that could potentially trigger solvency or liquidity problems) or aimed at protecting the interests of minority shareholders and creditors of subsidiaries (European Commission, 2010). In order to ring-fence the subsidiary from the rest of the group, the host country can target the subsidiary’s ability to transfer funds abroad directly or indirectly, through measures affecting the entire domestic banking system (e.g., stopping the distribution of dividends by all banks during a crisis).²

During 2008–09, many subsidiaries of European banking groups had to rely on their foreign parents for capital and liquidity support. There is some evidence that, *ex ante*, the CESE subsidiaries had expected that they could rely on parent banks in case of need (e.g., the average capitalization levels of the foreign-owned subsidiaries in most CESE countries were 1 to 2 percentage points lower than the banking system averages at the outset of the crisis (Appendix I). *Ex post*, these expectations were validated by the assistance provided by parent banks.³ Yet, in order to maintain financial and eco-

nomic stability, the regulators in many host countries tightened restrictions on intragroup cross-border transfers, limiting the ability of cross-border banking groups to reallocate funds from subsidiaries with excess capital (liquidity) to those that were in need of capital (liquidity).⁴

There are arguments both for and against ring-fencing. The arguments in favor of centralized cross-border bank structures and against ring-fencing rely on efficiency and financial stability considerations (e.g., benefits of diversification across country-specific shocks). From a cross-border bank’s perspective, the ability to freely reallocate funds across its affiliates is essential for achieving the most efficient outcome—a point emphasized in the 2010 report prepared by the Institute of International Finance (IIF, 2010). Centralized cross-border bank structures may yield benefits for the host country economies as well. De Haas and van Lelyveld (2010), for example, show that the ability of international banks to attract liquidity and raise capital allows them to operate an internal capital market, which provides their subsidiaries with better access to capital and liquidity than what they would have been able to achieve on a stand-alone basis and hence may help to reduce the pressure to scale back lending during economic downturns. For both home and host authorities, the absence of ring-fencing facilitates diversification and can thus make the group as a whole more stable, for example, against shocks in the home country.

However, there are also arguments in favor of ring-fencing. For a host country regulator, the decision to impose ring-fencing typically would be driven by macro-financial stability considerations, such as the need to protect the domestic banking system from negative spillovers from the rest of the group or, more generally, to increase reserves for the whole domestic banking system during a crisis when the magnitude of the impending output collapse and bank losses are uncertain. The possibility of contagion from a parent bank to subsidiaries in the European context was analyzed by Popov and

10/106, March 22, 2010. In the context of this initiative, the large bank groups with systemic presence in several CESE countries have committed to maintain their exposure and keep their subsidiaries well capitalized (<http://www.imf.org/external/np/sec/pr/2010/pr10106.htm>).

⁴ To name a few examples, bank regulators in Croatia, Poland, and Turkey recommended the nondistribution of profits by the subsidiaries of foreign banks despite relatively strong bank fundamentals. In the case of Croatia, the Croatian National Bank (CNB) governor, Dr. Željko Rohatinski, at a press conference held on February 18, 2009, said that “the CNB would not look favorably upon attempts to withdraw capital, deposits, or pay out total accumulated profits, because that would destabilize the domestic banking system. In such a case, the CNB would be forced to undertake protective measures, regardless of the connected risks.” In the case of Turkey, the head of the banking regulation agency stated in December 2009 that “it is our natural right to expect those profits generated in this country to be invested and used in credit extension again in this country.” Banks in Turkey were expected to consult the regulator before distributing any dividends in the two years preceding 2009. The IMF Article IV report on Poland (IMF, 2010b) classified the regulator’s recommendation for subsidiaries of foreign banks to refrain from paying out dividends, despite robust capital buffers, as a form of capital control.

¹ The interbank linkages in this region and their role in transmitting or mitigating shocks have been recently analyzed in Arvai, Driessen, and Otter-Robe (2009); Maechler and Ong (2009); and Hermann and Mihaljek (2010) but these analyses relied mainly on the country-level (not bank-level) data on cross-border bank exposures.

² The current initiatives at the European Union (EU) level aim to achieve the dual objective of (1) preventing the risk of insolvency that could potentially be generated by a disproportionate transfer of assets for the credit institution making the transfer; and (2) lifting restrictions on transfers of assets if such transfers can potentially limit the extent of a crisis (European Commission, 2010).

³ See, for example, the “Statement at the End of the European Bank Coordination Initiative’s Second Full Forum Meeting,” Press Release No.

Udell (2010), who showed that the contraction of banks' balance sheets caused by losses and/or a deterioration in bank solvency was transmitted across borders to Eastern Europe by Western European banking groups in the early stages of the 2007–08 crisis.

Moreover, the difficulties in resolving cross-border banking groups and the absence of agreements on burden-sharing mechanisms during the crisis triggered a discussion about the desirability of promoting greater self-sufficiency of banking groups' affiliates in normal and in crisis times. Hoelscher and others (2010) consider the pros and cons of the so-called stand-alone subsidiarization (SAS) approach, according to which a cross-border banking group should be set up as a network of fully self-sufficient national subsidiaries. The authors note that from a banking group's perspective, the SAS approach may be beneficial if it can provide additional incentives for subsidiaries to better manage liquidity and credit risk. From the host or home country perspective, the key benefits of SAS include limiting intragroup contagion and allowing selective resolution of problem parts of the group with minimal disruption for the rest of the group.

Leaving aside the question of the potential benefits of imposing greater autonomy on the banking group's affiliates operating in different jurisdictions, this chapter focuses on the costs of ring-fencing for cross-border banking groups under different forms of ring-fencing. The cost is measured in terms of the amount of external capital that is required to cover capital shortfalls faced by the affiliates of these groups as a result of a credit shock. More specifically, this study estimates the amount of additional capital that might be needed if the sample banking groups are restricted in their ability to reallocate excess profits and/or capital across jurisdictions following a credit shock that affects some of the affiliates within these groups.⁵

It should be noted that the transfers of excess profits/capital are not the only mechanisms through which banking groups could manage the level of capitalization of their affiliates. For example, the latter could also be achieved through capital injections via subordinated debt or by “shifting” assets (instead of capital) between different parts of the group.⁶ However, the empirical analysis of these alternative mechanisms is constrained by the lack of publicly available bank-level data on intragroup lending and asset transfers. That said, the conclusions that such exercises might yield are likely to be quite similar to the results regarding the transfers of excess capital/profit presented here.

Three different types of ring-fencing are considered in this chapter, ranging from partial ring-fencing to full ring-fencing. *Partial ring-fencing* assumes that only excess profits of subsidiaries but not their excess capital buffers can be re-allocated within a group. *Near-complete ring-fencing* assumes that only transfers from the parent to a subsidiary are allowed. *Full ring-fencing* corresponds to the strict *stand-alone subsidiarization* (SAS) model, where no intragroup transfers are allowed. The analysis presented here takes into account the parent banks' ownership stakes in their subsidiaries.

The sample of banks included in the analysis consists of 25 European banking groups and their 113 subsidiaries located in 18 countries in CESE. There are several reasons for using this sample: (1) most of these banks have a large network of subsidiaries operating in several countries in the CESE region; and (2) the fact that many countries in the region were severely hit by the crisis allows us to illustrate a range of outcomes under different ring-fencing assumptions, given a severe but realistic credit shock affecting parts of these banking groups. The individual bank-level data on branches are not used in the estimation because branches are not stand-alone entities, which makes it difficult for the host country authorities to ring-fence them. The CESE exposures via branches are analyzed as part of the total direct cross-border exposures of parent banks.⁷

Qualitatively, the results of the analysis are fairly intuitive: any type of restrictions on intragroup transfers would entail the need for additional, and possibly significant, capital buffers at the subsidiary and/or the parent bank level of cross-border banking groups. Quantitatively, the sample banks' capital needs resulting from a simulated credit shock affecting their CESE subsidiaries over the 2009–10 period are 1.5–3 times higher in the ring-fencing/SAS scenarios than those under no ring-fencing. These results are robust to variations in the methodology for computing capital needs, including the postshock adjustment in risk-weighted assets (standardized versus the Basel II Internal Ratings–Based [IRB] approach).

What are the policy implications of this analysis? First, the establishment of a credible framework for the resolution

⁵ The issue of intragroup liquidity transfers is not considered in this study. It is left for future research.

⁶ At the onset of the crisis, many European parent banks had direct cross-border loans on their books, which sometimes had been purchased from the subsidiaries in the boom years. There is anecdotal evidence that suggests that the reverse happened during the crisis; that is, in some cases, subsidiaries with large capital buffers bought back loans from the parent banks, thereby reducing their capital adequacy ratios (CARs).

⁷ The choice between branches and subsidiaries has been analyzed in the literature. Cerutti, Dell'Ariccia, and Martinez-Peria (2007), for example, found that cross-border banking groups are more likely to set up a branch than a subsidiary in host countries with relatively higher corporate taxes, because this makes it easier to transfer profits across borders. Other considerations in the choice between branches and subsidiaries include the following: (1) branches are more common when foreign operations are smaller in size and do not have a retail orientation; (2) branches are less common in countries with highly risky macroeconomic environments, where parent banks seem to prefer the “hard” shield of limited liability provided by subsidiaries; (3) foreign banks tend to specialize in one organizational form or the other, beyond what is explained by their home-country regulation; and (4) foreign banks are less likely to operate as branches in countries that limit their activities and where regulation makes it difficult to establish new banks.

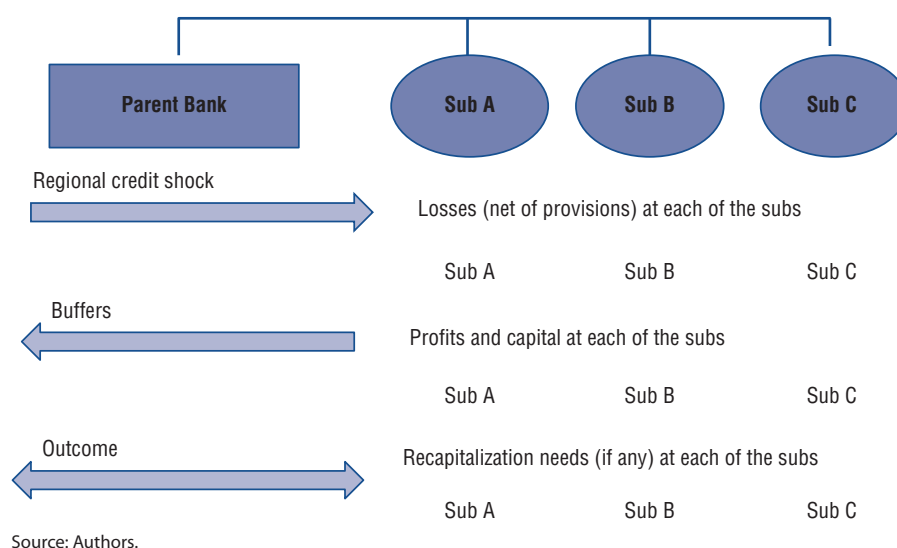


Figure 11.1 A Stylized Example of a Cross-Border Banking Group—Impact of a Regional Credit Shock on Subsidiaries A, B, and C

of cross-border banking groups would help to avoid unilateral and likely more costly solutions (in terms of capital requirements). This is because the existence of such a framework could reduce the incentives for and the incidence of ring-fencing by the home or host country authorities. Second, in the absence of such resolution and burden-sharing mechanisms, setting the minimum capital requirements for cross-border banking groups would have to take into account the potential presence of ring-fencing, especially in crisis times. Such a possibility may force cross-border banks to gravitate toward organizational structures that are more immune to ring-fencing (either SAS-type structures or branch structures). Third, should regulators decide to promote an SAS-like approach, its potential benefits would have to be carefully weighed against its potential costs.

The rest of the chapter is organized as follows. Section 1 provides a description of the exercise and the data. Section 2 explains the calibration of the credit shock affecting CESE subsidiaries. Section 3 presents the methodology for calculating capital needs, as well as the main results under different ring-fencing scenarios. Section IV draws conclusions and discusses policy implications.

1. CROSS-BORDER BANKING GROUPS

A. Description of the exercise

Consider a stylized cross-border banking group that has Subsidiaries A, B, and C operating in Countries A, B, and C respectively (Figure 11.1). Suppose that Countries A, B, and C are affected by a regional shock that leads to a significant deterioration in the credit quality of the loan books of subsidiaries operating in these countries. Suppose that losses resulting from this shock are offset by profits and capital buffers held by each of these subsidiaries (as a first line of

defense) and by funds transferred from the rest of the group (as a second line of defense).

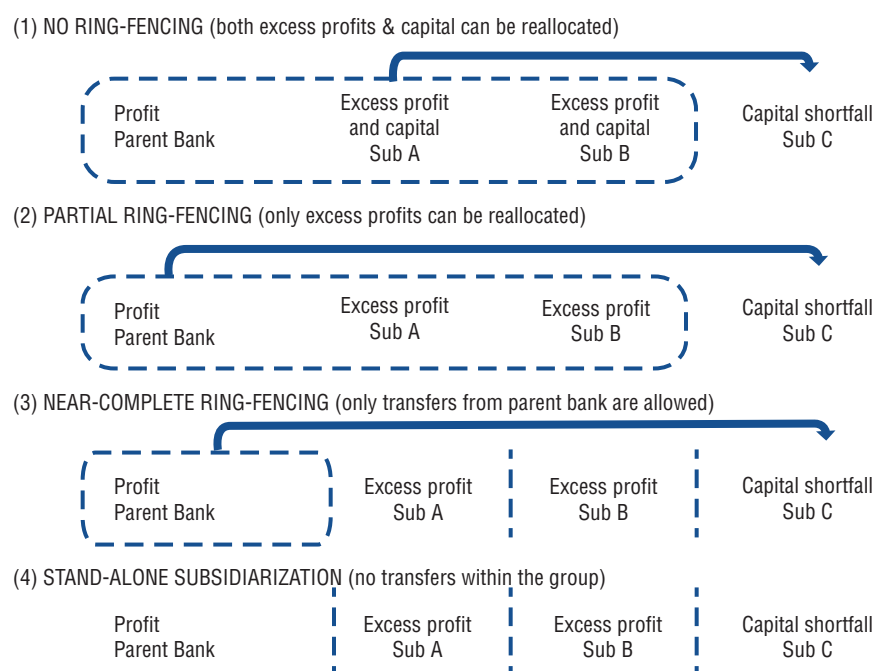
The capital needs resulting from the CESE credit shock are estimated in two steps:

1. For each subsidiary, the capital need is defined as the amount of capital required to bring its post-shock CAR back to either the country-specific (Basel II) regulatory minimum or to the subsidiary-specific preshock level.⁸ The latter is conservative in that it requires subsidiaries not to run down preshock buffers.
2. At the group level, total capital needs are computed by adding up all the capital needs of individual subsidiaries (and also losses on direct cross-border exposures of parent banks, in some simulations) and offsetting them against any other funds (i.e., excess profits and/or capital) that can be reallocated from other parts of the banking group.

Hence, the resulting total capital needs at the group level depend on the availability of excess profits and/or capital in the subsidiaries and parent bank, as well as on the degree to which these funds (excess profits and/or capital) can be reallocated within a group.

Suppose that as a result of the shock, one of the three subsidiaries (Sub C in Figure 11.2) experiences a capital shortfall (i.e., its regulatory capital falls below the national minimum capital requirement). Then, the extent to which this subsidiary can be recapitalized using the funds transferred from other parts of the group (i.e., without having to raise fresh capital) would depend on the existence of restrictions on such transfers (i.e., on the degree of ring-fencing).

⁸ The postshock CAR is estimated by taking into account actual or projected losses, provisions, capital buffers, and possible increases in risk-weighted assets.



Source: Authors.

Figure 11.2 A Stylized Example of a Cross-Border Banking Group—Reallocation of Funds within a Group to Cover Capital Shortfall at Subsidiary C

Four ring-fencing scenarios are analyzed in this chapter and illustrated in Figure 11.2:

1. The *no ring-fencing* scenario assumes that the parent bank's profits as well as subsidiaries' excess profits and excess capital buffers can be used to cover capital shortfall in any of the subsidiaries.
2. The *partial ring-fencing* scenario assumes that the parent bank's profits and only subsidiaries' excess profits but not excess capital can be reallocated within a group.
3. The *near-complete ring-fencing* scenario assumes that only transfers from the parent to any of the subsidiaries are allowed.
4. The *full ring-fencing*, that is, stand-alone subsidiarization (SAS), assumes that no transfers between any of the group's affiliates (including from the parent bank to subsidiaries) can take place.

B. Sample description

The analysis focuses on 25 European cross-border banking groups with parent banks domiciled in Austria, Belgium, Norway, France, Germany, Greece, Italy, the Netherlands, and Sweden that have significant presence in the CESE region (Table 11.1). The CESE subsidiaries of the European banking groups listed in Table 11.1 include the ones operating in Albania, Belarus, Bosnia and Herzegovina, Bulgaria, Croatia, Czech Republic, Estonia, Hungary, Latvia, Lithuania, Poland, Romania, Russia, Serbia, Slovakia, Slovenia, Turkey, and Ukraine.

For each banking group, only the CESE subsidiaries with total assets of at least \$1 billion, in which the parent bank has an ownership stake of at least 20 percent, are included in the sample.⁹ The final sample contains 25 banking groups and 113 CESE subsidiaries, covering the majority of European banking groups with a significant presence in the CESE region, as well as a substantial share of the banking sector assets in most CESE host countries. The total assets of the sample subsidiaries represent, on average, about 50 percent of the host country's total banking system assets, with significant coverage within EU countries (about 60 percent) (Figure 11.3).

That said, the total assets of subsidiaries included in the sample do not necessarily capture the full CESE exposures of these banking groups. This is because the latter could also include the parent banks' direct cross-border lending to CESE countries, as well as lending by the branches operating in the host countries alongside the subsidiaries.¹⁰ In order to capture these exposures, we used the aggregate Bank for International Settlements (BIS) data on foreign claims by reporting banks on the CESE countries to impute the residual exposures of the banking groups to

⁹ Data availability is another limiting factor. It should also be noted that for the majority of subsidiaries in the sample the ownership stake of the parent bank exceeds 50 percent.

¹⁰ There are some cross-border banking groups that conduct mainly direct cross-border lending instead of lending through subsidiaries/branches (see McCauley, McGuire, and von Peter, 2010).

TABLE 11.1

Sample Banking Groups and Their CESE Subsidiaries

Parent Bank	Parent bank's home country	Subsidiaries																	
		Central Europe				Southern and South-Eastern Europe				Southern Europe (Balkans)				Baltic Countries			CIS-3		
		Slovenia	Czech Rep.	Slovakia	Poland	Turkey	Bulgaria	Romania	Hungary	Albania	Bosnia	Croatia	Serbia	Estonia	Latvia	Lithuania	Russia	Ukraine	Belarus
Erste Group Bank	Austria		●	●				●	●			●	●					●	
RZB	Austria	●	●	●	●		●	●	●	●		●	●				●	●	●
Volksbank	Austria		●	●				●	●		●	●	●						
Bank Austria ¹	Austria	●	●	●		●	●	●	●		●	●	●		●		●	●	
Hypo Alpe Adria Group ²	Austria	●									●	●	●						
Dexia	Belgium					●													
KBC Bank	Belgium	●	●	●	●		●		●								●		
DNB NOR	Norway				●										●	●			
BNP Paribas	France		●		●	●											●	●	
Société Générale	France	●	●				●	●		●		●					●		●
Crédit Agricole	France				●	●	●	●										●	
Bayern LB	Germany							●											
Commerzbank	Germany				●		●	●		●							●		
Deutsche Bank	Germany				●	●			●										
Alpha Bank	Greece							●											
EFG Eurobank	Greece					●	●	●											
NBG	Greece					●	●	●											
Piraeus Bank	Greece						●			●									
Allied Irish Banks	Ireland				●		●												
Intesa Sanpaolo	Italy	●						●	●	●							●	●	
UniCredit SpA	Italy				●														
ING Bank	Netherlands				●	●											●	●	
Nordea Bank	Sweden				●														
SEB	Sweden													●	●	●	●		
Swedbank	Sweden													●	●	●	●	●	

Sources: Bankscope; and individual bank reports.

Note: A circle may indicate the presence of more than one subsidiary. See Appendix II for a detailed description of the sample.

1. Subsidiary of Unicredit.

2. Subsidiary of BayernLB.

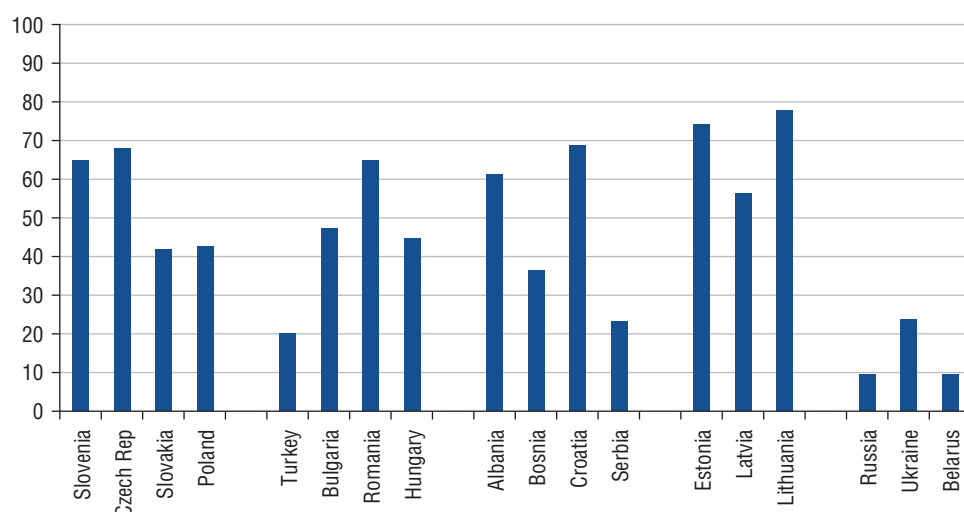
the CESE countries, including both direct cross-border exposures and exposures through branches (Figure 11.4). The data used for this purpose come from the BIS consolidated international banking statistics (see Appendix III for details).

For each banking group, end-2008 data on total assets, total customer loans, profits, nonperforming loans (NPLs), loan loss provisions, regulatory capital, Tier 1 capital, and risk-weighted assets were collected at the group level, at the parent bank level, and at the level of individual CESE subsidiaries (when available). The main data sources include Bankscope, Bloomberg, and individual bank reports.

2. CALIBRATION OF THE CESE REGIONAL SHOCK

A. Country-level data

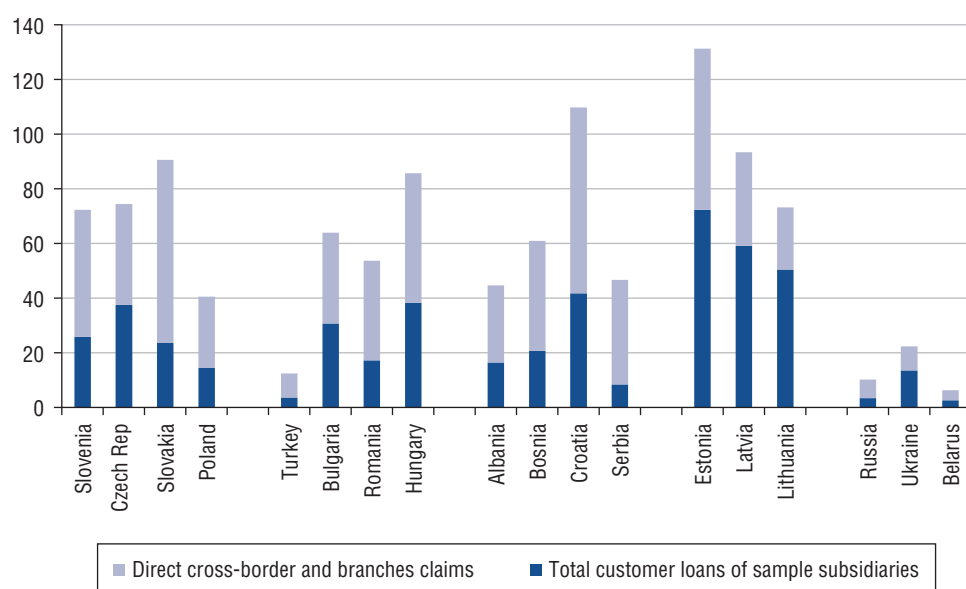
The CESE credit shock is modeled as deterioration in macroeconomic conditions during 2009–10, leading to an increase in NPLs and a decrease in the returns on assets (ROAs) of the CESE subsidiaries. The simulation of the shock relies largely on the actual data for 2009 and on projections for the CESE country-level NPLs and ROAs for 2010, which assume a slower economic recovery than the one envisaged



Sources: Authors; Bankscope; and national authorities.

Note: CESE = Central, Eastern, and Southern Europe.

Figure 11.3 Total Assets of Sample CESE Subsidiaries (in percent of total banking assets of the host country, December 2008)



Sources: Authors; Bank for International Settlements; Bankscope; and national authorities.

Note: Direct cross-border and branches claims were calculated as a residual from total foreign claims (see Appendix III).

CESE = Central, Eastern, and Southern Europe.

Figure 11.4 Total Foreign Claims of Sample Banking Groups on the CESE Countries (in percent of host country GDP, December 2008)

in the IMF's April 2010 *World Economic Outlook* (WEO) forecasts.

Changes in NPLs and ROAs are linked to the changes in macroeconomic conditions via panel regression models. The rationale behind this approach is to use consistent data across countries to come up with a specification that captures historical NPL and ROA patterns in the CESE region rather than fitting country-specific dynamics separately or extrapolating from past crises that occurred in other regions.

The panel regression analysis uses annual data for all CESE countries for the period of 1999–2009. The data for the dependent variable, aggregate NPLs at the country level, comes from Table 24 of the Statistical Appendix in the IMF's *Global Financial Stability Reports* (IMF, 2010a, table 24). The data are examined for possible structural breaks and inconsistencies in the NPL definitions across countries.¹¹

¹¹ See Appendix IV for details.

TABLE 11.2

Descriptive Statistics					
Variables (all in percent)	N	M	SD	Min.	Max.
NPLs	185	7.6	7.3	0.2	35.8
GDP growth	198	3.0	4.9	−18.5	12.2
Interest rate	187	10.5	12.9	1.4	93.2
ROA	179	1.2	1.3	−6.1	3.3

Sources: IMF (2010a); and international financial statistics.

Note: N denotes the number of observations. NPLs = nonperforming loans; ROA = return on assets.

TABLE 11.3

Dynamic Panel Regression Output for Nonperforming Loans			
Variable	Fixed Effects	Arellano-Bond	Arellano-Bover
NPL ($t-1$)	0.6116**	0.6342**	0.7198**
GDP (t)	−0.2633**	−0.2930**	−0.3576**
Interest (t)	0.1241**	0.1437**	0.0714*
Constant	2.1183**	1.8891**	2.1161**
Number of observations	170	143	161
Number of groups	18	18	18
R^2	0.69	N/A	N/A
Wald χ^2	N/A	361	424

Source: Authors.

Note: NPL = nonperforming loan; * $p < .05$. ** $p < .01$.

The GDP data come from the IMF's WEO database. Interest rate data come from different sources: when T-bill rates are not available, the Money Market Rates (Serbia, Ukraine, and Estonia) or other comparable interest rates (Bosnia and Herzegovina, Bulgaria, Romania, Slovakia, and Belarus) are used instead (Appendix V). Table 11.2 shows descriptive statistics for the variables used in the regression analysis.

B. Regression analysis

For the NPL model, from a large set of macroeconomic variables that could potentially influence the NPL dynamics, only two variables turned out to have reliable predictive power, namely, real annual GDP growth rates and short-term interest rates. Other potential explanatory variables (e.g., inflation rates, output gaps, private sector credit growth as a share of GDP, long-term nominal interest rates, the real effective exchange rate overvaluation, foreign currency debt as a share of GDP, government debt, and real government revenue) were either not significant or not available for the entire sample period. For the ROA model, the same variables (GDP growth, short-term interest rates) as well as the NPL ratios were statistically significant.

The estimated dynamic panel model for NPLs is as follows:

$$npl_{i,t} = \alpha + \beta_1 npl_{i,t-1} + \beta_2 \Delta GDP_{i,t} + \beta_3 interest_{i,t} + \varepsilon_{i,t} \quad (11.1)$$

where npl stands for nonperforming loans (as a share of total loans), ΔGDP for the real GDP growth on an annual basis,

and interest for the nominal short-term interest rate; ε is the error term, $i=1,2,\dots,N$ denotes the country, and $t=1,2,\dots,T$ denotes the time period.

Three specifications of model (11.1)—fixed effects, Arellano and Bond (1991), and Arellano and Bover (1995)—are considered, yielding similar results (Table 11.3). For the simulation of shocks, the Arellano-Bover dynamic panel specification is chosen, given its better asymptotic properties for small- t and large- N dynamic panels than fixed effects and the fact that it minimizes the data loss compared with the Arellano-Bond dynamic panel specification.

Unsurprisingly, NPLs display high persistence. The signs of the explanatory variables are also as expected, with a negative sign for GDP growth rates and a positive sign for interest rates. Real GDP growth rates have the most significant impact on NPLs among the macroeconomic variables included in the regression equation. Overall, the model-based NPLs match the historical NPL patterns quite well. The model works worst for Estonia, where the model-based NPLs are substantially higher than the observed ones, which is likely to be partly driven by the definition of NPLs (Appendix IV).

For the ROA model, all explanatory variables are highly significant in all specifications, except for the lagged ROAs in the fixed-effects and the Arellano-Bover specifications (Table 11.4). All signs are as expected and in line with other studies (e.g., Babihuga, 2007). The Arellano-Bover model is the preferred specification.

C. Calibration of NPLs and ROAs

The baseline scenario uses the actual 2009 NPL ratios for the banking sectors of the CESE countries and the 2010 NPL ratios projected using the Arellano-Bover regression estimation, the WEO forecast of GDP, CPI (consumer price index), and LIBOR six-month (in €) for 2010.

The adverse scenario uses the actual NPL data for 2009 (as in the baseline) and the 2010 NPL projections based on the assumption that for each of the CESE countries, the 2010 GDP growth rate is 2 percentage points lower than the 2010 April WEO GDP growth rate forecasts, and the 2010 interest rate is 200 basis points higher than in 2009. Given the dominant role of GDP in the regression specification that is used to calibrate the shock, the adverse scenario features a slow recovery and high NPLs in both years, with most of the NPL increase taking place in 2009. The NPLs estimated under the 2010 baseline scenario are broadly in line with the estimates in the IMF's *Global Financial Stability Report* (IMF, 2010a).¹² Overall, the adverse scenario can

¹² The adverse scenario used in this study is somewhat less severe than the adverse growth scenario in IMF (2010a), with NPLs that are 3–5 percentage points lower for the CEEs (Central and Eastern Europe), SEEs (Southeastern Europe), and the Baltic states. For the CIS, the figures are comparable, as the figures in this study include Belarus (with low NPLs), whereas IMF (2010a) does not.

TABLE 11.4

Dynamic Panel Regression Output for ROAs			
Variable	Fixed Effects	Arellano-Bond	Arellano-Bover
ROA ($t-1$)	0.0856	0.1326*	0.0834
GDP (t)	0.0723**	0.0840**	0.0768**
Interest (t)	-0.0523**	-0.0318**	-0.0230**
NPLs (t)	-0.0479**	-0.0514**	-0.0799**
Constant	1.7519**	1.5055**	1.7181**
Number of observations	157	139	157
Number of groups	18	18	18
R^2	0.52	N/A	N/A
Wald χ^2	N/A	146	232

Source: Authors.

Note: NPLS = nonperforming loans; ROA = return on assets; * $p < .05$. ** $p < .01$.

TABLE 11.5

Country-Specific NPL Assumptions				
Region	2008 Median	2009 Median ¹	2010 Baseline Median ²	2010 Adverse Median ³
Baltic countries	3.6	16.4	15.9	16.8
CEE-3	3.3	5.6	5.9	7.0
CIS	3.8	9.6	8.1	9.1
SEE	4.3	6.4	7.4	8.3

Sources: Authors; and IMF (2010a).

Note: CEE-3 = 3 Central and Eastern Europe sample countries; CIS = Commonwealth of Independent States; NPL = nonperforming loan; SEE = Southeastern Europe; WEO = *World Economic Outlook*.

1. The 2009 provisional data come from Table 24 in the Statistical Appendix of IMF (2010a).

2. Baseline scenario uses NPLs estimated via dynamic panel regression using CESE 1999–2008 data. WEO assumptions are used for out-of-sample forecasts.

3. Adverse scenario assumes a slow recovery, that is, 2010 GDP growth is 2 percentage points below the 2010 WEO GDP growth forecasts, and interest rates are 200 basis points higher than in 2009.

TABLE 11.6

Country-Specific ROA Assumptions				
Region	2008 Median	2009 Median ¹	2010 Baseline Median ²	2010 Adverse Median ³
Baltic countries	1.2	-0.1	0.0	-0.2
CEE-3	1.2	1.1	1.2	1.0
CIS-3	1.4	0.5	1.0	0.8
SEE	1.7	0.9	1.0	0.8

Sources: Authors; and IMF (2010a).

Note: CEE-3 = 3 Central and Eastern Europe sample countries; CIS = Commonwealth of Independent States; ROA = return on assets; SEE = Southeastern Europe; WEO = *World Economic Outlook*.

1. The 2009 provisional data come from Table 26 of the Statistical Appendix in IMF (2010a).

2. Baseline scenario uses ROAs estimated via dynamic panel regression using CESE 1999–2008 data.

3. Adverse scenario assumes a slow recovery, that is, 2010 GDP growth is 2 percentage points below the 2010 WEO GDP growth forecasts, and interest rates are 200 basis points higher than in 2009.

be characterized as relatively “mild” among the plausible adverse scenarios, not necessarily too far away from the baseline (Table 11.5).

For subsidiaries, 2009–10 profits are calculated by taking the actual 2008 preprovision profits of individual subsidiaries as a base and applying the same rate of change as that of the country-level ROAs calculated based on the regression model (Table 11.6). The regression-based estimates of ROAs are adjusted downward for Slovenia, Slovakia, Be-

larus, and Bosnia and Herzegovina and upward for Romania and Bulgaria.¹³

¹³ The adjustment accounts for the fact that the returns have been lower (first group of countries) or higher (Romania and Bulgaria) than on average in the sample in the past, which could be triggered by the level of competition, for example. The adjustment was 0.4 (Bulgaria) and 0.7 (Romania) in positive terms as well as 0.5 (Belarus and Slovakia) and 0.7 (Bosnia and Herzegovina and Slovenia) in negative terms.

For parent banks, the 2009 net profits are either the actual numbers or estimates based on market consensus forecasts.¹⁴ The 2010 profits are assumed to be equal to the 2009 profits, provided that the latter were positive, and zero otherwise. Although this assumption is ad hoc, it is fairly neutral and is unlikely to introduce an upward bias in the estimates of capital needs.

3. ASSESSING BANK CAPITAL NEEDS UNDER ALTERNATIVE RING-FENCING SCENARIOS

A. Methodology

This section presents the method applied to calculate capital adequacy under stress and capital requirements, respectively.

The *loan loss reserve (LLR)* for subsidiary k located in a CESE country i following a credit shock is as follows:

$$\text{Postshock LLR}_{k,i} = \text{NPL}_{k,i} * E_{k,i} * \text{LGD}_i \quad (11.2)$$

where $\text{NPL}_{k,i}$ is the postshock NPL ratio¹⁵ (nonperforming loans in percent of total exposure), $E_{k,i}$ is the total exposure (customer loans), and LGD_i is the loss given default (assumed to be the same for all subsidiaries operating in country i). Because bank-level end-2008 NPL data are not available for the majority of subsidiaries in the sample, the country-level end-2008 NPL ratios are used to proxy for the preshock bank-level NPL ratios. Country-specific LGDs come from the World Bank's Doing Business Web page. In order to account for the empirical finding that LGDs tend to increase during economic downturns, a formula proposed by the Board of Governors of the Federal Reserve (2006) to derive downturn LGDs is applied to country-specific LGDs.¹⁶

For each subsidiary, the *capital need* is defined as the amount of capital required to bring its postshock CAR back to the country-specific (Basel II) regulatory minimum level.¹⁷

The CAR of subsidiary k located in a CESE country i before the shock is

$$\text{Preshock CAR}_{k,i} = \text{Regulatory Capital}_{k,i} / \text{RWA}_{k,i} \quad (11.3)$$

The postshock CARs of subsidiaries reflect the impact of losses and possible underprovisioning (net of preshock LLR and preprovision profits at the subsidiary level) and an ad-

TABLE 11.7

Definitions of Capital Needs under Four Ring-Fencing Scenarios

Degree of Ring-Fencing	Capital Needs after a CESE Credit Shock (if positive)
No ring-fencing	CN(1) = sum of capital needs of all CESE subsidiaries – sum of excess profits and capital of all CESE subsidiaries – profits of the parent bank
Partial ring-fencing	CN(2) = sum of capital needs of all CESE subsidiaries – sum of excess profits of all CESE subsidiaries – profits of the parent bank
Near-complete ring-fencing	CN(3) = sum of capital needs of all CESE subsidiaries – profits of the parent bank
Stand-alone subsidiarization	CN(4) = sum of capital needs of all CESE subsidiaries

Source: Authors.

Note: CESE = Central, Eastern, and Southern Europe; CN = capital needs.

justment in risk-weighted assets (RWA) after the shock (denoted $\Delta \text{RWA}_{k,i}$). More specifically, the postshock CAR of subsidiary k located in a CESE country i is as follows:

$$\text{Postshock CAR}_{k,i} = \frac{(\text{Regulatory Capital}_{k,i} - \text{Additional Provisions}_{k,i})}{(\text{RWA}_{k,i} + \Delta \text{RWA}_{k,i})} \quad (11.4)$$

where

$$\text{Additional Provisions}_{k,i} = \begin{cases} \text{Postshock LLR}_{k,i} - \text{Preshock LLR}_{k,i} - \text{Profit}_{k,i}, & \text{if positive} \\ 0, & \text{otherwise} \end{cases}$$

Under the standardized approach, $\Delta \text{RWA}_{k,i}$ is negative and corresponds to written-off losses.^{18,19}

Then, the *capital need* (CN) at the group level is defined as the total amount of capital required to restore the CARs of all of the group's affiliates to their regulatory minimums. Clearly, the extent to which this can be done by reallocating excess profits or excess capital within a group would reduce the need for fresh capital at the group level. This, in turn, would depend on the degree of ring-fencing within a group.

The precise definitions of the banking groups' capital needs related to their CESE subsidiaries under different ring-fencing assumptions are presented in Table 11.7. *Excess profit*

¹⁴ Parent banks' net profits are used because parent bank' non-CESE-related losses are not explicitly included in the simulations. As explained at the end of the next section, some adjustments are needed to parent banks' net profits when including parent banks' direct cross-border CESE losses to avoid double counting.

¹⁵ The stock (rather than the flow) of NPLs is considered in order to account for both possible underprovisioning as well as provisions on additional NPLs.

¹⁶ See Appendix VI for details.

¹⁷ The regulatory minimum CARs for the CESE countries are presented in Appendix VII.

¹⁸ An alternative approach—the Basel II Internal Ratings-Based (IRB) approach—was considered as well. According to the IRB approach, $\Delta \text{RWA}_{k,i}$ takes into account the overall increase in the riskiness of performing loans that is likely to be associated with the increase in NPLs and hence can be positive despite the write-offs. The results based on the IRB approach are broadly similar to the standardized approach, with the estimated recapitalization needs being slightly higher under the IRB approach. The small difference between the two approaches results from the high level of credit risk. In "normal" times, the difference could be substantial.

¹⁹ Dividends are not explicitly modeled. It is implicitly assumed that most of the profits are retained.

refers to the residual profit of a given subsidiary after it covers its own capital needs, and *excess capital* refers to the capital cushion above the regulatory minimum. The excess profits/capital and losses of the subsidiaries are computed by taking into account the parent bank's *ownership stake* in each of these subsidiaries.

Note that under the SAS scenario, the capital needs of the banking group are equal to the sum of the recapitalization needs of all its CESE subsidiaries. Because the SAS approach presumes that all subsidiaries are self-sufficient, the capital needs under SAS would have to be covered by individual subsidiaries themselves, either by raising funds in the capital market or from other sources (such as the assistance from the local authorities).

As an additional exercise, the capital needs of sample banks are also computed by taking into account the parent bank's losses from direct cross-border exposures to the CESE countries. This is done in two steps: first, the 2009 cross-border losses are estimated as direct cross-border exposures * additional NPLs * LGD; second, the 2009 net profits of parent banks are adjusted by adding back 50 percent of the 2009 estimated cross-border losses to the actual or estimated 2009 net profits (50 percent provisioning assumption).²⁰ This conservative approach is used because the data on provisions on direct cross-border exposures of parent banks are not available.

B. Results

Using the framework and assumptions described above, we first computed the capital needs of the sample banking groups for their indirect CESE exposures via subsidiaries, assuming that subsidiaries have to restore their postshock CARs back to the regulatory minimum levels. Next, the banking groups' capital needs are computed for both indirect exposures via subsidiaries and direct cross-border exposures to the CESE region. Finally, the simulations are repeated using a different definition of capital needs for the subsidiaries, namely, the amount of capital required for the subsidiaries to restore the postshock CARs back to the preshock (end-2008) levels; because these precrisis CARs were generally above the regulatory minimums, capital needs computed under this alternative definition are higher.

Focusing only on indirect exposures via subsidiaries, the results are as follows (see Figure 11.5):

1. Eight out of 25 banking groups have no capital needs related to their CESE subsidiaries (i.e., $CN(4) = 0$).
2. Five out of 25 banking groups have significant capital needs related to their CESE subsidiaries (i.e., $CN(4) > 10$ percent of the banking group's regulatory capital). As expected, the capital needs of the bank-

ing groups to ensure adequate capitalization of all parts of the group after the shock are higher under near-complete/partial ring-fencing than under no ring-fencing, with the differences being larger for more diversified groups. For example, one of the banking groups (#24), which faces the CESE-related capital needs of over 20 percent of its regulatory capital under the SAS model ($CN(4)$), has zero capital needs under no ring-fencing ($CN(1)$). In the cases when the parent banks' profits are zero/negative (meaning that they cannot provide support for their subsidiaries), $CN(3) = CN(4)$. More generally, in the no ring-fencing scenario (which allows reallocation of both excess profits and capital), only 5 out of 25 banking groups would still face nonzero capital needs after reallocation, compared with 17 in the case of the SAS (where no transfers are allowed within a group).

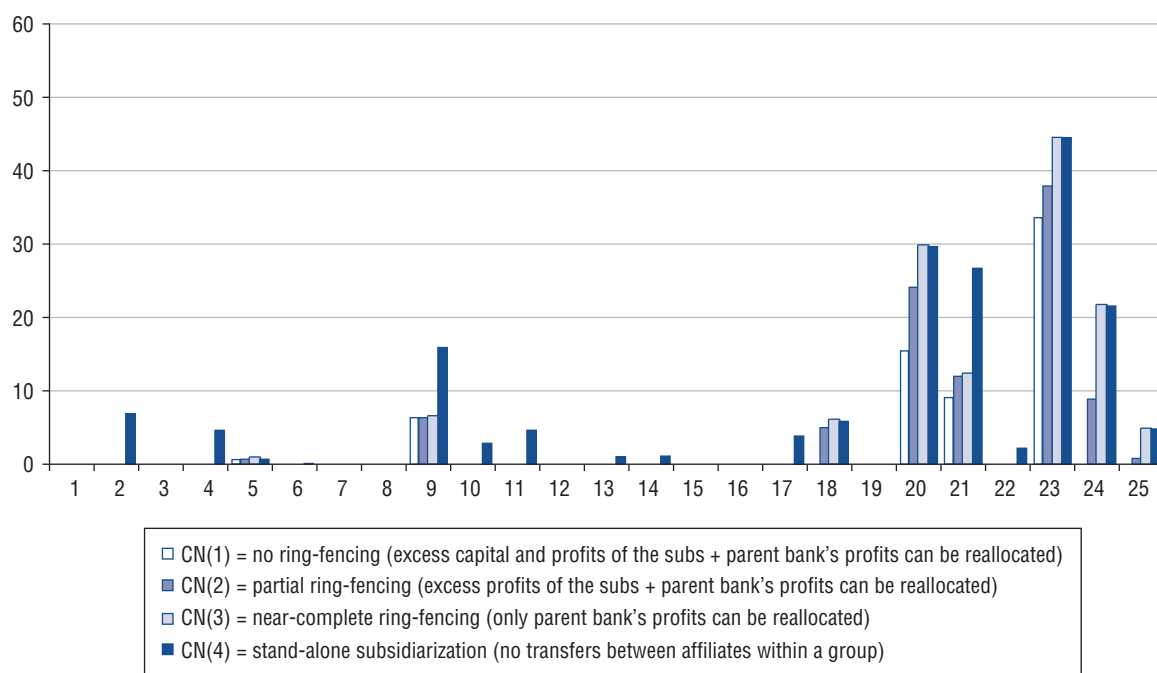
Figure 11.6 presents the estimated capital needs taking into account direct cross-border exposures and lending via branches, in addition to the exposures via subsidiaries in the sample. Although the CN measures in Figure 11.6 are notably higher than in Figure 11.5, the results are broadly similar, that is, more ring-fencing entails larger capital needs for most banking groups, with 9 banks (in the no ring-fencing scenario) to 22 banks (in the SAS scenario) in need of extra capital.

Recall that the simulations presented in Figures 11.5–11.6 use the definition of capital needs, which requires that the postshock CARs of all affiliates of the banking groups in the sample are restored to their respective country regulatory minimum levels. One could argue, however, that it may be prudent to maintain higher-than-regulatory minimum capital buffers, especially in crisis times, when uncertainty about credit quality is particularly high. To account for this possibility, capital needs are recomputed assuming that the postshock CARs of all affiliates of the banking groups have to be restored back to their preshock levels. Although the latter means that the overall capital needs are somewhat higher, the main result—that in the case of ring-fencing/SAS, the sample banks' aggregate capital needs are much higher than in the case of no ring-fencing—still holds (Figure 11.7).

To sum up, the results shown in Figures 11.5–11.6 suggest that in the case of ring-fencing/SAS, the sample banks' aggregate capital needs resulting from a CESE shock are 1.5–3 times higher than in the case of no ring-fencing (see Figure 11.7). These estimates appear to be within the range of the loss/recapitalization estimates provided by private analysts at end-2008 or early 2009 for different economic downturn scenarios in Eastern Europe during 2009–10.²¹

²⁰ The detailed discussion of how cross-border exposures are imputed from the Bank for International Settlements data is in Appendix III.

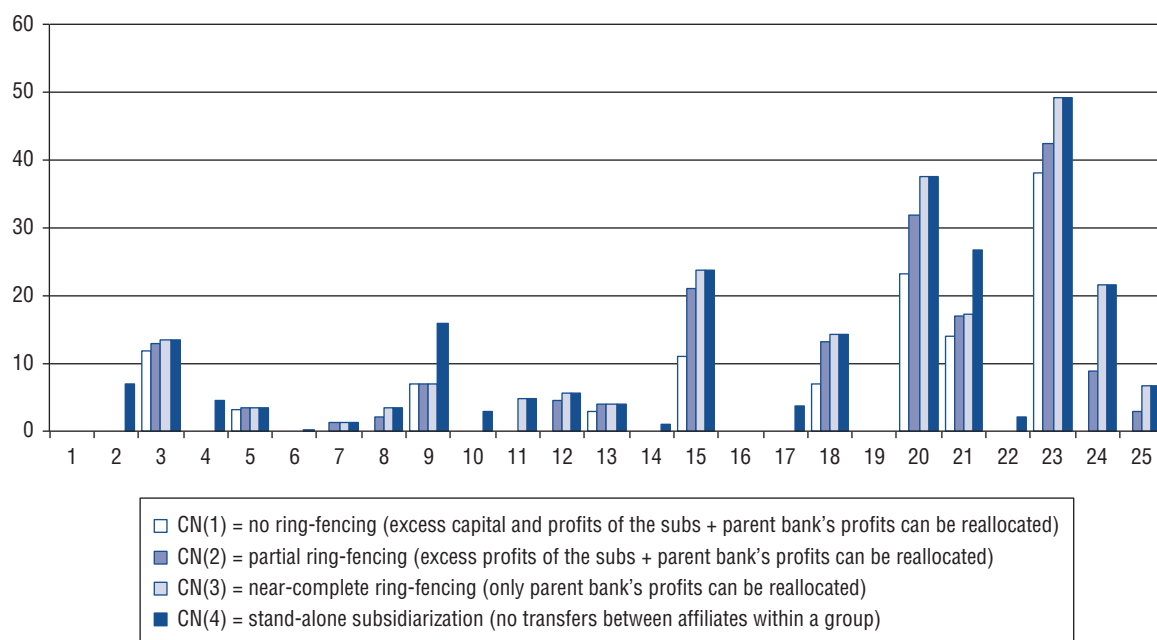
²¹ See Appendix VIII for details.



Source: Authors.

Note: CESE = Central, Eastern, and Southern Europe; CN = capital need.

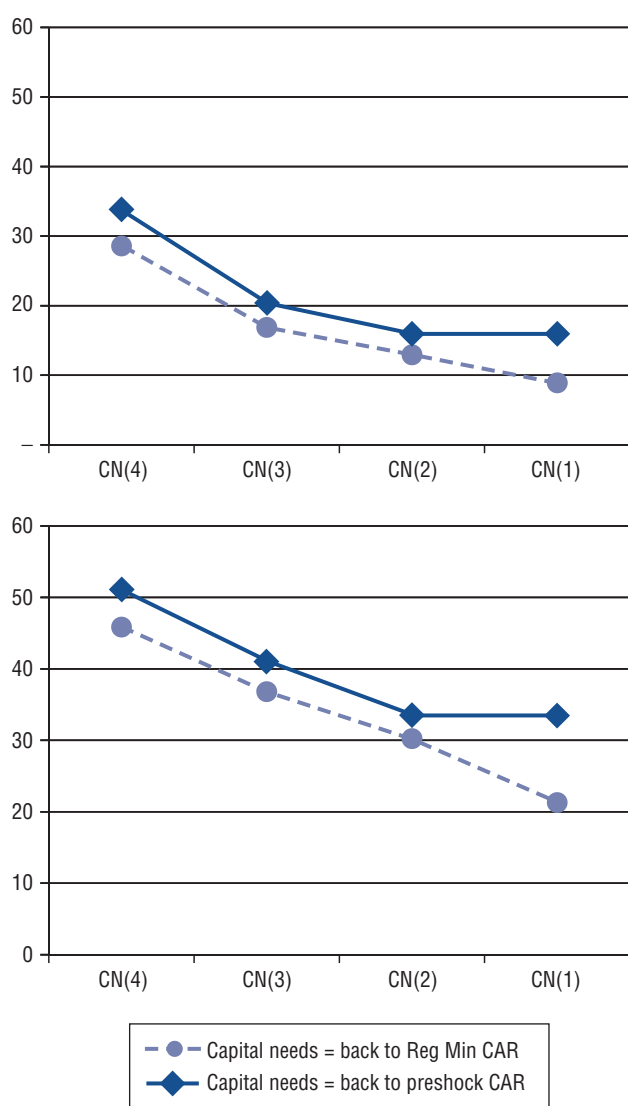
Figure 11.5 Estimated Capital Needs Resulting from a CESE Shock—Only Indirect Exposures via Subsidiaries in the Sample (as a percent of the group's regulatory capital)



Source: Authors.

Note: CESE = Central, Eastern, and Southern Europe; CN = capital need.

Figure 11.6 Estimated Capital Needs Resulting from a CESE Shock—Indirect Exposures via Subsidiaries and Direct Cross-Border Exposures (as a percent of the group's regulatory capital)



Source: Authors.

Note: Top panel shows only indirect exposures via subsidiaries in the sample; bottom panel shows indirect exposures via subsidiaries + direct cross-border exposures.

CAR = capital adequacy ratio; CESE = Central, Eastern, and Southern Europe; CN = capital needs; REG MIN = regulatory minimum.

Figure 11.7 Aggregate Capital Needs of Sample Banks Resulting from a CESE Shock (in billions of U.S. dollars)

4. CONCLUSION

The simulation of the capital needs of 25 large European banking groups resulting from a credit shock affecting their CESE subsidiaries, under different ring-fencing scenarios, shows that these groups would need to have substantially higher capital buffers at the parent and/or subsidiary level if they face a risk of being unable to transfer capital and/or profits across borders. The extent to which this would have a material impact on individual groups would depend on the significance of the CESE subsidiaries for each group as a whole.

As discussed above, the use of ring-fencing by regulators, as well as proposals, such as the SAS approach, have been motivated largely by the difficulty of resolving cross-border banking groups and the lack of agreed-upon principles on cross-border resolution and burden sharing. This highlights the policy dilemma that emerges from tensions between the increasingly international nature of banking activities and the national nature of regulatory/legal frameworks.

There seem to be two possible alternative paths going forward:

1. A credible international resolution and burden-sharing mechanism would reduce the need for and incidence of ring-fencing and allow greater scope for cross-border banking groups to manage their capital and liquidity in a centralized manner, provided that it is in line with the bank's business model.
2. In the absence of such burden-sharing mechanisms, regulators would need to consider setting minimum capital requirements for cross-border banking groups at a higher level, taking into account the possibility of ring-fencing, especially in crisis situations. In order to ensure that a banking group is resilient to all types of ring-fencing, all legally independent parts of the group (parent company and subsidiaries) would have to hold capital buffers that ensure self-sufficiency. The latter is likely to have implications for the banking group structures, the scope/scale of their cross-border activities, the supply of credit, and competition between local and foreign banks.

The choice between (1) and (2) would ultimately depend on the balance of potential benefits and costs, including those associated with higher capital requirements, as well as possible political constraints involved in choosing either one of the two paths. In particular, the European experience before the crisis showed that reaching an agreement on a credible resolution and burden-sharing mechanism may be challenging. However, the crisis provided additional impetus toward reaching an international consensus on a cross-border resolution framework, as evidenced by the ongoing work of the Financial Stability Board (on principles for cross-border resolutions), as well as of other international organizations.²²

REFERENCES

- Arellano, Manuel, and Stephen Bond, 1991, "Some Tests of Specification for Panel Data: Monte Carlo Evidence and an Application to Employment Equations," *Review of Economic Studies*, Vol. 58, No. 2, pp. 277–97.
- Arellano, Manuel, and Olympia Bover, 1995, "Another Look at the Instrumental Variable Estimation of Error-Components Models," *Journal of Econometrics*, Vol. 68, No. 1, pp. 29–51.
- Arvai, Zsofia, Karl Driessen, and Inci Otker-Robe, 2009, "Regional Financial Interlinkages and Financial Contagion within

²² See, for example, IMF (2010c).

- Europe,” IMF Working Paper 09/6 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=22569>
- Babihuga, Rita, 2007, “Macroeconomic and Financial Soundness Indicators: An Empirical Investigation,” IMF Working Paper 07/115 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=20651>
- Board of Governors of the Federal Reserve, 2006, “Basel II Capital Accord—Notice of Proposed Rulemaking” (Washington, September).
- Cerutti, Eugenio, Giovanni Dell’Ariccia, and Maria Soledad Martinez-Peria, 2007, “How Banks Go Abroad: Branches or Subsidiaries?” *Journal of Banking and Finance*, Vol. 31, No. 6, pp. 1669–92.
- Cerutti, Eugenio, Anna Ilyina, Yulia Makarova, and Christian Schmieder, 2010, “Bankers without Borders? Implications of Ring-Fencing for European Cross-Border Banks,” IMF Working Paper 10/247 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=24335>
- Czech National Bank, 2009, *Financial Stability Report 2009/2010* (Prague, March).
- De Haas, Ralph, and Iman van Lelyveld, 2010, “Internal Capital Markets and Lending by Multinational Bank Subsidiaries,” *Journal of Financial Intermediation*, Vol. 19, No. 1, pp. 1–25.
- Djankov, Simeon, Oliver Hart, Caralee McLiesh, and Andrei Shleifer, 2008, “Debt Enforcement around the World,” *Journal of Political Economy*, Vol. 116, No. 6, pp. 1105–49.
- European Commission, 2010, “Study on the Feasibility of Reducing Obstacles to the Transfer of Assets within a Cross Border Banking Group during a Financial Crisis,” Final Report (Brussels, April). Available via the Internet: http://ec.europa.eu/internal_market/bank/windingup/index_en.htm
- Hermann, Sabine, and Dubravko Mihaljek, 2010, “The Determinants of Cross-border Bank Flows to Emerging Markets: New Empirical Evidence on the Spread of Financial Crises,” BIS Working Paper No. 315 (Basel: Bank for International Settlements). Available via the Internet: <http://www.bis.org/publ/work315.htm>
- Hoelscher, David, Michael Hsu, Inci Otker-Robe, and Andre Santos, 2010, “Stand-Alone Subsidiarization Approach and Financial Stability” (unpublished manuscript; Washington: International Monetary Fund).
- Institute of International Finance, 2010, “A Global Approach to Resolving Failing Financial Firms: An Industry Perspective” (Washington, May).
- International Monetary Fund, 2010a, *Global Financial Stability Report*, World Economic and Financial Surveys (Washington, April). Available via the Internet: <http://www.imf.org/external/pubs/ft/gfsr/2010/01/index.htm>
- , 2010b, “Poland: 2010 Article IV Consultation,” IMF Country Report 10/118 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=23860>
- , 2010c, “Resolution of Cross-Border Banks—A Proposed Framework for Enhanced Coordination,” IMF Policy Paper (Washington, June 11). Available via the Internet: <http://www.imf.org/external/pp/longres.aspx?id=4462>
- Maechler, Andrea, and Li Lian Ong, 2009, “Foreign Banks in the CESE Countries: In for a Penny, in for a Pound?” IMF Working Paper 09/54 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=22781>
- McCauley, Robert, Patrick McGuire, and Goetz von Peter, 2010, “The Architecture of Global Banking: From International to Multinational?” *BIS Quarterly Review* (Basel, March). Available via the Internet: http://www.bis.org/publ/qtrpdf/r_qt1003e.htm
- Moody’s, 2010, “Moody’s Global Credit Policy—Corporate Default and Recovery Rates, 1920–2009,” Special Comment, February.
- Popov, Alexander, and Gregory Udell, 2010, “Cross-Border Banking and the International Transmission of Financial Distress during the Crisis of 2007–2008,” ECB Working Paper No. 1203 (Frankfurt: European Central Bank). Available via the Internet: <http://www.ecb.europa.eu/pub/scientific/wps/date/html/wps2010.en.html>

Appendix I

Capital Adequacy Rates by Country and Bank Type

TABLE 11.8

Average CARs of Foreign-Owned Sample Subsidiaries versus the Country-Level Average CARs

Country	CAR (Sample subsidiaries) ¹	CAR (Foreign-owned and domestic banks) ²
Albania	16.4	17.2
Belarus	14.3	21.8
Bosnia and Herzegovina	14.0	16.3
Bulgaria	13.6	14.9
Croatia	14.4	15.4
Czech Republic	11.0	12.3
Estonia	15.2	13.3
Hungary	9.4	11.1
Latvia	12.3	11.8
Lithuania	11.5	12.9
Poland	10.9	10.8
Romania	13.7	13.8
Russia	15.0	16.8
Serbia	18.6	21.9
Slovakia	10.0	11.1
Slovenia	11.7	11.7
Turkey	15.8	18.0
Ukraine	16.1	14.0
Cross-country average	13.5	14.7

Sources: Bankscope; *Global Financial Stability Report*, IMF; and individual bank reports (sample).

Note: CAR = capital adequacy ratio.

1. Authors' calculations.

2. *Global Financial Stability Report*, IMF (2010a).

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Appendix II

Sample of Banking Groups and Their Subsidiaries

TABLE 11.9

Sample Description				
	Parent Bank	Home Country	Subsidiary	Host Country
1	RZB	Austria	Raiffeisen Banka DD	Slovenia
2	Bank Austria	Austria	Unicredit Banka Slovenija DD	
3	Hypo Alpe Adria Group	Austria	Hypo Alpe-Adria-Bank DD	
4	KBC	Belgium	NLB DD-NOVA Ljubljanska Banka	
5	SocGen	France	SKB Banka DD	
6	Intesa	Italy	Banka Koper DD	
7	Erste Group	Austria	Ceska Sporitelna	Czech Republic
8	RZB	Austria	Raiffeisenbank Akciova Spolecnost	
9	Volksbank	Austria	Volksbank CZ	
10	Bank Austria	Austria	Unicredit Bank Czech Republic	
11	KBC	Belgium	Ceskoslovenska Obchodni Banka	
12	KBC	Belgium	Ceskomoravska Stavebni Sporitelna	
13	SocGen	France	Komerčni Banka	Slovakia
14	Erste Group	Austria	PRVA Stavebna Sporitelna AS	
15	Erste Group	Austria	Slovenska Sporitel'na AS	
16	RZB	Austria	Tatra Banka AS	
17	Volksbank	Austria	Volksbank Slovensko, AS	
18	Bank Austria	Austria	Unicredit Bank Slovakia AS	
19	KBC	Belgium	Ceskoslovenska Obchodna Banka	
20	RZB	Austria	Raiffeisen Bank Polska SA	Poland
21	KBC	Belgium	Kredyt Bank SA	
22	DNB Nor	Norway	DNB Nord	
23	BNP Paribas	France	Fortis Bank Polska SA	
24	Commerzbank	Germany	Bre Bank SA	
25	Deutsche	Germany	Deutsche Bank Polska SA	
26	Allied Irish Bks	Ireland	Bank Zachodni WBK SA	
27	Unicredit	Italy	Bank Polska Kasa Opieki Sa—Bank Pekao SA	
28	ING	Netherlands	ING Bank Slaski SA—Capital Group	
29	Nordea	Sweden	Nordea Bank Polska SA	
30	Bank Austria	Austria	Yapi VE Kredi Bankasi AS	Turkey
31	Dexia	Belgium	Denizbank AS	
32	BNP Paribas	France	Türk Ekonomi Bankasi AS	
33	Credit Agricole	France	Calyon Yatırım Bankası Türk AŞ	
34	Deutsche	Germany	Deutsche Bank AS	
35	Eurobank EFG	Greece	Eurobank Tekfen	
36	NBG	Greece	Finansbank AS	
37	ING	Netherlands	ING Bank AS	
38	RZB	Austria	Raiffeisenbank (Bulgaria) EAD	Bulgaria
39	Bank Austria	Austria	Unicredit Bulbank	
40	KBC	Belgium	CIBANK PLC	
41	SocGen	France	Societe Generale Expressbank	
42	Credit Agricole	France	Emporiki Bank—Bulgaria EAD	
43	Commerzbank	Germany	Procredit Bank (Bulgaria) AD	
44	Eurobank EFG	Greece	Eurobank EFG Bulgaria AD (Postbank)	
45	NBG	Greece	United Bulgarian Bank—UBB	
46	Allied Irish Bks	Ireland	Bulgarian-American Credit Bank	

(continued)

TABLE 11.9 (continued)

	Parent Bank	Home Country	Subsidiary	Host Country
47	Erste Group	Austria	Erste Bank Hungary NYRT	Hungary
48	RZB	Austria	Raiffeisen Bank ZRT	
49	Volksbank	Austria	Magyarországi Volksbank RT	
50	Bank Austria	Austria	Unicredit Bank Hungary ZRT	
51	KBC	Belgium	K&H Bank ZRT	
52	Bayern LB	Germany	MKB Bank ZRT	
53	Deutsche	Germany	Deutsche Bank Zártkörűen Működő Részvénytársaság	Romania
54	Intesa	Italy	CIB Bank ZRT	
55	Erste Group	Austria	Banca Comerciala Romana SA	
56	RZB	Austria	Raiffeisen Bank SA	
57	Volksbank	Austria	Volksbank Romania	
58	Bank Austria	Austria	Unicredit Tirioc Bank	
59	SocGen	France	BRD-Groupe Societe Generale SA	
60	Credit Agricole	France	Emporiki Bank—Romania SA	
61	Commerzbank	Germany	Procredit Bank SA	
62	Alpha	Greece	Alpha Bank Romania	
63	Eurobank EFG	Greece	Bancpost SA	
64	NBG	Greece	Banca Romaneasca SA	
65	Intesa	Italy	Intesa Sanpaolo Romania SA	
66	RZB	Austria	Raiffeisen Bank Albania	Albania
67	SocGen	France	Banka Popullore SHA	
68	Commerzbank	Germany	Procredit Bank (Albania) SHA	
69	Piraeus	Greece	Tirana Bank SA-Banka e Tiranës SHA	
70	Intesa	Italy	Intesa Sanpaolo Bank Albania	
71	Bank Austria	Austria	Unicredit Bank	Bosnia and Herzegovina
72	Hypo Alpe Adria Group	Austria	Hypo Alpe—Adria—Bank AD, Mostar	
73	Hypo Alpe Adria Group	Austria	Hypo Alpe—Adria—Bank AD Banja Luka	
74	Erste Group	Austria	Erste & Steiermärkische Bank DD	Croatia
75	RZB	Austria	Raiffeisenbank Austria DD, Zagreb	
76	Volksbank	Austria	Volksbank DD	
77	Bank Austria	Austria	Zagrebacka Banka DD	
78	Hypo Alpe Adria Group	Austria	Hypo Alpe—Adria—Bank DD	
79	Hypo Alpe Adria Group	Austria	Slavonska Banka DD, Osijek	
80	SocGen	France	Societe Generale—Splitska Banka	Serbia
81	Erste Group	Austria	Erste Bank AD Novi SAD	
82	RZB	Austria	Raiffeisenbank AD	
83	Bank Austria	Austria	Unicredit Bank Serbia JSC-Unicredit Bank Srbija AD	
84	Hypo Alpe Adria Group	Austria	Hypo Alpe—Adria—Bank AD Beograd	Estonia
85	SEB	Sweden	SEB Bank	
86	Swedbank	Sweden	Swedbank AS	
87	Bank Austria	Austria	Unicredit Bank AS	Latvia
88	DNB Nord	Denmark	DNB Nord	
89	SEB	Sweden	SEB Banka AS	
90	Swedbank	Sweden	Swedbank AS	Lithuania
91	DNB Nord	Norway	DNB Nord	
92	SEB	Sweden	SEB Bankas	
93	Swedbank	Sweden	Swedbank AS	Russia
94	RZB	Austria	ZAO Raiffeisenbank	
95	Bank Austria	Austria	Unicredit Bank ZAO	
96	KBC	Belgium	Absolut Bank	
97	BNP Paribas	France	BNP Paribas Vostok	
98	SocGen	France	Bank Société Générale Vostok	
99	SocGen	France	JSC Rosbank	
100	Commerzbank	Germany	Commerzbank (Eurasia)	
101	Intesa	Italy	KMB Bank/ Small Business Credit Bank	
102	ING	Netherlands	ING Bank (Eurasia) ZAO	
103	Nordea	Sweden	Nordea Bank	

TABLE 11.9 (continued)

	Parent Bank	Home Country	Subsidiary	Host Country
104	Swedbank	Sweden	Swedbank	Ukraine
105	Erste Group	Austria	Erste Bank OJSC	
106	RZB	Austria	Raiffeisen Bank Aval	
107	Bank Austria	Austria	Ukrsotsbank	
108	BNP Paribas	France	JSIB Ukrsibbank	
109	Intesa	Italy	Pravex Bank	
110	ING	Netherlands	ING Bank Ukraine	Belarus
111	Swedbank	Sweden	Public Joint Stock Company Swedbank	
112	RZB	Austria	Priorbank	
113	SocGen	France	Belrosbank	

Sources: Analysts' reports; Bankscope; and individual bank reports.

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Appendix III

Using BIS Data to Capture Remaining Banking Groups' Exposures to the CESE Countries

The portfolio of subsidiaries included in the sample does not necessarily capture the full CESE exposures of the sample banking groups, as it is missing their direct cross-border claims and lending through their branches in the host countries. In order to capture these exposures, the subsidiaries' balance sheet data are combined with the consolidated Bank for International Settlements (BIS) international banking statistics, which report, at the country level, the consolidated claims of BIS reporting banks on each CESE country.

The BIS total foreign claims data (on the immediate borrower's basis) include both the cross-border claims of foreign banks and all foreign affiliate—subsidiaries and branches—claims. In order to separate direct cross-border and branches' claims, we subtract the domestic lending to customers by sample subsidiaries from the total foreign claims (e.g., the customer lending by Greek-controlled subsidiaries operating in Turkey is subtracted from the consolidated foreign claims of the BIS reporting Greek banks on Turkish residents).

A few caveats need to be highlighted:

1. The residual labeled as “direct cross-border and branches claims” could also include nonlending claims by subsidiaries, such as government bonds held by a subsidiary. Nevertheless, the potential recap needs calculated for each subsidiary were based on their lending portfolio, and hence, the treatment of other nonlending claims as claims directly linked to the parent bank seems appropriate.
2. As documented in other papers, there are some discrepancies between the BIS data and other sources. In our case, we found that there would be a potential small downward bias in the BIS total claims of Italian banks on Ukraine and Swedish banks on Latvia, based on the subsidiaries balance sheet data. This is similar to the discrepancies found by Maechler and Ong (2009) when comparing the BIS data with the central bank data sources. These discrepancies could be due to the differences in the consolidation method and the group structure classification used by different internationally active banks.
3. The residual exposure identified at a parent bank's home-country level is distributed across the sample banking groups of each home country as a proportion of the group's assets. Because the sample of banking groups in this study was put together with the objective to capture all major European cross-border banking groups with significant presence in the CESE region, it is unlikely that the BIS data include large cross-border banks with operations in the CESE region that are not included already in the sample.

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Appendix IV

Panel Regression Analysis

BASIC SPECIFICATION

We estimated the fixed-effects and random-effects specifications of the following panel regression model:

$$npl_{i,t} = \alpha + \beta_1 \Delta GDP_{i,t} + \beta_2 interest_{i,t} + \varepsilon_{i,t} \quad (11.5)$$

where *npl* stands for nonperforming loans (as a share of total loans), ΔGDP for the real GDP growth on an annual basis, and *interest* for the short-term nominal interest rates (T-bill rate); ε is the error term, $i=1,2,\dots,N$ denotes the country, and $t=1,2,\dots,T$ denotes the period. The results are presented in Table 11.10. Based on the Hausman test, the random-effects specification is more appropriate than the fixed-effects specification.

ROBUSTNESS CHECKS

In order to investigate the *differences in NPL definitions* across countries, we compared the average level of bank provisions with NPLs over time. For most of the 18 countries, this ratio varies between 60 percent and 90 percent. In the case of Estonia, where the level of provisions ranges between 150 percent and 300 percent of NPLs during 2003–06 (the years for which the data are available), the level of NPLs seems particularly low, suggesting that the definition of NPLs in Estonia may be different from the one used in other countries. Estonia has, therefore, been excluded from some regression specifications as a means of a robustness check. In addition, the ratio of provisions to NPLs in Russia fluctuated between 120 percent and 160 percent during 2003–08, which can again be seen as a sign that the definition of NPLs is relatively narrow (i.e., not as conservative) as in other countries. For Romania and Ukraine, the opposite seems to be true, which indicates that NPLs may be based on a fairly broad definition of potential losses.

For countries with *structural breaks* (Latvia, Lithuania, Poland, Slovakia, Serbia, Ukraine), we also checked whether there were any major changes in the levels of NPLs observed around the timing of structural breaks in definitions. Because this was not the case, all NPL data from these countries were included in the regression models. Similarly, the regression results remained robust when excluding countries with structural breaks.

TABLE 11.10

Panel Regression Analysis (Dependent Variable: NPLs)		
Variable	Fixed Effects	Random Effects
GDP (<i>t</i>)	−0.1864**	−0.1797**
Interest (<i>t</i>)	0.2325**	0.2301**
Constant	5.8369**	5.7222**
Number of observations	178	178
Number of groups	18	18
<i>R</i> ²	0.23	0.23

Source: Authors.

Note: ***p* < .01.

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Appendix V

Interest Rates Used for Regression Analysis

TABLE 11.11

Overview on Data Sources for Short-Term Interest Rates		
Country	Explanation	Source
Albania	Treasury bill rate	IFS
Belarus	Data are scarce; adjusted deposit rates according to money market rates to deposit rates in Ukraine	IFS
Bosnia and Herzegovina	Interest rate is constructed as 2/3 times deposit rate + 1/3 times lending rate, starting in 2002 (other data are not available)	MBS
Bulgaria	Treasury bill rate when available (2 years); otherwise inferred from interbank rate (3 months, Bloomberg); otherwise from lending rate	MBS and Bloomberg
Croatia	Treasury bill rate from Bloomberg (from 2002 onward); before that, adjust money market rate weighted by the relative portion of T-Bill rates on money market rates from 2002 to 2008	IFS (for 1992 to 2001), Bloomberg (from 2002)
Czech Republic	Treasury bill rate	MBS
Estonia	Money market rate	IFS
Hungary	Treasury bill rate	MBS
Latvia	Treasury bill rate	MBS
Lithuania	Treasury bill rate (12 months) where possible; otherwise government bond yield (12 months, from Bloomberg), recalculated to 12 months level; otherwise interbank rate, recalculated to 12-month T-Bill rate	MBS and Bloomberg
Poland	Weighted average Treasury bill rate	MBS
Romania	Interbank short-term lending rate, similar to Treasury bill rate (91 days) (which is available only for some years)	MBS
Russia	Treasury bill rate for 2000–3; after 2003, average of money market rate and interbank rate (3 months); before 1999, money market rate	MBS, Bloomberg
Serbia	Money market rate	IFS
Slovakia	Average deposit rate (1993–99), government bond yield (2002–6), government bond yield (1 year, from Bloomberg, 2007–9)	MBS, Bloomberg
Slovenia	Treasury bill rate (from 1999); before that, money market rate adjusted to Treasury bill rate	MBS
Turkey	Treasury bill rate, if available; (before 1999: money market rate); after 2007, 1-year government bond yield	IFS, Bloomberg
Ukraine	Money market rate	IFS

Sources: Bloomberg; International Financial Statistics (IFS); and Money and Banking Statistics (MBS).

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Appendix VI

Loss-Given-Default Ratios in the CESE Countries

The country-specific loss-given-default ratios (LGDs) are taken from the World Bank Doing Business Web page and are based on work of Djankov and others (2008). The differences in LGDs across countries reflect the differences in bankruptcy codes, the duration of the proceedings, and legal costs. The CESE long-term average LGDs range from 51 percent (Lithuania) to 91 percent (Ukraine). The CESE average of 68 percent is above the LGD for senior unsecured credit under the Basel II Foundation Internal Ratings–Based approach (45 percent), which is often used as a benchmark for a through-the-cycle LGD.

In order to account for the empirical finding that LGDs increase during downturn periods, we use a formula proposed by the Board of Governors of the Federal Reserve (2006) to derive the downturn LGDs:

$$\text{Downturn LGD} = 0.08 + 0.92 * \text{Long-term average LGD}$$

The downturn LGDs for the CESE countries range from 55 percent (Lithuania) to 92 percent (Ukraine) (Table 11.12). The estimated CESE downturn LGDs are thus substantially higher than those observed for senior bank loans in the OECD countries, which range from 30 percent to 45 percent (see Moody's 2010; Doing Business database).

TABLE 11.12

Loss-Given-Default Ratios by Country

Country	Long-Term Average LGDs	Downturn LGDs
Albania ¹	N/A	70.0
Belarus	66.6	69.3
Bosnia and Herzegovina	64.1	67.0
Bulgaria	67.9	70.5
Croatia	69.5	71.9
Czech Republic ²	41.3	46.0
Estonia	62.5	65.5
Hungary	61.6	64.7
Latvia	71.0	73.3
Lithuania	50.6	54.6
Poland	70.2	72.6
Romania	71.5	73.8
Russia	71.8	74.1
Serbia	74.6	76.6
Slovakia	54.1	57.8
Slovenia	54.5	58.1
Turkey	79.8	81.4
Ukraine	90.9	91.6
Average	68.3	70.7

Sources: <http://www.doingbusiness.org> and local authorities.

Note: LGD = loss given default.

1. For Albania, the downturn LGD is assumed to be equal to its peer country average (70 percent).

2. For the Czech Republic, the average LGD is the one for corporate credit (Czech National Bank, 2009, p. 81).

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Appendix VII

Regulatory Minimum Capital Requirements by Country

TABLE 11.13

Regulatory Minimum Capital Requirements in the CESE Countries

Country	Minimum Capital Adequacy Requirements (in percent)
Albania	12
Belarus	10
Bosnia and Herzegovina	12
Bulgaria	12
Croatia	10
Czech Republic	8
Estonia	10
Hungary	8
Latvia	8
Lithuania	10
Poland	8
Romania	10
Russia	10
Serbia	12
Slovakia	8
Slovenia	8
Turkey	12
Ukraine	8

Source: IMF staff calculations.

Note: CESE = Central, Eastern, and Southern Europe.

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Appendix VIII

Analysts' Estimates of Losses and Capital Needs

TABLE 11.14

Private Analysts' Estimates of Losses and Capital Needs of Western European Banks on their CESE Exposures

Source	Publication Date	NPL Scenario (%)	LGD (%)	Period	Loss (billion EUR)	Capital Needs
JP Morgan	March 5, 2009	10 to 30 ¹	75	2 years	32	Average loss: 22 (in percent of equity)
Barclays	February 27, 2009	10/20/30 ²	N/A	2 years	67/133/200	EUR 100–150 billion
Deutsche Bank	February 23, 2009	1 scenario, no details available	N/A	2 years	34	Average loss: 17 (in percent of tangible common equity)
Goldman Sachs	January 29, 2009	10/20/30/ differentiated NPLs ^{1,2}	N/A	N/A	19/62/106/ 59	N/A
Merrill Lynch	October 27, 2008	7/11/20/33 ²	N/A	3 years	43	EUR 20 billion

Source: Analysts' reports.

Note: CESE = Central, Eastern, and Southern Europe; LGD = loss given default; NPL = nonperforming loan.

1. Different NPL assumptions for different asset "buckets."

2. Different NPL scenarios with uniform level of NPLs.

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Conducting Stress Tests of Defined Benefit Pension Plans

GREGORIO IMPAVIDO

This chapter describes the basic mechanics of defined benefit (DB) plan liability valuation and how to conduct simple stress tests of the solvency ratio. We first review the essential building blocks of liability valuation of DB plans and then, using an Excel-based template with institution-specific data, we take readers through the basics of liability valuation and stress testing of assets and liabilities of a typical DB plan. The template is by no means a substitute for a proper actuarial evaluation, but the simplifications introduced do not affect its usefulness to evaluate sensitivity of the funding ratio.

METHOD SUMMARY

Overview	The method implements various single-factor stress tests of the solvency ratio of a defined benefit plan. They include (1) changes of asset values for different asset classes; and (2) changes in actuarial assumptions for valuing liabilities.
Application	The method is appropriate for situations where data are limited and are of poor quality.
Nature of approach	• Full valuation approach (balance sheet). • Assets are valued at market prices. • Plan continuation pension liabilities are valued using a closed-group, aggregate, projected benefit obligation method with constant dollar prorating.
Data requirements	Portfolio distribution of assets under management, plan benefit formula, plan demographics, and actuarial assumptions.
Strengths	• The method focuses on the liability side of the balance sheet and is independent of solvency regulation. • It is very effective at assessing changes in solvency levels as a result of changes in asset values and actuarial assumptions. • The data requirement on assets is kept at a minimum, but more granular information on portfolio structure may be used if desired and available.
Weaknesses	The method focuses only on old age pension benefits, and it uses one actuarial valuation method to estimate old age pension liability levels. Hence, it is not a substitute for a full actuarial evaluation of the plan, which could cover (depending on the plan bylaws and relevant regulations) other benefit lines and valuation methodologies that can vary for funding or disclosure purposes.
Tool	The Excel spreadsheet macro is available in the toolkit, which is on the companion CD and at www.elibrary.imf.org/stress-test-toolkit .

This chapter develops a simple methodology for stress testing the funding ratio of defined benefit (DB) pension plans. Using an Excel-based template with institution-specific data, we take readers through the basics of liability valuation and

stress testing of assets and liabilities of a typical DB plan. The stress testing framework presented here uses a full asset and liability valuation approach, independent of the solvency regulation specific to any given country. Assets are valued at

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market prices. Plan continuation pension liabilities (i.e., excluding ancillary benefits) are valued using a closed-group, aggregate, projected benefit obligation method with constant dollar prorating.¹ The method developed here is independent of the solvency regulation of any given country as we compare the market value of assets with the value of actuarial liabilities and define the plan funded if the resulting funding ratio is above 100 percent.

The framework presented is no substitute for a full actuarial evaluation of the plan. The Excel-based template is ideally used in an operational context (such as the IMF's Financial Sector Assessment Program [FSAP] or for technical assistance) to inform: (1) the assessor on the nature of the plan actuarial assumptions; (2) the assessor on the sensitivity of the funding ratio to changes in actuarial assumptions and asset values; and (3) policy dialogue on the desirable actions needed to strengthen the solvency regulatory and supervisory framework, as well as the financial viability of the plan.

The remainder of this chapter is structured as follows. Section 1 introduces basic actuarial factors and functions involved in the estimation of actuarial liabilities of DB plans. Section 2 discusses the valuation method used in the accompanying template for the liabilities of retired and active members. Section 3 takes a real case of a DB pension plan and builds a model plan to conduct simple single-factor stress tests of assets and liabilities. Finally, conclusions and a brief discussion of possible extensions are contained in Section 4.

1. BASIC ACTUARIAL COST FACTORS AND FUNCTIONS

The calculation of DB pension liabilities relies on assumptions for several key actuarial costs considered in the calculation of liabilities. These are decrement assumptions (like mortality, termination, retirement, and disability) needed for the computation of survival probabilities, salary assumptions, inflation, and discount rate assumptions. We discuss these concepts in turn in the remainder of this section.

A. Decrement assumptions

Decrement assumptions describe how plan members' liabilities are affected by various risks/contingencies. For instance, retired members are exposed only to mortality risk, whereas active members are exposed to mortality, termination, disability, and retirement risk. In order to describe how members transit from one status to another, rates of decrement (given by available tables) are assumed for each contingency or risk. In other words, the rate of decrement refers to the proportion of the plan population in one status transiting to another status. In this chapter, we indicate with $q_x^{(k)}$ the rate at which members of age x transit from one status to another

within the period between x and $x+1$. The prime indicates we are working with rates (as opposed to probabilities), and k indicates the type of contingency; in this study, we use only mortality (m) and termination (t).

A DB plan would typically have other decrement assumptions. For instance, when disability benefits are provided for, the rate at which members become disabled and—depending on the nature of the disability benefits—the rates on death and continuance of disability could be relevant and would also need to be factored into the funding and costing calculations. Also, retirement decrement rates become relevant when the plan provides for early or delayed retirement. They affect only active members between the earliest and the latest permissible retirement ages, but they affect the level of benefits paid during retirement. In this study, we will not consider other decrement assumptions because doing so can greatly complicate the calculations but is unlikely to materially affect the outcome of stress tests performed in the context of an FSAP.

B. Contingent, composite survival probabilities

From the aforementioned decrement assumptions, it is possible to derive the composite survival probabilities. The idea is simple: All the various contingencies to which individuals are exposed will affect their probability of survival in the plan and, therefore, future liability cash flows. Composite survival probabilities are very important assumptions as they can affect cash flows into the very distant future, potentially more than 70 years away from the valuation date. Here, it is important to notice that we are moving from a decrement rate space to a decrement probability space. Therefore, we need to discuss how probabilities relate to rates in a multidecrement environment and how to construct composite survival probabilities.²

In a single-decrement environment, the rate of decrement is equal to the probability of decrement. In a multiple-decrement environment, the rate of decrement is higher than the probability of decrement. This is the case for active members who are exposed to multiple risks during the year. For them, the probability of dying (say) will be linked to the other decrement rates and will be smaller than the rate at which the population dies. More formally, if we indicate with $q_x^{(k)}$ the rate of decrement for cause k at age x and with $q_x^{(j)}$, then $q_x^{(k)} > q_x^{(j)}$. In practice, by assuming that rates are uniformly distributed over the period, decrement probabilities can be approximated with the following:

$$q_x^{(k)} \approx q_x^{(k)} \prod_{j \neq k} \left(1 - \frac{1}{2} q_x^{(j)} \right). \quad (12.1)$$

Composite survival probabilities give us the probability that an individual will survive from one period to the next. In a single-decrement environment, they are simply the complement of the decrement probability. In a multiple-decrement

¹ Given the summary nature of this chapter, we do not provide definitions here but refer the reader to Impavido (2011) and references contained therein for a more pedagogical text.

² The accompanying spreadsheet uses two decrement factors (mortality and termination) to derive composite survival probabilities for active members.

environment, they are “composite” in the sense that they are the complement of the sum of the relevant decrement probabilities. More concretely, for an individual aged x exposed to mortality, termination, and early retirement contingencies, the composite survival probability of surviving until age $x+1$ is given by

$$p_x^{(T)} = 1 - (q_x^{(m)} + q_x^{(t)} + q_x^{(r)}), \quad (12.2)$$

where $q_x^{(k)}$ is defined in equation (12.1) and (T) indicates that we are in a multiple-decrement world.

Contingent, composite survival probabilities can be used to derive the survival probability over more than one period. In particular, given the time independence of probabilities, for an individual aged x , the probability of surviving for the next n periods is simply given by

$${}_n p_x^{(T)} = \prod_{s=0}^{n-1} p_{x+s}^{(T)}, \quad (12.3)$$

C. Salary assumptions

The future level of salaries is also a key assumption affecting the liabilities of DB plans. Given that benefits accrual can depend on future salaries, there needs to be a way to estimate them. In addition, because such estimations can span many years (like for mortality assumptions), their impact on funding and costing can be important.

Future salary assumptions depend on three key factors. These are merit increase assumptions, productivity improvement assumptions, and inflation assumptions. Merit increase assumptions are intended to reflect the increased contributions of members to the employer as they progress in their careers. Productivity assumptions vary across sectors, but it is not uncommon to observe an assumption of 1 percent annual increases in productivity. Both merit and productivity assumptions are difficult to estimate as they can vary significantly from one employer to another and can change in response to changes in economic and labor market conditions. Finally, the most important assumption for future salaries is typically inflation.

Future wage assumptions are summarized by the wage function. Once the aforementioned assumptions are made, it is possible to estimate the future wage levels with the aid of the wage function. For any plan member of age x who joins the plan at age y with a beginning of the period wage $w_{y,y}$, future salaries can be calculated as

$$w_{x,y} = w_{y,y} \frac{ms_{x,y}}{ms_{y,y}} [(1 + \pi)(1 + pr)]^{(x-y)}, \quad (12.4)$$

where $w_{y,y}$ is the beginning of the period entry wage, $ms_{x,y}$ is the cumulative merit increase at age x , $ms_{y,y} = 1$ is the cumulative merit increase for service before age y (hence, $=1$ to indicate no increase), π is the inflation assumption,³ and pr is the

³ This does not need to be constant. In particular, when current levels of inflation are significantly different than long-term expectations, it is common to grade the assumptions over time.

productivity improvement assumption. The wage function $wf_{x,y}$ gives the wage $w_{x,y}$ as a multiple of the entry wage $w_{y,y}$. More formally:

$$wf_{x,y} = \frac{w_{x,y}}{w_{y,y}} = \frac{ms_{x,y}}{ms_{y,y}} [(1 + \pi)(1 + pr)]^{(x-y)}. \quad (12.5)$$

D. Discount rate assumptions

Discount rates, together with survival probabilities, are one of the most critical assumptions in valuing DB liabilities. The reason is that they are used to discount expected future cash flows, which are often very distant in time, to the present time. For instance, the last pension payment for a new entrant could be more than 80 years in the future.

There is a large debate on what the appropriate discount rate should be for pension liabilities. Traditionally, actuaries have argued that pension liabilities should be discounted at rates that reflect the expected long-term returns on a pension fund's assets, while financial economists (and increasingly some actuaries) have countered that pension liabilities should be discounted at market rates, either the risk-free government bond yields or the high-grade corporate bond yields. This debate is being superseded by a requirement for consistent (market) valuation of assets and liabilities and the introduction of countercyclical/dynamic solvency buffers.⁴ For our purpose, we allow for a very general formulation⁵ of the rate v^n that discounts the cash flow expected to occur at the beginning of period n to time zero and given by

$$v^n = \prod_{s=0}^{n-1} (1 + f_s)^{-s}, \quad (12.6)$$

where f_s is the forward interest rate assumed for the s th year. This, in turn, can be extracted from the government debt yield to maturity curve (ytm_s) from the following relationship:⁶

$$(1 + f_s) = \frac{(1 + ytm_s)^s}{(1 + ytm_{s-1})^{s-1}}. \quad (12.7)$$

2. ESTIMATION OF ACTUARIAL LIABILITIES

Actuarial liabilities are estimated for each category of plan member. The two most important categories are typically

⁴ In addition, there is a debate regarding the appropriate discount rate to be used for financial reporting versus funding standards. Regulations typically would require some form of market rate for financial reporting standards, while they would allow for a more stable rate for funding standards. See Rocha and Vitas (2010) for a discussion.

⁵ In other words, we do not enter in the debate of the appropriate discount rate. In our template, we simply allow for any type of rate, whether market determined or fixed in regulation.

⁶ The use of the forward rates, rather than the spot rates, in calculating discount rates implies an underestimation of the discount rate curve. This stems from the fact that yield rates typically contain an illiquidity premium to compensate for the higher risk involved in investing at longer maturities. Estimating such premium and correcting for it would be beyond the scope of this introductory text. For our purpose, we ignore the difference that is likely to be trivial in very liquid markets.

retirees and active members. Other categories can include active members with past service benefits, active but disabled members, and terminated members with deferred benefits. For the sake of simplicity, we will focus in this section on only retired and active members.

A. Individual actuarial liabilities for retired members

The actuarial liability for a retired member can be calculated through the use of annuity functions. Basically, this involves projecting the year-by-year contingent cash flows and discounting them through an annuity function that corresponds to the form of pension provided by the DB plan.

DB plans provide a variety of annuities. These aim at insuring retirees against risks like longevity and inflation. They can cover also spouses and other beneficiaries, insurance can be extended for a certain period or for life, and they sometimes provide for participation in investment and mortality experience. In our template, we focus exclusively on straight individual life annuities.

A straight life nominal annuity insures individuals against longevity and investment risks only. It provides periodical benefit payments fixed in nominal terms B starting with the normal retirement⁷ age $r=55$ until death. The present value of these expected cash flows is given by

$$B\ddot{a}_r = \sum_{s=0}^{\infty} B_s p_r^{(m)} v^s, \quad (12.8)$$

where $\ddot{a}_r$ is the present value of a straight life annuity due (i.e., payable at the beginning of each period), $p_r^{(m)}$ is the contingent survival probability for an individual retired at the normal retirement age r defined in equation (12.3), and v^s is the discount rate for period s defined in equation (12.6).

A straight life real annuity insures also against inflation risk. In this case, the value of the periodic benefit payment is not constant anymore. After the initial payment of B , it increases with inflation at a rate π .⁸ Its present value is given by

$$B\ddot{a}_r^\pi = \sum_{s=0}^{\infty} B(1+\pi)^s p_r^{(m)} v^s. \quad (12.9)$$

Initial periodic payments of real annuities are much lower than nominal annuities that have the same present value. Figure 12.1 reports the expected cash flows for a nominal straight life annuity due paying $B=1$ and a real annuity starting

at the normal retirement age $r=55$. The areas under the curves are nothing more than the sum of these expected cash flows and given by equations (12.8) and (12.9), respectively. Given that the present values of the two annuities must be the same,⁹ it follows that the initial payments from the real annuity must be lower than nominal payments (Figure 12.2). Of course, many more annuities can be imagined. However, these are not discussed here as this would be beyond the scope of this introductory presentation.

B. Individual actuarial liabilities for active members

The calculation of actuarial liabilities for active members is a considerably more complex affair than the calculation of actuarial liabilities for retired members. This depends on the plan benefit formula and on the actuarial cost method used. Unfortunately, many benefit formulae exist and many cost methods can be used. We discuss these two concepts in turn.

Benefit Functions

The accrued benefit function is used to determine the level of benefits paid at retirement. It sums the amount of benefits that individuals have accrued over time before retirement. Hence, if we define by $b_{x,y}$ the amount of benefits accrued between age x and age $x+1$ by a plan member who entered at age y (sometimes called “accrual factor”), the accrued benefit function is simply given by the sum of past accrued benefits:

$$B_{x,y} = \sum_{s=y}^{x-1} b_{s,y}. \quad (12.10)$$

Accrued benefit functions can take many forms, and in this study, we consider only the final average salary.¹⁰

Final average salary benefits are based on the average compensation over a period defined by the plan rules. In its simplest case, plan members are credited units of benefits during their career through a given accrual factor, and these will be applied to their average pensionable salary during a specified period (say, 1, or 3, or 5, or 10 years immediately prior to retirement or termination).¹¹ In a more complex case, the average pensionable salary is calculated over the years of highest compensation (rather than the last years prior to retirement), which might not need to be consecutive. Another variant is to average the highest compensation over n years in the final $m > n$ years of employment (rather than over the full career). If the

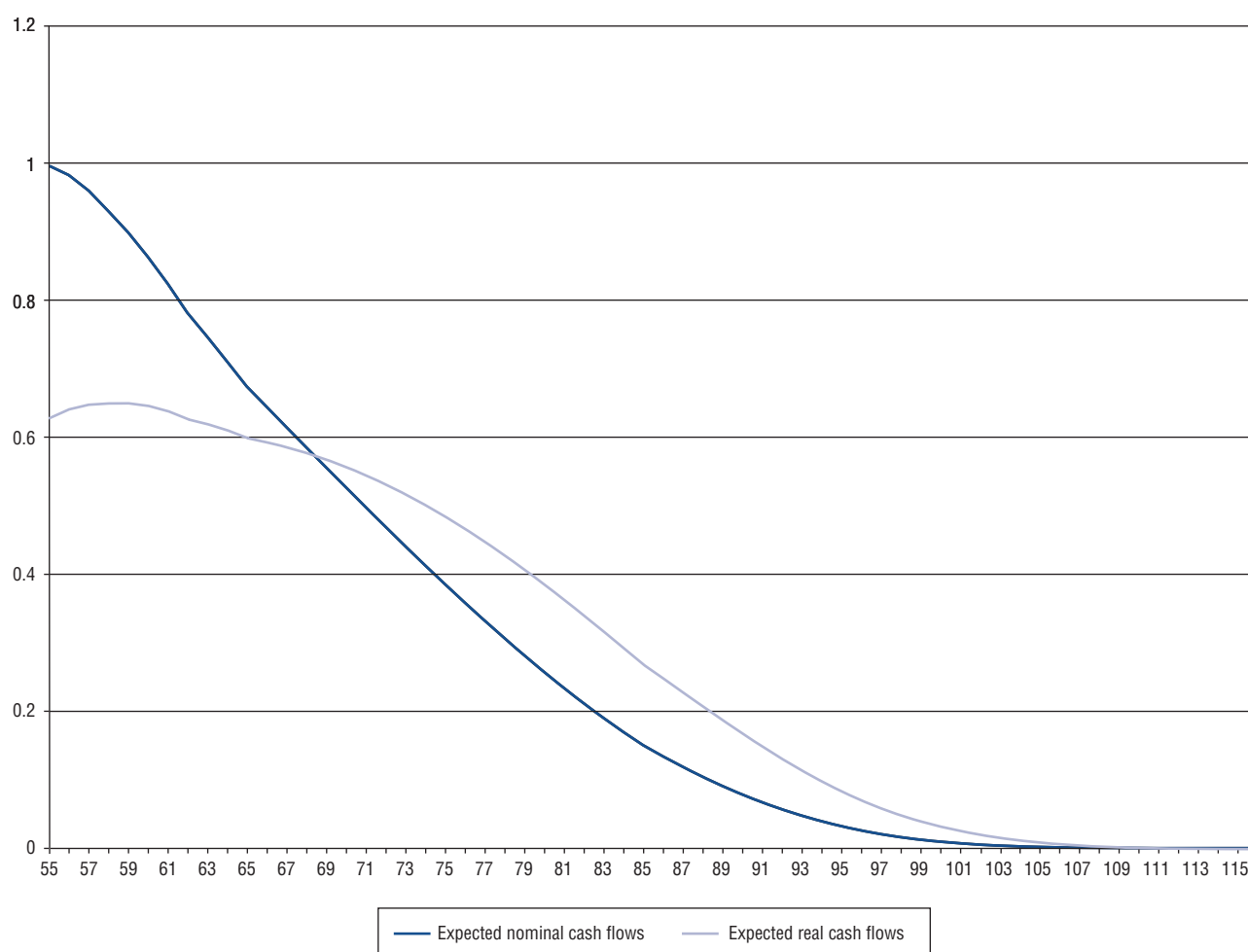
⁷ Throughout Sections 3 and 4, we calculate individual liabilities assuming that plan members all retire at the normal retirement age. In practice, for valuing liabilities, one needs to begin with the actual retirement age of each retiree. We discuss later the impact of this simplification, which is also made in the accompanying template Model.xls.

⁸ For example, the rate of increase might be one-half the inflation rate, or it may be discretionary or ad hoc. For instance, many pension plans that provide indexed pensions in the United States do not provide full indexation or automatic cost-of-living adjustment. As discussed in Section 4, we will be able to account for this possibility by assuming alternative inflation indexation rates for annuities in the accompanying template Model.xls.

⁹ Intuitively, if $\ddot{a}_r$ is the premium that would buy a straight nominal annuity that pays periodically $B=1$, then the same premium would buy a real annuity that pays less (initially, although increasing with inflation).

¹⁰ Again, Impavido (2011) and references contained therein include a more general discussion of benefit functions.

¹¹ For instance, with a 2 percent accrual factor, an average salary during the final five years of US\$100,000, and 40 years of service, the retired member would receive an initial pension worth US\$80,000.



Source: Author.

Note: Calculations assume: $r = 55$, $\pi = 0.035$, $B = 1$ for nominal annuity, $B = 0.6246$ for real annuity with same present value, mortality rates defined and discount rate defined in Table 12.1.

Figure 12.1 Straight Life Annuities—Expected Nominal and Real Cash Flows

average is calculated over the last five years immediately prior to retirement or termination, the accrued benefit function is given by

$$B_{x,y} = \sum_{s=y}^{x-1} b_{x,y} = b(x-y) \frac{\sum_{s=x-5}^{x-1} w_{s,y}}{5} = b(x-y) \overline{w_{x,y;5}}, \quad (12.11)$$

where $\overline{w_{x,y;5}}$ is the final five years' average salary. When the average is calculated only on the last year of service, the accrued benefit collapses to a final salary function, which we will be using in this study.

Projected Unit Credit Cost Method

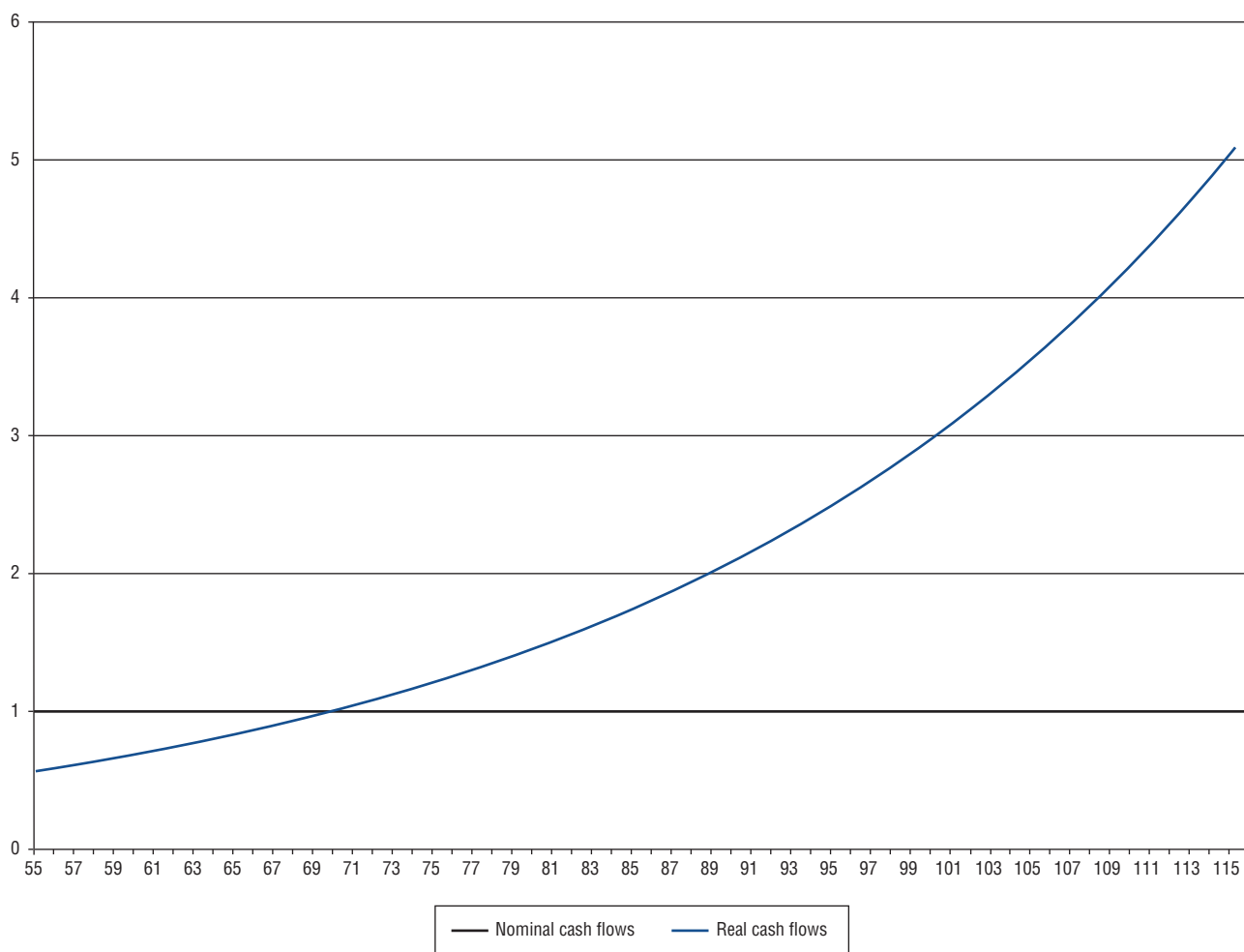
The second element needed for the calculation of liabilities for active members is the actuarial cost method. Actuarial cost methods define how expected costs of a pension plan are allocated over the period of active membership. These methods (also known as “funding methods”) generate an annual

normal cost ($NC_{x,y}$) representing the present value of the benefits earned under the plan that have been allocated to that year.¹² The $NC_{x,y}$ for a member for a particular year is given by

$$NC_{x,y}^* = (b_{x,y}^*) ({}_{r-x}p_x^{(T)}) (v^{r-x}) \ddot{a}_r, \quad (12.12)$$

where $b_{x,y}^*$ is the benefit accrual allocated to the period between attained age x and attained age $x+1$ for a member who joined

¹² Unfortunately, things are not as simple because many other types of costs also should be considered when calculating the actuarial liability. Three are more common: supplemental costs, past service costs, and ancillary costs. The first type of cost is generated when the experience of the plan deviates from the assumed cost factors creating actuarial gains or losses. The second type of cost is generated when the plan recognizes liabilities for past service prior to the introduction of the plan (or an amendment of the plan that applies improvements to past service). The third is associated with ancillary benefits like death and disability (in addition to retirement/pension benefits). For sake of simplicity, we will not discuss these types of costs, and we consider only normal costs associated with retirement benefits based on the normal retirement age r .



Source: Author.

Note: Calculations assume: $r = 55$, $\pi = 0.035$, $B = 1$ for nominal annuity, $B = 0.6246$ for real annuity with same present value, mortality rates defined and discount rate defined in Table 12.1.

Figure 12.2 Straight Life Annuities—Nominal and Real Cash Flows per Surviving Annuitant

the plan at age y and stemming from the use of a cost method yet to be defined (and therefore marked by the asterisk), ${}_{r-x}p_x^{(T)}$ is the composite probability that the individual aged x will survive until the normal retirement age r , v^{r-x} is the discount factor used to calculate the present value of the accrued benefits, and $\ddot{a}_r$ is the present value at the normal retirement age r of the annuity provided by the plan.

In general, the sum of the normal costs up to the attained age provides the actuarial liability under the given cost method. Hence, the actuarial liability for an individual with attained age x and entry age y can be defined as

$$AL_{x,y}^* = \sum_{s=y}^{x-1} NC_{s,y}^* = {}_{r-x}p_x^{(T)} v^{r-x} \ddot{a}_r \sum_{s=y}^{x-1} b_{s,y}^* = (B_{x,y}^*) ({}_{r-x}p_x^{(T)}) (v^{r-x}) \ddot{a}_r \quad (12.13)$$

where $B_{x,y}^*$ is the value of accrued benefit up to the attained age x for a member who joined the plan at age y and stemming

from the use of a cost method yet to be defined (and therefore marked by the asterisk).

Two general categories of cost methods are more common: (1) *benefit allocation methods*; and (2) *cost allocation methods*. The two methods are radically different and will produce very different values of actuarial liabilities for the same set of underlying data and assumptions, as shown later. In general:

- Benefit allocation methods define how to allocate the total benefits that will be earned by a person to the various years of service and then value the annual allocations.
- Cost allocation methods value the cost of the total benefits that will be earned by a person and then allocate the total cost among the various years of service.

As a result, they produce different values of actuarial liabilities, $AL_{x,y}^*$. In this study, we focus exclusively on benefit allocation methods.

Benefit allocation methods allocate units of benefits to the various plan years. With these methods, the actuary would calculate the present value of accrued benefits that all plan members may become entitled to (if they survive) either from past service (i.e., rendered up to the valuation date) or also adding future service (i.e., projected until retirement). Two types of benefit allocation methods are described here: the unit credit method and the projected unit credit method.¹³

The unit credit method considers only past service. The accrual function $b_{x,y}$ is determined by the benefit function/formula defined by the plan (typically, flat dollar). The accrued benefit function $B_{x,y}$ is simply the sum of the accruals between entry age y and attained age x . Finally, the actuarial liability is called accrued benefit obligation (ABO), which is equal to the present value of the benefits accrued up to the attained age x . Hence:

$$\begin{aligned} b_{x,y} &= b_{x,y} \\ B_{x,y} &= \sum_{s=y}^{x-1} b_{s,y} \\ ABO_{x,y} &= (B_{x,y})_{r-x} p_x^{(T)} v^{r-x} \ddot{a}_r \end{aligned} \quad (12.14)$$

The projected unit credit method considers also future service until the normal retirement age. The accrual function $b_{x,y}$ is determined by applying the benefit function/formula defined by the plan (flat dollar, career average salary, or final average salary). The accrued benefit function $B_{r,y}$ is simply the sum of the accruals between entry age y and the normal retirement age r . The method therefore produces an actuarial liability that is equal to the present value of the benefits that will be accrued at the normal retirement age evaluated with the accrued benefit function/formula defined by the plan. Given the definition, we will call this liability retirement benefit obligation (RBO).¹⁴ Hence:

$$\begin{aligned} b_{x,y} &= b_{x,y} \\ B_{x,y} &= \sum_{s=y}^{r-1} b_{s,y} = B_{r,y} \\ RBO_{x,y} &= (B_{r,y})_{r-x} p_x^{(T)} v^{r-x} \ddot{a}_r \end{aligned} \quad (12.15)$$

The RBO method is not used in practice as it breaks any relationship between costs and service rendered. The RBO method (also known as initial funding method) forces the sponsor to reserve for the present value of all possible benefits accruable until retirement right at entry age y , independently of whether the plan member will remain on payroll until then. This goes against rational actuarial and accounting practices, which attempt to allocate pension costs in reasonable relationship to the services rendered. Hence, it is not used in practice.

Nonetheless, it is a useful concept as all other accepted methods are essentially a weighted function of the RBO method.

For this reason, regulations allow fund managers to prorate the RBO obligation for funding purposes. Given that the RBO liability is often much larger than the ABO liability and because it is not certain that workers will remain on payroll until retirement, regulations allow plan managers to prorate the RBO liability over time. What proration achieves is to allocate projected benefits equally to all years of service. Notice that accrued benefit functions are very flat during the early years of an individual career and very steep toward retirement. Prorating reduces the amount of reserves that the sponsor has to allocate toward retirement by anticipating factors that are expected to increase the ultimate level of benefits and recognizing their cost throughout the working years. Proration of the RBO liability produces an actuarial liability called projected benefit obligation (PBO).

The most commonly used proration is the constant dollar method.¹⁵ With the constant dollar pro rata method, the accrual factor $b_{x,y}$ is constant and defined as a fixed share of retirement benefits—that is, $b_{x,y} = B_{r,y} / (r - y)$. Hence, the accrued benefits $B_{x,y}$ at age x are simply the product of the constant accrual benefit and the number of years between attained and entry age—that is, $B_{x,y} = (x - y) B_{r,y} / (r - y)$. Hence:

$$\begin{aligned} b_{x,y} &= \frac{B_{r,y}}{(r - y)} \\ B_{x,y} &= \frac{(x - y)}{(r - y)} B_{r,y} \\ PBOcd_{x,y} &= \frac{(x - y)}{(r - y)} B_{r,y} p_x^{(T)} v^{r-x} \ddot{a}_r = \frac{(x - y)}{(r - y)} RBO_{x,y} \end{aligned} \quad (12.16)$$

C. From individual to aggregate values

So far we have focused on the liabilities of a single individual—retired or in active service. Aggregate liabilities for the whole plan can be calculated in two different ways. Under the individual method, the actuary calculates the individual liabilities and other functions (as done in this section) and then sums them across all different individual members. Under the aggregate method, no reference is made to individuals, and values are calculated on an aggregate basis by averaging values across all individuals or specified cohorts. In the next section, we will use an aggregate method by annual cohorts as this allows us to be much more parsimonious in terms of data requirement. The template can also use average information on cohorts of more than one year.¹⁶

¹³ Again, Impavido (2011) and references contained therein include a more general discussion of alternative cost methods.

¹⁴ We owe this very intuitive RBO label to Milevsky (2006). This is also known as the present value of future benefits. Notice that the way we use “projected unit credit” in this text to indicate the general class of methods that project benefits until the normal retirement age can be misleading in the United States. In this country, this term is generally used to indicate the constant dollar variation that we will discuss later.

¹⁵ Indeed, in the United States, this is the only method that can be used for preparing financial statements, although many others can be used for funding purposes.

¹⁶ An aggregate approach is acceptable if the data necessary for individual calculations are unavailable or if the purpose of the valuation does not require much precision (such as a stress test done as part of an FSAP). However, individual calculations are greatly preferable.

2. STRESS TEST METHODOLOGY

This section refers to a real-life plan to discuss a possible stress test methodology implemented with the accompanying template Model.xls. The model plan uses the definitions developed in the previous sections, and the underlying data and assumptions are taken from a real DB plan. This section can be read in conjunction with the Appendix, which focuses on the mechanics of the accompanying Excel template.

Even for a simple plan like the one chosen, several simplifications are necessary in order to make the template easy to use in an operational context.

A. The real pension plan

The real plan has the following characteristics:

- **Benefits.** The plan provides for retirement, death, and disability benefits with various withdrawal options in the form of single gender-specific and/or joint straight life real inflation-indexed annuities or cash lump sums.
- **Retirement benefits.** Retirement benefits represent the largest liability of the plan and are based on a final five-year average salary formula with an accrual rate of 1 percent. Entry age varies from a minimum of $y=20$ to the normal retirement age, $r=55$. Early retirement is allowed starting with age $x=45$, and in a few cases delayed retirement was granted. We know that early retirement reductions are less than actuarially fair, but no information is available on benefit accrual in the case of delayed retirement.
- **Assets and liabilities.** The investment portfolio is reported by asset classes and economic sectors, as well as domestic and foreign. Total assets are expressed in local currency units (LCU) and amount to LCU3,773 million. The plan actuary uses a projected benefit obligation constant dollar (PBOcd) actuarial method to calculate active members' individual actuarial liabilities. These are added up to yield the plan aggregate liability. The latest actuarial report shows liabilities amounting to LCU4,000 million.
- **Solvency regulation.** No minimum solvency margin is required in the jurisdiction. The scant rules issued simply require that a fund has assets in excess of liabilities to be considered "funded." The reported solvency ratio is 94 percent.
- **Actuarial cost factors.** The plan actuary uses the male mortality and termination rates defined in Table 12.2. In addition, we could obtain inflation, salary increase, and discount rate assumptions reported in Table 12.1. We could not obtain information on other decrement factors as the actuarial report was not available.
- **Plan membership.** The relevant distributions are reported in Figure 12.3. The first panel reports the density distribution of the number of active workers with two peaks around ages 40–44 and 49–53. The second

panel reports the density distribution of the wage remuneration of active workers with two peaks around ages 40 and 52. The third and fourth panels report the density distributions of the number of retired workers and their pensions, with a large concentration soon after the normal retirement age of 55. Notice that the active distributions include individuals active beyond the normal retirement age and the retired distributions include individuals who retired before the normal retirement age. Actual data are reported in Table 12.1.

B. The model pension plan

On the basis of the information collected, we constructed a model plan with the following characteristics:¹⁷

- **Benefits.** We consider only retirement benefits in the form of a single life inflation-indexed annuity.
- **Retirement benefits.** Retirement benefits are calculated on the basis of a final salary formula with a constant accrual rate. Effective entry and retirement ages are assumed the same for all members and set at $y=20$ and $r=55$.
- **Assets and liabilities.** For assets, we use investment portfolio data reported by the real plan. Regarding liabilities, we use an aggregate (by annual cohorts) PBOcd method to calculate active member liabilities. Notice that we are not interested in the level of the actuarial liabilities per se but in their change for given shocks in the actuarial factors. Once the model plan's actuarial liabilities are estimated, they are re-scaled to coincide with the liabilities calculated by the plan actuary. Once the shock is applied to the actuarial factors and the change in the model plan's actuarial liability is calculated, this same change is applied to the real plan's actuarial liability to calculate the change in the funding ratio. The error made by moving from the change in the model liability to the change in the plan liability is immaterial for our purpose.
- **Actuarial cost factors.** Given that we could not obtain additional information on actuarial cost factors, we use only male mortality for all members and termination rates defined in Table 12.1. Wage projections are calculated using actual merit scale, inflation, and productivity assumptions reported in Table 12.2.
- **Plan membership.** We use real plan distributions reported in Figure 12.3. Given the cutoff date imposed by our normal retirement assumption at age $r=55$, early retirees are ignored in our calculations while late retirees are assumed to have retired at the normal retirement age; all members are assumed to have full career service.

¹⁷ The Appendix contains a detailed description on how the accompanying template works. Table 12.7 and Table 12.8 summarize the actuarial cost factors used.

TABLE 12.1

Actuarial Factors Used in the Model Plan (Series)									
x	$q_x^{(m)}$	$q_x^{(t)}$	ytm_s	ms_x	r_x	AN_x	RN_x	AW_x	RB_x
0	0.002080		0.09		0.02				
1	0.000815		0.09		0.02				
2	0.000454		0.09		0.02				
3	0.000367		0.09		0.02				
4	0.000321		0.09		0.02				
5	0.000291		0.09		0.02				
6	0.000270		0.09		0.02				
7	0.000257		0.09		0.02				
8	0.000294		0.09		0.02				
9	0.000325		0.09		0.02				
10	0.000350		0.09		0.02				
11	0.000371		0.09		0.02				
12	0.000388		0.09		0.02				
13	0.000402		0.09		0.019				
14	0.000414		0.09		0.019				
15	0.000425		0.09		0.019				
16	0.000437		0.09		0.019				
17	0.000449		0.09		0.019				
18	0.000463		0.09		0.019				
19	0.000480		0.09		0.019				
20	0.000499	0.246913	0.09	1	0.018				
21	0.000519	0.227261	0.09	1.0455535	0.017				
22	0.000542	0.209647	0.09	1.0921193	0.015				
23	0.000566	0.193147	0.09	1.1396974	0.013	1		13,051.2	
24	0.000592	0.177861	0.09	1.1882878	0.01				
25	0.000616	0.163588	0.09	1.2368782	0.006	2		26,645.7	
26	0.000639	0.154278	0.09	1.2874932	0.005	3		42,413.2	
27	0.000659	0.138179	0.09	1.3381082	0.005	1		9,350.3	
28	0.000675	0.126942	0.09	1.3887232	0.005	9		121,732.7	
29	0.000687	0.116617	0.09	1.4413628	0.005	14		194,015.8	
30	0.000694	0.107226	0.09	1.4929901	0.005	20		238,983.9	
31	0.000699	0.098598	0.09	1.5456297	0.005	17		195,757.2	
32	0.000700	0.090728	0.09	1.5992816	0.005	31		305,389.9	
33	0.000701	0.083717	0.09	1.6519212	0.005	34		373,748.8	
34	0.000702	0.077340	0.09	1.7055731	0.005	38		448,630.4	
35	0.000704	0.071678	0.09	1.7582127	0.005	44		640,414.5	
36	0.000719	0.066693	0.09	1.8118646	0.005	39		702,237.6	
37	0.000749	0.062155	0.09	1.8645042	0.006	41		882,063.2	
38	0.000796	0.058273	0.09	1.9171438	0.007	51		1,238,367.8	
39	0.000864	0.054765	0.09	1.9697834	0.008	54		1,061,260.2	
40	0.000953	0.051830	0.09	2.0203984	0.009	67		1,448,272.7	
41	0.001065	0.049299	0.09	2.0720257	0.01	49		1,140,185.0	
42	0.001201	0.047173	0.09	2.1216284	0.011	70		1,482,635.0	
43	0.001362	0.045351	0.09	2.1712311	0.012	69		1,589,866.5	
44	0.001547	0.043833	0.09	2.2188092	0.013	55		1,552,165.0	
45	0.001752	0.042618	0.09	2.265375	0.014	44	1	1,025,525.9	4,129.0
46	0.001974	0.041543	0.09	2.3109285	0.015	57		1,500,607.7	
47	0.002211	0.040694	0.09	2.3554697	0.016	48		1,144,279.7	
48	0.002460	0.039885	0.09	2.3979863	0.017	50		1,239,180.9	
49	0.002721	0.039277	0.09	2.4394906	0.018	75	1	1,779,703.7	1,391.7
50	0.002994	0.038670	0.09	2.477958	0.019	52	4	1,293,359.4	151.1
51	0.003279	0.038625	0.09	2.5154131	0.02	53	2	1,556,602.2	345.2
52	0.003576	0.037455	0.09	2.5508436	0.02	64	2	1,848,848.7	2,138.7
53	0.003884	0.036645	0.09	2.5842495	0.02	53	5	1,590,645.0	24,468.5
54	0.004203	0.035835	0.09	2.6156308	0.019	42	3	1,585,986.7	15,729.3
55	0.004534		0.09		0.018	8	40	225,327.3	38,941.3
56	0.004876		0.09		0.017		54		636,361.5
57	0.005228		0.09		0.016	3	29	105,313.1	279,947.7
58	0.005593		0.09		0.016	1	32	14,368.8	35,187.6
59	0.005988		0.09		0.016		13		114,217.0
60	0.006428		0.09		0.015		15		88,488.6
61	0.006933		0.09		0.015		8		72,969.6
62	0.007520		0.09		0.014		14		5,797.4

(continued)

TABLE 12.1 (continued)

x	$q_x^{(m)}$	$q_x^{(r)}$	ytm_s	ms_x	r_x	AN_x	RN_x	AW_x	RB_x
63	0.008207		0.09		0.014		15		1,592.2
64	0.009008		0.09		0.014		10		69,779.9
65	0.009940		0.09		0.013		4		13,472.2
66	0.011016		0.09		0.013		7		3,789.5
67	0.012251		0.09		0.014		7		38,979.3
68	0.013657		0.09		0.014		8		3,479.0
69	0.015233		0.09		0.015		6		3,249.4
70	0.016979		0.09		0.015		5		2,461.3
71	0.018891		0.09		0.015		3		1,627.4
72	0.020967		0.09		0.015		3		12,221.4
73	0.023209		0.09		0.015				
74	0.025644		0.09		0.014		4		1,182.3
75	0.028304		0.09		0.014				
76	0.031220		0.09		0.013		4		1,112.9
77	0.034425		0.09		0.012		1		1,534.3
78	0.037948		0.09		0.011				
79	0.041812		0.09		0.01				
80	0.046037		0.09		0.009		2		5,484.5
81	0.050643		0.09		0.008		2		299.5
82	0.055651		0.09		0.008				
83	0.061080		0.09		0.007		1		1,534.3
84	0.066948		0.09		0.007				
85	0.073275		0.09		0.007				
86	0.080076		0.09		0.006				
87	0.087370		0.09		0.005				
88	0.095169		0.09		0.005				
89	0.103455		0.09		0.004				
90	0.112208		0.09		0.004				
91	0.121402		0.09		0.003				
92	0.131017		0.09		0.003				
93	0.141030		0.09		0.003				
94	0.151422		0.09		0.002				
95	0.162179		0.09		0.002				
96	0.173279		0.09		0.002				
97	0.184706		0.09		0.001				
98	0.196946		0.09		0.001				
99	0.210484		0.09		0.001				
100	0.225806		0.09		0.000				
101	0.243398		0.09		0.000				
102	0.263745		0.09		0.000				
103	0.287334		0.09		0.000				
104	0.314649		0.09		0.000				
105	0.346177		0.09		0.000				
106	0.382403		0.09		0.000				
107	0.423813		0.09		0.000				
108	0.470893		0.09		0.000				
109	0.524128		0.09		0.000				
110	0.584004		0.09		0.000				
111	0.651007		0.09		0.000				
112	0.725622		0.09		0.000				
113	0.808336		0.09		0.000				
114	0.899633		0.09		0.000				
115	1.000000		0.09		0.000				

Source: Actual defined benefit plan.

C. Impact of simplifications made

As it is, we have had to make several simplifications in this very simple plan, but they do not reduce the usefulness of the accompanying template. The simplifications might result

in either an overestimation of real liabilities or an underestimation. Typically, however, they should have an ambiguous impact on liability estimation:

- **Simplifications that imply an underestimation of true liabilities.** We consider only pension benefits

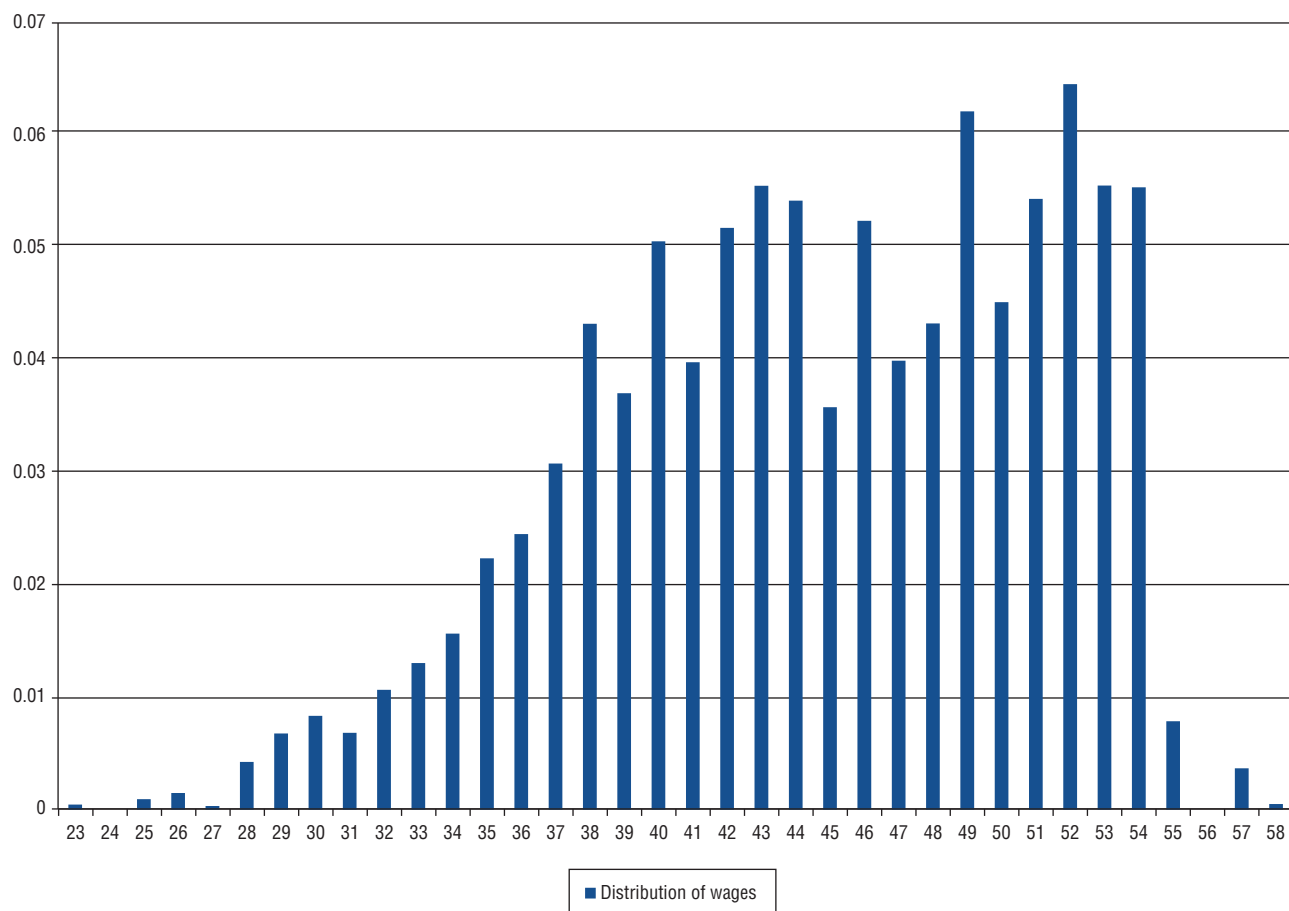
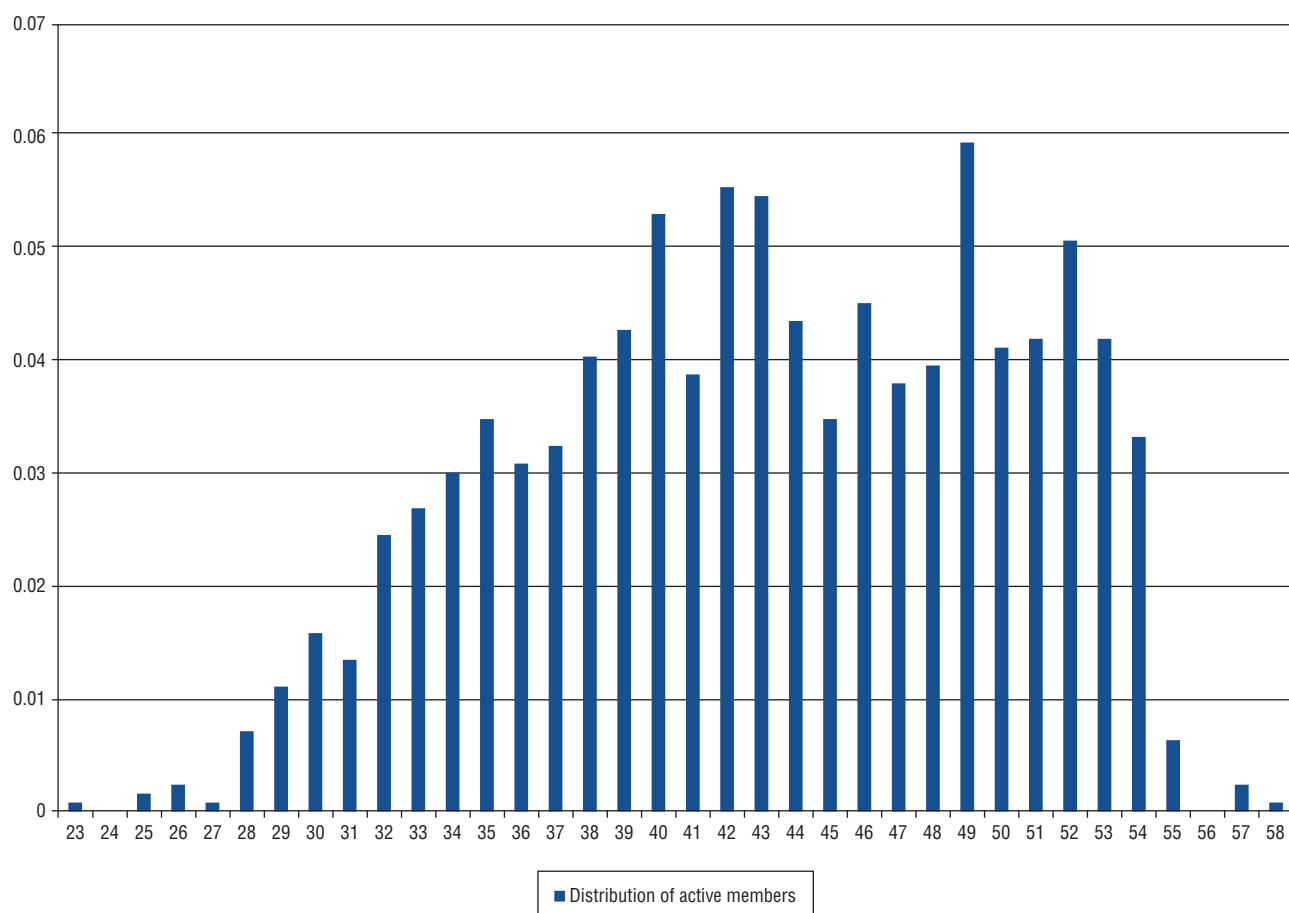
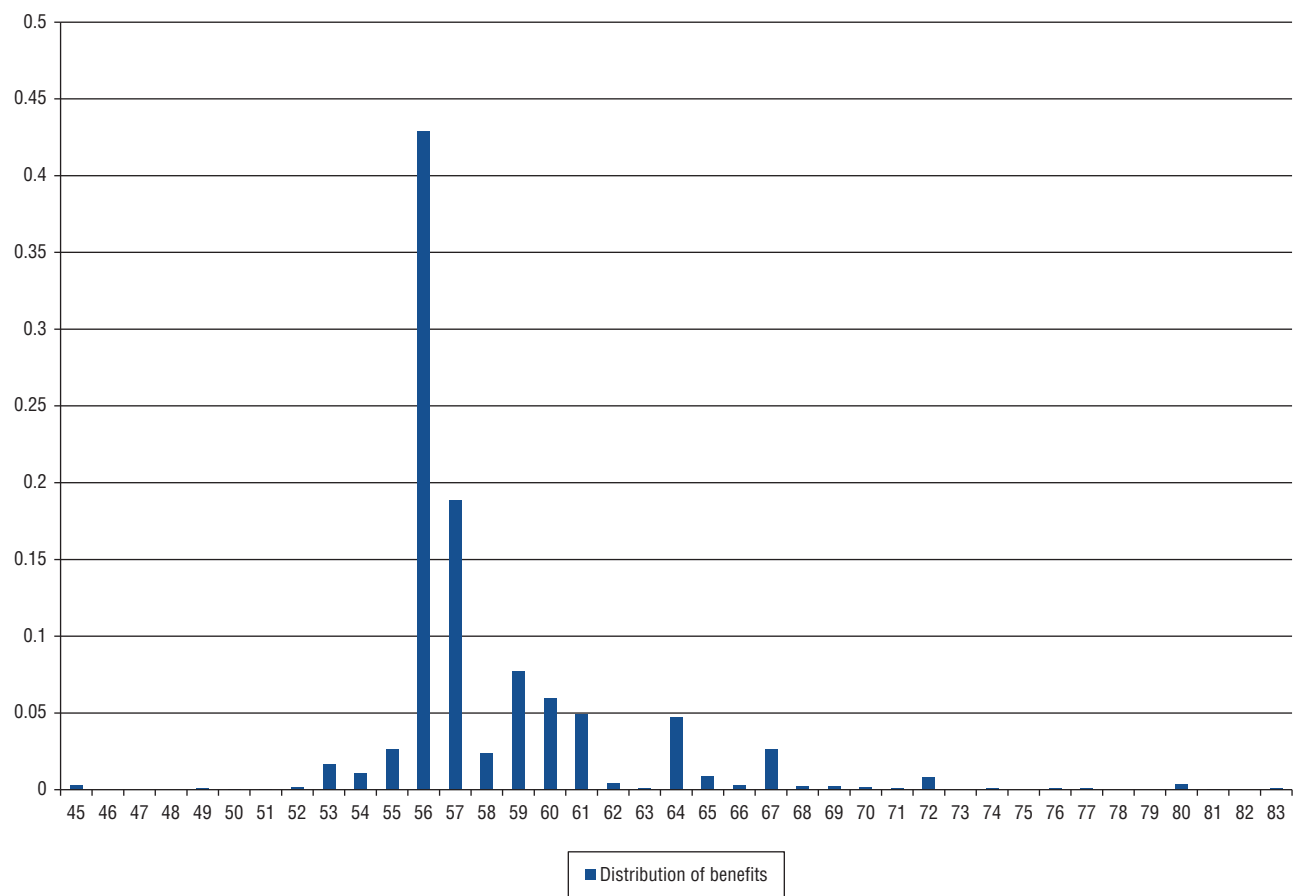
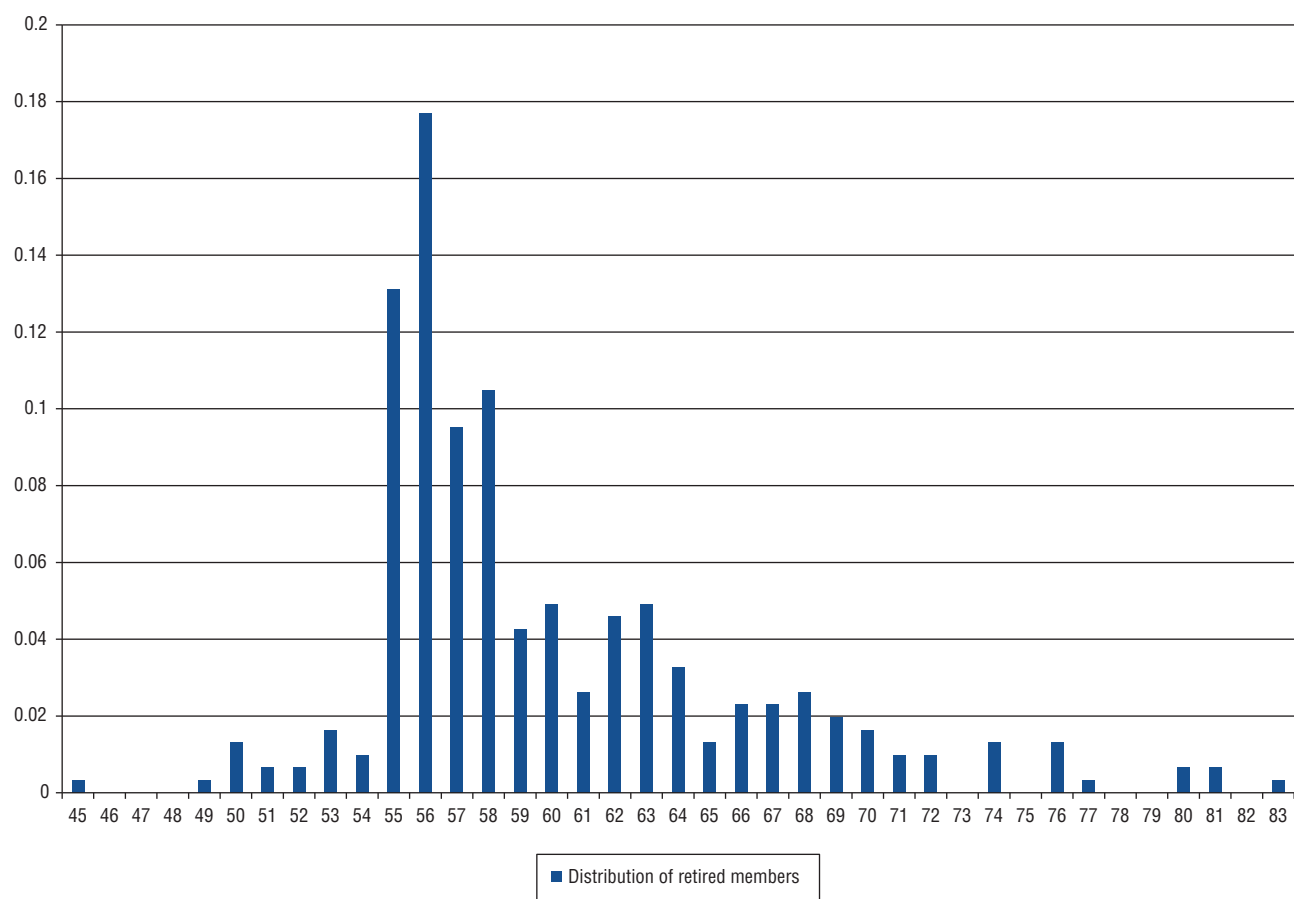


Figure 12.3 Active and Retired Member Distributions



Source: Actual DB plan.

Figure 12.3 (continued)

TABLE 12.2

Actuarial Factors Used in the Model Plan (Cells)			
Symbol	Value	Input Cells	Assumption
y	20	Input!D12	Entry age
r	55	Input!D13	Normal retirement age
x_{max}	115	Input!D14	Max age
p^r	1.0%	Input!D17	Labor productivity
π^w	3.5%	Input!D18	Inflation (wages)
π^a	3.5%	Input!D21	Inflation (annuities)
b	1.0%	Input!D22	Accrual rate

Source: Actual defined benefit plan.

and disregard ancillary benefits like death and disability. We also ignore lump sum commutation, which can result in large underestimation of retirees' liabilities.¹⁸ We also consider only single life annuities and disregard joint life annuities. This underestimates the annuity factor for active members and therefore produces a smaller valuation of their liabilities. Naturally, retired members' liabilities also are underestimated, and such underestimation could be significant if many retired members have joint life annuities and we value them as single life annuities.

- **Simplifications that imply an overestimation of true liabilities.** We assume that all plan members join the plan at entry age y with immediate vesting. However, in reality, some individuals would enter later and not necessarily with immediate vesting, therefore accruing smaller vested benefits by the time they terminate employment or retire.¹⁹ In addition, we consider only final salary pensions that imply an overestimation of true liabilities relative to the more common case of career-average salary pensions. Finally, we consider full indexation of pension rights to inflation and full longevity insurance, but in many jurisdictions, these indexations are more and more often conditional on the performance of the plan so that inflation, longevity, and investment risk are shared between providers and retirees. Where these forms of risk sharing are present, our model provides an overestimation of liabilities.
- **Simplifications that have an ambiguous impact.** We assume that active members all retire at the normal retirement age of r . Depending on the degree of actuarial fairness of early retirement provisions, the model estimation may be higher or lower than the

plan liabilities. We also assume that the pensionable salary (the base of the accrual rate) is equal to total remuneration. This overestimates (underestimates) liabilities when pensionable salary is in fact lower (higher) than total remuneration.

- **Plan termination or continuation valuation.** The choice of method (PBOcd) implies that we are conducting a plan continuation (ongoing concern) valuation as opposed to a plan termination valuation. This may differ from the regulatory requirements in the local jurisdictions where the template is used.

The simplifications used in our template do not affect the validity of the model for stress testing purposes. The simplifications made are immaterial for our purpose. Again, we are not interested in the absolute level of liabilities, the estimation of which is best left to the plan actuary. Here, we are merely interested in their rate of change for given shocks. Hence, we can rescale our model liabilities to coincide with the true liabilities and apply the model rate of change stemming from the shock to the true liabilities to obtain the change in the funding ratio.

D. Stress testing the funding position—asset shocks

This section uses the enclosed template Model.xls to conduct single-factor stress test of the plan funding ratio. The plan is reasonably healthy, with a support ratio of 413 percent and a funding ratio of 94 percent. However, we observe that the plan is exposed to concentration risk in key asset classes and suspect that the assumptions used by the actuary to calculate liabilities are overly optimistic. For instance, the level interest rate of 9 percent used for discounting cash flows appears excessively high in relation to market interest rates, and the mortality table used is static and not dynamic. In addition, discussions with the actuary have revealed that the termination rates used have not been updated for many years: the economy is booming and the sponsor is retaining far more staff than what it used to when the termination rates were estimated. Finally, we have initial signs that the economy is overheating and inflation expectations are increasing. We conduct both asset and actuarial factor shocks to see how sensitive the funding ratio is to market risk (on the asset side) and to the actuarial assumptions used for liability valuation purposes (on the liability side).

Asset stress tests focus on concentration risk. We collected information on the investment portfolio of the model plan in sheet BAL. The data are disaggregated by type of issuer (government, financial sector, and real sector; both domestic and foreign) and by type of investment vehicle. The analysis can design any type of scenario (macro, credit, or other) that has an impact on asset values that will translate into a change in the funding ratio. For the purposes of our analysis, we assume up to a 25 percent symmetrical shock in the value of assets, issued by the government, financial sector, and real sector, in the value of bonds, stocks, and foreign assets (amounting to

¹⁸ In the United States, for instance, the discount rate used to calculate lump sum commutation is neither the one used for financial reporting nor the one used for funding purposes. It is linked to the Fed rate and established year by year by the Internal Revenue Service. Such disconnection can result in very large unfunded liabilities, depending on the monetary policy stance.

¹⁹ Also, the calculations assume that all benefit payments are made annually, at the beginning of the year. On average, they will be made one-half year later.

TABLE 12.3

Model Plan—Stress Tests (Asset Concentration Risk)							
Market shock	Government	Financial	Real	Bonds	Stocks	FX	TOT
–25%	89.74	89.99	87.47	82.12	83.06	88.10	70.75
–20%	90.66	90.86	88.84	84.56	85.31	89.35	75.46
–15%	91.57	91.73	90.21	87.00	87.56	90.59	80.18
–10%	92.49	92.59	91.58	89.44	89.82	91.84	84.90
–5%	93.41	93.46	92.96	91.89	92.07	93.08	89.61
0%	94.33	94.33	94.33	94.33	94.33	94.33	94.33
+5%	95.25	95.20	95.70	96.77	96.58	95.57	99.05
+10%	96.17	96.06	97.07	99.21	98.84	96.82	103.76
+15%	97.08	96.93	98.44	101.65	101.09	98.07	108.48
+20%	98.00	97.80	99.82	104.10	103.35	99.31	113.19
+25%	98.92	98.67	101.19	106.54	105.60	100.56	117.91

Source: Author.

Note: FX = foreign exchange; TOT = total assets.

a change in the exchange rate).²⁰ Results are reported in Table 12.3. Plan assets are fairly diversified, with marginally higher concentration in real sector investments but almost equally divided among bonds and stocks. Foreign exchange (FX) risk is unhedged, and a 25 percent local currency appreciation (or a decrease in FX denominated assets by the same amount) would reduce the funding ratio from 94 percent to 88 percent.

E. Stress testing the funding position—liability shocks

Liability stress tests are more varied, and the results can differ significantly depending on the specific circumstances of the pension plan. The crucial issue about liabilities is that their valuation relies critically on assumptions that may not be validated *ex post* by experience. The mismatch between assumptions and reality creates unfunded liabilities that might develop very slowly over time and become evident only when it is too late for the sponsor to easily—if at all—remedy the situation. Hence, adequate liability measurement becomes critical even in the short term.²¹ We consider interest rate, inflation, longevity, and termination rate shocks.

Interest Rate Shock

We assume two types of shocks: (1) we substitute the level 9 percent interest rate used for discounting future cash flows

²⁰ This is a rough test. Flat percentage changes are more commonly used to test the effects of changes in equities and real estate, but for bonds and mortgages, it would be more common to calculate or estimate the average duration of the portfolio and then test the effects of a given change in interest rates. For this exercise, information on duration of fixed income instruments was not available.

²¹ According to a recent survey, MetLife Assurance Limited (2010) finds that the five risk factors ranked highest in importance in the United Kingdom were Measurement of Technical Provisions/Liabilities, Longevity Risk, Employer Covenant, Investment Management Style, and Funding Deficits, whereas in the United States, the “most important” risk factors were Liability Measurement, Underfunding of Liabilities, Plan Governance, Asset Allocation, and Advisor Risk.

TABLE 12.4

Model Plan—Stress Tests (Level Interest Rates)						
f_t	9%	8%	7%	6%	5%	4%
Funding ratio	94.33	80.76	68.33	57.11	47.08	38.25

Source: Author.

with more reasonable level rates; and (2) we derive market discount rates from a AAA government debt yield curve, which we then shock. In general, changes in the discount rate will affect the present value of retirees’ liabilities by changing v^r and $\ddot{a}_x^r$, in equation (12.18)²² and the present value of active members’ liabilities by changing v^{r-x} and $\ddot{a}_r^r$ in equation (12.22).²³ We expect liabilities for active members to be more sensitive to interest rate changes because of their higher duration: total liability duration is 15 years, versus 17 years for active members and 10 years for retired members.

Interest rate shocks have a large impact on liabilities. If the interest rate assumption decreases from 9 percent to 4 percent, the funding ratio decreases from 94 to 38 percent, or an average 11 percent for every percentage point change in the interest rate (Table 12.4).

With AAA market rates, the funding ratio is much smaller than reported. When we derive a market discount rate from a AAA government debt yield curve (in our case, the end of 2009 U.S. domestic debt curve²⁴), the funding ratio is around 33 percent (Table 12.5). This dramatic change in the solvency position underlines the importance of having a reasonable discount factor to value liabilities. The plan actuary has assumed that a 9 percent average rate of return on assets yields a 94 percent funding ratio. However, the risk-free rate at 30 years is around 4 percent, and even if the equity risk premium is realized over the full duration of liabilities, the plan will

²² Equation reported in the Appendix.

²³ Equation reported in the Appendix.

²⁴ This is done for illustrative purposes only: the relevant yield curve for the jurisdiction where the template is applied should be used.

TABLE 12.5

Model Plan—Stress Tests (Shifts in the Yield Curve)							
Shock	–150 bps	–100 bps	–50 bps	0	+50 bps	+100 bps	+150 bps
Funding ratio	26.52	28.51	30.59	32.70	34.86	37.04	39.24

Source: Author.

Note: bps = basis points.

TABLE 12.6

Model Plan—Stress Tests (Inflation Shocks)								
	–150 bps	–100 bps	–50 bps	π^a	+50 bps	+100 bps	+150 bps	ε_{FR,π^w}
–150 bps	118.44	112.85	107.36	101.97	96.70	91.53	86.49	–10.24
–100 bps	115.45	110.01	104.66	99.40	94.26	89.22	84.31	–10.24
–50 bps	112.49	107.19	101.97	96.86	91.84	86.93	82.14	–10.24
π^w	109.56	104.39	99.31	94.33	89.44	84.67	80.00	–10.25
+50 bps	106.66	101.63	96.68	91.83	87.07	82.42	77.87	–10.25
+100 bps	103.78	98.88	94.07	89.35	84.72	80.19	75.77	–10.25
+150 bps	100.94	96.17	91.49	86.90	82.39	77.99	73.69	–10.25
ε_{FR,π^w}	5.28	5.28	5.28	5.28	5.28	5.28	5.28	

Source: Author.

Note: bps = basis points.

never be able to meet its obligation. Finally, the plan termination funding ratio varies between 26 and 39 percent if we assume parallel shifts of the yield curve of up to 150 basis points²⁵ and derive corresponding discount rate assumptions (Table 12.5).

Inflation Shock

Inflation shocks affect the values of annuities and wage projections. Inflation affects $(1 + \pi^a)$ in equation (12.18)²⁶ and therefore the annuitization factor $\ddot{a}_r^\pi$ in equation (12.22).²⁷ In addition, it affects $(1 + \pi^w)$ in equation (12.20)²⁸ and therefore the value of accrued benefits $B_{r,x}$ in equation (12.22).²⁹ Table 12.6 reports the sensitivity of the funding ratios to changes in inflation assumptions used to project wages and to calculate the annuity factor. A 100 basis point (bps) increase in the inflation assumption for annuity valuation π^a reduces the funding ratio from 94 percent to 85 percent. A 100 bps increase in the inflation assumption for wage projections π^w reduces the funding ratio from 94 to 89 percent. A 100 bps increase in both inflation assumptions reduces the funding ratio from 94 to 80 percent.³⁰

The funding ratio is more sensitive to inflation assumptions used for the projection of increases in pension payments than for wage projections. Notice that the elasticity of the funding ratio, with respect to the inflation assumption for annuity valuation ε_{FR,π^a} , is much larger than the elasticity with respect to the inflation assumption used for wage projections ε_{FR,π^w} . This is due to the fact that π^a increases both $\ddot{a}_r^\pi$ and $B_{r,x}$ in equation (12.22).³¹

Longevity Shock

We model longevity shocks by projecting mortality improvements over a number of years. In order to capture longevity improvements not captured by the static table used by the plan actuary, we project over a number of improvement years, t , the period mortality rates.

The projected mortality rate in calendar year t , ${}^{PRJ}_t q_x^{t(m)}$, is the rate from the static table multiplied by $(1 - r_x)^t$. Thus:

$${}^{PRJ}_t q_x^{t(m)} = {}^{PER} q_x^{t(m)} (1 - r_x)^t, \quad (12.17)$$

where r_x represents annual rates of mortality the multiplicative improvement in longevity for each cohort is x . As an example, Figure 12.4 reports the change in the conditional survival probabilities for an individual aged 55 by applying the U.S. mortality improvement rates used for the 1994 Group Annuity Reserving Table and projecting the static rates for $t = 30$ years.³² The area between the two curves is the change in life expectancy at age 55 ($\Delta e_{55} = 2.70$).

Longevity improvements translate into higher values of life expectancy at different ages. For a given r_x , an increasing

²⁵ Negative shifts are not strictly parallel at shorter maturities as yield rates cannot become negative.

²⁶ Equation reported in the Appendix.

²⁷ Equation reported in the Appendix.

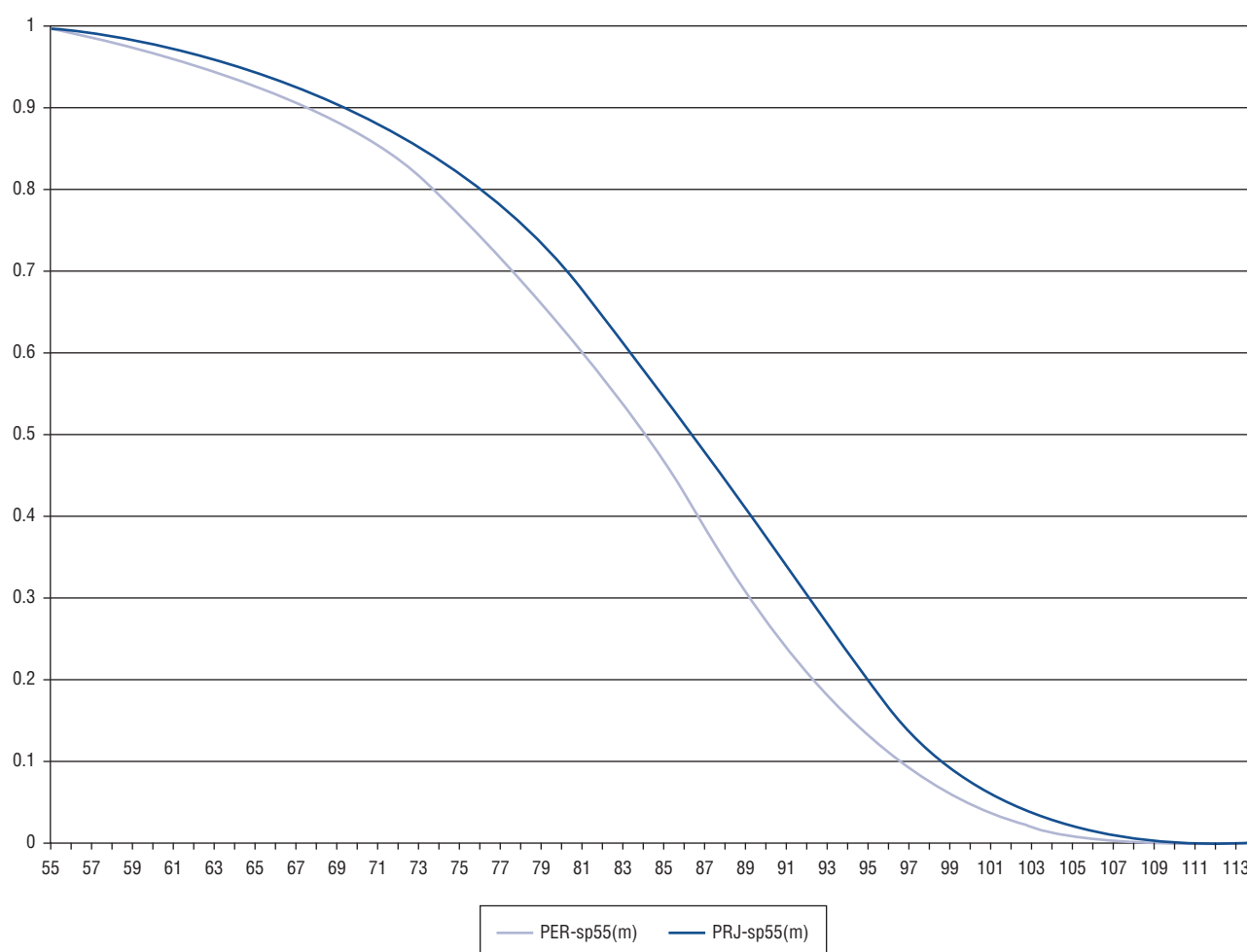
²⁸ Equation reported in the Appendix.

²⁹ Equation reported in the Appendix.

³⁰ In principle, there is no reason why we should have two different inflation assumptions for wage projections and annuity. The only reason why this is done here is that, typically, plans do not provide full inflation indexation. By disconnecting inflation assumptions for wages and annuities, we are able to test the impact of alternative inflation annuity guarantees, while maintaining full inflation indexation in wage projections.

³¹ Equation reported in the Appendix.

³² These rates are as good as any, and in any case, they should be discussed with the local actuary and amended as needed.



Source: Author.

Note: PER-sp55(m) = conditional survival probabilities (using the period table) for a male at age 55. PRJ-sp55(m) = projected (at 30 years) conditional survival probabilities for a male at age 55.

Figure 12.4 Impact of Longevity Improvements on Survival Probabilities

TABLE 12.7

Model Plan—Stress Tests (Longevity Shocks)						
t	0	30	40	50	60	70
e_{55}	28.39	31.09	31.88	32.62	33.31	33.96
Funding ratio	94.33	89.46	88.21	87.10	86.13	85.26

Source: Author.

number of improvement years translates into higher values of life expectancy. Table 12.7 reports the increase in life expectancy at the plan normal retirement age of 55 (e_{55}) from 28.39 to 33.96 years and the related decrease in the funding ratio between 94.33 to 85.26 for increasing number of improvement years between zero and 70.

Termination Rate Shock

Changes in termination rate assumptions will affect liabilities only of active members. A recent report of the personnel

TABLE 12.8

Model Plan—Stress Tests (Termination Shocks)						
$q_x^{(t)}$	–0%	–10%	–15%	–20%	–25%	–30%
Funding ratio	94.33	92.18	91.11	90.04	88.97	87.91

Source: Author.

department concludes that retention rates have increased by at least 10 percent across the board. On the basis of this information, we investigate the impact on the funding ratio of decreases in termination rates of between 10 and 30 percent. Results are reported in Table 12.8.

4. CONCLUSION

This chapter describes the basic mechanics of DB plan liability valuation and how to conduct simple stress tests of the solvency ratio, using the accompanying Excel template. We

review the essential building blocks of liability valuation of DB plans. We then construct an Excel template to analyze changes in funding ratio of DB plans for alternative values of actuarial factors and asset values. The accompanying template, Model.xls, uses a last salary DB formula and a PBOcd actuarial method to value liabilities. The template is by no means a substitute for a proper actuarial evaluation, but the simplifications introduced do not affect its usefulness to evaluate sensitivity of the funding ratio.

Section 3 provides a few examples of the stress tests that can be conducted with the template, and extensions are, of course, possible. The stress test methodology used in this note is parsimonious and aimed at quantifying key risk exposures by nonactuarial analysts. As already mentioned, it is not a substitute for a full actuarial evaluation. However, it can be extended in several ways:

- **Refinements.** On the asset side, it would be worthwhile to extend the analysis to identify sources of risk stemming from interest rate shocks at various maturities and credit risk shocks of large exposure, or of the sponsor (ability to pay contributions), so as to improve the connection of the stress tests with macro scenarios. On the liability side, the methodology could be refined by (1) improving the granularity of the age, wage, and pension distributions; (2) considering additional “decrement factors” beyond the mortality tables such as the distribution for entry into and exit from the labor force (e.g., retirement, disability, voluntary unemployment); (3) reflecting gender and types of pensions in the calculations; and (4) also considering the possibility of decreases in longevity that are due to health, famine, or natural catastrophe events. Finally, other tests could be conducted to assess the impact of plan changes. Indeed, the template lends itself to studying various parametric reforms such as changes in accrual rate, retirement ages, any actuarial assumption, and so forth.
- **Liquidity shocks.** Data on asset liquidity are not used in the template. Asset shocks should also include tests on the portion of assets that might need to be used to cover short-term liabilities. These types of shocks are potentially more severe than the ones considered, as they affect the ability of the plan to meet short-term liabilities (rather than long-term) and could force

plan managers to sell assets at (potentially) distressed prices, further undermining the ability to meet long-term liabilities. These shocks are very important for closed plans, with no active members accruing benefits or plans with very low support ratios.

- **Multi-asset shocks.** Multi-asset (factor) shocks have not been considered. This would require the estimation of asset classes return correlations.
- **Asset-liability correlations.** Asset shocks have been considered in isolation of liability shocks. When liabilities are discounted using a market yield curve, this becomes an unreasonable simplification. In such a case, the analyst should attempt to offset changes in liabilities, with changes in the value of the portion of assets that are interest-rate sensitive. This would, of course, require knowledge of the durations of these assets.
- **Expected cash flow analysis.** The analysis conducted in Section 3 is merely focused on changes in the funding ratio. An alternative, potentially appealing way to present the same results is to analyze the shocks in terms of impact on future cash flows to determine when assets (always on a termination basis) are enough to meet liabilities. The template already produces expected liability cash flows, and it would be possible to project asset cash flows, by assuming future rates of returns on assets, dispositions of assets, and the allocation of future cash flows to different types of assets.

REFERENCES

- Impavido, Gregorio, 2011, “Stress Testing for Defined Benefit Plans—A Primer,” IMF Working Paper 11/29 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=24623>
- MetLife Assurance Limited, 2010, *Comparing Pension Risk Attitudes and Aptitude in the United Kingdom and the United States* (London, December).
- Milevsky, Moshe, 2006, *The Calculus of Retirement Income* (New York: Cambridge University Press).
- Rocha, Roberto, and Dimitri Vitas, 2010, “Designing the Payout Phase of Pension Systems: Policy Issues, Constraints and Options,” World Bank Policy Research Working Paper No. 5289 (Washington: The World Bank). Available via the Internet: <http://elibrary.worldbank.org/doi/book/10.1596/1813-9450-5289>

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Appendix

The Accompanying Template (Model.xls) and Assumptions

The accompanying Model.xls template contains the following assumptions:

- **Basic Assumptions** (Table 12.2). Entry age is set at age $y=20$ (Input!D12) and normal retirement age at $r=55$ (Input!D13).
- **Decrement assumptions** (Table 12.1). Mortality rates $q_x^{(m)}$ (Input!I11:I136) are derived from the 1996 U.S. male annuitant table (t887). Termination rates $q_x^{(t)}$ (Input!J11:J136) are assumed between age 20 and 54. Mortality and termination rates are used to calculate the mortality probabilities $q_x^{(m)}$, the termination probabilities $q_x^{(t)}$, and the composite survival probabilities $p_x^{(T)}$ following the methodology discussed in Section 1.³³
- **Salary assumptions** (Table 12.1 and Table 12.2). Wage growth is set at about 7 percent and composed of 3 percent average merit increase (Input!T11:T136), 3.5 percent inflation ($\pi^w=0.035$ in Input!D18), and 1 percent productivity improvements ($pr=0.01$ in Input!D17).
- **Discount rate assumptions** (Table 12.1). A level 9 percent interest rate is assumed for the purpose of discounting liabilities. Nonetheless, the template allows for almost any type of discount curve derived from seven easily obtainable yield-to-maturity rates (Input!O11:O136) following the methodology discussed in Section 1, equations (12.6) and (12.7).
- **Retirement benefit assumptions** (Table 12.2). Retirement benefits are calculated on the basis of a final salary formula with constant accrual rate ($b=0.01$ in Input!D11). Effective entry and retirement ages are assumed the same for all members and set at $y=20$ and $r=55$. We consider only retirement benefits in the form of a single life inflation-indexed annuity, with the inflation assumption defined as $\pi^a=0.035$ in Input!D20.
- **Distributions of active and retired members** (Table 12.1). For each age cohort, we collected the number of workers (Input!V11:V136) and retirees (Input!W11:W136), as well as the cohort wage remuneration (Input!X11:X136) and cohort retirement benefit paid (Input!Y11:Y136). These data are used to calculate the total number of workers (Input!V6), retirees (Input!W6), remuneration (Input!X6), retirement benefits (Input!Y6), and their density distributions (Input!Z11:Z136 to Input!AC11:AC136).

ACTUARIAL LIABILITIES FOR RETIRED COHORTS

Actuarial liabilities for retired cohorts are calculated in sheet AL-R. This is implemented in two steps:

In the first step, we calculate for each cohort $x \in [r, \infty)$ (in row 'AL-R'!F8:CH8) the present value of \$1 real life annuity using equation (12.9). This is given by

$$\ddot{a}_x^\pi = \sum_{s=0}^{\infty} (1 + \pi^a)^s {}_s p_x^{(m)} v^s, \quad (12.18)$$

where the inflation term structure $(1 + \pi^a)^s$ is rendered in 'AL-R'!B11:B136, the discount factor curve v^s is derived using equation (12.6) and rendered in 'AL-R'!C11:C136, the conditional probabilities of survival ${}_s p_x^{(m)}$ are derived using equation (12.3) and rendered in matrix 'AL-R'!F11:CH136, and the present value $\ddot{a}_x^\pi$ is rendered in row 'AL-R'!F7:CH7.

In the second step, we aggregate the actuarial liabilities for all retired cohorts. This is given by

$$AL(R) = \sum_{x=r}^{\infty} [(RN)(RB)(RNd_x)(RBd_x)] \ddot{a}_x^\pi \quad (12.19)$$

and rendered in 'AL-R'!D1.

³³ $q_x^{(m)}$ and $q_x^{(t)}$ are calculated using equation (12.1) and rendered in Input!K11:K136 and Input!L11:L136, respectively. $p_x^{(T)}$ is calculated using equation (12.2) and rendered in Input!M11:M136.

ACTUARIAL LIABILITIES FOR ACTIVE COHORTS

Actuarial liabilities for active cohorts are calculated in sheets Brx and AL-A. Among the actuarial cost methods described in Section 3, we use the projected unit-credit, constant-dollar method (PBOcd). Again, there are two steps involved:

In the first step, we calculate in the Brx sheet the accrued benefit at retirement for each active cohort. This is done by first projecting wages until retirement for all active cohorts. These are needed because we are performing a projected benefit obligation constant-dollar evaluation as discussed in Section 1, equation (12.5). Hence, in matrix Brx!F11:CH136, we project for each cohort $x \in [y, r]$ (in row Brx!F8:CH8) the future wages for each period $s \in [x, r]$ (in column Brx!A11:A136) as a multiple of today's wage using the following formula:

$$wf_{s,x} = \frac{wf_{s,y}}{wf_{x,y}} = \frac{ms_{s,y}}{ms_{x,y}} [(1 + \pi^w)(1 + pr)]^{(s-x)}. \quad (12.20)$$

We then calculate for each cohort in row Brx!F1:CH1 the total accrued benefits at retirement, using the plan final salary benefit formula and the distributions of active members and salaries:

$$B_{r,x} = b(r-y)wf_{r,x}[(AN)(AW)(AND_x)(AWd_x)] \quad (12.21)$$

where $wf_{r,x} = \max(wf_{s,x})$.³⁴

In the second step, we aggregate in the sheet AL-A the actuarial liabilities for all active cohorts. As discussed in Section 3, these are given by the product of the present value of life annuity at retirement (${}_{r-x}p_x^{(T)}v^{r-x}\ddot{a}_r^\pi$), the accrued benefit at retirement ($B_{r,x}$), and the constant dollar method prorated $(x-y)/(r-y)$. The plan actuarial liabilities for active cohorts are nothing less than the sum of cohort liabilities across all active cohorts rendered in 'AL-A'!D1. Thus:

$$AL(A) = \sum_{x=y}^{r-1} \frac{(x-y)}{(r-y)} B_{r,x} ({}_{r-x}p_x^{(T)}v^{r-x}\ddot{a}_r^\pi) \quad (12.22)$$

where ${}_{r-x}p_x^{(T)}$ is the conditional composite probability of survival between age x and age r rendered in matrix 'AL-A'!E11:BM136 and row 'AL-A'!E3:BM3, v^{r-x} is the discount rate for $(r-x)$ periods rendered in column 'AL-A'!C11:C136 and row 'AL-A'!F4:CH4, and $\ddot{a}_r^\pi$ is the present value of a real life annuity at the normal retirement age rendered in cell 'AL-R'!F7.

Finally, total model plan liabilities are calculated in sheet AL-TOT. These are nothing less than the sum of the actuarial liabilities for active and retired members—that is, $AL(TOT) = AL(R) + AL(A)$.

³⁴ The template can be easily changed by substituting $\max(wf_{s,x})$ with $\sum_{j=0}^{n-1} \frac{wf_{r-j,x}}{n}$ to implement a final n years' average salary formula.

SECTION B

The Network Analysis Approach

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Introduction to the Network Analysis Approach to Stress Testing

MARCO A. ESPINOSA-VEGA • JUAN SOLÉ

The global financial crisis that erupted in 2007 prompted policymakers and financial stability practitioners to intensify their search for a better understanding of how and when financial linkages could pose systemic risks. Hence, it is not surprising that there has been a surge of work in this area seeking to address some of the key conceptual aspects of financial interconnectedness. However, an ongoing challenge for policymakers has been the difficulty in turning rather theoretical proposals into methodologies that can be applied to detect and assess important linkages on an ongoing basis.

One method that is receiving renewed attention is network analysis. This approach allows the design of contagion and vulnerability metrics that are relatively easy to implement and interpret, and it potentially can be combined with traditional stress testing techniques to study contagion effects. In an effort to develop and strengthen the IMF's financial surveillance capabilities, Espinosa-Vega and Solé (2011a) developed a network analysis tool that is capable of measuring direct and indirect systemic interconnectedness. We briefly discuss the rationale for using network analysis to conduct financial surveillance and provide a succinct review of the concept and some of the pertinent literature.¹

Identifying vulnerabilities arising from systemic linkages is important in an increasingly intricate and complex international financial system. In times of stress, metrics from this analysis can help policymakers decide which institutions could be too interconnected to fail. More important, a proac-

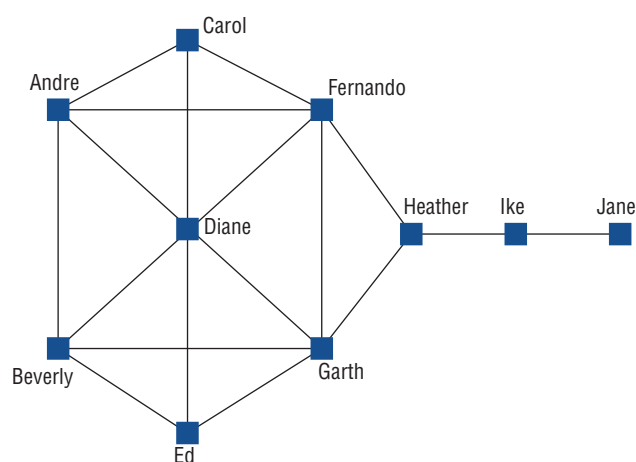
tive application of this analysis during tranquil times could help in regulating and containing specific institutions' *contribution* to systemic risk (see Espinosa-Vega and Solé, 2011b).

WHAT IS NETWORK ANALYSIS?

Network analysis is the mapping and measuring of relationships and flows within a group of agents. Thus defined, the nodes in a network represent agents (e.g., financial institutions, people, computers), while the links show relationships or flows between nodes (e.g., exchange of cash, messages, information). The main advantage of network analysis is that it provides both a visual and a mathematical analysis of relationships, from which answers to key questions about the characteristics and performance of the network can be obtained.

To illustrate, let us consider an early example from the *social networks* literature by David Krackhardt—the *kite network* (Figure 13.1). This network comprises a group of 10 friends (nodes) who communicate among themselves (exchange messages—represented by links). We could be interested, for example, in finding out which persons are more popular and whether some have friends outside the core group. Focusing individually on each agent (e.g., by interviewing him or her) may give us some information about his or her relationships. For instance, we may discover that the person called Diane talks to as many as six people in the group, whereas Heather talks only to three. This may lead us to conclude that Diane is more important than Heather in terms of guaranteeing that messages are delivered to all members of the group. But to fully exploit the information gathered—and to have a proper understanding of the group dynamics—we need a tool that can provide structure to these relationships.

¹ For an excellent Internet resource containing a wealth of information on network analysis, visit the Web site <http://www.financialnetworkanalysis.com>, maintained by Kimmo Soramäki.



Source: <http://www.orgnet.com>.

Figure 13.1 The Kite Network

Only then can we answer important questions about, for instance, an agent's relative importance within the group (e.g., his or her popularity); the speed at which a message can be delivered to all members in the group; or what possible communication failures there may be that could cause a message to never reach certain agents.

Applying network analysis to our example, it is clear that Diane is the agent with the higher number of connections. However, her failure to repeat a message would not imply that information stalls within the network. Consider instead the role played by Heather—if she is removed from the information flow, Ike and Jane's communication with the rest would break down. In technical terms, Diane has the highest *degree* in the network, whereas Heather has the highest *betweenness*. *Degree* is a measure that reflects a node's relative number of ties to other nodes, whereas *betweenness* measures the extent to which a node lies between other nodes, taking into account the connectivity of the node's neighbors and giving a higher value for nodes bridging clusters.²

The relevance of network analysis for financial stability analysis is immediately apparent if one reconsiders the above example imagining that each node is a bank receiving and/or giving funding to other banks. Thus, the application of network analysis to a banking system that had a kite structure would reveal that although Bank Diane is a central player in the network, Bank Heather is vital for the funding operations of Bank Ike and Bank Jane. Indeed, after the failure of Bank Heather, Bank Jane may not be able to survive even though Banks Heather and Jane do not have a direct link.

In sum, the application of network analysis to financial surveillance is a powerful tool for identifying both potentially systemic and vulnerable institutions in an objective manner, as well as for tracking potential contagion paths.

For example, we would see that the failure of Bank Heather would first lead to the failure of Bank Ike and then the failure of Bank Jane. It should be emphasized, however, that network analysis constitutes only one element in the financial stability tool kit. In fact, network analysis could be combined with regular stress testing exercises in order to gain a more comprehensive and clearer picture of vulnerabilities in a given banking system. For instance:

- Traditional stress tests could be run to identify the types of shocks that could have the largest first-round effects on individual institutions (Figure 13.2). Once these shocks are determined, they could be fed into a network model to assess second- (and higher) round effects.
- Conversely, network analysis could begin by applying a first set of shocks to the financial system in order to identify those institutions that are more systemic (i.e., whose failure would have the largest contagion impact) and those that are most vulnerable (i.e., most susceptible to failure) and then proceed with a more thorough stress test of some selected institutions.

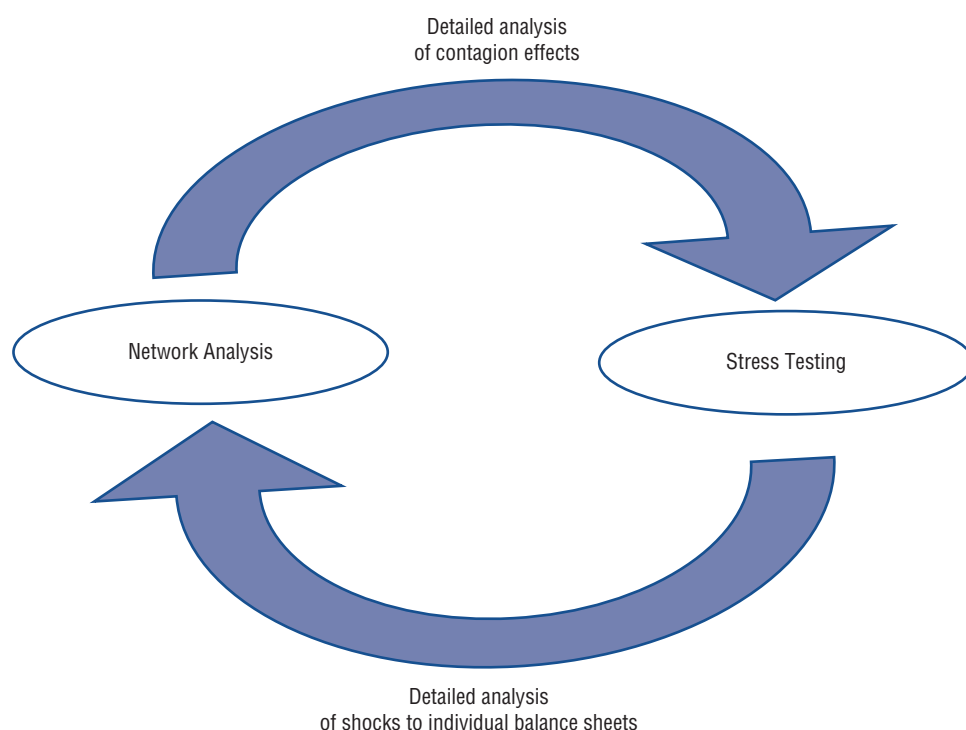
APPLICATIONS TO FINANCIAL STABILITY

Most of the initial applications of network analysis to financial surveillance were aimed at assessing the stability of domestic banking systems. In broad terms, the basis of these studies was the construction of a matrix of interbank exposures that identified gross lending and borrowing among institutions, in order to then simulate shocks to specific institutions and track domino effects to other institutions within that particular financial system.³ Some of the main findings of this literature are that (1) even though contagion owing to credit shocks seems likely to be rare, (2) whenever contagion takes place, it carries a high cost for the financial system. It should be said, however, that in part the low incidence of contagion in these papers could arise because most simulations ignore liquidity shocks and credit risk transfers (e.g., derivatives)—both of which have been shown to be important sources of potential contagion. Another potential problem with some of these results is the reliance on certain estimation techniques (i.e., maximum entropy) to complete data sets, which tend to ignore the role of relationship lending among financial institutions and thus underestimate the concentration of potential exposures.⁴

² A description of the computation of these (and other) network measures is beyond the scope of our introductory chapter. The interested reader can visit <http://www.orgnet.com>.

³ One of the main challenges faced in this line of study was the compilation of interbank exposures. This was attributable to a combination of factors, namely, that the data (1) may be available only to national supervisors; (2) are not collected in a systematic manner; and (3) are not publicly available. To circumvent these limitations, researchers often complemented available data with interpolations or estimations by different methods.

⁴ In broad terms, maximum entropy is akin to assuming the maximum possible diversification of exposures, thus minimizing the effect of concentration risk.



Source: Authors.

Figure 13.2 Interface between Network Analysis and Stress Testing

Largely triggered by the crisis, there has been a second wave of research applying network analysis to the international context.⁵ Most of these papers have relied on data from the Bank for International Settlements in order to study the topology of the international banking network as well as the potential cross-border contagion paths arising from shocks to a particular financial system. Taken as a whole, these papers provide important insights for international financial surveillance, as they reveal that (1) liquidity shocks—and not only credit shocks as mostly emphasized in the previous literature—can wreak havoc across the international financial network; (2) there are nowadays more (and smaller) financial systems (i.e., nodes) capable of triggering international contagion; and (3) the speed of propagation of contagion has increased in recent years.

The following two chapters in this volume on network analysis constitute an extension of the work begun in Chan-Lau and others (2009) and demonstrate how network analysis is used in IMF staff's work:

- Chapter 14 by Espinosa-Vega and Solé illustrates the application of network analysis to track the reverberation of credit and liquidity shocks throughout the global financial system and shows how

these techniques can be applied to draw global risk maps that reveal systemic and vulnerable spots, as well as potential contagion paths. In addition, the chapter shows the importance of incorporating off-balance risk transfers in the assessment of cross-border financial linkages, as these financial instruments can severely alter the risk profile of entire financial systems. The authors propose an enriched simulation algorithm that is able to account for risk transfers.

- Chapter 15 by Chan-Lau also shows the value of applying network techniques in a global setting to identify global sources of systemic risk, as well as regional ones. In addition, this chapter also illustrates a country case (Chile) using bank-specific disaggregated data to study potential intersectoral sources of risk. In exploiting supervisory data on individual banks' claims vis-à-vis other domestic and foreign banks, as well as other nonbank financial institutions, the study shows that the main sources of domestic risk in the Chilean banking system are shocks that affect banks' claims on households and domestic corporations, and to a lesser degree shocks emanating from Spanish and U.S. banks.

⁵ See McGuire and Tarashev (2008); Castrén and Kavonius (2009); Chan-Lau and others (2009); Degryse, Elahi, and Penas (2010); Murray and Rawcliffe (2010); and Garratt, Mahadeva, and Svirydenka (2011), among others, for papers that study international contagion channels using network techniques.

POLICY REFLECTIONS

The global financial crisis reminds us that financial interconnectedness leads (rapidly and forcefully) to spillovers not only

among banks but also to the nonbank financial sector, posing great risks to financial stability. Thus, financial regulators need to be equipped with the ability to improve their understanding and monitoring capability of direct and indirect linkages. Network analysis is one of the methods for doing so. That said, it is important to bear in mind that the application of network analysis to practical financial surveillance is still in a developmental stage and faces important challenges. Specifically:

- It is still difficult, if not impossible, to obtain comprehensive information at an institutional level. Some of the areas where better data are needed include bank-level information on banks' exposures and funding positions, with breakdowns by counterparty, currency, and remaining maturity. Fortunately, important progress is being made in this regard under the auspices of the Financial Stability Board (FSB) and IMF initiative on data gaps (FSB and IMF, 2009, 2011).
- In addition, network analysis still needs to be expanded to incorporate factors such as the imperfect integration of global money markets arising from heterogeneous resolution regimes; and potential problems with derivative and collateral clearing mechanisms; and it should include more information on off-balance-sheet and shadow banking activities.
- The existing literature does not yet include an adequate modeling of the endogenous response of institutions to shocks.

To conclude, globalization of financial activity means that it is close to impossible for a national regulator alone to undertake effective surveillance of potentially systemic linkages. Therefore, enhancing our understanding and monitoring of global systemic linkages requires a reliable method to assess financial connections, as well as strong information-sharing agreements on cross-market and cross-border linkages.

REFERENCES

- Castren, Olli, and Ilja Kristian Kavonius, 2009, "Balance Sheet Interlinkages and Macro-Financial Risk in the Euro Area," ECB Working Paper No. 1124 (Frankfurt: European Central Bank). Available via the Internet: <http://www.ecb.europa.eu/pub/scientific/wps/date/html/wps2009.en.html>
- Chan-Lau, Jorge, Marco Espinosa, Kay Giesecke, and Juan Solé, 2009, "Assessing the Systemic Implications of Financial Linkages," Chapter 2, *Global Financial Stability Report*, April (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2009/01/index.htm>
- Degryse, Hans, Muhammad Ather Elahi, and Maria Fabiana Penas, 2010, "Cross-Border Exposures and Financial Contagion," *International Review of Finance*, Vol. 10, No. 2, 209–40.
- Espinosa-Vega, Marco, and Juan Solé, 2011a, "Cross-Border Financial Surveillance: A Network Perspective," *Journal of Financial and Economic Policy*, Vol. 3, No. 3, pp. 182–205.
- , 2011b, "Regulating 'Too-Interconnected-to-Fail' Institutions," *Global Banking and Financial Policy Review 2011/12*, Euromoney Yearbooks, September, pp. 7–11.
- Financial Stability Board and International Monetary Fund, 2009, "The Financial Crisis and Information Gaps," Report to the G-20 Finance Ministers and Central Bank Governors (October 29). Available via the Internet: <http://www.financialstabilityboard.org>
- , 2011, "The Financial Crisis and Information Gaps," Implementation Progress Report (June). Available via the Internet: <http://www.financialstabilityboard.org>
- Garratt, Rodney J., Lavan Mahadeva, and Katsiaryna Svirydzenka, 2011, "Mapping Systemic Risk in the International Banking Network," Working Paper No. 413 (London: Bank of England). Available via the Internet: <http://www.bankofengland.co.uk/publications/Pages/workingpapers/2011/wp413.aspx>
- McGuire, Patrick, and Nikola Tarashev, 2008, "Global Monitoring with the BIS International Banking Statistics," BIS Working Paper No. 244 (Basel: Bank for International Settlements). Available via the Internet: <http://www.bis.org/publ/work244.htm>
- Murray, Andrew, and Gerry Rawcliffe, 2010, "International Financial Contagion—Easy to Define, Difficult to Measure," Fitch Ratings Global Special Report, October 6.

Cross-Border Financial Surveillance: A Network Perspective

MARCO A. ESPINOSA-VEGA • JUAN SOLÉ

Effective cross-border financial surveillance requires the monitoring of direct and indirect systemic linkages. This chapter illustrates how network analysis could make a significant contribution in this regard by simulating different credit and funding shocks to the banking systems of a number of selected countries. After that, we show that the inclusion of risk transfers could modify the risk profile of entire financial systems, and thus an enriched simulation algorithm able to account for risk transfers is proposed. Finally, we discuss how some of the limitations of our simulations are a reflection of existing information and data gaps and thus view these shortcomings as a call to improve the collection and analysis of data on cross-border financial exposures.

METHOD SUMMARY

Overview	The method is used to assess the total impact (after all rounds of contagion have taken place) on a group of financial institutions that are interconnected (i.e., have exposures to each other).
Application	The method is appropriate for assessing the financial stability conditions of individual and groups of institutions.
Nature of approach	Simulation based.
Data requirements	A combination of data sources, including at the institutional level (e.g., individual capital positions); interinstitutional (e.g., cross-institutional exposures); and market-level parameters (e.g., haircuts, funding rollover levels).
Strengths	• The network model allows identification of systemic and vulnerable institutions within a network of banks. • It also allows the tracking of contagion chains resulting from different financial shocks. • Model specification is very flexible.
Weaknesses	• Assumptions are required on the different parameters of the model. • Data may be very difficult to obtain (particularly interinstitutional exposures).
Tool	• The Excel-based program is available in the toolkit, which is on the companion CD and at www.elibrary.imf.org/stress-test-toolkit. • Contact author: M. Espinosa-Vega

Effective financial system surveillance requires the monitoring of direct and indirect financial *linkages*, whose disruption could have important implications for the stability of the entire financial system. Indeed, the recent financial crisis has underscored the need to go beyond the analysis of individual

institutions' soundness and assess whether the linkages across institutions may have systemic implications. Furthermore, it has become clear that, because of these financial interconnections, during stress events, even actions geared to enhance the soundness of a particular institution may undermine the

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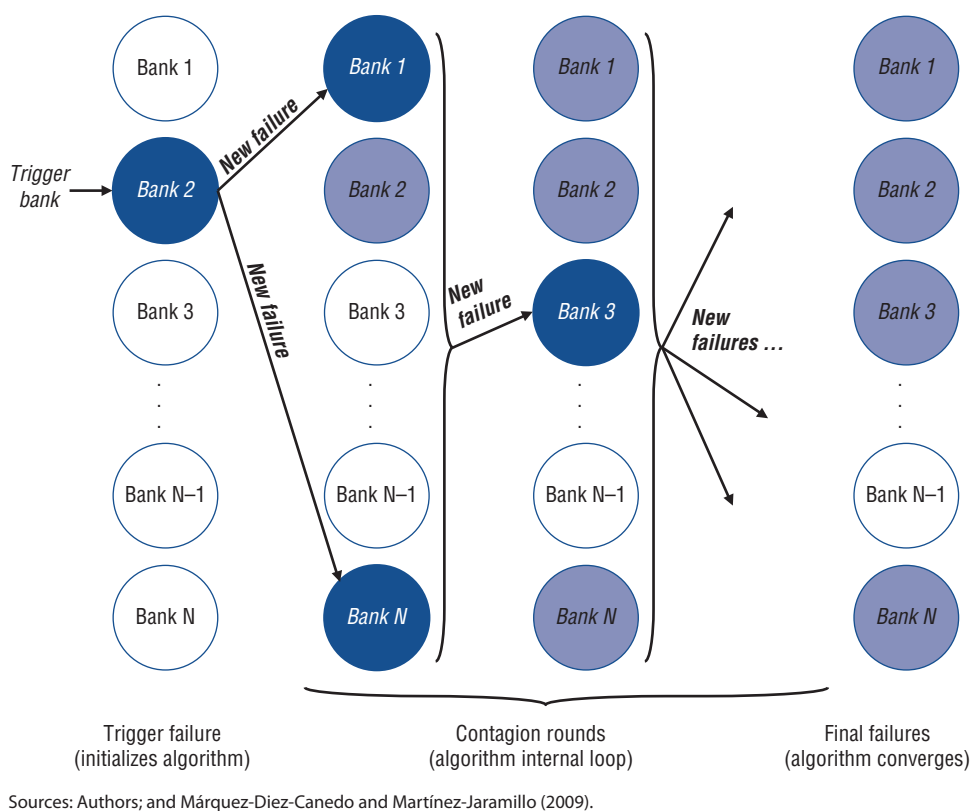


Figure 14.1 Network Analysis Based on Interbank Exposures

stability of other institutions. Consider, for instance, the depiction of the Northern Rock case by Morris and Shin (2008): “a prudent shedding of exposures from the point of view of Bank 2 is a run from the point of view of Bank 1. Arguably, this type of run is what happened to the U.K. bank Northern Rock, which failed in 2007” (italics added).¹

Therefore, policymakers and regulators worldwide have become increasingly aware of the importance of proactively tracking potential systemic linkages. As pointed out by, for instance, Allen and Babus (2009), network analysis is a natural candidate to aid in this challenge, as it allows regulators and policymakers to assess externalities to the rest of the financial system, by tracking the rounds of spillovers likely to arise from direct financial linkages.²

This chapter shows how network analysis can be used for cross-border financial sector surveillance by simulating different credit and funding shocks to the banking systems of a number of selected countries. In addition, the study illustrates how the inclusion of risk transfers (or contingent exposures) can alter the risk profile of an entire financial system. The basis of our analysis is the set of cross-country interbank exposures (including gross lending and borrowing and risk transfers) and estimates of a selected number of countries that report their banking data to the Bank for International Settlements (BIS).³ Among other things, our study tracks the systemic impact unleashed by the execution of cross-border interbank risk transfers following a credit event. The algorithm designed to track the domino effects triggered by hypothetical credit and funding shocks to each banking system in our sample is illustrated in Figure 14.1.

The rest of the chapter is organized as follows. Section 1 provides a detailed explanation of the methodology for the simulations and the data. Section 2 presents the results of our simulations. Section 3 concludes the chapter, and the

¹ Although the quote rightly points to the liquidity squeeze suffered by Northern Rock, it is important to note that the main source of the squeeze came from the wholesale market and not the interbank market.

² See Upper (2007) for an insightful survey of the network literature. Although most of the network literature that will be referenced in this chapter is of an applied nature, see Allen and Gale (2000) and Freixas, Parigi, and Rochet (2000) for some theoretical underpinnings to the network approach. In addition, Nier and others (2007) apply network theory to study contagion risk in simulated banking systems. There are also a number of papers that have focused on domestic banking systems—for example, Boss and others (2005) and Elsinger, Lehar, and Summer (2006) for Austria; Furfine (2003) for the United States; Márquez-Díez-Canedo and Martínez-Jaramillo (2009) for Mexico; Upper and Worms (2004) and Memmel and Stein (2008) for Germany; Sheldon and Maurer (1998) and Müller (2006) for Switzerland; and

Wells (2002) for the United Kingdom, among others. Finally, see Chan-Lau and others (2009) and Degryse, Elahi, and Penas (2010) for cross-border applications.

³ For purely illustrative purposes, we use BIS cross-country data. Our goal in carrying out this analysis is not to make specific pronouncements about particular countries but to illustrate the techniques described in the chapter as a useful tool for (macro-) financial surveillance.

Appendix presents some results obtained with an additional data set.

1. A SIMPLE INTERBANK EXPOSURE MODEL

The point of departure for our analysis is a stylized bank balance sheet that highlights the role of cross-border interbank exposures. The first set of simulations (Simulation 1) examines the domino effects triggered by the default of a banking system's interbank obligations—we call this a *credit* shock. The second set of simulations (Simulation 2) looks at the effects of a *credit-plus-funding* event, where the default of an institution also leads to a liquidity squeeze for those institutions funded by the defaulting institution (i.e., the credit shock is compounded by a funding shock and associated fire sale losses). After this, we look at the effects of incorporating risk transfers into our analysis (Simulations 3 and 4).

A. Network simulations of credit and liquidity shocks

To assess the potential systemic implications of interbank linkages, the study considers a network of N institutions. The point of departure is the following stylized balance sheet identity for bank i :

$$\sum_j x_{ji} + a_i = k_i + b_i + d_i + \sum_j x_{ij}, \quad (14.1)$$

where x_{ji} stands for bank i loans to bank j , a_i stands for bank i 's other assets, k_i stands for bank i 's capital, b_i are long-term and short-term borrowing (excluding interbank loans), d_i stands for deposits, and x_{ij} stands for bank i borrowing from bank j .

Transmission of Credit Shocks

To analyze the effects of a credit shock, the study simulates the individual default of each one of the 18 banking systems in the network, for different assumptions of loss given default (denoted by the parameter λ); it is assumed that banking system i 's capital absorbs the losses on impact; and then we track the sequence of defaults triggered by this event. For instance, after taking into account the initial credit loss stemming from the default of, say, institution h , the baseline balance sheet identity of bank i becomes

$$a_i + \sum_{j \neq h} x_{ji} + (1 - \lambda)x_{hi} = (k_i - \lambda x_{hi}) + b_i + d_i + \sum_j x_{ij}, \quad (14.2)$$

and bank i is said to fail if its capital is insufficient to fully cover its losses (i.e., if $k_i - \lambda x_{hi} < 0$). These losses are depicted in light gray in Figure 14.2.⁴

⁴ Subsequent rounds in the algorithm take into account the losses stemming from all failed institutions up to that point.

Preshock balance sheet		Postshock balance sheet	
$\sum_j x_{ji}$	k_i	λx_{hi}	λx_{hi}
	d_i	$\sum_j x_{ji}$	k_i
	b_i	a_i	d_i
	$\sum_j x_{ij}$		b_i
			$\sum_j x_{ij}$

Source: Authors.

Figure 14.2 Effect of a Credit Shock on a Bank's Balance Sheet

Transmission of Credit-Plus-Funding Shocks

The extent to which a bank is able to replace an unforeseen withdrawal of interbank funding will depend on money market liquidity conditions. During the 2007–09 crisis, for instance, we saw a gradual freezing of money markets that started with financial institutions becoming reluctant to roll over their funding of counterparties whose portfolios or business models were perceived to be similar to those of seemingly weak institutions.⁵ When liquidity is tight and in the absence of alternative sources of funding, a bank may be forced to sell part of its assets in order to restore its balance sheet identity. We study the situation where, as in the 2007–09 crisis, a bank is able to replace only a fraction of the lost funding and its assets trade at a discount (i.e., their market value is less than their book value). We exclude the possibility of institutions raising new capital and assume that the loss induced by a funding shortfall is absorbed by the bank's capital (Figure 14.3). Therefore, a bank's vulnerability not only stems from its direct credit exposures to other institutions but also from its inability to roll over (part of) its funding in the interbank market and thus having to sell assets at a discount in order to reestablish its balance sheet identity.

In terms of our stylized model, it is assumed that institutions are unable to replace all the funding previously granted

⁵ Indirect linkages among financial institutions may arise when banks hold the same type of asset in their balance sheets. These linkages can represent an important source of systemic risk, as the forced sale of assets by some institutions may trigger a decline in the market value of the other institutions' portfolios. Models with this type of portfolio linkages can be found, for instance, in Lagunoff and Schreft (2001), Cifuentes, Ferrucci, and Shin (2005), De Vries (2005), and Elsinger, Lehar, and Summer (2006). In addition, Furfine (2003), Müller (2006), and Nier and others (2007) also analyze liquidity shocks.

Preshock balance sheet		Postshock balance sheet	
$\sum_j x_{ji}$	k_i	$\sum_j x_{ji}$	$\delta \rho x_{ih}$
	d_i		k_i
a_i	b_i	a_i	d_i
	$\sum_j x_{ij}$		b_i
		$(1 + \delta) \rho x_{ih}$	$\sum_j x_{ij}$
			ρx_{ih}

Source: Authors.

Figure 14.3 Effect of a Funding Shock on a Bank's Balance Sheet

by the defaulted institutions, which, in turn, triggers a fire sale of assets. Thus, we study the situation where bank i is able to replace only a fraction $(1 - \rho)$ of the lost funding from bank h , and its assets trade at a discount, so that bank i is forced to sell assets worth $(1 + \delta)\rho x_{ih}$ in book value terms.⁶ The funding shortfall-induced loss, $\delta \rho x_{ih}$, is absorbed by bank i 's capital (Figure 14.3), and thus the new balance sheet identity for institution i is given by

$$a_i + \tilde{a}_j x_{ji} - (1 + \delta) \rho x_{ih} = (k_j - \delta \rho x_{ih}) + b_i + d_i + \tilde{a}_j x_{ij} - \rho x_{ih}. \quad (14.3)$$

Transmission of Shocks in the Presence of Risk Transfers

In addition to the type of direct exposures mentioned so far, it is important to consider the effect that contingent claims, in particular those stemming from credit guarantees or derivative products such as credit default swaps (CDSs), can have on the stability of a banking system. Contingent exposures deserve special consideration in times of stress because they activate dormant linkages across financial institutions and bring new exposures onto the balance sheet of an institution. As will be discussed, this study uses BIS data on risk transfers to illustrate a possible way in which risk transfers can be incorporated into financial system surveillance.

⁶ An alternative way to see this is the following. Let ρx be the amount of funding that cannot be replaced. Let p^1 be the current market price for assets, and let y be the quantity of assets sold. That is, $p^1 y = \rho x$. Suppose that these assets had been bought at a higher price p^0 ; thus, $\rho x = p^1 y < p^0 y \equiv \rho x (1 + \delta)$. Hence, it is possible to find a relationship between the parameter δ and the change in asset prices: $\delta = (p^0 - p^1)/p^1$; that is, δ is a parameter reflecting the degree of distress in asset markets. Higher δ reflects higher distress in markets.

Formally, let x_{ij} be the direct exposure of institution j to institution i . The effective exposure (i.e., net of risk transfers) of institution j to institution i consists of the original exposure x_{ij} , plus the amount that institution j needs to pay to all other institutions upon default of institution i (in other words, bank j 's sale of protection to other banks contingent on the default of bank i), minus the amounts that institution j receives from all other institutions upon default of institution i (i.e., bank j 's purchase of protection from all other banks against the default of bank i). Hence, the interbank exposure net of risk transfers, denoted by z_{ij} , can be expressed as

$$z_{ij} = x_{ij} + \sum_h \tau_{ih}^i - \sum_h \tau_{hj}^i, \quad (14.4)$$

where τ_{jh}^i denotes the risk transferred from institution h to institution j and referenced on institution i .

B. The simulation algorithms

The Algorithm without Risk Transfers

For each of the simulations, we programmed a MATLAB network algorithm of a system consisting of N nodes (each node representing a banking system) and with a structure of inter-node (or interbank) loans represented by the $N \times N$ matrix X , with a generic element denoted by x_{ij} —note that these loans are direct exposures across nodes. Let F_t be the set of failed institutions, and let NF_t be the set of not-failed institutions in round t of the simulations.

To initialize the credit shock simulation (Simulation 1), assume that institution h fails at $t=0$, and thus a fraction λ of its debts to the rest of institutions will not be repaid. Then, for each one of the not-failed institutions, $j \in NF_t$, the algorithm checks whether the amount of losses suffered by that institution is larger than the amount of capital of that particular institution. If that is the case, then that institution is also driven to bankruptcy. That is,

$$\text{if } \sum_{h \in F_t} \lambda x_{hj} > k_j \Rightarrow j \text{ defaults too: } j \in F_{t+1}. \quad (14.5)$$

The algorithm is said to converge once there are no further failures; that is, $F_t = F_{t+1}$.

For the credit-plus-funding shock simulation (Simulation 2), the previous shock is compounded by the funding shortfall-induced loss, $\delta \rho x_{ih}$. That is, at each stage of the simulation, an institution's capital may be negatively affected by the asset fire sale (recall Figure 14.3), and hence the default condition is given by

$$\text{if } \sum_{h \in F_t} \lambda x_{hj} + \sum_{h \in F_t} \delta \rho x_{jh} > k_j \Rightarrow j \text{ defaults too: } j \in F_{t+1}. \quad (14.6)$$

The Algorithm with Risk Transfers

We also programmed an algorithm that takes into account the effects of risk transfers on the transmission of both the credit and the credit-plus-funding shocks through the network (Simulations 3 and 4, respectively). For this, the previous algorithm

was modified so that the default condition for the credit shock becomes

$$\begin{aligned} & \text{if } \sum_{h \in F_t} \lambda x_{hj} - \sum_{h \in F_t, i \in NF_t} \lambda \tau_{ij}^h + \theta \sum_{h \in F_t, i \in NF_t} \lambda \tau_{ji}^h > k_j \\ & \Rightarrow j \text{ defaults too: } j \in F_{t+1}. \end{aligned} \quad (14.7)$$

The default condition for the credit-plus-funding shock becomes

$$\begin{aligned} & \text{if } \sum_{h \in F_t} \lambda x_{hj} + \sum_{h \in F_t} \delta p x_{jh} - \sum_{h \in F_t, i \in NF_t} \lambda \tau_{ij}^h + \theta \sum_{h \in F_t, i \in NF_t} \lambda \tau_{ji}^h > k_j \\ & \Rightarrow j \text{ defaults too: } j \in F_{t+1}, \end{aligned} \quad (14.8)$$

where the parameter θ captures the fraction of risk transfers that have not been provisioned for. Note that, in equations (14.7) and (14.8), outward and inward transfers are treated slightly differently: although both transfers are multiplied by λ , inward transfers are also multiplied by the parameter θ . This is to recognize the possibility that institutions may make provisions for those risk transfers that they take onto their balance sheets.⁷

After the default of an institution, the inward risk transfers of that institution are set to zero, because the defaulted institution will not be able to honor the guarantees it has extended. In terms of the notation above, this amounts to setting $\tau_{hj}^i = 0$, for all j , after the default of bank h . For instance, if $\tau_{hj}^i = 5$ billion USD, this implies that if institution i defaults, then h needs to make a transfer of \$5 billion to institution j .

C. The data

To illustrate the use of network analysis for cross-border financial surveillance, we focused on cross-country bilateral exposures at end-December 2007, published in the BIS's *International Consolidated Banking Statistics* database.⁸ The BIS compiles these data in two formats: (1) *immediate borrower basis* (IBB) and (2) *ultimate risk basis* (URB). Although both data sets consolidate the exposures of lenders' foreign offices (i.e., subsidiaries and branches) into lenders' head offices, the URB data set also consolidates by residency of the ultimate obligor (i.e., the party that is ultimately responsible for the obligation in case the immediate borrower defaults) and includes net risk transfers.⁹

Thus, for example, on an *immediate borrower basis*, a loan from Deutsche Bank's subsidiary in London to Banco Santander's subsidiary in London would appear as a loan from Germany to the United Kingdom in the IBB statistics (Figure 14.4). Conversely, on an *ultimate risk basis*, the borrower's balance sheet is also consolidated (as long as the parent entity explicitly guarantees the claim), and thus, the same loan described above would be accounted for as a loan from Germany to Spain.

From a financial stability perspective, one could be interested in using either set of data depending on the types of shocks and scenarios deemed relevant. In particular, IBB data provide information on the countries in which risks originate, whereas URB data allocate claims to the country that would ultimately bear the risks, as long as these have been explicitly guaranteed (see BIS, 2008, pp. 9 and 17). However, using URB data for network simulations raises the question of how to treat the risk transfers of failed institutions, because after each round of failures in the network, the risk transfers embedded in the URB data may not be actual any longer, as the counterparty may be among the failed institutions. Thus, in order to track more accurately where risks remain after each round of defaults, disaggregated data on risk transfers at an individual level need to be added to the simulation (Simulations 3 and 4 below show a way to do this).

Conversely, relying solely on IBB data may overestimate the systemic role played by the domestic banks of those countries with a large presence of foreign banks, as is the case with the United Kingdom, for instance. McGuire and von Peter (2009) show that non-U.K. banks' offices are a much bigger presence in the United Kingdom than are banks with U.K. headquarters. This fact is partly responsible for the important role played by the United Kingdom in the IBB simulations (Simulations 1 and 2). Nevertheless, as shown in the Appendix, the United Kingdom still plays a very key role in simulations based on URB data.

From an operational standpoint, because both data sets have limitations, our inclination would be to conduct simulations using both data sets and compare results. For expository purposes, however, and given that our simulations including risk transfers build from data on an *immediate borrower basis*, we present first the results based on IBB data and relegate the results obtained using URB data to the Appendix.¹⁰

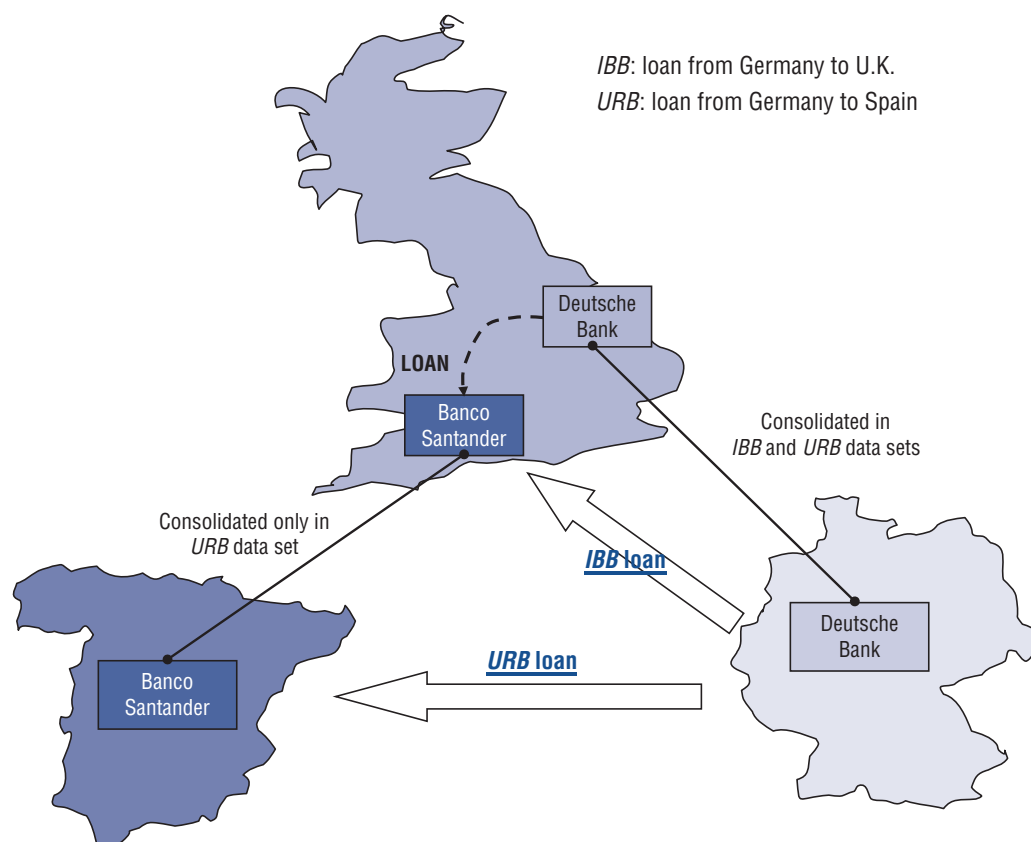
We also obtained data on cross-border risk transfers from the BIS. These data represent inward and outward

⁷ For comparison purposes, and given the lack of reliable estimates for its empirical value, this parameter was arbitrarily set equal to the loss-given-default parameter, $\theta = \lambda$, in order not to bias the results in favor of or against the simulations that include risk transfers: simulations with $\theta < \lambda$ would produce fewer defaults and vice versa. Factors that in practice would affect the numerical value of this parameter include the fraction of risk transfers that have been provisioned for or the cheapest-to-deliver option available to the protection seller at the time a transfer needs to be made.

⁸ We are indebted to Patrick McGuire and Götz von Peter for extensive discussions on the BIS statistics.

⁹ See McGuire and Wooldridge (2005) and BIS (2008) for a detailed description of these data, and McGuire and Tarashev (2008) for applications of the BIS statistics to monitor the international banking system. Hattori and Suda (2009) also use BIS data to study the network topology of the international banking system from a historical perspective. However, their study does not assess contagion patterns.

¹⁰ There is another difference between the IBB and the URB data sets that deserves mention: the sectoral split (e.g., banks vs. other sectors) is not consistent across the two data sets. In the IBB data, the sectoral split is available only for *international* claims (cross-border claims plus local claims in foreign currency), whereas there is no information on the sectoral split for *local claims in local currency*. In consequence, we applied the same sectoral fraction available for *international* claims to *local claims in local currency*. In contrast, in the URB data, the sectoral split applies to total *foreign claims* (equal to international plus local claims in local currency).



Source: Authors.

Figure 14.4 Cross-Border Claims on Immediate Borrower Basis (IBB) and Ultimate Risk Basis (URB)

transfers of risks by reporting banks and include credit guarantees and commitments, collateral, and credit derivatives in the banking book (see BIS, 2008, p. 16).¹¹ Unfortunately, it was not possible to obtain a further disaggregation, either by sector and nationality of the referenced entities (i.e., the entities on whose performance the risk transfer is contingent) or by sector of the party receiving the risk transfer.

Therefore, we had to assume a distribution of risk transfers across countries of residence of the referenced institution and that all risk transfers were interbank transfers. A generalized practice in the literature is to use maximum entropy to generate the unknown distribution. However, this procedure would be akin to assuming a uniform distribution of risk transfers across (referenced) countries, which would, in turn, spread the risk to the maximum amount possible and thus probably give an overly benign picture of systemic risk in the network. Alter-

natively, and in order to make a (perhaps more plausible) approximation of the distribution of risk transfers, we assumed that the distribution of referenced entities is the same as the distribution of direct credit exposures (Simulation 3). As a robustness check, we also conducted an additional set of simulations assuming a concentrated distribution of risk transfers for some countries (Simulation 4).

Data on country-specific bank capital at end-December 2007 were obtained from Bankscope. To match as closely as possible the number and type of banks in both data sets (i.e., BIS and Bankscope), we gathered data from only commercial and investment banks in Bankscope, excluding smaller entities (such as cooperative banks and savings banks), which, arguably, are less likely to engage in international transactions.

Finally, the 18 countries analyzed were Australia, Austria, Belgium, Canada, Finland, France, Germany, Greece, Ireland, Italy, Japan, the Netherlands, Portugal, Spain, Sweden, Switzerland, the United Kingdom, and the United States.

Before proceeding to the description of our simulation results, we would like to reiterate that because of the data constraints just described, the results of the study should be seen

¹¹ The BIS defines credit guarantees as contingent liabilities arising from an obligation to pay to a third party when a client fails to perform a contractual obligation; credit commitments are irrevocable obligations to extend credit at the request of a borrower (see McGuire and Tarashev, 2008).

TABLE 14.1

Results for Simulation 1 (Credit Channel)

	Failed Capital (% of Total Capital)	Induced Failures	Contagion Rounds	Absolute Hazard ¹	Hazard Rate ²
Australia	2.6	—	—	—	0.0
Austria	0.9	—	—	3	17.6
Belgium	1.6	—	—	4	23.5
Canada	3.5	—	—	1	5.9
Finland	1.2	1	1	—	0.0
France	6.6	—	—	3	17.6
Germany	3.6	—	—	3	17.6
Greece	0.7	—	—	—	0.0
Ireland	2.3	—	—	2	11.8
Italy	24.6	7	3	2	11.8
Japan	15.0	1	1	1	5.9
Netherlands	5.6	1	1	3	17.6
Portugal	0.5	—	—	2	11.8
Spain	4.2	—	—	2	11.8
Sweden	0.8	—	—	4	23.5
Switzerland	1.8	—	—	4	23.5
United Kingdom	45.0	11	3	1	5.9
United States	96.2	14	3	—	0.0

Source: Authors.

1. Number of simulations in which that particular country fails.

2. Percentage of failures as a percent of the number of simulations conducted.

as merely illustrative of the potential of network analysis as a tool for cross-border financial surveillance.

2. SIMULATION RESULTS

We begin this section by studying how shocks to a country's resident financial institutions unravel throughout the network before one considers risk transfers (Simulations 1 and 2). That is, we track shocks to financial institutions that are incorporated and operate in a given jurisdiction regardless of the residency of the parent company. Or to put it in terms of our earlier example, British regulators would be interested in tracking shocks not only to British banks but also to the subsidiaries of foreign entities operating in the United Kingdom; hence, a shock to Santander's subsidiary in London would be important to track.

After that, we present a second set of simulations that incorporate off-balance-sheet data on risk transfers (Simulations 3 and 4). Owing to the lack of granularity of the data, our analysis is subject to a number of caveats. In particular, it is probably unrealistic to view the entire external sector of a country's banking system as suffering a shock large enough to cause the entire banking system of another country to fail. Also, these data cover only a subset of bank exposures, namely, direct credit exposures, and do not include over-the-counter derivatives, specifically CDS contracts. However, our study provides a compelling illustration of the type of surveillance analysis that can be extracted from network analysis, including the identification

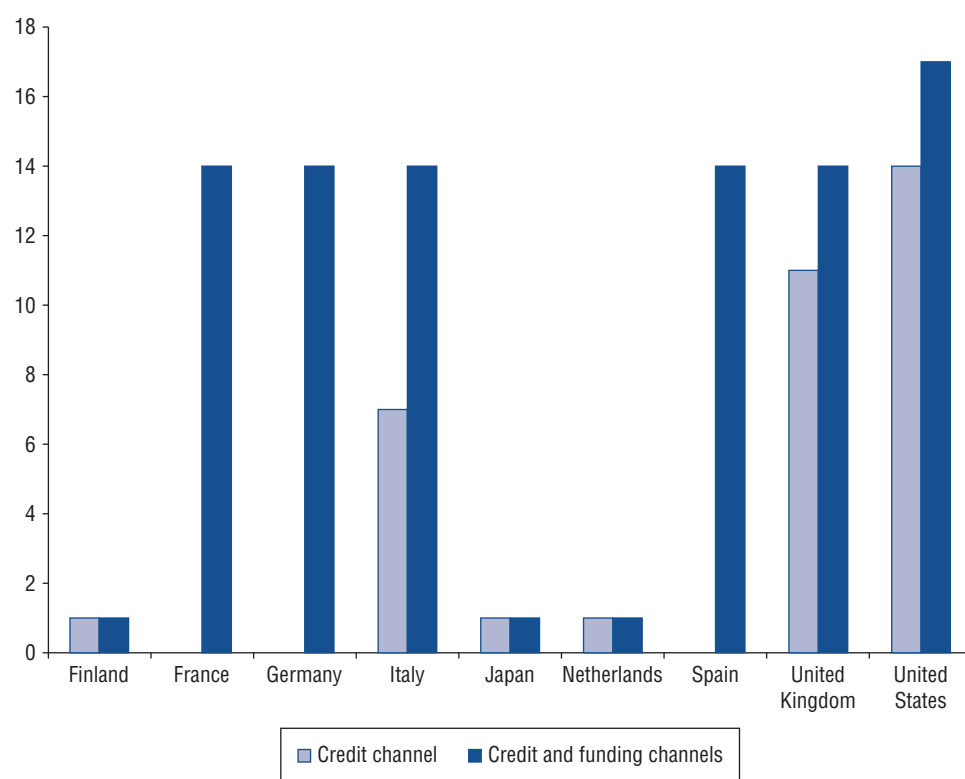
of systemic and vulnerable institutions, as well as the detection of data gaps for the identification of potentially systemic exposures.

A. Simulation 1: the transmission of a credit shock

The first simulation focuses on the transmission of a pure credit shock, thus assuming that institutions are able to roll over their funding sources and do not need to resort to fire sales of assets.¹² The credit shocks we analyze consist of the hypothetical default of a banking system's debts to foreign banks. The results of this simulation are reported in Table 14.1. Not surprisingly, what immediately emerges is that the U.K. and the U.S. banking systems are systemic players. Specifically, as of December 2007, the default of the U.K. and the U.S. systems would have led to losses—after all contagion rounds—of 45 and 96 percent, respectively, of the combined capital in our universe of banking systems.

The second and third columns in Table 14.1 indicate the number of induced failures and the number of contagion

¹² The simulations assume that the loss-given-default parameter equals 100 percent on impact. That is, when the credit event first materializes, banks are unable to recover any of their loans, as it takes time for secondary (and distress-debt) markets to price recently defaulted instruments. Thus, the simulation results should be interpreted as the on-impact transmission of systemic instability. In a similar vein, Wells (2002) argues that network studies should consider higher loss-given-default estimates than typically assumed, as banks tend to face substantial uncertainty over recovery rates in the short run.



Source: Authors.

Note: The figure shows the number of countries whose banking systems would fail as a result of the initial failure of each country (e.g., the initial failure of France would induce no failure under the credit shock scenario and 14 failures under the credit-plus-funding shock scenario).

Figure 14.5 Number of Induced Failures

rounds (the aftershocks) triggered by each hypothetical failure. The two countries that induce the highest number of contagion rounds are the United Kingdom and the United States (Figure 14.5). The failure of the U.K. banking system would trigger severe distress in 11 additional banking systems in three rounds of contagion. Similarly, the failure of the U.S. banking system would trigger the failure of 14 additional banking systems in three rounds of contagion.

It is important to emphasize that network analysis is useful not only in identifying potential failures but also in estimating the amount of impaired capital after all aftershocks have taken place. In other words, even when domino effects do not lead to systemic failures, network analysis provides a measure of the degree to which a financial system will be weakened by the transmission of financial distress across institutions (Table 14.2). For instance, the failure of Germany would produce a capital loss to Greek banks of only 1.4 percent of their initial capital, whereas the loss for Ireland would amount to 38 percent of initial capital.

In addition, this analysis facilitates the identification not only of institutions/systems whose stress poses systemic risks but also of “vulnerable” systems. For example, although the United Kingdom and the United States were identified as the most systemic systems (i.e., triggering the largest number

of contagion rounds and the highest capital losses), Belgium, Sweden, and Switzerland appear the most vulnerable banking systems, exhibiting the highest hazard rates in the sample (Table 14.1). In other words, the banking systems of these countries are severely affected in 4 out of the 17 simulations in which they were not the trigger country (Figure 14.6). This is not to say that a crisis in these countries is imminent and/or unavoidable. The point is that this type of analysis allows for the identification of the weakest points within the network.

As suggested by Figure 14.1, an additional advantage of network simulations is that the path of contagion can be tracked. To illustrate, consider the case of a hypothetical default of Italy’s cross-border interbank loans. Figure 14.7 features the ensuing contagion path: France would be affected in the first round; Belgium, Germany, and Switzerland in the second; the combination of these five defaults would be systemic enough to severely affect Austria, Sweden, and the Netherlands in the final round of contagion.

B. Simulation 2: the transmission of a credit-plus-funding shock

Next, the study considers the effects of a joint credit and liquidity shock assuming a 50 percent haircut in the fire sale of

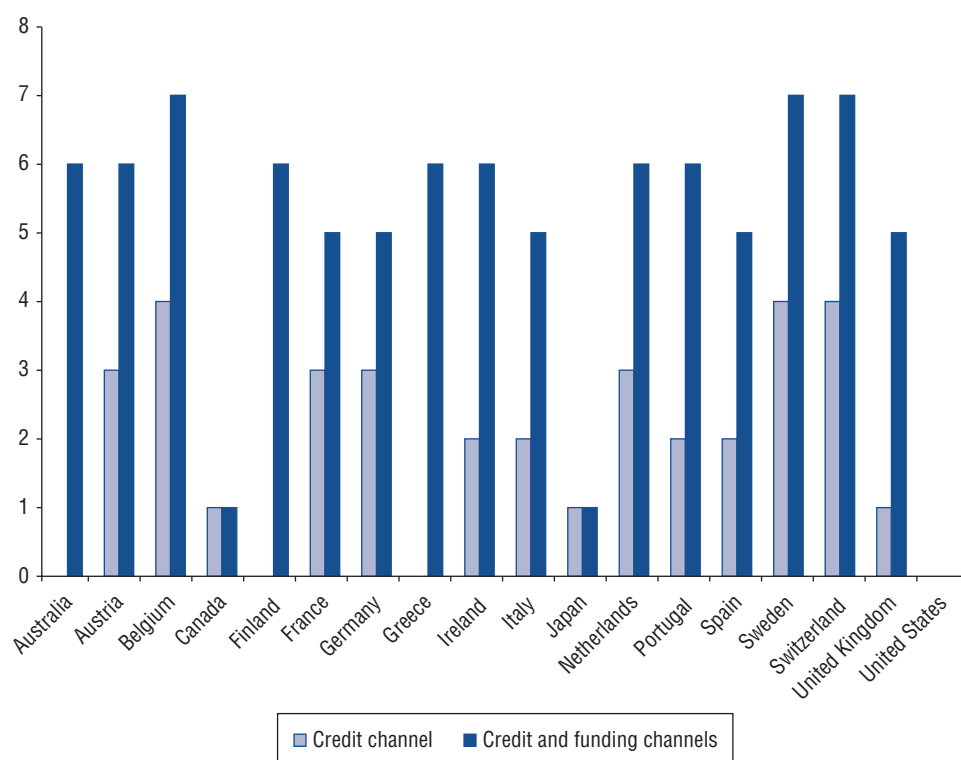
TABLE 14.2

Country-by-Country Capital Impairment (Credit Channel)
(capital impairment in percent of preshock capital)

	Postsimulation Capital Impairment																	
	Australia	Austria	Belgium	Canada	Finland	France	Germany	Greece	Ireland	Italy	Japan	Nether.	Portugal	Spain	Sweden	Switzer.	U.K.	U.S.
Trigger Country																		
Australia		-3.6	-5.7	-4.6	0.0	-12.3	-20.1	0.0	-4.0	-0.4	-4.8	-22.9	-1.1	-0.9	-5.2	-27.7	-13.6	-0.7
Austria	0.0		-8.3	-0.5	-0.5	-5.6	-35.5	-0.7	-6.5	-23.4	-0.6	-2.8	-1.7	-0.8	-3.2	-14.9	-0.9	-0.1
Belgium	-0.7	-5.6		-1.1	-0.7	-23.6	-17.2	-1.1	-3.3	-3.8	-2.7	-43.5	-3.9	-4.3	-6.1	-20.4	-3.9	-0.2
Canada	0.0	-1.9	-4.1		0.0	-5.9	-13.2	-0.1	-5.2	-0.4	-3.7	-10.3	-0.3	-0.6	-2.8	-20.8	0.0	-1.0
Finland	-0.1	-3.4	-4.6	-0.9		-5.4	-18.1	-0.1	-3.3	-1.1	-1.5	-5.0	-1.3	-1.3	Full	-10.7	-2.4	-0.2
France	-2.9	-12.2	-81.6	-2.9	-3.3		-70.3	-1.5	-14.2	-11.8	-8.9	-43.8	-12.1	-15.5	-14.0	-71.4	-22.3	-1.0
Germany	0.0	-70.2	-52.6	-3.8	-10.1	-54.2		-1.4	-38.0	-74.8	-11.1	-50.0	-15.6	-12.8	-96.1	-86.8	-11.0	-1.4
Greece	0.0	-6.9	-9.2	-0.2	-1.2	-13.6	-12.8		-4.2	-1.6	-0.6	-5.3	-13.8	-0.2	-0.6	-45.6	-1.2	-0.1
Ireland	-2.0	-11.3	-62.2	-4.6	-2.7	-20.9	-62.4	-1.0		-5.8	-3.3	-12.5	-8.9	-5.9	-7.3	-28.3	-17.5	-0.5
Italy	-6.1	Full	Full	-12.7	-19.6	Full	Full	-6.7	-98.2		-34.5	Full	-63.8	-58.2	Full	Full	-58.5	-4.5
Japan	-1.4	-11.7	-10.2	-2.0	0.0	-48.8	-47.9	-0.4	-7.7	-3.6		-19.1	-7.2	-1.5	-4.7	Full	-12.4	-2.0
Netherlands	-2.4	-25.4	Full	-3.3	-3.7	-53.2	-69.8	-2.3	-11.1	-10.2	-7.3		-15.6	-18.1	-23.2	-60.1	-13.0	-1.1
Portugal	0.0	-2.7	-6.0	-0.2	-0.3	-7.2	-14.5	0.0	-3.3	-1.5	-0.2	-3.9		-19.2	-0.7	-3.6	-1.4	-0.1
Spain	-0.6	-9.4	-30.3	-1.2	-2.7	-43.1	-89.9	-0.2	-16.9	-5.0	-2.3	-34.4	-59.1		-12.5	-18.0	-10.3	-0.6
Sweden	0.0	-1.9	-2.6	-0.6	0.0	-3.5	-13.1	0.0	-3.3	-0.5	-1.1	-3.8	-1.1	-0.8		-8.7	-1.8	-0.2
Switzerland	-0.8	-11.2	-7.3	-0.8	0.0	-13.5	-22.3	-0.4	-1.9	-2.6	-1.4	-5.6	-7.0	-1.3	-4.1		-2.5	-0.3
U.K.	-53.2	Full	Full	-48.0	-25.3	Full	Full	-17.8	Full	Full	-57.3	Full	Full	Full	Full	Full		-10.4
U.S.	-69.3	Full	Full	Full	-42.2	Full	Full	-21.0	Full	Full	Full	Full	Full	Full	Full	Full	Full	

Source: Authors.

Note: "Full" indicates that all capital is impaired.



Source: Authors.

Note: This figure depicts each country's absolute hazard level, defined as the number of simulations in which the banking system of the country failed as a result of another country's failure (e.g., under the first scenario Switzerland would be induced to fail in four simulations, whereas under the second scenario it would be induced to fail in seven simulations).

Figure 14.6 Country-by-Country Vulnerability Level

assets and a 65 percent rollover ratio of interbank debt.¹³ Table 14.3 summarizes the effects of this type of disturbance for the whole sample. From a financial stability perspective, it is also important to track the number of contagion rounds that each simulation yields, as these give an indication of the sequence of the aftershocks that reverberate throughout the entire network.

Considering scenarios that compound different types of distress allows regulators to identify new sources of systemic risk that were previously undetected. That is the case, for instance, for France, Germany, and Spain, where the combined shock increases the systemic role played by these countries as providers of liquidity in addition to their importance as recipients of funding: now they all induce 14 defaults compared with none under the credit shock scenario. Similarly, the United Kingdom and the United States also increase their systemic profile.

The addition of the funding channel also raises the vulnerability of all banking systems significantly, as measured by the hazard rate. This fact helps explain why numerous

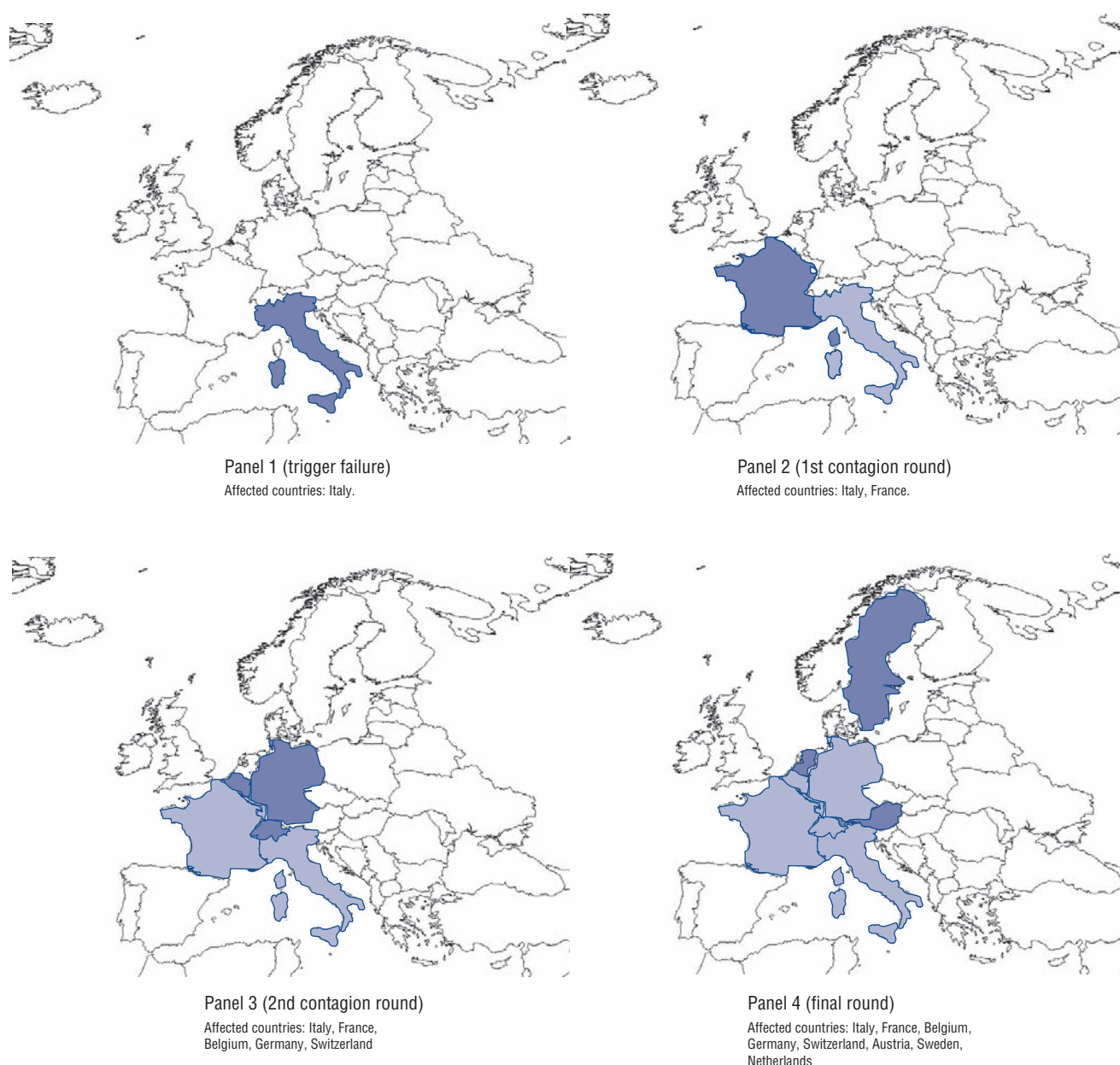
papers in the network literature—which focus only on credit events—have shown little contagion as a result of institutions' defaults. In other words, the combination of several channels produces a higher number of induced failures. Here, too, the most vulnerable banking systems continue to be the Belgian, Swedish, and Swiss systems. In addition, the hazard rate for most countries increases severalfold. Table 14.4 features the distribution of capital impairment, highlighting, once again, the fact that even when stress events do not bring down a banking system, they may significantly weaken it.

C. Simulations 3 and 4: transmission of shocks in the presence of risk transfers

We now present two simulations that take into account the presence of risk transfers among the banking systems of the network. The results of these two simulations are presented in Figures 14.8 to 14.11, along with results from the previous simulations for ease of comparison.

The point we want to make with these simulations is that the inclusion of risk transfers could change (sometimes even

¹³ Corresponding to parameter values of $\delta=1$ and of $\rho=0.35$.



Source: Authors.

Figure 14.7 Contagion Path Triggered by the Italian Failure under the Credit Shock Scenario

dramatically) the risk profile of the network. In particular, for the first distribution of risk transfer that we consider, the presence of risk transfers improves the resiliency of some countries to certain shocks (e.g., Belgium, the Netherlands, Sweden, and Switzerland; Figure 14.9).¹⁴

Given that we do not know the true distribution of risk transfers across referenced entities, and to illustrate that it is

¹⁴ Recall that the first distribution we assume is such that the distribution of risk transfers across referenced entities is the same as the distribution of direct credit exposures.

key to know this distribution, we assumed an alternative distribution with a strong concentration of transfers on specific countries. In particular, we assumed that Belgium and Switzerland have acquired risk transfers only that have France as the referenced entity and that the Netherlands has acquired risk transfers only that have Germany as the referenced entity. In other words, all the risk transfers that Belgium and Switzerland receive from all other countries are assumed to be referenced on France, and all the risk transfers that the Netherlands receives from all other countries are assumed to be referenced on Germany.

TABLE 14.3

Results for Simulation 2 (Credit and Funding Channel)

	Failed Capital (% of Total Capital)	Induced Failures	Contagion Rounds	Absolute Hazard ¹	Hazard Rate ²
Australia	2.6	—	—	6	35.3
Austria	0.9	—	—	6	35.3
Belgium	1.6	—	—	7	41.2
Canada	3.5	—	—	1	5.9
Finland	1.2	1	1	6	35.3
France	48.8	14	4	5	29.4
Germany	48.8	14	5	5	29.4
Greece	0.7	—	—	6	35.3
Ireland	2.3	—	—	6	35.3
Italy	48.8	14	4	5	29.4
Japan	15.0	1	1	1	5.9
Netherlands	5.6	1	1	6	35.3
Portugal	0.5	—	—	6	35.3
Spain	48.8	14	5	5	29.4
Sweden	0.8	—	—	7	41.2
Switzerland	1.8	—	—	7	41.2
United Kingdom	48.8	14	3	5	29.4
United States	100.0	17	3	—	0.0

Source: Authors.

1. Number of simulations in which that particular country fails.

2. Percentage of failures as a percent of the number of simulations conducted.

The results from assuming this distribution of risk transfers are presented in Figures 14.10 and 14.11 and show some differences from before. In particular, notice how, under the credit shock scenario, the inclusion of risk transfers increases the level of the systemic importance of France—now inducing 11 failures compared with none before. Similarly, Spain increases its systemic relevance under the credit-plus-funding shock—inducing now 14 failures compared with only one before. In terms of vulnerability levels (Figure 14.11), the results also point to an increased level of risk for almost all countries.

In sum, these examples indicate that having accurate information on risk transfers across their three relevant dimensions (i.e., the *originator* of the risk transfers, the *recipient*, and the *referenced* party or entity) is key to the assessment of financial stability, as different distributions can give rise to very different risk profiles, as shown by the difference between Figures 14.8–14.9 and 14.10–14.11. Thus, having data that include risk transfers enhances the knowledge of the risks actually present in a network—risks that could be realized if a distress event were to occur. In other words, we view the variation in results—which in turn depends on the assumed distribution of risk transfers—as a call to improve on the collection and analysis of data on risk transfers.

3. CONCLUSION

Our illustration of network analysis has highlighted its usefulness as a cross-border surveillance tool. We have also

stressed the need to give consideration to off-balance-sheet risks in network analysis and proposed a simulation algorithm for this purpose. In addition, we emphasized the need for on- and off-balance-sheet interbank linkages and cross-institutional linkages to be made increasingly available to those overseeing systemic stability. Going forward, financial regulators should continue to develop ways to systematically collect and analyze these data. Moreover, the global dimension of the 2007–09 crisis underscored the need to assess these exposures from a cross-border perspective, which would require further coordination and data sharing by national regulators.

The analysis of how shocks reverberate throughout the system is important to get a sense of how a crisis could unravel once the initial shocks have taken place, assuming the authorities fail, or are too slow, to respond. This is not to say that the analysis of interconnectedness will unequivocally reveal where the next crisis will arise, but including an analysis of interlinkages in the supervisors' repertoire may help identify institutions that need further scrutiny in terms of their vulnerability and/or level of systemic risk. If enough data are collected from various types of institutions, the perimeter of spillovers can be discerned, and this could help distinguish which types of firms should be under a regulatory net.

The 2007–09 crisis has proven that interconnectedness across institutions is present not only within the banking sector but, as important, with the nonbank financial sector (such as investment banking and hedge funds). In particular,

TABLE 14.4

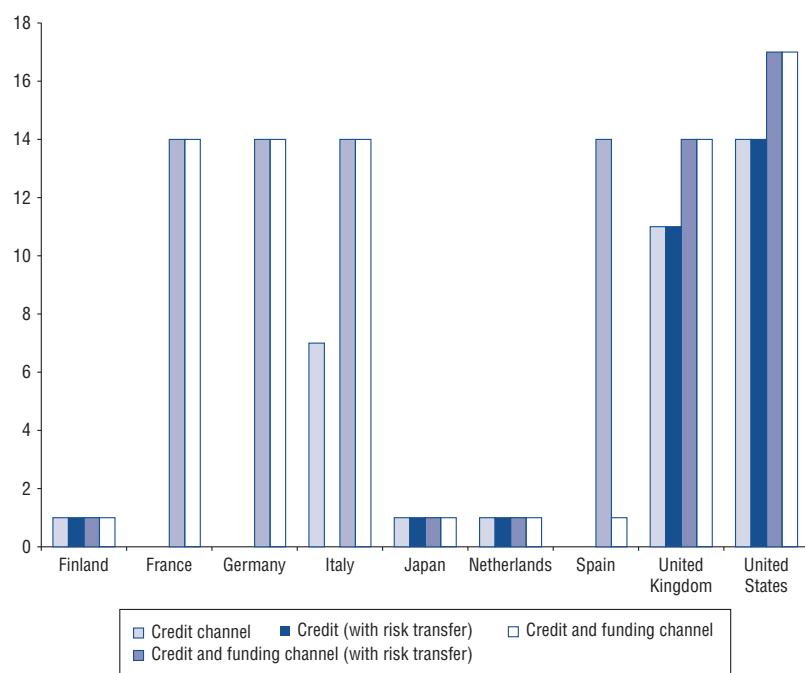
Country-by-Country Capital Impairment (Credit and Funding Channel)
(capital impairment in percent of preshock capital)

Postsimulation Capital Impairment

	Australia	Austria	Belgium	Canada	Finland	France	Germany	Greece	Ireland	Italy	Japan	Nether.	Portugal	Spain	Sweden	Switzer.	U.K.	U.S.
Trigger Country																		
Australia		-3.6	-6.0	-4.6	-0.1	-12.7	-20.1	-0.1	-4.8	-0.4	-4.8	-23.3	-1.1	-1.0	-5.3	-28.1	-16.5	-1.1
Austria	-0.5		-9.4	-0.6	-1.5	-6.2	-41.6	-3.6	-8.0	-25.2	-0.7	-4.4	-3.4	-1.5	-4.0	-16.9	-1.6	-0.4
Belgium	-1.9	-10.8		-1.8	-3.2	-30.6	-25.4	-8.2	-18.7	-7.5	-2.8	-66.4	-10.9	-8.4	-8.0	-22.7	-9.8	-2.0
Canada	-2.2	-2.5	-4.9		-0.7	-6.5	-14.5	-0.4	-7.6	-0.6	-3.8	-11.0	-0.6	-1.0	-3.8	-21.3	-2.7	-5.5
Finland	-0.6	-4.5	-5.7	-1.1		-6.0	-25.7	-0.6	-4.4	-1.4	-1.5	-6.3	-1.8	-2.2	Full	-11.3	-3.8	-0.8
France	Full	Full	Full	-72.4	Full		Full	Full	Full	Full	-83.7	Full	Full	Full	Full	Full	Full	-72.1
Germany	Full	Full	Full	-72.4	Full	Full		Full	Full	Full	-83.7	Full	Full	Full	Full	Full	Full	-72.1
Greece	0.0	-7.0	-9.4	-0.2	-1.2	-13.6	-12.9		-4.3	-1.6	-0.6	-5.3	-13.8	-0.2	-0.6	-45.7	-1.4	-0.1
Ireland	-3.3	-17.1	-63.9	-5.8	-2.7	-22.7	-70.7	-5.6		-9.3	-3.6	-14.1	-14.3	-9.2	-10.7	-29.2	-24.9	-1.4
Italy	Full	Full	Full	-72.4	Full	Full	Full	Full	Full		-83.7	Full	Full	Full	Full	Full	Full	-72.1
Japan	-16.7	-25.3	-25.8	-10.7	-7.1	-61.9	-76.8	-42.9	-22.0	-10.7		-30.5	-14.1	-6.7	-18.3	Full	-33.1	-26.2
Netherlands	-16.2	-35.0	Full	-8.1	-9.9	-69.6	-97.3	-19.5	-34.2	-24.3	-8.8		-33.6	-33.7	-32.0	-66.8	-29.6	-7.9
Portugal	-0.1	-3.1	-6.5	-0.2	-0.4	-7.5	-15.3	-3.3	-4.0	-1.9	-0.3	-4.4		-21.6	-0.9	-4.3	-1.6	-0.1
Spain	Full	Full	Full	-72.4	Full	Full	Full	Full	Full	Full	-83.7	Full	Full		Full	Full	Full	-72.1
Sweden	-0.6	-2.9	-3.6	-0.8	-83.2	-4.1	-20.2	-0.3	-4.2	-0.8	-1.1	-4.9	-1.4	-1.6		-9.3	-3.2	-0.7
Switzerland	-7.5	-21.5	-15.2	-4.5	-2.7	-20.2	-37.2	-39.3	-9.6	-6.2	-7.8	-11.8	-11.6	-4.0	-11.2		-17.4	-16.4
U.K.	Full	Full	Full	-72.4	Full	Full	Full	Full	Full	Full	-83.7	Full	Full	Full	Full	Full		-72.1
U.S.	Full	Full	Full	Full	Full	Full	Full	Full	Full	Full	Full	Full	Full	Full	Full	Full	Full	

Source: Authors.

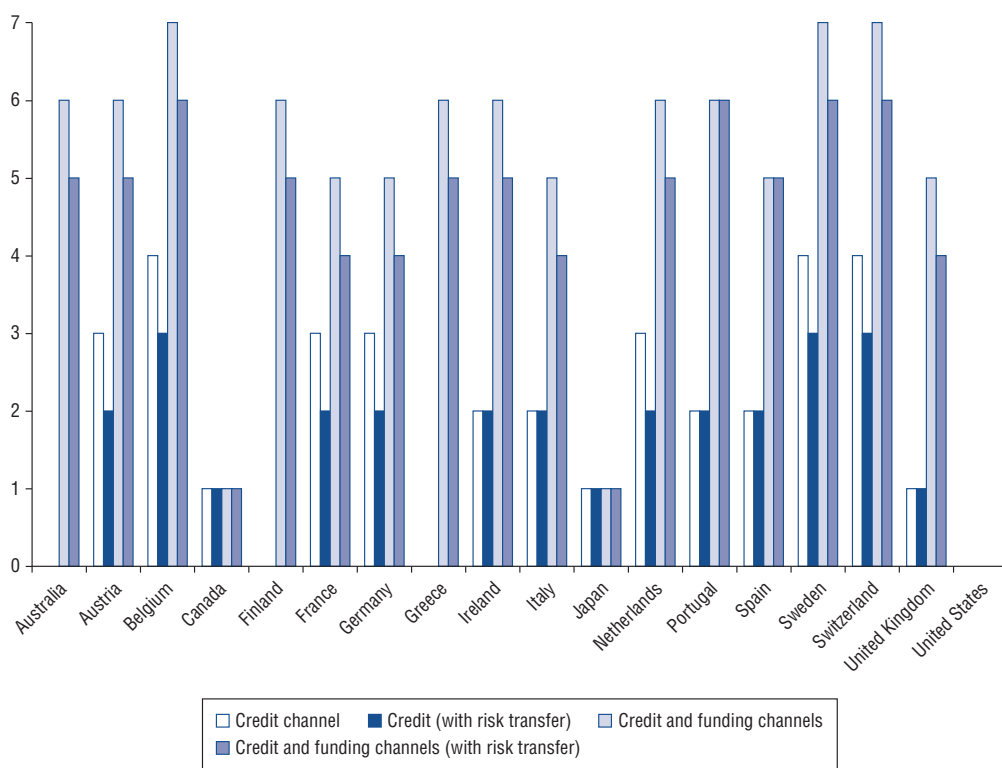
Note: "Full" indicates that all capital is impaired.



Source: Authors.

Note: The figure shows the number of countries whose banking systems would fail, as of December 2007, as a result of the initial failure of each country.

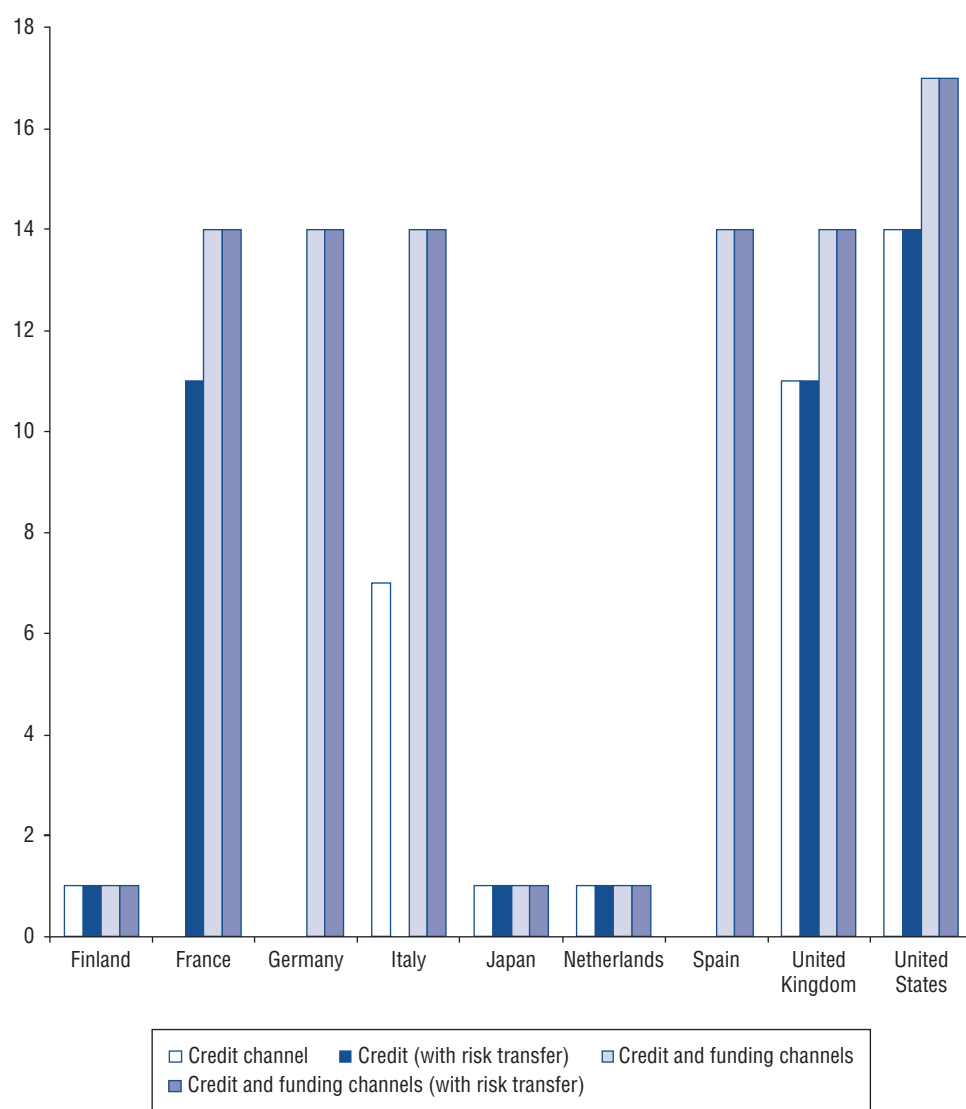
Figure 14.8 Number of Induced Failures—Uniform Distribution



Source: Authors.

Note: This figure depicts each country's absolute hazard level defined as the number of simulations in which the banking system of the country failed as a result of another country's failure.

Figure 14.9 Country-by-Country Vulnerability Level—Uniform Distribution



Source: Authors.

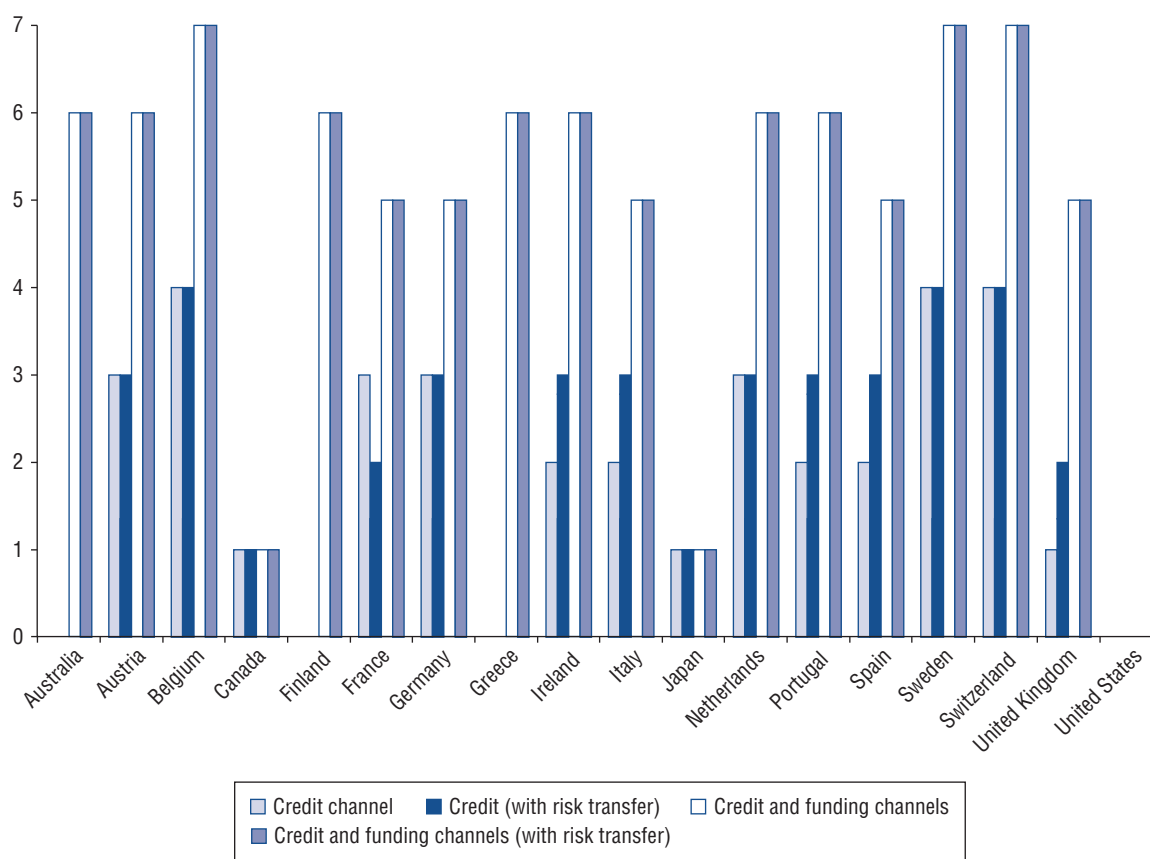
Figure 14.10 Number of Induced Failures—Biased Distribution

the liquidity problems have demonstrated that rollover risk can spill over to the whole financial system, thus requiring a better understanding and monitoring of both direct and indirect linkages.

This chapter also provides a potential approach to consider how to maintain an effective *perimeter of prudential regulation* without unduly stifling innovation and efficiency. It illustrates how network models should allow regulators to see which institutions are affected in subsequent rounds of spillovers and thus determine relative levels of supervision. Such an assessment would have to be conducted at regular intervals, as the structure of the network is likely to change over time. Similarly, network models can assist policymakers with their tough choices, such as how to design capital surcharges to lessen the too-connected-to-fail problem.

In sum, monitoring global systemic linkages undoubtedly will become increasingly relevant for financial regulators as well as for the IMF. Such monitoring can be enhanced in several ways:

- The development of reliable tools for this task should proceed expeditiously.
- Financial regulators need to strengthen their understanding of systemic linkages and improve their gathering of relevant data.
- New information-sharing agreements on cross-border financial exposures (including regulated and unregulated products and institutions) could strengthen the capacity of Fund members to provide it with the relevant data. In principle, such agreements could operate on a multilateral or bilateral basis and would



Source: Authors.

Figure 14.11 Country-by-Country Vulnerability Level—Biased Distribution

ideally address both the domestic and cross-border dimensions.

REFERENCES

- Allen, Franklin, and Ana Babus, 2009, "Networks in Finance," in *Network-Based Strategies and Competencies*, edited by P. Kleindorfer and J. Wind, pp. 367–82 (Philadelphia: Wharton School Publishing).
- Allen, Franklin, and Douglas Gale, 2000, "Financial Contagion," *Journal of Political Economy*, Vol. 108, No. 1, pp. 1–33.
- Bank for International Settlements, 2008, *Guidelines to the International Consolidated Banking Statistics*, Basel.
- Boss, Michael, Helmut Elsinger, Martin Summer, and Stefan Thurner, 2005, "Network Topology of the Interbank Market," *Quantitative Finance*, Vol. 4, No. 6, pp. 677–84.
- Chan-Lau, Jorge, Marco Espinosa, Kay Giesecke, and Juan Solé, 2009, "Assessing the Systemic Implications of Financial Linkages," Chapter 2, *Global Financial Stability Report*, April (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2009/01/index.htm>
- Cifuentes, R., G. Ferrucci, and H. Shin, 2005, "Liquidity Risk and Contagion," *Journal of the European Economic Association*, Vol. 3, Nos. 2–3, pp. 556–66.
- De Vries, Casper G., 2005, "The Simple Economics of Bank Fragility," *Journal of Banking and Finance*, Vol. 29, No. 4, pp. 803–25.
- Degryse, Hans, Muhammad Ather Elahi, and María Fabiana Penas, 2010, "Cross-Border Exposures and Financial Contagion," *International Review of Finance*, Vol. 10, No. 2, pp. 209–40.
- Elsinger, Helmut, Alfred Lehar, and Martin Summer, 2006, "Risk Assessment for Banking Systems," *Management Science*, Vol. 52, No. 9, pp. 1301–14.
- Espinosa-Vega, Marco, and Juan Solé, 2011, "Cross-Border Financial Surveillance: A Network Perspective," *Journal of Financial Economic Policy*, Vol. 3, No. 3, pp. 182–205.
- Freixas, Xavier, Bruno Parigi, and Jean-Charles Rochet, 2000, "Systemic Risk, Interbank Relations and Liquidity Provision by the Central Bank," *Journal of Money, Credit and Banking*, Vol. 32, No. 3, pp. 611–38.
- Furfine, Craig H., 2003, "Interbank Exposures: Quantifying the Risk of Contagion," *Journal of Money, Credit and Banking*, Vol. 35, No. 1, pp. 111–28.
- Hattori, Masazumi, and Yuko Suda, 2009, "Developments in a Cross-Border Bank Exposure 'Network,'" in *Research on Global Financial Stability: The Use of BIS International Financial Statistics*, CGFS Paper No. 29 (Basel: Committee on the Global Financial System). Available via the Internet: <http://www.bis.org/publ/cgfs29.htm>
- Lagunoff, Roger, and Stacey L. Schreft, 2001, "A Model of Financial Fragility," *Journal of Economic Theory*, Vol. 99, Nos. 1–2, pp. 220–64.
- Márquez-Diez-Canedo, Javier, and Serafin Martínez-Jaramillo, 2009, "Systemic Risk: Stress Testing the Banking System," *Inter-*

- national Journal of Intelligent Systems in Accounting, Finance and Management*, Vol. 16, No. 1, pp. 87–110.
- McGuire, Patrick, and Nikola Tarashev, 2008, “Global Monitoring with the BIS International Banking Statistics,” BIS Working Paper No. 244 (Basel: Bank for International Settlements). Available via the Internet: <http://www.bis.org/publ/work244.htm>
- McGuire, Patrick, and Götz von Peter, 2009, “The U.S. Dollar Shortage in Global Banking and the International Policy Response,” BIS Working Paper No. 291 (Basel: Bank for International Settlements). Available via the Internet: <http://www.bis.org/publ/work291.htm>
- McGuire, Patrick, and Philip Wooldridge, 2005, “The BIS Consolidated Banking Statistics: Structure, Uses and Recent Enhancements,” *BIS Quarterly Review*, September (Basel: Bank for International Settlements). Available via the Internet: http://www.bis.org/publ/qtrpdf/r_qt0509.htm
- Mommel, Christoph, and Ingrid Stein, 2008, “Contagion in the German Interbank Market,” Working Paper (Norfolk: European Financial Management Association).
- Morris, Stephen, and Hyun Song Shin, 2008, “Financial Regulation in a System Context,” Brookings Papers on Economic Activity, Fall 2008 (Washington: Brookings Institution).
- Müller, Jeanette, 2006, “Interbank Credit Lines as a Channel of Contagion,” *Journal of Financial Services Research*, Vol. 29, No. 1, pp. 37–60.
- Nier, Erlend, Jing Yang, Tanju Yorulmazer, and Amadeo Alentorn, 2007, “Network Models and Financial Stability,” *Journal of Economics Dynamics and Control*, Vol. 31, No. 6, pp. 2033–60.
- Sheldon, George, and Martin Maurer, 1998, “Interbank Lending and Systemic Risk: An Empirical Analysis for Switzerland,” *Swiss Journal of Economics and Statistics*, Vol. 134, No. 4, pp. 685–704.
- Stern, Gary H., 2008, “Repercussions from the Financial Shock,” Remarks by the President of the Federal Reserve Bank of Minneapolis (Three Forks, August 14). Available via the Internet: http://www.minneapolisfed.org/news_events/pres/stern08-14-08.cfm
- Upper, Christian, 2007, “Using Counterfactual Simulations to Assess the Danger of Contagion in Interbank Markets,” BIS Working Paper No. 234 (Basel: Bank for International Settlements). Available via the Internet: <http://www.bis.org/publ/work234.htm>
- Upper, Christian, and Andreas Worms, 2004, “Estimating Bilateral Exposures in the German Interbank Market: Is There a Danger of Contagion?” *European Economic Review*, Vol. 48, No. 4, pp. 827–49.
- Wells, Simon, 2002, “U.K. Interbank Exposures: Systemic Risk Implications,” *Financial Stability Review*, December, pp. 175–82 (London: Bank of England). Available via the Internet: <http://www.bankofengland.co.uk/archive/Pages/digitalcontent/historicpubs/fsr/fsr13.aspx>

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Appendix

Comparing Results Based on the IBB and the URB Data Sets

This appendix briefly summarizes the results from the simulations using *ultimate risk basis* (URB) data. As mentioned in Section 1.C, URB data are consolidated by residency of the ultimate obligor (i.e., the party that is ultimately responsible for the obligation in case the immediate borrower defaults) and includes net risk transfers. Therefore, the simulation results could potentially differ substantially from those using *immediate borrower basis* (IBB) data.

In order to assess whether the use of IBB or URB data would substantially alter our policy conclusions, we ran simulations using URB data. As Table 14.5 shows, the results of the simulations for the *credit* and the *credit-plus-funding* shocks retain some of their qualitative implications compared with our previous results: in both instances, the United Kingdom and the United States continue to be the two most systemic banking systems, and some other countries continue to maintain their (lower) profile as sources of contagion (e.g., Finland and Netherlands).

However, the comparison of IBB and URB data reveals some important differences for some other countries. For instance, Italy becomes less systemic with URB data in regard to *credit* shocks, going from seven to zero induced failures. Similarly, Germany, Italy, and Spain dramatically reduce their role under the *credit-plus-funding* shock scenario.

TABLE 14.5

Comparing Simulation Results Using IBB and URB Data Sets

Country	Simulations with IBB Data					Simulations with URB Data				
	Failed Capital (% of Total Capital)	Induced Failures	Contagion Rounds	Absolute Hazard ¹	Hazard Rate ²	Failed Capital (% of Total Capital)	Induced Failures	Contagion Rounds	Absolute Hazard ¹	Hazard Rate ²
Effects of Credit Shocks										
Australia	2.6	—	—	—	0.0	2.6	—	—	—	0.0
Austria	0.9	—	—	3	17.6	0.9	—	—	2	11.8
Belgium	1.6	—	—	4	23.5	1.6	—	—	3	17.6
Canada	3.5	—	—	1	5.9	3.5	—	—	—	0.0
Finland	1.2	1	1	—	0.0	1.2	1	1	—	0.0
France	6.6	—	—	3	17.6	6.6	—	—	2	11.8
Germany	3.6	—	—	3	17.6	3.6	—	—	2	11.8
Greece	0.7	—	—	—	0.0	0.7	—	—	—	0.0
Ireland	2.3	—	—	2	11.8	2.3	—	—	2	11.8
Italy	24.6	7	3	2	11.8	5.2	—	—	—	0.0
Japan	15.0	1	1	1	5.9	13.2	—	—	—	0.0
Netherlands	5.6	1	1	3	17.6	5.6	1	1	2	11.8
Portugal	0.5	—	—	2	11.8	0.5	—	—	—	0.0
Spain	4.2	—	—	2	11.8	4.2	—	—	2	11.8
Sweden	0.8	—	—	4	23.5	0.8	—	—	3	17.6
Switzerland	1.8	—	—	4	23.5	1.8	—	—	2	11.8
United Kingdom	45.0	11	3	1	5.9	39.3	9	3	1	5.9
United States	96.2	14	3	—	0.0	73.8	10	4	—	0.0

(continued)

TABLE 14.5 (continued)

Country	Simulations with IBB Data					Simulations with URB Data				
	Failed Capital (% of Total Capital)	Induced Failures	Contagion Rounds	Absolute Hazard ¹	Hazard Rate ²	Failed Capital (% of Total Capital)	Induced Failures	Contagion Rounds	Absolute Hazard ¹	Hazard Rate ²
Effects of Credit and Funding Shocks										
Australia	2.6	—	—	6	35.3	2.6	—	—	—	0.0
Austria	0.9	—	—	6	35.3	0.9	—	—	3	17.6
Belgium	1.6	—	—	7	41.2	1.6	—	—	4	23.5
Canada	3.5	—	—	1	5.9	3.5	—	—	1	5.9
Finland	1.2	1	1	6	35.3	1.2	1	1	3	17.6
France	48.8	14	4	5	29.4	46.2	13	6	2	11.8
Germany	48.8	14	5	5	29.4	4.4	1	1	3	17.6
Greece	0.7	—	—	6	35.3	0.7	—	—	3	17.6
Ireland	2.3	—	—	6	35.3	2.3	—	—	3	17.6
Italy	48.8	14	4	5	29.4	5.2	—	—	3	17.6
Japan	15.0	1	1	1	5.9	13.2	—	—	—	0.0
Netherlands	5.6	1	1	6	35.3	5.6	1	1	3	17.6
Portugal	0.5	—	—	6	35.3	0.5	—	—	3	17.6
Spain	48.8	14	5	5	29.4	4.2	—	—	3	17.6
Sweden	0.8	—	—	7	41.2	0.8	—	—	5	29.4
Switzerland	1.8	—	—	7	41.2	1.8	—	—	3	17.6
United Kingdom	48.8	14	3	5	29.4	46.2	13	3	2	11.8
United States	100.0	17	3	—	0.0	84.2	15	3	—	0.0

Source: Authors.

Note: IBB = immediate borrower basis; URB = ultimate risk basis.

1. Number of simulations in which that particular country fails.

2. Percentage of failures as a percent of the number of simulations conducted.

Balance Sheet Network Analysis of Too-Connected-to-Fail Risk in Global and Domestic Banking Systems

JORGE A. CHAN-LAU

The 2008–9 financial crisis highlighted the importance of evaluating vulnerabilities owing to interconnectedness, or too-connected-to-fail risk, among financial institutions for country monitoring, financial surveillance, investment analysis, and risk management purposes. This chapter illustrates the use of balance sheet–based network analysis to evaluate interconnectedness risk, under extreme adverse scenarios, in banking systems in mature and emerging market countries, and between individual banks in Chile, an advanced emerging market economy.

METHOD SUMMARY

Overview	This technique is used to assess the impact of credit and liquidity shocks on the balance sheets of a group of interconnected institutions.
Application	The method is appropriate for assessing how idiosyncratic shocks affecting one institution or group of institutions are propagated through the financial system.
Nature of approach	Simulation-based.
Data requirements	- Individual firm data on claims against and liabilities to other firms active in the financial system. - Users are required to make assumptions on market-level parameters (e.g., haircuts, funding rollover levels).
Strengths	- The model enables the ranking of firms by the systemic risk they pose to the system and their vulnerability to other firms' failures. - The model specification is flexible.
Weaknesses	- Assumptions are required on the different parameters of the model. - The data may be very difficult to obtain and aggregation issues may compromise the results.
Tool	Excel and MATLAB programs are available from the author upon request.

1. TOO-CONNECTED-TO-FAIL RISK

The recent financial crisis experienced in 2008–09 has raised concerns on the adverse consequences associated with externalities inherent in the financial system. One particular concern

associated with the increased pace of globalization and financial integration is interconnectedness risk, or the too-connected-to-fail (TCTF) problem. The TCTF problem, in turn, brings up issues on how best to regulate TCTF

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institutions and how to ensure they fall within the perimeter of regulation.

A rather simplistic characterization of the TCTF problem is that the failure of one institution could lead to successive rounds of failures of other institutions in the system. The failure cascade is prompted by interinstitution exposure from the existence of direct and indirect linkages between the institutions in the system. The direct linkages arise, for instance, from balance sheet claims that expose one institution to the default of other institutions or from the reliance on credit lines that can be withdrawn abruptly without enough advance notice. The source of indirect linkages could be through derivatives contracts and securities with market values linked to the failure of an otherwise unrelated institution.¹

Prudential regulation and systemic risk surveillance, therefore, call for the development of methods useful for assessing TCTF risk as emphasized recently by the IMF, the Bank for International Settlements (BIS), and the Financial Stability Board (FSB) (IMF/BIS/FSB, 2009a, 2009b). One such method, balance sheet–based network analysis, offers a practical way to analyze risks arising from linkages associated to direct exposures related to the balance sheet of financial institutions. The method is suitable for the type of data typically available to financial supervisors in most jurisdictions and the type of cross-country banking data available from institutions such as the BIS. When secondary market prices are available, network analysis complements assessments based on market-based methods.

This chapter first illustrates the use of balance sheet–based network analysis to evaluate interconnected risk from direct exposures across banking systems in advanced and emerging market economy countries. The analysis, based on BIS cross-country bank claim data for reporting jurisdictions, suggests that the main sources of risk in the global banking system are shocks that could impair the solvency of banks based in the United States and the United Kingdom. Among BIS reporting jurisdictions in Latin America, it appears that the banking system in Brazil is the most resilient to global shocks whereas the banking system in Chile may be more exposed to funding risk from Spanish banks.

The chapter then turns to the analysis of bank exposures within a jurisdiction, specifically Chile, and takes advantage of bank-specific disaggregated data on claims vis-à-vis domestic banks, foreign banks, and other nonbank financial institutions publicly reported by the supervisory agency. The country-specific analysis highlights that the main sources of domestic risk in the Chilean banking system are shocks that affect banks' claims on households and domestic corporations publicly reported by the supervisory agency. To a lesser extent, the banking system is exposed to shocks affecting banks domiciled in Spain and the United States. On the basis of balance sheet data, no bank operating in Chile appears to

be a TCTF institution. The results suggest that, from a financial stability perspective, the focus is better placed on linkages with foreign banks and the creditworthiness of claims on the nonbank financial sector.

Before proceeding to describe the method and results, it is worth bearing in mind that the TCTF problem is not necessarily related to the problems of too-big-to-fail (TBTF) or too-many-to-fail (TMTF). An institution considered to be TBTF is not necessarily TCTF. For instance, in a banking system where retail deposits are the main funding source, interbank exposures are negligible, and there is deposit insurance, any bank with a large share of deposits could be TBTF owing to the costs associated with deposit insurance in case the bank defaults. The failure of such a bank, however, may not have a direct negative impact on other banks. Moreover, stronger banks may benefit from a flight to quality from depositors. In contrast, the business and operational framework under which banks operate may be such that the failure of a relatively small institution may put all institutions in the system at risk. An example in point is the distress experienced by the payment and settlement system following the liquidation of Herstatt Bank in 1974.

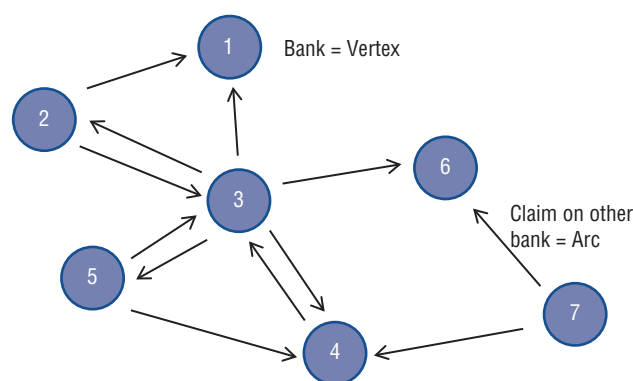
The TMTF problem is associated with the simultaneous failure of a large number of institutions owing to herd behavior (Acharya and Yorulmazer, 2007). The herd behavior is prompted by the observation that an individual institution, unless deemed TBTF, may be allowed to fail by the supervisory agency. It is in their best interest, therefore, for small institutions to coordinate their behavior such that, under adverse circumstances, a large number of them fail simultaneously. The TMTF problem, therefore, is different from the TCTF problem because the failure of one institution does not cause other institutions' failures.

2. BALANCE SHEET–BASED NETWORK ANALYSIS

A financial or banking system can be visualized as a network of institutions (or vertices) linked to each other through directed edges (or arcs). In graph theory, the combination of a set of vertices and a set of arcs constitutes a directed graph or digraph. Figure 15.1 depicts the digraph of a stylized banking system, consisting of seven banks that are represented by the vertices numbered from 1 to 7. Claims on a bank are represented by inbound arcs to the respective vertex. For instance, Bank 7 has claims on Banks 4 and 6.

The mapping of a financial system into a digraph is useful from a supervisory and financial stability perspective. The digraph provides a simple visualization of the different linkages across institutions that, for instance, could help guide the design of the perimeter of regulation. Once the mapping has been established, different analytical tools can be brought in to analyze the stability of the financial system. This chapter describes a simple set of tools, balance sheet–based network

¹ See Chan-Lau (2010b) and references therein.



Source: Author.

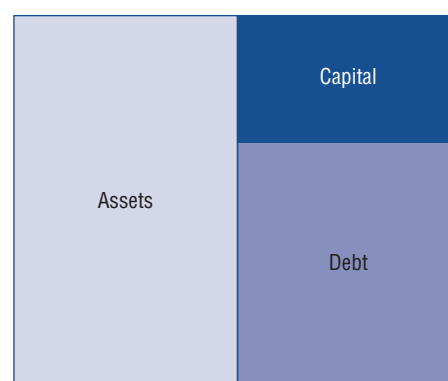
Figure 15.1 A Digraph Representation of a Banking System

analysis, which builds on simple balance sheet accounting identities and simulations to analyze different types of shocks affecting the banking system. A more sophisticated set of tools, graph theory-based models, are described in some detail in the Appendix.²

Balance sheet-based network analysis models rely on financial statements data to construct the matrix of cross-claims across institutions and use simulation to analyze how shocks affecting one institution are propagated through the banking system. Among others, balance sheet-based models include those of Sheldon and Maurer (1998), Furfine (2003), Upper and Worms (2004), Wells (2004), Elsinger, Lehar, and Summer (2006), and, more recently, Aikman and others (2009) and Chan-Lau and others (2009). Though definitely more rudimentary than network models based on graph theory, balance sheet-based models are relatively easy to implement, and the results have a clear economic interpretation, which facilitates communication with senior decision makers.³

A. The basic accounting identity

Balance sheet-based models start from the observation of the equality between the asset side and the liability side of the balance sheet of a bank or financial institution (Figure 15.2). On the asset side of the balance sheet, the bank records its claims on other financial institutions, corporations, and households, such as loans, receivables, equity shares, and debt securities. On the liability side, the bank records what it



Source: Author.

Figure 15.2 A Simplified Bank Balance Sheet

owes to its shareholders, that is, the bank's equity, and to different creditors, including depositors, plus other items, which together with the bank's equity constitutes the bank capital.⁴

The bank's capital is the first cushion against declines in the value of the assets of the bank. If the decline in asset value exceeds the value of the bank's capital, the bank defaults as its capital disappears.⁵ The default of the bank ripples through the banking system as the bank defaults on its liabilities or through the forced withdrawing of the funding it has extended to other banks in the system. Losses from claims on the defaulted banks erode the capital of the surviving banks. In addition, funding withdrawals, if not fully replaced, could force fire sale of assets at depressed values and reduce the asset and equity value of the selling bank while forcing mark-to-market losses in other banks.

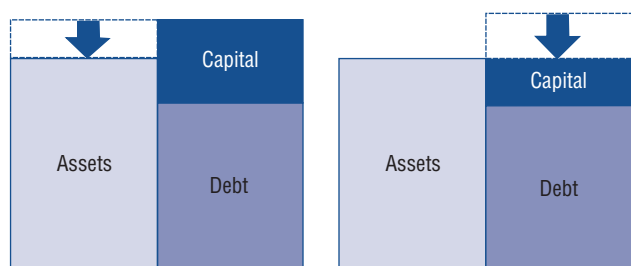
Balance sheet-based network analysis models can accommodate a number of different shocks, namely, credit shocks, funding shocks, risk-transfer shocks, and different combinations of the aforementioned shocks. The next paragraphs offer a diagrammatic explanation of the concepts underlying the framework presented in Chan-Lau and others (2009). Readers interested in the full details of the methodology, including the equations, should refer to this paper.

² It is important to distinguish between the usage of the term *network analysis* in economics and social sciences and the usage in other disciplines. In economics and social sciences, network analysis refers to the analysis of the properties of systems that can be mapped into graphs using *graph theory* and *combinatorics*, for example, Bollobas (1998) and Durrett (2007). In other disciplines, network analysis focuses on optimizing flows between different nodes (or vertices), a problem closely related to *optimization theory*, for example, Ahuja, Magnanti, and Orlin (1993). For an early application of the latter type of techniques to financial systems, see Nagurney and Siokos (1997).

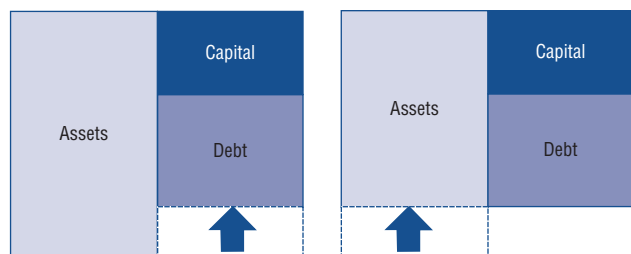
³ An alternative approach not described here is the use of agent-based computational modeling, as in Markose and others (2010).

⁴ The bank capital is equal to the sum of Tier 1 capital and Tier 2 capital. Tier 1 capital includes permanent shareholders' equity plus disclosed reserves, including retained earnings less goodwill. Tier 2 capital includes general provisions and loan loss reserves, revaluation reserves, hybrid capital instruments, subordinated term debt, and undisclosed reserves less investments in unconsolidated financial subsidiaries and in the capital of other financial institutions.

⁵ It can be argued, however, that the relevant event from a supervisory perspective is not the default of the bank per se but rather the event when the bank's capital falls below the regulatory minimum level. The analysis using the tools in this chapter, in this case, could be easily extended by assuming that a bank "default" occurs when capital is less than the regulatory capital rather than when capital vanishes. The simplest way to incorporate this situation into the analysis is by reducing the amount of capital in the bank by the minimum regulatory capital and keeping the original definition of default intact.



Source: Author.

Figure 15.3 Credit Shock

Source: Author.

Figure 15.4 Funding Shock under Normal Market Conditions

B. Credit shocks

Credit shocks are associated with losses on the asset side of the balance sheet.⁶ Claims on other banks are recorded as an asset. When other banks default, the value of the claim is reduced because it is unlikely that the creditor bank recovers the claim's full value, or in other words, the loss given default is strictly positive, or equivalently, the recovery ratio is less than 100 percent. In consequence, the capital of the bank declines (Figure 15.3).

The bank defaults if the losses exceed the value of its capital:

$$\text{Capital} - \sum \text{claims on defaulted banks} \times \text{Loss given Default} < 0. \quad (15.1)$$

A simple simulation can determine whether the default of a bank can trigger a cascade of failures. After setting the default of a specific bank, equation (15.1) is used to see whether losses to other banks could trigger their default. If another bank defaults, its defaulted claims are taken away from the capital of the surviving banks, and again, it is necessary to check whether another bank defaults as a consequence of the new default. The previous step is repeated until no other bank defaults. When only credit shocks are analyzed, it is assumed that other banks can replace funding from defaulted banks without major difficulties.

⁶ This is the case considered in Sheldon and Maurer (1998), Furfine (2003), and Wells (2004).

C. Funding shocks

Funding shocks are associated with the sudden withdrawal of funding and constitute a liability shock. Because assets need to be supported by liabilities, a sudden shortfall in funding sources leads to a reduction of the balance sheet of the bank if it cannot find alternative funding sources. Under normal market conditions, the amount of assets the bank needs to sell is equal to the loss of funding and the bank capital is not affected (Figure 15.4). Furthermore, the leverage of the bank declines, lowering the probability that it may become insolvent.

The bank, however, may be forced to liquidate assets at below face values under fire sale conditions, especially if the market is undergoing a liquidity shortage, the assets are very illiquid, or if the bank holds large positions in certain assets. In the last case, if other banks and market participants know that the affected bank needs to liquidate certain assets, they may collude to mark these assets down in order to take advantage of the bank's distress.⁷ In this case, the bank may need to liquidate assets in excess of the funding withdrawn from the bank. These losses are reflected in capital losses (Figure 15.5).

For some banks, funding losses could exceed their capital and cause their default. In this case, the initial funding shock leads to both subsequent credit and funding shocks as other banks default. The following inequality determines when a bank defaults:

$$\begin{aligned} \text{Capital} - \sum \text{Claims on defaulted banks} \times \text{loss given default} \\ - \sum \text{funding withdrawal} \times (1 - \text{replacement ratio}) \\ \times \text{fire sale loss ratio} < 0. \end{aligned} \quad (15.2)$$

As in the pure credit shock case, simulations can be used to determine how resilient the system is to funding shocks affecting the system banks.

D. Risk transfer shocks and off-balance-sheet exposures

Risk transfer shocks are associated with losses from off-balance-sheet (OBS) exposures. Analyzing these losses in a balance sheet framework requires additional information related to the size of the OBS exposure and the provisions held against it by the bank. Once this information is obtained, the analysis of a risk transfer shock can be analyzed in an extended balance sheet framework (Figure 15.6).

When the bank experiences losses in its OBS exposure, it first uses its OBS provisions as a buffer. When the loss exceeds the provisions, the bank suffers capital losses, and if the capital losses are large enough, the bank defaults:

$$\text{Capital} - \text{Off-balance-sheet exposure} + \text{provisions} < 0. \quad (15.3)$$

⁷ See Hagan (2009) for a vivid narrative of the problems related to fire sale of assets and the markdown of positions faced by Bear Stearns in 2008.

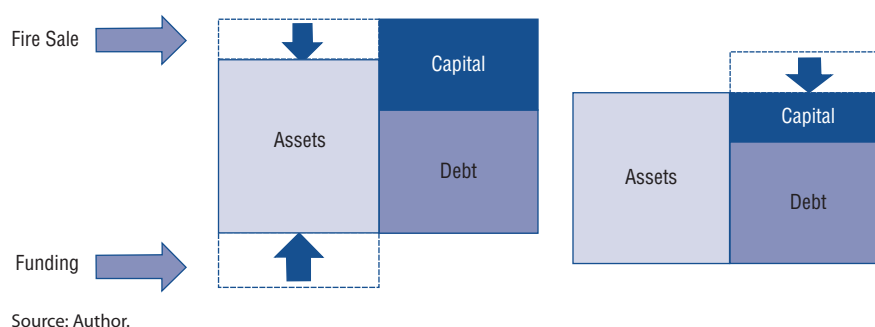


Figure 15.5 Funding Shock under Stressed Market Conditions

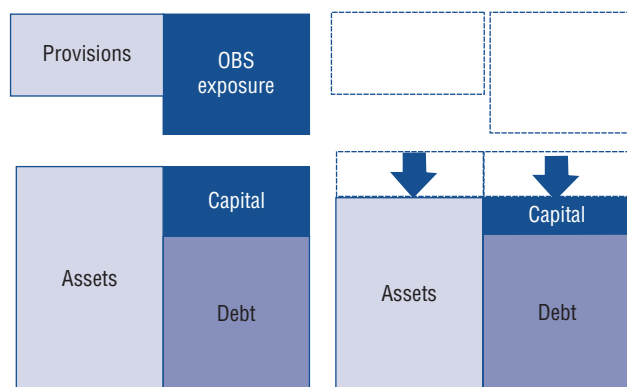


Figure 15.6 Risk Transfer Shock

It goes without saying that a bank default owing to an OBS exposure shock could trigger subsequent credit and funding shocks in the banking system and lead to the failure of other banks. As in the case of credit shocks and funding shocks, simple simulations can be used to analyze the impact of OBS losses on the banking system.

3. BALANCE SHEET-BASED NETWORK ANALYSIS IN PRACTICE

This section illustrates the use of balance sheet-based network analysis in the assessment of TCTF or interconnectedness risk in the global banking system and in the domestic banking system in Chile, an advanced emerging market economy closely integrated into the financial system.⁸

Only credit shocks, funding shocks, and simultaneous credit and funding shocks are considered in the analysis. The following assumptions are used to calculate the results: (1) the loss given default in the credit shock scenario is set equal to 70 percent; and (2) in the case of funding shocks, only 40

percent of the funding can be replaced, forcing the sale of assets at fire sale prices 50 percent below book value.

Risk transfer shocks are not considered in the analysis of the global banking system because the aggregate data approximate only roughly the interbank exposure across countries. Furthermore, up to date there are no publicly available data sources on risk transfers at the individual institution level. Given the data limitations, it may not be proper to add another layer of approximations to try to infer risk transfer between banks. In the analysis of the Chilean domestic banking system, credit risk transfers between institutions are negligible.

One important caveat in interpreting the results is that in all the calculations, it is assumed that either a bank or banking system fails, which is a very low-probability event under the current environment, or that a sudden withdrawal of funding takes place, which is also a low-probability event. Therefore, the results should be interpreted as those corresponding to very extreme adverse scenarios.

A. TCTF risk analysis in mature and emerging market countries' banking systems

The analysis of TCTF risk from the perspective of the global banking system relies on the availability of data on cross-country claims across different bank jurisdictions. Currently, the BIS is the only institution that compiles cross-border banking statistics. In particular, consolidated banking statistics, constructed using central bank reports in 42 countries, comprise a country's aggregate banking system financial claims on the rest of the world.⁹ These statistics have been widely used to assess the risk of exposures of lenders' national banking systems to other countries, but there is wide consensus that it is necessary to improve data collection on interbank exposures (FSB and IMF, 2009).

Consolidated banking statistics are reported on an immediate borrower and an ultimate borrower basis. Figures reported on an immediate borrower basis correspond to contractual

⁸ All calculations in this section were performed using Excel-Visual Basic for Applications programs based on Chan-Lau (2013).

⁹ The consolidated banking statistics are available at <http://www.bis.org/statistics/consstats.htm>. See BIS (2008) for a detailed guide to the statistics and several issues of the *BIS Quarterly Review*.

claims by the head office of a bank and all its branches and are assigned to the home country of the head office. Because banks have access to risk transfer or risk mitigation instruments and techniques, cross-country claims could actually differ from contractual claims. Figures reported on an ultimate borrowing basis account for the impact of risk transfers on contractual claims.

In principle, the difference between claims on an immediate borrower basis and ultimate borrower basis can be used to infer the risk transfer exposures. The analysis here refrains from performing such inference because the aggregate data are only an approximation to interbank exposures as they include, in addition to claims on banks, claims on nonbank financial institutions, nonfinancial institutions, and households. Under these circumstances, it was not considered appropriate to add another layer of approximation by inferring risk transfer exposures.¹⁰

Balance sheet–based network analysis was used to evaluate TCTF risk in banking systems with BIS consolidated claims data on an immediate borrower basis for 20 countries in the first and third quarters of 2009.¹¹ The country sample includes emerging market countries: Brazil, Chile, and Mexico; and mature market countries: Austria, Belgium, Canada, Denmark, France, Germany, Greece, Ireland, Italy, Japan, the Netherlands, Portugal, Spain, Sweden, Switzerland, the United Kingdom, and the United States. Data on total capital in the banking system in the countries analyzed were obtained from different sources, including the European Central Bank statistics and different issues of banking system supplements published by Moody's Investor Services.

Table 15.1 shows what banking systems may be at risk because of shocks affecting banks headquartered in other countries. For the purpose of the analysis, it is assumed that a banking system fails if losses from the shocks exceed the aggregate capital in the system because aggregated data do not allow identifying specific banks that may fail as a consequence of the shock. Imposing the above banking system failure condition is akin to assuming that claims and liabilities of failed banks are taken over by the surviving banks in the country. The results are complemented by Table 15.2, which shows capital losses in national banking systems because of shocks to other jurisdictions.

¹⁰ Readers interested in how to approximate risk transfers should refer to Chan-Lau and others (2009).

¹¹ The choice of consolidated banking statistics presumes that profits, as well as losses, are consolidated and ultimately borne by the head office of the bank. However, if foreign banks operate mainly as subsidiaries in host jurisdictions, the subsidiaries are required to operate as stand-alone institutions and to hold their own capital. In many instances, domestic supervisors would demand that foreign subsidiaries are ring-fenced, so that the head office or a related subsidiary in another country cannot have recourse to its capital. Under such circumstances, it could be argued that the analysis should be based on the BIS locational banking statistics. These statistics report claims and liabilities of bank offices in the countries in which they operate. Locational banking statistics are available at <http://www.bis.org/statistics/bankstats.htm>.

In general, if shocks originate in the banking systems in the United Kingdom and the United States, they may have had the potential to cause failures in the banking systems in most countries during 2009. By this standard, banks headquartered in the United Kingdom and the United States could be TCTF from a global perspective. Similarly, systemic risk from shocks originating in the banking system in Germany appears to have increased twofold in the six-month period from March 2009 to September 2009.

In Europe, shocks that originate in the French banking system could potentially have a large impact on banking systems in Belgium and the Netherlands. It is also apparent that Swedish banks could be exposed to negative shocks to Danish banks. Among the Latin American countries analyzed, the banking system in Chile is the most exposed to funding risks from Spanish banks. Interestingly, the Brazilian banking system appears robust enough to withstand the ripple effect from adverse shocks affecting major banking systems.

Some simple measures can be used to assess the vulnerability of a national banking system to external shocks and the risk it poses to the banking systems in other countries. For instance, the average capital loss experienced by a national banking system because of shocks in other countries is a measure of TCTF vulnerability. Similarly, the average losses that shocks in a national banking system induce in other countries is a measure of TCTF risk. These measures are reported in Table 15.3 for simultaneous credit and funding shocks. Banks headquartered in Austria, for instance, lose on average 23 percent of their capital because of shocks in other banking systems. Conversely, shocks to banks headquartered in Austria induce average capital losses of around 5 percent in other countries.

From a TCTF perspective, the most vulnerable banking systems are those of Belgium, the Netherlands, and Switzerland, while the riskiest are those based in Germany, the United Kingdom, and the United States. Both TCTF risks and vulnerabilities increased during 2009. From a financial stability perspective, note that the continuous updating of a matrix of TCTF risks and vulnerabilities such as the one presented in Table 15.3 could be a useful input for financial surveillance and the setup of a dynamic risk scoring system.

Finally, by recalling the digraph representation of a financial system network in Figure 15.1, it is clear that the absence of direct exposure between two banking systems does not preclude their simultaneous failure if both are connected to a banking system that fails. For instance, in Figure 15.1, the failure of Bank 1 could prompt the failure of Bank 4 if the failure of the former induces the failure of Bank 3. Therefore, the failure of one banking system can trigger a default cycle. Take, for instance, the hypothetical default cycle triggered by a credit shock originating in the banking system in the United Kingdom (Table 15.1, first column). In a first round, the banking systems of Ireland, the Netherlands, and Switzerland are affected. In a second round, the banking systems of Belgium and Germany are affected; in a third round, the banking system in France fails, and the cumula-

TABLE 15.1

Global Banking System: Number of Potential Failures

Panel A: First Quarter, 2009										
Banking system failure in	Type of Shock, and Originating Banking System									
	Credit		Funding	Credit and Funding						
	UK	United States		Denmark	France	Germany	Netherlands	Spain	UK	United States
Austria									Yes	Yes
Belgium	Yes	Yes			Yes	Yes	Yes		Yes	Yes
Brazil										
Canada		Yes							Yes	Yes
Chile			Yes					Yes	Yes	Yes
Denmark	Yes	Yes				Yes			Yes	Yes
France	Yes	Yes							Yes	Yes
Germany	Yes	Yes							Yes	Yes
Greece									Yes	Yes
Ireland	Yes	Yes				Yes			Yes	Yes
Italy									Yes	Yes
Japan		Yes							Yes	Yes
Mexico									Yes	Yes
Netherlands	Yes	Yes			Yes	Yes			Yes	Yes
Portugal		Yes							Yes	Yes
Spain		Yes							Yes	Yes
Sweden	Yes	Yes		Yes		Yes			Yes	Yes
Switzerland	Yes	Yes							Yes	Yes
UK		Yes								Yes
United States									Yes	
Total, excluding originating country	8	13	1	1	2	5	1	1	18	18

Panel B: Third Quarter 2009										
Banking system failure in	Type of Shock, and Originating Banking System									
	Credit			Funding	Credit and Funding					
	Denmark	UK	United States		Denmark	France	Germany	Spain	UK	United States
Austria			Yes				Yes		Yes	Yes
Belgium		Yes	Yes			Yes	Yes		Yes	Yes
Brazil										
Canada			Yes				Yes		Yes	Yes
Chile				Yes			Yes	Yes	Yes	Yes
Denmark		Yes	Yes				Yes		Yes	Yes
France		Yes	Yes				Yes		Yes	Yes
Germany		Yes	Yes						Yes	Yes
Greece							Yes		Yes	Yes
Ireland		Yes	Yes				Yes		Yes	Yes
Italy			Yes				Yes		Yes	Yes
Japan			Yes				Yes		Yes	Yes
Mexico							Yes		Yes	Yes
Netherlands		Yes	Yes			Yes	Yes	Yes	Yes	Yes
Portugal		Yes	Yes				Yes		Yes	Yes
Spain		Yes	Yes				Yes		Yes	Yes
Sweden	Yes	Yes	Yes		Yes		Yes		Yes	Yes
Switzerland		Yes	Yes				Yes		Yes	Yes
UK			Yes				Yes			Yes
United States							Yes		Yes	Yes
Total, excluding originating country	1	10	15	1	1	2	18	1	18	18

Sources: Author; Bank for International Settlements (BIS); European Central Bank; and Moody's Investors Services.

Note: The table shows potential banking system failures induced by shocks originating in banking systems in other countries. The calculations use BIS cross-border claims data on an immediate borrower basis.

TABLE 15.2

Global Banking System: Potential Capital Losses (in percent)

Panel A: First Quarter, 2009																				
Capital losses in	Austria	Belgium	Brazil	Canada	Chile	Denmark	France	Germany	Greece	Ireland	Italy	Japan	Mexico	Netherlands	Portugal	Spain	Sweden	Switzerland	UK	United States
Austria		3	0	1	0	3	26	75	3	4	37	2	0	13	1	5	1	9	100	100
Belgium	6		1	8	0	9	100	100	5	58	40	10	0	100	7	39	4	12	100	100
Brazil	1	0		1	1	1	5	5	0	0	1	2	0	2	1	21	0	3	58	58
Canada	0	4	1		0	2	18	26	0	2	1	12	0	12	0	1	1	5	100	100
Chile	0	1	5	0		1	22	33	0	0	3	4	1	14	0	100	1	1	100	100
Denmark	1	5	0	1	0		24	100	0	19	2	4	0	12	1	5	86	5	100	100
France	3	18	2	3	0	5		88	8	8	51	28	1	36	4	22	2	9	100	100
Germany	16	8	1	6	1	16	77		5	26	41	14	1	38	6	33	9	12	100	100
Greece	3	4	0	0	0	1	52	41		5	5	3	0	12	3	1	1	28	100	100
Ireland	5	20	0	6	0	11	65	100	5		31	15	0	36	7	21	4	5	100	100
Italy	20	5	0	1	0	1	54	82	2	7		3	0	15	1	8	1	3	100	100
Japan	1	2	1	4	0	2	25	28	0	3	4		1	8	0	3	1	5	100	100
Mexico	0	0	0	0	0	0	21	17	0	0	1	5		12	0	100	0	4	100	100
Netherlands	8	90	6	19	3	13	100	100	7	20	45	20	7		7	68	5	19	100	100
Portugal	2	6	7	0	0	2	45	72	7	22	7	1	0	22		69	1	6	100	100
Spain	1	5	20	1	8	2	40	51	0	5	10	2	23	18	16		1	2	100	100
Sweden	2	5	1	2	0	100	31	100	1	7	4	5	0	16	1	7		8	100	100
Switzerland	7	8	7	11	0	10	76	92	24	8	11	44	2	29	2	8	5		100	100
UK	2	7	3	8	0	5	50	77	1	22	8	13	2	20	2	21	3	10		100
United States	1	6	3	21	1	4	48	67	1	8	5	45	7	20	1	10	3	27	100	
Panel B: Third Quarter, 2009																				
Credit and funding shock originated in:																				
Capital losses in	Austria	Belgium	Brazil	Canada	Chile	Denmark	France	Germany	Greece	Ireland	Italy	Japan	Mexico	Netherlands	Portugal	Spain	Sweden	Switzerland	UK	United States
Austria		5	0	1	0	4	30	100	4	7	41	2	0	17	2	7	2	10	100	100
Belgium	9		1	6	0	10	100	100	7	64	46	11	0	100	8	43	3	12	100	100
Brazil	1	1		0	3	1	7	76	0	0	0	3	0	3	2	28	0	3	76	76
Canada	1	3	0		0	2	20	100	0	7	1	13	0	13	0	1	1	5	100	100
Chile	0	1	17	0		1	22	100	0	0	2	3	1	13	0	100	1	1	100	100
Denmark	1	5	0	1	0		25	100	0	19	3	4	0	11	1	4	98	8	100	100
France	3	18	3	3	0	6		100	8	10	56	29	1	37	4	24	3	9	100	100
Germany	18	10	1	5	1	16	85		6	27	44	16	1	41	6	35	9	13	100	100
Greece	4	5	0	0	0	1	56	100		6	5	5	0	13	6	1	1	36	100	100
Ireland	6	21	0	9	0	12	71	100	5		33	17	0	39	9	22	4	6	100	100
Italy	21	5	0	1	0	2	57	100	1	7		4	0	15	2	9	1	3	100	100
Japan	1	2	2	5	0	3	27	100	1	4	5		1	8	0	3	2	6	100	100
Mexico	0	0	0	0	0	1	17	100	0	0	1	5		9	0	100	0	4	100	100
Netherlands	8	87	7	23	3	12	100	100	6	21	46	20	5		9	72	6	20	100	100
Portugal	2	7	8	0	0	3	54	100	12	28	10	2	0	27		78	2	5	100	100
Spain	2	5	26	0	9	2	44	100	0	5	12	2	23	19	19		1	2	100	100
Sweden	2	4	1	3	0	100	34	100	1	7	4	7	1	16	1	7		8	100	100
Switzerland	9	9	8	11	0	14	77	100	31	9	13	55	2	32	2	9	5		100	100
UK	2	7	5	9	0	6	57	100	1	24	8	16	2	22	2	23	3	11		100
United States	1	6	5	23	1	5	55	100	1	9	6	50	7	22	1	11	3	24	100	

Sources: Author; Bank for International Settlements (BIS); European Central Bank; and Moody's Investors Services.

Note: The table shows capital losses induced by shocks affecting other countries as percent of banking system total capital. The calculations use BIS cross-border claims data on an immediate borrower basis.

TABLE 15.3

Global Banking System: Too-Connected-to-Fail Risk (TCTF) and Vulnerability Measures

	First Quarter 2009		Third Quarter 2009	
	TCTF Vulnerability	TCTF Risk	TCTF Vulnerability	TCTF Risk
Austria	20	4	23	5
Belgium	37	10	38	11
Brazil	8	3	15	4
Canada	15	5	19	5
Chile	20	1	24	1
Denmark	24	10	25	10
France	26	46	27	49
Germany	27	66	28	99
Greece	19	4	23	4
Ireland	28	12	29	13
Italy	21	16	23	18
Japan	15	12	19	14
Mexico	19	2	23	2
Netherlands	39	23	39	24
Portugal	25	3	28	4
Spain	21	29	25	30
Sweden	26	7	26	8
Switzerland	29	9	31	10
UK	19	98	21	99
United States	20	98	23	99

Sources: Author; Bank of International Settlements (BIS); European Central Bank; and Moody's Investors Service.

Note: The table shows TCTF vulnerability and risk measures due to joint credit and funding shocks. The TCTF vulnerability measure is the average capital loss suffered by the banking system in the country listed in the column due to shocks affecting banking systems in other countries. The TCTF risk measure is the average capital loss induced by the failure of the banking system listed in the first column on other banking systems.

tive impact affects the banking system in Denmark in the last round.

B. Chile: TCTF risk analysis in an advanced emerging market economy

This section uses balance sheet-based network analysis to evaluate the TCTF risk in Chile, an advanced emerging market economy. Data on claims and liabilities of individual banks vis-à-vis the central bank, foreign banks, and the corporate, nonbank financial, and household sectors are available from the *Monthly Report on Financial Institutions* published by the Banking Supervisory Agency/Superintendencia de Bancos e Instituciones Financieras with a one- to two-month lag.¹²

The monthly report covers 20 domestic banks and foreign bank subsidiaries and 5 foreign bank branches. For each supervised financial institution, the report states the total amount the institution owes to other banks in the system but not to specific institutions. Similarly, the report states the total claims

the institution has on other banks in the system but does not disaggregate claims by individual banks.

In the absence of more detailed interbank exposure data, the matrix of interbank exposures was constructed by assuming that the amount a bank owes in the system is divided equally among all banks reported as having claims on other banks. The above shortcut is analogous to assuming that banks attempt to diversify their interbank exposure as much as possible. This is not the only possible way to construct the matrix of interbank exposures. One alternative is to assign the amount owed by the bank proportionally to the size of the claimant banks. Another alternative is the use of entropy techniques, as done, for instance, by Sheldon and Maurer (1998) and Wells (2004).

Similarly, there is no publicly available detailed information on assets and liabilities vis-à-vis individual foreign banks or specific foreign banking systems. In the analysis presented here, foreign banks are modeled as a single sector. The assumption of a single foreign banking sector errs on the conservative side as it implies that all foreign banks will be simultaneously affected by the same shocks. For instance, a funding shock implies that all foreign banks withdraw funding to banks operating in Chile at the same time.

In addition to domestic and foreign banks, the banking system network was augmented to include the Central Bank of Chile/Banco Central de Chile (BCCCH), the nonbank financial institutions sector, the corporate sector, and the household sector. The BCCCH is included because it has been an important provider of liquidity and financing to the banking sector in the aftermath of the 2008–09 global financial crisis. The inclusion of the central bank in the network is relatively straightforward because there are bank-specific data on the claims and liabilities it holds vis-à-vis domestic banks.

The importance of nonbank financial institutions, especially pension funds and mutual funds, has increased in the wholesale funding market. In the case of nonbank financial institutions, the monthly report provides data on the deposits they held on individual banks, but there are no data on what they may owe to banks. So, in the analysis, nonbank financial institutions appear only as creditors to the banking system.

Corporations account for a substantial share of bank claims. Claims on corporates are constructed using data on corporate loans and receivables. In the analysis, the corporate sector appears only as a debtor because there are no data on corporates' claims on the banking system. Finally, households play a large role on the funding side, through deposits, as well as on the asset side, as borrowers. Data on household deposits and consumer loans are used to include the household sector in the network.

Tables 15.4 and 15.5 report the results of the analysis in January, July, and December 2009. Table 15.4 reports the number of potential defaults and Table 15.5 the average capital losses.

¹² See <http://www.sbif.cl/sbifweb/servlet/InfoFinanciera?indice=4.1&idCategoria=2151&tipocont=2359>.

TABLE 15.4

Chile: Potential Number of Induced Defaults

Number of bank failures when shock comes from	Type of Shock								
	Credit Shock			Funding Shock			Credit and Funding Shock		
	Jan. 2009	Jul. 2009	Dec. 2009	Jan. 2009	Jul. 2009	Dec. 2009	Jan. 2009	Jul. 2009	Dec. 2009
Bank 1	0	0	0	0	0	0	0	0	0
Bank 2	0	0	0	0	0	0	0	0	0
Bank 3	0	0	0	0	0	0	0	0	0
Bank 4	0	0	0	0	0	0	0	0	0
Bank 5	0	0	0	0	0	0	0	0	0
Bank 6	0	0	0	0	0	0	0	0	0
Bank 7	0	0	0	0	0	0	0	0	0
Bank 8	0	0	0	0	0	0	0	0	0
Bank 9	0	0	0	0	0	0	0	0	0
Bank 10	0	0	0	0	0	0	0	0	0
Bank 11	0	0	0	0	0	0	0	0	0
Bank 12	0	0	0	0	0	0	0	0	0
Bank 13	0	0	0	0	0	0	0	0	0
Bank 14	0	0	0	0	0	0	0	0	0
Bank 15	0	0	0	0	0	0	0	0	0
Bank 16	0	0	0	0	0	0	0	0	0
Bank 17	0	0	0	0	0	0	0	0	0
Bank 18	0	0	0	0	0	0	0	0	0
Bank 19	0	0	0	0	0	0	0	0	0
Bank 20	0	0	0	0	0	0	0	0	0
Bank 21	0	0	0	0	0	0	0	0	0
Bank 22	0	0	0	0	0	0	0	0	0
Bank 23	0	0	0	0	0	0	0	0	0
Bank 24	0	0	0	0	0	0	0	0	0
Bank 25	0	0	0	0	0	0	0	0	0
Central bank	0	0	0	0	0	0	0	0	0
Foreign banks	0	0	0	1	1	1	1	1	1
Nonbank financial institutions	0	0	0	2	1	4	2	1	4
Corporates	17	16	14	N/A	N/A	N/A	17	16	14
Households	14	12	12	17	17	16	17	17	16

Sources: Author; and Banking Supervisory Agency/Superintendencia de Bancos e Instituciones Financieras (SBIF).

Note: The table shows the number of induced bank defaults due to credit shocks, funding shocks, and the joint realization of credit and funding shocks.

Domestic interbank exposure is relatively unimportant as the default of a single bank would not lead to further defaults (Table 15.4). The average capital losses reported in Table 15.5 provide further support while pointing toward a reduction of interbank exposures during 2009. For instance, in January 2009, the hypothetical default of Bank 5 and Bank 3 would have triggered average capital losses of 10.2 percent and 7.4 percent, respectively, but in December 2009, these losses declined to 1.1 percent and 0.7 percent, respectively. Against this trend, risks from Bank 4 appear to have trended upward. Average capital losses induced by its hypothetical default have increased to 6.9 percent from 0.3 percent. The hypothetical capital losses are mainly concentrated on Bank 12 (35 percent), Bank 21 (54 percent), and Bank 25 (47 percent).

A close examination of the results in Table 15.4 and 15.5 suggests that domestic banks were reducing their claim exposure to the corporate and household sectors during 2009. The number of banks that could default following a negative credit shock from the systemwide collapse of the corporate

sector declined to 14 from 17 at the beginning of the year, while average capital losses declined to 76 percent from 82 percent. Similarly, average capital losses following defaults on household loans fell to 59 percent from 68 percent.

Domestic banks increased their reliance on wholesale funding from nonbank financial institutions in 2009. In December 2009, a hypothetical sudden withdrawal of nonbank financial deposits could lead to problems in four banks (Bank 16, Bank 17, Bank 18, and Bank 20) up from two in January 2009 (Bank 18 and Bank 20).

Another important funding source during 2009 was the central bank. On average, capital losses would be approximately 3 percent if central bank funding were not available. These losses, however, are heavily concentrated in a handful of institutions. As of December 2009, the institutions more affected by reduced funding from the central bank are Bank 4 (32 percent capital loss), Bank 11 (24 percent), and Bank 13 (22 percent).

Table 15.6 reports TCTF risk and vulnerability measures for Chilean financial institutions. Overall, in 2009, TCTF

TABLE 15.5

Chile: Average Capital Losses
(in percent)

Average bank capital losses when shock comes from	Type of Shock								
	Credit Shock			Funding Shock			Credit and Funding Shock		
	Jan. 2009	Jul. 2009	Dec. 2009	Jan. 2009	Jul. 2009	Dec. 2009	Jan. 2009	Jul. 2009	Dec. 2009
Bank 1	0.0	1.1	0.0	0.0	0.0	0.0	0.0	1.1	0.0
Bank 2	4.4	0.0	0.1	0.0	0.0	0.0	4.4	0.0	0.1
Bank 3	7.4	1.7	0.6	0.0	0.0	0.0	7.4	1.7	0.7
Bank 4	0.3	3.1	5.7	0.2	0.0	0.0	0.5	3.1	6.9
Bank 5	10.2	1.0	0.8	0.0	0.1	0.1	10.2	1.1	1.1
Bank 6	3.2	0.0	0.0	0.0	0.0	0.0	3.2	0.0	0.0
Bank 7	0.0	0.0	0.0	0.2	0.1	0.1	0.2	0.1	0.1
Bank 8	0.2	0.0	0.0	0.2	0.1	0.0	0.4	0.1	0.0
Bank 9	0.0	0.2	0.0	0.2	0.0	0.0	0.2	0.2	0.0
Bank 10	0.3	0.0	0.5	0.2	0.1	0.1	0.5	0.1	0.7
Bank 11	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1	0.0
Bank 12	0.0	0.0	0.0	0.2	0.1	0.1	0.2	0.1	0.1
Bank 13	0.5	0.0	0.0	0.0	0.0	0.0	0.5	0.0	0.0
Bank 14	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1	0.0
Bank 15	0.5	1.0	0.5	0.2	0.0	0.0	0.7	1.0	0.6
Bank 16	0.0	2.8	0.0	0.0	0.1	0.1	0.0	2.9	0.1
Bank 17	2.6	5.8	0.0	0.2	0.0	0.0	2.8	5.8	0.0
Bank 18	1.0	0.4	0.1	0.2	0.1	0.1	1.2	0.5	0.2
Bank 19	2.4	0.6	0.0	0.0	0.1	0.1	2.4	0.7	0.1
Bank 20	0.3	0.1	0.0	0.2	0.1	0.1	0.5	0.2	0.1
Bank 21	0.0	0.0	0.0	0.2	0.1	0.1	0.2	0.1	0.1
Bank 22	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Bank 23	0.0	0.1	0.0	0.2	0.1	0.0	0.2	0.2	0.0
Bank 24	0.0	0.0	0.0	0.2	0.1	0.0	0.2	0.1	0.0
Bank 25	0.0	0.0	0.0	0.2	0.1	0.1	0.2	0.1	0.1
Central bank	n.a.	n.a.	n.a.	2.9	3.9	3.5	5.0	7.0	4.7
Foreign banks	7.8	3.0	5.3	17.3	13.4	16.8	25.1	16.4	22.2
Nonbank financial institutions	0.0	0.0	0.0	43.1	41.7	43.7	43.1	41.7	43.7
Corporates	82.1	76.6	75.8	N/A	N/A	N/A	83.1	77.0	76.1
Households	68.0	64.5	58.5	86.5	85.9	85.4	86.5	85.9	85.5

Sources: Author; and SBIF.

Note: The table shows the average bank capital losses suffered due to credit shocks, funding shocks, and the joint realization of credit and funding shocks as percent of the institution's total capital.

vulnerability was higher for foreign bank branches owing to their large interbank exposures relative to their capital. TCTF risk related to Bank 5 and Bank 17 declined during the year, but it increased substantially for Bank 4.

Finally, Table 15.7 reports excess capital losses in the banking system, as percent of total aggregate capital, owing to shocks originating in banks operating in Chile. Excess capital losses measure the capital losses in the system excluding the losses of the bank that triggered the shocks. This measure is a TCTF risk measure closely related to the concepts of incremental contribution to risk introduced in Chan-Lau (2009) and of marginal contribution to risk introduced by Tarashev, Borio, and Tsatsaronis (2009).

In January 2009, Bank 3 accounted for 6.7 percent of the total capital in the banking system. In the hypothetical case that a credit shock affecting Bank 3 realizes, the banking system may suffer a capital loss of 7.9 percent, of which 6.7 percent corresponds to the capital of Bank 3 capital and an excess capital loss of 1.2 percent from banks with exposure

to Bank 3. Hence, it can be stated that the TCTF risk posed by Bank 3 is equivalent to 1.2 percent of the banking system capital. As of December 2009, excess capital losses were relatively small and at most equal to 1½ percent for shocks from Bank 4. The overall analysis indicates that TCTF risk in the domestic banking system in Chile is relatively small.

4. CONCLUSION

Interconnectedness risk and the externalities associated with too-connected-to-fail institutions were a major amplifying and transmission mechanism during the 2008–09 global financial crisis. In response, there have been renewed efforts to understand and measure interconnectedness and TCTF risk and to ensure that the risk is properly addressed, as advanced in the current proposals for reforming the regulatory and supervisory framework as suggested by the Basel Committee on Banking Supervision (2009) and by Brunnermeier and others (2009). For instance, recent proposals on capital

TABLE 15.6

Chile: Too-Connected-to-Fail Risk (TCTF) and Vulnerability Measures

	TCTF Vulnerability			TCTF Risk		
	January 2009	July 2009	December 2009	January 2009	July 2009	December 2009
Bank 1	0.0	0.1	0.0	0.0	1.1	0.0
Bank 2	0.2	0.0	0.0	4.4	0.0	0.1
Bank 3	0.3	0.1	0.0	7.4	1.7	0.7
Bank 4	0.1	0.1	0.2	0.5	3.1	6.9
Bank 5	0.2	0.1	0.1	10.2	1.1	1.1
Bank 6	0.8	0.0	0.0	3.2	0.0	0.0
Bank 7	1.3	0.6	0.7	0.2	0.1	0.1
Bank 8	1.2	0.5	0.0	0.4	0.1	0.0
Bank 9	3.8	0.1	0.0	0.2	0.2	0.0
Bank 10	0.4	0.2	0.2	0.5	0.1	0.7
Bank 11	0.0	4.7	0.0	0.0	0.1	0.0
Bank 12	4.3	1.9	2.2	0.2	0.1	0.1
Bank 13	0.4	0.0	0.0	0.5	0.0	0.0
Bank 14	0.0	1.3	0.0	0.0	0.1	0.0
Bank 15	3.5	0.5	0.4	0.7	1.0	0.6
Bank 16	0.0	0.1	0.0	0.0	2.9	0.1
Bank 17	0.2	0.1	0.0	2.8	5.8	0.0
Bank 18	0.7	0.3	0.3	1.2	0.5	0.2
Bank 19	0.4	0.4	0.4	2.4	0.7	0.1
Bank 20	0.2	0.1	0.1	0.5	0.2	0.1
Bank 21	6.5	2.9	3.3	0.2	0.1	0.1
Bank 22	0.0	0.0	0.0	0.0	0.0	0.0
Bank 23	5.1	2.3	0.0	0.2	0.2	0.0
Bank 24	0.6	0.3	0.0	0.2	0.1	0.0
Bank 25	6.0	2.6	2.9	0.2	0.1	0.1

Sources: Author; and SBIF.

Note: The table shows TCTF vulnerability and risk measures due to simultaneous credit and funding shocks. For the banks listed in the first column, the TCTF vulnerability is measured as the average capital loss the bank suffers from shocks to other banks in the system. The TCTF risk is measured as the average capital loss the bank induces on other banks in the system.

charges on interconnectedness risk are described in Adrian and Brunnermeier (2008); Tarashev, Borio, and Tsatsaronis (2009); Chan-Lau (2010c); and Gauthier, Lehar, and Souissi (2010).

This chapter has illustrated the use of a simple methodology, balance sheet–based network analysis, to capture interconnectedness and TCTF risk in domestic and international banking systems using publicly available data sources under extreme adverse scenarios. In contrast to graph theory–based network analysis, there is no complex mathematics involved in the analysis but just simple balance sheet accounting identities. In consequence, balance sheet–based network analysis can be easily implemented whenever balance sheet data are available.

In the case of the global banking system, the analysis relies on data on cross-country claims compiled by the BIS. The results point out that shocks affecting the solvency of banks based in the United Kingdom and the United States, and, to a lesser extent, in Germany, could be the main sources of risk to banking systems worldwide. The results, however, should be interpreted with caution because of the high level of aggregation of the data.

In the case of Chile, detailed data at the bank level on claims and liabilities vis-à-vis other banks in the system, for-

eign banks, and nonbank financial institutions are available from the banking supervisory agency. The results suggest that TCTF risk is low and that, from a financial stability perspective, financial surveillance is better focused on the linkages of domestic banks with foreign banks and nonbank financial institutions.

Finally, although there are advantages to the use of balance sheet data and accounting identities, there are also some disadvantages related to the reliability of accounting data, as these are subject to manipulation, and the reporting lags, which may render the analysis irrelevant, especially if inter-bank exposures are changing rapidly.¹³ Some of the disadvantages can be addressed by complementing network analysis with market-based measures, which use information from security prices and common economic factors to assess TCTF risk. Examples of recent work in this direction are

¹³ Even sophisticated financial institutions may have problems collecting accounting data on a timely basis. According to Paul Friedman, COO of fixed income at Bear Stearns in 2008: “We go through the cash position, and there’s lot of questions as to how accurate it is. . . . The firm was not really set up—most firms are not—to do real-time cash accounting” (as quoted in Cohan, 2009).

TABLE 15.7

Banking System Excess Capital Loss
 (in percent)

Banking system excess capital loss due to shocks from	Type of Shock								
	Credit Shock			Funding Shock			Credit and Funding Shock		
	Jan. 2009	Jul. 2009	Dec. 2009	Jan. 2009	Jul. 2009	Dec. 2009	Jan. 2009	Jul. 2009	Dec. 2009
Bank 1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0
Bank 2	0.7	0.0	0.1	0.0	0.0	0.0	0.7	0.0	0.1
Bank 3	1.2	0.2	0.7	0.0	0.0	0.0	1.2	0.2	0.7
Bank 4	0.1	0.5	6.9	0.2	0.0	0.0	0.2	0.5	6.9
Bank 5	1.8	0.2	1.0	0.0	0.1	0.1	1.8	0.2	1.1
Bank 6	0.5	0.0	0.0	0.0	0.0	0.0	0.5	0.0	0.0
Bank 7	0.0	0.0	0.0	0.2	0.1	0.1	0.2	0.1	0.1
Bank 8	0.0	0.0	0.0	0.2	0.1	0.0	0.2	0.1	0.0
Bank 9	0.0	0.0	0.0	0.2	0.0	0.0	0.2	0.0	0.0
Bank 10	0.0	0.0	0.7	0.2	0.1	0.1	0.2	0.1	0.7
Bank 11	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1	0.0
Bank 12	0.0	0.0	0.0	0.2	0.1	0.1	0.2	0.1	0.1
Bank 13	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0
Bank 14	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1	0.0
Bank 15	0.1	0.1	0.6	0.2	0.0	0.0	0.2	0.1	0.6
Bank 16	0.0	0.4	0.0	0.0	0.1	0.1	0.0	0.5	0.1
Bank 17	0.4	0.9	0.0	0.2	0.0	0.0	0.6	0.9	0.0
Bank 18	0.1	0.0	0.1	0.2	0.1	0.1	0.3	0.1	0.2
Bank 19	0.4	0.1	0.0	0.0	0.1	0.1	0.4	0.1	0.1
Bank 20	0.0	0.0	0.0	0.2	0.1	0.1	0.2	0.1	0.1
Bank 21	0.0	0.0	0.0	0.2	0.1	0.1	0.2	0.1	0.1
Bank 22	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Bank 23	0.0	0.0	0.0	0.2	0.1	0.0	0.2	0.1	0.0
Bank 24	0.0	0.0	0.0	0.2	0.1	0.0	0.2	0.1	0.0
Bank 25	0.0	0.0	0.0	0.2	0.1	0.1	0.2	0.1	0.1

Sources: Author; and SBIF.

Note: The table shows banking system excess capital losses, in percent of capital in the banking system, induced by the failure of one bank. Excess capital losses are the total banking losses excluding the losses from the bank that triggered subsequent shocks in the system.

Adrian and Brunnermeier (2008), Chan-Lau (2009), and Chan-Lau (2013), among others.¹⁴

REFERENCES

- Acharya, Viral V., and Tarun Yorulmazer, 2007, "Too Many to Fail—An Analysis of Time-Inconsistency in Bank Closure Policies," Working Paper No. 319 (London: Bank of England). Available via the Internet: <http://www.bankofengland.co.uk/publications/Pages/workingpapers/2007/wp319.aspx>
- Adrian, Tobias, and Markus K. Brunnermeier, 2008, "CoVaR," Staff Report No. 348 (New York: Federal Reserve Bank). Available via the Internet: http://www.newyorkfed.org/research/staff_reports/sr348.html
- Ahuja, Ravindra, Thomas L. Magnanti, and James B. Orlin, 1993, *Network Flows: Theory, Algorithms, and Applications* (New York: Prentice Hall).
- Aikman, David, Piergiorgio Alessandri, Bruno Eklund, Prasanna Gai, Sujit Kapadia, Elizabeth Martin, Nada Mora, Gabriel Sterne, and Matthew Willison, 2009, "Funding Liquidity Risk in a Quantitative Model of Systemic Stability," Working Paper No. 372 (London: Bank of England). Available via the Internet: <http://www.bankofengland.co.uk/publications/Pages/workingpapers/2009/wp372.aspx>
- Bank for International Settlements, 2008, *Guidelines to the International Consolidated Banking Statistics*, Basel.
- Basel Committee on Banking Supervision, 2009, "Strengthening the Resilience of the Banking Sector," Consultative Document (Basel, December). Available via the Internet: <http://www.bis.org/publ/bcbs164.htm>
- Bech, Morten, James Chapman, and Rod Garratt, 2008, "Which Bank Is the 'Central' Bank? An Application of Markov Theory to the Canadian Large Value Transfer System," Working Paper 2008-42 (Ottawa: Bank of Canada). Available via the Internet: <http://www.bankofcanada.ca/2008/10/publications/research/working-paper-2008-42>
- Bollobas, Bela, 1998, *Modern Graph Theory* (New York: Springer).
- Brunnermeier, Markus K., Andrew Crockett, Charles Goodhart, Avinash Persaud, and Hyun-Song Shin, 2009, "The Fundamental Principles of Financial Regulation," *Geneva Reports of the World Economy*, No. 11 (Geneva and London: International

¹⁴ One caveat about using security prices, however, is that prices may not fully reflect the fundamental value of the banks and/or their true default risk: prices may be capturing the effects of factors like liquidity and technical supply-and-demand drivers, such as regulatory changes, that affect the decisions of market participants.

- Center for Monetary and Banking Studies and Centre for Economic Policy Research).
- Chan-Lau, Jorge A, 2009, "Risk Codependence in the Global Financial System," in *The Banking Crisis Handbook*, edited by Greg Gregoriou (Boca Raton, London, New York: CRC Press).
- , 2010a, "Balance Sheet Network Analysis of Too-Connected-to-Fail Risk in Global and Domestic Banking Systems," IMF Working Paper No. 10/107 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=23809>
- , 2010b, "The Global Financial Crisis and Its Impact on the Chilean Banking System," IMF Working Paper 10/108 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=23781>
- , 2010c, "Regulatory Capital for Too-Connected-to-Fail Institutions: A Practical Proposal," *Financial Markets, Institutions and Instruments*, Vol. 19, No. 5, pp. 355–80.
- , 2013, *Systemic Risk Assessment and Oversight* (London: Risk Books).
- Chan-Lau, Jorge A, Marco Espinosa, Kay Giesecke, and Juan Solé, 2009, "Assessing the Systemic Implications of Financial Linkages," *Global Financial Stability Report*, Chapter 2 (International Monetary Fund, April). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2009/01/index.htm>
- Cohan, William, D., 2009, *House of Cards: A Tale of Hubris and Wretched Excess on Wall Street* (New York: Doubleday).
- Durrett, Richard, 2007, *Random Graph Dynamics* (New York: Cambridge University Press).
- Elsinger, Helmut, Alfred Lehar, and Martin Summer, 2006, "Risk Assessment for Banking Systems," *Management Science*, Vol. 52, No. 9, pp. 1301–66.
- Embre, Lana, and Tom Roberts, 2009, "Network Analysis and Canada's Large Value Transfer System," Discussion Paper 2009-13 (Ottawa: Bank of Canada). Available via the Internet: <http://www.bankofcanada.ca/2009/12/publications/research/discussion-paper-2009-13>
- Financial Stability Board and International Monetary Fund, 2009, "The Financial Crisis and Information Gaps," Report to the G-20 Finance Ministers and Central Bank (Basel and Washington, October). Available via the Internet: <http://www.imf.org/external/np/g20/pdf/102909.pdf>
- Furfine, Craig, 2003, "Interbank Exposures: Quantifying the Risk of Contagion," *Journal of Money, Credit and Banking*, Vol. 35, No. 1, pp. 111–28.
- Gai, Prasanna, and Sujit Kapadia, 2010, "A Network Model of Super-Systemic Crises," in *Financial Stability, Monetary Policy, and Central Banking*, edited by Rodrigo A. Alfaro, pp. 411–32 (Santiago: Banco Central de Chile).
- Gauthier, Celine, Alfred Lehar, and Moez Souissi, 2010, "Macroprudential Capital Requirements and Systemic Risk," Working Paper 2010-4 (Ottawa: Bank of Canada). Available via the Internet: <http://www.bankofcanada.ca/2010/01/publications/research/working-paper-2010-4>
- Hattori, Masazumi, and Yuko Suda, 2009, "Developments in a Cross-Border Bank Exposure 'Network,'" in *Research on Global Financial Stability: The Use of BIS International Financial Statistics*, CGFS Paper No. 29 (Basel: Committee on the Global Financial System). Available via the Internet: <http://www.bis.org/publ/cgfs29.htm>
- International Monetary Fund/Bank for International Settlements/Financial Stability Board, 2009a, "Guidance to Assess the Systemic Importance of Financial Institutions, Markets and Instruments: Initial Considerations," Report to the G-20 Finance Ministers and Governors (Washington and Basel, November). Available via the Internet: <http://www.imf.org/external/np/g20/pdf/100109.pdf>
- , 2009b, "Guidance to Assess the Systemic Importance of Financial Institutions, Markets and Instruments: Initial Considerations—Background Paper," Report to the G20 Finance Ministers and Governors (Washington and Basel, November). Available via the Internet: <http://www.imf.org/external/np/g20/pdf/100109a.pdf>
- Markose, Sheri, Simone Giansante, Mateusz Gatkowski, and Ali Shaghaghi, 2010, "Too Interconnected to Fail: Financial Networks of CDS and Other Credit Enhancement Obligations of U.S. Banks," Working Paper No. 33, Centre for Computational Finance and Economic Agents (Essex: University of Essex).
- Nagurney, Anna, and Stavros Siokos, 1997, *Financial Networks—Statics and Dynamics* (Heidelberg: Springer).
- Nier, Erlend, Jing Yang, Tanju Yorulmazer, and Amadeo Alentorn, 2007, "Network Models and Financial Stability," *Journal of Economic Dynamics and Control*, Vol. 31, No. 1, pp. 2033–60.
- Sheldon, George, and Martin Maurer, 1998, "Interbank Lending and Systemic Risk: An Empirical Analysis for Switzerland," *Swiss Journal of Economics and Statistics*, Vol. 134, No. 4, pp. 685–704.
- Soramaki, Kimo, Morten L. Bech, Jeffrey Arnold, Robert J. Glass, and Walter E. Beyeler, 2007, "The Topology of Interbank Payment Flows," *Physica A*, Vol. 379, No. 1, pp. 317–33.
- Tarashev, Nikola, Claudio Borio, and Kostas Tsatsaronis, 2009, "The Systemic Importance of Financial Institutions," *BIS Quarterly Review*, September, pp. 75–87 (Basel: Bank for International Settlements). Available via the Internet: http://www.bis.org/publ/qtrpdf/r_qt0909.htm
- Upper, Christian, and Andreas Worms, 2004, "Estimating Bilateral Exposures in the German Interbank Market: Is There a Danger of Contagion?" *European Economic Review*, Vol. 48, No. 4, pp. 827–49.
- Watts, Duncan, 2002, "A Simple Model of Global Cascades on Random Networks," *Proceedings of the National Academy of Sciences*, Vol. 99, No. 9, pp. 5766–71.
- Wells, Simon, 2004, "Financial Interlinkages in the U.K. Interbank Market and the Risk of Contagion," Working Paper No. 230 (London: Bank of England). Available via the Internet: <http://www.bankofengland.co.uk/publications/Pages/workingpapers/2004/wp230.aspx>

Appendix

Graph Theory–Based Network Models

Graph theory is the natural analytical framework for analyzing the properties of a financial system once it has been mapped into a digraph. In particular, it is possible to classify the different banks (vertices) into clusters and to evaluate how the system changes after the removal of a set of banks (vertices) and their respective linkages (arcs). In the latter case, the use of random graph theory makes it possible to add randomness either to the number of banks (vertices) in the system (digraph) or to the existence of linkages (arcs) between them. Random graph theory can, in principle, accommodate the observation that the number of participants in a connected financial system as well as the nature of their cross-linkages change over time.

Graph theory has been used extensively to analyze payment and settlement systems, as in Soramaki and others (2007); Bech, Chapman, and Garrat (2008); and Embree and Roberts (2009), among others. The focus of the analysis has been mostly on the topological properties of the system. Some of the topological properties include size, or the number of institutions in the system, connectivity, or the relative number of existing linkages to the maximum number of linkages, and the clustering coefficient, or the probability that two institutions “close” to a third one are also “close” to each other.

The insights gained from the application of graph theory to payment and settlement systems are difficult to translate to more complex systems, such as the interbank market, a domestic banking system, or the global financial system. The difficulty arises from the relative complexity that characterizes cross-claims across financial institutions, which stand in sharp contrast with the homogeneity of the transactions undertaken in the payment and settlement systems. Notwithstanding this difficulty, there have been some recent advances in extending graph theory to the analysis of complex banking systems in stylized models. Hattori and Suda (2009) use BIS banking data to analyze the topological properties of cross-border banking networks and their implications for banking stability. Nier and others (2007) use Erdos-Renyi graphs to explore how the different topological properties of a banking system affect the propagation of defaults. Gai and Kapadia (2010) use the small-world model of Watts (2002) to analyze contagion in a stylized network using numerical simulations.

Calibrating graph theory–based models with real data remains a major challenge for bringing these models to an operational level suitable for surveillance and supervisory purposes. As an alternative, applied research in policymaking institutions has opted for a simpler approach based on balance sheet data, which is described in the main text.

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PART II

The Market Price–Based Approach

SECTION A

The Equity Indicators–Based Approach

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Introduction to the Equity Indicators–Based Approach to Stress Testing

JORGE A. CHAN-LAU

For stress testing purposes, it is necessary to measure the default risk of individual financial and nonfinancial institutions and to assess how this risk changes under different scenarios. Once the default risk of individual institutions is measured, it is possible to analyze the credit exposures and potential losses and construct bottom-up measures of systemic risk under different stress scenarios.

The estimation of default risk is challenging because of limitations on data availability. One way around this challenge is to use income and balance sheet data to predict corporate bankruptcy, as is the case of the Altman Z-score (Altman, 1968). Although accounting data are readily available across jurisdictions, their use could be hampered by differences in standards, the risk that accounting figures may be manipulated, and, more important, the backward-looking nature of financial statements.

The shortcomings of using financial reporting data could be offset by using information extracted from security prices if secondary markets are judged adequately liquid and efficient. The preferred source of information should be the prices of securities with payoffs more directly related to the creditworthiness of the issuer, such as bonds and credit default swaps. In many instances, however, corporate bonds are rarely traded, so their prices and yields are unreliable. Credit default swaps, though liquid, cover only a relatively small subset of firms in advanced economies, but the coverage is increasing rapidly.

Equity prices are an alternative to bonds and credit default swaps. Unlike the markets in the last two instruments, stock markets tend to be relatively more liquid and cover a larger number of firms in a given jurisdiction. The usefulness

of equity prices for estimating probabilities of default is established by the observation, first made by Black and Scholes (1973) and Merton (1974), that corporate securities are contingent claims on the asset value of the issuing firm. The observation can be illustrated in the case of a firm that issues one unit of equity and one unit of a zero-coupon bond with face value D and maturity T . At expiration, the value of debt, B_T , and equity, E_T , are given by

$$B_T = \min(V_T, D) = D - \max(D - V_T, 0), \\ E_T = \max(V_T - D, 0),$$

where V_T is the asset value of the firm at maturity.

These equations state that bondholders get paid in full only if the firm's assets exceed the face value of debt. Otherwise, the firm is liquidated, and its assets are used to compensate debt holders (partially). Equity holders are thus residual claimants in the firm and hold a *call option* on the asset value of the firm. Hence, option pricing can be used to calculate the risk-neutral default probability in period t for a horizon of T years as

$$p_t = N \left[- \frac{\ln \frac{V_t}{D} + \left(r - \frac{\sigma_A^2}{2} \right) T}{\sigma_A \sqrt{T}} \right]$$

where N is the cumulative normal distribution, V_t is the value of assets in period t , r is the risk-free rate, and σ_A is the asset volatility.

The risk-neutral default probability then could be converted to a real-world default probability by using a number of techniques. For instance, Moody's KMV maps the numerator

of the previous equation, the *distance to default*, to default probabilities using historical data on defaults for a large panel of corporates. Alternatively, the risk-neutral default probability can be corrected for the risk premium obtained from an equity pricing model (Crouhy, Mark, and Galai, 2000). Empirical results by Moody's KMV have shown that the distance to default does a good job in predicting corporate defaults. Furthermore, work by Chan-Lau, Jobert, and Kong (2004) and by Gropp, Vesala, and Vulpes (2006) shows that distance to default predicts banks' downgrades in developed and emerging market countries.

Once default risk estimates are obtained, it is possible to use them to assess losses under different scenarios or to evaluate the level of systemic risk under stressed scenarios. There are several ways to accomplish this. For instance, standard linear and nonlinear econometric techniques could be used to evaluate how default risk responds to economic fundamentals, which could provide useful input for macro stress tests, as illustrated in the subpart of this book on contingent claims analysis.

The following chapters provide an example on how equity-based indicators can be used to evaluate the resilience of the banking system to a financial crisis and an example of the design of regulatory capital requirements for too-connected-to-fail institutions. The first example by Chan-Lau (Chapter 17) analyzes the impact of the 2008 global financial crisis on the Chilean banking system. The second example, also by Chan-Lau (Chapter 18), introduces the concept of the incremental contribution to systemic risk as a basic building block for the design of regulatory capital requirements. Both studies build from estimates of default probabilities from Moody's KMV and use CoRisk analysis (see also Chan-Lau, 2009; and Chan-Lau and others, 2009), a variation of CoVaR (Adrian and Brunnermeier, 2008), to evaluate default risk spillovers across financial institutions,

though the analyses can easily be extended using network analysis.

REFERENCES

- Adrian, Tobias, and Markus K. Brunnermeier, 2008, "CoVaR," FRBNY Staff Report No. 348 (New York: Federal Reserve Bank of New York). Available via the Internet: http://www.newyorkfed.org/research/staff_reports/sr348.html
- Altman, Edward I., 1968, "Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy," *Journal of Finance*, Vol. 23, No. 4, pp. 589–609.
- Black, Fischer, and Myron Scholes, 1973, "The Pricing of Options and Corporate Liabilities," *Journal of Political Economy*, Vol. 81, No. 3, pp. 637–54.
- Chan-Lau, Jorge A., 2009, "Default Risk Codependence in the Global Financial System: Was the Bear Stearns Bailout Justified?" in *The Banking Crisis Handbook*, edited by Greg N. Gregoriou (Boca Raton, London, New York: CRC Press).
- Chan-Lau, Jorge A., Marco Espinosa, Kay Giesecke, and Juan Solé, 2009, "Assessing the Systemic Implications of Financial Linkages," *Global Financial Stability Report*, Chapter 2 (International Monetary Fund, April). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2009/01/index.htm>
- Chan-Lau, Jorge A., Arnaud Jobert, and Janet Q. Kong, 2004, "An Option-Based Approach to Bank Vulnerabilities in Emerging Markets," IMF Working Paper 04/33 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=17059>
- Crouhy, Michel, Robert Mark, and Dan Galai, 2000, *Risk Management* (New York: McGraw-Hill, 1st ed.).
- Gropp, Reint, Jukka Vesala, and Giuseppe Vulpes, 2006, "Equity and Bond Market Signals as Leading Indicators of Bank Fragility," *Journal of Money, Credit and Banking*, Vol. 38, No. 2, pp. 399–428.
- Merton, Robert C., 1974, "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates," *Journal of Finance*, Vol. 29, No. 2, pp. 449–70.

The Global Financial Crisis and Its Impact on the Chilean Banking System

JORGE A. CHAN-LAU

This chapter explores how the global turmoil affected the risk of banks operating in Chile and provides evidence that could help strengthen work on vulnerability indicators and off-site supervision. In addition, it provides a background framework to assess default risk codependence, or CoRisk, between Chilean banks and global financial institutions; and assesses government measures aimed at reducing systemic risk in the domestic banking sector and the recommendations to allocate sovereign wealth fund assets to domestic banks.

METHOD SUMMARY

Overview	CoRisk analysis applies quantile regressions to determine how the default risk of an institution/sector/country is affected by the default risk of another, after controlling for common sources of risk.
Application	The method is used to measure interconnectedness risk across institutions/sectors/countries.
Nature of approach	Econometric analysis.
Data requirements	Security prices, risk measures.
Strengths	The method captures both direct and indirect sources of interconnectedness; it allows for the inclusion of information from other explanatory variables other than security prices and risk measures.
Weaknesses	The method has the same weaknesses as any econometric model, including model misspecification; statistical causality may not correspond to economic causality.
Tool	Standard econometrics package.

Financial systems in Latin America have not been immune to the ongoing financial turmoil. The situation has been brought about by the large presence of foreign-owned institutions and substantial cross-border claims. In particular, the crisis has highlighted the significant degree of interconnectedness and risk codependence among financial institutions, a development brought about by direct sources of exposure, such as interbank borrowing, and indirect sources of exposure, such as those related to common business practices like wholesale funding.

Chile, as other open emerging market economies with highly integrated financial systems and capital markets, has been affected by developments in global financial markets (Banco Central de Chile, 2009). Although the financial system has remained resilient to the significant shocks experienced since September 2008, the Chilean authorities took several measures to minimize domestic disruptions and preserve stable conditions in the domestic financial system. These measures included the flexibilization of reserve requirements, swap lines, as well as government auctions of foreign

This chapter was previously published as IMF Working Paper 10/108 (Chan-Lau, 2010). It has benefited from detailed comments by and discussions with Rodrigo Alfaro, Pedro de Beltran, Martin Cerisola, Daniel Oda, and Robert Rennhack as well as seminar participants at the Central Bank of Chile and the IMF.

currency denominated deposits for domestic banks. In addition, the Advisory Committee on Sovereign Wealth Funds has put forward a recommendation that domestic banks be eligible institutions for sovereign wealth fund (SWF) deposits, reflecting in part heightened risks in foreign financial institutions.

The analysis in this chapter explores how the global turmoil affected the risk of banks operating in Chile and provides evidence that could help strengthen work on vulnerability indicators and off-site supervision. In addition, it provides a background framework to assess government measures aimed at reducing systemic risk and default codependence (or CoRisk) as well as the recommendations to allocate SWF assets to domestic banks.

1. A FIRST GLIMPSE ON THE POTENTIAL EXPOSURE TO FOREIGN BANKS

Foreign bank claims on the domestic economy are one important source of interconnectedness and direct exposure to shocks affecting the global banking system. Bank for International Settlements (BIS, 2009) banking data show that foreign bank claims on Chile, measured as percent of the recipient country's GDP, are higher than those on other emerging market countries in Latin America and elsewhere, reflecting the openness of the country's financial system (Table 17.1). The exception are Eastern European countries, where foreign bank claims are substantial following the wave of acquisitions by Western European banks, especially those based in Austria and Germany.

Most of the claims on Chile are held by Spanish banks (Table 17.2). The dominant presence of Spanish banks is explained by the important role played by Banco Bilbao Vizcaya Argentaria (BBVA) and Banco Santander in the Chilean banking system. Between them, the two institutions account for 33 percent of the assets in the banking system, 24 percent of the nonderivatives financial instruments, and 40 percent of the notional outstanding amount of gross deriva-

TABLE 17.1

	Cross-Border ²		Consolidated ³	
	2007	2008	2007	2008
Czech Republic	24.5	22.9	89.6	81.1
Hungary	51.1	60.6	84.9	99.6
Poland	22.6	22.4	49.7	52.6
Chile	18.3	21.0	42.3	45.7
Brazil	11.2	9.7	20.2	17.1
Colombia	9.3	5.2	14.1	10.5
Mexico	10.2	9.8	34.7	29.1
South Korea	19.6	22.2	32.6	35.7
South Africa	12.7	12.5	39.1	35.1

Sources: Bank for International Settlements (BIS); and IMF staff calculations.

1. Calculated using end-2008 GDP figures.

2. Based on external asset positions, as of end-December 2008.

3. Based on consolidated claims on an immediate borrower basis, as of end-December 2008.

TABLE 17.2

Foreign Banks' Consolidated Claims on Chile by Nationality on an Immediate Borrower Basis			
	2002	2007	2008
Spain	48.0	56.2	52.0
United States	17.4	10.7	6.8
Germany	9.6	6.2	7.8
Netherlands	4.2	5.2	6.1
United Kingdom	3.8	0.0	0.0
Other	17.1	22.1	27.3
Total	100.0	100.0	100.0

Sources: Author; and Bank for International Settlements.

tives positions. Furthermore, because of the scale of its operations, measured in terms of assets, Banco Santander is required to hold regulatory capital in excess of 11 percent compared with 8 percent for other banks.

Spanish and U.S. banks have been losing market share to other European banks, and unallocated claims have become more important. U.S. bank lending has further declined because of the subprime crisis (Figure 17.1). Unallocated claims, as share of total claims, are up to 13 percent at end-2008 from 8.9 percent at end-2007. Assessing the risks from these claims is difficult because their geographic origin is not known.

Cross-country consolidated claims offer only a partial view on potential cross-border vulnerabilities in a country's banking system. In the specific case of Chile, they suggest that the domestic economy and banking system are vulnerable to adverse shocks to the Spanish banking system. Conversely, claims data suggest that Chile should be relatively insulated from adverse shocks to the British and Canadian banking systems, because cross-border claims ceased to exist from early 2003.

The claims data, however, may not be sufficient to identify risks associated with "second-round effects." In particular, Spanish banks are highly exposed to developments in the British economy because one-third of their claims are on the United Kingdom. Similarly, British banks hold one-tenth of their claims on Spain. The cross-border claims between Spain and the United Kingdom create exposures for the Chilean economy, as shocks in the United Kingdom will be transmitted via Spanish banks. Similarly, negative shocks to the Chilean economy can be transmitted to the United Kingdom with Spanish banks serving as conduits. The next section explains how to capture these second-round effects using market-based information.

2. ASSESSING THE IMPACT OF THE CRISIS ON CHILEAN BANKS USING MARKET INFORMATION: RISK MEASURES, DATA, AND EMPIRICAL METHOD

The assessment of the exposure of banks operating in any given jurisdiction is not a straightforward exercise owing to data limitations. Institutional data on cross-market, cross-

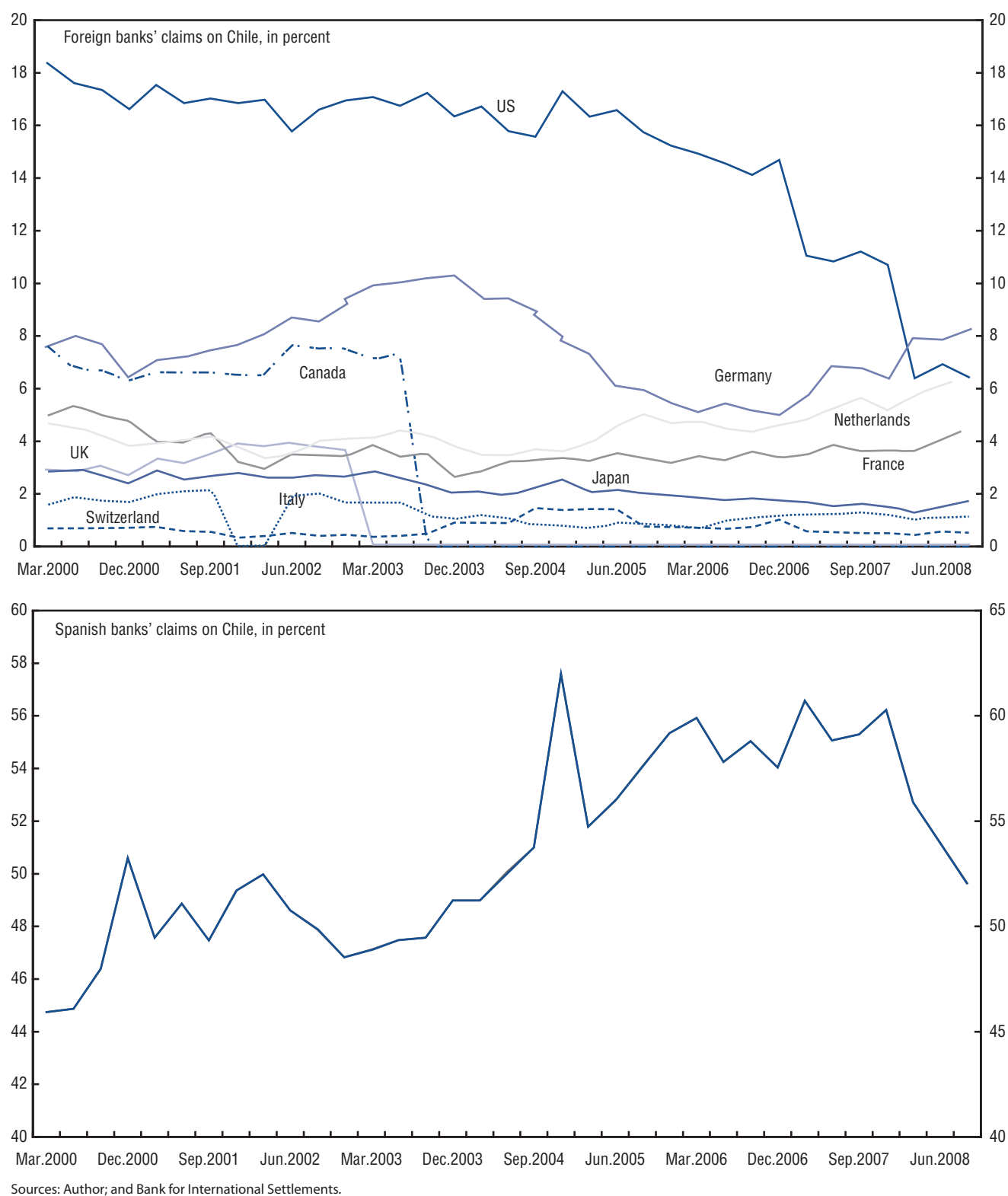


Figure 17.1 Chile: Foreign Bank Claims, in Percent of Total

currency, and cross-country linkages are, at best, not readily available. As the previous section discussed, aggregated consolidated claims data provide only a snapshot into potential cross-country exposures. In addition, the recent crisis has

shown that linkages are not restricted exclusively to direct exposures but can arise from indirect channels. One example of such a channel is the homogeneity of banks' balance sheets that make them vulnerable to a marked-to-market shock. In

the case of an open economy like Chile, where foreign banks play an important role in intermediating funds, indirect exposures may be more closely associated with the linkages of parent banks with other global financial institutions or the use of similar investment and lending strategies in domestic markets.¹

Information from security prices helps to deal with data limitations and imperfect knowledge about exposures across financial institutions. When market prices are available, it is possible to use different econometric and empirical methods to measure the CoRisk, or risk codependence across financial institutions, while accounting for potential nonlinear effects. This section starts by describing the market-based risk measure used in this study, describes the data used in the analysis, and concludes by explaining how to construct the CoRisk measures using quantile regressions.²

A. Expected default frequency measures

The market-based risk measure used here is the expected default frequency (EDF), a measure reported by Moody's KMV (MKMV) for a worldwide sample of banks and financial institutions. Although there are several risk measures based on market information, such as bond or credit default swap (CDS) spreads, EDFs offer some advantages over them. Foremost among them is that EDFs are objective or real-world default probabilities. In contrast, default probabilities measures extracted from bond or CDS spreads are risk neutral and tend to overstate their real-world counterparts because of the presence of a default risk premium.³ EDFs also combine both market information from equity prices and the nonmarket information from the liability side of the balance sheet of the firm. Finally, empirical studies have shown that EDF-like measures can explain default risk in equity returns and forecast financial institutions' failures.⁴

Conceptually, the EDF is based on the distance to default.⁵ The distance-to-default measure is built on the insight that the default of a firm occurs when the asset value of the firm is less than what the firm owes to its debtors. What the firm owes is commonly referred to as the default barrier, and in practice, it is calculated as the face value of short-term liabilities plus half of the face value of long-term liabilities. The wider the gap between the asset value and the default barrier, the safer the firm is. Similarly, the less volatile the asset value is, the safer the firm is because the likelihood of

TABLE 17.3

EDFs and Equivalent Moody's Credit Ratings			
Moody's Rating	EDF	Moody's Rating	EDF
Aaa	0.02	Ba1	0.408
Aa1	0.032	Ba2	0.544
Aa2	0.04	Ba3	0.848
Aa3	0.056	B1	1.323
A1	0.08	B2	2.064
A2	0.114	B3	4.168
A3	0.144	Caa1	8.418
Baa1	0.182	Caa2	17
Baa2	0.23	Caa3	17.946
Baa3	0.307	Ca,C	20

Source: Moody's KMV.

Note: The table shows the equivalence between five-year EDFs, in percent, and Moody's credit ratings scale. EDF = expected default frequency.

hitting the default barrier diminishes. The distance to default, therefore, can be expressed simply as

$$\text{Distance to default} = \frac{\text{Market value of assets} - \text{Default barrier}}{\text{Asset volatility}}. \quad (17.1)$$

The distance-to-default measure can be constructed using information from equity prices and balance sheet information. This information in turn helps to determine the market value of assets and the asset volatility of a firm. For a given firm, MKMV obtains the firm's EDF from its distance-to-default measure from the empirical calibration of various levels of distance to default to actual default probabilities based on a proprietary historical database. EDFs, therefore, are equivalent to objective default probabilities and can be associated with credit ratings, as shown in Table 17.3. The mapping between EDFs and Moody's ratings facilitates comparison with institutions rated by rating agencies but not yet included in the MKMV database.

B. Data

This study uses EDFs to measure the CoRisk induced by Latin American and global financial institutions on their Chilean counterparts. Table 17.4 shows the financial institutions included in the analysis. In addition to financial institutions operating in Chile, the sample includes financial institutions in Brazil, Colombia, and Peru; and major global banks in Canada, Europe, and the United States.⁶ For each of these institutions, weekly five-year EDF series were constructed from daily data for the period May 2, 2003 to February 27, 2009.⁷

¹ See Lagunoff and Schreft (2001), Cifuentes, Shin and Ferrucci (2005), and Chan-Lau (2008) among others.

² In addition to quantile regressions, as done in this study, it is possible to use extreme value theory and multivariate generalized autoregressive conditional heteroskedasticity (GARCH) and regime switching models.

³ Berndt and others (2005).

⁴ See Vassalou and Xing (2004) on default risk and equity returns; and Chan-Lau, Jobert, and Kong (2004) and Gropp, Vesala, and Vulpes (2006) on using the distance-to-default to forecast bank failures in advanced and emerging market economies respectively.

⁵ Crosbie and Bohn (2003).

⁶ Although it is not correct, institutions operating in a given country will be referred to interchangeably as "the country" or the "country's institutions" for simplicity.

⁷ Five-year EDFs correspond to the probability that the institution defaults sometime over a five-year horizon. The choice of the five-year horizon would facilitate contrasting the results presented herein with studies that use five-year CDS spreads, the latter being the most liquid CDS traded maturity in the market.

TABLE 17.4

Financial Institutions Included in the Analysis

<i>Chilean institutions</i>	<i>Dutch institutions</i>	<i>Spanish institutions</i>
Scotia Bank Chile	ABN Amro	BBVA
CorpBanca	ING Grope	Banco Santander
Banco de Credito e Inversiones		
Banco de Chile	<i>French institutions</i>	<i>Swiss institutions</i>
Banco BBVA Chile	Banque Nationale Paribas	Credit Suisse
	Credit Agricole	UBS
	Societe Generale	
<i>Brazilian institutions</i>	<i>German institutions</i>	<i>U.K. institutions</i>
Banco Bradesco	Commerzbank	HSBC
Banco Itau	Deutsche Bank	Lloyds
		Royal Bank of Scotland
<i>Colombian institutions</i>	<i>Peruvian institutions</i>	Standard Chartered
Banco BBVA Colombia	Banco Continental	
Banco de Bogota	Banco de Credito	<i>U.S. institutions</i>
BanColombia	ScotiaBank Peru	Bank of America
Banco de Occidente		Morgan Stanley
Banco Santander Colombia		Goldman Sachs
Corporacion Financiera		Citigroup
Grupo Aval	<i>Italian institutions</i>	Wells Fargo
	Banca Intesa	JP Morgan
<i>Canadian institutions</i>	MedioBanca	Bear Stearns
Bank of Nova Scotia	Unicredito	Lehman Brothers
CIBC		Merrill Lynch
Royal Bank of Canada		Wachovia

Source: Author.

The analysis includes a subset of Chilean institutions that represent a large share of the systemic core of the banking system. Only 6 Chilean institutions out of a total of 25 banks reporting to the Banking Supervisory Agency/Superintendencia de Bancos e Instituciones Financieras (SBIF) are included. But as of end-January 2009, the Chilean institutions analyzed accounted for 70 percent of the assets in the banking system, 56 percent of nonderivatives financial instruments positions, and 65 percent of the gross derivatives positions as measured by notional outstanding amounts (SBIF, 2009a, 2009b).⁸

Default risk increased rapidly because of problems in the U.S. banking system. Figure 17.2 shows how default risk in each country, measured as the average five-year EDF for the country's banks, has evolved since early 2006.⁹ In general, EDFs were rather compressed until early 2008 but started widening in 2008 following the failure of Bear Stearns (March 2008), the bankruptcy of Lehman Brothers, and the bailout of American International Group in September 2008. The data seem to show that the crisis originated in the United States, where EDFs widened earlier than in other countries with the exception of Germany. Note that the German banks included in the sample, as

the U.S. institutions, were reportedly important players in the structured credit market. Based on Table 17.3, at end-February 2009 the average implied rating of U.S. and German institutions was B3 but inching dangerously downward to Caa1.

Spanish institutions, among European ones, were among the least affected by the crisis. Arguably more conservative lending practices vis-à-vis other European banks, limited or no exposure to structured credit products referred to as "toxic assets" nowadays, and the use of regulatory dynamic provision requirements contributed to partly insulate Spanish banks from the stresses experienced by financial institutions in other countries.

The evolution of default risk in Latin America, where Spanish institutions have a substantial footprint, is rather similar to that of Spain. The average EDF remained in the range of 1 to 1½ percent by end-February 2009. Overall, banks in Latin America have little or no exposure to subprime assets and have small trading books relative to their banking book, which helps to insulate them from market shocks. Among Latin American countries, Chile has been the least affected by the global turmoil. The next section formally analyzes the comovements between individual institutions.

⁸ Calculated as the ratio of the sum of derivatives assets and liabilities positions for the banks in the sample to the corresponding total sum for the banking system.

⁹ Some caution is needed when interpreting the figure because the analysis includes only two institutions for some countries.

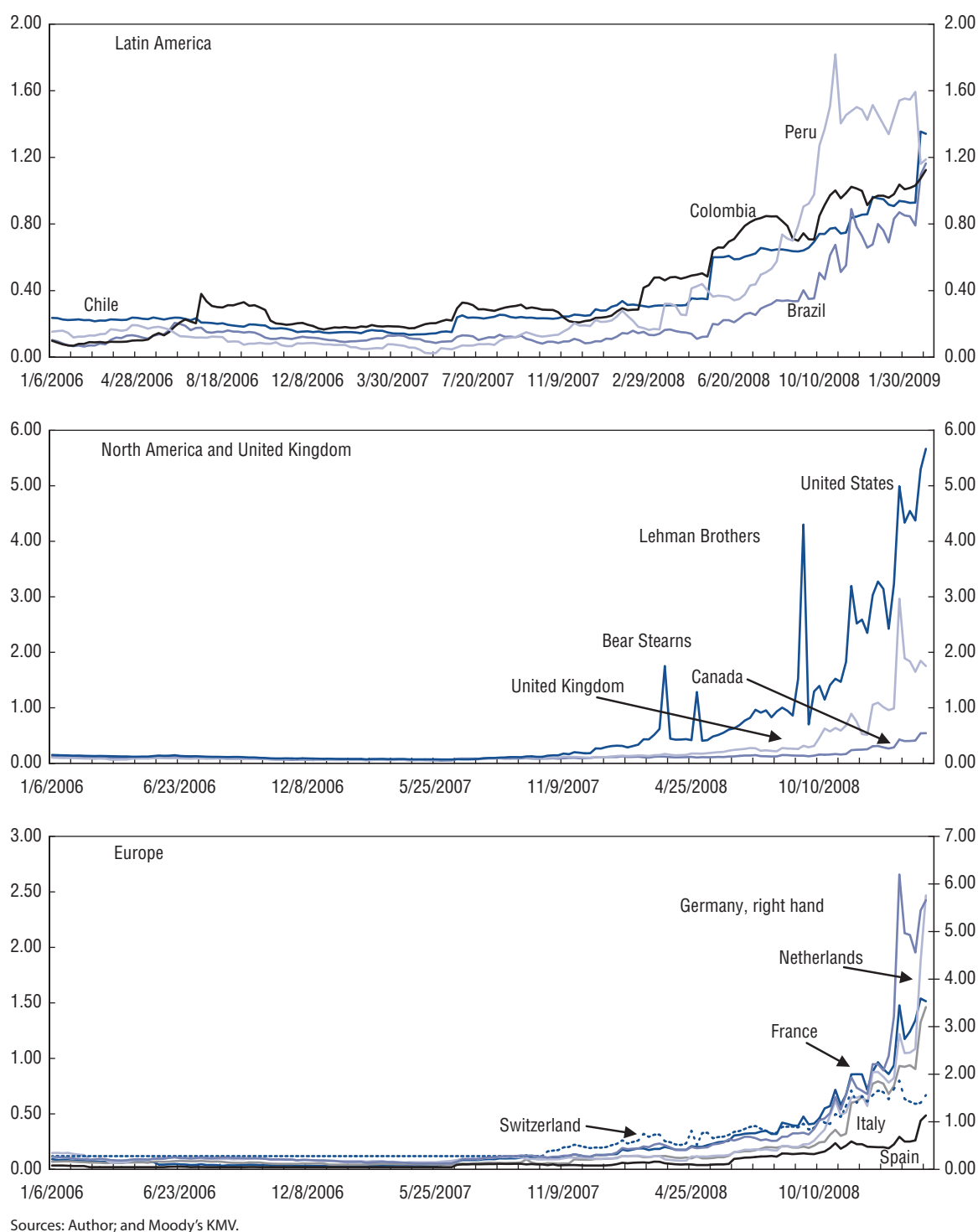


Figure 17.2 Five-Year EDFs, in Percent

C. Empirical method: CoRisk estimation using quantile regressions

CoRisk, or risk codependence, can be defined as the increase in the risk of one institution conditional on the risk of a peer institution. Although there is no unique or generally accepted method to measure CoRisk, this study uses CoRisk

analysis, a method based on quantile regressions motivated by the insights of Adrian and Brunnermeier (2008) and introduced in Chan-Lau (2009).¹⁰

¹⁰ See Chan-Lau and others (2009) for further specialization of this framework to the analysis of systemic risk and Chan-Lau (2013) for a comprehensive description.

The basic question to answer is how the default risk of an institution is affected by the default risk of another institution, after controlling for common sources of risk. In statistical terms, the goal is to learn $f(y|x, \theta)$, the conditional distribution of the default risk of institution y , given common drivers of default risk and the default risk of the other institution, which are denoted by x , where θ is a set of parameters that needs to be inferred from observed realizations of x . Ordinary least squares (OLS) is a useful technique to extract this information. However, OLS can provide information only about the *mean* relationship across institutions' default risk. Because this relationship is nonlinear, OLS may have some serious limitations.

Quantile regression is an alternative to the use of nonlinear models that can capture some of the nonlinearities of the relationship across institutions' default risk. Quantile regression, first introduced by Koenker and Bassett (1978), extends the OLS intuition beyond the estimation of the mean of the conditional distribution $f(y|x, \theta)$. It allows the researcher to "slice" the conditional distribution at the quantile (percentile) of interest, τ , and obtain the corresponding cross-section of the conditional distribution $f_\tau(y|x, \theta)$.

Quantile regression makes it possible to evaluate the response of the independent variable on particular segments of the conditional distribution. For instance, when a linear model is used to analyze systemic interlinkages, it is expected that the coefficients of the regressors change with the level of risk. In terms of model estimation, quantile regression does not consist of estimating a number of separate OLS regressions in nonoverlapping samples after sorting the data by quantiles (or percentiles), though the use of the term "quantile" may suggest so. Using nonoverlapping samples could introduce a small sample problem when dealing with lower and/or higher quantiles of the data, a problem often found when analyzing data with extreme value theory techniques.

Quantile regressions make the best use of all data available in the sample by weighting each available observation. In a quantile regression, the parameters are obtained by solving an optimization program that uses all the information contained in the data sample. The parameters are obtained from the minimization of the sum of residuals, $y - x\beta$, where the latter are weighted by a check function, ρ_τ , that depends on the quantile of interest, τ :

$$\min_{\beta} \sum_i^N \rho_\tau(y_i - \xi(x_i, \beta)), \quad (17.2)$$

where y is the dependent variable, $\xi(x, \beta)$ is a linear function of the parameters β and the exogenous variables x , and $\rho_\tau(\cdot)$ is a weighting function for each observation. More specifically, the function assigns a weight equal to the quantile τ if the residual is positive or a weight equal to $\tau - 1$ if the residual is negative. The minimization can be solved using standard linear programming methods. The covariance matrices are usually estimated using bootstrap techniques and remain

valid even if the residuals and explanatory variables are not independent (Koenker, 2005).

For analyzing CoRisk between Chilean and global banks, the following equation was estimated for τ set equal to the 95th quantile:

$$EDF_i = \alpha_\tau + \sum_k^K \beta_{\tau, i, k} R_k + \beta_{\tau, j} \text{Clean } EDF_j + \varepsilon_i, \quad (17.3)$$

where EDF_i is the EDF of institution i , R_k denotes the k th common aggregate risk factor, and $\text{Clean } EDF_j$ is the component of EDF_j that is orthogonal to the common aggregate risk factors R_k s, which is referred to here onward as the orthogonal EDF component. By using the orthogonal component, equation (17.3) isolates the idiosyncratic effect of institution j on institution i . The fitted values using equation (17.3) will be referred to as CoRisk EDF.

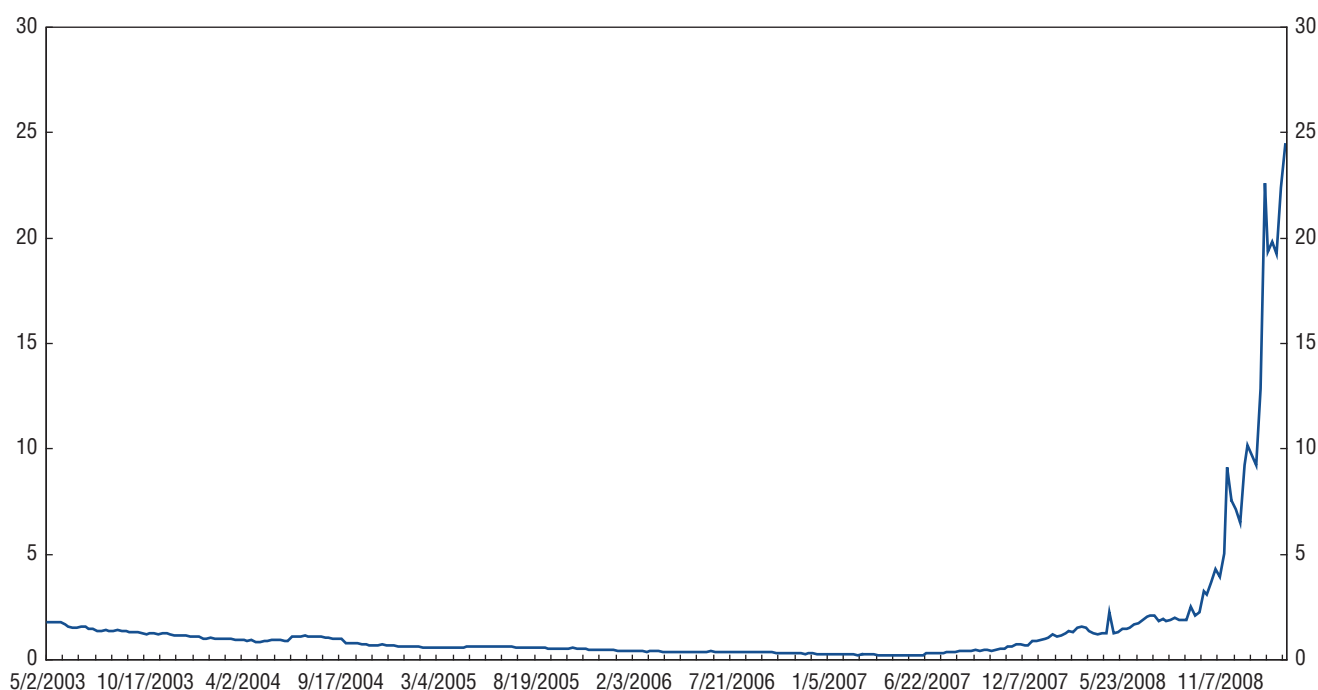
Economic theory can be used to guide the choice of aggregate risk factors. Usually, the common risk factors include variables such as the slope of the term structure of interest rates and the implied volatility index (VIX) as a proxy for investor sentiment. The aggregate risk factor in this study, however, was constructed by extracting the main principal components corresponding to the EDFs of all institutions in the sample excluding the Chilean institutions.¹¹ Only the first principal component was included as an aggregate risk factor because it accounted for close to 95 percent of the total variability in the data. Furthermore, as Figure 17.3 illustrates, the first principal component can be roughly identified with default risk in the global banking system.

3. RESULTS

Banks in Chile have been affected mainly by aggregate risk in the global financial system and to a lesser extent by idiosyncratic shocks affecting regional and international banks. The impact of changes in aggregate risk can be approximated roughly by the difference between the median EDF and the unconditional EDF measured at the 95th percentile (Table 17.5). For Chilean banks, the unconditional EDF is two to three times higher than the median EDF. The impact of idiosyncratic shocks can be gauged from the difference between the conditional EDF, or CoRisk EDF, obtained from equation (17.3) and the unconditional EDF. The median CoRisk EDF exceeds the unconditional 95th percentile EDF by 15 to 100 percent, depending on the institution analyzed.

Bank vulnerability to adverse idiosyncratic shocks affecting other banks seems related to leverage, external debt and/or obligations, and the strength of the parent institution in the case of foreign-owned banks. Higher unconditional and CoRisk EDFs are associated with highly levered banks and those with high external debt ratios, where the latter are

¹¹ Principal component analysis is a technique widely used to construct factors. For details, see, for instance, Timm (2002).



Sources: Author; and Moody's KMV.

Note: The aggregate risk factor corresponds to the first principal component extracted from the EDFs of all institutions in the sample excluding the Chilean institutions. EDF = expected default frequency.

Figure 17.3 Aggregate Risk Factor

measured as debt owed to foreign banks. This is the case, for instance, for Banco de Crédito e Inversiones. Among foreign-owned banks, the percent difference between the CoRisk EDF and the unconditional EDF risks are higher for Bank 1 and Bank 5, owned by Bank of Nova Scotia and Citibank, respectively, than for the Spanish-owned institutions. When the CoRisk EDF is measured as percent changes vis-à-vis the unconditional EDF, ScotiaBank and Banco de Chile are the two institutions most affected by institution-specific shocks.

Changes in implied ratings also highlight the relative importance of aggregate shocks vis-à-vis bank-idiosyncratic shocks. Table 17.6 shows the Moody's five-year credit ratings implied by the CoRisk and unconditional EDFs according to the mapping reported in Table 17.3.¹² Compared with the median rating, the unconditional 95th percentile EDF implies a downgrade of three to four notches, which can be attributed to the aggregate shock. In contrast, idiosyncratic shocks to foreign institutions induce, on average, at most one conditional rating downgrade on Chilean institutions from the rating implied by its unconditional 95th percentile EDF.

Santander appears more resilient to external shocks affecting the parent bank and its subsidiaries in the region

than BBVA. The rating of Santander Chile remains unchanged while BBVA experiences a one rating downgrade. Furthermore, Santander Chile ratings improve when BBVA and its subsidiaries are adversely affected by shocks. Arguably, this result suggests that Santander Chile gains market share at the expense of BBVA.

There are second-round effects on the Chilean banking system even in the absence of reported cross-border banking claims. For instance, since 2003 there have been no cross-border claims between Chile and the United Kingdom. Shocks affecting British banks, however, cause a one rating conditional downgrade in Chilean banks. This is also true, to a lesser extent, in the case of Canadian banks. Put together, these results suggest how information on direct exposures, such as consolidated claims, and market-based information, such as EDFs, complement each other and are useful for assessing risks in the financial sector.

Although Chilean banks are vulnerable to aggregate financial shocks, several factors may have contributed to make them relatively resilient to institution-specific shocks. First, reliance on external financing sources is limited because the domestic banking system funds its operations mainly through domestic deposits (60 percent of assets) and by issuing domestic securities (13 percent of assets) while external funding is small (5 percent of assets).¹³

¹² The reader should keep in mind that the ratings movements are those implied by the changes in the EDFs (or probabilities of default). Therefore, the analysis does not refer to actual upgrades or downgrades by credit rating agencies.

¹³ Figures in this section are estimated using publicly available data from the SBIF.

TABLE 17.5

CoRisk between Financial Institutions Abroad and Those Operating in Chile, Measured as Expected Default Frequency (in percent)

	Bank 1	Bank 2	Bank 3	Bank 4	Bank 5	Bank 6
Median EDF	0.71	0.12	0.08	0.20	0.12	0.32
95th percentile EDF	2.68	0.24	0.24	0.50	0.23	0.76
<i>Latin American institutions</i>						
Banco Bradesco, Brazil	5.94	0.27	0.31	0.71	0.30	0.91
Banco Itau, Brazil	5.75	0.27	0.30	0.68	0.10	0.88
Banco BBVA, Colombia	3.02	0.26	0.23	0.60	0.36	0.86
Banco de Bogotá, Colombia	5.87	0.32	0.30	0.70	0.33	0.83
Banco de Occidente, Colombia	3.51	0.30	0.27	0.59	0.39	0.85
Banco Santander, Colombia	3.84	0.31	0.27	0.55	0.46	0.73
BanColombia, Colombia	5.07	0.27	0.24	0.67	0.36	0.83
Corporacion Financiera, Colombia	5.86	0.27	0.32	0.67	0.35	0.92
Grupo Aval, Colombia	5.52	0.34	0.34	0.70	0.41	0.85
Banco Continental, Perú	5.29	0.32	0.31	0.68	0.31	0.89
Banco de Credito, Perú	5.30	0.31	0.25	0.68	0.37	0.90
Scotiabank, Peru	5.98	0.37	0.31	0.72	0.20	0.69
Minimum	3.02	0.26	0.23	0.55	0.20	0.69
Median	5.41	0.31	0.30	0.68	0.36	0.85
Maximum	5.98	0.37	0.34	0.72	0.46	0.92
<i>U.S. institutions</i>						
Bank of America	2.89	0.43	0.31	0.73	0.58	0.62
Morgan Stanley	4.02	0.31	0.31	0.68	0.33	0.90
Goldman Sachs	5.52	0.32	0.29	0.66	0.37	0.91
Citigroup	6.50	0.44	0.46	0.94	0.56	0.87
Wells Fargo	3.70	0.32	0.25	0.54	0.35	0.79
Bear Stearns	5.42	0.26	0.24	0.68	0.37	0.80
Lehman Brothers	6.14	0.43	0.42	0.76	0.45	1.01
Merrill Lynch	6.25	0.37	0.39	0.79	0.51	0.92
Wachovia	4.52	0.25	0.17	0.65	0.38	0.85
JP Morgan	5.44	0.32	0.34	0.69	0.35	0.88
Minimum	2.89	0.25	0.17	0.54	0.33	0.62
Median	5.43	0.32	0.31	0.69	0.38	0.88
Maximum	6.50	0.44	0.46	0.94	0.58	1.01
<i>Canadian institutions</i>						
Bank of Nova Scotia	3.44	0.34	0.29	0.54	0.31	0.80
CIB	3.33	0.40	0.35	0.55	0.37	0.84
Royal Bank of Canada	5.58	0.26	0.24	0.68	0.38	0.78
Minimum	3.33	0.26	0.24	0.54	0.31	0.78
Median	3.44	0.34	0.29	0.55	0.37	0.80
Maximum	5.58	0.40	0.35	0.68	0.38	0.84
<i>European institutions</i>						
BBVA, Spain	4.44	0.28	0.20	0.52	0.39	0.88
Banco Santander, Spain	4.25	0.29	0.28	0.56	0.37	0.92
Banque Nationale Paribas, France	3.82	0.33	0.29	0.59	0.30	0.89
Credit Agricole, France	5.73	0.28	0.33	0.69	0.39	0.89
Societe Generale, France	5.55	0.27	0.30	0.70	0.33	0.89
Commerzbank, Germany	6.08	0.27	0.33	0.72	0.30	0.91
Deutsche Bank, Germany	5.96	0.32	0.31	0.72	0.31	0.92
Banca Intesa, Italy	5.90	0.29	0.31	0.70	0.26	0.87
Mediobanca, Italy	5.99	0.29	0.32	0.70	0.30	0.90
Unicredit, Italy	5.54	0.30	0.31	0.70	0.42	0.87
Credit Suisse, Switzerland	5.51	0.29	0.19	0.60	0.19	0.84
UBS, Switzerland	4.49	0.32	0.27	0.71	0.35	0.84
Barclays, United Kingdom	3.73	0.31	0.28	0.63	0.38	0.84
HSBC, United Kingdom	4.29	0.31	0.28	0.62	0.37	0.86
Lloyds, United Kingdom	5.51	0.26	0.26	0.58	0.29	0.91
Royal Bank of Scotland, United Kingdom	4.41	0.36	0.31	0.62	0.36	0.84
Standard Chartered, United Kingdom	3.71	0.26	0.20	0.68	0.37	0.80
ABN Amro, Netherlands	4.24	0.27	0.20	0.68	0.37	0.87
ING, Netherlands	3.49	0.31	0.27	0.60	0.32	0.88
Minimum	3.49	0.26	0.19	0.52	0.19	0.80
Median	4.49	0.29	0.28	0.68	0.35	0.88
Maximum	6.08	0.36	0.33	0.72	0.42	0.92

Sources: Author; and Moody's KMV.

TABLE 17.6

CoRisk between Financial Institutions Abroad and Those Operating in Chile Measured as Moody's Credit Ratings

	Bank 1	Bank 2	Bank 3	Bank 4	Bank 5	Bank 6
Median rating	Ba2	A2	Aa3	Baa1	A2	Baa3
Unconditional rating, 95th percentile	B2	Baa2	Baa2	Ba1	Baa2	Ba2
<i>Latin American institutions</i>						
Banco Bradesco, Brazil	B3	Baa2	Baa3	Ba2	Baa2	Ba3
Banco Itau, Brazil	B3	Baa2	Baa2	Ba2	Baa3	Ba3
Banco BBVA, Colombia	B2	Baa2	Baa1	Ba2	Baa3	Ba3
Banco de Bogotá, Colombia	B3	Baa3	Baa2	Ba2	Baa3	Ba2
Banco de Occidente, Colombia	B2	Baa2	Baa2	Ba2	Baa3	Ba3
Banco Santander, Colombia	B2	Baa3	Baa2	Ba2	Ba1	Ba2
BanColombia, Colombia	B3	Baa2	Baa2	Ba2	Baa3	Ba2
Corporacion Financiera, Colombia	B3	Baa2	Baa3	Ba2	Baa3	Ba3
Grupo Aval, Colombia	B3	Baa3	Baa3	Ba2	Ba1	Ba3
Banco Continental, Perú	B3	Baa3	Baa3	Ba2	Baa3	Ba3
Banco de Credito, Perú	B3	Baa3	Baa2	Ba2	Baa3	Ba3
Scotiabank, Peru	B3	Baa3	Baa3	Ba2	Baa1	Ba2
Minimum	B2	Baa2	Baa1	Ba2	Baa1	Ba2
Median	B3	Baa2	Baa2	Ba2	Baa3	Ba3
Maximum	B3	Baa3	Baa3	Ba2	Ba1	Ba3
<i>U.S. institutions</i>						
Bank of America	B2	Ba1	Baa2	Ba2	Ba2	Ba2
Morgan Stanley	B2	Baa2	Baa2	Ba2	Baa3	Ba3
Goldman Sachs	B3	Baa3	Baa2	Ba2	Baa3	Ba3
Citigroup	B3	Ba1	Ba1	Ba3	Ba2	Ba3
Wells Fargo	B2	Baa3	Baa2	Ba2	Baa3	Ba2
Bear Stearns	B3	Baa2	Baa2	Ba2	Baa3	Ba2
Lehman Brothers	B3	Ba1	Ba1	Ba2	Ba1	Ba3
Merrill Lynch	B3	Baa3	Baa3	Ba2	Ba1	Ba3
Wachovia	B3	Baa2	A3	Ba2	Baa3	Ba3
JP Morgan	B3	Baa3	Baa3	Ba2	Baa3	Ba3
Minimum	B2	Baa2	A3	Ba2	Baa3	Ba2
Median	B3	Baa3	Baa2	Ba2	Baa3	Ba3
Maximum	B3	Ba1	Ba1	Ba3	Ba2	Ba3
<i>Canadian institutions</i>						
Bank of Nova Scotia	B2	Baa3	Baa2	Ba1	Baa2	Ba2
CIB	B2	Baa3	Baa3	Ba2	Baa3	Ba2
Royal Bank of Canada	B3	Baa2	Baa2	Ba2	Baa3	Ba2
Minimum	B2	Baa2	Baa2	Ba1	Baa2	Ba2
Median	B2	Baa3	Baa2	Ba2	Baa3	Ba2
Maximum	B3	Baa3	Baa3	Ba2	Baa3	Ba2
<i>European institutions</i>						
BBVA, Spain	B3	Baa2	Baa1	Ba1	Baa3	Ba3
Banco Santander, Spain	B3	Baa2	Baa2	Ba2	Baa3	Ba3
Banque Nationale Paribas, France	B2	Baa3	Baa2	Ba2	Baa2	Ba3
Credit Agricole, France	B3	Baa2	Baa3	Ba2	Baa3	Ba3
Societe Generale, France	B3	Baa2	Baa2	Ba2	Baa3	Ba3
Commerzbank, Germany	B3	Baa2	Baa3	Ba2	Baa2	Ba3
Deutsche Bank, Germany	B3	Baa3	Baa3	Ba2	Baa2	Ba3
Banca Intesa, Italy	B3	Baa2	Baa3	Ba2	Baa2	Ba3
Mediobanca, Italy	B3	Baa2	Baa3	Ba2	Baa2	Ba3
Unicredito, Italy	B3	Baa2	Baa3	Ba2	Ba1	Ba3
Credit Suisse, Switzerland	B3	Baa2	Baa1	Ba2	Baa1	Ba2
UBS, Switzerland	B3	Baa3	Baa2	Ba2	Baa3	Ba2
Barclays, United Kingdom	B2	Baa3	Baa2	Ba2	Baa3	Ba2
HSBC, United Kingdom	B3	Baa2	Baa2	Ba2	Baa3	Ba3
Lloyds, United Kingdom	B3	Baa2	Baa2	Ba2	Baa2	Ba3
Royal Bank of Scotland, United Kingdom	B3	Baa3	Baa3	Ba2	Baa3	Ba2
Standard Chartered, United Kingdom	B2	Baa2	Baa1	Ba2	Baa3	Ba2
ABN Amro, Netherlands	B3	Baa2	Baa1	Ba2	Baa3	Ba3
ING, Netherlands	B2	Baa3	Baa2	Ba2	Baa3	Ba3
Minimum	B2	Baa2	Baa1	Ba1	Baa1	Ba2
Median	B3	Baa2	Baa2	Ba2	Baa3	Ba3
Maximum	B3	Baa3	Baa3	Ba2	Ba1	Ba3

Sources: Author; and Moody's KMV.

Domestic funding, however, may be affected negatively going forward. Retail funding is being eroded by increased equity investment and the emergence of alternative investment vehicles targeted to retail customers, such as pension funds. Increased investment abroad by pension funds following the relaxation of foreign investment limits has gradually reduced the domestic investor base for bank securities (Saldías and others, 2008). These trends may foster increased external financing and, as the results show, lead to higher exposure to foreign banks' idiosyncratic shocks.

Second, market risk in the system is limited. The trading book, mostly in government securities, accounted for 4 to 5½ percent of assets, and securities available for sale for 7 to 8 percent of assets. Although derivatives are held mostly for trading purposes, the net open position in the system was at most 1½ percent of assets during the period. Furthermore, derivatives trading mainly involves trading forward and swaps, so notional amounts may certainly overstate losses due to market risk. A back-of-the-envelope calculation suggests that at most 13 percent of assets are affected directly by market risk.

Third, credit risk remains the main source of vulnerability in the banking system. Banks are somewhat insulated from problems affecting other banks because credit risk is driven predominantly by developments in the domestic

economy. According to aggregate balance sheet data, the banking book usually accounts for 67 percent of the assets in the banking system. Therefore, vulnerabilities are more likely to arise from the deterioration of creditworthiness in the corporate and household sectors.

Finally, counterparty risk within the domestic system appears limited and reduces exposures to international institutions through second-round-effects channels. Counterparty exposure can be estimated roughly as the sum of the trading exposure (4 to 5½ percent of assets), interbank lending (less than ½ percent of assets), and derivatives net open positions (1½ percent of assets). The reduced counterparty exposure translates into limited CoRisk exposure within Chilean banks, with banks experiencing at most a one rating downgrade conditional on other banks' increase in default risk (Table 17.7).

Spanish banks are the more resilient ones to risks from other Chilean banks. As noted by rating agencies repeatedly, a source of strength for subsidiaries of foreign banks operating in Chile is the implicit support from the parent bank. As a result, conditional EDFs for Santander Chile and BBVA remain mostly unchanged. In contrast, ScotiaBank is highly vulnerable partly because of the high exposure of its parent bank, Bank of Nova Scotia, to the global shocks. Finally, in the case of Santander Chile, the bank may not be affected as much by

TABLE 17.7

CoRisk between Chilean Banking Institutions Measured as Expected Default Frequency (EDF) and Implied Moody's Ratings

	Scotiabank	Corpbanca	Banco Santander	Banco de Credito e Inversiones	Banco de Chile	BBVA
Medium EDF	0.71	0.12	0.08	0.20	0.12	0.32
95th percentile EDF	2.68	0.24	0.24	0.50	0.23	0.76
Scotiabank	—	0.30	0.26	0.55	0.36	0.81
Corpbanca	5.41	—	0.26	0.60	0.32	0.90
Banco Santander	3.46	0.33	—	0.57	0.35	0.84
Banco de Credito e Inversiones	3.98	0.33	0.27	—	0.37	0.80
Banco de Chile	3.73	0.33	0.28	0.58	—	0.81
BBVA	4.66	0.32	0.31	0.62	0.38	—
Minimum	3.46	0.30	0.26	0.55	0.32	0.80
Median	3.98	0.33	0.27	0.58	0.36	0.81
Maximum	5.41	0.33	0.31	0.62	0.38	0.90
Median rating	Ba2	A2	Aa3	Baa1	A2	Baa3
Unconditional rating, 95th percentile	B2	Baa2	Baa2	Ba1	Baa2	Ba2
Scotiabank	—	Baa2	Baa2	Ba2	Baa3	Ba2
Corpbanca	B3	—	Baa2	Ba2	Baa3	Ba3
Banco Santander	B2	Baa3	—	Ba2	Baa3	Ba2
Banco de Credito e Inversiones	B2	Baa3	Baa2	—	Baa3	Ba2
Banco de Chile	B2	Baa3	Baa2	Ba2	—	Ba2
BBVA	B3	Baa3	Baa2	Ba2	Baa3	—
Minimum	B2	Baa2	Baa2	Ba2	Baa3	Ba2
Median	B2	Baa3	Baa2	Ba2	Baa3	Ba2
Maximum	B3	Baa3	Baa2	Ba2	Baa3	Ba3

Sources: Author; and Moody's KMV.

shocks to other Chilean banks because its loan portfolio is quite different from those of other banks and by being required to hold a larger capital requirement than other banks.¹⁴

A recent trend worth monitoring, however, is the sudden pickup in claims on foreign banks that started only in October 2008. Although still relatively small, at less than 1 percent of assets, they may reflect increased deposits with parent banks and suggest that subsidiaries could be propping up liquidity and/or capital in parent companies.

4. CONCLUSION

Nowadays, domestic banking systems are highly interconnected within the global financial system. The oft-quoted phrase from Donne (1624), “No man is an island, entire of itself . . . any man’s death diminishes me” rings true in light of the increased globalization and rapid pace of innovation experienced in the international financial system.

The CoRisk analysis in this chapter has shown that, even in the absence of direct exposures with other countries in the region, the domestic banking system in Chile is vulnerable to adverse idiosyncratic shocks affecting other banks in the region as well as in advanced economies. The empirical evidence on the interconnectedness between the domestic banking system and the global banking system makes it imperative to continue advancing the agenda on cross-border supervision and coordinated crisis management with countries in the region and advanced economies.

Notwithstanding the large presence of foreign-owned institutions and spillover risks, it is worth noting the resilience of the system to idiosyncratic shocks. The magnitude of the institution-specific spillovers is relatively constrained as adverse shocks abroad translate, at most, into an implied one rating downgrade for most institutions after accounting for aggregate shocks. Nevertheless, some caution is warranted because increased perceptions of risk can cause substantial increases in funding rates to domestic banks or lead to a loss of confidence by depositors.

Although a formal analysis was not conducted, measures enacted by the government may have contributed to offset the surge in risk in the banking system. These measures include the flexibilization of reserve requirements, swap lines, as well as government auctions of foreign currency denominated deposits for domestic banks. Indeed, as Figure 17.2 shows, the passage and implementation of the measures kept the average EDF in Chile mostly flat during most of the second half of 2008. In contrast, the average EDF in Brazil and Peru, which were below or in line with the EDF of Chile in the first half of 2008, widened rapidly in the second half of 2008. In this regard, the authorities should continue to monitor conditions in the market closely and to ascertain that government-provided liquidity remains onshore.

¹⁴ Pizarro (2009) reports that the loan portfolio of Santander Chile is tilted toward lower-income households and small to medium enterprises compared with other peer institutions.

The analysis provides some support to the recommendation by the Financial Advisory Committee to make domestic banks eligible for SWF deposits. Despite the spillover effects, the CoRisk analysis finds that Chilean institutions may be less vulnerable than banks abroad, especially those in advanced economies. Nevertheless, empirical analysis can offer only so much support, especially because it did not consider a number of relevant factors and alternatives. For instance, deposits with domestic banks may lead to “Dutch disease” problems. Also, the recommendation should be balanced against the alternative to invest in riskless assets, such as government bonds and bills. Finally, in the specific case of foreign-owned institutions, it is necessary to ensure that the domestic subsidiaries are effectively ring-fenced from weaker parent institutions to prevent the latter from draining resources from their subsidiaries.

Finally, going forward, risks in the domestic banking system may be more closely associated with credit risk in the household and corporate sectors rather than risk spillovers from the global financial system. Government support and intervention in the banking sector in advanced economies have helped stabilize the financial system and ease the liquidity problems experienced in the second half of 2008. Spillover risks from the financial system appear contained and have been reflected by a decline in the EDFs of Chilean banks from the high levels observed at end-February 2009, where the sample data end, to values closer to their median levels. This development suggests that in the short and medium term, bank vulnerabilities in Chile are more closely associated to stress scenarios that could affect the banks’ loan portfolio.

REFERENCES

- Adrian, Tobias, and Markus K. Brunnermeier, 2008, “CoVaR,” Staff Report No. 348 (New York: Federal Reserve Bank of New York). Available via the Internet: http://www.newyorkfed.org/research/staff_reports/sr348.html
- Banco Central de Chile, 2009, *Informe de Estabilidad Financiera*, Segundo Semestre 2008 (Santiago, January). Available via the Internet: <http://www.bcentral.cl/publicaciones/politicas/polit01.htm>
- Bank for International Settlements, 2009, *BIS Quarterly Review*, Statistical Annex (Basel, March). Available via the Internet: http://www.bis.org/publ/qtrpdf/r_qt0903.htm
- Berndt, Antje, Darrell Duffie, Rohan Douglas, Mark Ferguson, and David Schranz, 2005, “Measuring Default Risk Premia from Default Swap Rates and EDFs,” BIS Working Paper No. 173 (Basel: Bank for International Settlements). Available via the Internet: <http://www.bis.org/publ/work173.htm>
- Chan-Lau, Jorge A., 2008, “The Globalization of Finance and Its Implications for Financial Stability,” *International Journal of Banking, Accounting and Finance*, Vol. 1, No. 1, pp. 3–29.
- , 2009, “Default Risk Codependence in the Global Financial System: Was the Bear Stearns Bailout Justified?” in *The Banking Crisis Handbook*, edited by Greg Gregoriou (Boca Raton, London, New York: CRC Press).
- , 2010, “The Global Financial Crisis and Its Impact on the Chilean Banking System,” IMF Working Paper 10/108 (Wash-

- ington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=23781>
- , 2013, *Systemic Risk Assessment and Oversight* (London: Risk Books).
- Chan-Lau, Jorge A., Marco Espinosa, Kay Giesecke, and Juan Solé, 2009, “Assessing the Systemic Implications of Financial Linkages,” *Global Financial Stability Report*, Chapter 2 (International Monetary Fund, April). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2009/01/index.htm>
- Chan-Lau, Jorge A., Arnaud Jobert, and Janet Q. Kong, 2004, “An Option-Based Approach to Bank Vulnerabilities in Emerging Markets,” IMF Working Paper 04/33 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=17059>
- Cifuentes, Rodrigo, Hyun Shin, and Gianlugi Ferrucci, 2005, “Liquidity Risk and Contagion,” *Journal of the European Economic Association*, Vol. 3, Nos. 2–3, pp. 556–66.
- Crosbie, Peter J., and Jeffrey R. Bohn, 2003, “Modeling Default Risk,” Moody’s KMV (December 18). Available via the Internet: http://www.creditrisk.ru/publications/files_attached/modeling_default_risk.pdf
- Donne, John, 1624, *Devotions upon Emergent Occasions, Meditation XVII*.
- Gropp, Reint, Jukka Vesala, and Giuseppe Vulpes, 2006, “Equity and Bond Market Signals as Leading Indicators of Bank Fragility,” *Journal of Money, Credit and Banking*, Vol. 38, No. 2, pp. 399–428.
- Koenker, Roger, 2005, *Quantile Regression* (Cambridge: Cambridge University Press).
- Koenker, Roger, and Gilbert Bassett Jr., 1978, “Regression Quantiles,” *Econometrica*, Vol. 45, No. 1, pp. 33–50.
- Lagunoff, Roger, and Stacey Schreft, 2001, “A Model of Financial Fragility,” *Journal of Economic Theory*, Vol. 99, No. 1/2, pp. 220–64.
- Pizarro, Javier, 2009, *Sector Bancario: Actualización de Precios Objetivo y Proyecciones 2009–2010* (Santiago: Larraín Vial).
- Saldías, Carola, Eduardo Santibañez, Andres Martinez, Santiago Gallo, and Peter Shaw, 2008, “Chilean Banks—Annual Review and Outlook,” FitchRatings (New York, March 10).
- Superintendencia de Bancos e Instituciones Financieras, 2009a, *Reporte Mensual de Información Financiera* (Santiago, February). Available via the Internet: <http://www.sbif.cl/sbifweb/servlet/InfoFinanciera?indice=4.1&idCategoria=2151&tipocont=2359>
- , 2009b, *Reporte Mensual de Instrumentos Financieros No Derivados y Derivados* (Santiago, February). Available via the Internet: <http://www.sbif.cl/sbifweb/servlet/InfoFinanciera?indice=C.D.A&idContenido=10117>
- Timm, Neil, 2002, *Applied Multivariate Analysis* (New York: Springer Verlag).
- Vassalou, Maria, and Yuhang Xing, 2004, “Default Risk in Equity Returns,” *Journal of Finance*, Vol. 59, No. 2, pp. 831–68.

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Regulatory Capital Charges for Too-Connected-to-Fail Institutions: A Practical Proposal

JORGE A. CHAN-LAU

The recent financial crisis has highlighted once more that interconnectedness in the financial system is a major source of systemic risk. I suggest a practical way to levy regulatory capital charges based on the degree of interconnectedness among financial institutions. Namely, the charges are based on the institution's incremental contribution to systemic risk using a risk budgeting approach. The imposition of such capital charges could go a long way toward internalizing the negative externalities associated with too-connected-to-fail institutions and providing managerial incentives to strengthen an institution's solvency position, and avoid too much homogeneity and excessive reliance on the same counterparties in the financial industry.

METHOD SUMMARY

Overview	The method combines CoRisk analysis, a risk budgeting approach, and balance sheet network analysis.
Application	The method measures the systemic risk contribution of individual institutions.
Nature of approach	Econometric analysis; balance sheet network analysis; risk budgeting.
Data requirements	Security prices or risk measures; balance sheet data analysis.
Strengths	The method captures both direct and indirect sources of interconnectedness; and identifies the contribution of individual institutions to systemic risk.
Weaknesses	The method carries the same weaknesses as any econometric model, including model misspecification; statistical causality may not correspond to economic causality.
Tool	Standard econometrics package.

1. FINANCIAL CRISIS, INTERCONNECTEDNESS, AND REGULATORY CAPITAL

Problems experienced in the U.S. subprime mortgage market in mid-2007 rapidly escalated to a liquidity crisis in the financial system, leading to a full-blown financial crisis in 2008–09 and the worst global recession experienced since the

Great Depression in 1929. The economic consequences of the crisis have been substantial. Write-downs in the banking sector through the first half of 2009 amounted to \$1.3 trillion, and going forward, they may increase to \$2.8 trillion (IMF, 2009b). The financial crisis led to a large contraction of the real economy of around 1 percent in real terms in 2009. The contraction could have been larger without massive fiscal stimulus plans adopted by governments. For instance,

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discretionary government spending among the G20 economies would amount to 2 percent of GDP for 2009 and 1½ percent for 2010 relative to 2007 baselines (IMF, 2009c).

The substantial costs associated with the crisis, which originated in the banking and financial sectors, have raised concerns in regard to whether the financial regulatory framework provides adequate safeguards against externalities affecting the systemic financial stability. One such externality is associated with the too-big-to-fail problem, which motivated heavy government support of large financial institutions in advanced countries. Such a problem, in principle, could be prevented by imposing limits on the scope and scale of the activities undertaken by a financial institution at the expense of forgoing economies of scale.

A more insidious externality, which arises naturally from the increased interdependence and globalization of the financial system, is associated with the too-connected-to-fail (TCTF) problem. Arguably, the problems experienced in the payment and settlement system following the liquidation of Herstatt Bank in 1974 were an earlier manifestation of the TCTF problem. The arguments put forward by the Federal Reserve Bank of New York in 1998 for coordinating the rescue of Long Term Capital Management, a hedge fund, by a consortium of private institutions were also closely related to the TCTF problem. More recently, the TCTF problem was dramatically illustrated by the increased volatility and liquidity shortages affecting financial institutions in the aftermath of the demise of Lehman Brothers, an investment bank, on September 15, 2008. Partly motivated by these developments, American International Group (AIG), an insurance company, was bailed out the next day to prevent a disorderly unwinding of AIG's credit derivatives exposure and its potential negative consequences on the financial system.

In order to deal with the TCTF problem, the study proposes regulatory capital charges proportional to how interconnected an institution is within a financial system or to other systemic institutions. Until now, regulatory capital charges, especially those based on the Basel I and Basel II frameworks, focus mainly on the risks related to an individual institution's trading and loan books. In contrast, a *TCTF capital charge* penalizes institutions by their incremental contribution to the risk of other institutions. As one institution fails, the domino effect increases the risk of failure of other institutions and the potential costs that must be borne by the government and ultimately by the taxpayers.

One clear benefit of an interconnectedness capital charge is that it induces institutions to internalize the costs associated with TCTF. Therefore, the capital charge may provide managerial incentives to avoid too much homogeneity in the financial system and excessive reliance on a reduced number of counterparties. The adoption of a TCTF capital charge also goes along with calls to impose more and higher-quality capital requirements for systemically important institutions (Acharya and Richardson, 2009; Bernanke, 2009; Brunnermeier and others, 2009), and the capital

charges should reflect the market interaction between firms (Milne, 2009; Tarullo, 2009).¹ The TCTF capital charge could also complement the recent proposal by the IMF to make banks pay insurance fees to fund future rescues in the sector.²

The TCTF capital charge represents a step forward to meet calls on the reform of the regulatory framework with a practical methodology for allocating capital for systemic risk, in the same spirit as recent proposals advanced by Adrian and Brunnermeier (2008); Tarashev, Borio, and Tsatsaronis (2009); and Gauthier, Lehar, and Souissi (2010).³ Furthermore, the proposed capital charge methodology extends and integrates work on systemic risk indicators based on credit risk models (Chan-Lau and Gravelle, 2005; Huang, Zhou, and Zhu, 2009), CoRisk measures (Adrian and Brunnermeier, 2008; Chan-Lau, 2008), and network analysis (Aikman and others, 2009), along the lines proposed in the Basel II framework.

The rest of this chapter explains the TCTF capital charge methodology in detail. Section 2 explains the conceptual issues underlying the proposed TCTF capital charge, introduces the concept of the incremental portfolio and incremental contribution to systemic risk, and highlights the differences with earlier contributions. Section 3 introduces a basic toolkit for calculating the TCTF capital charges. The toolkit relies on three methods, CoRisk, network analysis, and one-factor credit risk portfolio models, to quantify the marginal contribution to risk arising from direct and indirect exposures. Some advantages of these methods are that they are relatively easy to implement and that the interpretation of the results is simple and intuitive. The usefulness of the methods is fully examined in Section 4, which illustrates the computation of the TCTF capital charge for a set of financial institutions using CoRisk analysis and data on expected default frequencies (EDFs) from Moody's KMV (MKMV). Finally, Section 5 concludes by discussing how the TCTF capital charge methodology bears on procyclicality, the imposition of capital charges, macro-financial linkages, and the perimeter of regulation.

¹ Although the reform proposals have been discussed mainly in the context of advanced economies, their implications for emerging market countries, such as those belonging to the G20, are important. For instance, see Rennhack and others (2009) for a timely and in-depth discussion of how prospective reforms may affect Latin America.

² "IMF Exploring Insurance Levy on Banks," November 8, 2009 (Reuters, online).

³ In particular, the interconnectedness capital charge complements and improves upon the CoVaR-based capital measures proposed by Adrian and Brunnermeier (2008), which are based on the dependence between asset returns. The TCTF capital charge, instead, focuses on the credit losses likely to be borne by taxpayers and addresses directly the costs default imposes on society. In contrast to Tarashev, Borio, and Tsatsaronis (2009), the TCTF accounts for the potential increase in the probability of default conditional on the failure of other institutions that can be calculated with the use of CoRisk analysis and network analysis, among other methods (Section 3). Also, the concept of incremental or marginal contribution to systemic risk is different (see Section 2 for details).

2. THE TOO-CONNECTED-TO-FAIL CAPITAL CHARGE: A SIMPLE BUT POWERFUL CONCEPT

As highlighted in IMF (2009a), the globalization of the financial system and the expansion of large complex financial institutions into an array of diverse activities that transcend national boundaries have led to stronger linkages and interconnectedness across institutions, whether regulated or not.⁴ This section first explains why societal loss is the right risk metric for measuring losses associated with TCTF, illustrates the concept of the TCTF capital charge by using a simple intuitive example, and shows how to use societal loss distributions for calculating the TCTF capital charges for an institution.

A. Societal losses

In a highly interconnected system, the failure of one institution is likely to trigger the failure of other institutions because of either direct or indirect sources of exposures. Direct exposure can arise from interbank claims and/or counterparty exposure from derivatives transactions. An institution also may have accumulated a substantial exposure through credit derivatives contracts to the credit risk of an otherwise unrelated institution, as was the case with AIG. Among indirect exposures, the similarities of investment portfolios and trading strategies subject institutions to substantial marked-to-market losses if the bankruptcy of one institution forces a disorderly fire sale of assets at drastically reduced prices.

Societal loss is the proper risk metric for assessing the TCTF capital charge. Owing to either direct or indirect exposures, the externalities associated with the failure of a too-big-to-fail or too-connected-to-fail institution force the government to partly assume some of the losses, which are ultimately borne out by taxpayers. It becomes natural, therefore, that the TCTF capital charge reflects the incremental contribution of the default of one institution to the potential societal loss from the surviving institutions.

The definition of societal loss is rather flexible and should be determined by the systemic regulatory agency. The operational definition depends on the dominant features of the financial system it oversees. In a country where the financial system comprises mostly deposit-taking institutions, the measure of societal loss could be associated to potential deposit losses. If pension funds are among the main sources of funding to domestic banks, it may be justified to include as societal losses potential losses faced by pension funds in case financial institutions fail. The recent experience in the United States indicates that societal losses could be roughly proxied

by the potential losses associated with the bank liabilities because many of the government measures effectively guarantee the claims of banks' senior creditors. Indeed, the implicit U.S. government guarantee was priced in the market as shown by the convergence of the U.S. sovereign and banks' credit default swaps following the implementation of support measures.

B. A two-bank example

The following two-bank example helps to clarify the intuition underlying the interconnectedness capital charge. Suppose that the two banks, A and B, are identical. Each bank holds \$100 million, and their unconditional probability of default is 5 percent. If one bank defaults, all deposits are lost and the probability of default of the surviving bank increases to 6 percent. Because deposits, in this example, are guaranteed by the government, the societal cost is equal to the losses suffered by the depositors. In isolation, the expected societal loss from the default of any of the two banks is \$5 million (5 percent times the amount of deposits, \$100 million). If Bank A defaults, the expected societal loss from the subsequent default of Bank B increases to \$6 million (6 percent times \$100 million).

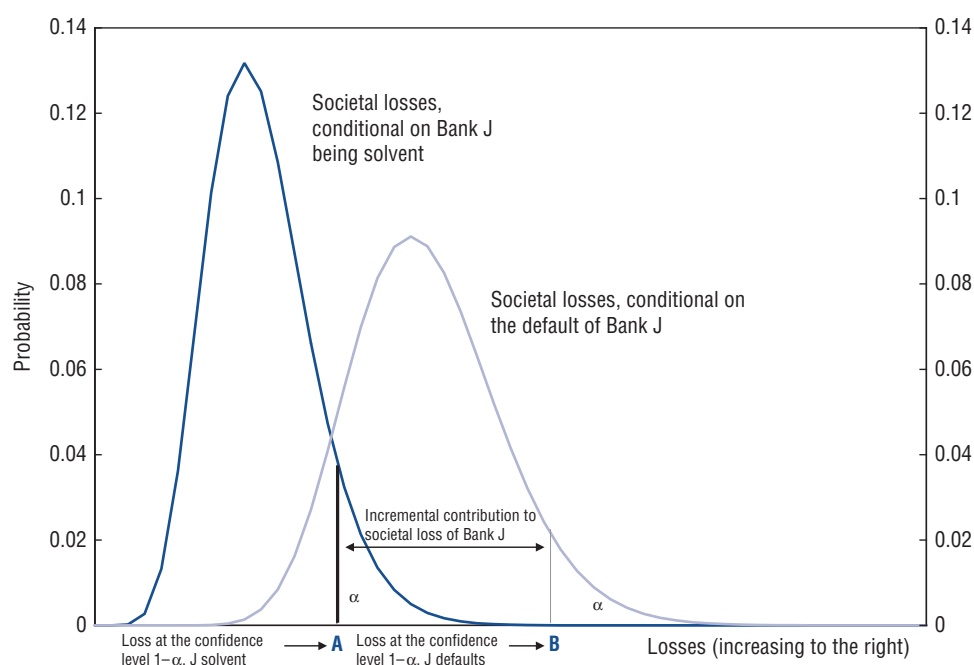
In this example, the incremental contribution to societal loss of the default of Bank A is \$1 million, or the increase in the expected losses of Bank B (\$6 million minus \$5 million). The TCTF capital charge should be proportional to the incremental contribution of \$1 million. For instance, if capital charges (or provisions, in this example) were set equal to the expected loss, Bank A would have to hold \$6 million in capital: \$5 million corresponding to its unconditional expected loss and an additional \$1 million corresponding to the expected losses its default induces on Bank B.

C. From societal losses to too-connected-to-fail capital charges: the incremental portfolio approach

More generally, the TCTF capital charge for a specific institution should be assessed based on the differences between the conditional societal loss distributions when the institution survives and defaults, respectively. Furthermore, in line with current practice established under the Basel Accord, the comparison between the loss distributions should be based on a predetermined confidence level set up by the regulatory authority. This is summarized in Figure 18.1. The figure depicts the hypothetical case of N banks or systemic financial institutions of interest to the regulator. The incremental contribution to societal loss of Bank J is determined by comparing changes in the loss distribution of the other $N-1$ banks when Bank J is solvent and when Bank J defaults.

The calculations to assess the incremental contribution of Bank J on the system, based on known techniques for risk budgeting (Litterman, 1996), are described next.

⁴ For a detailed account on the systemic risks posed by financial globalization and how they contribute to interconnectedness, see Chan-Lau (2008).



Source: Author.

Note: The figure shows the hypothetical loss distribution for a portfolio of $N-1$ banks (or financial institutions) that excludes Bank J conditional on whether Bank J defaults. It also shows the corresponding marginal contribution to societal loss of Bank J for a given confidence level α .

Figure 18.1 The Incremental Contribution of the Default of Institution J to Societal Loss

The starting point is a portfolio of $N-1$ banks that excludes Bank J. This portfolio is referred to as the N_J portfolio, or the *incremental portfolio of institution J*. The loss distribution over a fixed time horizon conditional on the survival of Bank J is calculated for the N_J portfolio using the probability of default, PD_J ; exposure at default, EAD_J ; and the loss given default, LGD_J . Depending on the loss distribution model used, it would also be necessary to specify the degree of association (or dependence) between the probabilities of default of the banks in the N_J portfolio. For instance, in the case of the one-factor Gaussian model (or Gaussian copula), the dependence parameter is the correlation of the asset values of the banks with a common factor. The loss distribution associated with the survival of institution J is referred to as the hypothetical *ex ante loss distribution*, that is, prior to the failure of the institution.

Similarly, the conditional loss distribution in the event that institution J defaults, or the hypothetical *ex post loss distribution*, can be calculated using the conditional probability of default, $PD_{N|J}$; the conditional exposure at default, $EAD_{N|J}$; the conditional $LGD_{N|J}$; and the conditional degree of association (or dependence). For instance, in the two-bank example in the previous section, it was assumed that the exposure at default, equal to the deposits, and the loss given default, 100 percent, remain the same regardless of whether a bank defaults. This is not necessarily the case, as the default of one bank could change the exposure at default of the surviving banks. For instance, in the case of Herstatt risk, or

settlement risk, the exposure at default depends on how much the defaulted institution owes to the surviving institutions. Furthermore, the dependence or association measure, such as the default correlation, could increase substantially following the failure of one institution as observed in the aftermath of the bankruptcy of Lehman Brothers on September 15, 2008. More generally, empirical studies have shown that dependence measures are higher during down-market periods and recessions and suggest that failures of financial institutions could be related to higher default correlation.

For the purpose of determining the incremental contribution to societal losses, it is appropriate to use the difference between the values of both loss distributions at a certain confidence level such that the probability of experiencing a large loss is relatively small. The use of the differences in the expected loss in both distributions is just the particular case of setting the confidence level at 50 percent. Furthermore, the use of a confidence level is also the standard practice for the calculation of economic and regulatory capital using value-at-risk (VaR) models, which facilitates the adoption of the interconnectedness capital charge by regulatory agencies. Alternatively, it is possible to use expected shortfall (ES) rather than VaR or other coherent risk measures.⁵ For illustration purposes only, the discussion that follows focuses on

⁵ See Artzner and others (1999) on coherent risk measures.

VaR, but extending the methodology to other risk measures is straightforward.

Consequently, as shown in Figure 18.1, for a confidence level of $1 - \alpha$, the incremental contribution to societal loss of Bank J is equal to $B - A$, the difference between the VaR_α values of the conditional and unconditional loss distributions, respectively:

$$\begin{aligned} \text{Incremental contribution to societal loss of Bank } J \\ &= VaR_\alpha(\text{Loss distribution when } J \text{ defaults}) \\ &\quad - VaR_\alpha(\text{Loss distribution when } J \text{ is solvent}). \end{aligned} \quad (18.1)$$

Once the incremental contribution to societal loss is calculated, the TCTF capital charge can be calculated simply as the product of the probability of default of Bank J times the incremental contribution to societal loss:

$$\begin{aligned} \text{Too-Connected-to-Fail capital charge for Bank } J \\ &= \text{Incremental contribution to societal loss of Bank } J \\ &\quad \times \text{Probability of default of Bank } J. \end{aligned} \quad (18.2)$$

In the two-bank example, the interconnectedness capital charge for any of the banks will be \$50,000 (5 percent times \$1 million). The magnitude of the TCTF capital charge of an institution increases with its incremental contribution to societal loss and with the probability it defaults. So imposing this charge provides banks with incentives to either change their business activities to reduce the risk they pose to other banks or to take measures aimed to strengthen their solvency.

Summarizing, the steps for calculating the interconnectedness capital charge for an institution J in relation to a group of connected institutions are as follows:

1. For each institution other than J, specify the probability of default of the remaining institutions in the events that institution J survives or defaults.
2. Similarly, for each one of the institutions in Step 1, determine the societal exposure at default and the societal loss given default for each of the two events. Typical choices are the nominal amount of potential losses incurred by the government in case that the institutions default.
3. Construct the societal loss distributions for the incremental portfolio that comprises all institutions excluding institution J under the events that J survives, ex ante loss distribution, or defaults, ex post loss distribution.
4. Pick up a given confidence level. Typical values correspond to those used for calculating VaR, that is, 95 percent, 99 percent, and 99.5 percent (alternatively, a tail-risk measure such as ES can be used).
5. Calculate the VaR in the ex ante and ex post societal loss distributions at the specified confidence level.
6. Calculate the incremental contribution to societal loss as the difference between the VaR conditional on the default of institution J (VaR of ex ante loss distribution) and the VaR conditional on the survival of institution J (VaR of ex post loss distribution).

7. Calculate the TCTF capital charge as the product of the probability of default of institution J and its incremental contribution to societal loss.

Among these steps, the most difficult ones are likely related to Step 1, the specification of the probability of default, but in particular to Step 2, the determination of the exposure at default and loss given default. Step 3 requires the specification of a loss distribution model and its associated characteristics. Foremost among them is the specification of the degree of association or dependence, that is, in layman's terms, comovement, between the probabilities of default of individual institutions. Steps 4 to 7 are relatively straightforward after the specification of the required elements in Steps 1 to 3.

Section 3 describes how the probabilities of default can be calculated using two methodologies, CoRisk analysis and network analysis. The calculation of the societal exposure at default and the corresponding loss given default is somewhat more complicated. To start with, the academic and practitioners' advances on these two topics, even for corporate credit risk, are still evolving and at a relatively primitive stage compared with advances on the calculation of the probability of default. Under these circumstances, it may be appropriate to use rough estimates of the exposure at default and loss given default based on the liabilities that could potentially be guaranteed by the government if a financial institution were to default. Although these liabilities should include at least deposits, policy responses during the recent crisis suggest that it would be justifiable to also include senior creditor claims.

Finally, it is worth mentioning that a rigorous calculation of the TCTF capital charge requires an iterative process, or finding a fixed point, as emphasized in Gauthier, Lehar, and Souissi (2010). Equation (18.2) points out that the capital charge depends on the probability of default. The addition of the TCTF capital charge increases the capital of the institution and reduces its leverage. Lower leverage usually contributes to lower asset volatility. As a result, the probability of default of the institution declines, and in turn, it reduces the TCTF capital charge. From a regulatory perspective, however, a single iteration of the TCTF capital charge errs on the conservative side and appears to be a second-best solution in a world where optimal solutions are unfeasible because of data quality and model risk.

D. Differences with other approaches

The step-by-step summary helps to highlight the difference between the TCTF capital charge and the systemic risk capital charge recently proposed by Tarashev, Borio, and Tsatsaronis (2009) and Gauthier, Lehar, and Souissi (2010). Although similar in spirit, there are two important differences between these two approaches and the one presented here.

The first difference is related to the measurement of an institution's contribution to systemic risk. The incremental contribution to risk of a specific institution, as explained in

the previous section, is proportional to the change in the risk metrics of a reference portfolio, that is, VaR or ES, induced by the failure of that institution. The reference portfolio in the TCTF capital charge excludes the failed institution both ex ante and ex post the failure event. In Tarashev, Borio, and Tsatsaronis (2009) and Gauthier, Lehar, and Souissi (2010), the capital charge for an institution is calculated in reference to two different portfolios. The first portfolio, or ex ante portfolio, does not include the institution, whereas the second portfolio, or ex post portfolio, does. So in both approaches, the capital charge is based on the marginal contribution to risk from adding one institution to the ex ante portfolio.⁶ Note that this portfolio-building approach cannot account for the impact of the failure of one institution. Or in other words, the ex ante and ex post portfolios suggested by these authors are portfolios conditional on the event that the institution survives.

The second difference is related to the probabilities of default used in the calculation of the loss distributions. Because the ex ante and ex post portfolios do not consider the failure of the institution, both Tarashev, Borio, and Tsatsaronis (2009) and Gauthier, Lehar, and Souissi (2010) do not factor in the increase in default risk triggered by the failure of one institution as it is done in the TCTF capital charge. Similarly, changes in other parameters, such as the societal exposure at default or the degree of association, that is, correlation, between institutions, are not accounted for. It is important to note that, as in the case of Tarashev, Borio, and Tsatsaronis (2009), the methodology to calculate the TCTF capital charge can be generalized to accommodate the simultaneous default of more than one institution, because this is not a limitation in the CoRisk and network analysis methods described in the next sections.

3. A BASIC TOOLKIT FOR CALCULATING TCTF CAPITAL CHARGES

As explained earlier, the first step in the calculation of the TCTF capital charge for a given institution requires knowing how the institution's survival or default affects the probability of default of other institutions. For systemic institutions, the calculation in the former case is relatively straightforward as probability-of-default estimates may be readily available from risk data providers such as MKMV. Alternatively, they can be obtained from market prices, such as credit default swap (CDS) and bond spreads, or equity prices, using reduced form and/or structural models of default risk and adjusting for the market price of risk. The calculation in the latter case is more complex, but fortunately some methodologies have been developed recently to address this task.

⁶ The calculations involved in obtaining the maximal contribution to risk in setting up the ex ante and ex post portfolios are relatively complex because they involve the use of the Shapley values for each institution. See Tarashev, Borio, and Tsatsaronis (2009) and Gauthier, Lehar, and Souissi (2010) for details and further references.

Once the individual probabilities of default have been calculated, it is necessary to use credit portfolio models to construct the loss distributions underlying the calculation of the TCTF capital charge. Credit portfolio models rely on a statistical technique, copulas, to construct the joint multivariate probability of defaults of the institutions included in the portfolio from the individual probabilities of default.

This section presents two useful methods for the estimation of conditional probabilities of default, the CoRisk analysis and network analysis. It also describes in detail the one-factor credit portfolio model as it is intuitive enough to understand what drives dependencies, that is, comovements, across individual probabilities of default. The choice of the methods presented in this section is guided by their ease of implementation, but more importantly by their transparency and tractability, the last two key characteristics to ensure model reliability and robustness. These characteristics facilitate the mapping of the model features to the underlying economic environment, identify model anomalies, suggest potential ways to deal with them, and, equally important, make it easier to communicate results and policy implications to senior decision makers.

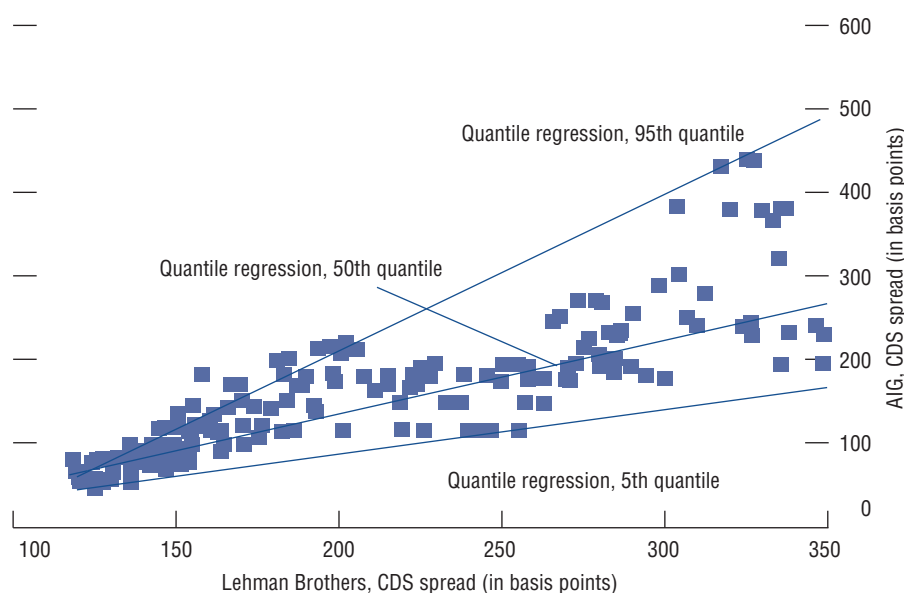
A. CoRisk analysis

As for the probabilities of default conditional on the failure of a specific institution, the use of CoRisk analysis provides a simple way to calculate them while accounting for market interaction that is due to either direct or indirect sources of exposure.

CoRisk analysis, based on the CoVaR analysis introduced by Adrian and Brunnermeier (2008) and extended in Chan-Lau (2009) to analyze interconnectedness risk between financial institutions, measures risk codependence, or how the risk of one institution reacts to changes in the risk of another institution after controlling for common risk factors. The risk codependence, or CoRisk, can be estimated using quantile regression.

In a quantile regression, first introduced by Koenker and Bassett (1978), the response of the independent variable to a number of covariates can be sorted in ascending order by quantiles. Roughly speaking, in a linear quantile regression, the response at different quantiles is represented by a linear regression with different slopes for each linear regression. If the dependent variable is the probability of default of an institution, each quantile regression can be identified with different risk regimes, that is, low risk to high risk. If among the covariates, in addition to aggregate risk factors, one includes the probability of default of another institution, the quantile regression estimates the CoRisk of the first institution to changes in the probability of default of the second institution at different quantile levels.⁷

⁷ The interpretation of quantile regression in the text assumes that the probability of default of the second institution "causes" the probability of default of the first institution, a typical problem in reduced-form econometric models.



Source: IMF (2009c).

Note: AIG = American International Group; CDS = credit default swap.

Figure 18.2 AIG and Lehman Brothers CoRisk

Figure 18.2 illustrates the use of quantile regressions to estimate CoRisk. In this illustration, CDS spreads are used to proxy probabilities of default. The scatter plot shows the relationship between the CDS spreads of Lehman Brothers and AIG CDS spreads for the period July 1, 2003–September 12, 2008. It also shows the quantile regression for the 5th, 50th, and 95th quantiles, where the Lehman Brothers CDS was included among the covariates in addition to other common risk factors.⁸ The slope of the regression steeps up as the quantile of interest increases and indicates that during high-risk regimes, that is, regimes corresponding to higher quantiles, the CoRisk of AIG to Lehman Brothers is higher than in low-risk regimes.

For the purpose of estimating the TCTF capital charge, the relevant quantile regression is

$$PD_i = \alpha_\tau + \sum_k \beta_{\tau,k} R_k + \beta_{\tau,j} PD_j, \quad (18.3)$$

where PD denotes the probability of default, R_k denotes the common factors, τ the quantile of interest, and β the coefficients associated with the covariates.⁹

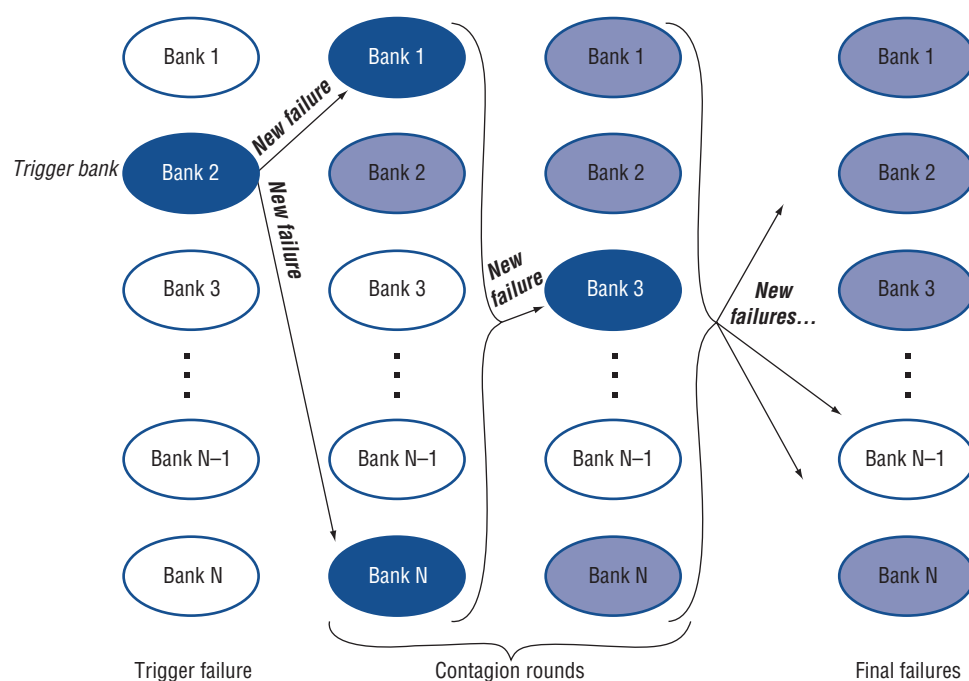
Two observations in regard to equation (18.3) are worth mentioning as they would ultimately affect the calculation of the TCTF capital charge. The first one corresponds to the choice of the quantile for the estimation of the probability of

default conditional on the failure of one institution. This choice is subject to the discretion of the regulatory agency as it will be affected by the agency's objectives and the inherent tradeoffs between the efficiency and safety of the financial system. For instance, in normal periods, it could be justifiable to use the 50th percent quantile regression. A more cautious approach would be to choose an upper quantile, such as the 95th or 99th quantiles. The tradeoff is that a higher quantile implies a higher TCTF capital requirement and a reduced supply of credit from the affected bank. It is worth keeping in mind, though, that the use of an upper quantile regression to set the TCTF capital charge is less vulnerable to procyclicality so it is better aligned with recent proposals for reducing the impact of procyclical capital requirements (Andritzky and others, 2009; Brunnermeier and others, 2009).

The second observation concerns the value of the probability of default of the institution assumed to fail in equation (18.3). The obvious choice is a value of one, as it indicates that the institution has already failed. Similarly to the choice of the quantile regression, the alternative to the use of a value of one is to use a value corresponding to the higher percentiles, that is, the 95th percentile, of the historical distribution of the probabilities of default. This alternative is viable if the data cover at least one business cycle or major crisis episode. Another alternative is to use a threshold “intervention” probability of default in the calculations, especially under a prompt corrective action framework. Under this scenario, no institution defaults because the systemic regulator intervenes well ahead of the default event and requires the implementation of corrective measures. If the threshold for intervention can be associated with a specific probability

⁸ In order to obtain a two-dimensional figure, it is necessary to specify constant values for the additional variables constant. Therefore, Figure 18.2 is an approximate two-dimensional representation of the quantile regression.

⁹ Because the probabilities of default are bounded between 0 and 1, it is necessary to use a Cox transformation before estimating the CoRisk quantile regressions.



Source: IMF (2009c).

Figure 18.3 Network Analysis: A Diagrammatic Representation of Cascading Bank Failures

of default, then the use of the latter in equation (18.3) is justified.

B. Network analysis

CoRisk analysis relies on market-based estimates of the probability of default of an institution. These estimates are reliable as long as the market is pricing correctly the risks to the institution, either those arising from direct or indirect sources. If information on the direct exposures across institutions is available, network analysis provides a complementary approach for calculating probabilities of default conditional on the failure of one institution and provides a cross-check to estimates based on CoRisk analysis. The use of network analysis could be very useful in jurisdictions where market data are either not available or not reliable.

Network analysis starts from the fact that the balance sheets of institutions are linked by the direct exposures across institutions, for example, interbank exposures.¹⁰ The liabilities of one institution constitute the assets of another institution. The default of the first institution has a negative impact on the asset value of the second institution and in turn raises its probability of default or ultimately causes its failure. The failure of the second institution could generate the failure of a third institution, and so on. Figure 18.3 illustrates the cascading effects prompted by the failure of one bank in a system where there are nonnegligible interbank exposures.

After performing the network analysis for a system of institutions, the calculation of the probabilities of default conditional on the failure of one institution using network analysis is relatively straightforward for banks that fail in the subsequent contagion rounds. Simply, their conditional probability of default is set equal to one, and their societal losses are added to the distribution loss.

For banks that survive the failure of the trigger bank, the calculation of their conditional probability of default relies on the use of balance sheet models of default risk, such as structural models, like that of Merton (1974) and its variations, or models based on financial ratios using balance-sheet information (Altman, 1968). In the case of structural models, the change in the probability of default can be related to the decline in the asset value of the institution.

In the context of the simple Merton model, the probability of default over a given time horizon, T , is directly proportional to the institution's distance-to-default, DD :

$$DD = \frac{\ln\left(\frac{A}{L}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}, \quad (18.4)$$

where A is the asset value of the institution, L is the face value of its liabilities, r is the risk-free rate, and σ is the volatility of the asset value.¹¹

¹⁰ See IMF (2009b) for details and references. The data requirements for implementing network analysis, however, could be too onerous.

¹¹ For a recent discussion on the distance to default and its possible variations, see Capuano and others (2009).

Building on the balance sheet linkages across institutions, network analysis pinpoints exactly the change in the asset values of the firm. Furthermore, network analysis can also capture second-round effects that result from the forced sale of assets due to the initial change in the asset value of the firm. The forced sale of assets may be prompted by margin calls and, in the case of regulated institutions, the need to rebuild capital ratios up to the minimum regulatory requirements. The sale of assets under distress would depress prices and further reduce the asset value of all surviving institutions.¹²

Continuing the example of the Merton model, the risk-neutral probability of default of institution I, conditional on the failure of institution J, PD_I^* , is given simply by the following:

$$PD_I^* = \Phi^{-1}(DD|J), \quad (18.5)$$

where Φ is the cumulative normal distribution, and $DD|J$ is the distance-to-default of institution I after accounting for negative impact on its balance sheet of the failure of institution J. The probability of default conditional on the default of institution J will likely increase vis-à-vis the value obtained when institution J survives, as the asset-liability ratio will deteriorate and the asset volatility will increase.

To transform the risk-neutral probabilities of default to real-world, or physical, probabilities of default, it is necessary to correct them by including a market risk premium. If the market risk premium is calculated using the capital asset pricing model (CAPM), Bohn (2000) has shown that the physical probability of default of institution I, PD_I , can be approximated by

$$PD_I = 2\Phi\left(\Phi^{-1}\left(\frac{PD_I^*}{2}\right) - \rho_{I,M}\beta T\right), \quad (18.6)$$

where T is the time horizon of the analysis, $\rho_{I,M}$ is the correlation between the asset returns and the market returns, and β is the market price of risk, or the market return in excess of the risk-free rate per unit of market volatility.¹³

C. The one-factor credit portfolio risk model

The one-factor credit portfolio model is the benchmark model for constructing the loss distribution of a portfolio of risky assets or, in the case of the TCTF capital charge,

of risky systemic institutions.¹⁴ The one-factor model offers a simple way to capture the joint behavior of default events for a portfolio of institutions. Furthermore, the model can be extended to account for several common factors (Hull and White, 2004), and the loss distribution can be computed using either Monte Carlo simulation or available fast and efficient numerical solution algorithms (Andersen, Sidenius, and Basu, 2003).

The one-factor model assumes that the normalized asset value of institution i , x_i , depends on a single common factor, M , and an idiosyncratic shock, Z_i :

$$x_i = a_i M + \sqrt{1 - a_i^2} Z_i, \quad (18.7)$$

where x_i , M , and Z_i are random variables with zero mean and unit variance and cumulative distribution functions F_i , G , and H , respectively. The coefficient a_i , or factor loading, is restricted to values between 0 and 1.¹⁵ Default occurs when the asset value falls below a threshold value $\bar{x}_i$. The threshold value depends on the current probability of default of the institution, PD_i , and is given by $\bar{x}_i = F_i^{-1}(PD_i)$. Finally, the probability of default conditional on the realization of the common factor is simply

$$\text{Prob}\{x_i < \bar{x}_i | M\} = PD_i(t|M) = H^{-1}\left(\frac{\bar{x}_i - a_i M}{\sqrt{1 - a_i^2}}\right). \quad (18.8)$$

Contrary to common perception, arguably misguided by the fixation with the special case of the one-factor Gaussian model, the one-factor model can easily accommodate skewness, fat tails, and the use of extreme value and empirical distributions by choosing the appropriate distribution functions for the asset value and the common and idiosyncratic shocks.

Equation (18.7) implies that the asset correlation between two institutions, i and j , is equal to $a_i a_j$. Hence, the practical problem of estimating multiple pairwise and/or groupwise correlations (or comovements) is reduced to, first, estimate the normalized asset value for each institution; second, identify the common factor; and third, estimate the factor loadings for each institution.

¹² The probability of default in equation (18.4) is risk neutral because it requires inverting an option pricing formula. In general, asset pricing formulas discount future cash flows using the risk-free rate rather than a discount rate equal to the risk-free rate plus a risk premium. To compensate for a lower discount rate, the probabilities of adverse events are overweight vis-à-vis those of positive events. The risk-neutral probabilities, therefore, reflect the risk aversion of the market.

¹³ For illustrative purposes, the Merton (1974) model is appropriate because it is simple and relatively well known in policymaking institutions and regulatory agencies. For a more complete taxonomy of models, see the textbook of Lando (2003).

¹⁴ See, among others, Gibson (2004) and Hull and White (2004) for a comprehensive description of the one-factor model. The Gaussian copula version of the one-factor credit portfolio model was originally introduced by Li (2000). Vasicek (1997) presents an earlier loss distribution model in the same spirit as Li's model. The special case of the one-factor Gaussian model came into the limelight in the aftermath of the 2008–9 subprime crisis. The model was the benchmark pricing model for collateralized debt obligation tranches and has been much maligned, unfairly, for contributing to the large losses observed in the structured credit market. The criticisms may have been overblown, as the model was used mainly as a pricing convention communication tool in the same spirit as traders used the Black-Scholes implied volatility to quote option prices.

¹⁵ A documented empirical observation is that correlations during stressful events are higher than during normal periods. This asymmetry can be captured by allowing the factor loading to be an increasing function of the common shock M , or of the probability of default of the institution. I thank Fuchun Li for discussions that led to the suggested modification.

TABLE 18.1

Average Increase in the Probability of Default Due to the Failure of Other Institution (in percent)

	United States	Canada	Spain	France	Germany	Italy	Switzerland	United Kingdom	Netherlands
United States	327	215	260	289	258	333	281	252	317
Canada	133	50	63	100	92	146	72	74	64
Spain	201	57	55	76	23	70	32	63	56
France	195	160	154	159	163	166	279	157	128
Germany	209	184	214	172	137	196	176	186	196
Italy	204	95	151	141	76	162	45	84	145
Switzerland	141	75	67	102	59	85	114	115	100
United Kingdom	146	75	112	123	119	128	134	104	122
Netherlands	181	140	120	78	42	84	136	85	152

Sources: Author; and Moody's KMV.

Note: The table reports the average increase, in percent, of the probability of default for institutions located in the countries listed in the first column conditional on the failure of one institution located in the countries listed as headings. The percent increase is calculated from the 95th quantile regression for pairwise combination of institutions and setting the explanatory variables at their 95th percentile.

Currently, there is no established methodology for performing these tasks, and the ones being used have their pros and cons (de Servigny and Renault, 2004). One simple procedure is to use a structural model to infer the normalized asset values from the estimated probabilities of default. Statistical methods like factor analysis or principal component analysis then are used to extract the common factor. Once the asset values and common factor have been estimated, it is straightforward to estimate the factor loadings. The loss distribution then can be computed using either Monte Carlo simulation or analytical algorithms.

4. AN ILLUSTRATIVE EXAMPLE

This section offers an example of how to use the toolkit. The data in the example consist of weekly time series of EDFs from MKMV for 26 global financial institutions for the sample period May 2, 2003–February 27, 2009.¹⁶ As of end-2008, these institutions held together US\$36 trillion in assets and US\$9¼ trillion in total debt according to balance sheet data reported by Primark Datastream and Google Finance.

As suggested in Section 3.A, CoRisk analysis was used to calculate the impact of the failure of a single institution on the probability of default of the remaining institutions. Equation (18.3) was estimated for the 95th quantile regression for all institutions for pairwise combinations of institutions. Rather than using common macro and financial risk

factors, the equations were estimated using the quick-and-dirty approximation that these common factors corresponded to the first two principal components of the probabilities of default for this illustrative example. The results of the estimation are reported in Table 18.1.

The results suggest that, on average, U.S. banks are the most vulnerable to shocks originating within the United States owing to the vulnerabilities of the three investment banks in the sample. Spanish banks and Canadian banks are the least vulnerable. The failures of U.S. banks, on average, are the most damaging to the financial system, but it is worth noting that this result is driven mostly by those corresponding to the largest U.S. commercial bank in the sample. The systemic risk created by failures of banks in countries other than the United States is roughly the same and slightly lower for German and Canadian institutions.

The calculation of the TCTF capital charge follows the steps outlined in Section 2.C and uses the one-factor Gaussian credit risk model described in Section 3.C. The probabilities of default conditional on the survival of an institution are set equal to the values observed in the last week of February 2009. The probabilities of default conditional on the default of an institution are calculated by adding the CoRisk percent increase caused by that institution to the probabilities corresponding to the survival event.

As a shortcut and just for illustration purposes, societal losses are set equal to the total debt as reported by the financial institutions in their end-2008 financial statements. The loss given default is assumed equal to 100 percent, an extreme case that implies that the institution is worthless after it fails and that there is an implicit full government guarantee for the institution's debt holders. Finally, the factor loading, that is, correlation of the asset returns of the institution with the common factor, is calculated according to Proposal II of the Basel Committee on Banking Supervision (BCBS, 2004).

Table 18.2 reports the TCTF capital charges corresponding to a confidence level of 95 percent for each institution. The capital charges were estimated using Monte Carlo simu-

¹⁶ The institutions are Citigroup, Goldman Sachs, JP Morgan, Morgan Stanley, and Wells Fargo (from the United States); the Bank of Nova Scotia, Canadian Imperial Bank of Commerce, and Royal Bank of Canada (from Canada); BBVA and Banco Santander (from Spain); Banque Nationale Paribas, Credit Agricole, and Societe Generale (from France); Commerzbank and Deutsche Bank (from Germany); Banca Intesa and MedioBanca (from Italy); Credit Suisse and UBS (from Switzerland); Barclays, Hongkong and Shanghai Banking Corporation, Lloyds, Royal Bank of Scotland, and Standard Chartered (from the United Kingdom); and ABN Amro and ING (from the Netherlands).

TABLE 18.2

Too-Connected-to-Fail Capital Charges for Global Financial Institutions

Institution	Probability of Default (in percent)	Assets (\$ billion)	Total Debt (\$ billion)	Incremental Portfolio Notional Amount (\$ billion)	Value at Risk, Incremental Portfolio				Too-Connected-to-Fail, Capital Charge			
					Event = Survival (in percent of incremental portfolio)		Event = Default (in percent of incremental portfolio)		(in \$ billion)		(in percent of assets)	
					Confidence Level	Confidence Level	Confidence Level	Confidence Level	Confidence Level	Confidence Level	Confidence Level	Confidence Level
					95	99.5	95	99.5	95	99.5	95	99.5
Morgan Stanley	5.38	659	289	8,929	18.9	20.1	25.1	26.7	29.8	31.9	4.52	4.83
Goldman Sachs	1.63	876	322	8,895	19.0	20.1	25.6	28.8	9.6	11.8	1.10	1.35
Citigroup	20.21	1,938	670	8,548	16.1	17.6	32.8	35.4	287.7	307.2	14.84	15.85
Wells Fargo	0.44	1,310	375	8,843	18.3	21.5	24.6	27.2	2.5	2.3	0.19	0.17
JP Morgan	1.85	2,175	633	8,584	19.7	21.0	27.8	29.3	12.9	13.2	0.59	0.61
Bank of Nova Scotia	0.39	414	44	9,174	18.4	20.4	23.7	26.1	1.9	2.1	0.46	0.50
Canadian Imperial Bank	0.43	289	38	9,180	17.7	18.4	24.5	26.4	2.7	3.1	0.93	1.08
Royal Bank of Canada	0.66	594	139	9,079	18.6	19.6	26.7	28.9	4.8	5.5	0.81	0.93
BBVA	0.57	543	121	9,097	17.0	18.8	27.3	30.3	5.4	6.0	0.99	1.10
Santander	0.40	1,050	39	9,179	17.9	18.7	29.1	30.3	4.1	4.3	0.39	0.41
Banque Nationale Paribas	0.71	2,880	420	8,798	19.2	21.0	23.6	25.7	2.7	2.9	0.09	0.10
Credit Agricole	0.76	2,292	487	8,731	19.4	22.7	30.6	32.8	7.5	6.7	0.33	0.29
Societe Generale	3.08	1,567	224	8,994	18.3	19.9	30.9	33.8	35.0	38.5	2.24	2.45
Commerzbank	7.80	861	429	8,789	19.2	19.5	29.3	31.2	69.2	79.9	8.04	9.29
Deutsche Bank	3.52	3,050	483	8,734	19.4	20.5	25.8	27.8	19.8	22.4	0.65	0.73
Intesa	0.29	878	388	8,830	19.1	20.9	31.3	32.7	3.1	3.0	0.35	0.34
Mediobanca	1.76	102	70	9,148	18.1	19.2	30.4	33.0	19.7	22.2	19.37	21.77
Credit Suisse	1.22	1,098	439	8,779	18.1	19.3	28.2	31.4	10.8	13.0	0.99	1.18
UBS	0.12	1,899	512	8,705	19.4	24.0	30.2	32.3	1.1	0.9	0.06	0.05
Barclays	4.39	2,948	693	8,525	17.3	19.8	24.1	27.4	25.4	28.3	0.86	0.96
HSBC	0.48	2,527	696	8,522	19.4	21.0	28.5	31.4	3.7	4.2	0.15	0.17
Lloyds	0.82	436	38	9,180	18.5	20.4	27.8	30.3	7.0	7.5	1.61	1.72
Royal Bank of Scotland	2.83	2,402	1,021	8,197	17.5	20.1	23.9	25.8	14.8	13.0	0.62	0.54
Standard Chartered	0.24	428	74	9,143	18.5	19.6	26.6	29.5	1.8	2.2	0.42	0.52
ABN Amro	0.06	940	176	9,042	18.7	19.7	26.5	29.0	0.4	0.5	0.05	0.05
ING	4.88	1,832	398	8,819	19.2	20.9	28.3	30.7	39.4	41.9	2.15	2.29

Sources: Author; Google Finance; Moody's KMV; and Primark Datastream.

Note: The table shows the too-connected-to-fail capital charges for the banks listed the first column for two confidence levels, 95 percent and 99.5 percent. The probability of default figures correspond to the average EDF reported by Moody's for the last week of February 2009. The societal loss in case the institution fails is assumed to be equal to the total debt of the institution, that is, the societal exposure at default is set equal to the amount of total debt, and the loss given default is set equal to 100 percent. The incremental portfolio includes all institutions excluding the one for which the capital charge is calculated.

EDF = expected default frequency.

lation with the number of simulations set to 5,000, and are reported in billions of U.S. dollars and as percent of the institutions' total assets.

Unsurprisingly, given the sample period analyzed and the specific point in time chosen for the probabilities of default, Citigroup stands out as the institution with the highest

TCTF capital charge, at around 15 percent of total assets. At the 95 percent confidence level, the VaR of Citigroup's incremental portfolio almost doubles to 32 percent of the nominal amount of the portfolio in case Citigroup defaults. But the main factor responsible for the high capital charge is Citigroup's high probability of default.

For comparison purposes, note that Spanish banks BBVA and Santander and the French banks Credit Agricole and Societe Generale have incremental portfolios roughly as large as Citigroup. For these institutions, their failure increases the VaR of the incremental portfolio by the same magnitude as in the case of Citigroup. But the probability of default of these institutions is one or two orders of magnitude lower than the probability of default of Citigroup and is reflected in substantially lower TCTF capital requirements.

The numerical example highlights some of the main issues discussed in Section 2.C, namely, that TCTF capital charges depend on both the risk of the individual institution and the risk its default poses to other institutions. The results, therefore, support current prudential measures aimed at reducing the risk of individual institutions deemed systemic to the financial system. For example, in Chile, Banco Santander is required to hold higher capital charges than other banks because of its dominance of the system.

One caveat to bear in mind is that the results reported in this section should not be taken at full face value. Because the purpose of the example was restricted to illustrate the methodology and how to use the different tools, some rough approximations and assumptions were used in the calculations. In particular, assuming normal distributions in the one-factor credit portfolio risk model could neglect tail risk. A more careful analysis would have analyzed whether the latent asset values associated to each institution were better fitted with a fat-tail distribution such as the Student-*t* or the Gumbel distributions. Also, the BCBS formula for correlation assigns lower correlations for higher probabilities of default and imposes an upper bound of 0.24. This shortcoming can be addressed with the use of structural models, as suggested in Section 3.C, assuming a constant, high-correlation value deemed consistent with a tail-risk scenario or specifying correlation as an increasing function of the probability of default. In any case, the two assumptions discussed here would bias down the TCTF capital charges reported in Table 18.2.

5. CONCLUSIONS

The failure of TCTF institutions has been acknowledged widely as a major source of systemic risk to the financial system, as witnessed in the failures of Herstatt Bank in 1974 and Long Term Capital Management in 1998. But only recently, and in response to the large losses in the financial system caused by the subprime crisis of 2008–09, have the policymaking community and regulatory agencies undertaken substantial efforts to revamp the regulatory framework to address the interconnectedness problem.

The proposed TCTF capital charge introduced in this study is an important contribution to the ongoing efforts toward the design of a modern regulatory framework. The proposed capital charge methodology has three important features. First, it builds on an intuitive principle: the capital charge must be proportional to the incremental contribution to societal losses (or risk) that result from the failure of the in-

stitution. Second, by relating the concept of incremental contribution to systemic risk to concepts such as VaR and ES, the TCTF capital charge is aligned with the spirit of Basel II. This alignment will facilitate its adoption and implementation by supervisory agencies and systemic risk regulators. Third, the measurement of the incremental contribution can be accomplished using a simple toolkit of models such as CoRisk analysis, network analysis, and portfolio credit risk models.¹⁷ The proposed TCTF capital charge methodology also may provide a practical way to define the proper perimeter of regulation (Carvajal and others, 2009). Although the methodology was illustrated here for financial institutions, it does not preclude the inclusion of nonfinancial and/or unregulated institutions. In some emerging market countries, nonbank institutions such as department stores are responsible for a substantial amount of financial intermediation and domestic credit to the household sector. In some countries, brokerage houses are important players in the repo market, so their failure could trigger a liquidity crunch in the domestic financial system.¹⁸

Determining what institutions should be covered by the perimeter of regulation can be based on each institution's incremental contribution to societal loss. If this contribution exceeds a predetermined threshold, the institution should be included in the perimeter. Under this procedure, some institutions that are not regulated currently may be subsequently included, while others may be dropped from it if they no longer present a systemic risk to society. The adoption of a dynamic perimeter of regulation not only responds to changes in market practices and business conditions but also helps to allocate scarce regulatory resources more efficiently by focusing them on the more vulnerable and/or riskier institutions from a societal vantage point.¹⁹

The methodology also contributes to the literature on macro-financial linkages in regard to the measurement of risk between the financial and nonfinancial institutions. For instance, it could well be the case that trade financing in one country relies on one domestic bank or, at best, on a reduced number of domestic banks. The TCTF capital charge for these banks should account for the potential impact of their failure on the export–import sector. Although there may not be information about the direct exposure between the banks and the export–import sector, the use of CoRisk analysis can help to compute the capital charges.

The discussion in this chapter assumes that the charge should be applied to the institution that fails. There are valid

¹⁷ The transparency of the methodology and its reliance on a simple toolkit of models facilitates its implementation and simplifies the design of the supporting technical infrastructure and its associated costs. For instance, in the numerical example presented in this chapter, all calculations were performed using standard econometric software and Microsoft Visual Basic.

¹⁸ Ana Carvajal, personal discussion.

¹⁹ During stress periods, however, there may be instances during which the authorities may need to support institutions not deemed too-big-to-fail or too-connected-to-fail to avoid a bank run triggered by rapidly deteriorating investor sentiment.

arguments, however, for applying the charges to the institutions most affected by the failure of the TCTF institution.²⁰ One argument is that the TCTF capital charge, if applied as suggested in the study, potentially may penalize efficient institutions. Their efficiency, arguably, is one reason why they have become TCTF. A second argument is that, from the perspective of a TCTF institution, the linkages with other institutions could depend on the business models of the latter. In this case, the TCTF institution would be punished unfairly for factors beyond its control. One potential way to partly address these concerns is to add up the individual TCTF capital charges and distribute them proportionally to each institution's probability of default.

As with any capital charge, there are some procyclical features in the TCTF capital charge as the probability of default of an interconnected institution increases during economic slowdowns and/or periods of generalized market underperformance. As discussed in Section 3, the procyclicality could be reduced to a certain extent by using estimates of the probability of default conditional on the realization of extreme events. In addition, to reduce the risks to the system, it may be possible to design a prompt corrective action framework where different thresholds for the TCTF capital charges trigger potential remedial actions. In this case, even if the TCTF capital charge is procyclical, its use within a prompt corrective action framework ensures a ceiling on the maximum probability of default of any given institution that partly offsets the procyclicality.

Finally, the imposition of a TCTF capital charge could go a long way toward internalizing the negative externalities associated with TCTF institutions and provide managers with the incentives to strengthen an institution's solvency position and avoid too much homogeneity and excessive reliance on the same counterparties in the financial industry.

But for the TCTF capital charge to have its intended effects, it would be necessary to harmonize regulatory practices, especially in regard to the measurement of interconnectedness risk, across different jurisdictions. The TCTF problem involves systemic bank and financial institutions that operate across the globe and subject to the different regulatory regimes in their home jurisdiction. Fortunately, the need to cope with future financial crisis has strengthened the political will to increase harmonization, as shown, for instance, by the establishment of the Financial Stability Board and the broadening of the surveillance mandate of the IMF. In this context, an intermediate step toward the global adoption of a TCTF charge would be its use as one input for constructing a scorecard for systemic institutions. To avoid conflicts of interest, the calculation of the charge could be performed by a neutral multilateral institution such as the Bank for International Settlements or the IMF.

²⁰ I thank Barry Johnston for raising these arguments.

REFERENCES

- Acharya, Viral V., and Matthew Richardson, 2009, "Government Guarantees: Why the Genie Needs to Be Put Back in the Bottle," *The Economists' Voice*, Vol. 6, No. 11, pp. 1–5.
- Adrian, Tobias, and Markus K. Brunnermeier, 2008, "CoVaR," Staff Report No. 348 (New York: Federal Reserve Bank of New York). Available via the Internet: http://www.newyorkfed.org/research/staff_reports/sr348.html
- Aikman, David, Piergiorgio Alessandri, Bruno Eklund, Prasanna Gai, Sujit Kapadia, Elizabeth Martin, Nada Mora, Gabriel Sterne, and Matthew Willison, 2009, "Funding Liquidity Risk in a Quantitative Model of Systemic Stability," Working Paper No. 372 (London: Bank of England). Available via the Internet: <http://www.bankofengland.co.uk/publications/Pages/workingpapers/2009/wp372.aspx>
- Altman, Edward, 1968, "Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy," *Journal of Finance*, Vol. 23, No. 4, pp. 589–609.
- Andersen, Leif, Jakob Sidenius, and Susanta Basu, 2003, "All Your Hedges in One Basket," *Risk Magazine* (London), November 1, pp. 67–72.
- Andritzky, Jochen, John Kiff, Laura Kodres, Pamela Madrid, Andrea Maechler, Aditya Narain, Noel Sacasa, and Jodi Scarlata, 2009, "Policies to Mitigate Procyclicality," Staff Position Note 09/09 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/ft/spn/2009/spn0909.pdf>
- Artzner, Philippe, Freddy Delbaen, Jean-Marc Eber, and David Heath, 1999, "Coherent Measures of Risk," *Mathematical Finance*, Vol. 9, No. 3, pp. 203–28.
- Basel Committee on Banking Supervision, 2004, "Modifications to the Capital Treatment for Expected and Unexpected Credit Losses" (Basel, January). Available via the Internet: <http://www.bis.org/publ/bcbs104.htm>
- Bernanke, Ben S., 2009, "Financial Regulation and Supervision after the Crisis—The Role of the Federal Reserve," speech at the Federal Reserve Bank of Boston 54th Economic Conference, Boston, October 23. Available via the Internet: <http://www.federalreserve.gov/newsevents/speech/bernanke20091023a.htm>
- Bohn, Jeffrey R., 2000, "A Survey of Contingent-Claims Approaches to Risky Debt Valuation," *Journal of Risk Finance*, Vol. 1, No. 3, pp. 53–70.
- Brunnermeier, Markus K., Andrew Crockett, Charles Goodhart, Avinash Persaud, and Hyun-Song Shin, 2009, "The Fundamental Principles of Financial Regulation," *Geneva Reports of the World Economy*, No. 11 (Geneva and London: International Center for Monetary and Banking Studies and Centre for Economic Policy Research).
- Capuano, Christian, Jorge A. Chan-Lau, Giancarlo Gasha, Carlos Medeiros, Andre O. Santos, and Marco R. Souto, 2009, "Recent Advances in Credit Risk Modeling," IMF Working Paper 09/162 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=23125>
- Carvajal, Ana, Randall Dodd, Michael Moore, Erlend Nier, Ian Tower, and Luisa Zanolini, 2009, "The Perimeter of Financial Regulation," Staff Position Note 09/07 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=22826>
- Chan-Lau, Jorge A., 2008, "The Globalization of Finance and Its Implications for Financial Stability: An Overview of the Issues,"

- International Journal of Banking, Accounting and Finance*, Vol. 1, No. 1, pp. 3–29.
- , 2009, “Default Risk Codependence in the Global Financial System: Was the Bear Stearns Bailout Justified?” in *The Banking Crisis Handbook*, edited by Greg Gregoriou (Boca Raton, London, New York: CRC Press).
- , 2010, “Regulatory Capital Charges for Too-Connected-to-Fail Institutions: A Practical Proposal,” *Financial Markets, Institutions, and Instruments*, Vol. 19, No. 5, pp. 355–80.
- , and Toni Gravelle, 2005, “The END Indicator: A New Indicator of Financial and Nonfinancial Corporate Sector Vulnerability,” IMF Working Paper 05/231 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=18654>
- de Servigny, Arnaud, and Olivier Renault, 2004, *Measuring and Managing Credit Risk* (New York: McGraw-Hill).
- Gauthier, Celine, Alfred Lehar, and Moez Souissi, 2010, “Macroprudential Capital Requirements and Systemic Risk,” Working Paper 2010-4 (Ottawa: Bank of Canada). Available via the Internet: <http://www.bankofcanada.ca/2010/01/publications/research/working-paper-2010-4>
- Gibson, Michael, 2004, “Understanding the Risk of Synthetic CDOs,” Working Paper 2004-36, Finance and Economics Discussion Series (Washington: Board of Governors of the Federal Reserve). Available via the Internet: <http://www.federalreserve.gov/pubs/feds/2004/200436/200436pap.pdf>
- Huang, Xin, Hao Zhou, and Haibin Zhu, 2009, “A Framework for Assessing the Systemic Risk of Major Financial Institutions,” *Journal of Banking and Finance*, Vol. 33, No. 11, pp. 2036–49.
- Hull, John C., and Alan White, 2004, “Valuation of a CDO and an Nth to Default CDS without Monte Carlo Simulation,” *Journal of Derivatives*, Vol. 12, No. 2, pp. 8–23.
- International Monetary Fund, 2009a, *Global Financial Stability Report*, World Economic and Financial Surveys (Washington, April). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2009/01/index.htm>
- , 2009b, *Global Financial Stability Report*, World Economic and Financial Surveys (Washington, October). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2009/02/index.htm>
- , 2009c, *World Economic Outlook*, World Economic and Financial Surveys (Washington, October). Available via the Internet: <http://www.imf.org/external/pubs/ft/weo/2009/02>
- Koenker, Roger, and Gilbert Bassett Jr., 1978, “Regression Quantiles,” *Econometrica*, Vol. 45, No. 1, pp. 33–50.
- Lando, David, 2003, *Credit Risk Modeling* (Princeton: Princeton University Press).
- Li, David X., 2000, “On Default Correlation: A Copula Function Approach,” *Journal of Fixed Income*, Vol. 9, No. 4, pp. 43–54.
- Litterman, Robert, 1996, “Hot Spots™ and Hedges,” *Journal of Portfolio Management*, Vol. 23 (December), pp. 52–75.
- Love, Brian, 2009, “IMF Exploring Insurance Levy on Banks,” Reuters (Paris, November 8). Available via the Internet: <http://www.reuters.com/article/2009/11/08/us-g20-brown-imf-exclusive-idUSTRE5A71HU20091108>
- Merton, Robert, 1974, “On the Pricing of Corporate Debt: The Risk Structure of Interest Rates,” *Journal of Finance*, Vol. 29, No. 2, pp. 229–70.
- Milne, Frank, 2009, “The Complexities of Financial Risk Management and Systemic Risks,” *Bank of Canada Review*, Summer, pp. 15–29. Available via the Internet: <http://www.bankofcanada.ca/2009/06/publications/periodicals/boc-review-article/complexities-financial-risk-management>
- Rennhack, Robert, and others, 2009, “Global Financial Regulatory Reform: Implications for Latin America and the Caribbean (LAC),” IMF Staff Position Note 09/19 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/ft/spn/2009/spn0919.pdf>
- Tarashev, Nikola, Claudio Borio, and Kostas Tsatsaronis, 2009, “The Systemic Importance of Financial Institutions,” *BIS Quarterly Review*, September, pp. 75–87 (Basel: Bank for International Settlements). Available via the Internet: http://www.bis.org/publ/qtrpdf/r_qtr0909.htm
- Tarullo, Daniel K., 2009, “Regulatory Reform,” Testimony before the Committee on Financial Services, U.S. House of Representatives (Washington, October 29). Available via the Internet: <http://www.federalreserve.gov/newsevents/testimony/tarullo20091029a.htm>
- Vasicek, Oldrich, 1997, “Probability of Loss on a Loan Portfolio,” Working Paper (San Francisco: KMV Corporation).

SECTION B

The Extreme Value Theory Approach

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Introduction to the Extreme Value Theory Approach to Stress Testing

SROBONA MITRA

As risks build up, systemic events could materialize very rapidly. The speed at which a financial crisis spreads depends on the degree of interconnectedness of financial institutions, within and across countries. This interconnectedness may be attributable to specific interinstitutional exposures, such as interbank loans and deposits and repo transactions, or to exposures to aggregate risk, such as investor confidence and real economic and financial cycles. Thus, it is important to try and gauge the extent of potential spillovers well in advance of an actual stress event.

There are two major obstacles to modeling spillovers: the lack of published data of actual exposures between institutions and the (in)frequency of large systemic events, often referred to as “tail-risk events.” Instead, data on market prices, such as equity prices, bond prices, and credit default swap (CDS) spreads, may be used to derive the dependence of a financial institution or a country on an event in another institution or a country.

There are various ways of modeling spillovers or distress dependence. Some are based on econometric techniques, which assume that the distribution of the shocks is known, or “parametric” methods. The “nonparametric” methods allow the data to speak for themselves without imposing restrictions on the distribution of the shocks:

- The CoRisk model provides a sense of the contribution of a financial institution to systemwide financial risk (e.g., Chan-Lau in Chapters 17 and 18). The Diebold-Yilmaz spillover matrix (Diebold and Yilmaz, 2009) is another parametric method that uses vector autoregressions and the associated variance decompositions to construct directional spillovers. The estimates from the latter model are based on central

moments and do not provide a sense of tail-risk movements.

- Nonparametric “copula-based” measures of spillover risk have become increasingly popular in recent years (e.g., Jobst and Gray in Chapter 26 and Segoviano and Goodhart in Chapter 32). The concept is based on the dissection of the joint multivariate distribution of returns (or other quantities) of several financial systems or institutions into the individual probability distribution and the dependence structure between the institutions. By “letting the data speak,” the spillover estimates from these methods are not tainted by the model. Moreover, these models provide several outputs including probability of default (PD) of an individual institution, the joint PD of the entire system, the PD of an institution conditional on one or more institutions defaulting, and the dependence structure of the institutions over time. However, these models are still in their infancy and are difficult to manipulate.

The *extreme value theory* (EVT) approach sits somewhere in between. It was introduced by Gropp and Moerman (2004) and Gropp, Lo Duca, and Vesala (2006) and provides a framework for analyzing extreme events (tail risks) that are rare but which could have a severe impact on the soundness of financial institutions or systems. It employs econometric and statistical analyses to estimate the potential for domino effects during a tail-risk event. The CoRisk method would be the closest comparator.

The EVT approach presented in this book first identifies all extreme events in the data by looking at, say, the 1st or the 5th percentile of the joint distribution of returns. These typically comprise weekly or daily observations of equity or

CDS prices or the market value of assets. All returns lying in the left tail, that is, the ones below the defined thresholds, are called “exceedances.” Distress dependence then is estimated by using a logit model to account for the fatness of the tails of the distribution of exceedances. The probability of an exceedance is estimated conditional on exceedances in other financial institutions or centers, after controlling for common shocks such as extreme conditions in the world equity markets, the country’s stock markets, and real sector indicators. The results are, in most cases, unchanged if we assume an extreme value distribution instead of logistic distribution.

The EVT approach is more versatile in capturing the extent of spillover risks from a particular financial system or institution. Take, for example, a country in which financial markets have been enjoying a calm period for a number of years. The average correlations in movements (given by the mean of the distribution) in CDS spreads or equity returns across financial institutions might appear to be very low. However, a lack of correlation (a linear concept) may not imply independence. Extreme movements on a day-to-day or week-on-week basis—for instance, negative weekly returns or volatility exceeding the 5th-percentile left tail of the joint distribution of weekly returns—could point to high spillover risks during distress episodes.

The EVT models applied by Duggar and Mitra (Chapter 20) and by Chan-Lau and others (Chapter 21) are easy to

maneuver and apply, with any standard econometric software package sufficient for analysis. Specific precrisis examples explored include (1) the risk of spillovers across borders for the Irish banking system; and (2) the direction of spillovers among the 25 largest banks in the world and that of the major banks within the European Union countries. In these examples, the distress-dependence matrices are largely static, and the sample periods are fairly long. The analyses can be made time varying by repeating the exercise over a rolling window, albeit one that is sufficiently long to provide an adequate number of observations of extreme movements.

REFERENCES

- Diebold, Francis X., and Kamil Yilmaz, 2009, “Measuring Financial Asset Return and Volatility Spillovers, with Application to Global Equity Markets,” *Economic Journal*, Vol. 119, No. 534, pp. 158–71.
- Gropp, Reint, Marco Lo Duca, and Jukka Vesala, 2006, “Cross-Border Bank Contagion in Europe,” ECB Working Paper No. 662 (Frankfurt: European Central Bank). Available via the Internet: <http://www.ecb.europa.eu/pub/scientific/wps/date/html/wps2006.en.html>
- Gropp, Reint, and Gerard Moerman, 2004, “Measurement of Contagion in Bank Equity Prices,” *Journal of International Money and Finance*, Vol. 23, No. 3, pp. 405–59.

External Linkages and Contagion Risk in Irish Banks

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Increasing financial integration makes the assessment of cross-country linkages crucial for effective financial surveillance. This study estimates contagion risk between large Irish banks and European and U.S. banks during 1994–2005, using distance-to-default measures and the methodology of extreme value theory. Employing an ordered logit model and controlling for Ireland-specific and global shocks, we find evidence of significant contagion risk coming from the United Kingdom, the United States, and the Netherlands toward Ireland. We also find that patterns of contagion to Irish banks have shifted over time, coming from the United Kingdom in the pre-euro period and from the United States in the post-2001 period.

METHOD SUMMARY

Overview	The logit model estimates conditional correlations between banking systems during extreme movements in distance-to-default indicators.
Application	The method is appropriate for modeling spillover risk.
Nature of approach	Parametric approach—conditional correlations are based on an econometric model.
Data requirements	Accounting information on bank liabilities; market price-based data on bank equity prices, interest rate (term) spreads, stock market indices.
Strengths	The results provide insights into the direction of spillovers across banking systems during crises; the methodology is transparent and can be easily replicated; data on actual exposures between institutions are not required in order to deduce spillovers.
Weaknesses	The model assumes that there are enough extreme movements in the data to provide insights about actual spillovers during a crisis, and it assumes that the movements in the market-based data are sufficient to summarize information on both common exposures and actual intra-institution exposures, when in reality the market does not have adequate information.
Tool	The EvIEWS program codes are available in the toolkit, which is on the companion CD and at www.elibrary.imf.org/stress-test-toolkit . Contact author: S. Mitra.

In the past decade, Ireland has become increasingly financially open. Global trends in financial liberalization, innovation, and banking consolidation activity, as well as greater bank reliance on wholesale funding, increasing competition from foreign-owned banks, and the development of the Irish Financial Services Center (FSC) have led to increasing inter-

national exposure for the banking system. At the same time, the Irish domestic banking system has remained relatively concentrated, with Bank of Ireland (BoI), Allied Irish Banks PLC (AIB), and Anglo Irish Bank Corp. PLC (Anglo IB) representing about 45 percent of total banking assets but nearly 80 percent of the domestic retail market.

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Box 20.1. Sources of Contagion—A Literature Survey

Sources of cross-country contagion risk can be grouped into the following:

- *Bank deposit runs.* Traditionally, the literature has focused on the implications of bank deposit runs for the payment system, the money supply, and financial intermediation. Liquidity shocks hitting one bank could cause depositors to also run on other solvent banks, in fear of lacking reserves of liquid assets in the banking system (Freixas, Parigi, and Rochet, 2000). The runs could be triggered by rumors on banking system fragility, reputation or operational risk in countries where banks may have subsidiaries, branches, and even representative offices.
- *Wholesale funding channels.* With the developments in technological change, deregulation, globalization, and the increased use of financial markets, the focus has shifted to systemic risks arising at the wholesale level (intermediation, investment banking, securities trading, asset management, business banking), and concentrating on the largest and most complex financial institutions.
- *Liquidity and credit risk in interbank markets.* One possible channel of contagion is through the interbank market (Allen and Gale, 2000; Freixas, Parigi, and Rochet, 2000) in the form of liquidity shocks when banks withdraw their deposits at other banks, or in the form of credit risk when deposits at other banks are not being repaid.
- *Unidentified channels.* There could also be contagion in the financial markets in the absence of explicit links, when in the presence of asymmetric information difficulties in one market are perceived as a signal of possible difficulties in others (Morgan, 2002).

In this study, following De Nicoló and Tieman (2005) and Gropp, Lo Duca, and Vesala (2006),¹ we explore whether international openness is accompanied by increasing financial risk for the three major Irish banks.² We explore trends in international interdependencies between Irish banks and banks in the major European countries and in the United States over the period 1994–2005 and analyze whether international linkages have led to possible contagion risk. We assess whether and to what extent large negative shocks in banks of these countries affected banks in Ireland. In particular, we measure the probability that a higher number of banks in Ireland will experience a large negative shock at the same time as banks in other countries (“co-exceedances”). The existence and the magnitude of these effects have implications for the monitoring of financial stability, as contagion risk is widely perceived to be an important element of banking crises and systemic risk. Understanding the direction of such risk is important for policymakers to better focus limited financial supervisory resources.

We use the term “contagion” to mean the transmission of an idiosyncratic shock affecting one bank or a group of banks to other banks or other banking sectors, using distance to default (DD) as a measure of bank risk. There could be a number of possible contagion channels (Box 20.1) and several ways to measure contagion risk: estimating autocorrelation and survival time tests using historical data on bank failures (Calomiris and Mason, 2000), using the interbank market lending exposure matrix and measuring domino ef-

fects of insolvency in one bank spreading to others (Furhine, 2003), or analyzing the correlations of stock returns of banks to measure interdependencies (De Nicoló and Kwast, 2002).

The study follows the approach employed in the recent literature, which estimates contagion risk by using the DD as a comprehensive measure of default risk (Gropp and Moerman, 2004; De Nicoló and Tieman, 2005; Gropp, Lo Duca, and Vesala, 2006). The DD represents the number of standard deviations away from the default point—the point at which the book value of liabilities of the bank is just equal to the market value of assets. Unlike unadjusted stock returns, the DD combines information about stock returns with leverage and asset volatility information, thus encompassing the most important determinants of default risk. The higher the DD, the greater the distance of the bank from default point and the lesser the risk or probability of default of the bank. The benefit of this approach is that comovements in DD can be analyzed without specifying a particular channel of contagion. Rather, these comovements reflect interdependencies between domestic and cross-border banks encompassing all potential channels of contagion, including that occurring in the absence of explicit links between banks.

We first use rolling correlations of changes in the DD of Irish banks with major European and U.S. banks to analyze trends in cross-country interdependencies. Then, we use an ordered logit model to estimate contagion risk: the probability of Irish banks experiencing a large shock on the same day (co-exceedances) as banks in other countries after controlling for Ireland-specific and global factors—common shocks that could affect all banks simultaneously. Large shocks are defined by the bottom 15th percentile of the weekly difference in the daily DD of all banks. The 15th percentile threshold was used, instead of (say) the bottom 5th percentile, because we wanted to include incidents that have a

¹ Gropp, Lo Duca, and Vesala (2006) find evidence of significant cross-border contagion in Europe (Ireland not included in the sample) during January 1994 to January 2003.

² The choice of the banks was guided by the availability of data on their equity prices.

higher probability of occurrence—those that could be associated with large shocks and not necessarily only with crisis scenarios.

Analyzing trends in rolling correlations in the percentage changes in the DD, we find evidence of increasing simultaneous occurrence of shocks across countries, suggesting increasing global interdependencies over the last decade. Further, following the approach of Gropp, Lo Duca, and Vesala (2006), we find evidence in favor of significant cross-border contagion from the United Kingdom, the United States, and the Netherlands to Ireland. Moreover, it seems that contagion risk has shifted from coming from Europe in the pre-euro period to coming from the United States in the post-euro period. These findings have policy implications for the monitoring of financial stability and could direct scarce supervisory resources in managing foreign exposures.

However, all results come with significant caveats: we are using equity prices of banking *groups* and yet mainly discussing international links typically associated with banking without explicitly stating links through securities and insurance (the top two Irish banks have links to insurance companies). We are using data available for only a small group of listed banks in each country—so it might not be representative of the system, and the banks in Ireland comprise less than 50 percent of the system (although a much larger share of the retail market). The number of observations experiencing large shocks is low; the results could, therefore, be driven by the large shocks associated with the tech-bubble burst in 2000.

The following section describes various external linkages of Irish banks. Trends in interdependencies using rolling correlations of DDs between the top three Irish banks and banks from other European countries and the United States are analyzed in Section 1. An ordered logit model, to estimate the probability of co-exceedances in Irish banks, is presented in Section 2, while Section 3 summarizes and concludes the discussion.

1. SOME EVIDENCE OF EXTERNAL LINKAGES OF THE IRISH BANKING SECTOR

This section aims to find the channels and the direction of potential contagion risk from other countries to Ireland, looking at external linkages of the Irish banks. The Irish financial system has been more closely integrated with the United Kingdom in the past, but links with continental Europe and the United States have strengthened. There are many external linkages of the Irish banking sector, stemming from direct equity exposures in cross-border banks; direct exposure through loan books in other countries; deposit and funding sources from other countries or from numerous foreign banks operating in Ireland; stock market

participation in other countries, through securities and asset management firms; and holding of credit risk transfer instruments written on assets located in another country, indirectly exposing Irish banks to international shocks. Each of these is explored in turn.³

First, Irish banks have foreign equity exposure through their expansion overseas. Both AIB and BoI have sizable operations in the United Kingdom and have been growing their niche wholesale international businesses. Both banks own universal banks in Northern Ireland. AIB has a large stake in a U.S. regional bank and majority-owns a Polish bank—the U.S. and Polish investments contributed a combined 16 percent to AIB's pretax profit in the first half of 2005 (Standard and Poor's, 2006), although the Polish operations also result in relatively high nonperforming loans.

Second, Irish banks have large loan-book exposures abroad. The two largest banks, AIB and BoI, are geographically diversified—each with almost equal share of domestic and foreign assets. Nearly 28 percent of AIB, 44 percent of BoI, and 41 percent of Anglo IB loan-book exposures were in the United Kingdom. Although AIB held over 20 percent stake in a U.S. bank, U.S. operations were only 2 percent of its loan book. Anglo IB, in contrast, has about 5 percent of its loan book exposed to the United States; without equity exposures in the United States, it operates through a representative office.

Third, the increased reliance on wholesale funding, including interbank borrowing and capital market issues, is another potential source of international interdependencies. The average loan-to-deposit ratio for Ireland exceeds 150 percent, one of the highest for industrial countries. All major Irish banks are dependent for funds on the interbank and securities markets—AIB and BoI fund about 40 percent of lending in the wholesale market, while the market funding requirement for Anglo IB is about 35 percent. The overwhelming bulk of both nonresident interbank borrowing (83 percent) and debt securities issued and held by non-residents (83 percent) in 2004 were vis-à-vis non-euro area residents.

Fourth, Irish banks buy risk protection mainly from banks of other countries. The underlying asset in structured credit risk transfer (CRT) products include mainly loans and bonds issued by financial and nonfinancial firms; mortgages (for asset-backed securities); and financial and nonfinancial firm debt as underlying asset for the more traditional CRT (e.g., mortgage indemnity guarantee). The United States and the United Kingdom are the main counterparty locations selling risk protection to Irish banks. Other countries

³ Some empirical evidence of foreign contagion risk in Ireland is provided by Gropp and Moerman (2004), who study the joint occurrence of both positive and negative extreme shocks in banks' DD among large European Union banks in the period January 1991 to January 2003 and identify AIB and BoI as among the systemically important banks in the European Union. Moreover, they find a contagion effect from the United Kingdom to Ireland.

include France, Germany, Canada, Switzerland, the Netherlands, Italy, and Poland. The major currencies of denomination are the euro and the dollar, along with the pound sterling. Irish banks also issue covered bonds—BoI has transferred the bulk of its domestic residential mortgage assets to a designated mortgage credit institution, which has a banking license to issue mortgage-covered securities. These covered bonds are used both for hedging interest rate risk and for generating additional funding. Almost 60 percent of these securities are held by other euro area members, while 25 percent are held in dollars by other countries.

Fifth, Irish banks are directly and indirectly exposed to property markets abroad. All the top three banks have loan-book exposures to the United Kingdom property market. At least AIB and Anglo IB sell mortgages in the United States—AIB through its U.S. subsidiary and Anglo IB through its representative office. The latter is more focused on commercial property lending in the United States. BoI has launched a new venture with a leading Spanish bank, La Caixa, to provide extra mortgage options for Irish people buying property in Spain, which includes equity release from existing BoI mortgages. Part of the real estate price risk is mitigated by the Irish banks buying risk protection against these exposures. Irish legislation on covered bonds broadens the scope of risk protection by making loans made from countries such as the United States, Canada, Switzerland, and Japan eligible for the collateral pool.⁴ However, Irish banks could be indirectly exposed to property markets by selling risk protection (buying of covered bonds, credit default swaps, and mortgage-backed securities) to other banks that are exposed to foreign property markets. From anecdotal evidence, some small FSC banks, exposed to international property markets, are selling credit default swaps to other domestic-oriented banks, indirectly exposing the latter to these property markets even with no loan-book exposure.

Sixth, Bank for International Settlements (BIS) data on banks resident in Ireland show their net asset positions vis-à-vis banks and nonbanks in various countries (Figure 20.1). During 2001–05, the Irish resident banks had a negative net asset position vis-à-vis banks and nonbanks in the United Kingdom; an overall positive net asset position vis-à-vis banks and nonbanks in the United States; and a large positive net asset position with Italy, Spain, and France. However, much of these positions could belong to foreign-owned banks operating through branches in the FSC⁵ in Dublin—

some of these banks operate almost exclusively with nonresidents and have some links, though limited, with the domestic economy.

Seventh, BIS-consolidated statistics report Irish banks' on-balance-sheet financial claims vis-à-vis the rest of the world. This type of data would give us an idea of the extent of exposure of the Irish banks during a credit event in these countries. Irish banks are exposed mostly to the United Kingdom and Germany among EU countries—and to the United States (although small in comparison) among non-EU countries.

Given the above evidence, we could expect the following possible channels of contagion risk: between Ireland and the United Kingdom and United States, on the basis of equity exposure, loan exposure, and exposure via the interbank and the securities markets; between Ireland and the Netherlands, on the basis of the presence of large Netherlands-owned banks in Ireland; and between Ireland and Italy, Spain, France, and Germany on the basis of interbank market exposure. The direction of contagion is more challenging to establish. However, evidence from BIS data (Figure 20.1) and the discussion in this section suggests the following:

- Countries with which Ireland has a net nonbank asset position would likely expose Ireland to credit risk in loans and other asset markets: Austria, France, Germany, Italy, the Netherlands, Spain, and the United States.
- Ireland's positive net bank exposure in Italy and Spain, and previously the United States, Austria, and France, possibly exposes Ireland to credit risk that could lead to liquidity risk.⁶
- Ireland has a negative net bank-asset position, which exposes it to liquidity risk if some problem were to arise in the country from which Ireland is borrowing: Germany, the United Kingdom, Belgium, the Netherlands, the United States, Austria, and France.

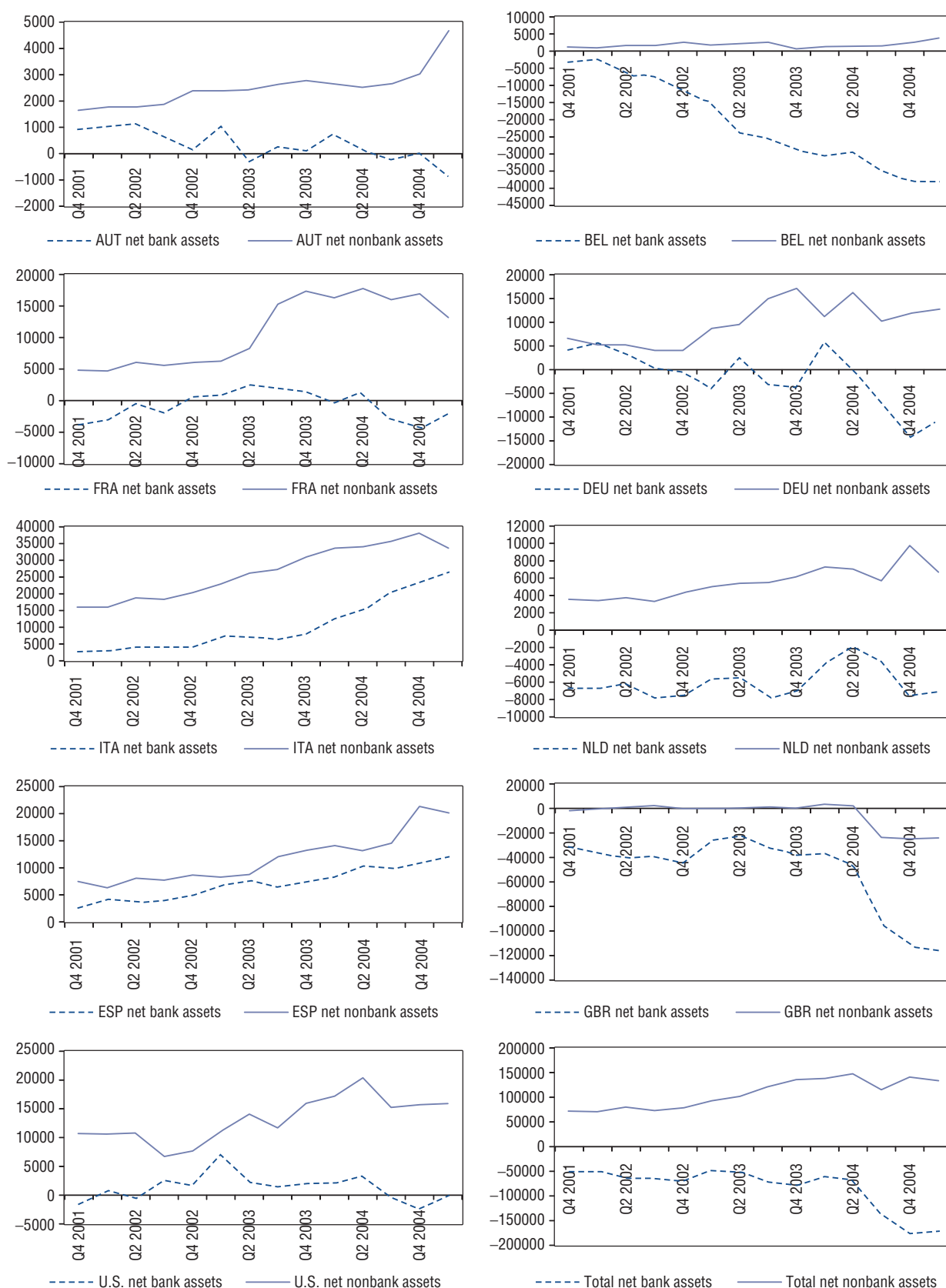
There could also be contagion in the absence of explicit links, when due to asymmetric information, difficulties in one country/market are perceived as a signal of possible difficulties in others. The interdependencies mentioned above may be overstated to the extent that they include banks resident in Ireland but operating in the FSC mainly with nonresidents; however, data were not sufficient to exclude these from the BIS data sample. We next look at only the three largest domestic banks—which have nearly 80 percent of the retail market—and their interlinkages and possible contagion risk channels with banks in other European countries and in the United States.

⁴ With the exception of Luxembourg, most European countries limit the asset pool to European Economic Area assets.

⁵ The FSC was established in Dublin in 1987 (known at that time as the International Financial Services Center, or the IFSC) to facilitate financial operations with nonresidents and was endowed with corporate tax benefits. These tax benefits have now been lifted. There are about 450 international institutions that have established offices in the FSC (accounting for about 40 percent of banking assets), including 50 percent of the 50 largest financial institutions in the world—including major banks from the United States, United Kingdom, the Netherlands, Italy,

and Germany. In turn, almost all domestic credit institutions also conduct business from the FSC.

⁶ The Irish banking supervisors have a specific requirement for banks on liquidity maintenance: 25 percent of deposits, and short-term liabilities have to be covered by liquid assets.



Source: Authors, based on Bank for International Settlements banking statistics, <http://www.bis.org/statistics/bankstats.htm>.

Note: AUT = Austria; BEL = Belgium; DEU = Germany; ESP = Spain; FRA = France; GBR = the United Kingdom; ITA = Italy; NLD = the Netherlands.

Figure 20.1 Ireland-Resident Banks Net Asset Position vis-à-vis Banks and Nonbanks in Various Countries (in millions of U.S. dollars)

Box 20.2. The Distance to Default Measure and Data Issues

The derivation of distance to default (DD) is described in detail in Gropp, Lo Duca, and Vesala (2006) and in Gropp and Moerman (2004), and in the case of a portfolio of bank assets in De Nicoló and Tieman (2005). The distance-to-default measure is based on the structural valuation model of Black and Scholes (1973) and Merton (1974), and is defined as follows:

$$DD_t = \frac{\ln\left(\frac{V_{A,t}}{X_t}\right) + \left(r - \frac{\sigma_A^2}{2}\right)T}{\sigma_A\sqrt{T}},$$

where $V_{A,t}$ is the firm's assets value with mean r and volatility σ_A , and X_t is the book value of the debt at time t , that has maturity equal to T . The market value of equity of the firm is viewed as a call option on the firm's assets, V_A , with time to expiration equal to T . The strike price of the call option is the book value of the firm's liabilities, X_t . Default occurs when the value of the firm's assets is less than the strike price—that is, when the ratio of the value of assets to debt is less than one. The distance to default tells us by how many standard deviations the log of this ratio needs to deviate from its mean in order for default to occur.

An estimation of distance to default requires knowing both the asset value and asset volatility of the firm. The required values, however, correspond to the forward-looking *economic* values rather than the accounting figures, and it is not appropriate to use balance-sheet data for estimating these two parameters. Instead, the asset value and volatility are estimated using equity data. The distance-to-default measures we use are taken from IMF's Monetary and Capital Markets Department Distance-to-Default Database, with the methodology described in Vassalou and Xing (2004), except that the value of assets is taken to be equal to the value of equity plus the book value of liabilities. At each date, the value of assets, the return on assets and its volatility are derived using the Black-Scholes option-pricing formula, using one year of daily equity return data preceding the estimation date, and the accounting value of liabilities for the relevant year.

Declines in the ($V_{A,t}/X_t$) ratio are equivalent to declines in capitalization. Thus, the distance-to-default measure combines information about equity returns with leverage and asset volatility information, hence encompassing the most important determinants of default risk. Empirical studies have shown that the distance to default is a good predictor of corporate defaults (Moody's KMV), and predicts banks' downgrades in developed and emerging market countries (Chan-Lau, Jobert, and Kong 2004; Gropp, Vesala, and Vulpes 2006).

We use a data set of daily distance-to-default data for 40 banks in eight countries: France (two banks), Germany (four banks), Ireland (three banks), Italy (six banks), the Netherlands (two banks), Spain (four banks), the United Kingdom (five banks), and the United States (14 banks), for the period January 1994 to November 2005 (Table 20.1). The data set includes all banks in these countries that are listed at a stock exchange and whose distances to default are available from the Distance-to-Default Database. We dropped four banks for which the distances to default were not available for the entire period (one bank for France, one bank for the United Kingdom, and two banks for Italy). In general, the banks in the sample are quite large relative to the population of banks in the European Union, and represent a high fraction of total assets of commercial banks in each country. For each bank, the sample contains 3,105 daily observations (except one U.K. bank with 2,522 observations and one bank from the Netherlands with 2,803).

2. TRENDS IN INTERDEPENDENCIES USING DISTANCE-TO-DEFAULT INDICATORS

This section describes correlations of banking risks between Ireland and other countries after a discussion of the data. We use the distance-to-default indicator to measure bank financial risk; the DD combines information about stock returns with leverage and asset volatility information, thus encompassing the most important determinants of default risk. The cross-country correlation of changes in the DD would indicate interdependencies arising from a broad set of channels, including contagion occurring in the absence of explicit links between banks.

The DD is based on the Black-Scholes option-pricing model (Black and Scholes, 1973) and is estimated using stock price data (Box 20.2). A bank's equity is viewed as a call option on the bank's assets, with strike price equal to the current book value of total liabilities. When the value of the bank's assets is less than the strike price, its equity value

is zero. The DD represents the number of asset value standard deviations that the bank is away from the default point, where the default point is defined as the point at which the liabilities of the bank are just equal to the market value of assets (alternatively, the point where the stock price is zero). The market value of assets is not observable but is estimated using equity values and accounting measures of liabilities.

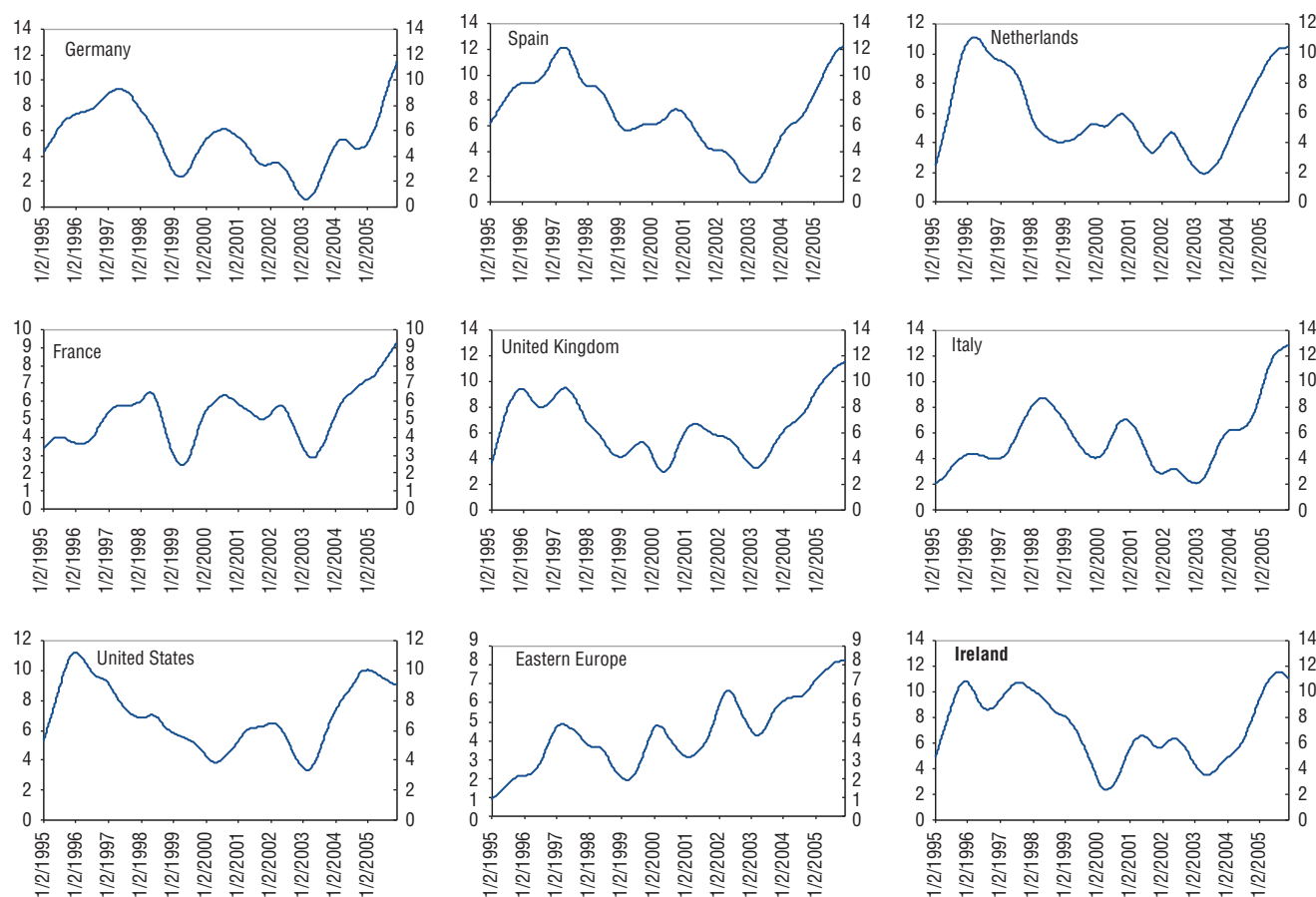
The banks in the sample are just about five standard deviations from the default point—the (pooled) average DD is 5.2; the median, 5.0. There is some variation among banks—the mean DD by bank ranges from 3.3 (Italy) to 7.7 (Spain). The three Irish banks have a mean DD of about 6 (median about 5.7). Figure 20.2 shows the trends in the systemwide distances to default by country, including an Eastern Europe average comprising Poland and Hungary.⁷ There seems to be a general increase in the DD starting in 2003 for all countries, indicating a global improvement in bank health in the

⁷ Weighted average DD of each listed bank, weighted by assets.

TABLE 20.1

Sample Banks					
	Bank	Country		Bank	Country
1	Allied Irish Banks	Ireland	21	Royal Bank of Scotland	U.K.
2	ANG.IR.BK	Ireland	22	Standard Chartered	U.K.
3	Bank of Ireland	Ireland	23	Bankgesellschaft Berlin	Germany
4	Citigroup	USA	24	Bayerische Hypo- und Vereins Bank	Germany
5	Bank of America	USA	25	Commerzbank	Germany
6	JP Morgan Chase & Co.	USA	26	Deutsche Bank	Germany
7	Wells Fargo & Co.	USA	27	BNP Paribas	France
8	Wachovia	USA	28	Societe Generale	France
9	US Bancorp	USA	29	Banco Espanol de Credito	Spain
10	Suntrust Banks	USA	30	Banco Popular Espanol	Spain
11	Nat. City	USA	31	Banco Santander Central Hispano	Spain
12	Bank of New York	USA	32	BBV Argentaria	Spain
13	BB&T	USA	33	ABN Amro Holding	Netherlands
14	Fifth Third Bancorp	USA	34	Fortis (AMS)	Netherlands
15	State Street Corp.	USA	35	Unicredito Italiano	Italy
16	Keycorp	USA	36	San Paolo IMI	Italy
17	PNC Financial Services Group	USA	37	Capitalia	Italy
18	Barclays	U.K.	38	Banca Intesa	Italy
19	Hongkong and Shanghai Banking Corporation Holdings (ORD \$0.50)	U.K.	39	Banca Intesa RNC	Italy
20	Lloyds TSB GP.		40	Unicredito Italiano RNC	Italy

Source: Authors.



Source: Authors.

Note: Each panel shows the aggregated DD of sample banks—weighted by assets—in a country. See Table 20.1 for the sample of banks from each country. The aggregates are HP-filtered. An increase shows a decline in systemic risks.

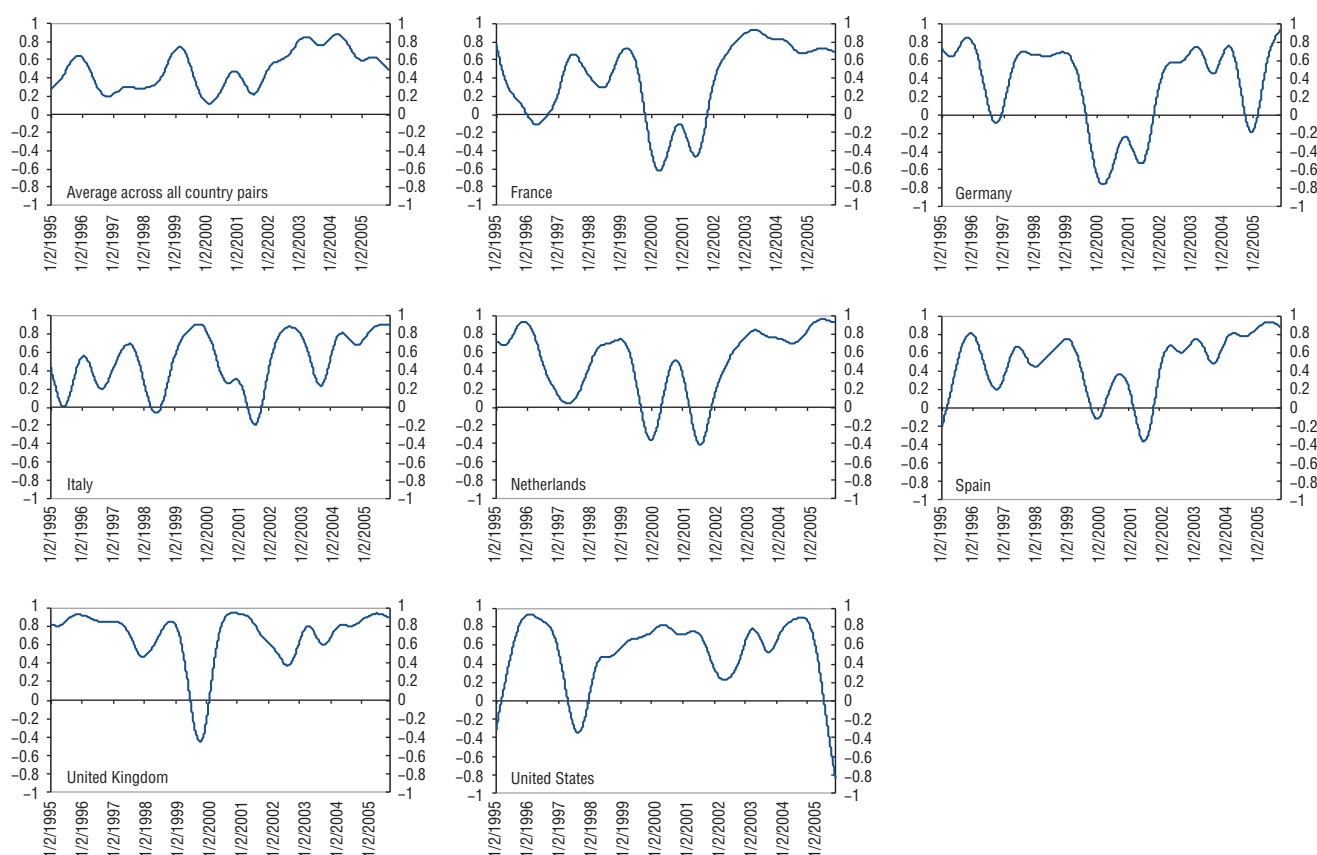
Figure 20.2 Hodrick-Prescott (HP) Trend of Banks' Aggregate Distance to Default (DD) by Country

TABLE 20.2
Correlations in Distance to Defaults

Ireland and	1994–2005	Pre-Euro 1994–98	Post-Euro 1999–2001	Post-September 11 2001–05
France	0.31	0.34	–0.53	0.89
Germany	0.64	0.61	–0.69	0.81
Italy	0.50	0.37	0.18	0.88
Netherlands	0.71	0.61	–0.39	0.94
United Kingdom	0.81	0.68	0.63	0.92
United States	0.73	0.45	0.83	0.76
Spain	0.76	0.74	–0.37	0.91
Eastern Europe	0.03	–0.12	–0.71	0.81

Source: Authors.

Note: Cross-country correlations in banking systems' distance to default. The sample covers January 3, 1994–November 25, 2005.



Source: Authors.

Note: The top-left corner chart shows the HP-trend of 249-day rolling average correlations of DDs across all country pairs. The rest of the charts show the same for correlations among Ireland and other countries. The HP-filter uses a lambda of 6812100.

Figure 20.3 Hodrick-Prescott Trend of One-Year Rolling Correlations of Distance to Default (DD)—Average of All Country Pairs, and between Ireland and Other Countries

last two years of the sample period. There also seems to be an upward trend over the past decade for France, Italy, and Eastern Europe. Irish banks suffered a trend decline in the DD around 1997 that continued until end-1999—this partly coincides with the experiences of the United States, the United Kingdom, and (partially) the Netherlands, Germany, and Spain, as does the continued recovery after 2003. They had

suffered another negative shock during 2002—likely the aftereffects of September 11, 2001, and possibly connected with the AIB's U.S. subsidiary scandal in 2002. Irish banks have recovered starting in 2003 and have DD levels comparable to the levels of the late 1990s.

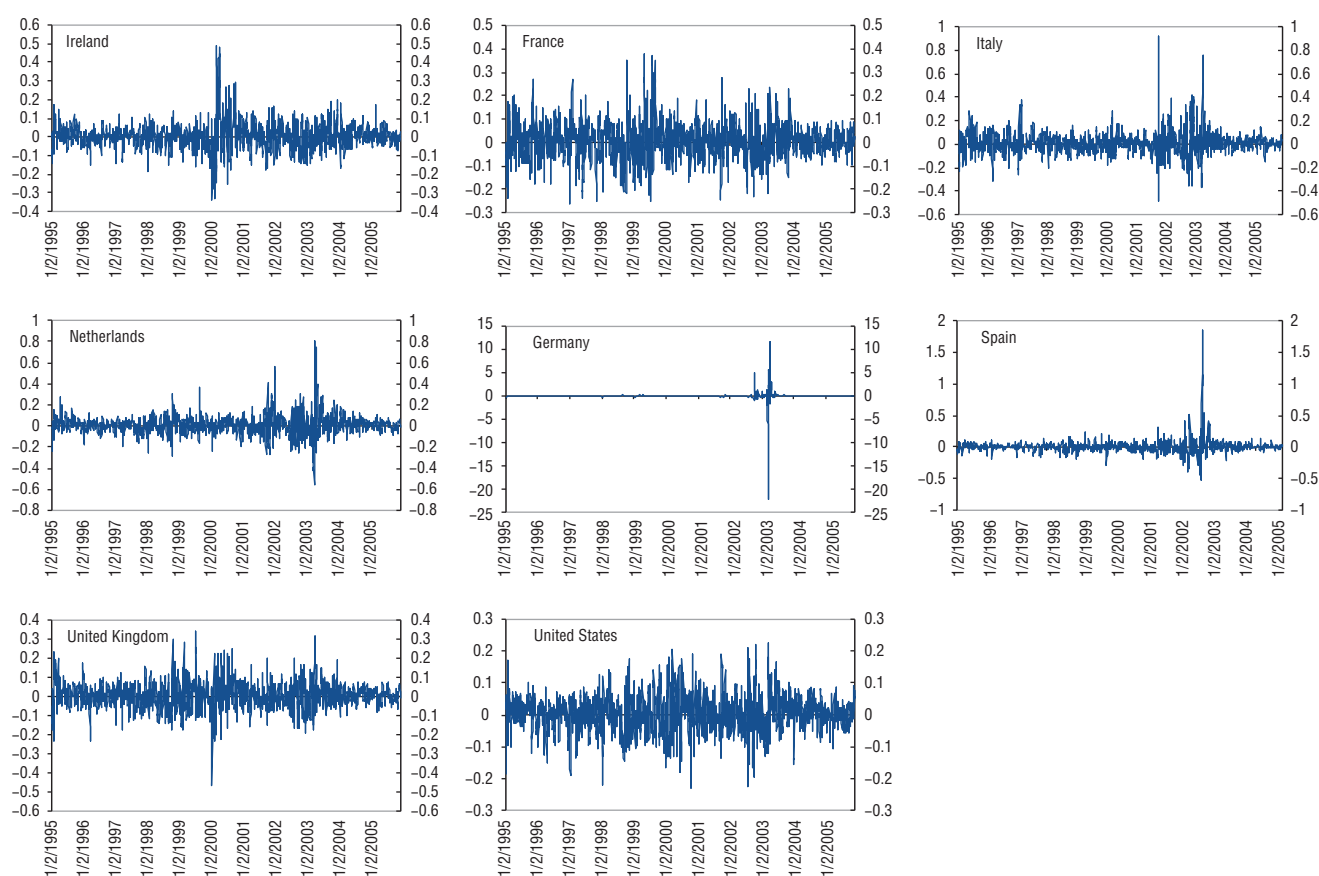
Table 20.2 shows the correlation of the distances to default between Ireland and the other countries. We find that

on average the correlations are positive and quite high for the Netherlands, the United Kingdom, the United States, and Spain. However, if we analyze the past decade in three separate periods—pre-euro period of January 3, 1994, to December 31, 1998; post-euro period of January 1, 1999, to September 11, 2001; and post–September 11 period of September 12, 2001, to November 25, 2005—we find that the correlations become generally much smaller or negative in the post-euro period compared with the pre-euro period and then increase to very high levels in the post–September 11 period.

Figure 20.3 shows the trends in one-year rolling correlations in the DD for all countries, including an average across all country pairs. The latter, which should be more indicative of global trends, seems to be increasing over time, being stronger in the 2003–04 period. It seems that the average pairwise correlation decreased in 1999 with the introduction of the euro and increased after 2001. It also seems that over the last couple of years of the sample period where bank health has been improving, the correlations seem to have diminished. The rolling correlations of Ireland and the United Kingdom seem to have been high dur-

ing the whole period, with the exception of 1999. The correlations of Ireland with France, Italy, and Spain seem to have increased over time. In general, the correlations of Ireland with continental Europe seem to have been negative during 1999–2001.

Occurrence of shocks, especially large shocks, in banks is captured by the weekly percentage change in the DD for each bank, $(\Delta DD_i / |DD_{it}|)$. The mean of the percentage change in the distances to default is zero, as expected, and the largest negative change is 237 percent, which represents a sizable shock. Figure 20.4 shows the trends in the percentage change in distances to default for all countries. The most volatile period for Ireland was 1999–2000, which was also volatile for the United Kingdom and France. Conversely, Italy, the Netherlands, Spain, and Germany experienced large shocks in 2001–03. Overall, for most countries, there were fewer shocks in 2003–05 than in the decade before. This could reflect a general drop in credit events across the world mainly due to benign macroeconomic conditions and credit cycle for the last couple of years of the sample period. For instance, FitchRatings (2005) reported a sharp drop in credit events to 37 in 2005 from 94 in 2003.



Source: Authors.

Note: The charts show the five-day weekly percent change in the banks' aggregated (weighted by assets) DDs for each country. See Table 20.1 for the sample of banks from each country. The percent changes are expressed as fractions.

Figure 20.4 Weekly Change in the Distance to Default (DD) by Country

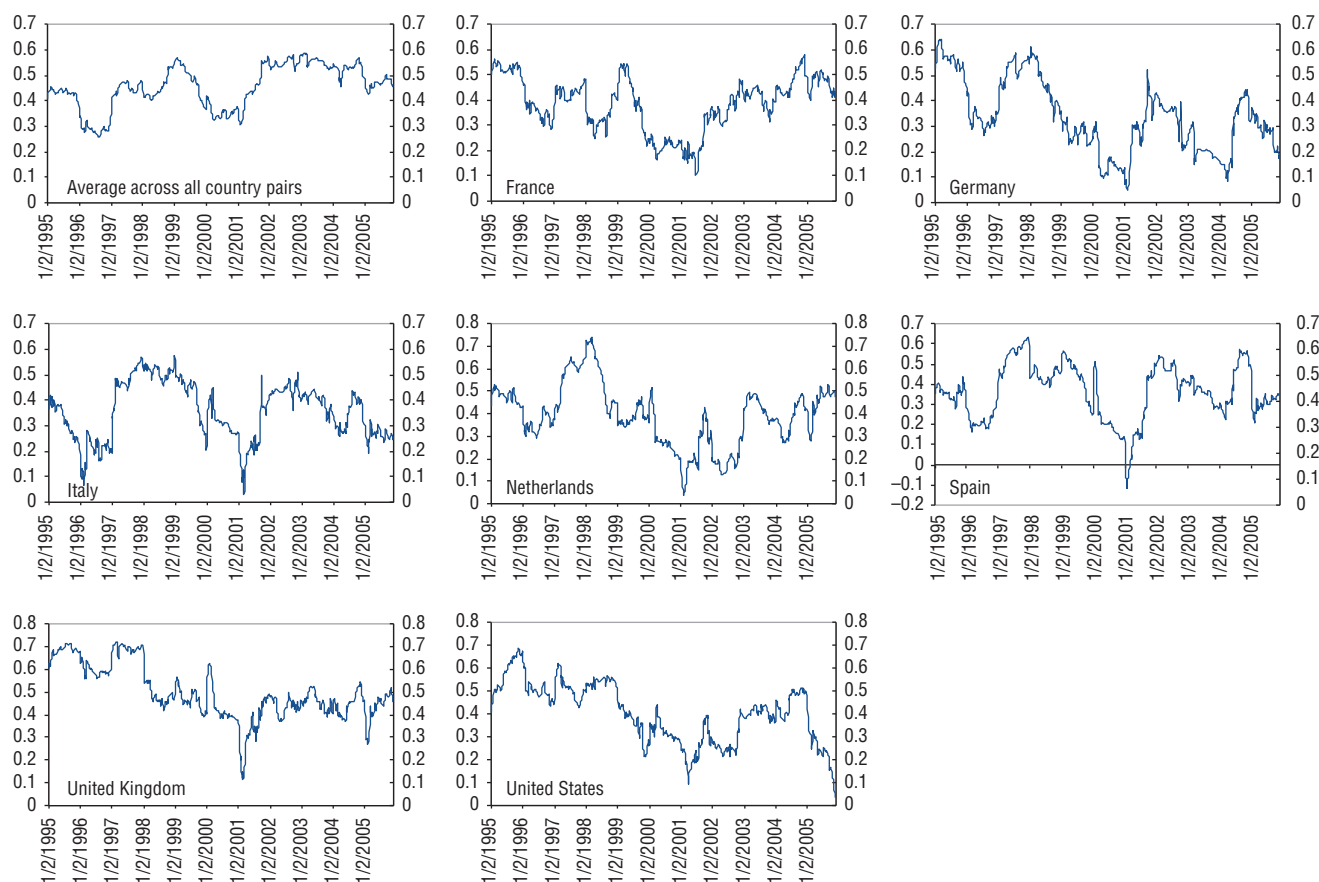
TABLE 20.3

Correlations in Changes in Distance to Defaults

Ireland and	1994–2005	Pre-Euro 1994–98	Post-Euro 1999–2001	Post-September 11 2001–05
France	0.32	0.43	0.17	0.42
Germany	0.08	0.45	0.17	0.12
Italy	0.31	0.36	0.26	0.37
Netherlands	0.31	0.47	0.23	0.32
United Kingdom	0.45	0.60	0.37	0.44
United States	0.36	0.50	0.28	0.34
Spain	0.27	0.41	0.19	0.33
Eastern Europe	0.02	0.03	0	0.26

Source: Authors.

Note: Cross-country correlations in banking systems' weekly changes in the distance to default. The sample covers January 3, 1994–November 25, 2005.



Source: Authors.

Note: Each chart shows the one-year rolling correlations between Ireland and another country's weekly change in the banking systems' distance to default; the last chart shows the average of these correlations. The percent changes are expressed as fractions.

Figure 20.5 Rolling Correlations of Weekly Percent Changes in Distance to Default—Between Ireland and Other Countries

In order to describe how interdependencies across countries have evolved over the decade, we try to capture the simultaneous occurrence of shocks across countries through simple correlations (Table 20.3) and the one-year rolling correlations (Figure 20.5) in the percentage change in the DD, between Ireland and the other countries. On

average, the correlations between the $(\Delta DD / |DD|)$ between Ireland and the other countries are positive. They seem to have a stronger magnitude for the United Kingdom and the United States than for the other European countries. Unlike the correlations in the DD, the correlations in the $(\Delta DD / |DD|)$ seem to be strongest in the pre-

euro period and weakest in the post-euro period (but still positive), suggesting that the joint improvement in the distances to default in the last couple of years of the sample period were accompanied by fewer simultaneous shocks. Over the decade, the rolling correlations seem to be decreasing over time for Germany, the United Kingdom, and the United States, with no clear trend for the other countries. However, the average of the rolling correlations across all country pairs has a positive trend over the decade, suggesting increasing global interdependencies.

3. CONTAGION DETERMINANTS: MULTIVARIATE ANALYSIS USING CO-EXCEEDANCES

The number of banks in Ireland that experience a large shock on the same day as banks in other countries, after controlling for Ireland-specific and global shocks, is labeled as “co-exceedances.” We use an ordered logit model to estimate such co-exceedances. Such methodology of extreme value theory to assess contagion risk was first proposed by Bae, Karolyi, and Stulz (2003) in the context of stock market returns in emerging markets. Bae, Karolyi, and Stulz (2003) and Gropp and Moerman (2004) show that it is useful to examine only the tails of the distributions of returns and of the distances to default, as the distributions exhibit fat tails, and the correlation among the observations is substantially higher for larger shocks. We use weekly changes in the DD to examine whether shocks in one bank or banking system appear to influence the DD of other banks, controlling for common shocks affecting all banks simultaneously.

A. Methodology and data

The dependent variable is the number of Irish banks simultaneously experiencing large shocks or tail events. We follow Gropp, Lo Duca, and Vesala (2006) in arguing that contagion is associated with extreme negative movements in banks’ default risk. These events can be identified from the negative tail of the distribution of the changes in the DD. We define large shocks as the negative 15th percentile of the common distribution of the percentage change in the DD across all banks. We compute the “co-exceedances” of banks in a given country as the number of banks in a given country that were simultaneously in the tail on the same day (Table 20.4).

Given the discrete ordinal nature of the dependent variable, we use an ordered logit model to estimate the probability of a number of Irish banks simultaneously in the tail as a function of the number of banks in the tail in the other countries, controlling for Ireland-specific and global shocks:

$$\Pr_c[Y = j] = \frac{e^{\alpha_j F_c + \beta_j C_{c,t-1} + \gamma_j C_{d,t}}}{\sum_k e^{\alpha_k F_c + \beta_k C_{c,t-1} + \sum_{d \neq c} \gamma_{d,k} C_{d,t}}},$$

where $j=0, 1, 2$ represents the number of banks in the tail simultaneously (“co-exceedances”) in country $c=0$ if no banks are in the tail, 1 if one bank is in the tail, and 2 if two or more banks are in the tail. The vector F_c comprises Ireland-specific and global shocks affecting Ireland. $C_{c,t-1}$ is the lagged number of co-exceedances in country c , and $C_{d,t}$ represents the co-exceedances in period t in country d . Table 20.5 summarizes such co-exceedances by country.

We perform three sets of estimations—the base model, the extended model, and the extended model for each of the three large Irish banks. The base model consists of a set of Ireland-specific and common global shocks, F_c , and persistence in Irish co-exceedances, $C_{c,t-1}$. We use four variables to represent F_c :

1. Systemic risk—we use a systemic risk indicator measuring the number of stock markets that are experiencing a large shock at time t . We construct this control similarly to the modeling of large shocks in banks: we use an indicator variable that we set equal to one if a stock market of a given country experienced a shock large enough to be in the bottom 15th percentile of the distribution of the weekly change in returns, and zero otherwise. We calculate such an indicator variable for the euro area stock market (Euronext Top 100 Index, or N100), the U.S. stock market (National Association of Securities Dealers Automated Quotations, or NASDAQ),⁸ and the Irish stock market (Irish Stock Exchange Overall Index, or ISEQ), from Bloomberg. The systemic risk control is the sum of the indicator variables measuring whether the European stock market, the Irish stock market, and the U.S. stock market were in the tail on a given day, and it ranges from 0 to 3. This control should be positively related to the number of co-exceedances in Ireland.
2. U.S. stock market volatility—we use the weekly change in the volatility of the U.S. stock market to control for volatility spillovers from the United States. There is a large presence of U.S. software firms located in Ireland—nearly all of the largest U.S. software companies use Ireland as their primary base of operations for servicing markets in Europe, Africa, and the Middle East. Also, Irish banks are directly and indirectly exposed to the U.S. stock market through their securities and asset management businesses and through their equity ownership, trade of credit risk transfer instruments, and wholesale funding activities. We estimated the volatility in NASDAQ using a generalized autoregressive conditional heteroskedasticity or GARCH(1,1) model of the form: $\sigma_t^2 = \alpha + \beta \sigma_{t-1}^2 + \gamma \varepsilon_{t-1}^2$, using maximum

⁸ The Dow Jones Industrial Average gives similar results.

TABLE 20.4

Data Description and Summary Statistics, January 3, 1994–November 25, 2005

Variable	Definition	Number of Observations	Mean	Median	SD	Min	Max
Bank specific							
dd_{it}	Distance to default (DD) of bank i at time t	1,23,304	5.19	5.00	2.33	−2.16	27.17
$\Delta dd_{it} / dd_{it-1} $	Percentage change in the distance to default	1,23,104	0.02	0.00	3.15	−236.53	895.32
Country specific							
$\Delta dd_{ct} / dd_{ct-1} $	Percentage change in the distance to default of banking system c at time t						
	France	2,845	0.00	0.01	0.08	−0.26	0.38
	Germany	2,845	0.01	0.00	0.60	−22.25	11.80
	Ireland	2,845	0.00	0.00	0.07	−0.34	0.49
	Italy	2,845	0.01	0.01	0.09	−0.49	0.92
	Netherlands	2,845	0.01	0.01	0.08	−0.57	0.81
	Spain	2,845	0.00	0.00	0.09	−0.52	1.84
	United Kingdom	2,845	0.00	0.01	0.07	−0.47	0.35
	United States	2,845	0.00	0.01	0.06	−0.23	0.23
	Eastern Europe average	2,845	0.01	0.01	0.10	−0.68	0.95
Co-exceedances IRL	Number of banks in the 15th percentile negative tail of $\Delta dd_{it} / dd_{it-1} $ in IRL	2,845	0.36	0.00	0.71	0.00	3.00
Co-exceedances US	Number of banks in the 15th percentile negative tail of $\Delta dd_{it} / dd_{it-1} $ in US	2,845	1.76	0.00	2.88	0.00	14.00
Co-exceedances UK	Number of banks in the 15th percentile negative tail of $\Delta dd_{it} / dd_{it-1} $ in UK	2,845	0.78	0.00	1.24	0.00	5.00
Co-exceedances GER	Number of banks in the 15th percentile negative tail of $\Delta dd_{it} / dd_{it-1} $ in GER	2,845	0.79	0.00	1.08	0.00	4.00
Co-exceedances FR	Number of banks in the 15th percentile negative tail of $\Delta dd_{it} / dd_{it-1} $ in FR	2,845	0.33	0.00	0.64	0.00	2.00
Co-exceedances SPA	Number of banks in the 15th percentile negative tail of $\Delta dd_{it} / dd_{it-1} $ in SPA	2,845	0.46	0.00	0.86	0.00	4.00
Co-exceedances NL	Number of banks in the 15th percentile negative tail of $\Delta dd_{it} / dd_{it-1} $ in NL	2,845	0.32	0.00	0.59	0.00	2.00
Co-exceedances IT	Number of banks in the 15th percentile negative tail of $\Delta dd_{it} / dd_{it-1} $ in IT	2,845	1.13	0.00	1.56	0.00	6.00
Memo items							
Negative 15th percentile of $\Delta dd_{it} / dd_{it-1} $ is equal to −0.070							

Source: Authors.

Note: IRL refers to Ireland, US to United States, UK to United Kingdom, GER to Germany, FR to France, SPA to Spain, NL to Netherlands, and IT to Italy. The banking system changes in DD for each country is calculated as follows: the DD for all the banks is aggregated (weighted by total assets) to get the system-wide DD; the summary statistics of the change in the systemwide DD are reported above.

likelihood. The regression results are reported in Table 20.6.

- Irish stock market volatility—similar to U.S. stock market volatility, the weekly change in the volatility of the ISEQ was calculated to account for Ireland-specific shocks.
- Interest rate shock—the fourth control includes the weekly change in the yield of the Irish 10-year govern-

ment bond to reflect interest rate shocks.⁹ We would expect the interest rate control to be positively related to the number of co-exceedances.

The extended model has, in addition, the number of co-exceedances in other countries in the sample—the United

⁹ This assumes that foreign interest rate shocks are reflected in the movement of Irish interest rates and in stock market movements.

TABLE 20.5

Description of the Sample by Country								
Country	Number of Banks	Maximum of Co-Exceedances	Co-Exceedances = 0		Co-Exceedances = 1		Co-Exceedances ≥ 2	
			Count	Percent	Count	Percent	Count	Percent
Ireland	3	3	2,329	75.01	497	16.01	279	8.99
United States	14	14	1,616	52.05	498	16.04	991	31.92
United Kingdom	5	5	1,909	61.48	556	17.91	640	20.61
Germany	4	4	1,718	55.33	755	24.32	632	20.35
France	2	2	2,355	75.85	465	14.98	285	9.18
Spain	4	4	2,238	72.08	460	14.81	407	13.11
Netherlands	2	2	2,330	75.04	563	18.13	212	6.83
Italy	6	6	1,594	51.34	611	19.68	900	28.99
Total	40	40						

Source: Authors.

Note: A “co-exceedance” refers to the simultaneous occurrence of an extreme shock in a number of banks. An extreme shock is defined by one when bank risk (weekly change in the distance to default) increases beyond a 15 percent threshold. This threshold is given by the 15th percentile left-tail cutoff of weekly changes in distance to default of the pooled sample of banks—when a bank’s weekly change in the distance to default ≤ -0.07 . Also refer to Table 20.4.

TABLE 20.6

Results from GARCH(1,1) Model: $\sigma_t^2 = \alpha + \beta\sigma_{t-1}^2 + \gamma\varepsilon_{t-1}^2$				
Constant	Coefficient	Standard Error	Z	p
U.S. stock market volatility—NASDAQ				
	7.52E-05	6.54E-06	11.5	0
ε_{t-1}^2	0.65	0.03	16.97	0
σ_{t-1}^2	0.38	0.01	24.02	0
Ireland stock market volatility—ISEQ				
	7.79E-05	4.46E-06	17.46	0
ε_{t-1}^2	0.67	0.04	15.34	0
σ_{t-1}^2	0.22	0.02	12.52	0

Source: Authors.

Note: “U.S. stock market volatility—NASDAQ” (“Ireland stock market volatility—ISEQ”) implies the weekly change in the volatility of the U.S. stock market or NASDAQ (Irish stock market, ISEQ), estimated using a GARCH(1,1) model using maximum likelihood, where σ_t^2 refers to volatility of weekly stock returns.

States, United Kingdom, Germany, France, Spain, Netherlands, and Italy. The model for each of the three main banks includes, in addition to the extended model regressors, co-exceedances from each of the other two banks.

The summary statistics and the descriptive statistics for the number of co-exceedances per country, that is, the number of banks simultaneously in the tail on a given day, are presented in Tables 20.1, 20.4, and 20.5. Even though the number of banks per country differs somewhat, there is at least one day on which all banks experienced a large adverse shock simultaneously. We limit the number of outcomes to 0, 1, and 2 or more co-exceedances.¹⁰ The time pattern of co-exceedances in Ireland is presented in Figure 20.6. The most turbulent period for Irish banks was the period 1999–

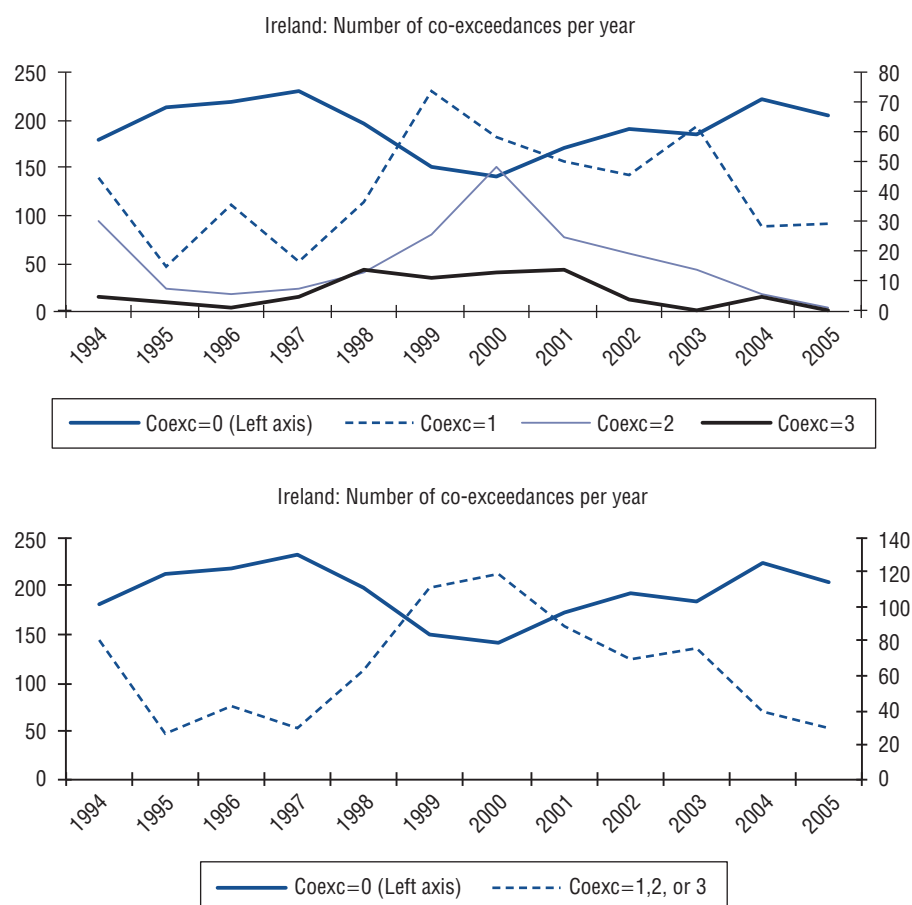
2000 and the calmest being the mid-1990s and the last couple of years of the sample period.

B. Econometric model results

The results for the basic and the extended models are given in Table 20.7. The dependent variable is the number of banks whose weekly percentage change in the DD was in the 15th percentile negative tail in a given day. In all countries with more than two banks, we limit the model to estimating three outcomes: 0, 1, and 2 or more banks simultaneously in the tail.

The basic regression results suggest that the probability of Irish banks being simultaneously in the bottom tail varies positively with systemic risk—real shocks experienced through large stock market movements in Ireland and elsewhere increase the probability of a higher number of Irish banks being simultaneously in the tail. However, controlling for global shocks and including that of the ISEQ, Irish co-exceedances still respond positively and significantly to changes in a long-term Irish interest rate. Long-term lending rates (especially mortgage rates) would likely follow movements in long-term

¹⁰ The number of co-exceedances depends on the number of banks included in the sample and may not necessarily reflect the strength of the banking system per se. Still, comparing countries with equal number of banks in the sample suggests that Spanish banks tended to experience fewer shocks compared with German banks and that Dutch banks tended to be somewhat less frequently subject to shocks compared with French banks.



Source: Authors.

Note: The figure shows the number of days in each year, n ($n = 0, 1 \dots 3$), banks simultaneously experienced an extreme shock ("co-exceedance"). The extreme event is defined by one when bank risk (weekly change in the distance to default) increased beyond a 15 percent threshold. This threshold is given by the 15 percent left-tail cutoff of weekly changes in distance to default of the pooled sample of banks—when a bank's weekly change in the distance to default ≤ -0.07 .

Figure 20.6 Time Pattern of Co-Exceedances in Ireland, January 3, 1994–November 25, 2005 (in number of days each year)

government bond yields. Given the large exposure of Irish banks to the real estate market and the prevalence of variable loan rates, sudden increases in interest rates could be associated with credit events that might have a negative impact on banks. In addition, the notion that the number of co-exceedances could be sticky is supported: the lagged (by one day) number of co-exceedances is positive and significant. The stock market volatility controls are not significant, suggesting that the systemic risk variable might be sufficiently capturing stock market spillover effects.

Next, we extend the model to include contagion or co-exceedances from other countries (Table 20.7). We measure contagion by including the first lag of the co-exceedances in the other seven countries. If, after controlling for common shocks, any of these variables turn out to be significant, we interpret this as contagion from that country. We find evidence of contagion from the United Kingdom, the United States, and the Netherlands (with a negative sign) toward Ireland. Adding foreign co-exceedances adds information to

the specification, which is reflected in the fact that the significance of the controls remains largely unchanged.

How does the size of the contagion from the United States to the Irish banks compare with that from the United States to other countries? Although the coefficient estimates should not be interpreted as marginal effects, the relative effect of one variable compared with another can be gauged by the relative sizes of the coefficient estimates. To get the relative U.S. effect, we benchmark the U.S. influence with that of the United Kingdom. That is, the relative size of the U.S. influence is the coefficient for the United States divided by the coefficient for the United Kingdom. For the Irish banks, the relative U.S. influence on the three Irish banks is $(0.27/0.13)$ 2.1 (from Table 20.7). Next we run similar regressions for each of the other countries (not reported here) and calculate the relative U.S. influence where both the United States and the United Kingdom influences are significant. Figure 20.7 shows a bubble chart for the four euro area countries with significant coefficients for both the United States and the United Kingdom.

TABLE 20.7

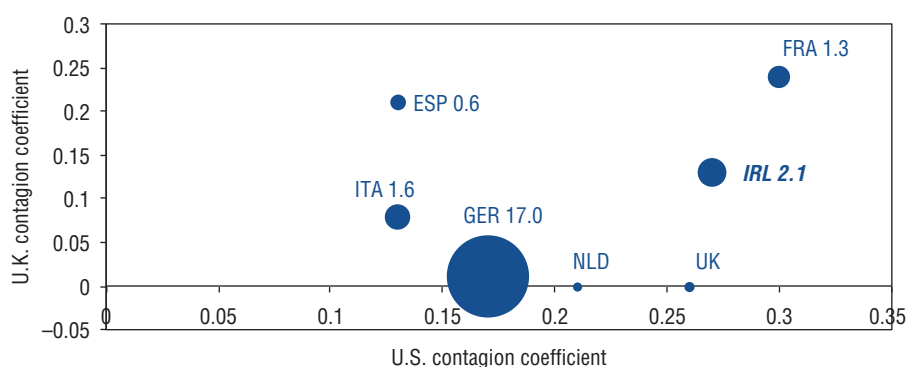
Ordered Logit Model: Contagion in Daily Co-Exceedances of the Weekly Change in Distances to Default, January 3, 1994–November 25, 2005

Regressors	Ireland				AIB				Anglo IB				BoI			
	Basic Model		Extended Model		Basic Model		Extended Model		Basic Model		Extended Model		Basic Model		Extended Model	
	Coeff.	SE	Coeff.	SE	Coeff.	SE	Coeff.	SE	Coeff.	SE	Coeff.	SE	Coeff.	SE	Coeff.	SE
Co-Exceedances own lagged	2.16**	0.07	2.08**	0.08	3.36**	0.14	3.11**	0.16	3.51**	0.14	3.36**	0.15	3.07**	0.14	2.77**	0.15
Systemic risk	0.81**	0.06	0.73**	0.06	0.87**	0.08	0.79**	0.09	0.62**	0.08	0.52**	0.09	0.81**	0.08	0.67**	0.08
Volatility ISEQ	−33.96	34.04	−34.41	33.97	37.56	53.89	11.23	55.89	5.9	47.39	−20.08	47.99	−19.56	41.21	−44.7	40.75
Volatility US	−18.66	11.74	−19.42*	11.52	−1.92	17.4	−0.96	16.96	−10	16.1	−10.29	16.03	−28.82*	16.67	−35.07**	15.91
Interest rate IRL	2.18**	0.41	1.92**	0.41	3.24**	0.58	3.03**	0.59	1.53**	0.59	1.15**	0.59	2.45**	0.54	2.03**	0.55
Contagion FR			0.11	0.08			0.12	0.11			−0.02	0.11			0.13	0.11
Contagion GER			0.01	0.07			−0.04	0.1			−0.02	0.1			0.006	0.09
Contagion IT			0.002	0.06			−0.13	0.09			0.05*	0.09			−0.05	0.09
Contagion NL			−0.21**	0.09			−0.36**	0.13			−0.17	0.13			0.019	0.12
Contagion SP			−0.04	0.07			0.21**	0.11			−0.07	0.11			−0.14	0.11
Contagion UK			0.13**	0.069			0.17*	0.09			0.15	0.1			0.06	0.09
Contagion US			0.27**	0.06			0.20**	0.08			0.23**	0.08			0.38**	0.08
AIB											0.03	0.2			0.51**	0.17
Anglo IB							0.19	0.19							0.34**	0.17
BoI							0.41**	0.18			0.39**	0.19				
Pseudo- R^2		0.31		0.32		0.393		0.408		0.351		0.362		0.327		0.347

Source: Authors.

Note: Dependent variable: number of Irish banks simultaneously in the tail ("co-exceedances"). "Co-exceedances own lagged" refers to one lag of the dependent variable. "Systemic risk" measures the number of stock markets that are experiencing a large shock at time t . It is constructed from indicators of three stock markets: the euro area stock market index (N100 or Euronext Top 100 Index), the U.S. (NASDAQ), and the Irish stock market (ISEQ or Irish Overall Index) indices. Each indicator is set equal to one if a stock market of a given country experiences a shock large enough to be in the bottom 15th percentile of the distribution of the weekly change in returns, and zero otherwise. Systemic risk is the sum of the indicator variables and it ranges from 0 to 3. "Volatility US" ("Volatility ISEQ") implies the weekly change in the volatility of the U.S. stock market or NASDAQ (Irish stock market, ISEQ), estimated using a GARCH(1,1) model of the form: $\sigma_t^2 = a + b\sigma_{t-1}^2 + \gamma \varepsilon_{t-1}^2$, using maximum likelihood, where σ_t^2 refers to volatility of weekly stock returns (also see Table 20.8). "Interest rate IRL" is the weekly change in the yield of the Irish 10-year government bond. "Contagion ..." refers to one lag of co-exceedances in France (FR), Germany (GER), Italy (IT), the Netherlands (ND), and the United States (US). The extreme movements in risk of the three Irish banks are referred to as AIB (Allied Irish Bank), Anglo IB (Anglo Irish Bank), and BoI (Bank of Ireland). The goodness of fit in LOGIT (and other binary) models is given by the "Pseudo- R^2 ." This statistic is the likelihood ratio index, computed as $R^2 = 1 - l(\hat{\beta}) / l(\bar{\beta})$ where $l(\bar{\beta})$ is the restricted log-likelihood or the maximized log-likelihood value when all slope coefficients are restricted to zero, which is equivalent to estimating the unconditional mean probability of an observation being in the tail.

* $p < .05$; ** $p < .01$.



Source: Authors.

Note: The black patches show the relative size of contagion from the United States (U.S.), benchmarked against contagion from the United Kingdom (UK). For example, for Ireland (IRL) it is $0.27/0.13=2.1$, derived from Table 20.7 ("extended model for Ireland" column). The patches for the other countries—France (FRA), Germany (GER), Italy (ITA), the Netherlands (NLD), Spain (ESP)—are derived from similar regressions that are not reported in the paper. For Italy, the Netherlands, and the United Kingdom, only the U.S. coefficient is shown, since the UK coefficient was not significant for Italy and the Netherlands, and not applicable for the UK.

Figure 20.7 Relative Size of Contagion from the United States to Selected European Countries

TABLE 20.8

Ordered Logit Model: Contagion in Daily Co-Exceedances of the Weekly Change in Distances to Default by Subsample

	Pre-Euro Period (Jan. 1994–Dec. 1998)		Post-Euro Period (Jan. 1999–Sept. 11, 2001)		Post-September 11 Period (Sept. 12, 2001–Nov. 2005)	
	Coeff.	SE	Coeff.	SE	Coeff.	SE
Ireland						
Co-Exceedances	1.73**	0.14	2.20**	0.14	2.05**	0.14
IRL lagged						
Systemic risk	1.17**	0.17	0.46**	0.12	0.76**	0.1
Volatility ISEQ	−120.12*	65.56	91.05	99.59	−27.86	60.73
Volatility US	33.86	48.71	−20.65*	12.81	−12.98	39.03
Interest rate IRL	2.52**	0.61	1.50*	0.91	−0.15	0.94
Contagion FR	0.12	0.14	−0.01	0.15	0.18	0.18
Contagion GER	0.22*	0.12	0.08	0.12	−0.14	0.13
Contagion IT	0.06	0.11	0.005	0.11	−0.002	0.12
Contagion NL	−0.24	0.16	−0.14	0.17	−0.03	0.17
Contagion SP	−0.01	0.13	−0.06	0.14	0.01	0.15
Contagion UK	0.29**	0.12	0.20*	0.12	−0.19	0.13
Contagion US	0.14	0.1	0.28**	0.11	0.18*	0.11
Pseudo- R^2		0.339		0.322		0.286

Source: Authors.

Note: Dependent variable: number of Irish banks simultaneously in the tail ("co-exceedances"). "Co-exceedances IRL lagged" refers to one lag of the dependent variable. "Systemic risk" measures the number of stock markets that are experiencing a large shock at time t . It is constructed from indicators of three stock markets: the euro area stock market index (N100 or Euronext Top 100 Index), the U.S. (NASDAQ), and the Irish stock market (ISEQ or Irish Overall Index) indices. Each indicator is set equal to one if a stock market of a given country experiences a shock large enough to be in the bottom 15th percentile of the distribution of the weekly change in returns, and zero otherwise. Systemic risk is the sum of the indicator variables, and it ranges from 0 to 3. "Volatility US" ("Volatility ISEQ") implies the weekly change in the volatility of the U.S. stock market or NASDAQ (Irish stock market, ISEQ), estimated using a GARCH(1,1) model of the form $\sigma_t^2 = a + \beta\sigma_{t-1}^2 + \gamma\varepsilon_{t-1}^2$, using maximum likelihood, where σ_t^2 refers to volatility of weekly stock returns. "Interest rate IRL" is the weekly change in the yield of the Irish 10-year government bond. "Contagion..." refers to one lag of co-exceedances in France (FR), Germany (GER), Italy (IT), the Netherlands (NL), Spain (SP), United Kingdom (UK), and the United States (US). The extreme movements in risk of the three Irish banks are referred to as AIB (Allied Irish Bank), Anglo IB (Anglo Irish Bank), and Bol (Bank of Ireland). The goodness of fit in LOGIT (and other binary) models is given by the "Pseudo- R^2 ." This statistic is the likelihood ratio index, computed as $R^2 = 1 - l(\hat{\beta})/l(\bar{\beta})$, where $l(\hat{\beta})$ is the restricted log-likelihood or the maximized log-likelihood value when all slope coefficients are restricted to zero, which is equivalent to estimating the unconditional mean probability of an observation being in the tail.

* $p < .05$; ** $p < .01$.

The relative contagion from the United States to the Irish banks is second only to Germany.

The final set of estimations involves estimating the extended model for each of the three individual Irish banks (Table 20.7). There is evidence of contagion from the United Kingdom (at the 10 percent significance level), the United States, Spain, and the Netherlands (negative coefficient) toward AIB; contagion from Italy (at the 10 percent significance level) and the United States toward Anglo IB; and from the United States toward BoI. We find evidence of a two-way contagion between AIB and BoI and between Anglo IB and BoI—these could reflect both interbank linkages and off-balance-sheet or derivative positions where one bank is the buyer and the other the seller of various risk protections.

The results for the extended model qualitatively survive several robustness checks. First, when the number of U.S. banks was reduced from 14 to 5 (the top 5), the U.S. influence on Irish banks remained the same. Second, taking monthly changes in DD, rather than weekly changes, does not change the U.S. and U.K. influence; the contagion from the Netherlands disappears. Third, changing the threshold that defines the negative tail of the distribution of large shocks does not qualitatively change the results for the United States and the United Kingdom. Finally, looking at the pattern of contagion risk over time, we find different linkages in the pre-euro, post-euro, and the post-September 11 periods (Table 20.8). We find evidence of contagion from the United Kingdom and Germany (at the 10 percent significance level) in the pre-euro period, from the United States and the United Kingdom (at the 10 percent significance level) in the post-euro period, and only from the United States (at the 10 percent significance level) to Ireland in the post-September 11 period. This suggests a changing pattern of contagion risk, from stronger linkages with Europe in the earlier periods to stronger linkages with the United States later on, consistent with the evidence presented earlier.

4. CONCLUSION

This chapter examines the external linkages of the Irish banking sector and estimates an indicator of potential contagion risk—arising from idiosyncratic shocks in other countries—that the three major Irish banks may be exposed to. Aggregate balance sheet data of Irish-resident banks suggest several channels of external interdependencies, including foreign equity exposures, loan-book exposures abroad, and wholesale funding through interbank and capital market issues. However, apart from these links, there could also be foreign exposures through credit risk transfers (e.g., Irish banks selling risk protection to cross-border banks that could be subject to credit events in other countries) and through operational risks that are difficult to measure but can quickly lead to large fluctuations in bank stock prices.

Given that Irish-resident banks include a large number of foreign banks operating in the Financial Services Center that have limited Irish linkages in the retail market, we focus on the three major listed banks—BoI, AIB, and Anglo IB—

which have nearly 80 percent of the domestic retail market. We proxy banking risk by distance-to-default measures constructed from bank equity prices and look at correlations of changes in the DD of Irish banks with banks in other countries. Following Gropp, Lo Duca, and Vesala (2006), we then define large changes in DD of each bank—co-exceedances—if the changes fall below the 15th percentile of the negative tail of the joint distribution of the distance-to-default changes across all banks of the sample. We use an ordered logit model to estimate the probability of the number of Irish banks being in the tail at the same time as banks in other countries, controlling for Irish-specific and global factors.

There is evidence suggesting contagion risk from the United States and the United Kingdom to Ireland, although the size of the U.S. influence dominates that of the United Kingdom in almost all regressions and across many robustness checks.¹¹ There are obvious balance sheet linkages of Irish banks with the United Kingdom through subsidiaries. The United States and the United Kingdom also remain the major countries selling risk protection to Irish banks. Aggregate data show that Irish-resident banks (but not necessarily the three banks in the empirical study) have positive net-asset exposures in the United States, which could give rise to credit risk from exposure to nonbank assets and liquidity risk from Irish banks' (net) interbank borrowing from the United States, from mid-2004. Conversely, Irish-resident banks have very large and *negative* net-asset exposure to the United Kingdom, suggesting that risks in Irish banks' on-balance-sheet exposures to the United Kingdom might have been mitigated by off-balance-sheet risk protection bought from United Kingdom banks. Still, both the United States and the United Kingdom have had booming property markets—Irish banks are exposed to them and could be affected in the event of a substantial downturn in any of these markets.

Some tentative policy lessons could be drawn from the results of this exercise. The Central Bank and Financial Services Authority of Ireland may want to stress test specific categories of exposures of Irish banks to both the United States and the United Kingdom. Even though linkages with the United States do not come out strongly from *aggregate consolidated* balance sheet exposures, there might be derivatives or other off-balance-sheet exposures that the bank supervisors may need to be vigilant of. The Irish authorities may need to collect more information about types and counterparties of derivative positions and risk transfers through structured products of Irish banks, as the use of these is likely to grow rapidly in the future.¹² This would especially

¹¹ Econometric estimates (not reported) suggest that although there is no evidence of contagion from Europe to the United States, the United States is a source of contagion not only for Ireland but also for all the other European countries. Germany seems to be mainly a receiver of contagion risk rather than a source, which is consistent with the evidence from BIS that Germany is mainly "a buyer" of risk.

¹² See Chan-Lau and Ong (2006) for the regulatory and supervisory initiatives taken by the U.S. Office of the Comptroller of the Currency and the U.K. Financial Services Authority in this regard.

be necessary if Irish banks are buying CRT products from foreign banks (that is, selling risk protection) that are in turn exposed to property markets or other loan products in the United States or the United Kingdom—thus exposing the Irish banks to these markets even though there is no direct loan exposure.

Finally, some caveats apply to the econometric results. We are using equity prices of banking *groups* and yet mainly discussing international links typically associated with banking without explicitly stating links through securities and insurance (the top two banks are also involved in insurance and securities business). We are using data available for only a small group of listed banks in each country—so it may not be representative of the system, and the banks in Ireland comprise less than 50 percent of the system in terms of total assets (although nearly 80 percent of the retail market). The number of observations experiencing large shocks is low; the results could therefore be driven by the large shocks associated with the tech-bubble burst of 2000.

REFERENCES

- Allen, Franklin, and Douglas Gale, 2000, “Financial Contagion,” *Journal of Political Economy*, Vol. 108, No. 1, pp. 1–33.
- Bae, Kee-Hong, G. Andrew Karolyi, and Rene M. Stulz, 2003, “A New Approach to Measuring Financial Contagion,” *Review of Financial Studies*, Vol. 16, No. 3, pp. 717–63.
- Black, Fisher, and Myron Scholes, 1973, “The Pricing of Options and Corporate Liabilities,” *Journal of Political Economy*, Vol. 81, No. 3, pp. 637–54.
- Calomiris, Charles W., and Joseph R. Mason, 2000, “Causes of U.S. Bank Distress during the Depression,” NBER Working Paper No. 7919 (Cambridge, MA: National Bureau of Economic Research). Available via the Internet: <http://www.nber.org/papers/w7919>
- Chan-Lau, Jorge A., Arnaud Jobert, and Janet Q. Kong, 2004, “An Option-Based Approach to Bank Vulnerabilities in Emerging Markets,” IMF Working Paper 04/33 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=17059>
- Chan-Lau, Jorge A., and Li Lian Ong, 2006, “The Credit Risk Transfer Market and Implications for Financial Sector Stability,” IMF Country Report No. 06/87, pp. 79–102 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=18986>
- De Nicoló, Gianni, and M. Kwast, 2002, “Systemic Risk and Financial Consolidation: Are They Related?” IMF Working Paper 02/55 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.cfm?sk=15680.0>
- De Nicoló, Gianni, and Alexander Tieman, 2005, “Economic Integration and Financial Stability: A European Perspective” (unpublished; Washington: International Monetary Fund).
- Duggar, Elena, and Srobona Mitra, 2009, “External Linkages and Contagion Risk: The Case of Irish Banks,” *IMF Staff Papers*, Vol. 56, No. 4, pp. 758–86.
- FitchRatings, 2005, “Global Credit Derivatives Survey: Risk Dispersion Accelerates,” Financial Institutions Special Report, November.
- Freixas, Xavier, Bruno Parigi, and Jean-Charles Rochet, 2000, “Systemic Risk, Interbank Relations and Liquidity Provision by the Central Bank,” *Journal of Money, Credit, and Banking*, Vol. 32, No. 3, pp. 611–38.
- Furhine, Craig H., 2003, “Interbank Exposures: Quantifying the Risk of Contagion,” *Journal of Money, Credit, and Banking*, Vol. 35, No. 1, pp. 111–28.
- Gropp, Reint, Mario Lo Duca, and Jukka Vesala, 2006, “Cross-Border Bank Contagion in Europe,” ECB Working Paper 662, November (Frankfurt: European Central Bank). Available via the Internet: <http://www.ecb.europa.eu/pub/scientific/wps/date/html/wps2006.en.html>
- Gropp, Reint, and Gerard Moerman, 2004, “Measurement of Contagion in Bank Equity Prices,” *Journal of International Money and Finance*, Vol. 23, No. 3, pp. 405–551.
- Gropp, Reint, Jukka Vesala, and Guiseppe Vulpes, 2004, “Equity and Bond Market Signals as Leading Indicators of Bank Fragility,” *Journal of Money, Credit and Banking*, Vol. 38, No. 2, pp. 399–428.
- Merton, Robert C., 1974, “On the Pricing of Corporate Debt: The Risk Structure of Interest Rates,” *Journal of Finance*, Vol. 29, No. 2, pp. 449–70.
- Morgan, Donald P., 2002, “Rating Banks: Risk and Uncertainty in an Opaque Industry,” *American Economic Review*, Vol. 92, No. 4, pp. 874–88.
- Standard and Poor’s, 2006, “Bank Industry Risk Analysis: Ireland (Republic of),” Ratings Direct (January).
- Vassalou, Maria, and Yuhang Xing, 2004, “Default Risk in Equity Returns,” *Journal of Finance*, Vol. 59, No. 2, pp. 831–68.

Identifying Spillover Risk in the International Banking System: An Extreme Value Theory Approach

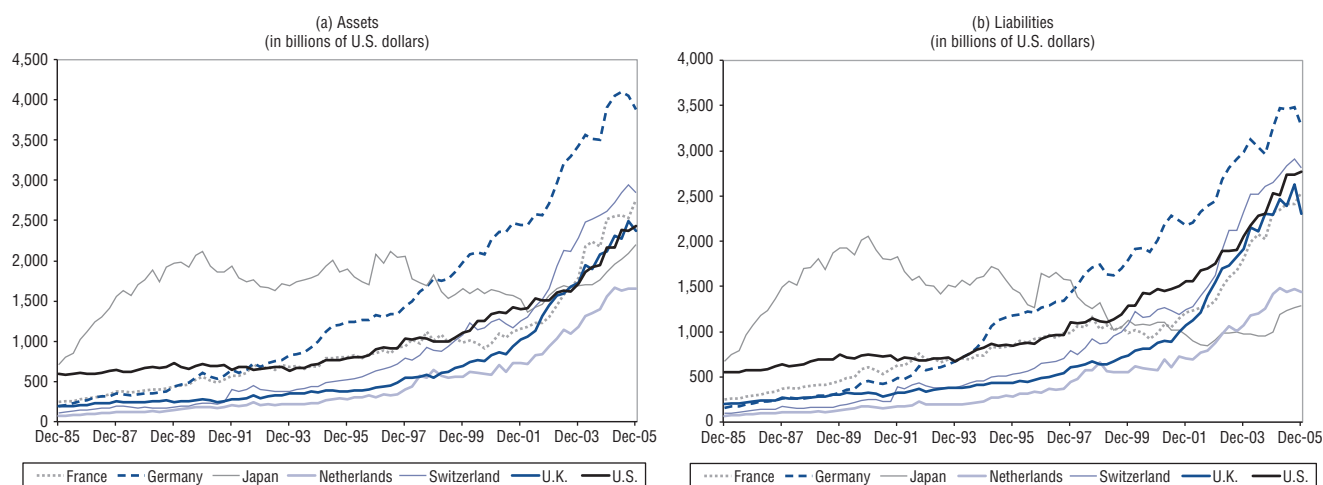
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We use the extreme value theory framework to analyze spillover risk across the international banking system. We test for the likelihood that an extreme shock affecting a major, systemic global bank would affect another large local or foreign counterpart, and vice versa. Our results reveal several key trends among major global banks: spillover risk among banks exhibits “home bias”; individual banks are affected differently by idiosyncratic shocks to their major counterparts; and banks are affected differently by common shocks to the real economy or financial markets. In general, bank soundness appears more susceptible to common (macro and market) shocks when the global environment is turbulent; this may have important implications for global financial stability especially during stressful periods. Not surprisingly, our findings also suggest that bank spillover risk has risen over time, which emphasizes the need for continuing collaboration on cross-border supervision and crisis management.

METHOD SUMMARY

Overview	The logit model estimates conditional correlations between banks during extreme movements in distance-to-default indicators.
Application	The method is appropriate for modeling spillover risk.
Nature of approach	Parametric approach—conditional correlations are based on an econometric model.
Data requirements	Accounting information on bank liabilities; market-price-based data on bank equity prices, interest rate (term) spreads, stock market indices.
Strengths	The results provide insights into the direction of spillovers across banking systems during crises; the methodology is transparent and can be easily replicated; data on actual exposures between institutions are not required in order to deduce spillovers.
Weaknesses	The model assumes that • there are enough extreme movements in the data to provide insights about actual spillovers during a crisis; • the movements in the market-based data are sufficient to summarize information on both common exposures and actual intra-institution exposures, when in reality the market does not have adequate information.
Tool	The EViews example program codes are available in the toolkit, which is on the companion CD and at www.elibrary.imf.org/stress-test-toolkit. Contact author: S. Mitra.

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Sources: Bank for International Settlements; and Table 8A of International Banking Statistics.

Note: BIS = Bank for International Settlements.

Figure 21.1 International Positions by Nationality of Ownership of BIS Reporting Banks

The international banking system has expanded rapidly since the late 1990s. In addition to the overall size of banking assets, the international positions of banks from many of the major banking systems have grown significantly, by several multiples, during this period (Figure 21.1).¹ Thus, the banking sector, which continues to be of key importance for global financial stability, represents a potentially important channel for spillovers through the international financial system.

Shocks to a major global or regional bank, originating from their operations in the home country or in a host country, potentially could affect the international banking system. Spillovers could occur through its linkages with other major banks or through its operations in global financial centers.² There are numerous specific ways in which this could occur. Notably, external linkages could stem from direct and indirect equity exposures of local banks to overseas banks or, conversely, from shareholdings of local banks by foreign banks; direct exposures through loan books; deposit and funding sources from overseas and/or from foreign banks operating in a particular country; payments and settlement systems; and holdings of credit risk transfer instruments written on assets held by local and/or overseas institutions. Spillovers could also occur without any explicit links between banks when a negative shock in one bank is misinterpreted by investors as a signal of diminished soundness in other banks, either in the same country or in a different country.

¹ Schoenmaker and van Laecke (2006) examine the internationalization of cross-border banking, the public policy issues, and the appropriate supervisory response.

² We define *contagion risk* as the transmission of an idiosyncratic shock from one banking group to another. In other words, by adopting this definition, we are allowing for normal linkages among banks to be a possible channel of contagion and not restricting our analysis to cover only changing cross-market linkages following a shock (referred to as “shift contagion” by Forbes and Rigobon, 2001).

Several studies have looked at the various transmission channels for spillovers across banks. These include liquidity shocks affecting one bank, causing deposit runs at other solvent banks (Freixas, Parigi, and Rochet, 2000); shocks via the interbank market when banks withdraw their deposits at other banks (Allen and Gale, 2000; Freixas, Parigi, and Rochet, 2000); or, in the absence of explicit links, difficulties in one market perceived as a signal of possible difficulties in others due to asymmetric information (Morgan, 2002).

Key to the financial stability debate is that banks also have become increasingly intertwined with other segments of the financial sector, both locally and across countries. For instance, banks provide insurers with liquidity facilities to pay current claims and letters of credit as evidence of their ability to pay future claims. The formation of *bancassurance* groups through the merger of banks with insurers is another example of cross-sector linkages. Banks are also increasingly providing services to investors such as hedge funds, mutual funds, and pension funds, in the form of devising, intermediating, and making markets for financial instruments. Cross-border lending activities also have increased strongly in recent years, especially with the sharp growth in private equity deals.

This chapter focuses on determining spillover risk among some of the largest, systemic banks in the world and within the European Union, using the extreme value theory (EVT) framework. Gropp and Moerman (2004) and Gropp, Lo Duca, and Vesala (2006) apply EVT in testing for spillovers across EU banking systems. It should be noted that the exact nature of the links between the financial institutions is not explored in this study. Rather, the results are intended to represent “maps” that could guide the allocation of limited surveillance and supervisory resources, so that more detailed links may then be identified as necessary. In addition to highlighting the relationships among the major global banks, it

could also focus cross-border collaboration among supervisors in these countries.³

Our aim is to identify potential risk concentrations among systemically important banks. Arguably, even if a crisis cannot ultimately be averted, early detection of vulnerabilities in the financial system could provide country authorities with additional time to prepare contingency plans and to focus their attention on likely stress points. To this end, understanding the interdependencies between individual banks with potential systemic impact and their exposure to economic and financial risks is crucial (Bank of England, 2006).

We attempt to answer the following questions on the global and EU banking systems: First, given the increasing internationalization of financial services, are all banks similarly affected by common shocks to the global economy or financial system? Alternatively, is “home bias” the most dominant factor, notwithstanding the effects of globalization? In other words, are banks predominantly influenced by domestic shocks, either because of their domestic focus or the local regulatory environment, despite being largely integrated into the global financial system? Or are different banks—irrespective of domicile—affected differently by shocks, because of their increasingly different business and geographic mixes?

Our results reveal several key trends among major global banks. Notably, “home bias” is an important factor in terms of spillovers. Banks also are affected generally by common shocks to the real economy or financial markets, although the global banking system as a whole tends to be more exposed to these shocks during more turbulent periods, compared with the more benign periods. Further, the risk of spillovers across the major global banks has risen in recent years. Several important relationships are also highlighted at a more specific, bank-by-bank level. Shocks to U.S. banks—especially to Morgan Stanley and Citigroup—appear to have become increasingly important for foreign banks over time, while some of the major U.S. banks appear largely insulated from foreign banks’ idiosyncratic shocks. In contrast, shocks to Japan’s major banks have limited impact on their counterparts and vice versa. Within the EU, ABN Amro and Fortis represent important cross-border spillover risks.

The remainder of this chapter is organized as follows: Section 1 describes the method and data. The empirical analysis of spillover risk across major global banks is presented in Section 2. The analysis is extended in Section 3 to focus specifically on the interlinkages among the largest banks from each EU country. Section 4 concludes the chapter.

1. EMPIRICAL METHOD

The EVT approach to identifying spillovers better captures the information that large, extreme shocks are transmitted

across financial systems differently than small shocks.⁴ Multivariate EVT techniques are used to quantify the joint behavior of external realizations (or “co-exceedances”) of financial prices or returns across different markets. The body of literature on EVT has grown in recent years. Recent empirical work using EVT to model spillovers in financial markets, including bond, equity, and currency markets, are Starica (1999); Forbes and Rigobon (2001); Longin and Solnik (2001); Quintos (2001); Bae, Karolyi, and Stulz (2003); Chan-Lau, Mathieson, and Yao (2004); Hartmann, Straetmans, and de Vries (2004); and Poon, Rockinger, and Tawn (2004). Gropp and Moerman (2004) subsequently apply this approach to changes in the distances to default of 67 individual EU banks.⁵ They use nonparametric tests of banks’ changes in distances to default to test for spillover risk between two banks, after adjusting for bank size. Separately, Gropp, Lo Duca, and Vesala (2006) use a multinomial logit model to the changes in the distances to default of European banks to determine spillovers among banking systems within the region.

Within this framework, we test for co-exceedances, that is, the likelihood that an extreme shock affecting a major, systemic global bank also would affect another large local or foreign counterpart. We assume that spillover risk is associated with extreme negative comovements in bank soundness measures. In other words, we try to determine whether extreme but plausible negative shocks to a particular bank’s stability could be associated with stresses experienced by other major banks in the international banking system. In our tests, we are able to identify the co-exceedances, or evidence of spillover risk, attributable to idiosyncratic shocks in the banking sector, because we factor out the impact of domestic and global shocks.

Our analysis differs from existing studies in several important ways:

- We test for spillover risk among individual, systemically important banks, rather than aggregating the effects for a particular country. Gropp and Moerman (2004) and Gropp, Lo Duca, and Vesala (2006) incorporate most listed banks in the EU, in their respective papers, including smaller banks that are likely nonsystemic. This could have the effect of overestimating the impact of certain banking systems on others. Indeed, Gropp and Moerman (2004) observe that “an unreasonable number of very small banks” appear to have systemic importance in their results.

³ For example, Duggar and Mitra (2009) demonstrate that the major Irish banks are also vulnerable to shocks emanating from the Netherlands and the United States, contrary to the focus of the Irish supervisory authorities largely on the United Kingdom.

⁴ For example, Granger causality tests may not accurately depict the common occurrence of extreme events between banks, because relationships between banks are likely to be very different across tranquil and turbulent periods. High correlations in bank soundness during normal times provide little information on the likelihood of contagion. Conversely, contagion—as we define it—during extreme events could well arise from interlinkages that also exist during tranquil times.

⁵ The distance to default (DD) is an indicator of default risk based on Merton (1974). Generally, empirical studies have shown that the DD is a good predictor of corporate defaults (Moody’s KMV, or MKMV) and is able to predict banks’ downgrades in developed and emerging market countries (Chan-Lau, Jobert, and Kong, 2004; Gropp, Vesala, and Vulpes, 2004).

- We select the 24 biggest banking groups in the world, by total assets, on the basis that these banks could individually pose systemic risk to the domestic or foreign banking systems. In particular, we include institutions from two of the biggest banking systems in the world—that is, Japan and the United States—which contribute significantly to international banking activity. The influence of big banks from Japan and the United States, which represent 9 out of the 24 biggest banks, is likely to be very important, especially given the sharp increase in international banking activity in recent years. The focus on major banks is very pertinent given that the financial authorities in our sample countries are looking to improve cross-border collaboration on supervision issues and are thus highly concerned about the impact of systemically important banks.
- As an extension, we map spillover risks among major EU banks. Much of the existing literature on the financial soundness of EU banks does not cover spillover effects and focuses instead on the impact from country-level and EU-level shocks.
- We incorporate local, regional (where relevant), and global market and real economy factors into our models. Given the global nature of our data set, we utilize regional and world stock market indices to capture shocks that are broader in nature, in addition to using individual local stock market indices and domestic interest rate yield spreads to reflect domestic developments.

A. Model

Distance to Default

First, we calculate distance to default (DD) as a comprehensive measure of a bank's default (solvency) risk. The DD measure is based on the structural valuation model of Black and Scholes (1973) and Merton (1974). The metric represents the number of standard deviations away from the point where the book value of a bank's liabilities is equal to the market value of its assets, that is, the default threshold. The DD is an attractive measure in that it measures the solvency risk of a bank by combining information from stock returns with information from leverage and volatility in asset values—key determinants of default risk. It does not require specification of a particular channel through which the transmission of shocks occurs. An increase in the DD implies greater stability/soundness, or a lower risk of default.⁶

Black and Scholes (1973) and Merton (1974) first drew attention to the concept that corporate securities are contingent claims on the asset value of the issuing firm. This insight is clearly illustrated in the simple case of a firm issuing

one unit of equity and one unit of a zero-coupon bond with face value D and maturity T . At expiration, the value of debt, B_T , and equity, E_T , are given by

$$B_T = \min(V_T, D) = D - \max(D - V_T, 0), \quad (21.1)$$

$$E_T = \max(V_T - D, 0), \quad (21.2)$$

where V_T is the asset value of the firm at expiration. The interpretation of equations (21.1) and (21.2) is straightforward. Bondholders get paid fully only if the firm's assets exceed the face value of debt; otherwise, the firm is liquidated and assets are used to partially compensate bondholders. Equity holders, thus, are residual claimants in the firm because they get paid only after bondholders.

Note that equations (21.1) and (21.2) correspond to the payoff of standard European options. The first equation states that the bond value is equivalent to a long position on a risk-free bond and a short position on a put option with strike price equal to the face value of debt. The second equation states that equity value is equivalent to a long position on a call option with strike price equal to the face value of debt. Given the standard assumptions underlying the derivation of the Black-Scholes option pricing formula, the default probability in period t for a horizon of T years is given by the following formula:

$$p_t = N \left[- \frac{\ln \frac{V_t}{D} + \left(r - \frac{\sigma_A^2}{2} \right) T}{\sigma_A \sqrt{T}} \right], \quad (21.3)$$

where N is the cumulative normal distribution, V_t is the value of assets in period t , r is the risk-free rate, and σ_A is the asset volatility.

The numerator in equation (21.3) is the DD. An examination of equation (21.3) indicates that estimating default probabilities requires knowing both the asset value and the asset volatility of the firm. The required values, however, correspond to the *economic* values rather than the accounting figures. It is thus not appropriate to use balance-sheet data for estimating these two parameters. Instead, the asset value and volatility can be estimated. It is possible to solve the following equations (21.4) and (21.5) for the asset value and volatility:

$$E_t = V_t N(d_1) - e^{-rt} D N(d_2), \quad (21.4)$$

$$\sigma_E = \frac{V_t}{E_t} \sigma_A N(d_1), \quad (21.5)$$

if E_t , the value of equity; σ_E , the equity price return volatility; and D , the face value of liabilities, are known; and d_1 and d_2 are given by equations (21.6) and (21.7):

$$d_1 = \frac{\ln \frac{V_t}{D} + \left(r + \frac{\sigma_A^2}{2} \right) T}{\sigma_A \sqrt{T}}, \quad (21.6)$$

$$d_2 = d_1 - \sigma_A \sqrt{T}. \quad (21.7)$$

⁶ Note that DDs are risk neutral, that is, they do not take into account that risk preferences may be different between volatile and benign periods.

We derive the parameters from market data in the following manner:

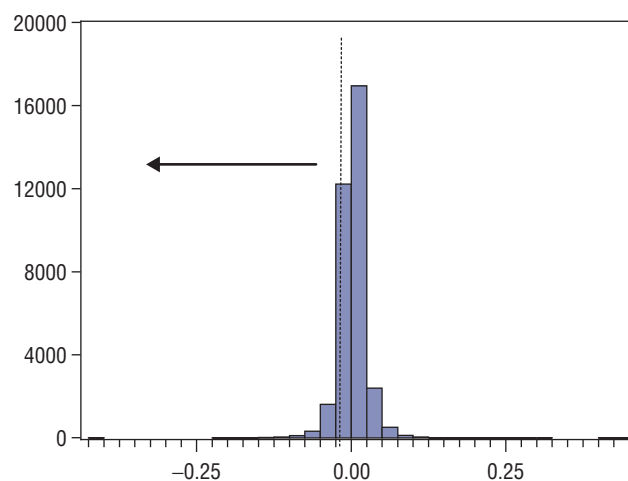
- The time horizon T is fixed at one year.
- The value of equity, E , corresponds to the market value of the firm. The data are obtained from Bloomberg by multiplying the number of shares outstanding for a firm by the closing share price on a particular day.
- The equity volatility, σ_E , corresponds either to historical equity volatility or implied volatility from equity options. This is calculated from the standard deviation of daily share price returns over a one-year period (which we define as 260 days).
- The face value of liabilities, D , usually is assumed equal to the face value of short-term liabilities plus half of the face value of long-term liabilities.⁷ This number represents the “default barrier.” The liability data are obtained from Bankscope. The item “Deposits and Short-Term Funding” is used to represent short-term liabilities, while the long-term liabilities are derived by deducting the short-term liabilities from the “Total Liabilities” item. To obtain daily liability data from annual balance sheets, the data are interpolated between two year-end balances.
- The risk-free rate, r , is the one-year government bond yield, in the same currency as those of the market and balance sheet data.

Once the asset value and volatility are estimated, the default probability of the firm, and thus the DD, could be derived from equation (21.3).

The Binomial Logit Model

We employ a binomial logit model to determine the likelihood that a large shock to one major bank would cause stress to a large counterpart. Specifically, we apply a model similar to that used by Gropp, Lo Duca, and Vesala (2006) to estimate the probability that the (percentage) change in the DD of one bank falls in a prespecified percentile in the negative tail, following large negative shocks to the DDs in the rest of the banks in the sample and after controlling for country-specific and global factors.

Defining the “Extreme Values.” First, we calculate the percentage change in the DD (which we denote ΔDD) from the generated series of DDs. The ΔDD is calculated over five trading days for the following reasons: (1) extreme events are more significant if they are prolonged; events that last for only a day are of little concern; and (2) the use of weekly changes reduces “noise” in the data.⁸ Corresponding ΔDD s between



Source: Authors.

Note: The 10th percentile left tail threshold for the stacked DD data of 18 banks is -0.018 .

Figure 21.2 Distribution of Changes in Distance to Default, 18-Bank Sample

banks reflect interdependencies that incorporate all potential channels for spillovers, thus precluding the need to define explicit links between banks or to specify a particular channel for potential spillovers. We define large shocks (or “extreme values”) as the 10th percentile left tail of the common distribution of the ΔDD s across all banks (Figure 21.2).⁹ We calculate the ΔDD , on a daily basis, as follows:

$$\Delta DD_{it} = \frac{DD_{it} - DD_{it-5}}{|DD_{it-5}|}. \quad (21.8)$$

We then stack all ΔDD_{it} observations from equation (21.8) and calculate the threshold, T_{10} , for the bottom 10 percent tail. For estimation purposes, we initially omit six banks—the three Japanese banks, Credit Agricole (France), Credit Suisse (Switzerland), and Halifax Bank of Scotland (HBOS) (United Kingdom)—because of the shorter periods for which their respective data are available. From the remaining 18 banks over the sample period May 30, 2000, through August 2, 2006, the threshold for the 10th percentile left tail is calculated at -0.018 . Observations that fall below this threshold, that is, in the bottom 10 percent tail, are the “extreme values.”

Applying the Econometric Model. A co-exceedance is defined as the probability that a particular bank will experience a large negative shock as a result of shock to another bank in the sample, after controlling for common shocks. That is, the observation of one extreme value as a result of another. The co-exceedances for each bank i at time t are defined as binary variables, y_{it} , such that

$$y_{it} = 1 \quad \text{if } \Delta DD_{it} < T_{10}, \text{ and } 0 \text{ otherwise,} \quad (21.9)$$

⁷ This is based on work done by MKMV (see Crosbie and Bohn, 2003).

⁸ For instance, stock price returns exhibit day-of-the-week effects (French, 1980; Keim and Stambaugh, 1984; Jaffe and Westerfield, 1985; Lakonishok and Smidt, 1988; Chang, Pinegar, and Ravichandran, 1993), while nonsynchronous trading effects related to the overnight or weekend non-trading periods affect the calculation of daily close-to-close returns (Rogalski, 1984), effects of which could be “smoothed” using weekly data.

⁹ Ideally, a 1st or even 5th percentile left tail would capture the very extreme events; however, either cutoff would have resulted in much too few observations for this period of data.

where T_{10} is the 10th percentile threshold in the left tail of the distribution.

We estimate the conditional probability that bank i will be in distress at time t conditional on bank j ($j \neq i$) being in distress, after controlling for other country-specific and global factors, as

$$\Pr(y_{it} = 1 | x, \beta) = \frac{e^{\alpha_i F_{it} + \sum_{s=1}^S \rho_{si} C_{it-s} + \gamma_j \sum_{j=1}^B C_{jt-1}}}{1 + e^{\alpha_i F_{it} + \sum_{s=1}^S \rho_{si} C_{it-s} + \gamma_j \sum_{j=1}^B C_{jt-1}}}, \quad (21.10)$$

which is based on the cumulative distribution function for the logistic distribution, where x represents the explanatory variables F and C , and β represents the slope coefficients α , ρ , γ . The parameter α represents the sensitivity of bank i to real and financial developments in its own country and in the global market, F_{it} ; ρ represents the sensitivity of bank i to extreme shocks it has experienced itself in the previous periods of up to s lags, C_{it-s} ,¹⁰ and γ represents the sensitivity of bank i to extreme shocks experienced by the rest of the banks in the sample during the previous period, C_{jt-1} (where $j \neq i$), or, in other words, the co-exceedance of bank i with other banks. All the C variables are lagged by one period to capture the impact on bank i from developments at the other banks, taking into account the differences in trading hours across the different time zones. The other explanatory variables are defined in the next subsection.¹¹

The goodness of fit in logit (and other binary) models is given by the McFadden R^2 . This statistic is the likelihood ratio index, computed as

$$R^2 = \frac{(1 - l(\tilde{\beta}))}{l(\tilde{\beta})}, \quad (21.11)$$

where $l(\tilde{\beta})$ is the restricted log likelihood—this is the maximized log likelihood value when all slope coefficients are restricted to zero and is equivalent to estimating the unconditional mean probability of an observation being in the tail.

Incorporating Nonbank Explanatory Variables

1. Country-specific market shocks

We use the local stock market return volatility to control for country-specific market shocks. We calculate the weekly (five trading-day) returns on each country-specific benchmark stock index by taking the weekly log-difference of the stock index in the local currency. The volatility of returns is proxied by the conditional variance estimated from a generalized autoregressive conditional heteroskedasticity or GARCH(1,1) model of the weekly returns,¹² such that

$$X_t = c + \varepsilon_t, \quad (21.12)$$

$$\sigma_t^2 = w + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (21.13)$$

where X_t is the weekly local currency return in the country's stock price index and σ_t^2 is the GARCH volatility, at time t . The autoregressive conditional heteroskedasticity or ARCH effect is captured by the lagged square residual, ε_{t-1}^2 . We predict this period's variance by forming the weighted average of a long-term mean (the constant, w), the forecast variance from the previous period (σ_{t-1}^2), and information about volatility observed in the last period (ε_{t-1}^2). This model is consistent with the volatility clustering associated with financial returns data, where large changes in returns are likely to be followed by further large changes. Lagrange multiplier tests show significant ARCH(1) effects for all the stock market returns used in this study.

2. Developments in the real economy

We use (five trading-day) changes in term structure spreads to represent expectations of changes in the business cycle in a bank's home country. Put another way, the changes in the spreads reflect the broader real economy developments in that country. The term structure spread is calculated as the difference between a long-term interest rate (the 10-year government bond yield) and a short-term rate (the one-year government bond yield) in any one country. Thus, the *change* in yield curve slope—our explanatory variable—is defined as follows:

$$\Delta y c_t = \frac{y c_t - y c_{t-5}}{|y c_{t-5}|}, \quad (21.14)$$

where $y c_t$ is the term structure spread at time t .

3. Global market shocks

We apply a global stock market return volatility variable to control for common shocks affecting global markets. This index is published in U.S. dollars but is converted to the currency of the country in which the bank associated with the dependent variable is located. We use the same method as that for the local stock markets and estimate the GARCH(1,1) volatility for the global stock market index.

B. Data

Our global data set includes the world's top 24 largest exchange-listed banking groups by total assets, as at end-2005, according to Bankscope.¹³ These comprise institutions from other major banking systems such as France, Germany, Japan, the Netherlands, Switzerland, the United Kingdom, and the United States; Spanish, Belgian, and Italian banking groups also make up the top 24, although banking activity in these three countries is much smaller by comparison. All

¹⁰ This operation adjusts for any serial correlation in the residuals, which may be induced by our use of overlapping weekly ΔDDs .

¹¹ The results are not significantly different when we apply the Gompertz distribution instead of the logistic distribution.

¹² This method was introduced by Ding and Engle (1991) and subsequently applied by De Santis and Gerard (1997, 1998); Bae, Karolyi, and Stulz (2003); and Ledoit, Santa-Clara, and Wolf (2003).

¹³ We originally selected the top 35 largest banks in the world but subsequently refined the sample to the 24 largest exchange-listed banks for which good quality and sufficient data are available. The list of banks in our data set and their corresponding data availability are presented in Appendix I, Table 21.5.

these banks have a presence in the major financial centers of the world.

Our EU data sample comprises the 33 largest listed EU banks, accounting for about a half of total EU banking system assets. We originally selected the top 50 largest banks in the EU and added the biggest bank in each EU country that would otherwise not have a representative in this category. We then refined the sample to include only banks for which good quality and sufficient data are available, which reduced the sample to 33 banks.¹⁴

We use separate control variables to account for common factors affecting local financial markets, the local real economy, and regional and global market developments. Specifically, we incorporate the following:

- The price return volatility of the local stock market index returns to capture local market influences.
- Changes in the slope of the local term structure (between one- and 10-year government bonds) to represent developments in the domestic real economy.¹⁵
- The price return volatility in the Morgan Stanley Capital International (MSCI) All-Country World Index returns to account for global market factors.
- For the EU-specific analysis, the price return volatility in the MSCI All-Country Europe Index returns to account for regional shocks as well.

Balance sheet data for the individual banks, used in all the DD calculations, are obtained from Bankscope, while the requisite market data are available from Bloomberg.¹⁶

Several caveats apply to our data. First, some of the banking groups in our sample represent important constituents in their respective country's stock market indices and may also be represented in the global and regional indices. This means that some of the stock market volatility effects captured in the results could be driven partly by the volatility in the individual bank stocks. This suggests that the impact of idiosyncratic shocks on interbank spillovers represents "conservative" estimates.¹⁷ Second, the balance sheet data on banks' long- and short-term liabilities are available only on an annual basis from Bankscope. Thus, our calculation of daily DDs requires extrapolation between two data points. In this case, we assume that the liabilities change proportionally each day between the two reporting dates. Finally, the DD risk measure does not factor in default risk arising from off-balance-

sheet exposures, which could be substantial especially for major international banks engaged in proprietary trading activities.¹⁸

2. DEFAULT AND SPILLOVER RISKS FOR THE GLOBAL BANKING SYSTEM

A. Analysis

The sample period covered in the analysis is May 30, 2000, to August 2, 2006. However, data for six banks—Credit Agricole (France), Credit Suisse (Switzerland), HBOS (United Kingdom), Mitsubishi UFJ (Japan), Mizuho (Japan), and Sumitomo Mitsui (Japan)—are available only from later dates. Thus, only 18 banks (the "main sample") are tested for the full sample period; the other banks are subsequently added to the main sample as their data become available, and we rerun the tests for each expanded sample.

An examination of banks' DD suggests that bank soundness broadly deteriorated across countries during the mid-2000 to mid-2003 period.¹⁹ The collective decline in DDs has coincided with the period following the bursting of the global information technology bubble; the slowdown in global economic growth; and the economic and financial difficulties experienced in some Latin American countries, such as Argentina and Brazil, where some of the major banks have direct business interests. The U.K. and U.S. banks appear to have been less affected by the general turbulence, relative to banks from other countries, as their DDs remained relatively stable during this time.

The stresses on the global banking system during the first half of the sample period are also evident from the number of negative extreme values, or left-tail events, across banks.²⁰ The health of the global banking system then improved vastly over the mid-2003 to end-2005 period. The DDs of all banks in our sample rose strongly during this period; correspondingly, the overall number of left-tail events fell substantially. However, the global banking system came under some pressure in 2006. The number of left-tail events increased across many banks in our sample during this period. Although the exact causes of the observed stress are unclear, it has coincided with the oil and commodity price shocks experienced in early 2006, as well as with the sharp corrections in global asset prices observed in the second quarter of 2006.

In the world's biggest banking market, we find that U.S. banks are vulnerable to spillover risk from banks from their own country as well as from overseas. Table 21.1 suggests that some of these banks also are susceptible to shocks affecting European banks; however, no European bank represents a consistently common spillover risk. Among U.S. banks, Goldman

¹⁴ The EU adopted a regulation requiring public companies to convert to International Financial Reporting Standards (IFRS) beginning in 2005. All publicly traded EU companies were required to prepare their consolidated accounts using IFRS from 2005. Thus, the Bankscope balance sheet data from 2005 onward incorporate IFRS requirements.

¹⁵ See, for example, Estrella and Hardouvelis (1991); Bernard and Gerlach (1998); and Estrella (2005).

¹⁶ Details and sources of the market data are presented in Appendix I, Table 21.6.

¹⁷ We test for robustness by omitting the local stock market variable and rerunning the binomial logit model. Our results show that the contagion effects remain largely the same; some of the local market effects are captured by the global market variable. However, the McFadden R^2 is slightly stronger for the existing model.

¹⁸ Notwithstanding this limitation, empirical studies have shown that the DD is still a good indicator of default risk in the banking sector (Chan-Lau, Jobert, and Kong, 2004; Gropp, Vesala, and Vulpes, 2004).

¹⁹ See Appendix II, Figure 21.3.

²⁰ See Appendix II, Figure 21.4.

TABLE 21.1

Contagion Risk among the World's Biggest Banking Groups (18 Banks), May 30, 2000, to August 2, 2006
(in percent level of significance)

Country	Bank	Contagion to						
		Belgium	France		Germany	Italy	Netherlands	
		Fortis	BNP Paribas	Societe Generale	Deutsche Bank	Unicredito	ABN Amro	ING
	Initial shock to							
Belgium	Fortis				5.0			1.0
France	BNP Paribas			5.0				
	Societe Generale							
Germany	Deutsche Bank							5.0
Italy	Unicredito							1.0
Netherlands	ABN Amro	5.0	1.0	1.0	1.0			
	ING							
Spain	Santander							
Switzerland	UBS			5.0				
United Kingdom	Barclays							
	HSBC				5.0			1.0
	RBS							
United States	Bank of America							
	Citigroup	1.0						
	Goldman Sachs							
	JP Morgan Chase							
	Merrill Lynch			5.0				
	Morgan Stanley						5.0	
Other factors	Constant	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	Local market volatility	1.0	1.0		1.0	1.0	1.0	1.0
	Change in term structure		1.0	1.0			1.0	
	Global market volatility							1.0
McFadden R ²		0.45	0.47	0.45	0.45	0.57	0.49	0.50

Source: Authors.

Note: Factors in the columns are dependent on factors in the rows.

Blank cells represent nonsignificance at any level below 10 percent.

■ represents coefficients on own lags, of up to 5 lags.

■ represents significance at the 1 percent level.

■ represents significance at the 5 percent level.

■ represents significance (negative sign) at the 5 percent level or lower.

□ represents the grouping of major banks in any one country, where 3 or more banks are represented in the sample.

Sachs appears to be largely insulated from external shocks, while shocks to U.S. banks have had some impact on their European counterparts. There appears to be little interaction between several U.S. banks with domestic stock market and interest rate shocks. This suggests that stresses to U.S. banks during this period have not necessarily been tied to develop-

ments in the local market or economy. Rather, bank soundness appears to be more closely related to volatilities in global markets, potentially reflecting the global nature of U.S. banking businesses.

Spillover risk appears quite significant among European banks. Shocks to ABN Amro (Netherlands) appear to affect

to spillover risk from Barclays; in turn, Barclays appears to be exposed to shocks from both of these banks as well.

Interestingly, our results thus far show some common threads with those of Gropp and Moerman (2004), notwithstanding the differences in time periods, model, and bank samples. Like us, the authors identify ABN Amro (Netherlands) and HSBC (United Kingdom) to be among the more systemically important banks for those outside their own country. They also find close links among banks within countries.

B. Robustness tests with subsamples

Next, we split the sample period into two subsamples, to determine the robustness of our initial findings. A natural structural break would be around mid-2003, which separates the more turbulent period in global economic and market conditions from the benign period that followed. Thus, we define the first subsample as May 30, 2000, to May 30, 2003, and the second as June 1, 2003 to August 2, 2006.

Our results reveal several broad trends among the major international banks. We find evidence of a “home bias” for spillover risk (Table 21.2). For example, spillovers from U.K. banks to local counterparts are more prevalent than to foreign banks. Similarly, shocks to U.S. banks also have a proportionately greater impact on their local counterparts. That said, individual banks are affected differently by idiosyncratic shocks to their major counterparts, possibly because of their different business and geographic mixes. Banks are also not similarly affected by common shocks to the domestic real economy or to financial markets, although the global banking system as a whole tends to be more exposed to these shocks during more turbulent periods, compared with the more benign times.

Importantly, spillover risk across the major global banks has risen in recent years (Table 21.3). The exposure of U.S. banks to each other appears to have intensified; the impact of shocks in U.S. banks on foreign banks has also increased, as have spillovers from U.K. to continental European banks. Individually, shocks to Societe Generale (France), HSBC (United Kingdom), and Morgan Stanley (United States) have had increasingly greater impact on foreign banks. Within Europe, spillover risk from the French banks appears to have increased over time.²¹

We subsequently add the remaining six banks to the sample of 18 banks as their data become available, and we rerun the tests for each expanded sample.²² The banks are added in the following order: Mitsubishi UFJ (Japan), HBOS (United Kingdom), Credit Agricole (France), Credit Suisse (Switzerland), Sumitomo Mitsui (Japan), and Mizuho (Japan).²³ Our

TABLE 21.2

Significant Co-Exceedances, 2000–2006
(in percent of total possible bank transmission channels)

Initial Shock to Banks in	Contagion to Banks in		
	Continental Europe	United Kingdom	United States
Continental Europe	17	4	9
United Kingdom	6	67	6
United States	6	6	23

Source: Authors.

TABLE 21.3

Change in the Number of Significant Co-Exceedances, 2000–2003 to 2003–2006 (in percent)

Initial Shocks to Banks in	Contagion to Banks in		
	Continental Europe	United Kingdom	United States
Continental Europe	20	0	–25
United Kingdom	400	0	0
United States	300	*	40

Source: Authors.

*The number increased from zero to 1.

analysis of each of the six sets of results reveals several notable trends:

- Among U.S. banks, Morgan Stanley and Goldman Sachs largely are insulated from spillovers. Morgan Stanley consistently represents the biggest spillover risk for foreign banks; Citigroup also has become increasingly important in recent years.
- In continental Europe, shocks to Societe Generale has had the widest impact over time, while banks such as Fortis (Belgium) and Santander (Spain) have become more exposed to shocks from elsewhere. In the United Kingdom, Barclays is the consistent risk factor for its local counterparts, while HSBC remains the most important U.K. spillover risk factor for foreign banks.
- Spillover risk for major Japanese banks has been limited. Japanese banks appear to pose little risk to the other major international banks, despite the size of the banking system, which is the fourth largest in the world. Similarly, these banks are largely insulated from shocks to foreign banks.

3. EXTENSIONS: SPILLOVER RISK WITHIN THE EU

We subsequently examine the spillover risks among major banks from EU countries. The sample consists of 33 banks.²⁴

²¹ The detailed results are available on request.

²² The 10th percentile left tail for each sample, expanded by one bank at a time, remains at –0.018.

²³ See Appendix III, Table 21.7, for the 24-bank results. Detailed results for the 19–23 bank samples are available on request.

²⁴ See Appendix IV, Tables 21.8 and 21.9, for lists of exchange-traded banks and associated market data.

TABLE 21.4

Significant Co-Exceedances among EU Banks,
May 2000–April 2007

Period	Number of Significant Links ¹	As Percent of All Possible Links ²
May 2000–April 2007		
Domestic	19	39.6
Cross-border	57	8.7
May 2000–November 2003		
Domestic	14	28.6
Cross-border	50	7.6
December 2003–April 2007		
Domestic	9	18.8
Cross-border	54	8.3
November 2005–April 2007		
Domestic	13	18.6
Cross-border	86	8.7

Source: Authors.

1. Number of bank pairs for which co-exceedances were significant at the 5 percent level in the given period.

2. Number of significant links in percent of all possible contagion channels (i.e., as a percentage of all possible domestic and cross-border pairings of banks, respectively).

The period covered is May 30, 2000, to April 30, 2007. However, the data for six banks—Bank Austria Creditanstalt (Austria), Credit Agricole (France), Deutsche Postbank (Germany), HBOS (United Kingdom), National Bank of Greece (Greece), and PKO (Poland)—are available only from later dates. Thus, only 27 banks are tested for the full sample period (the “main sample”); the other banks are added subsequently to the main sample as their data become available, and we rerun the tests for each expanded sample. The spillover risks among EU banks are summarized in Table 21.4.²⁵ We derive the following main observations from the results:²⁶

- *Consistent with the results for global banks, “home bias” is also evident within the EU.* Spillovers among banks in the same country appear to be relatively more frequent than among banks from different countries. For the whole sample period, significant spillovers were found in about 40 percent of all possible domestic links, compared with about 9 percent of all possible cross-border links. This result is significant (at the 5 percent level), and it also seems robust over time: for all the subperiods, the relative frequency of co-exceedances among domestic banks was higher than the relative frequency of co-exceedances among banks from different countries.
- *However, cross-border linkages remain an important consideration.* The absolute number of significant

cross-border spillovers in the EU sample was higher than the number of significant domestic spillovers. This is driven by the number of *potential* cross-border linkages among the large banks, which is much higher than the number of potential domestic linkages. So, even with the lower relative frequency, cross-border co-exceedances are more numerous than domestic co-exceedances (57 compared with 19 for the full sample). This finding may seem trivial, but it serves as a reminder that significant cross-border linkages, even if relatively less frequent than domestic linkages, may still be quite important and may require more attention (e.g., in terms of supervisory time) than suggested by the relative frequencies.

- *Moreover, the relative frequency of spillovers across borders has been increasing.* When we divide the sample period into two, we find the proportion of banks for which international spillovers is significant rising, from 7.6 percent in May 2000–November 2003 to 8.3 percent in December 2003–April 2007 and 8.7 in November 2005–April 2007; the proportion decreases for domestic linkages, from 28.6 percent in May 2000–November 2003 to 18.8 percent in December 2003–April 2007 and 18.6 in November 2005–April 2007.²⁷
- *The spillover risks posed by the large EU banks differ across institutions.* Some banks (e.g., OTP or Bank of Ireland) have no significant spillover impact on other banks, while others have significant impact on a number of domestic and foreign banks at the same time. Interestingly, the bank with the biggest potential for cross-border spillovers within the EU is Fortis (which ranks 19th in the EU in terms of total assets), which has significant impact on eight other banks (six cross-border and two domestic). HSBC is second, with six spillover links (five cross-border and one domestic). The largest number of banks (19) affect between one and three other cross-border banks.

4. CONCLUSION

This chapter uses market-based indicators to highlight potential interrelationships among the world’s biggest banks from three regions, as well as among the EU’s biggest banking groups, and their exposure to spillover risk from their counterparts. Specifically, the main objective is to identify the direction of spillovers among those banks through the international banking system. In doing so, our results also provide some information on areas where risks may be concentrated, thus highlighting relationships that may require closer supervision and surveillance and a more detailed understanding of linkages by the local authorities. Our findings could also help country

²⁵ See Appendix IV, Tables 21.10 and 21.11, for 27- and 33-bank results, respectively.

²⁶ As a side result, the tables also show the significance levels for the control variables. In many cases, the “common factors” turn out to be insignificant, but they are significant for some banks. Also, the number of significant cases is higher for the more recent subperiods.

²⁷ Detailed results are available upon request.

authorities focus their collaborative supervisory efforts on specific areas, given their limited resources.

Using an EVT framework, our results yield several clear trends in the interrelationships among the major banks. Overall, the risk of spillovers among local banks is important (“home bias”). Banks also are affected by common shocks to the real economy or financial markets, although they tend to be more vulnerable to these shocks during more turbulent periods, compared with more benign times. Meanwhile, spillover risk among these banks appears to have increased over time.

In light of these findings, ensuring sound risk management practices continues to be a key challenge, both for supervisors and for banks. Given the continuing growth and increased complexity of banks’ businesses, risk management techniques need to be continually enhanced and improved. In many countries, supervisors also are promoting greater use of stress testing as a key risk management tool.

Appropriately, greater emphasis is being placed on improving cooperation in cross-border financial crisis prevention and management. European regulators and supervisors and the European Commission support more efficient, risk-based, cross-border collaboration among supervisors. Internationally, the existing tripartite of Switzerland, the United Kingdom, and the United States is considered one of the most fully developed examples of home–host collaboration in supervision. In other collaborative efforts, the U.K. Financial Supervisory Authority and the New York Federal Reserve have worked closely and continuously with major participants in the credit risk transfer market to resolve the issue of backlogs in trade confirmations and assignments and continue to emphasize the need for “borderless” solutions in the oversight of the credit derivatives market (Geithner, McCarthy, and Nazareth, 2006). EU authorities have also signed the Memorandum of Understanding for crisis management, which includes performing crisis simulation exercises at the EU level. Nonetheless, country authorities largely acknowledge that there is a need for further work on cross-border coordination and information sharing between national authorities in promoting financial stability (Gieve, 2006).

REFERENCES

- Allen, Franklin, and Douglas Gale, 2000, “Financial Contagion,” *Journal of Political Economy*, Vol. 108, No. 1, pp. 1–33.
- Bae, Kee-Hong, G. Andrew Karolyi, and Rene M. Stulz, 2003, “A New Approach to Measuring Financial Contagion,” *Review of Financial Studies*, Vol. 16, No. 3, pp. 717–63.
- Bank of England, 2006, *Financial Stability Report*, No. 20 (London, July).
- Bernard, Henri, and Stefan Gerlach, 1998, “Does the Term Structure Predict Recessions? The International Evidence,” *International Journal of Finance and Economics*, Vol. 3, No. 3, pp. 195–215.
- Black, Fisher, and Myron S. Scholes, 1973, “The Pricing of Options and Corporate Liabilities,” *Journal of Political Economy*, Vol. 81, No. 3, pp. 637–54.
- Chan-Lau, Jorge A., Martin Čihák, Srobona Mitra, and Li Lian Ong, 2012, “Identifying Spillover Risk in the International Banking System: An Extreme Value Theory Approach,” *International Journal of Finance and Economics*, Vol. 17, No. 4, pp. 390–406.
- Chan-Lau, Jorge A., Arnaud Jobert, and Janet Q. Kong, 2004, “An Option-Based Approach to Bank Vulnerabilities in Emerging Markets,” IMF Working Paper 04/33 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=17059>
- Chan-Lau, Jorge A., Donald J. Mathieson, and James Y. Yao, 2004, “Extreme Contagion in Equity Markets,” *IMF Staff Papers*, Vol. 51, No. 2, pp. 386–408. Available via the Internet: <http://www.imf.org/External/Pubs/FT/staffp/2004/02/chanlau.htm>
- Chang, Eric C., J. Michael Pinegar, and Ravi Ravichandran, 1993, “International Evidence on the Robustness of the Day-of-the-Week Effect,” *Journal of Financial and Quantitative Analysis*, Vol. 28, No. 4, pp. 497–513.
- Čihák, Martin, and Li Lian Ong, 2007, “Estimating Spillover Risk among Large EU Banks,” IMF Working Paper 07/267 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=21459>
- Crosbie, Peter, and Jeffrey R. Bohn, 2003, “Modeling Default Risk,” Moody’s KMV (December 18). Available via the Internet: http://www.creditrisk.ru/publications/files_attached/modeling_default_risk.pdf
- De Santis, Giorgio, and Bruno Gerard, 1997, “International Asset Pricing and Portfolio Diversification with Time-Varying Risk,” *Journal of Finance*, Vol. 52, No. 5, pp. 1881–912.
- , 1998, “How Big Is the Premium for Currency Risk?” *Journal of Financial Economics*, Vol. 49, No. 3, pp. 375–412.
- Ding, Zhuanxin, and Robert Engle, 1991, “Large Scale Conditional Covariance Matrix Modeling, Estimation and Testing,” *Academia Economic Papers*, Vol. 29, No. 2, pp. 157–84.
- Duggar, Elena, and Srobona Mitra, 2009, “External Linkages and Contagion Risk: The Case of Irish Banks,” *IMF Staff Papers*, Vol. 56, No. 4, pp. 758–86.
- Estrella, Arturo, 2005, “Why Does the Yield Curve Predict Output and Inflation?” *Economic Journal*, Vol. 115, No. 5, pp. 722–44.
- Estrella, Arturo, and Gikas Hardouvelis, 1991, “The Term Structure as a Predictor of Real Economic Activity,” *Journal of Finance*, Vol. 46, No. 2, pp. 555–76.
- Forbes, Kristin, and Roberto Rigobon, 2001, “Measuring Contagion: Conceptual and Empirical Issues,” in *International Financial Contagion*, edited by Stijn Claessens and Kristin Forbes (Boston: Kluwer Academic Publishers).
- French, Kenneth R., 1980, “Stock Returns and the Weekend Effect,” *Journal of Financial Economics*, Vol. 8, No. 1, pp. 55–69.
- Freixas, Xavier, Bruno Parigi, and Jean-Charles Rochet, 2000, “Systemic Risk, Interbank Relations and Liquidity Provision by the Central Bank,” *Journal of Money, Credit, and Banking*, Vol. 32, No. 3, pp. 611–38.
- Geithner, Timothy, Callum McCarthy, and Annette Nazareth, 2006, “A Safer Strategy for the Credit Products Explosion,” *Financial Times* (London, September 27). Available via the Internet: <http://www.ft.com/intl/cms/s/1/ba807a22-4e4d-11db-bcbc-0000779e2340.html#axzz2wNtlxHs2>
- Gieve, John, 2006, “Practical Issues in Preparing for Cross-Border Financial Crises,” Financial Stability Forum Workshop: Planning and Communication for Financial Crises and Business Continuity Incidents (London, November 13).

- Gropp, Reint, Mario Lo Duca, and Jukka Vesala, 2006, "Cross-Border Bank Contagion in Europe," ECB Working Paper 662, November (Frankfurt: European Central Bank). Available via the Internet: <http://www.ecb.europa.eu/pub/scientific/wps/date/html/wps2006.en.html>
- Gropp, Reint, and Gerard Moerman, 2004, "Measurement of Contagion in Bank Equity Prices," *Journal of International Money and Finance*, Vol. 23, No. 3, pp. 405–59.
- Gropp, Reint, Jukka Vesala, and Guiseppe Vulpes, 2004, "Equity and Bond Market Signals as Leading Indicators of Bank Fragility," *Journal of Money, Credit and Banking*, Vol. 38, No. 2, pp. 399–428.
- Hartmann, Philipp, Stefan Straetmans, and Caspar G. de Vries, 2004, "Asset Market Linkages in Crisis Periods," *Review of Economics and Statistics*, Vol. 86, No. 1, pp. 313–26.
- Jaffe, Jeffrey, and Randolph Westerfield, 1985, "The Weekend Effect in Common Stock Returns: The International Evidence," *Journal of Finance*, Vol. 40, No. 2, pp. 433–54.
- Keim, Donald B., and Robert F. Stambaugh, 1984, "A Further Investigation of the Weekend Effects in Stock Returns," *Journal of Finance*, Vol. 39, No. 3, pp. 819–40.
- Lakonishok, Josef, and Seymour Smidt, 1988, "Are Seasonal Anomalies Real? A Ninety-Year Perspective," *Review of Financial Studies*, Vol. 1, No. 1, pp. 403–25.
- Ledoit, Olivier, Pedro Santa-Clara, and Michael Wolf, 2003, "Flexible Multivariate GARCH Modeling with an Application to International Stock Markets," *Review of Economics and Statistics*, Vol. 85, No. 3, pp. 735–47.
- Longin, Francois, and Bruno Solnik, 2001, "Extreme Correlation of International Equity Markets," *Journal of Finance*, Vol. 56, No. 2, pp. 649–76.
- Merton, Robert C., 1974, "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates," *Journal of Finance*, Vol. 29, No. 2, pp. 449–70.
- Morgan, Donald P., 2002, "Rating Banks: Risk and Uncertainty in an Opaque Industry," *American Economic Review*, Vol. 92, No. 4, pp. 874–88.
- Poon, Ser-Huang, Michael Rockinger, and Jonathan Tawn, 2004, "Extreme-Value Dependence in Financial Markets: Diagnostics, Models, and Financial Implications," *Review of Financial Studies*, Vol. 17, No. 2, pp. 581–610.
- Quintos, Carmela E., 2001, "Estimating Tail Dependence and Testing for Contagion Using Tail Indices," Working Paper, Simon Graduate School of Business (New York: University of Rochester).
- Rogalski, Richard J., 1984, "A Further Investigation of the Weekend Effect in Stock Returns: Discussion," *Journal of Finance*, Vol. 39, No. 3, pp. 835–37.
- Schoenmaker, Dirk, and Christiaan van Laecke, 2006, "Current State of Cross-Border Banking," Proceedings of the Eighth Annual International Banking Conference on International Financial Instability: Cross-Border Banking and National Regulation, Federal Reserve Bank of Chicago, Chicago, October 5–6.
- Starica, Catalin, 1999, "Multivariate Extremes for Models with Constant Conditional Correlations," *Journal of Empirical Finance*, Vol. 6, No. 5, pp. 515–53.

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Appendix I

Data Sets

TABLE 21.5

World's Largest Banking Groups: Exchange-Listed Institutions

Rank by Total Assets as at End-2005	Banking Group	Nationality	Bloomberg Ticker	Date from Which DD Data Are Available
1	Barclays PLC*	United Kingdom	BARC LN	May 17, 2000
2	UBS AG*	Switzerland	UBSN VX	May 30, 2000
3	Mitsubishi UFJ Financial Group, Inc.	Japan	8306 JP	April 3, 2002
4	HSBC Holdings PLC*	United Kingdom	HSBA LN	May 17, 2000
5	Citigroup, Inc.*	United States	C US	May 17, 2000
6	BNP Paribas*	France	BNP FP	May 30, 2000
7	ING Groep NV	Netherlands	INGA NA	May 30, 2000
8	Royal Bank of Scotland Group PLC*	United Kingdom	RBS LN	May 17, 2000
9	Bank of America Corporation*	United States	BAC US	May 17, 2000
10	Crédit Agricole SA	France	ACA FP	December 16, 2002
11	Mizuho Financial Group	Japan	8411 JP	March 12, 2004
12	JP Morgan Chase & Co.*	United States	JPM US	May 17, 2000
13	Deutsche Bank AG*	Germany	DBK GR	May 30, 2000
14	ABN Amro Holding NV*	Netherlands	AABA NA	May 30, 2000
15	Credit Suisse Group*	Switzerland	CSGN VX	January 1, 2003
16	Société Générale*	France	GLE FP	May 30, 2000
17	Banco Santander Central Hispano SA	Spain	SAN SM	May 30, 2000
18	HBOS PLC*	United Kingdom	HBOS LN	September 11, 2002
19	UniCredito Italiano SpA	Italy	UC IM	May 30, 2000
20	Morgan Stanley*	United States	MS US	May 17, 2000
21	Sumitomo Mitsui Financial Group, Inc.	Japan	8316 JP	December 3, 2003
22	Fortis	Belgium	FORB BB	May 30, 2000
23	Goldman Sachs Group, Inc.*	United States	GS US	May 17, 2000
24	Merrill Lynch & Co., Inc.*	United States	MER US	May 17, 2000

Source: Bloomberg.

Note: DD = distance to default.

*Identified by the Bank of England (BoE, 2006) as large, complex financial institutions (LCFIs), which carry out a diverse and complex range of activities in major financial centers.

TABLE 21.6

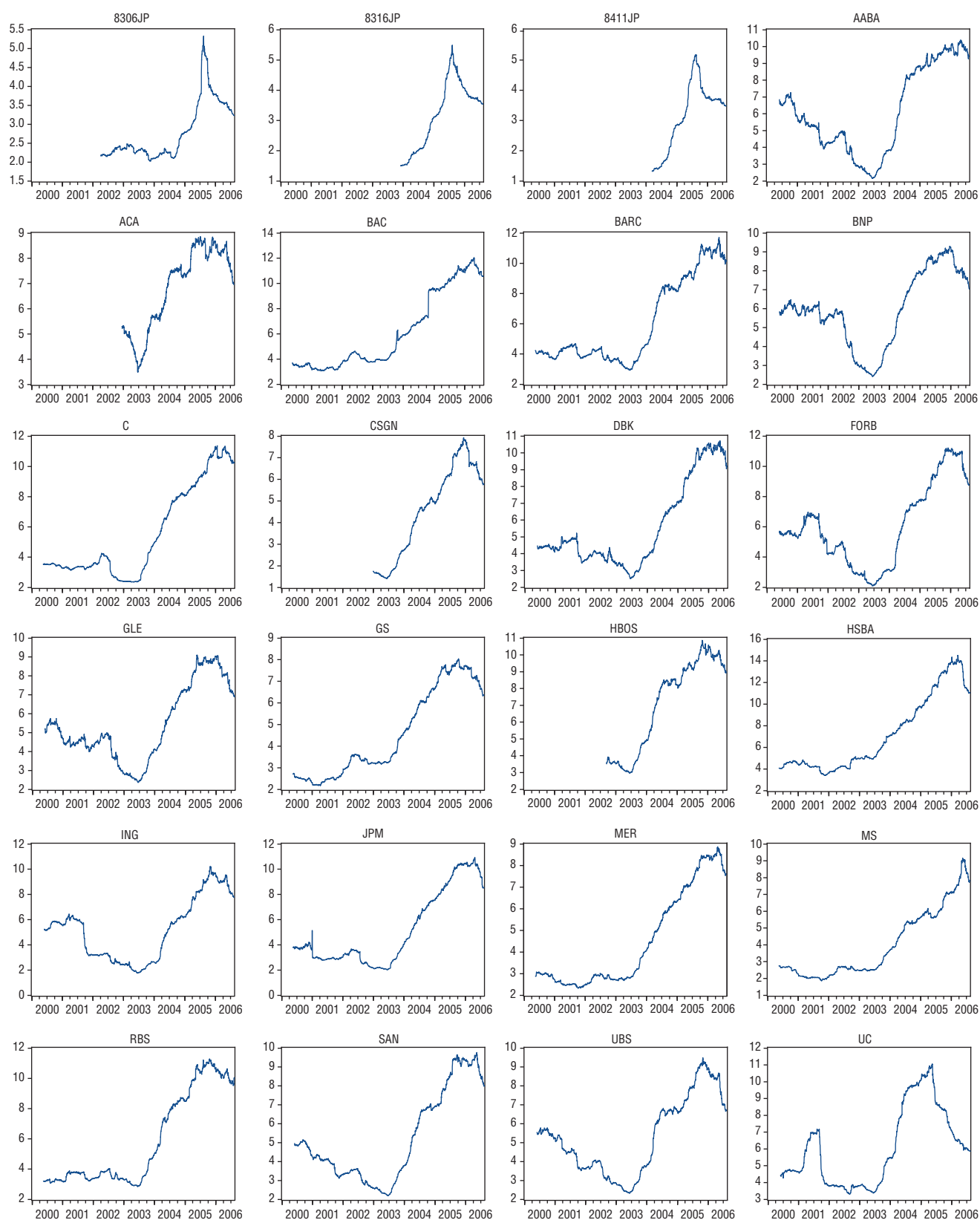
Stock Market Indices and Government Bond Yields

Country	Stock Market		Bond	
	Index	Bloomberg Ticker	Maturity	Bloomberg Ticker
Belgium	BEL 20	BEL20	EUR Belgium sovereign zero coupon yield, 1 year	I90001Y
			EUR Belgium sovereign zero coupon yield, 10 year	I90010Y
France	CAC 40	CAC	EUR France sovereign zero coupon yield, 1 year	I01401Y
			EUR France sovereign zero coupon yield, 10 year	I01410Y
Germany	DAX 30	DAX	EUR Germany sovereign zero coupon yield, 1 year	F91001Y
			EUR Germany sovereign zero coupon yield, 10 year	F91010Y
Italy	S&P MIB	SPMIB	EUR Italy sovereign zero coupon yield, 1 year	F90501Y
			EUR Italy sovereign zero coupon yield, 10 year	F90510Y
Japan	Nikkei 225	NKY	JPY Japan sovereign 10- to 30-year zero coupon yield, 1 year	F10501Y
			JPY Japan sovereign 10- to 30-year zero coupon yield, 10 year	F10510Y
Netherlands	AEX	AEX	EUR Netherlands sovereign zero coupon yield, 1 year	F92001Y
			EUR Netherlands sovereign zero coupon yield, 10 year	F92010Y
Spain	IBEX 35	IBEX	EUR Spain sovereign zero coupon yield, 1 year	F90201Y
			EUR Spain sovereign zero coupon yield, 10 year	F90210Y
Switzerland	SMI	SMI	CHF Switzerland sovereign zero coupon yield, 1 year	F25601Y
			CHF Switzerland sovereign zero coupon yield, 10 year	F25610Y
United Kingdom	FTSE 100	UKX	GBP United Kingdom zero coupon yield, 1 year	I02201Y
			GBP United Kingdom zero coupon yield, 1 year	I02210Y
United States	S&P 500	SPX	USD Treasury actives zero coupon yield, 1 year	I02501Y
			USD Treasury actives zero coupon yield, 10 year	I02510Y
World	MSCI All-Country World	MXWD		

Source: Bloomberg.

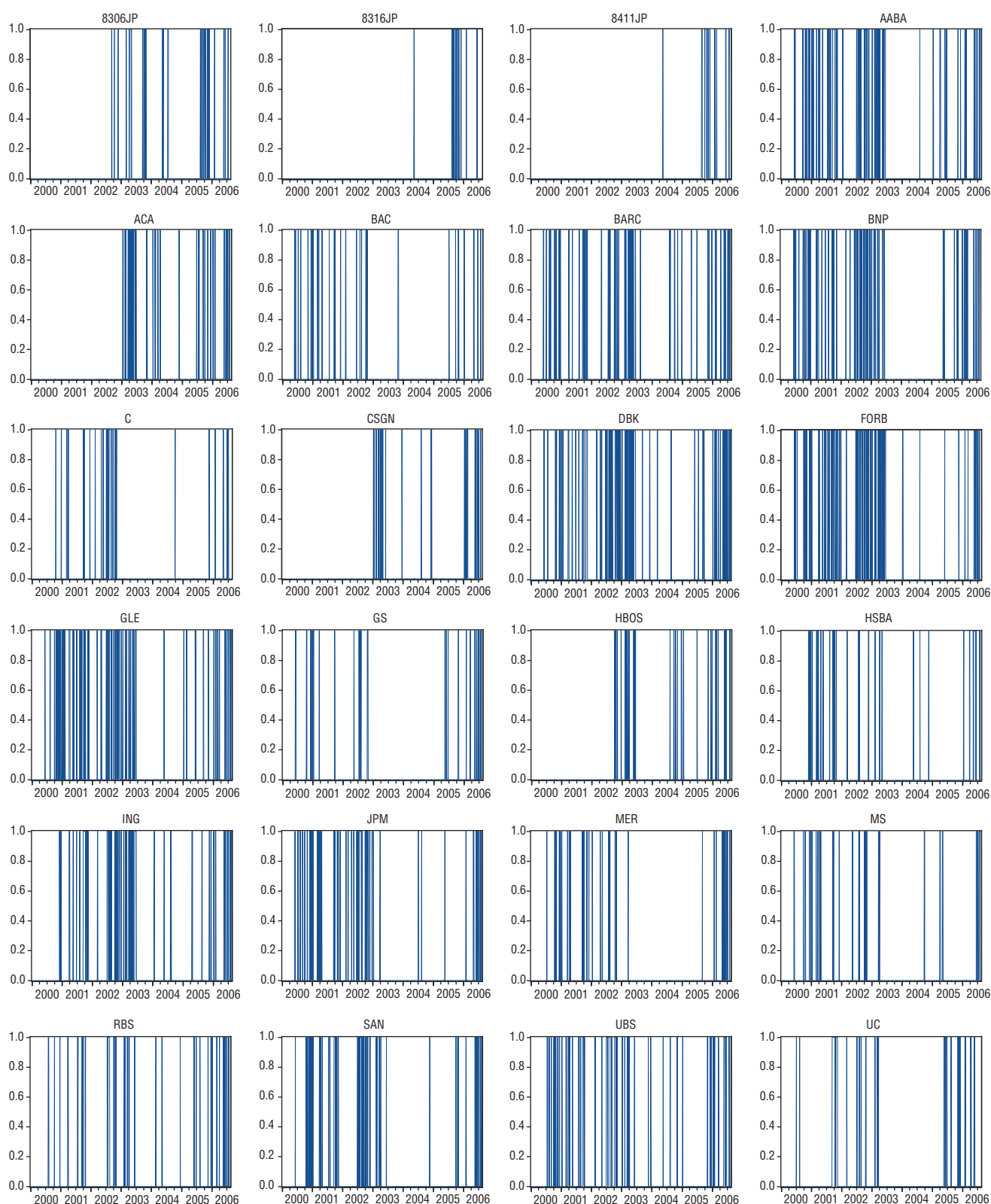
Appendix II

Results



Sources: Authors; Bankscope; and Bloomberg.

Figure 21.3 World's Largest Banking Groups: Distances to Default



Sources: Authors; Bankscope; and Bloomberg.

Figure 21.4 World's Largest Banking Groups: Binomial Logit Representations of the 10th Percentile Left Tail ("extreme values" or "exceedances")

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Appendix III

Binomial Logit Results

TABLE 21.7

Contagion Risk among the World's Biggest Banking Groups (24 Banks), March 12, 2004, to August 2, 2006
(in percent level of significance)

Country	Bank	Contagion to								
		Belgium	France			Germany	Italy	Japan		
		Fortis	BNP Paribas	Credit Agricole	Societe Generale	Deutsche Bank	Unicredito	Mitsubishi UFJ	Mizuho	Sumitomo
	Initial shock to									
Belgium	Fortis				5.0					
France	BNP Paribas					1.0				
	Credit Agricole		5.0			1.0				
	Societe Generale	1.0		5.0			5.0		1.0	5.0
Germany	Deutsche Bank									
Italy	Unicredito									
Japan	Mitsubishi UFJ								1.0	5.0
	Mizuho							5.0		
	Sumitomo							5.0		
Netherlands	ABN Amro									5.0
	ING									
Spain	Santander	5.0		1.0				1.0	1.0	
Switzerland	Credit Suisse									
	UBS									
United Kingdom	Barclays									
	HBOS									
	HSBC					1.0				
	RBS	1.0	1.0		5.0					
United States	Bank of America									
	Citigroup	5.0		5.0			5.0			
	Goldman Sachs									
	JP Morgan Chase									
	Merrill Lynch									
	Morgan Stanley	1.0								
Other factors	Constant	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	Local market volatility	1.0		5.0				5.0	1.0	5.0
	Change in term structure					1.0				
	Global market volatility									
McFadden R^2		0.81	0.57	0.60	0.58	0.56	0.53	0.54	0.70	0.55

Source: Authors.

Note: Factors in the columns are dependent on factors in the rows.

Blank cells represent nonsignificance at any level below 10 percent.

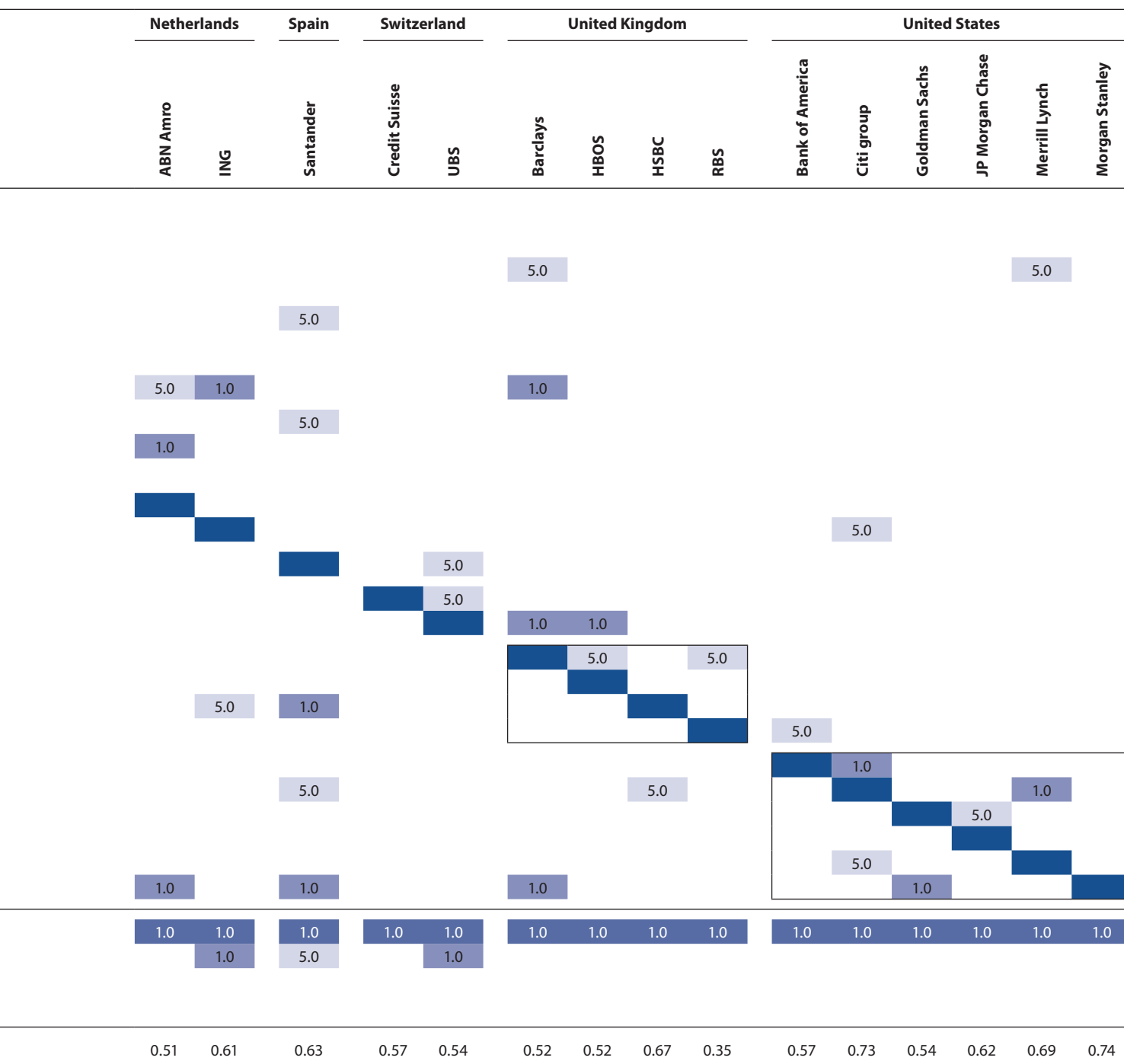
■ represents coefficients on own lags, of up to 5 lags.

■ represents significance at the 1 percent level.

■ represents significance at the 5 percent level.

■ represents significance (negative sign) at the 5 percent level or lower.

□ represents the grouping of major banks in any one country, where 3 or more banks are represented in the sample.



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Appendix IV

EU-Only Sample:

Data Sets and Results

TABLE 21.8

EU Banks: Exchange-Listed Institutions

Major Banking Groups	Nationality	Stock Ticker	Currency	Date of Availability				
				Stock Market		Risk-Free Rate	Financial Statement	Distance to Default
				Standard Deviation of Returns	Capitalization			
Erste Bank der Oesterreichischen Sparkassen AG	Austria	EBS AV	EUR	ρ	From May 30, 2000	ρ	§	From May 30, 2000
Bank Austria Creditanstalt AG	Austria	BACA AV	EUR	From July 8, 2004	From July 9, 2003	ρ	§	From 8 July, 2004
Fortis Group	Belgium	FORB BB	EUR	ρ	From May 30, 2000	ρ	§	From May 30, 2000
KBC Group—KBC Groep NV/ KBC Groupe SA	Belgium	KBC BB	EUR	ρ	From May 30, 2000	ρ	§	From May 30, 2000
Dexia SA	Luxembourg	DEXB BB	EUR	ρ	From May 30, 2000	ρ	1999–2005	From May 30, 2000
Danske Bank A/S	Denmark	DANSKE DC	DKK	ρ	From May 30, 2000	ρ	§	From May 30, 2000
BNP Paribas	France	BNP FP	EUR	ρ	From May 30, 2000	ρ	§	From May 30, 2000
Crédit Agricole SA	France	ACA FP	EUR	From December 16, 2002	From December 14, 2001	ρ	1999–2005	From December 16, 2002
Société Générale	France	GLE FP	EUR	ρ	From May 30, 2000	ρ	§	From May 30, 2000
Natixis	France	KN FP	EUR	ρ	From May 30, 2000	ρ	1999–2005	From May 30, 2000
Deutsche Bank AG	Germany	DBK GR	EUR	ρ	From May 30, 2000	ρ	§	From May 30, 2000
Commerzbank AG	Germany	CBK GR	EUR	ρ	From May 30, 2000	ρ	§	From May 30, 2000
Deutsche Postbank AG	Germany	DPB GR	EUR	From June 23, 2005	From June 23, 2004	ρ	§	From June 23, 2005
National Bank of Greece	Greece	ETE GA	EUR	ρ	From May 30, 2000	From August 1, 2000	§	From August 1, 2000
National Savings and Commercial Bank of Hungary (OTP Bank)	Hungary	OTP HB	HUF	ρ	From May 30, 2000	ρ	1999–2005	From May 30, 2000
Bank of Ireland	Ireland	BKIR ID	EUR	ρ	From May 30, 2000	ρ	§	From May 30, 2000
Allied Irish Bank PLC	Ireland	ALBK ID	EUR	ρ	From May 30, 2000	ρ	§	From May 30, 2000
UniCredito Italiano SpA	Italy	UC IM	EUR	ρ	From May 30, 2000	ρ	§	From May 30, 2000
Intesa Sanpaolo	Italy	ISP IM	EUR	ρ	From May 30, 2000	ρ	§	From May 30, 2000
ING Group—ING Groep NV	Netherlands	INGA NA	EUR	ρ	From May 30, 2000	ρ	§	From May 30, 2000
ABN Amro Holding NV	Netherlands	AABA NA	EUR	ρ	From May 30, 2000	ρ	§	From May 30, 2000
PKO BP	Poland	PKO PW	PLN	From November 9, 2005	From November 10, 2004	ρ	2003–6	From November 9, 2005
Santander Central Hispano Group—Banco Santander Central Hispano	Spain	SAN SM	EUR	ρ	From May 30, 2000	ρ	§	From May 30, 2000
Banco Bilbao Vizcaya Argentaria SA	Spain	BBVA SM	EUR	ρ	From May 4, 2000	ρ	§	From May 4, 2000
Nordea Bank AB/Nordea Group	Sweden	NDA SS	SEK	ρ	From May 30, 2000	ρ	§	From May 30, 2000
Skandinaviska Enskilda Banken AB	Sweden	SEBA SS	SEK	ρ	From May 30, 2000	ρ	§	From May 30, 2000
Svenska Handelsbanken	Sweden	SHBA SS	SEK	ρ	From May 30, 2000	ρ	§	From May 30, 2000
Barclays PLC	United Kingdom	BARC LN	GBP	ρ	From May 17, 2000	ρ	§	From May 17, 2000
HSBC Holdings PLC	United Kingdom	HSBA LN	GBP	ρ	From May 17, 2000	ρ	§	From May 17, 2000
Royal Bank of Scotland Group PLC	United Kingdom	RBS LN	GBP	ρ	From May 17, 2000	ρ	§	From May 17, 2000
HBOS PLC	United Kingdom	HBOS LN	GBP	From September 11, 2002	From September 10, 2001	ρ	§	From September 11, 2002
Lloyds TSB Group PLC	United Kingdom	LLOY LN	GBP	ρ	From May 17, 2000	ρ	§	From May 17, 2000
Standard Chartered PLC	United Kingdom	STAN LN	GBP	ρ	From May 17, 2000	ρ	§	From May 17, 2000

Sources: Bankscope and Bloomberg.

 ρ At least from May 2000.

§ At least from 1999.

TABLE 21.9

EU Countries: Stock Market Indices and Government Bond Yields

Country	Stock Market			Sovereign Bond	
	Index	Currency	Bloomberg Ticker	Maturity	Bloomberg Ticker
Austria	Austria Traded ATX Index	EUR	ATX	EUR Austria sovereign zero coupon yield, 1 year EUR Austria sovereign zero coupon yield, 10 year	F90801Y F90810Y
Belgium	BEL 20 Index	EUR	BEL20	EUR Belgium sovereign zero coupon yield, 1 year EUR Belgium sovereign zero coupon yield, 10 year	F90001Y F90010Y
Denmark	OMX Copenhagen 20 Index	DKK	KFX	DKK Denmark sovereign zero coupon yield, 1 year DKK Denmark sovereign zero coupon yield, 10 year	F26701Y F26710Y
France	CAC 40 Index	EUR	CAC	EUR France sovereign zero coupon yield, 1 year EUR France sovereign zero coupon yield, 10 year	I01401Y I01410Y
Germany	DAX Index	EUR	DAX	EUR Germany sovereign zero coupon yield, 1 year EUR Germany sovereign zero coupon yield, 10 year	F91001Y F91010Y
Greece	Athens Stock Exchange General Index	EUR	ASE	EUR Greece sovereign zero coupon yield, 1 year EUR Greece sovereign zero coupon yield, 10 year	F90401Y F90410Y
Hungary	Budapest Stock Exchange Index	HUF	BUX	HUF Hungary sovereign zero coupon yield, 1 year HUF Hungary sovereign zero coupon yield, 10 year	F11401Y F11410Y
Ireland	Irish Overall Index	EUR	ISEQ	EUR Ireland sovereign zero coupon yield, 1 year EUR Ireland sovereign zero coupon yield, 10 year	F91801Y F91810Y
Italy	S&P MIB Index	EUR	SPMIB	EUR Italy sovereign zero coupon yield, 1 year EUR Italy sovereign zero coupon yield, 10 year	F90501Y F90510Y
Netherlands	Amsterdam Exchanges Index	EUR	AEX	EUR Netherlands sovereign zero coupon yield, 1 year EUR Netherlands sovereign zero coupon yield, 10 year	F92001Y F92010Y
Poland	WSE WIG 20 Index	PLN	WIG20	PLN Poland sovereign zero coupon yield, 1 year PLN Poland sovereign zero coupon yield, 10 year	F11901Y F11910Y
Spain	IBEX 35 Index	EUR	IBEX	EUR Spain sovereign zero coupon yield, 1 year EUR Spain sovereign zero coupon yield, 10 year	F90201Y F90210Y
Sweden	OMX Stockholm 30 Index	SEK	OMX	SEK Sweden sovereign zero coupon yield, 1 year SEK Sweden sovereign zero coupon yield, 10 year	F25901Y F25910Y
United Kingdom	FTSE 100 Index	GBP	UKX	GBP United Kingdom zero coupon yield, 1 year GBP United Kingdom zero coupon yield, 1 year	I02201Y I02210Y
Region	MSCI All-Country Europe Index	EUR	MXER		
World	MSCI All-Country World	EUR	MXWD		

Source: Bloomberg.

TABLE 21.10

Contagion Risk among the Biggest Banking Groups in the EU (27 Banks), May 30, 2000, to April 30, 2007
(in percent level of significance)

Country	Bank	Contagion to									
		Austria	Belgium				Denmark	France			Germany
		Erste	Fortis	KBC	Dexia (Luxembourg)	Danske	BNP Paribas	Societe Generale	Natixis	Deutsche Bank	Commerzbank
	Initial shock to										
Austria	Erste			5.0	5.0						5.0
Belgium	Fortis				1.0		1.0	1.0		1.0	5.0
	KBC				5.0						
	Dexia (Luxembourg)			1.0							
Denmark	Danske								1.0		
France	BNP Paribas							1.0	1.0		
	Societe Generale										
	Natixis	1.0						5.0			
Germany	Deutsche Bank				1.0						
	Commerzbank									1.0	
Hungary	OTP										
Ireland	Bank of Ireland										
	Allied Irish					1.0					
Italy	Unicredito										
	Intesa Sanpaolo										
Netherlands	ING		5.0								
	ABN Amro						5.0	1.0			
Spain	Santander										
	Banco Bilbao Vizcaya										
Sweden	Nordea	5.0				5.0					
	SEB				5.0						
	Swedbank										
United Kingdom	Barclays							5.0			
	HSBC					5.0				1.0	
	RBS			1.0							
	Lloyds		1.0								
	Standard Chartered							1.0			
Other factors	Constant	5.0	5.0	5.0	5.0		5.0	5.0	5.0	5.0	5.0
	Change in term structure			5.0			5.0				
	Local market volatility										
	Regional market volatility										
	Global market volatility				5.0						
McFadden R^2		0.36	0.55	0.48	0.54	0.57	0.59	0.51	0.53	0.45	0.39

Source: Authors.

Note: Factors in the columns are dependent on factors in the rows.

Blank cells represent nonsignificance at any level below 10 percent.

■ represents coefficients on own lags, of up to 5 lags.

■ represents significance at the 1 percent level.

■ represents significance at the 5 percent level.

■ represents significance (negative sign) at the 5 percent level or lower.

□ represents the grouping of major banks in any one country, where 3 or more banks are represented in the sample.

TABLE 21.11

Contagion Risk among the Biggest Banking Groups in the EU (33 Banks), November 9, 2005, to April 30, 2007
(in percent level of significance)

		Contagion to												
		Austria		Belgium			Denmark	France				Germany		
Country	Bank	Erste	Bank Austria Creditanstalt	Fortis	KBC	Dexia (Luxembourg)	Danske	BNP Paribas	Credit Agricole	Societe Generale	Natixis	Deutsche Bank	Commerzbank	Deutsche Postbank
Austria	Initial shock to													
	Erste			5.0	5.0						5.0	5.0		
	Bank Austria Creditanstalt				5.0									
Belgium	Fortis								5.0					
	KBC			5.0		5.0								
	Dexia (Luxembourg)													
Denmark	Danske										5.0			
France	BNP Paribas													
	Credit Agricole													
	Societe Generale													1.0
	Natixis				5.0									
Germany	Deutsche Bank					5.0		5.0	5.0					
	Commerzbank													
	Deutsche Postbank							5.0				5.0		
Greece	National Bank of Greece			5.0	5.0					1.0				5.0
Hungary	OTP								5.0					
Ireland	Bank of Ireland													
	Allied Irish				5.0			5.0			1.0	5.0	1.0	
Italy	Unicredito													
	Intesa Sanpaolo				5.0									
Netherlands	ING							5.0				5.0		
	ABN Amro							5.0		1.0				
Poland	PKO													
Spain	Santander													
	Banco Bilbao Vizcaya													
Sweden	Nordea													
	SEB					5.0								
	Swedbank													
United Kingdom	Barclays													
	HSBC										1.0			
	RBS				5.0			5.0						
	HBOS													
	Lloyds													
	Standard Chartered									5.0				
Other factors	Constant	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
	Change in term structure			5.0	1.0	5.0		5.0	5.0	5.0				
	Local market volatility							5.0						
	Regional market volatility							5.0				5.0	5.0	
	Global market volatility	5.0										5.0		
McFadden R ²		0.85	0.58	0.82	0.80	0.63	0.68	0.83	0.73	0.76	0.76	0.81	0.73	0.70

Source: Authors.

Note: Factors in the columns are dependent on factors in the rows.

Blank cells represent nonsignificance at any level below 10 percent.

■ represents coefficients on own lags, of up to 5 lags.

■ represents significance at the 1 percent level.

■ represents significance at the 5 percent level.

□ represents significance (negative sign) at the 5 percent level or lower.

□ represents the grouping of major banks in any one country, where 3 or more banks are represented in the sample.

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SECTION C

The Contingent Claims Analysis Approach

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Introduction to the Contingent Claims Analysis Approach for Stress Testing

DALE F. GRAY • ANDREAS A. JOBST •
CHENG HOON LIM • YINGBIN XIAO

Contingent claims analysis (CCA) is a risk-adjusted balance sheet concept. It derives a coherent measure of credit risk based on the impact of uncertain changes in future asset values relative to promised payments on the valuation of debt obligations. It translates this uncertainty into quantifiable risk indicators that measure risk exposures to reveal whether balance sheet risks are building or subsiding. Such quantitative risk indicators also incorporate forward-looking information. CCA represents a generalization of the option pricing theory pioneered by Black and Scholes (1973) and Merton (1973). It is forward looking by construction, providing a consistent framework based on current market conditions rather than on historical experience.

CCA determines the risk-adjusted balance sheet based on three principles: (1) the values of liabilities (equity and debt) are derived from asset dynamics; (2) liabilities have different priority (i.e., senior and junior claims); and (3) assets (such as the present value of income flows and proceeds from asset sales) follow a stochastic process. Assets may be above or below promised payments on debt, which represent the default barrier. When there is a chance of default, the repayment of debt is considered “risky,” to the extent that it is not guaranteed in the event of default. Risky debt is decomposed into two components: (1) the default-free value of debt and deposits; and (2) the risk margin that compensates for the “expected loss to bank creditors” from default over a specific time horizon, which can be measured by the value of an implicit put option in the CCA model.¹ Because CCA combines market prices and balance sheet information to inform a risk-

adjusted measure of individual default risk, the evaluation of solvency conditions is linked inherently to investor perception as implied by the institution’s equity and equity option values (which determine the implied asset value of the firm conditional on its leverage and debt level).

CCA FOR INDIVIDUAL CORPORATE OR FINANCIAL INSTITUTIONS

Several widely used techniques have been developed to calibrate CCA. These methods use a combination of balance sheet information and forward-looking information from equity markets, which were first applied to individual corporate and extended to financial institutions (by Merton, Moody’s KMV, and others). The market value of assets of corporations and financial institutions cannot be observed directly but can be implied using financial asset prices. From the observed prices and volatilities of market-traded securities, one can estimate the implied asset values and volatilities of the underlying assets in financial institutions.

Most CCA models use market information plus balance sheet information. An advantage of CCA is that it provides high-frequency risk indicators and is very useful in capturing the nonlinearity between market capitalization, assets, and expected losses/default probabilities. Many stress test models can accommodate default probabilities calculated from CCA/Merton-type models as inputs into the calculation of loss distributions used for value-at-risk (VaR) and capital adequacy

¹ Note that if uncertainty in the risk-adjusted balance sheet is ignored, i.e., set to zero, the result is the accounting balance sheet where there is

no measure of credit risk (i.e., implicit put option goes to zero if volatility is zero) (Gray and Malone, 2008).

assessments. Thus, CCA represents an alternative method for analyzing financial stability when detailed prudential data on individual bank performance are not available. The caveat is that markets can overreact, and, in some cases, illiquidity can distort prices and affect CCA-based indicators. In Chapter 23, Ruiz-Arranz demonstrates the use of CCA to determine the risk of and expected losses from corporate sector defaults in the United Kingdom and the most vulnerable economic sectors. The analysis is complemented by accounting-based debt-at-risk analyses of corporate and household balance sheets.

EXTENSIONS OF THE BASIC CCA FRAMEWORK

CCA represents a flexible modeling approach that can be applied to an array of analyses. They can be extended to measure sovereign risk and to analyze the impact of banking system risk on the sovereign and feedback effects to banking risk as well as funding costs; to estimate the relationship between macroeconomic factors and the time pattern of bank assets or credit risk indicators, which is then applied to stress scenarios to project banking risks; and to analyze government implicit and explicit contingent liabilities using the Systemic CCA framework. Another application of CCA is to use short-term assets and liabilities in financial institutions to measure and analyze liquidity risk and systemic liquidity risk.

Measuring sovereign risk

CCA can be applied to determine the interaction between the risk-adjusted balance sheets of the government and monetary authorities. A marked-to-market balance sheet may be constructed to provide a structural framework that identifies balance sheet risks, incorporates uncertainty, and yields quantifiable risk indicators. The main outputs of this framework include sovereign credit risk indicators for foreign currency debt, sensitivity measures, and sovereign VaR. These sovereign risk indicators incorporate both forward-looking market prices and nonlinear changes in values and should consequently have greater predictive power in estimating sovereign credit risk than would traditional macroeconomic vulnerability indicators or accounting-based ratios. Gapen and others apply CCA to 12 emerging market economies in Chapter 24 to estimate the risk indicators for foreign currency debt. CCA also can be adapted to sovereigns that have only local currency debt—the calibration process uses debt data and the term structure of sovereign credit default swap (CDS) and bond spreads (Gray and Jobst, 2010a, 2010b).

Using the CCA approach to analyze sovereign risk has merits from a policy and stress testing perspective. The ability of the CCA approach to provide a structural interpretation of the sovereign balance sheet, unique to each economy, is a valuable contribution in the area of policy design and risk management, translating policy choices and changing economic conditions directly into quantitative indicators of financial soundness. It is well suited to a more robust analysis of

debt sustainability as compared with the widely used debt-to-GDP ratio, which is a static, backward-looking accounting indicator.

Linking banks to macroeconomic factors for stress testing

CCA can be used with satellite models that link macroeconomic factors to the bank CCA model. Econometric relationships can be estimated between macro factors and the implied bank assets, distance to distress, or default probability or expected losses. Then stress test scenarios of macroeconomic factors can be used to project changes in bank assets, which in turn affect the bank credit risk indicators and equity value. One application of this type of approach is described by Gray and Walsh in Chapter 25. The study estimates risk indicators for the major Chilean banks using CCA, which are linked to macroeconomic and financial factors. The scenarios derived in the chapter illustrate how stress testing can be used with the CCA bank model to assess the magnitude and persistence of responses of bank credit risk across banks in the system.

Jointly stress testing banking and sovereign risk

CCA can be applied to analyzing banking sector and sovereign risk feedbacks. It allows sovereign risk to be measured as an exposure (expected loss to holders of sovereign debt and sovereign spreads) instead of an accounting ratio (sovereign debt to GDP). CCA can capture nonlinear feedbacks that lead to bank and sovereign destabilization spirals. For example, the negative impact on banks of holding sovereign debt can lead to higher sovereign contingent liabilities leading to increased sovereign risk (higher spreads), which in turn further reduces the value of banks' sovereign debt holdings. The vicious circle eventually leads to a weakening in sovereign support, leading to higher bank spreads and funding costs (Gray and Jobst, 2010a, 2010b, 2011a, 2011b). Finally, the systemic risk dynamics are interlinked with important new measures of risk-adjusted economic output value via the CCA economy-wide balance sheets (Gray and Malone, 2010).

Measuring the size and impact of government support/guarantees

Relatedly, CCA facilitates measurement and analysis of implicit and explicit government guarantees and their impact on bank funding costs. The expected losses calculated for each financial institution from equity market and balance sheet information using CCA can be combined with information from CDS markets to estimate the government's contingent liabilities (Gray and Jobst, 2009). The price of insuring against default by a financial institution is reflected in its CDS spreads, which capture only the expected loss *retained* by the financial institution and borne by unsecured senior creditors. Thus,

the difference between the implied CDS and the actual CDS spreads reflects the explicit and implicit guarantees.

Measuring systemic risk (“Systemic CCA” and “Systemic Risk-Adjusted Liquidity”)

The Systemic CCA is a forward-looking, market data-based analytical framework for measuring joint default risk of financial institutions using an advanced form of CCA (Gray and Malone, 2010; Gray and Jobst, 2011c; Jobst and Gray in Chapter 26). It extends the risk-adjusted balance sheet approach underpinning CCA by modeling systemic risk of multiple institutions with “too-big-to-fail” properties falling into distress as a portfolio of individual expected losses, which are calculated from equity market and balance sheet information. Changes in market conditions (and their impact on the perceived risk profile of each firm via its equity price and volatility) establish market-induced linkages based on their sensitivity to common risk factors, which define the dependence structure of individual expected losses. Accounting for both linear and nonlinear dependence and its effect on joint expected losses can deliver important insights about the joint tail risk of multiple entities, and this method of measuring “tail dependence” is better suited to analyzing extreme linkages of multiple entities over time.

The Systemic CCA approach can be used to quantify the contribution of specific institutions to the dynamics of the components of systemic risk. The application of this methodology to the U.K. banking sector is discussed by Jobst and Gray in Chapter 26. In addition, CCA offers a promising framework for measuring and analyzing systemic liquidity risk by using the short-term risk-adjusted balance sheet with the default barriers calibrated to reflect rollover risk. Jobst applies the Systemic Risk-Adjusted Liquidity (SRL) dynamic factor model to U.S. banks in Chapter 27 and demonstrates its use within a stress testing framework.

REFERENCES

- Black, Fisher, and Myron Scholes, 1973, “The Pricing of Options and Corporate Liabilities,” *Journal of Political Economy*, Vol. 81, No. 3, pp. 637–54.
- Gray, Dale F., and Andreas A. Jobst, 2009, “Using the CCA Framework to Estimate the Cost of Government Guarantees,” appendix in “Crisis-Related Measures in the Financial System and Sovereign Balance Sheet Risks,” IMF Policy Paper, July 31, pp. 48–50 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pp/longres.aspx?id=4361>
- , 2010a, “New Directions in Financial Sector and Sovereign Risk Management,” *Journal of Investment Management*, Vol. 8, No. 1, pp. 23–38.
- , 2010b, “Risk Transmission between Sovereigns and Banks in Europe,” *Global Financial Stability Report*, World Economic and Financial Surveys, October (Washington: International Monetary Fund). Available via the Internet: <https://www.imf.org/external/pubs/ft/gfsr/2010/02>
- , 2011a, “Modeling Systemic and Sovereign Risk,” in *Lessons from the Financial Crisis*, edited by Arthur Berd, pp. 143–85 (London: RISK Books).
- , 2011b, “Modelling Systemic Financial Sector and Sovereign Risk,” *Sveriges Riksbank Economic Review*, No. 2, pp. 68–106. Available via the Internet: http://www.riksbank.se/upload/Rapporter/2011/POV_2/er_2011_2.pdf
- , 2011c, “Systemic Contingent Claims Analysis—A Model Approach to Systemic Risk,” in *Managing Risk in the Financial System*, edited by J. R. LaBrosse, R. Olivares-Caminal, and Dalvinder Singh, pp. 93–110 (London: Edward Elgar).
- Gray, Dale F., and Samuel Malone, 2008, *Macrofinancial Risk Analysis* (New York: Wiley).
- , 2010, “Quantifying Systemic Risk and Reconceptualizing the Role of Finance for Economic Growth,” *Journal of Investment Management*, Vol. 8, No. 2, pp. 90–110.
- Merton, Robert C., 1973, “Theory of Rational Option Pricing,” *Bell Journal of Economics and Management Science*, Vol. 4, No. 1, pp. 141–83.

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Vulnerabilities of Household and Corporate Balance Sheets in the United Kingdom and Risks for the Financial Sector

MARTA RUIZ-ARRANZ

This chapter estimates the impact of macroeconomic shocks to household and corporate balance sheets on banks. Two separate methods are employed. First, contingent claims analysis is used to assess the near-term prospects for the U.K. corporate sector. Specifically, this method attempts to determine the extent of the risk of corporate sector defaults and the most vulnerable sectors, as well as the size of expected losses from defaults. Second, a measure of debt at risk is used to quantify the size of the debt overhang in the private sector (corporates and households) and potential losses in the face of several plausible macroeconomic shocks. The results indicate that losses from defaults are expected to be limited, except for the corporate real estate sector. Separately, although U.K. households remain heavily indebted and vulnerable to shocks, the potential impact on banks appears to be manageable.

METHOD SUMMARY

Contingent Claims Analysis

Overview	The contingent claims analysis approach is based on a risk-adjusted balance sheet framework where the equity and risky debt of a corporation or financial institution derive their value from their assets.
Application	The method may be used to estimate expected losses from corporate defaults.
Nature of approach	Contingent claims analysis.
Data requirements	Balance sheet data of the nonfinancial corporate sector; equity prices.
Strengths	The method combines forward-looking and historical information.
Weaknesses	Only listed companies with sufficiently liquid trading should be included in the sample. The results should be interpreted with caution as they are sensitive to equity price volatility.
Tool	A modified version of the Balance Sheet Risk Analysis Excel spreadsheet is available from the author upon request.

Debt at Risk

Overview	Debt at risk is a method based on balance sheet data used to quantify the size of the debt overhang in the corporate and household sectors.
Application	The method could be used to estimate the impact of shocks on household and corporate sector balance sheets and, consequently, their impact on bank balance sheets.
Nature of approach	Balance sheet approach.
Data requirements	Microlevel data on household and nonfinancial corporate sector balance sheets.

Previously published as IMF Country Report No. 11/229 (IMF, 2011b). The author would like to thank counterparts at the Bank of England, HM Treasury, and the Financial Services Authority for helpful comments.

Strengths	The method is intuitive and easy to apply.
Weaknesses	The analysis requires firm-level and household-level data. Debt at risk does not correspond directly to nonperforming loans; rather, it is the debt that could come under financial strain or could potentially become nonperforming. As such, debt at risk likely overestimates credit losses.
Tool	Standard Excel spreadsheet.

The global financial crisis and postcrisis recovery profoundly affected household and corporate sector balance sheets in the United Kingdom. Although the process of balance sheet repair is well under way, pockets of vulnerability remain in both sectors, particularly in commercial real estate (CRE), unsecured household debt, and some segments of the mortgage debt market. Stress in these segments—triggered by a rapid increase in interest rates, an increase in unemployment, a sharp decline in real estate prices, or a combination thereof—could have a significant impact on the banking sector, both directly and indirectly through feedback loops between the real and financial sectors. Concentration of credit risks in some large banks and widespread exercise of lender forbearance aggravate these vulnerabilities.

This chapter explores these issues by reviewing the experience of the crisis, assessing the current state of the corporate and household balance sheets, and gauging the quantitative impact of risks by using sensitivity analysis and contingent claims analysis.

1. HOUSEHOLD AND FINANCIAL SECTOR LINKAGES

High household indebtedness raises concerns about mortgage credit risk and the stability of the financial system, particularly if interest rates or unemployment were to rise or house prices were to fall further. This section documents developments in the household sector during the crisis and recovery and assesses the potential risks to the banking sector emanating from household debt by using microlevel data.

A. Developments during and after the crisis

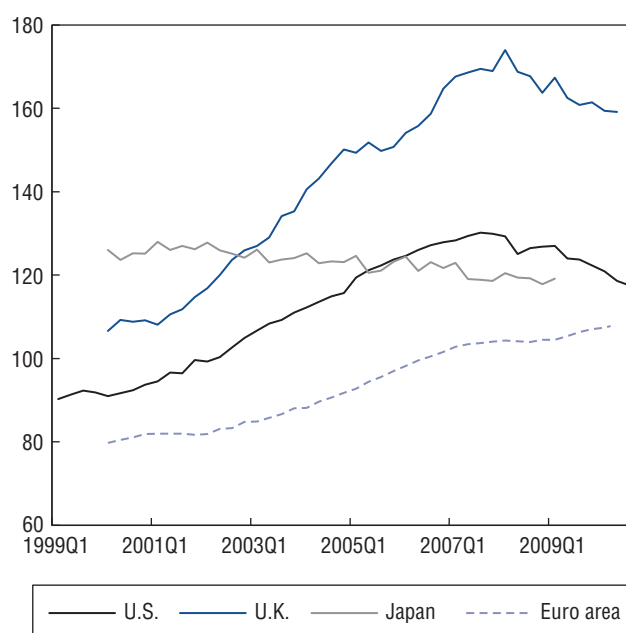
In the run-up to the crisis, household debt reached 175 percent of disposable income, fueled by rapid mortgage lending. Household indebtedness increased faster in the United Kingdom than in the United States or the euro area as a whole (Figure 23.1). The rise in debt was matched by an increase in the value of housing as well as of pension funds and other financial assets held by households (Figure 23.2). Indeed, the asset side of household balance sheets increased faster than total debt, leading to a rapid growth in net worth.

U.K. household balance sheets were hard hit by the crisis, and the deleveraging of the sector was a key feature of the downturn. Private consumption declined sharply as households responded to mounting economic uncertainty and

plunging asset prices. The fall in home values and equity prices resulted in a substantial decline in net worth. This loss of net worth helped stimulate increases in savings among households striving to rebuild their wealth. The gross saving rate surged to a peak of 7.5 percent in mid-2009—up 6 percentage points from its precrisis levels.

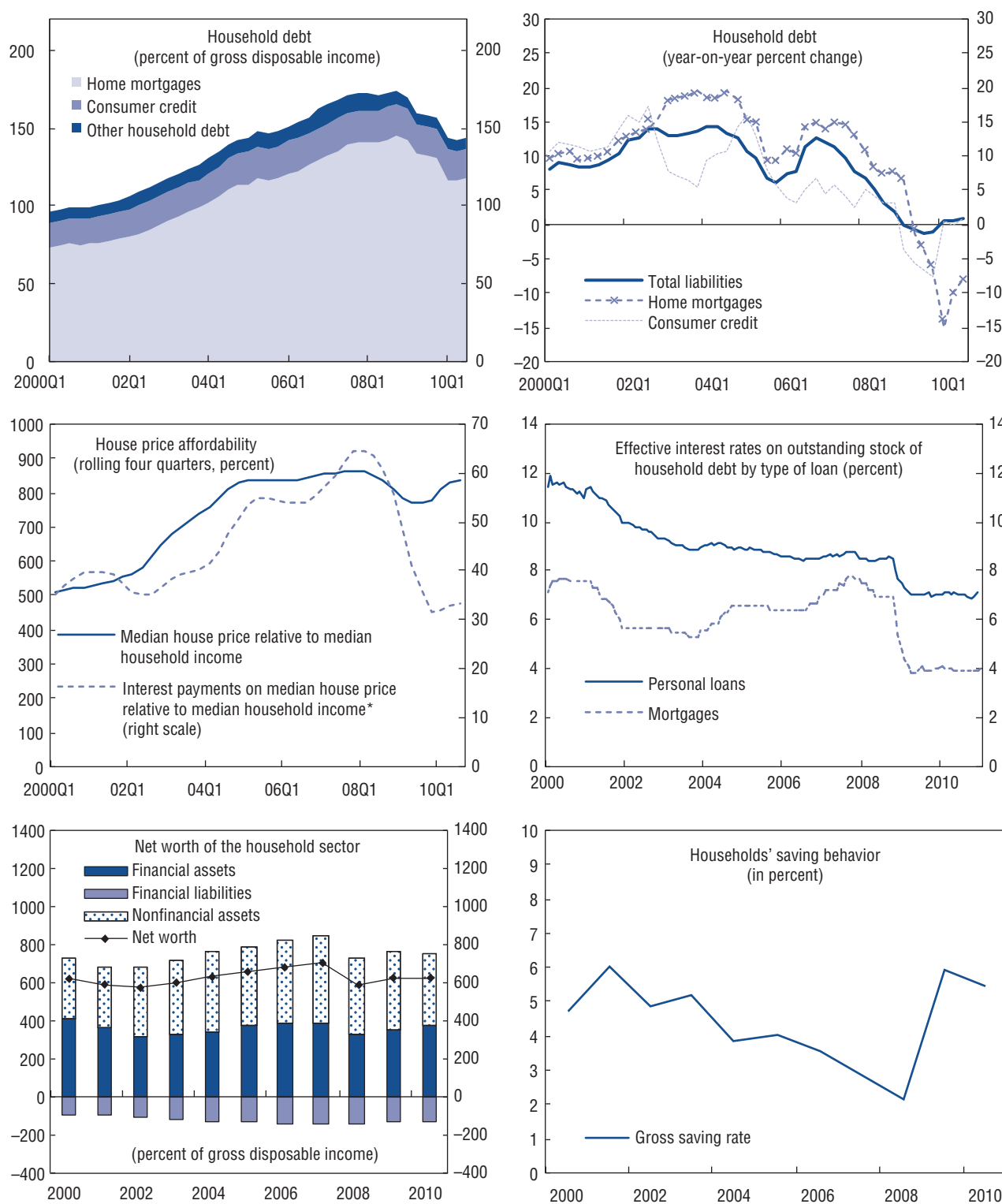
Several factors cushioned the impact of the crisis on household balance sheets, averting a more severe adjustment. These included (1) record low interest rates and dominance of variable mortgage rates, which boosted debt affordability; (2) forbearance and restructuring of loans by banks, which contained the increase in foreclosures (Figure 23.3); and (3) the relatively limited rise in unemployment compared with other countries and previous recessions.

Nonetheless, the stress on household balance sheets had an impact on banks' asset quality. Household lending accounts for three-quarters of U.K. banks' domestic lending, most of which is secured against collateral. Household write-off rates on secured lending increased a little, but write-off rates on unsecured lending are close to 7 percent, up 2.5 percentage points since 2008. Mortgage arrears and repossession



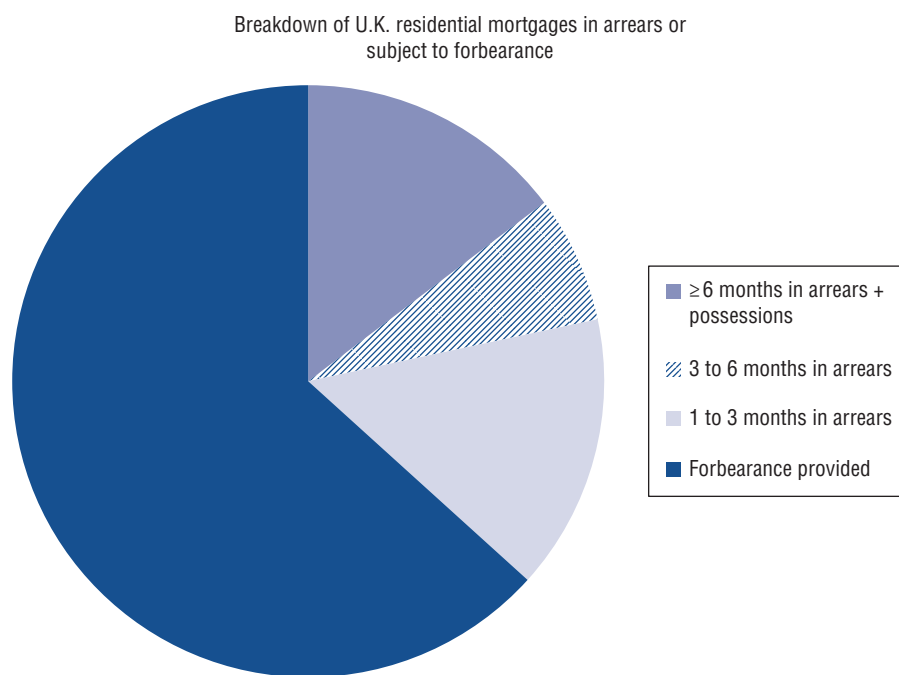
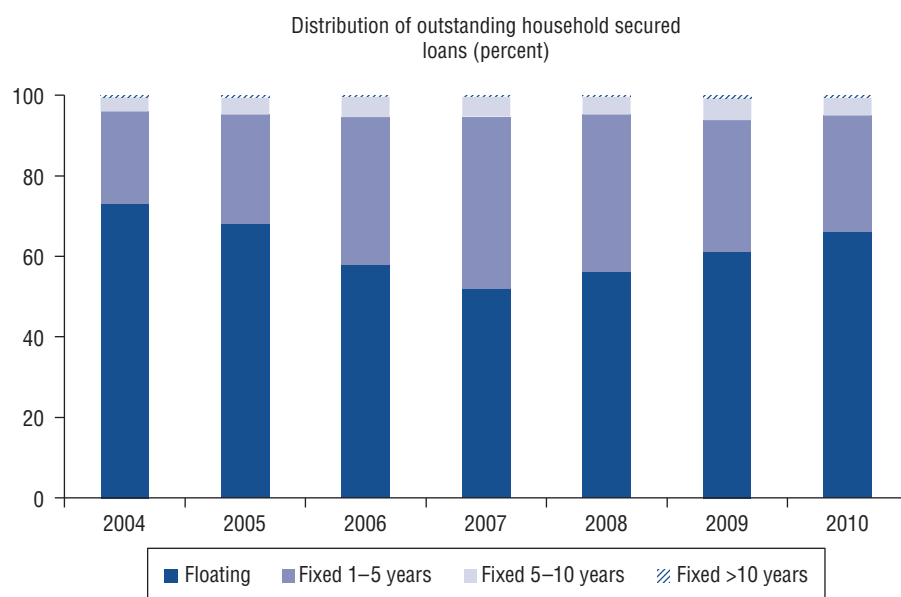
Sources: Author; and World Economic Outlook.

Figure 23.1 Major Countries: Household Debt-to-Disposable Income



Sources: Author; Department for Work and Pensions' HBAI Report; Haver Analytics; and Land Registry.
 *Based on standard variable rate mortgage.

Figure 23.2 United Kingdom: Household Sector, 2000–2010



Sources: Bank of England; and Financial Services Authority.

Figure 23.3 United Kingdom: Household Balance Sheets and Mitigating Factors during the Crisis

rates also increased, albeit by less than in previous crises and relative to the United States (Figure 23.4). Higher default rates on U.S. mortgages may partly reflect U.S. bankruptcy laws, which make mortgage defaults less costly to households than in the United Kingdom. Another possible explanation is the significantly lower use of mortgage securitization in the United Kingdom, which has facilitated loan restructuring, as loans have tended to remain in the banks' books.

B. Looking forward: household sector risks in the recovery

Despite significant deleveraging, household balance sheets remain fragile in the United Kingdom. Though off its peak, household debt remains high by historical and international standards. Its onerous impact has been mitigated to some extent by a buildup of household sector currency and bank deposits since the second half of 2010. Higher saving

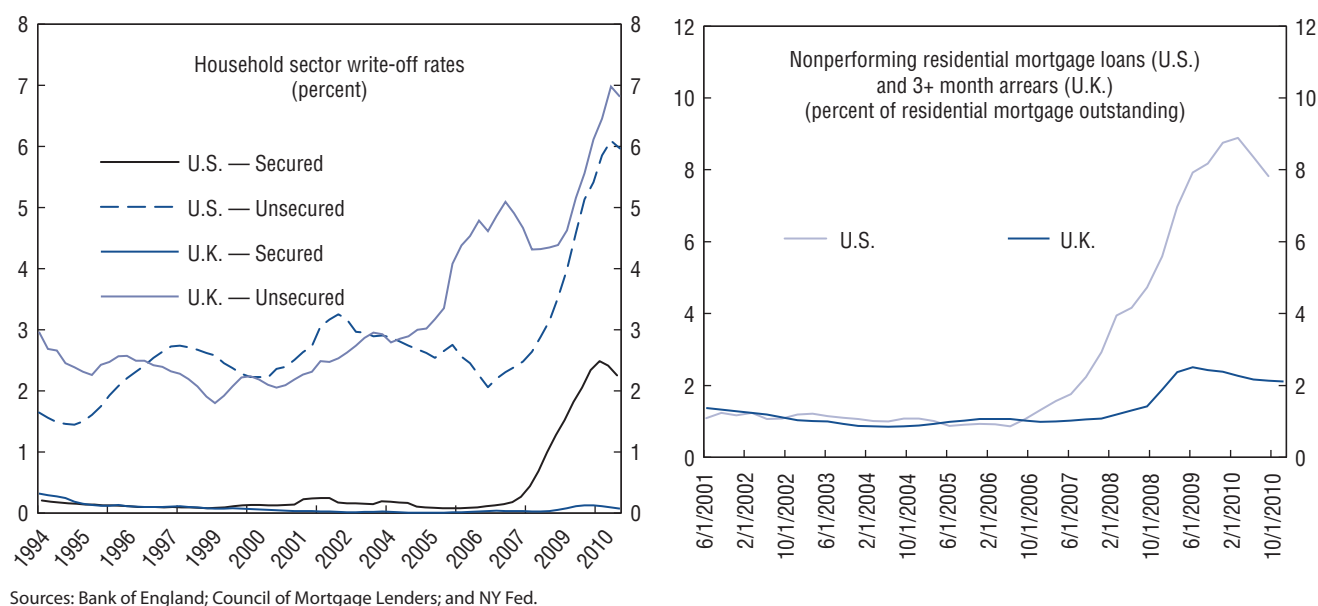


Figure 23.4 United Kingdom: Household Sector Write-off Rates and Nonperforming Loans

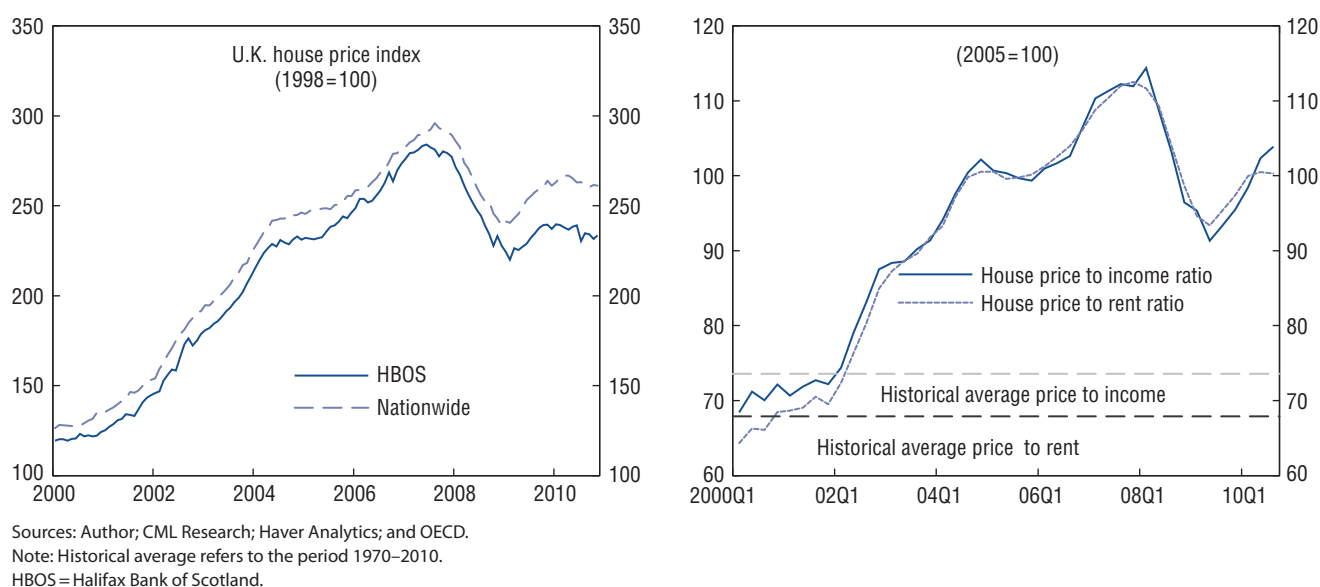


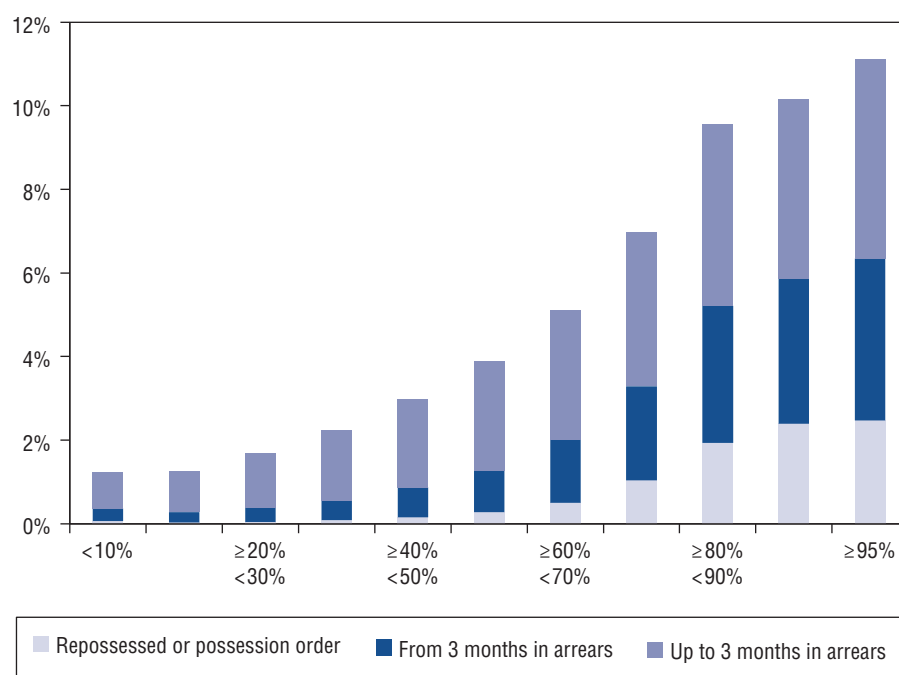
Figure 23.5 United Kingdom: Housing Developments

rates have the potential to generate enough financial surpluses to reduce household indebtedness. However, the saving rate has already fallen back to 5 percent from its peak of 7.5 percent in mid-2009, as recovering asset prices and a stabilizing labor market lifted confidence in 2010, slowing the pace of balance sheet repair.

Housing valuations continue to appear stretched relative to income and rents, leaving banks exposed to further falls in housing prices. After falling by 20 percent since their peak in 2007, house prices staged a faster-than-expected recovery in 2009 and the first half of 2010. The upward momentum

has since dissipated, with the three-month change in house prices turning negative, and surveys point to continued downward pressure on prices. Overall, prices remain about 15 percent below their peak in 2007. At this level, the price-to-income ratio and price-to-rent ratio remain 30 percent above historical averages (Figure 23.5). It is worth noting that supply constraints are likely to prevent house prices from fully falling back to their historical averages.¹

¹ Other factors affecting real house prices are discussed in Sá, Towbin, and Wieladek (2011).



Source: FSA Product Sales and Arrears Databases.

Note: Performance as of August 2009 of mortgages sold between April 2005 and March 2009.

Figure 23.6 Arrears and Repossessions by Loan-to-Value Band (percent of outstanding mortgages)

Some segments in the mortgage debt market appear vulnerable and represent yet another source of credit risk for banks. Although overall arrears and repossessions have so far remained contained (Figure 23.6), they are concentrated in some groups of risky borrowers. In particular, arrears rates are higher for borrowers with high loan-to-value (LTV) loans and with buy-to-let loans, as these may reflect a higher share of second mortgages than owner-occupied mortgages (Financial Services Authority, 2011). There are also regional variations: arrears in the North and in Wales, which experienced large house price declines and a slower housing market recovery, are significantly higher than in other regions.

Unsecured write-offs have kept rising, posing further risks to the banks. Although unsecured lending accounts for less than 10 percent of total loans, absolute losses from unsecured debt so far have surpassed losses from mortgage defaults. As reported in the 2010Q1 Bank of England (BoE) Credit Conditions Survey,² spreads on unsecured lending continue to widen (as banks appropriately price in the higher risk of default), and unsecured debt payments, notably on credit cards, have been rising as a share of household income despite the fall in policy rates. There has been, however, some recent improvement in the appetite for unsecured credit risk on the part of lenders. Evidence from the 2010 NMG Consulting Survey confirmed that households' financial position remains under strain. Households report that the burden of unsecured

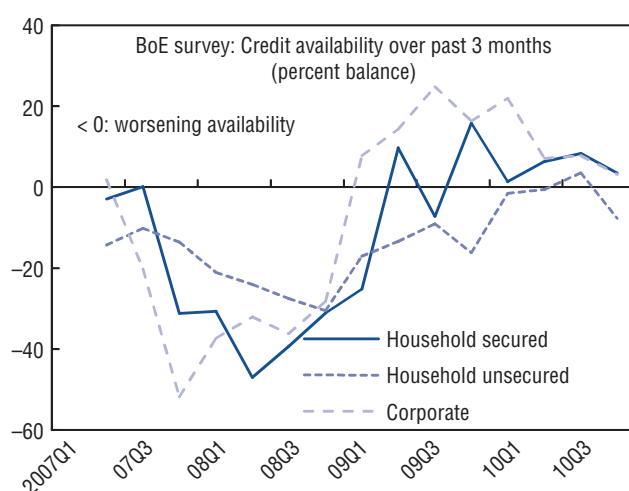
credit was higher, and concerns about the debt level have increased, particularly among high-LTV mortgagors. The survey also points to increased difficulties in keeping up with bills and credit commitments and a marked concern about the impact of fiscal consolidation measures.

Furthermore, banks' own deleveraging efforts and funding pressures could constrain credit conditions for households.³ Households remain dependent on banks for refinancing, but bank credit availability is still tight, as suggested by the BoE survey on credit availability (Figure 23.7). Mortgage approvals and net lending amounts have also stayed at low levels, and evidence from the 2010 NMG Consulting Survey shows an increase in the proportion of households reporting a tightening in credit conditions, especially among unsecured debtors and high-LTV mortgagors. Lending standards have appropriately strengthened: average LTV ratios on new first mortgages are about 75–80 percent, well below the ratios observed during the precrisis boom, which should contain potential losses given default.

The discussion above suggests continuing fragility that, under stress, may translate into credit losses in the banking system. In particular, banks would be exposed to higher defaults, were interest rates or unemployment to rise or house prices to fall. Write-offs on secured lending have so far remained contained, largely because of exceptionally low interest

² See <http://www.bankofengland.co.uk/publications/Pages/other/monetary/ccs/ccs1004.aspx>.

³ New regulatory requirements and the unwinding of policy support, including repayments to the Special Liquidity Scheme, may increase pressure on banks' balance sheets and affect credit availability.



Source: Bank of England.

Figure 23.7 United Kingdom: Bank Credit Availability

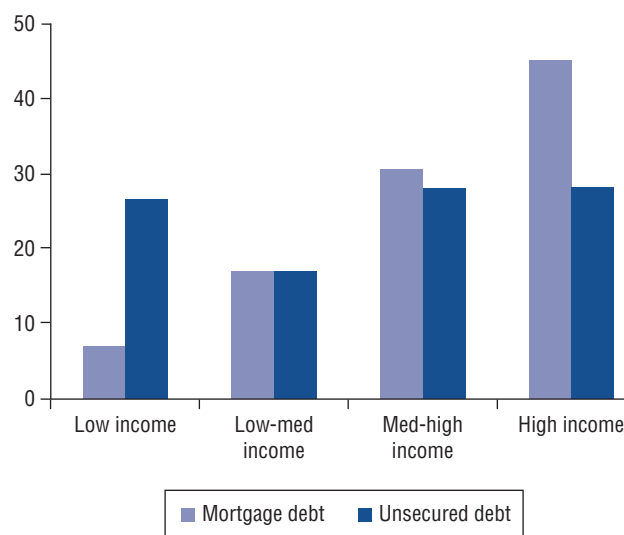
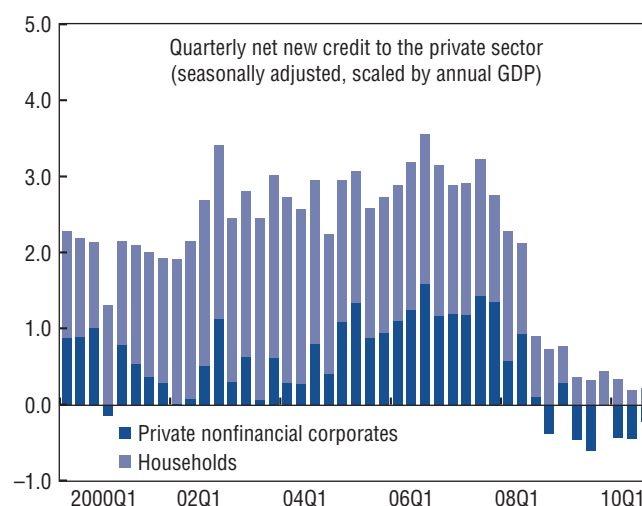
rates. Given the dominance of variable-rate mortgages, households' debt affordability could fall sharply if interest rates were to rise. These vulnerabilities are aggravated by the fact that risks are concentrated in some vulnerable banks with very large exposures.

C. Sensitivity analysis of household balance sheets

This section takes a more rigorous approach to the risks discussed and attempts to quantify some of the potential threats to financial and macroeconomic stability from household indebtedness. What happens to households' ability to pay if interest rates go up or unemployment rises? How many households will end up with negative equity if house prices fall further? What would be banks' credit losses in these scenarios?

This exercise uses microdata to analyze the stability risks arising from household indebtedness. The scope for using aggregate data from the financial and national accounts to evaluate these risks is limited, as such data do not provide information regarding the distribution and matching of debt, interest expenditures, income, and assets. More granular data regarding individual households may reveal pockets of vulnerability in the household sector. This study uses data from the September 2010 NMG Consulting Survey of U.K. households.⁴ The survey provides balance sheet information (secured and unsecured debt, housing, and financial assets), as well as data on income, debt expenditures, and self-reporting indicators of financial stress (e.g., ability to meet payments).

⁴ The survey covers about 2,000 households. Sampling techniques attempt to make the sample representative of the population. See <http://www.bankofengland.co.uk/publications/quarterlybulletin/qb100408.pdf>.



Sources: Author; and 2010 NMG Consulting Survey.
Note: med = medium.

Figure 23.8 United Kingdom: Share of Debt Held by Income Categories (in percent)

One rough measure of the risks in household lending is the distribution of household debt across income categories. In principle, the smaller the share of debt held by lower-income households, the lower the risks associated with household lending. As expected, high-income households hold the larger share of total household debt, especially mortgage debt (Figure 23.8). In contrast, low-income households' share of unsecured debt is more than commensurate to their income.⁵ The descriptive statistics in Table 23.1 also point to other

⁵ Income groups are defined as follows: low-income, gross annual income up to £17,500; low-medium income, £17,500–£35,000; medium-high income, £35,000–£60,000; and high income, >£60,000.

TABLE 23.1

United Kingdom: Descriptive Statistics by Income Category

Indicator	Low Income	Low-Medium Income	Medium-High Income	High Income
Mortgage debt-to-income ratio	481	297	223	170
Mortgage payments to income	36.9	23.0	17.6	12.9
Unsecured debt-to-income ratio	41	21	21	12
Unsecured debt payments to income	12.7	10.4	7.7	4.6

Sources: Author; 2010 NMG Consulting Survey.

Note: Income groups are defined as follows: low-income, gross annual income up to £17,500; low-medium income, £17,500–£35,000; medium-high income, £35,000–£60,000; and high income, >£60,000.

potential sources of vulnerability. The median low-income household has significantly higher mortgage debt-to-income and debt-mortgage-payments-to-income ratios than other income categories. This suggests that low-income households' ability to service their debt could be compromised under financial stress, leading to credit losses for the banking sector.

Three stress scenarios are considered: an increase in interest rates of 300 basis points (well above the 60–65 basis points rise in interest rates assumed in the solvency stress test of the banking sector),⁶ a decline in household income of 20 percent, and a decline in house prices of 20 percent (in line with the solvency stress test of the banking sector). These are severe (low probability/high impact) stress scenarios. To put this in perspective, it is worth noting that the Sterling Overnight Index Average futures suggest a rise of around 100 basis points in rates within a year. The IMF baseline scenario assumes a reduction in the house-to-income ratio of about 5 percent to 10 percent over the medium term.

To assess the impact of these shocks, a mortgagor household is defined to be financially stressed if its mortgage debt service-to-income ratio (DSI) is larger than a certain threshold. The stress threshold is defined as a DSI of 40 percent of income. This threshold is higher than many used in the literature, but the purpose is to identify the truly vulnerable households.⁷ In addition, two other thresholds were considered as a robustness check, but the results were essentially the same.⁸ The left panels of Figure 23.9 show the distribution of household DSI by income groups relative to each of the three stress thresholds under consideration. This gives an indication of how many households in each income category are below the stress threshold and how close the other

households are to it. In 2010, about 35 percent of low-income households had debt-service ratios above 40 percent, compared with 9 percent, 5 percent, and 0 percent in the higher-income categories.

The share of vulnerable households captures the impact of different stress scenarios on the household sector as a whole, but different indicators are needed to monitor possible bank losses. Two such indicators are used in this exercise:

- *Debt at risk* is defined as the debt of financially stressed mortgagors; default data are not available at the household level. Debt at risk does not correspond directly to nonperforming loans; rather, it is the debt that could come under financial strain or could potentially become nonperforming. As such, debt at risk likely overestimates credit losses.
- *Loss given default* (LGD) measures the share of debt at risk that is not covered by the household's assets (both real and financial). The intuition is that if a household defaults on its debt, the bank will incur losses only to the extent that the household's assets are not able to cover its debt. This measure, however, is likely to underestimate credit losses, as the banks may not be able to repossess all assets. In addition, the estimated LGD is based on the prevailing value of the assets. In a situation of macroeconomic stress, the value of both real and financial assets is likely to fall, which means that a smaller fraction of debt would be covered by assets. Finally, this LGD measure does not take into account the potentially large transaction costs involved in defaults. Formally,

$$\text{Debt at risk} = \frac{\sum_i S_i D_i}{\sum_i D_i}, \quad \text{where } S_i = \begin{cases} 1 & \text{if } DSI_i > 40 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{LGD} = \frac{\sum_i S_i L_i}{\sum_i D_i}, \quad \text{where } L_i = \begin{cases} -NW_i & \text{if } NW_i < 0 \\ 0 & \text{otherwise} \end{cases}$$

where D_i is the debt of household i , and NW_i is the net worth of household i , which is defined as the sum of household assets (real estate and financial) minus total debt (mortgage and unsecured debt). The survey provides information on households' assets and liabilities to estimate net worth.

The state of household balance sheets in September 2010 (date of the household survey) is the starting point for the sensitivity analysis.

⁶ See IMF (2011a).

⁷ May and Tudela (2005) find that a DSI of 20 percent or above is associated with higher probability of mortgage payment problems in England. For Austria, Beer and Schurz (2007) define financially distressed households as those that have a DSI above 30 percent. This chapter uses thresholds as in Karasulu (2008).

⁸ The two alternative thresholds are (1) two standard deviations of the average DSI within each income group; and (2) two standard deviations of the average DSI of mortgagors reporting difficulties in paying for their mortgage (a self-reporting measure of financial distress). Results from the sensitivity analysis are similar under the three alternative measures. However, the distributional implications are different, with the uniform threshold identifying a larger number of low-income financially stressed households than the other two measures.

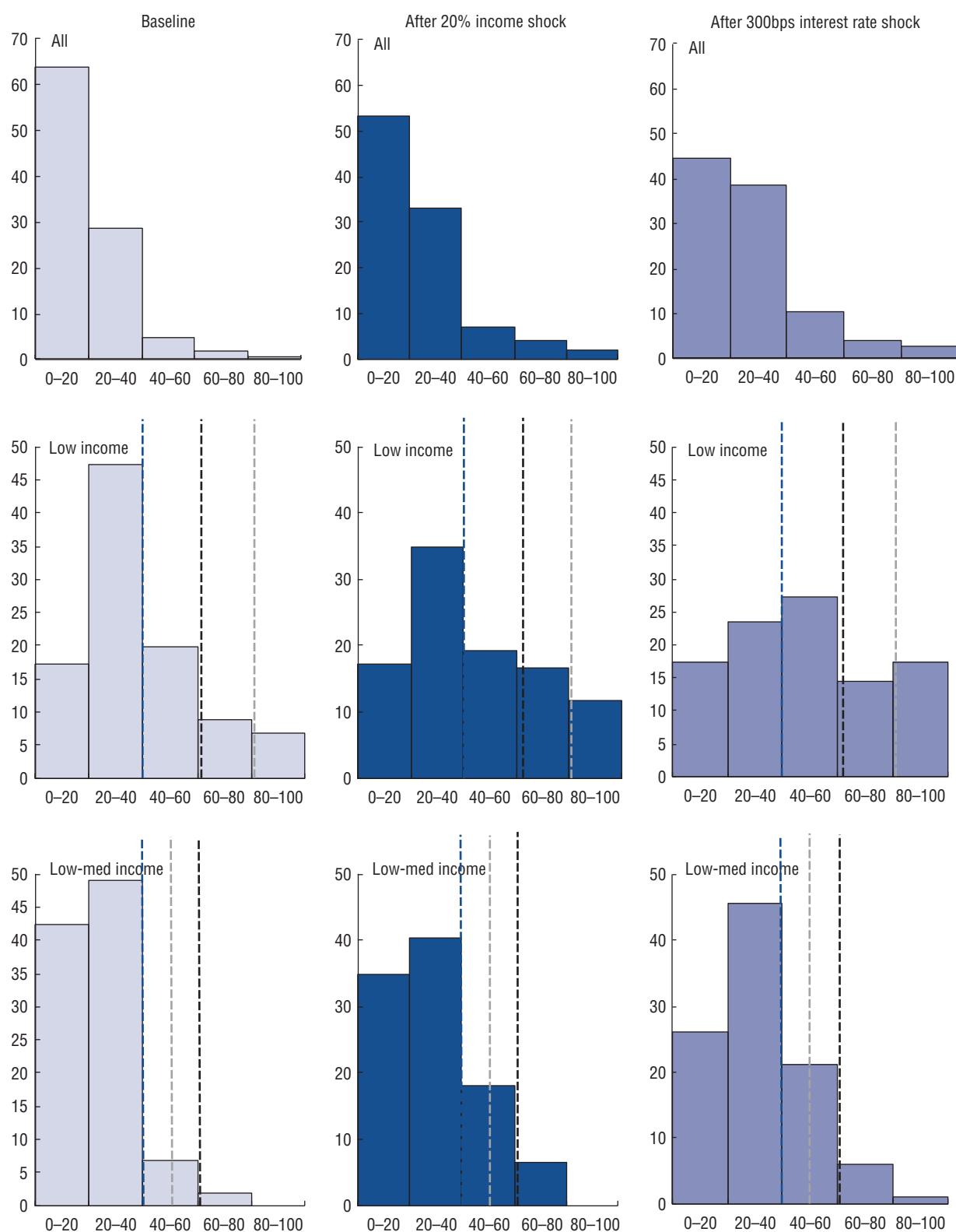
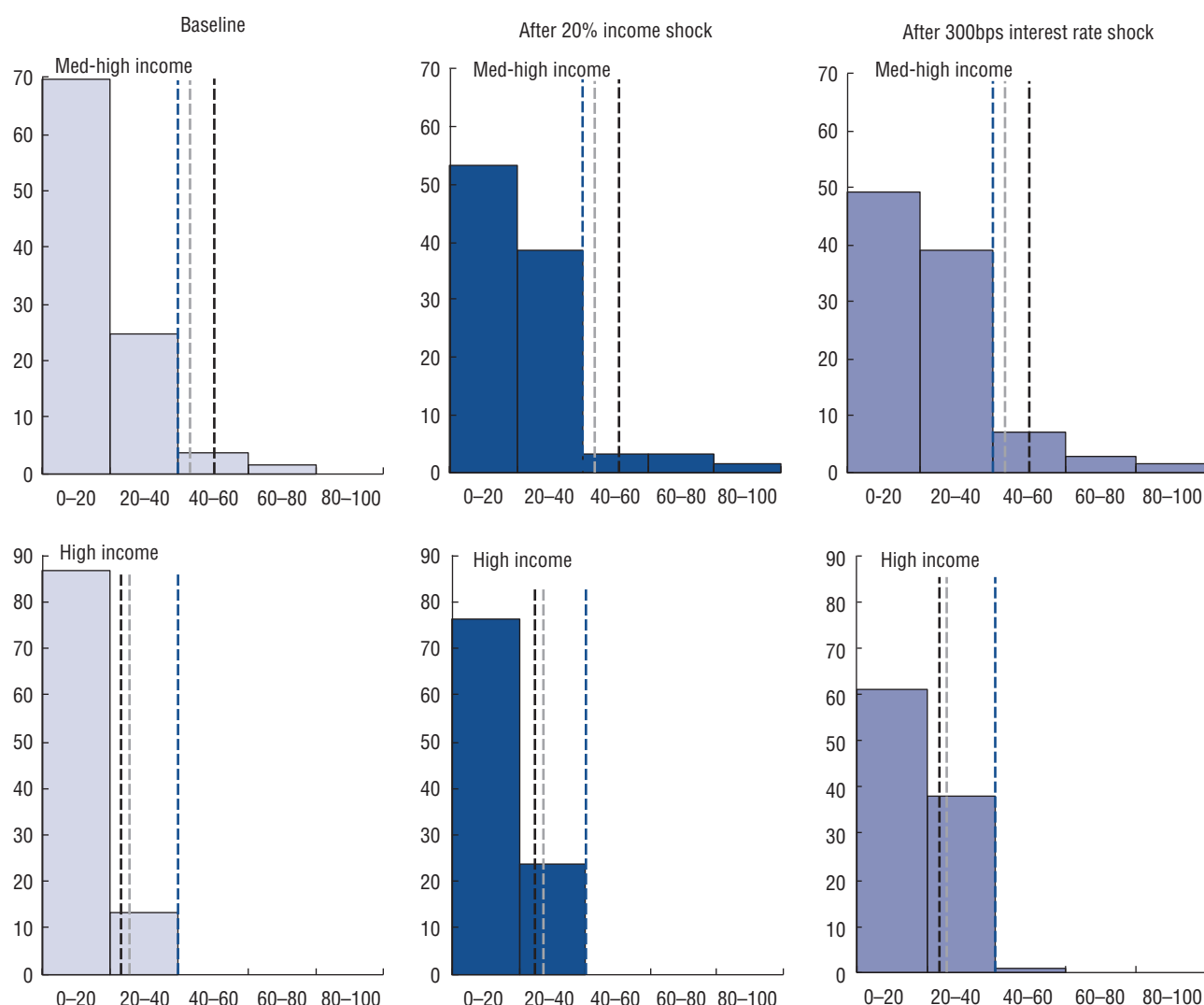


Figure 23.9 United Kingdom: Household Debt Service-to-Income Ratio by Income Groups (percent y-axis; DSI x-axis)



Sources: Author; and UK Household Survey.

Note: Each vertical dotted line represents a DSI threshold, i.e., T1 (blue); T2 (black); T3 (gray). Low income: T1 = 40, T2 = 62, T3 = 81.

Low-med income: T1 = 40, T2 = 61, T3 = 50.

Med-high income: T1 = 40, T2 = 50, T3 = 43.

High income: T1 = 40, T2 = 25, T3 = 26.

Figure 23.9 (continued)

- Eight percent of mortgagors were financially stressed in September 2010. The average DSI for financially stressed households was 58 percent, and their average mortgage debt was six times their average annual gross income (Table 23.2).
- Ten percent of total mortgage debt was at risk. With total household mortgage debt in the economy amounting to 96 percent of GDP, this means that about 10 percent of GDP worth of mortgages could come under financial strain (this is based on the presumption that the survey is representative of the population).⁹

⁹ These aggregate figures mask important differences across household income groups. The percentage of financially stressed households

- Estimates of LGD show that most mortgage debt is covered by household assets, mostly real estate. They suggest that if vulnerable households were to default on their debts, creditors would suffer losses corresponding to 0.4 percent of total household debt.

Sensitivity analyses of the impact of various shocks to household (and hence bank) balance sheets are conducted. They comprise shocks to interest rates, household income, and house prices. The results are summarized in Table 23.3 and are discussed in the rest of this section.

decreases with household income. Low-income households appear very vulnerable: over 35 percent of these households are financially stressed, and more than 50 percent of this group's total debt is at risk. However, this represents only 3.6 percent of total household sector debt.

TABLE 23.2

United Kingdom: Sensitivity Analysis of Household Sector—Baseline, September 2010

	Share of Stressed Households in Each Income Category	Share of Debt at Risk in Each Income Category	Debt at Risk (in Percent of Total Household Debt)	LGD (in Percent of Total Household Debt)
Low income	35.6	52.4	3.6	0.0
Low–medium income	8.7	14.1	2.4	0.3
Medium–high income	5.4	13.7	4.3	0.1
High income	0.0	0.0	0.0	0.0
All	8.2	10.3	10.3	0.4

Sources: Author; 2010 NMG Consulting Survey.

Note: Debt refers to mortgage debt only. LGD=loss given default.

TABLE 23.3

United Kingdom: Sensitivity Analysis of Household Sector—Summary of Shocks

Scenario	Share of Stressed Households in Each Income Category	Share of Debt at Risk (in Percent of Total Household Debt)	LGD (in Percent of Total Household Debt)
Baseline	8.2	10.3	0.4
Interest rate shock	17.8	21.5	0.9
Income shock	14.1	16.7	0.9
Housing price shock	15.6	22.1	4.0
Combined interest rate and income shock	28.4	36.4	0.9

Sources: Author; NMG Consulting Survey.

Note: LGD=loss given default.

Interest Rate Shock

The household sector is sensitive to increases in interest rates, but loan losses in the banking sector appear to be manageable—at least as long as the underlying assets maintain their value. A rise in interest rates by 300 basis points would increase the share of financially stressed households to 18 percent, from 8 percent in the baseline (Figure 23.10). Debt at risk would increase by over 10 percentage points, bringing total debt at risk to 20 percent of GDP. The impact would be most severe among low-income households, as they tend to have high DSIs to begin with: indeed, about three-fourths of low-income household debt would be considered at risk. However, given the relatively low share of bank loans to this income group, the impact on asset quality is likely to be small (Figure 23.11). This shock would more than double LGD (debt at risk not covered by assets) compared with the baseline, to almost 1 percent of total household sector loans (about 1 percent of GDP). This low LGD estimate reflects the partial-equilibrium nature of this exercise, which assumes that the value of the underlying asset is not affected. It is worth noting that this analysis suggests that even if household balance sheets were shocked with a 300 basis points interest

rate increase (well above the 60–65 basis points rise in interest rates assumed in the more severe scenarios in the solvency stress test of the banking sector), the impact on banks would be manageable.

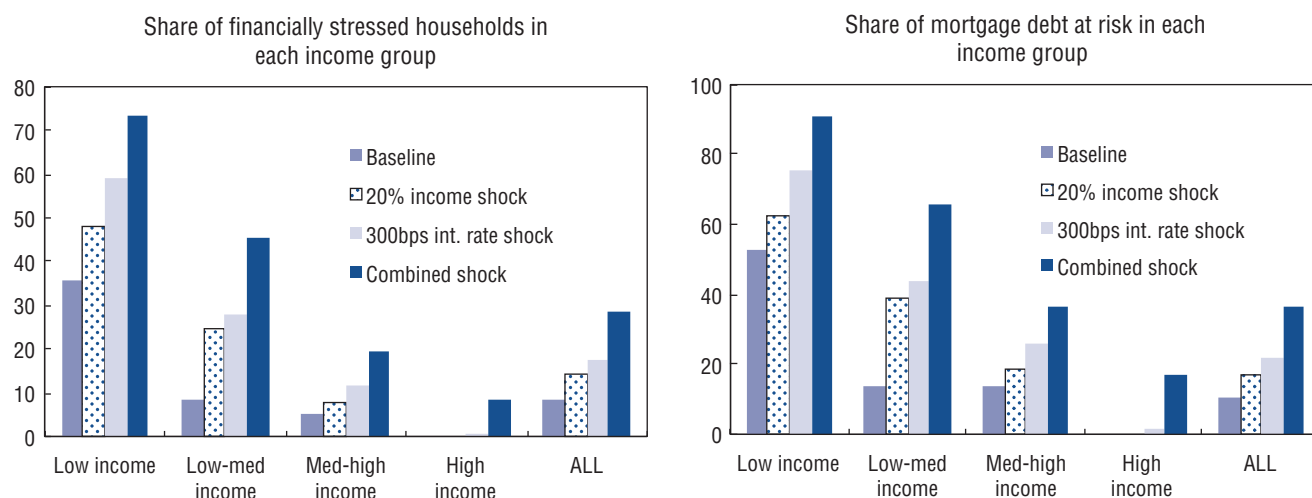
These estimates represent the long-run impact of the interest rate shock. They are based on the assumption that all mortgages are affected by the interest rate increase (as in the long run, even fixed-rate mortgages are affected because of renegotiation of interest rates). As such, they are likely to be an upper-limit estimate of financial distress. The short-run impact of the shock, affecting only variable-rate mortgages, would be half as large. This lower estimate should be interpreted with caution. First, the share of fixed-rate mortgages in our sample of mortgagors (almost 50 percent) is larger than the actual share in the population, which is about a third of outstanding mortgages. Second, mortgagors are currently moving from expiring fixed-rate mortgages to floating-rate mortgages (BoE, 2010), increasing the sensitivity of borrowers to interest rises.

Income Shock and Combined Interest Rate-Income Shock

Similarly, income shocks would significantly affect individual households but may not necessarily generate large bank losses. A drop in income by 20 percent would increase the share of vulnerable households to 14 percent (Figure 23.9). Debt at risk would rise by 6 percentage points of GDP, and LGD would reach almost 1 percent of total loans. Again, this shock would disproportionately affect low-income households. However, because they account for a relatively small proportion of total bank mortgage lending, the impact on bank asset quality is likely to be limited. This shock is indicative of the risks were unemployment to rise beyond current rates or the economy to slow down. In contrast, a combined interest rate and income shock could have a major impact on household balance sheets and thus the potential to inflict larger losses on the banks.

House Price Shock

Further declines in house prices would affect households' net worth and carry implications for bank LGDs. With less home equity available, homeowners would tend to have fewer

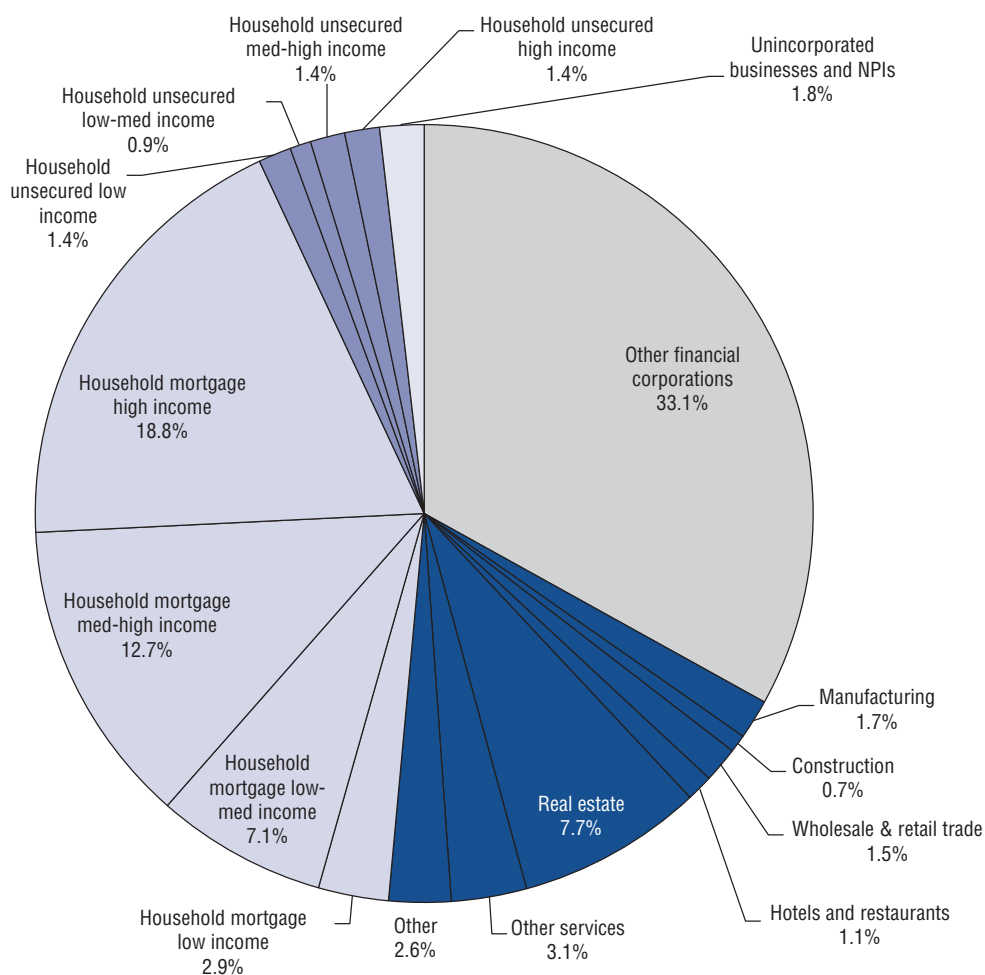


Sources: Author; and 2010 NMG Consulting Survey.

Note: Financially distressed households are defined to have mortgage debt service above 40 percent of income.

bps = basis points; int. = interest; med = medium.

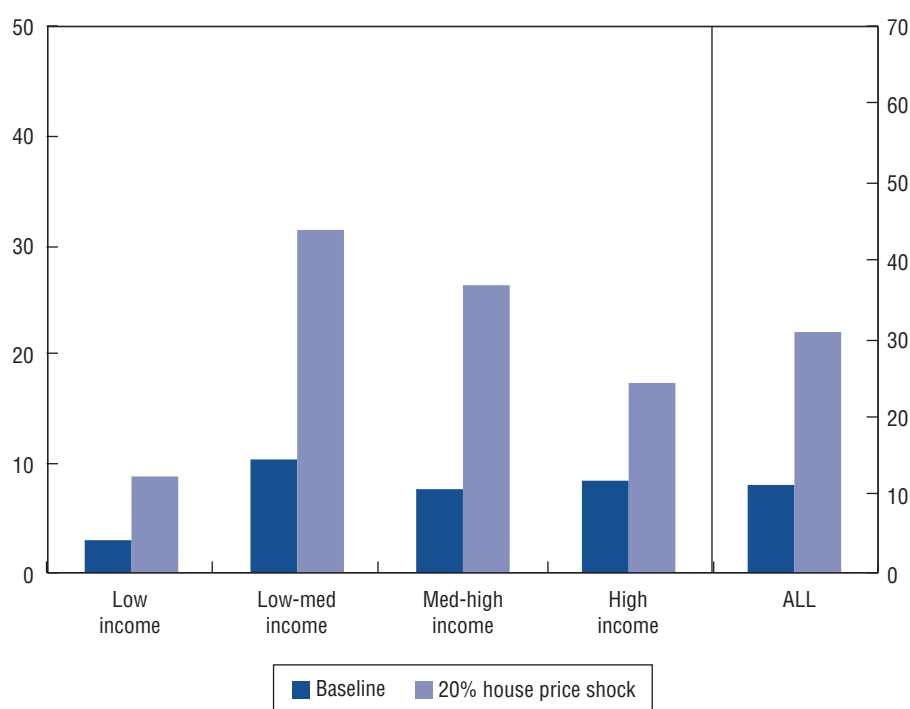
Figure 23.10 United Kingdom: Sensitivity Analysis of the Household Sector



Source: Author.

Note: med = medium; NPI = nonprofit institutions.

Figure 23.11 United Kingdom: Composition of Bank Loans to the Economy



Sources: Author; and 2010 NMG Consulting Survey.
Note: med = medium.

Figure 23.12 United Kingdom: Impact of a House Price Shock (in percentage of household debt with negative net worth)

options to refinance existing mortgage debt or take on new debt that could be used to pay off other debt contracted at higher rates. Given that house price shocks do not directly affect household debt service payments, an alternative measure of financial stress is needed to assess the sensitivity of households to falls in house prices. One such measure is based on household net worth. Households whose net worth becomes negative are considered financially stressed. By associating financial stress to wealth, this measure attempts to capture the ability of households to liquidate assets (or pledge real estate assets) in order to service debt before default:

- By late 2010, less than 6 percent of households, accounting for about 8 percent of total household debt (mortgage and consumer debt) had negative net worth. After a 20 percent house price drop, about 16 percent of households would have negative net worth, pushing debt at risk to 22 percent of the total (Figure 23.12). Potential losses for banks as measured by LGD would climb to 4 percent of all household loans.
- Tellingly, arrears in the North and Wales, which experienced large house-price declines and a slower housing market recovery, are higher than in other regions. This also suggests that a combined shock that affects both household debt service capacity—such as an increase in unemployment—and the collateral value of secured debt could have a material impact on asset quality.

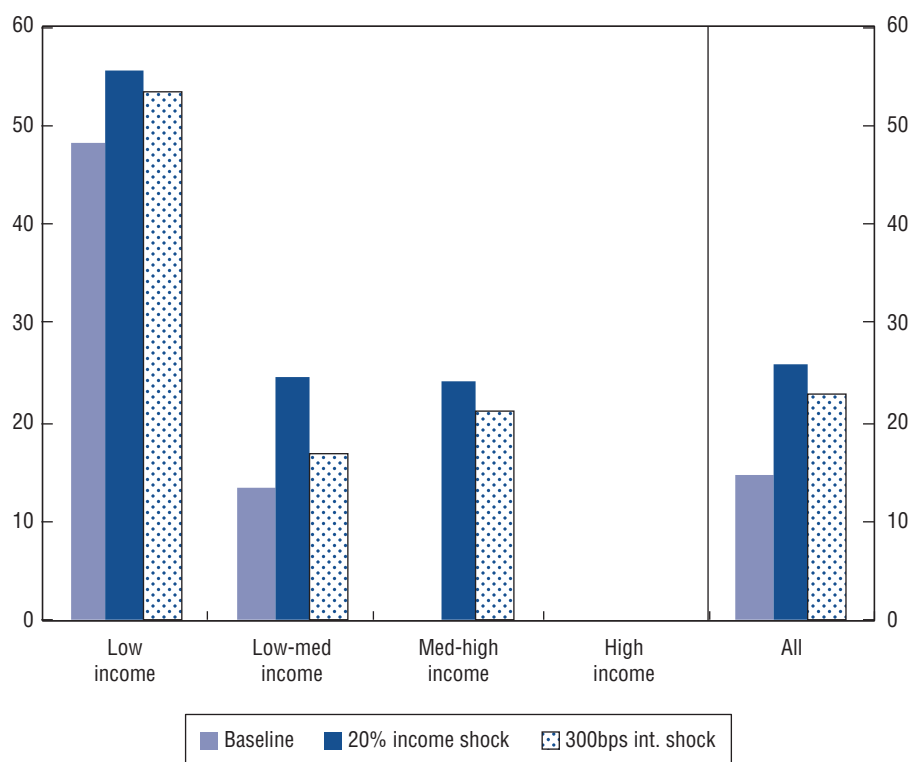
- A mitigating factor, nonetheless, is that, unlike in the United States, mortgagors in the United Kingdom historically have tended to service their mortgages, even during the worst of times, in part because of a greater willingness on the part of banks to restructure loans in order to maintain debt affordability.

Unsecured Debt

The burden of unsecured credit could increase further were interest rates to rise or income to decline. About 26 percent and 23 percent of total unsecured debt would become at risk following a 20 percent negative income shock and a 300 basis point interest rate rise, respectively (Figure 23.13). This would represent about 5 percent and 4.5 percent of GDP, respectively, of potential default losses, given that total unsecured debt was 20.5 percent of GDP at end-2010. These estimates assume a stress threshold of unsecured debt service payments to income of 20 percent.

2. CORPORATE AND FINANCIAL SECTOR LINKAGES

This section discusses how the nonfinancial corporate sector was affected by the crisis and the recovery. It goes on to assess how vulnerable the corporate sector is, going forward, using two different methodologies. Contingent claims analysis



Sources: Author; and NMG Consulting Survey.

Note: Financially stressed households have unsecured debt service payments above 20 percent of income.
bps = basis points; int. = interest; med = medium.

Figure 23.13 United Kingdom: Share of Unsecured Debt at Risk in Each Income Group (in percent)

(CCA) is used to estimate expected losses from corporate defaults one year ahead. And sensitivity analysis is used to assess how the corporate balance sheets would be affected by interest rate and profitability shocks. The results suggest that the corporate sector as a whole remains relatively resilient to shocks, but there is significant variation across subsectors.

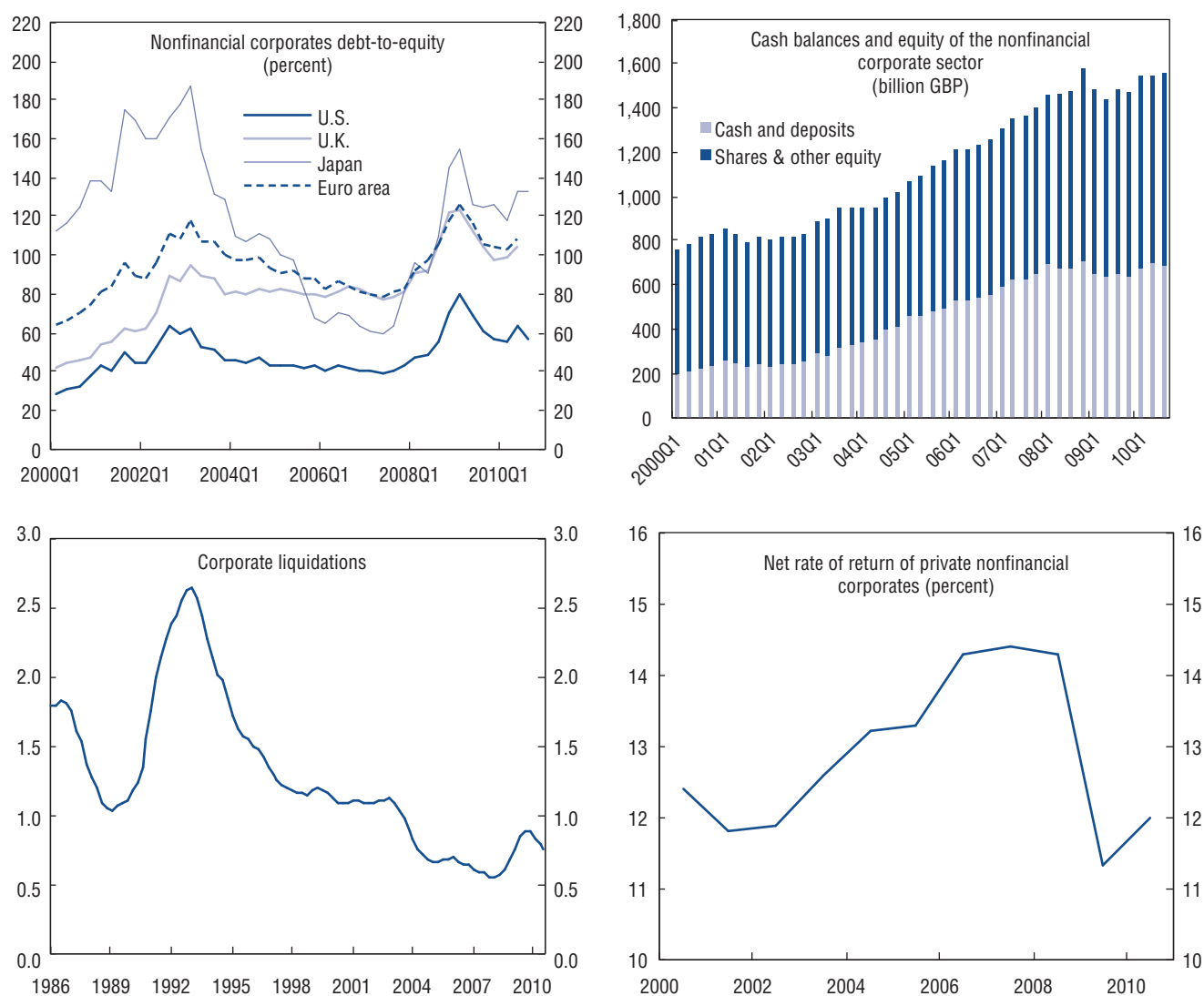
A. Developments during and after the crisis

The U.K. nonfinancial corporate sector weathered the financial crisis relatively well. Firms entered the crisis with relatively solid financial balances, despite high debt levels. As demand and profits contracted and the outlook deteriorated, firms cut back on investment to preserve cash flow. Still, the increase in the number of company liquidations during the crisis was moderate compared with previous crises and has been declining since late 2009. Low interest rates and bank restructuring have helped companies avoid default (Figure 23.14).

Corporate sector balance sheets have strengthened over the last year of this analysis, along with the rest of the economy. Debt represents about half of the total sources of funding for U.K. nonfinancial corporate (Figure 23.15). The crisis triggered deleveraging and changes in the composition of debt financing away from bank debt. Since the crisis, leverage ratios

have continued to trend downward, corporate profitability has increased, and investment has rebounded. Corporate bond spreads are back to precrisis levels, and equity prices have been rising. A key indicator of firms' viability, the interest cover ratio (ICR), has also recovered. The ICR measures the degree to which cash flows are sufficient to cover interest on debt. Firms where earnings before interest and taxes are less than interest payments due (i.e., the ICR is less than one) are vulnerable, and their debt is considered "at risk." They can survive by selling assets to meet their obligations, but this is not sustainable over the long term. Just before the crisis, one-third of listed companies in the Worldscope database had ICRs of less than one. But most of these were small, accounting for 13 percent of corporate sector debt. In 2009, the share of firms with ICRs less than one rose to 45 percent, accounting for 16.5 percent of total corporate debt. Preliminary data suggest that at the end of 2010, total debt at risk was slightly below precrisis levels (Figure 23.16).¹⁰

¹⁰ Data coverage is limited for 2010, as not all firms had posted financial results in Worldscope. It is worth noting that owing to data availability limitations, firm coverage in this analysis is smaller than in the firm-level analysis conducted at the BoE (only listed firms are included). This implies that the estimates in this analysis may not match similar estimates in BoE publications. Nonetheless, the thrust of our results and qualitative assessment are in line with the authorities' own analysis.



Sources: Author; Bank of England; Bloomberg; and Office of National Statistics.

Figure 23.14 United Kingdom: Nonfinancial Corporate Sector

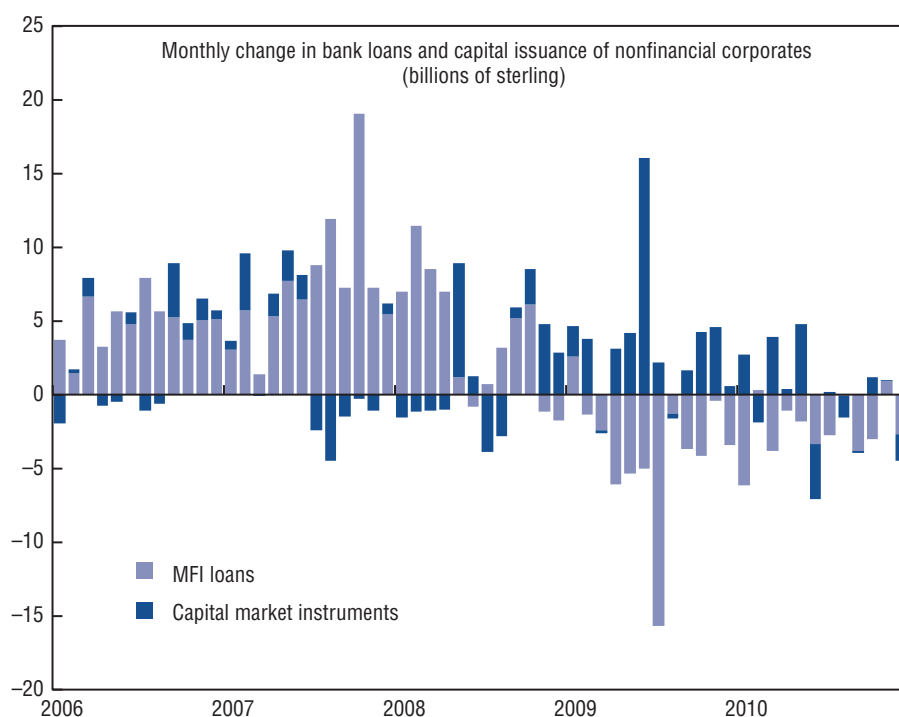
These aggregate figures mask substantial variation across sectors:

- Many real estate firms had ICRs of close to one before the crisis and so quickly crossed over the viability threshold when earnings suffered. By 2009, about 60–70 percent of total debt in this sector was at risk. Although this sector has since recovered somewhat, the share of debt at risk remains very high.
- Services and manufacturing were also hit by the crisis, albeit to a lesser extent. The number of financially stressed firms has fallen significantly and the share of debt at risk is now close to precrisis levels.
- The ongoing restructuring and deleveraging of transportation and communication translated into a continuous decline of debt at risk, which was barely dented by the crisis.

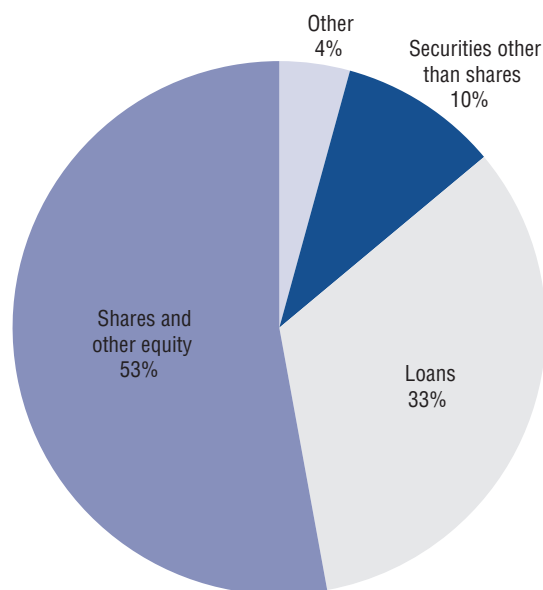
- In contrast, the debt at risk in retail kept increasing in 2010 relative to 2009. As a result, the share of distressed debt is still significantly above precrisis levels.

The hardest-hit sectors represent only about 20 percent of total corporate sector debt. Despite the increase of debt at risk, the total debt of the real estate, construction, and retail sectors is only about 20 percent of total (bank and nonbank) corporate debt (Figure 23.17), while the financial performance of the most leveraged sector at the beginning of the crisis (transportation and communication) strengthened significantly, pushing the aggregate level of debt at risk down.

However, these sectors represent a disproportionately high share of total bank claims on the corporate sector. Over 40 percent of total outstanding bank loans are to the real estate sector (Figure 23.18). Real estate companies remain exposed to further declines in property prices. Despite the rebound



Nonfinancial corporate sector: Sources of funding

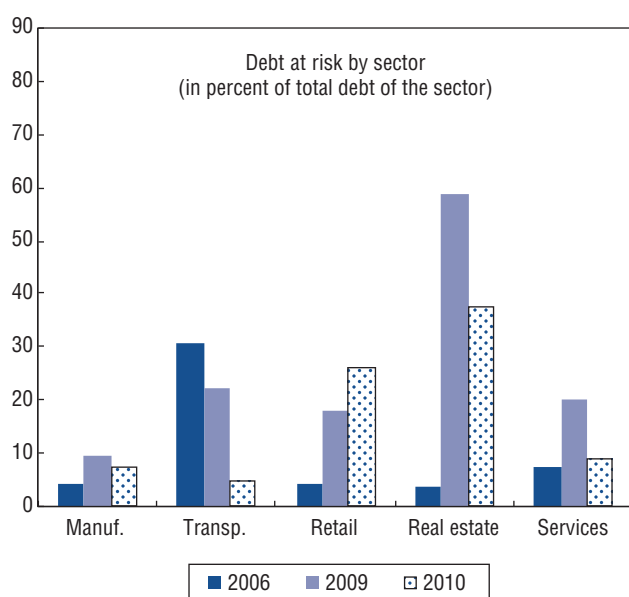
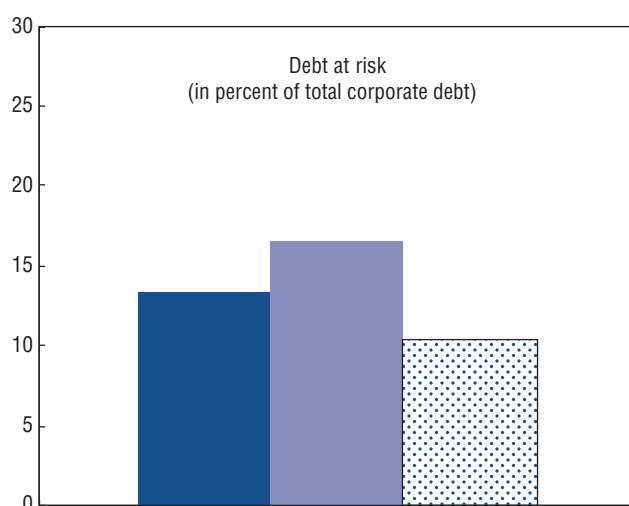


Sources: Bank of England; and Office of National Statistics.
Note: MFI = monetary and financial institutions.

Figure 23.15 Funding of U.K. Nonfinancial Corporates

in commercial property values, prices remain substantially below their peak, leaving a number of borrowers in negative equity or in breach of LTV covenants. So far, defaults have been contained by banks' forbearance and low interest rates. Lenders are exercising forbearance by ignoring breaches in LTV covenants, when borrowers continue to service their debt, and by evergreening loans. But a large share of loans is due

to be refinanced in the next few years, and a large funding gap may arise if banks are less willing or able to roll over or restructure loans (Figure 23.19). In addition, CRE lending is very concentrated, with a few big banks accounting for a large share of the outstanding debt. Banks with large exposures would thus be hard-hit if economic and market conditions were to deteriorate sharply. Indeed, it is already the



Sources: Author; and Worldscope.

Note: Manuf. = manufacturing; Transp. = transportation.

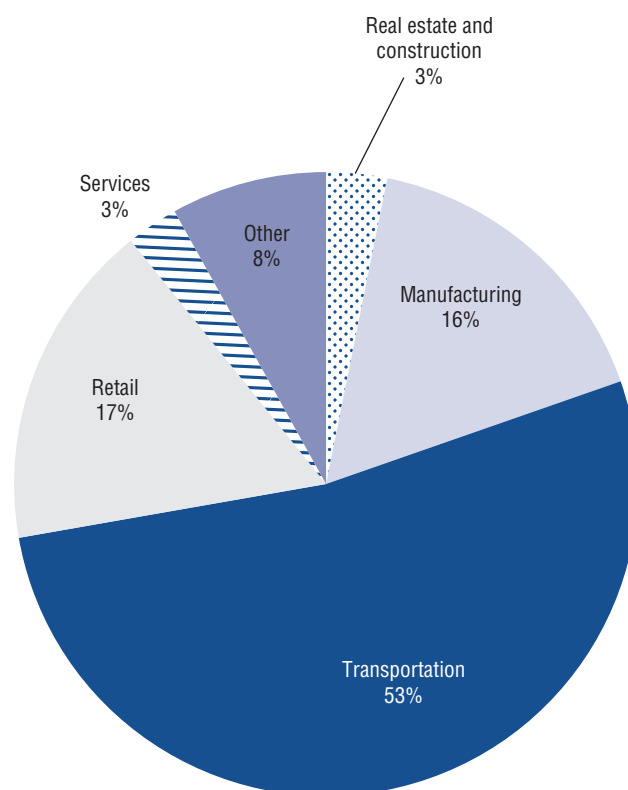
Figure 23.16 Debt at Risk of U.K. Nonfinancial Corporates

case that some lenders are facing arrears rates in excess of 30 percent in their CRE loan portfolios.

B. Contingent claims analysis of the corporate sector

This section assesses the near-term prospects for the U.K. corporate sector. It attempts to answer two questions: First, how high is the risk of corporate sector defaults, and what sectors are most vulnerable? Second, how large are the expected losses from defaults?

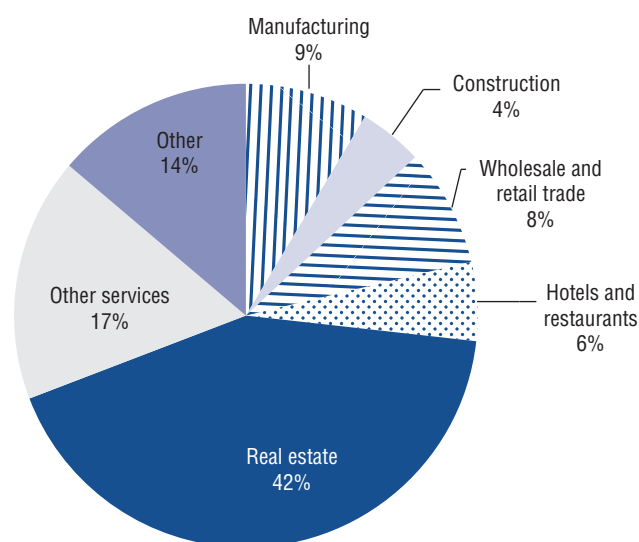
To answer these questions, this chapter uses CCA to estimate risk indicators for the nonfinancial corporate sector. The CCA approach combines balance sheet accounting



Sources: Author; and Worldscope.

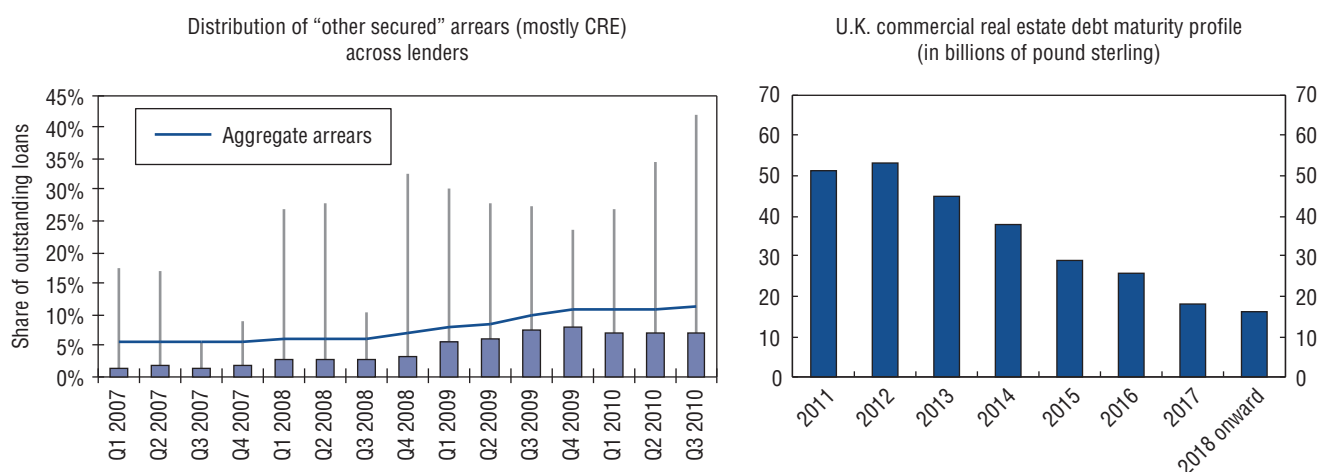
Note: Manuf. = manufacturing; Real est. & constr. = real estate and construction; Transp. = transportation.

Figure 23.17 United Kingdom: Corporate (Bank and Nonbank) Debt by Sector



Sources: Author; and Bank of England.

Figure 23.18 United Kingdom: Breakdown of Bank Lending to the Resident Nonfinancial Corporate Sector



Source: CB Richard Ellis; and Financial Services Authority.

Note: The data is for the 40 largest lenders. Spikes and bars show minimum, maximum, and interquartile range.

CRE = commercial real estate.

Figure 23.19 United Kingdom: Commercial Real Estate Sector

information with equity prices prevailing in the financial market to obtain forward-looking measures of the risk of defaults and potential losses (Box 23.1). Under this approach, the risk of default is related to the probability that the value of a firm's assets will fall below the value of its liabilities. This in turn depends on two factors: firm leverage and uncertainty about the value of firm assets, which reflects the expected value of future profits.

Expected default probabilities increased markedly after September 2008 but have fallen back to near precrisis levels. The median expected one-year default probability for listed corporates in the United Kingdom, derived from Moody's KMV (MKMV) implied credit default swap (CDS) spreads, increased from 0.3 percent in early 2008 to 2.8 percent in March 2009, an amount equivalent to a 15-standard deviation increase relative to the 2004–08 average (Figure 23.20). At the height of the crisis, the risk of default exceeded 13.6 percent for a quarter of firms (75th percentile series in the chart) and 30 percent for 10 percent of firms (90th percentile series in the chart). The increase in default probabilities was triggered by the collapse of share prices and rising volatility as well as by tighter financing conditions and increased rollover risks. Default probabilities are now back to about 0.5 percent. Similarly, before the crisis (July 2007), only a small fraction of firms (accounting for less than 1 percent of total assets) had a default risk one year ahead that exceeded 1 percent. By January 2010, this proportion had increased to 10.5 percent, and it has narrowed to 3.25 percent in February 2011.

However, default risks vary widely across sectors. At the height of the crisis, default risk in some sectors (real estate, business products, leisure and entertainment) increased sharply above the median for the overall corporate sector. Currently, these sectors continue to display higher default risk. In particular, the median real estate firm has a probability of default twice as large as the corporate sector as a whole. This

is also the case for the leisure and entertainment sector, which was severely affected by the sharp decline in consumption, the slow turnaround during the recovery, and the cold winter weather.

Expected losses from corporate defaults appear contained relative to GDP. Based on default probabilities inferred from equity price data as of February 2011, the listed corporate sector is estimated to incur losses amounting to about 0.5 percent of GDP (Figure 23.13). This loss calculation is based on historical LGD rates, which, in the case of U.K. listed companies, is about 60 percent. With a lower LGD rate of 20 percent, corporate losses could be just 0.2 percent of GDP.

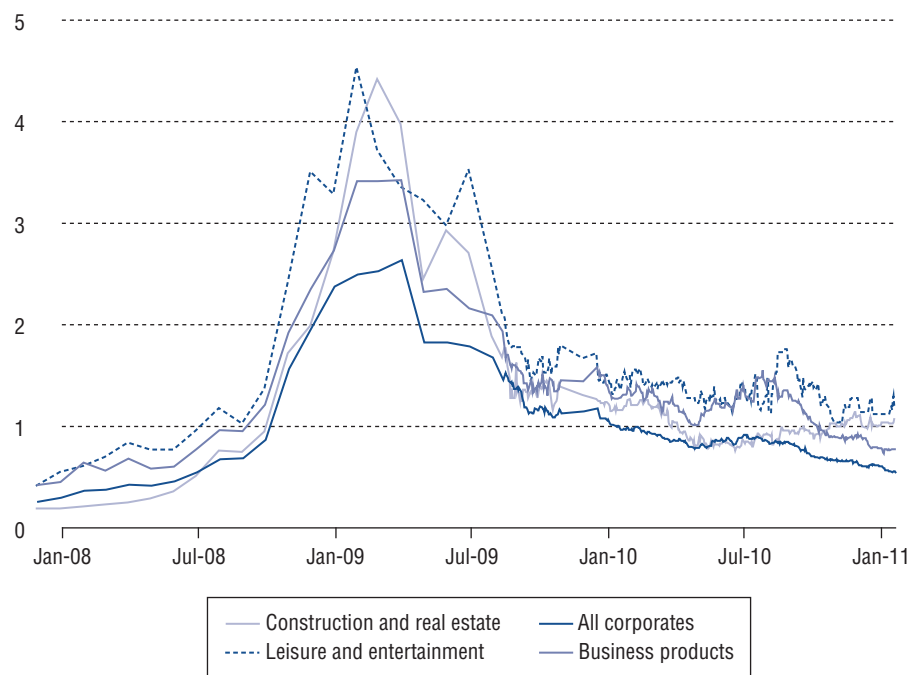
Corporate losses are expected to be the largest in the real estate sector. With higher than average default probabilities and historical LGD rates, construction and real estate is the most vulnerable sector, followed by business products and services (Figure 23.21). Overall, losses are expected to be larger in the nontradable sector. Breaking down the sample of listed companies by size shows that about three-quarters of losses are expected to fall on small- and medium-sized firms.

C. Sensitivity analysis of the corporate sector

This section examines how different types of shocks to the corporate sector would affect firms' viability. Firms where earnings before interest and taxes are less than interest payments due (i.e., ICR less than one) are considered unviable and their debt "at risk." The objective of this section is to analyze how many firms would face financial difficulties when affected by interest rate and profitability shocks. The results are expressed in terms of the debt of firms with ICR less than one or, in other words, the debt at risk as a percentage of total corporate sector debt. The sensitivity analysis uses firm-level data from Worldscope. As in the sensitivity analysis exercise for households, corporate balance sheets at

Box 23.1. Contingent Claims Analysis

Contingent claims approach is a risk-adjusted balance sheet framework where equity and risky debt of a corporation or financial institution derive their value from assets. In this framework, first proposed by Merton (1973) and by Black and Scholes (1973), the total market value of assets of a firm or bank at any time is equal to the market value of equity and risky debt. Asset values are uncertain and, in the future, may decline below the point where debt payments on scheduled dates cannot be made. Debt is “risky” because there is a chance of default.



Source: Moody's KMV.

United Kingdom: Expected Default Probabilities by Industry (median, in percent)

In CCA, the equity and risky debt can be valued using finance techniques, that is, valued using formulas for implicit call and put options, whose values are derived from assets, uncertainty of assets, and the promised debt payments. The value of risky debt is equivalent to the default-free debt minus the expected loss due to default. In CCA, the value of the equity is computed as the value of an implicit call option, and the value of the expected loss due to default can be modeled with an implicit put option. The risk-adjusted balance sheet components can be calibrated by using forward-looking information from the equity market and information from the balance sheet to define the default barrier. The implied market value of assets and implied asset volatility are inferred from the market and balance sheet information; credit risk indicators can then be calculated—for example, default probabilities and credit spreads. See Gray and Malone (2008) for a comprehensive analysis of the CCA framework.

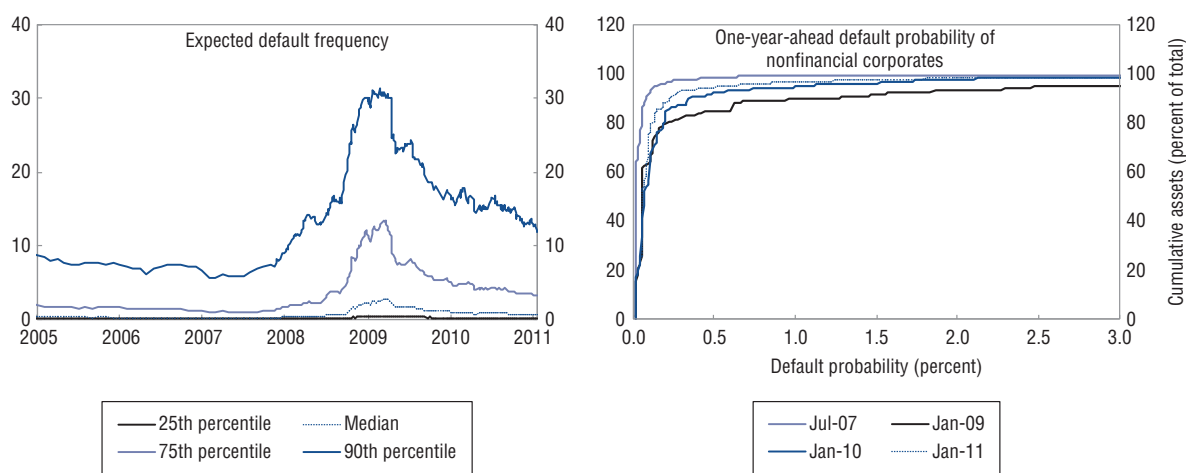
The value of expected loss due to default can be modeled with an implicit put option and can be estimated using information from MKMV implied CDS spreads and expected default probabilities inferred from market and balance sheet information. $Loss = LGD * RNDP * B \exp(-rt)$, where LGD is the loss given default (which is one minus the recovery rate), $RNDP$ is the risk-neutral default probability, B is the default barrier, and r is the risk-free rate.

end-2010 are the starting point for the analysis. The results that follow are based on a partial equilibrium exercise and, as such, should be treated with caution.

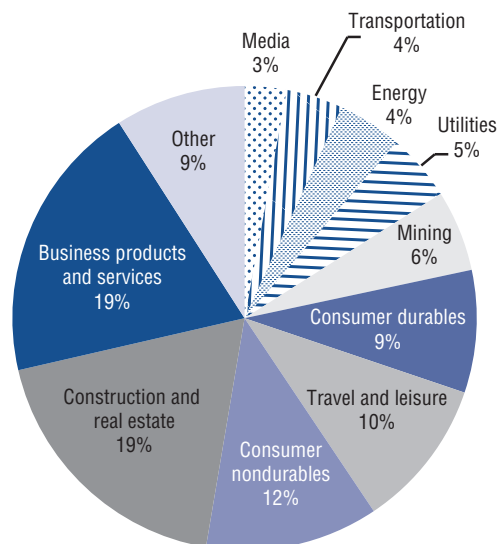
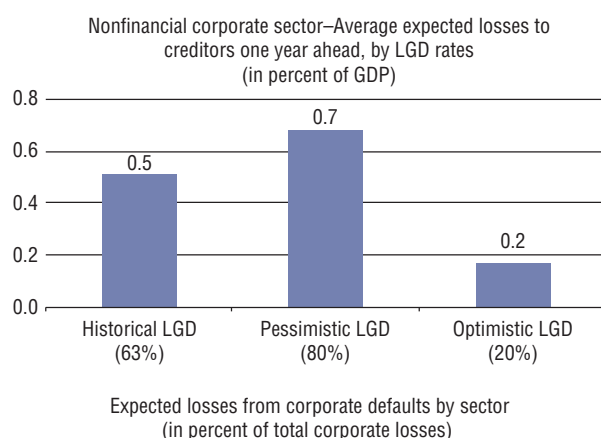
The results suggest that the corporate sector as a whole is vulnerable to profit and interest rate shocks. Nevertheless, the overall picture is not alarming as the increase in debt at risk is modest (Figure 23.22). In particular, a 300-basis points increase in interest rates would increase debt at risk by 3 percentage points, which is about the same increase observed during the crisis. In turn, a 30 percent decline in prof-

its would increase debt at risk by 2 percentage points. The combined shock would have a more than proportional impact, pushing up debt at risk by 6½ percentage points.

Macroeconomic shocks would exacerbate sectoral differences and further deteriorate the financial position of real estate companies, increasing the probability of default of those firms whose debt is at risk. The main concern is that the large portion of CRE firms that are already facing financial difficulties go into actual bankruptcy. Furthermore, a big enough shock, such as the combined shock under



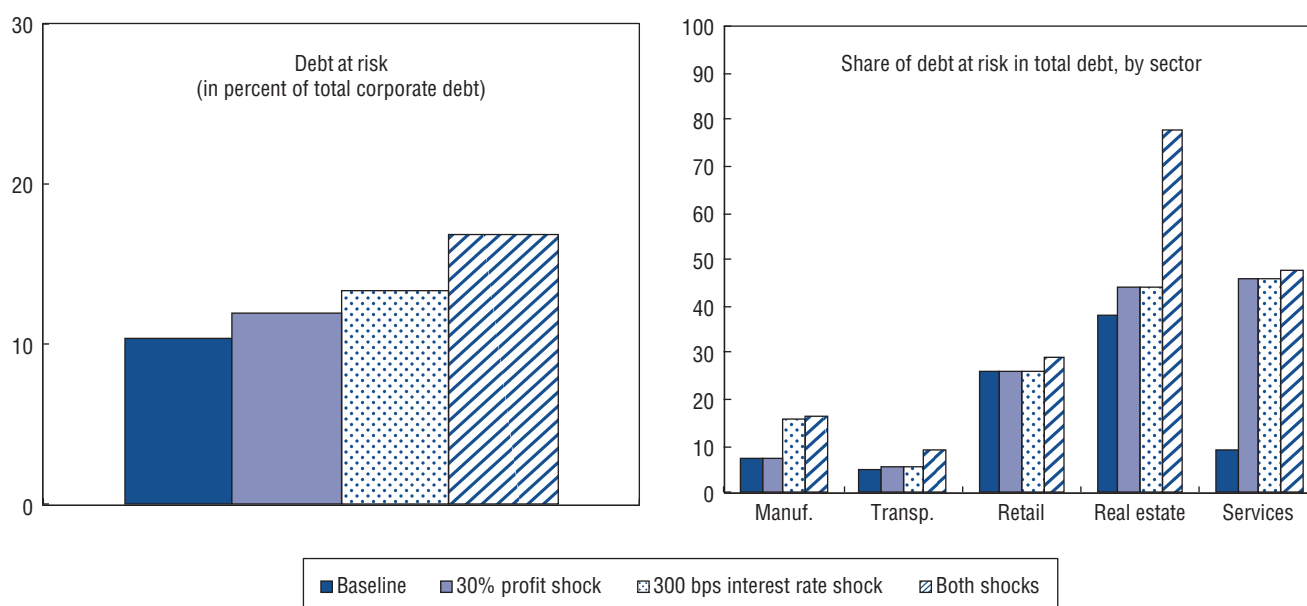
Source: Moody's KMV.

Figure 23.20 United Kingdom: Default Risk of U.K. Nonfinancial CorporatesSources: Author; and Moody's KMV.
Note: LGD = loss given default.**Figure 23.21** Contingent Claims Analysis of the U.K. Corporate Sector

consideration, would result in over three-quarters of construction and real estate companies becoming unviable. This would have a significant impact on the banking sector, given that these firms make up a disproportionate share of total bank loans. Indeed, the most vulnerable firms (real estate and construction) have mainly bank debt, while the stronger companies (manufacturing) have more nonbank debt. In contrast to what happened during the crisis, the retail sector appears relatively resilient to both interest rate and profit shocks, but the services sector would be particularly affected. The shocks would push the share of debt of firms with ICR less than one to almost 50 percent from 10 percent currently. Some sectors, such as manufacturing, are more vulnerable to interest rate increases than to profit shocks.

3. CONCLUSION

U.K. households remain heavily indebted and vulnerable to shocks, but on the whole, the potential impact on banks appears to be manageable. Though off its precrisis peak, household debt relative to disposable income remains high by both international and U.K. historical standards. This makes households vulnerable to income and interest rate shocks. Indeed, a very rapid normalization of interest rates to precrisis levels would have a large impact on household debt affordability, with negative implications for the economy. But the potential knock-on effect of these shocks on banks, albeit nonnegligible, would be mitigated, as most household bank debt is secured. Furthermore, low-income households, which are most sensitive to interest rate and income shocks, account for a relatively small proportion of bank mortgage lending. These results are based on a partial equilibrium exercise and hinge on the assumption that the underlying asset value is not affected by interest rate and income shocks and, as such, should be treated with caution. If house prices fell simultaneously, banks' LGD would be much higher, as discussed below.



Sources: Author; and Worldscope.

Note: bps = basis points; Manuf. = manufacturing; Transp. = transportation.

Figure 23.22 Sensitivity Analysis of the U.K. Nonfinancial Corporate Sector

At a more granular level, however, there are important vulnerabilities. First, although unsecured debt is a relatively small share of bank lending, absolute losses from unsecured debt have so far surpassed losses from mortgage defaults, which have remained low relative both to those in the United States and to previous crises. Given that low-income households account for a relatively large share of banks' unsecured lending, bank losses could jump if interest rates were to increase rapidly. Second, because the ratio of house prices to income remains significantly above the historical average, further sharp declines in house prices could have a major impact on household net worth and thus have the potential to inflict significant losses on the banks. Finally, some segments in the mortgage debt market appear vulnerable, and negative shocks would especially affect borrowers with high LTV ratios and buy-to-let loans.

The corporate sector weathered the crisis relatively well, and leverage has subsequently declined and profitability strengthened. The increase in the number of company liquidations during the crisis was moderate compared with previous crises and has been declining since late 2009, supported by the low interest rate environment. The sector as a whole appears relatively resilient to profitability and interest rate shocks. CCA, which combines balance sheet and equity market information to obtain forward-looking measures of the risk of default, also suggests that losses from corporate defaults one year ahead are expected to be limited.

One important exception is the corporate real estate sector. Both sensitivity analysis and CCA suggest that macroeconomic shocks would affect disproportionately this sector, which is still recovering from the impact of the crisis. Al-

though the share of this sector in total corporate sector liabilities is relatively contained, it represents almost half of total bank loans to the nonfinancial corporate sector and thus is an important potential risk for the banking system. Furthermore, a large share of loans is due to be refinanced in the next few years, and a large funding gap may arise if banks are less willing or able to roll over or restructure loans. The CRE companies remain exposed to further declines in CRE prices. Despite the recent rebound, these prices remain substantially below their peak, and many borrowers are in negative equity or in breach of LTV covenants.

Financial distress and credit risks may be larger than suggested by the headline figures of write-offs and liquidations. Two important factors are at play: lender forbearance and concentration of risks in some banks. Lenders' forbearance has played a material role in containing defaults to date, particularly in the CRE sector, and may be disguising the true extent of risks. Lenders are exercising forbearance by ignoring breaches in LTV covenants when borrowers continue to service their debt and by evergreening loans. Although some degree of restructuring is acceptable for a reasonable period of time, it is unclear how widespread this practice is and whether it is masking true impairment and whether restructured assets are properly accounted for and monitored. In addition, CRE lending is very concentrated, with a few big banks accounting for a large share of the outstanding debt. Banks with large exposures would thus be hard-hit if economic and market conditions were to deteriorate sharply.

Furthermore, banks are exposed to the household and corporate sectors not just through loan portfolios but also through holdings of securities. In the latter case, markets have tended to

react much more quickly to changes in risk factors—higher interest rates, unemployment, changes in house prices—well before these factors have had any kind of impact on loan portfolios. This was, in fact, a key feature of the recent crisis, with its impact on residential and commercial mortgage-backed securities and consumer asset-backed securities.

Finally, credit risks may also arise from banks' exposures to non-U.K. household and corporate sectors. In particular, some banks have significant exposures to the residential and commercial property markets in the United States and some countries in Asia. Interest rate and house price developments in these markets could have implications for major U.K. banks' financial positions.

REFERENCES

- Bank of England, 2010, *Financial Stability Report* (London, December).
- Beer, Christian, and Martin Schurz, 2007, "Characteristics of Household Debt in Austria: Does Household Debt Pose a Threat to Financial Stability?" *Monetary Policy and the Economy*, 2nd Quarter, pp. 58–79.
- Black, Fisher, and Myron Scholes, 1973, "The Pricing of Options and Corporate Liabilities," *Journal of Political Economy*, Vol. 81, No. 3, pp. 637–54.
- Financial Services Authority, 2011, *Prudential Risk Outlook* (London, March). Available via the Internet: <http://www.fsa.gov.uk/pubs/other/pro.pdf>
- Gray, Dale F., and Samuel Malone, 2008, *Macrofinancial Risk Analysis* (New York: Wiley).
- International Monetary Fund, 2011a, "United Kingdom: Stress Testing the Banking Sector—Technical Note," IMF Country Report 11/227 (Washington, July). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=25119>
- , 2011b, "United Kingdom: Vulnerabilities of Household and Corporate Balance Sheets and Risks for the Financial Sector—Technical Note," IMF Country Report 11/229 (Washington, July). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=25121>
- Karasulu, Meral, 2008, "Stress Testing Household Debt in Korea," IMF Working Paper 08/255 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=22458>
- May, Orla, and Merxe Tudela, 2005, "When Is Mortgage Indebtedness a Financial Burden to British Households? A Dynamic Probit Approach," Working Paper No. 277 (London: Bank of England). Available via the Internet: <http://www.bankofengland.co.uk/publications/Pages/workingpapers/2005/wp277.aspx>
- Merton, Robert C., 1973, "Theory of Rational Option Pricing," *Bell Journal of Economics and Management Science*, Vol. 4, No. 1, pp. 141–83.
- Sá, Filipa, Pascal Towbin, and Tomasz Wieladek, 2011, "Low Interest Rates and Housing Booms: The Role of Capital Inflows, Monetary Policy and Financial Innovation," Working Paper No. 411 (London: Bank of England). Available via the Internet: <http://www.bankofengland.co.uk/publications/Pages/workingpapers/2011/wp411.aspx>

Measuring and Analyzing Sovereign Risk with Contingent Claims

MICHAEL T. GAPEN • DALE F. GRAY •
CHENG HOON LIM • YINGBIN XIAO

This chapter develops a comprehensive new framework to measure and analyze sovereign risk. Given that traditional macroeconomic vulnerability indicators and accounting-based measures do not address risk in a comprehensive and forward-looking way, the contingent claims approach is used to construct a marked-to-market balance sheet for the sovereign and derive a set of credit risk indicators that serve as a barometer of sovereign risk. Applications to 12 emerging market economies show the risk indicators to be robust and highly correlated with market spreads. The framework can help policymakers design risk mitigation strategies and rank policy options using a calibrated structural model unique to each economy.

METHOD SUMMARY

Overview	The risk-adjusted sovereign balance sheet model enables the valuation of sovereign foreign currency debt.
Application	The model calculates credit spreads and expected losses on sovereign foreign currency debt calibrated from value and volatility of base money and local currency debt (in foreign currency terms) and foreign currency debt level, giving implied sovereign asset value, volatility, and credit risk indicators.
Nature of approach	Contingent claims analysis approach (risk-adjusted balance sheet).
Data requirements	Time series value of base money, local currency debt, exchange rate, and long- and short-term foreign currency debt.
Strengths	This model estimates spreads on foreign currency debt, which can be used for stress testing changes in sovereign foreign currency debt value, given changes in exchange rates, fiscal policy, government contingent liabilities, and reserves.
Weaknesses	• This model works best with flexible exchange rate regimes and where the stock of foreign currency debt is large. • Data on local currency debt are needed. • This particular model cannot be applied to countries that have only local currency debt.
Tool	The Balance Sheet Risk Analysis Excel spreadsheet is available in the toolkit, which is on the companion CD and at www.elibrary.imf.org/stress-test-toolkit . Contact author: D. Gray.

As economies become more reliant on private capital flows, they have also become more vulnerable to the volatility of capital flows, prices, and other shocks. A comprehensive framework is needed to analyze—and, one hopes, help prevent—large-scale capital account crises and associated financial distress. A useful approach that has been gaining

popularity since the Asian crisis is to assess the risk posed by potentially unstable positions in sectoral balance sheets, including in the corporate, financial, and public sectors. Shocks to interest rates, exchange rates, or market sentiment that bring about a deterioration in the value of a sector's assets compared with its liabilities lead to a reduction of its net

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worth. In the extreme case, net worth turns negative, and the sector may become insolvent. In these cases, risks are transferred across balance sheets, triggering widespread distress. Risk transfer can be “bottom-up” from the corporate sector to the banking system and ultimately to the sovereign balance sheet, as was the case during the Asian crisis, or it can be “top-down,” as was seen more recently in Latin America. Developing an effective approach to detect and assess balance sheet vulnerabilities before they become severe is essential in minimizing risks, thereby protecting the stability of the overall economy. In this chapter, the contingent claims analysis (CCA) approach is used to measure and analyze risk on the public sector, or sovereign, balance sheet.¹ Estimating risk using such an approach has a long tradition in modern financial theory and has been applied widely in the analysis of corporate sector credit risk.² It is being used increasingly to estimate risk in the financial sector but has yet to be applied broadly at the sovereign level.³ This study represents a first step in this direction.

Effective risk analysis must meet three objectives. First, it needs to identify existing balance sheet mismatches. Second, it must incorporate uncertainty inherent in balance sheet components because uncertain changes in future asset value relative to promised payments on debt obligations ultimately drive default risk. Third, effective risk analysis must translate uncertainty into quantifiable risk indicators that measure risk exposures to reveal whether balance sheet risks are building or subsiding. Such quantitative risk indicators should also incorporate forward-looking information.

The CCA approach meets all three objectives. It uses the basic structure of a balance sheet, adding market prices and uncertainty as key inputs, to derive simple risk indicators that are forward looking. In effect, this framework provides a marked-to-market balance sheet for the sovereign. In measuring sovereign risk, the CCA approach derives estimates for sovereign asset value and asset volatility—which are not directly observable—from the value and volatility of sovereign liabilities that are observable. These values are then weighed against existing contractual liabilities to provide a market-based assessment of sovereign default risk. Unlike traditional macroeconomic vulnerability indicators and accounting-based measures, which cannot address risk or uncertainty in a comprehensive or forward-looking manner and rely mainly on static ratios, the CCA approach provides a richer, dynamic way to measure and analyze risk.

This study develops a set of key *credit risk indicators* to measure sovereign balance sheet risk. These include distance to distress, probability of default, credit spreads, and the market value of risky foreign currency denominated debt.

These indicators are closely related because they all are derived from the core contingent claims relationships. Associated with these risk indicators are *sensitivity measures*, which report how responsive the credit risk indicators are to changes in underlying model parameters, such as changes in the value of sovereign assets and volatility. Importantly, the sensitivity measures capture nonlinear changes in value, which are often observed during crisis periods.

To illustrate the usefulness of the credit risk indicators as a collective barometer of sovereign risk, we subjected them to robustness tests using observed market data for a sample of emerging market countries. The tests suggest a high degree of correlation between the credit risk indicators and the observed market data on spreads. As market credit spreads were not used as inputs in deriving the risk indicators, the high correlation suggests that the risk indicators can be confidently used as reasonable measures of sovereign credit risk, thus lending support to the contingent claims structural model developed in this study. The risk indicators can be examined in individual country cases to evaluate whether market expectations of sovereign vulnerabilities are increasing or decreasing over time, or they can be examined across countries to rank relative riskiness.

As a further demonstration of the applicability of the CCA approach in evaluating sovereign risk, the study uses the model calibrated to market data to evaluate how risk indicators change given specific scenarios. Through scenario analysis, policymakers can observe the extent to which negative economic shocks could worsen sovereign financial soundness through capital outflows, a depreciating exchange rate, or slower economic growth. As an additional step, Monte Carlo simulations are used in conjunction with the CCA approach to yield probability distributions and confidence intervals for the set of sovereign credit risk indicators. Because simulations allow for the assessment of many potential market scenarios, they provide for a more comprehensive risk analysis that includes probability distributions and value-at-risk (VaR) measures. Policymakers can use these tools to help them design and implement risk mitigation strategies to reduce balance sheet risk and to rank competing policy choices.

Finally, the study points to two promising areas in which the CCA approach can be usefully applied: reserve management and debt sustainability. On reserve management, the CCA approach can be used to derive an appropriate target for reserve adequacy, where an adequate level of reserves could be defined as the level of reserves that keep the credit risk indicators above a specified threshold. On debt sustainability, the CCA approach offers several advantages over the traditional debt sustainability analysis, which has tended to focus on ratios of debt-to-GDP as the primary criterion for deciding whether public debt is on a sustainable path. In particular, the approach provides a structural framework that relates debt payments with the capacity to pay as well as threshold levels for sovereign credit risk.

The chapter is structured as follows: Section 1 defines sovereign risk and introduces the CCA approach of measuring the value and volatility of sovereign assets. Section 2 con-

¹ The contingent claims approach was applied to estimate balance sheet risk in the aggregated corporate sector in Gapen and others (2004). The analysis also provided estimates of risk transfer across the corporate, financial, and public sectors.

² Black and Scholes (1973) and Merton (1973, 1998).

³ Merton (1977); Gray, Merton, and Bodie (2002, 2003); Draghi, Giavazzi, and Merton (2003); and Chan-Lau, Jobert, and Kong (2004).

Box 24.1. From Corporate to Sovereign Risk Analysis

It is useful to point out the key similarities between the sovereign balance sheet and balance sheets of individual firms. On the asset side,

- **Firms** have main assets consisting of cash and the present value of future earnings (stream of revenues minus expenditures). The firm has other assets, including property, plant, equipment, and inventory. The firm also may have contingent assets from a parent company or implicit guarantees to subsidiaries.
- The **sovereign** has, on its consolidated balance sheet, international reserves and present value of the net fiscal surplus (stream of revenues minus expenditures). Sovereign assets may increase because of contingent financing arrangements with multilateral or other sources, or they may be reduced by the cost of implicit or explicit guarantees to financial institutions or other too-important-to-fail entities. The sovereign has land and other assets, which are unlikely to be sold, so they do not enter into the expected government revenue stream and thus are not included in this definition of asset.

On the liability side, firm liabilities may include senior debt, subordinated debt, and equity. Market capitalization of the firm is equal to price of equity multiplied by the number of shares issued. Sovereign liabilities include foreign currency debt. The sovereign also has local currency debt and base money, which when multiplied by the exchange rate yields the foreign currency value of domestic currency liabilities.

Default risk in the corporate and sovereign sectors is also similar from a valuation perspective:

- **Corporate default.** Corporate sector defaults trigger a bankruptcy process, well defined in some countries and less well defined in others, whereby creditors are assigned their claim to firm assets based on the legally defined seniority of liabilities in the capital structure. Because debt is senior to equity in the firm liability structure, bondholders have senior legal claims to remaining firm assets in the event of default. A review of corporate sector defaults reveals that senior bondholders exercise their control in various ways postdefault. In some cases, bondholders choose to liquidate remaining assets to obtain cash payment. In other cases, bondholders choose to replace management while receiving new claims, oftentimes in the form of equity.
- **Sovereign default.** Although sovereign defaults do not trigger a well-defined bankruptcy process that applies equally across countries, instances of sovereign default trigger a restructuring process whereby predefault liabilities are exchanged for postdefault liabilities. In this restructuring process, holders of sovereign liabilities do not receive similar legal claims to ownership of sovereign assets in the event of a sovereign default (e.g., bondholders cannot assume control of the policy apparatus, possess public sector entities, or liquidate assets). Instead, the holders of sovereign debt have a claim on restructured debt of lower value in the event of default.

From a valuation perspective, default risk in a sovereign setting is similar to default risk in the corporate sector. The present value of the expected loss in the sovereign setting is associated with receiving restructured debt of lower value after default while in the corporate setting it is associated with postdefault cash payouts or new claims at lower value. Bondholders in both cases value their claims at their default-free value minus the present value of expected loss, which can be estimated using an implicit put option.

structs the contingent claims balance sheet. Section 3 shows how the credit risk indicators are developed. Section 4 applies several robustness checks to the credit risk indicators to assess their correlation with actual market data. Section 5 presents a calibrated baseline balance sheet for a hypothetical sovereign. Scenario and simulation analysis is conducted on the baseline balance sheet to assess the impact on sovereign risk. The section discusses briefly how this approach can be used to evaluate potential policy choices. Section 6 details the next steps in the application of this approach to evaluating reserve management and debt sustainability. Section 7 concludes the chapter. Further details on the use of option pricing techniques to derive the credit risk indicators are provided in the Appendix.

1. A PRACTICAL APPROACH TO SOVEREIGN RISK

Of the different types of sovereign risk, one of the most important is the risk of default. Sovereign default is often a culmination of accumulated distress, where the risk of sovereign default is effectively driven by the interplay of three main elements: the value of sovereign assets, asset volatility, and leverage. Sovereign asset value is defined as the combined *market value* of all sovereign assets. Asset value is an aggregate

of different components that are dependent on the country's future economic prospects and policy decisions. Given that future economic prospects are uncertain, asset volatility captures the inherent uncertainty, or variability, of future sovereign asset value. Leverage measures the size of the sovereign's contractual liabilities. Contractual liabilities are measured in *book value* terms because these are the amounts that the sovereign is obligated to pay.

The approach to sovereign risk outlined in this chapter closely mirrors a similar process that has successfully been applied to estimate firm credit risk.⁴ There are sufficient similarities between individual firm risk and sovereign risk to suggest a reasonable transfer of the CCA approach from corporate to sovereign risk analysis. These similarities are examined in more detail in Box 24.1. In the next section, the concept of sovereign distress is defined, and the interplay between the sovereign asset, its volatility, and debt obligations in the determination of sovereign distress is discussed.

⁴ The contingent claims approach has been applied widely by financial market participants to measure the default probability of corporations and banks based on the market prices of the equity and book values of debt. See McQuown (1993); Crouhy, Galai, and Mark (2000); Cossin and Pirotte (2001); and Crosbie and Bohn (2003) for the contingent claims approach to individual firm credit risk.

A. Defining sovereign distress: the concept

Sovereign distress increases when the market value of sovereign assets declines relative to its contractual obligations on debt. Default ultimately occurs when the sovereign assets fall below the contractual liabilities. Contractual liabilities, therefore, constitute a *distress barrier*, and sovereign distress is measured by the relationship between sovereign assets relative to this distress barrier. Default risk increases when the value of sovereign assets declines toward the distress barrier or when asset volatility increases such that the value of sovereign assets becomes more uncertain and the probability of the value falling below the distress barrier becomes higher.⁵

Evidence from the universe of corporate defaults indicates that the market value of firm assets can sometimes trade below the book value of total liabilities for a significant period of time.⁶ This is most often the case when the majority of liabilities are long-term, allowing the firm to continue servicing debt payments while undertaking steps to improve the financial health of the firm. A similar argument can be applied to sovereign credit risk, whereby the probability of distress is increased when most of the liabilities are short-term, or when rollover risk is highest. Therefore, the approach adopted in this study follows the well-established procedure in estimating corporate default risk, namely, that the value of sovereign assets that triggers an incidence of sovereign distress lies somewhere in between the book value of total liabilities and short-term liabilities. This adjusted value of liabilities is defined as the distress barrier and is commonly denoted as the sum of short-term debt, interest payments for one year, and half of long-term debt.⁷

The market value of sovereign assets in relation to the distress barrier is illustrated in Figure 24.1. The uncertainty in future sovereign asset value is represented by a probability distribution at the time horizon. At the end of the period, the value of sovereign assets may be above the distress barrier, indicating that debt service can be made, or below the distress barrier, leading to default. The probability that sovereign assets will fall below the distress barrier is simply the area of the distribution that lies below the distress barrier. The bottom panels in Figure 24.1 detail the effects of a decline in the value of sovereign assets and an increase in uncertainty over future sovereign asset value. In the first case, the lower expected sovereign asset value means that more of the prob-

ability distribution lies below the unchanged distress barrier, which results in a higher probability of default. In the second case, the lower expected sovereign asset value is also accompanied by an increase in asset volatility. The increased volatility widens the probability distribution, leading to an even higher probability of default as more of the area under the probability distribution now lies below the distress barrier.

B. Estimating the value of sovereign assets

Given the conceptual definition of sovereign distress, how does one go about estimating it? The main challenge would be deriving an accurate estimate for the market value and volatility of sovereign assets. Although the levels and amounts of contractual liabilities are relatively easy to determine from balance sheet information, the same is not true when measuring the value of sovereign assets or its volatility.⁸ The market value of sovereign assets is not directly observable and must therefore be estimated. With this in mind, there are several ways to value an asset:

1. determining value from observed market prices of all or part of the asset. This can be from a market price quote, direct observation, bid-ask quote, or other similar direct measures;
2. determining value by a comparable or adjusted comparable. A sophisticated version of obtaining a comparable value is the present value of discounted expected cash flows—such as the primary surplus—with an appropriate discount rate;
3. determining value from an implied value where the balance sheet relationships between assets and liabilities allow the observed prices of liabilities to be used to obtain the implied value of the assets.

The three methods have different advantages and disadvantages. The first method is straightforward but difficult to apply because only a few components of sovereign assets have directly observable market prices. International reserves are both observable and have a market value, yet the remaining items lack observable market prices.⁹ The second method using comparables is commonly used but also has shortcomings. These are related to the difficulty of projecting future cash flows, deciding the appropriate discount rate, and determining all of the relevant components that underlie the cash flow projections for tangible and intangible items included in the asset value estimation. For example, determining the present value of the net fiscal asset requires estimates

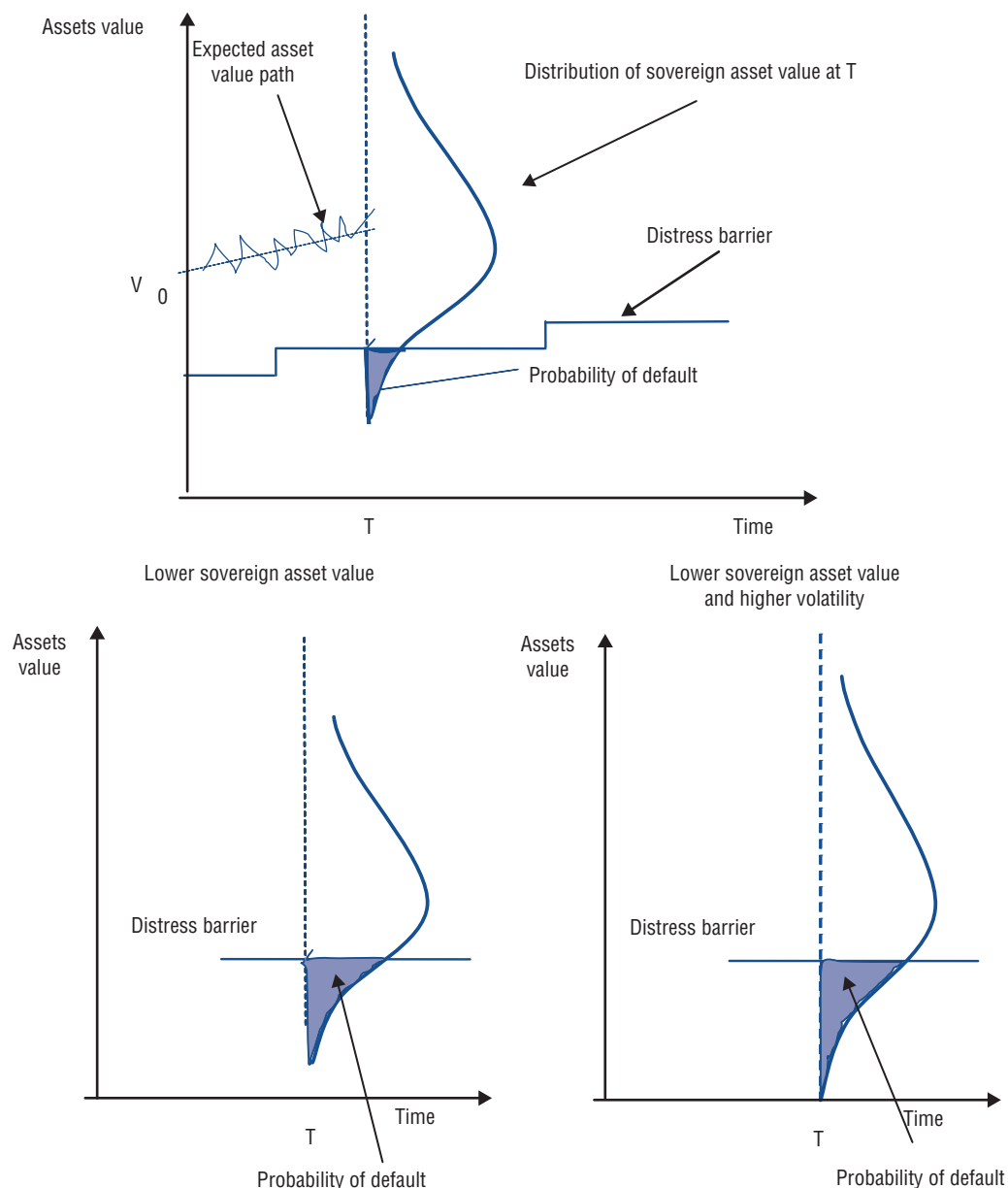
⁵ Volatility of sovereign assets can differ across countries for many reasons, including but not limited to the level of international reserves on the government's balance sheet, the exchange rate, and variations in government revenue and expenditures. Countries with lower asset volatility are generally able to use larger amounts of leverage with relative comfort while countries with higher asset volatility would be better off taking on less leverage.

⁶ Crosbie and Bohn (2003). Moody's KMV (MKMV) maintains a database with over 250,000 company-years of data and 4,700 incidents of default or bankruptcy.

⁷ This definition of the distress barrier is identical to that used by MKMV in corporate sector default risk analysis (Crosbie and Bohn, 2003). *Short-term* is defined as one year or less by residual maturity.

⁸ Foreign currency debt in global markets is predominantly fixed-rate, "bullet" maturity debt, which results in easily defined contractual flows. Some global debt is amortizing, but these payments are usually well specified. The main difficulties in estimating debt payments arise when the debt payments are linked to changes in interest rates, exchange rates, or inflation. These forms are more often found in domestic as opposed to global capital markets.

⁹ Buiter (1993) discusses in detail the many items on the balance sheet of the public sector, including nonmarketable items, such as social overhead capital.



Source: Authors.

Figure 24.1 Distribution of Sovereign Asset Value and the Distress Barrier

of future economic performance, the political commitment to a variety of programs, including social security and other entitlement programs, and the use of an appropriate discount rate. Estimates for the value of other assets like the value of the public sector monopoly on money issuance run into similar problems. Furthermore, it is unclear how asset volatility should be best measured under the first two methods.

The third method, which is the approach adopted in this study, circumvents the problems in the first two methods by estimating sovereign asset value and volatility indirectly with information on observable values of the liability side of the balance sheet. This approach relies on the relationship between assets and liabilities. Given that liabilities are claims on current

or future assets, this approach is often referred to as “contingent claims” analysis and yields an “implied” estimate for sovereign assets. The calculation of implied values is a very common technique in the finance world. The collective view of many market participants is incorporated in the observable market prices of liabilities, and the change in the market price of these liabilities will determine its volatility. This contingent claims approach implicitly assumes that market participants’ views on prices incorporate forward-looking information about the future economic prospects of the sovereign. This does not imply that the market is always right about its assessment of sovereign risk but that it reflects the best available collective forecast of the expectations of market participants.

Implementing CCA to derive the implied sovereign asset value and volatility requires several steps and assumptions. These are discussed in the next section.

2. CONTINGENT CLAIMS ANALYSIS OF THE SOVEREIGN BALANCE SHEET

The contingent claims sovereign balance sheet is constructed from the basic accounting balance sheet of the government and monetary authorities. Figure 24.2 shows the balance sheets of the government and monetary authorities as two segregated yet linked balance sheets. Government liabilities include foreign currency debt, domestic currency debt, and obligations owed by the government to the monetary authorities and the guarantees to “too-important-to-fail” entities. Government assets include a claim on a portion of the foreign currency reserves held by the monetary authority and other public sector assets such as the present value of the primary fiscal surplus. The balance sheet of the monetary authority in Figure 24.2 has assets consisting of international reserves (net foreign assets) and credit to government (net domestic assets). Liabilities of the monetary authority are base money and a claim of the government on a portion of foreign currency reserves.

In order to use the contingent claims balance sheet to estimate the value and volatility of sovereign assets, three steps are needed:

- First, a sovereign balance sheet needs to be constructed with the liability side containing only elements with observable quantities and market prices and in a common currency.
- Second, assumptions on the seniority of sovereign liabilities need to be defined to use standard contingent claims relationships.

GOVT ASSETS <i>Claim on a Portion of International Reserves (of Monetary Authority)</i> Other Public Sector Assets	GOVT LIABILITIES <i>Credit to Government (from Monetary Authority)</i> Domestic Currency Debt Foreign Currency Debt Implicit and Explicit “Too-Important-to-Fail” Guarantees
MONETARY AUTHORITY ASSETS International Reserves <i>Credit to Government</i>	MONETARY AUTHORITY LIABILITIES Base Money <i>Government Claim on Portion of International Reserves</i>

Source: Authors.

Note: Govt = government.

Figure 24.2 Segregated Balance Sheets of the Government and Monetary Authority

ASSETS International Reserves Domestic Currency Assets, in Foreign Currency Terms [= Other Assets – Guarantees]	LIABILITIES Value of Domestic Currency Liabilities, in Foreign Currency Terms [= Domestic Currency Debt + Base Money] Foreign Currency Debt
--	--

Source: Authors.

Figure 24.3 The Consolidated Contingent Claims Public Sector Balance Sheet

- Third, option pricing techniques are used to estimate the value and volatility of sovereign assets from the observable market value and volatility of sovereign liabilities.

A. Consolidating the sovereign balance sheet

The two segregated balance sheets of the government need to be consolidated so that every entry on the liability side can be traced to observable data, and the entire balance sheet is denominated in a common currency. Balance sheets for the country case studies presented in this chapter are measured in U.S. dollars for ease of comparison, but the analysis holds even if they are valued in domestic currency.¹⁰ Through the consolidation process, the government claim on foreign currency reserves and credit to government net out and guarantees to too-important-to-fail entities are subtracted from the sovereign asset.¹¹ Figure 24.3 shows the consolidated sovereign balance sheet denominated in a common foreign currency. All the entries on the liability side of the contingent claims sovereign balance sheet in Figure 24.3 are now directly observable from market prices.

B. Seniority of consolidated balance sheet liabilities

Seniority of sovereign liabilities is not defined through legal status as in the corporate sector but may be inferred from examining the behavior of government policymakers during

¹⁰ Measuring the balance sheet in U.S. dollars results in variable sovereign assets versus a fixed distress barrier. Measuring the balance sheet in domestic currency will result in both variable sovereign assets and a variable distress barrier. In either configuration, the contingent claims formulas will produce the same results.

¹¹ The implicit guarantees to the financial sector, or other entities, could remain on the liability side of the consolidated public sector balance sheet and be modeled as implicit put options. For more details, see Merton (1977); Gray, Merton, and Bodie (2002, 2003); Gapen and others (2004); and van den End and Tabae (2005). These papers link the sovereign to the contingent claims balance sheets of the banking or corporate sectors. The detailed analysis of the links to other sectors is beyond the scope of this study.

periods of stress. In times of stress, governments often make strenuous efforts to remain current on their foreign currency debt, efforts that effectively make such debt senior to domestic currency liabilities.¹² The payment of foreign currency debt requires the acquisition of foreign currency, which the government has a more limited capacity to produce. In contrast, the government has much more flexibility to issue, repurchase, and restructure local currency debt. For this reason, governments sometimes introduce capital controls to prevent convertibility and preserve remaining international reserves to service sovereign external debt obligations. In other instances, governments have insisted on the mandatory rollover or restructuring of domestic currency debt during periods of distress without simultaneously engaging foreign currency creditors. In these circumstances, holders of domestic currency liabilities will see the value of their claim greatly reduced because sovereign distress is often accompanied by instances of exchange rate depreciation, which reduces the value of the domestic currency liabilities in terms of its value in foreign currency.

Three recent examples of sovereign debt restructuring illustrate this implicit seniority structure. Russia in 1998–1999 introduced capital controls, forced a lengthening of maturities on domestic currency government debt, and declared a unilateral moratorium on private sector external debt obligations while still publicly stating its intention to honor sovereign external debt.¹³ In March 1999, Ecuador froze all checking, savings, and time deposits to limit further exchange rate depreciation.¹⁴ In August 1998, Ukraine imposed convertibility restrictions in the foreign exchange market and selectively restructured domestic debt held by banks.¹⁵ (Other examples include government restructuring of debt held by domestic banks or pension funds, thereby reducing their present value, prior to the restructuring of foreign currency denominated external debt.)

For these reasons, this study models foreign currency debt as a senior claim and domestic currency liabilities as junior claims.¹⁶ Default in this study, therefore, means default on foreign currency debt, and the distress barrier defines the level at which payments on foreign currency debt cannot be made. The distress barrier is assumed to be equal to the book value

of short-term external debt plus interest and one-half of long-term external debt.¹⁷ Default is assumed to occur when the value of sovereign assets declines below the distress barrier. As the junior claim, the value of domestic currency liabilities is dependent on the level of sovereign assets above and beyond what is necessary to service senior foreign currency debt. Senior claims, however, are risky because asset value may not be sufficient to meet promised payments. The value of senior claims, therefore, can be seen as having two components, the *default-free value* (promised payment value) and the *expected loss* associated with default when the assets are insufficient to meet the promised payments. The value of junior claims is the residual value of sovereign assets after the promised payments to senior claims have been made. Thus, in financial terminology, the value of domestic currency liabilities can be modeled as an implicit call option on sovereign assets, while the value of *risky* foreign currency debt can be modeled as default-free value of debt—equivalent to the distress barrier—minus the expected loss in the event of default.

The next section discusses how changes in the values of observed market variables—the value and volatility of sovereign liabilities—are used to infer changes in unobserved variables—the value of sovereign assets based on the contingent claims balance sheet and seniority structure above.

C. Calculating implied sovereign asset value and volatility

Given that the value of domestic currency liabilities can be modeled as an implicit call option on sovereign assets, standard option pricing techniques can be applied to derive implied estimates for sovereign asset value and volatility. The balance sheet states that the value of assets is the sum of the value of domestic and foreign currency liabilities, $V_A = V_{DCL} + V_{FCL}$, where V_{FCL} represents the value of risky foreign currency liabilities, and V_{DCL} represents the value of domestic currency liabilities.¹⁸ The option pricing formulas used to estimate sovereign asset value and volatility rely on a few select variables: the value and volatility of domestic currency liabilities (V_{DCL} and Vol_{DCL} , respectively), the distress barrier (DB), the risk-free interest rate (r_f), and time (t). As shown in more detail in the Appendix, these variables can be combined into two equations:

$$V_{DCL} = \text{Function}(V_A, Vol_A, DB, t, r_f), \quad (24.1)$$

$$Vol_{DCL} = \text{Function}(V_A, Vol_A, DB, t, r_f), \quad (24.2)$$

which can be solved simultaneously to derive the two unknowns, which are the implied market value (V_A) and volatility (Vol_A) of sovereign assets. Thus, the information embedded in the value and volatility of domestic currency liabilities (in

¹² Support for viewing foreign currency debt as senior can be found in the literature on “original sin” in Eichengreen, Hausmann, and Panizza (2002). Support for modeling domestic currency liabilities as junior claims can be found in Sims (1999), who argues that local currency debt has many similarities to equity issued by firms. He models domestic currency debt as “equity,” and in this setting, domestic currency debt becomes an important absorber of fiscal risk, just as equity is a cushion and risk absorber for firms. As long as there is some probability that the government will run a primary surplus in the future and/or will engage in the repurchase of domestic currency debt, then such debt has value.

¹³ Ariyoshi and others (2000).

¹⁴ IMF (2002); Gulde and others (2003).

¹⁵ IMF (1999, 2002).

¹⁶ It should be noted that this ordering can be flexible, and the contingent claims framework can be adapted to any number of different seniority structures. In future work, the seniority assumption will be relaxed to take into account multiple layers of liabilities, described in more detail in the Appendix.

¹⁷ An alternative procedure would be to use all of short-term external debt and interest, plus long-term external debt discounted by the risk-free rate.

¹⁸ The value of senior foreign currency liabilities can also be obtained using the implicit put option in risky debt (Gray, Merton, and Bodie 2003; Gapen and others, 2004; Gray 2004).

units of foreign currency) and the distress barrier derived from the book value of foreign currency debt yield estimates of implied sovereign asset value and implied asset volatility over a given time horizon. The volatility of the domestic currency liabilities (domestic currency debt and base money) come from a variety of sources, including the volatility of the exchange rate and of the quantities issued. The volatility of the exchange rate process is relatively more important in a floating exchange rate environment, while the quantities of domestic currency liabilities may vary substantially under a fixed or heavily managed exchange rate system.

3. SOVEREIGN CREDIT RISK INDICATORS

Having derived the estimates of implied asset value and volatility, this section details how they can be used to develop useful indicators of sovereign risk. These risk indicators are the distance to distress, the risk-neutral probability of default, the value of senior foreign currency debt, and the sovereign credit risk premium or sovereign risk-neutral credit spread.¹⁹ Although price and spread information may be easily observable from the market, the market information itself does not reveal the rationale underlying the risk premium, nor does it reveal what is often the most valuable piece of information in risk analysis—how much risk exposures could change as the health of the sovereign improves or declines on the margin. The CCA approach links the credit risk premium to the balance sheet framework, allowing for an evaluation of the structural determinants of credit risk.

A. Distance to distress and risk-neutral probability of default

The implied value and volatility of sovereign assets can be combined with the distress barrier to produce an indicator of default risk, referred to here as the *distance to distress*. This measure computes the difference between the implied forward market value of sovereign assets and the distress barrier scaled by a one standard deviation move in sovereign assets. The distance to distress is defined conceptually as follows:

$$\frac{\text{Implied market value of sovereign assets} - \text{Distress barrier}}{\text{Implied market value of sovereign assets} * \text{Sovereign asset volatility}}.$$

The numerator measures the distance between the expected one-year-ahead market value of sovereign assets and the distress barrier. This amount is then scaled by a one standard

deviation move in sovereign assets. The distance to distress therefore yields the number of standard deviations sovereign asset value is from distress. Lower market value of sovereign assets, higher levels of foreign currency debt, and higher levels of sovereign asset volatility all serve to decrease the distance to distress.

In formula representation:

$$\begin{aligned} \text{distance to distress} &= d_2 \\ &= \text{function}(V_A, DB, r_f, Vol_A, t), \end{aligned} \quad (24.3)$$

where d_2 is from the Black-Scholes option pricing formula (Appendix). Distance to distress for a hypothetical sovereign is illustrated in Figure 24.1.

The option pricing relationships in equations (24.1) and (24.2) also yield a measure of probability of default, commonly referred to as the *risk-neutral default probability*. The probability of default is simply the probability that future sovereign asset value will fall below the distress barrier. The option pricing formula used in this analysis assumes that future sovereign asset value is distributed log-normally, and the risk-neutral probability of default is therefore the shaded area that lies below the distress barrier as shown in Figure 24.1. The risk-neutral default probability (RNDP) is

$$\begin{aligned} \text{RNDP} &= N(-d_2), \\ \text{where } d_2 &= \text{function}(V_A, DB, r_f, Vol_A, t), \end{aligned} \quad (24.4)$$

and $N(\cdot)$ is the cumulative normal distribution at the distance to distress, d_2 .

B. Value of foreign currency liabilities and sovereign credit risk premium

The other two useful sovereign risk indicators that can be obtained using the CCA approach are the sovereign credit spread, or credit risk premium, and the market value of foreign currency liabilities. The value of risky senior foreign currency liabilities (V_{FCL}) can be derived using the implied value and volatility of sovereign assets, equations (24.1) and (24.2), and the balance sheet identity noted above—that is, the value of assets is the sum of the value of domestic and foreign currency liabilities. Using these relationships together yields the value of risky foreign currency liabilities:²⁰

$$V_{FCL} = \text{function}\left(\left(\frac{V_A}{DBe^{-r_f t}}\right), Vol_A, t, r_f\right). \quad (24.5)$$

The term $DBe^{-r_f t}$ is the distress barrier discounted to the present by the risk-free rate over the time horizon t . Given that the distress barrier is based on the book value of foreign currency debt, it is equivalent to the default-free value of foreign currency liabilities. If the ratio of sovereign assets to the default-free value of foreign currency liabilities rises or the volatility of sovereign assets declines, the value of risky foreign currency liabilities increases. Conversely, as the ratio of sovereign assets over the discounted distress barrier falls or asset volatility

¹⁹ Risk-neutral valuation is an important factor underlying the derivation of the Black-Scholes option pricing formula whereby the value of the option can be derived by forming a riskless hedge portfolio. Thus, option values do not depend on the investor's or decision maker's attitude toward risk, which is a major benefit of this approach. Alternative balance sheet approaches based on discounted cash flows are subject to serious error not only from errors in cash flow projections but from errors in choosing the discount rate. See Hull (1993, pp. 221–22) and Chriss (1997, pp. 190–93) for additional discussion of risk-neutral valuation.

²⁰ See the Appendix.

risks, the market value of risky debt will decline, possibly falling below its default-free value. In other words, if the sovereign becomes more wealthy and the stream of its income less uncertain, the market value of its foreign currency debt will become more valuable, and vice versa. Therefore, the value of foreign currency liabilities is a useful indicator of the expected gain/loss in asset value that the sovereign is likely to experience if the market prices of its debt increase/decrease.

Manipulating equation (24.5) results in an estimate of the risk-neutral credit spread (RNS) of

$$RNS = y - r_f = \text{function} \left(\left(\frac{V_A}{DBe^{-rt}} \right), Vol_A, t, r_f \right), \quad (24.6)$$

where $y = -(1/t)\ln(V_{FCL}/DB)$. The left-hand side of equation (24.6) represents the yield to maturity on risky foreign currency debt less the risk-free rate of interest and is therefore equivalent to a credit risk premium, or risk-neutral credit spread. In addition to the risk-free rate and time, the sovereign risk premium is a function of only two variables: the volatility of sovereign assets and the ratio of the value of sovereign assets to the distress barrier. Both increases in the ratio of sovereign assets to foreign currency liabilities and decreases in sovereign asset volatility reduce the sovereign risk premium. Conversely, as the ratio of sovereign assets to foreign currency liabilities decreases or sovereign asset volatility increases, the risk premium widens. The intuition is similar as before. The sovereign's credit risk declines if it has a cushion of assets to protect it from negative shocks and the cushion is relatively stable.

It is useful to note that no market information on foreign currency denominated debt, namely, bond spreads or credit default swap spreads, have been used while computing the value of risky foreign currency liabilities and credit risk premium in the model. Only information on the book value of payments on existing foreign currency debt is used in construction of the distress barrier. This is combined with market information from domestic currency liabilities and the exchange rate to estimate the value of foreign currency liabilities and the credit risk premium. This is noteworthy because the model output can be compared then with readily available market information to evaluate the robustness of this approach.

C. Sensitivity measures

Associated with the sovereign risk indicators are sensitivity measures, which reveal how responsive the set of risk indicators are to changes in model parameters, namely, changes in the value of sovereign assets and asset volatility. This chapter focuses on eight relevant sensitivity measures. The first four are the changes in distance to distress, risk-neutral default probability, risk-neutral credit spreads, and value of foreign currency debt from a 1 percent change in the value of sovereign assets. The second four are changes in the same risk indicators from a 1 percent change in sovereign asset volatility. These sensitivity measures are critical in risk analysis because

they capture nonlinear changes in value, and equally important, they look beyond the current level of distance to distress, spreads, or probability of default. In other words, they provide an indication of the potential *risk exposure* of the sovereign. The sensitivity measures are highest when sovereign asset value is in the neighborhood of the distress barrier, reflecting magnified default risk. In this instance, small changes in underlying asset value in either direction will have proportionately larger impacts on the balance sheet risk indicators. In sum, while the credit risk indicators yield current estimates of sovereign balance sheet risk, the sensitivity measures point to how sovereign risk could further change if the balance sheet improves or weakens on the margin.

4. ROBUSTNESS OF SOVEREIGN CREDIT RISK INDICATORS

The degree to which the contingent claims risk indicators closely parallel actual market data will indicate their usefulness as early warning indicators of sovereign risk. To this end, a historical time series of the risk indicators is compared with actual market data for a number of emerging market countries. The historical data for the sovereign risk indicators in equations (24.3)–(24.6) were obtained from the Macro-Financial Risk (MfRisk) model, while the historical market data were obtained from credit default swap and external debt markets.²¹ Robustness of the sovereign credit risk indicators is examined through their correlation and relationship with actual data and through regression analysis.

A. Correlation with market data

If the model output is robust, distance to distress should be negatively correlated with actual sovereign credit spreads. As distance to distress increases, credit risk should decline and be reflected in lower credit default swap spreads. Figure 24.4 displays the relationship between the distance-to-distress indicator for 12 emerging market sovereign balance sheets versus that country's observed credit default swap (CDS) spread.²² Table 24.1 reports the correlation of the risk indicators with the observed sovereign credit default swap spreads and Emerging Market Bond Index (EMBI+) spreads.²³ As

²¹ The MfRisk model was developed by Macro Financial Risk, Inc., and applied to 17 countries under a joint research effort between Moody's and Macro Financial Risk, Inc. Access to MfRisk is available only through subscription.

²² In Figure 24.4, the distance to distress scale is inverted. Although this visually depicts a positive correlation between distance to distress and credit spreads, this implies that increases in distance to distress should result in lower spreads on credit default swaps.

²³ The reported correlations in Table 24.1 were computed using Spearman's rank correlation instead of conventional correlation. Conventional correlation is inappropriate in this case because it implicitly assumes linear relationships among variables, an assumption that contradicts the nonlinear relationship between variables as found in this study. Spearman's rank correlation is a less restrictive measure to gauge relationships among variables because it does not impose any linearity assumptions.

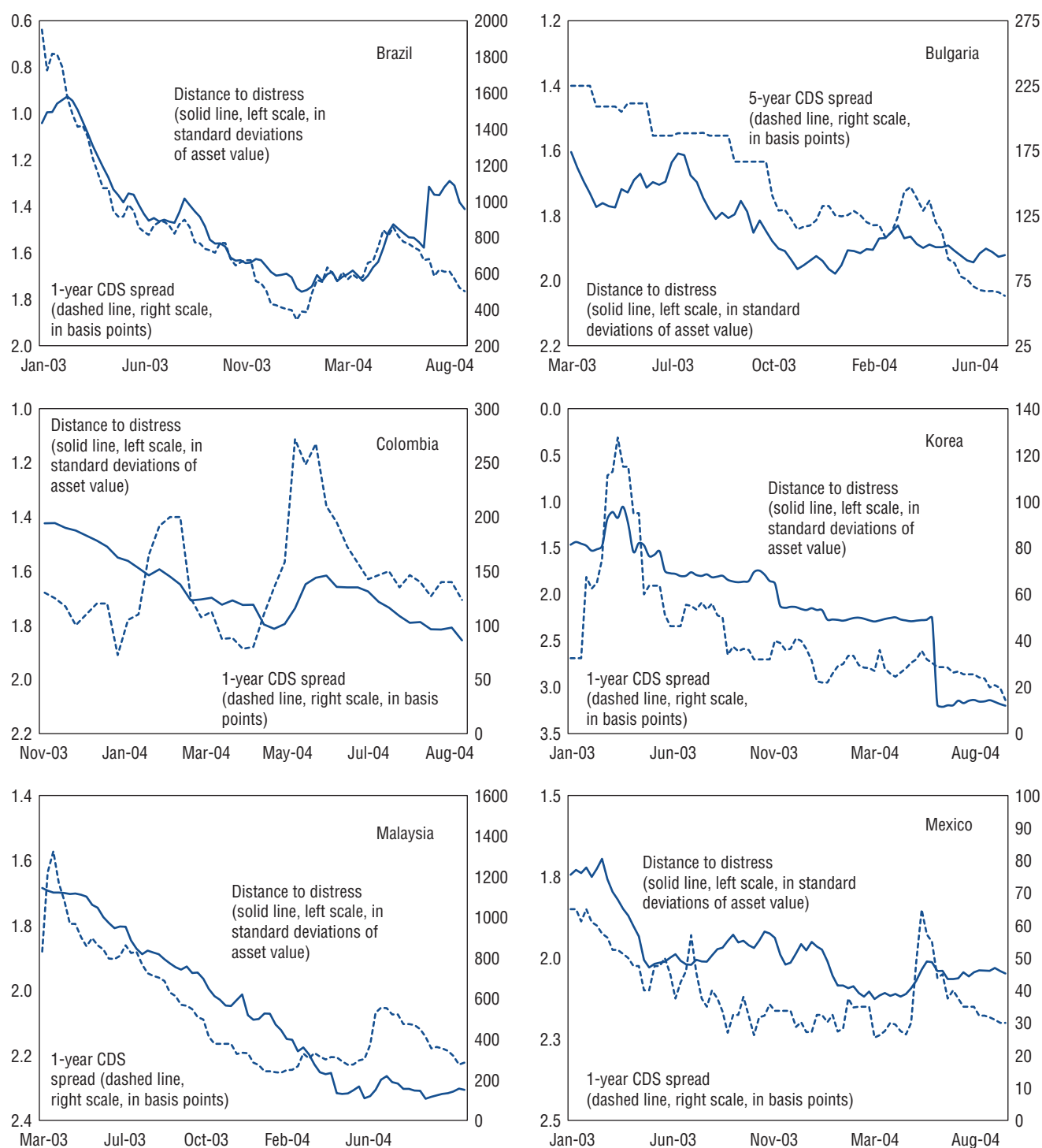
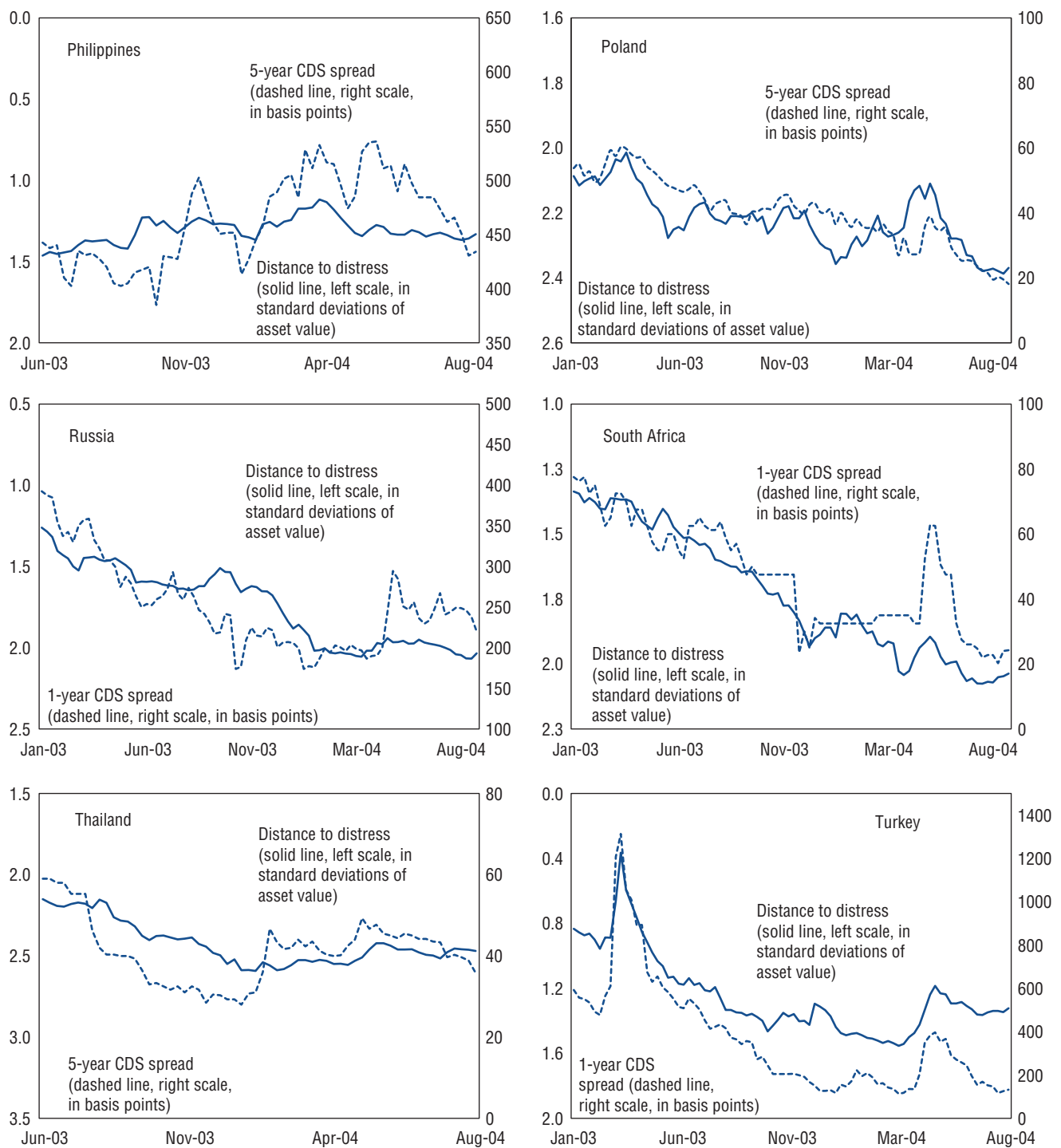


Figure 24.4 Distance to Distress and Credit Default Swaps (CDSs)

can be seen, there is a very high correlation for most countries between the two risk indicators—distance to distress and risk-neutral spread—and the observed CDS spreads and EMBI+ spreads from January 2003 to August 2004. The reported correlations confirm the expected negative relationship between distance to distress and both CDS and EMBI+ spreads. The correlations also display a high degree of significance, as 29 of the 34 reported correlations be-

tween distance to distress and CDS spreads are significant at the 95 or 99 percent level. In many cases, correlation is highest with the five-year CDS spread that likely reflects the greater liquidity in this market relative to the shorter-maturity CDS market. A similar level of significance is found between distance to distress and country EMBI+ spreads, with 8 of the 9 reported correlations significant at the 99 percent level.



Sources: Bloomberg; and Moody's MfRisk.

Figure 24.4 (continued)

As a second check on robustness, the risk-neutral sovereign credit spread for each country is compared with the EMBI+ spread and CDS spread.²⁴ Figure 24.5 displays the expected positive relationship between the risk-neutral sov-

ereign credit spread and each EMBI+ country spread for nine emerging markets for the sample period from January 2003 to August 2004. The correlation between the risk-neutral sovereign credit spread and the respective EMBI+ spread

²⁴ The Emerging Market Bond Index Global (EMBIG) has replaced the EMBI+ as the preferred index for tracking emerging market credit

spreads, but historical EMBIG data were not available at the time of the writing of this research.

TABLE 24.1

Spearman Rank Correlation: Sovereign Risk Indicators and Actual Data

Country	Distance to Distress and				Risk Neutral Spread and			
	1-Year CDS Spread	3-Year CDS Spread	5-Year CDS Spread	Country EMBI+ Spread	1-Year CDS Spread	3-Year CDS Spread	5-Year CDS Spread	Country EMBI+ Spread
Brazil	−0.68**	−0.79**	−0.80**	−0.81**	0.70**	0.82**	0.82**	0.83**
Bulgaria	N/A	−0.72**	−0.91**	N/A	N/A	0.72**	0.83**	N/A
Korea	−0.83**	−0.85**	−0.88**	N/A	0.84**	0.85**	0.89**	N/A
Malaysia	−0.72**	−0.73**	−0.14	−0.36**	0.72**	0.73**	0.15	0.39**
Mexico	−0.44**	−0.62**	−0.73**	−0.72**	0.44**	0.62**	0.73**	0.73**
Philippines	−0.33*	−0.43**	−0.53**	−0.20	0.33*	0.43**	0.54**	0.17
Poland	−0.16	−0.68**	−0.69**	−0.44**	0.06	0.67**	0.69**	0.45**
Russia	−0.29**	−0.54**	−0.66**	−0.47**	0.30**	0.54**	0.67**	0.47**
South Africa	−0.80**	−0.76**	−0.75**	−0.47**	0.86**	0.77**	0.75**	0.64**
Thailand	−0.29	N/A	−0.28*	N/A	0.41*	N/A	0.27*	N/A
Turkey	−0.83**	−0.84**	−0.84**	−0.85**	0.82**	0.83**	0.83**	0.85**
Venezuela	−0.29*	−0.22	−0.20	−0.89**	0.33*	0.27	0.22	0.90**

Source: Author.

Note: CDS = credit default swap; EMBI = Emerging Market Bond Index.

*Denotes significance at 5 percent level.

**Denotes significance at 1 percent level.

during the same time period is also reported in Table 24.1. The correlations show the expected positive relationship between the risk-neutral credit spread and EMBI+ country and CDS spreads. The correlations between the risk-neutral credit spread and the CDS spread display a high degree of significance at the 95 and 99 percent levels for 30 out of 34 reported correlations. The correlations between the risk-neutral credit spread and EMBI+ spread display significance at similar levels in 8 out of 9 cases.

B. Regression analysis

Two fixed-effects panel regressions are used to estimate the relationship between risk-neutral spreads and EMBI+ country spreads and CDS spreads. The mapping from risk-neutral spreads to actual market spreads is important because the risk-neutral spreads tend to underestimate actual market credit spreads. The fixed-effects model treats differences across countries in the sample as parametric shifts of the regression function (i.e., differences across countries are captured in differences in the constant term). This approach yields the following relationships:

- **Risk-neutral credit spreads and observed CDS spreads.** The relationship between risk-neutral credit spreads and observed CDS spreads is estimated by applying a fixed-effects panel regression to a combined cross-country sample of 981 observations from April 2002 to August 2004.²⁵ Results, which are reported in Table 24.2, indicate that the coefficient and constants are highly significant at all confidence intervals and the *R*-squared from the panel regression is 88 percent.

- **Risk-neutral credit spreads and EMBI+ spreads.** The relationship between risk-neutral credit spreads and EMBI+ spreads is estimated by applying a similar fixed-effects panel regression to the same cross-country sample of 981 observations from April 2002 to August 2004. Results, which are reported in Table 24.3, indicate that the coefficient and constants are highly significant at all confidence intervals and the *R*-squared from the panel regression is 96 percent.

Given the goodness of fit of the preceding regressions, the individual country panel equations can be used to map sovereign risk-neutral credit spreads into (1) actual CDS spreads and (2) actual EMBI+ spreads. For example, the estimated equation for Mexico used to map sovereign risk-neutral credit spread into the actual spread on credit default swaps is

$$\ln(CDS_t) = 1.72 + 0.52 * \ln(RNS_t). \quad (24.7)$$

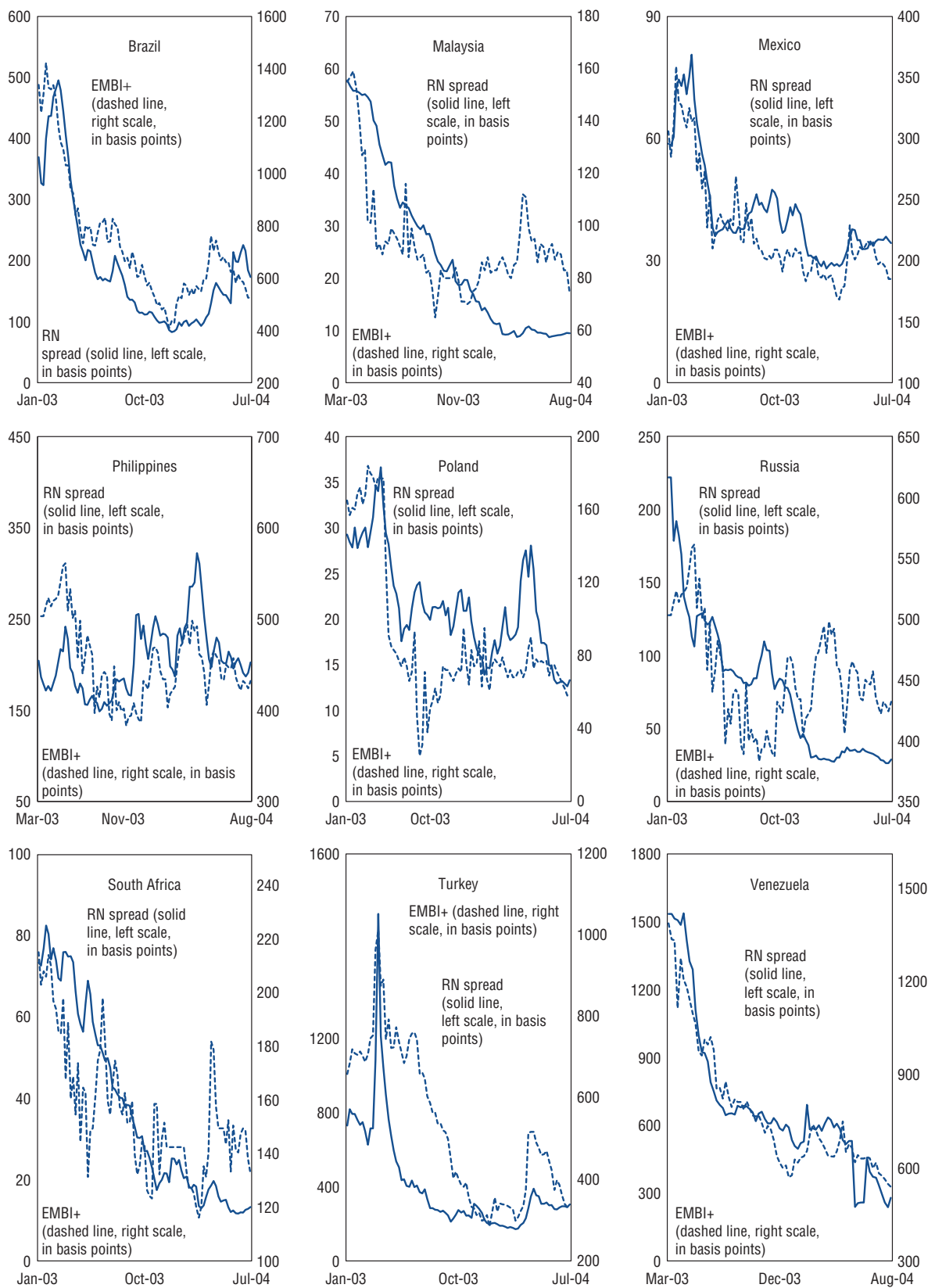
The similar estimated equation used to map risk-neutral credit spreads into actual EMBI+ spreads for Mexico is

$$\ln(EMBI_t) = 4.78 + 0.15 * \ln(RNS_t). \quad (24.8)$$

As a numerical example, suppose that application of equation (24.6) results in risk-neutral spreads on foreign currency debt for Mexico of 200 basis points. Inserting this value into equations (24.7) and (24.8) results in a credit default swap spread of 88 basis points and an EMBI+ spread of 263 basis points.

The relationship between sovereign risk-neutral default probability and estimated actual default probability can also be determined. This procedure is necessary because risk-neutral default probabilities overstate the actual probability of default. To implement this comparison, some estimate of actual default probability is needed. In the application of the CCA approach to corporate credit risk, the standard adjustment mechanism is to map firm risk-neutral default

²⁵ The countries in the sample include Brazil, Bulgaria, Colombia, Malaysia, Mexico, Philippines, Poland, Russia, South Africa, Turkey, and Venezuela.



Sources: Bloomberg; and Moody's MfRisk.

Note: EMBI+ = Emerging Market Bond Index; RN = risk-neutral.

Figure 24.5 Model Credit Spread and JP Morgan EMBI+ Country Index

TABLE 24.2

Regression Output: Risk-Neutral Spreads and CDS Spreads

Country	Independent Variable				
	Constant	Coefficient	Standard Error	t	Probability
ln(RNS)		0.52	0.02	24.53	0.00
Brazil	3.43		0.12	28.09	0.00
Colombia	2.54		0.11	22.34	0.00
Korea	2.59		0.07	38.50	0.00
Malaysia	1.54		0.08	18.57	0.00
Mexico	1.72		0.09	18.42	0.00
Philippines	2.53		0.12	20.50	0.00
Poland	1.20		0.08	14.55	0.00
Russia	2.72		0.10	26.94	0.00
South Africa	2.09		0.09	23.63	0.00
Turkey	2.98		0.13	23.04	0.00
Venezuela	2.94		0.15	20.14	0.00

Source: Authors.

Note: adjusted $R^2 = 0.88$; $F = 640.78$; log likelihood = -645.02; Prob(F) = 0.00. $R^2 = 0.88$; RNS = risk-neutral credit spread.

TABLE 24.3

Regression Output: Risk-Neutral Spreads and EMBI + Spreads

Country	Independent Variable				
	Constant	Coefficient	Standard Error	t	Probability
ln(RNS)		0.15	0.02	6.30	0.00
Brazil	5.69		0.11	50.86	0.00
Colombia	4.68		0.08	60.98	0.00
Korea	5.41		0.11	48.94	0.00
Malaysia	4.04		0.07	60.30	0.00
Mexico	4.78		0.08	58.74	0.00
Philippines	5.29		0.12	42.88	0.00
Poland	3.79		0.08	47.83	0.00
Russia	5.05		0.09	54.28	0.00
South Africa	4.52		0.07	61.41	0.00
Turkey	5.26		0.12	43.85	0.00
Venezuela	5.61		0.14	39.41	0.00

Source: Authors.

Note: adjusted $R^2 = 0.96$; EMBI = Emerging Market Bond Index; $F = 1,548.00$; log likelihood = 340.48; Prob(F) = 0.00; $R^2 = 0.96$; RNS = risk-neutral credit spread.

probabilities against a database of actual corporate defaults.²⁶ However, a sufficiently large data set of sovereign defaults is not available, meaning that some other approach is necessary. A second best approach is to use estimates of actual default probability, or market implied default probabilities (MIDP), which can be obtained from credit default swap spreads assuming a specific loss given default and time horizon (a recovery rate of 30 percent was used in this analysis).²⁷

²⁶ For example, MKMV utilizes 30 years of historical data over 6,000 public and 70,000 private company default events to derive the firm-specific probability of default, which is referred to as the Expected Default Frequency.TM

²⁷ Market implied default probabilities (MIDP) can be obtained from CDS spreads through the following equation:

$$MIDP = \frac{1 - \exp(-spread * t)}{1 - R},$$

where *spread* is the net one-year credit default swap spread, *t* is the time horizon (equal to 1 in this case), and *R* = 30 percent is the recovery rate.

Using this approach, a fixed-effects panel regression is applied to a cross-country sample of 935 observations from January 2003 to August 2004 to examine the relationship between risk-neutral default probabilities and market implied default probabilities.²⁸ The fixed-effects panel regression displays high explanatory power, with an *R*-squared of 93 percent. Results are reported in Table 24.4.

However, there is a problem with the panel regression results. As can be seen through close examination of Table 24.4, the regression equations for Korea, Malaysia, Mexico, Poland, and South Africa result in market implied default probabilities that are higher than risk-neutral default probability, which is a contradiction. This problematic result is likely

If the one-year CDS spread is 180 basis points, the implied default probability is 2.5 percent. See the Appendix.

²⁸ The countries in the sample include Brazil, Colombia, Korea, Malaysia, Mexico, Philippines, Poland, Russia, South Africa, Turkey, and Venezuela.

TABLE 24.4

Regression Output: Default Probability

Country	Independent Variable				Probability
	Constant	Coefficient	Standard Error	t	
ln(RNDP)		0.23	0.03	7.38	0.00
Brazil	0.61		0.08	7.94	0.00
Colombia	0.24		0.07	3.37	0.00
Korea	-0.77		0.04	-21.63	0.00
Malaysia	-1.44		0.03	-49.84	0.00
Mexico	-0.90		0.04	-25.61	0.00
Philippines	0.50		0.09	5.52	0.00
Poland	-1.65		0.04	-38.67	0.00
Russia	0.16		0.04	3.82	0.00
South Africa	-0.79		0.05	-17.13	0.00
Turkey	0.64		0.08	8.39	0.00
Venezuela	1.08		0.08	12.76	0.00

Source: Authors.

Note: adjusted $R^2=0.93$; $F=770.00$; log likelihood=-157.18; Prob(F)=0.00. $R^2=0.93$; RNDP=risk-neutral default probability.

due to two factors: (1) the assumption of a constant loss given default for all countries regardless of credit risk; and (2) lack of a sufficiently long time series for credit default swaps. In practice, loss given default may change as probability of default and credit default swap spreads change. More sophisticated methods of estimating MIDP from credit default swap data are therefore needed, including methods that allow the recovery rate to vary with probability of default. These advanced methods are beyond the scope of this study and are left for future research.

An alternative approach is to pool the individual country data into one regression to increase the number of observations, while maintaining the assumption of a constant loss given default.²⁹ The estimated equation using data from the pooled countries in the sample is

$$\ln(\text{MIDP}_i) = -1.24 + 1.01 * \ln(\text{RNDP}_i), \quad (24.9) \\ (-15.54) \quad (23.23) \quad R^2 = 0.735$$

where the numbers in parentheses represent the relevant t -statistic. Although the explanatory power of this pooled regression falls slightly relative to the panel regression reported in Table 24.4 (R^2 declines from 0.93 to 0.74), the level of explanatory power remains high and the relationship is highly significant. Furthermore, equation (24.9) produces a market implied probability of default that is lower than the risk-neutral probability of default. For example, suppose that application of equation (24.4) results in a risk-neutral probability of default equal to 8 percent. Inserting this value into equation (24.9) results in a market implied default probability of 2.3 percent. In other words, actual probability of default is approximately one-third of risk-neutral probability of default.

The sensitivity of sovereign spreads in response to a change in the distance to distress is a nonlinear relationship. Figure 24.6 plots observed market spreads (one-year CDS spreads)

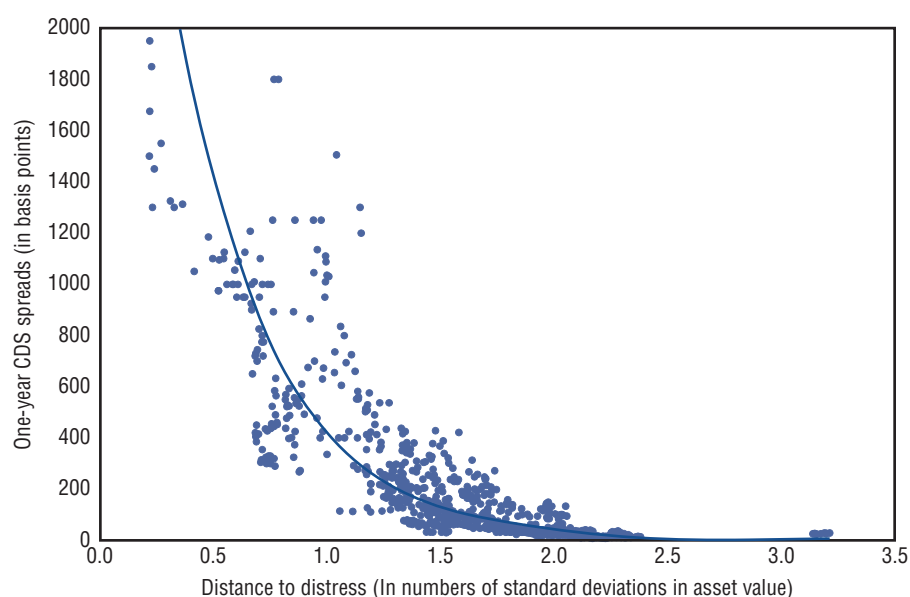
versus model distance to distress for all 12 countries in the sample (889 data pairs from the period mid-2002 to mid-2004). The figure shows the relationship aggregated across countries. The solid line is a best-fit regression, with an R -squared of 0.80. It shows the spread going up exponentially as the distance to distress declines. The sensitivity of changes in spreads, for a given change in distance to distress, is much lower for countries with a high distance to distress. As the distance to distress declines from 1.5 to 1.4, the spread increases on average by 35 basis points. However, if the distance to distress drops from 0.5 to 0.4, the spread increases on average by 375 basis points.

The robustness checks in this section suggest that the distance to distress, risk-neutral credit spread, and risk-neutral probability of default are useful for evaluating sovereign vulnerabilities. The evidence indicates that the book value of foreign currency liabilities along with market information from domestic currency liabilities and the exchange rate contain important information about changes in the value of foreign currency liabilities and credit risk premium. The nonlinearities and inclusion of volatility in the option pricing relationship used in this analysis contribute to the high degree of explanatory power and correlation with actual data. Finally, the estimated relationships in equations (24.7)–(24.9) allow for straightforward transformation of model outputs into estimates of observable market data.

5. SCENARIO AND SIMULATION ANALYSIS: HYPOTHETICAL SOVEREIGN

With robustness verified, the structural models calibrated using the contingent claims framework and unique to each economy can be used with scenario and simulation analysis to evaluate shocks and policies. The goal is to estimate the potential effects of changes in economic conditions and

²⁹ Crouhy, Galai, and Mark (2000) also assume constant loss given default.



Source: Author.
Note: CDS = credit default swap.

Figure 24.6 Market Observed Spreads and Model Distance to Distress

impact of government policies on sovereign credit risk and sensitivity indicators. To begin with, a baseline balance sheet for a hypothetical sovereign is calibrated (Table 24.5), and the resulting baseline risk indicators and sensitivity measures are reported (Table 24.6). Scenario analysis is then conducted using two capital flow examples, and the resulting point estimates for the credit risk indicators and sensitivity measures are compared with the baseline set of indicators. Next, the scenario analysis is extended using Monte Carlo simulation, which provides one way of generating a large number of market outcomes and produces probability distributions for each risk indicator. Unlike the scenario analysis, which is intended to investigate the effects of a specific market outcome, the Monte Carlo simulation draws randomly from sample interest rate and exchange rate distributions to compute probability distributions and confidence intervals for a set of market outcomes. This process allows for the “stress testing” of the risk levels of the sovereign risk indicators to derive what are commonly known as VaR measures.

A. The baseline

The starting point is the baseline balance sheet as displayed in Table 24.5:

- **Calibrated values.** The distress barrier is assumed to be US\$100 billion, comprising short-term foreign currency debt plus interest of US\$40 billion and one-half of long-term debt of US\$60 billion. The value and volatility of domestic currency liabilities in dollar terms are US\$82 billion and 0.76 (76%), respectively. Using equations (24.1) and (24.2), the implied

TABLE 24.5

Example Contingent Claims Sovereign Balance Sheet Risk Indicators

Contingent Claims Sovereign Balance Sheet	(US\$ billion, unless indicated)
Value of sovereign assets (implied)	175
Foreign reserves (observed value)	40
Sovereign asset less reserves (implied)	135
Value of risky foreign currency debt	94.5
Distress barrier ¹	100
PV of distress barrier ¹	96
PV of expected losses (= implicit put option)	1.5
Value of local currency liabilities ¹	80.5
Volatility of assets (implied)	38%
Credit Risk Indicators	
Distance to distress ²	1.4
Risk-neutral default probability (RNDP)	8%
Risk-neutral spread (RNS) ³	115
Sensitivity Measures⁴	
Change in distance to distress / 1% change in assets ²	−0.03
Change in distance to distress / 1% change in asset vol. ²	−0.05
Change in RNDP / 1% change in assets	0.41
Change in RNS / 1% change in assets ³	7
Change in RNS / 1% change in asset vol. ³	16
Change in PV expected loss / 1% change in assets	0.07
Change in PV expected loss / 1% change in asset vol.	0.15

Source: Authors.

1. Model inputs. Remainder are model outputs.

2. In standard deviation of sovereign asset value.

3. Spread in basis points.

4. Based on a 1 percent change in sovereign asset value (e.g., from 175 to 176.75) and sovereign asset volatility (e.g., from 38 percent to 39 percent).

TABLE 24.6

Alternative Scenarios and Contingent Claims Sovereign Balance Sheet Risk Indicators

	Scenario 1	Baseline	Scenario 2
Contingent Claims Sovereign Balance Sheet (US\$ billion, unless indicated)			
Value of sovereign assets	155	175	195
Foreign reserves ¹	35	40	45
Sovereign asset less reserves	120	135	150
Value of risky foreign currency debt	93	94.5	96
Distress barrier ¹	100	100	100
PV of distress barrier ¹	96	96	96
PV of expected losses (= implicit put option)	3	1.5	0.5
Value of local currency liabilities ¹	62	80.5	99
Volatility of asset	43%	38%	37%
Credit Risk Indicators			
Distance to distress ²	1.0	1.4	2.1
Risk-neutral default probability (RNDP)	17%	8%	4%
Risk-neutral spread (RNS) ³	325	115	60
Sensitivity Measures⁴			
Change in distance to distress / 1% decrease in assets ²	-0.02	-0.03	-0.03
Change in distance to distress / 1% increase in asset vol. ²	-0.03	-0.05	-0.06
Change in RNDP / 1% decrease in assets	0.63	0.41	0.23
Change in RNS / 1% decrease in assets ³	16	7	3
Change in RNS / 1% increase in asset vol. ³	28	16	9
Change in PV expected loss / 1% decrease in assets	0.15	0.07	0.03
Change in PV expected loss / 1% increase in asset vol.	0.26	0.15	0.08

Source: Authors.

1. Model inputs for baseline.

2. In standard deviation of sovereign asset value.

3. Spread in basis points.

4. Based on a 1 percent change in sovereign asset value (e.g., from 175 to 176.75) and sovereign asset volatility (e.g., from 38 percent to 39 percent).

value of sovereign assets is US\$175 billion, and the implied volatility of sovereign assets is 0.38 (38%). Foreign currency reserves are assumed to make up US\$40 billion of implied sovereign assets.

- **Credit risk indicators.** From equation (24.3), the resulting distance to distress is 1.4 standard deviations. The distance to distress results in a risk-neutral probability of default of 8 percent in equation (24.4). Equations (24.5) and (24.6) yield the value of risky foreign currency debt as US\$95 billion and risk-neutral credit spread of 115 basis points.³⁰ The market value of risky foreign currency debt implies a present value of expected loss of US\$1 billion. This value is derived from the difference between the discounted present value of the distress barrier (using a risk-free rate of 4 percent yields a present value distress barrier of US\$96 billion) and the implied market value of foreign currency debt.
- **Sensitivity measures.** Sensitivity measures are calculated from a 1 percent change in sovereign asset value and volatility. For example, when the value of sovereign assets decreases by 1 percent, Table 24.5 shows that the distance to distress falls by 0.03 stan-

dard deviations (i.e., from 1.4 to 1.37 standard deviations), risk-neutral default probability increases by 0.41 percent, risk-neutral credit spreads increase by 7 basis points, and the expected loss on foreign currency debt increases by US\$70 million. Sensitivity measures also are reported for a 1 percent change in sovereign asset volatility.

As previously discussed, the sensitivity measures capture the nonlinearity present within the contingent claims relationships. An example of the presence of nonlinearities is shown in Figure 24.6, which plots how risk-neutral spreads change in response to changes in sovereign asset value. As the value of sovereign assets approaches the distress barrier, a 1 percent reduction in sovereign assets results in a 25 basis point increase in credit spreads compared with the baseline calibration, where the same 1 percent reduction in sovereign asset value leads only to a 7 basis point increase in credit spreads. The nonlinearity implies that the drop in sovereign assets has a proportionately larger impact on credit risk when sovereign assets are close to the distress barrier. The converse is true when the implied value of sovereign assets is well above the distress barrier.

B. Scenario analysis

Two scenarios are examined and compared with the baseline. Scenario 1 represents the potential negative effects associated

³⁰ At this point, the risk-neutral default probability and spread then would be mapped into market implied probability of default and market spreads using equations (24.7)–(24.9), but this step is ignored here because this exercise is only for a hypothetical sovereign balance sheet.

with capital outflows, and Scenario 2 illustrates the positive effects from capital inflows. First, suppose that economic conditions deteriorate so that capital outflows occur. Capital outflows normally are associated with some combination of an exchange rate depreciation, a drop in domestic debt prices (possibly associated with a rise in domestic interest rates), and an increase in volatility of both debt prices and the exchange rate. The impact of capital outflows on the sovereign balance sheet risk indicators depends in part on the response of policymakers. The assumption in this example is that policymakers accommodate some but not all of the shock. This would include some loss of international reserves, tighter interest rate policy, and increase in the net fiscal asset. Under this scenario, sovereign asset value is assumed to fall by US\$25 billion to US\$155 billion, with international reserves falling from US\$40 billion to US\$35 billion. Sovereign asset volatility increases from 38 percent to 43 percent. The left column in Table 24.6 (Scenario 1) displays the new contingent claims sovereign balance sheet, balance sheet risk indicators, and sensitivities after capital outflows.

In sum, capital outflows worsen the credit risk indicators, and risk exposure of the sovereign has increased relative to the baseline. Distance to distress falls from 1.4 to 1.0 standard deviations, and risk-neutral probability of default increases from 8 to 17 percent. Risk-neutral spreads on foreign currency debt rise to reflect the increased risk of nonrepayment as the expected loss has increased from US\$1.5 billion to US\$3 billion. In addition to a worsening of the credit risk indicators, the sensitivity measures have increased because implied sovereign asset value is fewer standard deviations from the distress barrier. For example, a 1 percent decline in sovereign asset value under the baseline scenario increased risk-neutral default probability by 0.41 (from 8 to 8.41 percent) and risk-neutral spreads by 7 basis points, while in this capital outflow scenario these values are now 0.63 and 16, respectively. The higher sensitivities reflect the higher degree of nonlinearity within the option pricing formula as sovereign assets move closer to the distress barrier. This is indicative of observed nonlinear value changes in actual credit events.

A similar procedure can be applied to illustrate the opposite effects of capital inflows. Sustained capital inflows typically result in some exchange rate appreciation, improvement in domestic debt prices, and lower financial market volatility. Capital inflows may also provide space for an increase in international reserves, which may necessitate sterilization operations. Based on this scenario, the value of sovereign assets is assumed to rise US\$195 billion while its volatility drops to 37 percent. Also, international reserves are assumed to rise by US\$5 billion, and the increase in the dollar value of domestic currency liabilities is a reflection of both sterilization and exchange rate appreciation. The right column of Table 24.6 (Scenario 2) displays the contingent claims sovereign balance sheet, credit risk indicators, and sensitivities after capital inflows. The increase in sovereign asset value and reduction in volatility yield the expected decrease in

credit risk and sensitivity relative to the baseline. Distance to distress rises above two standard deviations, and risk-neutral probability of default decreases by half to 4 percent. Risk-neutral spreads on foreign currency debt decline as the value of risky foreign currency debt approaches its default-free value. Each of the sensitivity measures decreases relative to the baseline from the improved sovereign asset value and volatility with respect to the distress barrier.

C. Monte Carlo simulations

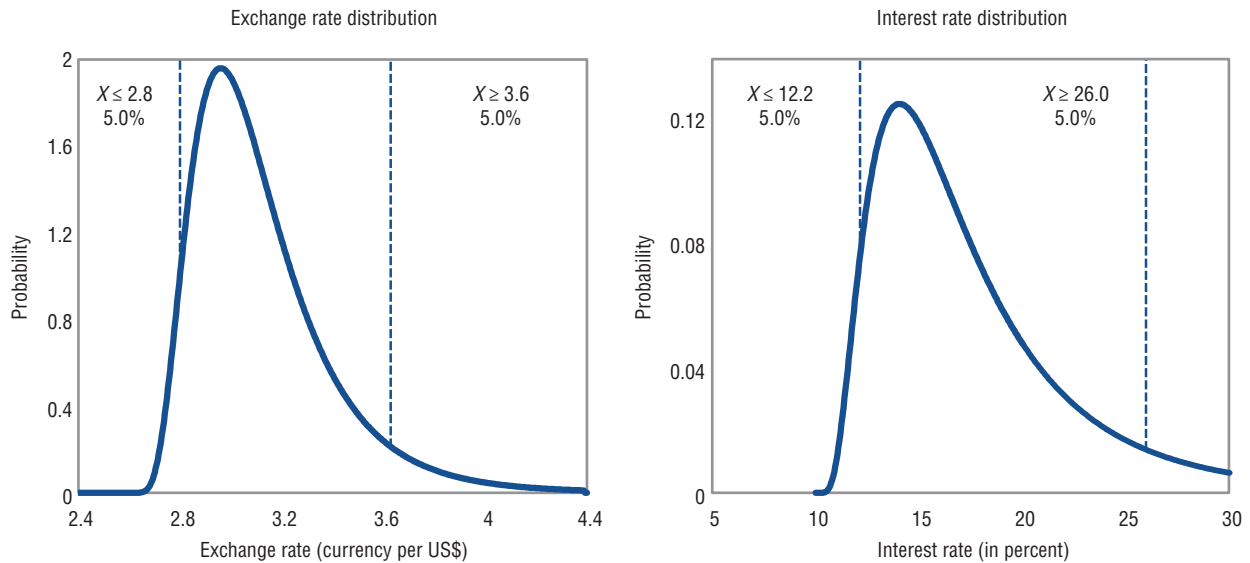
The preceding scenario analysis yields three related point estimates for the credit risk indicators, one from the baseline calibration and two from a negative and positive shock. Although such scenario analysis may be useful in examining a specific event, it reveals only a very small view of the possible set of market disturbances. Scenario analysis to re-create a specific event is always subject to the criticism that market stress scenarios, in fact, rarely repeat themselves. Monte Carlo simulation methods can be used to systematically deal with multiple scenarios, yielding probability distributions for risk indicators and VaR measures.³¹ The Monte Carlo procedure implemented in this section takes random draws from hypothetical forward distributions for domestic interest rates and the exchange rate. Following the process outlined in Box 24.2, Monte Carlo simulations are conducted on the baseline sovereign balance sheet presented in the previous section.

Probability distributions for distance to distress, risk-neutral default probability, risk-neutral spreads, and the value of sovereign assets resulting from the simulation are reported in Figure 24.7. Although the mean distance to distress remains 1.4 standard deviations, the same value as reported in Table 24.4, the distribution reveals a confidence interval for distance to distress based on the sample exchange rate and interest rate distributions. For example, from Figure 24.7, the lower 5 percent probability for distance to distress is 0.9 standard deviations, the upper 5 percent probability for risk-neutral default probability is 18 percent, and the upper 5 percent probability for risk-neutral spreads is 387 basis points. In other words, given the assumed exchange rate and interest rate distributions and correlation, distance to distress remains above 0.9 standard deviations, risk-neutral default probability remains below 18 percent, and risk-neutral spreads remain below 387 basis points 95 percent of the time. Finally, the 5 percent lower bound on sovereign assets is US\$160 billion, making the implied sovereign asset VaR equal to US\$15 billion.

³¹ Although Monte Carlo simulations are able to handle many thousands of possible events, they produce a random set of outcomes based on the market characteristics assumed, which may or may not predict potential shocks. The simulation process will produce only as many extreme events as dictated by the distribution assumption of the market variables. To be comprehensive, simulation procedures should be combined with various scenario assumptions to produce a set of stress outcomes.

Box 24.2. Implementing the Monte Carlo Simulation

The Monte Carlo simulation procedure applied in this section takes random draws from hypothetical forward distributions for both domestic interest rates and exchange rates and calculates the effects of these variables on balance sheet values and risk indicators. The one-year forward exchange rate is assumed to be 3 units of the domestic currency to 1 U.S. dollar, and the one-year forward interest rate is assumed to be 17 percent. Log-normal distributions for each were constructed based on recently observed market patterns in several emerging market economies, as shown in the figure below. The correlation between exchange rates and interest rates was set at 0.6, meaning that the Monte Carlo simulation conducts sample draws such that exchange rate depreciations are generally associated with higher interest rates, and vice versa.



The Monte Carlo simulation procedure then selects random draws from these hypothetical distributions. The sample forward exchange rate is applied to the contingent claims sovereign balance sheet in the translation of domestic currency assets and liabilities into their respective U.S. dollar values. In contrast to exchange rate variations, simulating the effect of interest rate changes requires additional assumptions. Broad money is assumed to comprise half of domestic currency liabilities, with the remainder in interest rate linked domestic debt. The interest rate draw is applied to the existing domestic currency debt for a period of three years and then is assumed to return to 17 percent. If the realization of interest rates in the random draw is above the assumed 17 percent forward interest rate, the discounted marginal increase in interest costs is subtracted from the value of sovereign assets to reflect higher debt service costs. Alternatively, if the interest rate draw is below the assumed forward interest rate, then this discounted decrease in debt service costs is added to the value of sovereign assets.

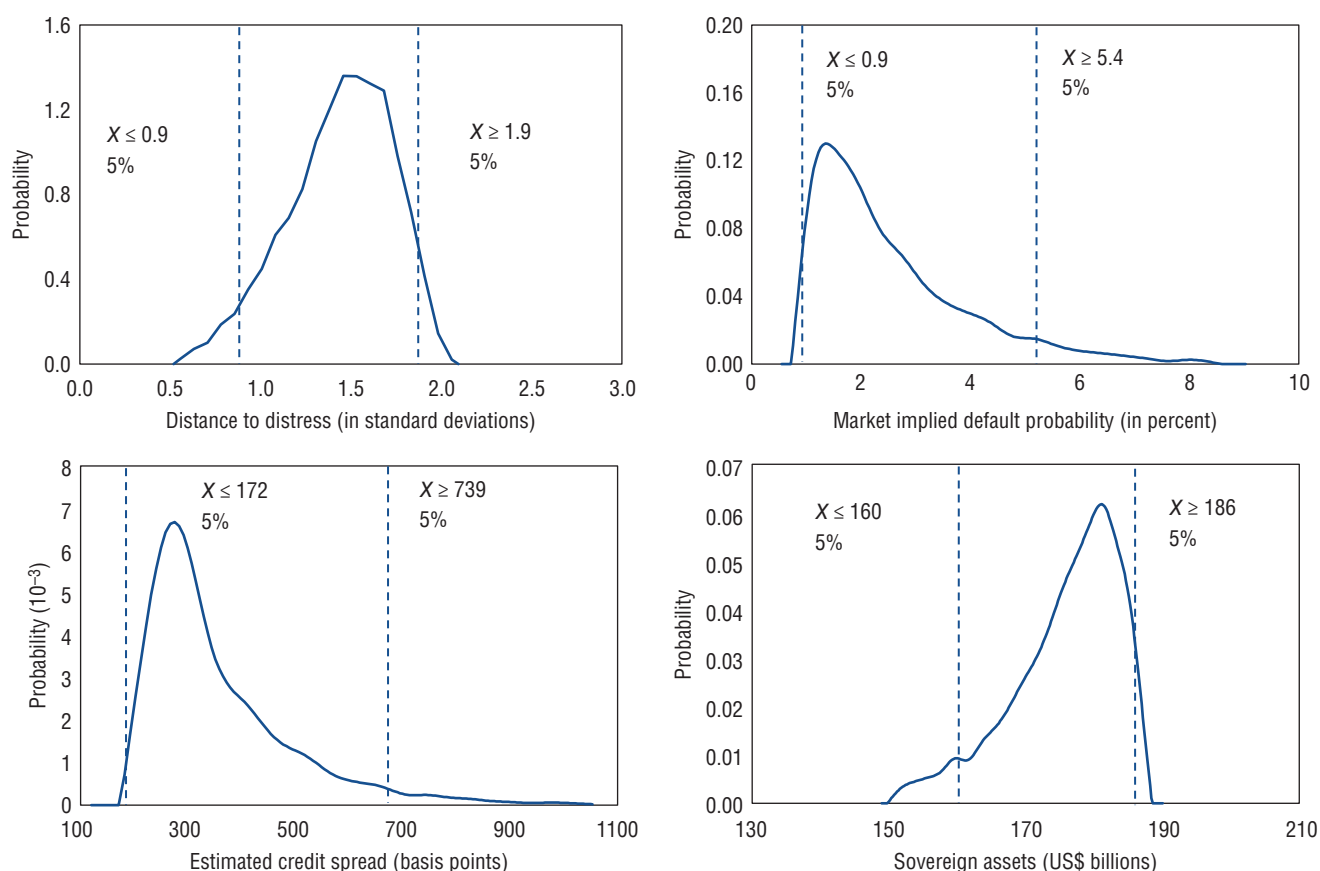
The resulting sovereign balance sheet values from each random draw are then used to compute the new set of risk indicators. In contrast to the point estimates for the balance sheet risk indicators that result from scenario analysis, the process of conducting random samples from distributions of exchange rates and interest rates results in probability distributions for the relevant risk indicators.

VaR measures often are used to evaluate both market and credit risk in the financial sector.³² In the financial sector, VaR typically defines a level of capital that, for a high degree of confidence, is an upper bound on the amount of gains or losses to a portfolio from market or credit risk. On the sovereign balance sheet, VaR by corollary, could be defined as the upper bound on the amount of gains or losses to implied

sovereign asset value from market risk.³³ Just as a bank or asset manager is required to hold capital in reserve to protect

³² See Jorion (2000). VaR models estimate the exposure of a portfolio, or the equivalent set of positions, to market risk. The measure captures the expected maximum loss and is usually expressed within a confidence interval.

³³ Two other sovereign VaR measures can be calculated. The first, sovereign capital at risk, is an extension of sovereign VaR for the central bank. The probability distribution of the residual value of "capital" or junior claim of the monetary authorities is calculated and a confidence level attached to the risk that the monetary authorities cannot meet its commitments. Blejer and Schumacher (1999) use a similar construction. The second, sovereign credit at risk, is the upper bound on gains or losses due to credit risk, which in this case is the value of the guarantee to the banking system. See Gapen and others (2004) for an example of how this could be modeled.



Source: Author.

Figure 24.7 Monte Carlo Simulations: Hypothetical Sovereign

against market or credit loss, governments often identify a need to acquire sufficient levels of foreign currency reserves or insurance arrangements to protect against adverse market developments. The sovereign VaR measure can be used as a tool to gauge whether the level of reserves is sufficient to protect against the risk of “sudden stops,” or to maintain debt sustainability against adverse economic shocks.

D. Evaluating policy design

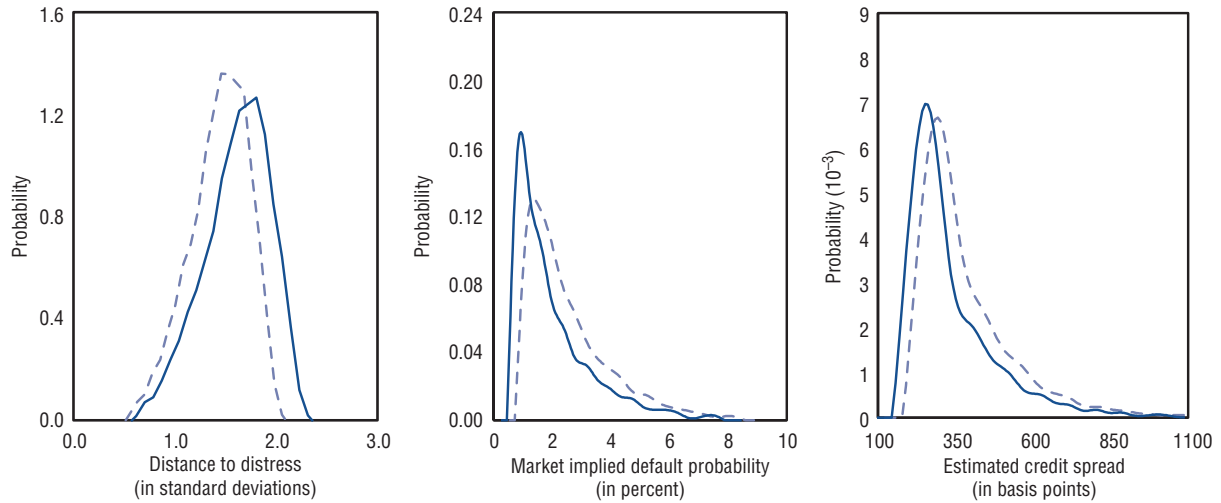
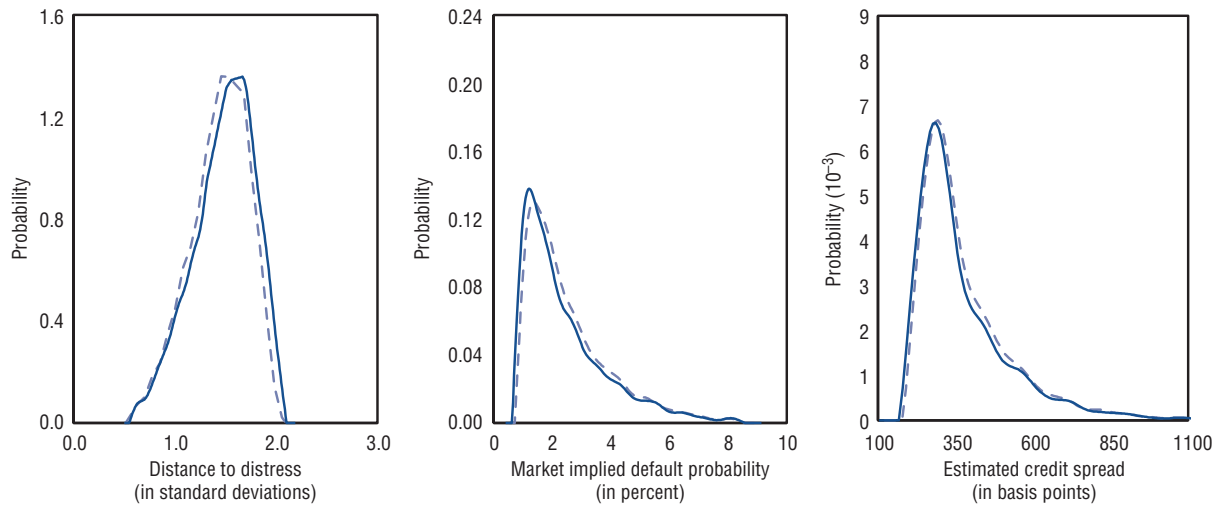
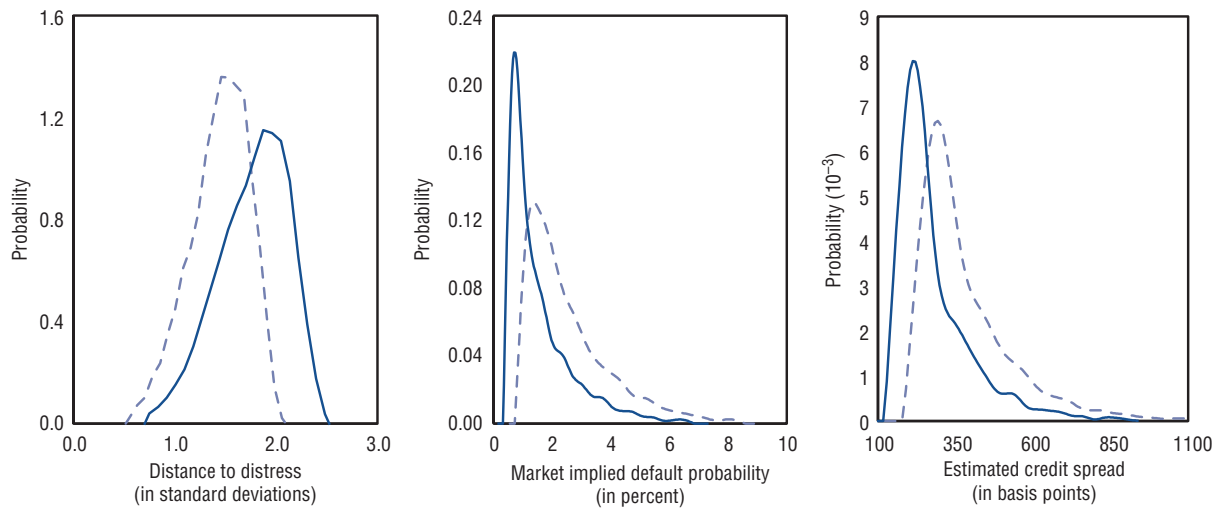
Using the Monte Carlo baseline simulation as a starting point, potential policy choices can be evaluated. For example, changes in the level of reserves, alternative debt structures, or the use of risk mitigation instruments like insurance contracts can be tested. The new policy option modifies the sovereign balance sheet and simulations using draws from the same interest rate, and exchange rate distributions will reveal new distributions of risk indicators that can be evaluated against the original baseline configuration.³⁴ The example of debt man-

agement with alternative debt structures is examined first, followed by a strategy for reserve accumulation. Finally, a combination of debt and reserve management is considered.

- **Debt management.** Panel A in Figure 24.8 illustrates an example of liability management, whereby US\$10 billion in foreign currency debt is replaced with an equal amount of interest rate linked domestic currency debt to examine the impact of reduced exchange rate exposure. As a result, the distress barrier falls to US\$90 billion while domestic currency liabilities increase by US\$10 billion. The new Monte Carlo simulations on this adjusted balance sheet yield improvements in the risk indicators. The mean values and confidence intervals for distance to distress, risk-neutral default probability, and mean risk-neutral spreads all improve.
- **Reserve management.** Panel B of Figure 24.8 illustrates the example of reserve accumulation financed with an equal amount of domestic currency debt such

³⁴ Simulating the adjusted sovereign balance sheet under the same exchange rate and interest rate distributions and correlation is subject to the Lucas critique—namely, that these distributions and correlations are derived from market expectations, which are likely to change with the shift in policy. In this debt management example, the expected fu-

ture exchange rate could be more appreciated with lower volatility given the lower levels of foreign currency debt. Consequently, the simulations conducted in this study should be viewed only as illustrative of potential impacts from policy changes. See Best (2000) for additional discussion on the limits of stress testing.

Panel A. Debt Management Risk Indicators (solid) versus Baseline Risk Indicators (dashed).**Panel B. Reserve Management Risk Indicators (solid) versus Baseline Risk Indicators (dashed).****Panel C. Debt and Reserve Management Risk Indicators (solid) Versus Baseline Risk Indicators (dashed).**

Source: Author.

Figure 24.8 Evaluation of Policy Options

that both the level of sovereign assets and interest rate linked domestic currency liabilities increase by US\$10 billion. This scenario could be viewed as a proactive strategy to accumulate reserves or reflect capital inflows, and therefore, the increase in domestic currency debt is the result of sterilization. The operation yields improvements in the risk indicators—as seen in the mean values and lower 5 percent probability distributions—although the margin of improvement is less than that found in the example on debt management.

- **Asset and liability management.** The preceding reserve and debt management strategies also can be implemented simultaneously, as shown in Panel C of Figure 24.8. The distress barrier declines to US\$90 billion, the amount of domestic currency debt increases by US\$20 billion, and the level of foreign currency reserves increases by US\$10 billion. The effect of simultaneously enacting both strategies yields improvements in the risk indicators by an amount equal to more than the sum of the two strategies individually, reflecting model nonlinearity. Combining the two strategies is advantageous because the debt management operation reduces the distress barrier while the reserve accumulation strategy leaves the mean value of sovereign assets nearly unchanged relative to the baseline.

Deciding on the efficacy of any of the preceding strategies involves a systematic weighting of the trade-offs inherent in each case. There are clear elements in each of the three alternative strategies that are beneficial from a policy perspective (the reduction in exchange rate exposure and increase in reserves), which need to be balanced against the clear negatives from a balance sheet perspective (the increase in domestic currency interest bearing obligations). The contingent claims balance sheet risk indicators therefore can be useful in guiding policy design given their ability to compare different policy options using quantifiable risk indicators.

6. NEXT STEPS

A. A robust framework for reserve management

The application of CCA and sovereign VaR to reserve management is a stark departure from accounting indicators commonly used for reserve management. One widely used indicator of reserve adequacy is the ratio of foreign currency reserves to total public and private short-term foreign currency debt. Both public and private sector debt are included because reserves of the public sector must facilitate transactions related to economy-wide financing requirements. However, the simple accounting ratio of reserves to total short-term foreign currency debt is deficient when it comes to risk analysis because it does not take uncertainty of balance sheet risks into account. Applying a broad-based rule for an appropriate ratio of reserve coverage uniformly across countries implicitly assumes that all sovereign balance sheet risks are

similar and neglects cross-country differences in sovereign balance sheet risk.³⁵

In contrast, an appropriate target for reserve adequacy could be based on a level of reserves that minimizes instances of distress using the contingent claims risk indicators. For example, an adequate level of reserves could be defined as the level of reserves that keeps distance to distress above a desired standard deviation 95 percent of the time based on the likely exchange rate process. Adequate reserve coverage could also target a basket of credit risk indicators by setting reserve levels to maintain the combined set of indicators at target levels for a specific confidence interval. In sum, reserve management using this framework examines the impact of the level and volatility of reserves as a component of the wider sovereign asset value and volatility with a link to the balance sheet risk indicators. The application of contingent claims in analyzing sovereign credit risk can be adapted to include many different aspects of reserve management, including the currency composition of reserves, or various other risk mitigation techniques.³⁶

B. A robust framework for debt sustainability

In addition to providing a framework for reserve management, the use of contingent claims to analyze sovereign risk is well suited for robust debt sustainability analysis. Traditional debt sustainability analysis has focused on ratios of current and forecasted debt-to-GDP as the primary criterion for deciding whether the public sector debt remains on a sustainable path, usually without explicitly incorporating uncertainty in a systematic, coherent framework. The following elements indicate why the approach in this study could provide the basis for a more robust framework for debt sustainability analysis:

- The contingent claims sovereign balance sheet translates balance sheet risks into quantifiable risk indicators. In this framework, debt sustainability could be defined as the debt structure that keeps key credit risk indicators below (or above) certain threshold levels for a given confidence level. In contrast, the debt-to-GDP ratio identifies an element of sovereign risk but is not part of a structural framework that measurably relates debt payments with the capacity to pay. For example, the contingent claims structural framework is able to assess the impact of changes in the level of reserves on sovereign risk, whereas the debt-to-GDP ratio remains invariant to such changes.
- The quantitative sovereign credit risk indicators described in this chapter incorporate uncertainty and

³⁵ IMF (2000) examines three ratios: reserves to imports, reserves to monetary aggregates, and reserves to public and private short-term foreign currency debt by residual maturity. The report concludes that reserves to short-term foreign currency debt is a superior measure and recommends that a ratio of 1 be a lower bound for adequate reserve coverage.

³⁶ Caballero and Panageas (2005) examine various instruments and risk mitigation strategies that policymakers could implement in addition to traditional reserve accumulation in a model of sudden stops in capital flows.

volatility. Higher market uncertainty often is translated into higher interest rate and exchange rate volatility, widening the forward distributions on both variables and increasing the volatility of sovereign assets. Distance to distress will fall, probability of default will rise along with spreads on foreign currency debt, and the expected loss on risky foreign currency debt will increase. However, the debt-to-GDP ratio does not change with an increase in sovereign asset volatility and would therefore miss an important component of risk analysis.

- The contingent claims sovereign balance sheet includes an assessment of maturity or rollover risk through construction of the distress barrier.³⁷ The debt-to-GDP ratio does not change if a decrease in short-term foreign currency debt is matched by an equal book value increase in long-term foreign currency debt. The use of the contingent claims sovereign balance sheet reflects this change by signaling a decrease in sovereign risk due to the more favorable debt profile.
- Finally, the CCA approach incorporates nonlinear value changes. The use of nonlinear modeling in a structural framework captures complex relationships and more accurately conveys the nonlinear nature of credit events. During periods of stress, small changes in interest rates, exchange rates, and/or volatility can result in large changes in sovereign risk on the margin. An accounting ratio like debt-to-GDP is not capable of this level of complexity, nor is it released with enough frequency to enable its use during periods of stress where vulnerabilities may build or subside rapidly.

Using contingent claims to model sovereign credit risk therefore offers several advantages over the traditional debt-to-GDP analysis. Additional research in this direction could prove useful and would require extension of the framework to a medium-term setting while incorporating the outlook for relevant economic and policy variables.

7. CONCLUSION

This study develops a comprehensive new framework to measure and analyze sovereign risk by applying the CCA approach to the balance sheet of the combined government and monetary authorities. A marked-to-market balance sheet is constructed that provides a structural framework that

identifies balance sheet risks, incorporates uncertainty, and yields quantifiable risk indicators. The main outputs of this framework include sovereign credit risk indicators, sensitivity measures, and sovereign VaR. These sovereign risk indicators incorporate both forward-looking market prices and nonlinear changes in values and should consequently have greater predictive power in estimating sovereign credit risk than would traditional macroeconomic vulnerability indicators or accounting-based ratios.

Application to 12 emerging market economies shows the risk indicators to be robust and significant when compared with market observed credit spreads on foreign currency debt, even though the spreads were not used as inputs. This lends support for the approach and illustrates that the level and variation of forward exchange rates and other market variables contain valuable information for analyzing sovereign credit risk.

Using the CCA approach to analyze sovereign risk has several merits from a policy perspective. The ability of the CCA approach to provide a structural interpretation of the sovereign balance sheet, unique to each economy, is a valuable contribution in the area of policy design and risk management, translating policy choices and changing economic conditions directly into quantitative indicators of financial soundness. This chapter describes how the framework can be used with scenario and simulation analysis to evaluate shocks and the impact of corrective policies. Policymakers can observe how negative economic shocks worsen sovereign financial soundness through capital outflows, a depreciating exchange rate, or slower economic growth. Policymakers can use this tool to design and implement risk-mitigation strategies to reduce sovereign balance sheet risk. Equally important, the set of tools available to policymakers has direct links to the assets and liabilities in the sovereign balance sheet and can influence market expectations. In response to changing economic conditions, active policy decisions to alter the primary fiscal surplus, level of interest rates, structure of debt, or reserve intervention policy can mitigate or offset shocks to the government balance sheet. The ability of the contingent claims balance sheet risk to measurably assess the potential policy mix is an important element of strategic planning and offers policymakers the valuable opportunity to rank policy options.

The contingent claims framework can be adapted and extended in several important directions. It is well suited for a more robust analysis of debt sustainability as compared with the widely used debt-to-GDP ratio, which is a static, backward-looking indicator. The framework can also be used to estimate an appropriate target for reserve adequacy, where adequacy could be defined as the level of reserves that minimizes the probability of distress.

REFERENCES

Ariyoshi, Akira, Karl Habermeier, Bernard Laurens, Inci Ötker-Robe, Jorge Iván Canales-Kriljenko, and Andrei Kirilenko,

³⁷ This is true whether one uses the simplified distress barrier in this study (short-term foreign currency debt and interest plus one-half long-term foreign currency debt) or a more sophisticated approach (short-term foreign currency debt and interest plus the present discounted value of long-term foreign currency debt and interest). Both approaches would reflect an increase in sovereign risk if long-term foreign currency debt was traded for equal book value amounts of short-term foreign currency debt. The distress barrier under the second approach, however, would be more sensitive to near-term repayment humps, which would carry a higher weight in the distress barrier than a similar payment profile further out on the maturity scale.

- 2000, "Capital Controls: Country Experiences with Their Use and Liberalization," IMF Occasional Paper No. 190 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/ft/op/op190/pdf/appendix.pdf>
- Baxter, Martin, and Andrew Rennie, 1996, *Financial Calculus: An Introduction to Derivative Pricing* (Cambridge: Cambridge University Press).
- Best, Philip, 2000, "Stress Testing," in *The Professional's Handbook of Financial Risk Management*, edited by Marc Lore and Lev Borodovsky (Oxford: Reed Educational and Professional Publishing).
- Black, Fischer, and Myron Scholes, 1973, "The Pricing of Options and Corporate Liabilities," *Journal of Political Economy*, Vol. 81, No. 3, pp. 637–54.
- Blejer, Mario I., and Liliana Schumacher, 1999, "Central Bank Vulnerability and the Credibility of Its Commitments: A Value-at-Risk Approach," *Journal of Risk*, Vol. 2, No. 1, pp. 37–55.
- Buiter, Willem H., 1993, "Measurement of the Public Sector Deficit and Its Implications for Policy Evaluation and Design," in *How to Measure the Fiscal Deficit*, edited by Mario I. Blejer and Adrienne Cheasty (Washington: International Monetary Fund).
- Caballero, Ricardo J., and Stavros Panageas, 2005, "Insurance and Reserves Management in a Model of Sudden Stops," paper presented at the International Monetary Fund Institute seminar "Emerging Markets Macroeconomics: An Insurance Perspective," Washington, February 7–9.
- Chan-Lau, J., Arnaud Jobert, and Janet Q. Kong, 2004, "An Option-Based Approach to Bank Vulnerabilities in Emerging Markets," IMF Working Paper 04/03 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=17059>
- Chriss, Neil A., 1997, *Black-Scholes and Beyond* (New York: McGraw-Hill).
- Cossin, Didier, and Hugues Pirotte, 2001, *Advanced Credit Risk Analysis* (New York: Wiley).
- Crosbie, Peter J., and Jeffrey R. Bohn, 2003, "Modeling Default Risk," Moody's KMV, December 18. Available via the Internet: http://www.creditrisk.ru/publications/files_attached/modeling_default_risk.pdf
- Crouhy, Michel, Dan Galai, and Robert Mark, 2000, "A Comparative Analysis of Current Credit Risk Models," *Journal of Banking and Finance*, Vol. 24, Nos. 1–2, pp. 59–117.
- Draghi, Mario, Francesco Giavazzi, and Robert C. Merton, 2003, "Transparency, Risk and International Financial Fragility," NBER Working Paper No. 9806 (Cambridge: National Bureau of Economic Research).
- Eichengreen, Barry, Ricardo Hausmann, and Ugo Panizza, 2002, "Original Sin: The Pain, the Mystery, and the Road to Redemption," paper presented at the Inter-American Development Bank conference, "Currency and Maturity Matchmaking: Redeeming Debt from Original Sin," Washington, November 21–22.
- Gapen, Michael T., Dale F. Gray, Cheng Hoon Lim, and Yingbin Xiao, 2004, "The Contingent Claims Approach to Corporate Vulnerability Analysis: Estimating Default Risk and Economy-Wide Risk Transfer," IMF Working Paper 04/121 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=17465>
- , 2008, "Measuring and Analyzing Sovereign Risk with Contingent Claims," *IMF Staff Papers*, Vol. 55, No. 1, pp. 113–48.
- Available via the Internet: <http://www.imf.org/external/pubs/ft/staffp/2008/01/gapen.htm>
- Gray, Dale F., 2004, "Modeling Sovereign Default Risk and Country Risk Using Moody's-MfRisk Framework with Specific Country Applications," MfRisk Working Paper No. 1-04.
- Gray, Dale F., Robert C. Merton, and Zvi Bodie, 2002, "A New Framework for Analyzing and Managing Macrofinancial Risks," presentation at the NYU CV Starr/RED Conference, "Finance and the Macroeconomy," New York, October 11–12.
- , 2003, "A New Framework for Analyzing and Managing Macrofinancial Risks of an Economy," MfRisk Working Paper No. 1-03. Available via the Internet: <http://www.moody's-mfrisk.com>
- Gulde, Anne-Marie, David Hoelscher, Alain Ize, Alfredo Leone, David Marston, and Marina Moretti, 2003, "Dealing with Banking Crises in Dollarized Economies," in *Managing Financial Crises: Recent Experiences and Lessons for Latin America*, edited by Charles Collins and G. Russell Kincaid, IMF Occasional Paper No. 217 (Washington: International Monetary Fund).
- Hull, John C., 1993, *Options, Futures, and Other Derivative Securities* (Englewood Cliffs, NJ: Prentice Hall).
- International Monetary Fund, 1999, "Ukraine: Recent Economic Developments," IMF Country Report 99/42 (Washington, April).
- , 2000, "Debt- and Reserve-Related Indicators of External Vulnerability," IMF policy paper (Washington, March 23). Available via the Internet: <http://www.imf.org/external/np/pdr/debtres>
- , 2002, "Sovereign Debt Restructurings and the Domestic Economy Experience in Four Recent Cases," IMF policy paper (Washington, February 21). Available via the Internet: <http://www.imf.org/external/NP/pdr/sdrm/2002/022102.pdf>
- Jorion, Philippe, 2000, *Value at Risk: The New Benchmark for Managing Financial Risk* (New York: McGraw-Hill).
- McQuown, J. A., 1993, "A Comment on Market vs. Accounting-Based Measures of Default Risk," mimeograph, KMV Corporation.
- Merton, Robert C., 1973, "Theory of Rational Option Pricing," *Bell Journal of Economics and Management Science*, Vol. 4, No. 1, pp. 141–83.
- , 1974, "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates," *Journal of Finance*, Vol. 29, No. 2, pp. 449–70.
- , 1977, "An Analytic Derivation of the Cost of Loan Guarantees and Deposit Insurance: An Application of Modern Option Pricing Theory," *Journal of Banking and Finance*, Vol. 1, No. 1, pp. 3–11.
- , 1998, "Applications of Option-Pricing Theory: Twenty-Five Years Later," *American Economic Review*, Vol. 88, No. 3, pp. 323–49.
- Sims, Christopher A., 1999, "Domestic Currency Denominated Government Debt as Equity in the Primary Surplus," paper presented at the Latin American meetings of the Econometric Society, Cancun, August 19.
- van den End, Willem, and M. Tabbae, 2005, "Measuring Financial Stability: Applying the MfRisk Model to the Netherlands" DNB Working Paper No. 30 (Amsterdam: De Nederlandsche Bank). Available via the Internet: <http://www.dnb.nl/publicatie/publicaties-dnb/dnb-working-papers-reeks/dnb-working-papers/auto53249.jsp>

Appendix

Black-Scholes Option Pricing Formula in Contingent Claim Analysis

The Black-Scholes option pricing formula originally was applied to the valuation of options on traded equity and quickly spread to a variety of applications.³⁸ The original paper by Merton (1974) extended the option pricing formula from explicit option to implicit options in the context of corporate liabilities. Merton pointed out how equity can be modeled as a call option on firm assets and that risky debt has an embedded put option. This work has been applied extensively (McQuown, 1993; Crosbie and Bohn, 2003) and extended in many different directions to include multiple layers of debt and integrated with interest rate models (Cossin and Pirotte, 2001), to basket assets in multiple currencies.

Implied sovereign asset value and volatility

In this study, the Black-Scholes option pricing formula is used to relate the value and volatility of domestic currency liabilities to the value and volatility of sovereign assets. The value of domestic currency liabilities as a call option on sovereign assets is

$$V_{DCL} = V_A N(d_1) - DB e^{-r_f t} N(d_2), \quad (24.10)$$

where V_A is the value of sovereign assets, V_{DCL} is the value of domestic currency liabilities, DB is the distress barrier or value of default-free debt, r_f is the risk-free rate of interest, and t is the time to maturity on a default-free bond in years. $N(d)$ is the cumulative probability distribution function for a standard normal variable (i.e., the probability that a random draw from a standard normal distribution will be below d), where

$$d_1 = \frac{\ln\left(\frac{V_A}{DB}\right) + \left(r_f + \frac{1}{2}\sigma_A^2\right)t}{\sigma_A \sqrt{t}},$$

$$d_2 = d_1 - \sigma_A \sqrt{t}, \quad (24.11)$$

and σ_A is the standard deviation of return on sovereign assets.

The Black-Scholes formula contains two unknowns, sovereign assets and volatility of sovereign assets. The relationship between volatility of sovereign assets and volatility of domestic currency liabilities is given by

$$V_{DCL} = \frac{\sigma_A}{\sigma_{DCL}} V_A N(d_1), \quad (24.12)$$

where σ_{DCL} is the standard deviation of domestic currency liabilities.³⁹ Here, $N(d_1)$ is the change in the price of domestic currency liabilities with respect to a change in sovereign assets, or $\partial V_{DCL} / \partial V_A$. This ratio is also referred to as the option *delta*. However, the main implication of the above relationship is that the standard deviation of domestic currency liabilities can be derived from historical data, including exchange rate data, and used to solve for sovereign asset volatility. Using standard iterative techniques, equations (24.10) and (24.12) can be solved simultaneously for the implied value of sovereign assets and sovereign asset volatility. Using this output, the precise measure of distance to distress is d_2 in equation (24.11):

$$d_2 = \frac{\ln\left(\frac{V_A}{DB}\right) + \left(r_f - \frac{1}{2}\sigma_A^2\right)t}{\sigma_A \sqrt{t}} = \frac{\ln\left(V_A * \exp\left(\left(r_f - \frac{1}{2}\sigma_A^2\right)t\right)\right) - \ln(DB)}{\sigma_A \sqrt{t}}.$$

³⁸ For readers interested in a more explicit derivation of the Black-Scholes option pricing formula, see Black and Scholes (1973) and Merton (1973, 1974). Although the derivations in these studies use continuous-time mathematics, Hull (1993) and Baxter and Rennie (1996) detail how binomial models can be used to develop discrete-time representations.

³⁹ See Hull (1993, p. 38).

Probability of default and sovereign risk premium

The probability of default is the likelihood that future sovereign asset value will fall below the distress barrier. Therefore, computing probability of default requires calculating the cumulative normal distribution function, $N(\cdot)$. This can be done using numerical methods or polynomial approximation. Tables that compute $N(\cdot)$ also are found in many financial and econometric texts. Using one of these methods will yield the probability of default as

$$\text{Risk-Neutral Probability of Default} = N(-d_2). \quad (24.13)$$

The face value of senior foreign currency debt can be derived from equation (24.10) and the balance sheet relationship, $V_A = V_{DCL} + V_{FCL}$, where V_{FCL} represents the value of foreign currency liabilities.⁴⁰ Using these relationships together yields the value of foreign currency liabilities as

$$V_{FCL} = V_A(1 - N(d_1)) + DBe^{-\pi t}N(d_2), \quad (24.14)$$

which is also equal to

$$V_{FCL} = DBe^{-\pi t} - [DBe^{-\pi t}N(-d_1) - V_A N(-d_2)], \quad (24.15)$$

when modeled as the default-free value minus the implicit put option (present value of expected loss). The term $DBe^{-\pi t}$ is the distress barrier discounted to the present by the risk-free rate.

Equation (24.14) can also be expressed in terms of a credit risk premium:

$$y_t - r_f = \frac{-1}{t} \ln \left\{ \frac{V_A}{DBe^{-\pi t}} N(-d_1) + N(d_2) \right\}, \quad (24.16)$$

where $y_t = -(1/t)\ln(V_{FCL}/DB)$. The left-hand side of the equation represents the yield to maturity on risky debt less the risk-free rate of interest and is therefore equivalent to a risk premium. In addition to the risk-free rate and time, examination of equation (24.16) reveals that sovereign risk premium is a function of only two variables: the volatility of sovereign assets and the ratio of the value of sovereign assets to the present value of the promised payments on foreign currency liabilities, discounted by the risk-free rate. Both increases in the ratio of sovereign assets to foreign currency liabilities and decreases in sovereign asset volatility decrease the sovereign risk premium.⁴¹

As described in the body of the chapter, there is a strong relationship of the sovereign risk-neutral default probabilities with the market implied default probabilities (MIDP). The risk-neutral probability of default is $N(-d_2)$. Its relationship with the estimated default probability (EDP) is

$$N(-d_2) = N(N^{-1}(\text{MIDP}) + \lambda\sqrt{T}), \text{ where } \text{MIDP} = \frac{1 - e^{-st}}{1 - R} \approx \text{EDP}, \quad (24.17)$$

where λ is the market price of risk, s is the observed spread, and R is the assumed recovery rate. If we use the MIDPs implied from observed sovereign credit default swap (CDS) spreads as a proxy for the estimated default probability (EDP), then

$$N^{-1}(N(-d_2)) - N^{-1}\left(\frac{1 - e^{-st}}{1 - R}\right) = \lambda\sqrt{T} = \frac{\mu_{\text{sov}} - r}{\sigma_{\text{sov}}} \sqrt{T}, \quad (24.18)$$

where μ_{sov} is the return on sovereign assets, r is the risk-free rate, and σ_{sov} is the volatility of sovereign assets.

Extensions to multiple layers of liabilities

Contingent claims analysis (CCA) can be extended to multiple layers of liabilities. Instead of one distress barrier, there can be multiple distress barriers tied to the different layers of debt. With three layers of liabilities, the implicit options that make up the liabilities are shown in Table 24.7.⁴²

The Moody's–Macro-Financial Risk model uses two layers of sovereign liabilities: local currency liabilities as junior claims and foreign currency denominated debt as the senior claim. This assumption appears to be a reasonable approxima-

⁴⁰ Merton (1974) derives similar measures for the pricing of corporate debt.

⁴¹ Although the Banking Stability Measure model assumes constant volatility, the Macro-financial Risk model is more realistic, including adjustments to volatility and deviations from strictly log-normal distributions of asset value and other values, such as exchange rate.

⁴² See Cossin and Pirotte (2001) for a discussion on how the framework can handle multiple layers of liabilities or default sequences.

TABLE 24.7

Liabilities, Distress Barriers, and Implicit Options		
Type of Liability	Distress Barrier	CCA Implicit Options
Most junior (equity-like)	Senior plus subordinated debt default barrier (DB_{Sr+Sub})	Call Option 1 (Assets, DB_{Sr+Sub} , r , t , Asset volatility)
Subordinated debt or preferred equity	Senior debt default barrier (DB_{Sr})	Call Option 2 (Assets, DB_{Sr} , r , t , Asset volatility) minus Call Option 1 (Assets, DB_{Sr+Sub} , r , t , Asset volatility)
Senior debt		Assets minus Call Option 2 = DB_{Sr} minus Put Option (Assets, DB_{Sr} , r , t , Asset volatility)
Total		Sum Equals Assets

tion from anecdotal evidence, from the observed robustness of the model, and from the behavior of spreads during periods of stress.⁴³

CCA balance sheets with three or more layers, as described above, can be constructed, calibrated, and tested to refine the model further. For a consolidated CCA sovereign balance sheet with three layers, the seniority structure could be foreign currency debt as senior, part of local currency debt as subordinated, and the rest of local currency liabilities as junior. Alternatively, the priority could be part of local currency debt as senior, foreign currency debt as subordinated, followed by the remainder of local currency liabilities as the most junior claim. The three-layer model could be used to analyze the segregated government balance sheet. For example, the most senior claim on the government balance sheet could be foreign currency denominated debt, the subordinated claim could be local currency debt, and the most junior claim could be government obligations to the monetary authorities.

⁴³ Assuming that all money and local currency debt are senior and that all foreign currency debt is junior leads to inconsistencies. Crises resulting in depreciation of the exchange rate cause the “foreign currency junior claim” to grow large compared with domestic currency debt. This is inconsistent with the observation that CDS spreads on foreign currency debt increase with sharp depreciations. In situations of large exchange rate appreciation, usually considered beneficial from a credit risk perspective, the value of the “foreign currency debt junior claim” would be very small relative to domestic currency debt, indicating that a large expected loss is associated with the domestic currency debt. Observed spreads on local and foreign currency debt during periods of stress are not consistent with assuming that all local currency debt and money are senior to foreign currency debt.

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Factor Model for Stress Testing with a Contingent Claims Model of the Chilean Banking System

DALE F. GRAY • JAMES P. WALSH

This chapter derives risk indicators for the major Chilean banks based on contingent claims analysis, an extension of Black-Scholes-Merton option-pricing theory. These risk indicators are tied clearly to macroeconomic and financial developments in Chile and outside, but bank responses are highly heterogeneous. To reduce the number of variables linked to the banks' risk to a tractable number, we apply principal component analysis. Vector autoregressions of risk indicators with the most significant factors show strong ties from financial markets and regional developments. Impulse response functions from these factors are derived, which allow for scenario testing. The scenarios derived in the study illustrate how the magnitude and persistence of responses of bank credit risk can vary across banks in the system.

METHOD SUMMARY

Overview	The risk-adjusted contingent claims bank balance sheet model for estimation of distance to distress (DtD) for banks and banking system risk indicators is combined with a macroeconomic model and may be used for stress testing.
Application	The model calculates risk indicators for individual banks, including DtD, credit spreads, and expected losses. Macroeconomic relationships are estimated bank by bank, and then the principal components framework with vector autoregression is estimated and used for stress testing.
Nature of approach	Contingent claims analysis (CCA) approach (risk-adjusted balance sheet).
Data requirements	Time series value equity, equity volatility, and default barrier used in the CCA model to get bank's implied assets, asset volatility, and risk indicators (DtD, expected losses, spreads). Macro factors include GDP, unemployment, interest rate, and other domestic and foreign factors.
Strengths	The model estimates forward-looking bank risk indicators linked to macro factors for stress testing. In general, there are a number of ways to use the econometric equations—principal component analysis is one as it has the advantage of reducing the number of factors to a manageable number.
Weaknesses	• The principal component is an aggregate factor that cannot be specifically tied to one macro factor for stress testing. • The model depends on econometric results, and different adaptations are possible. • Bank models work best where the banks have traded equity, although it is possible to adapt the framework to banks without traded equity.
Tool	Standard econometrics package.

The vulnerability of financial to macroeconomic and financial shocks is an area of important and growing interest to policymakers, especially in emerging markets. However, estimating the effect of such shocks on the risk of banks in a coherent manner requires both a model of banking sector

risk and a tractable methodology for simulating shocks and estimating their effect on various risk measures.

This chapter uses contingent claims analysis (CCA) tools developed in finance to estimate the riskiness of banks. The basis of the CCA methodology is the estimation of the

This chapter is an abridged version of IMF Working Paper 08/89 (Gray and Walsh, 2008).

probability that an entity (in our case banks, but also corporations or even governments) will default on its obligations. Additional measures include the expected loss to the holders of the bank's debt and also a measure of the financial fragility of the bank based on the distance, measured in the standard deviation of asset returns, that a bank's assets exceed its liabilities. The CCA is widely used by financial practitioners such as Moody's KMV (MKMV) to estimate the creditworthiness of corporates and is highly correlated both with the ratings assigned by such agencies as well as with historic probabilities of default.

Given its explicit focus on risk and probability and its basis on market-sourced data on equity prices, debt, and interest rates, CCA has many advantages. Equity data by nature incorporate the forward-looking expectations of the market in a way that static indicators of bank risk, such as nonperforming loan ratios and provisioning, cannot, and the high frequency of observations, at least for equity and interest rate data, allows for much faster updating of risk measures than data available only at monthly or quarterly frequencies.

After deriving the risk measures provided under the CCA model, this study assesses how various macroeconomic and financial variables in Chile affect them. One immediate result is that banks even in a relatively small emerging market such as Chile display a high degree of heterogeneity: the correlation of estimated risk measures with historic values of macroeconomic variables differs not only in magnitude but even in sign across the various banks, limiting the usefulness of a systemic approach that constrains the manner in which these shocks are propagated to be similar across banks. Beyond this, some variables that appear strongly correlated with the riskiness of one bank will have little or no correlation with the riskiness of others, necessitating the inclusion of a large number of variables in order to assess adequately the riskiness of all the banks in our sample.

As tractability is reduced by including such an exhaustive sample of variables, we reduce the size of this state space through principal component analysis (PCA). As some variables comove in expected ways, we are able to drop those variables with minor fluctuations only marginally related to some of the macro variables and retain four significant factors that account for the largest share of the variance of all the macro variables.

The risk measures (or, more specifically, the implied asset output of the CCA that underlies these measures) and the derived data from the principal components analysis are then plugged into a standard vector autoregression (VAR) and impulse response function (IRF) model that allows for scenario analysis. The IRFs thus calculated yield various insights about how various macroeconomic factors affect the default risk of banks and how the timing of both the maximum impact of shocks as well as the recovery from them can differ across banks.

The study assesses the impact of shocks of magnitudes similar to those observed in Chile over the past 10 years on the risk measures of the banks in the sample, showing that the riskiness of Chilean banks has declined to historically low levels and that

even under shocks of magnitude similar to those seen in recent periods of financial turmoil, banks appear unlikely to reach the high levels of risk associated with the 1998 financial crisis.

Section 1 presents the background of the CCA model, and discusses the data used in the analysis. Section 2 presents the results of the CCA analysis and its extension via principal components analysis to macro and financial shocks. Section 3 presents possible extensions in this line of research. Section 4 concludes.

1. RISK MEASURES FROM CONTINGENT CLAIMS ANALYSIS

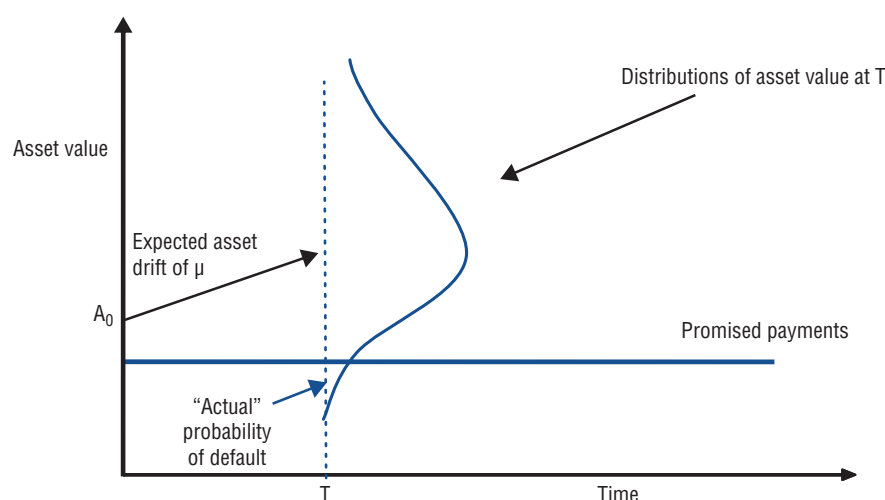
A. Background

The CCA approach provides a methodology for combining balance sheet information with widely used finance and risk management tools to construct marked-to-market balance sheets that better reflect underlying risk. The risk-adjusted balance sheets use option pricing tools to value the liabilities, which are modeled as claims on stochastic assets. It can be used to derive a set of risk indicators that can serve as barometers of risk for firms, financial sector vulnerability, and sovereign risk.

A contingent claim is any financial asset whose future payoff depends on the value of another asset. The prototypical contingent claim is an option—the right to buy or sell the underlying asset at a specified exercise price by a certain expiration date. A call is an option to buy, a put is an option to sell, and the value of each is contingent on the price of the underlying asset to be bought or sold. CCA is a generalization of the option pricing theory pioneered by Black and Scholes (1973) and Merton (1973). Since 1973, option pricing methodology has been applied to a wide variety of contingent claims. In this study, we focus on its application to the analysis of credit risk and guarantees against the risk of default and their links to macroeconomic and financial developments.

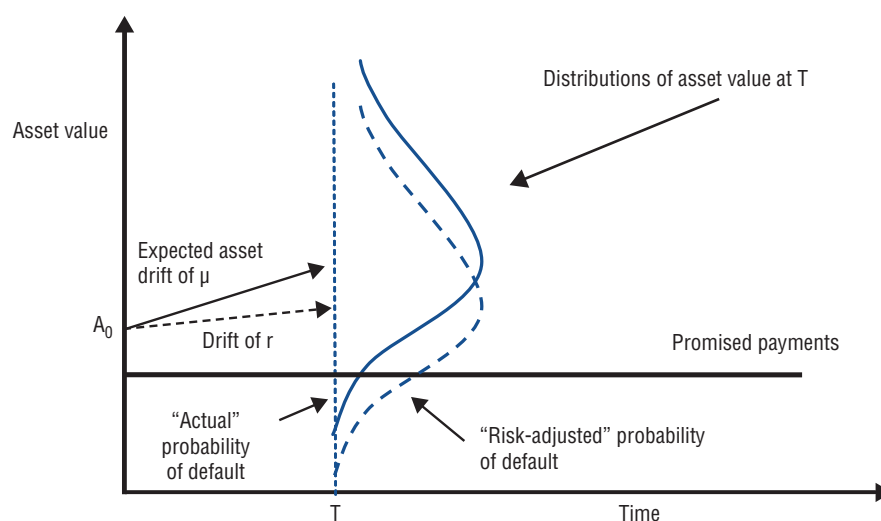
The CCA approach is based on three principles: (1) the values of liabilities are derived from assets; (2) liabilities have different priority (i.e., senior and junior claims); and (3) assets follow a stochastic process. The liabilities consist of senior claims (such as senior debt), subordinated claims (such as subordinated debt), and the junior claims (equity or the most junior claim). For a bank, as the value of its total assets declines, the debt that it owes to other institutions becomes riskier, and its value declines, while credit spreads on its risky debt rise.

Balance sheet risk is the key to understanding credit risk and crisis probabilities. Default happens when assets cannot service debt payments, that is, when assets fall below a distress barrier comprising the total value of the firm's liabilities. Uncertain changes in future asset value, relative to promised payments on debt, is the driver of default risk. Figure 25.1 illustrates the key relationships. The uncertainty in asset value is represented by a probability distribution at time horizon T . At the end of the period, the value of assets may be above the promised payments, indicating that debt service can



Source: Authors.

Figure 25.1 Distribution of Asset Value



Source: Authors.

Figure 25.2 Estimated Probability of Default

be made, or below the promised payments, leading to default. The area below the distribution in Figure 25.1 is the “actual” probability of default. The asset-return probability distribution used to value contingent claims is not the “actual” one but the “risk-adjusted” or “risk-neutral” probability distribution, which substitutes the risk-free interest rate for the actual expected return in the distribution. This risk-neutral distribution is the dashed line in Figure 25.2 with expected rate of return r , the risk-free rate. Thus, the “risk-adjusted” probability of default calculated using the “risk-neutral” distribution is larger than the actual probability of default for all assets that have an actual expected return (μ) greater than the risk-free rate r (i.e., a positive risk premium).¹

¹ See Merton and Bodie (1992).

The calculation of the actual probability of default is outside the CCA/Merton model but such a probability can be calculated by combining the CCA/Merton model with an equilibrium model of underlying asset expected returns to produce estimates that are consistent for expected returns on all derivatives, conditional on the expected return on the asset. One does not have to know expected returns to use the CCA/Merton models for the purpose of value or risk calculations, but for calibration into actual probabilities, such data are necessary. The value of assets at time t is $A(t)$. The asset return process is $dA/A = \mu_A dt + \sigma_A \varepsilon \sqrt{t}$, where μ_A is the drift rate or asset return, σ_A is equal to the standard deviation of the asset return, and ε is normally distributed, with zero mean and unit variance. The probability distribution at time T is shown in Figure 25.1.

Default occurs when assets fall to or below the promised payments, B_t . The probability of default is the probability that $A_t \leq B_t$, which is

$$\begin{aligned}\text{Prob}(A_t \leq B_t) &= \text{Prob}\left(A_0 \exp\left[(\mu_A - \sigma_A^2/2)t + \sigma_A \varepsilon \sqrt{t}\right] \leq B_t\right) \\ &= \text{Prob}(\varepsilon \leq -d_{2,\mu}).\end{aligned}$$

Given that $\varepsilon: N(0,1)$, the “actual” probability of default is $N(-d_{2,\mu})$, where $d_{2,\mu} = \frac{\ln(A_0/B_t) + (\mu_A - \sigma_A^2/2)t}{\sigma_A \sqrt{t}}$. This is distance to distress (DtD) with drift of μ . $N(\cdot)$ is the cumulative standard normal distribution.

Shown in Figure 25.2 is the probability distribution (dashed line) with drift of the risk-free interest rate, r . Risk-adjusted probability of default is $N(-d_2)$, where $d_2 = \frac{\ln(A_0/B_t) + (r - \sigma_A^2/2)t}{\sigma_A \sqrt{t}}$. This is DtD with drift of r , the risk-free rate.²

Financial fragility is intimately related to probability of default. Shocks to prices or liquidity frequently end up being converted into credit risk in a crisis as banks’ debtors’ income flows weaken and they run into difficulties servicing their loans to banks. Default is hard to handle in traditional macro models in part because of assumptions that usually exclude the possibility. In addition, flow of funds and accounting balance sheets cannot provide measures of risk exposures that are forward-looking estimates of losses. CCA, conversely, is a framework that explicitly includes and estimates the probability of default.

Given that there is a nonzero chance of default, the value of debt is risky and therefore less than the value of risk-free debt:

$$\text{Risky debt} + \text{Guarantee against default} \equiv \text{Risk-free debt}$$

The value of “risky” debt can therefore be modeled as the default-free value of the debt less the expected loss:

$$\text{Risky debt} \equiv \text{Risk-free debt} - \text{Guarantee against default}$$

Given that this guarantee is an asset of uncertain value, the debt can be thought of and modeled as a contingent claim.

This identity holds both conceptually and in terms of value. If the debt is collateralized by a specific asset, then the guarantee against default can be modeled as a put option on the asset with an exercise price equal to the face value of the debt. The debt holder is offering an implicit guarantee as it is obligated to absorb the losses if there is default. However, often a third party is the guarantor, as is the case when government guarantees the deposit liabilities of banks or the pension-benefit promises of firms.³

Using the Black-Scholes-Merton differential equation for pricing contingent claims, the value of risky debt is a func-

tion of the default-free value of debt (i.e., distress barrier) at time 0, asset level at time 0, volatility of the asset, the time horizon until the expiration date of the claim, and the risk-free interest rate. Since 1973, the Merton model methodology has been applied to a wide variety of contingent claims.

B. Calculating risk indicators for individual banks or financial institutions

Domestic equity markets provide pricing and volatility information for the calculation of implied assets and implied asset volatility in corporate, bank, and nonbank financial institutions. The simplest method solves two equations for two unknowns, asset value and asset volatility. Details are shown in Merton (1974) and Crouhy, Mark, and Galai (2000). Levonian (1991) used explicit option prices on bank equity to measure equity volatility and calibrate Merton models for banks. MKMV has successfully applied its version of the CCA model to measure the implied assets values and volatilities and to calculate expected default frequencies (EDFs) for over 35,000 firms and financial institutions in 55 countries around the world (Crosbie and Bohn, 2003).

For unlisted corporates and banks, the relationship between the accounting information and the risk indicators, of companies with traded equity, can be used as a guide to map accounting information of companies without traded equity to default probabilities and risk indicators for institutions that do not have traded equity. (An example is Moody’s RiskCalc for corporate sectors in many countries and for banks in the United States.)

The CCA model for banks and financial institutions uses a time series of the daily market capitalization, the volatility of the market capitalization, and the distress barrier (derived from book values of deposits and debt) to estimate a time series of the implied market value of bank assets and asset volatility. Several useful risk indicators can be calculated for each bank or institution: (1) DtD; (2) the risk-adjusted and actual probabilities of default; (3) the expected losses (put option) to depositors and debt holders; (4) potential size of financial guarantees of the public sector; and (5) sensitivity of risk indicators to changes in underlying bank assets, asset volatility, or other factors. The steps used to calculate the implied assets and asset volatility of the individual bank or financial institution and the risk indicators is shown in Figure 25.3.

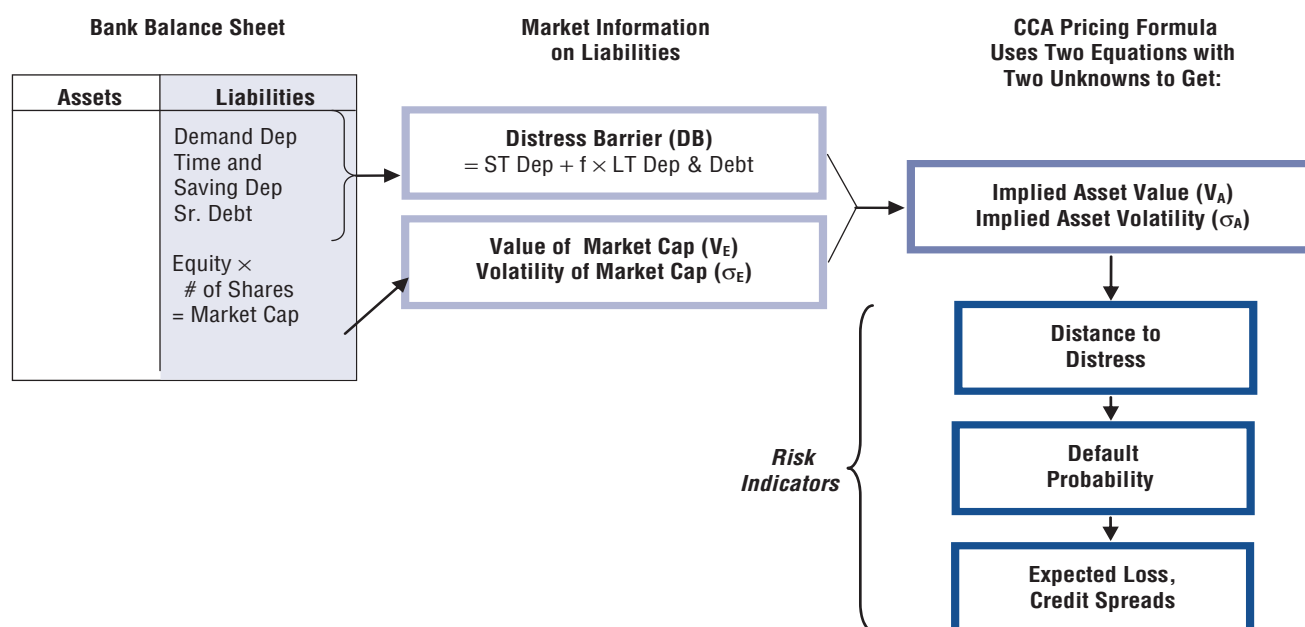
C. Data

Derivation of CCA risk indicators can be done at any frequency, but this study uses daily data when available. Daily market capitalization data for the various banks were obtained by the Central Bank of Chile from the Bolsa de Santiago. Bank debt was obtained from the Central Bank of Chile’s database.

Some judgment was used in estimating the default barrier and interest rate used in the calculations. Practitioners

² See the Appendix on how to link actual and risk-adjusted probabilities of default and extensions of the CCA model.

³ The CCA framework is an extension of Merton’s models of risky debt (1974) and deposit insurance (1977); see Gray, Merton, and Bodie (2006, 2007).



Source: Authors.

Note: CCA = contingent claims analysis; LT Dep = long-term deposits; Sr. Debt = senior debt; ST Dep = short-term deposits.

Figure 25.3 Calibrating Bank CCA Balance Sheets and Risk Indicators

do not generally set the default barrier equal to the total book value of a company's debt. Given that longer-term debt can often be restructured, the default barrier is most frequently calculated as the sum of short-term debt and a share of long-term debt, in this study, 50 percent. The choice of this parameter naturally has some impact on the results, but this effect is generally minor.

A more serious concern about the choice of default barrier is whether the threshold of interest is in fact default. Banks rarely default, and regulators are likely to be interested less in the probability of such an event than they are in the possibility that bank assets will fall below a level at which the authorities might be expected to intervene or at which depositors might panic. However, such a "distance-to-distress" measure would require another assumption about what level assets would have to reach to warrant distress. One such assumption would be to estimate the level of assets implied by the CCA consistent with a minimum level of regulatory capital. This issue is discussed in Section 3.

The choice of interest rate is also not straightforward. The assumption under the Black-Scholes model is of a risk-free rate. In the Chilean context, where the sovereign is not AAA and thus no bank is either, the choice of risk-free rate is not straightforward. For this study, a constant fixed rate of 5 percent is assumed.

This fixed-rate assumption, though not realistic, is not uncommon in financial-market contexts. Moreover, for relatively low-risk situations, such as Chilean banks during all but high-volatility periods, the sensitivity of default measures to interest rate shifts is minor. During high-volatility

periods, moreover, although the sensitivity to the value of the interest rate does rise, other components (such as rising equity volatility) have an equally large impact, and the misspecification of the interest rate will not fundamentally affect the conclusion of higher bank risk. Figure 25.4 shows the default probability of one of the banks in our sample at three different points.

Although the estimated (risk-neutral) default probabilities vary strongly based on the value of the risk-free rate for the high-risk period around the Long-Term Capital Management (LTCM)/Russia crisis in 1998, even the lowest default value based on a relatively high risk-free rate is quite high. During less active periods, such as late 2006, the interest rate makes less of a difference.

Finally, financial practitioners use various methods for estimating the volatility of daily asset returns. Two frequently used methods model daily volatility either as a generalized autoregressive conditional heteroskedasticity, or GARCH(1,1), or as a moving average process. This study uses the GARCH(1,1) methodology for all banks in the sample, but the results of the moving-average model are similar. Figure 25.5 shows the daily estimated volatility for Bank 2 under both volatility assumptions.

Section 2.C discusses other ways in which both the interest rate and the volatility may be modeled more accurately.

D. Banking risk indicators

As noted earlier, using daily price data and debt data (quarterly or monthly debt data are used to calculate the distress barrier) for major banks or financial institutions, the CCA

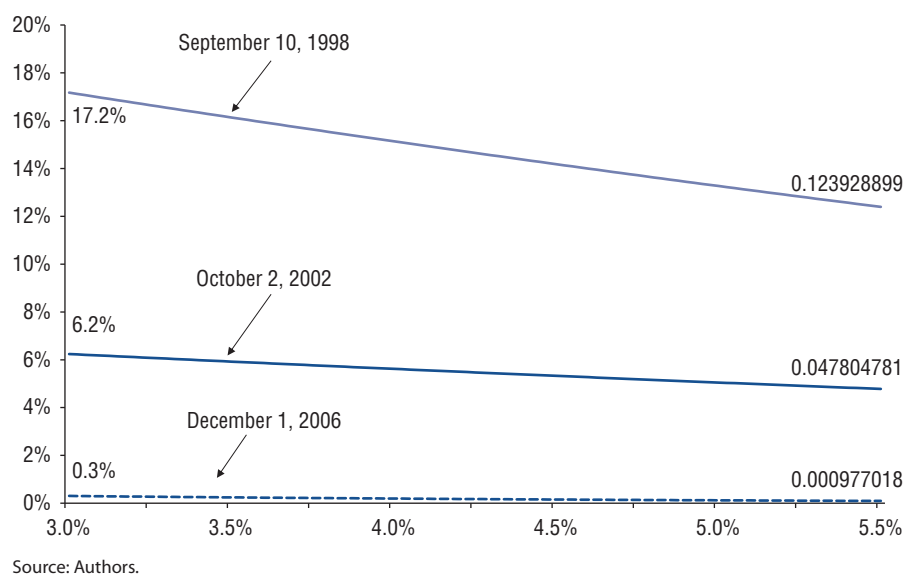


Figure 25.4 Estimated Risk-Neutral Default Probabilities for Bank 2 under Varying Risk-Free Rate Assumptions

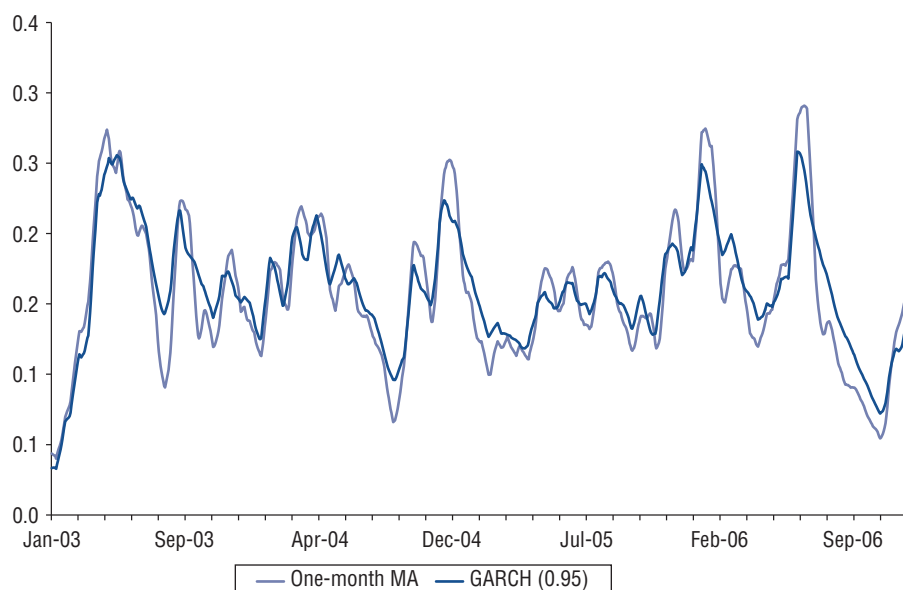


Figure 25.5 10-Day Moving Averages of Volatility Measures for Changes in Stock Price of Bank 2

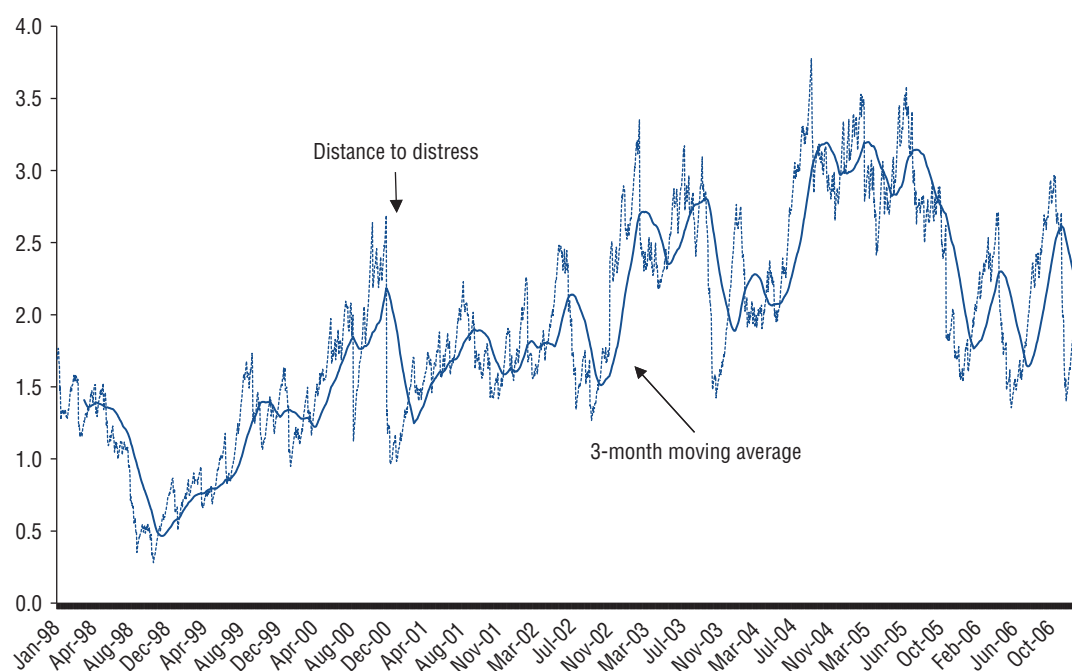
model can be used to derive a time series of CCA risk indicators for individual banks and financial institutions and time series of more aggregate systemic risk indicators.

Distance to Distress

A useful indicator of banking or financial sector risk over time is a graph of average DtD, where $DtD = d_2$ from Figure 25.1. Figure 25.6 shows the estimated time pattern of DtD

for the Chilean banking system from 1998 to 2007, along with a three-month moving average. It was calculated by treating the portfolio of banks in the system as one “big bank” and estimating implied assets, implied asset volatility, and DtD. This procedure used historical volatility of market capitalization calculated with GARCH(1,1).

It is clear that the highest risk period was in late 1998 and early 1999, during the fallout from the LTCM/Russia crisis. Since then, the Chilean banking system has gradually re-



Sources: Authors; and Bolsa de Santiago.

Figure 25.6 Distance-to-Distress Indicator of Financial Fragility

duced its risk, though this trend appears to have leveled off in late 2005.⁴ Other periods where markets assessed suddenly higher risk for Chilean banks are easily discerned, for example, the decline in world stock markets following the collapse of the Internet bubble in 2000 and the period preceding the Brazilian presidential elections in the third quarter of 2002.

Risk-Neutral Default Probabilities

The Merton model also can produce the risk-neutral default probability for each of the financial institutions analyzed. Using the methodology laid out earlier, the model produces the cumulative probability over the time period T that the institution will default, for which we assume one year. As noted, this default probability is not the same as the real-world probability of default. That can be calculated using the extensions laid out in Section 3 but is beyond the scope of this study. However, the relative shifts in default probability over time are illustrative.

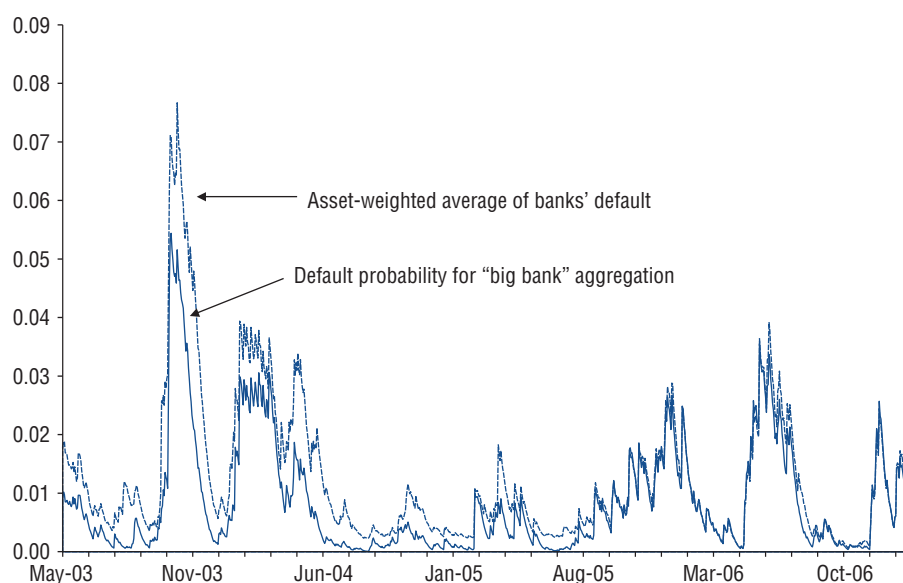
As is clear from Figure 25.7, the default probability associated with a given rating is surprisingly low. The overall risk of the Chilean banks in our sample peaked around the time of the 1998 crisis and did not approach that level during more recent periods of financial volatility, such as those preceding the 2002 Brazilian election or associated with the 2005 unrest in emerging markets.

In addition, Figure 25.7 presents not only the default probabilities for the aggregation of the banking system into one large bank, as in Figure 25.6, but also the asset-weighted average of the various banks' default probabilities. This measure shows slightly higher risks during crisis periods as relatively small banks, which are lost in the portfolio aggregation of the "big bank" measure, have a larger impact on a default-probability weighted average. This asset-weighted average thus indicates better whether an individual bank might run into risks, while the "big bank" aggregation shows a broader systemwide concern. The difference between the two is a rough indicator of the size of the banks most affected.

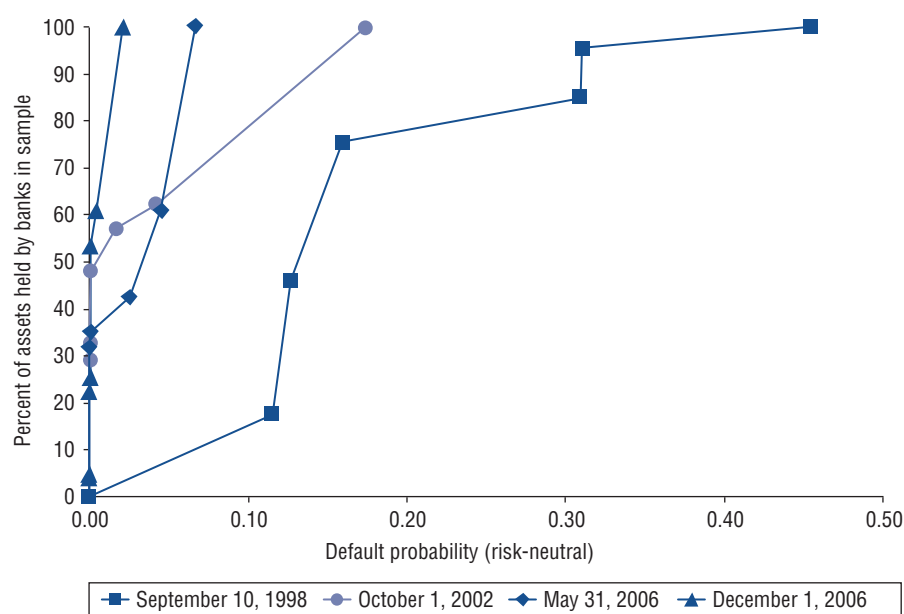
MKMV estimates actual default probabilities by using historical default information, which are "bucketed" and compared with modified CCA risk indicators. MKMV then estimates the risk-neutral default probability of default using the market price of risk technique, which is described in the Appendix. Such an extension of CCA-derived indicators is beyond the scope of this study, but the magnitude of the default probabilities is illustrative: a rating of AAA is consistent under the Moody's methodology with a rating of about only 0.01 percent, while a Ba1 rating is associated with a default risk close to 0.13 percent. By the time an empirical default probability reaches 1 percent, the rating is generally in the Ca range.

Turning to individual banks, Figure 25.8 shows, for each of four dates in our sample, the percentage of banking sector assets (held by banks in our sample) with default probabilities at or below a given level. The closer the line for a given date is to the y -axis, the higher the share of banking-sector

⁴ As we will see, this leveling off has occurred at a very low level of risk.



Source: Authors.

Figure 25.7 Risk-Neutral Probability of Default for Aggregated Banks

Sources: Authors; and Moody's KMV.

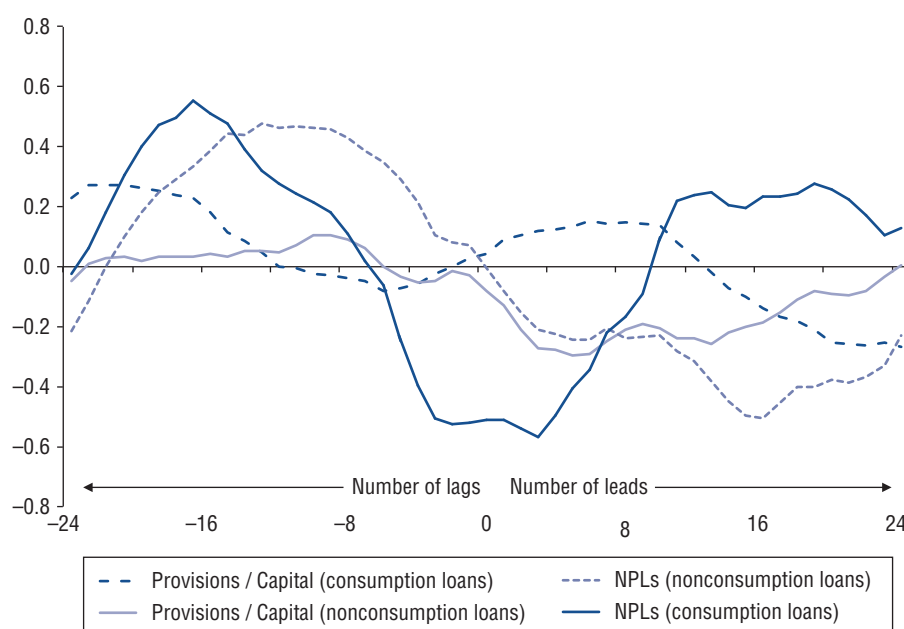
Figure 25.8 Default Probabilities Relative to Asset Size for Chilean Banks

assets held by banks associated with a given probability of default, and thus the safer the banking system. For the first date in our sample, September 10, 1998, during the LTCM crisis, only about 20 percent of banking sector assets were held by banks with default probability below 0.12 percent, and 15 percent of assets were held by banks with default probabilities greater than or equal to 0.3 percent.

Figure 25.8 shows snapshots of how banking sector risk has evolved in Chile over time. During the height of the

1998 crisis, a fairly large portion of the assets held by Chile's main banks were held by banks with default probabilities greater than 10 percent, and only a small share (less than 20 percent) was held by banks with default probabilities at very high levels exceeding 30 percent.

Over time, as the health of the banking system has improved, these risks have fallen. Interestingly, the period just before the 2002 Brazilian presidential elections in late 2002 is associated with generally lower risks of default than the



Source: Luna and Gomez (2007).

Note: NPL = nonperforming loan.

Figure 25.9 Correlations between Banking Sector Risk Measures and Leads and Lags of Systemwide Distance to Default

May 2006 “mini-crisis,” which appears to have affected more Chilean banks, even though no banks show the more extreme values observed here for late 2002.

Correlations with Other Measures of Banking Risk

Risk measures derived from CCA also have been shown to be significantly correlated with other measures of banking risk. Figure 25.9, sourced from Luna and Gómez (2007), presents the correlations between various measures of systemwide bank risk and the leads and lags of system DtD. For example, the measure of nonperforming loans (NPLs) for consumption purposes used in the study are strongly correlated with declining DtD estimates from a few months previous; that is, falling DtD predicts rising NPLs a few months later.

2. RELATING DEFAULT RISK TO MACROECONOMIC VARIABLES

A. Motivation

Monitoring the risk of default of financial institutions is an important task for policymakers, and estimating the effect of forecast changes in the macroeconomic environment can provide useful insights, as can estimating the impact of shocks to macroeconomic or financial variables.

There is a wide range of variables that observers might reasonably expect to be correlated with banking sector risk. We focus on those variables with data available at a monthly frequency and, to keep the dimensionality of the regressions

down, on variables with relatively low correlation.⁵ Although full orthogonality cannot, obviously, be assured, we tried to focus on relatively independent variables.

The variables chosen are presented in Table 25.1. These variables cover domestic developments (unemployment, consumer price index (CPI), and the IMACEC or Indicador Mensual de Actividad Económica), foreign spillovers (U.S. interest rates and CPI), and financial market developments (the VIX volatility measure, returns on the S&P 500, and the Chilean blue-chip Índice de Precios Selectivo de Acciones, or IPSA index). In each case, monthly log returns are used in the VARs reported.⁶

Macro Variables and Asset Returns

Given bank heterogeneity, no small sample of variables appears to be sufficient to explain the variation in bank-asset returns across all the banks. We run stepwise regressions as a diagnostic for the asset returns of each of the banks in our sample against the full sample of macroeconomic variables. Each step of the stepwise regression process eliminates the variable with the least significance, until all remaining variables are significant at a specified level. Choosing the 5 percent level of significance, the variables laid out on Table 25.2 are those that are jointly significant.

⁵ All financial and macroeconomic data used in this section were obtained from the Haver economic database.

⁶ For interest rates, the level and the change are used rather than log returns.

TABLE 25.1

Variables Used in Analysis			
Variable	Chile	United States	International
Financial variables			
Stock market returns	IPSA	S&P 500	
Volatility measure		VIX	
Prices	CPI	CPI	
Interest rates	¹	1-year Treasury bill (Level and change) 10-year Treasury bond (Level and change) Difference (yield curve)	
Exchange rates			Peso / USD Peso / real
Real variables	IMACEC ² Unemployment		Oil price Copper price

Source: Authors.

Note: CPI=Consumer price index; IMACEC=Indicador Mensual de Actividad Económica; IPSA=Indice de Precios Selectivo de Acciones; VIX=Chicago Board Options Exchange Market Volatility Index (implied volatility of the S&P 500 index).

1. Domestic interest rate is incorporated into contingent claims analysis modeling.

2. IMACEC is a monthly index of economic activity highly correlated to GDP growth.

TABLE 25.2

Results of Stepwise Regressions							
Variable (t-statistics in parentheses)	Returns on Implied Bank Assets						
	Bank 1	Bank 2	Bank 3	Bank 4	Bank 5	Bank 6	Bank 7
Peso- <i>real</i> exchange rate	-0.10 (-2.43)		-0.35 (-3.85)				
U.S. CPI	2.78 (2.72)		-10.14 (-3.65)				
IPSA return	0.24 (6.67)		0.22 (2.15)			0.19 (3.15)	
U.S. yield curve slope	0.01 (2.76)	0.09 (6.66)	-0.06 (-2.29)	0.22 (4.17)			0.03 (2.58)
S&P 500 return		0.25 (3.60)	-1.09 (-6.39)	-8.14 (-4.69)		0.47 (4.27)	
IMACEC		1.05 (3.33)	1.56 (2.30)	-20.90 (-4.12)	2.16 (8.00)	1.92 (5.43)	0.93 (3.14)
U.S. 1-year Treasury bill rate (level)		0.06 (6.18)	-0.04 (-2.12)			-0.01 (-2.38)	0.02 (2.37)
Oil price			0.35 (4.72)			0.20 (5.33)	
Copper price				1.28 (2.92)			
VIX				-1.36 (-3.27)		0.13 (3.47)	
Peso- <i>real</i> exchange rate					-0.19 (-2.79)		
U.S. 1-year Treasury bill rate (change)						-0.16 (-5.68)	
Constant	0.01 (0.33)	-0.29 (-5.50)	0.37 (3.36)	0.22 (1.60)	0.04 (5.92)	0.04 (2.51)	-0.13 (-2.52)
Adjusted R ²	0.38	0.42	0.50	0.42	0.41	0.53	0.13
F	16.09	18.69	13.30	6.68	35.16	16.62	5.84

Source: Authors.

Note: CPI=Consumer price index; IMACEC=Indicador Mensual de Actividad Económica; IPSA=Indice de Precios Selectivo de Acciones; VIX=Chicago Board Options Exchange Market Volatility Index (implied volatility of the S&P 500 index).

TABLE 25.3
Factor Loadings and Variable Interpretation

Variable	Factor 1 (Financials)	Factor 2 (Interest rates)	Factor 3 (Cyclicals)	Factor 4 (Regional/ domestic)
U.S. yield curve slope	0.00	-0.93	-0.10	0.00
VIX	-0.76	0.28	0.07	-0.07
IMACEC	-0.02	0.01	0.32	0.58
Unemployment	0.19	0.09	-0.06	-0.86
CPI Chile	-0.38	0.73	-0.14	0.08
Copper	0.34	-0.21	0.64	0.22
Oil	0.06	-0.10	0.87	-0.13
S&P 500	0.86	0.23	-0.12	-0.11
CPI U.S.A.	-0.16	0.23	0.82	0.14
U.S. 10-year rate (change)	0.73	0.00	0.40	-0.05
U.S. 1-year rate (change)	0.51	0.21	0.61	0.18
IPSA	0.73	-0.20	0.16	0.00
Peso- <i>real</i> rate	-0.49	0.30	-0.23	-0.25
Peso- <i>real</i> /CLP-BRL rate	0.18	0.12	-0.06	0.81
U.S. 1-year rate (level)	0.09	0.95	0.02	-0.02

Source: Authors.

Note: Bold numbers indicate factors greater than 0.5 in magnitude.

CPI = consumer price index; IMACEC = Indicador Mensual de Actividad Económica; IPSA = Índice de Precios Selectivo de Acciones; VIX = Chicago Board Options Exchange Market Volatility Index (implied volatility of the S&P 500 index).

Although some variables, such as the returns on the S&P 500, the level of U.S. short-term rates, and the slope of the U.S. yield curve, are significant for four of the seven banks, only one variable (the IMACEC) is significant in six. The stepwise regressions produce models with *R*-squared statistics in the 40–60 percent range, while smaller models using only those variables that are significant in at least three regressions (returns on the IPSA and S&P 500, the U.S. yield curve slope and one-year rate, and the IMACEC) produce adjusted *R*-squared statistics only in the 20–40 percent range. This is not a particularly bad result, but clearly significant information on the results of each bank is also available using a more tailored set of variables.⁷

To reduce the dimensionality of the state space—and also simplify the estimations and produce IRFs that are easier to interpret—we apply PCA to these variables.

B. Scenario testing using principal component analysis

Principal Component Analysis

PCA can be used to extract useful common information from a number of time series and synthesize it into a smaller number of factors. In our case, we are interested in distilling the information contained in our macroeconomic variables into a more manageable data set, which we can then use for analysis.

⁷ Also, these variables exhibit some cross-correlation. The principal component analysis results in uncorrelated factors, which make analysis and estimation easier.

We apply PCA to the system of 15 macro variables. The procedure outputs 15 orthogonal factors, each of which is a linear combination of each of the macro variables but of which the first four account for 70 percent of the variation of all the underlying macro variables.⁸

We apply a varimax rotation procedure to these four factors, a transformation that chooses factor loadings to maximize the largest of them and minimize the smallest. This procedure does not change the orthogonality of the factors, but by setting the factor loadings as close to either one or zero as possible, it allows us to interpret the factors as something akin to a sum of a small number of variables, rather than only a combination of 15 variables.

The factor loadings of the macro variables onto the four significant factors are presented in Table 25.3, with the high loadings for the macro variables in bold. We categorize each of the factors according to the variables they weigh most strongly following the rotation. Based on the macro variables each of the factors loads most strongly, we interpret Factors 1–4 as representing developments in financial markets, U.S. interest rates, cyclical variables, and Chile- or Latin American-specific factors, respectively.

The factors themselves are shown in Figure 25.10. Factor 1, the “financials” factor, commoves very strongly with returns on the IPSA and S&P 500, which together have a correlation of 0.6. The factor is negatively correlated both with the VIX volatility measure (the highest peak of the VIX in 1998 appears as the low point in the financials factor) and the peso-dollar exchange rate. Finally, changes in the U.S.

⁸ These are the four factors with eigenvalues greater than one.

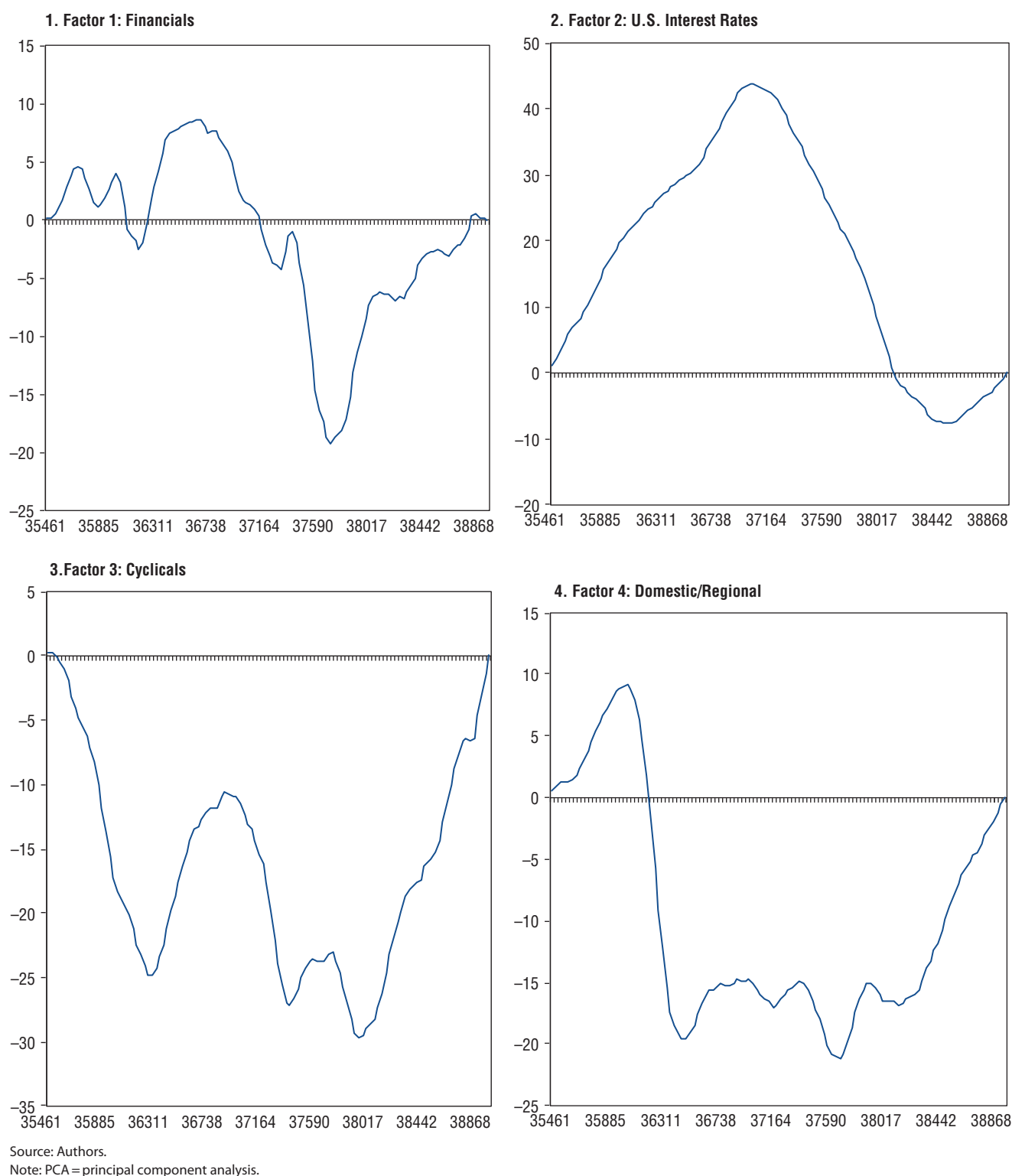


Figure 25.10 Significant Factor Output from PCA

10-year Treasury bond rate, which also commoves with the S&P 500 and the VIX, are linked to this factor.

The second factor shown in Figure 25.10, which we call the interest rate factor, mirrors the level of the U.S. one-year Treasury bond rate and also (negatively) the slope of the U.S.

yield curve. This factor rises from 1997 to 2000 as (short-term) interest rates rose in the United States, and the yield curve correspondingly flattened, then reverses direction in 2000, as the yield curve slowly began to steepen again as short-term rates fell.

Most of the variation in the third factor comes from changes to oil and copper prices and the U.S. CPI. There are also weaker loadings on the changes in U.S. interest rates and the Chilean IMACEC index. As copper, oil, and, generally, inflation tend to commove with the business cycle, we interpret this variable as representing cyclical factors in the world economy.

Finally, the fourth factor weighs most strongly Chilean factors such as the IMACEC index and unemployment but also the exchange rate with the Brazilian *real*. The graph of this factor in Figure 25.10 shows that its variation stems mostly from a large decline in 1998 and a subsequent recovery beginning in 2004. This tracks quite closely the evolution of most Chilean macro variables during the 1997–2006 period. However, the exchange rate between the peso and the *real* is not a domestic factor, and the 1998 crash was not unique to Chile. For this reason, we interpret this factor as a domestic/regional factor: many variables in Latin America, but very few elsewhere, exhibit this pattern.

As these four factors account for more than 70 percent of the variation in all 15 of the macroeconomic variables laid out earlier, the relationship between the risk measures derived and our factors should be quite similar to their relationship with those macro variables surviving in the stepwise regressions but allow us to use a common model across all seven banks, rather than make do with a small set of variables that are useful for the largest number of banks while neglecting other variables that might help with the results of only a few.

Vector Autoregressions

We next estimate VARs for the factors and the implied assets extracted earlier. We estimate one VAR for each bank, and each VAR includes five variables: the bank's implied assets and the four factors. The ordering of the system is to some extent trivial: although each of the factors may Granger-cause changes in the bank's implied assets, the factors cannot, by construction, contemporaneously affect each other, as they are orthogonal. However, they can be autocorrelated and can Granger-cause changes in other factors.

Using asset returns rather than distances to default (DDs) or EDFs is not an obvious choice. The results of VARs run using DDs and EDFs are similar to those run using assets, but the fewer coefficients are significant. Given that the risk measures are merely transformations of the asset data, no information is lost by using assets instead of risk measures, and we derive DD measures from simulated asset series.

The asset returns are all stationary under the Dickey-Fuller criterion at the 5 percent level of significance, as are the log changes in macro variables that we use. Each VAR is specified in two lags. The Schwarz information criterion (SIC) and Akaike information criterion (AIC) were calculated for each bank at one, two, three, and four lags. The results differed across banks, but the AIC and SIC point to two lags more frequently than to any other number of lags.

In the VARs, returns on the assets of each of the banks show some persistence: the first lag of the bank's assets is significantly positive in all regressions, and the second lag is significant in two cases. This similarity breaks down when we look at factors, however. Each of the macroeconomic factors is significant for at least one of the banks, though the financials and domestic/regional factors are significant for four of the variables, while the interest rate and cyclical factors are significant for only one of the banks.

Our sample can be divided into Banks 1–3, which are the three largest private banks in the system, and Banks 4–7, which are smaller. The first three banks cumulatively account for 52 percent of the assets of the financial system (or 59.9 percent of the nonstate financial system), while the last four account for 20.1 percent (23.2 percent of the nonstate system). The financials variable is significant for the first three, large, banks but not for the smaller ones, while the regional factor is significant for the smaller banks but not for the larger ones.⁹

The reason for this is not clear. Both the larger banks and the smaller banks focus on corporations: 73 percent of the loan portfolio of the larger banks is to companies, while 67 percent of the small banks' portfolios are. It is possible that the larger banks focus on larger, more internationally connected companies such as exporters, while the smaller banks focus on smaller corporations with fewer links to financial markets, but without company-level data on bank lending, this cannot be verified.

An improvement in financial conditions (i.e., a decline in the VIX or a rise in the IPSA or S&P 500) is tied to significantly higher asset returns at two of the three largest banks and one of the smaller ones but lower returns at one of the larger banks. This is somewhat counterintuitive: it seems unlikely that a bank can focus its lending portfolio on countercyclical companies. The signs of the coefficients on the regional factor are more intuitive: in each case where the variable is significant and three of the five where it is not, an improvement in Chilean domestic conditions (such as a decline in unemployment, an improvement in IMACEC, or a depreciation of the peso relative to the Brazilian *real*) leads to an increase in the assets of the banks.

The interest rate and cyclical factors are less significant in our regressions. Our stepwise regressions also showed that cyclical variables such as copper and oil prices were related to banking sector risk at least for some banks and that at least one cyclical variable could explain a significant share of variation in banking sector–implied assets for six of the seven banks. Interest rate variables were even more useful and significant contributors in four of the stepwise regressions. However, once we distill macroeconomic developments into orthogonal factors and estimate a consistent model across the different banks, the picture changes slightly, and each is significant for only one bank.

⁹ All figures for bank assets and concentration are as of July 2007.

In both cases, this is counterintuitive. The effect of rising U.S. interest rates (and thus a flattening of the U.S. yield curve) on liquidity in Latin American financial markets is widely recognized, as is the impact of commodity prices and cyclical developments in the United States. In this case, however, it appears that the effect of interest rates and such cyclical variables is weaker than simultaneous innovations in other financial-market variables, and those are the coefficients that here have the most significance, with interest rate and cyclical developments relegated to a secondary position.

Impulse Response Functions

From the VARs presented, we derive impulse response functions (IRFs) that we use to show how a shock to each of the factors affects the implied assets of each of the banks. IRFs created from the factors have an advantage above IRFs calculated from traditional macro variables: we cannot separate the effect of a change in the VIX from a change in the S&P 500 index, for example, but the factors derived from the PCA conducted are by construction orthogonal; the impulse response to the financials factor shows the impact of a shock to that factor (itself a linear combination of macro variables) on the assets of each of the banks keeping all the other factors constant.

In interpreting these results, it should be borne in mind that each factor is merely a linear combination of the underlying macro variables, and although we talk about the impact of the interest rate factor or the financials factor, this is only shorthand for a grouping of shocks that tends to affect a subset of the macro variables simultaneously, with minimal (if not necessarily zero) effect on other macro variables. That is, a jump in the VIX is correlated with negative returns to the IPSA and the S&P 500 indices, which we interpret as a shock to the financials factor, even though not every shock to the VIX will have the same transmission to the IPSA and S&P 500, nor will it in real life be independent of the other macro variables.

The impulse responses are graphed in Figure 25.11. The first set is each bank's response to a shock to its own asset level. The IRF thus measures the persistence of shocks to banks. As discussed earlier in this section, these variables were highly significant in regressions, and the impulse responses are also quite large. In each case, a shock to the assets of a bank results in continued increase of assets for a few months, with half the shock dissipated within four months for all banks and 90 percent of the shock dissipated within eight months for all the banks but one.

A shock to the financials factor causes a strongly positive impulse response for four of the banks but a negative one for another bank, with two close to zero. Presumably for the same reasons posited earlier, the implied assets of one of the banks in the sample declines as a result of a (positive) financial markets shock, while the results for the other four banks could intuitively be attributable to an improvement in financial conditions.

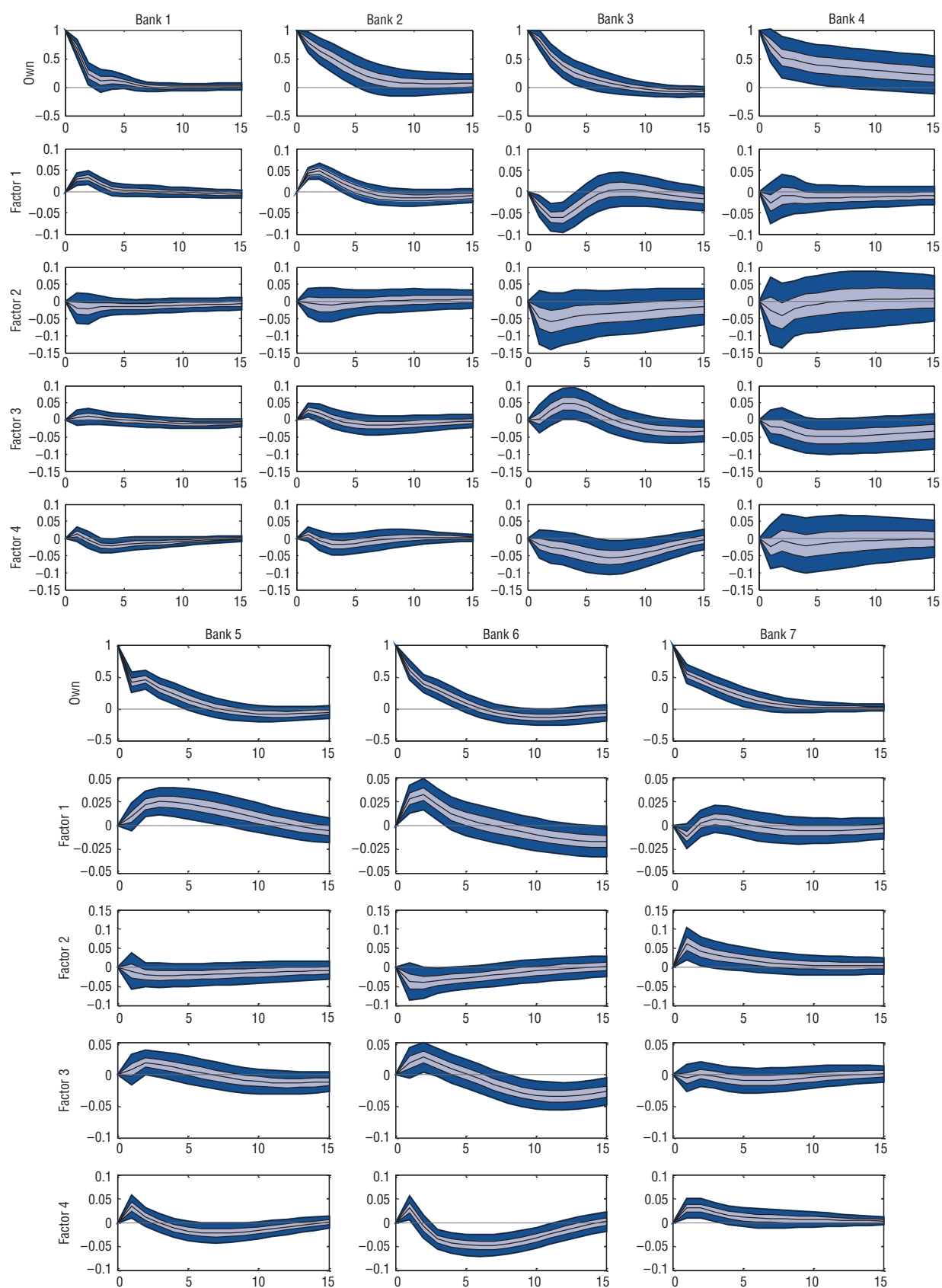
The banks also differ in the timing of their response to the financials shock. Three of the four banks with the intuitively signed response, as well as the bank showing a negative response, have a maximum response to the shock two to three months afterward. These are the same four banks for which the financials variable was significant in the VARs. The final bank (number 5) shows a maximum response to the shock a few months later, and the shock is also more persistent. The VAR for this bank did not have a significant coefficient on the lags of the first factor, but it did on the fourth factor. As there are nontrivial feedback loops between the factors, the impulse response from the fourth factor back to the first results in a delayed but significant impulse response for this bank as well. In this case, financial shocks may not affect Bank 5 as severely as others, but the feedback from a regional shock to the financials factor can be significant enough to cause a delayed response.

A shock to the interest rate factor produces a significant impulse response only for the seventh bank, though here again the result is somewhat counterintuitive: a positive shock to the interest rate factor (associated with an increase in short-term U.S. interest rates or a flattening of the yield curve) results in an increase in the implied assets of Bank 7, with the other banks showing the intuitive, if insignificant, opposite result.

The cyclicals factor displays a pattern similar to the interest rate factor. The impulse response is significant for the two variables showing a significant coefficient in the VAR but not for the others. In both cases, the sign of the result is in the expected direction: an improvement in cyclical conditions (i.e., rises in oil or copper prices, increases in inflation or short-term rates) results in an increase in banks' implied assets, which peaks after a few months and then, the factor itself being cyclical, turns negative. The banks with significant impulse responses in this case, numbers 2 and 5, are one of the large ones and one of the small ones. The other banks showing insignificant responses display a broadly similar pattern, with the exception of Bank 4.

Finally, the domestic and regional factor provides significant responses for three of the banks. In this case, the impulse response is significant for the small banks, numbers 5 through 7, the same banks that showed significant coefficients in the VARs. In each of these cases, as well as in two of the less significant ones, the impulse responses are of the expected sign. For two of the banks, the shocks are not persistent: within six months, the effect of a positive shock to the local economy has dissipated. For the final bank, Bank 7, the effect is more persistent, with half the effect dissipated within six months.

The impulse responses thus reflect the results shown in the VARs. Each of the banks responds quite strongly to either the domestic/regional or the financial factors, while the responses to changes in the interest rate or cyclical factors have a significant impact on only one bank each. In general, the larger banks are more responsive to developments in financial markets, while the smaller banks respond more strongly



Source: Authors.

Note: Bands indicate 50 and 95 percent confidence intervals.

Figure 25.11 Impulse Response Functions

to developments in the domestic economy. In both cases, shocks are relatively short-lived but for some banks can be more persistent than for others.

Scenario Testing

Deriving impulse responses allows us to extend this analysis further by testing scenarios. Here we derive CCA risk measures under scenarios calibrated to approximate recent shocks affecting Chile, to estimate the impact such a shock would have on banking sector risk today.

For each scenario, we estimate the DD of each of the banks under the given scenario. Each series represents the simulated DD following a shock to the cited variable. The DD formula requires as inputs assets, asset volatility, as well as the parameters indicated earlier. We derive assets from our impulse responses, with the initial value based on the values observed for the end of our sample in February 2007. The parameters we use are the constant ones used for the derivation of the historic risk measures. For asset volatility, we use the final observed value. As each of the subsequent cases represents a shock to economic conditions, we might expect that asset volatility, which is closely related to equity volatility, would rise. A higher asset volatility in all cases implies a lower DD; thus the scenarios laid out likely overestimate the actual DD observed. Some alternatives are presented as robustness tests.

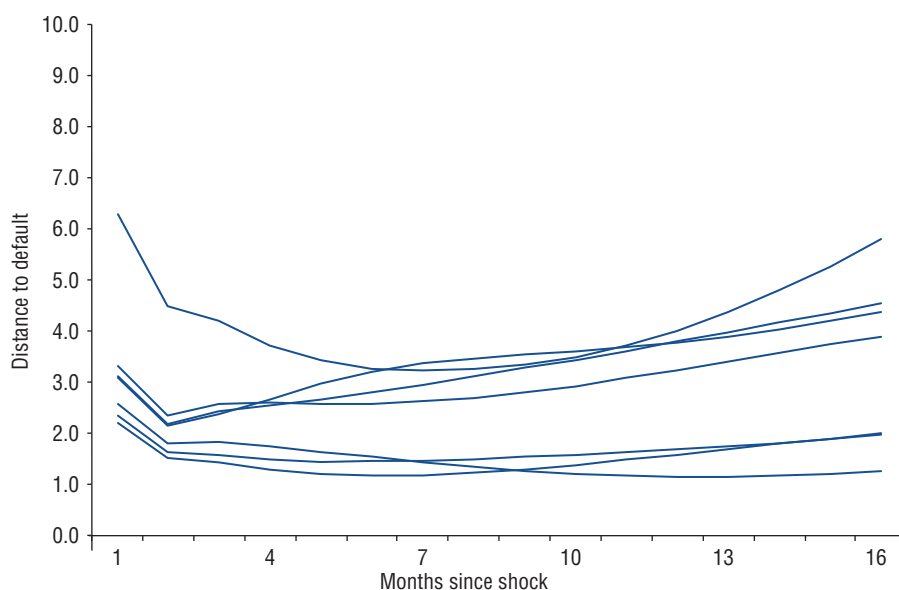
The scenario presented in Figure 25.12 is intended to be similar to the financial conditions at the beginning of the Russia/LTCM crisis in late 1998. At that time, the VIX jumped by about two standard deviations, while stock markets around the world, including in the United States and Chile, fell dramatically. Multiplying the log changes in those

variables by the factor loadings derived earlier gives an average innovation to the financial factor slightly less than one.

For each of the banks, DtD falls in the months immediately following the shock. As DD measures the number of standard deviations of implied assets between the bank's current position and its default barrier, a decline of one in DD represents a statistically significant increase in the probability of a bank's asset level falling beneath the default barrier. If we apply a slightly more rigorous standard such as DtD (a level below which the government would intervene and bail out the bank), this amount would be even lower, depending on the threshold for government intervention.

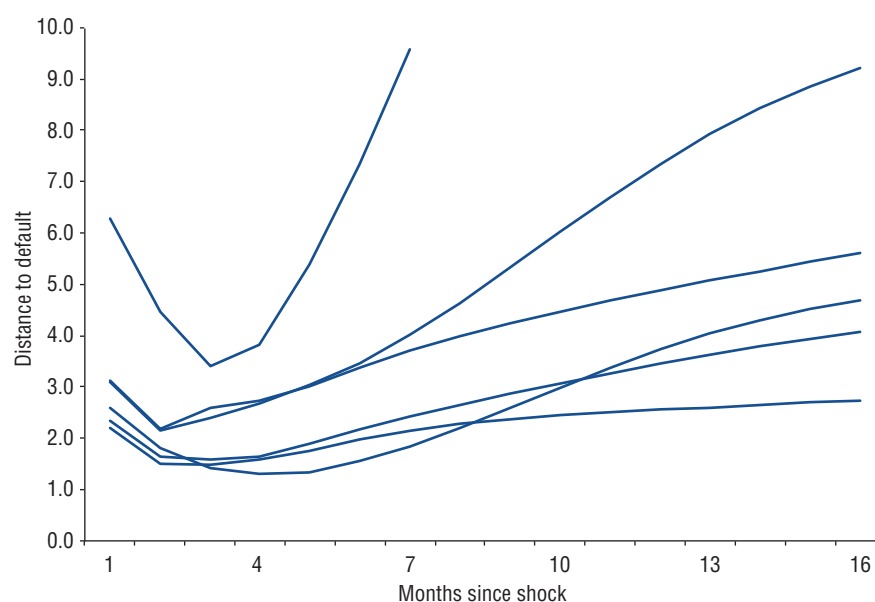
Four of the five banks show recoveries beginning a few months after the shock. In two cases, the recovery is rapid, and within 10 months the banks' assets have risen above the original level. In the other cases, the recovery from financial turmoil is slower. In both cases, one of the two banks is one of the large ones, and the other is a smaller bank. To a regulator, this different pace of recovery should imply that although some banks have historically been able to bounce back quickly following a crisis, others have taken a longer time to recover and should be monitored more closely during the aftermath. The most extreme case we observe here is of the fifth bank, which does not begin to revert until a year after the shock, showing that some Chilean banks (here one of the smaller ones) may take a long time to bounce back from episodes of financial turmoil.

In the second scenario (Figure 25.13), we look at the response of the variables to a domestic shock. We calibrate the shock to be comparable to the increase in unemployment observed during 1999 as Chile began a long period of adjustment following the 1998 financial crisis. Here, as in the first scenario, there is a decline in DD following the shock of



Source: Authors.

Figure 25.12 Scenario 1: Shock to Financials Factor Comparable to 1998



Source: Authors.

Figure 25.13 Scenario 2: Shock to Domestic Factor

between one and two standard deviations (except for one outlying bank with idiosyncratic results). This shock is less persistent than the financials shock, however, and recovery in all cases begins after a period of less than six months.

There is some heterogeneity: one of the banks' implied assets bottoms out a few months later than the others but then recovers more quickly. But overall a single domestic shock appears to have less of a long-term impact than the financials factor. Although it is true that financial turmoil generally lasts less time than a recession, shocks of the magnitude we use in both of these scenarios are not common. A domestic factor shock of this magnitude hit Chile only during the 1999 crisis; the subsequent recession was marked more by a stagnation in employment and the IMACEC than by continued large increases in unemployment or shrinking of production. Thus, the short time period (one month) of each of these shocks does not distort the effect of their impact.

C. Robustness tests

Volatility and Interest Rates

As discussed earlier, two areas where the parameterization of the model differs from reality are in the selection of the risk-free interest rate incorporated into the model and the estimation of volatility. The results presented in Section 2.A show that utilizing another widely used volatility measure would not have much impact on results. Allowing for a variable interest rate raises the question of which interest rate might most accurately be used, a nontrivial question. The impact on results, however, is too small to be highly significant during those periods (low risk) in which small changes are important and large only during those periods (high

risk) in which resulting probabilities will be very high under any assumption.

Estimation Period

Clearly much of our analysis is affected by the financial turmoil of 1998. To see how much of the analysis is dependent on this unusual period, we reestimate the stepwise regressions, the PCA, and the VARs by dropping all 1998 observations.

The stepwise regressions, intended to show which of the macro variables used in our sample have the greatest explanatory power for each of the banks' asset returns, are not substantively changed. In each case, dropping the 1998 observations does not change the variables that survive the stepwise procedure and does not materially change the estimates of the parameters.

Beyond this naive point, the PCA, though largely unchanged, does have one significant difference. Following the rotation of the covariance matrix, four factors accounting for 70 percent of the macro variables' variance had eigenvalues greater than one; these were the variables we preserved for the VARs. Dropping 1998 observations increases the explanatory power of the first four factors, which now account for 73 percent of overall variance, but a fifth factor has an eigenvalue greater than one. Applying our earlier selection criterion, we would also include this factor in the regressions.

The interpretation of the factors is largely unchanged. Each of the first four factors weighs most of the macro variables in a similar fashion to the PCA, but the fifth factor most heavily weighs the U.S. exchange rate, copper prices, and changes to U.S. interest rates. Thus, although the qualitative interpretation of the first four factors is essentially unchanged, the fifth factor is difficult to place.

Despite including five factors in the VARs, the results are not dramatically different from the four-factor VARs. The first, financial developments, factor is significant for the same banks (the large ones and number 6), while the fourth, domestic, factor, is significant for the smaller banks. The new factor is significant for two banks, comparable to the interest rate and cyclical factors.

Overall, these results are sufficiently similar to the main results that it does not appear that the 1998 observations are substantively changing our results. Including them thus does not appear to be unwarranted. In fact, these observations, as they represent the largest-scale financial crisis in our estimation period, are likely to overestimate the riskiness of banks, which for our purposes, is not a major drawback.

3. AREAS FOR FURTHER RESEARCH

A. Modeling bank risk with CCA

Extensions of Current Analysis

The analysis in this chapter models banks individually and does not allow feedback between banks. This restriction may be justified in a case in which interbank lending is quite small, banks are highly competitive, and banks are quite heterogeneous. These assumptions may not be met in every case, however, and estimation of a broader VAR allowing feedback effects from, for example, larger banks to smaller ones might improve the fit. The data in this study extend back only to 1998, unfortunately, and increasing the number of parameters through estimation of such a high-dimensionality VAR would be problematic.

The constant interest rate assumption in this study is also unrealistic, though the resulting distortion of results should be minor. Using a series of risk-free interest rates would improve the realism of the estimation. The Merton model has been extended to include stochastic interest rates as well (see the Appendix).

In addition, the Black-Scholes-Merton assumption of normally distributed asset returns is highly unrealistic for financial assets. The widely recognized skewness of financial sector returns, with large drops far more likely than the normal distribution would predict, reduces the usefulness of the normal model. This lack of realism can be addressed, however, by modifying the formulas underlying the CCA. One method used is rather than calibrating returns to a normal distribution to calibrate them to a combination of two (or more) log-normals, which allows both for the nonzero skew exhibited by financial returns and also for a higher kurtosis than assumed under basic option pricing formulas. Another, related, way to model fat tails or skewed asset distribution is by having the asset volatility vary with the level of the asset. This is the constant elasticity of variance approach. Market data determine how the volatility changes as the asset varies.

Some of the results presented here are puzzling and call for a more detailed analysis of the banks in the sample. The

result that improvements in financial market conditions affect most banks positively but affect one of the main Chilean banks adversely is quite counterintuitive. The differences between small and large banks are also striking, specifically, the difference in responses to domestic and financial market developments. A more detailed analysis of the lending portfolios of the banks might shed light on these differences and would provide more detailed insights into which banks might be most exposed to various shocks.

Stress Testing and Assessing Capital Adequacy Using CCA Models

One major goal of financial sector stress testing is to assess capital adequacy of various institutions under different potential shocks. Shocks to financial institution assets and asset volatility and/or interest rates and other parameters can be used in the CCA model to measure the impact on capital adequacy. An advantage of using CCA models for financial institutions is that the capital adequacy can be related to asset level, asset volatility, and default probability on the institution's liabilities and other factors. This way to calculate capital adequacy has been extended to include interest rates, interest rate volatility, and correlation of asset return with interest rates by van Deventer and Imai (1997, 2003) and Belmont (2004) using the Merton-STV model.

The financial stress testing process commonly used by central banks and banking supervisors uses various models to measure the change in expected default probability of the obligors, usually representative corporates or corporate sub-sectors. The default probabilities are then used with estimates of exposure and loss given default in a model of bank credit losses (e.g., Credit Risk+) to estimate the impact on economic capital.¹⁰ Calibrated CCA models of financial institutions can be used to estimate capital adequacy without the need for detailed data on default probabilities or loss given default of obligors. According to van Deventer and Imai (2003), they did not use the probability of default or the loss given default in allocating capital in the capital allocation using the Merton-STV model because the probability of default and loss given default are both implied by the STV model and the value of asset volatility and interest rate and correlation. This makes CCA a potentially useful tool when detailed data on obligor exposures, default probabilities, or loss given default are not available.

B. Incorporating CCA into macroeconomic models

Given that the economy and interest rates affect financial sector credit risk, and the financial sector affects the economy, an important issue is whether credit risk indicators (CRIs)

¹⁰ Work at the European Central Bank uses MKMV median default probability for various corporate sectors with a VAR or global VAR; see Pesaran, Schuermann, and Weiner (2004).

should be included in monetary policy models and, if so, how. Including an aggregate credit risk CRI in the GDP gap equation and testing whether the coefficient is significant is an important first step to get a better understanding of how the financial sector credit risk affects GDP. The next step could be to add a fifth equation relating the CRIs to GDP and interest rates (this could draw on analysis from the previous section relating financial risk indicators to macroeconomic variables).

Using past data, it might be interesting to include the CRI in the policy rate reaction function to examine whether financial stability appears to have been taken into account when setting interest rates in the past. A variation of this approach is being investigated in the research department of the Central Bank of Chile.¹¹ The approach taken in the Central Bank of Chile is to first estimate the DtD for the banking system (each individual bank's DtD from a CCA model is weighted by bank implied assets). The DtD for the banking system is included in the GDP gap equation and in the policy rate reaction function. The model parameters are then estimated using historical data, including the DtD indicator. The approach can be used to examine the trade-offs between GDP, inflation, and DtD for the banking system.¹²

4. CONCLUSION

CCA can be a useful tool for assessing the risk of financial companies. No risk measure is perfect, and CCA has both advantages and disadvantages. For example, by being based on market data, the CCA accumulates the information of market participants in a forward-looking way that other measures of bank risk do not do. But markets themselves do not always foresee risky events, and the CCA also faces this limitation.

In the case of Chile, CCA risk measures show the gradual improvement in the health of Chilean banks over the past decade while also clearly indicating those periods when risk rose because of international shocks. The measures derived in this study are correlated both with other measures of bank risk (in some cases leading them) as well as with macroeconomic and financial market variables.

The inclusion of these risk measures in a VAR with macroeconomic variables is the chief achievement of this study. Chilean banks differ drastically in their short- and medium-term reactions to various macroeconomic shocks, both domestic and international. This makes it difficult to arrive at a parsimonious model of economic shocks that can be applied across banks. This study takes a wide variety of Chilean and international variables and distills the large majority of their variation into four macro factors, which are then used to as-

sess the interaction between bank risk and these broader macro variables. We interpret these factors as representing developments in financial markets, U.S. interest rates, cyclical variables, and Latin American- or Chilean-specific factors.

The VAR results show that innovations in these factors are tied quite strongly to the evolution of bank risk. Each of the factors is significant in at least one of the VARs, and some are significant for most of the banks. IRFs derived from the VARs show a wide variety of significant effects as well. We find that developments in financial markets are most significant in explaining the variation in banks' risk during our sample, with positive developments in financial markets being strongly related to a reduction in the risk assessment of most of the banks. Interest rate changes in the United States (the second factor) are less strongly tied to the riskiness of Chilean banks. The riskiness of most of the banks moves procyclically, at least as measured by relation to the cyclical factor, which is intuitive. However, some banks do not react strongly to this factor at all. Finally, the regional/domestic factor appears to have a much stronger effect on the smaller banks than on the larger banks. This is in opposition to the result that the financials factor appears to be tied more strongly to changes among the larger banks.

On the basis of these results, it would appear that in assessing the importance of various shocks to banking sector risk in Chile, regulators should focus their attention on larger, more internationally connected banks in assessing the possible impact of financial market shocks and alternately on smaller banks in assessing the risk of regional or domestic shocks. At least when isolated from other events, U.S. interest rates themselves do not appear to strongly affect banking sector risk, and the evolution of cyclical variables, while affecting most banks intuitively, appears to affect some banks more strongly than others, and indeed, one bank appears to be almost countercyclical.

Another crucial difference among banks is the persistence of shocks. In some cases, the increase in risk associated with an innovation in one of the factors dissipates relatively quickly, and the bank returns to its baseline level of risk in a few months. In some cases, however, the reaction of one or a few of the banks begins later and lasts longer than in the case of the other banks. In assessing the effect of a world shock on the Chilean banking system, this difference in reaction times should be taken into mind: the damage caused to most banks is not immediate but can take a few months to appear. Moreover, the time lag varies among banks, and as some banks are emerging from the shock, the risk associated with others might still be rising.

Numerous extensions to this analysis can be envisioned. Monetary and macroeconomic models can attain greater realism by explicitly including measures of financial risk, and CCA provides an easily calculated option for this. More broadly, the realism of macroeconomic analysis can be greatly increased by incorporating risk assessment tools from finance, and CCA can be a building block in this work as well.

¹¹ See Gray and others (2011).

¹² A related issue is whether an indicator of market risk appetite such as the VIX should be included in monetary policy models along with the CRI. This could help estimate the impact of the CRI on the GDP gap, adjusted for changes in risk appetite.

REFERENCES

- Belmont, David, 2004, *Value Added Risk Management in Financial Institutions* (Singapore: Wiley).
- Black, Fischer, and John C. Cox, 1976, "Valuing Corporate Securities: Some Effects on Bond Indenture Provisions," *Journal of Finance*, Vol. 31, No. 2, pp. 351–68.
- Black, Fischer, and Myron Scholes, 1973, "The Pricing of Options and Corporate Liabilities," *Journal of Political Economy*, Vol. 81, No. 3, pp. 637–54.
- Cossin, Didier, and Hugues Pirotte, 2001, *Advanced Credit Risk Analysis* (New York: Wiley).
- Crosbie, Peter J., and Jeffrey R. Bohn, 2003, "Modeling Default Risk," Moody's KMV, December 18. Available via the Internet: http://www.creditrisk.ru/publications/files_attached/modeling_default_risk.pdf
- Crouhy, Michel, Robert Mark, and Dan Galai, 2000, *Risk Management* (New York: McGraw-Hill; 1st ed.).
- Gray, Dale F., Carlos Garcia, Leonardo Luna, and Jorge E. Restrepo, 2011, "Incorporating Financial Sector into Monetary Policy Models: Application to Chile," IMF Working Paper 11/228 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=25268>
- Gray, Dale F., Robert C. Merton, and Zvi Bodie, 2006, "A New Framework for Analyzing and Managing Macrofinancial Risks of an Economy," NBER Working Paper 12637 (Cambridge, MA: National Bureau of Economic Research). Available via the Internet: <http://www.nber.org/papers/w12637>
- , 2007, "New Framework for Measuring and Managing Macrofinancial Risk and Financial Stability," NBER Working Paper 13607 (Cambridge, MA: National Bureau of Economic Research). Available via the Internet: <http://www.nber.org/papers/w13607>
- Gray, Dale F., and James P. Walsh, 2008, "Factor Model for Stress-Testing with a Contingent Claims Model of the Chilean Banking System," IMF Working Paper 08/89 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=21805>
- Hull, John, Izzy Nelken, and Alan White, 2004, "Merton's Model, Credit Risk and Volatility Skews," *Journal of Credit Risk*, Vol. 1, No. 1, pp. 3–28.
- Jain, Sameer, 2005, *Credit and Market Risk Arbitrage*. Lulu.com.
- Levonian, Mark, 1991, "Have Large Banks Become Riskier? Recent Evidence from Option Markets," *Economic Review* (Fall), Federal Reserve Bank of San Francisco, pp. 3–17.
- Longstaff, Francis A., and Eduardo S. Schwartz, 1995, "A Simple Approach to Valuing Risky Fixed and Floating Rate Debt," *Journal of Finance*, Vol. 50, pp. 789–819.
- Luna, Leonardo, and G. Gomez. (2007) "Riesgo de insolvencia en la Banca: Un analisis tecnico." Unpublished document. Central Bank of Chile. Santiago, Chile.
- Merton, Robert C., 1973, "Theory of Rational Option Pricing," *Bell Journal of Economics and Management Science*, Vol. 4, No. 1, pp. 141–83.
- , 1974, "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates," *Journal of Finance*, Vol. 29, No. 2, pp. 449–70.
- , 1977, "An Analytic Derivation of the Cost of Loan Guarantees and Deposit Insurance: An Application of Modern Option Pricing Theory," *Journal of Banking and Finance*, Vol. 1, No. 1, pp. 3–11.
- Merton, Robert C., and Zvi Bodie, 1992, "On the Management of Financial Guarantees," *Financial Management Journal*, Vol. 21 (Winter), pp. 87–109.
- Pesaran, M. Hashem, Til Schuermann, and Scott M. Weiner, 2004, "Modeling Regional Interdependencies Using a Global Error-Correcting Macroeconometric Model," *Journal of Business and Economic Statistics*, Vol. 22, No. 2, pp. 129–62.
- Saunders, Anthony, and Linda Allen, 2002, *Credit Risk Measurement: New Approaches to Value at Risk and Other Paradigms* (New York: Wiley; 2nd ed.).
- Shimko, David C., Naohiko Tejima, and Donald van Deventer, 1993, "The Pricing of Risky Debt When Interest Rates Are Stochastic," *Journal of Fixed Income*, Vol. 3, No. 2, pp. 58–65.
- van Deventer, Donald, and Kenji Imai, 1997, *Financial Risk Analytics: A Term Structure Model Approach for Banking, Insurance and Investment Management* (New York: McGraw-Hill).
- , 2003, *Credit Risk Models and the Basel Accords* (Singapore: Wiley).

Appendix

Extensions of the Merton Model

Numerous extensions of the original Merton model have been developed that relax certain assumptions in the original model. Restrictions of the model include the assumptions that (1) default can occur only at the maturity date of the debt; (2) there is a fixed default barrier; (3) there is a constant risk-free rate; and (4) asset volatility is constant. Cossin and Pirotte (2001) and Jain (2005) provide a good summary of extensions of the Merton model.

Black and Cox (1976) extended the Merton model to relax assumptions (1) and (2) by introducing a “first passage time” model where default can occur prior to the maturity of the debt if the asset falls below a specified barrier function for the first time.

Although the strict theoretical condition in the Merton model for default is that the value of assets is less than the required payments due on the debt, in the real world, default typically occurs at much higher asset values, either because of a material breach of a debt covenant or because assets cannot be sold to meet the payments (“inadequate liquidity”) or because the sovereign decides to default and induce a debt renegotiation rather than sell assets. To capture these real-world conditions for default in the model, we specify a market value of total assets at which default occurs. We call this level of assets that triggers default the *default barrier*. This barrier can be viewed as the present value of the promised payments discounted at the risk-free rate. The approach used in the KMV model sets the barrier level equal to the sum of the book value of short-term debt, promised interest payments for the next 12 months, and half of long-term debt (see Crouhy, Mark, and Galai, 2000; Saunders and Allen, 2002; Crosbie and Bohn, 2003).

In the 1990s, the model was called the VK model (Vasicek and Kealhofer), and it has multiple layers of liabilities and several confidential features. MKMV’s expected default frequency (EDF) credit measure is calculated using an iterative procedure to solve for the asset volatility. This distance to distress (DtD) was then mapped to actual default probabilities, called cumulative expected default frequencies (CEDFs), using a database of detailed real-world default probabilities for many firms. The MKMV DtD and the CEDF are calculated as follows:

$$DtD_{KMV} = f\left(\frac{\ln(A_0/B_t) + (\mu_A - \sigma_A^2/2)t}{\sigma_A \sqrt{t}}\right),$$

$$CEDF_t = f(DtD_{KMV}(t)).$$

Note that the definition of DtD_{KMV} includes the real drift of the asset, μ_A , whereas the DtD from the Merton approach has r for the asset drift. Given that MKMV estimates the actual default probabilities, the risk-neutral default probabilities are calculated from the correlation of the implied asset with the market, the market Sharpe Ratio, and time (see Crouhy, Mark, and Galai, 2000; Crosbie and Bohn, 2003).

The original RiskMetrics CreditGrades model (2002) included a diffusion of a firm’s assets and a first passage time default with a stochastic default barrier. The model was modified to incorporate equity derivatives. Recent research has studied the relationship between the volatility skew implied by equity options and CDS spreads (Hull, Nelken, and White, 2004). They establish a relationship between implied volatility of two equity options, leverage and asset volatility. This approach is, in fact, another new way of implementing Merton’s model to get spreads and risk-neutral default probabilities directly from the implied volatility of equity options.

The Merton model has been extended to include stochastic interest rates as well. Van Deventer and Imai (2003) include a Vasicek interest rate term structure model, allowing the risk-free interest rate to change and including the correlation of asset return with the interest rate. There are two stochastic factors, the asset and the interest rate, and this model is frequently called the STV model. This closed-form model is a very useful extension by including the impact of changing interest rate term structures. This allows not only for a realistic interest rate series but also for scenario testing with varying assumptions or calibrations for how macroeconomic shocks affect both short- and long-term rates. Longstaff and Schwartz (1995) take the Black and Cox model and add in stochastic interest rates, similar to the way STV includes interest rates.

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Systemic Contingent Claims Analysis

ANDREAS A. JOBST • DALE F. GRAY

This chapter introduces a new framework for macroprudential analysis using a risk-adjusted balance sheet approach that supports policy efforts aimed at mitigating systemic risk from linkages between institutions and the extent to which they precipitate or amplify general market distress. In this regard, a forward-looking framework for measuring systemic solvency risk using an advanced form of contingent claims analysis (CCA), known as “Systemic CCA,” is presented for the systemwide capital assessment in top-down stress testing. The magnitude of joint default risk of multiple financial institutions falling into distress is modeled as a portfolio of individual market-implied expected losses (with individual risk parameters) calculated from equity market and balance sheet information. An example of Systemic CCA applied to the U.K. banking sector delivers useful insights about the magnitude of systemic losses in the context of macroprudential stress testing. In addition, the framework also helps quantify the individual contributions of institutions to systemic risk of the financial sector during times of stress.

METHOD SUMMARY

Overview	The Systemic Contingent Claims Analysis (Systemic CCA) model extends the traditional risk-adjusted balance sheet model (based on contingent claims analysis) to analyze systemic risk from the interlinkages among institutions, including time-varying dependence of default risk. This approach helps quantify the contribution of individual institutions to systemic (solvency) risk and to assess spillover risks from the financial sector to the public sector (and vice versa). In addition, Systemic CCA can be applied to macroprudential stress testing using the historical dynamics of market-implied expected losses.
Application	The method is appropriate in situations where access to prudential data is limited but market information is readily available.
Nature of approach	Model-based (option pricing, multivariate parametric/nonparametric estimation, credit default swap [CDS] curve pricing).
Data requirements	• Accounting information (amounts/maturities) of outstanding liabilities. • Market data on equity and equity option prices. • Various market rates and macro data for satellite model underpinning the stress test. • Sovereign CDS term structure/debt level and debt repayment schedule (if financial sector-sovereign spillover risk is tested).
Strengths	• The model integrates market-implied expected losses (and endogenizes loss given default) in a multivariate specification of joint default risk. • The approach is highly flexible and can be used to quantify an individual institution’s time-varying contribution to systemic solvency risk under normal and stressed conditions.
Weaknesses	• Assumptions are required regarding the specification of the option pricing model (for the determination of implied asset and asset volatility of firms). • Technique is complex and resource intensive.
Tool	The Excel VBA-based tool is available from the authors upon request. Contact author: A. Jobst.

Large parts of the material in this chapter are published in the *Sveriges Riksbank Economic Review*, No. 2, pp. 68–106 (Gray and Jobst, 2011b) and IMF Working Paper 13/54 (Jobst and Gray, 2013).

In the wake of the global financial crisis, there has been increased focus on systemic risk as a key aspect of macroprudential policy and surveillance (MPS) with a view toward enhancing the resilience of the financial sector.¹ MPS is predicated on (1) the assessment of systemwide vulnerabilities and the accurate identification of threats arising from the buildup and unwinding of financial imbalances, (2) shared exposures to macro-financial shocks, and (3) possible contagion/spillover effects from individual institutions and markets owing to direct or indirect connectedness. Thus, it aims to limit, mitigate, or reduce systemic risk, thereby minimizing the incidence and impact of disruptions in the provision of key financial services that can have adverse consequences for the real economy (and broader implications for economic growth). Systemic risk refers to individual or collective financial arrangements—both institutional and market-based—which could either lead directly to systemwide distress in the financial sector or significantly amplify its consequences (with adverse effects on other sectors, in particular capital formation in the real economy). Typically, such distress manifests itself in disruptions to the flow of financial services because of an impairment of all or parts of the financial system that are deemed material to the functioning of the real economy.²

Although there is still no comprehensive theory of MPS related to the measurement of systemic risk, existing approaches can be distinguished broadly on the basis of their conceptual underpinnings regarding several core principles. There are two general approaches: (1) a particular activity causes a firm to fail, whose importance to the system imposes marginal distress on the system owing to the nature, scope, size, scale, concentration, or connectedness of its activities with other financial institutions (“contribution approach”), or (2) a firm experiences losses from a single (or multiple) large shock(s) because of a significant exposure to the commonly affected sector, country, and/or currency (“participation approach”).³ In the case of the

former, the contribution to systemic risk arises from the initial effect of direct exposures to the failing institution (e.g., defaults on liabilities to counterparties, investors, or other market participants), which could also spill over to previously unrelated institutions and markets as a result of greater uncertainty or the reassessment of financial risk (i.e., changes in risk appetite and/or the market price of risk). Table 26.1 summarizes the distinguishing features of both approaches.

The growing literature on systemic risk measurement has responded to greater demand placed on the ability to develop a better understanding of the interlinkages between firms and their implications for financial stability (see Table 26.2). The specification of dependence between firms helps identify common vulnerabilities to risks as a result of the (assumed) collective behavior under common distress (Jobst, 2013). Most of the prominent institution-level measurement approaches have focused on the contribution approach by including an implicit or explicit treatment of statistical dependence in determining joint default risk or expected losses. For instance, Acharya, Pedersen, and others (2009, 2010, 2012) estimate potential losses as marginal expected shortfall (MES) of individual banks in the event of a systemic crisis, which is defined as the situation when the aggregate equity capital of sample banks falls below some fraction of aggregate assets. Brownlees and Engle (2011) apply the same definition for a systemic crisis and formulate a capital shortfall measure (“SRISK index”) that is similar to MES; however, they provide a closed-form specification of extreme value dependence underpinning MES by modeling the correlations between the firm and market returns using the dynamic conditional correlation (DCC)-GARCH model (Engle and Sheppard, 2001; Engle, 2002). Also, Huang, Zhou, and Zhu (2009, 2010) derive correlation of equity returns via DCC-GARCH as statistical support to motivate the specification of the dependence structure for default probabilities within a system of firms. Both CoVaR and CoRisk (Adrian and Brunnermeier, 2008; Chan-Lau, 2010) follow the same logic of deriving a bivariate measure of dependence between a firm’s financial performance and an extreme deterioration of market conditions (or that of its peers). For instance, the CoVaR (CoRisk) for a certain firm is defined as the value at risk (VaR) (as a measure of extreme default risk) of the whole sector (firm) conditional on a particular institution being in distress.

However, only a few models estimate multivariate (firm-by-firm) dependence through either a closed-form specification or the simulation of joint probabilities using historically informed measures of association. None of those include a structural definition of default risk. A chapter of a recent issue of the *Global Financial Stability Report* provides comparative analysis of several of these systemic risk measures in the context of designing early-warning indicators for MPS (IMF,

institution being part of it, which determines the institution’s marginal contribution to systemic risk.

¹ In a progress report to the G20 (Financial Stability Board, International Monetary Fund, and Bank for International Settlements [FSB, IMF, and BIS], 2011a), which followed an earlier update on macroprudential policies (FSB/IMF/BIS, 2011b), the FSB takes stock of the development of governance structures that facilitate the identification and monitoring of systemic financial risk as well as the designation and calibration of instruments for macroprudential purposes aimed at limiting systemic risk. The Committee on the Global Financial System (2012) published a report on operationalizing the selection and application of macroprudential policies, which provides guidance on the effectiveness and timing of banking sector–related instruments (affecting the treatment of capital, liquidity, and assets for the purposes of mitigating the cyclical impact of shocks and enhancing systemwide resilience to joint distress events). See Acharya, Cooley, and others (2010) as well as Acharya, Santos, and Yorulmazer (2010) for the implications of systemic risk in MPS in the U.S. context.

² For a discussion of ways to assess the systemic importance of financial institutions and markets, see IMF, FSB, and BIS (2009).

³ Drehmann and Tarashev (2011) refer to this as a “bottom-up approach,” whereas as a “top-down approach” would be predicated on the quantification of expected losses of the system, with and without a particular

TABLE 26.1

General Systemic Risk Measurement Approaches		
	Contribution approach (“Risk Agitation”)	Participation approach (“Risk Amplification”)
Concept	Systemic resilience to individual failure	Individual resilience to common shock
Description	A contribution to systemic risk conditional on individual failure due to knock-on effect	Expected loss from systemic event due to common exposure and risk concentration
Risk transmission	“Institution-to-institution” Economic significance of asset holdings (“size”)	“Institution-to-aggregate” Claims on other financial sector participants (credit exposure)
	Intra- and intersystem liabilities (“connectedness”)	Market risk exposure (interest rates, credit spreads, currencies)
Risk indicators	Degree of transparency and resolvability (“complexity”)	Risk-bearing capacity (solvency and liquidity buffers, leverage)
	Participation in system-critical function/service, e.g., payment and settlement system (“substitutability”)	Economic significance of asset holdings, maturity mismatches debt pressure (“asset liquidation”)
Policy objectives	Avoid/mitigate contagion effect (by containing systemic impact upon failure)	Maintain overall functioning of system and maximize survivorship
	Avoid moral hazard	Preserve mechanisms of collective burden sharing

Sources: Drehmann and Tarashev (2011); FSB, IMF, and BIS (2011a, 2011b); Jobst (2014); Tarashev, Borio, and Tsatsaronis (2009); and Weistroffer (2011).

Note: The policy objectives and different indicators to measure systemic risk under both contribution and participation approaches are not exclusive to each concept. Moreover, the availability of certain types of balance sheet information and/or market data underpinning the various risk indicators varies between different groups of financial institutions, which requires a certain degree of customization of the measurement approach to the distinct characteristics of a particular group of financial institutions.

2011d).⁴ There also are several studies on network analysis and agent-based models that are more closely related to the participation approach by modeling how interlinked asset holdings matter in the generation and propagation of systemic risk (Allen, Babus, and Carletti, 2010; Espinosa-Vega and Solé, 2011; Organization for Economic Cooperation and Development, 2012). Haldane and Nelson (2012) underscore this observation by arguing that networks can produce nonlinearity and unpredictability with the attendant extreme (or fat-tailed) events.⁵

We propose a forward-looking, analytical, market data-based framework for estimating systemic risk in order to fill this gap in the literature. The suggested approach (“Systemic Contingent Claims Analysis” or “Systemic CCA”) extends the risk-adjusted (or economic) balance sheet approach to generate estimates of the joint default risk of multiple institutions. Under this approach, the magnitude of systemic risk depends on the firms’ size and interconnectedness and is defined by the multivariate density of combined expected losses within a given system of financial institutions.

Systemic CCA identifies market-implied linkages affecting joint expected losses during times of stress, which can deliver important insights about the joint tail risk of multiple entities. A sample of firms (as proxy for an entire financial system or parts thereof) is viewed as a portfolio of individual expected losses (with individual risk parameters) whose sensitivity to common risk factors is accounted for by including the statistical dependence of their individual expected losses (“dependence structure”). Like other academic proposals, this approach helps assess individual firms’ contributions to systemic solvency risk (at different levels of statistical confidence). However, by accounting for the time-varying dependence structure, this method links the market-based assessment of each firm’s risk profile with the risk characteristics of other firms that are subject to common changes in market conditions. Based on the expected losses arising from the variation of each individual firm’s expected losses, the joint probability of all firms experiencing distress simultaneously can be estimated.

In addition, the Systemic CCA framework can generate closed-form solutions for market-implied estimates of capital adequacy under various stress test scenarios. By modeling how macroeconomic conditions and bank-specific income and loss elements (net interest income, fee income, trading income, operating expenses, and credit losses) have influenced the changes in market-implied expected losses (as measured by implicit put option values), it is possible to link a particular macroeconomic path to financial sector performance in the future and derive estimates of joint capital needed to maintain current capital adequacy. As a result, the market-implied joint capital need arises from the amount of expected losses relative to current (core) capital levels as well as its escalation

⁴ A comparison of three measures of institution-level systemic risk exposure (Sedunov, 2012) suggests that CoVaR shows good forecasting power of the within-crisis performance of financial institutions during two systemic crisis periods (Long-Term Capital Management in 1998 and Lehman Brothers in 2008). In contrast, Systemic Expected Shortfall (SES) is based on MES after considering different degrees of firm leverage (Acharya, Pedersen, and others, 2012) and Granger causality does not seem to forecast the performance of firm performance reliably during crises.

⁵ One empirical example of the network literature is based on a Federal Reserve (Fed) data set, which allowed for the mapping of bilateral exposures of 22 global banks that accessed Fed emergency loans in the period 2008 to 2010 (Battiston and others, 2012). The authors find that size is not a relevant factor to determine systemic importance.

TABLE 26.2

Selected Institution-Level Systemic Risk Models							
Description	Conditional Value at Risk (CoVaR)	Conditional Risk (CoRisk)	SRISK	Systemic Expected Shortfall	Distress Insurance Premium	Joint Probability of Default	Systemic Contingent Claims Analysis
Systemic risk measure	Value at risk	Value at risk	Expected shortfall	Expected shortfall	Expected shortfall	Conditional probabilities	Expected shortfall
Conditionality	Percentile of individual return	Percentile of joint default risk	Threshold of capital adequacy	Threshold of capital adequacy	Percentage threshold of system return	Various (individual or joint expected losses)	Various (individual or joint expected losses)
Dimensionality	Multivariate	Bivariate	Bivariate	Bivariate	Bivariate	Multivariate	Multivariate
Dependence measure	Linear, parametric	Linear, parametric	Parametric	Empirical	Parametric	Nonlinear, nonparametric	Nonlinear, nonparametric
Method	Panel quantile regression	Bivariate quantile regression	Dynamic conditional correlation (DCC GARCH) and Monte Carlo simulation	Empirical sampling and scaling; Gaussian and power law	Dynamic conditional correlation (DCC GARCH) and Monte Carlo simulation	Empirical copula	Empirical copula
Data source	Equity prices and balance sheet information	CDS spreads	Equity prices and balance sheet information	Equity prices and balance sheet information	Equity prices and CDS spreads	CDS spreads	Equity prices and balance sheet information
Data input	Quasi-asset returns	CDS-implied default probabilities	Quasi-asset returns	Quasi-asset returns	Equity returns and CDS-implied default probabilities	CDS-implied default probabilities	Expected losses ("implicit put option")
Reference	Adrian and Brunnermeier (2008)	Chan-Lau (2010)	Brownlees and Engle (2011)	Acharya, Pedersen, and others (2009, 2010, 2012)	Huang, Zhou, and Zhu (2009, 2010)	Segoviano and Goodhart (2009)	Gray and Jobst (2010, 2011b)

Source: Authors.

during extreme market stresses at a very high statistical confidence level (expressed as “tail risk”).⁶

Because its measure of systemic risk is derived from the joint distribution of individual risk profiles, Systemic CCA addresses the general identification problem of existing approaches. It does not rely on stylized assumptions affecting asset valuation, such as linearity and normally distributed prediction errors, and it generates point estimates of systemic risk without the need of reestimation for different levels of desired statistical confidence. None of the recently published systemic risk models, such as Adrian and Brunnermeier (2008), Acharya, Pedersen, and others (2009, 2010, 2012), and Huang, Zhou, and Zhu (2009, 2010), apply multivariate density estimation, which allows the determination of the marginal contribution of an individual institution to concurrent changes of both the severity of systemic risk and the dependence structure across any combination of sample institutions for any level of statistical confidence.

The chapter is structured as follows: Section 1 explains the general concept of the Systemic CCA framework and provides a detailed description of how joint default risk is modeled via a portfolio-based approach using CCA. Section 2 introduces various extensions to the Systemic CCA framework, including the estimation of contingent claims to the financial sector and the assessment of capital adequacy using market-implied measures of systemwide solvency in the context of macroprudential stress testing. Section 3 presents the estimation results of the Systemic CCA framework for the U.K. banking sector from the IMF’s 2011 Financial Sector Assessment Program (FSAP). The chapter concludes with possible extensions to the presented methodology (and its scope of application) and provides some caveats on systemic risk measurement.

1. METHODOLOGY

A. Overview

Systemic CCA models joint solvency risk by combining the multivariate extension to CCA with the concept of extreme value theory (EVT).⁷ The magnitude of systemic risk jointly posed by multiple firms falling into distress is modeled as a portfolio of individual expected losses (with individual risk parameters) calculated from equity market and balance sheet information. More specifically, these expected losses and the dependence between them are combined to generate a multi-

variate distribution that formally captures the potential of extreme realizations of joint expected losses as a conditional tail expectation (CTE) as a result of systemwide effects on solvency conditions.

The model comprises two sequential estimation steps in order to measure joint market-implied expected losses. First, each firm’s expected losses (and associated change in existing capital levels) are estimated using an enhanced form of CCA, which has been applied widely to measure and evaluate credit risk at financial institutions.⁸ Second, these expected losses are assumed to follow a generalized extreme value (GEV) distribution, which are combined with a multivariate solution by utilizing a novel application of a nonparametric dependence measure in order to derive the amount of joint expected losses (and changes in corresponding capital levels).

In order to understand individual risk exposures, first, CCA is applied to construct risk-adjusted (economic) balance sheets of financial institutions and estimate their expected losses. In its basic concept, CCA quantifies default risk on the assumption that owners of corporate equity in leveraged firms hold a call option on the firm value after outstanding liabilities have been paid off. The pricing of these state-contingent contracts is predicated on the risk-adjusted valuation of the balance sheet of firms whose assets are stochastic and may be above or below promised payments on debt over a specified period of time. When there is a chance of default, the repayment of debt is considered “risky”—to the extent that it is not guaranteed in the event of default—and thus generates expected losses conditional on the probability and the degree to which the future asset value could drop below the contemporaneous debt payment. Higher uncertainty about changes in future asset value, relative to the default barrier, increases default risk, which occurs when assets decline below the barrier. Thus, the expected loss from default can be valued as an implicit put option.

Thus, the marginal distributions of individual expected losses are combined then with their time-varying dependence structure to generate the multivariate distribution of joint expected losses of all sample firms. This approach of analyzing extreme linkages of multiple entities links the univariate marginal distributions in a way that formally captures both linear and nonlinear dependence and its impact on joint tail risk behavior over time. Given that the simple summation of implicit put option values would presuppose perfect correlation, that is, a coincidence of defaults, the correct estimation of aggregate risk requires knowledge about the dependence structure of individual balance sheets and associated expected

⁶ See also Gray and Jobst (2011a, 2011b, 2011c) for a more general application of this approach to the integrated balance sheets of an entire economy. For first versions of the model, see Gray and Jobst (2009a, 2009b, 2010); Gray and Malone (2010); as well as González-Hermosillo and others (2009). An application of the Systemic CCA approach in the context of Financial Sector Assessment Program stress tests and spillover analysis are published in IMF (2010, 2011a, 2011c, 2011d, 2011e, 2011f, 2012a, 2012b). For an earlier application of the methodology underpinning this framework, see also IMF (2009a, 2009b).

⁷ EVT is a useful statistical concept to study the tail behavior of heavily skewed data, which specifies residual risk at high percentile levels through a generalized parametric estimation of order statistics.

⁸ The CCA is a generalization of option pricing theory pioneered by Black and Scholes (1973) and Merton (1973, 1974). It is based on three principles that are applied in this chapter: (1) the values of liabilities are derived from assets; (2) assets follow a stochastic process; and (3) liabilities have different priorities (senior and junior claims). Equity can be modeled as an implicit call option, while risky debt can be modeled as the default-free value of debt less an implicit put option that captures expected losses. In the Systemic CCA model, advance option pricing is applied to account for biases in the Black-Scholes-Merton specification.

losses. The expected losses of each firm are assumed to be “fat-tailed” and fall within the domain of the GEV distribution, which identifies asymptotic tail behavior of normalized extremes.⁹ The joint distribution is estimated iteratively over a prespecified time window with the frequency of updating determined by the periodicity of available data. The specification of a closed-form solution can then be used to (1) determine point estimates of joint expected losses as a multivariate CTE and (2) quantify the contribution of specific institutions to the dynamics of systemic risk.

Systemic CCA accounts for changes in firm-specific and common factors determining individual default risk, their implications for the market-implied linkages between firms, and the resulting impact on the overall assessment of systemic risk. Thus, it accomplishes the essential goal of measuring the extent to which an institution contributes to systemic risk (in keeping with the contribution approach as shown in Table 26.1).¹⁰ The systemic dimension of the model is captured by three properties:

1. *Drawing on the market’s evaluation of a firm’s risk profile.* Given that the framework combines market prices and balance sheet information to inform a risk-adjusted measure of individual default risk, the evaluation of solvency conditions is inherently linked to investor perception as implied by the institution’s equity and equity options (which determine the implied asset value of the firm conditional on its leverage and debt level).
2. *Controlling for common factors affecting the firm’s solvency.* The implied asset value of a firm (which underpins the estimation of implied expected losses) is modeled as being sensitive to the same markets as the implied asset value of every other institution but by varying degrees because of a particular capital structure (and its implications for the market’s perception). Changes in market conditions (and their impact on the perceived risk profile of each firm via its equity price and volatility) establish market-induced linkages among sample firms. Thus, measuring the joint expected losses via option prices links institutions implicitly to the markets in which they obtain equity capital and funding.
3. *Quantifying the chance of simultaneous default via joint probability distributions.* The probability that multiple firms experience a realization of expected losses simultaneously is made explicit by computing their joint probability distributions (which also account for differences in the magnitude of individual expected losses). Hence, the default risk—that is, the likelihood that the implied asset value (including avail-

able cash flows from operations and asset sales) falls below the amount of required funding to satisfy debt payment obligations—is assessed not only for individual institutions but for all firms within a system in order to generate estimates of systemic risk.

B. Model specification and estimation steps

The Systemic CCA framework follows a two-step estimation process. First, the changes to default risk causing a bank to fail (i.e., future debt payments exceeding the asset value of the firm) are modeled as a put option in order to estimate the market-implied expected losses over a certain time horizon (consistent with the application of CCA). Second, these individually estimated expected losses are combined into a multivariate distribution that determines the probabilistic measure of joint expected losses at a systemwide level for a certain level of statistical confidence. Finally, the sensitivity of systemic solvency risk to changes in individual default risk of a single firm (and its implications for the dependence structure of expected losses across all firms) quantifies the degree to which a particular firm contributes to total default risk.

Calculating Expected Losses from CCA Using Option Pricing

CCA is a generalization of the option pricing theory pioneered by Black and Scholes (1973) as well as by Merton (1973, 1974) to the corporate capital structure context. When applied to the analysis and measurement of credit risk, it is commonly called the Black-Scholes-Merton (BSoM) model.¹¹ In its basic concept, it quantifies default risk on the assumption that owners of equity in leveraged firms hold a call option on the firm value after outstanding liabilities have been paid off. The concept of a risk-adjusted balance sheet is instrumental in understanding default risk. The total market value of firm assets, A , at any time, t , is equal to the sum of its equity market value, E , and its risky debt, D , maturing at time T . The asset value follows a random, continuous process and may fall below the bankruptcy level (“default threshold” or “distress barrier”), which is defined as the present value of promised payments on risky debt B , discounted at the risk-free rate. The variability of the future asset value relative to promised debt payments is the driver of credit and default risk. Default happens when assets are insufficient to meet the amount of debt owed to creditors at maturity. Thus, the market-implied

⁹ The choice of the empirical distribution function of the underlying data to model the marginal distributions avoids problems associated with using specific parameters that may or may not fit these distributions well (Gray and Jobst, 2009a, 2009b).

¹⁰ See also Tarashev, Borio, and Tsatsaronis (2009).

¹¹ The Merton approach assumes that the firm’s debt consists of a zero-coupon bond B with a notional value F and a maturity term of T periods. The firm’s outstanding liabilities constitute the bankruptcy level, whose standard normal density defines the “distance to default” relative to the firm value. This capital structure–based evaluation of contingent claims on firm performance implies that a firm defaults if its asset value is insufficient to meet the amount of debt owed to bondholders at maturity. Conversely, if the distance to default is positive, and the asset value of the firm exceeds the bankruptcy level, the call option held by equity holders on firm value has intrinsic value (in addition to its time value until the maturity of debt).

expected losses associated with outstanding liabilities can be valued as an implicit put option (and its cost reflected in a credit spread above the risk-free rate that compensates investors for holding risky debt). The put option value is influenced by the duration of the total debt claim, the leverage of the firm, and the volatility of its asset value.

In the traditional definition of bank balance sheets, a change in accounting assets entails a one-for-one change in book equity. Assets comprise cash and cash equivalents, investments, loans, mortgages, and other cash claims on counterparties as well as noncash claims (derivatives and contingent assets), and liabilities consist of book equity and book value of debt and deposits. When assets change, the full change affects book equity (see Table 26.3). In the conventional definition of credit risk, the concept of “expected losses” refers to impairments arising from exposures on the asset side of the bank’s balance sheet. In this context, expected loss is calculated as the probability of default (PD) times a loss-given-default (LGD) times the exposure at default. The expected losses of different exposures are aggregated (using certain assumptions regarding correlation, etc.) and used as an input into loss distribution calculations, which are in turn used for the estimation of risk-weighted assets (RWAs) and the assessment of regulatory capital adequacy.

CCA constructs a risk-adjusted (economic) balance sheet by transforming accounting identities into exposures so that changes in the value of assets are linked directly to changes in the market value of equity and expected losses in an *integrated* framework. A decline in the value of assets increases expected losses to creditors and leads to less than one-to-one decline in the market value of equity; the amount of change in equity depends on the severity of financial distress (i.e., the perceived impact of anticipated operating—or accounting—losses on equity returns), the degree of leverage, and the volatility of asset returns over time. The underlying “exposure” is represented by the default-free value of the bank’s total debt and deposits after accounting for the (now) higher probability of assets creating insufficient cash flows to meet debt payments (without compromising the integrity of the deposit base) (see Table 26.4). Thus, the integrated modeling of default risk via CCA can quantify the impact on bank borrowing costs of higher (or lower) levels of equity and the market-implied capital assessment. A lower market value of equity would increase the probability of a bank to generate capital losses, which is directly related to higher funding costs.

TABLE 26.3**Traditional Accounting Bank Balance Sheet**

Assets	Liabilities
Accounting assets (i.e., cash, reserves, loans, credits, and other exposures)	Debt and deposits Book equity

Source: Authors.

TABLE 26.4**Risk-Adjusted (CCA) Bank Balance Sheet**

Assets	Liabilities
(Implied) market value of assets (A) (i.e., cash, reserves, and implied market value of “risky” assets)	“Risky” debt (D) (= default-free value of debt and deposits minus expected losses to bank creditors) (Observable) market value of equity (E) (= market capitalization)

Source: Authors.

Note: CCA = contingent claims analysis.

First, the amount of expected loss is modeled as a put option based on the individual risk-adjusted balance sheet. The expected loss can be quantified then by viewing default risk as if it were a put option written on the amount of outstanding liabilities, where the present value of debt (i.e., the default barrier) represents the “strike price,” with the value and volatility of assets determined by changes in the equity and equity option prices of the firm. The value of the put option increases the higher the probability of the implied asset value falls below the default barrier over a predefined horizon. Such probability is influenced by changes in the level and the volatility of their implied asset value reflected in the institution’s equity and equity option prices conditional on its capital structure. Thus, the present value of market-implied expected losses associated with outstanding liabilities in keeping with the traditional BSM model can be valued as an implicit (European) put option value:

$$P_E(t) = Be^{-r(T-t)}\Phi\left(-\left(d - \sigma_A\sqrt{T-t}\right)\right) - A(t)\Phi(-d), \quad (26.1)$$

with the present value of debt B as strike price on the asset value $A(t)$ and asset return volatility:

$$\sigma_A = \frac{E(t)\sigma_E}{A(t)\Phi(d)} = \left(1 - \frac{Be^{-r(T-t)}\Phi\left(d - \sigma_A\sqrt{T-t}\right)}{A(t)\Phi(d)}\right)\sigma_E, \quad (26.2)$$

over time horizon $T-t$, where σ_E is the (observable) equity volatility, $E(t)$ is the equity price, r is the risk-free discount rate, and $\Phi(d)$ is the cumulative probability of the standard normal density function of the leverage:

$$d = \left(\ln\left(\frac{A(t)}{B}\right) + \left(r + \frac{\sigma_A^2}{2}\right)(T-t)\right) / \sigma_A\sqrt{T-t}, \quad (26.3)$$

changes in $A(t)$ relative to outstanding debt claims. This specification of option price-based expected losses, however, does not incorporate skewness, kurtosis, and stochastic volatility, which can account for implied volatility smiles of equity prices. These shortcomings of the conventional Merton model can be addressed with more advanced valuation models, such as the Gram-Charlier expansion of the density function of asset changes (Backus, Foresi, and Wu, 2004) or a jump diffusion process of asset value changes in order to achieve

more robust and reliable estimation results.¹² Some of these approaches have been applied in the empirical use of the Systemic CCA framework in this chapter.¹³

Given that the asset value is influenced by the empirical irregularities contained in the Merton model (which also affects the model-based calibration of implied asset volatility), it is estimated directly from observable equity option prices.¹⁴ The state-price density (SPD) of the implied asset value is estimated from the risk-neutral probability distribution of the underlying asset price at the maturity date of equity options with different strike prices, without any assumptions on the underlying asset diffusion process (which is assumed to be log-normal in the Merton model). The implied asset value is defined as the expectation over the empirical SPD by adapting the well-established Breeden and Litzenberger (1978) method, together with a semiparametric specification of the Black-Scholes option pricing formula (Ait-Sahalia and Lo, 1998). More specifically, this approach uses the second derivative of the call pricing function (on European options) with respect to the strike price (rather than option price). Estimates are based on option contracts with identical time to maturity, assuming a continuum of strike prices.¹⁵ The SPD of the equity price includes the risk-neutral expectation of the implied asset value and removes the impact of differences of higher moments of both the equity price and the implied asset value dynamics over the chosen time horizon. Thus, the current implied asset value can be transposed from the SPD of the equity price conditional on the contemporaneous leverage of the firm.

Note, however, that the presented valuation model is subject to varying degrees of estimation uncertainty and parametric assumptions. The option pricing model (given its specific distributional assumptions, the derivation of both implied as-

sets and asset volatility, and assumptions about the default barrier) could fail to capture some relevant economics that are needed to fully understand default risk and thus could generate biased estimators of expected losses. Moreover, equity prices might not only reflect fundamental values because of both shareholder dilution and trading behavior that obfuscate proper economic interpretation.¹⁶

Estimating the Joint Expected Losses from Default Risk

Second, the individually estimated expected losses are combined to determine the magnitude of default risk on a systemwide level. The distribution of expected losses is assumed to be “fat-tailed” in keeping with EVT. This specification of individual loss estimates then informs the estimation of the dependence function. As part of a three-step subprocess, we define the univariate marginal density functions of all firms, which are then combined with their dependence function in order to generate an aggregate measure of default risk. This multivariate setup underpinning the Systemic CCA framework formally captures the realizations of joint expected losses. We can then use tail risk estimates, such as the conditional VaR (or expected shortfall, or ES), in order to gauge systemic solvency risk in times of stress.

1. **Estimating the Marginal Distributions of Individual Expected Losses.** We first specify the statistical distribution of individual expected losses (based on the series of put option values obtained for each firm) in accordance with EVT. This is a general statistical concept of deriving a limit law for sample maxima. Let the vector-valued series

$$\mathbf{X}_m^n = P_{E,1}^n(t), \dots, P_{E,m}^n(t) \quad (26.4)$$

denote independent, identically distributed random observations of expected losses (i.e., a total of n -number of daily put option values $P_{E,m}^n(t)$ up to time t), each estimated according to equation (26.1) over a rolling window of τ observations with periodic updating (e.g., a daily sliding window of 120 days) for $j \in m$ firms in the sample. The individual asymptotic tail behavior is modeled in accordance with the Fisher-Tippett-Gnedenko theorem (Fisher and Tippett, 1928; Gnedenko, 1943), which defines the attribution of a given distribution of normalized maxima (or minima) to be of extremal type (assuming that the underlying function is continuous on a closed interval). Given

$$\mathbf{X}^n = \max(P_{E,1}^n(t), \dots, P_{E,j}^n(t)), \quad (26.5)$$

there exists a choice of normalizing constants $\beta^n > 0$ and α^n , such that the probability of each ordered n -sequence of normalized sample maxima $(\mathbf{X}^n - \alpha^n)/\beta^n$ converges to the nondegenerate limit distribution $H(x)$ as $n \rightarrow \infty$ and $x \in \mathbb{R}$, so that

¹² The suggested approach takes into account the nonlinear characteristics of asset price changes, which deviate from normality and are frequently influenced by stochastic volatility. The BSM model has been shown to consistently understate spreads (Jones, Mason, and Rosenfeld, 1984; Ogden, 1987), with more recent studies pointing to considerable pricing errors because of its simplistic nature.

¹³ Other extensions are aimed at imposing more realistic assumptions, such as the introduction of stationary leverage ratios (Collin-Dufresne and Goldstein, 2001) and stochastic interest rates (Longstaff and Schwartz, 1995). Incorporating early default (Black and Cox, 1976) does not represent a useful extension in this context given the short estimation and forecasting time window.

¹⁴ Given that the implied asset value is derived separately, this approach avoids the (problematic) “two-equations-two-unknowns” approach to derive implied assets and asset volatility based on Jones, Mason, and Rosenfeld (1984), which was subsequently extended by Ronn and Verma (1986) to a single equation to solve two simultaneous equations for asset value and volatility as two unknowns. Duan (1994) shows that the volatility relation between implied assets and equity could become redundant if the equity volatility is stochastic. An alternative estimation technique for asset volatility introduces a maximum likelihood approach (Ericsson and Reneby, 2004, 2005).

¹⁵ Because available strike prices always are discretely spaced on a finite range around the actual asset value, interpolation of the call pricing function inside this range and extrapolation outside this range are performed by means nonparametric (local polynomial) regression of the implied volatility surface (Rookley, 1997).

¹⁶ For instance, during the financial crisis, rapid declines in market capitalization of firms were not only a signal about future solvency risk but also reflected a “flight to quality” motive that was largely unrelated to expectations about future firm earnings or profitability.

$$F_n^{[\beta^n x + \alpha^n]}(x) = \lim_{n \rightarrow \infty} \Pr\left(\frac{\mathbf{X}^n - \alpha^n}{\beta^n} \leq y\right) \\ = [F(\beta^n y + \alpha^n)]^n \rightarrow H(x) \quad (26.6)$$

falls within the maximum domain of attraction (MDA) of the GEV as limiting distribution of maxima of dependent random variables (Coles, Heffernan, and Tawn, 1999; Poon, Rockinger, and Tawn, 2003; Jobst, 2007) with the cumulative distribution function

$$H_{\mu, \sigma, \xi}(x) = \begin{cases} \exp\left(-\left(1 + \frac{\xi(x-\mu)}{\sigma}\right)^{-1/\xi}\right) & \text{if } 1 + \frac{\xi(x-\mu)}{\sigma} \geq 0 \\ \exp\left(-\exp\left(-\frac{x-\mu}{\sigma}\right)\right) & \text{if } x \in \mathbb{R}, \xi = 0 \end{cases} \quad (26.7)$$

and differencing equation (26.7) above as $H'_{\mu, \sigma, \xi}(x) = \frac{d}{dx} H_{\mu, \sigma, \xi}(x)$ yields the probability density function,

$$h_{\mu, \sigma, \xi}(x) = \frac{1}{\sigma} \left(1 + \xi \frac{(x-\mu)}{\sigma}\right)^{(-1/\xi)-1} \\ \times \exp\left(-\left(1 + \xi \frac{(x-\mu)}{\sigma}\right)^{-1/\xi}\right)_+. \quad (26.8)$$

Thus, the j th univariate marginal density function of each expected loss series converging to GEV in the limit is defined as

$$y_j(x) = \left(1 + \hat{\xi}_j \frac{(x - \hat{\mu}_j)}{\hat{\sigma}_j}\right)^{-1/\hat{\xi}_j}_+ \quad (\text{for } j = 1, \dots, m) \quad (26.9)$$

where $1 + \hat{\xi}_j (x - \hat{\mu}_j)/\hat{\sigma}_j > 0$, scale parameter $\hat{\sigma}_j > 0$, location parameter $\hat{\mu}_j \in \mathbb{R}$, and shape parameter $\hat{\xi}_j \neq 0$.¹⁷ These moments are estimated concurrently by means of numerical iteration via maximum likelihood (ML), which identifies possible limiting laws of asymptotic tail behavior, that is, the likelihood of even larger extremes as the level of statistical confidence approaches certainty. The ML estimator in the GEV model is evaluated numerically by using an iteration procedure (e.g., over a rolling window of τ observations with periodic updating) to maximize the likelihood $\prod_{i=1}^n h_{\mu, \sigma, \xi}(x_i | \theta)$ over all three parameters $\theta = (\mu, \sigma, \xi)$ simultaneously, where the *linear combinations of ratios of spacings* estimator serves as an initial value (Jobst, 2014).¹⁸

2. Estimating the Dependence Structure of Individual Expected Losses. Second, we define a nonparametric, multivariate dependence function between the marginal distributions of expected losses by expanding the bivariate logistic method proposed by Pickands (1981) to the multivariate

case and adjusting the margins according to Hall and Tajvidi (2000) so that

$$\Upsilon(\omega) = \min\left\{1, \max\left\{n \left(\sum_{i=1}^n \bigwedge_{j=1}^m \frac{y_{i,j}/\hat{y}_{\cdot,j}}{\omega_j}\right)^{-1}, \omega, 1-\omega\right\}\right\}, \quad (26.10)$$

where $y_{\cdot,j} = \sum_{i=1}^n y_{i,j}/n$ reflects the average marginal density of all $i \in n$ put option values and $0 \leq \max(\omega_1, \dots, \omega_{m-1}) \leq \Upsilon(\omega_j) \leq 1$ for all $0 \leq \omega_j \leq 1$.¹⁹ $\Upsilon(\bullet)$ represents a convex function on $[0,1]$ with $\Upsilon(0) = \Upsilon(1) = 1$, that is, the upper and lower limits of $\Upsilon(\bullet)$ are obtained under complete dependence and mutual independence, respectively. It is estimated iteratively (over a rolling window of τ observations with periodic updating at a frequency that is consistent with that in equation (26.5), subject to the optimization of the $(m-1)$ -dimensional unit simplex:

$$S_m = \left\{(\omega_1, \dots, \omega_{m-1}) \in \mathbb{R}_+^m; \omega_j \geq 0, 1 \leq j \leq m-1; \sum_{j=1}^{m-1} \omega_j \leq 1 \text{ and } \omega_m = 1 - \sum_{j=1}^{m-1} \omega_j\right\}, \quad (26.11)$$

which establishes the degree of coincidence of multiple series of cross-classified random variables similar to a chi-statistic that measures the statistical likelihood that observed values differ from their expected distribution. This specification stands in contrast to a general copula function that links the marginal distributions using only a single (and time-invariant) dependence parameter.

3. Estimating the Joint Distribution and a Tail Risk Measure of Joint Expected Losses. We then combine the marginal distributions of these individual expected losses with their dependence structure to generate a *multivariate extreme value distribution* over the same estimation period as earlier.²⁰ The resultant cumulative distribution function is specified as

$$G_{t,m}(x) = \exp\left\{-\left(\sum_{j=1}^m y_{t,j}\right) \Upsilon_t(\omega)\right\}, \quad (26.12)$$

¹⁹ Note that the marginal density of a given extreme relative to the average marginal density of all extremes is minimized (" $\wedge$ ") across all firms $j \in m$, subject to the choice of factor ω_j . A bivariate version of this approach has been implemented in Jobst and Kamil (2008).

²⁰ The analysis of dependence in this presented approach is completed separately from the analysis of marginal distributions and thus differs from the classical approach, where multivariate analysis is performed jointly for marginal distributions and their dependence structure by considering the complete variance-covariance matrix, using techniques like the Multivariate Generalized Autoregressive Conditional Heteroskedasticity approach. To obtain a multivariate distribution function, the dependence function combines several marginal distribution functions in accordance with Sklar's (1959) theorem on constructing joint distributions with arbitrary marginal distributions via copula functions (which completely describes the dependence structure and contains all the information to link the marginal distributions). For the formal treatment of copulas and their properties, see Hutchinson and Lai (1990); Dall'Aglia, Kotz, and Salinetti (1991); and Joe (1997).

¹⁷ The upper tails of most (conventional) limit distributions (weakly) converge to this parametric specification of asymptotic behavior, irrespective of the original distribution of observed maxima (unlike parametric VaR models).

¹⁸ Note that the ML estimator fails for $\xi \leq -1$ since the likelihood function does not have a global maximum in this case. However, a local maximum close to the initial value can be attained.

with corresponding probability density function

$$g_{t,m}(x) = \frac{1}{\hat{\sigma}_{t,m}} \left(\left(\sum_{j=1}^m y_{t,j} \right) \Upsilon_t(\omega) \right)^{-1} \exp \left\{ - \left(\sum_{j=1}^m y_{t,j} \right) \Upsilon_t(\omega) \right\}, \quad (26.13)$$

at time $t = \tau + 1$ by maximizing the likelihood $\prod_{j=1}^m g_{t,m}(x | \theta)$ over all three parameters $\theta = (\mu, \sigma, \xi)$ simultaneously so that the ML estimate of the true values θ_0 is

$$\hat{\theta}_{MLE} = \arg \max_{\theta \in \Theta} \hat{\ell}(\theta | x) \rightarrow \theta_0. \quad (26.14)$$

Finally, we obtain the joint ES (or conditional VaR) as a measure of CTE. ES defines the probability-weighted residual density (i.e., the average aggregate expected losses) beyond a prespecified statistical confidence level (“severity threshold”) over a given estimation time period. Given the dependence structure defined in equation (26.10), we can calculate ES in a multivariate context as

$$ES_{t,\tau,m,a} = \mathbb{E} \left[z_\tau | z_\tau \geq G_{t,\tau,m}^{-1}(a) = VaR_{t,a} \right], \quad (26.15)$$

where $G^{-1}(a)$ is the quantile corresponding to probability $0 < a < 1$ (say, $a = 0.95$), $z \in \mathbb{R}$ and $G^{-1}(a) = G^{\leftarrow}(a)$ with $G^{\leftarrow}(a) \equiv \inf \{x | G(x) \geq a\}$ so that

$$VaR_{t,a} = \sup \{ G_{t,\tau,m}^{-1}(a) | \Pr[z_\tau > G_{t,\tau,m}^{-1}(a)] \geq a = 0.95 \}, \quad (26.16)$$

with the point estimate of joint potential losses of m firms at time t defined as

$$G_{t,\tau,m}^{-1}(a) = \hat{\mu}_{t,m} + \hat{\sigma}_{t,m} / \hat{\xi}_{t,m} \left(\left(- \frac{\ln(a)}{\Upsilon_t(\omega)} \right)^{-\frac{1}{\hat{\xi}_{t,m}}} - 1 \right)^{21} \quad (26.17)$$

The contribution of each firm can be determined by calculating the cross-partial derivative of the joint distribution of expected losses. The joint ES also can be written as a linear combination of individual ES values, $ES_{t,\tau,j,a}$, where the relative weights $\psi_{\tau,m,a}$ (in the weighted sum) are given by the second order cross-partial derivative of the inverse of the joint probability density function $G_{t,\tau,m}^{-1}(\bullet)$ to changes in both the dependence function $\Upsilon_t(\bullet)$ and the individual marginal severity $y_{i,j}$ of expected losses. Thus, the contribution can be derived as the partial derivative of the multivariate density to changes in the relative weight of the univariate marginal distribution of expected losses and its impact on the dependence function (of all expected losses of sample firms) at the specified percentile. By rewriting $ES_{t,\tau,m,a}$ in equation (26.15), we obtain the following:

$$ES_{t,\tau,m,a} = \sum_j^m \psi_{\tau,m,a} \mathbb{E} \left[z_{\tau,j} | z_{\tau,m} \geq G_{t,\tau,m}^{-1}(a) = VaR_{t,a} \right], \quad (26.18)$$

²¹ ES is an improvement over VaR, which, in addition to being a pure frequency measure, is “incoherent,” that is, it violates several axioms of monotonicity, subadditivity, positive homogeneity, and translation invariance found in coherent risk measures. For the application of this kind of coherent risk measure, we refer to Artzner and others (1999) as well as Wirth and Hardy (1999).

where the relative weight of institution j is defined as the marginal contribution,

$$\psi_{\tau,m,a} = \frac{\partial^2 G_{t,\tau,m}^{-1}(a)}{\partial y_{t,j} \partial \Upsilon_t(\omega)} \quad \text{s.t.} \quad \sum_j^m \psi_{\tau,j,a} = 1 \text{ and } \psi_{\tau,j,a} z_{\tau,j} \leq z_{\tau,m}, \quad (26.19)$$

attributable to the joint effect of both the marginal density and the change of the dependence function owing to the presence of institution $j \in m$ in the sample.

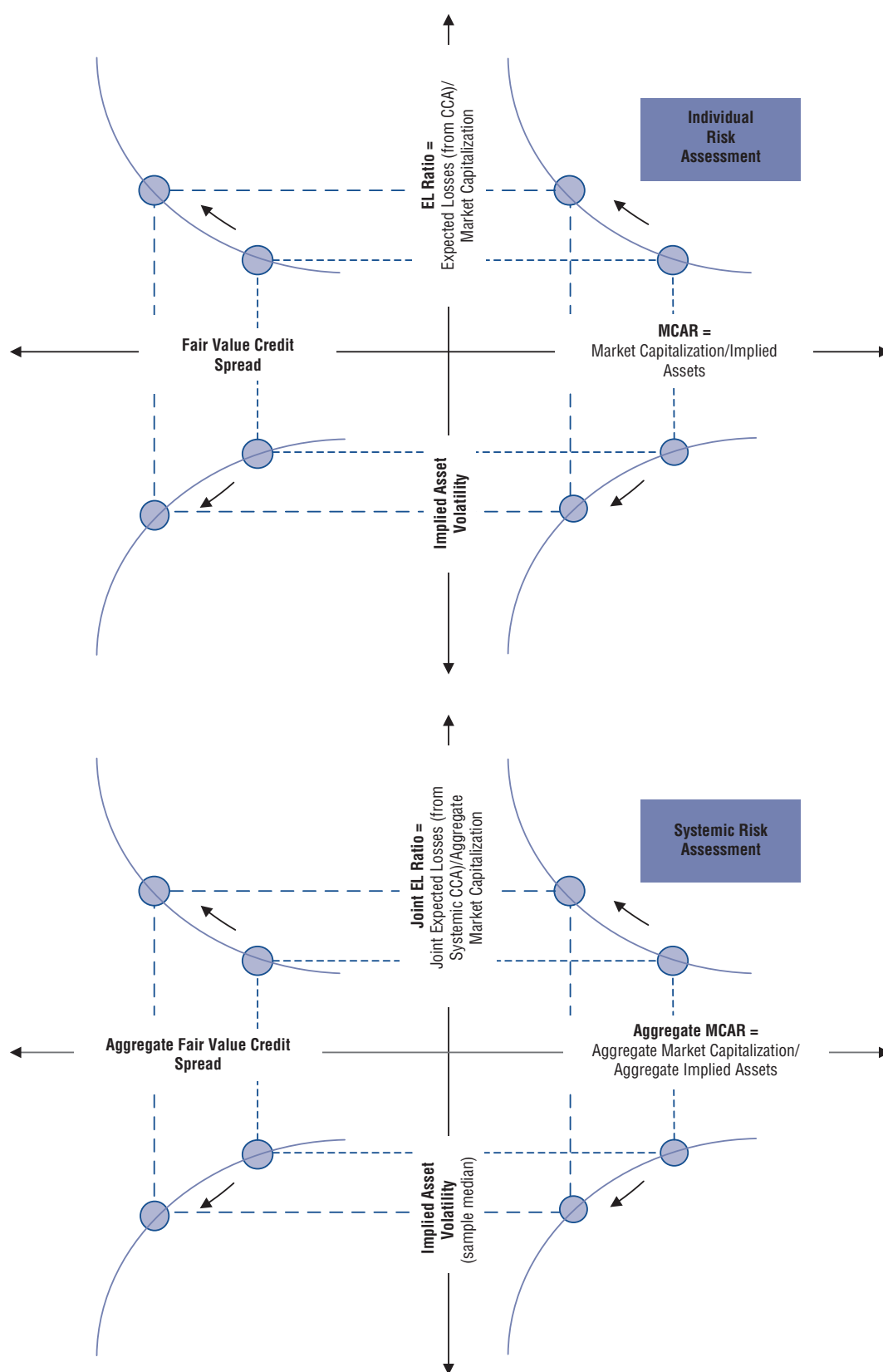
2. EXTENSIONS OF THE SYSTEMIC CCA FRAMEWORK

A. Integrated market-implied capital assessment using CCA and Systemic CCA

Given that the evaluation of default risk is inherently linked to perceived riskiness as implied by the changes in equity and equity option prices, the risk-adjusted (economic) balance sheet approach—and its multivariate extension in the form of Systemic CCA—provides an integrated analytical framework for a market-based assessment of individual and systemwide solvency. Higher expected losses to creditors lead to less than a one-to-one decline in the market value of equity, depending on the severity of the perceived (and actual) decline in the value of assets, the degree of leverage, and the volatility of assets. Thus, the interaction between these constituent elements of the risk-adjusted (economic) balance sheet gives rise to a market-based capital assessment, which links changes in the value of assets directly to the market value of equity.

As opposed to the regulatory definition of capitalization (based on the discrete nature of accounting identities), the CCA-based capital assessment hinges on the historical dynamics of implied asset values and their effect on the magnitude of expected losses. Two broad indicators serve to illustrate this point: (1) the *market-implied capital adequacy ratio* (MCAR), which is defined as the ratio between the amount of market capitalization divided by the implied asset value of a firm; and (2) the *expected loss ratio* (“EL ratio”) between individual expected losses (at a defined percentile of statistical confidence) and the amount of market capitalization (see Figure 26.1, first panel). Both ratios also can be generated on a systemwide basis, with inputs defined as sample aggregates of input variables to the multivariate specification of the Systemic CCA. In this case, the EL ratio is defined by magnitude of joint expected losses (at the 95th percentile, for instance) relative to aggregate market capitalization of all sample firms (see Figure 26.1, second panel).

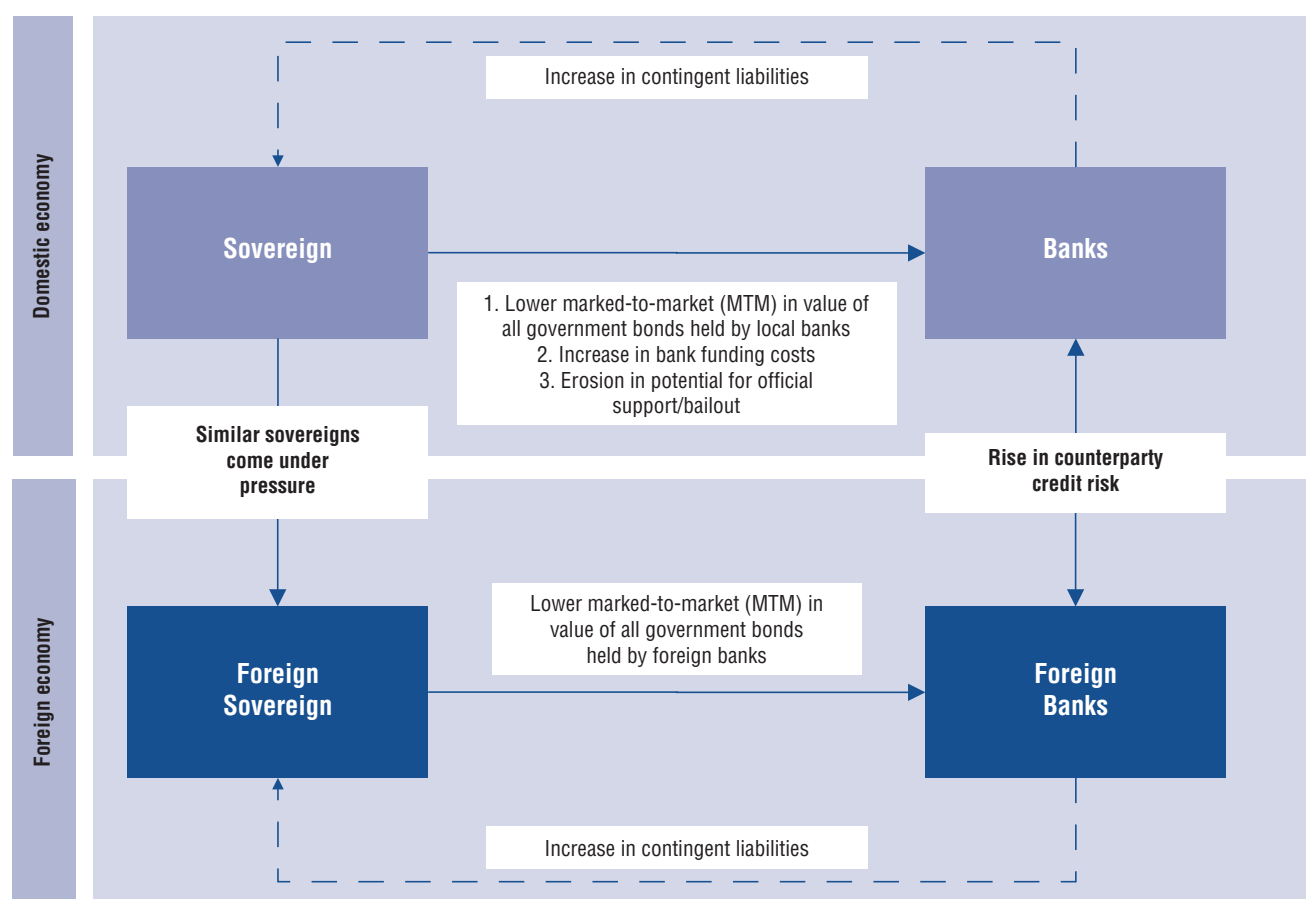
The market-implied capital assessment also can help identify a capital shortfall. The MCAR represents an important analogue to the regulatory definition of capital adequacy, which reflects the market’s perception of solvency (based on the implied value of assets relative to the prevailing default risk) and is completely removed from prudential determinants of default risk affecting a bank’s capital assessment,



Source: Authors.

Note: CAR = capital adequacy ratio; CCA = contingent claims analysis; EL = expected losses; MCAR = market-implied CAR.

Figure 26.1 Integrated Market-Based Capital Assessment Using CCA and Systemic CCA (based on nonlinear relation between CCA-based CAR, EL ratio, fair value credit spread, and implied asset volatility)



Source: Authors.

Figure 26.2 Valuation Linkages between the Sovereign and Banking Sector

such as risk-weighted assets. In the empirical section of the chapter, the relation between the MCAR and regulatory capital adequacy is investigated further, which establishes the possibility of defining a systemically based capital shortfall using CCA-based measures of solvency (see Section 3).

B. Estimating the contingent liabilities to the public sector

Large implicit government financial guarantees (valued as contingent liabilities) create significant valuation linkages between sovereigns and financial sector risks. These linkages could give rise to a destabilization process that increases the susceptibility of public finances to the potential impact of distress in an outsized financial sector (see Figure 26.2). The financial crisis that started in 2007 is a stark reminder of how public sector support measures provided to large financial institutions can result in considerable risk transfer to the government, which places even greater emphasis on long-term fiscal sustainability. A negative feedback loop between the financial sector risks and fiscal policy arises because a decline in the market value of sovereign debt not only weakens the implicit sovereign guarantees to systemically important finan-

cial institutions (and large banks in particular) but also raises default risk in the financial sector overall (which increases the cost of borrowing). There is also a higher likelihood of downgrades to financial institutions following downgrades of sovereign credit ratings, which establish an upper ceiling to their unsecured funding ratings. This linkage between the credit-worthiness of the sovereign and the financial system (in particular, the banking sector) is prone to perpetuate a negative feedback effect that is accentuated by the lender–borrower channel in situations where financial institutions hold large exposures to public sector entities.

Based on CCA, the market-implied expected losses calculated from equity market and balance sheet information of financial institutions can be combined with information from their credit default swap (CDS) contracts to estimate the government's contingent liabilities.²² Because systemically important institutions enjoy an implicit government guarantee, their creditors have an expectation of public sector support in the

²² See Gray and Jobst (2009b). See also Gapen (2009) for an application on CCA to the measurement of contingent liabilities from government-sponsored enterprises in the United States.

case of failure.²³ Conversely, the cost of insuring the nonguaranteed debt (as reflected in the senior CDS spread) should capture only the expected loss *retained* by such financial institutions (and borne by unsecured senior creditors), after accounting for any implicit public sector support that is due to their systemic significance. Thus, the market-implied government guarantee for large banks (as prime candidates for the status of systemically important financial institutions) can be derived heuristically as the difference between the total expected loss (i.e., the value of a put option $P_E(t)$ derived from the bank's equity price) and the value of an implicit put option $P_{CDS}(t)$ derived from the bank's CDS spread—assuming that equity values remain unaffected (which is very likely when banks are close to the default barrier, as the call option value from the residual claim of equity to future profits after debt payment converges to zero).

By replacing individual expected losses with this measure of contingent liabilities, the Systemic CCA framework can be used to derive an estimate of systemic risk from joint contingent liabilities. Given that the put option value $P_{CDS}(t) \leq P_E(t)$, it reflects the expected losses associated with default net of any financial guarantees, that is, residual default risk on unsecured senior debt. $P_{CDS}(t)$ can be written as

$$P_{CDS}(t) = \left(1 - \exp\left(-\left(\frac{s_{CDS}(t)}{10,000}\right) \times \left(\frac{B}{D(t)} - 1\right)(T-t)\right) \right) Be^{-r(T-t)}, \quad (26.20)$$

after rearranging the specification of the CDS spread (in basis points), $s_{CDS}(t)$, under the risk-neutral measure as

$$s_{CDS}(t) = -\frac{1}{T-t} \ln\left(1 - \frac{P_{CDS}(t)}{Be^{-r(T-t)}}\right) \times \left(\frac{B}{D(t)} - 1\right) \times 10,000, \quad (26.21)$$

assuming a default probability,

$$1 - \exp\left(-\int_0^t \varphi(u) du\right) = 1 - \exp(-\varphi t) \equiv \frac{P_{CDS}(t)}{Be^{-r(T-t)}}, \quad (26.22)$$

at time t , constant hazard rate $s_{CDS}(t) = \varphi$, and the implied yield to maturity defined by

$$D(t) = Be^{-y(T-t)} \Leftrightarrow e^{-y(T-t)} = \frac{D(t)}{B} = \frac{Be^{-r(T-t)} - P_E(t)}{B}, \quad (26.23)$$

so that $(1 - s_{CDS}(t))Be^{-r(T-t)} = PD \times LGD$ over one period $T-t=1$. A linear adjustment of $B/D(t) - 1$ is needed in order to control for the impact of any difference between the market price and the repayable face value of outstanding debt on the fair pricing of credit protection. If outstanding debt trades above par, $D(t) > B$, the recovery of the nominal value under the CDS contract ("recovery at face value") drops below the recovery rate implied by the market price of debt ("recovery at market value"). As a result, the CDS spread, $s_{CDS}(t)$, drops

below the bond spread.²⁴ Conversely, a below-par bond price pushes up the implied recovery rate of CDS, causing a markup of CDS spreads over comparable bond spreads for the same estimated default risk.

Estimated expected losses derived from both equity prices and credit spreads can be combined to derive the market-implied degree to which potential public sector support depresses the cost of credit. Given $P_{CDS}(t)$, the ratio

$$\alpha(t) = 1 - \frac{P_{CDS}(t)}{P_E(t)}, \quad (26.24)$$

defines the share of expected loss covered by implicit (or explicit) government guarantees that depress the CDS spread below the level that would be warranted by the equity-implied default risk.²⁵ The following equation,

$$\alpha(t)P_E(t) = \left(1 - \frac{P_{CDS}(t)}{P_E(t)}\right)P_E(t) = P_E(t) - P_{CDS}(t), \quad (26.25)$$

represents the fraction of default risk covered by the government (and no longer reflected in the market-implied default risk), and $(1 - \alpha(t))P_E(t)$ is the risk retained by an institution and reflected in the senior CDS spread. Thus, the time pattern of the government's contingent liabilities then can be used to replace the marginal densities of expected losses in equation (26.9) in order to derive the systemic risk from the implicit public sector support to the banking sector according to the specification in equation (26.12).

Although this definition of market-implied contingent liabilities provides a useful indication of possible sovereign risk transfer, the presented estimation method depends on a variety of assumptions that influence the assessment of the likelihood of public sector support, especially at times of extreme stress. The extent to which the equity put option values differ from the one implied by CDS spreads might reflect distortions stemming from the modeling choice (and the breakdown of efficient asset pricing in situations of illiquidity), changes in market conditions, and the capital structure impact of government interventions, such as equity dilution in the wake of capital injections, beyond the influence of explicit or implicit guarantees.

Even though the equality condition of default probabilities derived from equity prices and CDS spreads implies that positive $\alpha(t)$ values cannot exist in absence of arbitrage, empirical

²³ As a result, the cost of funding for these institutions remains below the level commensurate to their actual default risk.

²⁴ The difference in recovery values (often referred to as "basis") implied by a divergence of CDS and bond spreads is approximated by using the fair value CDS (FVCDS) spread and the fair value option adjusted spread (FVOAS) reported by Moody's KMV (MKMV) *CreditEdge*. Both FVOAS and FVCDS represent credit spreads (in basis points) for the bond and CDS contract of a particular firm based on risk horizon of t years (where $t=1-10$ years). Both spreads imply an LGD determined by the industry category. In practice, this adjustment factor is very close to unity for most of the cases, with a few cases where the factor falls within a 20 percentage point range (0.9 to 1.1).

²⁵ Note that the estimation assumes a European put option, which does not recognize the possibility of premature execution. This might overstate the actual expected losses inferred from put option values in comparison with the put option derived from CDS spreads.

evidence during times of stress suggest otherwise.²⁶ Carr and Wu (2007) show that for many firms the put option values from equity options and CDS are, indeed, closely related.²⁷ In stress situations, however, the implicit put options from equity markets and CDS spreads can differ in their capital structure impact and thus should be priced differently. Besides guarantees, there are several distortions that could set apart put option values derived from CDS and equity prices, even if the risk-neutral default probability (RNDP) implied by the CDS spread (based on an exponential hazard rate) were the same as the RNDP component of the equity put option value. Some of these factors include (1) the recovery-at-face-value assumption underlying CDS spreads, and (2) different risk horizons of put option values derived from CDS and equity prices. We address these two potential sources of distortion via an adjustment for “basis risk” in equation (26.20).

C. Systemic CCA and stress testing

The Systemic CCA framework also can be used to project the dynamics of systemic solvency risk for the purposes of assessing capital adequacy under stress scenarios. Based on the historical distribution of firm-specific market-implied expected losses (which can be generated using conventional or more advanced option pricing techniques, future changes in market-implied default risk of each firm can be estimated over the selected forecast horizon (based on their macro-financial linkages under different stress scenarios) and finally combined to define systemwide solvency risk within the Systemic CCA framework.

First, a suitable univariate input is defined. Individual CCA-based estimates of market-implied expected losses can generate the following measures that support a firm-specific assessment of capital adequacy (and which have been applied empirically already):

1. capital shortfall based on the amount of expected losses in excess of existing common (core) equity Tier 1 capital above the regulatory minimum;²⁸ and

2. capital shortfall based on the MCAR generated from the change in market capitalization relative to the asset value under the impact of expected losses.

The first measure is presented in the empirical section of this chapter (see Section 3), which illustrates the practical application of the Systemic CCA framework for macroprudential stress testing. The second measure of MCAR is presented in the descriptive part of the section but is not pursued further in the application of the stress test scenarios.

Second, empirical and theoretical (endogenous) models can be used to specify the macro-financial linkages of expected losses. By modeling how the impact of macroeconomic conditions on income components and asset impairment (such as net interest income, fee income, trading income, operating expenses, and credit losses) has influenced a bank’s market-implied expected losses in the past, it is possible to link individual estimates of expected losses (and their implications for potential capital shortfall) to a particular macroeconomic path (and associated financial sector performance) in the future (IMF, 2010, 2011a, 2001b, 2011d, 2012a, 2012b). Alternatively, the historical sensitivity of the main components of the option pricing formula themselves (i.e., the implied asset value and asset volatility) can be determined by applying stress conditions directly to the theoretical specification of market-implied expected losses. Also, the implied asset value underpinning the put option value of each sample bank could be adjusted by the projected profitability under different stress scenarios in order to determine the corresponding change in the level of market-implied expected losses (IMF, 2011c, 2011e).

Third, the joint potential capital shortfall is derived from estimating the multivariate density of each bank’s individual marginal distribution of forecast expected losses (if any) and their dependence structure. If the number of forecast expected losses is insufficient because of the low frequency of macroeconomic variables chosen for the specification of macro-financial linkages (e.g., quarterly values over a five-year forecast horizon), the time series of expected losses could be supplemented by historical expected losses to a point when the number of observations are sufficient for the estimation of joint capital shortfall.

3. EMPIRICAL APPLICATION: STRESS TESTING SYSTEMIC RISK FROM EXPECTED LOSSES IN THE U.K. BANKING SECTOR

This section summarizes the findings from applying the Systemic CCA framework as a market-based top-down (TD) solvency stress testing model as part of the IMF’s FSAP stress

²⁶ Arbitrage trading between both prices shows that synthetic replication of credit protection on guaranteed bonds using equity can be obtained from combining a long position in an equity option “straddle” with a short CDS position.

²⁷ Carr and Wu (2007) find that equity options used in a modified CCA seem to produce risk-neutral default probabilities (RNDP) that closely match RNDPs derived from CDS (sometimes higher, sometimes lower, and differences seem to predict future movements in both markets). Yu (2006) uses a less sophisticated model based on *CreditGrades*, which contains some simplifying assumptions.

²⁸ The capital shortfall of each firm is defined as the amount of capital required to preserve prestress solvency conditions, that is, the existing capital level above the regulatory minimum. It is measured as the marginal change of expected losses over the forecast horizon relative to the expected losses measured at the end point of the historical sample of observations. Note that the market value of equity is considered equivalent to common equity Tier 1 capital because CCA does not specify reported capital tiers but implicitly assumes that any potential loss first affects the most junior claims on firm assets. However, cross-sectional differences in the quality of capital held by firms will affect changes in

valuation and, thus, individual estimates of market-implied capital shortfall. In the empirical application of Systemic CCA for capital assessment, Tier 1 capital was used for consistency purposes (because of divergent regulatory treatments of more junior capital tiers).

test of the U.K. banking sector (IMF, 2011e).²⁹ The model was estimated based on market and balance sheet information of the five largest commercial banks, the largest building society, and the largest foreign retail bank using daily data between January 3, 2005, and end-March 2011.³⁰ The individual market-implied expected losses were treated as a portfolio, whose combined default risk over a prespecified time horizon generated estimates of individual firms' contributions to systemic risk in the event of distress ("tail risk").

A. Illustrating the nonlinearity of historical estimates of individual and joint expected losses

Based on the historical estimates of expected losses, the risk-based valuation of default risk within the Systemic CCA framework supports an integrated market-based capital assessment (see Section 2). Figure 26.3 illustrates the nonlinear relation between the daily estimates of the MCAR, the EL ratio, the fair value credit spread, and the implied asset volatility of one selected sample firm in accordance with the stylized example shown in Figure 26.1. As expected, the rising default risk during the financial crisis establishes a high sensitivity of MCAR to a precipitous increase of expected losses at a level consistent with the regulatory measure of capital adequacy (see Figure 26.3, second panel). However, the growing gap between MCAR and the Core Tier 1 ratio toward the end point of the estimation time period implies that capitalization of the selected bank was perceived by investors as much lower than what regulatory standards would otherwise suggest.

Similarly, the interaction between these variables also can be shown on a systemwide basis (see Figure 26.4). For all firms in the sample, the firm-specific variables underpinning the presented measure were replaced with systemwide measures, such as the joint EL ratio (defined as the sum of 95 percent

VaR values of individual expected losses or the joint 95 percent ES divided by the aggregate market capitalization) and the aggregate MCAR. The empirical results suggested an even higher sensitivity of MCAR to changes in systemwide expected losses than in the case of the single-firm example shown in Figure 26.3. In addition, the comparison of MCAR to the regulatory benchmark (Core Tier 1) indicates that the improvement of solvency conditions after the peak of the financial crisis in March 2009 appeared more protracted across all sample firms (see Figure 26.4, bottom chart).

B. Estimating expected losses under stress conditions based on macro-financial linkages

Individual market-implied expected losses under stress conditions were derived from their historical sensitivity to changes in macro-financial conditions and bank performance. Expected losses under different stress scenarios were forecast using two different methods for the specification of the macro-financial linkages of expected losses as measured by the implicit put option values of sample firms (see Section 2.B):

- In the first model ("IMF satellite model"), expected losses were estimated using a dynamic panel regression specification with several macroeconomic variables (short-term interest rate [+], long-term interest rate [–], real GDP [–], and unemployment [+]) and several bank-specific output variables (net interest income [+], operating profit before taxes [–], credit losses [+], leverage [+], and funding gap [+]).³¹
- In the second model ("structural model"), the value of implied assets of each bank as at end-2010 is adjusted by forecasts of operating profit and credit losses generated in the Bank of England's Risk Assessment Model for Systemic Institutions (RAMSI) in order to derive a revised put option value (after reestimating implied asset volatility), which determines the market-implied capital loss.³²

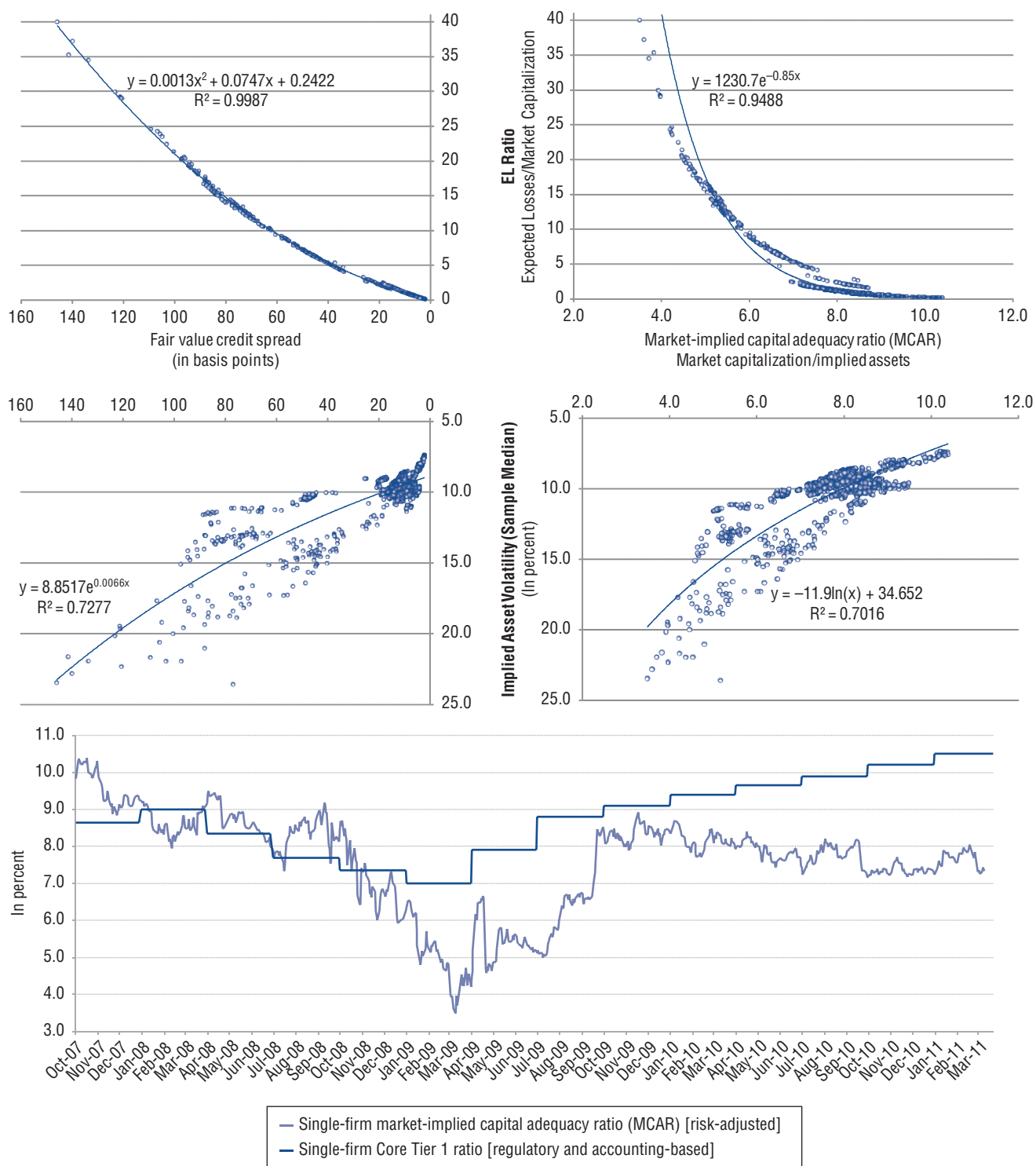
The individually estimated expected losses then were transposed into a capital shortfall. The potential capital loss was determined on an individual basis as the marginal change of expected losses over the forecast horizon (2011–2015) and evaluated for each macro-financial approach and under each of the specified adverse scenarios. As an alternative, the market-implied CAR (see Figures 26.3 and 26.4) could have been used as a basis for assessing capital adequacy and the extent to which expected losses result in potential capital shortfall. However, in the context of this exercise, the CCA-based results were transposed into conventional (regulatory)

²⁹ The Systemic CCA framework also served as a means of cross-validating the stress testing results from the bottom-up exercise conducted jointly with the Financial Services Authority (FSA).

³⁰ The sample comprised the following firms: Hongkong and Shanghai Banking Corporation (HSBC), Barclays, Royal Bank of Scotland (RBS), Lloyd's Banking Group, Standard Chartered Bank, Nationwide, and Santander U.K. Key inputs used were the daily implied asset values derived as the expectation over the state-price density (SPD) using equity option data (from Bloomberg), the default barrier estimated for each firm based on quarterly financial accounts (from *MKMV CreditEdge*), the risk-free rate of interest ($r=3.0$ percent), a one-year time horizon $T=1$, and one-year CDS spreads on senior debt (from *MarkIt*). The default barrier was set equal to total short-term debt plus one-half of long-term debt, consistent with the approach taken by *MKMV CreditEdge*. More specifically, the SPD is calculated using daily European-style equity call option price data over the sample time period with a time to maturity of three months. Besides the maturity tenor, option price data also included information on strike prices as well as actual and corrected spot prices (with implied dividends taken into account). The multivariate distribution of expected losses (i.e., the implicit put option value over a one-year horizon) across all sample firms was estimated over a rolling window of 60 working days (i.e., three months) with daily updating, consistent with the maturity tenor of equity option prices used for the estimation of the implied asset value of each firm.

³¹ The plus or minus signs indicate whether the selected variable exhibited a positive or negative regression coefficient. To be included in the model, the variables needed to be statistically significant at least at the 10 percent level.

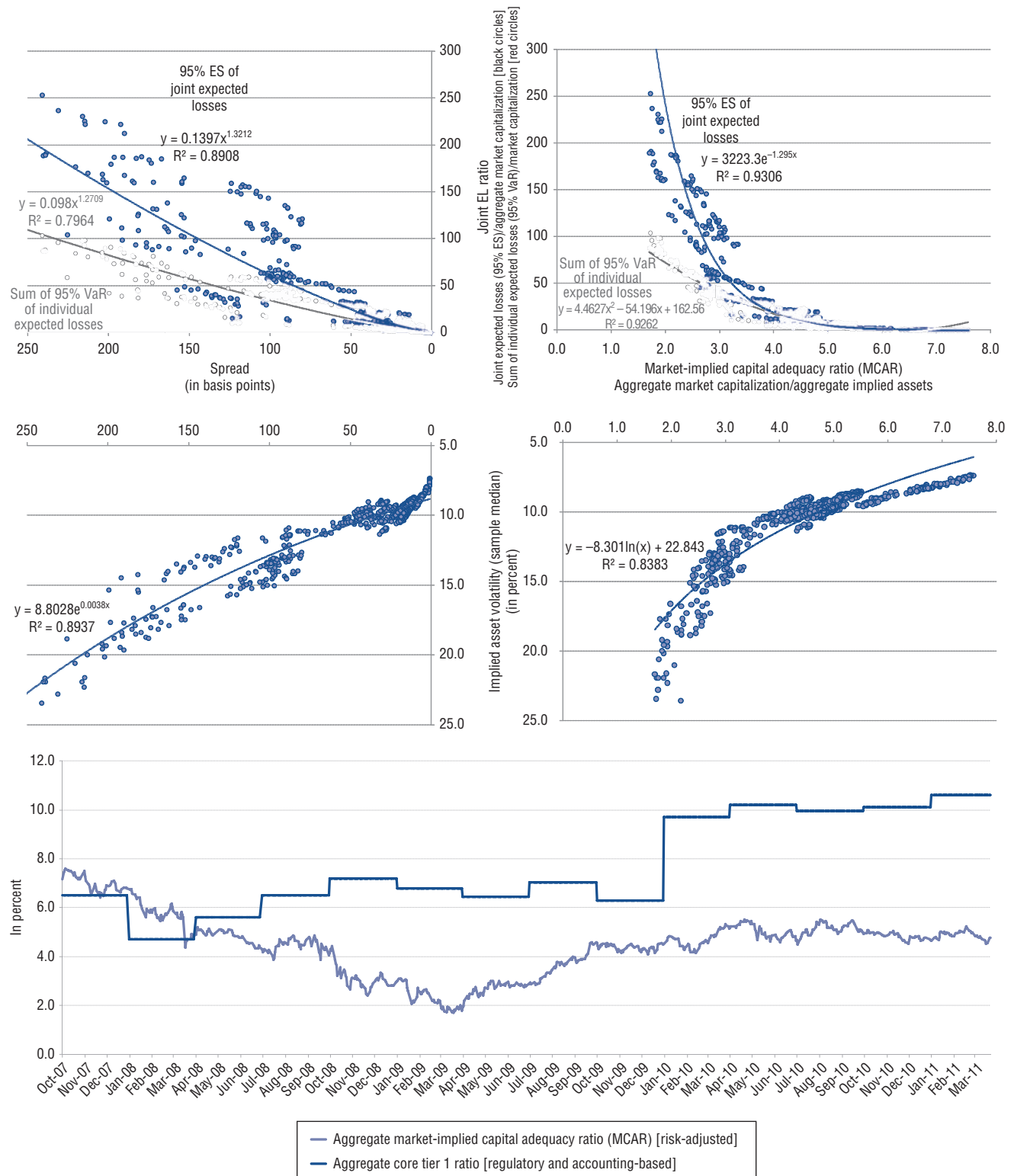
³² This approach also assumed a graduated increase of the default barrier consistent with the transition period to higher capital requirements for the most junior levels of equity under Basel III standards (Basel Committee on Banking Supervision, 2012).



Source: Authors.

Note: CCA = contingent claims analysis; EL = expected losses.

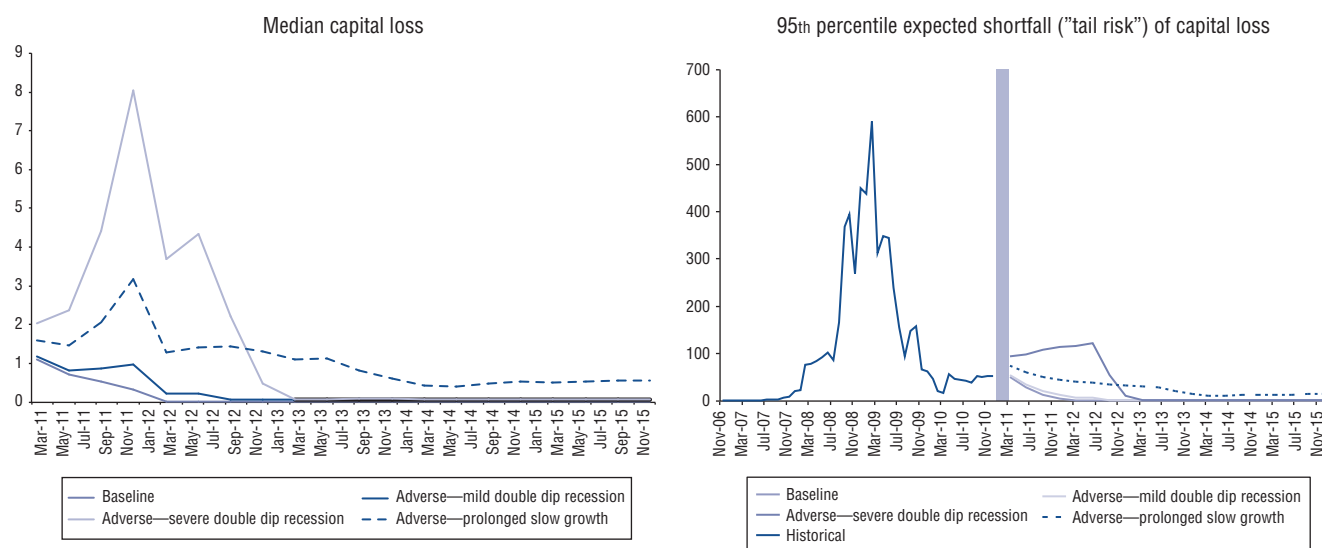
Figure 26.3 United Kingdom: Integrated Individual Market-Based Capital Assessment Based on CCA-Derived Estimates of Expected Losses



Source: Authors.

Note: CCA = contingent claims analysis; ES = expected shortfall; VaR = value at risk.

Figure 26.4 United Kingdom: Integrated Aggregate Market-Based Capital Assessment Based on Systemic CCA-Derived Estimates of Joint Expected Losses



Source: IMF (2011b).

Sample period: 01/03/2005–03/29/2010, monthly observations of historical and forecast joint capital losses of seven sample firms.

Note: The multivariate density is generated from univariate marginals that conform to the GEV distribution and a non-parametrically identified time-varying dependence structure. The marginal severity and dependence measure are estimated (i) over a rolling window of 120 working days (with daily updating) for the historical measure (up to a sample cut-off at end-2010) and (ii) over a rolling window of 60 months (with monthly updating) for the generation of forecast values (using the historical dynamics of capital losses as statistical support).

CCA = contingent claims analysis.

Figure 26.5 United Kingdom: Systemic CCA Estimates of Market-Implied Joint Capital Losses, with IMF Satellite Model (in billions of pound sterling)

measures of solvency, assuming that any increase in expected losses in excess of existing capital buffers constitutes a shortfall of Tier 1 capital (which was chosen as the closest approximation of common equity in CCA).

Finally, individual capital losses then were combined using the Systemic CCA framework in order to determine the joint capital shortfall relative to the current solvency level over the forecast horizon. The univariate density functions of individual capital shortfalls and their dependence structure were combined to a multivariate probability distribution of joint capital shortfall (with a five-year sliding window and monthly updates). The estimated distribution allowed for an assessment of *joint* capital adequacy under stress conditions at different levels of statistical confidence (e.g., 50th and 95th percentiles) as opposed to the summation of individual institutions' capital adequacy (see Section 1).³³ Any capital shortfall arose when the level of Tier 1 capital after accounting for joint capital losses fell below the hurdle rate of regulatory capital defined by the transition period in Basel III (and compared against the FSA's interim capital regime requirements).

Estimation results of the joint capital shortfall suggested that any impact from the realization of systemic solvency risk

would have been limited even in a severe recession. The joint capital loss at the 50th percentile would have been contained under all adverse scenarios. Existing capital buffers were sufficient to absorb the realization of central case (median) joint solvency risks. The severe double-dip recession scenario had the biggest impact on the banking system, but there would have been no capital shortfall (see Figure 26.5). Overall, both satellite models, that is, the IMF satellite model and the structural model, yielded similar and consistent results despite their rather different specifications. They suggested robustness of estimates under either a panel regression approach or a structural approach via updates of model parameters using forecast changes in profitability.

However, the Systemic CCA framework also allowed for the examination of more extreme outcomes amid a rapidly deteriorating macroeconomic environment. The estimation results, which show relatively benign outcomes, pointed to the existing challenges of incorporating the "extreme tail risk" of such a situation. Given that the Systemic CCA framework considers the stochastic nature of risk factors, we examined the realization of a 5 percent "tail of the tail" risk event (as the average density beyond the 95th percentile statistical confidence level) for multiple firms experiencing a dramatic escalation of losses. Under the adverse scenarios, the market-implied capital shortfall would have been significantly increased, likely as a result of significantly lower profitability in conjunction with a sharp deterioration in asset quality and weaker fee-based income reducing the implied asset value of sample banks.

³³ It should be noted that the application of the Systemic CCA framework for the TD stress test of the U.K. FSAP also took into account further declines in the market value of sovereign and bank debt (in the form of valuation haircuts applied to banks' debt holdings of all non-AAA rated sovereigns and banks in both the trading and banking books), which were added to market-implied expected losses.

TABLE 26.5

United Kingdom: Systemic CCA Estimates of Market-Implied Individual Contributions of Sample Banks to Systemic Risk—Market-Implied Joint Capital Loss
(average per time period, in percent of joint capital shortfall)

Range	Precrisis: End-June 2005– End-June 2007	Subprime Crisis: Jul 1, 2007– Sep 14, 2008	Crisis Period 1: Sep 15–Dec 31, 2008	Crisis Period 2: Jan 1–Sep 30, 2009	Crisis Period 3: Oct 1, 2009– Feb 28, 2010	Sovereign Crisis: Mar 1, 2010– Mar 2011	Average
Expected Joint Capital Shortfall (Median)¹							
Minimum	0.1	0.3	0.8	0.6	0.2	1.0	0.4
25th percentile	2.4	1.5	1.8	1.5	2.1	2.6	2.0
Median	5.8	4.9	3.1	4.5	6.8	8.1	5.7
75th percentile	15.3	16.0	6.8	20.0	21.8	22.1	17.2
Maximum	58.9	59.9	79.0	51.9	45.2	41.4	55.3
Extreme Joint Capital Shortfall (95th Percentile)¹							
Minimum	0.1	0.4	1.4	1.0	0.6	1.1	0.6
25th percentile	3.6	2.9	4.3	3.2	2.9	4.8	3.6
Median	7.4	7.3	6.2	6.5	4.9	9.5	7.5
75th percentile	14.0	19.1	13.4	14.0	16.0	20.3	16.2
Maximum	57.2	48.4	56.9	58.0	56.7	39.3	52.2

Source: IMF (2011b).

Note: Each bank's percentage share is based on its time-varying contribution to the multivariate density of capital losses at the 50th (median) and 95th percentile.

CCA = contingent claims analysis.

1. Sample period: 01/03/2005–03/29/2010.

The dispersion of individual contributions to the realization of the central case and extreme tail systemic risk suggests that a few banks contributed disproportionately to joint solvency risk under stress. Table 26.5 shows the percentage share of the minimum, maximum, and interquartile range of the individual banks' time-varying contribution to the multivariate density of potential losses at the 50th (median) and 95th percentiles. With the exception of the first phase of the European sovereign debt crisis (during most of 2010 and the beginning of 2011), one bank, represented by the maximum of the distribution of all banks, seemed to have accounted for more than half of the solvency risk in the sample.

4. CONCLUSION

The identification of systemic risk is an integral element in the design and implementation of MPS with a view toward enhancing the resilience of the financial sector. However, assessing the magnitude of systemic risk is not a straightforward exercise and has many conceptual challenges. In particular, there is danger of underestimating practical problems of applying systemic risk management approaches to a real-life situation as complex modeling risks lose transparency, with conclusions being highly dependent on the assumptions, the differences in business models of financial institutions, and differences in regulatory and supervisory approaches across countries.

This chapter presented a new modeling framework, Systemic CCA, which can help in the measurement and analysis of systemic solvency risk by estimating the joint expected losses of financial institutions based on multivariate EVT—with practical applications for stress testing. Based on its re-

cent application to the banking sector in the context of the IMF's FSAP exercise for the United Kingdom (2011), this chapter demonstrates how the individual contributions to systemic risk can be quantified under different stress scenarios and how such systemic relevance of financial institutions can be used to design macroprudential policy instruments, such as systemic risk surcharges. The aggregation model underlying the Systemic CCA framework also was used to illustrate the systemic risk of the financial sector implicitly taken on by governments and the potential (nonlinear) destabilizing feedback processes between the financial sector and the sovereign balance sheet.

Some limitations of the Systemic CCA framework need to be acknowledged and have already been reflected as caveats in the interpretation of the findings. For instance, the market-based determination of default risk might be influenced by the validity of valuation models. Economic models are constructed as general representations of reality that, at best, capture the broad outlines of economic phenomena in a steady state. However, financial market behavior might defy statistical assumptions of valuation models, such as the option pricing theory, extreme value measurement, and nonparametric specification of dependence between individual default probabilities in the Systemic CCA approach. For instance, during the credit crisis, financial market behavior was characterized by rare and nonrecurring events and not by repeated realizations of predictable outcomes generated by a process of random events that exhibits stochastic stability. Thus, the statistical apparatus underlying conventional asset pricing theory fails to fully capture sudden and unexpected realizations beyond historical precedent.

Going forward, the Systemic CCA framework could be expanded in scale and scope to include other types of risks. For instance, most recently, the Systemic CCA framework also has been adapted to measure systemic liquidity risk by transforming the net stable funding ratio as standard liquidity measure under Basel III into a stochastic measure of aggregate funding risk (see Chapter 27). In addition, given the flexibility of this framework, the financial sector and sovereign risk analysis could be integrated with macro-financial feedbacks in order to design monetary and fiscal policies. Such an approach could inform the calibration of stress scenarios of banking and sovereign balance sheets and the appropriate use of macroprudential regulation. Using an economy-wide Systemic CCA for all sectors, including the financial sector, non-financial corporates, households, and the government, also can provide new measures of economic output—the present value of risk-adjusted GDP (Gray, Jobst, and Malone, 2010).

REFERENCES

- Acharya, Viral V., Thomas F. Cooley, Matthew P. Richardson, and Ingo Walter, eds., 2010, *Regulating Wall Street: A New Architecture for Global Finance* (Hoboken, NJ: Wiley).
- Acharya, Viral V., Lasse H. Pedersen, Thomas Philippon, and Matthew P. Richardson, 2009, “Measuring Systemic Risk,” Working Paper presented at the Federal Reserve Bank of Cleveland and the National Bureau of Economic Research “Research Conference on Quantifying Systemic Risk,” Cambridge, Massachusetts, November 6. Available via the Internet: <http://www.clevelandfed.org/research/conferences/2009/11-6-2009/index.cfm>
- , 2010, “Measuring Systemic Risk,” FRBC Working Paper 10-02 (Cleveland: Federal Reserve Bank of Cleveland). Available via the Internet: <http://www.clevelandfed.org/research/workpaper/2010/wp1002.pdf>
- , 2012, “Measuring Systemic Risk,” CEPR Discussion Paper 8824, February. Available via the Internet: <http://www.cepr.org/pubs/dps/DP8824.asp>
- Acharya, Viral V., Joao A. C. Santos, and Tanju Yorulmazer, 2010, “Systemic Risk and Deposit Insurance Premiums,” *Economic Policy Review* (New York: Federal Reserve Bank of New York, August).
- Adrian, Tobias, and Markus K. Brunnermeier, 2008, “CoVaR,” Staff Report No. 348 (New York: Federal Reserve Bank of New York). Available via the Internet: http://www.newyorkfed.org/research/staff_reports/sr348.html
- Aït-Sahalia, Yacine, and Andrew W. Lo, 1998, “Nonparametric Estimation of State-Price Densities Implicit in Financial Asset Prices,” *Journal of Finance*, Vol. 53, No. 2, pp. 499–547.
- Allen, Franklin, Ana Babus, and Elena Carletti, 2010, “Financial Connections and Systemic Risk,” NBER Working Paper No. 16177 (Cambridge, MA: National Bureau of Economic Research). Available via the Internet: <http://www.nber.org/papers/w16177>
- Artzner, Philippe, Freddy Delbaen, Jean-Marc Eber, and David Heath, 1999, “Coherent Measures of Risk,” *Mathematical Finance*, Vol. 9, No. 3, pp. 203–28.
- Backus, David, Silverio Foresi, and Liuren Wu, 2004, “Accounting for Biases in Black-Scholes,” Working Paper, New York University, Goldman Sachs Group, and Baruch College.
- Basel Committee on Banking Supervision, 2012, “Results of the Basel III Monitoring Exercise as of 30 June 2011” (Basel, April 12). Available via the Internet: <http://www.bis.org/publ/bcb217.htm>
- Battiston, Stefano, Michelangelo Puliga, Rahul Kaushik, Paolo Tasca, and Guida Caldarelli, 2012, “DebtRank: Too Central to Fail? Financial Networks, the Fed and Systemic Risk,” *Nature*, August 2. Available via the Internet: <http://www.nature.com/srep/2012/120802/srep00541/full/srep00541.html>
- Black, Fischer, and John C. Cox, 1976, “Valuing Corporate Securities: Some Effects on Bond Indenture Provisions,” *Journal of Finance*, Vol. 31, No. 2, pp. 351–68.
- Black, Fischer, and Myron Scholes, 1973, “The Pricing of Options and Corporate Liabilities,” *Journal of Political Economy*, Vol. 81, No. 3, pp. 637–54.
- Breeden, Douglas T., and Robert H. Litzenberger, 1978, “Prices of State-Contingent Claims Implicit in Option Prices,” *Journal of Business*, Vol. 51, No. 4, pp. 621–51.
- Brownlees, Christian T., and Robert F. Engle, 2011, “Volatility, Correlation and Tails for Systemic Risk Measurement,” Working Paper (New York: Stern School of Business, New York University). Available via the Internet: <http://vleda.stern.nyu.edu/src/?p=2241>
- Carr, Peter, and Liuren Wu, 2007, “Theory and Evidence on the Dynamic Interactions between Sovereign Credit Default Swaps and Currency Options,” *Journal of Banking and Finance*, Vol. 31, No. 8, pp. 2383–403.
- Chan-Lau, Jorge, 2010, “Regulatory Capital Charges for Too-Connected-to-Fail Institutions: A Practical Proposal,” IMF Working Paper 10/98 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=23753>
- Coles, Stuart G., Janet Heffernan, and Jonathan A. Tawn, 1999, “Dependence Measures for Extreme Value Analyses,” *Extremes*, Vol. 2, pp. 339–65.
- Collin-Dufresne, Pierre, and Robert S. Goldstein, 2001, “Do Credit Spreads Reflect Stationary Leverage Ratios?” *Journal of Finance*, Vol. 56, No. 5, pp. 1929–57.
- Committee on the Global Financial System, 2012, “Operationalising the Selection and Application of Macroprudential Instruments,” CGFS Papers No. 48, December (Basel: Committee on the Global Financial System). Available via the Internet: <http://www.bis.org/publ/cgfs48.htm>
- Dall’Aglia, Giorgio, Samuel I. Kotz, and Gabriella Salinetti, eds., 1991, *Advances in Probability Distributions with Given Marginals (Beyond the Copulas)* (Dordrecht: Kluwer Academic Publishers).
- Drehmann, Mathias, and Nikola Tarashev, 2011, “Systemic Importance: Some Simple Indicators,” *BIS Quarterly Review*, pp. 25–37 (Basel: Bank for International Settlements). Available via the Internet: http://www.bis.org/publ/qtrpdf/r_qt1103.htm
- Duan, Jin-Chuan, 1994, “Maximum Likelihood Estimation Using Price Data of the Derivative Contract,” *Mathematical Finance*, Vol. 4, No. 2, pp. 155–67.
- Engle, Robert F., 2002, “Dynamic Conditional Correlation: A Simple Class of Multivariate Generalized Autoregressive Conditional Heteroskedasticity Models,” *Journal of Business & Economic Statistics*, Vol. 20, pp. 339–50.
- Engle, Robert F., and Kevin Sheppard, 2001, “Theoretical and Empirical Properties of Dynamic Conditional Correlation Multivariate GARCH,” NBER Working Paper No. 8554 (October). Available via the Internet: <http://www.nber.org/papers/w8554>

- Ericsson, Jan, and Joel Reneby, 2004, "An Empirical Study of Structural Credit Risk Models Using Stock and Bond Prices," *Journal of Fixed Income*, Vol. 13, No. 4, pp. 38–49.
- , 2005, "Estimating Structural Bond Pricing Models," *Journal of Business*, Vol. 78, No. 2, pp. 707–35.
- Espinosa-Vega, Marco A., and Juan Solé, 2011, "Cross-Border Financial Surveillance: A Network Perspective," *Journal of Financial Economic Policy*, Vol. 3, No. 3, pp. 182–205.
- Financial Stability Board, International Monetary Fund, and Bank for International Settlements, 2011a, "Macroprudential Policy Tools and Frameworks," Progress Report to the G-20 Finance Ministers and Central Bank Governors (Basel, October 26). Available via the Internet: <http://www.bis.org/publ/othp17.pdf>
- , 2011b, "Macroprudential Policy Tools and Frameworks," Update to the G-20 Finance Ministers and Central Bank Governors (Basel, February 14). Available via the Internet: <http://www.bis.org/publ/othp13.pdf>
- Fisher, Ronald A., and Leonard H. C. Tippet, 1928, "Limiting Forms of the Frequency Distribution of the Largest or Smallest Member of a Sample," *Proceedings of the Cambridge Philosophical Society*, Vol. 24, pp. 180–90.
- Gapen, Michael T., 2009, "Evaluating the Implicit Guarantee to Fannie Mae and Freddie Mac Using Contingent Claims," *International Finance Review: Credit, Capital, Currency, and Derivatives: Instruments of Global Financial Stability or Crisis?*, Vol. 10, pp. 329–52.
- Gnedenko, Boris V., 1943, "Sur la Distribution Limite du Terme Maximum d'Une Série Aléatoire," *Annals of Mathematics*, Vol. 44, pp. 423–53.
- González-Hermosillo, Brenda, Christian Capuano, Dale F. Gray, Heiko Hesse, Andreas A. Jobst, Paul Mills, Miguel Segoviano, and Tao Sun, 2009, "Methods to Identify Systemic Financial Risks," *Vox*, April 23. Available via the Internet: <http://www.voxeu.org/index.php?q=node/3495>
- Gray, Dale F., and Andreas A. Jobst, 2009a, "Higher Moments and Multivariate Dependence of Implied Volatilities from Equity Options as Measures of Systemic Risk," *Global Financial Stability Report*, Chapter 3, pp. 128–31 (Washington, April). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2009/01/index.htm>
- , 2009b, "Using the CCA Framework to Estimate the Cost of Government Guarantees," in "Crisis-Related Measures in the Financial System and Sovereign Balance Sheet Risks," Board paper, edited by Adrienne Cheasty and Udaibir S. Das, Fiscal Affairs and Monetary and Capital Markets Departments, July 31 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/np/sec/pn/2009/pn09118.htm>
- , 2010, "New Directions in Financial Sector and Sovereign Risk Management," *Journal of Investment Management*, Vol. 8, No. 1, pp. 23–38.
- , 2011a, "Modeling Systemic and Sovereign Risk," in *Lessons from the Financial Crisis*, edited by Arthur Berd, pp. 143–85 (London: RISK Books).
- , 2011b, "Modelling Systemic Financial Sector and Sovereign Risk," *Sveriges Riksbank Economic Review*, No. 2, pp. 68–106. Available via the Internet: http://www.riksbank.se/upload/Rapporter/2011/POV_2/er_2011_2.pdf
- , 2011c, "Systemic Contingent Claims Analysis—A Model Approach to Systemic Risk," in *Managing Risk in the Financial System*, edited by John R. LaBrosse, Rodrigo Olivares-Caminal, and Dalvinder Singh, pp. 93–110 (London: Edward Elgar).
- Gray, Dale F., and Samuel Malone, 2010, "Quantifying Systemic Risk and Reconceptualizing the Role of Finance for Economic Growth," *Journal of Investment Management*, Vol. 8, No. 2, pp. 90–110.
- Haldane, Andrew G., and Benjamin Nelson, 2012, "Tails of the Unexpected," paper presented at "The Credit Crisis Five Years On: Unpacking the Crisis," University of Edinburgh Business School conference, Edinburgh, June 8–9. Available via the Internet: <http://www.bankofengland.co.uk/publications/Documents/speeches/2012/speech582.pdf>
- Hall, Peter, and Nader Tajvidi, 2000, "Distribution and Dependence-Function Estimation for Bivariate Extreme Value Distributions," *Bernoulli*, Vol. 6, pp. 835–44.
- Huang, Xin, Hao Zhou, and Haibin Zhu, 2009, "A Framework for Assessing the Systemic Risk of Major Financial Institutions," *Journal of Banking and Finance*, Vol. 33, No. 11, pp. 2036–49.
- , 2010, "Assessing the Systemic Risk of a Heterogeneous Portfolio of Banks during the Recent Financial Crisis," Working Paper No. 296 (Basel: Bank for International Settlements). Available via the Internet: <http://www.bis.org/publ/work296.htm>
- Hutchinson, Timothy P., and Chin D. Lai, 1990, *Continuous Bivariate Distributions, Emphasizing Applications* (Sydney: Rumsby Scientific Publishing).
- International Monetary Fund, 2009a, "Assessing the Systemic Implications of Financial Linkages," *Global Financial Stability Report*, Chapter 2 (Washington, April). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2009/01/index.htm>
- , 2009b, "Detecting Systemic Risk," *Global Financial Stability Report*, Chapter 3 (Washington, April). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2009/01/index.htm>
- , 2010, "United States: Stress Testing—Technical Note," IMF Country Report 10/244 (Washington, July 29). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=24101>
- , 2011a, "Germany: Stress Testing—Technical Note," IMF Country Report 11/371 (Washington, December 23). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=25461>
- , 2011b, "Israel: Selected Issues," IMF Country Report 11/23 (Washington, January 24). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=24587>
- , 2011c, "Sweden: Contingent Claims Analysis Approach to Measure Risk and Stress Test the Swedish Banking Sector—Technical Note," IMF Country Report 11/286 (Washington, September 1). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=25241>
- , 2011d, "Toward Operationalizing Macroprudential Policies: When to Act?," *Global Financial Stability Report*, Chapter 3 (Washington, September). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2011/02/index.htm>
- , 2011e, "United Kingdom: Stress Testing the Banking Sector—Technical Note," IMF Country Report 11/227 (Washington, July 1). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=25119>
- , 2011f, "The United States: Spillover Report—2011 Article IV Consultation," IMF Country Report 11/203 (Washington, July 22). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=25083>
- , 2012a, "Israel: Stress Test of the Banking, Insurance and Pension Sectors—Technical Note," IMF Country Report 12/88

- (Washington, April). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=25850>
- , 2012b, “Spain: Financial System Stability Assessment,” IMF Country Report 12/137 (Washington, May 30). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=25977>
- International Monetary Fund, Financial Stability Board, and Bank for International Settlements, 2009, “Guidance to Assess the Systemic Importance of Financial Institutions, Markets and Instruments: Initial Considerations,” Report to the G-20 Finance Ministers and Central Bank Governors, October. Available via the Internet: <http://www.imf.org/external/np/g20/pdf/100109.pdf>
- Jobst, Andreas A., 2007, “Operational Risk—The Sting Is Still in the Tail but the Poison Depends on the Dose,” *Journal of Operational Risk*, Vol. 2, No. 2, pp. 156.
- , 2013, “Multivariate Dependence of Implied Volatilities from Equity Options as Measure of Systemic Risk,” *International Review of Financial Analysis*, Vol. 28 (June), pp. 112–29.
- , 2014, “Measuring Systemic Risk-Adjusted Liquidity (SRL)—A Model Approach,” *Journal of Banking and Finance*, Vol. 45, pp. 270–87.
- Jobst, Andreas A., and Dale F. Gray, 2013, “Systemic Contingent Claims Analysis—Estimating Market-Implied Systemic Risk,” IMF Working Paper 13/54 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=40356>
- Jobst, Andreas A., and Herman Kamil, 2008, “Stock Market Linkages between Latin America and the United States during ‘Tail Events,’” *Regional Economic Outlook*, Western Hemisphere Department (Washington: International Monetary Fund).
- Joe, Harry, 1997, *Multivariate Models and Dependence Concepts*, Monographs on Statistics and Applied Probability 73 (London: Chapman and Hall).
- Jones, E. Philip, Scott P. Mason, and Eric Rosenfeld, 1984, “Contingent Claims Analysis of Corporate Capital Structures: An Empirical Investigation,” *Journal of Finance*, Vol. 39, No. 3, pp. 611–25.
- Longstaff, Francis A., and Eduardo S. Schwartz, 1995, “A Simple Approach to Valuing Risky Fixed and Floating Rate Debt,” *Journal of Finance*, Vol. 50, pp. 789–819.
- Merton, Robert C., 1973, “Theory of Rational Option Pricing,” *Bell Journal of Economics and Management Science*, Vol. 4, No. 1, pp. 141–83.
- , 1974, “On the Pricing of Corporate Debt: The Risk Structure of Interest Rates,” *Journal of Finance*, Vol. 29, No. 2, pp. 449–70.
- Ogden, Joseph P., 1987, “Determinants of the Ratings and Yields on Corporate Bonds: Tests of the Contingent Claims Model,” *Journal of Financial Research*, Vol. 10, No. 4, pp. 329–39.
- Organization for Economic Cooperation and Development, 2012, “Systemic Financial Risk,” OECD Reviews of Risk Management Policies (Paris, June).
- Pickands, James, 1981, “Multivariate Extreme Value Distributions,” *Proceedings of the 43rd Session of the International Statistical Institute*, Vol. 49, pp. 859–78.
- Poon, Ser-Huan, Michael Rockinger, and Jonathan Tawn, 2003, “Extreme Value Dependence in Financial Markets: Diagnostics, Models, and Financial Implications,” *Review of Financial Studies*, Vol. 17, No. 2, pp. 581–610.
- Ronn, Ehud I., and Avinash K. Verma, 1986, “Pricing Risk-Adjusted Deposit Insurance: An Option-Based Model,” *Journal of Finance*, Vol. 41, No. 4, pp. 871–95.
- Rookley, Cameron, 1997, “Fully Exploiting the Information Content of Intra Day Option Quotes: Applications in Option Pricing and Risk Management,” mimeo, University of Arizona.
- Sedunov, John, III, 2012, “What Is the Systemic Risk Exposure of Financial Institutions?” mimeo, Villanova University, Department of Finance, September 1. Available via the Internet: <http://ssrn.com/abstract=2143633>
- Segoviano, Miguel A., and Charles A. E. Goodhart, 2009, “Banking Stability Measures,” IMF Working Paper 09/04 (Washington, International Monetary Fund).
- Sklar, Abe, 1959, “Fonctions de Répartition à n Dimensions et Leurs Marges,” *Publications de l’Institut de Statistique de L’Université de Paris*, Vol. 8, pp. 229–31.
- Tarashev, Nicola, Claudio Borio, and Kostas Tsatsaronis, 2009, “The Systemic Importance of Financial Institutions,” *BIS Quarterly Review*, pp. 75–87 (Basel, September). Available via the Internet: http://www.bis.org/publ/qtrpdf/r_qt0909.htm
- Weistroffer, Christian, 2011, “Identifying Systemically Important Financial Institutions (SIFIs),” International Topics—Current Issues, Deutsche Bank Research (August 11). Available via the Internet: <http://ideas.repec.org/p/ess/wpaper/id4383.html>
- Wirch, Julia L., and Mary R. Hardy, 1999, “A Synthesis of Risk Measures for Capital Adequacy,” *Insurance: Mathematics and Economics*, Vol. 25, No. 3, pp. 337–47.
- Yu, Fan, 2006, “How Profitable Is Capital Structure Arbitrage?” *Financial Analysts Journal*, Vol. 62, No. 5, pp. 47–62.

Measuring Systemic Risk-Adjusted Liquidity

ANDREAS A. JOBST

The Systemic Risk-Adjusted Liquidity (SRL) model combines option pricing with market information and balance sheet data to generate a probabilistic measure of multiple entities experiencing a liquidity event. This measure links a firm's maturity mismatch between assets and liabilities, its overall risk profile, and the stability of its funding with those characteristics of other firms that are subject to common changes in market conditions. This approach can then be used (1) to quantify an individual institution's time-varying contribution to expected losses from systemwide liquidity shortfalls and (2) to price contingent liquidity support during times of stress within a macroprudential framework that provides incentives for liquidity managers to internalize the negative externalities from systemic risk implications of individual funding decisions. The model also can accommodate a stress testing approach for institution-specific and/or general funding shocks that generate estimates of systemic liquidity risk (and associated charges) under adverse scenarios.

METHOD SUMMARY

Overview	The Systemic Risk-Adjusted Liquidity model generates a risk-adjusted, market price-based measure of multiple entities experiencing a liquidity event under stress conditions.
Application	The method is appropriate in situations where prudential data are limited but market information is available; the approach relies on predefined prudential specification of liquidity risk (e.g., the net stable funding ratio, or NSFR) to assess the impact of maturity mismatches.
Nature of approach	Model-based (option pricing, multivariate parametric/nonparametric estimation).
Data requirements	- Accounting information on all constituent balance sheet elements underpinning the NSFR. - Market data on equity and equity option prices. - Various market rates (short-term sovereign rate; with maturities ranging from 3 to 12 months), long-term sovereign rates (with maturity ranging from 3 to 10 years), total equity market returns (domestic market and other equity markets), financial bond rates (investment grade, both medium- and long-term), local currency London interbank offered rate (ranging from 3 to 12 months), and the domestic short-term currency overnight indexed swap.
Strengths	The model combines a prudential measure of liquidity risk with an advanced version of contingent claims analysis as inputs to a multivariate specification of joint liquidity risk; approach can be used to: - quantify an individual institution's time-varying contribution to systemwide liquidity shortfalls; and - price contingent liquidity support during times of stress.
Weaknesses	Assumptions are required regarding the prudential measure of liquidity risk and the specification of the option pricing model; there is no explicit treatment of the impact of solvency risk on liquidity risk; technique is complex and resource intensive.
Tool	Under development.

This chapter is an abridged version of IMF Working Paper 12/209, which was also published as “Measuring Systemic Risk-Adjusted Liquidity (SRL)—A Model Approach,” *Journal of Banking and Finance*, Vol. 45, pp. 270–87 (Jobst, 2014). The model is based on previous analytical work in the context of the October 2009, October 2010, and April 2011 issues of the *Global Financial Stability Report* (IMF, 2009, 2010, 2011; Jobst, 2011).

A defining characteristic of the recent financial crisis was the simultaneous and widespread dislocation in funding markets, which revealed that the lack of sound liquidity risk management (and suitable policy responses) can have a significant impact on financial stability. In particular, banks' common asset exposures and their increased reliance on short-term wholesale funding in tandem with high leverage levels helped propagate rising counterparty risk. The implications from liquidity risk management decisions made by some institutions spilled over to other markets and other institutions, contributing to others' losses, amplifying solvency concerns, and exacerbating overall liquidity stress as a result of these negative dynamics.

Little progress has been made so far in addressing—in a comprehensive way—the externalities caused by impact of the interconnectedness within financial institutions and financial markets on systemic liquidity risk, that is, the risk that multiple institutions may face simultaneous difficulties in rolling over their short-term debts or in obtaining new short-term funding (much less long-term funding). Although systemic solvency risk already has entered the prudential debate in the form of additional capital rules to apply to systemically important financial institutions, proposals aimed at measuring and regulating systemic liquidity have been few and far between. This is largely owed to the rarity of systemwide liquidity risk events, the variegated interactions between institutions and funding markets within countries and across national boundaries, and the conceptual challenges in modeling liquidity conditions affecting institutions and transactions separately or jointly.

The policy objective of such efforts would be to minimize the possibility of systemic liquidity disruptions that necessitate costly public sector support. Disruptions to the flow of financial services due to an impairment of all or parts of the financial system give rise to systemic risk if there is the potential of financial instability to trigger serious negative spillovers to the real economy.¹ The recent financial crisis demonstrated that higher perceived counterparty risk precipitates escalating funding constraints that further depress asset prices and eventually overwhelm individual liquidity provisions. Because every financial institution's risk-taking affects the operation of the financial system as a whole, the magnitude of market disruptions increases with the level of asymmetric information as coordination failure accentuates the impact of common shocks. One case in point is the wholesale funding market, where these network externalities and the structural complexity across financial and nonfinancial institutions during the recent crisis has amplified aggregate vulnerabilities to shocks from counterparty risk. As the economic environ-

ment or asset allocation behavior changed, lenders were more likely to increase haircuts on repo financing, limit eligibility of collateral, or stop rolling over short-term funding altogether in order to offset an asset shock by means of deleveraging their balance sheets. As such a behavior occurred collectively, it caused liquidation of assets under fire sale conditions, which resulted in a negative confidence-induced downward liquidity spiral, increasing funding pressures as deteriorating counterparty risk eventually weighs on solvency (Shin, 2009; Shleifer and Vishny, 2010).

A number of reforms and initiatives are under way to address shortcomings in financial institutions' liquidity practices. They have resulted in more stringent supervisory liquidity requirements without directly targeting systemwide implications. Under the postcrisis revisions of the existing Basel Accord, known as Basel III, the Basel Committee on Banking Supervision (BCBS, 2010a, 2010b) has issued two quantitative liquidity standards to be applied at a global level and published qualitative guidance to strengthen liquidity risk management practices in banks.² Individual banks will have to maintain higher and better-quality liquid assets as measured by the liquidity coverage ratio and the net stable funding ratio (NSFR).³ This approach assumes that sufficient liquidity would limit knock-on effects on solvency conditions in distress situations and complement the risk absorption role of capital—but without considering systemwide effects and liquidity buffers. These standard measures do not address the additional risk of such simultaneous shortfalls arising out of the interconnectedness of various institutions across a host of financial markets and jurisdictions.

In contrast, a macroprudential approach to systemic liquidity risk should offset procyclical tendencies of liquidity risk and account for changes to an institution's risk contribution, which might not necessarily follow cyclical patterns, in order to inform countercyclical risk mitigants to joint liquidity risk. It would also motivate a risk-adjusted pricing scheme so that institutions that contribute to systemic liquidity risk through their interconnectedness or through their impact are assigned a proportionately higher charge (while the opposite holds true for firms that help absorb systemwide shocks from sudden increases in liquidity risk). In this regard, several proposals are under discussion. They include the internalization of public sector cost of liquidity risk via insurance schemes (Goodhart, 2009; Gorton and Metrick, 2009; Perotti and Suarez, 2009, 2011), capital charges (Brunnermeier and Pedersen, 2009), taxation (Acharya, Cooley, and others, 2010; Acharya, Santos, and Tanju, 2010), investment requirements

¹ Impairment to the flow of financial services occurs where certain financial services are temporarily unavailable, as well as in situations where the cost of obtaining the financial services is sharply increased. It would include disruptions owing to shocks originating outside the financial system that have an impact on it, as well as shocks originating from within the financial system.

² See also BCBS (2009).

³ The NSFR represents a structural ratio, which limits the stock of unstable funding by encouraging longer-term borrowing in order to restrict liquidity mismatches from excessive maturity transformation. Based on existing proposals, it would require banks to establish a stable funding profile over the short term, that is, the use of stable (long-term and/or stress-resilient) sources to continuously fund cash flow obligations that arise from lending and investment activities inside a one-year time horizon.

(Cao and Illing, 2009; Farhi, Golosov, and Tsyvinski, 2009), as well as arrangements aimed at mitigating the systemwide effects from the fire sale liquidation of assets via collateral haircuts (Valderrama, 2010) and modifications of resolution regimes (Roe, 2009; Acharya and Oncu, 2010).

In this chapter, we propose a combined price/quantity-based mechanism—the SRL model—for the assessment and stress testing of systemic liquidity risk. This methodology complements the current Basel III liquidity framework by extending the prudential assessment of liquidity using the NSFR to a systemwide approach, which can help substantiate different types of macroprudential liquidity regulation, such as a capital surcharge, a fee, a tax, or an insurance premium for contingent liquidity access. The SRL model combines both perspectives by assessing how the size and interconnectedness of individual institutions (with varying degrees of leverage and maturity mismatches defining their risk profile) can create short-term vulnerabilities to liquidity risk on a systemwide level and under distress conditions (Davis, 2011).

The SRL model is based on an options pricing concept to gauge the general level of liquidity risk for a portfolio of institutions in a three-step process. First, using an “expected loss” notion to evaluate the level of liquidity shortfall, the model combines balance sheet information and market data (equity/equity options) in order to generate a risk-adjusted measure of stable funding. In this way, the probability of falling below the joint NSFR boundary translates into a market-implied analogue to the NSFR. Given the historical variation in the underpinnings of the NSFR and the extent to which this measure links institutions implicitly to the markets, it is then possible to derive a joint distribution of expected losses and determine when firms simultaneously fall below a funding threshold defined by the required stable funding (RSF). The joint probability function then is used to identify an individual institution’s contribution to systemic liquidity risk to price contingent liquidity support within a macroprudential framework that provides incentives for liquidity managers to internalize the systemic risk of their decisions. The contribution to systemic liquidity risk depends on an institution’s funding and asset structure and its interconnectedness with other institutions, which informs the calculation of a firm-specific capital charge and/or insurance premium.

The SRL model accounts for changes in common factors determining individual funding conditions, their implications for the market-implied linkages between financial institutions, and the resulting impact on systemic liquidity risk. Thus, it accomplishes two essential goals of risk measures in this area: (1) to measure the extent to which an institution contributes to systemic liquidity risk; and (2) to use this to indirectly price the liquidity assistance that an institution would need to receive in cases of severe funding problems in order to head off further escalation toward insolvency. The systemic dimension of the model is captured by three properties:

1. *Drawing on the market’s evaluation of the riskiness of a firm (including the liquidity risk that the institution will be unable to offset continuous cash outflows).* That eval-

uation, in turn, is based on perceived riskiness as implied by the institution’s equity and equity options in the context of the economic and financial environment present at the time of measurement.

2. *Controlling for the firm’s sources of stable funding.* The sources are modeled as being sensitive to the same markets as the funding sources of every other institution but at varying degrees. Changes in common funding conditions (and their impact on the perceived risk profile of each firm) establish market-induced linkages among institutions. The proposed framework thus combines market prices and balance sheet information to inform a risk-adjusted measure of systemic liquidity risk. That measure links institutions implicitly to the markets in which they obtain equity capital and funding.
3. *Quantifying the chance of simultaneous liquidity shortfalls via joint probability distributions.* After obtaining a market-based measure of individual liquidity risk, the probability that banks will experience liquidity shortfall simultaneously—that is, the likelihood that the amount of available stable funding (ASF) for each institution falls below the amount of RSF as defined by the NSFR⁴—is made explicit by computing joint probability distributions (which also account for difference in the magnitude of individual liquidity shortfall). Hence, the liquidity risk resulting from a particular funding configuration is assessed not only for individual institutions but for all firms within a system in order to generate estimates of systemic risk.

1. METHODOLOGY

The innovation of the SRL approach is its use of contingent claims analysis (CCA) to measure individual liquidity risk consistent with proposed prudential standards in order to quantify its systemwide effects. So far, the CCA methodology⁵ been applied widely to measure and evaluate solvency risk and credit risk at financial institutions. In the SRL model, however, CCA is used to derive a forward-looking measure of liquidity risk that helps determine the probability of an individual institution experiencing liquidity shortfall and the associated expected loss when the shortfall indeed occurs. For a sample of financial institutions, these individual estimates are aggregated to a joint probability of simultaneous liquidity

⁴ The NSFR is defined as the ratio, for a bank, of its “available amount of stable funding” divided by its “required amount of stable funding.”

⁵ The CCA is a generalization of option pricing theory pioneered by Black and Scholes (1973) and Merton (1973, 1974). It is based on three principles that are applied in this study: (1) the values of liabilities are derived from assets; (2) assets follow a stochastic process; and (3) liabilities have different priorities (senior and junior claims). Equity can be modeled as an implicit call option, while risky debt can be modeled as the default-free value of debt less an implicit put option that captures expected losses. In the SRL model, advance option pricing is applied to account for biases in the Black-Scholes-Merton specification.

shortfall, which also quantifies the marginal contribution of an institution to systemic liquidity risk.⁶

First, the SRL model transposes the current regulatory proposal aimed at limiting term structure transformation into a market-based measure of individual liquidity risk. In keeping with existing microprudential measures, *individual liquidity risk* is defined as the effective maturity mismatch between short-term liabilities, weighted by their susceptibility to market fragilities (rollover risk), and short-term assets, weighted by their market liquidity value (funding risk),⁷ after controlling for the maturity of off-balance-sheet hedging transactions, such as foreign exchange swaps.⁸ In the SRL model, the proposed Basel III liquidity standard aimed at limiting maturity transformation—the NSFR—is a regulatory benchmark in this regard and serves as the starting point for the quantification of individual liquidity risk.⁹ The components of the NSFR—ASF and RSF—are transposed into a time-varying measure of the NSFR at market prices, where RSF and ASF values reflect differences between the balance sheet and actual market values of total assets and liabilities of each firm (see Figure 27.1). The actual balance sheet measures of ASF and RSF values are rescaled by (1) the ratio of the book value of total liabilities to the present value of total liabilities B (which can be observed) and (2) the ratio of the book value of total assets to market-implied value of total assets A which is obtained as a risk-neutral density from equity option prices, respectively.¹⁰ Doing so generates the values A_{RSF} and B_{ASF} , which transform the NSFR ratio into a market-based measure of an institution's liquidity risk (“market-based NSFR”).

Given that A cannot be observed, it is estimated directly from option prices prior to the estimation of expected losses arising from the market-based NSFR. More specifically, the state-price density (SPD) of the implied total asset value underpinning A_{RSF} is estimated from equity option prices without any assumptions on the underlying diffusion process. This avoids the calibration error of using two-equations-two-unknowns in the traditional Merton model, which contains empirical

irregularities that can influence solving both implied asset value and asset volatility simultaneously. Using equity option prices, we can derive the risk-neutral probability distribution of the underlying equity price at the maturity date of the options. We determine the implied asset value based on the empirical SPD of the equity price by adapting the Breeden and Litzenberger (1978) method, together with a semiparametric specification of the Black-Scholes option pricing formula (Aït-Sahalia and Lo, 1998). Estimates are based on option contracts with identical time to maturity (and consistent with the residual maturity assumed in the application of the BSM model), assuming a continuum of strike prices. The SPD of the equity price includes the risk-neutral expectation of the implied asset value and removes the impact of differences of higher moments of both the equity price and the implied asset value dynamics over the chosen time horizon. Thus, the current implied asset value can be transposed from the SPD of the equity price conditional on the contemporaneous leverage of the firm (see equations (27.5) and (27.9)).

Second, the net exposure arising from individual liquidity risk is modeled as a put option in order to derive a risk-adjusted version of the market-based NSFR (see Figure 27.2). The aggregate cash flow implications of changes to liquidity risk then can be quantified as expected losses arising from insufficient stable funding of each bank. Thus, liquidity risk is viewed as a put option written on the NSFR, where the present value of the RSF represents the “strike price,” with the short-term volatility of all assets underpinning the RSF determined by the implied volatility of observable equity option prices of a firm.¹¹ The value of the ASF is assumed to follow a random walk with intermittent jumps that create sudden and large changes in the valuation of the liabilities. The volatility of these liabilities included in the ASF is computed as a weighted average of the observed volatilities of latent factors derived from a set of market funding rates deemed relevant for banks. These two time-varying elements provide the basis for computing a put option, which has intrinsic value (i.e., is “in-the-money”) when the discounted value of ASF falls below that of RSF over the same time horizon, constituting an expected loss owing to liquidity shortfall. The value of this derived put option can be shown to result in significant hypothetical cash losses for an individual firm as the risk-adjusted NSFR declines.

The value of the put option increases the higher the probability of ASF falling below RSF over a one-year risk horizon so that NSFR is smaller than one. This would breach the lower boundary that banks will be mandated to maintain under the current Basel III proposal. Such probability is influenced by changes to the firm's funding pattern, risk profile, and market perceptions of risk, which can be derived from changes to implied asset value and volatility reflected in the institution's equity option prices and from its asset and lia-

⁶ This method uses publicly available information. Although the focus here is on banks given data limitations, the methodology is sufficiently flexible to be used for nonbank institutions that contribute to systemic liquidity risk. Indeed, the proposal builds on several strands of recent research that focus on the interactions between financial institutions and markets in the context of systemic liquidity risk.

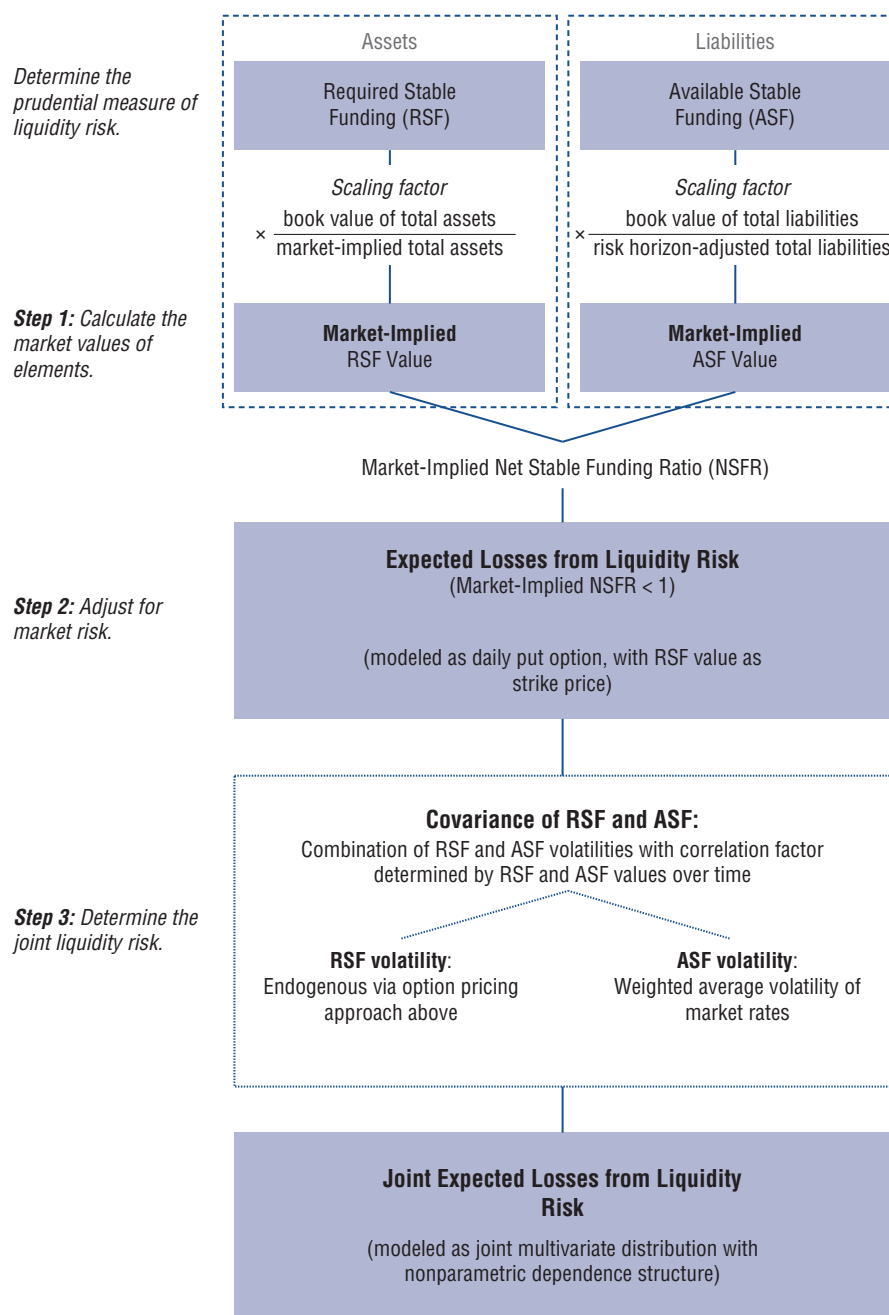
⁷ This will be reflected in the liquidity of the market, i.e., the ability to trade an asset without affecting the price. We will introduce a parameter that denotes the cost of early liquidation from imperfect market depth.

⁸ This approach represents a significant improvement over the concept of systemic liquidity insurance presented in Perotti and Suarez (2009). The implementation of the model purports to be a market-driven regulation of a quantity-based restriction on liquidity transformation as opposed to a variable Pegovian tax (Perotti and Suarez, 2011).

⁹ The NSFR measures the amount of longer-term, stable sources of funding employed relative to the liquidity profiles of the assets funded and the potential for contingent calls on funding liquidity arising from off-balance-sheet commitments and obligations.

¹⁰ Estimations of these scaling factors, and the subsequent covariance and the joint expected losses, are computed over a rolling window of 120 working days to reflect their changing characteristics.

¹¹ The NSFR reflects the impact of funding shocks as an exposure to changes in market prices in times of stress. The procedure also can be applied to other measures of an individual firm's liquidity risk.



Source: Author.

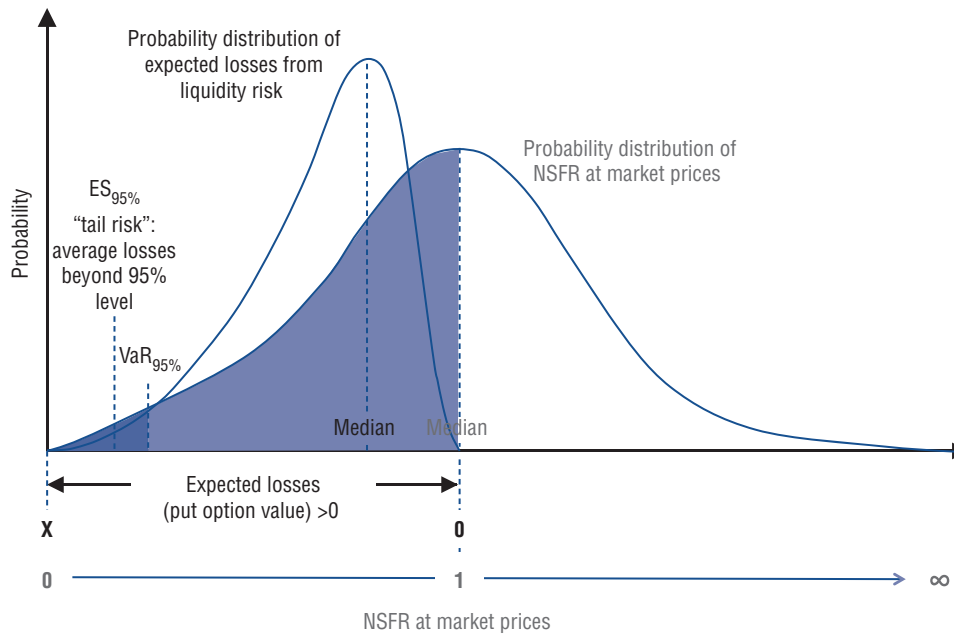
Figure 27.1 Methodology to Compute Joint Expected Losses from Systemic Liquidity Risk

bility structure. Thus, the present value of market-implied expected losses associated with the current liquidity position of a single institution can be valued as a modified implicit put option of B_{ASF} on the present value of A_{RSF} as strike price in keeping with the traditional Black-Scholes-Merton (BSM) model as

$$P_{NSFR}(t) = A_{RSF}(t)e^{-r_{RSF}(T-t)}\Phi\left(-\left(d_{NSFR} - \sigma_{A_{RSF}, B_{ASF}}\sqrt{T-t}\right)\right) - B_{ASF}\Phi(-d_{NSFR}), \quad (27.1)$$

over time horizon $T-t$ at the general risk-free rate r and asset growth (or drift) rate $r_{A_{RSF}}$, where $\Phi(\bullet)$ denotes the standard normal cumulative distribution function and

$$d_{NSFR} = \left(\ln\left(\frac{B_{ASF}}{A_{RSF}(t)}\right) + \left(r_{A_{RSF}} + \frac{\sigma_{A_{RSF}, B_{ASF}}^2}{2}\right)(T-t) \right) / \sigma_{A_{RSF}, B_{ASF}}\sqrt{T-t}, \quad (27.2)$$



Source: Author.

Note: This figure illustrates the relation between the net exposure from liquidity risk and the NSFR at market prices distribution functions (based on multiple observations of each over a certain period of time). Expected losses, which are modeled as a put option that approximates the cash flow profile over a given risk horizon, arise once there is some probability that the NSFR drops below the regulatory requirement to be greater than one. The greater the potential for funding distress projected by a declining NSFR, the greater are these losses. The tail risk of individual expected losses from a liquidity shortfall is represented by the ES at the 95th percentile, which is the area under the curve beyond the threshold value set by the VaR. ES = expected shortfall.

Figure 27.2 Conceptual Relation between the Net Stable Funding Ratio (NSFR) at Market Prices and Expected Losses from Liquidity Risk

subject to the duration of liabilities (or debt claims) B underpinning ASF, the total funding mismatch of the firm $A_{RSF}(t)/B_{ASF}$ under NSFR assumptions, and joint asset-liability volatility

$$\sigma_{A_{RSF}, B_{ASF}} = \sqrt{\sigma_{A_{RSF}}^2 + \sigma_{B_{ASF}}^2 - 2\rho_{A_{RSF}, B_{ASF}}\sigma_{A_{RSF}}\sigma_{B_{ASF}}}. \quad (27.3)$$

The volatility of ASF, $\sigma_{A_{RSF}}$, draws on the *general* asset volatility

$$\begin{aligned} \sigma_{A_{RSF}} &\equiv \sigma_A = \frac{E(t)}{A(t)\Phi(d)} \sigma_E \\ &= \left(1 - \frac{Be^{-r_A(T-t)}\Phi(d - \sigma_A\sqrt{T-t})}{A(t)\Phi(d)}\right) \sigma_E, \end{aligned} \quad (27.4)$$

where E represents the equity value at time t , and σ_E is the equity volatility. The volatility, $\sigma_{A_{RSF}}$, is endogenous to the CCA approach and conditional on changes in leverage $A(t)/B$, given

$$d = \left(\ln\left(\frac{A(t)}{B}\right) + \left(\left(r_A + \frac{\sigma_A^2}{2} \right) (T-t) \right) \right) / \sigma_A \sqrt{T-t}, \quad (27.5)$$

where r_A is the risk-free asset rate. The short-term volatility of all liabilities underpinning the ASF, $\sigma_{B_{ASF}}$, is identifiable but not readily observable for each source of funding. Thus, it is derived as a weighted average of the observed volatilities of

latent factors of significant market interest rates affecting the funding components of the RSF determined by a dynamic factor model (DFM).¹² The correlation $\rho_{A_{RSF}, B_{ASF}}$ between both volatilities can be constant or estimated using the historical relation between ASF and RSF at levels over a rolling time window.

This specification of option price-based expected losses, however, does not incorporate skewness, kurtosis, and stochastic volatility, which can account for implied volatility smiles of equity prices (Backus, Foresi, and Wu, 2004). Thus, we mitigate the shortcomings of the BSOM approach

¹² A DFM of the ASF is specified based on one principal component extracted from each group of observed market rates (at different maturities) as explanatory variables: (1) short-term sovereign rates (with maturities ranging from three to nine months) and long-term sovereign rates (with maturity ranging from 3 to 10 years); (2) total equity market returns (domestic market and Morgan Stanley Capital International [MSCI] index); (3) financial bond rates (investment grade, both medium- and long-term); and (4) domestic currency London Interbank Offered Rates (LIBOR, ranging from three to nine months) and the domestic currency overnight indexed swap (OIS) rates (ranging from three to nine months). In the empirical analysis of the chapter, the volatility of the ASF is calculated as the average volatility of these daily market rates weighted by the regression coefficient of each principal component, estimated over a five-year period from January 3, 2005, to December 14, 2010.

and enhance equation (27.1), without altering its general analytical closed-form form, by means of a jump diffusion that follows a standard Poisson process:

$$\Pr(N(t) = k) = (\lambda t)^k / k! e^{-\lambda t}, \quad (27.6)$$

where λ is the average number of jumps per unit time (i.e., the number of jump events up to time t). The jump size follows a log-normal distribution,

$$J \sim \varphi \exp(-(\nu^2/2 + \nu\Phi(0,1))), \quad (27.7)$$

with average random jump size φ and volatility ν of the jump size calibrated over a rolling estimation period of τ observations with periodic updating of daily log returns (which sets the time horizon for the subsequent estimation steps of the model).¹³ The k th term in this series corresponds to the scenario where k jumps occur over observation period. Hence, the put option value can be written as follows:¹⁴

$$P_{NSFR_k}(t) = \sum_{k=0}^{\infty} \frac{\exp(-\varphi\lambda t)(\varphi\lambda t)^k}{k!} A_{RSF}(t) e^{-r_{ASF}(T-t)} \Phi\left(-\left(d_{NSFR} - \sigma_{ASF, BASF} \sqrt{T-t}\right)\right) - B_{ASF} \Phi(-d_{NSFR}) \quad (27.8)$$

and asset volatility of RSF with jumps $\sigma_{A_{RSF_k}} \equiv \sigma_{A_k} = \sqrt{\sigma_A^2 + k\nu^2/(T-t)}$ as well as the risk-free interest rate $r_{A_{RSF_k}} = r_{A_{RSF}} - \lambda(\varphi - 1) + k \ln(\varphi)/(T-t)$ are updated accordingly,¹⁵ so that

$$d_{NSFR_k} = \left(\ln\left(\frac{B_{ASF}}{A_{RSF}(t)}\right) + \left(\left(r_{A_{RSF_k}} + \frac{\sigma_{A_{RSF_k}, BASF}^2}{2} \right) (T-t) \right) \right) / \sigma_{A_{RSF_k}, BASF} \sqrt{T-t}, \quad (27.9)$$

and

$$\sigma_{A_{RSF_k}, BASF} = \sqrt{\sigma_{A_{RSF_k}}^2 + \sigma_{B_{ASF}}^2 - 2\rho_{A_{RSF_k}, BASF} \sigma_{A_{RSF_k}} \sigma_{B_{ASF}}}. \quad (27.10)$$

¹³ Further refinements of this option pricing model are possible, including various simulation approaches, which might come at the expense of losing analytical tractability. The ad hoc model of Dumas and others (1998) is designed to accommodate the implied volatility smile and is easy to implement, but requires a large number of market option prices. The pricing models by Heston (1993) and Heston and Nandi (2000) allow for stochastic volatility, but the parameters driving these models can be difficult to estimate. Many other models have been proposed, to incorporate stochastic volatility, jumps, and stochastic interest rates. Bakshi and others (1997), however, suggest that most of the improvement in pricing comes from introducing stochastic volatility. Introducing jumps in asset prices leads to small improvements in the accuracy of option prices. Other option pricing models include those based on copulas, Levy processes, neural networks, GARCH models, and nonparametric methods. Finally, the binomial tree proposed by Cox and others (1979) spurred the development of lattices, which are discrete-time models that can be used to price any type of option—European or American, plain-vanilla or exotic.

¹⁴ All parameters are calibrated over the entire sample period.

¹⁵ The value of derivatives with convex payoff (which includes this put option specification) increases when jumps are present (i.e., when $\lambda > 0$)—regardless of the average jump direction.

Thus, the volatility of ASF, $\sigma_{A_{RSF_k}}$, would be the same as the revised general asset volatility σ_{A_k} (consistent with the above specification of a jump diffusion process),¹⁶ so that

$$\sigma_{A_{RSF_k}} \equiv \sigma_{A_k} = \frac{E_k(t)}{A_k(t)\Phi(d_k)} \sigma_{E_k} = \left(1 - \frac{Be^{-r_{A_k}(T-t)}\Phi(d_k - \sigma_{A_k}\sqrt{T-t})}{A_k(t)\Phi(d_k)} \right) \sigma_{E_k} \quad (27.11)$$

where

$$d_k = \left(\ln\left(\frac{A_k(t)}{B}\right) + \left(\left(r_{A_k} + \frac{\sigma_{A_k}^2}{2} \right) (T-t) \right) \right) / \sigma_{A_k} \sqrt{T-t}, \quad (27.12)$$

given the general asset volatility with jumps $\sigma_{A_k} = \sqrt{\sigma_A^2 + k\nu^2/(T-t)}$ and the risk-free interest rate $r_{A_k} = r_A - \lambda(\varphi - 1) + k \ln(\varphi)/(T-t)$.

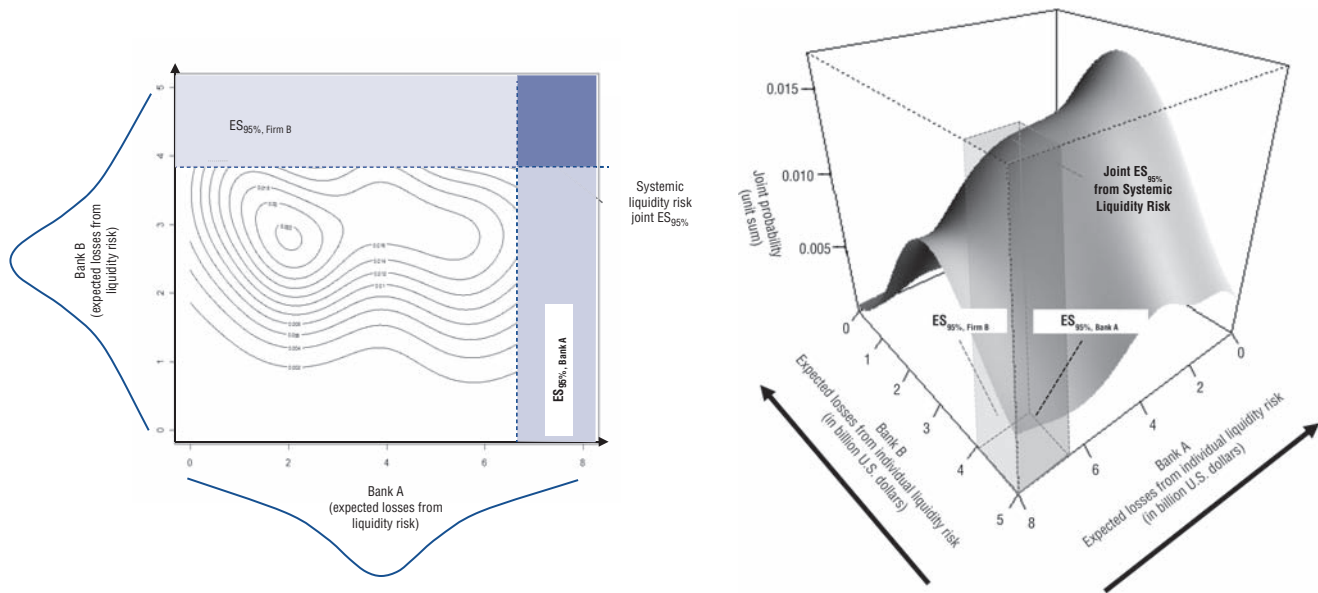
Third, individually estimated net exposures to liquidity risk are aggregated to determine the magnitude of liquidity shortfalls on a systemwide level (see Figure 27.3). The expected losses arising from the variation of each individual firm's risk-adjusted NSFR over time are treated as a portfolio for which we calculate the joint probability of all firms experiencing a liquidity shortfall simultaneously. In this way, each firm's maturity mismatch between assets and liabilities, its implication on the market-based assessment of its risk profile, and the stability of its funding are linked with those characteristics at other firms that are subject to common changes in market conditions.

More specifically, we adapt the Systemic CCA framework to generate a *multivariate extreme value distribution* that formally captures the realizations of joint liquidity shortfalls. We define a nonparametric dependence function of individual expected losses, which is combined with the marginal distributions of these individual loss estimates to generate a joint distribution that defines an aggregate measure of liquidity risk for a selected sample of firms. Based on the conditional tail expectation of this multivariate distribution, such as the conditional value at risk (VaR) (or expected shortfall [ES]), we can derive point estimates of systemic liquidity risk in times of stress at a statistical confidence level of choice.

We first specify the individual asymptotic tail behavior of individual expected losses in keeping with extreme value theory (EVT) as a general statistical concept of deriving a limit law for sample maxima, where the Fisher-Tippett-Gnedenko theorem (Fisher and Tippett, 1928; Gnedenko, 1943) defines the attribution of a given distribution of normalized maxima (or minima) to be of extremal type. Let the vector-valued series $X_m^n = P_{NSFR_k, 1}^n(t), \dots, P_{NSFR_k, m}^n(t)$ denote independent and identically distributed random observations of expected losses (i.e., put option values) each estimated over the given time horizon (e.g., a daily sliding window of $n=120$ days) for a total m number of j firms. Given

$$X^n = \max(P_{NSFR_k, j}^1(t), \dots, P_{NSFR_k, j}^n(t)),$$

¹⁶ Given that the implied asset value is derived separately, this approach avoids the traditional “two-equations-two-unknowns” approach applied by Jones, Mason, and Rosenfeld (1984) to derive implied assets and asset volatility.



Source: Author.

Note: This figure illustrates the bivariate case of aggregating expected losses in order to determine the joint probability of two sample firms experiencing a liquidity shortfall at the same time, using the estimation results for individual institutions. The left panel shows the kernel density function of two firms (Bank A and Bank B). The probability of systemic liquidity risk is captured by combining the individual bank estimates (depicted by the green and blue panels), which generates the joint expected losses at a defined level of statistical confidence, such as the 95th percentile (red cube). The left panel can also be shown in two dimensions as a so-called “contour plot” (see right panel).

ES = expected shortfall.

Figure 27.3 Conceptual Relation between Expected Losses from Liquidity Risk: Two-Firm (Bivariate) Case

there exists a choice of normalizing constants $\beta^n > 0$ and α^n , such that the probability of each ordered n -sequence of normalized sample maxima $(\mathbf{X}^n - \alpha^n)/\beta^n$ converges to the nondegenerate limit distribution $H(x)$ as $n \rightarrow \infty$ and $x \in \cdot$, so that

$$\begin{aligned} F_n^{[\beta^n x + \alpha^n]}(x) &= \lim_{n \rightarrow \infty} \Pr\left(\frac{\mathbf{X}^n - \alpha^n}{\beta^n} \leq y\right) \\ &= [F(\beta^n y + \alpha^n)]^n \rightarrow H(x) \end{aligned} \quad (27.13)$$

falls within the maximum domain of attraction (MDA) of the generalized extreme value (GEV) distribution with probability density function

$$\begin{aligned} h_{\mu, \sigma, \xi}(x) &= \frac{1}{\sigma} \left(1 + \frac{\xi(x - \mu)}{\sigma}\right)^{-1/\xi - 1} \\ &\times \exp\left(-\left(1 + \frac{\xi(x - \mu)}{\sigma}\right)_+^{-1/\xi}\right), \end{aligned} \quad (27.14)$$

which is obtained from differentiating the cumulative distribution function:

$$H_{\mu, \sigma, \xi}(x) = \begin{cases} \exp\left(-\left(1 + \frac{\xi(x - \mu)}{\sigma}\right)_+^{-1/\xi}\right) & \text{if } 1 + \frac{\xi(x - \mu)}{\sigma} \geq 0 \\ \exp\left(-\exp\left(-\frac{x - \mu}{\sigma}\right)\right) & \text{if } x \in \mathbb{R}, \xi = 0. \end{cases} \quad (27.15)$$

Thus, in the context of multiple series, the j th univariate marginal density function based on GEV is defined as

$$y_j(x) = \left(1 + \frac{\hat{\xi}_j(x - \hat{\mu}_j)}{\hat{\sigma}_j}\right)_+^{-1/\hat{\xi}_j} \quad (\text{for } j = 1, \dots, m), \quad (27.16)$$

where $1 + \xi_j(x - \mu_j)/\sigma_j > 0$, scale parameter $\sigma_j > 0$, location parameter $\mu_j \in \mathbb{R}$, and shape parameter $\xi_j \neq 0$.¹⁷ The moments of the univariate density function are estimated via the *linear combinations of ratios of spacings* method (Gray and Jobst, 2010), which identifies possible limiting laws of asymptotic tail behavior of normalized extremes (Coles, Heffernan, and Tawn, 1999; Poon, Rockinger, and Tawn, 2003; Jobst, 2007).

Second, we define the comovement of expected losses as a nonparametric, multivariate dependence function. It is estimated iteratively on a unit simplex that optimizes the coincidence of multiple series of cross-classified random variables—similar to a chi-statistic that measures the statistical likelihood of observed values to differ from their expected distribution. We specify dependence function $A(\omega_1, \dots, \omega_{m-1})$ by expanding the bivariate logistic method proposed by Pickands (1981) to the multivariate case and adjusting the margins according to Hall and Tajvidi (2000) so that

$$Y(\omega) = \min\left\{1, \max\left\{n \left(\sum_{i=1}^n \bigwedge_{j=1}^m \frac{y_{i,j}/\hat{y}_{\cdot j}}{\omega_j}\right)^{-1}, \omega, 1 - \omega\right\}\right\}, \quad (27.17)$$

¹⁷ The upper tails of most (conventional) limit distributions (weakly) converge to this parametric specification of asymptotic behavior, irrespective of the original distribution of observed maxima (unlike parametric VaR models). The higher the absolute value of shape parameter, the larger the weight of the tail and the slower the speed at which the tail approaches its limit.

where $\hat{y}_j = \sum_{i=1}^n y_{i,j} / n$ and $0 \leq \max(\omega_1, \dots, \omega_{m-1}) \leq \Upsilon(\omega_j) \leq 1$ for all $0 \leq \omega_j \leq 1$, subject to the optimization of the $(m-1)$ -dimensional unit simplex:

$$S_m = \left\{ (\omega_1, \dots, \omega_{m-1}) \in \mathbb{R}_+^m : \omega_j \geq 0, 1 \leq j \leq m-1; \sum_{j=1}^{m-1} \omega_j \leq 1 \text{ and } \omega_m = 1 - \sum_{j=1}^{m-1} \omega_j \right\}. \quad (27.18)$$

S_m establishes the degree of coincidence of multiple series of cross-classified random variables similar to a chi-statistic that measures the statistical likelihood of observed values to differ from their expected distribution. $\Upsilon(\bullet)$ represents a convex function on $[0,1]$ with $\Upsilon(0) = \Upsilon(1) = 1$, that is, the upper and lower limits of $\Upsilon(\bullet)$ are obtained under complete dependence and mutual independence, respectively. This specification stands in contrast to a general copula function that links the marginal distributions using only a single (and time-invariant) dependence parameter.

Finally, we combine the marginal distributions of these individual expected losses with their dependence structure to determine a multivariate distribution (over the estimation time horizon defined earlier) by following the aggregation mechanism of the Systemic CCA approach (Gray and Jobst, 2009, 2010, 2011; Jobst and Gray, 2013).^{18,19} The resultant multivariate cumulative distribution function is specified as

$$G_{t,m}(x) = \exp \left\{ - \left(\sum_{j=1}^m y_{t,j} \right) \Upsilon_t(\omega) \right\} \quad (27.19)$$

with corresponding probability density function

$$g_{t,m}(x) = \left(\left(\frac{1}{\hat{\sigma}_{t,m}} \sum_{j=1}^m \left(1 + \frac{\hat{\xi}_{t,j}(x - \hat{\mu}_{t,j})}{\hat{\sigma}_{t,j}} \right)_+^{-1/\hat{\xi}_{t,j}-1} \right) \Upsilon_t(\omega) \right) \times \exp \left(- \left(\sum_{j=1}^m \left(1 + \frac{\hat{\xi}_{t,j}(x - \hat{\mu}_{t,j})}{\hat{\sigma}_{t,j}} \right)_+^{-1/\hat{\xi}_{t,j}} \right) \Upsilon_t(\omega) \right) \quad (27.20)$$

at time $t = \tau + 1$.

We then obtain the general expression of joint ES (or conditional VaR) as the probability-weighted residual density beyond a prespecified statistical confidence level (say, $a = 0.95$) of maximum losses. ES defines the conditional average value $z \in \mathbb{R}$ of aggregate potential losses in excess of the statistical confidence limit (“severity threshold”) based on observations over estimation days τ . Thus, we can obtain continuous densities from $G(\bullet)$ and compute ES as

$$ES_{t,\tau,m,a} = \frac{1}{a} \int_0^a G_{t,\tau,m}^{\leftarrow}(x) dx = \frac{1}{a} \int_0^a G_{t,\tau,m}^{-1}(x) dx = \mathbb{E} \left[z_\tau \mid z_\tau \geq G_{t,\tau,m}^{-1}(a) = VaR_{t,a} \right], \quad (27.21)$$

where $G^{-1}(a)$ is the quantile corresponding to probability $0 < a < 1$ and $G^{-1}(a) = G^{\leftarrow}(a)$ with $G^{\leftarrow}(a) \equiv \inf \{ x \mid G(x \geq a) \}$ and threshold quantile value

$$VaR_{t,a} = \sup \{ G_{t,\tau,m}^{-1}(\bullet) \mid \Pr[z_\tau > G_{t,\tau,m}^{-1}(\bullet)] \geq a = 0.95 \}, \quad (27.22)$$

with the point estimate of joint potential losses of m firms at time t defined as follows:²⁰

$$G_{t,\tau,m}^{-1}(a) = \hat{\mu}_{t,m} + \hat{\sigma}_{t,m} / \hat{\xi}_{t,m} \left(\left(- \frac{\ln(a)}{A_t(\omega)} \right)^{-\hat{\xi}_{t,m}} - 1 \right). \quad (27.23)$$

ES also can be written as a linear combination of individual ES values, $ES_{t,\tau,j,a}$, where the relative weights (in the weighted sum) are given by the second-order cross-partial derivatives of the inverse of the joint probability density function $G_{t,\tau,m}^{-1}(\bullet)$ to changes in both the dependence function and the individual marginal severity of expected losses.

Thus, by rewriting $ES_{t,\tau,m,a}$, we obtain

$$ES_{t,\tau,m,a} = \sum_j^m \psi_{\tau,j,a} \mathbb{E} \left[z_{\tau,j} \mid z_{\tau,j} \geq G_{t,\tau,m}^{-1}(a) = VaR_{t,a} \right], \quad (27.24)$$

where the relative weight of institution j statistical confidence level a is defined as the marginal contribution

$$\psi_{\tau,j,a} = \frac{\partial^2 G_{t,\tau,m}^{-1}(a)}{\partial y_{t,j} \partial \Upsilon_t(\omega)} \quad \text{s.t.} \quad \sum_j^m \psi_{\tau,j,a} = 1 \quad \text{and} \quad \psi_{\tau,j,a} z_{\tau,j} \leq z_{\tau,m}, \quad (27.25)$$

attributable to the joint effect of both the marginal density $y_{t,j}(\bullet)$ and the change of the dependence function $\Upsilon_t(\bullet)$ absent institution j .

The resulting measure of the contribution of each individual firm to aggregate expected losses can be used as a basis for practical macroprudential tools designed to create incentives that help mitigate liquidity risk from both individual and systemwide perspectives.²¹ We use this amount (as a result of the quantity-based restriction of NSFR > 1) to develop two price-based risk mitigation mechanisms—a capital surcharge and an insurance premium—that take into account the contingent support that banks would receive from a central bank in times of systemic liquidity stress and thus represent the potential public sector cost arising from two or more institutions experiencing a significant liquidity shortfall.

¹⁸ See also Gray and Malone (2010) for a more general application of this approach to the integrated balance sheets of an entire economy. An application of the Systemic CCA approach in the context of IMF stress tests and spillover analysis can be found in Chapter 26.

¹⁹ The analysis of dependence is completed independently from the analysis of marginal distributions and thus differs from the classical approach, where multivariate analysis is performed jointly for marginal distributions and their dependence structure by considering the complete variance–covariance matrix, such as the Multivariate Generalized Autoregressive Conditional Heteroskedasticity approach.

²⁰ ES is an improvement over VaR, which, in addition to being a pure frequency measure, is “incoherent,” that is, it violates several axioms of monotonicity, subadditivity, positive homogeneity, and translation invariance found in coherent risk measures. For example, subadditivity, which is a mathematical way to say that diversification leads to less risk, is not satisfied by VaR.

²¹ Note that this approach also could be used to identify the effectiveness of closer supervisory monitoring in response to identified liquidity problems of a particular bank. That can be done if remedial actions decrease the bank’s contribution to overall systemic risk from liquidity shortfalls up to the point where it closely matches the individual liquidity risk.

For the capital surcharge, the method follows the current bank supervisory guidelines for market risk capital requirements (BCBS, 2009), in which the VaR is calculated each day and compared with three times the average quarterly VaRs over the last four quarters. The maximum of these two numbers becomes the required amount of regulatory capital for market risk. In a similar way, each firm would need to meet an additional capital requirement, c_{SLR} (in currency units), at time t , to offset the net exposure from either its individual liquidity risk or its contribution to systemic liquidity risk at a statistical confidence level of $\alpha=0.95$. First, we choose the higher of (1) the previous quarter's ES at percentile α , $\overline{ES}_{t-1, \tau, j, \alpha}$, averaged over daily observations associated with individual liquidity risk of firm j ; and (2) the average of this quarterly measure over the preceding four quarters, multiplied by an individual multiplication factor κ_j . This amount would be compared with the greater of (1) the last available quarterly marginal contribution, $\overline{\psi}_{t-1, \tau, j, \alpha}$, and multiplied by the total ES, $\overline{ES}_{t, \tau, \alpha}$, over the same time period; and (2) the average of this quarterly measure over the preceding four quarters, multiplied by a general multiplication factor κ . The higher of the two maximums would then be the surcharge. Therefore, based on an estimation window of τ days in each quarter t , the capital surcharge c_{SLR} would be

$$c_{SLR} = \max \left\{ \begin{array}{l} \max \left\{ \overline{ES}_{t-1, \tau, j, \alpha}; \kappa_j \times \frac{1}{4} \sum_{t=-4}^0 \overline{ES}_{t, \tau, j, \alpha} \right\}; \\ \max \left\{ \overline{\psi}_{t-1, \tau, j, \alpha} \times \overline{ES}_{t-1, \tau, \alpha}; \kappa \times \frac{1}{4} \sum_{t=-4}^0 \left(\overline{\psi}_{t, \tau, \alpha} \times \overline{ES}_{t, \tau, \alpha} \right) \right\} \end{array} \right\}. \quad (27.26)$$

An alternative method is to require individual firms to pay an insurance premium for liquidity support based on their likelihood to experience expected losses from liquidity risk that is shared by other firms as well. The conditional probability of expected losses from individual shortfall risk to coincide with higher systemwide liquidity risk can be used to calculate a fair value price for the necessary insurance coverage specific to each firm.

To illustrate this, we calculate the ratio of the potential systemically based dollar losses of each firm to its RSF—the probabilistic proportion of underfunding (relative to the existing funding level), akin to a probability of distress for a certain risk horizon. More specifically, we estimate the average marginal contribution of each firm to the ES (with statistical significance α) as a measure of systemic liquidity, averaged over the previous four quarters, which is then divided by the average of the discounted present value of the asset value underpinning RSF over the previous four quarters.

Assuming that this *conditional* probability is constant over time and can be expressed as an exponential function, the fair value of a risk-based insurance premium can be obtained as the natural logarithm of 1 minus the preceding ratio and multiplied by the negative inverse of the time period under consideration. Thus, the cost f_{SLR} of insuring stable funding over the short term against possible liability runoffs can be

calculated by multiplying the insurance premium with the value of average uncovered short-term liabilities $\overline{L}_{t, j}^{ST}$ (i.e., excluding secured deposits and investments) over the previous four quarters as a nominal base:

$$f_{SLR} = -\frac{1}{T} \ln \left(1 - \frac{\sum_{t=-4}^0 \left(\overline{\psi}_{t, \tau, j, \alpha} \times \overline{ES}_{t, \tau, \alpha} \right)}{\sum_{t=-4}^0 RSF_{t, j} \times \exp(-r(T-t))} \right) \times \frac{1}{4} \sum_{t=-4}^0 \overline{L}_{t, j}^{ST}, \quad (27.27)$$

where r is the risk-free rate and $T-t$ is the time horizon (i.e., residual maturity). This amount would compensate for the individual firm's cost of future systemic liquidity support.

The SRL approach also can be used within a stress testing framework to examine the vulnerabilities of individual institutions—and the system as a whole—to shocks affecting the valuation of assets and liabilities that underpin the NSFR. Volatility shocks to both asset returns and funding costs as well as the joint dynamics (i.e., dependence structure) between them can alter significantly the net exposure to liquidity shortfall reflected in a market-based measure of liquidity risk. Thus, stress scenarios can be determined based on the historical calibration of market factors affecting the valuation of both ASF and RSF, the constituent components of NSFR—as an approximation of net cash flows in stress situations. The stresses can be based on firm-specific shocks, common shocks, or both:

- *Firm-specific shocks* are modeled by modifying the jump diffusion process (with the frequency, average size, and volatility of jumps calibrated to past stress scenarios) of assets underpinning RSF, which could result in unbounded and random liabilities.
- *Common shocks* are modeled by means of stressing the historical variance–covariance matrix of latent factors of several market interest rates (see Box 27.1). In adverse conditions, we observe higher volatilities of all market interest rates but lower correlation between those rates that are more indicative of changes in short-term asset returns and those rates that better explain short-term funding costs.

2. EMPIRICAL APPLICATION: U.S. BANKING SECTOR

The SRL model was applied to the 13 largest commercial and investment banks in the United States using firm-level data obtained from financial statements and markets. The five-year time period covers January 1, 2005, to December 14, 2010 (1,442 observations).²² The variations in the components of the NSFR—that is, in the ASF and RSF as weighted

²² The sample covers Bank of America, JP Morgan, Citigroup, Morgan Stanley, Goldman Sachs, Merrill Lynch, Wells Fargo, US Bancorp, PNC Financial Services, Bank of New York Mellon, Sun Trust, BB&T, State Street, and Regions Bank.

Box 27.1. Stress Testing within the SRL Model Framework

In adverse conditions, shocks to the volatilities of market funding rates and the correlation between asset returns and funding rates can be imposed mechanically in the model to better examine short-term funding vulnerabilities. The following five-step process offers a useful way to apply market-informed common shocks to the liquidity position of sample firms:¹

1. *Estimation of latent variables from market rates affecting ASF and RSF.* We derive latent variables of multiple market rates by extracting the principal component of each of the following six general categories: (a) short-term government bond yields (at maturities of 3, 5, 7, and 10 years), (b) long-term government bond yields (at maturities of 3, 6, and 12 months), (c) total equity market returns (total return of domestic stock market, MSCI Total Return Index), (d) home currency LIBOR (at maturities of 3, 6, and 12 months), (e) home currency OIS rates (at maturities of 3, 6, and 12 months), and (f) financial bond yields (1–5 years, 15+ years, and rated AAA).
2. *Definition of baseline ASF volatility as a composite measure.* Under baseline conditions, the volatility of ASF is defined as a weighted average volatility of these latent variables whose explanatory power on ASF—on levels—defines the relative size of these weights. The weights are derived as dynamic factor model regression coefficients of ASF on these latent variables and—once rescaled to unity—define the individual contribution of each latent variable (measured as the 120-day rolling window standard deviation) to the composite ASF volatility. The volatility of RSF is endogenous to the model specification, while the covariance between RSF and ASF is based on the 120-day rolling window correlation of their levels.
3. *Definition of shocks to volatility—stressed volatility and covariance of ASF and RSF.* The historical return series of each latent variable are multiplied by the product of the empirical variance–covariance matrix and a “shocked” variance–covariance matrix² (defined by some change to the historical correlation and volatility of all elements) in order to derive stressed series of each latent variable and its volatility under stress.³
4. *Estimation of stressed ASF and RSF volatility.* Like in the baseline case, the stressed ASF volatility is a composite measure derived as a weighted average of the stressed individual volatility series of each latent variable. The stressed volatility series are multiplied by corresponding coefficient values estimated in a dynamic factor regression of the baseline composite ASF volatility on the baseline volatility of these latent variables. The same process is repeated for the stressed RSF volatility based on the endogenously generated RSF volatility and the “shocked” variance–covariance matrix of the same latent variables. The stressed covariance between ASF and RSF is an explicit result of the “shocked” variance–covariance matrix.
5. *Updating the individual and joint measure of liquidity risk using the stressed input values.* The joint probability distribution of liquidity shortfalls is reestimated after recalculating individual expected losses from liquidity shortfall using the stressed ASF and RSF volatility measure and aggregating these results in the same way as in the baseline case.

¹ Also, various liquidity stress testing methodologies could be incorporated in this approach, such as van den End (2008), Wong and Hui (2009), and Aikman and others (2009).

² The shock to the variance–covariance matrix does not need to be uniform, such as a 20 percentage point shock to volatility and/or correlation, but also can accommodate asymmetric changes based on each particular market rate, with some experiencing greater shocks than others.

³ This is done technically by multiplying the inverted, lower-triangular variance–covariance matrix of historical market returns with an inverted variance–covariance matrix defined by “shocked” assumptions on correlation and individual volatility of latent variables.

sums of their constituent liabilities and asset positions (see Table 27.1)—were used to compute the expected losses owing to liquidity shortfalls under extreme conditions.^{23,24}

The results suggest that the current liquidity standards in Basel III (whether as an accounting measure or a risk-adjusted measure) are not able to capture the potential liquidity shortfall under stressed conditions. The median of the market-based NSFR for the 13 banks stays above one and has continued to improve since the credit crisis (see Figure 27.4). In contrast, the median expected losses generated by the SRL model would suggest that banks have become more vulnera-

ble to extreme liquidity shocks and that their expected losses arising from potential liquidity shortfalls were higher during some timeframes, namely, in the run-up to the March 14, 2008, Bear Stearns rescue and around year-end 2008. Those results apply especially to firms dependent on funding sources that are more susceptible to short-term (and more volatile) market interest rates; that dependency, in combination with their relatively higher exposure to maturity mismatches, accentuates their vulnerability to liquidity risk. Because the SLR model takes into account the joint dynamics between the ASF and RSF via their covariance structure, it provides a far deeper analysis of the liquidity risk to which a firm is exposed than does looking at them separately or with only accounting data.

Using the results for individual banks, we estimate systemwide liquidity risk under extreme stress. We derive the joint ES at the 95th percentile for all sample banks and find that accumulated expected losses associated with individual banks’ risk-adjusted NSFR would have underestimated systemwide liquidity shortfall for the period from mid-2009 to mid-2010, where the dashed line exceeds the solid line in

²³ Extreme conditions were defined to be those that occur with a probability of 5 percent or less.

²⁴ Further specifications of the model include three-month equity call prices for sample banks, put option time horizon of $t=1$, risk-free rate=0.03, and an estimation window for jump parameters over 120 days. Price changes would qualify as jumps if equity daily returns were outside an acceptable range of –10 and +10 percent and represented a deviation of more than 50 percent on a normalized scale over a 120-day rolling window (i.e., observations were in the top decile over a half-year observation period).

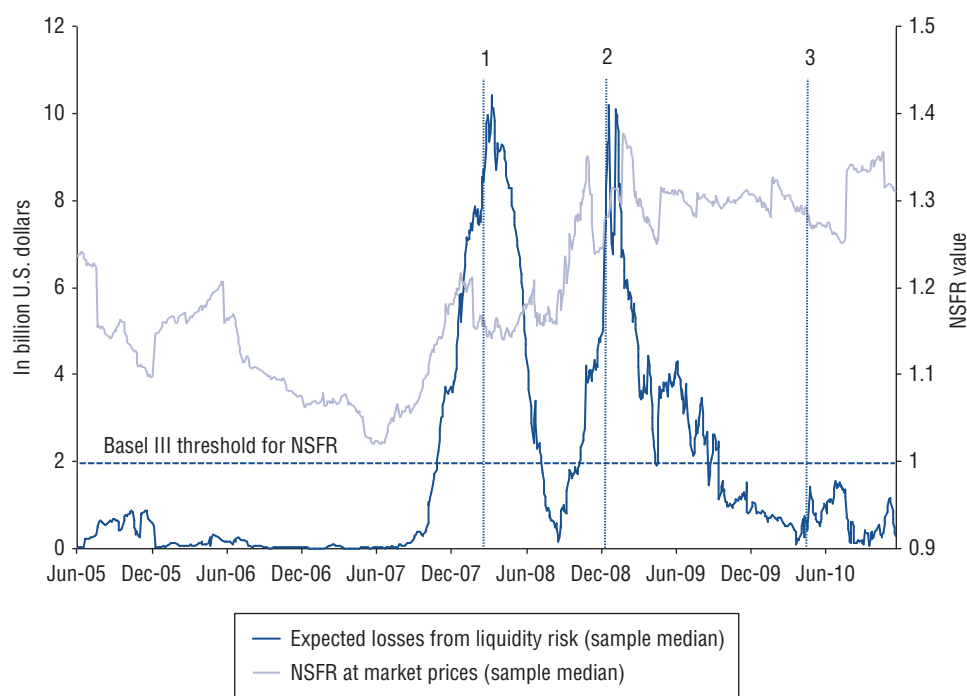
TABLE 27.1

Factors Used in Estimations

Available Stable Funding		Required Stable Funding	
Equity	1.00	Cash	0.00
Tier 2	1.00	<i>Customer loans</i>	0.75
<i>Demand deposits</i>	0.80	<i>Commercial loans</i>	0.85
<i>Saving and term deposits</i>	0.85	<i>Advances to banks</i>	0.00
<i>Bank deposits</i>	0.00	<i>Other commercial and retail loans</i>	0.85
<i>Other deposits and short-term borrowing</i>	0.00	<i>Other loans</i>	1.00
<i>Derivative liabilities</i>	0.00	<i>Derivative assets</i>	0.90
<i>Trading liabilities</i>	0.00	<i>Trading securities</i>	0.15
Senior debt maturing after one year	1.00	<i>Available for sale securities</i>	0.15
Other non-interest-bearing liabilities	0.00	At-equity investments in associates	1.00
Other reserves	1.00	Other earning assets	1.00
		Insurance assets	1.00
		Residual assets	1.00
		Reserves for nonperforming loans	1.00
		Contingent funding	0.05

Sources: Author; and Bankscope.

Note: Categories in italics are staff judgements; others more closely approximate Basel classifications.



Sources: Author; Bankscope; and Bloomberg.

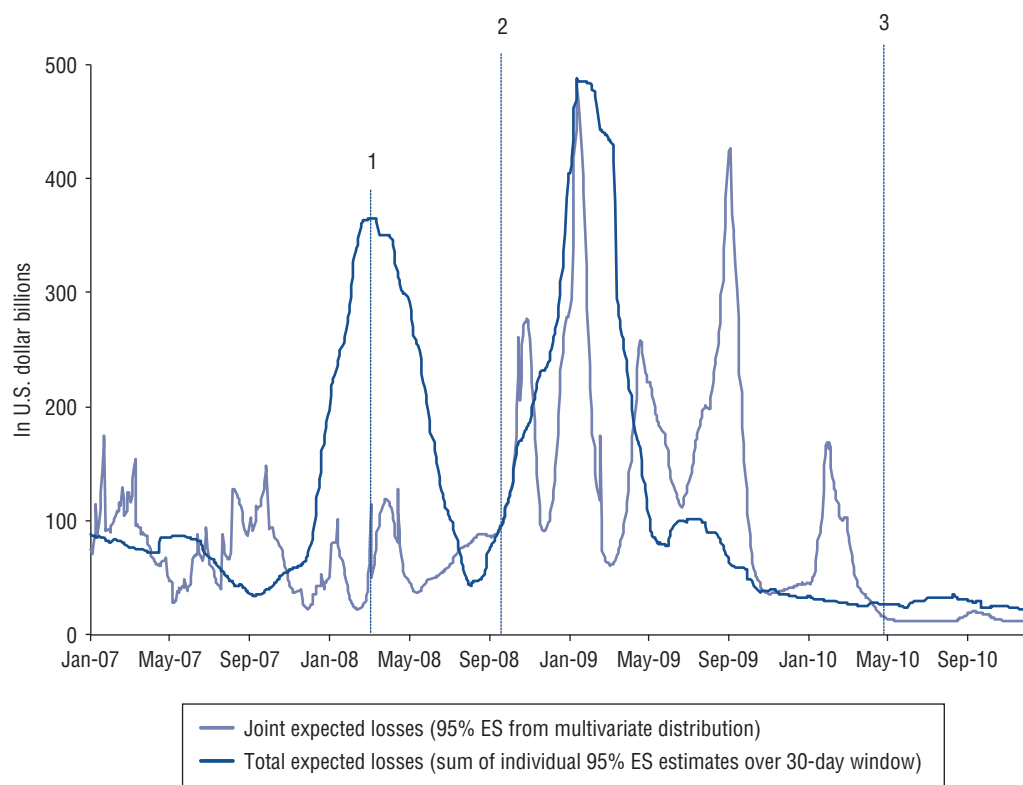
Note: Dates of vertical lines are as follows: (1) March 1, 2008—Bear Stearns rescue; (2) September 14, 2008—Lehman Brothers failure; and (3) April 27, 2010—Greek debt crisis. Expected losses are at the 95 percent confidence level.

Figure 27.4 Individual Market-Based Net Stable Funding Ratio (NSFR) and Associated Expected Losses Using Option Pricing (sample median)

Figure 27.5.²⁵ It would have failed to take into account the interlinkages in banks' funding positions and their common

²⁵ In Figure 27.5, the solid line represents the daily sum of individual, market-implied expected losses, and the dashed line indicates the joint tail risk of these individual expected losses. Both tail risks are measured so that the chances of such events are 5 percent or less.

exposure to the risk of funding shocks—that is, the systemic component—which raises joint liquidity risk beyond the simple composition of individual net exposures if estimates are derived at a high level of statistical confidence (at percentile levels far removed from average outcomes). Conversely, during the first half of 2008, the sum of individual expected losses associated with individual banks' risk-adjusted NSFR



Sources: Author; Bankscope; and Bloomberg.

Note: Dates of vertical lines are as follows: (1) March 1, 2008—Bear Stearns rescue; (2) September 14, 2008—Lehman Brothers failure; and (3) April 27, 2010—Greek debt crisis. Expected losses are at the 95 percent confidence level.

ES = expected shortfall.

Figure 27.5 Joint Expected Losses from Systemic Liquidity Risk Using Option Pricing
(expected shortfall at the 95th percentile, in billions of U.S. dollars)

would have overestimated systemwide liquidity shortfall under extreme stress conditions. The results suggest that if liquidity shortfalls happen simultaneously, accounting for their interdependence is imperative for a more accurate representation of systemic liquidity risk, and contagion risk from this interdependence gets accentuated during times of extreme stress in markets.

The results imply that some banks contribute to systemic liquidity risk beyond their individual exposure to liquidity shortfalls in times of distress, which underscores the usefulness of a systemwide assessment of liquidity risk. During the height of the credit crisis, the average contribution of the largest U.S. banks to the tail risk from simultaneous liquidity shortfalls was higher than their individual liquidity risk (with the latter accounting for only the average net exposure to liquidity risk). In 2010, expected losses from the likelihood of rising systemic liquidity risk in the wake of extreme disruptions to funding markets (and their impact on asset prices and volatility) exceeded US\$31 billion (see Table 27.2), which is considerable but far lower than during the peak of the financial crisis (at US\$150 billion). Appreciable differences among banks indicate varying sensitivity to common changes in funding conditions and the extent to which these translate

into externalities affecting general liquidity risk in the overall sample.

We also calculate both a capital surcharge and an insurance premium, which take into account the support that U.S. banks would receive in times of systemic liquidity stress. It thus represents the public cost of marginal expected losses when two or more institutions face significant liquidity shortfalls. Table 27.2 presents the distribution of the capital charges over selected U.S. banks, and Table 27.3 does so for the value of the insurance premium that would compensate for the contribution to joint expected losses caused by each bank. We find that the application of the SRL model would lead to a potential 5 percent increase of total capital to include net exposures arising mainly from the individual contribution to systemic liquidity risk. This number, however, would exceed 25 percent for the worst bank in the sample (for which individual liquidity risk dominates) and could more than triple on average for all banks if stress periods, like the recent credit crisis, were to be included in the historical estimation (see Table 27.2). Alternatively, we calculate the commensurate fair value insurance premium to compensate for the liquidity support that would be needed to bring the NSFR above one during stressful times. For most banks, the premium

TABLE 27.2

Capital Charge for Individual Liquidity Risk and Contributions to Systemic Liquidity Risk (in billions of U.S. dollars)									
Individual Liquidity Risk				Contribution to Systemic Liquidity Risk			Economic Significance		
	<i>Stress Period:</i> Sept. 14, 2008– Dec. 31, 2009	Last quarter (2010 Q4)	Average of 2010 Q1–Q4	<i>Stress Period:</i> Sept. 14, 2008– Dec. 31, 2009	Last quarter (2010 Q4)	Average of 2010 Q1–Q4	Capital charge (maximum of (1)–(4))	Share of total capital (in percent)	Share of total assets (in percent)
	(1)	(2)		(3)	(4)				
Minimum	0.00	0.00	0.03	1.55	0.08	0.27	0.27	1.57	0.20
Median	1.46	0.74	1.18	6.42	0.66	2.05	2.05	4.82	0.73
Maximum	33.32	8.53	9.86	13.51	3.09	5.96	9.86	3.25	0.44

Sources: Author; Bankscope; and Bloomberg.

Note: This exercise was run on a selected set of 13 large U.S. commercial and investment banks. The last column matches the distributions of the individual capital charges and reported total capital of all sample institutions. In this case, the maximum capital charge for the worst bank in 2010 coincides with a disproportionately higher total capital amount, which reduces the percentage share of the capital add-on for systemic liquidity from 4.82 percent (median) to 3.25 percent (maximum). The expected losses during the stress period (covering the height of the recent credit crisis) are listed here for illustrative purposes (indicating a potential extension of the presented approach consistent with the revised market risk estimation under the new regulatory framework for banks (BCBS, 2010b).

TABLE 27.3

Summary Statistics of Individual Contributions to Systemic Liquidity Risk and Associated Fair Value Insurance Premium				
Range	Precrisis: end-June, 2006–end-June 2007	Subprime Crisis: July 1, 2007–Sept. 14, 2008	Credit Crisis: Sept. 14, 2008–Dec. 31, 2009	Sovereign Crisis: Jan. 1–Dec. 31, 2010
Individual contribution to systemic liquidity risk (at 95th percentile; in percent)				
Minimum	1.2	0.6	1.0	1.7
Median	6.8	4.5	8.3	7.6
Maximum	13.4	35.1	16.7	14.5
<i>Total</i>	<i>100.0</i>	<i>100.0</i>	<i>100.0</i>	<i>100.0</i>
Insurance cost based on reported exposure: Fair value insurance premium uninsured short-term liabilities (in billion U.S. dollars)				
Minimum	0.7	0.1	0.7	0.1
Median	1.9	1.4	3.9	0.8
Maximum	7.8	17.2	11.3	1.9

Sources: Bankscope; Bloomberg; and IMF staff calculations.

Note: This exercise was run on a number of selected U.S. banks. Each bank's percentage share reflects its contribution to total expected losses from systemic liquidity risk. Insured deposits here are defined as 10 percent of demand deposits reported by sample banks. Note that the share of deposits covered by guarantees varies by country and could include time and savings deposits. Robustness checks reveal that reducing the amount of uninsured short-term liabilities does not materially affect the median and maximum.

would amount to US\$0.8 billion for most banks, or less than half the additional capital needed to offset expected losses from liquidity risk at any given point in time over a one-year risk horizon (see Table 27.3).

3. CONCLUSION

The SRL model offers several potential benefits in areas that are largely unaddressed in the current Basel III liquidity framework. It is a price-based approach that treats liquidity risk endogenously as a dynamic exposure via the risk-adjusted value of stable funding rather than a combination of discrete accounting identities by drawing on the market-based evaluation of the riskiness of a firm. This approach generates a forward-looking measure of liquidity risk (subject to different

degrees of leverage and maturity mismatches defining the risk profile of institutions), which helps determine the probability of an individual institution experiencing liquidity shortfall and incurring associated expected loss (as an approximation of the economic cost of being unable to service ongoing debt payments, resulting in net cash outflows).

Overall, the SLR model provides a tractable framework for the assessment of systemwide valuation effects arising from joint liquidity risk. The impact of a particular funding configuration is not assessed individually but in concert with all banks to generate estimates of systemic liquidity risk. As such, it takes the systemic components of liquidity risk into account by estimating the joint sensitivity of assets and liabilities (and their volatilities) of multiple entities to common changes in market prices. In this way, the SRL model helps assess how

the size and interconnectedness of individual institutions can create short-term, systemwide vulnerabilities to cases of considerable liquidity risk shared by several entities. Thus, it seems better suited than existing prudential approaches to identify, quantify, and mitigate systemic risk because it (1) measures the marginal contribution of each institution to total systemic liquidity risk and (2) can be used to construct a supervisory charge for the institution's contribution to systemic liquidity risk that provides incentives for the internalization of the cost of contingent liquidity support in times of stress.

REFERENCES

- Acharya, Viral V., Thomas F. Cooley, Matthew P. Richardson, and Ingo Walter, eds., 2010, *Regulating Wall Street: A New Architecture for Global Finance* (Hoboken, NJ: John Wiley & Sons, Inc.).
- Acharya, Viral V., and Sabri Oncu, 2010, "The Repo Market," in: Acharya, Viral V., Cooley, Thomas F., Richardson, Matthew P., and Ingo Walter (eds.) *Regulating Wall Street: A New Architecture for Global Finance* (Hoboken, NJ: John Wiley & Sons, Inc.).
- Acharya, Viral V., Joao A. C. Santos, and Yorulmazer Tanju, 2010, "Systemic Risk and Deposit Insurance Premiums," *Economic Policy Review* (New York: Federal Reserve Bank of New York).
- Aikman, David, Alessandro Piergiorgio, Bruno Eklund, Prasanna Gai, Sujit Kapadia, Elizabeth Martin, Nada Mora, Gabriel Sterne, and Matthew Willison, 2009, "Funding Liquidity Risk in a Quantitative Model of Systemic Stability," Working Paper No. 372 (London: Bank of England). Available via the Internet: <http://www.bankofengland.co.uk/publications/Pages/workingpapers/2009/wp372.aspx>
- Ait-Sahalia, Yacine, and Andrew W. Lo, 1998, "Nonparametric Estimation of State-Price Densities Implicit in Financial Asset Prices," *Journal of Finance*, Vol. 53, No. 2, pp. 499–547.
- Backus, David, Silverio Foresi, and Liuren Wu, 2004, "Accounting for Biases in Black-Scholes," Working Paper, New York University, Goldman Sachs Group, and Baruch College.
- Bakshi, Gurdip, Cao, Charles, and Zhiwu Chen, 1997, "Empirical Performance of Alternative Option Pricing Models," *Journal of Finance*, Vol. 52, No. 5, pp. 2003–49.
- Basel Committee on Banking Supervision, 2009, "Revisions to the Basel II Market Risk Framework" (Basel, July). Available via the Internet: <http://www.bis.org/publ/bcbs158.htm>
- , 2010a, "Basel III: International Framework for Liquidity Risk Measurement, Standards and Monitoring" (Basel, December). Available via the Internet: <http://www.bis.org/publ/bcbs188.pdf>
- , 2010b, "Results of the Comprehensive Quantitative Impact Study" (Basel, December). Available via the Internet: <http://www.bis.org/publ/bcbs186.htm>
- Black, Fischer, and Myron Scholes, 1973, "The Pricing of Options and Corporate Liabilities," *Journal of Political Economy*, Vol. 81, No. 3, pp. 637–54.
- Breeden, Douglas T., and Robert H. Litzenberger, 1978, "Prices of State-Contingent Claims Implicit in Option Prices," *Journal of Business*, Vol. 51, No. 4, pp. 621–51.
- Brunnermeier, Markus, and Lasse Pedersen, 2009, "Market Liquidity and Funding Liquidity," *Review of Financial Studies*, Vol. 22, No. 6, pp. 2201–38.
- Cao, Jin, and Gerhard Illing, 2009, "Endogenous Systemic Liquidity Risk," CESIFO Working Paper No. 2627 (Munich: Ludwig-Maximilians-Universitaet and the Ifo Institute for Economic Research).
- Coles, Stuart G., Janet Heffernan, and Jonathan A. Tawn, 1999, "Dependence Measures for Extreme Value Analyses," *Extremes*, Vol. 2, pp. 339–65.
- Cox, John C., Ross, Stephen A., and Mark Rubinstein, 1979, "Option Pricing: A Simplified Approach," *Journal of Financial Economics*, Vol. 7, No. 3, pp. 229–63.
- Davis, Ellen, 2011, "IMF Modelling Work on Liquidity Risk Points to Capital Hike," *Risk Magazine*, October 10. Available via the Internet: <http://www.risk.net/risk-magazine/news/2114972/imf-modelling-liquidity-risk-capital-hike-jobst#>
- Dumas, Bernard, Fleming, Jeff, and Robert E. Whaley, 1998, "Implied Volatility Functions: Empirical Tests," *Journal of Finance*, Vol. 53, No. 6, pp. 2059–106.
- Farhi, Emmanuel, Mikhail Golosov, and Aleh Tsyvinski, 2009, "A Theory of Liquidity and Regulation of Financial Intermediation," *Review of Economic Studies*, Vol. 76, No. 3, pp. 973–92.
- Fisher, Ronald A., and Leonard H. C. Tippett, 1928, "Limiting Forms of the Frequency Distribution of the Largest or Smallest Member of a Sample," *Proceedings of the Cambridge Philosophical Society*, Vol. 24, pp. 180–90.
- Gnedenko, Boris V., 1943, "Sur la Distribution Limite du Terme Maximum d'Une Série Aléatoire," *Annals of Mathematics*, Vol. 44, pp. 423–53.
- Goodhart, Charles, 2009, "Liquidity Management," presented at "Financial Stability and Macroeconomic Policy: A Symposium," Jackson Hole, Wyoming, August 20–22.
- Gorton, Gary B., and Andrew Metrick, 2009, "Securitized Banking and the Run on the Repo," NBER Working Paper No. 15223 (Cambridge, MA: National Bureau of Economic Research). Available via the Internet: <http://www.nber.org/papers/w15223>
- Gray, Dale F., and Andreas A. Jobst, 2009, "Higher Moments and Multivariate Dependence of Implied Volatilities from Equity Options as Measures of Systemic Risk," *Global Financial Stability Report*, Chapter 3, pp. 128–31 (Washington, April). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2009/01/index.htm>
- , 2010, "New Directions in Financial Sector and Sovereign Risk Management," *Journal of Investment Management*, Vol. 8, No. 1, pp. 23–38.
- , 2011, "Modelling Systemic Financial Sector and Sovereign Risk," *Sveriges Riksbank Economic Review*, No. 2, pp. 68–106. Available via the Internet: http://www.riksbank.se/upload/Rapporter/2011/POV_2/er_2011_2.pdf
- Gray, Dale F., and Samuel Malone, 2010, "Quantifying Systemic Risk and Reconceptualizing the Role of Finance for Economic Growth," *Journal of Investment Management*, Vol. 8, No. 2, pp. 90–110.
- Hall, Peter, and Nader Tajvidi, 2000, "Distribution and Dependence-Function Estimation for Bivariate Extreme Value Distributions," *Bernoulli*, Vol. 6, pp. 835–44.
- Heston, Steven L., 1993, "A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options," *Review of Financial Studies*, Vol. 6, No. 2, pp. 327–43.
- Heston, Steven L., and Saikat Nandi, 2000, "A Closed-Form GARCH Option Valuation Model," *Review of Financial Studies*, Vol. 13, No. 3, pp. 585–625.

- International Monetary Fund, 2009, *Global Financial Stability Report*, Chapter 3, World Economic and Financial Surveys (Washington, April). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2009/01/index.htm>
- , 2010, *Global Financial Stability Report*, Chapter 2, World Economic and Financial Surveys (Washington, October). Available via the Internet: <http://www.imf.org/external/pubs/ft/gfsr/2010/02/index.htm>
- , 2011, *Global Financial Stability Report*, Chapter 2, World Economic and Financial Surveys (Washington, April). Available via the Internet: <http://www.imf.org/external/pubs/ft/gfsr/2011/01/index.htm>
- Jobst, Andreas A., 2007, "Operational Risk—The Sting Is Still in the Tail but the Poison Depends on the Dose," *Journal of Operational Risk*, Vol. 2, No. 2, pp. 1–56.
- , 2011, "Technical Description of the Systemic Risk-Adjusted Liquidity Model," *Global Financial Stability Report*, Chapter 3, Annex 2.2, World Economic and Financial Surveys, pp. 101–5 (Washington, April). Available via the Internet: <http://www.imf.org/external/pubs/ft/gfsr/2011/01/index.htm>
- , 2014, "Measuring Systemic Risk-Adjusted Liquidity (SRL)—A Model Approach," *Journal of Banking and Finance*, Vol. 45, pp. 270–87. Also published as IMF Working Paper 12/209 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/ft/wp/2012/wp12209.pdf>
- Jobst, Andreas A., and Dale F. Gray, 2013, "Systemic Contingent Claims Analysis—Estimating Market-Implied Systemic Risk," IMF Working Paper 13/54 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=40356>
- Jones, E. Philip, Scott P. Mason, and Eric Rosenfeld, 1984, "Contingent Claims Analysis of Corporate Capital Structures: An Empirical Investigation," *Journal of Finance*, Vol. 39, No. 3, pp. 611–25.
- Merton, Robert C., 1973, "Theory of Rational Option Pricing," *Bell Journal of Economics and Management Science*, Vol. 4, No. 1, pp. 141–83.
- , 1974, "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates," *Journal of Finance*, Vol. 29, No. 2, pp. 449–70.
- Perotti, Enrico, and Javier Suarez, 2009, "Liquidity Risk Charges as a Macroprudential Tool," CEPR Policy Insight No. 40 (London). Available via the Internet: http://www.cepr.org/pubs/PolicyInsights/CEPR_Policy_Insight_040.asp
- , 2011, "A Pigovian Approach to Liquidity Regulation," *International Journal of Central Banking*, Vol. 7, No. 4, pp. 3–41. Available via the Internet: <http://www.ijcb.org/journal/ijcb11q4a1.htm>
- Pickands, James, 1981, "Multivariate Extreme Value Distributions," *Proceedings of the 43rd Session of the International Statistical Institute*, Vol. 49, pp. 859–78.
- Poon, Ser-Huan, Michael Rockinger, and Jonathan Tawn, 2003, "Extreme Value Dependence in Financial Markets: Diagnostics, Models, and Financial Implications," *Review of Financial Studies*, Vol. 17, No. 2, pp. 581–610.
- Roe, Mark, 2009, "End Bankruptcy Priority for Derivatives, Repos and Swaps," *Financial Times*, December 16.
- Rookley, Cameron, 1997, "Fully Exploiting the Information Content of Intra Day Option Quotes: Applications in Option Pricing and Risk Management," mimeo, University of Arizona.
- Shin, Hyun Song, 2009, "Reflections on Modern Bank Runs: A Case Study of Northern Rock," *Journal of Economic Perspectives*, Vol. 23, No. 1, pp. 101–19.
- Shleifer, Andrei, and Robert Vishny, 2010, "Fire Sales in Finance and Macroeconomics," NBER Working Paper No. 16642 (Cambridge, MA: National Bureau of Economic Research). Available via the Internet: <http://www.nber.org/papers/w16642>
- Valderrama, Laura, 2010, "Macroprudential Regulation under Repo Funding," IMF Working Paper 10/220 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.cfm?sk=24249.0>
- van den End, Jan Willem, 2008, "Liquidity Stress-Tester: A Macro Model for Stress-Testing Banks' Liquidity Risk," DNB Working Paper No. 175 (Amsterdam: De Nederlandsche Bank). Available via the Internet: <http://www.dnb.nl/en/publications/dnb-publications/dnb-working-papers-series/dnb-working-papers/auto175528.jsp>
- Wong, Eric, and Cho-Hoi Hui, 2009, "A Liquidity Risk Stress-Testing Framework with Interaction between Market and Credit Risks," HKMA Working Paper 06/2009, March (Hong Kong SAR: Hong Kong Monetary Authority). Available via the Internet: <http://www.hkma.gov.hk/eng/publications-and-research/research/working-papers/2009>

PART III

The Macro-Financial Approach

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Introduction to the Macro-Financial Approach to Stress Testing

ANDREA M. MAECHLER

During the global financial crisis, the world witnessed the near collapse of major financial institutions that many had believed to be impervious to the sort of panic that brought the global financial system to its knees and led to a worldwide recession. A key lesson from these events is that a purely micro-prudential approach to financial regulation and supervision, focused on the conditions of individual financial institutions and markets, could fail to detect important systemic or cross-cutting risks (Bernanke, 2011). Since then, there has been a generalized push globally to make the financial sector as a whole more robust by taking a broader macro-financial approach to oversight. The key feature of such an approach is to recognize that financial stability depends not only on the safety and soundness of individual financial institutions but also on how they interact with each other and with the real economy.

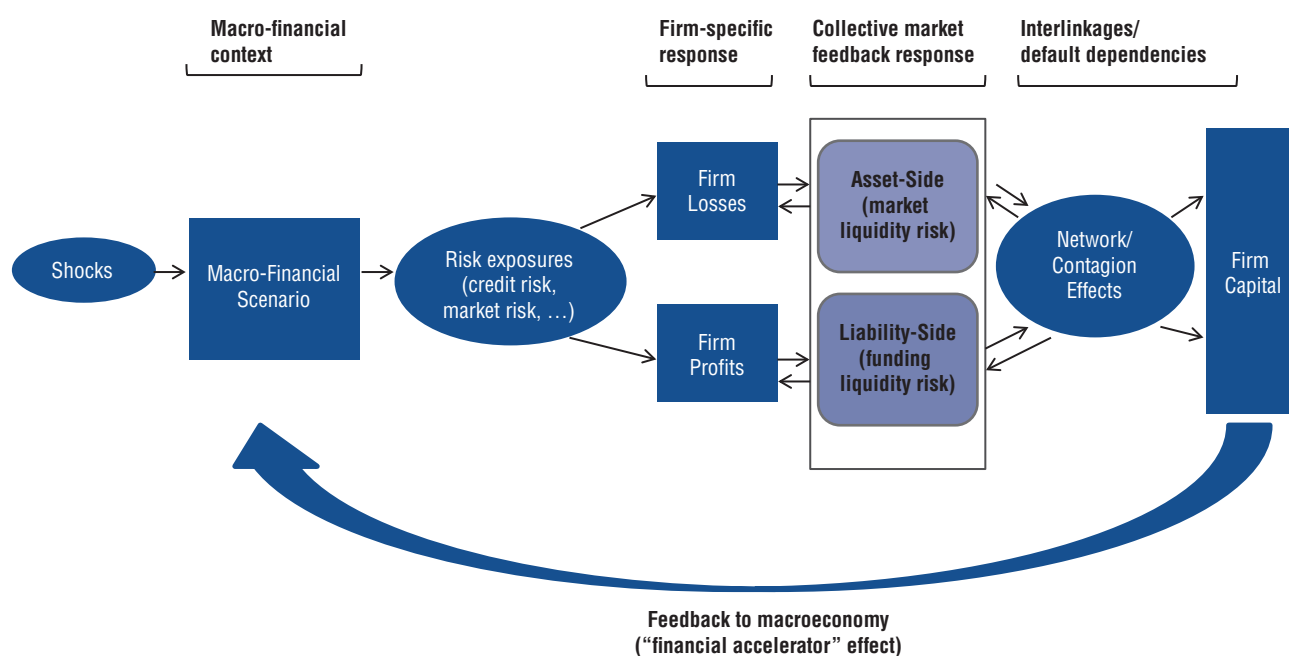
A macro-financial approach to financial oversight requires a holistic understanding of the sources of systemic risk. This includes monitoring the inherently procyclical nature of risk buildups (e.g., through leverage and rising asset prices), both in significantly important financial institutions and within the financial system more generally.¹ It also means assessing risk correlations among individual market participants and understanding how they may be interlinked with other market participants. Finally, there is a need to explicitly incorporate macro-financial linkages in countries' financial oversight frameworks, something that was poorly done before the global financial crisis (Bernanke, 2011).

In this regard, macro-financial stress tests have proven particularly useful in overcoming key shortcomings in tradi-

tional financial supervision. They have provided a consistent framework for assessing firms' common exposures to a set of risks. Moreover, by linking key projected macroeconomic and financial variables to firms' performance, they have also helped to improve the understanding of how shocks are transmitted through the financial system to the real economy and to provide a forward-looking assessment of financial stability.

Although not new, macro-financial stress tests have gained prominence over the last few years. Macro-financial stress tests have been an integral part of the IMF's Financial Sector Assessment Program (FSAP) since the late 1990s, although they typically were not integrated into countries' regular surveillance framework (Moretti, Stolz, and Swinburne, 2008). One noteworthy exception is the Bank of England, which has been implementing macro-financial stress tests since the early 2000s (Bunn, Cunningham, and Drehmann, 2005). In 2009, the U.S. authorities applied macro-financial stress tests under the Supervisory Capital Assessment Program (SCAP) to assess how much capital their largest bank holding companies needed to absorb potential losses and meet the needs of creditworthy borrowers under a more adverse scenario (Board of Governors of the Federal Reserve System, 2009a, 2011). In Europe, macro-financial stress tests were used to assess banks' capital adequacy in the context of the sovereign debt crisis. At the IMF, they were used to conduct a regular capital shortfall exercise that would feed into their global financial stability assessment (IMF, 2008a, 2008b, 2009a, 2009b, 2010a). A stylized macro-financial stress test framework is presented in Figure 28.1 (see Bunn, Cunningham, and Drehmann, 2005; Jenkinson, 2008; and Constâncio, 2011, for discussions on macro-financial stress testing frameworks).

¹ See, for example, Adrian and Shin (2008) on how leveraged investment can fuel an asset and/or credit bubble.



Source: Author.

Figure 28.1 A Stylized Macro-Financial Stress Test Framework

There are two stages to the process:

- In the first stage, an initial shock needs to be translated into a macro-financial scenario that maps the projection of key macroeconomic and financial variables (e.g., GDP, potential output, unemployment, house prices, bond yields). For example, market fears about an unsustainable fiscal position could result in higher sovereign yields, which could lead to a dampening in aggregate demand, higher unemployment, and lower potential output. This type of translation can be done using a variety of tools, ranging from large-scale in-house macroeconomic models (e.g., structural macroeconomic models, dynamic stochastic general equilibrium models) to smaller partial equilibrium time series models and vector autoregressive-driven impulse response functions.
- In the second stage, the macro-financial scenario must be translated into firms' financial conditions (e.g., firm losses and earnings potential). In an accounting world, this is done by mapping the shock onto firms' balance sheet exposures. In a risk-based world, part of the adjustment takes place through changes in financial firms' risk profiles (e.g., probability of default, loss given default, credit ratings). Market price-based risk approaches, including some of those presented in Part II of this volume, have the advantage of computing the distribution of possible outcomes, rather than single point estimates.

The strength of macro-financial stress tests is to produce a consistent set of impact assessments, although each bank

will show a different vulnerability depending on its own risk exposures:

- Credit risk models are the most developed. They can be based on accounting data or on more detailed portfolio-based or market-based data. Models linking nonperforming loans to macroeconomic variables are presented in Vazquez, Tabak, and Souto (Chapter 29) and Wezel, Canta, and Luy (Chapter 30). Sevogiano and Padilla (Chapter 31) present methodologies that incorporate the effects of macroeconomic shocks into credit risk, providing robust estimators when only short time series of loans exist and when only partial information about borrowers is available.
- Market risk models, on the other hand, are still largely in their infancy. Different approaches have been used to capture market risk. The IMF's U.S. FSAP capital shortfall exercise (IMF, 2010b), for example, focused on modeling the marked-to-market repricing of structured products (e.g., asset-backed securities, including mortgage-backed securities, credit default obligations). Among the high-profile stress testing exercises, the European Banking Authority stress test applied a fixed haircut to specific sovereign paper, while the U.S. SCAP exercise conducted a separate test to stress banks' trading book.

Another challenge is to model the amplification of systemic risk through network effects. A severe strain at, or even default in, a systemically important financial institution, for example, is likely to affect the soundness of similar firms. This requires accounting for the inherently nonlinear nature of financial

instability, where risk correlations can rise sharply in tail-risk scenarios. Promising work has been done in the area, including the approaches presented by Segoviano and Goodhart in Chapter 32 and by Jobst and Gray earlier in Chapter 26, where the entire nonparametric time-varying dependence structure among financial institutions is derived. Nonetheless, much remains to be done to fully integrate high-frequency, market-based approaches into the more traditional and slow-moving macroeconomic models.

Most macro-financial stress tests focus on the first-round effects, namely, on financial firms' ability to absorb an adverse shock and provide credit to the economy. The impacts of severe macro shocks typically are modeled as if they occur on a one-off basis, leading to an underestimation of the possible impacts. The macro-financial shock is then mapped onto firms' capital needs and credit growth potential. One example is the forward-looking balance sheet approach, such as the one used in the U.S. SCAP exercise and presented in a simplified fashion by Keim and Maechler in Chapter 33, which models the impact of a projected path for key macroeconomic and financial variables on banks' capital buffers and other balance sheet positions, including their ability to provide credit. The crisis also has highlighted the challenge of aggregating risk factors across individual institutions and modeling firms' profitability (see Board of Governors of the Federal Reserve System, 2009b), with stress testing revenues—especially for stressed conditions—largely seen to be a “black box” (Schuermann, 2012).

Thus, one of the biggest challenges for stress testing is to adequately model the systemic impact of a particular shock, taking into account the full cycle of macroeconomic and financial interactions and feedback effects. Individual bank responses to a given shock can lead to a broader market response, with further feedback effects to banks' exposures. For example, a slowdown in issuance or a pullback of credit by banks following an initial economic or market shock could further affect economic activity (second-round effect), which would have implications for the banks' ability to additionally absorb the consequences of further deterioration in the macroeconomic environment (third-round effect) and so forth. This requires macroeconomic models with richer real-financial linkages, something that largely was lacking prior to the global financial crisis (Roger and Vlcek, 2011). Tieman and Maechler (Chapter 34) take a step toward closing this gap in the literature by estimating the short-run feedback effects from financial sector risk to the real economy through the credit channel.

REFERENCES

- Adrian, Tobias, and Hyun Song Shin, 2008, “Financial Intermediaries, Financial Stability, and Monetary Policy,” FRBNY Staff Report No. 346 (New York: Federal Reserve Bank of New York). Available via the Internet: http://www.newyorkfed.org/research/staff_reports/sr346.html
- Bernanke, Ben S., 2011, “Implementing a Macroprudential Approach to Supervision and Regulation,” Remarks at the Federal Reserve Bank of Chicago, 47th Annual Conference on Bank Structure and Competition, Chicago, May 5. Available via the Internet: <http://www.federalreserve.gov/newsevents/speech/bernanke20110505a.htm>
- Board of Governors of the Federal Reserve System, 2009a, “The Supervisory Capital Assessment Program: Design and Implementation” (Washington, April 24). Available via the Internet: <http://www.federalreserve.gov/bankinforeg/scap.htm>
- , 2009b, “The Supervisory Capital Assessment Program: Overview of Results” (Washington, May 7). Available via the Internet: <http://www.federalreserve.gov/bankinforeg/scap.htm>
- , 2011, “Comprehensive Capital Analysis and Review: Objectives and Overview” (Washington, March 18). Available via the Internet: www.federalreserve.gov/newsevents/press/bcreg/bcreg20110318a1.pdf
- Bunn, Philip, Alastair Cunningham, and Mathias Drehmann, 2005, “Stress Testing as a Tool for Assessing Systemic Risks,” *Financial Stability Review*, June, pp. 116–26 (London: Bank of England). Available via the Internet: <http://www.bankofengland.co.uk/publications/Pages/fsr/2005/fsr18.aspx>
- Constâncio, Vitor, 2011, “Macro-Prudential Policy: Strengthening the Foundations—Enhancing the Toolkit and Taking Action,” keynote address at the First Conference of the Macro-Prudential Research Network, Frankfurt, October 5. Available via the Internet: <http://www.ecb.europa.eu/press/key/date/2011/html/sp111005.en.html>
- International Monetary Fund, 2008a, *Global Financial Stability Report*, World Economic and Financial Surveys (Washington, April). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2008/01/index.htm>
- , 2008b, *Global Financial Stability Report*, World Economic and Financial Surveys (Washington, October). Available via the Internet: <http://www.imf.org/external/pubs/ft/gfsr/2008/02/index.htm>
- , 2009a, *Global Financial Stability Report*, World Economic and Financial Surveys (Washington, April).
- , 2009b, *Global Financial Stability Report*, World Economic and Financial Surveys (Washington, October).
- , 2010a, *Global Financial Stability Report*, World Economic and Financial Surveys, (Washington, April).
- , 2010b, “United States: Stress Testing—Technical Note,” IMF Country Report 10/244 (Washington, July). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=24101>
- Jenkinson, Nigel, 2008, “Developing a Framework for Stress Testing of Financial Stability Risks,” comments to the ECB High Level Conference on Simulating Financial Instability, Frankfurt, July 12–13. Available via the Internet: <http://www.bis.org/review/r070716g.pdf>
- Moretti, Marina, Stephanie Stolz, and Mark Swinburne, 2008, “Stress Testing at the IMF,” IMF Working Paper 08/206 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=22275>
- Roger, Scott, and Jan Vlcek, 2011, “Macro-Financial Modeling at Central Banks: Recent Developments and Future Directions,” IMF Working Paper 11/103 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=24829>
- Schuermann, Til, 2012, “Stress Testing Banks,” Wharton Financial Institutions Center Working Paper No. 12-08 (Philadelphia: University of Pennsylvania).

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A Macro Stress Test Model of Credit Risk for the Brazilian Banking Sector

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This chapter proposes a model to conduct a macro stress test of credit risk for the banking sector based on scenario analysis. We employ an original bank-level data set that splits bank credit portfolios into 21 granular categories, covering household and corporate loans. The results corroborate the presence of a strong procyclical behavior of credit quality and show a robust negative relationship between the logistic transformation of NPLs and GDP growth, with a lag response up to three quarters. The results also indicate that the procyclical behavior of loan quality varies across credit types. This is novel in the literature and suggests that banks with larger exposures to highly procyclical credit types and economic sectors would tend to undergo sharper deterioration in the quality of their credit portfolios during an economic downturn. Lack of sufficient portfolio granularity in macro stress testing fails to capture these effects and thus introduces a source of bias that tends to underestimate the tail losses stemming from the riskier banks in a system.

METHOD SUMMARY

Overview	The model enables macro stress testing of credit risk based on scenario analysis.
Application	- The stress test framework comprises three components that are integrated in sequence: (1) a macro vector autoregression model; (2) a micro logit panel; and (3) a credit value-at-risk (VaR) model based on the CreditRisk+ formulation. - It is appropriate for situations where granular data at the portfolio level are available.
Nature of approach	Econometric analysis.
Data requirements	- Bank-level nonperforming loan (NPL) data that split bank credit portfolios in granular categories, covering household and corporate loans. - Macro variables such as GDP growth, credit growth, and interest rates. - Information on distressed probabilities of default is estimated from the credit VaR model.
Strengths	The model allows the simulation of credit risk at the bank level under distressed macroeconomic scenarios, while minimizing portfolio aggregation bias.
Weaknesses	- The model does not capture potential feedback effects between the banking system and the economy under distressed scenarios. - In addition, the model implicitly assumes a linear, symmetric response of credit quality to macroeconomic conditions throughout the cycle, which may not be appropriate. - The implementation requires granular data on NPLs at the bank level, split by sectors for corporate loans and by size (small, medium, and large) for household loans.
Tool	- For the panel econometrics, the Stata script is available from the authors upon request. - For the credit VaR, the CreditRisk+ model is available upon request. - Contact authors: F. Vazquez and M. Souto.

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There has been a growing literature on stress testing in recent years. The importance of these exercises has been highlighted by the recent crisis and the cascade of bank failures in many countries. A deep understanding of the resilience of a banking sector to adverse macroeconomic scenarios is of crucial importance for the proper evaluation of systemic risk and has a direct connection with the development of new regulatory and prudential tools.

This chapter describes a model to conduct a macro stress test of credit risk for the Brazilian banking sector based on scenario analysis. The proposed framework comprises three independent, yet complementary modules that are combined in sequence. The first module uses time series econometrics to estimate the relationship between selected macroeconomic variables and uses the results to simulate distressed, internally consistent, macroeconomic scenarios spanning two years. The second module uses panel data econometrics to estimate the sensitivity of nonperforming loans (NPLs) to GDP growth and uses the results to simulate the evolution of credit quality for individual banks and credit types under distressed scenarios.¹ This module exploits a rich database that tracks the evolution of NPLs for 78 individual banks and 21 categories of credit for the 2001–09 period.² The third module uses the predicted NPLs as a proxy for distressed probabilities of default (PDs) and combines this information with data on the exposures and concentration of bank credit (gross loans) portfolios to estimate tail credit losses, using a credit value-at-risk (VaR) framework.

This study makes three main contributions to the literature on stress testing. First, it exploits a rich partition of bank credit portfolios by borrower types (i.e., consumer versus corporates) and economic sectors and assesses the extent of differences in the sensitivity of credit quality to macroeconomic conditions across credit types. Second, it illustrates that macro stress test models based on insufficiently granular data on banks' credit portfolios may be biased in a material way. In particular, macroeconomic stress test models based on undifferentiated credit data may tend to underestimate the credit losses stemming from the highly procyclical credit types (and overestimate the losses associated with the relatively safer credit types). To the extent that the composition of bank credit portfolios varies across institutions, the use of insufficiently granular credit data would tend to underestimate the tail losses of riskier banks, which runs against prudent principles. Third, we present and discuss the results for the Brazilian banking system, which is one of the largest banking systems in Latin America.

The results corroborate the presence of a strong procyclical behavior of credit quality, as indicated by a robust nega-

tive relationship between (the logit transformation of) NPLs and GDP growth, with a lag response up to three quarters. Comparative static exercises indicate that a 2 percentage point drop in yearly GDP growth, which is akin to the maximum drop observed in Brazil during 1996–2008, would cause a two-time increase in NPLs from their March 2009 levels, to about 7 percent. In addition, credit quality displays a strong inertial behavior across all credit types, with autoregressive coefficients implying that a 1 percentage point increase in NPLs in a given quarter produces a 0.4 percentage increase in NPLs in the next quarter. Credit to individuals, vehicles, and retail commerce was found to be relatively more sluggish.

The models also indicate substantial variations in the cyclical behavior of NPLs across credit types, with no statistically significant differences across state-owned and private banks, suggesting that the results are not due to likely differences in credit origination practices across these two types of banks.³ At the same time, some credit types appeared to be more sensitive to changes in economic activity, particularly agriculture, sugar and alcohol, livestock, small consumer credit, and textiles. Consequently, the quality of these credit types would likely undergo more severe erosion under a protracted drop in economic activity. Banks with higher exposures to these credit types may need to be followed up more closely.

Overall, the stress tests suggest that the Brazilian banking sector is well prepared to absorb the credit losses associated with a set of distressed macroeconomic scenarios without threatening financial stability. Four alternative macroeconomic scenarios, each one projected over two years, were analyzed. These comprised a baseline reflecting the expected path of GDP growth and three distressed scenarios that were deemed to be extreme but nevertheless likely under current circumstances. Overall, the results of the baseline scenario indicate that NPLs peak at 6.7 percent in the fourth quarter of 2010, before recovering. The simulated NPLs for the distressed scenarios are higher than for the baseline. The more severe deterioration in credit quality is associated with a slowdown in GDP growth akin to two standard deviations below its 2001–9 mean, with overall NPLs reaching a maximum of 8.5 percent, which is about a two-time increase from their starting levels.

The remainder of the chapter is structured as follows: Section 1 presents a brief literature review, whereas Section 2 discusses the methodology. Section 3 presents the empirical results. Finally, Section 4 concludes the chapter.

1. LITERATURE REVIEW

Since the seminal works of Wilson (1997a, 1997b), which present a framework to examine credit risk under distressed macroeconomic conditions, several papers have applied macro

¹ NPLs for each credit type are computed as the ratio of loans past due in excess of 90 days relative to the total loans in the corresponding category.

² The data come from information reported by the supervised institutions to the credit registry of the Central Bank of Brazil. In general, the credit portfolios analyzed in this study cover virtually all the bank credit to the private sector under market conditions. This represents about two-thirds of total bank credit, owing to the exclusion of credit operations granted under statutory conditions (the so-called directed lending).

³ Recent literature for the Brazilian banking system suggests that there may be important differences across banks because of ownership (Tabak and Staub, 2007; Staub, da Silva e Souza, and Tabak, 2010; Tecles and Tabak, 2010; Tabak, Fazio, and Cajueiro, 2011).

stress test tools to assess the resilience of various banking systems to adverse macroeconomic shocks (Berkowitz, 1999; Pesola, 2001, 2005; Frøyland and Larsen, 2002; Hoggarth and Whitley, 2003; Gerlach, Peng, and Shu, 2004; Sorge, 2004; Virolainen, 2004; Barnhill, Souto, and Tabak, 2006; van den End, Hoeberichts, and Tabbac, 2006; Boss and others, 2007; and Misina and Tessier, 2008, among others).⁴ In this literature, the main objective is to gauge the vulnerability of a portfolio (market, credit, or both) to adverse macroeconomic scenarios or to extreme but plausible events or shocks. The objective of such tests is to make risks more transparent, assessing the potential losses of a given portfolio under abnormal markets. These tools are commonly used by financial institutions as part of their internal models and management systems and to inform decisions regarding risk taking and capital allocation. Besides, these tools have become used increasingly by financial regulators to evaluate the soundness of the financial systems under their control.

Typically, macro stress tests of credit risk involve three major tasks: first, the development of a model to capture the interrelationships between selected macroeconomic and financial variables; second, the calibration of parameter vectors linking macroeconomic and financial variables to specific measures of loan performance; third, the design of adverse macroeconomic scenarios and the computation of their impact on credit quality and banks' solvency. Usually, the macroeconomic variables used in stress test models include measures of economic activity (i.e., GDP growth, the output gap, and unemployment) and measures of monetary conditions and key prices (i.e., interest rate, exchange rate, inflation, money growth, and property prices).

The investigation of how adverse scenarios may affect asset quality and solvency in the banking sector can be done using two approaches: top-down or bottom-up. The first one builds on aggregated data on bank credit portfolios, sometimes split by credit types or economic sectors, and simulates evolution of aggregated credit quality under distressed macro scenarios with the help of time-series analysis (see, e.g., Virolainen, 2004; and Wong, Choi, and Fong, 2006). A key shortcoming of this approach is its limited capacity to assess the financial conditions of individual institutions, which are frequently the focus of the analysis. The bottom-up approach addresses this shortcoming by resorting to the use of bank-level data.⁵ Typically, models based on this approach use panel data econometrics to gauge the evolution of asset quality under dis-

tressed macroeconomic scenarios, and the results are then mapped into banks' solvency and aggregated to get a systemic picture. Possibly because of data constraints, however, bottom-up models fail to exploit granular data on the characteristics of individual banks' credit portfolios (i.e., portfolio concentration and loan performance by credit types).⁶

Our study contributes to this literature by presenting a macro stress test model of credit risk that combines the use of bank-level information with a granular partition of banks' credit portfolios between consumer and corporate loans, classifying the former by size and the latter by economic sectors. In particular, we assess the sensitivity of credit quality to macroeconomic conditions using a bank-level data set that keeps track of 21 credit categories during 2001–9. The estimated parameter vectors are used to simulate the evolution of credit quality for individual banks and specific credit types, under adverse macroeconomic scenarios. This information is then combined using a credit portfolio approach to estimate the bank-specific capital needs conditional on the realization of the adverse macroeconomic scenarios.

Overall, the results suggest that the procyclical behavior of credit quality varies across credit types. By failing to account for these differences, current macro stress test models may be biased in a material way, underestimating the riskiness of banks that are more heavily exposed to highly procyclical credit types and economic sectors. We illustrate this bias by running parallel simulations of bank-level NPLs under adverse macroeconomic scenarios, using two approaches. The first, akin to typical macro stress test models of credit risk, uses bank-level data on credit quality, without allowing for differences in the behavior of credit quality across credit types. The second, following the approach presented in this study, exploits granular information on the characteristics of bank credit portfolios. The findings provide strong evidence of a data aggregation bias that tends to underestimate the impact of macro shocks on the quality of bank credit portfolios. Papers reporting that banking systems were resilient to adverse macroeconomic scenarios may have been influenced partly by this underestimation of credit risk.

2. METHODOLOGY

A. Overview of the methodology

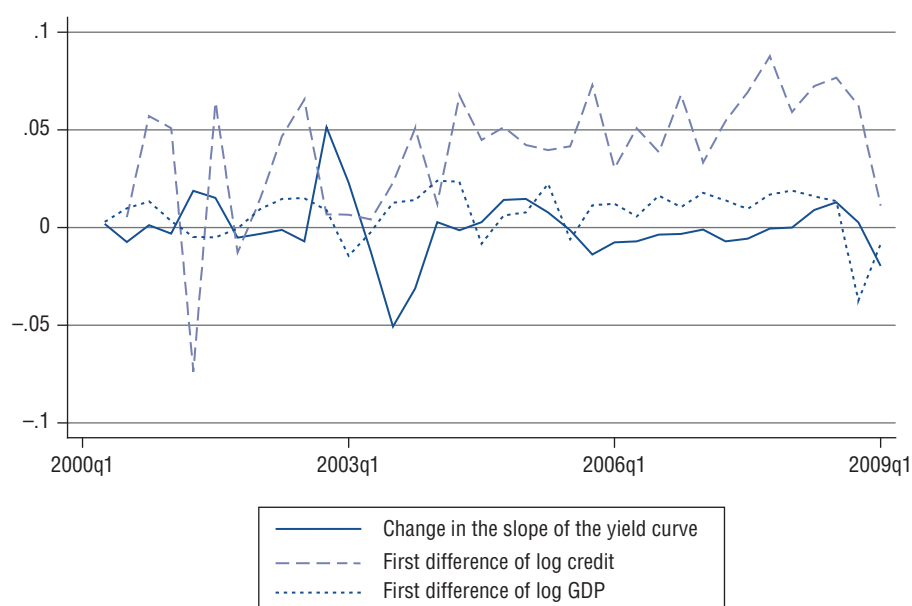
The stress test framework presented in this chapter comprises three components that are integrated in sequence:

- A macroeconomic model to estimate the relationship between selected macroeconomic variables with the help of times-series analysis. This model is used to simulate distressed, internally consistent, macroeconomic scenarios, projected over two years.

⁴ See Sorge and Virolainen (2006) for an overview of stress test methodologies. See also Illing and Liu (2006); Rodriguez and Trucharte (2007); Blank, Buch, and Neugebauer (2009); Castrén, Dées, and Zaher (2010); and Cardarelli, Elekdag, and Lall (2011). Foglia (2009) provides a very interesting review of current approaches to stress testing employed by supervisory authorities.

⁵ An example is Duellmann and Erdelmeier (2009), who stress test the credit portfolio of German banks using a different approach from ours (Merton-type multifactor credit risk model). The authors show that it is crucial to capture credit risk dependencies between sectors. The focus is on the automobile sector (key sector) and its interdependencies with other sectors.

⁶ There is to date little research using the bottom-up approach. Interesting examples are the works of Coffinet and Martin (2009) and Coffinet and Lin (2010), which perform a bottom-up stress test for French banks profitability and income subcomponents, respectively.



Source: Authors.

Figure 29.1 Selected Macroeconomic Variables, First Differences, 2000–2009

- A microeconomic model to assess the sensitivity of loan quality to macroeconomic conditions with the help of dynamic panel econometrics. The model is based on bank-level data, using separate equations for 21 credit types. The results are used to simulate the path of NPLs for each bank and for each of the 21 categories of credit, under the distressed macroeconomic scenarios produced in the previous stage.
- A credit VaR model to estimate the banks' capital needs to cover tail credit losses under the distressed scenarios. The model uses the simulated distributions of NPLs for each bank and credit type as a proxy for the distribution of distressed PDs and combines this information with data on the credit exposures of individual banks using the CreditRisk+ approach with the programs developed by Avesani and others (2006).

B. The macro model

The model outlined in this section uses times-series econometrics to capture the relationship between selected macroeconomic variables. As mentioned before, the results are used to build distressed scenarios projected over two years.

Macroeconomic data on key target series are available at a quarterly frequency, from the first quarter of 2001 to the first quarter of 2009.⁷ Although the length of the time series is somewhat short, the period covers some important macro

events, including a substantial shock in 2002–3, when the referential interest rate shot up by almost 10 percentage points to 26.5 percent and the exchange rate depreciated from 2.3 to almost 4 Brazilian real (BRL) per U.S. dollar (USD). The memory of this shock is important to help model the dynamics of the global financial crisis, which also affected Brazil, particularly since the third quarter of 2008. The substantial contraction in GDP is an important consideration for the vector autoregression (VAR) specification as it will, mechanically, force the factor to rebound in a way that may not be completely consistent with macroeconomic dynamics going forward. We present credit growth, GDP growth, and changes in the yield curve for the Brazilian economy in Figure 29.1.

The selected specification captures linkages between GDP growth, credit growth, and changes in the slope of the domestic yield curve. We choose a parsimonious specification given the relatively short length of the time series. The variables were selected after exploring the relationships between a larger set of macroeconomic variables restricting the factors to those that were statistically more relevant to the VAR specification, also yielding tighter error bands.⁸ The selected variables are defined as follows: (1) GDP growth, *GDP*, computed by taking the first difference to the natural log of the seasonally

⁷ Before 2001, we had the peg regime in exchange rate and a transition to the floating rate regime. After 2001, floating rate regime was in permanent regime.

⁸ The set of variables used in the selection of the specification include: the short-term policy rate (i.e., SELIC), the spread between bank lending and deposit rates, the U.S. yield curve (measured by the difference between the seven-year and three-month Treasury bill rates, the Chicago VIX index, the Emerging Market Bond Index (EMBI) spreads, a commodity price index (proxied by the Commodity Research Bureau Index), the unemployment rate, and the exchange rate. We have estimated the correlation for slope between estimations with the 7 years and 3 months and 10 years and 3 months periods, and it is above 99 percent.

TABLE 29.1

Summary Statistics of Selected Variables					
Variable	Obs	Mean	SD	Min	Max
D.yc	35	-0.0005	0.0164	-0.0507	0.0514
D.Lncr	35	0.0399	0.0318	-0.0736	0.0878
D.Lngdp	35	0.0075	0.0124	-0.0372	0.0241

Source: Authors.

Note: Obs = observations.

adjusted GDP series; (2) credit growth, *Credit*, computed by taking the first difference to the natural log of total (gross) loans in bank credit portfolios; and (3) the slope of the domestic yield curve, *YC*, measured by the difference between the monetary policy rate (i.e., the Sistema Especial de Liquidação e Custódia, or SELIC) and the long-term interest rate. Summary statistics of the selected variables are presented in Table 29.1. In order to control for the impact of the global financial crisis in the system, we add a dummy variable that equals one for the last two quarters of the sample (i.e., 2008:Q4 and 2009:Q1) and zero otherwise.⁹ This variable is treated as exogenous. Unit root tests indicate that GDP growth and credit growth are stationary but fail to reject the null for the slope of the yield curve, probably because of the short size of the sample. We therefore use the first difference of the series to achieve stationarity. All variables are end of period and in real terms. The model is of the form

$$y_t = c + \sum_{s=1}^p A_s y_{t-s} + Bx_t + \varepsilon_t, \quad (29.1)$$

where $y = \begin{bmatrix} D.yc \\ D.Ln(Credit) \\ D.Ln(GDP) \end{bmatrix}$, D represents the first difference,

$Ln(\cdot)$ represents the natural logarithm of the variables, and x stands for the exogenous regressors.

The ordering of the variables in equation (29.1) reflects the conjecture that credit markets play a role in the transmission of interest rate shocks to economic activity. The number of lags is set to four, taking into account the frequency of the data and the results of alternative lag order selection criteria (which indicate two to five lags).

The estimated coefficients are consistent with a priori expectations on the relationship between the selected variables. The results of an unrestricted VAR are presented in the first three data columns of Table 29.2. According to these, a tightening in monetary policy is associated with a drop in credit

growth and GDP growth, and there is a strong positive relationship between the last two variables. There is also evidence that the decline of GDP growth during the last quarter of 2008 and the first quarter of 2009 was larger than otherwise explained by the interaction between the endogenous variables included in the model, as indicated by the coefficient of the dummy variable, which is negative and statistically significant. The results also indicate that the domestic credit markets were somehow isolated from the effects of the global financial crisis, which is likely attributable to the strong expansion of credit by state-owned banks to compensate for the collapse of credit growth by private banks during this period. Similar conclusions can be extracted from the results of a restricted VAR, presented in the right three columns. Postestimation tests (not reported to save space) indicate that the models are stable and that the errors are not autocorrelated and pass standard normality tests. The impulse response functions, together with 95 percent confidence error bands, are presented in Figure 29.2.

The first difference of the slope of the yield curve would represent a change in the yield curve slope from one period to the next. Changes in yield curve slope are associated with investors' perception about future monetary policy, vis-à-vis current interest rates. For example, if GDP decreases from one period to the next, investors may expect the interest rates to go down in the future, causing a drop in the slope of the yield curve.

C. The microeconomic model

Data on credit portfolios were gathered from the credit registry of the Central Bank of Brazil, which contains rich information on individual credit operations granted by the supervised banks. The registry covers the bulk of credit in the system, leaving aside operations lower than a minimum reporting threshold and credits granted by unsupervised entities (such as nonfinancial corporations).¹⁰ The data used in this exercise, however, focus on lending granted with non-earmarked resources, which accounts for about 70 percent of total credit, as information on directed lending was not available.¹¹ For the purposes of the analysis, the data were aggregated at the level of individual banks and classified in 21 categories (Table 29.3). For each one, we have (1) total (gross) loans; (2) NPLs; (3) number of loan operations; (4) number of loan operations in default; and (5) (specific) loan-loss provisions.

Overall, the database covers the credit operations of 78 banks, at a quarterly frequency, between 2003:Q1 and 2009:Q1. The size of the credit portfolios included in the

⁹ We tried a number of variables to capture external effects (GDP growth in the United States and the European Union, commodity prices, EMBI, VIX) with poor (insignificant) t -statistics. Therefore, this may suggest that a decline in foreign demand was not the main reason affecting Brazil. One likely cause is the sharp exchange rate depreciation following the collapse of Lehman Brothers, partly triggered by the combination of heavy capital outflows with the sudden unwinding of corporate positions in foreign exchange rate derivatives, which prompted a number of monetary policy responses by the Central Bank.

¹⁰ It is important to highlight that the credit registry covers operations that represent more than 80 percent of the total volume of credit. Also, in Brazil most credit operations are performed within the regulated banking system. Therefore, the database is highly representative of the credit operations in Brazil.

¹¹ Non-earmarked resources are credit granted by financial institutions without implicit or explicit subsidies from the government.

TABLE 29.2

Macro Model Specification						
Variable	Unrestricted Model			Restricted Model		
	D_yc	D_Incr	D_ingdp	D_yc	D_Incr	D_ingdp
LD.yc	0.594*** (0.000)	-0.575** (0.022)	-0.263*** (0.004)	0.618*** (0.000)	-0.595*** (0.007)	-0.259*** (0.004)
L2D.yc	-0.027 (0.885)	-0.16 (0.580)	-0.135 (0.205)			-0.054 (0.536)
L3D.yc	-0.089 (0.605)	0.178 (0.511)	-0.207** (0.038)			-0.269*** (0.002)
L4D.yc	-0.03 (0.868)	0.316 (0.261)	-0.059 (0.566)			
LD.Incr	0.209** (0.013)	-0.391*** (0.003)	0.148*** (0.002)	0.239*** (0.001)	-0.306** (0.014)	0.159*** (0.000)
L2D.Incr	0.167* (0.054)	0.051 (0.705)	0.177*** (0.000)	0.180*** (0.006)	0.197* (0.074)	0.197*** (0.000)
L3D.Incr	-0.119 (0.162)	0.212 (0.112)	0.079 (0.106)	-0.137* (0.074)	0.315*** (0.008)	0.065 (0.135)
L4D.Incr	-0.264*** (0.001)	0.261** (0.032)	0.04 (0.371)	-0.304*** (0.000)	0.230** (0.028)	
LD.Ingdp	0.039 (0.856)	1.100*** (0.001)	-0.514*** (0.000)		0.918*** (0.000)	-0.504*** (0.000)
L2D.Ingdp	0.182 (0.557)	1.129** (0.020)	-0.524*** (0.003)		0.656* (0.089)	-0.482*** (0.002)
L3D.Ingdp	-0.001 (0.997)	0.779* (0.097)	-0.425** (0.014)			-0.436*** (0.001)
L4D.Ingdp	-0.107 (0.696)	0.606 (0.159)	-0.279* (0.078)			-0.304** (0.019)
dummy_crisis	-0.01 (0.280)	-0.004 (0.788)	-0.044*** (0.000)			-0.044*** (0.000)
Constant	-0.002 (0.674)	0.009 (0.264)	0.008*** (0.005)	-0.001 (0.802)	0.014* (0.061)	0.009*** (0.001)
Observations	31	31	31	31	31	31
R ²	0.63	0.63	0.63	0.56	0.56	0.56
AIC	-16.9	-16.9	-16.9	-16.5	-16.5	-16.5
HQIC	-16.2	-16.2	-16.2	-15.8	-15.8	-15.8
SBIC	-14.9	-14.9	-14.9	-14.5	-14.5	-14.5

Source: Authors.

Note: p -values in parentheses. *** $p < .01$; ** $p < .05$; * $p < .1$.

AIC = Akaike's information criterion; HQIC = Hannan-Quinn information criterion; SBIC = Schwarz Bayesian Information Criterion.

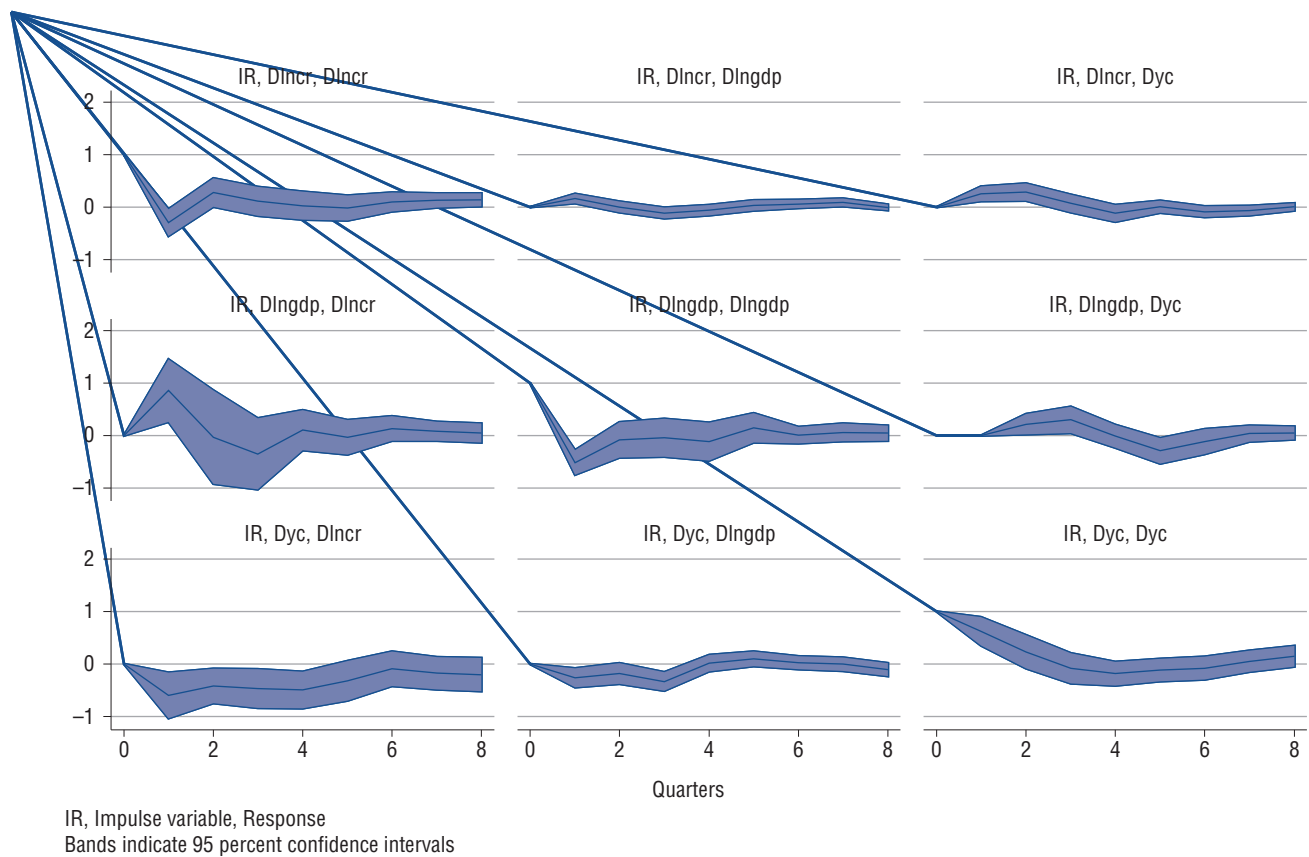
analysis is rather continuous throughout the sampled period (Table 29.4). The sample, however, is unbalanced because of the exit or merging of some banks and the incorporation of new ones. As of March 2009, the sample included 49 banks jointly accounting for about 85 percent of total bank credit.¹² The time coverage was dictated by data availability. In particular, the construction of time series going further back in time was not possible because of a change in accounts and data reporting definitions introduced in 2002. The quality of the data was deemed to be good. Several filters were applied to identify potential inconsistencies, and a few data reporting issues were found to be (generally) associated with a specific subgroup of banks.

A look at the bank-level data indicates that credit quality has been relatively poor and extremely heterogeneous across credit types. Overall, NPLs averaged 3.6 percent during the

sampled period, which is relatively high considering the favorable macroeconomic environment and the rapid expansion of credit portfolios. Furthermore, credit quality has been dispersed across banks and throughout time, as indicated by the size of the standard deviations of NPLs, which are generally two to three times larger than their corresponding mean values (Table 29.5). The extent of the dispersion of credit quality and the severity of loan nonperformance in some institutions is also illustrated by the NPL ratios of banks in the 90th percentile of the distribution, which exceeded 10 percent in many sectors. Across credit types, the higher average rates of NPLs have been associated with credit to individuals (particularly small- and medium-sized loans), firms operating in the services sector, producers of livestock, and electric and electronic equipment.

The evolution of NPLs was also diverse across bank types. Overall, state-owned banks displayed better loan quality during the sampled period, interrupted only by a sharp increase in NPLs on exposures to the petrochemical and food indus-

¹² Credit is highly concentrated in Brazil, with the largest five banks accounting for approximately 70 percent of total credit.



Source: Authors.

Note: This figure presents the impulse-responses of the VAR model described in equation (29.1).

Dlnr is the first difference in the natural logarithm of bank's credit growth, where credit is estimated as the total loans in the aggregate banking sector portfolio, at the end of the period, and growth is estimated quarter-on-quarter.

Dlngdp is the first difference in the natural logarithm of GDP growth, where GDP growth is computed as the natural logarithm of the seasonally adjusted GDP series, quarter-on-quarter, using end of the period numbers for GDP.

Dyc is the first difference in the yield curve slope, measured by the difference between the monetary policy yield curve (i.e., SELIC), and the long-term interest rate.

Figure 29.2 Macro Model Impulse Response Functions

tries in 2005–06 (Figure 29.3). Remarkably, the segments of private and foreign banks experienced a moderate but sustained increase in NPL ratios after 2005, despite rapid credit growth and the supportive economic environment. More recently, since the third quarter of 2008, credit quality deteriorated rapidly and across the board, reflecting the impact of the global financial crisis on the macroeconomic and financial environment. As mentioned before, however, these aggregate figures mask large differences in loan quality across individual banks and credit types. In general, the smaller banks have tended to underperform, also displaying higher concentration in their loan exposures to specific credit types.

The model discussed in this section analyzes the sensitivity of NPLs to macroeconomic conditions with the help of dynamic panel econometric techniques. The specification was selected after exploring the sensitivity of NPLs to a combination of candidate macroeconomic and bank-level variables encompassing, inter alia, GDP growth, the unemployment rate, credit growth (both aggregated and bank-

specific), long-term and short-term interest rates, bank lending spreads, and the change of the exchange rate (both in nominal and real terms). The inclusion of credit growth in the exploratory specifications was motivated by the observation that credit quality tends to improve in the early stages of an episode of accelerating credit expansion. However, after exploring with various lag structures and using both aggregate and bank-specific credit growth, this variable turned out to be not significant in the regressions. Similarly, and more in line with expectations, the real exchange rate was also not significant, likely reflecting the lack of material dollarization in the credit portfolios of Brazilian banks.

The main criterion guiding model selection was the precision of the parameter estimates and the robustness of the results, reflecting the purpose of the exercise (i.e., simulating loan quality under alternative macroeconomic scenarios). In particular, we postulate that the logit-transformed NPLs of each credit type of bank i follow an AR(1) process and are influenced by past GDP growth, with up to S lags:

TABLE 29.3

Structure of Loan Portfolios across Bank Ownership, March 2009, in Percent

Sector	Nonperforming Loans			Share in Loan Portfolio		
	Private Domestic	Public	Foreign	Private Domestic	Public	Foreign
Consumer (large)	2.9	2.0	3.3	1.4	5.8	2.5
Consumer (medium)	6.5	2.0	7.1	7.5	13.4	10.7
Consumer (small)	8.9	2.9	7.2	28.3	20.3	25.8
Agriculture	2.7	1.0	3.8	2.0	2.2	2.7
Food	3.2	1.4	2.8	2.2	2.7	2.5
Livestock	2.4	1.2	3.9	3.0	3.9	3.2
Vehicles	4.4	2.3	5.1	3.0	2.6	2.5
Electrical and electronic	6.8	2.9	5.0	1.4	1.5	1.5
Electricity and gas	0.0	0.0	1.1	3.2	3.0	3.7
Wood and furniture	2.9	2.5	2.8	8.8	6.0	8.8
Recreation services	4.7	3.3	4.8	1.8	1.6	1.8
Petrochemicals	2.3	0.7	2.4	3.1	5.6	2.6
Chemicals	3.8	1.6	2.3	1.5	1.6	2.2
Health services	2.7	1.9	2.6	1.9	1.6	2.5
Other services	3.9	3.0	4.0	3.8	1.9	3.2
Metal products	1.3	0.4	1.5	3.2	4.4	2.6
Sugar and alcohol	1.2	1.4	1.4	3.8	1.5	3.1
Textile	6.5	3.1	5.5	2.5	3.3	3.0
Transportation	1.8	1.0	2.2	6.5	3.2	4.2
Retail trade	3.8	1.8	2.9	2.7	3.4	2.6
Other	1.4	0.8	1.3	8.5	10.3	8.2

Sources: Authors; and Central Bank of Brazil.

TABLE 29.4

Brazilian Banking System, Sample Coverage, 2003–09

Year: Quarter	Number of Sampled Banks				Total Loans (in million BRL)
	Public	Private	Foreign	Total	
2003:Q1	6	37	25	68	214,838
2003:Q2	7	39	23	69	214,368
2003:Q3	7	39	22	68	219,499
2003:Q4	7	38	20	65	239,102
2004:Q1	6	38	20	64	242,760
2004:Q2	6	38	19	63	258,230
2004:Q3	6	37	19	62	268,066
2004:Q4	6	37	20	63	277,670
2005:Q1	6	37	21	64	291,032
2005:Q2	6	36	21	63	303,805
2005:Q3	6	36	21	63	316,163
2005:Q4	6	35	21	62	343,966
2006:Q1	5	36	21	62	357,901
2006:Q2	5	35	20	60	380,806
2006:Q3	5	35	19	59	401,241
2006:Q4	5	35	19	59	438,637
2007:Q1	5	35	19	59	456,863
2007:Q2	5	34	18	57	490,680
2007:Q3	5	33	18	56	533,389
2007:Q4	3	27	15	45	533,458
2008:Q1	5	33	18	56	619,536
2008:Q2	5	32	18	55	676,095
2008:Q3	4	32	17	53	733,894
2008:Q4	4	32	16	52	767,665
2009:Q1	4	29	16	49	779,501

Sources: Authors; and Central Bank of Brazil.

TABLE 29.5

Selected Statistics of NPLs across Loan Types and Bank Ownership, 2003:Q1–2009:Q1, in Percent

Sector	Private Domestic			Public Banks			Foreign Banks			Total Sample		
	Mean	SD	Pct. 90	Mean	SD	Pct. 90	Mean	SD	Pct. 90	Mean	SD	Pct. 90
Consumer (large)	4.6	14.7	7.3	4.0	6.2	14.0	1.5	4.7	2.7	3.6	11.8	7.1
Consumer (medium)	7.4	12.2	17.6	3.3	3.9	7.9	4.3	7.6	9.9	6.1	10.6	14.7
Consumer (small)	6.9	9.0	14.0	3.0	1.7	4.8	4.7	8.2	10.2	5.9	8.4	12.9
Wood and furniture	5.0	11.1	12.7	3.6	4.8	7.4	1.3	4.1	2.8	3.8	9.1	8.5
Transportation	4.7	13.6	8.9	5.5	11.5	12.2	1.5	6.9	2.1	3.8	11.8	7.7
Petrochemicals	3.9	10.1	9.6	9.7	23.6	26.8	0.7	2.4	1.8	3.6	11.4	7.4
Metal products	2.9	12.4	4.2	2.8	6.8	8.8	0.3	1.6	0.8	2.1	10.0	2.9
Electricity and gas	1.8	7.9	3.1	1.3	5.9	1.5	0.6	3.9	0.7	1.3	6.6	1.5
Livestock	5.4	16.7	8.0	5.8	11.4	17.2	1.4	4.5	2.6	4.2	13.8	6.9
Other services	6.3	14.6	19.3	5.7	8.4	13.6	1.8	5.2	3.1	5.0	12.3	12.7
Sugar and alcohol	0.5	2.9	0.5	0.3	1.2	0.7	0.8	6.7	0.2	0.6	4.3	0.5
Retail trade	4.5	13.0	9.0	5.2	8.9	15.5	1.4	7.5	2.3	3.7	11.3	7.1
Textile	4.2	10.1	10.1	5.3	9.2	11.5	2.8	10.7	4.4	3.9	10.2	9.2
Vehicles	3.8	11.5	7.2	3.2	9.1	6.0	0.9	2.1	2.5	3.0	9.6	5.5
Food	4.0	11.7	8.2	14.0	27.2	60.3	1.2	3.9	2.7	4.3	13.5	7.7
Agriculture	2.2	8.9	4.0	2.3	7.3	4.0	0.6	2.5	1.0	1.7	7.3	2.9
Health services	3.9	12.2	6.7	2.2	3.8	5.2	1.8	7.5	2.1	3.2	10.5	5.0
Chemicals	2.5	9.6	4.3	3.3	4.1	8.8	0.9	3.2	2.3	2.2	7.8	4.1
Recreation services	5.4	14.6	15.3	4.7	5.4	10.0	2.4	7.1	5.3	4.5	12.4	9.9
Electrical and electronic equipment	5.9	16.1	13.3	5.4	6.1	16.6	2.2	7.1	3.4	4.9	13.4	11.1
Other	3.6	10.7	7.2	4.7	10.5	11.5	1.3	6.2	1.4	2.9	9.5	6.6

Sources: Authors; and Central Bank of Brazil.

Note: Pct. 90 = 90th percentile.

$$\ln\left(\frac{NPL_{i,t}}{1-NPL_{i,t}}\right) = \mu_i + \alpha \ln\left(\frac{NPL_{i,t-1}}{1-NPL_{i,t-1}}\right) + \sum_{s=0}^S \beta_{t-s} \Delta \ln(GDP)_{t-s} + \varepsilon_{i,t}, \quad (29.2)$$

where $NPL_{i,t}$ stands for the ratio of NPLs to total gross loans of each credit type of bank i in period t , and GDP_t stands for GDP in quarter t .¹³ The inclusion of the lagged dependent variable is motivated by the persistence of NPLs. The term μ_i refers to the bank-level fixed effects, which are treated as stochastic, and the idiosyncratic disturbances $\varepsilon_{i,t}$ are assumed to be independent across banks and serially uncorrelated (i.e., after the inclusion of the lagged dependent variable).¹⁴ The coefficient α is expected to be positive but less than one, and the β coefficients are expected to be negative, reflecting deteriorating loan quality during the economic downturn.

¹³ Because the NPL ratio is bounded in the interval $[0, 1]$, the dependent variable was subject to the logit transform $\log(NPL/(1-NPL))$, to avoid problems associated with non-Gaussian errors.

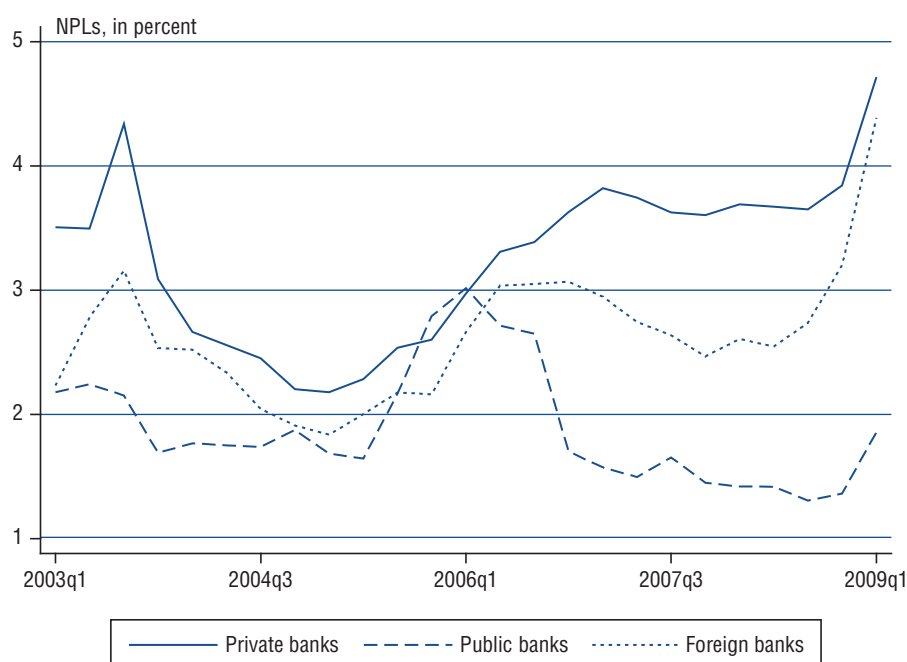
¹⁴ Therefore, the model assumes that the (positive) correlation of NPLs across individual banks originates exclusively from their common exposure to macroeconomic conditions. It also assumes that the effect of macroeconomic conditions on loan quality is symmetric during the upturn and the downturn of the economic cycle and neglects possible nonlinear dynamics and potential feedback effects running from credit markets to macroeconomic activity. A set of alternative specifications (available upon request) was estimated, exploring for nonlinear effects and for potential differences in the sensitivity of loan quality to economic activity through the cycle, with nonsignificant results.

Under this specification, the short-term effect of a change in quarter-on-quarter GDP growth on the logit of NPLs is given by the sum of the estimated β coefficients. By the chain rule, the effect of a shock to GDP growth on the untransformed NPL ratios, evaluated at the sample mean of NPLs is given by equations (29.3) and (29.4):

$$\text{Short-term effect: } \frac{\Delta NPL}{\Delta \ln(GDP)} = \overline{NPL} \times (1 - \overline{NPL}) \times \sum_s \beta_{t-s}, \quad (29.3)$$

$$\text{Long-term effect: } \frac{\Delta NPL}{\Delta \ln(GDP)} = \frac{1}{1 - \alpha} \times \overline{NPL} \times (1 - \overline{NPL}) \times \sum_s \beta_{t-s}. \quad (29.4)$$

As a first approximation, we estimate equation (29.2) for the overall NPL ratios of individual banks, without distinguishing between credit types, which is the typical approach used in macro stress test models. The estimation was carried out using several alternative methods to assess the robustness of the results. We then selected a preferred estimation method and reestimated equation (29.2) for each of the 21 credit types. All the models were computed over the entire sample of banks and separately for state-owned, private domestic, and foreign banks with the help of interacting dummies. The last were used to explore for differences in the sensitivity of loan quality to macroeconomic conditions across bank types, possibly induced by systematic differences in loan origination practices and bank clientele across state-owned, private, and foreign



Source: Authors.

Note: NPL = nonperforming loan.

Figure 29.3 Evolution of NPLs across Bank Types

banks. However, because the results showed no evidence of systematic differences across bank types, the final specification was computed over the entire sample to increase efficiency.

The results of the exploratory regressions were consistent with expectations and extremely robust under alternative estimation methods, including pooled ordinary least squares (OLS), within-group estimation, and two alternative applications of generalized method of moments (GMM) estimators, treating GDP growth as either predetermined or strictly exogenous for the panel variables (Table 29.6). After exploring with various lag structures, we selected four lags of GDP growth, also reflecting the frequency of the data. Overall, the coefficients of the lagged dependent variable are around 0.6, reflecting the strong persistence of NPLs. In turn, the coefficients of the lagged GDP growth are negative, as expected, and significant for up to three lags, falling within a relatively narrow interval.

On the basis of a comparison across estimation methods, we select the specification presented in column (4) of Table 29.6 as the preferred model. In particular, the estimation in column (1) uses OLS in levels, which produce upward-biased estimates of the coefficients associated with the lagged dependent variable (the $\alpha_{i,s}$) due to the positive correlation between the latter and the fixed effects. The within-groups estimator in column (2) eliminates the fixed effects by subtracting the mean from the series but introduces a downward bias stemming from negative correlation between the lagged dependent variable and the transformed errors. Therefore, the consistent

estimator of α is expected to fall between the OLS and the within-groups estimators. This is, in fact, the case for all the models that follow, which are based on GMM estimators. The results presented in columns (3) and (4) use the Arellano-Bond GMM estimator in first differences, treating GDP growth as strictly exogenous in the first case and as predetermined in the second (see Arellano and Bond, 1991; Blundell and Bond, 1998). The latter seems to be the preferred treatment, as indicated by the results of the Hansen test presented at the bottom, which fail to reject the null of orthogonality between the instruments and the error term. In turn, the results presented in columns (5) and (6) use the Arellano-Bover System GMM estimator, which exploits additional information from the equations in levels but requires the additional assumption that GDP growth is uncorrelated with the bank-level fixed effects, which may not be realistic. In all the GMM estimations, the number of instruments was limited by setting a maximum of six lags, to avoid problems associated with instrument proliferation.

The estimates of a full set of parallel regressions, one for each credit type, also are consistent with expectations and broadly robust. All the coefficients of the lagged dependent variable are positive in the interval $[0, 1]$ as expected and statistically significant at conventional levels (Table 29.7). The average value across all credit types is 0.4, which is slightly below the estimate obtained for the entire loan portfolios, likely reflecting the stronger sluggishness of the loan portfolios induced by diversification. The results also indicate that the AR(1) specification is adequate to eliminate the autocor-

TABLE 29.6

Results of Exploratory Panel Regressions

	(1)	(2)	(3)	(4)	(5)	(6)
Variable	Pooled OLS	Within Groups	Difference GMM GDP Exog.	Difference GMM GDP Pred.	System GMM GDP Exog.	System GMM GDP Pred.
L.Logit (NPL)	0.905*** (0.024)	0.569*** (0.064)	0.589*** (0.124)	0.597*** (0.123)	0.602*** (0.088)	0.631*** (0.082)
D.LnGDP	-7.481*** (2.032)	-7.853*** (1.903)	-9.529*** (2.198)	-8.804*** (2.132)	-7.767*** (1.927)	-6.928*** (1.939)
LD.LnGDP	-2.569 (2.282)	-4.544** (1.935)	-6.081*** (2.254)	-5.729*** (1.990)	-3.922* (2.026)	-3.086 (2.023)
L2D.LnGDP	-7.482** (3.197)	-6.877** (3.081)	-10.675*** (3.627)	-9.152*** (3.361)	-8.123** (3.138)	-5.971* (3.077)
L3D.LnGDP	1.597 (3.273)	1.067 (3.172)	0.423 (3.433)	-0.734 (3.130)	1.225 (3.337)	0.828 (3.322)
Observations	1201	1201	1121	1121	1201	1201
R ²	0.83	0.341				
Hansen test (p)			0.02	0.13	0.04	0.11
AR(1) (p)			0.00	0.00	0.00	0.00
AR(2) (p)			0.184	0.175	0.184	0.191
Number of instruments			11	17	13	17
Number of banks		70	69	69	70	70

Source: Authors.

Note: Robust standard errors are in parentheses. Exog. = exogenous; GMM = Generalized Method of Moments; NPL = nonperforming loan; OLS = ordinary least squares; Pred. = predetermined. * $p < .1$; ** $p < .05$; *** $p < .01$.

relation of the errors, as the tests of autocorrelation of order 2 in the first-differenced errors fail to reject the null in all cases. In turn, the sums of the coefficients of lagged GDP growth are negative in all cases, with the exception of credit to transport and “other credits” categories, and statistically significant at the 5 percent level in about one-half of the cases. The largest autoregressive coefficients are obtained for small credits to consumers, retail, textiles, and vehicles, indicating higher sluggishness in loan quality to these sectors. In turn, the largest coefficients for GDP growth are obtained for agriculture, sugar and alcohol, and energy. In order to gauge the sensitivity of NPLs to economic activity, however, these coefficients have to be rescaled by the average NPLs of the corresponding credit types, as shown in equations (29.3) and (29.4).

Using these results, we compute “rule-of-thumb” estimates of the impact of a change in GDP growth on NPLs (Table 29.8). For the overall sample, displayed at the bottom of Table 29.8, we go back to the regression presented in column (4) of Table 29.7, where the coefficients of GDP growth add up to -24.4. Plugging this into equation (29.2), and using the average NPLs (2.8 percent), we find that a 2 percentage point drop in GDP growth (which is akin to the maximum drop observed between 1996 and 2008) would cause a 1.3 percentage point increase in NPLs in the short term (i.e., $0.028 \times (1 - 0.028) \times 24.4 \times 2$). In turn, using equation (29.3), the predicted long-term increase in NPLs would be 3.3 percentage points (i.e., $1.3 \div (1 - 0.6)$), entailing a distressed NPL level of 7.2 percent, which is almost two times higher than their March 2009 levels. Carrying out similar calculations for each credit type gives a range of results. The higher NPL ratios are obtained for consumer credit, which reaches 7.6 percent

for medium-sized loans and 10.4 percent for small loans. Among lending to firms, the sectors reaching the highest NPL levels include textile, electric and electronic equipment, retail trade, and vehicles. In relative terms, the distressed NPL ratios are generally between 1½ and 2 times higher than their March 2009 values, with the most sensitive sectors being electricity and gas, livestock, agriculture, food, sugar and alcohol, and retail trade.

3. STRESS TESTS

This section summarizes the results of stress test exercises of credit risk based on scenario analysis. It describes the criteria used in the construction of the scenarios and provides a brief comparison of their evolution. It also discusses the main characteristics of the out-of-sample simulations of NPLs under selected scenarios and presents an illustration of the bias that can result from inadequate granularity in the credit portfolio data. Finally, the section presents the results of a credit VaR calculation based on these projections.

A. Simulation of NPLs under alternative scenarios

The exercises to assess credit risk are based on four macroeconomic scenarios, including a baseline that reflects the expected path of GDP growth and three distressed scenarios. Designing relevant stress scenarios is not a trivial issue. One can use history as guidance to construct the shocks, but history hardly repeats and the circumstances surrounding the shocks are almost always different, which raises questions about the validity of the scenarios. Alternatively, the shocks can also

TABLE 29.7

Results of the Dynamic Panel Regressions for Individual Credit Types, 2003:Q1–2009:Q1

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Consumer (Large)	Consumer (Medium)	Consumer (Small)	Wood and Furniture	Transportation	Petrochemicals	Metal Products	Electricity and Gas	Livestock
L.Dependent	0.351 (0.000)	0.379 (0.000)	0.665 (0.000)	0.335 (0.000)	0.380 (0.000)	0.398 (0.000)	0.483 (0.000)	0.423 (0.000)	0.498 (0.000)
D.Ingdp	−6.129 (0.136)	−4.186 (0.008)	−5.906 (0.025)	−3.235 (0.070)	3.759 (0.296)	−2.008 (0.385)	−1.526 (0.581)	−16.453 (0.003)	−7.283 (0.144)
LD.Ingdp	−7.951 (0.071)	−2.931 (0.032)	−2.168 (0.148)	−8.643 (0.005)	−4.182 (0.045)	−8.169 (0.010)	−6.355 (0.044)	−3.671 (0.609)	−14.080 (0.271)
L2D.Ingdp	−2.797 (0.602)	−6.578 (0.011)	−1.730 (0.377)	−4.565 (0.097)	3.592 (0.379)	−2.753 (0.282)	−6.047 (0.383)	−17.539 (0.008)	−0.978 (0.876)
L3D.Ingdp	−8.132 (0.258)	−0.023 (0.992)	0.333 (0.887)	−2.059 (0.498)	−3.089 (0.538)	1.284 (0.705)	−3.086 (0.486)	−23.542 (0.006)	8.514 (0.245)
Observations	376	889	983	726	561	570	412	287	477
Number of banco	37	58	61	54	43	41	35	25	38
Hansen test (<i>p</i>)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
AR(1) (<i>p</i>)	0.00	0.00	0.00	0.00	0.00	0.00	0.10	0.00	0.02
AR(2) (<i>p</i>)	0.19	0.25	0.95	0.03	0.14	0.57	0.90	0.39	0.92
Number of instruments	146	146	146	146	146	146	146	146	146
Sum of GDP coefficient	−25.009	−13.72	−9.47	−18.50	0.08	−11.65	−17.01	−61.21	−13.83
<i>p</i>	0.04	0.02	0.09	0.01	0.99	0.06	0.06	0.00	0.55
Long-term effect	−38.535	−22.090	−28.272	−27.823	0.129	−19.346	−32.909	−106.075	−27.544
<i>p</i>	0.03	0.01	0.14	0.02	0.52	0.07	0.09	0.02	0.25

Source: Authors.

Note: The numbers in the parentheses are the *p*-values, that is, the probability of incorrectly rejecting the null hypothesis.

be constructed more arbitrarily, considering current conditions and incorporating forward-looking considerations. In this study, we abstract from this discussion and use a mix of history and current conditions to shock the framework. The idea is to illustrate the model sensitivity to these various scenarios.

The evolution of GDP growth under the four scenarios considered was determined as follows:

- **Baseline scenario.** This scenario is taken as reference and aims at capturing the expected evolution of economic activity. Under this, the results of the VAR model are projected onward, without shocking the system. The resulting GDP growth drops from 5.1 percent in 2008 to −0.6 percent in 2009, followed by a resumption to above 3 percent in the subsequent two years.
- **Scenario 1.** This scenario uses the results of the VAR to simulate the effect of an 11.6 percentage point increase in the slope of the yield curve in 2009:Q2. The shock is akin to the mean of the slope during 2001–09 plus 2 standard deviations.
- **Scenario 2.** This scenario uses the results of the VAR to simulate the effects of a negative shock to credit growth equal to 2.4 percentage points in 2009:Q2. The shock is akin to the mean quarterly credit growth during 2001–09 minus 2 standard deviations.
- **Scenario 3.** This scenario uses the results of the VAR to simulate the effects of a negative shock to GDP growth equal to 1.9 percentage points in 2009:Q2. The shock is akin to the mean quarterly GDP growth during 2001–09 minus 2 standard deviations.

A comparison of the evolution of GDP growth under these four scenarios is provided in Figure 29.4.

Using the results of the panel estimations, we conduct an out-of-sample simulation of NPLs for each bank and credit type under each of the four scenarios. The results indicate deteriorating loan quality during 2009 (Figure 29.5), followed by a relatively quick and steady recovery in 2010. For the baseline scenario, NPLs peak to 6.7 percent in the fourth quarter of 2010, before recovering. This out-of-sample simulation tracks reasonably well the ex post observed data on NPLs on reference credit operations during the second and third quarters of 2009 (NPLs reached 5.8 percent in September 2009). The simulated NPLs for the distressed scenarios are higher than for the baseline but follow a qualitatively similar dynamic. The more severe deterioration in credit quality is associated with Scenario 3, with NPLs reaching a maximum of 8.5 percent, which is high at about two times the maximum observed during the sample period. Across credit types (not shown), the higher levels of NPLs are associated with credit to consumers, sugar and alcohol, textiles, electricity and gas, and vehicles, which is roughly consistent with the results of the static exercise.

B. Portfolio aggregation bias

As mentioned before, the use of granular data on bank credit portfolios is a key contribution of this study. Existing macro stress test models try to map the impact of the macroeconomic environment on credit quality using two approaches. The first one exploits aggregate data on credit quality, sometimes split

(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)
Other Services	Sugar and Alcohol	Retail Trade	Textile	Vehicles	Food	Agriculture	Health Services	Chemicals	Recreation Services	Electrical and Electronic Equipment	Other
0.409	0.340	0.628	0.543	0.522	0.465	0.451	0.443	0.468	0.172	0.352	0.287
(0.000)	(0.004)	(0.000)	(0.003)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.042)	(0.000)	(0.000)
-2.102	-6.134	-0.798	-0.684	-2.170	-7.343	-11.447	-1.207	-1.794	-3.641	-1.032	-0.957
(0.265)	(0.485)	(0.722)	(0.855)	(0.370)	(0.005)	(0.006)	(0.743)	(0.442)	(0.162)	(0.786)	(0.769)
-3.057	-11.806	-6.651	-10.950	-2.901	-6.508	-11.446	-2.730	-3.751	-5.318	-5.744	0.913
(0.143)	(0.117)	(0.004)	(0.000)	(0.116)	(0.032)	(0.000)	(0.372)	(0.188)	(0.148)	(0.004)	(0.704)
-0.498	-24.826	-4.095	-7.673	-1.249	-0.124	-5.723	-2.194	0.632	-2.213	1.932	1.008
(0.897)	(0.005)	(0.239)	(0.026)	(0.689)	(0.971)	(0.213)	(0.584)	(0.863)	(0.632)	(0.629)	(0.854)
-2.879	-24.430	-3.719	-2.667	-5.113	-1.821	-6.605	-2.649	1.060	-3.243	-9.148	2.825
(0.442)	(0.122)	(0.189)	(0.557)	(0.040)	(0.628)	(0.029)	(0.512)	(0.670)	(0.381)	(0.003)	(0.450)
659	184	502	577	509	549	377	515	443	521	469	711
51	18	41	44	36	42	30	37	39	41	42	51
1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.00
0.84	0.18	0.11	0.88	0.42	0.35	0.08	0.54	0.85	0.42	0.92	0.82
146	144	146	146	146	146	146	146	146	146	146	146
-8.54	-67.20	-15.26	-21.97	-11.43	-15.80	-35.22	-8.78	-3.85	-14.42	-13.99	3.79
0.26	0.01	0.01	0.01	0.07	0.03	0.00	0.33	0.43	0.07	0.08	0.69
-14.443	-101.812	-41.030	-48.083	-23.918	-29.525	-64.155	-15.763	-7.242	-17.409	-21.593	5.314
0.18	0.12	0.08	0.26	0.20	0.00	0.00	0.44	0.17	0.17	0.42	0.88

by economic sectors, and applies econometric techniques to compute (elasticity) parameters linking macroeconomic conditions to credit quality. This approach is frequently used to assess systemic financial stability, but it has important shortcomings, as the profiles of banks' credit portfolios and the cushions to absorb credit losses are likely to differ across banks. The validity of the results under this approach becomes weaker as bank sizes, solvency, and risk profile of their credit portfolios depart from the population mean.

The second approach exploits bank-by-bank data on credit quality, albeit without differentiating between types of credit and without taking into account information on large exposures and other measures of portfolio concentration. Although this approach allows the assessment of bank solvency at the level of individual institutions, it has its own shortcomings and potential sources of bias. In particular, the estimated elasticities linking macroeconomic conditions to credit quality reflect average values, mixing divergent elasticities between types of credit.

Arguably, the latter approach would tend to lead to biased estimations, as the sensitivity of credit quality to macroeconomic conditions is likely to differ between economic sectors. A bank with larger exposures to highly cyclical sectors would tend to be more vulnerable to credit losses under an adverse economic scenario. Therefore, simulations based on this approach would underestimate the potential losses of riskier banks, which is contrary to a prudent principle. In addition, the lack of information on portfolio concentration is a critical shortcoming, as concentration plays a major role in the risk profile of banks' credit portfolios.

To illustrate the bias stemming from the use of insufficient granularity in banks' credit portfolios, we use the results of the previous section to estimate the bank-specific NPLs under two approaches. The first one, akin to typical macro stress test models of credit risk, simulates the evolution of bank-level NPLs without exploiting information on specific credit types (joint portfolios), while the second exploits a partition of banks' credit portfolios in specific credit categories (granular portfolios). The two approaches share the same estimation techniques and the two-year macroeconomic scenarios described earlier.

The results are consistent with the presence of a bias stemming from inadequate granularity in the credit portfolios. In particular, the weighted average of the simulated NPLs using joint portfolios is always lower than the simulated NPLs under the granular portfolios (Figure 29.6). The kernel densities of the simulated NPLs under the two approaches also illustrate this portfolio aggregation bias (Figure 29.7). In particular, the mean and the median NPLs under the granular approach exceed those of the joint approach in all cases (Table 29.9). Similarly, the probability mass at the (right) tail of the distributions is also thicker under the granular approach.¹⁵ On the other hand, the skewness and kurtosis of the granular approach are lower than those of the joint approach, reflecting the bias induced by the latter. Paired *t*-tests

¹⁵ The maximum value under the granular approach, however, was smaller than the joint approach because of a small bank that started the simulation period with very low credit quality.

TABLE 29.8

Effect of a 2 Percentage Point Drop in GDP Growth on NPLs, by Credit Types (in percent unless indicated)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Sector	Estimates of Panel Regressions						Increase in NPLs		Stressed NPLs	
	Average NPLs 2003–9	NPLs March 2009	Coef. Lagged NPLs	Sum Coef. GDP Growth	Long-Term Effect ¹	Scale Factor ²	Short-Term (Percentage Points) ³	Long-Term (Percentage Points) ⁴	Level ⁵	Times Increase
Consumer (large)	3.6	2.5	0.4	−25.0	−38.5	0.035	1.7	2.7	5.2	2.1
Consumer (medium)	6.1	5.0	0.4	−13.7	−22.1	0.057	1.6	2.5	7.6	1.5
Consumer (small)	5.9	7.3	0.7	−9.5	−28.3	0.055	1.0	3.1	10.4	1.4
Wood and furniture	3.8	2.8	0.3	−18.5	−27.8	0.036	1.3	2.0	4.8	1.7
Transportation	3.8	1.7	0.4	0.1	0.1	0.037	0.0	0.0	1.7	1.0
Petrochemicals	3.6	1.7	0.4	−11.6	−19.3	0.035	0.8	1.3	3.0	1.8
Metal products	2.1	1.0	0.5	−17.0	−32.9	0.021	0.7	1.4	2.4	2.4
Electricity and gas	1.3	0.3	0.4	−61.2	−106.1	0.013	1.6	2.8	3.1	10.0
Livestock	4.2	2.4	0.5	−13.8	−27.5	0.041	1.1	2.2	4.6	2.0
Other services	5.0	3.7	0.4	−8.5	−14.4	0.047	0.8	1.4	5.1	1.4
Sugar and alcohol	0.6	1.3	0.3	−67.2	−101.8	0.006	0.8	1.2	2.5	1.9
Retail trade	3.7	3.0	0.6	−15.3	−41.0	0.035	1.1	2.9	5.9	2.0
Textile	3.9	5.2	0.5	−22.0	−48.1	0.038	1.7	3.6	8.8	1.7
Vehicles	3.0	4.0	0.5	−11.4	−23.9	0.029	0.7	1.4	5.4	1.3
Food	4.3	2.6	0.5	−15.8	−29.5	0.041	1.3	2.4	5.0	1.9
Agriculture	1.7	2.6	0.5	−35.2	−64.2	0.017	1.2	2.2	4.7	1.8
Health services	3.2	2.5	0.4	−8.8	−15.8	0.031	0.5	1.0	3.5	1.4
Chemicals	2.2	2.8	0.5	−3.9	−7.2	0.021	0.2	0.3	3.1	1.1
Recreation services	4.5	4.4	0.2	−14.4	−17.4	0.043	1.2	1.5	5.9	1.3
Electrical equipment	4.9	5.3	0.4	−14.0	−21.6	0.046	1.3	2.0	7.3	1.4
Other	2.9	1.2	0.3	3.8	5.3	0.029	−0.2	−0.3	0.9	0.7
Overall sampled credit	2.8	3.9	0.6	−24.4	−60.6	0.027	1.3	3.3	7.2	1.8

Source: Authors.

Note: Change in yearly GDP growth:

1. Computed as (5) = (4)/(1−(3)).

2. The scale factor is computed as: (6) = (1)/(1 − (1)) (i.e., NPL * (1−NPL)).

3. Assuming a 2 pp drop in GDP growth, the short-term increase in NPLs is computed as: (7) = (4) * (6) * (−2).

4. The long-term increase in NPLs is computed as: (8) = (7)/(1−(3))

5. The stressed PD are computed as: (9) = (2) + (8)

Coef. = coefficient; NPL = nonperforming loan.

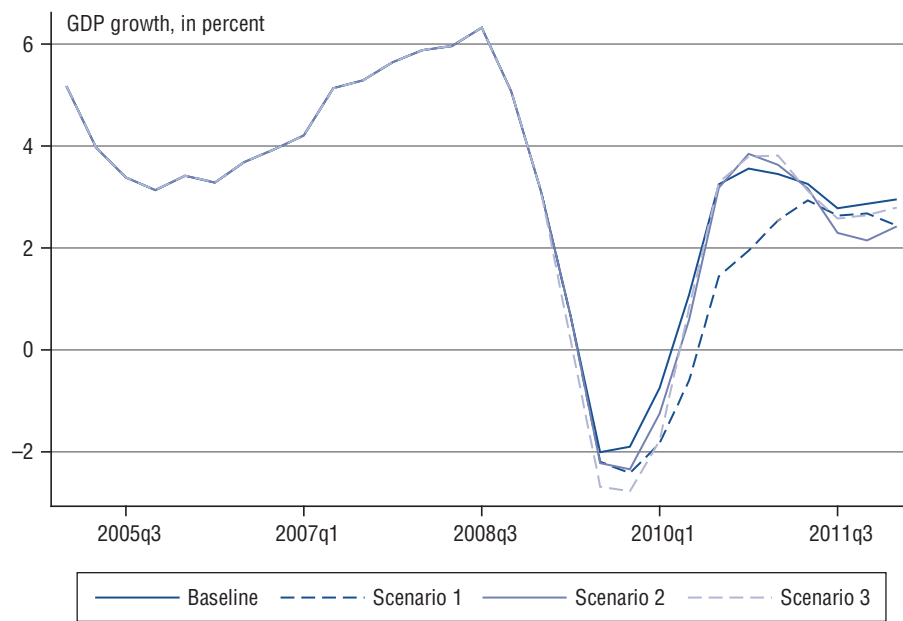
of mean differences confirm that the population mean under the granular approach exceeds those resulting from the joint approach in all cases, although differences in their population variances are not statistically significant (Table 29.10).

C. Credit VaR

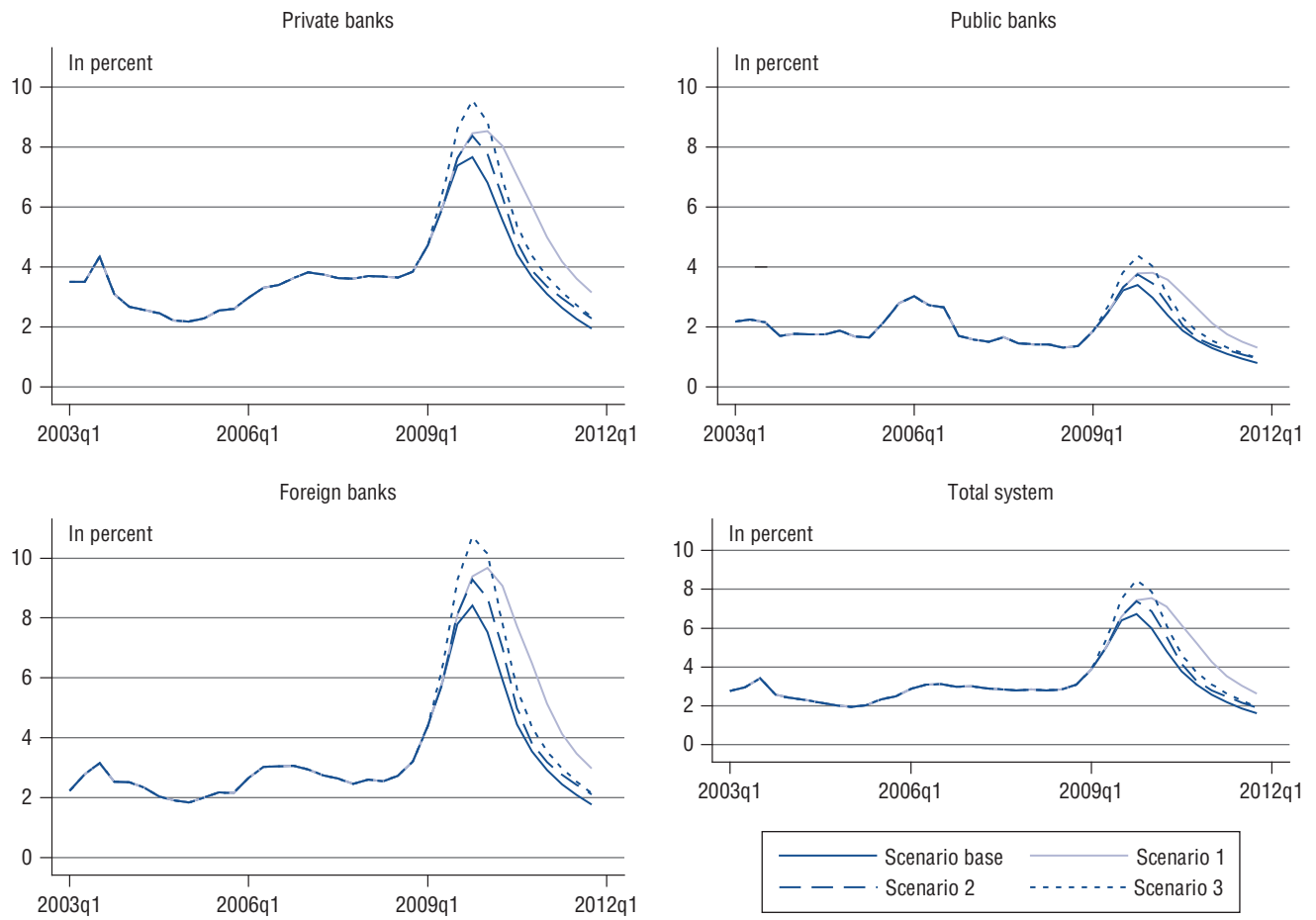
This section uses the previous results to compute a credit VaR and come up with an estimate of banks' (unexpected) credit losses under an adverse macroeconomic environment. For each bank, we model the distribution of credit losses using the CreditRisk+ (Credit Suisse Financial Products, 1997) formulation and the exposures of each bank as of March 2009. Under CreditRisk+ and assuming that default probabilities are random, the probability-generating function $G(z)$ of the normalized total expected losses of a portfolio of n credit types can be written as (Avesani and others, 2006)

$$G(z) = \exp \left\{ - \sum_{k=1}^n \frac{1}{\sigma_k^2} \ln [1 - \sigma_k^2 \cdot P_k(z)] \right\}, \quad (29.5)$$

where default rates are assumed to be distributed according to a gamma distribution, $\Gamma(\alpha_k, \beta_k)$ with $\alpha_k = \mu_k^2 / \sigma_k^2$ and $\beta_k = \sigma_k^2 / \mu_k$. In turn, μ_k is estimated as $\mu_k = \varepsilon_k / v_k$, where ε_k is the expected loss and v_k the exposure to credit type k . We use the bank-specific estimates of NPLs for each credit type under Scenario 1 as a proxy for distressed PDs. In particular, we take the average of the out-of-sample simulation of NPLs for each bank and credit type over the first simulated year of Scenario 3 (the more severe) as a proxy for distressed PDs of the corresponding credit categories. To account for uncertainty on the true value of the PDs (σ_k), we use the standard deviation of the NPLs over the corresponding first year of the out-of-sample simulation. Admittedly, NPLs are an imperfect proxy for PDs because of their “backward-looking” nature. While PDs are intended to capture the likelihood of borrowers' default within a given horizon (i.e., one year ahead), NPLs typically measure the proportion of loans that are more than 90 days past due in total credit portfolios. Therefore, PDs would vary in response to changes in the repayment capacity of a given borrower, which may not immediately translate

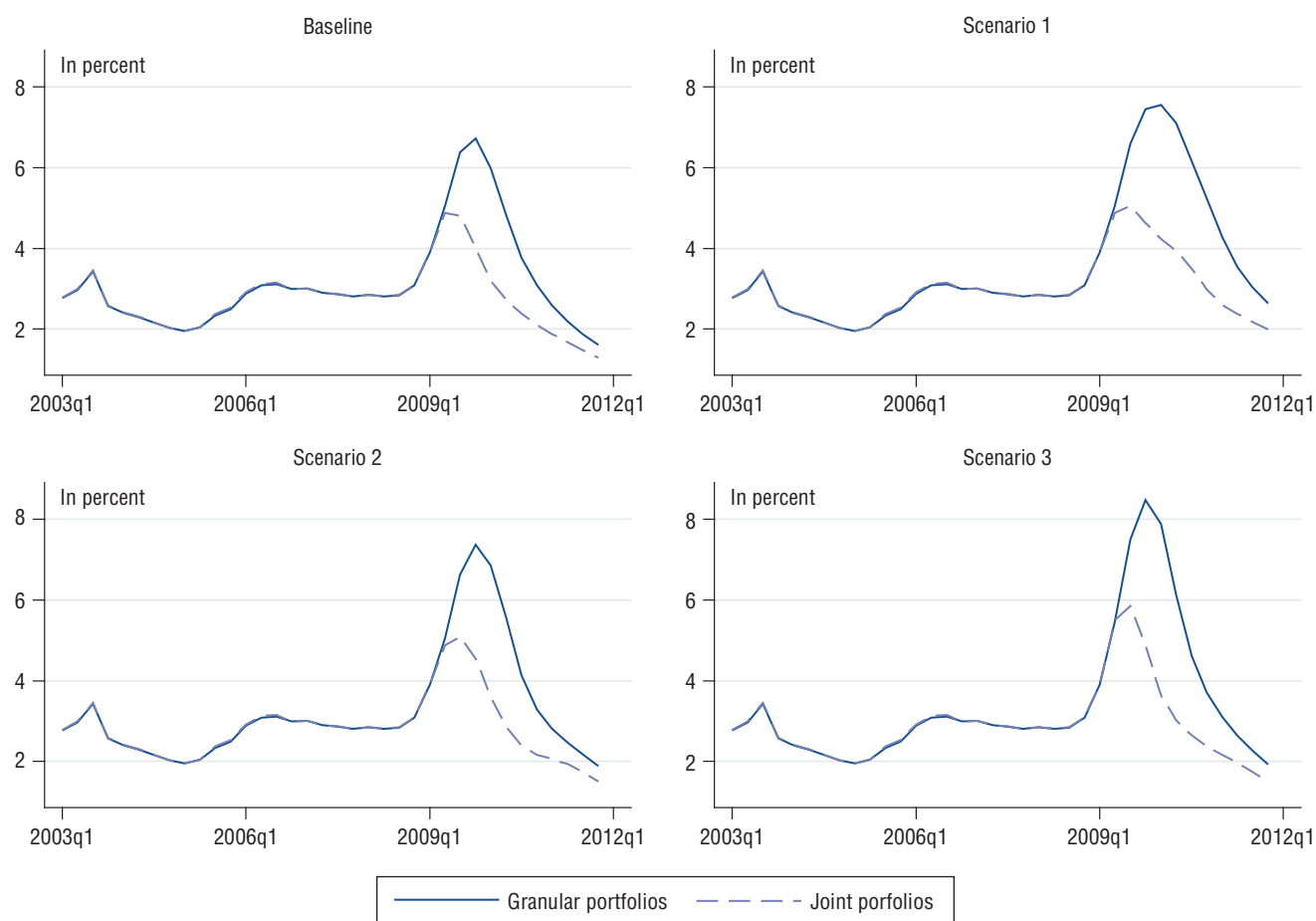


Source: Authors.

Figure 29.4 Evolution of GDP Growth under Alternative Scenarios (in percent year-on-year)

Source: Authors.

Figure 29.5 Evolution of NPLs under Alternative Scenarios



Source: Authors.

Note: NPL = nonperforming loan.

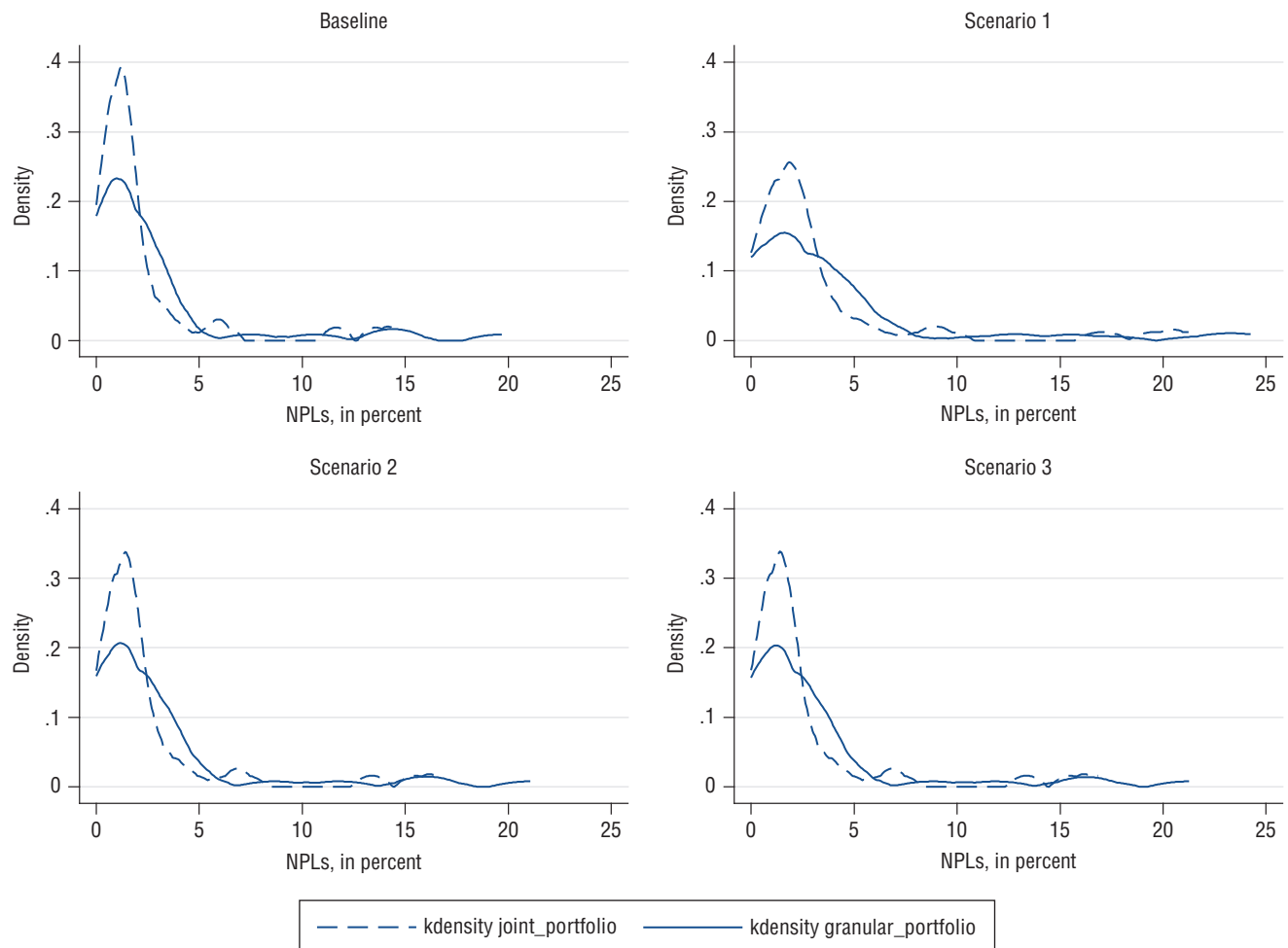
Figure 29.6 Evolution of NPLs by Scenarios and Computation Methods

into changes in NPLs. Using NPLs as a proxy for PDs is prone to various sources of bias that may operate in opposite directions. During the upturn of the economic cycle, which tends to be associated with strong credit growth, NPLs may underestimate the risk profile of banks' credit portfolios. In contrast, NPLs may overestimate credit risk during the downturn, as banks defer loan write-offs.

For each credit type, we conduct an initial calculation of the average exposures to individual borrowers, by dividing the corresponding total exposures over the number of loan operations. This treatment, however, may underestimate portfolio concentration and therefore the results of the credit VaR. We thus consider an alternative exercise by assuming that 80 percent of the exposures under each credit category are concentrated in 20 percent of the number of credit operations (and the remaining 20 percent of the exposures correspond to 80 percent of the number of credit operations). Because we do not have information on losses given default (LGDs), we choose a generic value of 50 percent for all credit types. We recognize that this is also a critical assumption in our assessment of banks' solvency, as higher LGD would mechanically lead to higher unexpected losses in the credit

VaR calculation. Moreover, LGDs are likely to vary across credit types, also depending on the existence of collateral on individual credit operations, which we cannot observe. There is also evidence that LGDs tend to vary throughout the cycle, increasing during the economic downturn. Conversely, to the extent that potential differences in the sensitivity of LGDs to the economic cycle across credit types go in tandem with the dynamics of their respective PDs, which is likely the case, our relative assessment of individual banks' solvency would not be biased. Furthermore, our assumption of a 50 percent LGD is likely to be conservative.

The parameters used in the credit VaR are bank specific. Although the sensitivities of credit quality to macroeconomic conditions are the same for each credit type, the quality of individual banks' credit portfolios in each of the distressed scenarios would vary according to their starting conditions. Overall, consumer loans and credit to the textile industry, agriculture, and sugar and alcohol are the most procyclical and concentration to the last more pronounced (Table 29.11). Using these parameters, the distribution of credit losses is computed using the probability-generating function defined in equation (29.5), and the unexpected



Source: Authors.
Note: NPL = nonperforming loan.

Figure 29.7 Kernel Densities of Simulated NPLs in 2011Q4, by Estimation Methods

TABLE 29.9

Summary Statistics of Simulated NPLs by Scenarios and Estimation Methods

Statistic	Baseline		Scenario 1		Scenario 2		Scenario 3	
	Granular	Joint	Granular	Joint	Granular	Joint	Granular	Joint
Mean	5.690	4.630	7.225	5.943	6.116	5.010	6.572	5.300
SD	6.643	6.168	7.616	7.361	6.954	6.567	7.313	6.939
Minimum	0.007	0.014	0.013	0.021	0.008	0.016	0.009	0.016
Median	3.358	2.520	4.963	3.521	3.756	2.699	4.011	2.880
90th Percentile	14.412	11.114	17.452	13.910	15.468	11.476	16.335	12.623
95th Percentile	19.694	18.488	22.108	23.804	21.083	19.914	21.858	21.009
99th Percentile	33.272	32.054	36.733	35.435	35.474	34.621	37.529	36.631
Maximum	36.171	41.186	38.682	42.489	37.170	42.671	39.367	46.367
Skewness	2.241	2.975	2.049	2.666	2.182	2.923	2.122	2.937
Kurtosis	8.529	12.784	7.545	10.138	8.281	12.418	8.090	12.670

Source: Authors.
Note: NPL = nonperforming loan.

TABLE 29.10

Comparison of Simulated Distributions of NPLs				
Test	Baseline	Scenario 1	Scenario 2	Scenario 3
Mean paired t-test				
(1) Granular (Mean)	5.690	7.225	6.116	6.572
(2) Joint (Mean)	4.630	5.943	5.010	5.300
Difference (1)–(2)	1.060	1.282	1.105	1.273
H1: Mean Diff<0 (<i>p</i>)	1.000	1.000	1.000	1.000
H1: Mean Diff~0 (<i>p</i>)	0.000	0.000	0.000	0.000
H1: Mean Diff>0 (<i>p</i>)	0.000	0.000	0.000	0.000
Variance ratio test				
(1) Granular (<i>SD</i>)	6.643	7.616	6.954	7.313
(2) Joint (<i>SD</i>)	6.168	7.361	6.567	6.939
Difference (1)–(2)	0.475	0.254	0.387	0.374
H1: Mean Diff<0 (<i>p</i>)	0.933	0.754	0.876	0.856
H1: Mean Diff~0 (<i>p</i>)	0.134	0.493	0.247	0.288
H1: Mean Diff>0 (<i>p</i>)	0.067	0.246	0.124	0.144

Source: Authors.

Note: Diff = difference; NPL = nonperforming loan.

TABLE 29.11

Results of the Credit VaR (in million BRL, unless otherwise indicated)					
Bank Number	Credit VaR	Credit VaR/Net Exposure (Percent)	Credit VaR/Gross Exposure (Percent)	Gross Exposure	Share of Loans in Sample (Percent)
1	4,689	5.0	2.5	1,89,052	24.3
2	11,095	12.8	6.4	1,73,172	22.2
3	6,865	10.1	5.1	1,35,276	17.4
4	8,181	14.0	7.0	1,16,957	15.0
5	1,479	9.3	4.7	31,798	4.1
6	906	5.8	2.9	31,378	4.0
7	1,152	10.9	5.5	21,118	2.7
8	551	11.4	5.7	9,642	1.2
9	286	6.3	3.2	9,046	1.2
10	347	10.2	5.1	6,839	0.9
11	20	0.6	0.3	6,348	0.8
12	506	16.3	8.1	6,216	0.8
13	185	10.0	5.0	3,675	0.5
14	161	9.1	4.5	3,538	0.5
15	200	11.4	5.7	3,502	0.4
16	71	4.6	2.3	3,113	0.4
17	309	20.7	10.4	2,985	0.4
18	116	9.1	4.5	2,563	0.3
Total	37,118	9.8	4.9	7,56,218	97.0

Source: Authors.

Notes: Net exposures are computed by subtracting the estimated recovery values from the gross credit exposures. Loss given default = 0.5; model = Poisson defaults/fast Fourier transform; VaR = value at risk. Parameters: VaR level = 0.99.

losses are estimated as the 99th percentile of the expected losses distribution.

The results suggest that the banking sector is well prepared to undergo the credit losses associated with the distressed scenarios considered without threatening financial stability. The

(unexpected) credit losses associated with a 99 percent credit VaR for the 18 banks with the largest credit portfolios in the sample amount to about BRL37 billion, or 4.9 percent of their gross exposures (Table 29.12). As a reference, these losses are roughly equivalent to about 19 percent of the joint tangible capital of these banks. Our measure of tangible capital equals regulatory capital minus the sum of specific loan loss provisions included in banks' own resources, deferred taxes, and goodwill. Therefore, the capital cushions of the largest banks appear sufficient to absorb the credit losses associated with the scenarios considered without threatening financial stability.

4. CONCLUSION

The econometric estimations presented in this study provide strong evidence of cyclical behavior of loan quality. The estimations substantiate the existence of a robust inverse relationship between GDP growth and NPLs, with the effects operating with up to three quarter lags. The results also indicate differences in the persistence of NPLs across credit types, and in their sensitivity to economic activity. Loan quality in Brazil appears to be more sensitive to GDP growth for small consumer loans, credit to agriculture, sugar and alcohol, live-stock, and textiles. In addition, credit for vehicle acquisition and electric and electronic equipment displayed a high level of NPLs under distressed macroeconomic scenarios. Banks with relatively higher exposures to these sectors are likely to experience larger credit losses under a macroeconomic downturn.

Although intuitive, the modeling of differences in the sensitivity of loan quality to macroeconomic conditions at the level of individual banks is novel to the macro stress testing literature. Existing models based on bank-level data do not allow for a differential response of credit quality to macroeconomic conditions across credit types, possibly owing to lack of data availability. Conversely, existing macro stress test models that exploit variations in the sensitivity of loan quality to macro-

TABLE 29.12

Credit Risk Parameters Used to Estimate Credit VaR						
Sector	Distressed PDs			Credit Exposures		
	Mean	Median	Max	Average Size (Million BRL)	Number of Operations	Average Credit (Million BRL)
Agriculture	0.14	0.09	0.86	915	1,802	0.508
Food	0.03	0.02	0.11	999	4,344	0.230
Livestock	0.09	0.03	0.69	1,360	2,513	0.541
Vehicles	0.06	0.05	0.21	1,185	7,728	0.153
Electrical and electronic equipment	0.11	0.06	0.51	624	4,944	0.126
Electricity and gas	0.03	0.00	0.44	1,376	175	7.867
Wood and furniture	0.05	0.05	0.12	3,369	17,700	0.190
Recreation services	0.04	0.03	0.09	742	9,708	0.076
Other	0.02	0.01	0.17	3,755	7,098	0.529
Petrochemicals	0.09	0.04	0.42	1,559	4,121	0.378
Consumer (large)	0.07	0.04	0.29	1,103	1,782	0.619
Consumer (medium)	0.11	0.12	0.22	4,114	66,188	0.062
Consumer (small)	0.10	0.09	0.23	10,814	814,219	0.013
Chemicals	0.02	0.01	0.07	701	3,760	0.187
Health services	0.05	0.04	0.20	809	5,741	0.141
Other services	0.07	0.05	0.29	1,318	11,036	0.119
Metal products	0.03	0.02	0.17	1,421	1,770	0.803
Sugar and alcohol	0.22	0.20	0.77	1,351	172	7.860
Textile	0.18	0.13	0.67	1,184	12,264	0.097
Transportation	0.02	0.02	0.08	2,173	7,222	0.301
Retail trade	0.10	0.05	0.81	1,213	11,631	0.104

Source: Authors.

Note: PD = probability of default; VaR = value at risk.

economic conditions across credit types are based on aggregated data, and are therefore less suited to assess the adequacy of bank capital at the level of individual institutions.

The results presented in this study show that the lack of sufficiently granular data on the composition of bank credit portfolios can bias the results in a way that is contrary to a prudent criterion. To the extent that the sensitivity of credit quality to macroeconomic conditions varies between different credit types, the lack of differentiation would tend to underestimate the deterioration of credit quality for the highly procyclical credit types and sectors under a distressed macroeconomic environment (and overestimate the deterioration of credit quality for the relatively safer credit types). These biases would translate in a systematic way into the assessment of bank risk profiles, leading to an underestimation of tail losses in the more vulnerable institutions in the banking system under analysis.

The model presented in this study represents an improvement over existing literature but is still subject to several important caveats. First, the model assumes a linear relationship between loan quality and macroeconomic conditions, which may fail to capture potential nonlinear relationships during periods of severe macroeconomic distress. Second, the model assumes that historic correlations between loan quality and macroeconomic conditions are symmetric during the upturn and the downturn of the economic cycle and remain valid during periods of severe distress. Third, the model fails to capture potential feedback effects between credit quality and economic growth, as it does not fully integrate the macro and microeconomic modules. In particular, the macroeconomic

module allows credit volumes to vary over time, while the microeconomic module assumes that individual banks maintain constant credit portfolios. To the extent that credit quality tends to deteriorate during periods of slow credit growth, the model presented in this study may underestimate banks' loan losses. All these caveats may likely bias the results in the same direction during periods of financial distress, causing a potential underestimation of bank credit losses. Further analysis is needed to address these shortcomings.

REFERENCES

- Arellano, Manuel, and Stephen Bond, 1991, "Some Tests of Specification for Panel Data: Monte Carlo Evidence and an Application to Employment Equations," *Review of Economic Studies*, Vol. 58, No. 2, pp. 277–97.
- Avesani, Renzo, Kexue Liu, Alin Mirestean, and Jean Salvati, 2006, "Review and Implementation of Credit Risk Models of the Financial Sector Assessment Program," IMF Working Paper 06/134 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=19111>
- Barnhill, Theodore, Marcos R. Souto, and Benjamin M. Tabak, 2006, "An Analysis of Off-Site Supervision of Banks' Profitability, Risk and Capital Adequacy: A Portfolio Simulation Approach Applied to Brazilian Banks," Working Paper No. 117 (Brasilia: Central Bank of Brazil). Available via the Internet: <http://www.bcb.gov.br/pec/wps/port/wp117.asp?idiom=I>
- Berkowitz, Jeremy, 1999, "A Coherent Framework for Stress-Testing," Working Paper No. 29, Finance and Economics Discussion Series (Washington: Board of Governors of the Federal Reserve

- System). Available via the Internet: <http://www.federalreserve.gov/pubs/feds/1999/199929/199929abs.html>
- Blank, Sven, Claudia M. Buch, and Katja Neugebauer, 2009, "Shocks at Large Banks and Banking Sector Distress: The Banking Granular Residual," *Journal of Financial Stability*, Vol. 5, No. 4, pp. 353–73.
- Blundell, Richard, and Stephen Bond, 1998, "Initial Conditions and Moment Restrictions in Dynamic Panel Data Models," *Journal of Econometrics*, Vol. 87, No. 1, pp. 115–43.
- Boss, Michael, Gerald Krenn, Claus Pühr, and Markus S. Schwaiger, 2007, "Stress Testing the Exposure of Austrian Banks in Central and Eastern Europe," *Financial Stability Report*, No. 13, pp. 115–34.
- Cardarelli, Roberto, Selim Elekdag, and Subir Lall, 2011, "Financial Stress and Economic Contractions," *Journal of Financial Stability*, Vol. 7, No. 2, pp. 78–97.
- Castrén, Olli, Stéphane Déas, and Fadi Zaher, 2010, "Stress-Testing Euro Area Corporate Default Probabilities Using a Global Macroeconomic Model," *Journal of Financial Stability*, Vol. 6, No. 2, pp. 64–78.
- Coffinet, Jérôme, and Surong Lin, 2010, "Stress Testing Banks' Profitability: The Case of French Banks," Working Paper No. 306 (Paris: Banque de France). Available via the Internet: <http://www.banque-france.fr/en/economics-statistics/research/working-paper-series/document/306-1.html>
- Coffinet, Jérôme, and Clément Martin, 2009, "Stress Testing French Banks' Income Subcomponents," Working Paper No. 242 (Paris: Banque de France). Available via the Internet: <http://www.banque-france.fr/en/economics-statistics/research/working-paper-series/document/242-1.html>
- Credit Suisse Financial Products, 1997, "CreditRisk+: A Credit Risk Management Framework," Technical Report (London: Credit Suisse First Boston). Available via the Internet: <http://www.csfb.com/institutional/research/assets/creditrisk.pdf>
- Duellmann, Klaus, and Martin Erdelmeier, 2009, "Crash Testing German Banks," *International Journal of Central Banking*, Vol. 5, No. 3, pp. 139–79.
- Foglia, Antonella, 2009, "Stress Testing Credit Risk: A Survey of Authorities' Approaches," *International Journal of Central Banking*, Vol. 5, No. 3, pp. 9–45.
- Froyland, Espen, and Kai Larsen, 2002, "How Vulnerable Are Financial Institutions to Macroeconomic Changes? An Analysis Based on Stress Testing," *Economic Bulletin*, Vol. 3, No. 11, pp. 127–69.
- Gerlach, Stefan, Wensheng Peng, and Chang Shu, 2004, "Macroeconomic Conditions and Banking Performance in Hong Kong: A Panel Study," Working Paper (Hong Kong SAR: Hong Kong Monetary Authority). Available via the Internet: <http://www.hkma.gov.hk/eng/publications-and-research/research/working-papers/pre-2007.shtml>
- Hoggarth, Glenn, and John Whitley, 2003, "Assessing the Strength of U.K. Banks through Macroeconomic Stress Tests," *Financial Stability Review*, No. 14 (London: Bank of England). Available via the Internet: <http://www.bankofengland.co.uk/publications/Pages/fsr/2003/fsr14.aspx>
- Illing, Mark, and Ying Liu, 2006, "Measuring Financial Stress in a Developed Country: An Application to Canada," *Journal of Financial Stability*, Vol. 2, No. 3, pp. 243–65.
- Misina, Miroslav, and David Tessier, 2008, "Non-Linearities, Model Uncertainty and Macro Stress Testing," Working Paper 2008-30 (Ottawa: Bank of Canada). Available via the Internet: <http://www.bankofcanada.ca/2008/09/publications/research/working-paper-2008-30/>
- Pesola, Jarmo, 2001, "The Role of Macroeconomic Shocks in Banking Crises," Discussion Paper 6/2001 (Helsinki: Bank of Finland). Available via the Internet: http://www.suomenpankki.fi/en/julkaisut/tutkimukset/keskustelualoitteet/Pages/dp2001_6.aspx?hl=The%20Role%20of%20Macroeconomic%20Shocks%20in%20Banking%20Crises
- , 2005, "Banking Fragility and Distress: An Economic Study of Macroeconomic Determinants," Discussion Paper 13/2005 (Helsinki: Bank of Finland). Available via the Internet: http://www.suomenpankki.fi/en/julkaisut/tutkimukset/keskustelualoitteet/Pages/dp2005_13.aspx?hl=Banking%20Fragility%20and%20Distress%20A%20An%20Economic%20Study%20of%20Macroeconomic%20Determinants
- Rodriguez, Adolfo, and Carlos Trucharte, 2007, "Loss Coverage and Stress Testing Mortgage Portfolios: A Non-Parametric Approach," *Journal of Financial Stability*, Vol. 3, No. 4, pp. 342–67.
- Sorge, Marco, 2004, "Stress-Testing Financial Systems: An Overview of Current Methodologies," BIS Working Paper No. 165 (Basel: Bank for International Settlements). Available via the Internet: <http://www.bis.org/publ/work165.htm>
- Sorge, Marco, and Kimmo Virolainen, 2006, "A Comparative Analysis of Macro Stress-Testing Methodologies with Application to Finland," *Journal of Financial Stability*, Vol. 2, No. 2, pp. 113–51.
- Staub, Roberta B., Geraldo da Silva e Souza, and Benjamin M. Tabak, 2010, "Evolution of Bank Efficiency in Brazil: A DEA Approach," *European Journal of Operational Research*, Vol. 202, No. 1, pp. 204–13.
- Tabak, Benjamin M., Dimas M. Fazio, and Daniel O. Cajueiro, 2011, "The Effects of Loan Portfolio Concentration on Brazilian Banks' Return and Risk," *Journal of Banking and Finance*, Vol. 35, No. 11, pp. 3065–76.
- Tabak, Benjamin M., and Roberta B. Staub, 2007, "Assessing Financial Instability: The Case of Brazil," *Research in International Business and Finance*, Vol. 21, No. 2, pp. 188–202.
- Tecles, Patricia Langsch, and Benjamin M. Tabak, 2010, "Determinants of Bank Efficiency: The Case of Brazil," *European Journal of Operational Research*, Vol. 207, No. 3, pp. 1587–98.
- van den End, Jan Willem, Marco Hoeberichts, and Mostafa Tabbæ, 2006, "Modelling Scenario Analysis and Macro Stress-Testing," DNB Working Paper No. 119 (Amsterdam: De Nederlandsche Bank). Available via the Internet: <http://www.dnb.nl/en/publications/dnb-publications/dnb-working-papers-series/dnb-working-papers/auto101896.jsp>
- Vazquez, Francisco, Benjamin M. Tabak, and Marcos Souto, 2012, "A Macro Stress Test Model of Credit Risk for the Brazilian Banking Sector," *Journal of Financial Stability*, Vol. 8, No. 2, pp. 69–83.
- Virolainen, Kimmo, 2004, "Macro Stress Testing with a Macroeconomic Credit Risk Model for Finland," Discussion Paper 18/2004 (Helsinki: Bank of Finland). Available via the Internet: http://www.suomenpankki.fi/en/julkaisut/tutkimukset/keskustelualoitteet/Pages/dp2004_18.aspx?hl=Macro%20Stress%20Testing%20with%20a%20Macroeconomic%20Credit%20Risk%20Model%20for%20Finland
- Wilson, Thomas, 1997a, "Portfolio Credit Risk (i)," *Risk*, Vol. 10, No. 9, pp. 111–17.
- , 1997b, "Portfolio Credit Risk (ii)," *Risk*, Vol. 10, No. 10, pp. 56–61.
- Wong, Jim, Ka-fai Choi, and Tom Fong, 2006, "A Framework for Stress Testing Bank's Credit Risk," Research Memorandum 15/2006 (Hong Kong SAR: Hong Kong Monetary Authority). Available via the Internet: <http://www.hkma.gov.hk/eng/publications-and-research/research/working-papers/pre-2007.shtml>

A Practical Example of the Nonperforming Loans Projection Approach to Stress Testing

TORSTEN WEZEL • MICHEL CANTA • MANUEL LUY

Using a case study approach, this chapter illustrates the process of developing a balance sheet stress testing model for a relatively large banking sector of regional importance. The chapter first discusses the importance of ensuring consistent data for stress testing and of choosing the appropriate econometric setup. Given the unique characteristics of the data set—short time period, large number of banks—the model applies the System Generalized Method of Moments estimator that also deals with so-called dynamic panel bias. The setup consists of credit risk models for projecting the impact of macroeconomic shocks on the delinquency ratios of loans to seven main economic sectors, as well as a satellite model for credit growth to determine the absolute increase in nonperforming loans (NPLs). On the basis of projections for additionally required provisions, preprovision net income and change in risk-weighted assets (RWA), the expected change in capital adequacy ratios is then calculated. The stress test results for the country at hand illustrate that severe shocks in the stress scenarios cause a considerable increase in NPL ratios, whereas the average capitalization ratio does not fall by much. This discrepancy is attributable to banks' high preprovision net income absorbing the cost of additional loan losses and the relative inelasticity of RWA under the Basel I framework applied in this country.

METHOD SUMMARY

Overview	The model estimates the nonperforming loan (NPL) ratio under stress as a function of macroeconomic and financial variables in a panel data setting and determines the change in regulatory capital given assumptions about credit growth, NPL transition, preprovision net income, and dividend payout.
Application	The method is appropriate in situations where information on loan portfolios and macroeconomic data is reliable and available at quarterly frequency.
Nature of approach	Balance sheet-based.
Data requirements	• Accounting information on capital, loans and risk-weighted assets (RWA). • Supervisory data on classified loans and provisions. • Macroeconomic data.
Strengths	The model is based on financial statements and accounting rules. It explicitly takes into account banks' profits as first line of defense.
Weaknesses	• Assumptions are required on the amounts of transitioning NPLs and corresponding loan loss provisions from one category to another. Estimates of expected credit growth are essential for determining the changes in NPLs and RWA. • Primarily intended for banking systems operating under the Basel I framework.
Tool	Standard econometrics package.

This chapter describes the features of a stress testing model that was developed for a small open economy with a relatively large banking sector. As most banks in the sector follow a traditional intermediation model focused on lending and bank

regulation as governed by the Basel I Accord, a balance sheet approach to stress testing was deemed appropriate. The approach presented in this chapter is relatively straightforward compared with modeling techniques used to assess more

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sophisticated and complex banking systems, particularly those using the internal ratings–based (IRB) approach of Basel II (e.g., Schmieder, Puh, and Hasan, 2011). The stress testing framework presented here comprises two main elements: (1) an econometric model that statistically relates past-due loans to macroeconomic variables; and (2) a template to predict nonperforming loans (NPLs), provisioning, and capital adequacy ratios (CARs). Specifically, the stress testing module consists of the following:

- A set of dynamic panel data models estimated to ascertain the relationship between an indicator of loan quality for the main economic sectors of the economy and selected macroeconomic and financial variables. The results of these models are combined with projections of those variables found to affect credit quality to project the path of NPLs for each of the loan categories under a baseline and two stress scenarios.
- A template specifically designed to aggregate the impact of the postshock increase in NPLs from the sectoral models and calculate the amount of additionally required provisioning, which, combined with projections of other income items and some auxiliary calculations, yields the predicted change in banks' CARs.

The purpose of this study is to describe both the options in designing a credit risk model and the reasons for the methodology eventually chosen. Given that commonly agreed best practices in stress testing are just now emerging (see Basel Committee on Banking Supervision [BCBS], 2009) from among the numerous existing empirical approaches (see Sorge, 2004), the stress tester faces a set of choices when building a credit risk model. These choices concern, inter alia, the econo-

metric setup of the credit risk model, the definition of the dependent variable, methods for projecting explanatory and other noncredit risk variables, and ways to convert a stress-induced increase in loan delinquencies into impaired capital positions. To this end, the chapter discusses the pros and cons of the panel estimation approaches commonly used in stress testing and also gives reasons for the method chosen for the country case at hand.

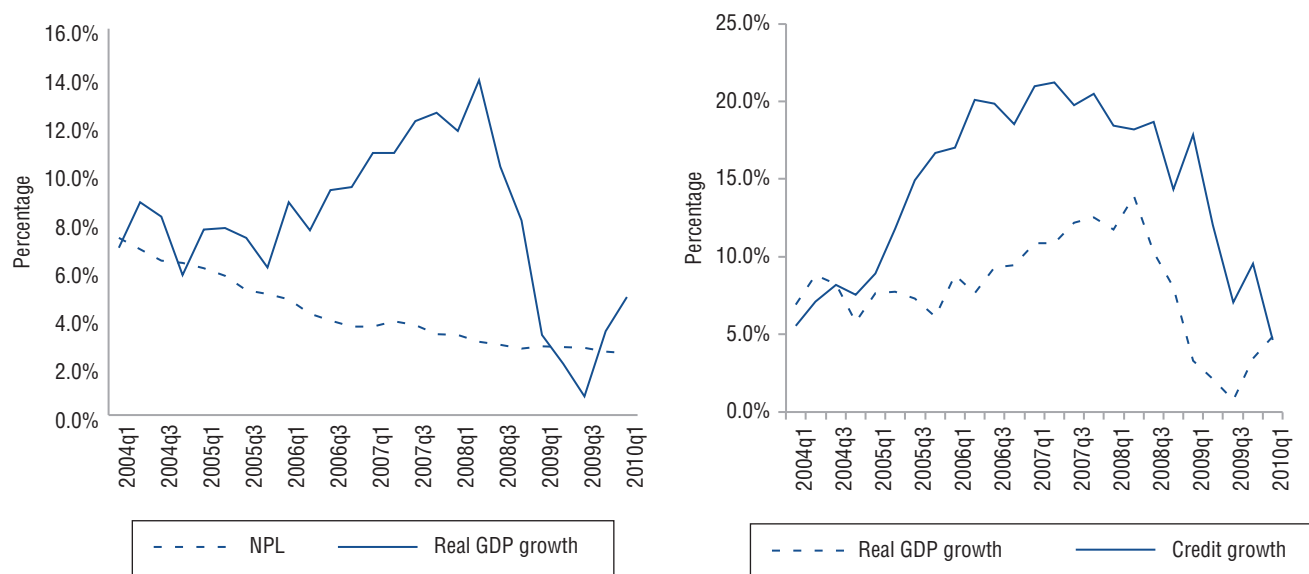
The chapter is structured as follows. Section 1 describes the structure and quality of the data supplied. Section 2 discusses the characteristics of the credit risk model and set of variables and why this particular approach was chosen among several alternatives. Section 3 describes the selection of stress test scenarios and the method for projecting NPLs. Section 4 shows how projected loan losses affect bank capital and presents the stress test results. Section 5 concludes the chapter.

1. BACKGROUND

A. Recent macroeconomic and financial developments

The country for which the stress testing model was developed can be characterized as a small open economy with a relatively sizable banking sector. The openness of the economy makes the banking system susceptible to external shocks, which were accounted for in the selection of explanatory variables for the credit risk model. The outward orientation also coincides with a significant presence of foreign-owned banks.

Bank credit evolved broadly in line with economic developments, but NPLs remained fairly stable (Figure 30.1). Credit growth was rapid during the expansionary phase pre-



Source: National Statistical Office.

Figure 30.1 Evolution of GDP Growth, Credit Growth, and the Overall Nonperforming Loan (NPL) Ratio

ceding the global financial crisis—even exceeding the buoyant GDP growth—but then slowed down considerably as banks became more cautious amid the volatile international trade and financial environment. Banks also restricted their lending in anticipation of softening demand for housing and lower tourism receipts. However, these expectations largely did not materialize during the downturn as ongoing projects in commercial real estate and infrastructure helped sustain economic growth. The seeming stability of NPLs was, however, partly attributable to write-offs during the downturn. Overall, NPLs—defined as the sum of substandard, doubtful, and loss loans—steadfastly declined by 5 percentage points between 2004 and 2010 despite volatile economic and credit growth.

The soft landing of the economy and tight supervision of banks during the downturn helped preserve financial stability. Banks remained adequately capitalized and highly liquid throughout, which was also the result of the strict regulatory regime mandating exposure limits, timely recognition of loan losses, and ample liquidity buffers. The application of the Basel I framework meant that most loans carried a 100 percent risk weight.

B. Stress testing data

Although overall credit developments were consistent with the general soundness of the financial system, the data supplied by the supervisory authority showed that the composition and riskiness of loan portfolios differed widely at the level of individual banks. This was to be expected for a regional financial center hosting investment banks, corporate banks, and consumer credit institutions. The variation in loan quality within and certainly across banks provided a basis for estimating a credit risk model in a panel setting.

For any credit risk model, high-quality data are an essential precondition. Considerable time and effort were necessary in ensuring that the data inputs for the stress testing system were sufficiently complete and consistent. This process encompassed checking the data series submitted by the authorities for consistency as well as for missing or illogical values. Presumed inconsistencies or inaccuracies prompted a resubmission of data until the data set as a whole was deemed sufficiently accurate and reliable for estimation purposes.

Series of key macroeconomic and financial data were available at the quarterly frequency. Macroeconomic data are published by the national statistical office with a lag of one quarter, while bank-specific data become available with about a one-month lag. Data for some aggregated sectors of the economy were available but were somewhat lacking in granularity. Although the time series were rather short, the available period covered important macro events, including the credit boom preceding the global financial crisis and the crisis period itself.

Data on banks' credit portfolios were obtained from the supervisory authority. The official database contains information on bank loans extended to the main economic sectors.

The supervisory authority also collects data at the debtor level, but these are not exploited and thus not readily available for analytical purposes. Consistent information on bank balance sheets and income statements was available for 41 banks from 2003:Q1 to 2010:Q2. Loans at the individual banks' level were aggregated and classified into seven sectoral categories on the basis of similarity of characteristics and their contributions to GDP, namely, the primary sector, manufacturing, construction, commerce, services, as well as mortgage and consumer loans.

Notwithstanding the wide scope of available data, doubts about data integrity initially remained. For most of the banks, the data were continuous and consistent throughout the sample period. Some data gaps were found for the smaller banks, which were subsequently dropped from the sample in order not to skew the regression outcome. The sample period was also restricted by the availability of consistent data, as there had been a structural change in data reporting in 2002. In the sample, inconsistencies were identified and purged before conducting the estimations. Still, these inconsistencies were transmitted to the supervisory authority to help identify the sources of data issues so that appropriate remedial action could be taken to improve data quality.¹

2. ECONOMETRIC CREDIT RISK MODEL

Most credit risk models in stress testing involve econometric estimations of the determinants of loan impairment. Although empirical approaches clearly differ in their econometric designs (see Foglia, 2009), all of them seek to establish a robust long-term relationship between certain measures of loan delinquency and the underlying macroeconomic and financial drivers of loan quality. Available specification options include (1) the use of panel data comprising individual banks or aggregated data for the entire system; (2) various definitions of the dependent variable denoting loan impairment and estimation with ordinary least squares (OLS) or bank-specific fixed effects; and (3) setups with or without dynamic elements, including instrument variable and co-integration techniques.

In the country case at hand, a dynamic panel data model was deemed appropriate. This type of model accounts for inertia in the dependent variable by including its lagged value among the explanatory variables. The particular setup chosen—the Generalized Method of Moments (GMM)—also deals appropriately with the bias arising from including this lagged variable as well as the possible endogeneity of explanatory variables (see Section 2.B). Given that the dependent variable displayed a unit root, a co-integration approach also was considered but ultimately discarded because some of the explanatory variables turned out to be integrated at different orders.

¹ A long series of consecutive zeros in the data, although seemingly implausible, was not considered an inconsistency per se.

A. Dependent variable

The stress tester typically faces the choice of the most appropriate measure of loan quality. Many models use either the absolute amount of NPLs as the dependent variable or the NPL ratio relating impaired loans to total loans. The projection of this variable itself can be considered an intermediate indicator in the stress testing exercise, and in some cases, stress testers deduct the increment in NPLs—somewhat erroneously—directly from banks' capital position. An alternative measure of loan impairment is either the stock of loan loss provisions or the provisioning flows, perhaps set in relation to total loans. This measure has the clear advantage of not having to translate the projected increase in NPLs into additional provisioning. In practice, however, provisioning is not driven fully by credit risk because banks resort to over- (or under-) provisioning to smooth income, and therefore the variable typically reacts less to changes in the explanatory variables, rendering the statistical relationship less robust (as in this case). Other models use as dependent variable Moody's KMV expected default frequency (EDF), such as Åsberg and Shahnazarian (2008); or other measures denoting the probability of default (PD), as discussed in Moretti, Stolz, and Swinburne (2008); or even loan transition rates (Bank of Japan, 2007).

This study used as dependent variable the share of past-due loans in total loans, including special mention loans. This measure of loan quality produced a better fit than provisioning measures, and estimates of PDs were not available. The ratio of past-due loans to total loans was preferred over the absolute amount because it is a widely applied indicator of loan quality (including by the supervisory authority in this case). Also, the ratio did not systematically trend up compared with the absolute amount of NPLs by virtue of the underlying positive loan growth. However, use of the NPL ratio then required the projection of credit growth in order to obtain a projection of the ratio's denominator (see Section 4.B).

A wide definition was chosen for the measure of delinquent loans. It included not only the usual definition of NPLs that ranged from substandard (Category 3) to loss loans (Category 5) but also the special mention loans in Category 2. This inclusion made sense because such loans also require provisioning, and, although the corresponding rate was low at 2 percent, the large volume of loans in this category still required a significant amount of provisions. It also had a better econometric fit as the migration from Category 1 to 2 was more pronounced than in the lower categories. As we will show, the projected increase in NPLs that the credit risk model produced needed to be distributed across the nonnormal loan categories, and including special mention loans in NPLs avoided an underestimation of additional provisions that would have invariably arisen if these loans had been omitted.

This NPL variable was augmented by an estimate of loan write-offs. This estimate accounts for the significant increase in write-offs during the global financial crisis amounting to approximately 0.5 percent of total loans during 2008:Q4–

2009:Q4. This add-on was deemed necessary to correct for the benign effect that write-offs had on the NPL ratio. Some banks actually experienced falling NPL ratios during the crisis on account of higher-than-usual write-offs. However, as data on loan write-offs had not been collected by the supervisory authority, the write-offs needed to be proxied by taking the observed negative change in the category of irrecoverable (defaulted) loans from one quarter to the next.² Although admittedly an imperfect proxy,³ it helped to account for abnormally large drops in defaulted loans that likely represented write-offs. Considering that a certain amount of write-offs is normal in good times, only the difference between this proxied amount and the average quarterly average write-off (between 2003:Q1 and 2008:Q3) was added back onto NPLs.⁴

In addition, the NPL variable underwent a standard logistic transformation. Given that the dependent variable was bounded between zero and one by construction, the logit-transformed value was used to create an unrestricted variable in the regression and thus avoided nonnormality of the error term. It also accounted for nonlinearities in the sense that larger shocks to the explanatory variables may have caused a large, nonlinear response in the transformed dependent variable. Specifically, the NPL ratio was transformed as follows:

$$\text{logit NPL} = \ln \left(\frac{\text{NPL}}{1 - \text{NPL}} \right).$$

The estimated logit NPL ratios were later appropriately retransformed to the normal measure.

B. Independent variables

Given the country's high degree of openness, a wide range of explanatory variables potentially affecting loan quality were considered. It was imperative that the economic meaning of the macroeconomic factors used be clear, with no counterintuitive relationship to the dependent variable. In view of this, the set of domestic macro variables considered initially included real GDP growth, changes in the indices of sectoral activity, and changes in employment, as well as—at the bank level—interest rates charged on loans to a particular sector and profitability measures. External variables included GDP growth of the main trade partners, exports of goods to these markets (and, separately, their prices), prices of principal commodity imports such as oil and cement, exchange rates, and the London interbank offered rate as a principal international interest rate. These variables were deemed to affect NPLs by improv-

² To account only for significant amounts, only differences were taken that were greater than 10 percent of loans in that category.

³ It could well be that part of that difference is owed to upward loan reclassifications rather than write-offs. In the absence of information on the migration of loans, one has to make the assumption that the bulk of loans disappearing from that category indeed represented write-offs.

⁴ To avoid jumps in the NPL ratio after the crisis period (i.e., in 2010:Q1), the additional write-offs were maintained in the subsequent NPLs for the first half of 2010.

ing or worsening the capacity of borrowers to service their debt with banks. As a secondary effect, swings in activity also affected credit growth and therefore the NPL ratio via the changes in the denominator.

Consequently, the specification of any of the sectoral credit risk models can be represented in the following form:

$$\ln\left(\frac{NPL_{i,t}}{1-NPL_{i,t}}\right) = \mu + \alpha \ln\left(\frac{NPL_{i,t-1}}{1-NPL_{i,t-1}}\right) + \sum_{j=1}^n \beta X_{j,t} + \varepsilon_{i,t},$$

where in addition to the logit-transformed NPL for bank i in period t , $X_{j,t}$ represents the j th explanatory variable selected for a given specification, μ is a constant, and $\varepsilon_{i,t}$ is the idiosyncratic disturbance term assumed to be independent across banks and serially uncorrelated (after the inclusion of the lagged dependent variable).

C. Estimation procedure

To account for the different business focus of banks, we specified a set of sectoral panel data models. The use of dedicated sectoral models has the advantage of first identifying the industry-specific drivers of credit risk and then weighing these exposures according to their shares in the loan portfolio of each bank. This proper identification of credit risk exposures was difficult with an economy-wide model that basically assumed that banks were equally exposed to the macro factors in that model.

The seven sectoral panel models used quarterly data for the sectoral exposures of between 31 and 41 banks⁵ during 2003:Q1–2010:Q2. As several banks entered the system toward the end of the sample period, the panel was unbalanced, but the incomplete time series, even if rather short, nonetheless may have provided valuable information and could be accommodated by the estimation procedure. In the case of bank mergers and acquisitions, the data of the acquired banks were added to the acquiring bank, which in individual cases led to jumps in the total loans of the merged bank but not necessarily in the NPL ratio.

Given the relatively short time dimension of the panel, a lagged dependent variable in a conventional estimation setup causes so-called dynamic panel bias (Nickell, 1981). Regular OLS estimation, with or without fixed effects, produces biased coefficient estimates for short time series (of only 30 periods in this case). This distortion is caused by the correlation of the lagged observations with the unobserved fixed effect when nonmodeled shocks lead to a bias in the fixed effects. Under a simple OLS estimation, the coefficient would be biased upward, because individual negative shocks that are not modeled enter the error term and skewed both the error and the dependent variable downward. Removing the fixed effects by applying the within-group estimator that regresses deviations from the respective means would not eliminate this

bias but rather would lead to an underestimation of the coefficient of the lagged dependent variable. These models, though clearly not correct, are instructive nonetheless as they establish a ceiling and a floor for the estimated coefficient.⁶

Appropriately specified dynamic panel models thus were applied to remove dynamic panel bias. Both the Difference GMM and the System GMM approach were considered (see Holtz-Eakin, Newey, and Rosen, 1988; Arellano and Bond, 1991; Arellano and Bover, 1995). These setups use past observations uncorrelated to the fixed effects as instruments in order to arrive at plausible and unbiased coefficient estimates for the lagged dependent and independent variables. Essentially, Difference GMM uses past levels as proxies for current differences, whereas System GMM uses past differences as instruments for current levels, adding a transformed equation to the original one and so allowing for the inclusion of time-invariant regressors that would disappear in Difference GMM.⁷ Although potentially the superior estimator, System GMM assumes that changes in any instrument are uncorrelated with the fixed effects and that the errors are not serially correlated. It also poses the potential problem that the instrument count doubles (as two instruments per observation are used), which in short samples tends to cause the Sargan test⁸ for the validity of instruments to break down. Mindful of these caveats, when applied to the different panels of sectoral NPLs, System GMM produced coefficient estimates for lagged NPLs that were generally within the aforementioned credible range, whereas this was not the case in any panel regression under Difference GMM.

For each panel, a System GMM specification was chosen, while considering optimal lags. For one thing, unit root tests determined the optimal number of lags as instruments to remove autocorrelation in the NPL series,⁹ thereby limiting the instrument count.¹⁰ For another, using lags of the covariates

⁶ See Roodman (2009).

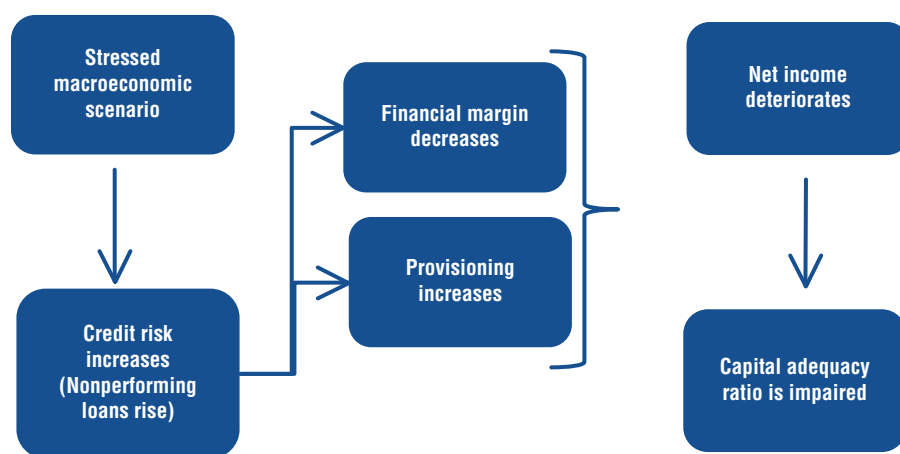
⁷ Difference GMM may perform poorly if the dependent variable is close to a random walk, because past levels provide little insight into future changes and thus become weak instruments. In this case, System GMM may be superior as it utilizes both difference and level information (see Roodman, 2009).

⁸ See Sargan (1958).

⁹ Here, the criterion was that the lag length chosen would remove autocorrelation for 80, in some cases 90, percent of banks. In the event, the lag length ranged from one in most panels to four in the mortgage panel.

¹⁰ As a general rule, the number of instruments should not exceed the number of units included in the sample if the GMM is to be applied appropriately to a panel and sensible statistics are to be obtained from the Sargan test for overidentifying restrictions. It was borderline for this panel of banks as the instrument count (for four of the seven panels using one or two lags of the lagged NPL variable between 54 and 81 instruments) slightly exceeded the number of banks in the sample (between 30 and 38 in these panels) such that validity could still be assumed. However, even in these cases, and certainly in the panel using the maximum number of 177 instruments, the Sargan test tended to produce implausibly high values—near one—so that the validity of the instruments could not be proved beyond doubt. By contrast, the Arellano-Bond AR(2) test did not find any evidence of second-order correlation in the differences of the idiosyncratic error term (which would indicate first-order serial correlation in the levels).

⁵ Offshore banks were omitted as they are prohibited from dealing with residents and have no formal relationship with the onshore banking sector.



Source: Authors.

Figure 30.2 Transmission Channel from Macroeconomic Shocks to Capital Adequacy Ratio

generally produced more significant estimates than the contemporaneous observations, which was plausible as it took time for deteriorating economic conditions to affect debtors' repayment capacity and subsequently for loans to be classified as delinquent. Clearly, the explanatory variables differed in how a shock affected loan quality, and therefore each commanded a unique lag.

For illustrative purposes, the Appendix shows the estimation results, which suggest that a decline in economic activity had the most significant impact on loan impairment:

- In five out of seven sectoral panels, a decline in real GDP growth—lagged by one period for personal loans and three periods for corporate loans—turned out to be highly significant.¹¹
- Loans to the commerce sector reacted to the economic growth of the main trading partner more than to domestic growth.
- For the loans to the primary sector, changes in the lagged activity indicator for the agricultural sector (being the largest primary sector) gave a better link to loan impairment than overall GDP growth.
- In five of the seven panels, a supplementary variable other than economic growth depicting idiosyncratic risk germane to the sector was found to have explanatory power. Specifically, in three models (mortgage loans, services, and the primary sector), the interest rate on loans to that sector charged by each bank—the only fairly robust firm-level variable—was significant at the 5 and 10 percent levels.

- Furthermore, the percentage change in exports to the main trading partner was found to be a good predictor for delinquencies in loans to the commerce sector.
- NPLs in consumer loans responded additionally to swings in the growth rate of employment.¹²

3. STRESS TEST DESIGN

This section summarizes the design of the credit risk stress test based on scenario analysis. First, it describes the transmission of shocks emanating from a deterioration in macroeconomic variables to key banking variables (i.e., NPLs, loan loss provisions, capital ratio). It then presents the rationale for different stress test scenarios, providing a brief explanation of the underlying assumptions. Finally, it explains the method used to deal with prediction bias in order to seamlessly project the resulting expected NPL ratios of each bank.

A. Transmission of shocks

The assumed economic shock increases NPLs and ultimately affects banks' CARs through increased provisioning and lower net operating margins. As shown in Figure 30.2, these two adverse effects reduce bank profits or even cause losses and consequently affect the CAR, other things being equal. In building the stress testing module, various worksheets in Excel were created to flesh out these relationships between NPLs and key explanatory variables.

¹¹ The quarterly rate of growth of the seasonally adjusted GDP series (using the Census X12 method in EViews with default settings) was used rather than the year-on-year growth rate, because it produced a better fit econometrically.

¹² Quarterly employment growth was taken as a proxy for changes in the unemployment rate, which is collected once a year but not quarterly as the panel estimation calls for.

B. Macroeconomic scenarios

In the stress testing literature, the use of extreme but plausible scenarios for the drivers of credit risk is advocated. As pointed out in BCBS (2009, p. 2), “a stress test is commonly described as the evaluation of the financial position of a bank under a severe but plausible scenario to assist in decision making within the bank.” In that sense, the selection of an appropriately grave yet realistic scenario is central to the validity of stress testing. Indeed, during the global financial crisis, most credit risk models failed to forecast the severe stress experienced by many banking systems, precisely because the assumptions about macroeconomic developments turned out to be much too benign (Alfaro and Drehmann, 2009). It obviously is difficult to gauge *ex ante* whether a certain scenario meets the “extreme but plausible” criterion, but it would be fair to say that many such stress test scenarios had not appropriately considered tail risks or hard-to-model feedback effects from a banking sector under stress onto the economy and back (see Jones, Hilbers, and Slack, 2004).

Against this background, our simulations considered three macroeconomic scenarios of varying severity: (1) a baseline scenario reflecting the expected path of macroeconomic and financial variables, mostly based on IMF projections and some additional expert judgment; (2) a scenario of severe stress depicting the most extreme historical changes observed in each of the explanatory variables; and (3) a moderate stress scenario incorporating variations centered between the baseline and the severe scenarios.

The baseline scenario used the IMF forecast for the key macroeconomic variables at the time. Assumptions about exports to the main trading partner and interest rates reflected also the (pessimistic) expert judgment of the supervisory authority. By contrast, the most extreme historical variations for each of the variables were used for the severe stress scenario. Under this scenario, the worst deterioration in both macroeconomic and financial variables during the sample period was taken (quarterly data between 2003:Q1 and 2010:Q2; for local GDP growth starting in 1996:Q1), which corresponded to the 97th percentile or about two standard deviations. Finally, the moderate stress scenario considered the midpoint in projected variations between the baseline and the severe stress scenarios.

The flexibility of the stress testing module meant that the preceding choice of scenarios did not preclude other, more extreme, scenarios. In principle, simulated scenarios not based on historical values easily could be applied by adjusting the corresponding template.

C. Projecting nonperforming loans

Using the results of the econometric estimations, the increase in the wide definition of NPLs (including special mention loans) for each bank and economic sector under the three

scenarios was projected. The methodology was divided into three steps, as follows:

$$\begin{aligned} \text{Observed Data: } & NPL_{t-30}, NPL_{t-29}, \dots, NPL_t \\ \text{Projected Data: } & NPL_{t+1}, NPL_{t+2}, \dots, NPL_{t+6} \\ \text{Projection Method: } & NPL_t = \alpha + \beta_1 NPL_{t-1} + \beta_2 X_t + \beta_3 X_t + \varepsilon_t \\ & NPL_{t+1} = \alpha + \beta_1 NPL_t + \beta_2 X_{t+1} + \beta_3 X_{t+1} + \varepsilon_{t+1} \\ & \dots \\ & NPL_{t+6} = \alpha + \beta_1 NPL_{t+5} + \beta_2 X_{t+6} + \beta_3 X_{t+6} \\ & \quad + \varepsilon_{t+6}, \end{aligned}$$

where NPL is the ratio of NPLs to total loans (wide definition), α is the constant term, β is a vector of estimated parameters, and X represents a vector of explanatory variables depicting macroeconomic and financial conditions. The predicted NPL ratios were determined as follows:

- First, the value for NPL_t (given X_t , which in some cases meant using the lagged values) and NPL_{t+1} (given projected X_{t+1}) was calculated using the econometric model.
- Next, the difference between these two predicted values was determined.
- Finally, the estimated difference was added to the observed NPL_t to obtain the projected NPL_{t+1} .

This recursive methodology was employed to minimize the deviation of the predicted values of NPL ratios in 2010:Q3 from the observed values one quarter earlier. This approach represents a practical way to deal with out-of-sample prediction error that stems from unexplained deviations of the observed current NPL ratio from the one that the model would project based on the historical patterns. Put differently, idiosyncratic shocks that are not modeled cause the model to over- or underestimate the NPL ratios, which in some cases can lead to a large difference between the two numbers. The method described is able to remove a large part of this prediction error, although abnormally high NPL ratios are projected to return to its longer-term levels over time in an error-correction fashion.

4. STRESS TEST OUTCOME

A. Projected nonperforming loans and loan loss provisions

As could be expected from the differing degree of severity of the assumed shocks, the increase in the projected NPL ratio under severe stress six quarters out is a multiple of the change in the baseline scenario (Table 30.1). Across credit types, the NPL ratios rise was found to be strongest in loans to the commercial sector (not reported).

Obtaining the projected increase in loan loss provisions required an out-of-sample forecast of the provisions-to-total-loans ratio that predicted both variables separately. Credit growth was estimated using satellite models that computed the comovement of GDP growth with credit growth for each bank in the sample. Specifically, linear univariate models

TABLE 30.1

Projected NPL Ratios (in percent)							
Scenario	Jun 2010	Sep 2010	Dec 2010	Mar 2011	Jun 2011	Sep 2011	Dec 2011
Baseline	5.4	5.3	5.5	5.6	5.6	5.8	6.0
Moderate	5.4	5.4	5.6	5.8	6.0	6.4	7.0
Severe	5.4	5.4	5.8	6.1	6.5	7.2	8.3

Source: Authors.

TABLE 30.2

Projected Provisions-to-Total-Loans Ratios (in percent)							
Type	Jun 2010	Sep 2010	Dec 2010	Mar 2011	Jun 2011	Sep 2011	Dec 2011
Baseline	0.9	0.9	0.9	1.0	1.1	1.2	1.3
Moderate	0.9	0.9	1.0	1.1	1.2	1.3	1.6
Severe	0.9	0.9	1.0	1.2	1.3	1.6	1.9

Source: Authors.

regressing past changes in total loans of each bank on GDP growth were estimated in order to calculate this variation. For banks whose total loans turned out not to be robustly related to GDP, the average historical change in total loans during the sample period was taken. In addition, the effective provisioning rates (as defined by total loan loss provisions divided by total loans) for each bank and risk category were used to forecast the increase in the provisioning flow under each of the scenarios.¹³ In order to project the share of impaired loans in each of the loan classification categories, similar satellite models were estimated.¹⁴

The difference in both the projected provisioning ratios between the severe stress scenario and baseline scenario was less pronounced than for the projected NPL ratio (Table 30.2). Also, the change in the ratio during the projection period was smaller because delinquent loans in the less affected categories (special mention to doubtful) did not require full provisioning.

B. Projected capital adequacy ratios

The final step of the stress test exercise consisted of forecasting banks' CARs by determining separately the projected capital and risk-weighted assets (RWA). RWA were calculated in line with the country's bank regulation that was based on the Ba-

sel I Accord. Therefore, each type of loan (i.e., commercial loans and consumer loans) was weighted 100 percent except mortgage loans, which carried a risk weight of 50 percent.

Thus, the RWA were calculated as follows:

$$RWA_t = RWA_{t-1} + \Delta Non_{mort,debt_t} + 50\% * \Delta Mort_{Debt_t}.$$

That is, the RWA of the current period equaled the RWA of the previous period plus the entire change in the volume of nonmortgage loans and 50 percent of the change in mortgage loans. For the CAR, the following formula was used:

$$CAR_{t+1} = (Capital_t + After-Tax Profits_{t+1} * Profit Retention Rate) / RWA_{t+1},$$

where

$$After-Tax Profits_{t+1} = Net Revenue_t - \Delta Provisioning_{t+1} - \Delta Interest Income_{t+1},$$

where

$$\Delta Interest Income_{t+1} = Implicit Interest Rate * \Delta(\Sigma Substandard, Doubtful, Loss Loans).$$

As can be seen from these formulas, the capital ratio was calculated by dividing the forecast capital by the projected amount of risk-weighted assets. The projection of capital itself was calculated by adding to current capital the projected increase in retained earnings, that is, the share of expected after-tax profits that was not distributed to shareholders. In turn, to estimate after-tax profits in $t+1$, the following components were added up:

- the observed net revenue in the previous period;
- the projected stress-induced increase in provisioning; and
- the projected decline in interest income calculated using implicit interest rates and the increase in loans that did not generate earnings. This variation in impaired loans was calculated using the share of loans in each loan category and the estimate of credit growth, as explained.

¹³ The effective provisioning rates rather than the statutory ones were taken because loans in the system were amply collateralized (total guarantees accounted for about 60 percent of loans on average), and even when applying a haircut to the value of the collateral, which would be prudent in a stress scenario, its remaining value would still be large enough to require provisioning well below that using statutory rates.

¹⁴ For three categories of impaired loans (special mention, doubtful, loss) fixed-effects panel data models with GDP growth as explanatory variable were estimated. The separate models project the share of normal, special mention, doubtful, and loss loans in total projected credit. The remaining fifth loan category, substandard loans, is then backed out as the difference between 100 percent and the share of the four other loan categories.

TABLE 30.3

Projected Capital Adequacy Ratios (in percent)							
Scenario	Jun 2010	Sep 2010	Dec 2010	Mar 2011	Jun 2011	Sep 2011	Dec 2011
Baseline	16.3	16.1	15.9	15.7	15.4	15.2	14.9
Moderate	16.3	16.1	15.9	15.7	15.4	15.1	14.8
Severe	16.3	16.1	15.9	15.7	15.4	15.1	14.7

Source: Authors.

Assuming that some banks remain profitable even under stress, the increased provisioning costs should be channeled through the income statement instead of deducting it directly from capital. Bank profits act as a first line of defense in cushioning the cost push associated with rising loan losses. Indeed, as Čihák (2007) observes, banks normally are profitable—many even under stress—and therefore it is necessary to account for such profits that will need to be exhausted before the regulatory capital is reduced.

The stress test results show that the considerable increase in NPLs under stress and provisions affects capital positions to a lesser degree. Under the baseline scenario, the CAR decreases from 16.3 to 14.9 percent (Table 30.3). This decline in the CAR is explained by (1) an increase in RWA as a result of solid credit growth (under the assumption that bank owners do not inject capital to keep it constant); (2) a decline in revenues (due to a predicted reduction in the net operating margin of institutions); and (3) a conservative profit retention rate of only 25 percent based on historical evidence and judgment by the supervisory authority. Compared with the baseline scenario, the system's CAR declines by only another 0.2 percentage points under severe stress.

There are two principal reasons for this muted response in the CARs. Banks generally remained profitable even under stress, which cushioned the provisioning shock, and RWA grew less strongly owing to slower credit growth in the stress scenarios.¹⁵ This lack of any significant impact on the CAR is not uncommon in stress testing credit risk under the Basel I Accord, as the risk weights of loans stay constant under stress and move only with projected credit growth; under Basel II, they increase and sometimes dramatically so.

5. CONCLUSION

This chapter presented key elements of a credit risk model comprising individual sectoral panel data models and the steps needed to translate the projected increase in past-due loans into a change in the CAR. Given the numerous choices available in designing the credit risk model and the needs of the particular supervisory authority, this study put forward the reasons for applying each of the key features of the model chosen. In doing so, emphasis was placed on explaining the operational details

of the stress testing module, namely, the procedures for projecting loan delinquencies, provisioning, and credit growth.

Our model applied the “NPL projection approach” whereby banks' NPL ratios as a measure of loan impairment were regressed on a set of explanatory macroeconomic and financial variables in dynamic panel models. To account for the different business models of the large number of banks in the sample, seven sectoral credit risk models were estimated. On the basis of three scenarios spanning expected to extreme-but-plausible developments, the NPLs and additional loan loss provisions first were projected using the models and then channeled through banks' projected income statements to obtain the stress-induced variation in the CARs.

The stress test outcome highlighted an important finding that is particularly relevant for banking systems under the Basel I Accord. The projected drop in the CAR under stress was much less pronounced than the stronger increase in NPLs would suggest. Indeed, the considerable deterioration in loan quality caused by the assumed macroeconomic shocks was not sufficient to severely affect banks' capital positions because of the beneficial effect of strong bank profits as the primary shock absorber and the slower growth of RWA under stress, allowing the denominator of the CAR to rise less than in normal times.

REFERENCES

- Alfaro, Rodrigo, and Mathias Drehmann, 2009, “Macro Stress Tests and Crises: What Can We Learn?” *BIS Quarterly Review*, December, pp. 29–41 (Basel: Bank for International Settlements). Available via the Internet: http://www.bis.org/publ/qrtrpdf/r_qt0912e.htm
- Arellano, Manuel, and Stephen Bond, 1991, “Some Tests of Specification for Panel Data: Monte Carlo Evidence and an Application to Employment Equations,” *Review of Economic Studies*, Vol. 58, No. 2, pp. 277–97.
- Arellano, Manuel, and Olympia Bover, 1995, “Another Look at the Instrumental Variables Estimation of Error-Components Models,” *Journal of Econometrics*, Vol. 68, No. 1, pp. 29–51.
- Åsberg, Per, and Hovick Shahnazarian, 2008, “Macroeconomic Impact on Expected Default Frequency,” Sveriges Riskbank Working Paper No. 219 (Stockholm: Sveriges Riksbank). Available via the Internet: <http://www.riksbank.se/en/Press-and-published/Published-from-the-Riksbank/Other-reports/Working-Paper-Series/2008/No-219-Macroeconomic-Impact-on-Expected-Default-Frequency>
- Bank of Japan, 2007, “The Framework for Macro Stress-Testing of Credit Risk Incorporating Transitions in Borrower Classifications,” *Financial System Report*, pp. 44–45 (Tokyo, September).

¹⁵ To run more meaningful economic stress tests under Basel I (or the standardized approach of Basel II), Schmieder, Puhr, and Hasan (2011) propose calculating “quasi-IRB risk weighted assets” based on implied credit risk parameters (PDs and loss given default, or credit losses).

- Available via the Internet: <https://www.boj.or.jp/en/research/brp/fsr/fsr07b.htm>
- Basel Committee on Banking Supervision, 2009, "Principles for Sound Stress Testing Practices and Supervision" (Basel, May). Available via the Internet: <http://www.bis.org/publ/bcbs155.htm>
- Čihák, Martin, 2007, "Introduction to Applied Stress Testing," IMF Working Paper 07/59 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=20222>
- Foglia, Antonella, 2009, "Stress Testing Credit Risk: A Survey of Authorities' Approaches," *International Journal of Central Banking*, Vol. 5, No. 2, pp. 9–45.
- Holtz-Eakin, Douglas, Whitney Newey, and Harvey S. Rosen, 1988, "Estimating Vector Autoregressions with Panel Data," *Econometrica*, Vol. 56, No. 6, pp. 1371–95.
- Jones, Matthew T., Paul Hilbers, and Graham Slack, 2004, "Stress Testing Financial Systems: What to Do When the Governor Calls," IMF Working Paper 04/127 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=17517>
- Moretti, Marina, Stephanie Stolz, and Mark Swinburne, 2008, "Stress Testing at the IMF," IMF Working Paper 08/206 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=22275>
- Nickell, Stephen J., 1981, "Biases in Dynamic Models with Fixed Effects," *Econometrica*, Vol. 49, No. 6, pp. 1417–26.
- Roodman, David M., 2009, "How to Do xtabond2: An Introduction to Difference and System GMM in Stata," *Stata Journal*, Vol. 9, No. 1, pp. 86–136.
- Sargan, John D., 1958, "The Estimation of Economic Relationships Using Instrumental Variables," *Econometrica*, Vol. 26, pp. 393–415.
- Schmieder, Christian, Claus Puhr, and Maher Hasan, 2011, "Next Generation Balance Sheet Stress Testing," IMF Working Paper 11/83 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=24798>
- Sorge, Marco, 2004, "Stress-Testing Financial Systems: An Overview of Current Methodologies," BIS Working Paper No. 165 (Basel: Bank for International Settlements). Available via the Internet: <http://www.bis.org/publ/work165.htm>

Appendix

System GMM Estimation: Impact of Macroeconomic Variables on Loan Impairment

Appendix table

Variable	Construction	Commerce	Manufacturing	Primary Sector	Services	Mortgage	Consumer
NPL ratio (–1)	0.726 (12.95)***	0.795 (14.03)***	0.605 (8.72)***	0.782 (11.93)***	0.428 (7.82)***	0.628 (6.39)***	0.764 (9.16)***
Real GDP growth (–1) or (–3)		–0.031 (–2.69)***	–0.046 (–2.03)**		–0.072 (–2.30)**	–0.083 (–2.02)**	–0.021 (–3.43)***
Real GDP growth trading partner (–5)	–0.072 (–2.06)**						
Growth of exports to trading partner (–2)		–0.006 (–2.82)***					
Employment growth (current period)							–0.036 (–3.24)***
Change in activity index/ agriculture (–4)				–0.022 (–1.97)**			
Lending rate sector (–1) or (–3)				0.048 (1.67)*	0.037 (2.07)***	0.115 (1.75)*	
Constant	–0.957 (–4.00)***	–0.525 (–2.86)***	–1.641 (–6.34)***	–1.145 (–2.95)***	–2.744 (–7.83)***	–1.898 (–3.23)***	–0.753 (–2.33)**
Observations	677	769	762	589	712	601	921
Number of banks	36	38	38	33	36	30	41
Instruments	99	81	177	54	56	56	111
Lags dependent variable	3	2	4	1	1	1	1
Lags independent variable	1	1	1	1	1	1	2
Arellano-Bond (1)	0.00	0.00	0.00	0.00	0.00	0.02	0.00
Arellano-Bond (2)	0.14	0.94	0.23	0.22	0.18	0.54	0.78
Sargan Test	1.00	1.00	1.00	0.98	0.98	0.99	1.00

Source: Authors.

Note: GMM = Generalized Method of Moments; NPL = nonperforming loan.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

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Portfolio Credit Risk and Macroeconomic Shocks: Applications to Stress Testing under Data-Restricted Environments

MIGUEL A. SEGOVIANO • PABLO PADILLA

Portfolio credit risk measurement is greatly affected by data constraints, especially when focusing on loans given to unlisted firms. Standard methodologies adopt convenient but not necessarily properly specified parametric distributions, or they simply ignore the effects of macroeconomic shocks on credit risk. Aiming to improve the measurement of portfolio credit risk, we propose the joint implementation of two new methodologies, namely the Conditional Probability of Default (CoPoD) and the Consistent Information Multivariate Density Optimizing (CIMDO) methodologies. The CoPoD methodology incorporates the effects of macroeconomic shocks into credit risk, recovering robust estimators when only short time series of loans exist. The CIMDO methodology recovers portfolio multivariate densities (on which portfolio credit risk measurement relies) with improved specifications, when only partial information about borrowers is available. Implementation is straightforward and can be very useful in stress testing exercises, as illustrated by that carried out in the International Monetary Fund's Denmark Financial Sector Assessment Program.

METHOD SUMMARY

Overview	The framework can be used to estimate extreme portfolio losses for banks (consistent with stressed macro-financial scenarios, which account for simultaneous shocks) under data-restricted environments (that would not allow proper estimation of parametric structural models for credit risk).
Application	The framework is appropriate in situations where 1. data are limited; and 2. the aim is to assess the impact of simultaneous macro-financial shocks on banks' portfolios.
Nature of approach	Structural portfolio approach, which takes into account concentration-diversification effects.
Data requirements	• Accounting data on capital adequacy ratio, provisions, and loan exposures. • Statistics on loan default frequency (or data on number of loans in default and total loans). • Macroeconomic and financial variables.
Strengths	• The framework allows measurement of portfolios' unexpected losses (UL) that account for portfolio/diversification effects, incorporating changes in such effects at different stages of the business cycle. • Moreover, UL are estimated consistently with simultaneous shocks assumed under stressed macroeconomic scenarios; thus, the framework allows for a consistent view of the evolution of banks' solvency under specific scenarios. • The framework can be estimated under restricted data environments by which stress testers are usually confronted.

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Weaknesses	It is necessary to have robust estimations of probabilities of default (PD) by loan type. If PDs are not available, it is necessary to make assumptions of this variable. However, it is possible to make robust estimations with shorter time series than what would be needed under standard parametric econometric approaches.
Tool	Proprietary estimation codes are available from the authors upon request. Contact author: M. Segoviano.

A sound and strong financial system is critical for a nation's macroeconomic stability, for the development of national savings, and for the efficient allocation of resources to investment opportunities. The strength of the financial system is dependent on the strength of its constituent financial institutions, which, in turn, depends on the individual institution's portfolio credit risk relative to its economic capital (EC). The EC of a bank is a measure of the resources that a bank should hold in order to withstand the extreme losses, for example, unexpected losses (ULs), that its portfolio could experience. Thus, a comparison of the capital base that a bank actually holds with the EC that the bank should hold—given the riskiness of its portfolio—under different macroeconomic scenarios provides a measure of the solvency of the bank. The EC requires the estimation of the bank's portfolio loss distribution (PLD) which, in turn, requires the estimation of the bank's portfolio multivariate density (PMD).¹

Unfortunately, portfolio credit risk assessment suffers from a major problem, namely, the lack of adequate data. This problem affects the measurement of portfolio credit risk at any point in time and also through time. Earlier attempts to deal with this problem unfortunately have resulted in the use of measurement methodologies that adopted convenient but not necessarily realistic statistical assumptions or that simply ignored the potential effects of macroeconomic and financial developments in an economy on the portfolio credit risk of financial institutions. Recent evidence indicates that such an approach may not be appropriate. Goodhart, Hoffmann, and Segoviano (2004), for example, show that, during the 1992 Norwegian and the 1994 Mexican crises, estimates of annual and quarterly bank portfolio ULs increased on average by 48 percent and 70 percent, respectively, from the levels recorded before the crises.²

Given the importance of portfolio credit risk, the ability to quantify it adequately in data-restricted environments becomes highly desirable. Given that such quantification is our objective, we present the joint implementation of the

Conditional Probability of Default (CoPoD) methodology (Segoviano, 2006a) and the Consistent Information Multivariate Density Optimizing (CIMDO) methodology (Segoviano, 2006b) within the framework of a stress testing exercise. The CoPoD methodology is designed to incorporate the effects of macroeconomic and financial developments into credit risk and recover robust estimators in settings of short time series, thereby aiming to improve the measurement of credit risk through time. The CIMDO methodology avoids reliance on possibly unrealistic statistical assumptions in settings of partial information, thus improving the measurement of PMDs and, therefore, of portfolio credit risk at any point in time.³ Moreover, because the information requirements necessary for the implementation of these methodologies are less stringent than the information requirements necessary for the proper calibration and implementation of standard credit risk methodologies, the implementation of the CoPoD and CIMDO methodologies is feasible and straightforward, even under the data constraints that usually bind credit risk modeling.⁴

Information constraints affecting the measurement of credit risk usually arise because either (1) credit risk modelers have arm's-length relationships with the firms,⁵ and as a consequence those modelers do not have access to the market and/or financial information that is necessary for the firms' risk assessment; or (2) there are certain variables that do not exist for the type of firms in which modelers are interested. The latter is the case when the modeling interest lies in the credit risk of loans granted to small- and medium-sized enterprises (SMEs) and unlisted firms, that is, closely held firms. Henceforth, we will refer indistinguishably to firms

¹ ULs are defined by the Basel II Accord (Basel Committee on Banking Supervision, 2001) as the 99.5 value at risk (VaR) of the PLD. The PMD describes the joint likelihood of changes in the credit risk quality of the loans that make up the loan portfolio (see also Goodhart and Segoviano, 2004).

² Similarly, Caprio and Klingebiel (1996) show that there have been numerous episodes in which bank portfolio credit losses have nearly, or completely, exhausted a banking system's capital (see also Krugman, 1979; Kaminsky and Reinhart, 1996; and Kaminsky and Schmukler, 2003).

³ In Segoviano (2006a), we show that CoPoD recovers estimators that in the setting of finite samples are superior to ordinary least squares (OLS) estimators under the mean-square-error criterion. In Segoviano (2006b), we show that CIMDO-recovered distributions outperform widely used distributions in portfolio credit risk modeling under the Probability Integral Transformation criterion.

⁴ The CoPoD allows approximation of the macroeconomic and financial estimators of probabilities of default with shorter time series than the ones required by OLS. The PDs of each type of loan making up a portfolio are the only data that are necessary to estimate the PMDs with the CIMDO methodology. Information of the dependence structure of a multivariate distribution or underlying factors indicating credit quality changes are not necessary to recover the latter.

⁵ Those modelers are usually authorities or institutions that do not have direct relationships with the analyzed firms, for example, International Monetary Fund economists trying to assess credit risk in Financial Stability Assessment Programs, financial regulators, central banks, and risk consultants.

that are subject to these information constraints as arm's-length or closely held firms. Portfolio credit risk modeling under any of these information constraints constitutes the focus of our modeling interest. This is because in many countries, SMEs and unlisted firms represent the backbone of the economy, making a significant contribution to GDP and to the sustainability of employment levels. Moreover, loans granted to these types of firms usually represent an important percentage of the assets held by commercial banks in most developed and developing economies.⁶ Therefore, improvements in the methodologies used to measure the portfolio credit risk of these types of financial assets can have important implications for individual banks' risk management and for a system's financial stability.⁷

In an attempt to improve financial stability surveillance, stress testing exercises have become a popular tool to assess the resilience of financial systems to a variety of relevant risks. Given that stress testing is one of the tools used in Financial Sector Assessment Plans (FSAPs) carried out by the IMF, the presentation in this chapter of the joint implementation of the CoPoD and CIMDO methodologies within a stress testing framework is of particular interest and importance. We illustrate this procedure by presenting the stress testing exercise that was conducted for the Denmark FSAP.

We reiterate that the probabilities of default (PDs) of loans grouped by sectors or by ratings are the only data that are necessary to implement these methodologies. Information of the dependence structure (correlations) of the loans making up the loan portfolio or assumptions of underlying factors indicating such dependencies are not necessary to implement these methodologies. The possibility of carrying out stress tests, irrespective of how loans are grouped, represents a valuable feature, which provides great modeling flexibility under data-constrained environments.

The models used in this framework are consistent with economic theory and empirical evidence. This feature allows us to ensure the economic consistency of the estimated risk measurements; it also facilitates dialogue with the relevant entities and policy recommendations. We consider these features to be extremely relevant in stress tests because, if we are interested in making explicit links between economic shocks and risks and vulnerabilities in financial systems, it is much easier to conduct discussions and negotiations and to make policy recommendations when the results of a risk model can be backed by economic theory and empirical evidence and when the relevant explanatory variables are identifiable. This approach constitutes a clear improvement over purely

statistical or mathematical models that are economically atheoretical or models that use underlying nonobservable variables as explanatory variables.

The rest of the chapter is structured as follows: Section 1 illustrates the concepts of EC, PLD, and PMD. Information constraints that usually bind credit risk modeling of closely held firms also are discussed. Such constraints have motivated the development of the CoPoD and the CIMDO methodologies, which are presented in Section 2. In Section 3, we describe the proposed procedure for stress testing, which involves a four-step process: (1) the definition of extreme but plausible macroeconomic scenarios; and (2) the incorporation of the macroeconomic shocks—implied by the previously defined scenarios—into the PDs of loans grouped by sector or ratings. For this purpose, we use the CoPoD methodology, thus obtaining CoPoD-PDs. (3) By making use of the latter and the CIMDO methodology, we recover the PMD, that is, the CIMDO-PMD. (4) The CIMDO-PMD is employed to simulate the possible losses/gains that the loan portfolio can experience. Such losses/gains are used to construct the PLD, from which the bank's EC is defined. This procedure is repeated under each of the macroeconomic scenarios defined in Step 1 in order to estimate the EC that a bank should hold—given the riskiness of its portfolio—under different macroeconomic scenarios. This course of action allows us to link macroeconomic shocks to an estimation of EC and therefore evaluate how vulnerable the banks in a banking system might be under different macroeconomic shocks; therefore, this estimation is indispensable in stress tests. In Section 4, we illustrate step by step the stress testing procedure developed under the aegis of the Danish FSAP. In Section 5, we present an analysis of the obtained results. Section 6 concludes the chapter.

1. PORTFOLIO CREDIT RISK

A. Portfolio loss distribution and economic capital

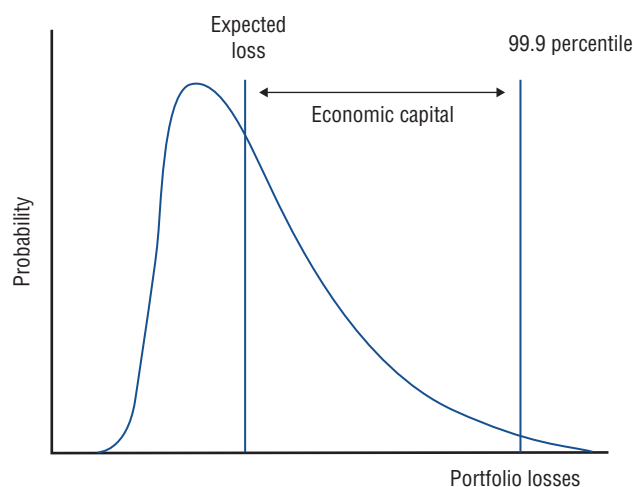
The credit risk of a bank's portfolio of loans (and any other financial asset) is summarized by its PLD, presented in Figure 31.1. This distribution shows the possible gains and losses in the value of the portfolio and the related likelihood of such events.

The possible gains and losses in the value of a loan portfolio are generated by changes in the credit risk quality of the loans that make up the portfolio. The PMD describes the joint likelihood of changes in the credit quality of the loans that make up the portfolio. Therefore, the PMD is used to simulate the possible gains and losses that the loan portfolio can experience given the type of loans in the portfolio. The gains and losses obtained from such simulation are used to construct the PLD of the portfolio.⁸ The PMD should incorporate the effects of

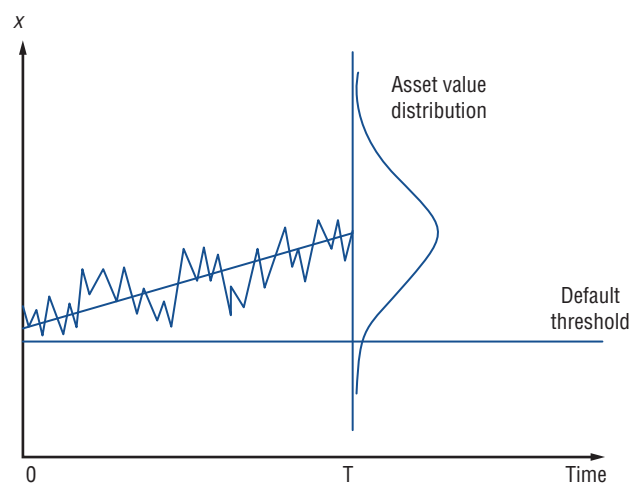
⁶ For example, Saurina and Trucharte (2004) report that, in Spain, exposures to SMEs represent, on average, 71.4 percent of total bank exposures. Similar results have been reported in the case of Germany. The Comisión Nacional Bancaria y de Valores (Mexico) reports that in Mexico, exposures to SMEs are about 85 percent.

⁷ Although we restrict our attention to loans, the proposed procedure easily can be extended to measure the portfolio credit risk of baskets, credit derivatives, or any other synthetic instrument that holds underlying assets with similar data constraints to those of closely held and arm's-length firms.

⁸ The PLD is the histogram of the simulated gains and losses of the portfolio.



Source: Authors.

Figure 31.1 Portfolio Loss Distribution of a Loan Portfolio

Source: Authors.

Figure 31.2 The Structural Approach

changing economic conditions, given that the loans' credit risk quality is affected by the economic conditions in the system.

The proper estimation of the PLD and consequently the correct modeling of the PMD are key objectives in financial risk management, because EC decisions and risk management strategies are heavily reliant on it. The structural approach is one of the most common approaches to modeling portfolio credit risk and therefore to estimating PLDs.⁹ Under the structural approach, the change in the value of the assets of a borrowing company is related to the change in its credit risk quality.¹⁰ The basic premise of this approach is that a borrowing firm's underlying asset value evolves stochastically over time and default is triggered by a drop in the firm's asset value below a threshold value, the latter being modeled as a function of the firm's financial structure. Thus, the likelihood of the firm's asset value falling below the default threshold, that is, defaulting, is summarized by the PD of the firm (Figure 31.2).¹¹

The distributions behind the stochastic processes driving the firm's asset values under the structural approach usually are assumed to be parametric.¹² For any of these parametric

assumptions, the availability of variables, or proxies, indicating the evolution of the firm's underlying asset value is of crucial importance for the adequate calibration of the distributions.¹³

B. Information restrictions binding portfolio credit risk measurement

Although, in recent years, credit risk measurement has been improving rapidly, especially for market-traded instruments, information constraints still impose severe limitations when trying to measure the portfolio credit risk of loans granted to arm's-length and closely held firms. This is because the approaches that have been developed for the measurement of portfolio credit risk, the structural and the reduced-form approaches, rely on parametric assumptions that, for proper calibration, need data that are nonexistent for these types of firms. It is not possible to observe variables indicating the evolution of the underlying asset value of such firms, nor, at the portfolio level, is it possible to observe the joint likelihood of credit risk quality changes in the loans making up a portfolio.

⁹ Widely known applications of the structural approach include the Credit Metrics framework (Gupton, Finger, and Bhatia, 1997) and the KMV framework (Crosbie and Bohn, 2003).

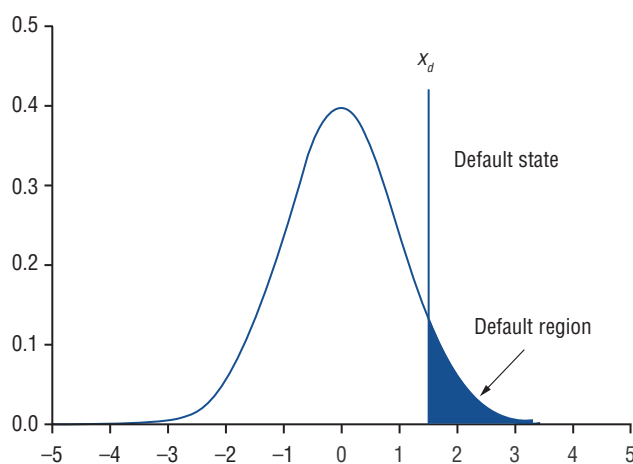
¹⁰ It is assumed that the credit risk quality of a given loan is equivalent to the credit risk quality of the firm requesting the loan. This might not always be the case, because different types of collateral, covenants, and so on, can make the firm's and the loan's credit risk quality differ.

¹¹ The generalization of this approach includes, in addition to the default state, different credit risk quality states (ratings), and thus, changes in credit risk quality also are triggered by changes in the firm's asset value with respect to threshold values.

¹² In its most basic version, it is assumed that the firms' logarithmic asset values are distributed normally (Merton, 1974). The normality assumption has been justified by its simplicity and tractability. From a theoretical point of view, its justification is given by the Central Limit Theorem. However, it has long been observed that market returns exhibit systematic deviations from normality, across virtually all markets. Empirical

returns show heavier tails than would be predicted by the normal distribution. Therefore, in most recent versions of the structural approach, alternative parametric distributions have been proposed, including t -distributions, historical simulation, and mixture models. Empirical support for modeling univariate returns with t -like tails can be found in Danielsson and de Vries (1997) and Hosking, Bonti, and Siegel (2000). Glasserman, Heidelberger, and Shahabuddin (2002) present the multivariate modeling framework. Historical simulation for portfolio credit risk is presented by Mina and Xiao (2001) and by Butler and Schachter (1998). For mixture models, see McLachlan and Basford (1987) and Zangari (1996).

¹³ The reduced-form approach (RA) represents an alternative to the structural approach. Reduced-form approach models have been used mainly to model the behavior of credit spreads. This approach treats default as a jump process with exogenous intensity (Duffie and Singleton, 1999). These models rely on variables that are available only for market-traded companies (e.g., bond yield spreads) to calibrate the intensity function for each obligor. A widely known application is the CreditRisk+ framework.



Source: Authors.

Figure 31.3 The Region of Default

For arm's-length and closely held firms, it usually is possible only to observe the default frequencies of loans grouped under different characteristics of the firms (borrowers) to whom the loans have been granted; for example, economic sector activities or credit-risk quality classes (ratings).¹⁴

If the default frequency of a given type of loan is interpreted as the PD of that type of loan and one assumes that the structural approach holds, the PD provides information on one region of the asset value distribution of the firm to whom the loan has been granted; we call this the *region of default*.

The region of default of a specific type of firm can be appreciated in Figure 31.2 as the region where its asset value (x) is lower than its default threshold (X_d). Accordingly, if the distribution of asset values in Figure 31.2 is turned around, the region of default for the firm is defined in Figure 31.3 as the region where $x \geq X_d$ under an assumed distribution.

Thus, the PD of each type of loan in a portfolio represents partial information of the (marginal) asset value distribution of each type of firm to whom the loans of a portfolio have been granted. Using partial information of these marginal distributions, the modelers' aim is to recover the PMD that describes the joint likelihood of asset value changes of the firms to whom the loans of a portfolio have been granted and, equivalently, to recover the PMD that describes the joint likelihood of changes in the credit quality of the loans that make up a portfolio.

Undertaking such a task is complicated by the fact that the problem is underidentified. There are infinitely many solutions to such a density, and a basis for selecting a particular solution is needed. The traditional route taken is to impose convenient distributional assumptions, representing information that is not available, so that the problem can be analyzed with familiar classical tools. For the specific case of credit

risk modeling of arm's-length and closely held firms, parametric assumptions may not provide an appropriate framework. The lack of adequate data may make it impossible to calibrate the assumed parametric distributions, and these distributions may not be consistent with the analyzed assets' data-generating processes. If these shortcomings are manifested, erroneous statistical inferences may result and the economic interpretations would be incorrect, undermining banks' profitability or even threatening their sustainability.

An additional data limitation is that the time series of PDs are very short, in both developed and developing countries. This lack of adequate data through time represents a significant problem for any risk managers who attempt to evaluate the impact of specific macroeconomic and financial events on the credit risk of their respective portfolios. In many models, the number of observations in the time series of PDs barely exceeds the number of parameters to be estimated. In such situations, although the parameters are determinate, they possess large standard errors, placing a limit on the level of analysis of credit risk through time that can be undertaken by financial institutions.

2. PROPOSAL TO IMPROVE PORTFOLIO CREDIT RISK MEASUREMENT

In order to estimate PMDs that incorporate the effects of changing economic conditions, a procedure based on the joint implementation of the CoPoD methodology and the CIMDO methodology has been developed. These methodologies are easily implementable in data-constrained environments and improve the measurement of portfolio credit risk. The CoPoD methodology is developed in Segoviano (2006a) and is based on the principle of maximum entropy (Jaynes, 1957). The CIMDO methodology is developed in Segoviano (2006b) and is based on applications of the minimum cross-entropy approach (Kullback, 1959).¹⁵

The CoPoD methodology allows the modeling of PDs as functions of macroeconomic and financial variables and produces efficient (smaller variance) estimators with few PD observations. Once the PDs are modeled, the CIMDO methodology allows inferring PMDs from the estimated PDs. In comparison with standard portfolio credit risk models, the CIMDO methodology relaxes the reliance on potentially unrealistic statistical assumptions in the measurement of portfolio credit risk. As discussed earlier, this is particularly important for portfolios comprising loans granted to SMEs and nonlisted firms. Rather than applying convenient parametric distributional assumptions, representing the information that is not available, the CIMDO methodology defines a selection criterion for choosing one of the infinite number of density specifications that are possible under the underidentified credit risk problem. The CIMDO methodology

¹⁴ For these types of loans, sometimes it also is possible to obtain frequencies of credit risk quality changes. However, for an outsider to the firm (a bank or a regulator) the quality of these statistics at a given point in time is difficult to verify. Therefore, we chose to develop our models assuming the lowest level of data availability.

¹⁵ A summary of the maximum and minimum cross-entropy approaches is presented in the Appendix.

does not need stock prices or market or financial variables of individual firms to model the credit risk. Nor is information on the loans' dependence structure (correlations) needed; it requires only the PDs of each type of loan making up a portfolio to recover its PMD.

A. The Conditional Probability of Default methodology

The CoPoD methodology models the empirical frequencies of loan defaults, PDs, as functions of (identifiable) macroeconomic and financial variables. The detailed development of this methodology is presented in Segoviano (2006a). A summary of the CoPoD methodology is presented in Box 31.1.

PDs of loans grouped by economic sector (e.g., manufacturing, agriculture, fishing, and tourism) or by risk rating—for example, AAA and AA—are usually the only statistic available for modeling the credit risk of SMEs and of unlisted and arm's-length firms. The process of modeling PDs represents a challenging task, because the time series of PDs usually contain few observations, thus making ordinary least squares (OLS) estimation imprecise or unfeasible.¹⁶

Short time series of PDs are the norm in most instances. This is because the PDs of unlisted firms or arm's-length firms are statistics that are usually recorded with low frequencies (most of the time, annually, quarterly, or monthly). Moreover, PDs have been recorded from the middle/end of the 1990s in most cases. Although there are exceptions to this norm, even if longer time series exist, it might not be possible to use them if such series belong to economies that have gone through economic structural changes.^{17,18}

The CoPoD methodology improves the measurement of the impact of macroeconomic variables on PDs and consequently the measurement of loans' credit risk through time, thereby making a twofold contribution. First, econometrically, it recovers estimators that show greater robustness than OLS estimators in finite sample settings under the mean squared error (MSE) criterion. Second, economically, on the basis of economic theory and empirical evidence, the CoPoD methodology incorporates a procedure to select a relevant set of (identifiable) macroeconomic explanatory variables that have an impact on the PDs.

CoPoD: Intuition behind Efficiency Improvements in Small Sample Settings

When dealing with time series of PDs that contain few observations, modeling PDs as functions of macroeconomic

variables represents a challenging task, because OLS estimation becomes imprecise or unfeasible. We have claimed that in these circumstances the CoPoD methodology recovers estimators that show greater robustness than OLS estimators under the MSE criterion. Theoretical proofs, as well as a Monte Carlo experiment, that demonstrate the efficiency gains achieved by CoPoD estimators are presented in Segoviano (2006a). In this section, we aim to provide an intuitive explanation of this result.

It is well known that for the general linear model, maximum likelihood (ML) estimators are equivalent to OLS estimators (under the assumption that the Y_i are normally and independently distributed). When ML estimation is computed, first, implicitly, equal weights to each sample observation are assigned. Second, the normal distribution is assumed to be the parametric distributional form that represents the stochastic behavior of the Y_i . Then, the unknown parameters are solved by maximizing the probability of observing the Y_i , for example, by maximizing the Y_i likelihood function. In order to provide structure to these ideas, we present the analysis developed in Box 31.2.

Equivalently, given an ex ante assumed (fixed) parametric distribution, we adjust its parameters (with information provided by the sample) in such a way that the distribution is maximized. Alternatively, the CoPoD criterion would seem more appealing because (1) it weights sample observations using different estimates of the probabilities of the Y_i , which are based on observed data information rather than on fixed uniform weights; and (2) the distribution of probability weights that is employed is inferred from data information rather than ex ante fixed. Therefore, assuming that the information contained in the moment-consistency constraints is valid, one would anticipate the CoPoD estimators to be more efficient because they are calculated with respect to probability weights inferred from a more efficiently used information base.

CoPoD: Selection of Explanatory Variables

The selection of an initial set of macroeconomic variables that are analyzed to determine the significant explanatory variables that define the PDs' model specifications is based on an economic hypothesis presented as an integral part of the CoPoD methodology. The initial set of macroeconomic variables is consistent with theoretical arguments and empirical evidence.¹⁹

The economic hypothesis that is adopted in the CoPoD methodology implies that fluctuations in key macroeconomic and financial variables have the potential to generate endogenous cycles in credit, economic activity, and asset prices. These cycles, in turn, appear to involve and indeed may amplify financial imbalances, which can place great stress on the financial system. During the upturn of the cycle, banks may increase their lending excessively, in some part, as prices of assets held as collateral increase, and the state of confidence in the system

¹⁶ It is well known that in settings of short time series, OLS estimators possess large variances and can be very sensitive to small changes in the data. These characteristics represent important problems for risk modelers when trying to evaluate the impact of economic shocks in PDs.

¹⁷ Common proxies for PoDs are nonperforming loan (NPL) ratios. Short time series of NPLs are usually common; thus, the CoPoD can be applied exactly in the same way to model NPLs.

¹⁸ See also Kullback and Leibler (1951).

¹⁹ See Minsky (1982), Obstfeld and Rogoff (1995), and McKinnon and Pill (1996).

Box 31.1. The Conditional Probability of Default Methodology

In this box, a summary of the CoPoD methodology econometric setting is presented. Readers interested in the detailed development may refer to Segoviano (2006a).

CoPoD: Rationale

Under the Merton (1974) model, a borrower is assumed to default at time $T > t$, if, at that time, the value of his or her assets, S_t^i , fall below a prespecified barrier, a_t^i , which is modeled as a function of the borrower's leverage structure. Therefore, default can be characterized by $s(T) \leq a_t^i$. Thus, at time t , the PD at time T is given by

$$PD_t = \Phi(a_t^i), \quad (31.1)$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function *cdf*.

If we group the PDs of loans classified under the sectoral activity of the borrower to whom the loans are granted or their risk rating category in a T -dimensional vector, $\mathbf{PD}$, each observation in the vector of frequencies of **loan defaults PD** represents the empirical measure of **PD** for the i th type of borrower at each point in time t . Given that each observation in the vector of PDs is restricted to lie between 0 and 1, we make the following transformation: $\mathbf{a}^i = \Phi^{-1}(\mathbf{PD})$, where $\Phi(\cdot)$ is the inverse standard normal *cdf*.

We are interested in modeling the PDs as functions of identifiable macroeconomic and financial developments X ; therefore, we can formalize the problem as

$$\mathbf{a} = X\beta + \mathbf{e}, \quad (31.2)$$

where $\mathbf{a}$ is a T -dimensional vector of noisy observations (transformation of the PDs), X is a known $(T \times K)$ matrix of macroeconomic and financial series and β is a K -dimensional vector of unknown coefficients that we are interested in estimating. Consequently, we know X , observe $\mathbf{a}$, and wish to determine the unknown and unobservable parameter vector β .

CoPoD: Formulation

As detailed in Segoviano (2006a), we reformulate the model set in equation (31.2) as follows: We express our uncertainty about β and $\mathbf{e}$ by viewing them as random variables on supports $\mathbf{Z}$ and $\mathbf{V}$, respectively, where $\mathbf{p}(\lambda)$ and $\mathbf{w}(\lambda)$ represent families of probability density functions for these random variables. Accordingly, the model set in equation (31.2) can be written in terms of random variables, and the estimation problem is to recover the probability distributions for β and $\mathbf{e}$; for example, $\mathbf{p}(\lambda)$ and $\mathbf{w}(\lambda)$.

As a result, we treat each β_k as a discrete random variable with a compact support $2 \leq M < \infty$ possible outcomes. For example if $M = 2$, z_{k1} and z_{kM} represent the plausible extreme values (upper and lower bounds) of β_k . Therefore, we can express as β_k a convex combination of these two points. That is, there exists $p_k \in [0, 1]$ such that, for $M = 2$, $\beta_k = p_k z_{k1} + (1 - p_k) z_{kM}$. We can do this for each element of β . However, $M \geq 2$ may be used to express the parameters in a more general fashion. In this case, we also restrict the weights p_k to be strictly positive and to sum to 1 for each k . Therefore, we are in a position to rewrite each β_k as a convex combination of M points. These convex combinations may be expressed in matrix form as $\beta = \mathbf{Zp}$. We also can reformulate the vector of disturbances, $\mathbf{e}$. Accordingly, we represent our uncertainty about the outcome of the error process by treating each e_t as a finite and discrete random variable with $2 \leq J < \infty$ possible outcomes.

We also suppose that there exists a set of error bounds, v_{it} and v_{jt} , for each e_t . For example, for $J = 2$, each disturbance may be written as $e_t = w_t v_{i2} + (1 - w_t) v_{j2}$ for $w_t \in [0, 1]$. However, $J \geq 2$ may be used to express the parameters in a more general fashion. As before, we restrict the weights so as to be strictly positive and to sum to 1 for each t . The T unknown disturbances may be written in matrix form as $\mathbf{e} = \mathbf{Vw}$. Using the reparameterized unknowns, $\beta = \mathbf{Zp}$ and $\mathbf{e} = \mathbf{Vw}$, we can rewrite the model presented in equation (31.2) as

$$\mathbf{a} = \mathbf{XZp} + \mathbf{Vw}. \quad (31.3)$$

This model incorporates the effects of macroeconomic and financial developments into the PDs at each point in time. It also accounts for possible noise in the data. The information represented by this set of equations is incorporated into the entropy decision rule that is used to recover the distributions $\mathbf{p}$ and $\mathbf{w}$. Such information is formulated as a set of moment-consistency constraints that has to be fulfilled by the distributions $\mathbf{p}$ and $\mathbf{w}$. The objective of the entropy decision rule is to choose the set of relative frequencies, $\mathbf{p}$ and $\mathbf{w}$, that could have been generated in the greatest number of ways consistent with what is known—for example, the moment-consistency constraints. Thus, following Segoviano (2006a), the probability vectors $\mathbf{p}$ and $\mathbf{w}$ are recovered by maximizing the functional:

$$L = - \left[\sum_{k=1}^K \sum_{m=1}^M p_m^k \ln p_m^k \right] - \left[\sum_{t=1}^T \sum_{j=1}^J w_j^t \ln w_j^t \right] + \sum_{t=1}^T \lambda_t \left[a_t - \sum_{k=1}^K \sum_{m=1}^M x_{tk} z_{km} p_m^k - \sum_{j=1}^J v_{jt} w_j^t \right] + \sum_{k=1}^K \theta_k \left[1 - \sum_{m=1}^M p_m^k \right] + \sum_{t=1}^T \tau_t \left[1 - \sum_{j=1}^J w_j^t \right] \quad (31.4)$$

where $\gamma \in^{-T}$ represents the Lagrange multipliers corresponding to the moment-consistency constraints that embed the information provided by the data and $\theta \in^{-K}$ and $\tau \in^{-T}$ and represent the Lagrange multipliers corresponding to the additivity restrictions that are imposed on $\mathbf{p}$ and $\mathbf{w}$, because those distributions represent densities; thus, they must add to one. The entropy solution is given by:

$$\hat{p}_m^k(\hat{\lambda}) = \frac{\exp \left[- \sum_{t=1}^T \hat{\lambda}_t x_{tk} z_{km}^k \right]}{\sum_{m=1}^M \left[\exp \left[- \sum_{t=1}^T \hat{\lambda}_t x_{tk} z_{tm}^k \right] \right]} \quad (31.5)$$

(continued)

Box 31.1. continued

and by

$$\hat{w}_j^t(\hat{\lambda}) = \frac{\exp[-\hat{\lambda}_t v_j^t]}{\sum_{j=1}^J \exp[-\hat{\lambda}_t v_j^t]}. \quad (31.6)$$

Once we recover the optimal probability vector $\hat{p}$, we are in a position to form point estimates of the unknown parameter vector $\hat{\beta}$ as follows: $\hat{\beta} = Z\hat{p}$. Equally, the optimal probability vector $\hat{w}$ may also be used to form point estimates of the unknown disturbance $\mathbf{e} = \mathbf{V}\hat{w}$.

Thus, the CoPoD methodology provides a rationale for choosing a particular solution vector $\mathbf{p}$, which is the density that could have been generated in the greatest number of ways consistent with what is known (without imposing arbitrary distributional assumptions). Because we do not want to assert more of the distribution $\mathbf{p}$ than is known, we choose the $\mathbf{p}$ that is closest to the uniform distribution and also consistent with the available information, expressed as constraints in equation (31.4).

Box 31.2. Conditional Probability of Default Efficiency

This box presents a heuristic explanation of the robustness of the CoPoD estimators in small sample settings. However, this is intended to be an intuitive exposition. The large and small sample properties of the CoPoD estimators and a Monte Carlo experiment that proves their robustness are presented in Segoviano (2006a).

In the Appendix, we present the Kullback (1959) minimum cross-entropy distribution (MXED) approach to recover densities. The MXED objective is to minimize the cross-entropy distance between the posterior $\mathbf{p}$ distribution and the prior $\mathbf{q}$ distribution. The cross-entropy objective function is defined as $C[p_k, q_k] = \sum_k p_k \ln \left[\frac{p_k}{q_k} \right]$. Note that the MXED can be interpreted as the expectation of a log-likelihood ratio.

This interpretation is useful to illustrate the efficiency gains provided by the CoPoD estimation framework. Let θ_1 and θ_2 be two alternative values of the parameter vector defining two alternative discrete probability distribution characterizations $f(y; \theta_2)$ and $f(y; \theta_1)$ for the distribution of $\mathbf{y}$. The MXED information in $f(y; \theta_1)$ relative to $f(y; \theta_2)$ is given by

$$C[f(y; \theta_1), f(y; \theta_2)] = \sum_y \left[\ln \left[\frac{f(y; \theta_1)}{f(y; \theta_2)} \right] \right] f(y; \theta_1) = E_{\theta_1} \left[\ln \left[\frac{L(y; \theta_1)}{L(y; \theta_2)} \right] \right], \quad (31.7)$$

where E_{θ_1} denotes an expectation taken with respect to the distribution $f(y; \theta_1)$. It is clear that the MXED functional in equation (31.7) can be interpreted as a mean log-likelihood ratio, where averaging is done using weights provided by the distribution $f(y; \theta_1)$.

If $\left[\frac{f(y; \theta_1)}{f(y; \theta_2)} \right] = \ln \left[\frac{L(y; \theta_1)}{L(y; \theta_2)} \right]$ is interpreted as the information in the data outcome $\mathbf{y}$ in favor of the hypothesis θ_1 relative to θ_2 , then equation (31.7) can be interpreted as the *mean information* provided by sample data in favor of θ_1 relative to θ_2 under the assumption that $f(y; \theta_2)$ is actually true.

Now, consider the MXED criterion when the distribution $f(y; \theta_1)$ is some discrete distribution $p(\mathbf{y})$ having discrete and finite support $\mathbf{y} \in \Gamma$ and the distribution $f(y; \theta_2)$ is $n^{-1} \forall \mathbf{y} \in \Gamma$. In this context, the MXED is given by

$$C[n^{-1}, p] = \sum_{i=1}^n \frac{1}{n} \left[\ln \left[\frac{1}{n} \frac{1}{p(y_i)} \right] \right] \quad (31.8)$$

$$= -n^{-1} \sum_{i=1}^n \ln(p_i) - \ln(n) \quad (31.9)$$

where $\mathbf{1}$ represents a $(n \times 1)$ vector of ones so that $n^{-1} \mathbf{1} = [n^{-1}, \dots, n^{-1}]'$ denotes a discrete uniform distribution, n is the number of elements in Γ , $\mathbf{p} = (p_1, p_2, \dots, p_n)$, and $p_i \equiv p(y_i)$. It is evident that minimizing equation (31.8) is equivalent to maximizing the scaled (by n^{-1}) function $\ln(p_i)$ in equation (31.9), equivalently, maximizing the likelihood function in equation (31.9).

(continued)

Box 31.2. continued

If we reverse the roles of the distributions, the MXED function is defined as

$$\begin{aligned} C[p, n^{-1}] &= \sum_{i=1}^n p_i \left[\ln \left[\frac{p_i}{\frac{1}{n}} \right] \right] \\ &= \sum_{i=1}^n p_i \ln(p_i) + \ln(n), \end{aligned} \quad (31.10)$$

where $\mathbf{p} = (p_1, p_2, \dots, p_n)'$ denotes probability weights on the sample observations. Thus, minimizing equation (31.10) is equivalent to maximizing $-\sum_{i=1}^n p_i \ln[p_i]$, which is precisely Shannon's (1948) entropy measure, defined in equation (31.17). Thus, the difference between minimizing equation (31.8) and equation (31.10) lies in the calibration of the mean information provided by sample data.

In the former case, $C[n^{-1}, p]$, averaging is performed with respect to predata weights n^{-1} for each Y_i . Then, assuming the uniform distribution to be true, the mean information is interpreted in the context of the expected information in favor of the uniform distribution relative to the postdata estimated probability distribution $\mathbf{p}$. Equivalently, we choose the distribution $p(\mathbf{y})$ that minimizes mean information in favor of the empirical distribution weights, n^{-1} (e.g., assuming that weights n^{-1} are actually true). This interpretation is consistent with the idea of drawing the distribution $p(\mathbf{y})$ as closely as possible to the uniform distribution, which will concomitantly minimize the expected log-likelihood ratio. Note that this is what we implicitly do when computing ML estimation. From this generic representation, we can see that when we perform standard maximum likelihood estimation, we implicitly give equal weights n^{-1} to each Y_i . Then, given an ex ante fixed (normal) distribution, we choose the distribution's parameters in a way that the probability of observing the given Y_i is as high as possible; that is, we maximize the likelihood function.

In the latter case, $C[p, n^{-1}, 1]$, that is, the CoPoD approach, averaging is performed with respect to the postdata probability distribution $\mathbf{p}$, where the $\mathbf{p}$ estimates of the probabilities of the Y_i are based on observed data information rather than on fixed predata uniform weights. Assuming that the information contained in the moment-consistency constraints is valid, one would anticipate that the expected log-likelihood ratio is estimated more efficiently in the CoPoD approach because it is calculated with respect to probability weights inferred from data information. Then, assuming the $\mathbf{p}$ distribution to be true, the mean information is interpreted in the context of the expected information in favor of the $\mathbf{p}$ distribution relative to the predata weights, n^{-1} for each Y_i .

is positive. It is also during this stage that the seeds of financial imbalances are sown and the financial vulnerability (risk) of the economy increases, as do the levels of leverage in the banking system.

This hypothesis is consistent with theoretical models with credit constraints and a financial accelerator²⁰ and with theories that emphasize the importance of the incentive structures created under financial liberalization that can exacerbate the intensity of such cycles. The relevant economic theory includes second-generation models in the currency crisis literature, which stress the role of self-fulfilling expectations and herding behavior in determining the intensity of the cycles; models that point out that under financial liberalization the scope for risk taking is increased; and theories that call attention to the creation of perverse mechanisms, such as moral hazard lending and carry trades, that under financial liberalization can exacerbate banking and currency crises.²¹ This hypothesis also is consistent with empirical evidence.²² Thus,

significant information on systemic vulnerabilities, which have the potential to increase financial risk and thus affect the empirical frequencies of loan defaults experienced by companies in the economy, may be obtained from an analysis of key macroeconomic and financial variables that exhibit regularities when systemic vulnerabilities and macroeconomic imbalances are being created and before they unwind. Equivalently, this hypothesis implies that frequencies of default can be explained partially by lagged values of relevant macroeconomic and financial explanatory variables.²³ The initial set of variables that are analyzed is presented in Table 31.1.

B. The Consistent Information Multivariate Density Optimizing methodology

The CIMDO methodology allows the modeler to recover the multivariate distributions that describe the joint likelihood of credit risk quality changes in the loans making up a portfolio. The CIMDO methodology recovers these multivariate distributions without imposing unrealistic parametric assumptions and by using only partial information, characterized by the realized frequencies of default of each type of loan making up the portfolio. Equivalently, multivariate

²⁰ See Kiyotaki and Moore (1997).

²¹ See Obstfeld (1995), Calvo (1998), and Flood and Marion (1999) for the first. See Allen and Gale (2001) for the second and Dooley (1997) for the third.

²² There is a growing literature documenting this empirical evidence. See Heiskanen (1993), Frankel and Rose (1996), Mishkin (1997), Demirgüç-Kunt and Detragiache (1998), Kaminsky, Lizondo, and Reinhart (1998), Kaminsky and Reinhart (1999), Goldstein, Kaminsky, and Reinhart (2000), Eichengreen and Areta (2000), Reinhart and Tokatlidis (2001), Borio and Lowe (2002), and Goodhart, Hofmann, and Segoviano (2004).

²³ Further evidence is presented in Segoviano (2006b). In the latter, the model specifications that were consistent with economic theory, empirical evidence and that showed the best goodness of fit, contained explanatory variables that were lagged values of macroeconomic and financial variables.

TABLE 31.1

Conditional Probability of Default: Initial Set of Variables

Code	Variable	Source
REPROPRI	Residential property prices	National sources as per detailed documentation and BIS calculations based on national data
SHAPRI	Share price index	IMF International Financial Statistics
AGGASPRI	Aggregate asset price index	National sources as per detailed documentation and BIS calculations based on national data
NEER	Nominal foreign exchange	IMF International Financial Statistics
M2	M2 monetary aggregate	IMF International Financial Statistics
REER	Real foreign exchange	IMF International Financial Statistics
RESER	International reserves	IMF International Financial Statistics
REINT	Real interest rates	IMF International Financial Statistics
SHINT	Short interest rates	IMF International Financial Statistics
LINT	Long interest rates	IMF International Financial Statistics
MTG	Mortgage bond interest rates	National sources
REGDP	Real GDP	IMF International Financial Statistics and OECD
RECRE	Real aggregate credit	IMF International Financial Statistics and OECD
CON	Consumption aggregate	IMF International Financial Statistics
CA	Current account balance	IMF International Financial Statistics
FDI	Foreign direct investment	IMF International Financial Statistics
REINVE	Investment aggregate	IMF International Financial Statistics
RECON	Real consumption	Authors' calculations based on national data
CREOVGDP	Ratio of aggregate credit in the financial system to GDP	Authors' calculations based on national data
INVOVGDP	Ratio of investment to GDP	Authors' calculations based on national data
CONOVGDP	Ratio of consumption to GDP	Authors' calculations based on national data
RECUAOVREINV	Ratio of real current account to real investment	Authors' calculations based on national data
M2OVRES	Ratio of M2 to international reserves	Authors' calculations based on national data
LOMISH	Difference of long minus Short interest rates	Authors' calculations based on national data
INREVO	Realized volatility of money market rates	Authors' calculations based on national data

Source: Authors.

Note: BIS = Bank for International Settlements; OECD = Organization for Economic Cooperation and Development.

distributions can be inferred even if only partial information on the marginal distributions describing the credit quality changes of each type of loan in a portfolio is available.²⁴ Consequently, information on the dependence structure of the loans in a portfolio is not needed in order to recover their multivariate distribution. However, if information is available, it can be incorporated. A summary of the CIMDO methodology is presented in Box 31.3; however, its detailed development is presented in Segoviano (2006b). In this study, it is also shown that the multivariate distributions recovered with the CIMDO methodology outperform the most common parametric multivariate distributions used for the modeling of portfolio credit risk (i.e., the standard and conditional normal distributions, the t -distribution, and the mixture of normal distributions) under the Probability Integral Transformation (PIT) criterion.²⁵

The implementation of the CIMDO methodology is feasible and straightforward, even for arm's-length and closely held firms, because it depends only on PD or NPL statistics, which are generally the only readily observable variables for such loans.

CIMDO: Intuition of Improvements in Density Specification

Intuitively, the optimization procedure embedded in the CIMDO methodology updates the shape of the PMD (at each period of time) such that the posterior multivariate distribution is consistent with the empirically observed probabilities of default of each type of borrower to whom the loans in the portfolio have been given; that is, the posterior multivariate distribution fulfills a set of restrictions imposed on its marginal distributions. Therefore, the shape of the posterior distribution is varied according to the empirical frequencies of default of each of the borrowers that at each period of time change because of numerous factors affecting the underlying asset value of the firm—for example, macroeconomic shocks and variations in the business cycle. When recovering multivariate distributions through the CIMDO approach, the available partial information is used in a more efficient manner than when parametric distributions, under data-restricted environments, are calibrated through the use of a fixed set of parameters. Detailed expositions of the PIT criterion and Monte Carlo studies used to evaluate improvements in the specifications of CIMDO densities are presented in Segoviano (2006b). However, in Box 31.4, we provide an intuitive explanation of the reasons behind such improvements.

²⁴ See Golan and Judge (1992).

²⁵ This criterion for multivariate density's evaluation is presented in Diebold, Hahn, and Tay (1999).

Box 31.3. The Consistent Information Multivariate Density Optimizing Methodology

The detailed formulation of CIMDO is presented in Segoviano (2006b). CIMDO is based on the Kullback (1959) minimum cross-entropy approach (MXED) presented in the Appendix. For illustration purposes, we focus on a portfolio containing loans given to two different classes of borrowers, whose logarithmic returns are characterized by the random variables x and y , where $x, y \in \mathbb{I}$ s.t. $i = 1, \dots, M$. Therefore, the objective function can now be defined as $C[p, q] = \int \int p(x, y) \ln \left[\frac{p(x, y)}{q(x, y)} \right] dx dy$, where $q(x, y)$ the prior distribution and $p(x, y)$ the posterior distribution $\in \mathbb{R}^2$.

It is important to point out that the initial hypothesis is taken in accordance with economic intuition (default is triggered by a drop in the firm's asset value below a threshold value) and with theoretical models (structural approach) but not necessarily with empirical observations. Thus, the information provided by the frequencies of default of each type of loan making up the portfolio is of prime importance for the recovery of the posterior distribution. In order to incorporate this information into the recovered posterior density, we formulate moment-consistency constraint equations of the form

$$\int \int p(x, y) \chi_{(x_d^x, \infty)} dx dy = PoD_i^x, \quad \int \int p(x, y) \chi_{(x_d^y, \infty)} dy dx = PoD_i^y, \quad (31.11)$$

where $p(x, y)$ is the posterior multivariate distribution that represents the unknown to be solved. PoD_i^x and PoD_i^y are the empirically observed probabilities of default (PDs) for each borrower in the portfolio, and $\chi_{(x_d^x, \infty)}, \chi_{(x_d^y, \infty)}$ are the indicating functions defined with the default thresholds (Figure 31.3) for each borrower in the portfolio. In order to ensure that $p(x, y)$ represents a valid density, the conditions that $p(x, y) \geq 0$ and the probability additivity constraint, $\int \int p(x, y) dx dy = 1$, also need to be satisfied. Imposing these constraints on the optimization problem guarantees that the posterior multivariate distribution contains marginal densities that in the region of default, as defined in Figure 31.3, are equalized to each of the borrowers' empirically observed probabilities of default. The CIMDO density is recovered by minimizing the functional:

$$\begin{aligned} L[p, q] = & \int \int p(x, y) \ln p(x, y) dx dy - \int \int p(x, y) \ln q(x, y) dx dy \\ & + \lambda_1 \left[\int \int p(x, y) \chi_{(x_d^x, \infty)} dx dy - PoD_i^x \right] + \lambda_2 \left[\int \int p(x, y) \chi_{(x_d^y, \infty)} dy dx - PoD_i^y \right] + \mu \left[\int \int p(x, y) dx dy - 1 \right], \end{aligned} \quad (31.12)$$

where λ_1 and λ_2 represent the Lagrange multipliers of the moment-consistency constraints and the Lagrange multiplier of the probability additivity constraint. By using the calculus of variations, the optimization procedure is performed. The optimal solution is represented by the following posterior multivariate density as

$$\hat{p}(x, y) = q(x, y) \exp \{ -[1 + \hat{\mu} + (\hat{\lambda}_1 \chi_{(x_d^x, \infty)}) + (\hat{\lambda}_2 \chi_{(x_d^y, \infty)})] \}. \quad (31.13)$$

Box 31.4. Consistent Information Multivariate Density Optimizing Methodology: Improvements in Density Specification

Analyzing the functional defined in equation (31.11), it is clear that the consistent multivariate density optimizing methodology generates the distribution that minimizes the probabilistic divergence, that is, "entropy distance," from the prior distribution and that is consistent with the information embedded in the moment-consistency constraints. Thus, out of all the distributions satisfying the moment-consistency constraints, the proposed procedure provides a rationale by which we select the posterior that is closest to the prior, thereby solving the underidentified problem that was faced when trying to determine the unknown multivariate distribution from the partial information provided by the PDs.

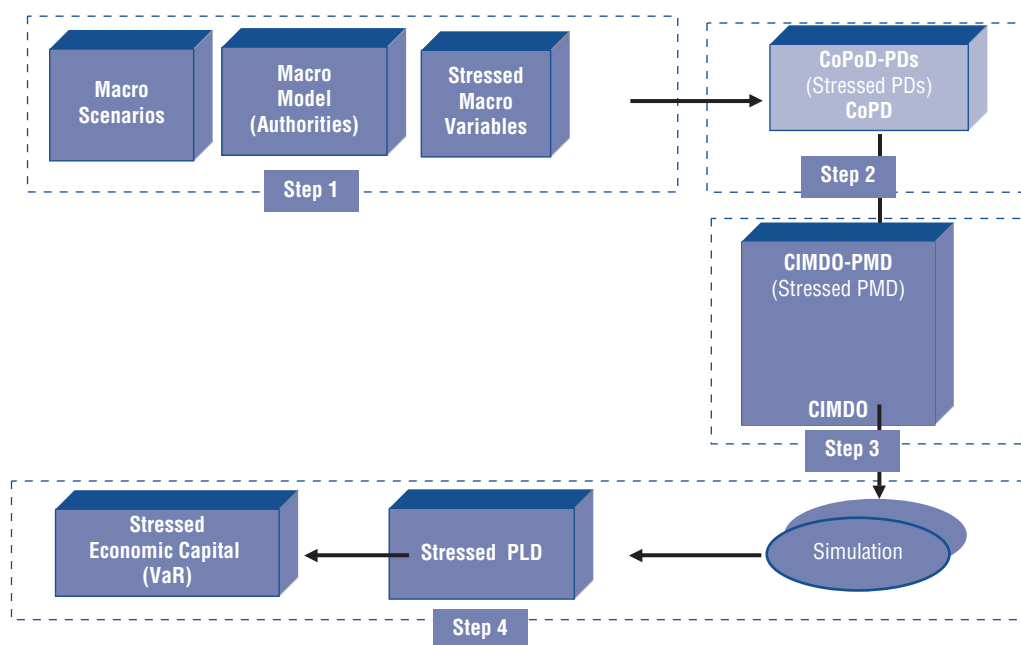
The intuition behind this optimization procedure can be understood by analyzing the development of the cross-entropy formulation (MXED) in the Appendix. The MXED objective function presented in equation (31.20) is an extension of the pure-entropy function in equation (31.17), which is a monotonic transformation of the multiplicity factor shown in equation (31.16). The multiplicity factor indicates the number of ways that a particular set of frequencies can be realized (i.e., this set of frequencies corresponds to the frequencies of occurrence attached to specific values of a random variable; Shannon, 1948).

Therefore, when maximizing the entropy function subject to a set of constraints, we obtain the set of frequencies (frequency distribution) that can be realized in the greatest number of ways and that is consistent with the constraints (Jaynes, 1957). However, if an initial hypothesis of the process driving the behavior of the stochastic variable can be expressed in the form of a prior distribution, now, in contrast to the maximum entropy pure inverse problem, the problem can be reformulated to minimize the probabilistic divergence between the posterior and the prior. Out of all the distributions of probabilities satisfying the constraints, the solution is the posterior closest to the prior (Kullback, 1959). Although a prior distribution is based on economic intuition and theoretical models, it is usually inconsistent with empirical observations. Thus, using the cross-entropy solution, one solves this inconsistency, reconciling it in the best possible way by generating the distribution that is closest to the prior but consistent with empirical observations.

(continued)

Box 31.4. continued

When we use CIMDO to solve for $p(x,y)$, the problem is converted from one of deductive mathematics to one of inference involving an optimization procedure; that is, instead of assuming parametric probabilities to characterize information contained in the data, this approach uses the data information to infer values for the unknown probability density. Thus, the estimated probability values can be interpreted as inverse probabilities. Using this procedure, we look to make the best possible predictions from the scarce information that we have. This feature of the methodology not only makes implementation simple and straightforward, it also seems to reduce model and parameter risks of the generated distribution, as indicated by the PIT criterion (Segoviano, 2006b). This is because in order to estimate the posterior density, only variables that are directly observable for the type of loans that are the subject of our interest are needed, and by construction, the estimated posterior is consistent with the empirically observed probabilities of default. Thus, the proposed methodology represents a more flexible approach to modeling multivariate densities, making use of the limited available information in a more efficient manner.



Source: Authors.

Note: CIMDO = Consistent Information Multivariate Density Optimizing; CoPoD = Conditional Probability of Default; PD = probability of default; PLD = portfolio loss distribution; PMD = portfolio multivariate density; VaR = value at risk.

Figure 31.4 Stress Test Procedure

3. PROPOSED PROCEDURE FOR STRESS TESTING

The proposed stress test procedure involves four steps (Figure 31.4):

1. The definition of macroeconomic scenarios. These should be extreme but plausible and should be thought to capture the main risks to an economy. The macroeconomic scenarios usually are defined by the mission and the authorities.
2. The incorporation of macroeconomic shocks into the PDs of the loans making up the portfolios. The effects of the macroeconomic shocks implied by the previously defined scenarios are incorporated into the probabilities of default of loans grouped

by sector or rating. For this purpose, the CoPoD econometric framework is used, thus obtaining CoPoD-PDs.

3. The modeling of the portfolios' multivariate densities (PMDs). With the use of the CIMDO methodology and the CoPoD-PDs (as exogenous variables), we recover the PMDs, that is, the CIMDO-PMDs.
4. The simulation of portfolio loss distributions (PLDs) and the estimation of their EC. The CIMDO-PMD of each bank in the system is used to simulate the possible losses/gains that a bank's loan portfolio can experience, given its characteristics, for example, the type of loans held in the portfolio. The losses/gains obtained from this simulation are used to construct the PLD of the portfolio. From this distribution, the

EC that a bank should put aside can be determined. This procedure is repeated under each of the macroeconomic scenarios defined in Step 1 in order to estimate the EC that a bank should hold—given the riskiness of its portfolio—under different macroeconomic scenarios.

The stress testing framework employed allows explicit linkage of macroeconomic shocks to estimation of EC and therefore the evaluation of the banks' vulnerability under different shocks. It ensures the economic consistency of the estimated risk measurements.

4. STRESS TESTING: EMPIRICAL IMPLEMENTATION IN DENMARK

A. Definition of macroeconomic scenarios

On the basis of the current macroeconomic situation and vulnerabilities in the Danish economy, extreme but plausible single-factor shocks and macroeconomic scenarios were defined. In all cases, the sizes of the shocks draw from historic information, taking into account a time frame large enough to cover at least an entire economic cycle, or encompassing episodes of financial distress.

Economic Outlook and Vulnerabilities

The Danish economy has performed strongly in recent years, and 2005 was no exception. Sound macroeconomic policies, adequate regulatory environments, and flexible labor markets underpin this performance, although temporary factors such as revenue from oil exports and shipping services have further reinforced the positive outcome. After 2000, GDP growth has increased noticeably, reaching 3.0 percent, and continues to expand more rapidly than its potential rate. The unemployment rate has fallen below 4.8 percent, a historical low. Private consumption, exports, and investment are all expanding, and consumer confidence is at its highest level for several years. At the same time, inflation remains low although there are signs that the labor market is becoming increasingly tight, but asset inflation is high.

With the economy approaching capacity limits, the immediate challenge is to avoid overheating in the economy. An accommodative monetary policy and rising house prices are two underlying factors that represent a challenge for the authorities (Figure 31.5). Each provides an added stimulus to private consumption and the overall economy.

Monetary policy, however, is constrained because Denmark operates a strict fixed exchange-rate link to the euro—which has served Denmark well—so monetary policy is de facto set by the European Central Bank (ECB). House prices have grown strongly and added about half a percentage to consumption growth each year via wealth effects and mortgage equity withdrawal. Experience from other small open economies shows that an overheating scenario can then set

in very quickly. The recent experience of the Netherlands is one example (Organization for Economic Cooperation and Development, 2006) that could hold some important lessons for Denmark. Therefore, as stated during discussion of the economic hypothesis, special attention should be given to preventing fluctuations in key macroeconomic and financial variables from becoming endogenous cycles, which have the potential to develop and amplify financial imbalances, thus placing stress on the financial system.

Recent economic and financial developments in the Danish economy suggest the emergence of risks that could undermine financial stability. Over the last year, the rate of house price increases has been particularly strong, with prices rising at double-digit rates, reaching over 21 percent in the fourth quarter of 2005, the largest increase in nearly 20 years. Conversely, credit growth has been boosted in parallel, reaching over 20 percent in 2005. Moreover, the housing finance structure has changed quickly in recent years. The levels of total debt to disposable income have increased greatly, as has the share of adjustable rate loans, which have become prominent in the mortgage markets. These facts make households more vulnerable to fluctuations in interest rates, but also, because of the importance of mortgage loans in the economy, the new structure might have the potential to amplify interest rate effects on the economy.

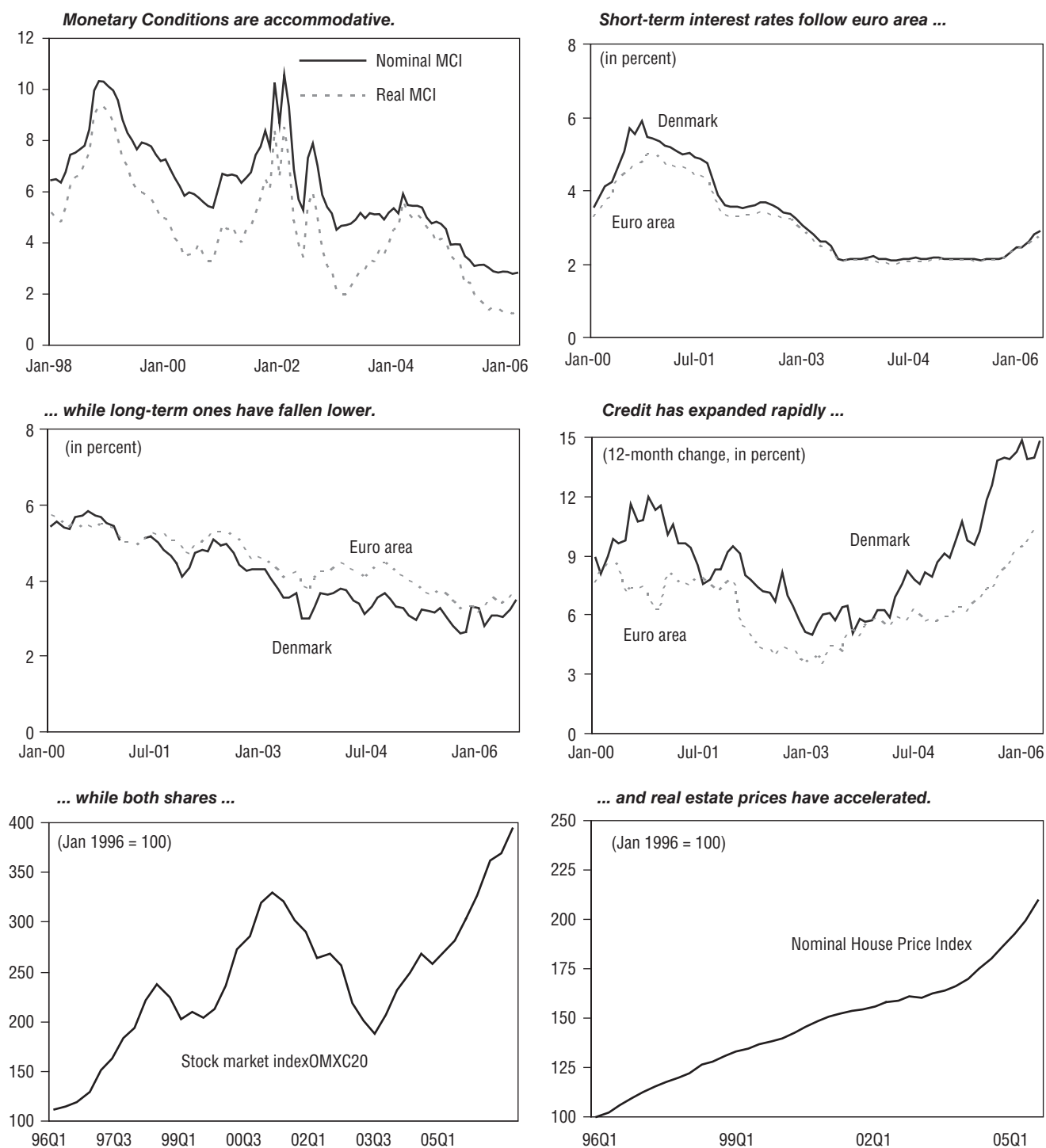
In addition, the profitability of the banking sector should be carefully assessed. An increasing proportion of the gross income of Danish banks has been derived in recent years from noninterest income, some sources of which are unlikely to be sustainable over the longer term (in particular, refinancing of houses, securities and foreign exchange income, or income attributable to equity investments in associates and unconsolidated subsidiaries). Growth in fee and commission income has been moderate in both gross and net terms, and the share of this income in total assets has been broadly stable. This ratio is similar to that of banks operating in other European Union countries (Figure 31.6).

Thus, like any small open economy, Denmark seems prone to vulnerabilities imported from larger partner economies. Given its openness, the restrictions on the conduct of its monetary policy, and the importance of net exports in GDP, the Danish economy is affected seriously by developments in foreign economies.

Macroeconomic Scenarios

Three macroeconomic scenarios were defined relative to a baseline scenario (that is consistent with current macroeconomic perspectives), capturing the main risks to the Danish economy. The assumed shocks under each scenario are presented in Table 31.2. The scenarios are the following:

Scenario 1: Boom-Bust in Real Estate Prices and Credit. A low-interest-rate environment has increased the value of collateralizable assets (stocks and real estate). As the borrowing

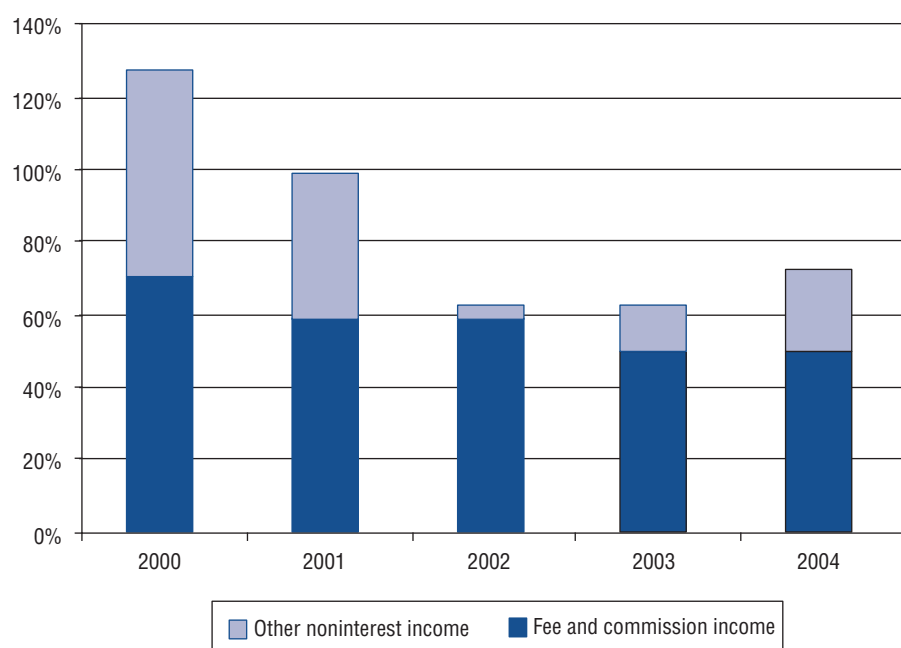


Sources: Danmarks Nationalbank; and International Financial Statistics.
Note: MCI = Monetary Conditions Index.

Figure 31.5 Recent Economic and Financial Developments in Denmark

capacity of borrowers depends on the value of their collateralizable assets, higher asset values have contributed to higher lending, which in turn has further fueled economic activity and asset prices. The increased value of assets has increased wealth and thus boosted consumption. Higher consumption

has increased aggregate demand and decreased unemployment, with further knock-on effects on asset prices, competitiveness, and economic activity. In this way, the levels of indebtedness in the economy have increased quickly. Eventually, decreases in asset returns—resulting from the sharp



Source: Authors.

Figure 31.6 Fee Income as a Percentage of Pretax Income

TABLE 31.2

Macroeconomic Scenarios: Percentage Deviations from Baseline

Variable	Baseline			Scenario 1			Scenario 2			Scenario 3		
	2006	2007	2008	2006	2007	2008	2006	2007	2008	2006	2007	2008
GDP % y/y	2.4	2.2	2.2	-1.8	-5	-6.8	-1.4	-3.2	-4.1	-2.4	-6.7	-9.2
Unemployment ¹	5.1	5	5	0.5	2.2	3.8	0.6	2	3.2	0.7	2.9	5.1
Private consumption % y/y	2.5	2.2	2.2	-2.8	-6.9	-9.1	0.4	-0.5	-1.2	-3.5	-8.8	-11.7
House price % y/y	7	3	3	-7.2	-17.8	-27	-0.6	-2.9	-5.5	-15.6	-29.8	-41.9
Business investment % y/y	5.9	4	4	-5.7	-12.6	-17.1	-1.8	-4.9	-7.7	-8	-19.5	-26.5
Exports % y/y	4.6	5.5	5.5	-2.4	-3.4	-3.1	-5.9	-8.3	-8.6	-3.3	-3.8	-3.1
Manufacturing output % y/y	3.5	3.2	3.2	-3.5	-9.7	-12.9	-3.6	-7.9	-9.9	-4.6	-13	-17.5
Consumer price % y/y	2.1	1.9	1.9	-0.1	-0.4	-0.8	-2	-3.1	-4	0.2	0	-0.4
Hourly wage % y/y	3.7	3.8	3.8	-0.1	-0.9	-2.7	-0.3	-1.5	-3.2	-0.1	-1.1	-3.5
Oil price \$/barrel	56.4	56.4	56.4	0	0	0	0	0	0	30	30	30
DKK/USD	638	638	638	0	0	0	-29	-29	-29	0	0	0
Bond yield % points	3.9	3.9	3.9	0	0	0	0	0	0	2.5	2.5	2.5
Discount rate % points	2.2	2.2	2.2	0	0	0	0	0	0	2.5	2.5	2.5
Share price % y/y	8	5	5	-0.7	-6.5	-11.5	-1.9	-9.6	-14.6	-23.7	-31.6	-30.8
Loans to private sector % y/y	8	5	5	-1.5	-7.0	-14.2	0	-1	-2.7	-1.8	-9.5	-19.5

Sources: Danish authorities; and IMF staff calculations.

Note: The shocks are reported as percentage deviations from the baseline scenario and thus need to be interpreted with caution. For example, a shock to GDP in 2008 under Scenario 3 is equal to -9.2 percent from the baseline scenario in 2008. However, this shock corresponds to -2.4 percent from the baseline scenario level in 2005. This result is equal to the accumulated percentage increase in GDP under the baseline scenario from 2005 to 2008 (6.8 percent) minus the percentage deviation from the baseline scenario in 2008 (-9.2 percent).

y/y = year-on-year.

1. Percent of labor force national definition.

increase in asset prices and high levels of indebtedness in the economy—can make economic agents revise their expectations about asset prices, and the boom can turn into a bust. Asset prices then start falling and credit growth decreases, suppressing economic activity. Expectations of wealth de-

crease, consumption falls, aggregate demand falls, GDP falls, and unemployment increases. Depressed collateral prices, higher unemployment, and decreasing GDP have a negative impact on banks' assets, increasing the proportion of nonperforming loans (NPLs).

Scenario 2: Foreign Shock Owing to a Correction in U.S. Imbalances. Investors around the world become nervous about global imbalances and withdraw their assets, looking for a “safe haven” in Europe. This results in a reduction of European net foreign investment (NFI). A downward shift in European NFI reduces the demand for loans and therefore reduces the equilibrium interest rate. The reduced level of NFI in Europe reduces the supply of euros in the market for foreign exchange; therefore, the euro appreciates. The Danish kroner, pegged to the euro, appreciates against the U.S. dollar and other currencies as well, leading to a loss of competitiveness and lower demand for Danish goods. The trade balance deteriorates, GDP falls, and unemployment rises in Denmark. There is a shift in expectations, negatively affecting private consumption and depressing economic activity further. These effects have a negative impact on banks’ NPLs, increasing banks’ losses.

Scenario 3: Boom-Bust in Real Estate Prices and Credit Plus an Increase in European Interest Rates. The background leading up to this scenario is similar to that described in Scenario 1, that is, a boom in asset prices and credit. However, in this scenario, we also assume that the continuation of high oil prices increases inflationary pressures in Europe, prompting the ECB to increase interest rates. Denmark’s Nationalbank follows suit and increases the policy interest rate.

The reassessment of asset values is reinforced by higher interest rates and higher debt servicing, making the boom turn into a bust. As in Scenario 1 but only more so, asset prices start falling, consumption decreases, aggregate demand contracts, GDP falls, credit growth decreases, and unemployment increases. Higher interest rates, depressed collateral prices, higher unemployment, and decreasing GDP have a negative impact on banks’ assets and NPLs.

B. Necessary data for models’ implementation

In this exercise, we had access to two data sets. In one data set, the loan books of the five largest banks were aggregated by economic sectors (Table 31.3).²⁶ For each of these sectors, we had annual observations of NPL ratios²⁷ from 1991 to 2004 (14 observations) and the percentage loan exposure to each economic sector. The second data set comprised information of loans grouped by risk rating classifications.²⁸ For each of these RRs, we had annual observations of empirical frequencies of default²⁹ from 1996 to 2003 (8 observations) and the percentage loan exposures to each risk rating category.

²⁶ The economic sectors were the public sector, agriculture, fisheries, manufacturing, building, trade, transport, finance and insurance, property administration, and households.

²⁷ The NPL ratios were defined as the nonperforming loan exposure in a given sector over the total loan exposure in the sector.

²⁸ The RRs are based on the De Nederlandsche Bank (2005) failure-rate model described in the Financial Stability Overview.

²⁹ The empirical frequencies of default were defined as the number of loans that went into default in a given risk rating over the total number of loans in the risk rating category.

TABLE 31.3

Necessary Data for Model Implementation

Variable	Number of Observations
Annual NPLs of loans grouped by economic sectors (10 sectors)	14 annual observations (1991–2004) per sector
Percentage loan exposure to each economic sector	Average between 1991 and 2004
Macroeconomic variables (described in Table 31.1)	From 1970 to 2004

Source: Authors.

Note: NPL = nonperforming loan.

Both variables, NPL ratios and frequencies of default, appeared to be adequate proxies for PDs. Therefore, initially, we developed our analysis using both data sets. However, because we were interested in capturing in the best possible way the effects of the business cycle (and, therefore, macroeconomic shocks), we decided to base our estimations on the variable that comprised the longest time series. Thus, all the results that are reported from now on in this section are derived using NPL ratios as proxies for PDs.

Note that the time series that was available and used was very small in statistical terms (14 observations). Note also that information of the dependence structure (correlations) of the borrowers’ PDs was nonexistent, and stock prices or any other market risk indicator of the borrowers was unknown. This certainly represents a highly restricted data set that would make impossible the proper calibration of standard credit risk models.³⁰ However, this data set was enough to properly implement the CoPoD and CIMDO methodologies.

C. Incorporation of macroeconomic shocks into the probability of defaults

Using the CoPoD methodology and the macroeconomic variables described in Table 31.1 (as explanatory variables), we modeled NPLs (dependent variables) as functions of macroeconomic variables, obtaining estimating equations for 10 categories of NPLs (e.g., for each economic sector). Model specifications were selected based on consistency with economic theory and empirical evidence (as indicated in Section 2.A). Specifications are summarized in Table 31.4. Figure 31.7 shows estimated NPL values for a selected group of economic sectors.

In all these cases, credit over GDP (CREOVGDP) and house prices (HOUP) are significant. It is clear that owing to the impact of the value of collateral assets on the level of credit, wealth effects, and economic activity, the evolution of these variables has a significant effect on the relevant economic sectors. It is also noteworthy that there are specific variables that are more relevant to specific sectors because of the nature of their economic activity. For example, interest rates (AVGR) and unemployment (UNEM) are highly significant

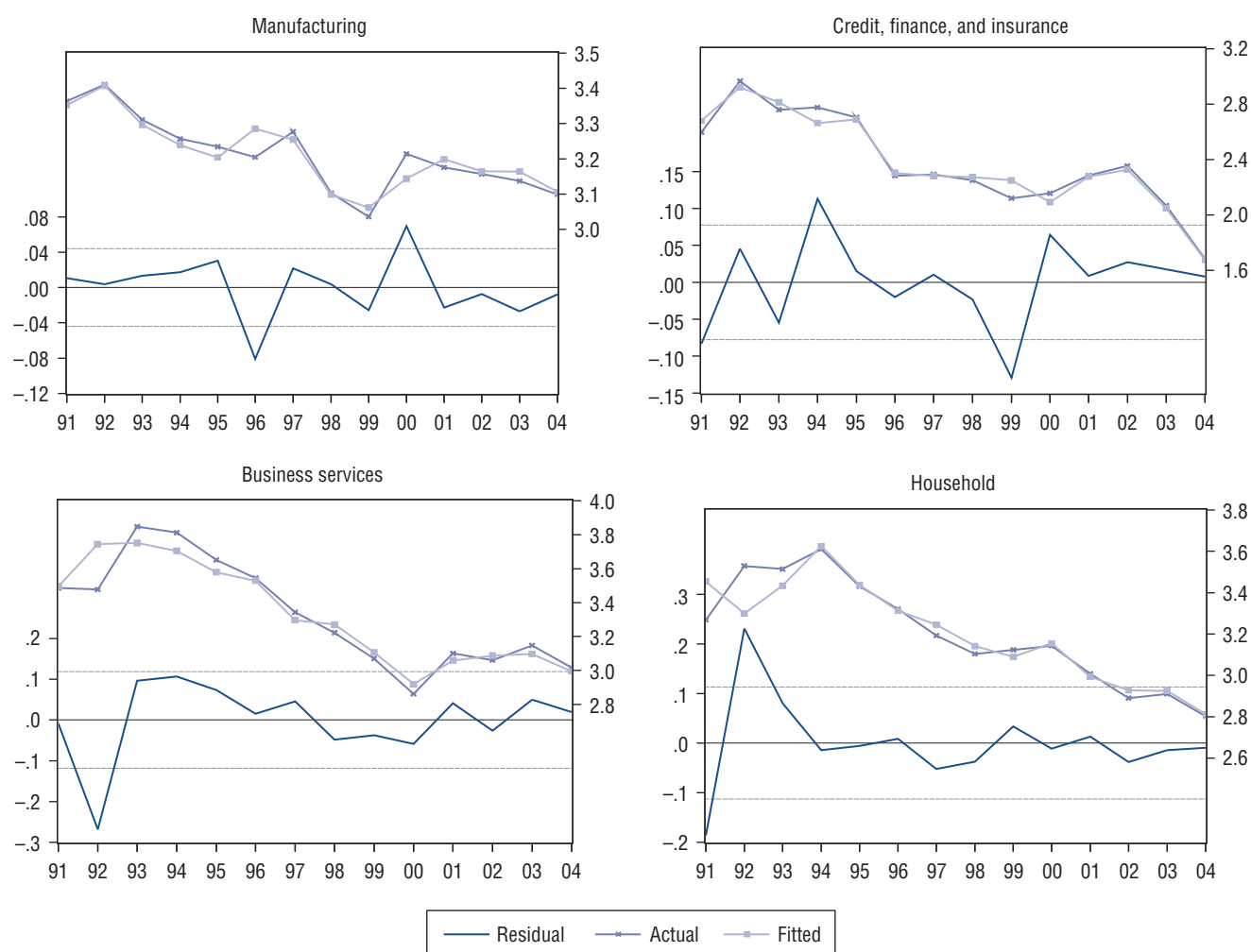
³⁰ In Segoviano (2006b), we discuss in detail the data requirements that are necessary for the proper calibration of standard credit risk models.

TABLE 31.4
Significant Explanatory Variables

Sector	CREOVGDP		HOUP		GDP		UNEM		AVGR		MTGR		FXDU		A-R ²
	Sign	Lag	Sign	Lag	Sign	Lag	Sign	Lag	Sign	Lag	Sign	Lag	Sign	Lag	
Public sector	+	3	+	2			+	2					–	0	0.83
Agriculture	+	5	+	4	–	3							–	3	0.81
Fisheries	+	5	+	4			+	2	+	1					0.78
Manufacturing	+	4	+	1	–	1			+	0			–	2	0.74
Building	+	5	–, +	0, 5									–	3	0.68
Trade	+	4	–	0	–	3					+	0	–	0	0.72
Transport	–	1	–	0	–	0			+	0					0.69
Finance and insurance	–	0	+	3					+	0					0.73
Business services	+	6	–	0	–	3									0.74
Households	+	4	–	0			+	1	+	1			–	2	0.71

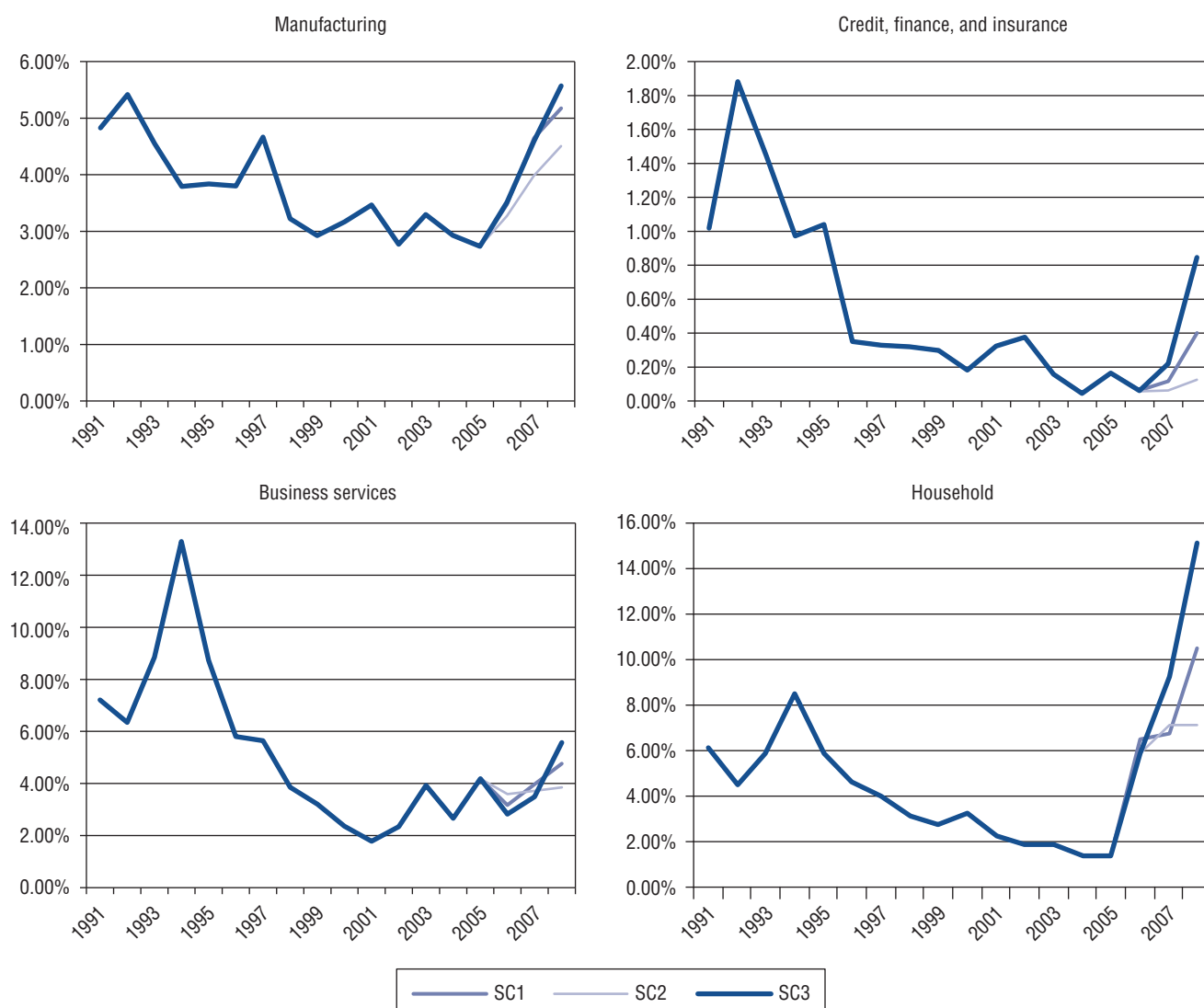
Source: Authors.

Note: AVGR=interest rates; CREOVGDP=credit over GDP; FXDU=foreign exchange rate; HOUP=house prices; MTGR=mortgage bond interest rate; UNEM=unemployment.



Source: Authors.

Figure 31.7 Probability of Defaults as Functions of Macroeconomic Variables



Source: Authors.
Note: SC = scenario.

Figure 31.8 Conditional Probability of Default—Probability of Defaults

to explain the default behavior of the “Credit, Finance, and Insurance” sector and households, respectively. In addition, the foreign exchange rate (FXDU) is highly significant to manufacturing, given that, in Denmark, this sector is strongly focused on export markets, while GDP is highly important for business and services, which is a sector that is focused mainly on the domestic economy.

With the selected model specifications (for each economic sector) and the values assumed to be taken by the significant explanatory variables, under each of the macroeconomic scenarios, we computed annual forecasts of NPL ratios for 2006, 2007, and 2008, under three different macroeconomic scenarios. Therefore, for each economic sector, we estimated 9 NPL ratio forecasts, which we denominated CoPoD-PDs. Note that CoPoD-PDs correspond to stressed values of PDs (stressed PDs). Figure 31.8 shows the paths taken by selected PDs under the three scenarios that were considered.

As an illustration of the efficiency gains attained with the CoPoD methodology in the presence of small samples, in Table 31.5 the standard errors and variances of the CoPoD estimators vis-à-vis the OLS estimators (for the manufacturing sector) are presented.

From these results, it is possible to observe that the CoPoD estimators show the same signs as the OLS estimators; thus, the CoPoD estimators remain consistent with economic theory and empirical evidence. However, the CoPoD estimators show greatly reduced variances.

D. Modeling of the portfolio multivariate density

The CIMDO methodology recovers PMDs period by period. It requires PDs of each type of loan comprising the loan portfolio at each period of time. Therefore, in order to recover PMDs for 2006, 2007, and 2008, under each of the

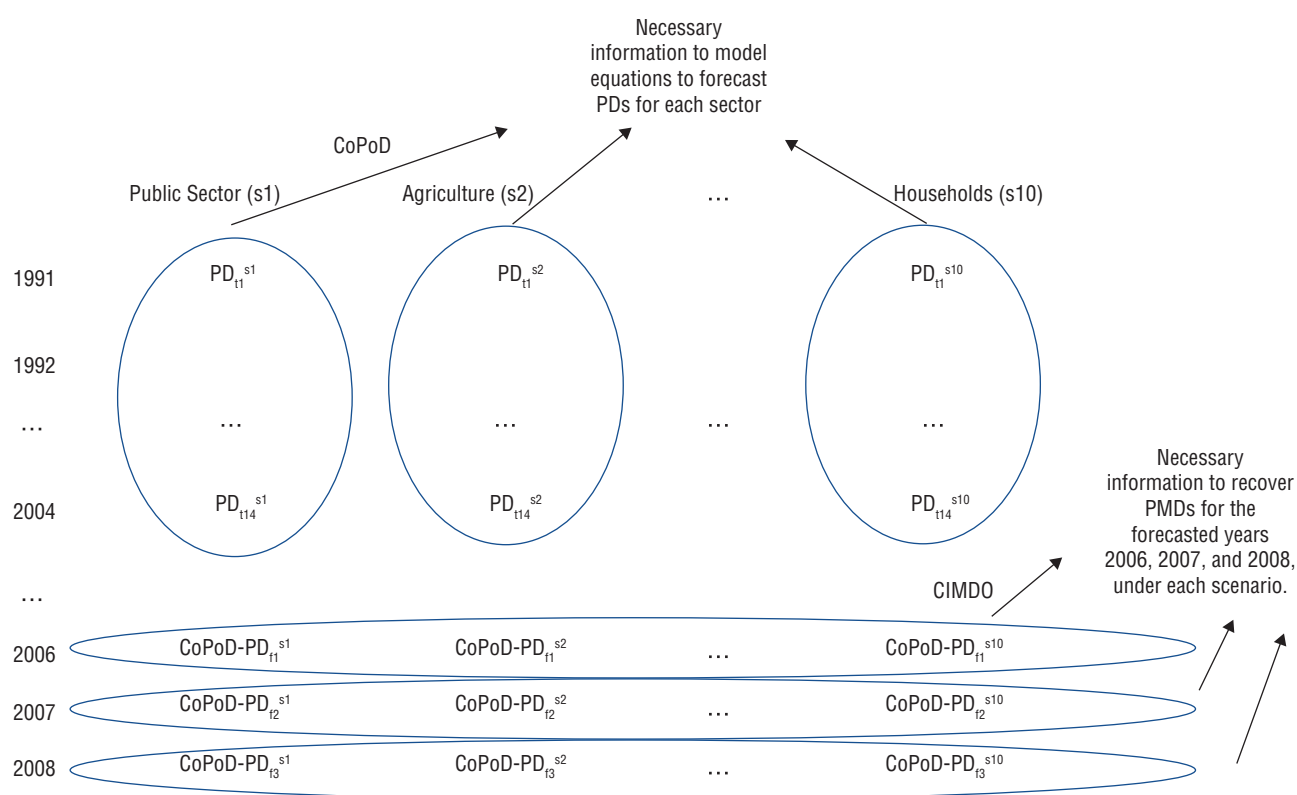
TABLE 31.5
Conditional Probability of Default Efficiency Gains

Variable	OLS	p	OLS SE	CoPoD	CoPoD SE	Variance Difference (in percent) ¹
C	3.2163	0.0000	0.0130	3.2164	0.0081	155.04
CREOVGDP(-4)	0.9010	0.0000	0.1614	0.9001	0.1312	51.18
HOUPL(-1)	0.5215	0.0200	0.1811	0.5189	0.1309	91.55
GDP(-1)	-1.8838	0.0000	0.3938	-1.8884	0.2920	81.82
AVGR(-0)	0.9701	0.0200	0.3408	0.9712	0.2513	83.87
FXDU(-2)	-0.3140	0.0600	0.1548	-0.3398	0.1116	92.31
Adjusted R ²		0.74				

Source: Authors.

Note: AVGR=interest rates; C=constant; CoPoD=Conditional Probability of Default; CREOVGDP=credit over GDP; FXDU=foreign exchange rate; HOUPL=house prices; OLS=ordinary least squares.

1. This indicator is estimated as Variance OLS – Variance CoPoD/Variance CoPoD.



Source: Authors.

Note: Subscripts denote time period; e.g., $t_1, \dots, t_{14}$ and forecasts; e.g., $f_1, \dots, f_3$.

Superscripts denote economic sector; e.g., s_1 and s_{10} correspond to the public and household sectors, respectively.

CIMDO=Consistent Information Multivariate Density Optimizing; CoPoD=Conditional Probability of Default; PD=probability of default; PMD=portfolio multivariate density; s =sector.

Figure 31.9 Necessary Data to Incorporate Macroeconomic Shocks into PDs and PMDs

three scenarios, we used the estimated CoPoD-PDs of each economic sector (Figure 31.9). The recovered PMDs describe the joint likelihood of changes in the credit risk quality of the loans that make up banks' portfolios. As indicated in Box 31.3, the CIMDO-PMD is recovered by minimizing a function of the form described in equation (31.11).

For example, with the 2007 CoPoD-PDs (stressed PDs) of each economic sector under Scenario 3 (Table 31.6), moment-consistency constraints were formulated as indicated in Box 31.3. Consequently, the CIMDO multivariate density in 2007 under Scenario 3, corresponding to the analyzed portfolio, is recovered by minimizing the following functional:

TABLE 31.6

Conditional Probability of Default—Probability of Defaults in 2007: Scenario 3	
Sector	2007
Public sector	0.0055
Agriculture	0.0490
Fisheries	0.1059
Manufacturing	0.0351
Building	0.0516
Trade	0.0495
Transport	0.0287
Finance and insurance	0.0006
Property administration	0.0282
Households	0.0582

Source: Authors.

$$L[p, q] = \int \dots \int p(x_1, \dots, x_{10}) \ln p(x_1, \dots, x_{10}) dx_1, \dots, dx_{10} \\ - \int \dots \int p(x_1, \dots, x_{10}) \ln q(x_1, \dots, x_{10}) dx_1, \dots, dx_{10} \quad (31.13)$$

$$\begin{aligned} & \text{Public sector} + \lambda_1 \left[\int \dots \int p(x_1, \dots, x_{10}) \chi_{(x_d^1, \infty)} dx_1, \dots, dx_{10} - .0055 \right] \\ & \text{Agriculture} + \lambda_2 \left[\int \dots \int p(x_1, \dots, x_{10}) \chi_{(x_d^2, \infty)} dx_1, \dots, dx_{10} - .0490 \right] \\ & \text{Fisheries} + \lambda_3 \left[\int \dots \int p(x_1, \dots, x_{10}) \chi_{(x_d^3, \infty)} dx_1, \dots, dx_{10} - .1059 \right] \\ & \text{Manufacture} + \lambda_4 \left[\int \dots \int p(x_1, \dots, x_{10}) \chi_{(x_d^4, \infty)} dx_1, \dots, dx_{10} - .0351 \right] \\ & \text{Building} + \lambda_5 \left[\int \dots \int p(x_1, \dots, x_{10}) \chi_{(x_d^5, \infty)} dx_1, \dots, dx_{10} - .0516 \right] \\ & \text{Trade} + \lambda_6 \left[\int \dots \int p(x_1, \dots, x_{10}) \chi_{(x_d^6, \infty)} dx_1, \dots, dx_{10} - .0495 \right] \\ & \text{Transport} + \lambda_7 \left[\int \dots \int p(x_1, \dots, x_{10}) \chi_{(x_d^7, \infty)} dx_1, \dots, dx_{10} - .0287 \right] \\ & \text{Finance} + \lambda_8 \left[\int \dots \int p(x_1, \dots, x_{10}) \chi_{(x_d^8, \infty)} dx_1, \dots, dx_{10} - .0006 \right] \\ & \text{Property} + \lambda_9 \left[\int \dots \int p(x_1, \dots, x_{10}) \chi_{(x_d^9, \infty)} dx_1, \dots, dx_{10} - .0282 \right] \\ & \text{Household} + \lambda_{10} \left[\int \dots \int p(x_1, \dots, x_{10}) \chi_{(x_d^{10}, \infty)} dx_1, \dots, dx_{10} - .0582 \right] \\ & + \mu \left[\int \int p(x_1, \dots, x_{10}) dx_1, \dots, dx_{10} - 1 \right], \end{aligned}$$

where, $\lambda_1, \dots, \lambda_{10}$ represent the Lagrange multipliers of the moment-consistency constraints and μ represents the Lagrange multiplier of the probability additivity constraint.³¹

The optimal solution is represented by a posterior multivariate density of the form

$$\hat{P}(x_1, \dots, x_{10}) = q(x_1, \dots, x_{10}) \exp \left\{ - \left[1 + \hat{\mu} + \left(\hat{\lambda}_1 \chi_{(x_d^1, \infty)} \right) \right. \right. \\ \left. \left. + \dots + \left(\hat{\lambda}_{10} \chi_{(x_d^{10}, \infty)} \right) \right] \right\}. \quad (31.14)$$

E. Simulation of portfolio loss distribution and estimation of economic capital

The expected losses (ELs) of the analyzed portfolio under each scenario were estimated. ELs are defined as follows:

TABLE 31.7

Expected Losses as a Ratio of Risk-Weighted Assets (in percent)			
Year	Expected Losses		
	Scenario 1	Scenario 2	Scenario 3
2006	1.41	1.31	1.30
2007	1.61	1.60	2.00
2008	2.30	1.70	3.01

Source: Authors.

EL = PD × Exposure at Default × Loss Given Default. For this purpose, CoPoD-PDs (stressed PDs) for each economic sector, under each scenario, were used. Information on losses given default (LGDs) was not available. As a result, the Basel II guidelines were followed by setting this variable equal to 50 percent. Table 31.7 shows the ELs as a ratio of risk-weighted assets (RWA), that is, the EL ratio.

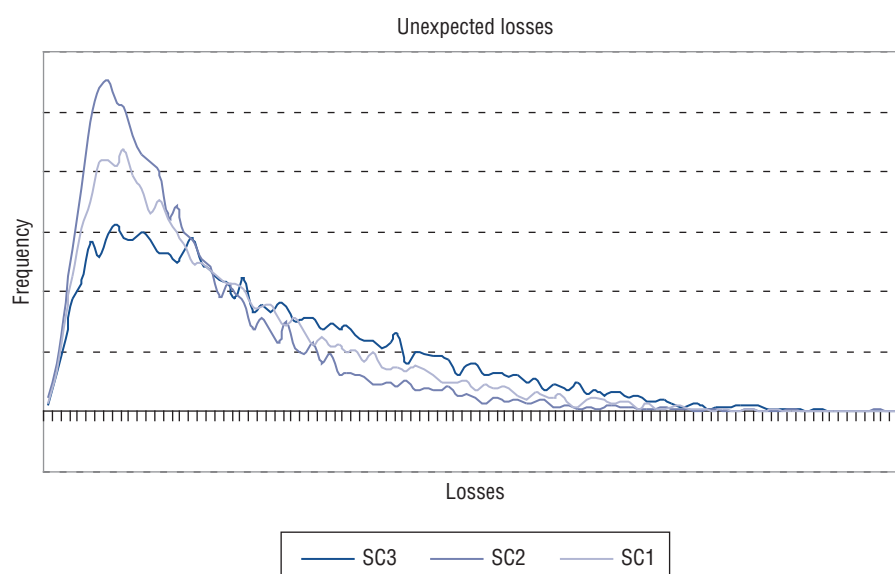
The potential losses that can be experienced by a bank (given the composition of its portfolio) are simulated using the PMD. Such losses are used to construct the PLD from which the ULs are estimated—thus, the EC defined. The joint likelihood of credit quality migrations of the loans making up a portfolio is embedded in the PMD. This density is used to simulate the different credit qualities that loans in the portfolio can take on. Thus, if a macroeconomic shock causes the quality of the loans in the portfolio to decrease (making loans riskier), losses are incurred. The losses obtained from such a simulation process are used to build a histogram, which summarizes the distribution of the portfolio losses under each scenario (Figure 31.10).³² The EC is set to be equal to the loss level corresponding to the 99.5 percentile of the PLD. Equivalently, ULs that could be suffered by a bank under a specific macroeconomic scenario will be covered 99.5 percent of the time; however, with a probability of 0.5 percent, there will be (extreme) ULs higher than the EC. Table 31.8 shows the total ULs as a ratio of RWA, that is, UL ratio.

5. STRESS TESTING RESULTS' ANALYSIS

Banks' financial performance in Denmark has been very solid, especially during the favorable macroeconomic environment enjoyed over the last two years of our analysis. However, results of the stress test show that under changing macroeconomic conditions, credit risk could materialize, causing a drastic deterioration in banks' results. Credit portfolios are highly concentrated in loans to the sectors of credit, finance and insurance; household; manufacturing; and business services. Together, loans to these sectors constituted over 82 percent in 2004 and between 1991 and 2004, they averaged over 73 percent. Note that "loans to the private sector," "house prices," "unemployment," and "GDP"

³¹ Like all optimization problems, the Lagrange multipliers reflect the change in the objective function's value as a result of a marginal change in the constraint set.

³² See Segoviano and Lowe (2002) and Vassalou and Xing (2004).



Source: Authors.
Note: SC = scenario.

Figure 31.10 Portfolio Loss Distribution under the Three Macroeconomic Scenarios

TABLE 31.8

Unexpected Losses as a Ratio of Risk-Weighted Assets (in percent)

Year	Unexpected Losses		
	Scenario 1	Scenario 2	Scenario 3
2006	10.94	10.89	10.82
2007	11.16	11.12	11.39
2008	12.76	11.28	13.71

Source: Authors.

are the macroeconomic variables that deteriorate faster under the assumed scenarios (Table 31.2). Deterioration is especially drastic under Scenario 3 (followed by Scenarios 1 and 2, respectively). These are also the variables that are significant in explaining the default behavior of the sectors to which the banks are more exposed. These effects are captured in the forecasts of the sectors' PDs and passed on to the estimated ELs and ULs (Tables 31.7 and 31.8).

A buffer large enough to absorb ELs should be generated through adequate pricing and provisioning. Thus, the expected loss buffer (EL buffer) was defined as the ratio of pretax income plus provisions to RWA. The capital adequacy ratio (CAR) represents the buffer to cover ULs, given that the realized level of credit losses experienced under a specific shock could be significantly higher than the expected level. The EL buffer and the CAR constitute the banking institutions' total buffer to withstand losses.

Overall, the Danish banking institutions' total buffer as a ratio of RWA has fallen from 2004, and this trend is likely to continue. The reduction in the total buffer is the result of a

fall in accumulated provisions and lower solvency ratios, both in absolute levels and as a ratio of RWA. The new International Accounting Standards/International Financial Reporting Standards, effective from January 2005, have had a big impact on the EL buffer via large reductions in provisions. This is seen in Table 31.9, where the total buffer and its components are presented. Note that from 2004, pretax income increased; however, the increase in pretax income was lower than the decrease in provisions.

Under the current regulatory guidelines and the Basel II proposal, a minimum CAR of 8 percent should always be kept in order to guarantee the regular operations and solvency of a bank. In order to check for this requirement to be fulfilled, we estimated the CAR after the shocks (CAR-S).

The decrease that the CAR can suffer after the shocks (CAR decrease) is defined as the difference between the UL ratio under each scenario minus the CAR at the end of 2004 (i.e., 10.21 percent). Results are presented in Table 31.10.

The CAR adjusted after the shocks (CAR-S) is estimated as the difference between the CAR at the end of 2004 minus the CAR decrease. Results are presented in Table 31.11. From this table, we see that the required minimum CAR of 8 percent would be bridged under Scenarios 1 and 3 in 2008. In these situations, the solvency of the analyzed banks would be threatened.

Individual institutions could suffer from much higher losses than those estimated in the stress test. The results presented here represent the results of the "average effect" of the shocks to a group of banks. However, the effects of the shocks are likely to be worse for specific banks. Unfortunately, we were unable to assess individual banks' portfolio credit risk because of the use of bank-grouped data.

TABLE 31.9

Expected Loss Buffer and Capital Adequacy Ratio

Item	2000	2001	2002	2003	2004
Total provisions + pretax income	14,799.00	19,314.00	19,057.00	23,551.00	22,485.00
Total capital (CAR)	110,460.47	117,205.30	122,495.97	125,752.22	126,958.58
Total risk-weighted assets	1,135,287.00	1,141,915.00	1,174,840.00	1,168,680.00	1,243,096.00
EL buffer/risk-weighted assets (in percent)	1.30	1.69	1.62	2.02	1.81
CAR (in percent)	9.73	10.26	10.43	10.76	10.21

Source: Authors.

Note: CAR = capital adequacy ratio; EL = expected loss.

TABLE 31.10

CAR Decrease (in percent)

Year	CAR decrease		
	Scenario 1	Scenario 2	Scenario 3
2006	-0.72	-0.68	-0.61
2007	-0.95	-0.91	-1.17
2008	-2.55	-1.07	-3.50

Source: Authors.

Note: CAR = capital adequacy ratio.

TABLE 31.11

CAR-S (in percent)

Year	CAR-S		
	Scenario 1	Scenario 2	Scenario 3
2006	9.49	9.54	9.60
2007	9.26	9.31	9.04
2008	7.66	9.14	6.71

Source: Authors.

Note: CAR-S = capital adequacy ratio after the shock.

The credit portfolios of other bank groups seem to be riskier, and therefore, the impact of the assumed shocks on those banks' CAR might be higher. The stress test did not include smaller banks; however, the exposure of lending to agriculture and fisheries is the highest among those banks. Some of these banks have grown rapidly. During the challenging times in the late 1980s and early 1990s, it was typically the banks that had grown fastest that also experienced the largest losses. Thus, it is very likely that the potential losses (ELs and ULs) of the assumed shocks might be larger among smaller banks.

The effects of shocks to unemployment and interest rates might be larger than they appeared. Lending by banking institutions to households has increased significantly during recent years. In January 2005, the annual growth rate exceeded 20 percent. Growth in this sector has mainly been attributable to the introduction of adjustable-rate bank mortgage loans to homeowners as an alternative to traditional capital-market financing via mortgage-credit institutions. The introduction of mortgage-credit loans with an option to defer amortization has had a significant impact on the composition of loan portfolios. The total debt of households continued to

rise at a higher rate than their disposable income; thus, the burden of households has increased. However, low interest rates in recent periods have temporarily neutralized the budget impact of the growing indebtedness. Nonetheless, all these factors will likely make households more vulnerable to fluctuations in interest rates and unemployment.

The increase in PDs, ELs, and ULs is consistent across the scenarios (Scenario 3 is the most severe, followed by Scenarios 1 and 2) and over time (the cumulative effects of the macroeconomic shocks over the three-year horizon are reflected in the increasing PDs, ELs, and ULs over the three-year period). These results indicate that credit risk could materialize quickly, if the accelerated and acute increase in house prices and levels of leverage in the economy developed into macroeconomic imbalances, leading to a boom-bust in real estate prices and credit. Moreover, results clearly indicate that current developments in the market, which seem to increase the vulnerability of households to interest rate fluctuations, as well as foreign shocks (as indicated by Scenario 3), could reinforce the stress to the banking system. The macroeconomic shocks that were considered (and, as a result, the simulated increases in PDs, ELs, and ULs) may be extreme, but they are plausible. Thus, credit risk clearly remains a major risk factor for the Danish banking system.

6. CONCLUSION

The information requirements (presented in Table 31.3) necessary for the implementation of the CoPoD and the CIMDO methodologies are less stringent than the information requirements necessary for the proper calibration and implementation of standard credit risk methodologies. Therefore, their implementation is feasible and straightforward, even under the data constraints that usually bind portfolio credit risk modeling:

- The CoPoD methodology is designed to recover robust estimators in settings of short time series, incorporating the effects of changing macroeconomic and financial developments into PDs. The CIMDO methodology recovers PMDs, avoiding reliance on possibly unrealistic statistical assumptions in settings of partial information.
- The CIMDO methodology can be implemented using CoPoD-PDs, thus recovering PMDs that also reflect the impact of changing economic conditions.

Consequently, if CIMDO-PMDs are used to simulate the PLDs of bank portfolios and therefore estimate their EC, these EC estimations will embed the effect of changing economic conditions.

As a result, the joint implementation of the CoPoD and CIMDO methodologies provides a useful tool that leads to earlier recognition of banks' risks as macroeconomic conditions change, making a contribution to the challenging task of portfolio credit risk measurement through time. Moreover, if the joint implementation of the CoPoD and CIMDO methodologies is performed within the framework of stress testing, it becomes possible to quantify the impact of pre-specified macroeconomic shocks on banking systems' estimates of EC and therefore quantify the systems' resilience to specific shocks. Equivalently, this course of action allows us to identify explicit links between economic shocks and risks and vulnerabilities in financial systems.

The risk measurements produced by these methodologies are consistent with economic theory and empirical evidence. We consider these features to be extremely relevant in the stress test exercises performed as part of the FSAPs. This is because it is much easier to conduct discussions and negotiations and to make policy recommendations when the results of a risk model can be backed by economic theory and empirical evidence.

REFERENCES

- Allen, Franklin, and Douglas Gale, 2001, "Bubbles and Crises," *The Economic Journal*, Vol. 110, No. 460, pp. 236–55.
- Basel Committee on Banking Supervision, 2001, "Basel II: The New Basel Capital Accord—Second Consultative Paper" (Basel, January). Available via the Internet: <http://www.bis.org/bcb/bcbcp2.htm>
- Borio, Claudio, and Philip Lowe, 2002, "Asset Prices, Financial and Monetary Stability: Exploring the Nexus," BIS Working Paper No. 114 (Basel: Bank for International Settlements). Available via the Internet: <http://www.bis.org/publ/work114.htm>
- Butler, J., and Barry Schachter, 1998, "Estimating Value-at-Risk with a Precision Measure by Combining Kernel Estimation with Historical Simulation," *Review of Derivatives Research*, Vol. 1, No. 4, pp. 371–90.
- Calvo, Guillermo A., 1998, "Varieties of Capital-Market Crises," in *The Debt Burden and Its Consequences for Monetary Policy*, edited by Guillermo A. Calvo and Mervyn King (New York: Macmillan Press).
- Caprio, Gerard, and Daniela Klingebiel, 1996, "Bank Insolvency: Cross-Country Experiences," Policy Research Working Paper No. 1625 (Washington: World Bank). Available via the Internet: <http://elibrary.worldbank.org/content/workingpaper/10.1596/1813-9450-1620>
- Crosbie, Peter J., and Jeffrey R. Bohn, 2003, "Modeling Default Risk," Moody's KMV, December 18. Available via the Internet: http://www.creditrisk.ru/publications/files_attached/modeling_default_risk.pdf
- Danielsson, Jon, and Casper G. de Vries, 1997, "Tail Index and Quantile Estimation with Very High Frequency Data," *Journal of Empirical Finance*, Vol. 4, No. 2, pp. 241–57.
- De Nederlandsche Bank, 2005, *Overview of Financial Stability in the Netherlands*, No. 3 (Amsterdam, December). Available via the Internet: <http://www.dnb.nl/en/publications/dnb-publications/overview-of-financial-stability/>
- Demirgüç-Kunt, Asli, and Enrica Detragiache, 1998, "The Determinants of Banking Crises: Evidence from Developing and Developed Countries," *IMF Staff Papers*, Vol. 45, No. 1, pp. 89–109. Available via the Internet: <http://www.imf.org/external/Pubs/FT/staffp/1998/03-98/demirguc.htm>
- Diebold, Francis X., Jin Hahn, and Anthony S. Tay, 1999, "Multivariate Density Forecast Evaluation and Calibration in Financial Risk Management: High-Frequency Returns on Foreign Exchange," *Review of Economics and Statistics*, Vol. 81, No. 4, pp. 661–73.
- Dooley, Michael P., 1997, "A Model of Crises in Emerging Markets," NBER Working Paper No. 6300 (Cambridge: National Bureau of Economic Research). Available via the Internet: <http://www.nber.org/papers/w6300>
- Duffie, Darrell, and Kenneth J. Singleton, 1999, "Modeling Term Structures of Defaultable Bonds," *Review of Financial Studies*, Vol. 12, No. 4, pp. 687–720.
- Eichengreen, Barry, and Carlos Areta, 2000, "Banking Crises in Emerging Markets: Presumptions and Evidence," CIDER Working Paper C00-115 (Berkeley: Center for International and Development Economics Research).
- Flood, Robert, and Nancy Marion, 1999, "Perspectives on the Recent Currency Crisis Literature," *International Journal of Finance and Economics*, Vol. 4, No. 1, pp. 1–26.
- Frankel, Jeffrey A., and Andrew K. Rose, 1996, "Currency Crashes in Emerging Markets: An Empirical Treatment," *Journal of International Economics*, Vol. 41, No. 3–4, pp. 351–66.
- Glasserman, Paul, Philip Heidelberger, and Perwez Shahabuddin, 2002, "Portfolio Value-at-Risk with Heavy-Tailed Risk Factors," *Mathematical Finance*, Vol. 12, No. 3, pp. 239–69.
- Golan, Amos, and George G. Judge, 1992, "Recovering Information in the Case of Ill-Posed Inverse Problems with Noise," mimeograph (Berkeley: University of California, Berkeley).
- Goldstein, Morris, Graciela Kaminsky, and Carmen Reinhart, 2000, *Assessing Financial Vulnerability: An Early Warning System for Emerging Markets* (Washington: Institute for International Economics).
- Goodhart, Charles A. E., Boris Hofmann, and Miguel A. Segoviano, 2004, "Bank Regulation and Macroeconomic Fluctuations," *Oxford Review of Economic Policy*, Vol. 20, No. 4, pp. 591–615.
- Goodhart, Charles A. E., and Miguel A. Segoviano, 2004, "Basel and Procyclicality: A Comparison of the Standardized and IRB Approaches to an Improved Credit Risk Method," Discussion Paper No. 524 (London: Financial Markets Group, London School of Economics). Available via the Internet: <http://eprints.lse.ac.uk/24821>
- Gupton, Greg M., Chris C. Finger, and Mickey Bhatia, 1997, *Creditmetrics: The Benchmark for Understanding Risk* (New York: JP Morgan)
- Heiskanen, Reijo, 1993, "The Banking Crisis in the Nordic Countries," *Kansallis Economic Review*, Vol. 2, pp. 13–19.
- Hosking, Jonathan R. M., Gabriel Bonti, and Dirk Siegel, 2000, "Beyond the Lognormal," *Risk*, Vol. 13, No. 5, pp. 59–62.
- Jaynes, Edwin T., 1957, "Information Theory and Statistical Mechanics," *Physical Review*, Vol. 106, No. 4, pp. 620–30.
- , 1984, "Prior Information and Ambiguity in Inverse Problems," in *Inverse Problems*, SIAM-AMS Proceedings, edited by D. W. McLaughlin, Vol. 14, pp. 151–66 (Providence: American Mathematical Society).

- Kaminsky, Graciela, Saul Lizondo, and Carmen Reinhart, 1998, "Leading Indicators of Currency Crises," *IMF Staff Papers*, Vol. 45, No. 1, pp. 1–48. Available via the Internet: <http://www.imf.org/external/Pubs/FT/staffp/1998/03-98/kaminsky.htm>
- Kaminsky, Graciela, and Carmen Reinhart, 1996, "Banking and Balance of Payments Crises: Models and Evidence," Working Paper No. 544, International Finance Discussion Papers (Washington: Board of Governors of the Federal Reserve System). Available via the Internet: <http://www.federalreserve.gov/pubs/ifdp/1996/544/default.htm>
- , 1999, "The Twin Crisis: The Causes of Banking and Balance of Payments Problems," *American Economic Review*, Vol. 89, No. 3, pp. 473–500.
- Kaminsky, Graciela, and Sergio Schmukler, 2003, "Short-Run Pain, Long-Run Gain: The Effects of Financial Liberalization," NBER Working Paper No. 9787 (Cambridge, MA: National Bureau of Economic Research). Available via the Internet: <http://www.nber.org/papers/w9787>
- Kiyotaki, Nobuhiro, and John Moore, 1997, "Credit Cycles," *Journal of Political Economy*, Vol. 105, No. 2, pp. 211–48.
- Krugman, Paul, 1979, "A Model of Balance of Payments Crises," *Journal of Money, Credit and Banking*, Vol. 11, No. 3, pp. 311–25.
- Kullback, Solomon, 1959, *Information Theory and Statistics* (New York: John Wiley).
- Kullback, Solomon, and R. Leibler, 1951, "On Information and Sufficiency," *Annals of Mathematical Statistics*, Vol. 22, No. 1, pp. 79–86.
- McKinnon, Ronald I., and Huw Pill, 1996, "Credible Liberalizations and International Capital Flows: The Over Borrowing Syndrome," in *Financial Deregulation and Integration in East Asia*, edited by Takatoshi Ito and Anne O. Krueger, pp. 7–45 (Chicago: University of Chicago Press).
- McLachlan, Geoff J., and Kaye E. Basford, 1987, *Mixture Models: Inference and Applications to Clustering* (New York and Basel: Marcel Dekker).
- Merton, Robert C., 1974, "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates," *Journal of Finance*, Vol. 29, No. 2, pp. 449–70.
- Mina, Jorge, and Jerry Yi Xiao, 2001, "Return to RiskMetrics: The Evolution of a Standard" (New York: RiskMetrics).
- Minsky, Hyman P., 1982, *Can "It" Happen Again? Essays on Instability and Finance* (Armonk, NY: M. E. Sharpe).
- Mishkin, Frederic S., 1997, "Understanding Financial Crises: A Developing Country Perspective," in *Annual World Bank Conference on Development Economics 1996*, edited by Michael Bruno and Boris Pleskovic, pp. 29–61 (Washington: World Bank).
- Obstfeld, Maurice, 1995, "Models of Currency Crises with Self-Fulfilling Features," NBER Working Paper No. 5285 (Cambridge, MA: National Bureau of Economic Research). Available via the Internet: <http://www.nber.org/papers/w5285>
- Obstfeld, Maurice, and Kenneth Rogoff, 1995, "The Mirage of Fixed Exchange Rates," NBER Working Paper No. 5191 (Cambridge, MA: National Bureau of Economic Research). Available via the Internet: <http://www.nber.org/papers/w5191>
- Organization for Economic Cooperation and Development, 2006, "Economic Survey of Denmark, 2006" (Paris, May).
- Reinhart, Carmen, and Ioannis Tokatlidis, 2001, "Before and After Financial Liberalization" (unpublished paper; College Park: University of Maryland).
- Saurina, Jesus, and Carlos Trucharte, 2004, "The Impact of Basel II on Lending to Small and Medium-Sized Firms: A Regulatory Policy Assessment Based on Spanish Credit Register Data," *Journal of Financial Services Research*, Vol. 26, No. 2, pp. 121–44.
- Segoviano, Miguel A., 2006a, "The Conditional Probability of Default Methodology," Discussion Paper No. 558 (London: Financial Markets Group, London School of Economics). Available via the Internet: <http://eprints.lse.ac.uk/24512/>
- , 2006b, "The Consistent Information Multivariate Density Optimizing Methodology," Discussion Paper 557 (London: Financial Markets Group, London School of Economics). Available via the Internet: <http://eprints.lse.ac.uk/24511/>
- Segoviano, Miguel A., and Philip Lowe, 2002, "Internal Ratings, the Business Cycle and Capital Requirements: Some Evidence from an Emerging Market Economy," BIS Working Paper No. 117 (Basel: Bank for International Settlements). Available via the Internet: <http://www.bis.org/publ/work117.htm>
- Segoviano, Miguel A., and Pablo Padilla, 2006, "Portfolio Credit Risk and Macroeconomic Shocks: Applications to Stress Testing under Data-Restricted Environments," IMF Working Paper 06/283 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=20064>
- Shannon, Claude E., 1948, "A Mathematical Theory of Communication," *Bell System Technical Journal*, Vol. 27, pp. 379–423.
- Vassalou, Maria, and Yuhang Xing, 2004, "Default Risk in Equity Returns," *Journal of Finance*, Vol. 59, No. 2, pp. 831–68.
- Zangari, P., 1996, "An Improved Methodology for Measuring VaR," *RiskMetrics Monitor*, No. 2, pp. 7–25.

Appendix

Entropy in a Nutshell

Given nonexperimental data generation processes in economics and finance, and the corresponding data available for applied econometric analysis, economists face the problem of how to make predictions for the population or how to construct an image of it, using the information at hand. Unfortunately, in many cases, the models that economists use for analysis contain unknowns that are unobservable and indeed that are not accessible to direct measurement. Consequently, in order to recover the unknown parameters representing the economic system of interest, economists are faced with an inverse problem that can be formalized as follows. Our interest focuses on the unknown and unobservable frequencies $\mathbf{p} = (p_1, p_2, \dots, p_K)$ that represent the data-generating process. However, it is the outcome of an economic variable $\mathbf{y}$ that is observable. Thus, in order to recover $\mathbf{p}$, the indirect measurements on the observable $\mathbf{y}$ must be used. Equivalently, from partial information $\mathbf{y}$, we try to define $\mathbf{p}$. In this context, consider the following finite, discrete, linear problem:

$$\mathbf{Y} = \mathbf{X}\mathbf{p}, \quad (31.15)$$

where $\mathbf{Y} = (y_1, y_2, \dots, y_T)'$ is a T dimensional vector of observations (data), $\mathbf{p}$ is an unobservable K dimensional vector of frequencies, and $\mathbf{X}$ is a known ($T \times (K > T)$) noninvertible matrix. Thus, out of all the probability distributions that satisfy equation (31.15) and fulfill the conditions $\sum_k p_k = 1$ and $p_k \geq 0$, we are asked to recover or choose unambiguous probabilities p_k . Given this specification, the information contained in equation (31.15) does not appear adequate to determine the unknown probabilities $\mathbf{p}$. That is because the number of data points are less than the number of unknowns. In its present form, the problem appears underspecified; that is, the problem is ill posed, and the basis for assigning a probability is, at this point, unresolved. In these circumstances, creative assumptions are usually employed to induce a well-posed problem that is amenable to solution. However, the imposition of inconsistent assumptions may lead to erroneous interpretations and conclusions.

Because usually we must cope with partial-incomplete information and ill-posed inverse problems, we would like to use a principle or formalism that provides us with the “best” conclusions possible based on the data available. In looking for such a principle, or formalism, two requirements appear essential (Jaynes, 1984). We know something, but we do not know everything, or perhaps not enough to proceed in a traditional way, and we do not want to claim any more or any less than we know. These are the guidelines in which the entropy principle is based.

The entropy formalism seeks to make the “best” predictions possible from the information that is available and provides a basis for transforming the latter into a distribution of probabilities describing our state of knowledge.

ENTROPY: MODELING FOUNDATIONS

The “entropy of the distribution of probabilities” is an axiomatic method used by Shannon (1948) to define a unique function to measure the uncertainty of a collection of events.³³ In developing this approach, Shannon (1948) supposed that an experiment with N trials (repetitions) was carried out. This experiment had K possible outcomes (states). He assumed that $N_1, N_2, \dots, N_K$ represented the number of times that each outcome occurs in the experiment of length N , where $\sum_k N_k = N$, $N_k \geq 0$, and $k = 1, 2, \dots, K$.

In this setting, there are N trials and each trial has K possible outcomes; therefore, there are K^N conceivable outcomes in the sequence of N trials. Of these, a particular set of frequencies,

$$p_k = \frac{N_k}{N} \quad \text{or} \quad N_k = Np_k \quad \text{for } k = 1, 2, \dots, K,$$

can be realized in a given number of ways as measured by the multiplicity factor (possible permutations). Thus, the number of ways that a particular set of $N_{\{k\}}$ is realized can be represented by the multinomial coefficient

$$W = \frac{N!}{Np_1! Np_2! \dots Np_K!} = \frac{N!}{\prod_k N_k!},$$

³³ The origins of the entropy concept go back to the 19th century with the work developed by Boltzman and continued subsequently by Maxwell, Gibbs, Bernoulli, and Laplace.

$$\text{or its monotonic function, } \ln W = \ln N! - \sum_{k=1}^K \ln N_k! \quad (31.16)$$

Shannon (1948) manipulated equation (31.16), as presented in Segoviano (2006b), to obtain the Shannon (1948) entropy measure, which is defined as

$$H(p) = -\sum_{k=1}^K p_k \ln p_k, \text{ where } p_k \ln p_k = 0 \text{ for } \ln p_k = 0. \quad (31.17)$$

Jaynes (1957) proposed to make use of the entropy concept to choose an unknown distribution of probabilities when only partial information is available. He proposed to maximize the function presented in equation (31.17), subject to the limited available data, in order to obtain the probability vector $\mathbf{p}$ that can be realized in the greatest number of ways consistent with the known data.

The rationale provided by Jaynes (1957) for choosing a particular solution, that is, probability vector $\mathbf{p}$ from partial information, is known as the principle of maximum entropy. Let

$$L = -\sum_k^K p_k \ln p_k + \sum_{t=1}^T \lambda_t \left[y_t - \sum_k^K p_k f_t(x_k) \right] + \mu \left[1 - \sum_k^K p_k \right] \quad (31.18)$$

be the Lagrange function. Then, the problem of maximum entropy is to maximize L . In this function, the information contained in the data has been formalized in $1 \leq t \leq T$ moment-consistency constraints of the form $\sum_{k=1}^K p_k f_t(x_k) = y_t$. These moment-consistency constraints are formulated with T functions $\{f_1(x), f_2(x), \dots, f_T(x)\}$ representing the information contained in the data and with a set of observations (averages or aggregates) $\{y_1(x), y_2(x), \dots, y_T(x)\}$ that are consistent with the distribution of probabilities $\{p_1, p_2, \dots, p_k\}$. In this function, the additivity restriction $\sum_{k=1}^K p_k = 1$ has to be fulfilled as well, because $\mathbf{p}$ represents a probability distribution. Note also that λ_t represents the Lagrange multiplier of each of the $1 \leq t \leq T$ moment-consistency constraints and represents the Lagrange multiplier of the probability additivity constraint. Using the method of Lagrange multipliers, the maximum entropy solution is given by

$$\hat{p}_k = \frac{1}{\sum_{k=1}^K \exp \left[-\sum_{t=1}^T \hat{\lambda}_t \hat{f}_t(x_k) \right]} \exp \left[\sum_{t=1}^T \hat{\lambda}_t \hat{f}_t(x_k) \right]. \quad (31.19)$$

An extension to the rationale provided by Jaynes (1957) is the minimum cross-entropy distribution (MXED) developed by Kullback (1959). Under the MXED, it is assumed that, in addition to the T moment constraints, some form of conceptual knowledge exists about the properties of the system and that this knowledge can be expressed in the form of a prior probability vector $\mathbf{q}$. In contrast to the principle of maximum entropy pure inverse problem framework, the objective may be reformulated as being to minimize the cross-entropy distance between the posterior $\mathbf{p}$ and the prior $\mathbf{q}$. The cross-entropy objective function is defined as follows:

$$C[p_k, q_k] = \sum_k^K p_k \ln \left[\frac{p_k}{q_k} \right], \quad (31.20)$$

subject to T moment-consistency constraints $\sum_{k=1}^K p_k f_t(x_k)$ and the additivity restriction $\sum_{k=1}^K p_k$.

Consequently, the probability vector $\mathbf{p}$ may be recovered by minimizing the Lagrangian function:

$$L = \sum_k^K p_k \ln \left[\frac{p_k}{q_k} \right] + \sum_{t=1}^T \lambda_t \left[y_t - \sum_k^K p_k f_t(x_k) \right] + \mu \left[1 - \sum_k^K p_k \right], \quad (31.21)$$

where λ and μ represent Lagrange multipliers. Using the method of Lagrange multipliers, the optimal solution to the cross-entropy problem is

$$\hat{p}_k = \frac{q_k}{\sum_{k=1}^K q_k \exp \left[\sum_{t=1}^T \hat{\lambda}_t \hat{f}_t(x_k) \right]} \exp \left[\sum_{t=1}^T \hat{\lambda}_t \hat{f}_t(x_k) \right]. \quad (31.22)$$

Shannon's entropy measure $\Sigma(p_k)$, presented in equation (31.17), can be interpreted as a measure of the degree of uncertainty in the distribution of probabilities, representing the extent to which the distribution is concentrated on specific points as opposed to

being dispersed over many points. $\Sigma(p_k)$ takes its maximum value $\ln K$ when the distribution is maximally dispersed and thus uniformly distributed on the range of the distribution, in other words, when $p_1 = p_2 = \dots = p_k = \frac{1}{K}$. $\Sigma(p_k)$ takes its minimum value 0 when the distribution is maximally informative in that $\mathbf{p}$ degenerates on one particular x_k value, in other words, $p_k = 1$ and $p_j = 0$, $\forall k \neq j$. The objective of maximizing $\Sigma(p_k)$ in the absence of any constraints, other than the additivity restriction $\sum_{k=1}^K p_k = 1$, can be interpreted as choosing p_k to be the maximally uniform distribution. This is because, as mentioned, $\Sigma(p_k)$ takes its maximum value when $p_1 = p_2 = \dots = p_k = \frac{1}{K}$.

However, in the presence of T moment constraints, the objective can be interpreted as choosing the p_k s to be as maximally uninformative as the moment constraints allow. This objective is consistent with the goal of not wanting to assert more of the distribution p_k than is known. In other words, irrelevant information is “minimized out.” When, in addition to the T moment constraints, supplementary information in the form of a prior probability is incorporated into the optimization framework, the Kullback cross-entropy framework estimates the density that could have been recovered in the greatest number of ways consistent with these constraints and that has the smallest entropic distance from the prior distribution. The entropic distance between p_k and q_k is not a metric distance,³⁴ but it does satisfy

$$C[p_k, q_k] = 0 \quad \text{for } p_k = q_k; \quad \text{and} \quad C[p_k, q_k] > 0, \quad \text{whenever } p_k \neq q_k.$$

Note that the definition of $C[p_k, q_k]$ does not contain a negative sign, so cross-entropy is minimized rather than maximized. Note also that equation (31.22) reduces to equation (31.19) when $q(x)$ is constant, that is, a uniform distribution indicating no prior information about X .

³⁴ This is because $C[p(x), q(x)] \neq C[q(x), p(x)]$; however, for our objective, this property is not particularly important because the prior distribution is taken to be fixed in estimation problems. Our interest centers on choosing the posterior distribution that solves the moment equations and that is as close as possible to the prior distribution.

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Banking Stability Measures

MIGUEL A. SEGOVIANO • CHARLES A. E. GOODHART

This chapter defines a set of banking stability measures that take account of distress dependence among the banks in a system, thereby providing a set of tools to analyze stability from complementary perspectives by allowing the measurement of (1) common distress of the banks in a system; (2) distress between specific banks; and (3) distress in the system associated with a specific bank. Our approach defines the banking system as a portfolio of banks and infers the Banking System's (Portfolio) Multivariate Density (BSMD) from which the proposed measures are estimated. The BSMD embeds the banks' default interdependence structure that captures linear and nonlinear distress dependencies among the banks in the system and its changes at different times of the economic cycle. The BSMD is recovered using the Consistent Information Multivariate Density Optimizing approach, a new approach that in the presence of restricted data improves density specification without explicitly imposing parametric forms that, under restricted data sets, are difficult to model. Thus, the proposed measures can be constructed from a very limited set of publicly available data and can be provided for a wide range of both developing and developed economies.

METHOD SUMMARY

Overview	Banking stability measures embed banks' linear (correlation) and nonlinear distress dependence and their changes through the economic cycle. The approach defines the banking system as a portfolio of banks and infers its multivariate density from which the proposed measures are estimated. These can be provided for developed and developing economies and can be estimated under data-restricted environments.
Application	The proposed stability measures incorporate changes in distress dependence that are consistent with the economic cycle; this is a key advantage over traditional risk models that typically incorporate only linear dependence (correlation structure) and assume it to be constant throughout the economic cycle. The framework is appropriate in situations where 1. data are limited; and 2. distress dependence among financial institutions in a financial system needs to be quantified.
Nature of approach	Statistical nonparametric density estimation.
Data requirements	Probabilities of default (PDs): these can be market based (credit default swap spreads, bond spreads, structural models); they also can be estimated with supervisory information. Stock prices of financial institutions.
Strengths	The framework allows the estimation of financial stability measures that provide complementary perspectives on risk, with very limited data requirements. These measures can be updated with daily frequency.
Weaknesses	It is necessary to have robust estimations of PDs. If PDs are not available, it is necessary to make assumptions about this variable.
Tool	Proprietary estimation codes are available from the authors upon request. Contact author: M. Segoviano.

Economics is a quantitative science. Macroeconomics depends on data for national income, expenditure, and output variables. Macromonetary policy requires measures of inflation. Microeconomics is based on data for prices and quanti-

ties of inputs and outputs. Even when the variables of concern are difficult to measure, such as the output gap, expectations, and "happiness," economists use various techniques, for example, survey data, to provide quantitative proxy variables—

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often according these data more weight than is consistent with their inherent measurement errors. Without such quantification, comparisons over time, and on a cross-sectional basis, cannot be made; nor would it be easy to provide a quantified analysis of the determinants of such variables.

However, there is currently no such widely accepted measure, quantification, or time series for measuring either financial or banking stability. What is most often used instead is an on/off (1/0) assessment of whether a “crisis” has occurred.¹ This then has been used to review whether there have been common factors preceding, possibly even causing, such crises and to assess what official responses have best mitigated such crises (see Hoelscher, 2006). Although much useful research employing a crisis on/off dichotomy has been done, this approach has several inherent deficiencies.² In particular, the lack of a continuous scale makes it impossible to measure with sufficient accuracy either (1) the relative riskiness of the system in noncrisis mode; or (2) the intensity of a crisis once it has started. If the former could be measured, it may be easier to take early remedial action as the danger of a systemic crisis increases, while measurement of the latter would facilitate decision making on the most appropriate measures to address the crisis.

In our view, a precondition for improving the analysis and management of financial (banking) stability is to be able to construct a metric for it.³ The purpose of this study therefore is to present a method for estimating a set of stability measures for the banking system, or banking stability measures (BSMs). There is no unique, best way to estimate BSMs, any more than there is a unique best way to measure such concepts as “output” or “inflation,” but we hope to demonstrate that our approach is reasonable. Moreover, there are very few alternative available measures that might be used.

Briefly, and as described in far more detail in the subsequent discussion, we conceptualize the banking system as a portfolio of banks comprising the core, systemically important banks in any country. Thus, we infer the Banking System’s (Portfolio) Multivariate Density (BSMD), from which we construct a set of BSMs. These measures embed the banks’ distress interdependence structure, which captures not only linear (correlation) but also nonlinear distress dependencies among the banks in the system. Moreover, the structure of linear and nonlinear distress dependencies changes as banks’ probabilities of distress (PoDs) change; hence, the proposed stability measures incorporate changes in distress dependence that are consistent with the economic cycle. This is a key advantage over traditional risk models that most of the time incorporate only linear dependence (correlation structure) and assume it constant throughout the economic

cycle.⁴ Consequently, the proposed BSMs represent a set of tools to analyze (define) stability from four different yet complementary perspectives, by allowing the quantification of (1) “common” distress in the banks of the system, (2) distress between specific banks, and (3) distress in the system associated with a specific bank.

In this chapter, we estimate the proposed BSMs using publicly available information from 2005 up to the beginning of October 2008. These estimations are to illustrate the methodology rather than to make an assessment of the conjunctural financial stability of any particular system. We examine relative changes in stability over time and among different banks’ business lines in the U.S. banking system. We also analyze cross-region effects between American and European banking groups. Last, we show how our technique can be extended to incorporate the effect of foreign banks on sovereigns with banking systems with cross-border institutions. For this purpose, we estimate the BSMs for major foreign banks and sovereigns in Latin America, Eastern Europe, and Asia. This implementation flexibility is of relevance for banking stability surveillance, because cross-border financial linkages are growing and becoming significant, as has been highlighted by the financial market turmoil of recent years. Thus, surveillance of banking stability cannot stop at national borders.

We show how these BSMs can be constructed from a very limited set of data, for example, empirical measurements of distress of individual banks. Such measurements can be estimated using alternative approaches, depending on data availability; thus, the data set that is necessary to estimate the BSMs is available in most countries. Consequently, such measures can be provided for a wide group of developing as well as developed economies. Establishing such a set of measures with a minimum of basic components makes it feasible to undertake a wider range of comparative analysis, both time series and cross section. It is important to note that when measurements of distress of nonbanking financial institutions (NBFIs), that is, insurance companies, hedge funds, and others, are available, our methodology can be extended easily to incorporate the effects of such institutions in the measurement of stability, hence allowing us to estimate a set of stability measures for the *financial system*. This could be of relevance for countries where NBFIs have systemic importance in the financial sector.

¹ For a review of the literature on financial crises, see Bordo and others (2001).

² See Kaminsky and Reinhart (1999).

³ See Kiyotaki and Moore (1997) and Krugman (1979).

⁴ In contrast to correlation, which captures only linear dependence, copula functions characterize the whole dependence structure—that is, linear and nonlinear dependence, embedded in multivariate densities (Nelsen, 1999). Thus, in order to characterize banks’ distress dependence, we employ a novel nonparametric copula approach, that is, the CIMDO copula. In comparison to traditional methodologies to model parametric copula functions, the CIMDO copula avoids the difficulties of explicitly choosing the parametric form of the copula function to be used and calibrating its parameters, because CIMDO copula functions are inferred directly (implicitly) from the joint movements of the individual banks’ PoDs.

Economically, our approach is based on the microfounded, general equilibrium theoretical framework of Goodhart, Suniand, and Tsomocos (2006), which indicates that financial instability can arise through systemic shocks, contagion after idiosyncratic shocks, or a combination of both. In this respect, our proposed banking system stability measures represent a clear improvement over purely statistical or mathematical models that are economically atheoretical and therefore difficult to interpret.

Over the past nearly two decades, we have completed a research agenda that has allowed us to gain important insights into the analysis of financial stability. For example, in Segoviano (1998), Segoviano and Lowe (2002), and Goodhart and Segoviano (2004), we present simple approaches that vary in the degree of sophistication to quantify portfolio credit risk (all of these approaches were based on parametric models and accounted only for correlations that were fixed through the cycle) and analyze the procyclicality of banking regulation and its implications for financial stability. In Segoviano and Padilla (2006), we present a framework for macroeconomic stress testing combined with a model for portfolio credit risk evaluation, which accounts for linear and nonlinear dependencies among the assets in banks' portfolios and their changes across the economic cycle; however, in all these cases, we focus on individual banks' portfolios or on the overall aggregate banking system. In Goodhart, Hofmann, and Segoviano (2004, 2006) and Aspachs and others (2006), we focus on systemic risk, making use of average measurements of distress of the system, which do not incorporate banks' distress dependencies or their changes across the economic cycle.

Thus, we hope that the proposed BSMs will allow us to complement our previous research and gain further insights into our understanding of financial stability. We therefore are extending our research as follows in order to achieve specific aims:

1. We examine relative changes in the BSMs over time and between countries, in order to identify the occasion and determinants of changes in the riskiness of the banking system.
2. We try to predict future movements of the BSMs for use as an early-warning mechanism.
3. We explore the significant macroeconomic and financial factors and shocks influencing the BSMs, in order to identify macro-financial linkages.
4. We explore the factors that can limit and reverse tendencies toward instability, so as to discover what instruments may be available (and under what conditions) to control such instability.

In Section 1, we explain the importance of incorporating banks' distress dependence in the estimation of the stability of the banking system and describe the modeling steps followed in our framework. In Section 2, we present the Consistent Information Multivariate Density Optimizing (CIMDO) methodology to infer the BSMD from which the

proposed BSMs are estimated. These are defined in Section 3. In Section 4, we present empirical estimates of the proposed BSMs as described. Finally, conclusions are presented in Section 5.⁵

1. DISTRESS DEPENDENCE AMONG BANKS AND STABILITY OF THE BANKING SYSTEM

The proper estimation of distress dependence among the banks in a system is of key importance for the surveillance of stability of the banking system. Financial supervisors recognize the importance of assessing not only the risk of distress—that is, large losses and possible default of a specific bank—but also the impact that such an event would have on other banks in the system. Clearly, the event of simultaneous large losses in various banks would affect a banking system's stability and thus represents a major concern for supervisors. Banks' distress dependence is based on the fact that banks are usually linked—either directly, through the interbank deposit market and participations in syndicated loans, or indirectly, through lending to common sectors and proprietary trades. Banks' distress dependence varies across the economic cycle and tends to rise in times of distress because the fortunes of banks decline concurrently through either contagion after idiosyncratic shocks, affecting interbank deposit markets and participations in syndicated loans (direct links), or through negative systemic shocks, affecting lending to common sectors and proprietary trades (indirect links). Therefore, in such periods, the banking system's joint probability of distress (JPoD)—that is, the probability that all the banks in the system experience large losses simultaneously, which embeds banks' distress dependence—may experience larger and nonlinear increases than those experienced by the probabilities of distress (PoDs) of individual banks.

Consequently, it becomes essential for the proper estimation of the banking system's stability to incorporate banks' distress dependence and its changes across the economic cycle. Quantitative estimation of distress dependence, however, is a difficult task. Information restrictions and difficulties in modeling distress dependence arise due to the fact that distress is an extreme event and can be viewed as a tail event that is defined in the “distress region” of the probability distribution that describes the implied asset price movements of a bank (Figure 32.1).⁶ The fact that distress is a tail event makes the often used correlation coefficient inadequate to capture bank distress dependence and the standard approach to model parametric copula functions difficult to implement. Our methodology embeds a *reduced-form* or *nonparametric*

⁵ See Goodhart (1995); Areta and Eichengreen (2000); Altman, Elitzondo, and Segoviano (2002); and Goodhart and Hofmann (2003).

⁶ See Black and Scholes (1973); Merton (1974); Butler and Schachter (1998); and Crosbie and Bohn (2003).

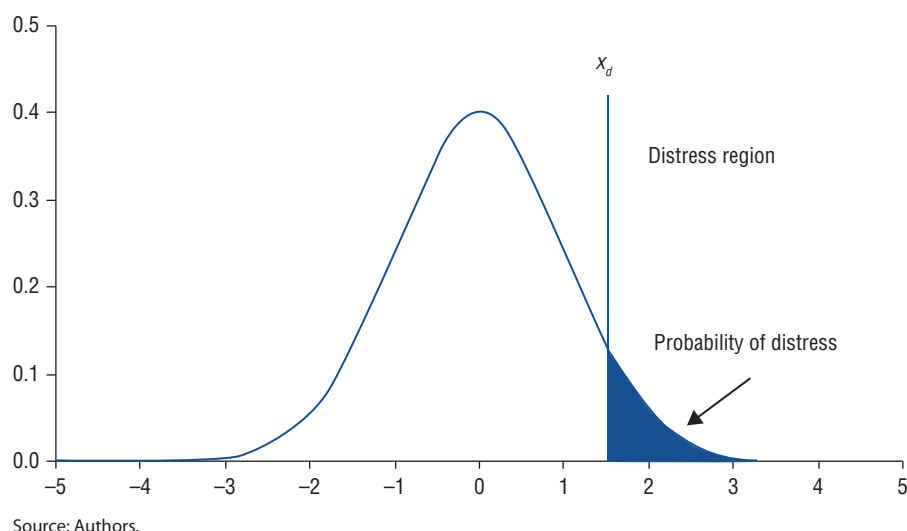
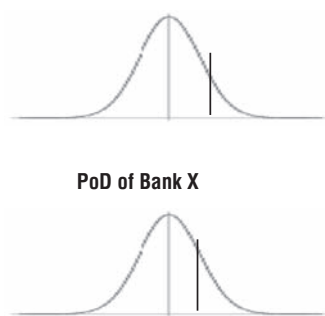


Figure 32.1 The Probability of Distress

Step 1:
View the financial system as a portfolio of banks.



Step 2:
Estimate individual FIs' Probabilities of Distress (PoDs)

Step 3:
Recover the Financial System Multivariate Density
It characterizes the implied asset values of the portfolio of FIs and the distress dependence among these FIs.

Step 4:
Estimate Financial Stability measures.
Based on conditional Probabilities of distress

Source: Authors.
Note: FI = financial institution.

Figure 32.2 The Banking System's Multivariate Density

approach to model copulas that seems to adequately capture default dependence and its changes at different points of the economic cycle. This methodology is easily implementable under the data constraints affecting bank default dependence modeling and produces robust estimates under the Probability Integral Transformation (PIT) criterion.⁷

In our modeling of banking systems' stability and distress dependence, we follow four steps (Figure 32.2):

Step 1: We conceptualize the banking system as a *portfolio* of banks.

Step 2: For each of the banks included in the portfolio, we obtain empirical measurements of PoDs.

Step 3: Making use of the CIMDO methodology, presented in Segoviano (2006b) and summarized in subsequent discussion, and taking as input variables the individual banks' PoDs estimated in the previous step, we recover the BSMD.

Step 4: On the basis of the BSMD, we estimate the proposed BSMs.

Section 2 describes the procedure to recover the BSMD.

2. BANKING SYSTEM'S MULTIVARIATE DENSITY

The BSMD characterizes both the individual and joint asset value movements of the portfolio of banks representing the banking system. The BSMD is recovered using the

⁷ The PIT criterion for multivariate density's evaluation is presented in Diebold, Hahn, and Tay (1999).

CIMDO methodology (Segoviano, 2004, 2006a, 2006b). The BSMD embeds the banks' distress dependence structure—characterized by the CIMDO copula function (Segoviano and Goodhart, 2009)—that captures linear and nonlinear distress dependencies among the banks in the system and allows for these to change throughout the economic cycle, reflecting the fact that dependence increases in periods of distress. These are key technical improvements over traditional risk models, which usually account only for linear dependence (correlations) that are assumed to remain constant over the cycle or a fixed period of time. In order to show such improvements in the modeling of distress dependence—thus, in our proposed measures of stability—in what follows, we (1) model the BSMD using the CIMDO methodology; and (2) illustrate the advantages embedded in the CIMDO copula to characterize distress dependence among the banks in the banking system.

A. The CIMDO approach: modeling the banking system's multivariate density

We recover the BSMD by employing the CIMDO methodology and empirical measures of PoDs of individual banks. There are alternative approaches to estimate individual banks' PoDs. For example, we analyzed (1) the structural approach; (2) credit default swaps (CDS); and (3) out-of-the-money (OOM) option prices. These are discussed further in Section 4.⁸ It is important to emphasize that individual banks' PoDs are exogenous variables in the CIMDO framework; thus, the framework can be implemented with any alternative approach to estimate PoDs. Consequently, this provides great flexibility in the estimation of the BSMD.

The CIMDO methodology is based on the minimum cross-entropy approach (Kullback, 1959). Under this approach, a posterior multivariate distribution p —the CIMDO density—is recovered using an optimization procedure by which a prior density q is updated with empirical information via a set of constraints. Thus, the posterior density satisfies the constraints imposed on the prior density. In this case, the banks' empirically estimated PoDs represent the information used to formulate the constraint set. Accordingly, the CIMDO density—the BSMD—is the posterior density that is closest to the prior distribution and that is consistent with the empirically estimated PoDs of the banks making up the system.⁹

In order to formalize these ideas, we proceed by defining a banking system—portfolio of banks—comprising two banks, Bank X and Bank Y, whose logarithmic returns are characterized by the random variables x and y . Hence, we define the CIMDO objective function as follows:¹⁰

$$C[p, q] = \iint p(x, y) \ln \left[\frac{p(x, y)}{q(x, y)} \right] dx dy, \\ \text{where } q(x, y) \text{ and } p(x, y) \in \mathbb{R}^2.$$

It is important to point out that the prior distribution follows a parametric form q that is consistent with economic intuition (e.g., default is triggered by a drop in the firm's asset value below a threshold value) and with theoretical models (i.e., the structural approach to modeling risk). However, the parametric density q is usually inconsistent with the empirically observed measures of distress. Hence, the information provided by the empirical measures of distress of each bank in the system is of prime importance for the recovery of the posterior distribution. In order to incorporate this information into the posterior density, we formulate consistency-constraint equations that have to be fulfilled when optimizing the CIMDO objective function. These constraints are imposed on the marginal densities of the multivariate posterior density and are of the form

$$\iint p(x, y) \chi_{(x_d^x, \infty)} dx dy = PoD_i^x, \\ \iint p(x, y) \chi_{(x_d^y, \infty)} dy dx = PoD_i^y, \quad (32.1)$$

where $p(x, y)$ is the posterior multivariate distribution that represents the unknown to be solved. PoD_i^x and PoD_i^y are the empirically estimated PoDs of each of the banks in the system, and $\chi_{(x_d^x, \infty)}, \chi_{(x_d^y, \infty)}$ are indicating functions defined with the distress thresholds x_d^x, x_d^y , estimated for each bank in the portfolio. In order to ensure that the solution for $p(x, y)$ represents a valid density, the conditions that $p(x, y) \geq 0$ and the probability additivity constraint $\iint p(x, y) dx dy = 1$, also need to be satisfied. Once the set of constraints is defined, the CIMDO density is recovered by minimizing the functional:

$$L[p, q] = \iint p(x, y) \ln p(x, y) dx dy - \iint p(x, y) \ln q(x, y) dx dy \\ + \lambda_1 \left[\iint p(x, y) \chi_{(x_d^x, \infty)} dx dy - PoD_i^x \right] \\ + \lambda_2 \left[\iint p(x, y) \chi_{(x_d^y, \infty)} dy dx - PoD_i^y \right] \\ + \mu \left[\iint p(x, y) dy dx - 1 \right], \quad (32.2)$$

where λ_1, λ_2 represent the Lagrange multipliers of the consistency constraints, and μ represents the Lagrange multiplier of the probability additivity constraint. By using the calculus of variations, the optimization procedure can be performed. Hence, the optimal solution is represented by a posterior multivariate density that takes the following form:

$$\hat{p}(x, y) = q(x, y) \exp \left\{ - \left[1 + \hat{\mu} + \left(\hat{\lambda}_1 \chi_{(x_d^x, \infty)} \right) \right. \right. \\ \left. \left. + \left(\hat{\lambda}_2 \chi_{(x_d^y, \infty)} \right) \right] \right\}. \quad (32.3)$$

Intuitively, imposing the constraint set on the objective function guarantees that the posterior multivariate distribution—the BSMD—contains marginal densities that satisfy the PoDs observed empirically for each bank in the

⁸ See Kullback and Leibler (1951) and Golan and Judge (1992).

⁹ See Jaynes (1957, 1984).

¹⁰ A detailed definition and development of the CIMDO objective function and constraint set, as well as the optimization procedure that is followed to solve the CIMDO functional, is presented in Segoviano (2006b).

banking portfolio. CIMDO-recovered distributions outperform the most commonly used parametric multivariate densities in the modeling of portfolio risk under the PIT criterion.¹¹ This is because when recovering multivariate distributions through the CIMDO approach, the available information embedded in the constraint set is used to adjust the “shape” of the multivariate density via the optimization procedure described. This appears to be a more efficient manner of using the empirically observed information than under parametric approaches, which adjust the “shape” of parametric distributions via fixed sets of parameters. A detailed development of the PIT criterion and Monte Carlo studies used to evaluate specifications of the CIMDO density are presented in Segoviano (2006b).

B. The CIMDO copula: distress dependence among banks in the system

The BSMD embeds the structure of linear and nonlinear default dependence among the banks included in the portfolio that is used to represent the banking system. Such dependence structure is characterized by the copula function of the BSMD, that is, the CIMDO copula, which changes at each period of time consistently with changes in the empirically observed PoDs. In order to illustrate this point, we heuristically introduce the copula approach to characterize dependence structures of random variables and explain the particular advantages of the CIMDO copula.

The Copula Approach

The copula approach is based on the fact that any multivariate density, which characterizes the stochastic behavior of a group of random variables, can be broken into two subsets of information: (1) information of each random variable, that is, the marginal distribution of each variable; and (2) information about the dependence structure among the random variables. Thus, in order to recover the latter, the copula approach sterilizes the marginal information of each variable, consequently isolating the dependence structure embedded in the multivariate density. Sterilization of marginal information is done by transforming the marginal distributions into uniform distributions: $\mathbf{U}(0,1)$, which are *uninformative* distributions.¹² For example, let x and y be two random variables with individual distributions $x \sim F$, $y \sim H$ and a joint distribution $(x, y) \sim G$. To transform x and y into two random variables with uniform distributions $\mathbf{U}(0,1)$, we define two new variables as $u = F(x)$, $v = H(y)$ both distributed as $\mathbf{U}(0,1)$ with joint density $c[u, v]$. Under the distribution of

transformation of random variables, the copula function $c[u, v]$ is defined as

$$c[u, v] = \frac{g[F^{-1}(u), H^{-1}(v)]}{f[F^{-1}(u)]h[H^{-1}(v)]}, \quad (32.4)$$

where g , f , and h are defined densities. From equation (32.4), we see that copula functions are multivariate distributions, whose marginal distributions are uniform on the interval $[0,1]$. Therefore, because each of the variables is individually (marginally) uniform—that is, their information content has been sterilized—their joint distribution will contain only dependence information. Rewriting equation (32.4) in terms of x and y , we get

$$c[F(x), H(y)] = \frac{g[x, y]}{f[x]h[y]}. \quad (32.5)$$

From equation (32.5), we see that the joint density of u and v is the ratio of the joint density of x and y to the product of the marginal densities. Thus, if the variables are independent, equation (32.5) is equal to one.

The copula approach to model dependence possesses many positive features when compared with correlations (see Box 32.1). In comparison to correlation, the dependence structure as characterized by copula functions describes linear and nonlinear dependencies of any type of multivariate densities and along their entire domain. In addition, copula functions are invariant under increasing and continuous transformations of the marginal distributions. Under the standard procedure, first, a given parametric copula is chosen and calibrated to describe the dependence structure among the random variables characterized by a multivariate density. Then, marginal distributions that characterize the individual behavior of the random variables are modeled separately. Last, the marginal distributions are “coupled” with the chosen copula function to “construct” a multivariate distribution. Therefore, the modeling of dependence with standard parametric copulas embeds two important shortcomings:

1. It requires modelers to deal with the choice, proper specification, and calibration of parametric copula functions—that is, the Copula Choice Problem (CCP). The CCP is in general a challenging task, because results are very sensitive to the functional form and parameter values of the chosen copula functions (Frey and McNeil, 2001). In order to specify the correct functional form and parameters, it is necessary to have information on the joint distribution of the variables of interest, in this case, joint distributions of distress, which are not available.
2. The commonly employed parametric copula functions in portfolio risk measurement require the specification of correlation parameters, which usually are specified to remain fixed through time.¹³ Thus, the

¹¹ The standard and conditional normal distributions, the t -distribution, and the mixture of normal distributions.

¹² For further details, proofs, and a comprehensive and didactical exposition of copula theory, see Embrechts, McNeil, and Straumann (1999) and Nelsen (1999), where also properties and different types of copula functions are presented.

¹³ See Appendix I of Segoviano and Goodhart (2009).

Box 32.1. Drawbacks to the Characterization of Distress Dependence of Financial Returns with Correlations

Interdependencies of financial returns traditionally have been modeled based on correlation analysis (de Bandt and Hartmann, 2001). However, the characterization of financial returns with correlations presents important drawbacks, the most relevant of which are the following.

Financial Returns and Gaussian Distributions

The popularity of linear correlation stems from the fact that it can be easily calculated, easily manipulated under linear operations, and is a natural scalar measure of dependence in the world of multivariate normal distributions. However, empirical research in finance shows that distributions of financial assets are seldom in this class.¹ Thus, using multivariate normal distributions and, consequently, linear correlations, might prove very misleading for describing bank distress dependence (Embrechts, McNeil, and Straumann, 1999). Moreover, when working with heavy-tailed distributions—which usually characterize financial asset returns—their variances might not be finite; hence, correlation becomes undefined.²

Linear and Nonlinear Dependence

Another problem associated with correlation is that the data may be highly dependent, while the correlation coefficient is zero. Equivalently, the independence of two random variables implies they are uncorrelated, but zero correlation does not imply independence. A simple example where the covariance disappears despite strong dependence between random variables is obtained by taking $X \sim N(0,1)$, $Y = X^2$ given that the third moment of the standard normal distribution is zero.

Nonlinear Transformations

In addition, linear correlation is not invariant under nonlinear strictly increasing transformations. For two real-valued random variables, we have in general $\rho(T(X), T(Y)) \neq \rho(X, Y)$. This is relevant when modeling dependence among financial assets. For example, suppose that we have a copula function describing the dependence structure among bank percentage returns. If we decide to model the dependence among logarithm returns, the copula will not change; only the marginal distributions will (Embrechts, McNeil, and Straumann, 1999).

Dependence of Extreme Events

Furthermore, correlation is a measure of dependence in the center of the distribution, which gives little weight to tail events, that is, extreme events, when evaluated empirically. Hence, because distress is characterized as a tail event, correlation is not an appropriate measure of distress dependence when marginal distributions of financial assets are nonnormal (de Vries, 2005).

¹ Empirical support for modeling financial returns with t-distributions can be found in Danielsson and de Vries (1997); Bonti, Hoskin, and Siegel (2000); and Glasserman, Heidelberg, and Shahabuddin (2002).

² Even for jointly elliptical distributed random variables, there are situations where using linear correlation does not make sense. If we modeled asset values using heavy-tailed distributions, for example, t_2 -distributions, the linear correlation is not even defined because of infinite second moments.

dependence structure that is characterized with parametric copula functions, although improving the modeling of dependence versus correlations, still embeds the problem of characterizing dependence that remains fixed through time.¹⁴

The CIMDO Copula

Our approach to model multivariate densities is the inverse of the standard copula approach. We first infer the CIMDO density as explained in Section 2.A. The CIMDO density embeds the dependence structure among the random variables that it characterizes; therefore, once we have inferred the

CIMDO density, we can extract the copula function describing such dependence structure, that is, the CIMDO copula. This is done by estimating the marginal densities from the multivariate density and using Sklar's theorem (Sklar, 1959).

The CIMDO copula maintains all the benefits of the copula approach:

1. It describes linear and nonlinear dependencies among the variables described by the CIMDO density. Such dependence structure is invariant under increasing and continuous transformations of the marginal distributions.
2. It characterizes the dependence structure along the entire domain of the CIMDO density. Nevertheless, the dependence structure characterized by the CIMDO copula appears to be more robust in the tail of the density, where our main interest lies, that is, to characterize distress dependence.

¹⁴ Note that even if correlation parameters are dynamically updated using rolling windows, correlations remain fixed within such rolling windows. Moreover, the choice of the length of such rolling windows remains subjective most of the time.

However, the CIMDO copula avoids the drawbacks implied by the use of standard parametric copulas:

1. It circumvents the CCP. The explicit choice and calibration of parametric copula functions is avoided because the CIMDO copula is extracted from the CIMDO density; therefore, in contrast with most copula models, the CIMDO copula is recovered without explicitly imposing parametric forms that, under restricted data sets, are difficult to model empirically and frequently wrongly specified. It is important to note that under such information constraints, that is, when only information of marginal PoDs exists, the CIMDO copula is not only easily implementable, it outperforms the most common parametric copulas used in portfolio risk modeling under the PIT criterion. This is especially on the tail of the copula function, where distress dependence is characterized.¹⁵
2. The CIMDO copula avoids the imposition of constant correlation parameter assumptions. It updates “automatically” when the PoDs are employed to infer the CIMDO density change. Therefore, the CIMDO copula incorporates banks’ distress dependencies that change, according to the dissimilar effects of shocks on individual banks’ PoDs, and that are consistent with the economic cycle.

In order to formalize these ideas, note that if the CIMDO density is of the form presented in equation (32.3), the CIMDO copula, $c_c(u, v)$, is represented by

$$c_c(u, v) = \frac{q[F_c^{-1}(u), H_c^{-1}(v)] \exp\{-[1 + \hat{\mu}]\}}{\int_{-\infty}^{+\infty} q[F_c^{-1}(u), y] \exp\{-\hat{\lambda}_2 \chi_{x_d^y}(y)\} dy \int_{-\infty}^{+\infty} q[x, H_c^{-1}(v)] \exp\{-\hat{\lambda}_1 \chi_{y_d^x}(x)\} dx}, \quad (32.6)$$

where $u = F_c(x) \Leftrightarrow x = F_c^{-1}(u)$, and $v = H_c(y) \Leftrightarrow y = H_c^{-1}(v)$.¹⁶

Equation (32.6) shows that the CIMDO copula is a nonlinear function of λ_1, λ_2 , and μ , the Lagrange multipliers of the CIMDO functional presented in equation (32.2). Like all optimization problems, the Lagrange multipliers reflect the change in the objective function’s value as a result of a marginal change in the constraint set. Therefore, as the empirical PoDs of individual banks change at each period of time, the Lagrange multipliers change, the values of the constraint set change, and the CIMDO copula changes; consequently, the default dependence among the banks in the system changes.

Thus, as already mentioned, the default dependence gets updated “automatically” with changes in empirical PoDs at each period of time. This is a relevant improvement over most risk models, which usually account only for linear depen-

dence (correlation) that is also assumed to remain constant over the cycle or a fixed period of time.

3. BANKING STABILITY MEASURES

The BSMD characterizes the PoD of the individual banks included in the portfolio, their distress dependence, and changes across the economic cycle. This is a rich set of information that allows us to analyze (define) banking stability from three different, yet complementary, perspectives. For this purpose, we define a set of BSMs to quantify

1. common distress in the banks of the system;
2. distress between specific banks; and
3. distress in the system associated with a specific bank.

We hope that the complementary perspectives of financial stability brought by the proposed BSMs represent a useful tool set to help financial supervisors to identify how risks are evolving and where contagion might most easily develop.

For illustration purposes, and to make it easier to present definitions, we proceed by defining a banking system—portfolio of banks—comprising three banks, whose asset values are characterized by the random variables x and y and r . Hence, following the procedure described in Section 2.A, we infer the CIMDO density function, which takes the following form:

$$\hat{p}(x, y, r) = q(x, y, r) \exp\left\{-\left[1 + \hat{\mu} + \left(\hat{\lambda}_1 \chi_{[x_d^r, \infty)}\right) + \left(\hat{\lambda}_2 \chi_{[x_d^y, \infty)}\right) + \left(\hat{\lambda}_3 \chi_{[x_d^x, \infty)}\right)\right]\right\}, \quad (32.7)$$

where $q(x, y, r)$ and $p(x, y, r) \in \mathbb{R}^3$.

A. Common distress in the banks of the system

In order to analyze common distress in the banks comprising the system, we propose using the JPoD and the Banking Stability Index (BSI).

Joint Probability of Distress

The JPoD represents the probability of all the banks in the system (portfolio) becoming distressed, that is, the tail risk of the system. The JPoD embeds not only changes in the individual banks’ PoDs; it also captures changes in the distress dependence among the banks, which increases in times of financial distress; therefore, in such periods, the banking system’s JPoD may experience larger and nonlinear increases than those experienced by the (average) PoDs of individual banks. For the hypothetical banking system defined in equation (32.7), the JPoD is defined as $P(XI YI R)$, and it is estimated by integrating the density (BSMD) as follows:

$$\int_{x_d^r}^{\infty} \int_{x_d^y}^{\infty} \int_{x_d^x}^{\infty} \widehat{p}(x, y, r) dx dy dr = JPoD. \quad (32.8)$$

¹⁵ See Appendix III of Segoviano and Goodhart (2009) for a summary of this evaluation criterion and its results.

¹⁶ See Appendix II of Segoviano and Goodhart (2009) for an exposition.

TABLE 32.1

Distress Dependence Matrix			
Bank	Bank X	Bank Y	Bank R
Bank X	1	$P(X/Y)$	$P(X/R)$
Bank Y	$P(Y/X)$	1	$P(Y/R)$
Bank R	$P(R/X)$	$P(R/Y)$	1

Source: Authors.

Banking Stability Index

The BSI is based on the conditional expectation of the default probability measure developed by Huang (1992).¹⁷ The BSI reflects the expected number of banks becoming distressed given that at least one bank has become distressed. A higher number signifies increased instability.

For example, for a system of two banks, the BSI is defined as follows:

$$BSI = \frac{P(X \geq x_d^x) + P(Y \geq x_d^y)}{1 - P(X < x_d^x, Y < x_d^y)}. \quad (32.9)$$

The BSI represents a probability measure that conditions on any bank becoming distressed, without indicating the specific bank.¹⁸

B. Distress between specific banks

Distress Dependence Matrix

For each period under analysis, for each pair of banks in the portfolio, we estimate the set of pairwise conditional PoDs, which are presented in the Distress Dependence Matrix (DiDe). This matrix contains the PoD of the bank specified in the row, given that the bank specified in the column becomes distressed.

Although conditional probabilities do not imply causation, this set of pairwise conditional probabilities can provide important insights into interlinkages and the likelihood of contagion between the banks in the system. For the hypothetical banking system defined in equation (32.7), at a given date, the DiDe is represented in Table 32.1.

For example, the PoD of Bank X conditional on Bank Y becoming distressed is estimated by

$$P(X \geq x_d^x | Y \geq x_d^y) = \frac{P(X \geq x_d^x, Y \geq x_d^y)}{P(Y \geq x_d^y)}. \quad (32.10)$$

Distress in Specific Banks/Groups of Banks Associated with Distress in Other Banks/Groups of Banks

Note that the BSMD allows us to estimate any conditional probability of distress, including conditional probabilities of

groups or specific banks. This feature provides great flexibility to analyze linkages among diverse groups of banks. For example, we can estimate conditional probabilities between groups or individual banks in different business lines or geographical zones.

C. Distress in the system associated with a specific bank

The probability that at least one bank becomes distressed (PAO) given that a specific bank becomes distressed characterizes the likelihood that one, two, or more institutions, up to the total number of banks in the system, become distressed. Therefore, this measure quantifies the potential “cascade” effects in the system given distress in a specific bank. Consequently, we propose this measure as an indicator to quantify the systemic importance of a specific bank if it becomes distressed. Again, it is worth noting that conditional probabilities do not imply causation; however, we consider that the PAO can provide important insights into systemic interlinkages among the banks comprising a system.

For example, in a banking system with four banks, X, Y, Z, and R, the PAO given that Bank X becomes distressed corresponds to the probability set marked in the Venn diagram (Figure 32.3). In this example, the PAO can be defined as follows:

$$PAO = P(Y/X) + P(Z/X) + P(R/X) - [P(Y|R/X) + P(Y|Z/X) + P(Z|R/X)] + P(Y|R|Z/X). \quad (32.11)$$

4. BANKING STABILITY MEASURES: EMPIRICAL RESULTS

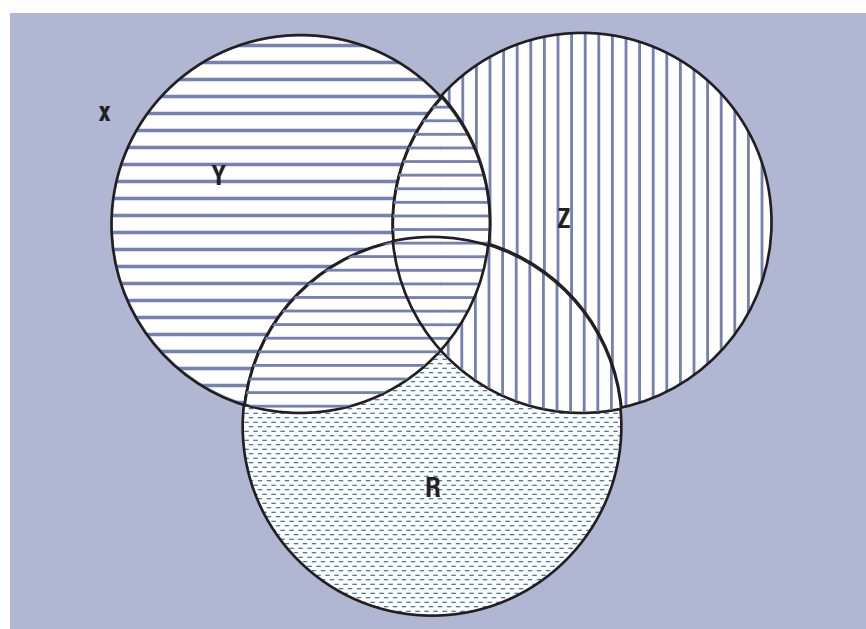
To illustrate the methodology, in this section we estimate the proposed BSMs to

1. examine relative changes in stability over time and among different banks' business;
2. analyze cross-regional effects between different banking groups; and
3. analyze the effect of foreign banks on sovereigns with banking systems with cross-border institutions.

Our estimations are performed from 2005 up to October 2008 using only publicly available data and include major American and European banks and sovereigns in Latin America, Eastern Europe, and Asia. Implementation flexibility in our approach is of relevance for banking stability surveillance, because cross-border financial linkages are growing and becoming increasingly significant, as has been highlighted by the financial market turmoil of recent years. Thus, surveillance of banking stability cannot stop at national borders. An important feature of this methodology is that it can be implemented with alternative measures of PoDs of individual banks, which we will describe. We continue by presenting the estimated BSMs and analyzing them.

¹⁷ This function is presented in Huang (1992). For empirical applications, see Hartmann, Straetmans, and de Vries (2001).

¹⁸ Huang (1992) shows that this measure also can be interpreted as a relative measure of *banking linkage*. When the BSI = 1 in the limit, banking linkage is weak (asymptotic independence). As the value of the BSI increases, banking linkage increases (asymptotic dependence).



Source: Authors.

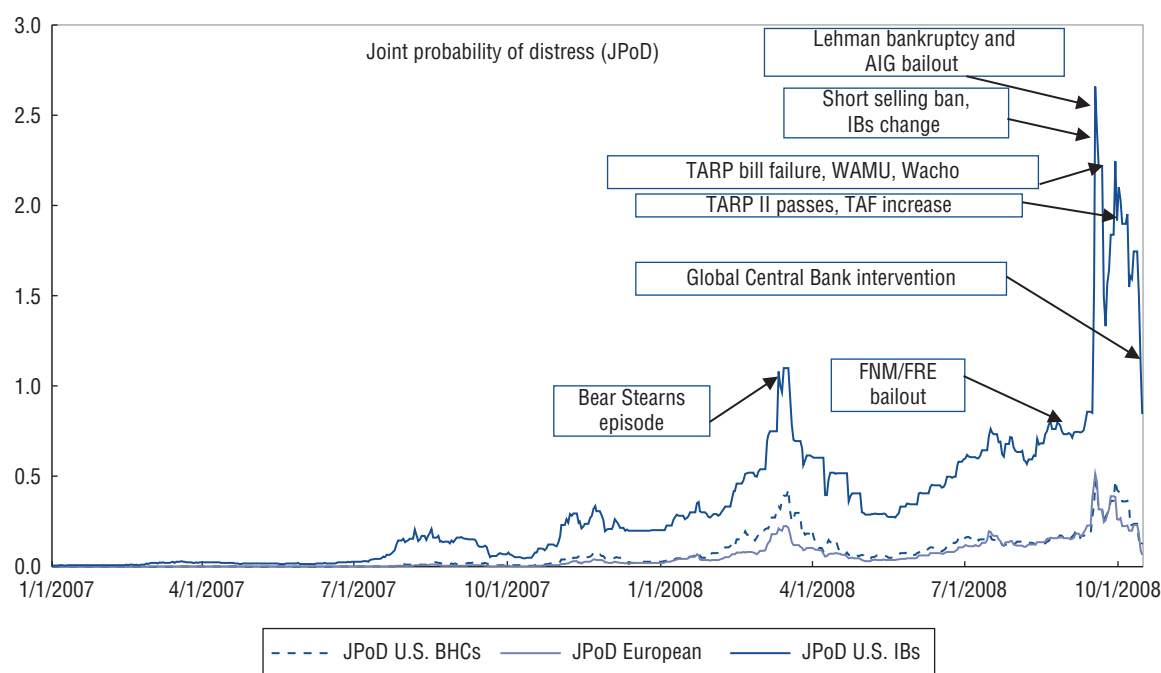
Figure 32.3 Probability That at Least One Bank Becomes Distressed

A. Estimation of probabilities of distress of individual banks

There are alternative approaches by which PoDs of individual banks can be empirically estimated. The most well known include the structural approach, PoDs derived from CDS spreads (CDS-PoDs), or PoDs derived from OOM option prices. These alternative approaches present diverse advantages and disadvantages, in terms of availability of data necessary for their implementation, parameterization of quantitative techniques, and consistency of empirical estimations. We performed an extensive empirical analysis of these approaches. The structural approach presented significant difficulties for the proper parameterization of its quantitative framework. It also produced estimates that appeared inconsistent. The OOM approach suffered from the latter problem, in addition to data restrictions for its implementation across time. Nor were CDS-PoDs free of problems. There are arguments against the trustworthiness of the CDS spreads as a reliable barometer of firms' financial health. In particular, CDS spreads may exaggerate a firm's "fundamental" risk when there is (1) lack of liquidity in the particular CDS market and (2) generalized risk aversion in the financial system. Although such arguments might be correct to some degree, these factors can become self-fulfilling if they affect the market's perception and therefore have a real impact on the market's willingness to fund a particular firm. Consequently, this can cause a real effect on the firm's financial health, as has been seen in the recent financial turmoil. Moreover, although CDS spreads may overshoot at times, they do not

generally stay wrong for long. Rating agencies have mentioned that CDS spreads frequently anticipate rating changes. Though the magnitude of the moves may at times be unrealistic, the direction is usually a good distress signal. For these reasons, and because of the problems encountered with the other approaches (which we consider more serious), we decided to use CDS-PoDs to estimate the proposed BSMs. Although we consider that CDS-PoDs represent reasonable input variables to estimate the proposed BSMs, we keep in mind their potential shortcomings when drawing conclusions in our analysis. Furthermore, because none of these estimators represents a "first best" choice, we continue performing empirical research to improve the estimation of individual banks' PoDs and to investigate which of the alternative approaches (already investigated or to be investigated) is the most appropriate for specific countries and types of banks. Thus, if we found a better approach, it would be straightforward to replace the chosen PoD approach in the estimation of the BSMs, because PoDs are exogenous variables in the CIMDO framework.

Finally, we would like to explain our definition of "distress" risk. Assessing at what point "liquidity risk" becomes solvency risk, that is, credit risk, is difficult, and disentangling these risks is a complex issue. In addition, CDS often covers not only the event of default of an underlying security but a wider set of "credit events," that is, downgrades. We consider the combined effects of these factors, which are embedded in CDS spreads, to be "distress" risk: large losses and the possible default of a specific bank. Thus, our definition of "distress" risk is broader than "default," "credit," or "liquidity" risks.



Source: Authors.

Note: AIG = American International Group; BHC = bank holding company; FNM/FRE = Fannie Mae/Freddie Mac; IB = investment bank; TAF = Term Auction Facility; TARP = Troubled Asset Relief Program; WaMu = Washington Mutual.

Figure 32.4 Joint Probability of Distress: January 2007–October 2008

B. Examination of relative changes of stability over time

The analysis of risks among banks in specific countries and among different business lines is illustrated by estimating our proposed measures of stability for a set of large U.S. banks as it was up to October 2008 using only publicly available data.¹⁹ For this purpose, we focus on the largest U.S. banking groups. The bank holding companies (BHCs) that are included are Citigroup, Bank of America, JP Morgan, and Wachovia. The investment banks included are Goldman Sachs, Lehman Brothers, Merrill Lynch, and Morgan Stanley.²⁰ In addition to the major U.S. banks, we included Washington Mutual (WaMu) and American International Group (AIG; a thrift and an insurance company both under intense market pressure in September 2008). The results can be summarized as follows:

Perspective 1. Common Distress in the Banks of the System: BSI and JPoD

- U.S. banks are highly interconnected, with distress in one bank associated with high PoD elsewhere. This is clearly indicated by the JPoD. Moreover,

movements in the JPoD and BSI coincide with events that were considered relevant by the markets on specific dates (Figures 32.4 and 32.5).

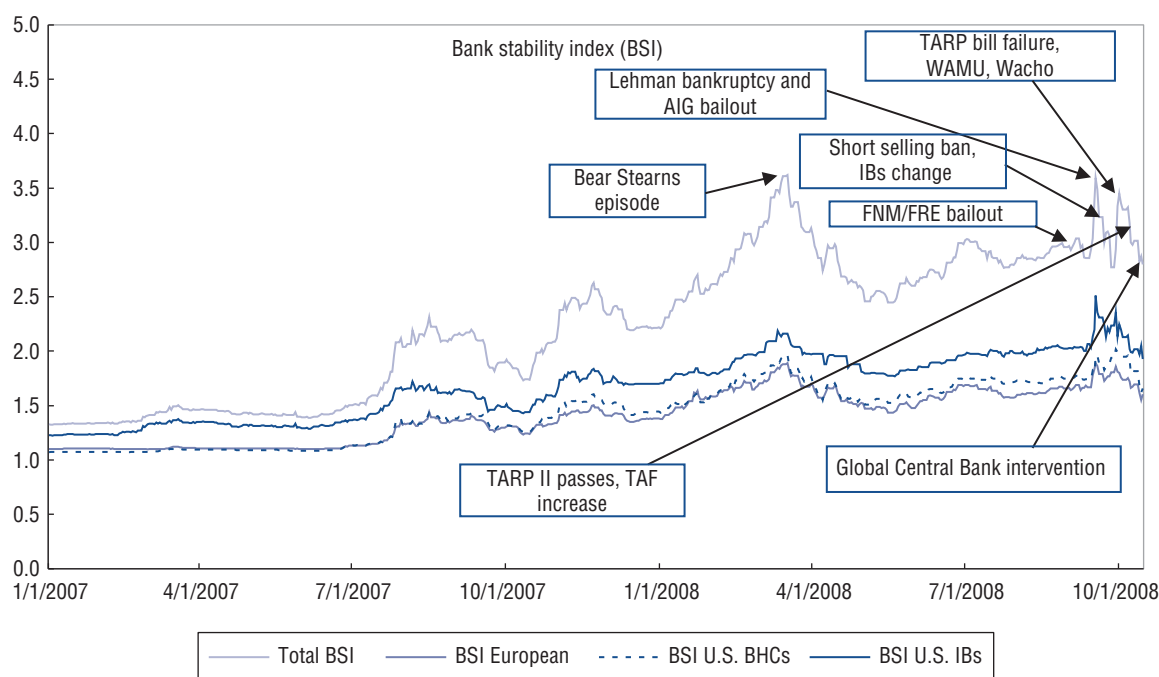
- Distress dependence across banks rises during times of crisis, indicating that systemic risks, as implied by the JPoD and the BSI, rise faster than idiosyncratic risks. The JPoD and the BSI not only take account of individual banks' PoDs, but these measures also embed banks' distress dependence. Therefore, these measures may experience larger and nonlinear increases than those experienced by the PoDs of individual banks. Figure 32.6 shows that daily percentage changes of the JPoD are larger than daily percentage changes of the individual (average) PoDs. This empirical fact provides evidence that in times of distress, not only do individual PoDs increase, but so does distress dependence.
- Risks vary by the business line of the banks. Figures 32.4 and 32.5 show that IBs' JPoD and BSI are larger than those for BHCs. This chart also shows that for IBs, risks were higher at the time of Lehman's collapse.

Perspective 2. Distress between Specific Banks: Distress Dependence Matrix

The DiDe presented in Table 32.2 shows the (pairwise) conditional PoDs of the bank in the column, given that the bank in the row falls into distress. The DiDe is estimated daily. For purposes of analysis, we have chosen July 1, 2007,

¹⁹ The authors would like to thank Tamim Bayoumi for insightful discussions and contributions in the analysis of these empirical results.

²⁰ Although investment banks (IBs) have changed their status recently to BHCs, we keep referring to this group of banks as IBs for the purpose of differentiating their risk profiles in the analysis.



Source: Authors.

Note: AIG = American International Group; BHC = bank holding companies; FNM/FRE = Fannie Mae/Freddie Mac; IB = investment bank; TAF = Term Auction Facility; TARP = Troubled Asset Relief Program; WaMu = Washington Mutual.

Figure 32.5 Banking Stability Index: January 2007–October 2008

and September 12, 2008; thus, we can show how conditional PoDs have changed from a precrisis date to the day before Lehman Brothers filed for bankruptcy. We have also broken these matrices into four quadrants, top left (quadrant 1), top right (quadrant 2), bottom left (quadrant 3), and bottom right (quadrant 4), to make explanations clearer. From these matrices we can observe the following:

- Links across major U.S. banks have increased greatly. This is clearly shown by the conditional PoDs presented in Quadrant 1 of the DiDe presented in Table 32.2. On average, if any of the U.S. banks fell into distress, the average probability of the other banks being distressed increased from 27 percent on July 17, 2007, to 41 percent on September 12, 2008.
- In September, Lehman was the bank under highest stress. This is revealed by Lehman's large PoD conditional on any other bank falling into distress, which on September 12 reached on average 56 percent (row-average Lehman). Moreover, a Lehman default was estimated on September 12 to raise the chances of a default elsewhere by 46 percent. In other words, the PoD of any other bank conditional on Lehman falling into distress went from 25 percent on July 17, 2007, to 37 percent on September 12, 2008 (column-average Lehman).
- AIG's connections to the other major U.S. banks were similar to Lehman's. This can be seen by comparing the chances of each one of the U.S. banks being affected by distress in AIG and Lehman (column

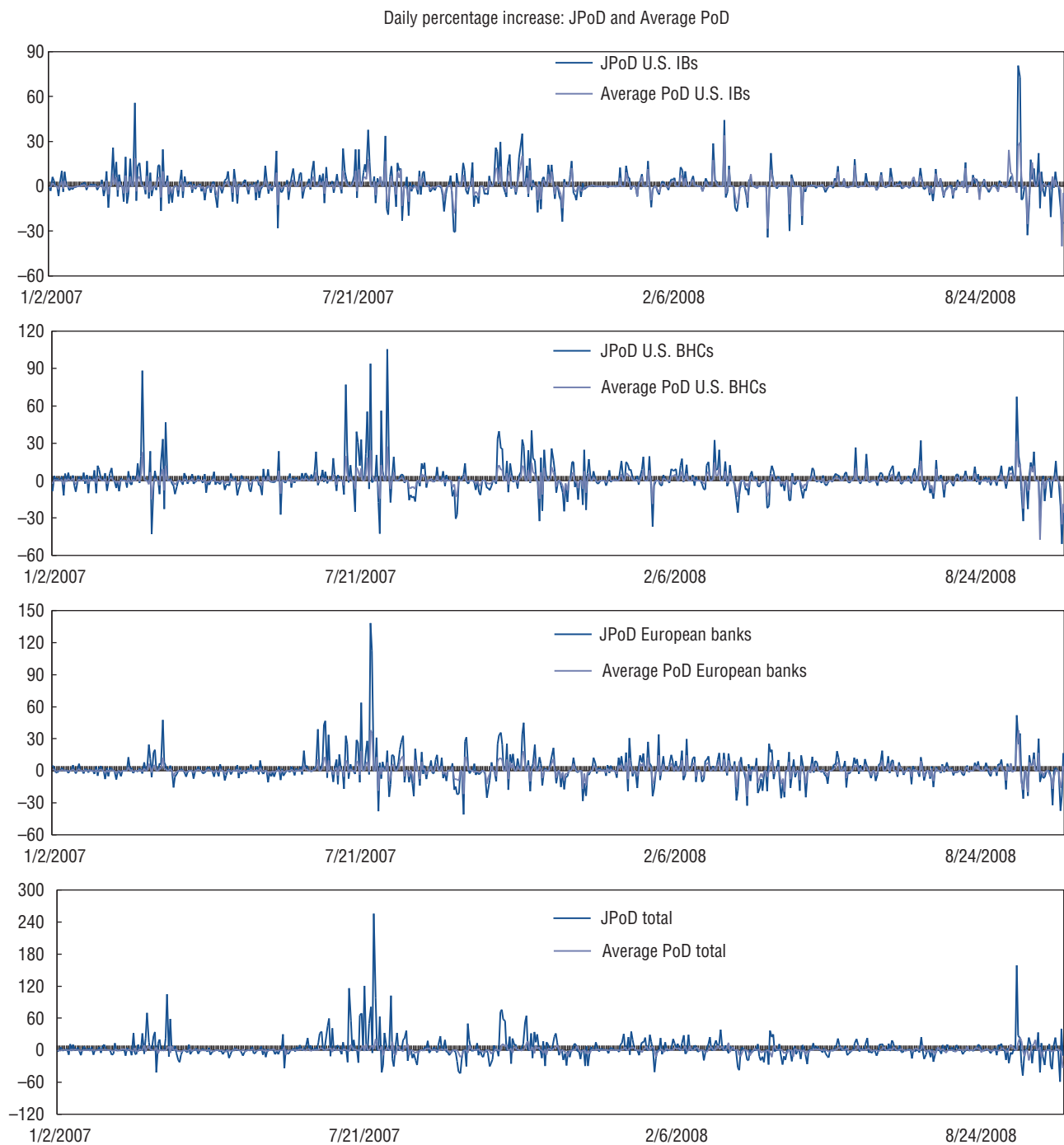
AIG vs. column Lehman) on September 12. Links were particularly close between Lehman, AIG, WaMu, and Wachovia, all of which were particularly exposed to housing. On September 12, a Lehman bankruptcy implied chances of 88, 43, and 27 percent that WaMu, AIG, and Wachovia, respectively, would fall into distress.

Perspective 3. Distress in the System Associated with a Specific Bank: Probability That at Least One Bank Becomes Distressed

On September 12, using equation (32.11), we estimated the probability that one or more banks in the system would become distressed, given that Lehman became distressed. This reached 97 percent. Thus, the possible “domino” effect observed in the days after its collapse were signaled by the PAO (Figure 32.7). This analysis (Perspective 3) is in line with the insights brought by the DiDe (Perspective 2), which indicated that Lehman's distress would be associated with distress in several institutions.

C. Analysis of cross-region effects among different banking groups

In order to gain insight into the cross-region effects between American and European banking groups, we included five major European banks: Barclays (BARC) and Hongkong and Shanghai Banking Corporation (HSBC) from the United



Source: Authors.

Note: BHC = bank holding company; IB = investment bank; JPoD = joint probability of distress; PoD = probability of distress.

Figure 32.6 Daily Percentage Increase: Joint and Average Probability of Distress

Kingdom, UBS and Credit Suisse (CSFB) from Switzerland, and Deutsche Bank from Germany.

Perspective 1. Common Distress in the Banks of the System: BSI and JPoD

- The JPoD indicates that risks among European banks are highly interconnected, with distress in one

bank associated with a high PoD elsewhere in Europe. The European JPoD and BSI move in tandem with movements in the U.S. indicators, coinciding also with relevant market events (Figures 32.3 and 32.4).

- Distress dependence among banks in Europe also rises during times of crisis, indicating that systemic risks, as implied by the JPoD and the BSI, rise faster

TABLE 32.2

Distress Dependence Matrix: American and European Banks, July 2007 and September 2008

Bank	Citi	BAC	JPM	Wacho	WaMu	GS	LEH	MER	MS	AIG	Row average	BARC	HSBC	UBS	CSFB	DB	Row average
July 17, 2007																	
Citigroup	1.00	0.14	0.11	0.11	0.08	0.09	0.08	0.09	0.09	0.08	0.19	0.07	0.07	0.08	0.06	0.07	0.07
Bank of America	0.12	1.00	0.27	0.27	0.11	0.11	0.10	0.12	0.12	0.15	0.24	0.08	0.07	0.09	0.06	0.10	0.08
JP Morgan	0.15	0.42	1.00	0.31	0.13	0.19	0.16	0.19	0.18	0.17	0.29	0.10	0.08	0.12	0.09	0.14	0.10
Wachovia	0.12	0.33	0.24	1.00	0.11	0.12	0.10	0.12	0.12	0.14	0.24	0.07	0.05	0.07	0.05	0.08	0.07
Washington Mutual	0.16	0.28	0.21	0.23	1.00	0.12	0.12	0.16	0.13	0.15	0.26	0.09	0.08	0.09	0.06	0.09	0.08
Goldman Sachs	0.17	0.25	0.28	0.21	0.11	1.00	0.31	0.28	0.31	0.17	0.31	0.13	0.11	0.15	0.12	0.18	0.14
Lehman	0.22	0.32	0.32	0.26	0.15	0.43	1.00	0.35	0.33	0.20	0.36	0.14	0.12	0.15	0.14	0.22	0.15
Merrill Lynch	0.19	0.32	0.33	0.25	0.17	0.33	0.31	1.00	0.31	0.20	0.34	0.15	0.15	0.19	0.15	0.21	0.17
Morgan Stanley	0.19	0.31	0.28	0.24	0.14	0.35	0.28	0.30	1.00	0.16	0.33	0.14	0.12	0.14	0.12	0.18	0.14
AIG	0.07	0.14	0.10	0.10	0.05	0.07	0.06	0.07	0.06	1.00	0.17	0.05	0.06	0.07	0.04	0.06	0.06
Column average	0.24	0.35	0.31	0.30	0.21	0.28	0.25	0.27	0.26	0.24	0.27	0.10	0.09	0.11	0.09	0.13	0.11
Barclays	0.04	0.05	0.04	0.04	0.02	0.04	0.03	0.04	0.04	0.04	0.04	1.00	0.18	0.18	0.12	0.12	0.32
HSBC	0.04	0.04	0.03	0.02	0.02	0.03	0.02	0.03	0.03	0.04	0.03	0.16	1.00	0.13	0.09	0.11	0.30
UBS	0.04	0.05	0.04	0.03	0.02	0.04	0.03	0.04	0.03	0.04	0.04	0.17	0.13	1.00	0.21	0.15	0.33
CSFB	0.05	0.06	0.05	0.04	0.03	0.05	0.05	0.06	0.05	0.05	0.05	0.19	0.15	0.36	1.00	0.21	0.38
Deutsche Bank	0.05	0.09	0.08	0.06	0.03	0.07	0.06	0.07	0.06	0.06	0.06	0.17	0.16	0.22	0.19	1.00	0.35
Column average	0.05	0.06	0.05	0.04	0.02	0.05	0.04	0.05	0.04	0.05	0.04	0.34	0.32	0.38	0.32	0.32	0.34
September 12, 2008																	
Citigroup	1.00	0.20	0.19	0.14	0.07	0.17	0.13	0.14	0.16	0.11	0.23	0.15	0.17	0.17	0.15	0.16	0.16
Bank of America	0.14	1.00	0.31	0.18	0.05	0.16	0.10	0.13	0.15	0.11	0.23	0.12	0.13	0.13	0.11	0.15	0.13
JP Morgan	0.13	0.29	1.00	0.16	0.05	0.19	0.11	0.14	0.16	0.09	0.23	0.11	0.10	0.12	0.11	0.15	0.12
Wachovia	0.34	0.60	0.55	1.00	0.17	0.36	0.27	0.31	0.34	0.29	0.42	0.27	0.23	0.27	0.25	0.31	0.27
Washington Mutual	0.93	0.97	0.95	0.94	1.00	0.91	0.88	0.92	0.91	0.89	0.93	0.87	0.86	0.86	0.83	0.86	0.86
Goldman Sachs	0.15	0.19	0.24	0.13	0.06	1.00	0.18	0.20	0.27	0.11	0.25	0.14	0.13	0.15	0.15	0.19	0.15
Lehman	0.47	0.53	0.58	0.43	0.25	0.75	1.00	0.59	0.62	0.37	0.56	0.39	0.37	0.40	0.42	0.52	0.42
Merrill Lynch	0.32	0.41	0.47	0.30	0.16	0.53	0.37	1.00	0.48	0.26	0.43	0.31	0.33	0.35	0.35	0.39	0.35
Morgan Stanley	0.21	0.28	0.29	0.19	0.09	0.40	0.22	0.27	1.00	0.14	0.31	0.18	0.18	0.18	0.18	0.23	0.19
AIG	0.50	0.66	0.59	0.53	0.29	0.54	0.43	0.49	0.47	1.00	0.55	0.49	0.53	0.53	0.49	0.53	0.52
Column average	0.42	0.51	0.52	0.40	0.22	0.50	0.37	0.42	0.46	0.34	0.41	0.30	0.30	0.32	0.31	0.35	0.32
Barclays	0.10	0.11	0.10	0.08	0.04	0.10	0.07	0.09	0.09	0.07	0.08	1.00	0.36	0.31	0.30	0.28	0.45
HSBC	0.06	0.06	0.05	0.03	0.02	0.05	0.04	0.05	0.05	0.04	0.05	0.20	1.00	0.16	0.16	0.17	0.34
UBS	0.11	0.11	0.11	0.07	0.04	0.11	0.07	0.10	0.09	0.08	0.09	0.32	0.30	1.00	0.47	0.34	0.48
CSFB	0.07	0.07	0.07	0.05	0.03	0.07	0.05	0.07	0.06	0.05	0.06	0.20	0.20	0.31	1.00	0.26	0.40
Deutsche Bank	0.06	0.08	0.09	0.05	0.03	0.09	0.06	0.07	0.07	0.05	0.06	0.18	0.20	0.21	0.24	1.00	0.36
Column average	0.08	0.09	0.08	0.06	0.03	0.09	0.06	0.07	0.07	0.06	0.07	0.38	0.41	0.40	0.43	0.41	0.41

Source: Authors.

Note: The table shows the probability of distress of the bank in the row conditional on the bank in the column becoming distressed. AIG=American International Group; BAC=Bank of America; BARC=Barclays; Citi=Citigroup; CSFB=Credit Suisse; DB=Deutsche Bank; GS=Goldman Sachs; HSBC=Hongkong and Shanghai Banking Corporation; JPM=JP Morgan; LEH=Lehman; MER=Merrill Lynch; MS=Morgan Stanley; Wacho=Wachovia; WaMu=Washington Mutual.

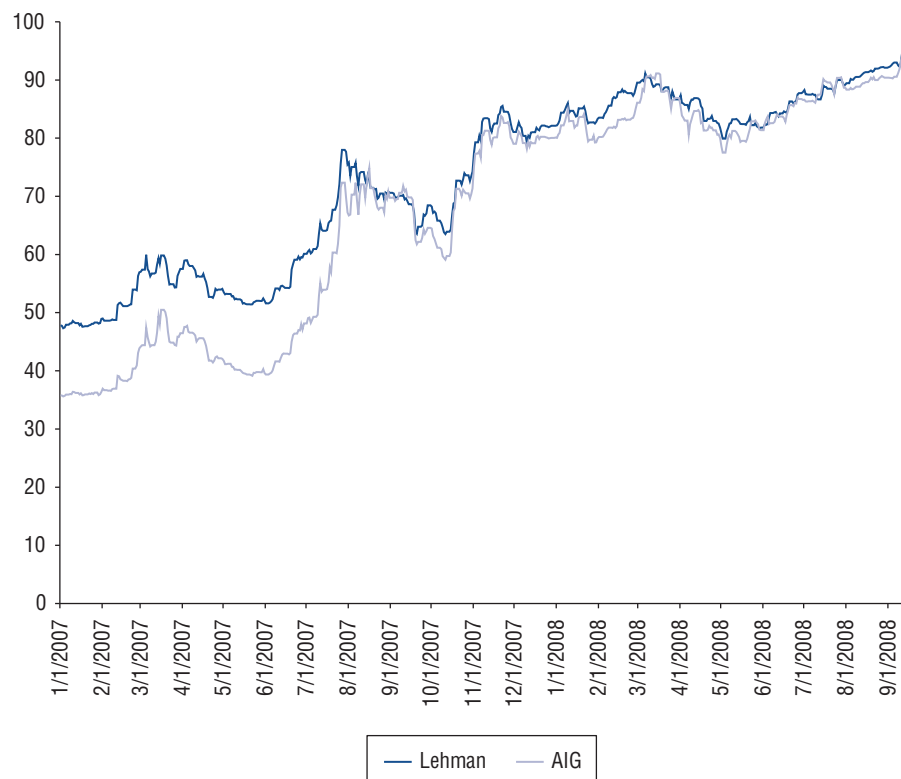
than idiosyncratic risks. In Figure 32.5, it is clear that also for European banks, the daily percentage changes of the JPoD are larger than those of the individual (average) PoDs.

- Risks for European banks as measured by JPoD and the BSI appear lower than those for U.S. IBs and very similar to those for U.S. BHCs across time. Figures 32.3 and 32.4 also show that risks among European banks were similar at the time of the Bear Stearns debacle (March 17) and Lehman's collapse (September 15). This is in contrast to American banks, for which risks appear larger at

the time of Lehman's collapse. However, the situation in Europe appeared to be deteriorating fast in mid-September.

Perspective 2. Distress between Specific Banks: Distress Dependence Matrix

- Links across major European banks have increased significantly (Table 32.2). This is clearly shown by the conditional PoDs presented in Quadrant 4 of the DiDe. On average, if any of the European banks appeared in distress, the probability of the other banks



Source: Authors.

Note: AIG = American International Group.

Figure 32.7 Probability That at Least One Bank Becomes Distressed: Lehman

- being distressed increased from 34 percent on July 17, 2007, to 41 percent on September 12, 2008.
- Among the European banks under analysis, UBS appeared to be the bank under highest stress on September 12, 2008. It showed the largest PoD conditional on any other bank falling into distress, reaching on average 48 percent (row-average UBS). UBS's distress would also be associated with high stress on Barclays, whose PoD conditional on UBS becoming distressed was estimated to reach 31 percent on September 12, 2008. This was a significant increase from 18 percent estimated on July 17, 2007.
- Among the European banks under analysis, distress at CSFB would be associated with the highest stress on other European banks on September 12, 2008. The (average) PoD of European banks conditional on CSFB falling into distress reached 43 percent (quadrant 4, column-average CSFB). However, the European bank that would be associated with the highest distress among American banks is Deutsche Bank. The (average) PoD of American banks conditional on Deutsche Bank falling into distress reached 35 percent (Quadrant 2, column-average Deutsche Bank). This might be related to the high integration of Deutsche Bank in some markets at the global level.
- On September 12, 2008, whereas failure of one of the U.S. banks implied (on average) chances of distress of one European bank of 7 percent (Quadrant 3), the (average) PoD of one American bank, conditional on a European bank becoming distressed is above 30 percent (Quadrant 2). This is possibly because a European default would imply more generalized problems, including in U.S. markets.

Even though distress dependence does not imply causation, these results help explain why the Lehman bankruptcy led to a global crisis. The bankruptcy of Lehman appears to have sealed the fate of AIG and WaMu, while putting greatly increased pressure on Wachovia, as indicated by the DiDe. In market terms, this was equivalent to the failure of a major U.S. institution, with significant reverberations on both sides of the Atlantic.

D. Analysis of foreign banks' risks to sovereigns with banking systems with cross-border institutions

We extend our methodology to analyze how rising problems in advanced market banking systems are linked with increasing risks to emerging markets. For this purpose, we use CDS spreads written on sovereign and banks' bonds to derive

PoDs of banks and sovereigns. Therefore, such PoDs represent markets' views of risks of distress for these banks and countries. Although absolute risks are discussed, the focus is largely on cross-distress dependence of risks and what they can say about emerging vulnerabilities (Perspective 2). More precisely, using publicly available data, we estimate cross-vulnerabilities between Latin American, Eastern European, and Asian emerging markets and the advanced market banks with larger regional presences in these regions. The countries and banks analyzed are

- **Latin America.** Countries included were Mexico, Colombia, Brazil, and Chile; banks included were Banco Bilbao Vizcaya Argentaria, Santander, Citigroup, Scotia Bank, and HSBC.
- **Eastern Europe.** Countries included were Bulgaria, Croatia, Hungary, and Slovakia; banks included were Intesa, Unicredito, Erste, Societe Generale, and Citigroup.
- **Asia.** Countries included were China, Korea, Thailand, Malaysia, the Philippines, and Indonesia; banks included were Citigroup, JP Morgan Chase, HSBC, Standard and Chartered, BNP, Deutsche Bank, and Development Bank of Singapore.

The detailed results are presented in Segoviano and Goodhart (2009).

5. CONCLUSION

The purpose of this study was to seek to provide a set of quantitative measures of the financial stability of the main banks in any country or region, so that this portfolio of banks' relative stability as a group can be tracked over time and compared in a cross section of comparative groupings. To this end, we have developed a new framework that has several advantages:

- It provides measures that allow us to analyze (define) stability from three different yet complementary perspectives.
- It can be constructed from a very limited set of data, that is, the empirical measurements of default probabilities of individual banks. Such measurements can be estimated using alternative approaches, depending on data availability; thus, the data set that is necessary for the estimation is available in many countries, with either developed or developing economies, as long as there are reasonable data to reflect individual banks' PoDs.
- It embeds the banks' default interdependence structure (copula function), which captures linear and non-linear default dependencies among the main banks in a system.
- It allows the quantification of changes in the banks' default interdependence structure at specific points in time; hence, it can be useful to quantify the empirically observed increases in dependencies in periods of distress and relax the commonly used assumption in risk measurement models of fixed correlations across time.

The empirical part of the chapter applied this methodology to a number of country and regional examples by using publicly available information up to October 2008. This implementation flexibility is of relevance for banking stability surveillance, because cross-border financial linkages are growing and becoming significant, as has been highlighted by the financial market turmoil of recent years. Thus, surveillance of banking stability cannot stop at national borders.

The proposed measures will allow us to complement our previous research and expand our research agenda; thus, we hope to gain further insights in our understanding of financial stability: by trying to predict future movements of the BSMs for use as an early-warning mechanism; by exploring the significant macroeconomic and financial factors and shocks influencing the BSMs, in order to identify macro-financial linkages; and by exploring the factors that can limit and reverse tendencies toward instability, so as to discover what instruments may be available (and under what conditions) to control such instability.

REFERENCES

- Altman, Edward, Alan Elizondo, and Miguel Segoviano, 2002, *Medicion Integral del Riesgo de Credito* (Mexico, Limusa).
- Areta, Carlos, and Barry Eichengreen, 2000, "Banking Crises in Emerging Markets: Presumptions and Evidence," CIDER Working Paper C00-115 (Berkeley: Center for International and Development Economics Research).
- Aspachs, Oriol, Charles A. E. Goodhart, Miguel A. Segoviano, Dimitrios Tsomocos, and Lee Zicchino, 2006, "Searching for a Metric for Financial Stability," Special Paper No. 167 (London: Financial Markets Group, London School of Economics). Available via the Internet: <http://www.lse.ac.uk/fmg/workingPapers/specialPapers/2000s.aspx>
- Black, Fischer, and Myron Scholes, 1973, "The Pricing of Options and Corporate Liabilities," *Journal of Political Economy*, Vol. 81, pp. 637–59.
- Bonti, Gabriel, Jonathan R. M. Hoskin, and Dirk Siegel, 2000, "Beyond the Lognormal," *Risk*, Vol. 13, No. 5, pp. 59–62.
- Bordo, Michael, Barry Eichengreen, Daniela Klingebiel, and Maria Soledad Martinez-Peria, 2001, "Financial Crises: Lessons from the Last 120 Years," *Economic Policy*, Vol. 32, pp. 53–75.
- Butler, J. S., and Barry Schachter, 1998, "Estimating Value-at-Risk with a Precision Measure by Combining Kernel Estimation with Historical Simulation," *Review of Derivatives Research*, Vol. 1, No. 4, pp. 371–90.
- Crosbie, Peter J., and Jeffrey R. Bohn, 2003, "Modeling Default Risk," Moody's KMV, December 18. Available via the Internet: http://www.creditrisk.ru/publications/files_attached/modeling_default_risk.pdf
- Danielsson, Jon, and Casper G. de Vries, 1997, "Tail Index and Quantile Estimation with Very High Frequency Data," *Journal of Empirical Finance*, Vol. 4, pp. 241–57.
- de Bandt, Olivier, and Phillipp Hartmann, 2001, "Systemic Risk: A Survey" in *Financial Crisis, Contagion and the Lender of Last Resort: A Book of Readings*, edited by C. Goodhart and G. Illing, pp. 249–98 (Oxford: Oxford University Press).
- de Vries, Casper G., 2005, "The Simple Economics of Bank Fragility," *Journal of Banking and Finance*, Vol. 29 (April), pp. 803–25.

- Diebold, Francis X., Jin Hahn, and Anthony S. Tay, 1999, "Multivariate Density Forecast Evaluation and Calibration in Financial Risk Management: High-Frequency Returns on Foreign Exchange," *Review of Economics and Statistics*, Vol. 81, No. 4, pp. 661–73.
- Embrechts, Paul, Alexander J. McNeil, and Daniel Straumann, 1999, "Correlation and Dependence Properties in Risk Management: Properties and Pitfalls," Working Paper (Zurich: ETH Zurich, RiskLab).
- Frey, Rudiger., and Alexander J. McNeil, 2001, "Modelling Dependent Defaults," Working Paper (Zurich: ETH Zurich, Department of Mathematics).
- Glasserman, Paul, Philip Heidelberger, and Perwez Shahabuddin, 2002, "Portfolio Value-at-Risk with Heavy-Tailed Risk Factors," *Mathematical Finance*, Vol. 12, No. 3, pp. 239–69.
- Golan, Amos, and George G. Judge, 1992, "Recovering Information in the Case of Ill-Posed Inverse Problems with Noise," mimeo (Berkeley: University of California, Berkeley).
- Goodhart, Charles, 1995, "Price Stability and Financial Fragility," in *Financial Stability in a Changing Environment*, edited by K. Sawamoto, Z. Nakajima, and H. Taguchi (New York: St. Martin's Press).
- Goodhart, Charles, and Boris Hofmann, 2003, "Deflation, Credit, and Asset Prices," in *The Anatomy of Deflation*, edited by Pierre Siklos and Richard Burdekin (Cambridge: Cambridge University Press).
- Goodhart, Charles, and Miguel A. Segoviano, 2004, "Bank Regulation and Macroeconomic Fluctuations," *Oxford Review of Economic Policy*, Vol. 20, No. 4, pp. 591–615.
- , 2004, "Basel and Procyclicality: A Comparison of the Standardized and IRB Approaches to an Improved Credit Risk Method," Discussion Paper No. 524 (London: Financial Markets Group, London School of Economics). Available via the Internet: <http://eprints.lse.ac.uk/24821>
- , 2006, "Default, Credit Growth, and Asset Prices," IMF Working Paper 06/223 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=19902.0>
- Goodhart, Charles A. E., Pojanart Sunirand, and Dimitrios Tsomocos, 2006, "A Model to Analyse Financial Fragility," *Economic Theory*, Vol. 27, No. 1, pp. 107–42.
- Hartmann, P., S. Straetmans, and C. de Vries, 2001, "Asset Market Linkages in Crisis Periods," Working Paper No. 71 (Frankfurt: European Central Bank). Available via the Internet: <http://www.ecb.europa.eu/pub/scientific/wps/date/html/wps2001.en.html>
- Hoelscher, David S., 2006, *Bank Restructuring and Resolution* (Washington: International Monetary Fund).
- Huang, Xin, 1992, "Statistics of Bivariate Extreme Values" (Ph.D. thesis, No. 22, Tinbergen Institute Research Series; Rotterdam, the Netherlands: Erasmus University).
- Jaynes, Edwin T., 1957, "Information Theory and Statistical Mechanics," *Physical Review*, Vol. 106, No. 4, pp. 620–30.
- , 1984, "Prior Information and Ambiguity in Inverse Problems," in *Inverse Problems*, SIAM-AMS Proceedings, edited by D. W. McLaughlin, Vol. 14, pp. 151–66 (Providence: American Mathematical Society).
- Kaminsky, Graciela, and Carmen Reinhart, 1999, "The Twin Crisis: The Causes of Banking and Balance of Payments Problems," *American Economic Review*, Vol. 89, No. 3, pp. 473–500.
- Kiyotaki, Nobuhiro, and John Moore, 1997, "Credit Cycles," *Journal of Political Economy*, Vol. 105, No. 2, pp. 211–48.
- Krugman, Paul, 1979, "A Model of Balance of Payments Crises," *Journal of Money, Credit and Banking*, Vol. 11, No. 3, pp. 311–25.
- Kullback, S., 1959, *Information Theory and Statistics* (New York: John Wiley).
- Kullback, S., and R. Leibler, 1951, "On Information and Sufficiency," *Annals of Mathematical Statistics*, Vol. 22, No. 1, pp. 79–86.
- Merton, Robert C., 1974, "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates," *Journal of Finance*, Vol. 29, No. 2, pp. 449–70.
- Nelsen, Roger B., 1999, "An Introduction to Copulas," *Lecture Notes Statistics*, 2nd Edition, Vol. 139.
- Segoviano, Miguel A., 1998, "A Structural Approach for Portfolio Credit Risk vs. Creditmetrics" (unpublished; London: London School of Economics).
- , 2004, "A System for Measuring Credit Risk When Only Limited Data Is Available," Credit Risk International, London, October 2004.
- , 2006a, "The Conditional Probability of Default Methodology," Discussion Paper No. 558 (London: Financial Markets Group, London School of Economics). Available via the Internet: <http://eprints.lse.ac.uk/24512/>
- , 2006b, "The Consistent Information Multivariate Density Optimizing Methodology," Discussion Paper No. 557 (London: Financial Markets Group, London School of Economics). Available via the Internet: <http://eprints.lse.ac.uk/24511/>
- Segoviano, Miguel A., and Charles A. E. Goodhart, 2009, "Banking Stability Measures," Working Paper 09/4 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=22554.0>
- Segoviano, Miguel A., and Philip Lowe, 2002, "Internal Ratings, the Business Cycle and Capital Requirements: Some Evidence from an Emerging Market Economy," BIS Working Paper No. 117 (Basel: Bank for International Settlements). Available via the Internet: <http://www.bis.org/publ/work117.htm>
- Segoviano, Miguel A., and Pablo Padilla, 2006, "Portfolio Credit Risk and Macroeconomic Shocks: Applications to Stress Testing under Data Restricted Environments," Working Paper 06/283 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=19902.0>
- Sklar, Abe, 1959, "Fonctions de répartition à n dimensions et leurs marges," *Publications de l'Institut de Statistique de l'Université de Paris*, Vol. 8, pp. 229–31.

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A Forward-Looking Macroprudential Stress Test for U.S. Banks

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This chapter presents a forward-looking macroprudential stress test methodology for 53 U.S. bank holding companies to assess how changes in the operating environment may affect the medium-term health of individual banks and the system as a whole. By forecasting key elements of the banks' balance sheets, it is possible to capture the interactions between banks' earnings potential, their losses and capital positions, and their ability to support economic growth through lending. Scenario analysis then is used to explore banks' resilience to different stresses, from worse-than-anticipated macroeconomic conditions to more targeted shocks, such as large-scale up-front principal write-down on their impaired residential mortgages. Our results indicate that banks' balance sheets may not be as strong as their capital buffers would suggest.

METHOD SUMMARY

Overview	The forward-looking macroprudential stress test is an exercise that models the impact of a projected path for key macroeconomic and financial variables on banks' capital buffers and other balance sheet positions, including their ability to provide credit.
Application	The method has proven particularly useful in assessing banks' ability to withstand a severe but plausible adverse scenario over a forecast period, while maintaining an adequate capital base to provide credit to the real economy.
Nature of approach	Accounting based (balance sheet and financial statements), forward looking (over a two- to four-year horizon), and with macro-financial linkages.
Data requirements	Bank-specific balance sheet data and financial statements, historical based projections of key economic and financial variables.
Strengths	If needed, the model can be run solely on publicly available data or supplemented by supervisory information where available; the flexibility of the framework allows capturing a wide range of shocks and scenarios.
Weaknesses	The framework is based on low-frequency balance sheet data; it does not capture default dependencies and other cross-firm asset correlations and requires judgment about tail-risk calibrations; it tends to focus on the first-round impact, although the GDP forecast accounts for banks' asset growth in an iterative way.
Tool	Standard econometrics package. The Excel spreadsheet macro is available in the toolkit, which is on the companion CD and at www.elibrary.imf.org/stress-test-toolkit . Contact author: G. Keim.

To assess the resilience of the banking system to changes in the U.S. macroeconomic environment, the IMF Financial Sector Assessment Program (FSAP) team conducted a forward-

looking balance sheet-based analysis. This analysis is similar to the Supervisory Capital Assessment Program (SCAP) exercise, conducted by the authorities in early 2009, in the sense

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that it forecasts banks' capital needs into the future based on particular macroeconomic projections.¹ In contrast to the SCAP, however, the present exercise is based entirely on publicly available information as of end-March 2010.² This approach is also related to the capital adequacy analysis presented in the *Global Financial Stability Report* (GFSR) with regard to aggregate loss estimates; however, it differs from the GFSR in its bank-specific "bottom-up" focus on earnings, losses, and capital positions.

The exercise covers 53 bank holding companies (BHCs), representing 85.2 percent of all bank BHC assets, including a number of regional and smaller banks with less widely tracked information (Table 33.1).³ To capture differences across sizes and business strategies, we grouped the sample into seven subcategories, namely, the "top 4" (accounting for 46.7 percent of sample assets); the 2 former investment banks (10.3 percent of sample assets); 9 regional banks (7.7 percent of sample assets); 3 processing banks (2.7 percent of sample assets); 3 consumer banks (3.2 percent of sample assets); 21 "small" banks (3.4 percent of sample assets); and 11 foreign banks (11.1 percent of sample assets).⁴ The rest of the system was grouped into a residual category, which accounted for 14.8 percent of sample assets.⁵

The purpose of the stress tests is to assess the soundness of BHCs, including under "worse-than-anticipated" macroeconomic conditions. By forecasting key elements of the banks' balance sheets, the exercise captures the interactions between banks' earnings potential, their capital positions, and their ability to absorb losses. All else equal, it also provides some insights into banks' ability to support the economic recovery through lending. Last, the analysis can be used to explore the vulnerability of BHCs to a wide range of stresses, from broad macro scenarios to more targeted shocks.

The results of the stress test should be interpreted with caution. These are subject to uncertainty from a number of sources, including the specification of our statistical models, the level of bank-level detail, the possibility of even more severe and unforeseen events, and the validity of our assumptions on banks' future business practices. Many of these fac-

tors would be present under any forecasting exercise. Moreover, historical correlations that were observed in the past may not be indicative of the relationship that can be expected going forward in light of the substantial economic shock experienced during the crisis, the many and varied associated policy responses that have followed, and the more recent vulnerabilities in Europe. Thus, when interpreting the results, it is important to appreciate that the results are point estimates, and there is uncertainty around them, which is not quantified. Nonetheless, where appropriate, standard tests and alternative scenarios provide some indication of the sensitivity of the results to underlying assumptions.

The results contain both upside and downside risks. The largest downside risk pertains to banks' earnings outlook, which is assumed to recover around their 1990–99 historical average, with an average annualized return on assets (ROA) of 2 percent for the 2010–14 sample periods. These estimates are likely to be on the high side, as they do not incorporate the impact of the upcoming financial regulatory reforms, both domestically and internationally, banks' greater risk retention in the absence of a return to precrisis securitization levels, and lower credit growth in line with the relatively weak economic outlook. Another important downside risk surrounds loss estimates, particularly due to the large uncertainties regarding the new phenomenon of strategic defaults in the case of "underwater" mortgages and banks' recovery rates, given the potentially long-lasting depressed collateral values. On the upside, banks' ability to raise private capital or reduce their dividend policy could significantly strengthen their capacity to absorb losses and support economic growth when demand recovers.⁶

The assessment of BHCs' capital adequacy over the forecast period employed several capital measures. To assess the quality of capital, while allowing comparability both across countries and to the SCAP, three capital metrics were used; (1) the ratio of Tier 1 capital to risk-weighted assets with 6 and 8 percent thresholds; (2) the SCAP's Tier 1 common capital/risk-weighted assets ratio with 4 and 6 percent thresholds; and (3) a tangible common equity to tangible assets ratio with 4 and 6 percent thresholds. The thresholds were not ambitious relative to historical norms (SCAP institutions maintained an average 10 percent Tier 1 capital ratio during the crisis, and their Tier 1 common capital ratio was on average 7.4 percent over 1997–2007) (Figure 33.1).⁷

The exercise spans over a seven-year horizon. It used realized quarterly data from end-2007 to end-March 2010 and

¹ See Board of Governors of the Federal Reserve System (2009a, 2009b).

² An earlier version of this framework was used to estimate capital shortfalls in U.S. banks in the context of the *Global Financial Stability Report* (IMF 2008a, 2008b, 2009a, 2009b) and the 2009 U.S. Article IV Consultation Report (IMF, 2009c). Benefiting from constructive comments from the authorities in the context of the U.S. FSAP, the framework has been revamped to cover a wider range of institutions, better capture institution-specific idiosyncrasies, and account for various regulatory measures and other onetime events during the crisis.

³ The institutions included in the stress tests all were operating as of end-March 2010. Since then, however, one of the regional banks (W Holding) failed, with its deposits and a portion of its assets acquired by another regional bank (Popular). No adjustment was made for this event, as it would have been difficult to assess its impact, which would depend on the terms and conditions of the takeover.

⁴ The asset size of the "small" banks ranges from \$10 billion to \$60 billion, and market share of assets is as of end-March 2010.

⁵ Excluding BHCs with assets smaller than \$80 million, which are not reported in the bank-specific database SNL Financial.

⁶ We assume that banks would not raise capital over the sample horizon or that, under the baseline, profit-making financial firms would not reduce their dividends policy in anticipation of a future capital need. The latter assumption was relaxed under the adverse scenario, where banks were expected not to pay out common stock dividends, in line with the authorities' SCAP exercise.

⁷ Tier 1 common capital deducts all "noncommon" elements from Tier 1 capital (i.e., qualifying minority interest in consolidated subsidiaries, qualifying trust preferred securities, and qualifying perpetual preferred stock).

TABLE 33.1

List of Institutions, Total Assets as of End-March 2010

Institution	Number of Institutions	Total Assets (in billions)	In Percent of Sample
Total system	54	16,483.9	100.0
Top 4 banks	4	7,702.3	46.7
Bank of America Corporation	1	2,340.7	14.2
JPMorgan Chase & Co.	1	2,135.8	13.0
Citigroup, Inc.	1	2,002.2	12.1
Wells Fargo & Company	1	1,223.6	7.4
Investment banks	2	1,700.4	10.3
Goldman Sachs Group, Inc.	1	880.7	5.3
Morgan Stanley	1	819.7	5.0
Bear Stearns	—	—	—
Lehman	—	—	—
Regional banks	9	1,273.9	7.7
PNC Financial Services Group, Inc.	1	265.4	1.6
U.S. Bancorp	1	282.4	1.7
SunTrust Banks, Inc.	1	171.8	1.0
BB&T Corporation	1	163.7	1.0
Regions Financial Corporation	1	137.3	0.8
Fifth Third Bancorp	1	112.7	0.7
KeyCorp	1	95.3	0.6
Popular, Inc.	1	33.8	0.2
W Holding Company, Inc. ¹	1	11.5	0.1
Washington Mutual	—	—	—
Processing banks	3	450.2	2.7
Bank of New York Mellon Corporation	1	221.0	1.3
State Street Corporation	1	152.9	0.9
Northern Trust Corporation	1	76.3	0.5
Consumer banks	3	522.4	3.2
Capital One Financial Corporation	1	200.7	1.2
American Express Company	1	142.3	0.9
Ally Financial ²	1	179.4	1.1
CIT	—	—	—
Ameriprise	—	—	—
Small banks	21	564.9	3.4
Comerica Incorporated	1	57.2	0.3
Marshall & Ilsley Corporation	1	56.6	0.3
Zions Bancorporation	1	51.7	0.3
Huntington Bancshares Incorporated	1	51.9	0.3
Synovus Financial Corp.	1	32.4	0.2
New York Community Bancorp, Inc.	1	42.4	0.3
First Horizon National Corporation	1	25.9	0.2
BancorpSouth, Inc.	1	13.2	0.1
Associated Banc-Corp	1	23.1	0.1
BOK Financial Corporation	1	23.5	0.1
First BanCorp.	1	18.9	0.1
Webster Financial Corporation	1	18.0	0.1
Commerce Bancshares, Inc.	1	18.0	0.1
TCF Financial Corporation	1	18.2	0.1
First Citizens BancShares, Inc.	1	21.2	0.1
First National of Nebraska, Inc.	1	16.6	0.1
City National Corporation	1	20.1	0.1
Fulton Financial Corporation	1	16.4	0.1
New York Private Bank & Trust Corporation	1	13.1	0.1
Susquehanna Bancshares, Inc.	1	13.8	0.1
South Financial Group, Inc.	1	12.4	0.1
Foreign banks	11	1,822.0	11.1
M&T Bank Corporation	1	68.4	0.4
Harris Financial Corp.	1	65.5	0.4
BancWest Corporation	1	75.2	0.5
UnionBanCal Corporation	1	85.5	0.5
Barclays Group US, Inc.	1	427.8	2.6
BBVA USA Bancshares, Inc.	1	65.2	0.4

(continued)

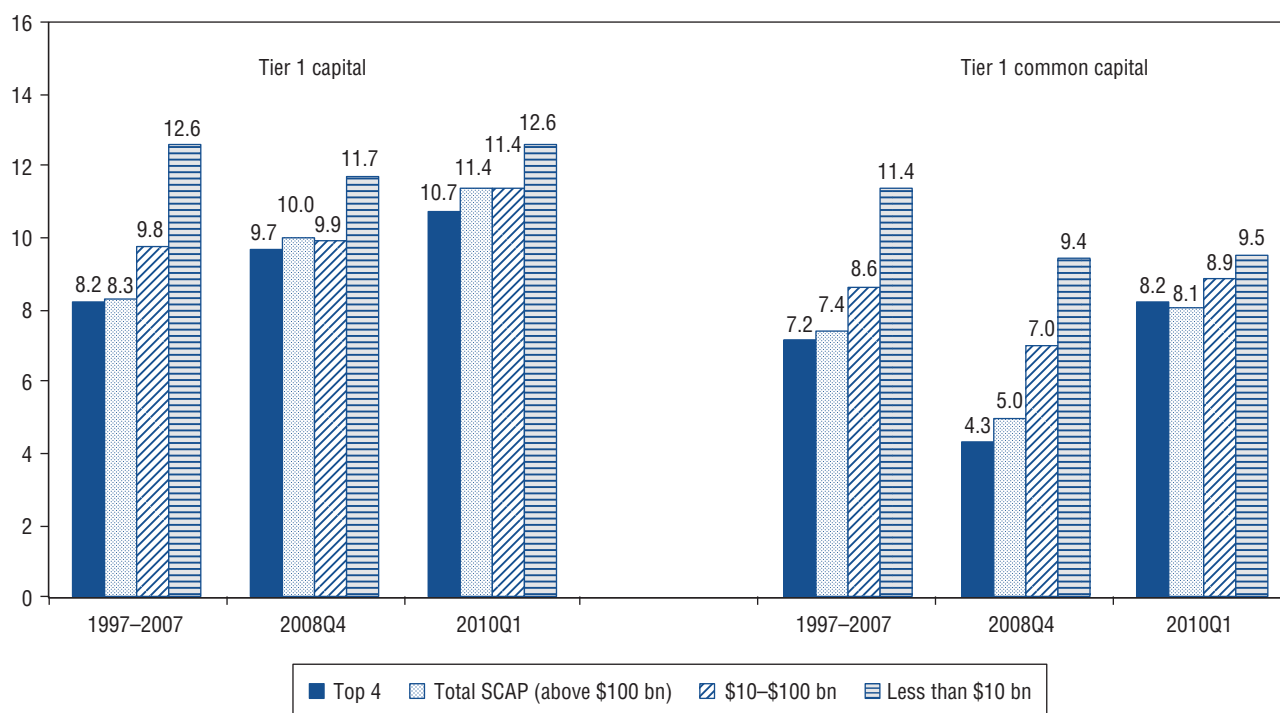
TABLE 33.1 (continued)

Institution	Number of Institutions	Total Assets (in billions)	In Percent of Sample
HSBC North America Holdings, Inc.	1	345.4	2.1
RBC Bancorporation (USA)	1	26.2	0.2
Taunus Corporation	1	364.1	2.2
TD Banknorth, Inc.	1	154.7	0.9
Citizens Financial Group, Inc.	1	144.0	0.9
Other	1	2,447.8	14.8

Source: SNL Financial.

1. Acquired by Banco Popular de Puerto Rico in April 2010.

2. Operating under the name of GMAC, Inc., prior to May 2010.



Source: SNL Financial.

Note: bn = billion; SCAP = Supervisory Capital Assessment Program.

Figure 33.1 Capital Position of Bank Holding Companies, 1997–2010 (in percent of risk-weighted assets)

produced quarterly forward-projections until end-2014. Most of the bank-specific data came from the publicly available Y-9C reports that bank holding companies file with the Fed and were obtained from SNL Financial's database. The regulatory data were augmented further with Securities and Exchange Commission data for nonbanks before their BHC conversion in 2008, *Bloomberg* for capital-raising measures and securities write-downs, and the U.S. Treasury Department's Web site <http://www.FinancialStability.gov> for Troubled Asset Relief Program (TARP) repayments and dividends.

1. BASELINE SCENARIO

A. Framework

The analysis projected firms' net revenues, losses, and balance sheet expansion to assess banks' potential capital shortfalls over a five-year period. A bank's capital shortfall (if any) was computed based on its lowest capital position over the horizon. Consideration was given to firm-specific differences in earnings and losses, based on portfolio composition and historical performance. An attempt also was made to cap-

ture a number of specific postcrisis factors that would affect BHCs' asset growth, including banks' efforts to deleverage and de-risk their balance sheets, the new Financial Accounting Standard 166/167 accounting rules requiring banks to on-board certain assets previously held off-balance-sheet, and greater risk retention due to impaired securitization. Moreover, the calculations also incorporated firms' ability to accumulate tax assets in loss-making quarters that could be used to offset future tax liabilities.

Macro-financial linkages are built into the stress test framework. By modeling how macroeconomic variables have influenced historically the behavior of specific financial variables, it is possible to link a particular macroeconomic path to financial sector developments and their related impact on financial firms' capital position. The nominal GDP growth forecast, for example, drives asset growth; loan loss rates reflect movements in the path for real GDP, real consumption, the unemployment rate, and the output gap. Other macroeconomic variables critical for the loss estimates, such as lending standards and house prices, are forecasted separately (Appendix I). Although the link between macroeconomic variables and financial variables is the centerpiece of this exercise, it should be noted that there is no universal, consensus view as to how these variables should relate to each other and that judgmental adjustments may be needed.

The baseline scenario was taken from the IMF's April 2010 *World Economic Outlook* (WEO). In particular, the output gap closed over the medium term from a negative level in 2009, with the unemployment rate remaining elevated (above 8 percent) until end-2011 before dropping to 5.5 percent by end-2014. Real GDP growth was expected to peak at 3.1 percent in 2010 and to stabilize around 2.5 percent by 2012. House prices were expected to rise over the forecast horizon, albeit at a very slow pace (peaking at 4.1 percent in 2011) (Appendix I).

B. Underlying assumptions and forecasting methodology

Five categories of loan charge-off rates were estimated on an industry-wide basis from regression analysis (Appendix II). These include losses on commercial real estate (CRE) loans, residential real estate (RRE) loans, commercial and industrial (C&I) loans, and consumer loans (Figure 33.2).⁸ To capture historical cross-firm differences, BHC-specific charge-off rates were projected for each type of loan, given by $COR_{i,q}^j$, where i denotes firms (with I being the industry average), j indexes the loan type, and q denotes time. The forecasted rates were computed recursively taking as their base the previous quarter's value, to which the change in the industry-wide charge-off rate for that class of loan ($\Delta \overline{COR}_{I,q}^j$) was applied:

$$COR_{i,q}^j = COR_{i,q-1}^j + \Delta \overline{COR}_{I,q}^j.$$

No adjustment was made to account for much stricter underwriting standards post-2009. In practice, stricter lending standards should help reduce future loss estimates and particularly their sensitivity to adverse shocks. In the near term, this omission should not play a large role in the context of falling credit growth rates. For the outer years of the forecast, however, it could lead to an upward bias in the loss estimates.

Only limited account was taken of the mergers and acquisitions that took place in 2008 among several large banks. In the SCAP exercise, adjustments for losses already taken on impaired loans acquired through mergers (i.e., purchase accounting adjustments) reduced estimated losses by \$64 billion. No such adjustment was made in the current exercise, largely because of the difficulty of assessing the performance of the acquired loans relative to expectations in the absence of detailed loan-by-loan information. Furthermore, it is reasonable to assume that the impact of such purchasing accounting adjustments would diminish over time, including through amortization.

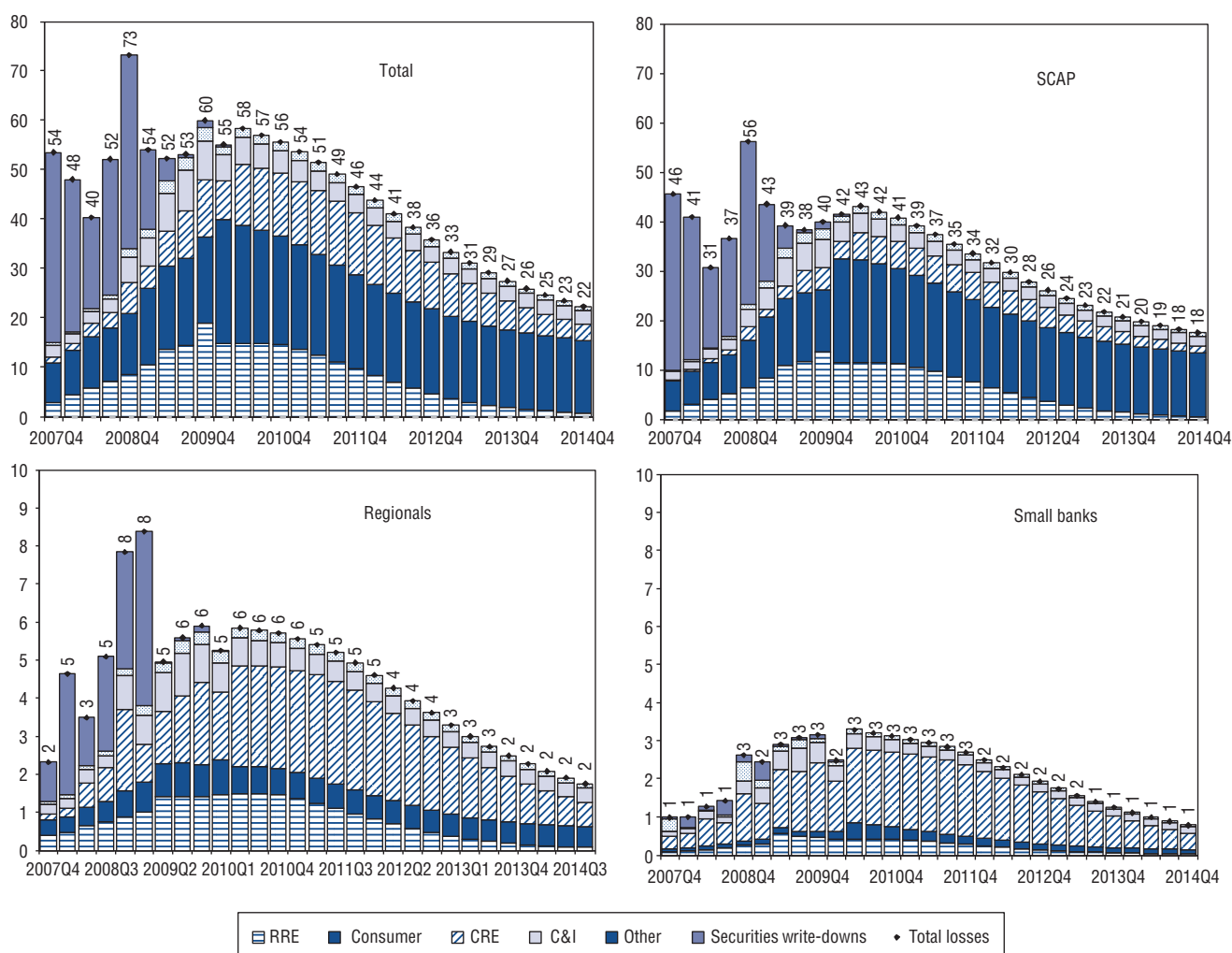
Minor adjustments were made for the investment banks that converted to bank holding companies in late 2008. Because of their recent conversion, focusing on historical prudential data would have put an unreasonably large weight on their (poor) performance during the crisis. In the case of Morgan Stanley, the calculations omit the company's abnormally high loan loss rates during the fourth quarter of 2008 from the moving average used to forecast its future losses. The company's earnings path was raised by adjusting it to the estimated average of the fixed effects of the top six firms (Figure 33.3).⁹

The results suggest that credit risk is likely to remain a source of concern for some time, with some loss rates not peaking before mid-2011 (Table 33.2). Some of the highest loss rates may have already peaked, such as consumer loans, which reached 6.5 percent at end-January 2010, and RRE, which rose to 2.7 percent at end-2009. Losses on CRE loans, however, are expected to continue to rise until mid-2011, as they are generally expected to peak later than in the RRE sector. The high loss rate on consumer credit reflects partly the aggressive charge-off policy of the consumer banks in an effort to clean their balance sheets, possibly in anticipation of a pickup in demand following the sharp credit line contractions since 2008. The relatively low C&I loan loss rate, which peaked at 2.6 percent at end-September 2009, reflects the relatively healthy financial position of the corporate sector, which was able to either use its cash buffers to pay down debt or benefit from advantageous refinancing terms in the corporate bond market.

While showing signs of stabilization, losses on CRE loans are projected to remain high over the forecast horizon. Unless

⁸ We also computed a charge-off rate for "other" loans as a simple average of the other four categories.

⁹ The bank-specific fixed effects were obtained from regression analysis.



Sources: Authors; and SNL Financial.

Note: C&I = commercial and industrial; CRE = commercial real estate; RRE = residential real estate; SCAP = Supervisory Capital Assessment Program.

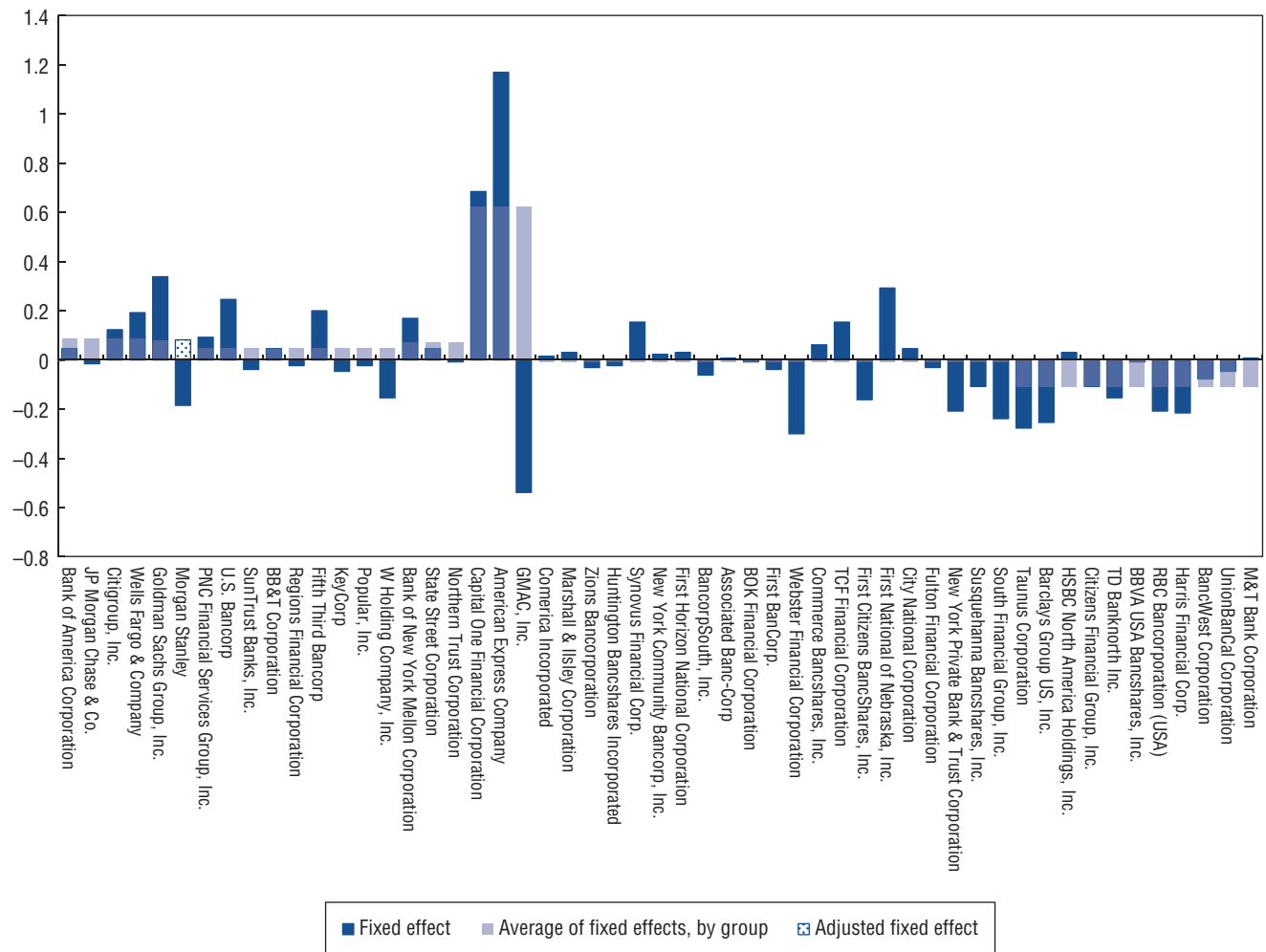
Figure 33.2 Baseline Scenario: Quarterly Loss Profiles, 2007–14 (in billions of dollars)

commercial property prices start recovering from their 30 percent fall since mid-2006, banks are likely to face heavy losses, given the large volume of underwater mortgage borrowers and upcoming adjustable-rate mortgage resets. Furthermore, the weak economy continues to hammer rents and occupancy rates in many markets, with negative consequences for defaults, foreclosures, and losses. The Congressional Oversight Panel (2010), for example, estimated that about \$1.4 trillion in loans will mature in 2010–14, nearly half of which are already seriously delinquent (90 days or more past due) or “underwater” (with a loan value exceeding the property value).

Real estate loan quality is vulnerable to downside risks. Although the loss rate on RRE loans is expected to decline going forward, the rising gap between delinquencies and foreclosures suggests a large volume of pent-up supply of houses

for sale through foreclosures, which could put further downward pressure on house prices and exacerbate strategic defaults for some time to come. As of end-March 2010, seriously delinquent loans (90 days or more overdue) accounted for 5 percent of total mortgages, in contrast to actual foreclosures, which accounted for only 1.25 percent of total loans. Loan modifications could help mitigate the risk of a pickup in foreclosures; however, redefault may be high, in which case modifications simply could postpone losses further into the future.

In the baseline, total cumulative loan losses are expected to reach \$802 billion by end-2014 (\$592 billion for SCAP firms). This represents a 6.5 percent cumulative loss for 2010–11 (12.3 percent for 2010–14) (Table 33.3). Although the two-year loss rates are below the 9.1 percent 2009–10 loss rate assumed in the SCAP stress test, they amount to an annual average of 3.3 percent for 2010–11 and 2.5 percent



Source: Authors.

Figure 33.3 Estimated Bank-Specific Effects and Group Averages in Return on Asset Regressions**TABLE 33.2**

Peaks for Loan Loss Charge-off Rates, 2009–14 (in percent)

Sector	Baseline Scenario		Adverse Scenario		Alternative Scenario	
	Max.	Period	Max.	Period	Max.	Period
RRE	2.7	2009: Q4	3.4	2011: Q4	3.5	2012: Q1
Consumer	6.5	2010: Q1	6.5	2010: Q1	6.5	2010: Q1
CRE	3.4	2011: Q2	4.6	2011: Q3	5.1	2011: Q4
C&I	2.6	2009: Q3	2.6	2009: Q3	2.6	2009: Q3
Other	3.4	2009: Q4	3.8	2011: Q2	3.6	2011: Q3

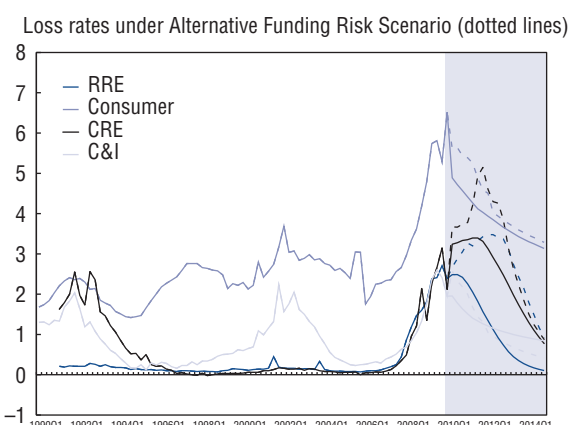
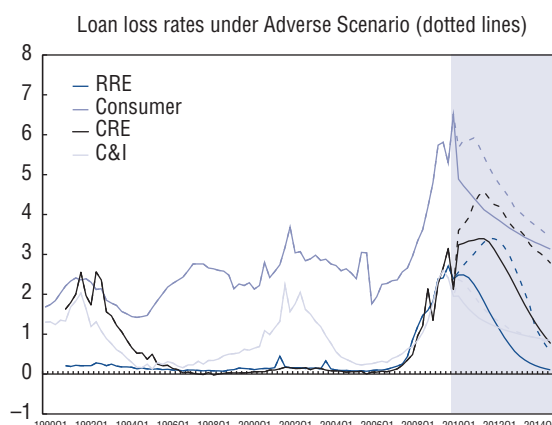
Sources: Authors; Bloomberg; and SNL Financial.

Note: C&I = commercial and industrial; Cons = construction; CRE = commercial real estate; RRE = residential real estate.

TABLE 33.3

Cumulative Loss Rates, 2010–14 (in percent)

Institution	Baseline				Adverse Scenario				Alternative Scenario			
	Loan Losses		Total Losses		Loan Losses		Total Losses		Loan Losses		Total Losses	
	2010–11	2010–14	2010–11	2010–14	2010–11	2010–14	2010–11	2010–14	2010–11	2010–14	2010–11	2010–14
Total	6.4	12.1	6.4	12.1	7.7	15.9	8.7	17.4	7.6	15.2	8.2	15.8
Top four	8.1	15.3	8.1	15.3	9.8	19.8	10.8	21.3	9.7	19.5	10.3	20.2
Regional bank	5.1	9.4	5.1	9.4	6.3	13.4	7.0	14.5	6.3	12.4	6.7	12.8
Consumer banks	8.9	18.7	8.9	18.7	10.7	22.2	11.5	23.4	10.2	20.3	10.6	20.8
Small bank	5.5	9.7	5.5	9.7	6.5	13.3	7.0	14.1	6.5	12.0	6.8	12.3
Foreign banks	6.3	10.9	6.3	10.9	7.7	15.3	8.7	16.7	7.6	14.8	8.2	15.4
Annual average	3.2	2.4	3.2	2.4	3.8	3.2	4.4	3.5	3.8	3.0	4.1	3.2



Source: Authors.

for 2010–14. Consumer banks face the largest two-year loss rate, followed by the top four banks. Small and regional banks face lower but still material loss rates (9.7 and 9.4 percent, respectively), reflecting their heavy exposure concentration to CRE.

C. Securities write-downs

Write-downs on securities were measured as declines in market valuations based on the methodology developed and updated in a recent GFSR.¹⁰ Under the baseline, no additional securities write-down on available-for-sale securities was expected, and the framework did not envisage any shocks to marked-to-market trading account securities (Table 33.4). Since the beginning of the crisis in end-2007, BHCs reported a cumulative \$385 billion of realized marked-to-market securities write-downs (not shown), relative to the \$296 billion of write-downs estimated by the model. To be on the conservative side, we made no allowance for write-ups to banks' securities holdings.

¹⁰ See IMF (2008a, 2008b) for a description of the methodology used for U.S. securities. This methodology was further revised to better capture losses in non-U.S. countries, particularly Europe and Asia.

D. Earnings profiles

One of the most challenging elements of this exercise was to forecast banks' earnings. The analysis focused on preprovision, pretax, and predividend net revenues as a percentage of total assets, henceforth referred to as ROA. The ultimate regression, which covers 53 BHCs, was run using a fixed-effects panel specification. It included a vector X_t of three macroeconomic variables (real GDP quarterly growth rate, output gap, and lagged quarterly unemployment growth rate), one bank-specific variable y_{it-1} (the lagged loan-to-asset ratio) to capture banks' different business strategies, and one financial market variable, z_{it} (the spread between the three-month London interbank offered rate [LIBOR] and the Treasury bill of similar maturity), as a proxy for financial market conditions; u_{it} is the unit-specific residual, and ε_{it} is the usual residual:

$$ROA_{it} = \alpha + \beta X_t + \delta y_{it-1} + \lambda z_{it} + u_i + \varepsilon_{it}$$

The model was estimated using quarterly frequency data from 1990:Q1 to 2010:Q1. The macroeconomic variables were adjusted seasonally, whereas the bank-specific and financial variables were not. For the forecast period, the estimated coefficients were applied to the forecasted explanatory variables, allowing the resulting retained earnings to feed back into total assets each quarter. No attempt was made to model subgroups of institutions, although the fixed effect from

TABLE 33.4

Securities Write-down Projections

Securities	Estimated Holdings	January 2010 Cumulative Losses	January 2010 Cumulative Loss Rate (Percent)	Share of Total (Percent)
Residential mortgage	1,472	166	11.3	56.2
Agency (prime conforming)	492	0	0.0	0.0
ABS (home and multifamily)	980	166	17.0	56.2
of which: nonagency prime MBS	530	0	0.0	0.0
of which: ABS (CDOs, other MBS)	450	166	37.0	56.2
Consumer	142	0	0.0	0.0
Commercial mortgage	196	48	24.5	16.3
Corporate	1,115	17	1.5	5.6
Governments	580	0	0.0	0.0
Foreign	975	66	6.7	22.2
Total for securities	6,932	296	6.6	100.0

Sources: Authors; and Bloomberg.

Note: ABS = asset-backed security; CDO = collateralized debt obligation; MBS = mortgage-backed security.

TABLE 33.5

Summary Results for Return on Asset Regressions

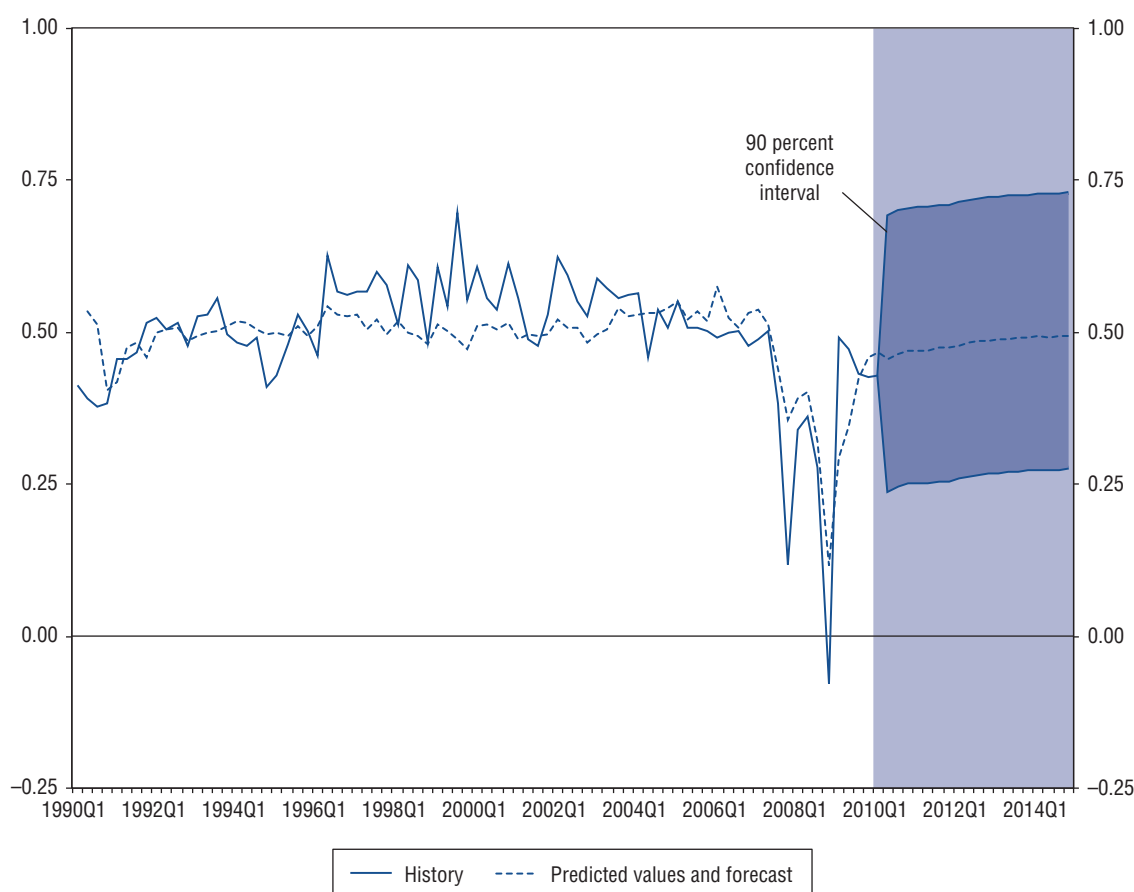
	Macro Data		Fixed-effects Panel			Mixed-effects Panel	
	(1)	(2)	(3)	(4)	(5)	Bank and Group Level (6)	Bank Level (7)
Loan-to-asset (lagged)	0.584*** <i>-1.06E-04</i>	1.792*** <i>-8.83E-08</i>	0.276* <i>-0.0714</i>	0.307** <i>-0.0433</i>	0.292* <i>-0.0508</i>	0.324*** <i>0.00E+00</i>	0.317*** <i>0.00E+00</i>
Log of total assets (lagged)	— <i>-6.07E-06</i>	0.0684*** <i>-0.138</i>	-1.65E-02 <i>-0.151***</i>	— <i>-0.150***</i>	— <i>-0.103***</i>	— <i>-0.150***</i>	— <i>-0.150***</i>
Real GDP quarterly growth	2.432 <i>-0.15</i>	2.704** <i>-0.0363</i>	1.363** <i>-0.0471</i>	1.687** <i>-0.014</i>	1.356** <i>-0.0449</i>	1.690*** <i>-0.00238</i>	1.708*** <i>-0.00213</i>
3-month LIBOR to 3-month T-Bill	-0.201*** <i>-7.42E-08</i>	-0.188*** <i>-4.54E-07</i>	-0.151*** <i>-1.28E-09</i>	-0.150*** <i>-1.15E-09</i>	-0.103*** <i>-1.28E-05</i>	-0.150*** <i>0.00E+00</i>	-0.150*** <i>0.00E+00</i>
Output gap	0.0129* <i>-0.051</i>	-0.0104 <i>-0.225</i>	0.0240*** <i>-0.000185</i>	0.0197*** <i>-0.00332</i>	0.0204*** <i>-0.00232</i>	0.0196*** <i>-1.24E-10</i>	0.0194*** <i>-1.65E-10</i>
Unemployment quarterly growth (lagged)	0.296 <i>-0.101</i>	0.0319 <i>-0.848</i>	0.138 <i>-0.153</i>	0.0205 <i>-0.857</i>	0.0262 <i>-0.817</i>	0.0215 <i>-0.805</i>	0.0225 <i>-0.795</i>
Dummy for 2007–08	— <i>-0.000193</i>	—	—	— <i>-0.0929***</i>	— <i>-0.000193</i>	—	—
Constant	0.269*** <i>-2.08E-04</i>	-1.925*** <i>-9.85E-05</i>	0.698*** <i>-0.00199</i>	0.393*** <i>-1.64E-04</i>	0.389*** <i>-1.63E-04</i>	0.447*** <i>-1.24E-10</i>	0.396*** <i>0.00E+00</i>
Number of observations	79	79	3,401	3,401	3,401	3,401	3,401
Number of banks	—	—	53	53	53	—	—
R ²	0.650	0.736	—	—	—	—	—
Adjusted R ²	0.626	0.714	0.136	0.132	0.139	—	—
Within = R ²	—	—	0.138	0.134	0.140	—	—
Standard deviation residual error (<i>e_{it}</i>)	—	—	0.176	0.176	0.176	—	—
Number of groups	—	—	—	—	—	7	53
Random effect at bank level	—	—	—	—	—	-1.613*** <i>0.00E+00</i>	-1.478*** <i>0.00E+00</i>
Random effect at group level	—	—	—	—	—	-1.948*** <i>-1.90E-05</i>	—
Standard deviation of overall error term	—	—	—	—	—	-1.734*** <i>0.00E+00</i>	-1.734*** <i>0.00E+00</i>

Source: Authors.

Notes: Independent variable is return on assets. Robust *p*-values indicated in italics below coefficient: **p* < 0.1; ***p* < 0.05; ****p* < 0.01. LIBOR = London interbank offered rate; T-Bill = Treasury bill.

the panel regression allowed introducing bank-specific differences (Figure 33.3). Key results are shown in Table 33.5. The final model specification is highlighted in Column (4). The results for the macroeconomic regressions are presented in

Columns (1) and (2), whereas the fixed-effects regressions are presented in Columns (3) to (5), and those for the single and multilevel mixed effects regressions are presented in Columns (6) and (7).



Sources: Authors; and SNL Financial.

Figure 33.4 Return on Assets, Historical and Model Forecast, 1990–2014

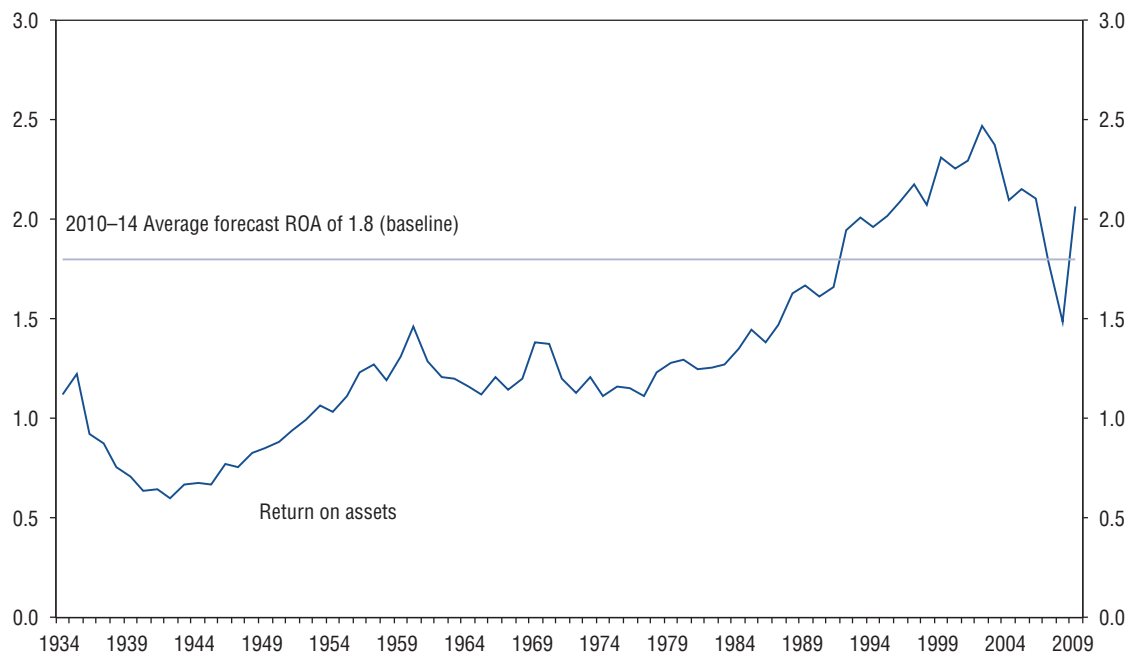
A wide range of model specifications were tested to estimate earnings, including macroeconomic (industry-wide) models and different bank-specific panel models. Given the objective of linking the earnings forecast to various macroeconomic scenarios, the choice of explanatory variables was restricted to those that could be linked directly to the macroeconomic model used for our scenario analysis (e.g., GDP, output gap, unemployment, real consumption) or for which there was an in-house forecasting model (e.g., house prices, yields on the LIBOR, or Treasury bills) that could be linked to our scenario analysis. Real personal consumption, expenditures, growth, and house prices (both on an unadjusted and detrended basis) were found to be statistically insignificant. The lagged log of total assets (*L.lnta*), which was used to capture differences associated with size, was not found to be statistically significant. The inclusion of different financial market variables yielded limited differences if using different spreads, including a 10-year high yield bond spread to 10-year treasuries and a 10-year to 3-month term structure. The three-month LIBOR to Treasury bill spread was chosen for convenience, as these variables were part of our macroeconomic modeling forecast.

Given the macroprudential focus of the exercise, model specifications do not claim to be definitive in forecasting bank

earnings. Instead, the analysis is designed to capture the sensitivity of banks' earnings to changes in the macroeconomic variables used in our stress scenarios. Our estimates would not be appropriate for forecasting earnings of individual institutions or groups of institutions, given that no attempt was made to model different business lines. It is possible that the crisis and the subsequent changes in the financial landscape (e.g., new patterns of competition, impact of regulatory reform or limited securitization) have changed the underlying relationships between banks' earnings and their determinants. Not surprisingly, there are uncertainties over the earnings outlook over the forecast period, with significantly reduced accuracy of the estimates in the outer years (Figure 33.4).¹¹ Thus, parameter estimates may not be applicable, and the forecasts would contain a bias. Nonetheless, the estimates provide some indication of the earnings capacity of the system as a whole (with group-specific variances) and the sensitivity of these earnings to various shocks.

A range of common statistical tests were used to select the final model specification. In particular:

¹¹ A 90 percent confidence interval could yield quarterly return on asset estimates anywhere between 0.25 and 0.75 percent.



Source: Federal Deposit Insurance Corporation, Historical Statistics on Banking.
Note: ROA = return on assets.

Figure 33.5 Pretax, Preprovision Net Revenue of Commercial Banks, 1984–2009 (in percent of total assets)

- The Akaike and Bayesian information criteria were helpful in narrowing down the choice of explanatory variables.¹²
- The Hausman specification and likelihood-ratio tests were used to differentiate across model specifications.¹³
- Both the Durbin-Watson *d*-statistic and the Breusch-Godfrey serial correlation Lagrange multiplier test indicate that serial correlation would disappear with the inclusion of bank-specific variables.¹⁴
- The random-effects model was rejected consistently based on the Breusch-Pagan Lagrange multiplier test as an appropriate model specification.¹⁵
- Multilevel mixed-effects panel models also were run. Dummies were introduced to account for the dramatic downturn in 2007 and 2008. They were not included in the final specification, as they did not help improve the model as the effects of the crisis would be captured through real GDP.

According to our baseline, industry-wide bank earnings would remain modest, closer to mid-1990s levels, with large variations across subgroups (Figure 33.5). Return on asset is expected to average 1.96 percent on an annualized basis over the forecast horizon. This is substantially higher than the SCAP exercise, which assumed that banks' ROA would remain almost 15 percent below the past 20-year

average for 2009–10, or around 1.6 percent on an annualized basis.

The larger banks are expected to yield larger-than-average earnings, in line with historical experience. By category, the projected ROA for the top four bank holding companies is expected to average an annualized 2.17 percent over the forecast horizon, or around its 1990–99 historical average of 2.2 percent. By comparison, the average ROA forecast is 2.33 percent for regional banks, 1.75 percent for small banks, and 1.36 percent for foreign banks (Figure 33.6). As with the larger banks, earnings are well below the levels observed in the years immediately preceding the crisis, which were 2.37 percent for regional banks, 2.26 percent for small banks, and 1.70 percent for foreign banks. The projected ROA is slightly lower over the next two years, or 1.9 percent for the system relative to 2 percent over 2010–14. In the adverse scenario, the system is expected to average an annualized 1.6 percent until end-2012, or 1.7 percent over the forecast horizon.

E. Retained earnings

To estimate retained earnings, we paid close attention to firms' tax profiles, including their ability to defer tax assets in loss-making periods.¹⁶ Broadly, the framework applied a simple 30 percent flat tax rate on banks' corporate income. In addition,

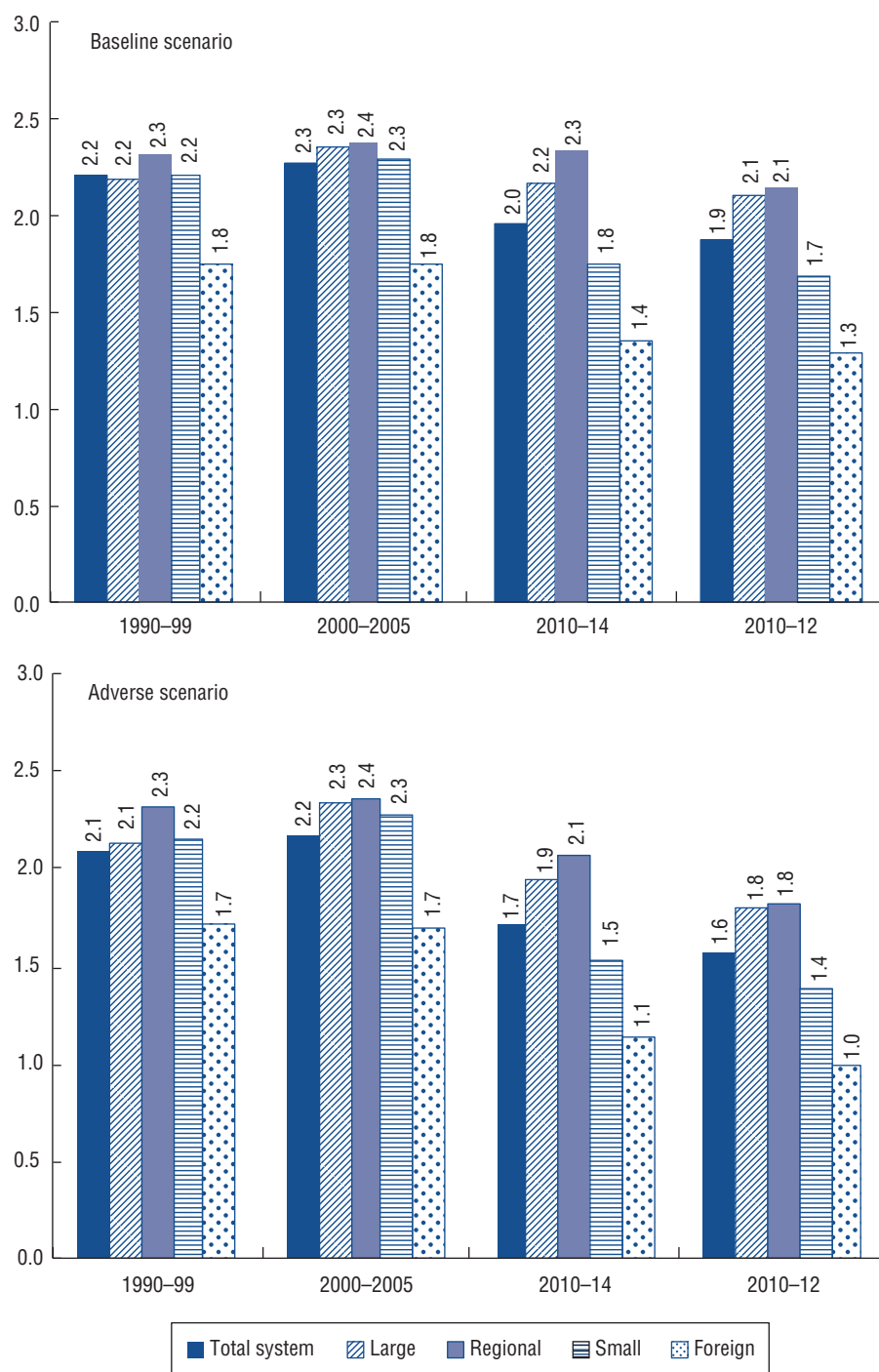
¹² See Akaike (1974) and Schwarz (1978).

¹³ See Hausman (1978).

¹⁴ See Durbin and Watson (1950, 1951), Godfrey (1978), and Breusch (1979).

¹⁵ See Breusch and Pagan (1979).

¹⁶ Retained earnings, which are defined as the preprovision net revenue minus loan charge-offs, write-downs on securities, taxes, and dividends, can be thought of as the net profits that are returned to capital at the end of each quarter.



Sources: Authors; and SNL Financial.

Note: Pretax, preprovision, predividend net revenues to total assets.

Figure 33.6 Baseline and Adverse Scenarios: Annualized Return on Asset, by Subgroups, 1990–2014

it accounted for banks' ability to accumulate deferred tax assets (DTAs), which could be used later to pay for future tax liabilities.¹⁷ In particular, when a BHC would make losses

following periods when it had taxable income, it would be allowed to carry back operating losses for two years to recover income taxes previously paid and accumulate tax benefits against future (positive) income going forward. These carry-backs are DTAs as they could be used to pay down future tax liabilities. A fraction of these DTAs would qualify as Tier

¹⁷ For details on the regulatory treatment of DTAs, see Schedules HC-R and HC-F of the FR Y-9C financial statements.

1 capital (we assumed up to 10 percent of Tier 1 capital).¹⁸ When the institution would make profits again, it would draw down its accumulated DTAs to pay for its tax liabilities, thereby boosting retained earnings and hence organic capital growth.

According to our baseline results, the large institutions would benefit materially from DTAs over the forecast horizon. Cumulative DTAs peaked at end-March 2009 at \$143 billion for the system, 68 percent of which was accounted for by the top 4 institutions (Figure 33.7). By end-2014, DTAs would help reduce future tax liabilities by 82 percent. In the baseline, 21 institutions would not have to pay income tax over the sample horizon, including one of the top 4 institutions and 5 regional banks (in the adverse scenario, the number of firms would rise to 31 institutions, including 3 of the top 4 institutions).

A straightforward dividend rule was applied to all financial firms in the baseline. In particular, when net aftertax income was positive, the calculations assumed a 5 percent annualized dividend rate for TARP preferred shares, 8 percent for other preferred shares (relative to an average of 5 percent over 1990–99), and 15 percent for common equity (relative to an average of 22 percent over 1990–99). This resulted in an 11.6 percent annualized average dividend rate for common equity and 2.6 percent annualized average dividend rate for preferred shares (Figure 33.8). This is significantly lower than historical dividend rates. In the downside risk scenarios, however, banks were not expected to pay out dividends on their common shares, in line with the assumption underlying the authorities' SCAP exercise.

Under the baseline, banks' retained earnings would be sufficient to cover losses over the forecast period. For the industry as a whole, retained earnings (defined as preprovision pretax net revenue minus loan charge-offs, securities write-downs, taxes, and dividends) would remain positive, although low (slightly above \$20 billion on average) until end-2011, at which point they would start rising (Figure 33.9). Regional banks follow a similar pattern (with average quarterly retained earnings of less than \$1 billion until end-2011), while small banks face negative retained earnings until the first quarter of 2012. Over the full 2010–14 forecast horizon, retained earnings for the system would average \$43 billion on a quarterly basis (\$34 billion for SCAP firms, \$4 billion for regional banks, and less than \$1 billion for small banks).

F. Balance sheet expansion

Over the 2010–14 horizon, asset growth is slightly weaker than nominal GDP growth due to deleveraging (Figure 33.10).

The calculations control for a number of factors that affect balance sheet expansion. In particular, weak securitization markets, which could lead banks to retain a larger-than-ordinary share of assets on their balance sheets, are predicted to add \$195 billion to total system assets. The introduction of the FAS 166/167 accounting rules in 2010, which require banks to bring on balance sheet a significant amount of assets previously held off-balance-sheet, are also assumed to expand banks' balance sheets by \$375 billion.

Furthermore, retained earnings were added back into total assets, adding \$670 billion to banks' balance sheets over the sample horizon (64 percent of which was generated by the top 6 firms). Factors tempering growth of total system assets included asset sales, which subtracted \$375 billion, and asset maturities without rollovers, which reduced assets by \$496 billion. Except for retained earnings, which were estimated on a bank-by-bank basis, the balance sheet expansion factors were distributed across firms according to their share of total system assets. Total assets can be decomposed as follows:

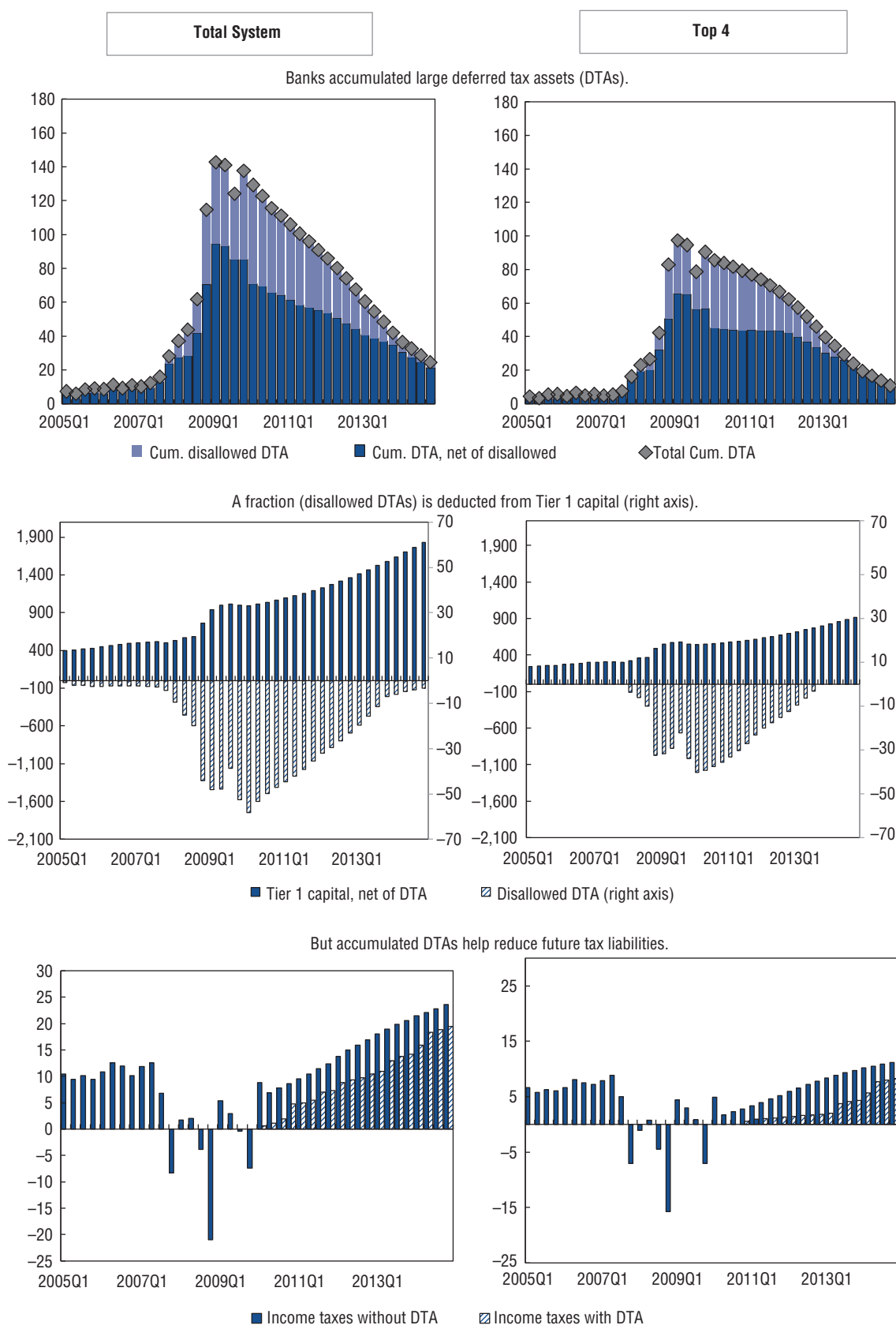
$$TA_{it} = (rgdp_qg * TA_{t-1}) + \left(\frac{TA_{t-1}}{TA_{t-1}} \right) * BSE_{it} + RE_{it} - co_{it} - wd_{it},$$

where TA_{it} is total assets for bank i at time t (I refers to the BHC sample); $rgdp_qg$ is the quarterly growth rate of real GDP; BSE_{it} and RE_{it} are, respectively, the projected balance sheet expansion and the estimated retained earnings; and co_{it} and wd_{it} are, respectively, the estimated loan charge-offs and securities write-downs.

The path for risk-weighted assets also was modeled carefully. Since end-2007, the ratio of risk-weighted assets to total assets has fallen by over 5 percentage points to 61 percent, the lowest point recorded since the introduction of risk-weighted assets (Figure 33.11). This falling trend reflects banks' substantial efforts to "de-risk" their balance sheets since the onset of the crisis. However, it would not be likely for banks to maintain such a low ratio, especially as they expand their lending activities. Thus, it was assumed that, under the baseline, banks' risk-weighted to total asset ratios would return progressively back to their 2000–2005 average by mid-2011 (the adverse scenarios assumed that the ratio would remain constant at the low end-March 2010 level).

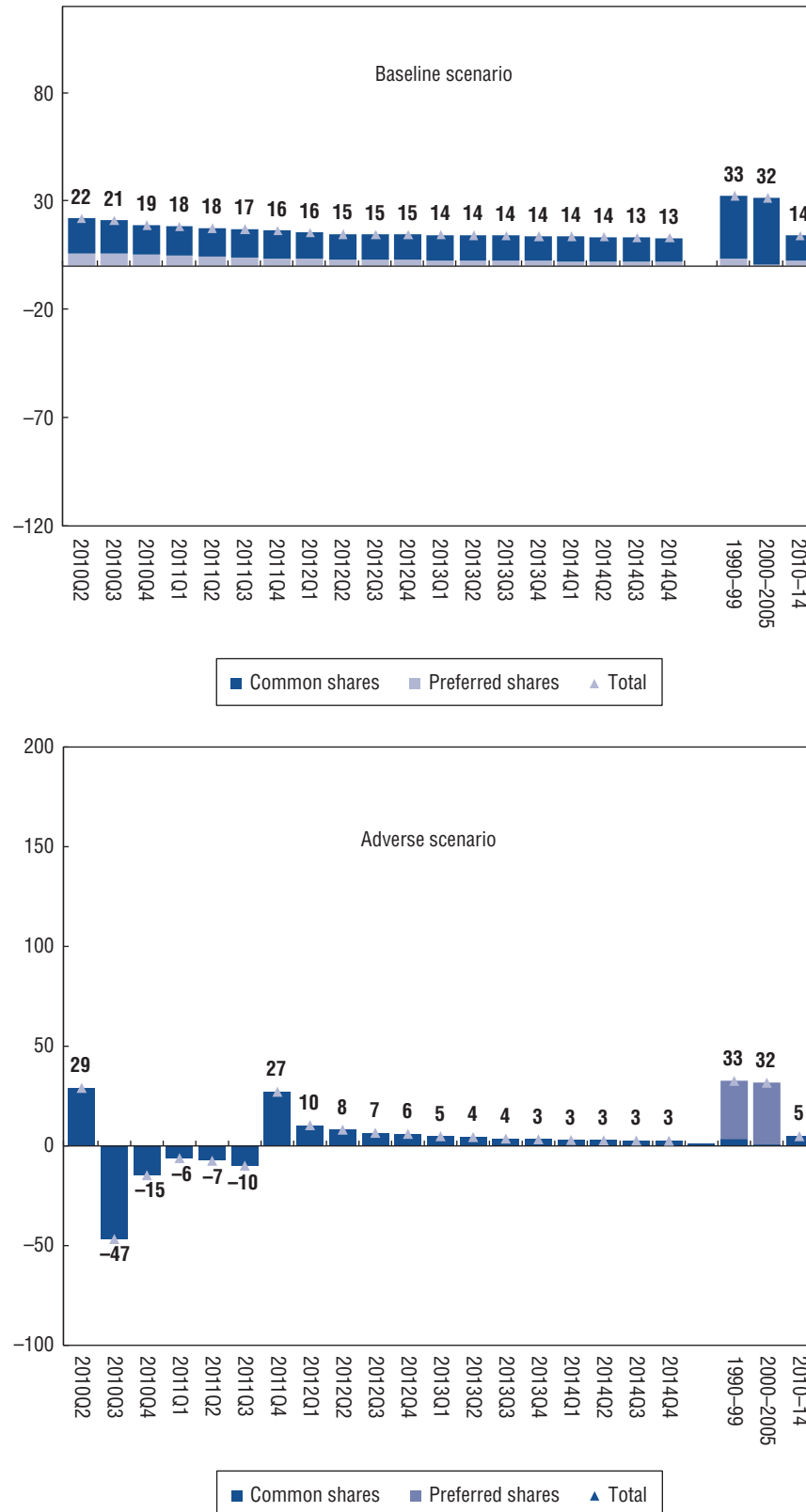
Furthermore, the composition of BHCs' balance sheets was allowed to adjust to ensure maximum room for credit expansion. Broadly, the framework allowed banks' loan portfolios to grow in proportion to their asset growth, assuming a constant loan-to-asset ratio. However, this could have materially underestimated credit growth, given banks' record low loan-to-asset ratio at end-March 2010. Instead, BHCs also were allowed to expand their loan portfolios by drawing down up to 5 percent of their "other assets" for eight consecutive quarters (or until "other assets" reached 20 percent of total assets). As a result, the loan-to-asset ratio was raised by 7 percentage points to 50 percent by the end of the forecast horizon, although still well below historical averages. The path for total loans can be decomposed as follows:

¹⁸ According to U.S. BHC prudential requirements, "allowed DTAs" are to be equal to the lesser of 10 percent of Tier 1 capital (before DTA adjustments) or the amount of DTAs expected to be realized within one year, based on the BHC's projection of future taxable income.



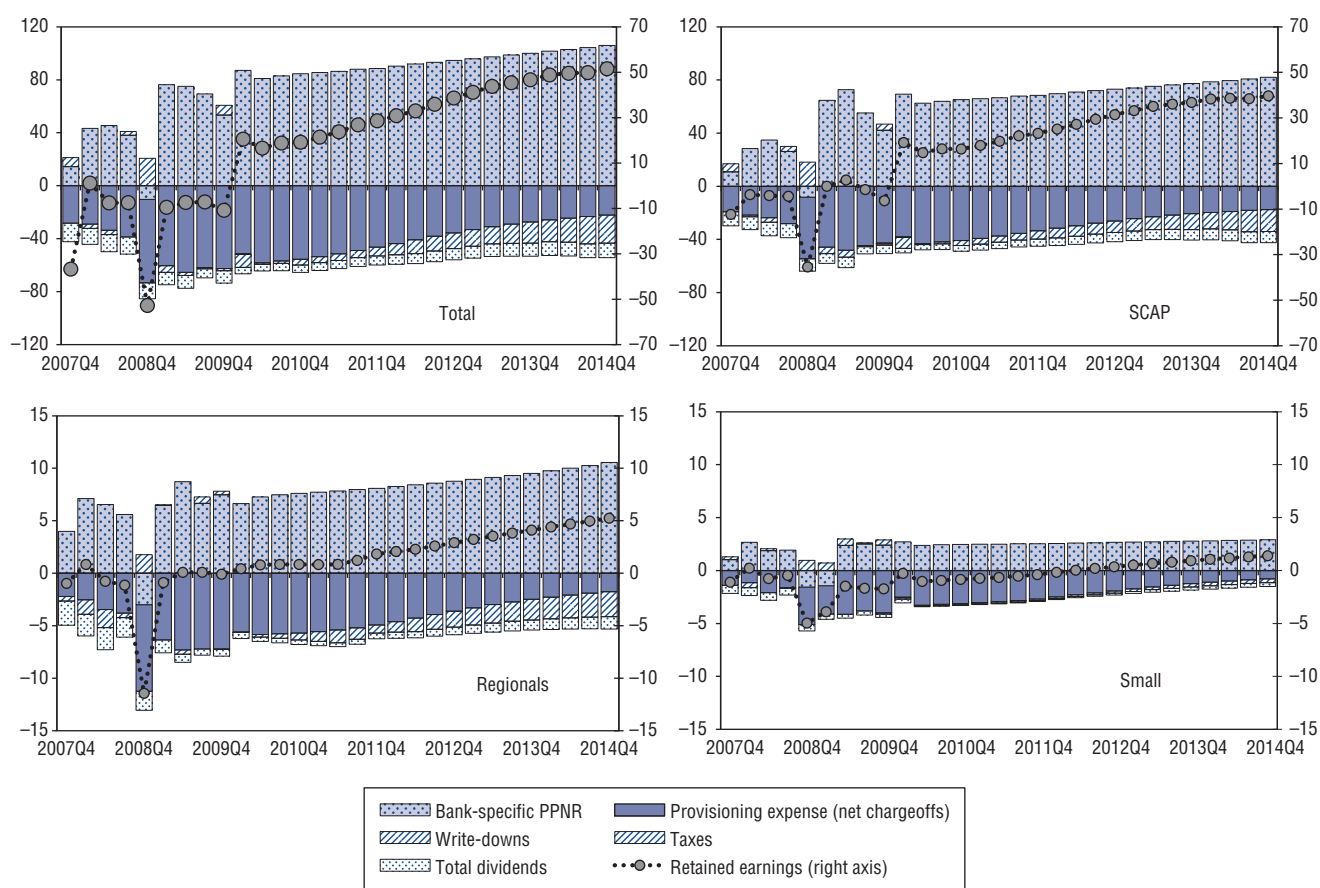
Sources: SNL Financial; and IMF staff calculations.
Note: Cum. = cumulative.

Figure 33.7 Baseline Scenario: Impact of Deferred Tax Assets, 2005–14



Sources: Authors; and SNL Financial.

Figure 33.8 Baseline and Adverse Scenarios: Historical and Projected Dividends, 1990–2014 (in percent of pre-tax income)



Sources: Authors; and SNL Financial.

Note: PPNR=preprovision pretax net revenue; SCAP=Supervisory Capital Assessment Program.

Figure 33.9 Baseline Scenario: Retained Earnings, 2007–14 (in billions of dollars)

$$TL_{it} = (rgdp_qg * TL_{t-1}) + \left(\frac{TL_{t-1}}{TL_{t-1}} \right) * BSE_{it} \left(\frac{TL_{t-1}}{TA_{t-1}} \right) RE_{it} - co_{it} - wd_{it} + \alpha_i \left(\frac{QA_{t-1}}{TA_{t-1}} \right),$$

where TL_{it} stands for total loans, OA_{it} for other (nonloan, nonsecurity) assets, and α for the fraction by which securities can be substituted for loans.

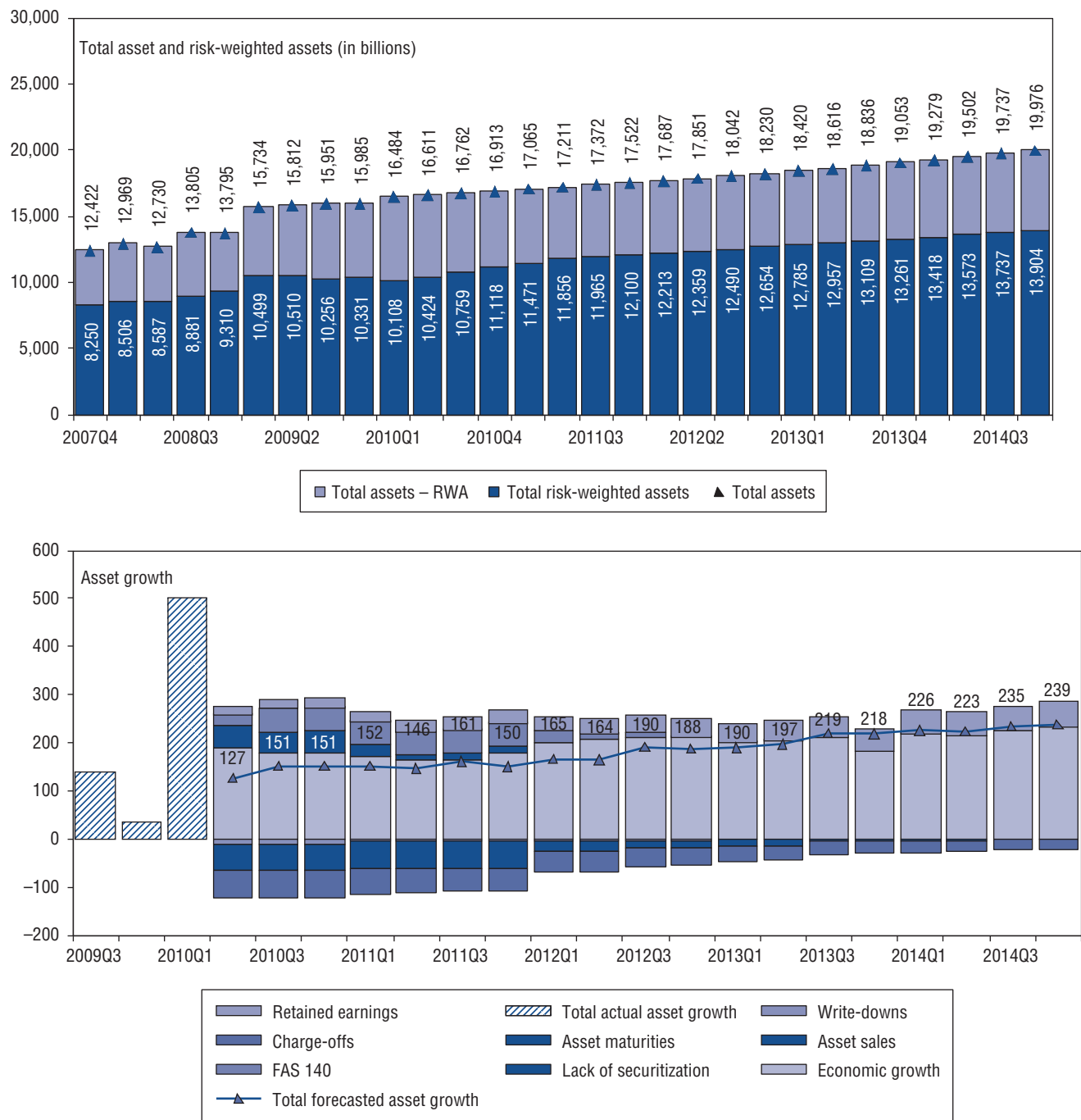
G. Capital shortfall estimates

Under the baseline scenario, bank capital would be adequate on an industry-wide basis. Notwithstanding weak growth, high unemployment, and record high charge-off rates, the top four BHCs and the former broker dealers are expected to maintain a 6 percent Tier 1 common equity ratio over 2010–14 (bottom panel in Table 33.5). However, three SCAP institutions would require an addition of \$7.4 billion in capital to maintain the same ratio. Because of high CRE exposure, four regional banks (including two SCAP institutions) may require \$1.3 billion in additional capital, while seven smaller institutions would likely require an additional \$6.3 billion.

Subsidiaries of foreign banks, which tend to be lightly capitalized and rely on parental support, may require up to \$26.3 billion. Overall, the system would require \$40.5 billion in additional capital. The top four institutions would need to raise \$40.4 billion in additional capital if required to maintain a 5.9 percent tangible common equity to tangible assets ratio (or 17 times leverage).

Our results suggest that weak financial institutions should be encouraged to raise capital as current conditions do not allow them to grow out of their problems. The picture of capital shortfall does not change materially when focusing on a shorter two-year forecast horizon (upper panel in Table 33.6). Overall, the capital shortfalls would affect the same institutions. This suggests that weak institutions are not able to rely on organic growth to improve their financial condition. The system as a whole could require as much as \$33.6 billion in additional capital to maintain a 6 percent Tier 1 common capital ratio, most of which borne by the foreign banks (\$26.3 billion).

The estimated capital shortfall of foreign banks is difficult to interpret. The current exercise stresses foreign insti-

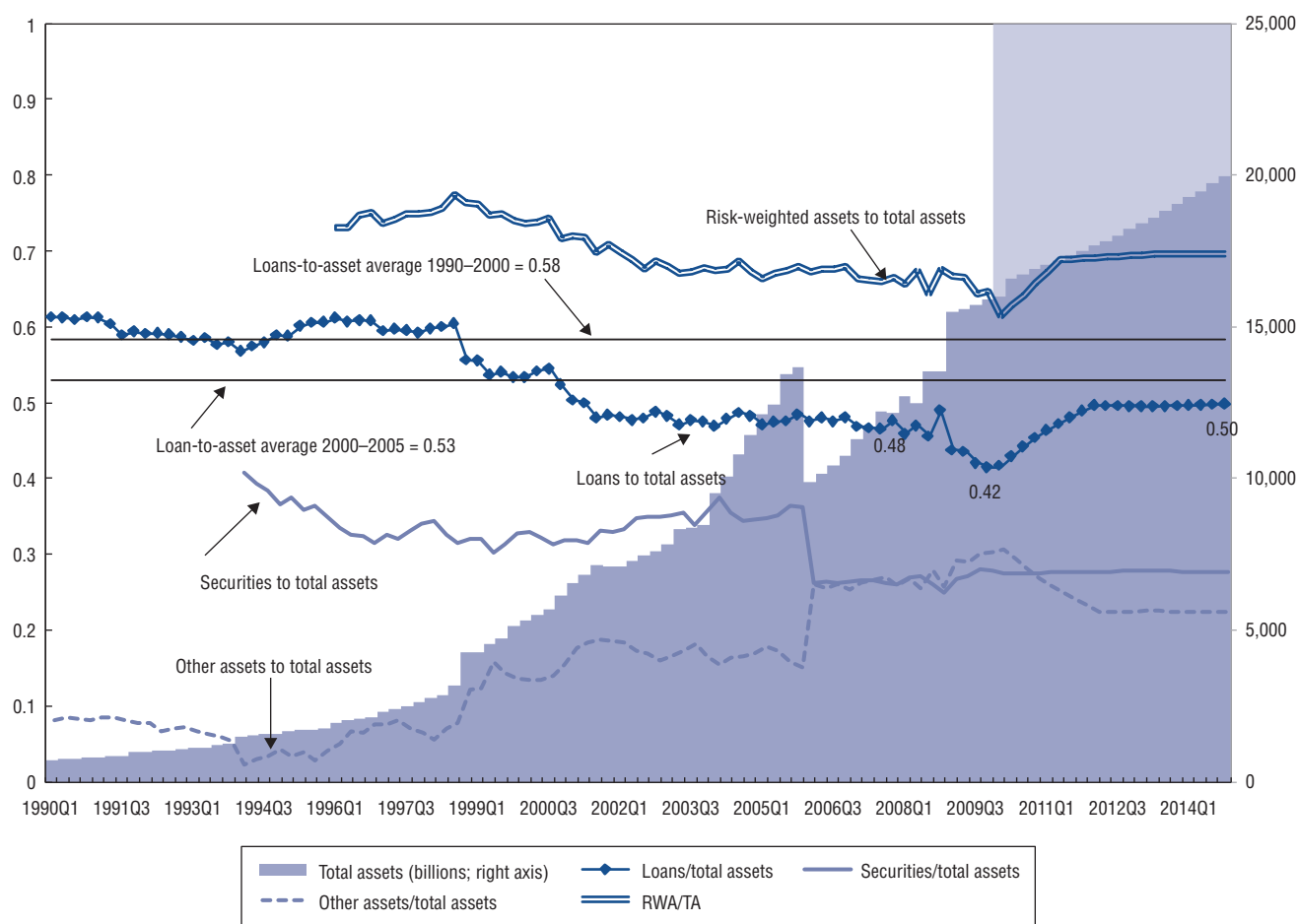


Sources: SNL Financial; and IMF staff calculations.

Figure 33.10 Balance Sheet Expansion (in billions of dollars)

tutions in the same way as it does domestic ones as a way of assessing the broader shock absorption capacity of the U.S. banking system. In normal times, foreign holding companies tend to operate with lower capital buffers than their domestic peers, as they are not required to comply with the U.S. regulatory capital requirements, provided their par-

ents are deemed well capitalized and well managed. Under a global adverse shock, however, it could be particularly difficult for regulators to require higher capital buffers when parent banks could be equally strained. Although the resulting retrenchment or closure of foreign banks would likely not have systemic consequences from a financial stability



Sources: Authors; and SNL Financial.

Note: RWA = risk-weighted assets; TA = total assets.

Figure 33.11 Baseline Scenario: Bank Holding Company Asset Composition, 1990–2014 (in percent)

perspective, it may have broader macroprudential implications depending on the operations of the affected institutions.¹⁹ Since end-2007, foreign BHCs reduced their loan market share by 5 percentage points to 9 percent.

Credit growth could remain limited for some time (Figure 33.12). Although the financial system appears stable from a financial stability perspective, its relatively low level of retained earnings, combined with banks' recent efforts to deleverage and de-risk their balance sheets, may result in limited credit expansion, even after accounting for internal asset substitution away from cash and other assets into loans. Our results suggest that, in the absence of additional capi-

tal injections, credit growth could average around 8 percent for 2010–14, which is substantially lower than historical levels. For example, credit growth rates averaged around 16.1 percent in 1993–96 (following the savings and loan crisis) and 16.8 percent in 2004–07 (after the 2002–03 recession). In the adverse scenario, the average credit growth could fall by another 2 percentage points for the forecast horizon. In reality, banks will have various ways to meet credit demand, including by raising new capital, curbing dividend rates, or managing to generate higher retained earnings than anticipated.

2. ALTERNATIVE SCENARIOS

To test banks' shock absorption capacity under worse-than-anticipated macroeconomic conditions, we considered downside risk scenarios. These included an adverse macroeconomic scenario and an alternative funding risk scenario. The assumed values were consistent with historical distress episodes, and

¹⁹ The possible retrenchment of foreign banks following a shock in the home country is well documented in the literature, including in Peek and Rosengren (1997) regarding the behavior of Japanese banks after the stock market shock in Japan in the early 1990s and in Martinez-Peria, Powell, and Vladkova-Hollar (2005) regarding foreign banks in Latin America.

TABLE 33.6

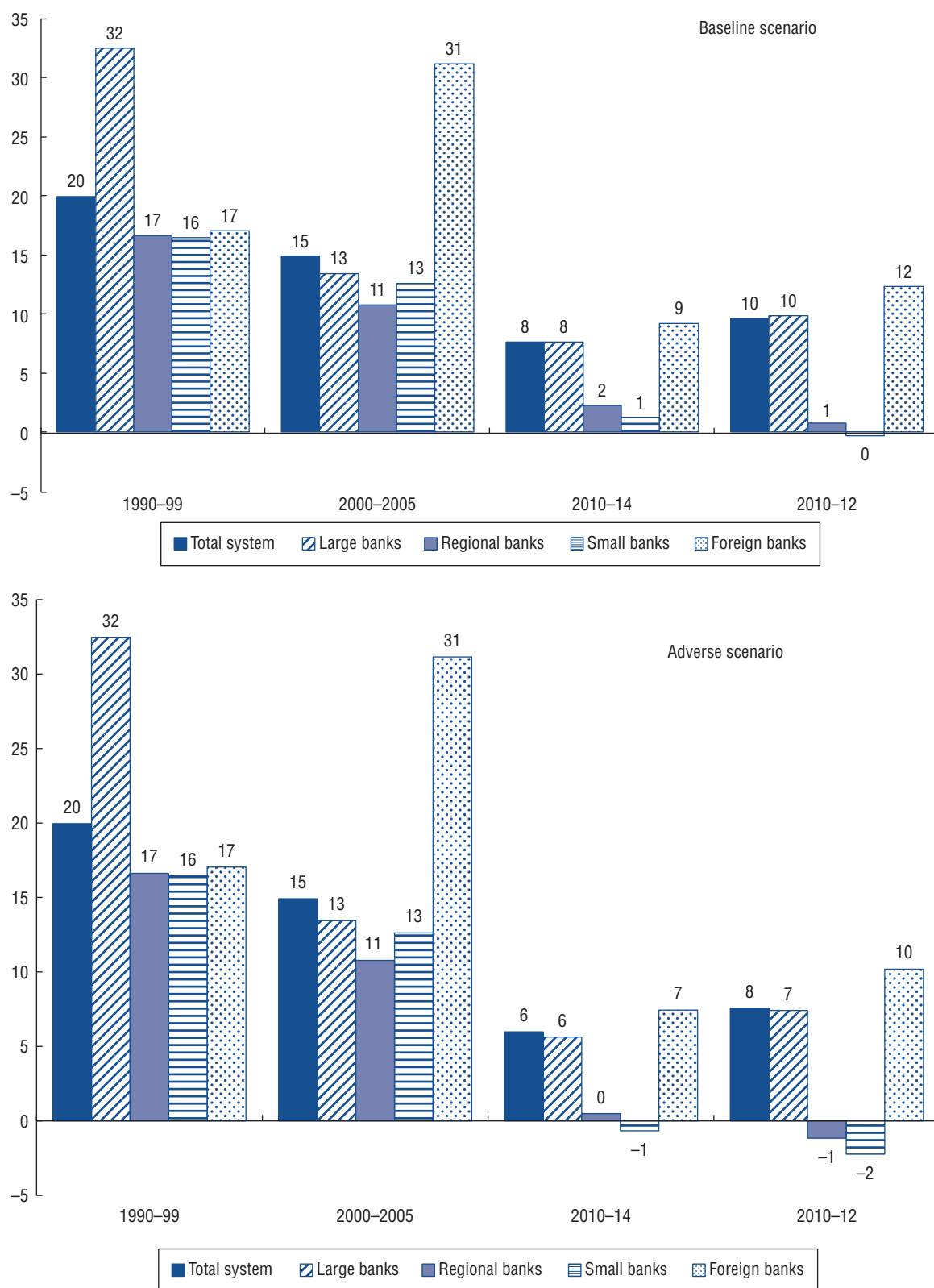
Baseline Scenario: BHC Capital Needs, 2010–14 (in billions of dollars, unless otherwise noted)

Scenario	Top Four	Investment	Regional	Processing	Consumer	Small	Foreign	Other	Total	U.S. Only
2010:Q2–2011:Q4 (cumulative)										
Pretax, preprovision net revenue	351.0	97.6	68.1	21.3	38.8	22.4	45.9	92.2	737.3	691.4
Loan losses	276.6	0.2	51.2	0.9	26.4	27.7	40.3	62.3	485.5	445.2
Securities losses	1.51	0.06	0.19	0.00	0.00	0.13	0.00	0.0	1.9	1.9
Taxes	2.16	16.84	5.00	5.97	0.61	0.41	–0.80	0.8	31.0	31.8
Dividends	18.9	13.7	4.2	2.1	3.8	1.2	2.3	9.8	56.0	53.7
Addition to retained earnings	53.3	66.4	7.6	12.1	6.3	–6.9	4.5	23.5	166.8	162.3
Capital injection end-2011 to reach Tier 1 capital/risk-weighted assets ratio										
6 percent	0.0	0.0	0.0	0.0	0.0	1.6	26.6	0.0	28.2	1.6
8 percent	0.0	0.0	0.0	0.0	0.0	2.6	37.8	0.0	40.4	2.6
Number of banks requiring injection										
6 percent	0	0	0	0	0	3	4	N/A	7	3
8 percent	0	0	0	0	0	3	4	N/A	7	3
Tier 1 common capital/risk-weighted assets ratio¹										
Capital injection end-2011 to reach										
4 percent	0.0	0.0	0.0	0.0	0.0	2.7	15.8	0.0	18.5	2.7
6 percent	0.0	0.0	0.5	0.0	2.2	4.5	26.3	0.0	33.6	7.2
Number of banks requiring injection										
4 percent	0	0	1	0	0	4	4	N/A	9	5
6 percent	0	0	4	0	1	7	4	N/A	16	12
Tangible common equity/tangible assets ratio										
Capital injection end-2011 to reach										
4 percent (25 times leverage)	0.0	2.7	0.4	0.0	2.9	3.4	32.2	0.0	41.6	9.5
5.9 percent (17 times leverage)	40.4	18.3	4.4	3.2	7.0	6.9	53.4	0.0	133.4	80.0
Number of banks requiring injection										
4 percent (25 times leverage)	0	1	2	0	1	5	4	N/A	13	9
5.9 percent (17 times leverage)	2	1	5	1	2	11	6	N/A	28	22
2010:Q2–2014:Q4 (cumulative)										
Pretax, preprovision net revenue	895.0	262.5	179.7	56.7	99.5	55.2	121.6	244.0	1914.3	1792.6
Loan losses	496.3	0.3	87.2	1.5	51.4	46.2	66.0	111.9	860.9	794.9
Securities losses	1.5	0.1	0.2	0.0	0.0	0.1	0.0	0.0	1.9	1.9
Taxes	43.8	66.3	26.3	15.5	12.4	3.5	6.9	30.6	205.4	198.5
Dividends	68.5	34.3	14.6	5.9	8.0	4.1	8.8	23.8	168.2	159.4
Addition to retained earnings	286.2	161.1	51.5	33.6	25.9	1.3	40.3	81.8	681.8	641.5
Capital injection at lowest point for Tier 1 capital/risk-weighted assets ratio										
6 percent	0.0	0.0	0.0	0.0	0.0	2.8	26.6	0	29.5	2.8
8 percent	0.0	0.0	0.0	0.0	0.0	3.8	37.8	0	41.6	3.8
Number of banks requiring injection										
6 percent	0	0	0	0	0	3	4	N/A	7	3
8 percent	0	0	0	0	0	3	4	N/A	7	3
Tier 1 common capital/risk-weighted assets ratio¹										
4 percent	0.0	0.0	0.0	0.0	3.8	4.1	15.8	0.0	23.7	7.9
6 percent	0.0	0.0	1.3	0.0	6.6	6.3	26.3	0.0	40.5	14.2
Number of banks requiring injection										
4 percent	0	0	1	0	1	4	4	N/A	10	6
6 percent	0	0	4	0	1	7	4	N/A	16	12
Tangible common equity/tangible assets ratio										
4 percent (25 times leverage)	0.0	2.7	0.6	0.0	7.7	4.8	32.2	0.0	47.9	15.7
5.9 percent (17 times leverage)	40.4	18.3	5.3	3.2	12.1	9.0	53.5	0.0	141.6	88.2
Number of banks requiring injection										
4 percent (25 times leverage)	0	1	2	0	1	5	4	N/A	13	9
5.9 percent (17 times leverage)	2	1	5	1	2	11	6	N/A	28	22
Memo: Percent of total system assets	46.5	10.4	8.0	2.8	2.9	3.5	11.3	14.6	100.0	88.7

Sources: Authors; and SNL Financial.

Note: BHC = bank holding company.

1. Tier 1 common capital deducts all “noncommon” elements of Tier 1 capital (i.e., qualifying minority interest in consolidated subsidiaries, qualifying trust preferred securities, and qualifying perpetual preferred stock).



Sources: Authors; and SNL Financial.

Figure 33.12 Baseline and Adverse Scenarios: Credit Growth, 1990–2014 (year-on-year in percent)

the magnitudes of the shocks are broadly in the ranges analyzed in other FSAPs (details on the alternative scenario are presented in Appendix I). These scenarios also help demonstrate the sensitivity of the baseline results to underlying assumptions.

A. Adverse scenario

Under the adverse scenario, loan losses continue increasing appreciably. Residential and CRE loan losses continue to rise until 2011 (peaking at 3.4 percent and 4.6 percent, respectively), while losses on consumer and C&I loans rise further, without reaching their earlier peaks (Table 33.7). Cumulative loan losses are expected to reach \$1.1 trillion for the system as a whole by end-2014, representing a 7.7 percent cumulative loss rate for 2010–11 (15.9 percent for 2010–14). In addition, the institutions with securities portfolios are also expected to write down \$100 billion of marked-to-market securities, resulting in a total cumulative loss rate of 17.4 percent for 2010–14 (8.7 percent for 2010–11).

On aggregate, BHCs no longer would be able to absorb their losses through earnings in the near term. Retained earnings would remain negative until 2012 for the system as a whole and until 2014 for the smaller banks. The SCAP firms would fare slightly better with retained earnings turning positive by end-2011. Retained earnings for the system would record an average quarterly loss of \$2.4 billion for 2010–11 (\$1.2 billion for the regional banks and \$1.8 billion for the small institutions).

Almost half of the U.S. BHCs would experience some capital shortfall under the adverse scenario (Table 33.7). U.S. BHCs would require a total of \$31.8 billion capital to maintain a 4 percent Tier 1 common capital ratio until end-2014 (\$53.6 billion including the foreign BHCs). In particular, 4 regional banks would require \$8.1 billion, 10 smaller institutions another \$14.9 billion. Three SCAP banks would face a shortfall of \$14.5 billion. One of the top 4 institutions would need to raise \$15.2 billion to maintain a 4 percent tangible common equity to tangible assets ratio by end-2014. Over the 2010–11 horizon, 10 U.S. BHCs (including 2 SCAP institutions) would be expected to face a capital shortfall of \$8.9 billion to maintain a 4 percent Tier 1 common capital ratio.

B. Alternative scenario

Rollover risk warrants careful surveillance. Market liquidity risks appear to have declined, thanks to effective and powerful policy response by the authorities during the crisis. Financial institutions, however, remain vulnerable to the potential risk posed by the large volume of CRE loans that are expected to mature between 2010 and 2014 (many of which with negative equity) when real estate prices may not have yet recovered, together with the rising stock of seriously delinquent mortgages on banks' balance sheets.

The alternative scenario tests the sensitivity of banks' capital shortfall estimates to a further small deterioration in the CRE sectors in 2010–11. Under this scenario, the macroeconomic conditions are broadly similar to those in the adverse scenario for the first two years but return faster to the baseline beyond 2011 (Appendix I). CRE prices are expected to fall by another 8 percent by end-2012 (as opposed to 3.3 percent in the adverse), while house prices are expected to fall by 4.1 percent in 2010 and another 2.6 percent in 2011. Banks' assumed difficulty in rolling over maturing debt leads to higher losses on CRE loans, which peak at 5.1 percent at end-2011.

Our results suggest that, except for banks already heavily exposed to CRE, macroeconomic conditions are currently the key determinant to banks' financial soundness. Broadly, our results under the alternative scenario (Table 33.8) are not materially different from those in the adverse scenario. Overall, 14 U.S. BHCs would require \$20.5 billion capital to maintain a 4 percent Tier 1 common capital ratio over the 2010–14 period (\$7.4 billion over the 2010–11 period). This suggests that banks that are heavily exposed to the CRE losses will find it difficult to earn their way out of their problems under worse-than-expected macroeconomic conditions, but this result is not highly sensitive to a further small deterioration in real estate prices or recovery rates on delinquent real estate loans. Clearly, a broader shock to banks' funding conditions that leads, for example, to a substantive rise in short-term spreads would likely have a more dramatic impact on banks' earnings and hence on their ability to absorb losses.

3. CONCLUSION

The results confirm that BHCs' asset quality and capital positions are closely interlinked with developments in the housing sector and the broader macroeconomy. There is much uncertainty about the shape and height of the loss profiles, although they are expected to be a drag on retained earnings and credit growth. Identified fragilities in regional and smaller institutions do not appear systemic but could hamper economic recovery in local communities with broader repercussions on bank loss rates. To mitigate this risk, the authorities intend to allocate \$30 billion of TARP money to community banks. Another potential vulnerability is the low capitalization of foreign-owned BHCs.

Despite the significant improvement in BHCs' capital buffers, our results suggest that additional capital may be needed to create room for meaningful credit growth. Since the crisis, BHCs have managed to almost double their holdings of "high-quality" capital. Nonetheless, the current combination of record low risk-weighted to total asset ratio, outlook for a protracted period of high loss profiles, limited risk transfer through securitization, and general recognition that financial institutions need to hold higher capital buffers than in precrisis means that banks' balance sheets may not be as strong as their capital buffers would suggest.

TABLE 33.7

Adverse Scenario: BHC Capital Needs, 2010–14 (in billions of dollars, unless otherwise noted)

Scenario	Top Four	Investment	Regional	Processing	Consumer	Small	Foreign	Other	Total	U.S. Only
2010:Q2–2011:Q4 (cumulative)										
Pretax, preprovision net revenue	307.5	88.2	60.7	18.6	35.5	19.3	36.0	80.5	646.3	610.3
Loan losses	327.2	0.2	60.9	1.1	31.1	32.2	48.5	69.5	570.6	522.2
Securities losses	32.70	0.17	6.47	4.05	2.03	2.47	5.32	16.5	69.7	64.4
Taxes	–2.18	13.98	1.20	6.89	0.46	0.00	–1.73	0.4	19.1	20.8
Dividends	5.9	3.6	1.8	0.4	1.5	0.7	0.3	3.8	18.0	17.7
Addition to retained earnings	–54.8	69.8	–9.6	6.1	–1.2	–16.0	–16.0	–8.5	–30.2	–14.2
Capital injection end-2011 to reach Tier 1 capital/risk-weighted assets ratio										
6 percent	0.0	0.0	0.3	0.0	0.0	2.8	19.4	0.0	22.5	3.1
8 percent	0.0	0.0	0.9	0.0	0.0	4.6	28.9	0.0	34.3	5.5
Number of banks requiring injection										
6 percent	0	0	1	0	0	4	4	N/A	9	5
8 percent	0	0	2	0	0	5	5	N/A	12	7
Tier 1 common capital/risk-weighted assets ratio¹										
Capital injection end-2011 to reach										
4 percent	0.0	0.0	1.5	0.0	2.2	5.2	14.6	0.0	23.5	8.9
6 percent	0.0	0.0	5.5	0.0	5.2	9.5	24.4	0.0	44.6	20.2
Number of banks requiring injection										
4 percent	0	0	3	0	1	6	4	N/A	14	10
6 percent	0	0	5	0	1	11	5	N/A	22	17
Tangible common equity/tangible assets ratio										
Capital injection end-2011 to reach										
4 percent (25 times leverage)	8.6	2.3	3.2	0.7	4.5	6.9	36.5	0.0	62.8	26.2
5.9 percent (17 times leverage)	106.3	17.9	10.9	4.6	8.8	12.6	62.7	0.0	223.8	161.1
Number of banks requiring injection										
4 percent (25 times leverage)	1	1	4	1	1	11	5	N/A	24	19
5.9 percent (17 times leverage)	4	1	7	1	2	12	7	N/A	34	27
2010:Q2–2014:Q4 (cumulative)										
Pretax, preprovision net revenue	770.8	238.9	156.1	49.5	89.6	46.3	94.9	208.7	1654.8	1559.8
Loan losses	633.3	0.4	121.4	2.2	60.5	61.5	90.8	143.2	1113.4	1022.6
Securities losses	47.2	0.2	9.4	5.9	3.0	3.6	7.8	24.1	101.1	93.3
Taxes	6.3	59.1	10.1	13.4	7.0	1.4	0.8	3.0	101.2	100.4
Dividends	12.6	7.6	3.6	0.4	1.5	0.9	0.4	7.5	34.4	34.0
Addition to retained earnings	72.8	171.1	11.8	27.4	15.9	–21.0	–4.4	32.1	305.7	310.1
Capital injection at lowest point for Tier 1 capital/risk-weighted assets ratio										
6 percent	0.0	0.0	2.8	0.0	0.0	9.1	25.5	0	37.4	11.9
8 percent	0.0	0.0	6.4	0.0	0.6	12.9	37.7	0	57.5	19.9
Number of banks requiring injection										
6 percent	0	0	3	0	0	6	6	N/A	15	9
8 percent	0	0	4	0	1	10	7	N/A	22	15
Tier 1 common capital/risk-weighted assets ratio¹										
4 percent	0.0	0.0	8.1	0.0	8.8	14.9	21.8	0.0	53.6	31.8
6 percent	0.0	0.0	12.8	0.0	12.1	19.7	31.7	0.0	76.4	44.6
Number of banks requiring injection										
4 percent	0	0	4	0	1	10	4	N/A	19	15
6 percent	0	0	5	0	1	11	6	N/A	23	17
Tangible common equity/tangible assets ratio										
4 percent (25 times leverage)	15.2	2.3	9.8	0.7	11.2	16.4	43.4	0.0	99.1	55.7
5.9 percent (17 times leverage)	119.1	17.9	18.4	4.6	15.7	22.5	72.0	0.0	270.3	198.2
Number of banks requiring injection										
4 percent (25 times leverage)	1	1	4	1	1	11	5	N/A	24	19
5.9 percent (17 times leverage)	4	1	7	1	2	13	8	N/A	36	28
Memo: Percent of total system assets	46.5	10.4	8.0	2.8	2.9	3.5	11.3	14.6	100.0	88.7

Sources: Authors; and SNL Financial.

Note: BHC = bank holding company.

1. Tier 1 common capital deducts all “non-common” elements of Tier 1 capital (i.e., qualifying minority interest in consolidated subsidiaries, qualifying trust preferred securities, and qualifying perpetual preferred stock).

TABLE 33.8

Alternative Scenario: BHC Capital Needs, 2014–14 (in billions of dollars, unless otherwise noted)

Scenario	Top Four	Investment	Regional	Processing	Consumer	Small	Foreign	Other	Total	U.S. Only
2010:Q2–2011:Q4 (cumulative)										
Pretax, preprovision net revenue	313.3	89.3	61.7	19.0	35.9	19.7	37.6	82.6	659.1	621.5
Loan losses	326.5	0.2	61.0	1.1	29.8	32.3	48.0	69.5	568.3	520.3
Securities losses	18.91	0.12	3.70	2.26	1.13	1.44	2.97	9.2	39.7	36.8
Taxes	–2.18	14.35	1.61	7.35	0.46	0.08	–1.73	0.4	20.4	22.1
Dividends	6.4	3.6	1.9	0.4	1.5	0.7	0.3	5.5	20.3	20.0
Addition to retained earnings	–35.0	70.6	–6.3	7.8	1.4	–14.7	–11.5	–0.8	11.4	22.9
Capital injection end-2011 to reach										
Tier 1 capital/risk-weighted assets ratio										
6 percent	0.0	0.0	0.2	0.0	0.0	2.5	18.4	0.0	21.1	2.7
8 percent	0.0	0.0	0.7	0.0	0.0	4.0	26.3	0.0	31.1	4.7
Number of banks requiring injection										
6 percent	0	0	1	0	0	3	4	N/A	8	4
8 percent	0	0	2	0	0	5	5	N/A	12	7
Tier 1 common capital/risk-weighted assets ratio¹										
Capital injection end-2011 to reach										
4 percent	0.0	0.0	0.9	0.0	1.7	4.8	13.4	0.0	20.8	7.4
6 percent	0.0	0.0	4.3	0.0	4.7	8.4	22.0	0.0	39.4	17.4
Number of banks requiring injection										
4 percent	0	0	3	0	1	5	3	N/A	12	9
6 percent	0	0	4	0	1	11	5	N/A	21	16
Tangible common equity/tangible assets ratio										
Capital injection end-2011 to reach										
4 percent (25 times leverage)	0.0	2.5	2.3	0.4	3.8	5.9	34.1	0.0	49.0	14.9
5.9 percent (17 times leverage)	78.4	18.0	8.7	4.3	8.1	11.5	59.7	0.0	188.6	128.9
Number of banks requiring injection										
4 percent (25 times leverage)	0	1	3	1	1	11	5	N/A	22	17
5.9 percent (17 times leverage)	4	1	6	1	2	12	7	N/A	33	26
2010:Q2–2014:Q4 (cumulative)										
Pretax, preprovision net revenue	812.6	247.3	163.5	52.3	92.7	49.4	106.4	223.6	1747.7	1641.4
Loan losses	625.3	0.4	112.9	2.1	55.7	56.0	88.2	128.4	1068.8	980.7
Securities losses	21.1	0.1	4.1	2.5	1.3	1.6	3.3	10.4	44.6	41.2
Taxes	8.9	61.7	14.6	15.4	9.2	2.1	3.2	16.1	131.3	128.0
Dividends	14.5	7.6	4.1	0.4	1.5	1.5	0.4	9.2	39.1	38.7
Addition to retained earnings	144.2	177.1	27.9	31.7	23.3	–11.6	11.6	60.8	464.9	453.3
Capital injection at lowest point for										
Tier 1 capital/risk-weighted assets ratio										
6 percent	0.0	0.0	0.8	0.0	0.0	5.9	21.4	0	28.1	6.7
8 percent	0.0	0.0	2.7	0.0	0.0	8.0	31.9	0	42.7	10.8
Number of banks requiring injection										
6 percent	0	0	1	0	0	5	5	N/A	11	6
8 percent	0	0	3	0	0	7	5	N/A	15	10
Tier 1 common capital/risk-weighted assets ratio¹										
4 percent	0.0	0.0	4.0	0.0	6.4	10.0	17.8	0.0	38.3	20.5
6 percent	0.0	0.0	8.2	0.0	9.7	14.5	27.6	0.0	60.0	32.4
Number of banks requiring injection										
4 percent	0	0	4	0	1	9	4	N/A	18	14
6 percent	0	0	4	0	1	11	5	N/A	21	16
Tangible common equity/tangible assets ratio										
4 percent (25 times leverage)	0.0	2.5	5.9	0.4	9.0	11.6	39.3	0.0	68.6	29.3
5.9 percent (17 times leverage)	79.3	18.0	12.6	4.3	13.5	17.3	65.7	0.0	210.5	144.8
Number of banks requiring injection										
4 percent (25 times leverage)	0	1	4	1	1	11	5	N/A	23	18
5.9 percent (17 times leverage)	4	1	6	1	2	12	8	N/A	34	26
Memo: Percent of total system assets	46.5	10.4	8.0	2.8	2.9	3.5	11.3	14.6	100.0	88.7

Sources: Authors; and SNL Financial.

Note: BHC = bank holding company.

1. Tier 1 common capital deducts all “non-common” elements of Tier 1 capital (i.e., qualifying minority interest in consolidated subsidiaries, qualifying trust preferred securities, and qualifying perpetual preferred stock).

REFERENCES

- Akaike, Hirotugu, 1974, "A New Look at the Statistical Model Identification," *IEEE Transactions on Automatic Control*, Vol. 19, No. 6, pp. 716–23.
- Board of Governors of the Federal Reserve System, 2009a, "The Supervisory Capital Assessment Program: Design and Implementation" (Washington, April 24). Available via the Internet: <http://www.federalreserve.gov/bankinfo/scap.htm>
- , 2009b, "The Supervisory Capital Assessment Program: Overview of Results" (Washington, May 7). Available via the Internet: <http://www.federalreserve.gov/bankinfo/scap.htm>
- Breusch, Trevor S., 1979, "Testing for Autocorrelation in Dynamic Linear Models," *Australian Economic Papers*, Vol. 17, pp. 334–55.
- Breusch, Trevor S., and Adrian R. Pagan, 1979, "Simple Test for Heteroscedasticity and Random Coefficient Variation," *Econometrica*, Vol. 47, No. 5, pp. 1287–94.
- Congressional Oversight Panel, 2010, "February Oversight Report: Commercial Real Estate Losses and the Risk to Financial Stability" (Washington, February). Available via the Internet: <http://cop.senate.gov/reports/library/report-021110-cop.cfm>
- Durbin, James, and Geoffrey S. Watson, 1950, "Testing for Serial Correlation in Least Squares Regression, I," *Biometrika*, Vol. 37, Nos. 3–4, pp. 409–28.
- , 1951, "Testing for Serial Correlation in Least Squares Regression, II," *Biometrika*, Vol. 38, Nos. 1–2, pp. 159–79.
- Godfrey, Leslie G., 1978, "Testing against General Autoregressive and Moving Average Error Models When the Regressors Include Lagged Dependent Variables," *Econometrica*, Vol. 46, No. 6, pp. 1293–302.
- Hausman, Jerry A., 1978, "Specification Tests in Econometrics," *Econometrica*, Vol. 46, No. 6, pp. 1251–71.
- International Monetary Fund, 2008a, *Global Financial Stability Report*, World Economic and Financial Surveys (Washington, April). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2008/01/index.htm>
- , 2008b, *Global Financial Stability Report*, World Economic and Financial Surveys (Washington, October). Available via the Internet: <http://www.imf.org/external/pubs/ft/gfsr/2008/02/index.htm>
- , 2009a, *Global Financial Stability Report*, World Economic and Financial Surveys (Washington, April). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2009/01/index.htm>
- , 2009b, *Global Financial Stability Report*, World Economic and Financial Surveys (Washington, October). Available via the Internet: <http://www.imf.org/External/Pubs/FT/GFSR/2009/02/index.htm>
- , 2009c, "United States—Staff Report for the 2009 Article IV Consultation," Country Report 09/187 (Washington, March 18). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=23025>
- , 2010, "United States of America: Stress Testing—Technical Note," IMF Country Report 10/244 (Washington, July).
- Kohn, Donald L., 2010, "Focusing on Bank Interest Rate Risk Exposure," speech at the Federal Deposit Insurance Corporation's Symposium on Interest Rate Risk Management, Arlington, January 29. Available via the Internet: <http://www.federalreserve.gov/newsevents/speech/kohn20100129a.htm>
- Kumhof, Michael, and Douglas Laxton, 2007, "A Party without a Hangover? On the Effects of U.S. Government Deficits," IMF Working Paper 07/202 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=21269>
- Lunn, David J., Andrew Thomas, Nicky Best, and David Spiegelhalter, 2000, "WinBUGS—A Bayesian Modelling Framework: Concepts, Structure, and Extensibility," *Statistics and Computing*, Vol. 10, No. 4, pp. 325–37.
- Martinez-Peria, Maria Soledad, Andrew Powell, and Ivanna Vladkova-Hollar, 2005, "Banking on Foreigners: The Behavior of International Bank Claims on Latin America, 1985–2000," *IMF Staff Papers*, Vol. 52, No. 3, pp. 430–61.
- Peek, Joe, and Eric S. Rosengren, 1997, "The International Transmission of Financial Shocks: The Case of Japan," *American Economic Review*, Vol. 87, No. 4, pp. 495–505.
- Schwarz, Gideon E., 1978, "Estimating the Dimension of a Model," *Annals of Statistics*, Vol. 6, No. 2, pp. 461–64.
- Swiston, Andrew, 2008, "A U.S. Financial Conditions Index: Putting Credit Where Credit Is Due," IMF Working Paper 08/161 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=22077>

Appendix I

Stress Test Scenarios and Shocks for the U.S. Financial Sector Assessment Program

The stress tests in the U.S. Financial Sector Assessment Programs (FSAPs) included a baseline scenario and alternative scenarios. These included an adverse macroeconomic scenario and an alternative funding risk scenario.²⁰ The assumed values were consistent with historical distress episodes, and the magnitudes of the shocks are broadly in the ranges analyzed in other FSAPs.

BASELINE SCENARIO

The baseline was the scenario from the IMF's April 2010 *World Economic Outlook* update. The output gap closes over the medium term from a negative level in 2009, while inflation is well anchored and stabilizes at about 2¼ percent. Ten-year government bond yields continue to rise moderately from 3.3 percent to 6.6 percent by 2015, reflecting the increasing government debt-to-GDP ratio.

ADVERSE SCENARIO

The adverse scenario was generated using a simple closed-economy business cycle model for the United States, with standard monetary channels (Taylor rule and nominal rigidities) and fiscal channels (a fiscal rule and a link between the real interest rate and government debt).²¹ The scenario was calibrated to illustrate the combined impact of four adverse shocks: (1) a sizable and persistent shock to the growth rate of potential output, reflecting continued difficulties in the financial system and very weak investment; (2) an additional short-term demand shock, reflecting high unemployment, weak credit, and continued fall in housing prices; (3) further near-term fiscal stimulus to support near-term growth; and (4) rising inflation expectations, reflecting concerns over medium-term fiscal risks and renewed higher oil prices.

Reflecting this combination of shocks, the output gap falls by another 2.3 percentage points in 2011 relative to the baseline, and the unemployment rate remains close to 10 percent in 2010–11. House prices fall by another 2.2 percent in 2010 and 2.1 percent in 2011, before starting a modest upward trend in 2012. Reflecting the weaker macroeconomic environment, banks' annualized return on assets falls slightly over the forecast horizon. Inflation and government bond yields rise, and the government debt-to-GDP ratio increased by almost 10 percentage points compared with the baseline in 2013 (Table 33.9).

ALTERNATIVE SCENARIO

The alternative funding risk scenario was conducted to test banks' resilience to a further small deterioration in the real estate sectors, including difficulties in rolling over their commercial real estate (CRE) maturing debt and continuing to accumulate seriously delinquent mortgages on their balance sheets. The fact that nearly half of the \$1.4 trillion in CRE loans expected to mature between 2010 and 2014 have negative equity (see Congressional Oversight Panel, 2010), together with the rising stock of seriously delinquent mortgages (many of which are "underwater"), suggests that banks could face difficulties in refinancing a large volume of loans and face larger real estate loan losses if economic conditions do not improve and real estate prices do not rebound.

²⁰ Kohn (2010) highlighted the potential upward push on interest rates if the rising trajectory of U.S. debt to GDP is not curbed in the future and the impact of higher interest rates on financial intermediaries. Furthermore, the federal banking agencies released policy statements highlighting their expectations for sound practices in managing interest rate risk (<http://www.federalreserve.gov/newsevents/press/bcreg/20100107a.htm>) and funding and liquidity risk (<http://www.federalreserve.gov/newsevents/press/bcreg/20100317a.htm>).

²¹ See Kumhof and Laxton (2007).

TABLE 33.9

Macroeconomic Assumptions (percent change, unless otherwise noted)										
Scenario	2010		2011		2012		2013		2014	
Baseline										
Real GDP	3.1		2.6		2.4		2.5		2.4	
Real personal consumption expenditures	2.4		2.1		2.0		2.0		2.0	
Nominal GDP	3.9		4.0		4.2		4.4		4.3	
Output gap (percent)	-2.0		-1.0		-0.6		-0.3		-0.1	
Unemployment rate (percent)	9.8		8.9		7.0		5.8		5.5	
Case-Shiller 10-city house prices	2.1		2.0		2.9		2.5		1.5	
Spread of 3-month LIBOR to 3-month T-Bill	0.2		0.4		0.4		0.4		0.4	
Return on assets (annualized; percent)	1.7		1.8		1.8		1.8		1.9	
Adverse										
Real GDP	2.3	-0.8	0.8	-1.7	2.6	0.2	2.6	0.1	2.2	-0.2
Real personal consumption expenditures	1.9	-0.6	0.6	-1.6	1.6	-0.4	1.5	-0.5	1.3	-0.6
Nominal GDP	3.8	-0.2	3.4	-0.7	4.1	-0.2	4.6	0.2	4.5	0.2
Output gap (percent)	-3.0	-1.0	-3.3	-2.3	-2.1	-1.5	-1.1	-0.8	-0.6	-0.4
Unemployment rate (percent)	10.0	0.2	9.9	1.0	8.9	1.9	7.7	1.9	6.9	1.5
Case-Shiller 10-city house prices	-2.2	-4.3	-2.1	-4.1	2.2	-0.7	2.5	0.0	1.8	0.2
Spread of 3-month LIBOR to 3-month T-Bill	0.3	0.1	0.6	0.3	0.7	0.3	0.6	0.2	0.6	0.2
Return on assets (annualized; percent)	1.6	-0.2	1.4	-0.5	1.5	-0.4	1.6	-0.3	1.7	-0.3
Alternative funding risk										
Real GDP	2.4	-0.6	0.8	-1.8	1.6	-0.8	2.5	0.0	2.4	0.0
Real personal consumption expenditures	2.3	-0.2	0.2	-1.9	0.0	-2.0	1.3	-0.7	1.7	-0.2
Nominal GDP	3.3	-0.7	1.9	-2.1	3.1	-1.1	4.4	0.0	4.4	0.1
Output gap (percent)	-3.3	-1.3	-2.6	-1.6	-0.8	-0.3	-0.3	0.0	-0.1	0.0
Unemployment rate (percent)	10.6	0.8	9.9	1.0	7.2	0.1	5.8	0.0	5.5	0.0
Case-Shiller 10-city house prices	-4.1	-6.1	-2.6	-6.7	3.1	0.3	2.4	0.0	1.5	0.0
Spread of 3-month LIBOR to 3-month T-Bill	0.4	0.1	0.5	0.1	0.5	0.1	0.4	0.0	0.4	0.0
Return on assets (annualized; percent)	1.5	-0.3	1.6	-0.3	1.7	-0.2	1.8	-0.1	1.9	-0.1

Source: Authors.

Note: Shaded numbers denote deviations from baseline. LIBOR = London interbank offered rate.

Under this scenario, the output gap falls more sharply in 2010 (3.3 percent relative to 3 percent), and the unemployment rate rises faster (10.6 percent relative to 10 percent). House prices are expected to fall by 4.1 percent in 2010 and another 2.6 percent in 2011, while CRE prices fall by another 8 percent by end-2012 (as opposed to 3.3 percent in the adverse scenario). Short-term market spreads react slightly more than under the adverse scenario in 2010 but return faster to the baseline in the outer years, allowing banks to earn higher profits over the forecast horizon. Importantly, banks' assumed difficulty in rolling over maturing debt leads to higher losses on CRE loans, which peak at 5.1 percent at end-2011.

SINGLE-FACTOR SHOCKS

In addition to the scenarios, a range of single-factor shocks were employed to examine resilience of the financial system with respect to individual risk factors. The calibration of these shocks was based on long-term U.S. historical data as well as experience from other countries.

Appendix II

Industry-Wide Loan Loss Projections for Bank Holding Companies

Charge-off rates for different loan types are modeled as dependent on a set of economic and financial variables.²² The general methodology was developed initially and subsequently refined in the context of the systemwide capital shortfall estimates presented in the *Global Financial Stability Report* (see IMF, 2008b, 2009a, for a description of the methodology). This appendix outlines the methodology for forecasting bank charge-off rates, lending standards, and house prices.

In order to better capture future turning points in the charge-off patterns, levels and log levels (rather than growth rates) were used for the explanatory variables. Because a decline in bank lending standards indicates a slower rate of tightening, the use of cumulative net balances for lending standards was warranted (referred to as “cumulative lending standards” hereafter). This was to reflect that, for example, charge-offs can continue to rise despite a slowdown in house price declines and a deceleration in the pace of tightening in lending standards. Similarly, to capture deterioration in economic conditions amid a slowdown in negative growth rates, we used the output gap (which can be viewed as the detrended level of real GDP), instead of GDP growth.

The underlying historical data on loan loss rates and lending standards were obtained from the Fed, while macroeconomic and financial data came from Haver Analytics. Where available, forecast data were taken from the *World Economic Outlook* (WEO). Housing prices and lending standards were modeled separately. The sample was composed of quarterly data from 1991 to 2009 so as to incorporate the last two recessions.

CHARGE-OFF RATES

To deal with nonstationarity in the variables, the empirical Bayesian approach was employed. The estimation was carried out by running 10,000 Markov Chain Monte Carlo simulations using the Gibbs sampler package WinBUGS (Lunn and others, 2000). Convergence was obtained within 1,000 burn-in runs. The estimated coefficients in the resented equations were statistically significant at 5 percent. Lending standards were particular to each type of loan.

Real estate charge-off rates

Charge-off rates for real estate loans followed a two-step approach. First, the percentage of loans that would become delinquent was estimated. Second, the “hazard rate” or the transition rate was estimated to capture the percentage of delinquent loans that would transition into actual charge-offs. The exact specification for commercial and residential charge-off rates looked as follows:

Commercial real estate charge-off rate (C_CRE)

$$C_CRE_t = D_CRE_t * T_CRE_t$$

The first term D_CRE_t is the delinquency rate, which is modeled as a function of the cumulative lending standards LS_CRE and CRE prices CP :

$$\ln(D_CRE_t) = 0.00131 * LS_CRE_t - 2.3137 * \ln(CP_t) + 11.45.$$

The second term is the transition rate, which is modeled as a function of the cumulative lending standards:

$$\ln(T_CRE_t) = 0.00117 * LS_CRE_t + 1.37.$$

²² This appendix was prepared jointly with Sergei Antoshin (IMF).

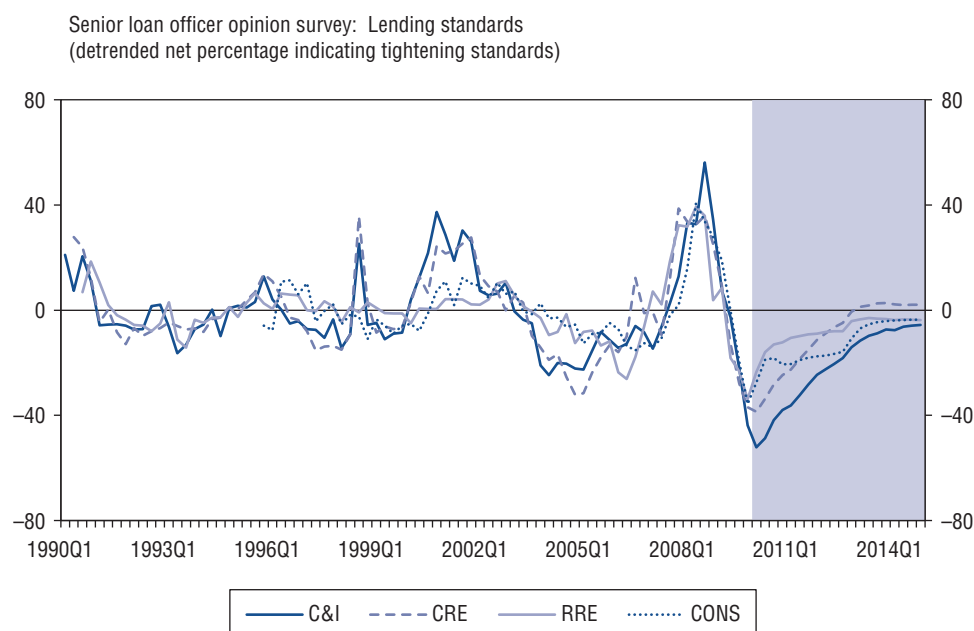


Figure 33.13 Senior Loan Officer Opinion Survey: Lending Standards (de-trended net percentage indicating tightening standards)

Residential real estate charge-off rate (C_{CRE})

$$C_{RRE_t} = D_{RRE_t} * T_{RRE_t}$$

The delinquency rate is a function of the cumulative lending standards and level of RRE prices:

$$\ln(D_{RRE_t}) = 0.00308 * LS_{RRE_t} - 0.0025 * HP_t + 0.912.$$

The transition rate is a function of the cumulative lending standards and unemployment rate:

$$\ln(T_{RRE_t}) = 0.00284 * LS_{RRE_t} + 0.12848 * UR_{t+4} + 0.997.$$

Consumer loans charge-off rate (C_{CONS})

The consumer loans charge-off rate is modeled as a function of the cumulative lending standards and the output gap:

$$\ln(C_{CL_t}) = 0.00079 * LS_{CL_t} - 0.0925 * GAP_t + 0.705.$$

Commercial and industrial loans charge-off rate (C_{CI})

The commercial and industrial loans charge-off rate is modeled as a function of the detrended cumulative lending standards and the output gap:

$$\ln(C_{CI_t}) = 0.00341 * LS_{CI_DT_t} - 0.1398 * GAP_{t+1} - 0.5752.$$

LENDING STANDARDS

The lending standards reported in the Board of Governors of the Federal Reserve System's ("Fed") Senior Loan Officer Opinion Survey of Bank Lending Practices (SLOOS) appear as an independent variable in each of the charge-off and delinquency models. The respondents to the survey, which is usually conducted quarterly, include large U.S. commercial banks and large branches and agencies of foreign banks. For each loan category, the respondents report whether over the past three months they have tightened, loosened, or maintained the criteria used to determine whether a borrower is creditworthy. The Fed reports these responses as the percentage of respondents indicating that their bank had tightened standards less the percentage indicating that their bank had loosened standards for a number of different kinds of loans. Whenever these measures are positive, it is generally more difficult for borrowers to obtain loans.

To ground forecasts of these lending standards, we estimated four vector autoregression models. The variables in each system included real GDP growth, CPI inflation, the change in corporate bond spreads, the change in the Federal Funds Target Rate, oil prices, and the growth of total bank loans. Further, each model contained one loan category's lending standards. Given that SLOOS asks banks to report on their lending standards over the previous three months, we lagged the lending standards one quarter so that the change in standards would be contemporaneous with the changes in the other variables. We obtained forecasts for each class of

lending standards from the estimated systems of equations, treating the macro variables as exogenous, similar to the approach discussed by Swiston (2008). Output from these models estimated with detrended lending standards augurs a loosening in lending standards for all types of loans over the forecast horizon (Figure 33.13). These results are in line both with recent developments in the SLOOS and with the baseline scenario, which calls for continued macroeconomic growth and financial stabilization. Nevertheless, to align the forecasts with the historical behavior of lending standards in previous business cycles, we adjusted model-based projections with judgmental forecasts.

HOUSE PRICES

Prices for residential real estate (RRE) and commercial real estate (CRE) enter the equations used to forecast RRE and CRE loan delinquency rates. Neither the WEO nor the macro simulation models used in the scenario analysis contain a forecast for prices of housing or CRE. Thus, to generate the forecasts required for the delinquency rate models, we used a model for house prices and an observation about the house prices–CRE prices to generate the CRE price forecast.

We tried a number of specifications for the S&P/Case-Shiller 10-city house price index, which included variables, both considered key determinants of house prices and forecasted in the WEO and simulation models. The specification that we used in this exercise was the following:

$$\begin{aligned}\Delta^2 HP = & -0.7449 * \Delta^2 HP_{t-2} + 0.8184 * \Delta^2 RGDP_t \\ & (0.0944)*** \quad (0.2445)*** \\ & + 0.4605 * \Delta^2 RGDP_{t-1} - 0.0154 * \Delta^2 UR_{t-1} \\ & (0.2392)* \quad (0.0076)**\end{aligned}$$

In the equation, HP is the 10-city Standard and Poor's Case-Shiller house price index in log levels, $RGDP$ is real GDP, UR is the unemployment rate, and Δ^2 denotes the second

TABLE 33.10

Correlations between Commercial and Residential Real Estate Prices (Sample: 1987:Q1–2009:Q4)

S&P Case-Shiller House Prices (10-city)	Commercial Real Estate Prices					
	t	$t+1$	$t+2$	$t+3$	$t+4$	$t+5$
Levels	0.92	0.94	0.95	0.96	0.96	0.95
Growth	0.43	0.33	0.31	0.41	0.49	0.34

Sources: Authors; and Haver Analytics; MIT Center for Real Estate; S&P Macro Markets, LLC.

difference.²³ Standard errors are reported underneath the coefficient estimates, and ***, **, and * denote statistical significance at the 1, 5, and 10 percent levels, respectively. Under the baseline scenario, using output and unemployment forecasts from WEO, house prices rise moderately but are still 24 percent below their 2006:Q2 peak by end-2014.

The calculations use the observation that developments in house prices generally have fed into CRE prices with a lag when forecasting CRE prices. Consistent with the *Global Financial Stability Report's* methodology, the Massachusetts Institute of Technology Center for Real Estate's Transactions–Based Index of Institutional Commercial Property Investment Performance was used. Since the late 1980s, this price index has closely tracked the Case-Shiller index, with about a four-quarter lag (Table 33.10). Taking this relationship into account, we took actual and forecasted growth rate for the Case-Shiller index and used it to project CRE prices for the same period in the following year. As with house prices, CRE prices increase steadily but are 39 percent off their 2007:Q2 peak and the end of the forecast period.

²³ The second difference was used after performing an Augmented Dickey-Fuller test on the Case-Shiller 10-city house price inflation rate series, which provided evidence of a unit root.

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The Real Effects of Financial Sector Risk

ALEXANDER F. TIEMAN • ANDREA M. MAECHLER

This chapter estimates the magnitude of key effects on the real economy from financial sector stress. We focus on the short-term feedback effect from market-based indicators of financial sector risk to the real economy through the credit channel and estimate this effect on an economy-wide (macro) level, as well as on the level of individual large banks. Both estimates yield significant feedback effects of substantial magnitude. The estimates are consistent with other work in this area. Our results suggest that prudential supervision could be enhanced by taking into account the feedback effects of financial instability in the real economy. They also propose a way to integrate feedback effects into stress tests in order to improve realism and accuracy on macroeconomic stress scenarios, as well as a metric to interpret stress testing results.

METHOD SUMMARY

Overview	The feedback loop model estimates the short-term feedback effects from market-based indicators of financial sector risk to the real economy through the credit channel and estimates this effect at an economy-wide (macro) level, as well as at the level of individual large banks.
Application	• The model estimates the effect on growth of financial sector stress (perception) for a sample of Western European countries and the major banks in those countries. • The model results may be used to incorporate the feedback loop from financial risk to the real economy in macro-prudential stress tests, as well as in regular macroeconomic models.
Nature of approach	Regression analysis.
Data requirements	• Stock market data or expected default frequencies (EDFs); accounting information on loans; macro data on credit growth, GDP, interest rates, housing prices. • All are public data, although EDFs are subscription based.
Strengths	• The model estimations are independent of the exact model specification. • The model may be applied at the micro and macro levels, and the results are robust.
Weaknesses	The strength of the feedback effects could change over time—the model is estimated over the period 1991–2007, prior to the global financial crisis.
Tool	Standard econometrics package.

This chapter estimates the effects on the real economy from financial sector stress. Motivating the topic of inquiry of this research has progressively become easier. The crisis has shown that real effects of financial sector turmoil can be large and, quite possibly, prolonged. In addition, international contagion from financial distress in one country to that in another,

and its adverse feedback effect to that country's real economy, is evident.

The real effects of financial sector stress have been recognized at least since Bagehot (1873). They have been studied extensively on the back of the 1930s depression (comprehensively discussed in, for example, Bernanke, 1983) and the more

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recent Asian crisis of the late 1990s (e.g., books by Noland and others, 1998; and Hunter, Kaufman, and Krueger, 1999; and the references therein). These studies have led to a better understanding of the nature of financial sector risk. One of the recurring conclusions of the Asian crisis episode has been that financial systems, and banks in particular, did not adequately manage the risks in their portfolios. At the same time, supervisors and other policymakers failed to identify the buildup of risks in these institutions. Finally, bank-based financial crises, in contrast to, say, balance of payment or currency crises, tend to have a much more severe and prolonged effect on the real economy.

To date, a large part of financial stability work has focused on identifying the factors that contribute to financial sector risk and assessing and monitoring changes in financial institutions' risk profiles. The subsequent feedback loop, in which financial sector risk influences the real macro environment through the credit channel, usually remains unmodeled. To the extent research has focused on the influence of financial variables on the real economy, this mostly has been confined to variables related to stock market performance and the yield curve. Explicit financial sector risk variables, or risk variables pertaining to individual institutions, normally are not taken into account. This study tries to help fill this gap by examining how financial stress affects credit growth to the private sector and GDP.

The objective of this study is to model the feedback loop from financial system stress to the real economy and empirically estimate its magnitude. On the basis of a sample of mature European economies, we provide in-sample and out-of-sample predictions of declines in credit growth using observed market-based indicators of increases in financial sector fragility. We focus on the short-term feedback effect from market-based indicators of financial sector risk to the real economy through the credit channel. This short-term feedback effect is estimated on an economy-wide (macro) level, as well as on the level of credit supplied by key individual large banks. Both estimates yield significant feedback effects of substantial magnitude.

A better understanding of the adverse feedback loops between financial stress and the broader economy is the first step toward a "macroprudential" approach to financial supervision. One of the key lessons of the current global financial crisis is the need to better internalize the cost of systemic risk resulting from financial strains. This requires a novel approach that helps mitigate the procyclical tendencies of asset prices and risk and takes better account of various types of spillovers, while continuing to monitor the health of individual institutions. The current model contributes to this effort, by shedding light on how financial risk is linked to economic activity. The model can be employed in several ways. For instance, it allows for examining how financial sector strain may spill over to other countries, thus providing a richer cost-benefit analysis of cross-border systemic risk. In addition, it would allow a stress test in which the usual assumption of unchanged behavior by financial institutions is replaced by a feedback

effect from financial sector strain to credit and output. This way, the model would add realism to macroeconomic stress tests.

To our knowledge this is among one of the first studies focusing on the feedback effects from financial sector fragility variables in this way, and on Europe in particular. The academic and empirical literature on the credit channel thus far has focused mainly on transmission from monetary policy to bank lending. In addition, most of the literature on the credit channel looks at the United States (see, e.g., Bernanke and Blinder, 1988; and Bernanke and Gertler, 1995, for an overview). Studies that do focus specifically on Europe, such as Altunbas, Fazylov, and Molyneux (2002) or Angeloni and Ehrmann (2003), have not generated unambiguous conclusions. A recent study by Čihák and Koeva-Brooks (2008) models real effects from losses in the banking system and stock price developments in the euro area. It does find real effects from financial sector turmoil in the second half of 2007 to the euro area's real economy, on the order of magnitude of 0.2–0.3 percentage points of GDP. Other studies have modeled the cross-border comovement between output and financial variables (see, e.g., Bayoumi and Swinston, 2007; Esponiza, Fornari, and Lombardi, 2008). However, they generally are confined to financial variables related to stock market performance (and volatility) and the yield curve.

The structure of this chapter is as follows: Section 1 describes the data. Section 2 describes the methodology we use and the two different models we estimate using panel regressions. Section 3 describes the regression outcomes and discusses these. Section 4 places the results in a broader context and concludes.

1. DATA

We focus on major Western European economies for which both macroeconomic and financial sector data are readily available. Our sample includes seven countries—France, Germany, Italy, Spain, Sweden, Switzerland, and the United Kingdom—and focuses on the largest banks in each of these countries (a total of 26 banks). The period covered is 1991–2007, over which we perform regression analysis on quarterly data.

For each country, we first construct several economy-wide and bank-specific financial sector risk variables. These variables all are based either on a simple Merton-type distance-to-default (DD) (Box 34.1) model or on Moody's KMV (MKMV) expected default frequency (EDF) (Figures 34.1 and 34.2).¹ In both cases, we construct economy-wide risk measures by averaging the DDs and EDFs of individual large banks in the specific country. Although there is considerable variation across indicators, we find that, in both cases, the

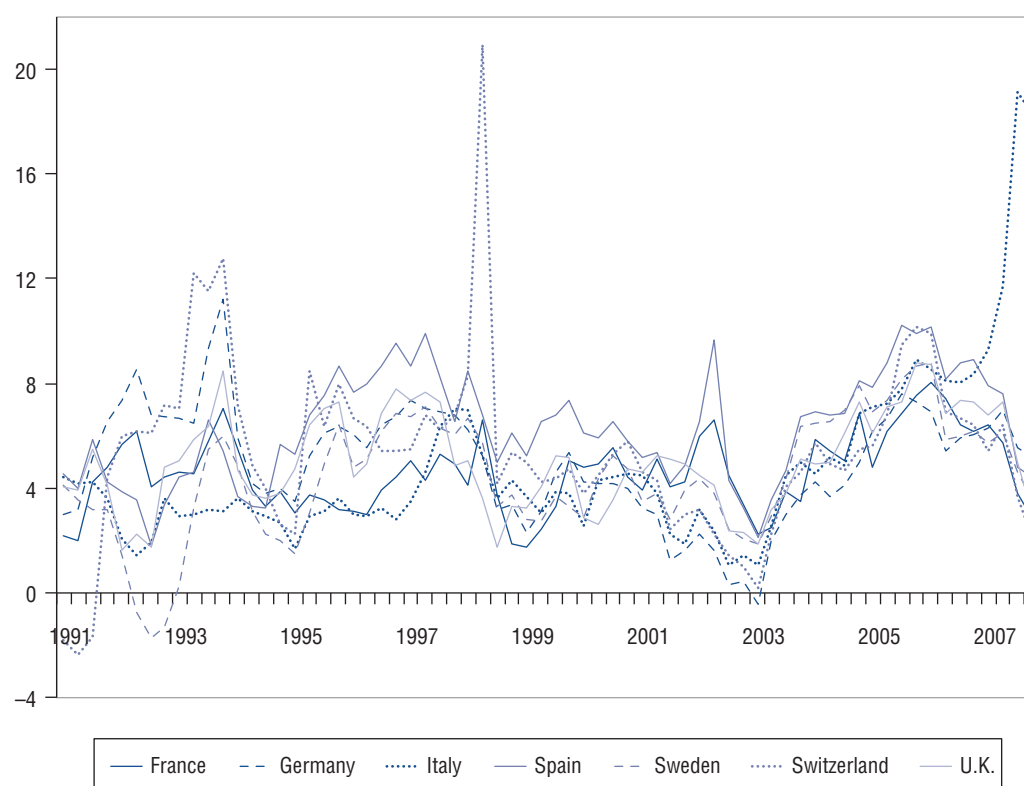
¹ Moody's KMV EDFs are also based on Merton-like calculations, after which a database of historic defaults is used to translate the model-implied risk-neutral probabilities to real-world (i.e., risk-averse) EDFs (i.e., probabilities of default).

Box 34.1. The Distance-to-Default Measure

The basic structural valuation model by Black and Scholes (1973) and Merton (1974)—hereafter BSoM—underpins the DD measure used in this study. In the BSoM model, equity is viewed as a call option on a company's assets, with strike price equal to the current book value of total liabilities. When the value of assets is less than the strike price, its equity value is zero. The market value of assets is not observable but can be estimated using equity prices and accounting measures of liabilities. The DD measures used here are estimated with the methodology described in Vassalou and Xing (2004) using daily equity data and annual accounting data. The formula for DD is

$$DD_t = \frac{\ln(V_t/L_t) + \mu}{\sigma},$$

where V_t and L_t are, respectively, the (market-based) value of assets and (accounting) value of liabilities, and μ and σ are the mean and variance of the company's stock price, respectively. As a proxy for implied volatility σ , we employ the rolling 12-month historical volatility.



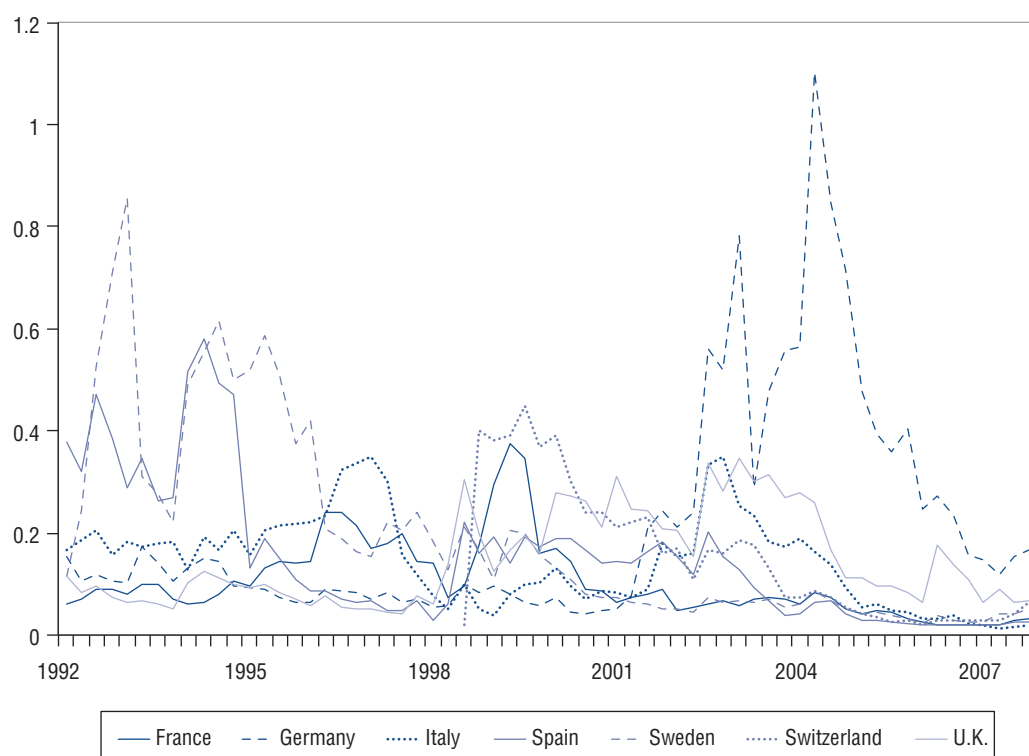
Sources: Authors; and Datastream.

Figure 34.1 Average Distance to Default in the Sample (number of standard deviations)

period 2004 to mid-2007 is characterized by low risk, as reflected by (almost) uniformly high DD indicators or, conversely, low EDFs. We also construct a systemwide DD, based on a hypothetical superbank comprising all banks in the system, as well as a DD index, based on the Datastream banking sector index for a particular country. For the EDFs, we use both the one-year EDF and the five-year EDF. In the regressions for individual banks, their respective individual DDs and EDFs, rather than economy-wide aggregates, are used.

We use different variations of our risk estimates. First, in addition to bank-specific DDs, we construct several system-wide indicators. In particular, we compute an asset-weighted

DD (DD-aw), a simple averages DD (DD-av), a system-wide DD based on the Datastream banking sector index (DD-index), and a system or portfolio DD (DD-system). The DD-system is constructed by creating a hypothetical “superbank” comprising all the banks in the system of a particular country. For these firms, accounting and equity values simply are added up, as if they are all part of a single superfirm. Subsequently, the DD for this imaginary superfirm is calculated as shown in Box 34.1. The DD-system can be viewed as a risk profile measure tracking the evolution of the joint risks of failure of the firms comprising a portfolio (see also De Nicoló and others, 2005).



Source: Moody's KMV.

Figure 34.2 Average One-Year Expected Default Frequency (in percent)

The macroeconomic data we collect consists of real GDP and real private sector credit, calculated as nominal variables deflated by the GDP deflator, and economy-wide interest rates (interbank rates and marginal rates of new lending). For the regressions on individual banks, we use total loans from banks' balance sheets as the equivalent to private sector credit and total net interest income divided by total loans as a proxy for the average interest rate each bank charges. Note that this implies that off-balance-sheet credit extension is not taken into account. Off-balance-sheet items do, however, influence our risk variables to the extent that they influence the market value of assets. Real private sector credit growth is depicted in Figure 34.3. Credit growth is clearly higher on average in the post-2000 years of the sample (at 6.3 percent per year) compared with the 1990s (2.8 percent), even when abstracting from the crisis-related decline in Sweden in the early 1990s (3.5 percent vs. 6.0 percent).

In addition to the pure macroeconomic data, we employ several control variables. To control for changes in house prices, we use the house price indices from Global Insight. As a control for stock market valuations and volatility, we use the Dow Jones Stoxx 600 stock index of the largest European companies; the VDAX new index of 30-day implied volatility in the Deutscher Aktien Index (DAX), that is, the European equivalent of the Chicago Board of Exchange (CBOE) Market Volatility Index (VIX); and the iTraxx Crossover Credit Default Swap (CDS) index of European subinvestment grade

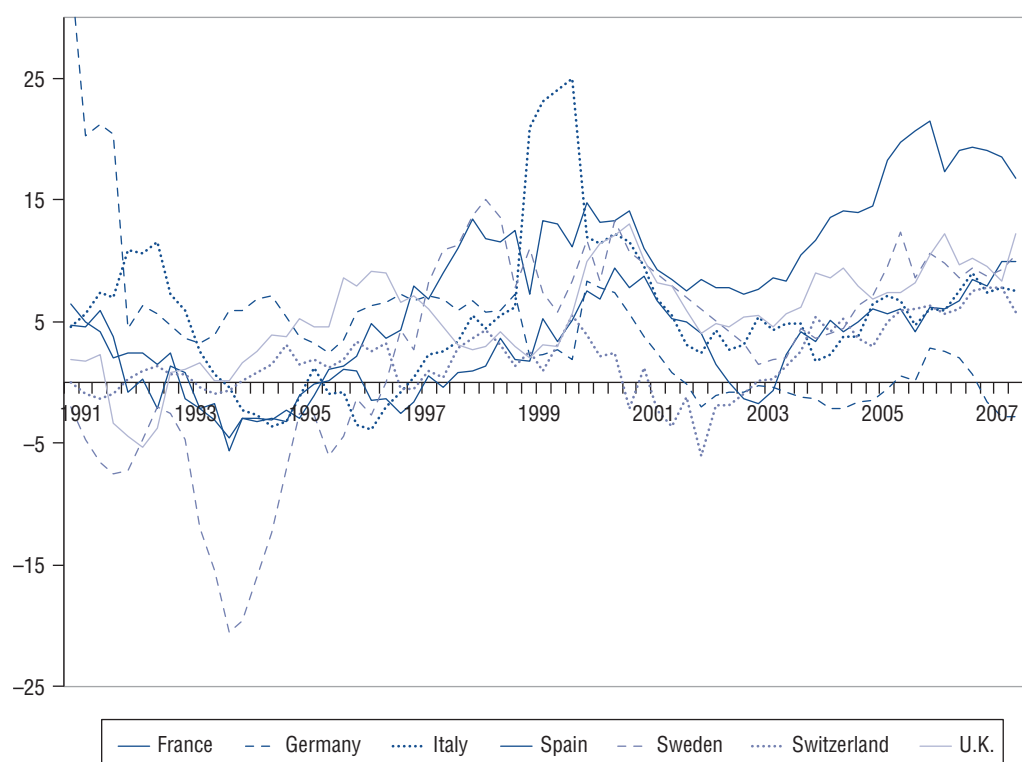
names. We also construct a control variable that indicates the relative size of the financial sector, by dividing total financial sector assets by GDP.

All macroeconomic data is available on a quarterly frequency. Financial market data is available daily, while accounting variables are available annually. Our data sources are the IMF's International Financial Statistics (IFS), Datastream, and Bloomberg. In order to get to quarterly frequencies for all data, the daily data was collapsed by taking the last observation of the quarter, while the annual accounting data was assumed to remain constant over the year. A complete overview of the data, including summary statistics and sources, can be found in Appendix I.

2. METHODOLOGY

In our analysis of the feedback effect from financial sector risk to the real economy, we focus solely on the credit channel. As argued in a substantial body of research on this transmission channel, we believe credit to be the main variable through which financial risk affects the broader economy. This focus is justified by the fact that in many real-life cases in the past, banks under stress have curtailed their supply of credit, with often destabilizing real economy effects.

We first estimate a vector error-correction model to establish whether a cointegrating relationship exists between credit and GDP (Appendix II). This allows us to confirm



Source: International Financial Statistics.

Figure 34.3 Real Private Sector Credit Growth

that financial risk plays a significant role in the short-term dynamics around the cointegrating relationship between credit and GDP. Given these results, our main regression models employ two equations: credit as a function of financial sector risk variables (and controls, including lagged GDP growth) and GDP growth as a function of (lagged) credit. This central regression model is employed on macroeconomic data in a panel regression across countries, as well as on bank-specific data in a panel of large banks from the same countries.

A. Country-specific regression model

Next, we address the nonstationary nature of several variables. Specifically, stationarity is rejected for credit, GDP, the DJ Stoxx 600, housing prices, and our measure of the relative size of the financial sector, whereas it is not rejected for the risk variables and interest rates. Except for the asset size variable, nonstationarity would be expected. Figure 34.3 shows how over the period 1991–2007 credit growth was on average considerably above GDP growth, whereas financial sector assets generally have grown relative to GDP, confirming the nonstationary nature of this variable. We difference the nonstationary variables and use lags of the differenced series in our subsequent analysis. We use four quarter differences to account for seasonality in the data. Moreover, we lag the differenced variables so as to prevent simultaneity problems.

We proceed to estimate a system of simultaneous equations, which captures the impact of financial risk on bank lending, while accounting for the endogenous relationship between GDP and credit:

$$\Delta C_t = \alpha_1 + \beta_1 R_t + \beta_2 I_t + \gamma_1 \Delta_4 GDP_{t-1} + \gamma_2 \Delta_4 DJ_t + \gamma_3 \Delta_4 H_{t-1} + \gamma_4 \Delta_4 S_{t-1} + \varepsilon_t$$

$$\Delta_4 GDP_t = \alpha_2 + \delta \Delta_4 C_{t-1} + \eta_t$$

where

$\Delta_u C_t$ = the growth rate of credit to the private sector at time t over the last u quarters,

R_t = a financial risk measure at time t ,

I_t = interest rates at time t ,

$\Delta_u GDP_t$ = the growth rate of real gross domestic product at time t over the last u quarters,

$\Delta_u DJ_t$ = the growth rate of the DJ Stoxx 600 index at time t over the last u quarters,

$\Delta_u H_t$ = the growth of the Global Insights housing price index at time t over the last u quarters,

$\Delta_u S_t$ = the growth of the relative size of the financial sector at time t , defined as total financial sector assets divided by GDP, over the last u quarters.

The subscript u is omitted for $u=1$, that is, ΔC_t is the first difference of credit, or the quarterly credit growth rate.

We employ a panel consisting of the seven countries in the sample. All variables are expressed in natural logarithms,

and all nominal variables have been deflated using the GDP deflator. Various other control variables have been employed. Specifically, indices of implied volatility,² as well as national stock exchange indices, were used. The estimated coefficients for these control variables either yielded insignificant results or were dropped due to strong collinearity with our risk measures. For example, implied volatility is a major driver of DD. In addition, in periods of market turbulence, the volatility of an individual stock will often display a high correlation with volatility of the index. Volatility and national stock market indices were omitted subsequently from the regressions.

Macroeconomic regressions are run for all combinations of the dependent variable private sector credit with the different financial sector fragility variables (EDF1, EDF5, DD, DD index, DD-system, weighted average DD, weighted average EDF1, weighted average EDF5). They incorporate different control variables (house price, DJ Stoxx 600, size of financial sector) individually and with several combinations of the controls. The results are discussed in Section 3.

B. Individual bank model

The same basic model specification is used on a bank-specific level. The panel for the individual bank regressions consists of 26 large banks from the seven countries in the sample. On this panel, we estimate the same model (1) as previously. All variables are indexed by both time and an index $j = 1, \dots, 26$, indicating each individual bank. The dependent variable ΔC_{jt} is the growth rate of total loans on bank j 's balance sheet. This is a close equivalent to private sector credit, which is available for individual institutions. The interest rate variable I_{jt} is the average interest rate bank j charges on its loans, calculated by dividing total interest income by total loans. In some of the regressions, we added a "competition" variable that captures how a bank in a given country reacts to changes in the risk profile of its peers in the same country. It consists of the average fragility of a bank's domestic peers, measured as the average-weighted EDF1, EDF5, and DD for the other banks in the same country. All other variables are the same as in the macro regressions, except that by construction we use only the bank-specific risk measures (DD and EDF; i.e., no weighted averages or system-wide indicators). The results are discussed in Section 3.³

3. REGRESSION RESULTS AND DISCUSSION

A. Macroeconomic regressions

The macroeconomic panel regressions show that financial sector fragility has a significant effect on credit and GDP growth.

This holds for all different specifications of the risk variables, mostly at the 1 percent uncertainty level. For Moody's KMV EDFs, the relation is negative and significant at the 1 percent level (i.e., a higher probability of default is related to lower credit growth). This result holds for both the one-year EDF and the five-year EDF and for the asset-weighted EDF as well as the simple average EDF. For the DD risk measures, much the same holds: the estimated coefficients show significance at the 1 percent level, for the simple average DD, the asset-weighted DD, as well as the DD based in the banking index (DD-index), and the DD based on the portfolio consisting of all banks in a country (DD-system), suggesting that credit growth is lower when banks are closer to their default barrier (i.e., when DD is lower). These results are for the precrisis period up to 2007 and would likely come out even stronger if data on the ongoing crisis would be included.

The estimated magnitude of the coefficient is stable across the risk variables and specifications. For the EDFs, the coefficient estimate on the five-year EDF is slightly higher compared with the one-year EDF, which might be explained by the higher relative variance of the one-year EDF compared with the five-year EDF. Estimates show little sensitivity to using simple average rather than asset-weighted average EDFs. For the DDs, results are similar, with estimated coefficients stable across the different specifications of DD (DD-aw versus DD-av, versus the DD-index and the DD-system).

The interest rate variable is significant at the 5 percent level in most specifications, with a negative coefficient. This holds both for the interbank rate and for the IFS countrywide rate on new loans. Higher interest rates hence are associated with significantly lower credit growth, as expected from theory. More important, a price effect of loans thus can be distinguished clearly from a volume effect, and the volume effect found remains after price effects have been taken into account.

A number of macroeconomic and financial sector control variables are found generally to be significant. Controlling for GDP in the credit equation mostly yields a highly significant coefficient estimate. In the equations where EDF is used as the financial sector risk variable, the significance of the GDP growth variable in the credit equation varies from 1 percent to an insignificant coefficient, depending on the number and nature of other control variables employed. Using DD as the risk variable, GDP growth is always a significant control variable, mostly at the 1 percent significance level. House prices are found generally to be significant at the 1 percent level. However, including the control for house prices in several cases results in lower significance for the GDP growth control variable. This is unsurprising, as, over the sample period, strong GDP and house price growth tend to occur simultaneously. The relative size of the financial sector, measured as total financial sector assets over GDP, is a significant control variable (at the 1 or 5 percent level) in the estimations where the EDFs are used as risk variables. In the DD regressions, however, it is generally not significant at even the 10 percent level. The DJ Stoxx 600 index of large European corporates does not come up with a coefficient significantly different

² The German VDAX new index of 30-day implied volatility in the DAX and the U.S. CBOE VIX index of 30-day implied volatility in the S&P 500.

³ See Appendix Tables 3–11 in Tieman and Maechler (2009) for detailed panel regression results.

from zero. Other control variables, such as the VDAX new volatility index or the iTraxx Crossover CDS index, also yield insignificant coefficient estimates (these results have been omitted in the tables).

The signs of the estimated coefficients for the controls are as expected. Credit growth is positively associated with GDP growth. Higher growth in house prices is associated with higher credit growth, which would be explained by increased mortgage lending. A bigger expansion of the financial sector relative to GDP also is associated positively with credit growth. Here, the explanation might partially be tautological, as many of the financial sector assets that define the size of the sector actually consist of loans, which would show up in credit figures.

The results are generally robust. A change in the exact specification of the regression equation does not alter the results much. We have employed different lag structures for both the dependent variable as well as the various explanatory variables, including the controls, and found that the size of the coefficient estimate does not differ much and that it generally remains significant.⁴

Our preferred regression equation for the macro regression panel is the one including controls for GDP and house prices but excluding the DJ Stoxx 600. The control for the relative size of the financial sector could be included in the EDF regressions, where it is significant, but we prefer not to include it in the DD regressions, where it has no significance. We generally prefer the regressions using the interbank interest rate instead of the long rate. Even though both rates generally are significant and of the correct sign, the interbank rate is more responsive to changes and most loans would be based on the (short) interbank rate rather than the long-term rate. Our preferred regressions are highlighted in the regression tables. The fit of our preferred regression equations is reasonable, with a coefficient of determination (R^2) between 0.14 and 0.21 depending on the exact specification. For models in first difference, this is well within range of normal values for the fraction of variance explained by the regression equation.

Our results allow us to compute the model-implied reduction in credit growth associated with financial distress. To get a better feel for the implications of the coefficient estimate, we calculated the increase in EDFs and decrease in DDs between July 1 and December 31, 2007 (i.e., in sample), and between July 1, 2007, and July 1, 2008 (i.e., out of

sample), and the associated model-implied reduction in real credit growth (Table 34.1). Between July and end-2007, the increased financial sector risk as perceived by the market would lead real credit growth to decrease by 0.4 percentage point in real terms on average in the countries in our sample. Set against an average real credit growth of 4.4 percent over the period 1991–2006, this implies a decrease of some 10 percent. Over the period July 2007 to July 2008, credit growth would decrease by 1.7 percentage points on average across the banks in the sample (based on the DD regression).⁵ This translates into a 32 percent average model-implied reduction in credit growth across countries (see Table 34.1). For some countries, the estimated reductions in credit growth are much steeper, at about 60 percent for France and Sweden and 200 percent (i.e., a credit contraction equivalent to 100 percent of average annual growth, or some 1.9 percent) in Switzerland.⁶

According to our estimates, such reductions in credit growth would affect GDP growth substantially. The average decrease in DD between July and end-2007 would reduce real GDP growth by about 0.2 percentage points, while a decrease in DD as seen between July 2007 and July 2008 would lower GDP growth by about 1.0 percentage point (see Table 34.1).⁷ For Switzerland, our estimates predict financial stress to reduce GDP growth by some 2.2 percentage points, enough to bring it into negative territory by most estimates.

These estimated reductions in credit and GDP growth, on average, still are smaller than what has been experienced in the recent “Great Recession.” The discrepancy can be explained by the unusual severity of the global financial crisis compared with the relatively benign economic environment in Western Europe during the last two decades. In addition, several caveats surround our estimates. Still, our estimates indicate significant and pronounced effects, and illustrate the importance of financial sector stability on credit extension and growth.

Our estimates must be interpreted with caution, as they capture only the short-term direct effects of increased financial sector risk on credit and GDP growth. Several multiplier effects, through, for example, lower investment, may increase these effects in the medium to longer term. In addition, although the period over which the relationships are estimated contains economic downturns, it does not contain a major

⁴ We have also assessed the robustness of the results against a change in the dependent variable. When using domestic credit instead of private sector credit, we find that the results are fairly robust. The same relationship between risk and credit holds, and coefficient estimates are of the correct sign and the same order of magnitude. However, many coefficient estimates lose significance. As domestic credit is a much broader category than private sector credit, and most bank credit in the countries under study is provided to the private sector, we interpret this result as indicating that private sector credit is the more relevant variable influenced by financial sector health. The domestic credit estimation results are omitted for this reason.

⁵ For these calculations, we employ the coefficient value from our preferred specification of the regression equation, which is 0.00648.

⁶ A caveat that applies here is that the decreases in DD seen over the period from July 2007 are steeper than experienced in-sample. Hence, these decreases take the estimates outside of the range of values for which the model is validated. In particular, nonlinearities related to the estimation of relationships in natural logarithms increase when DD estimates reach values close to zero.

⁷ These results are based on regressions of financial risk variables directly on GDP. Equivalent to our credit regressions, our preferred regression equation employs long-term interest rates and house prices as controls. Table 11 in Tieman and Maechler (2009) presents the regression outcomes. The coefficient value used for these calculations is 0.00368.

TABLE 34.1

Macroeconomic Regressions: Distance-to-Default Indicators and Implied Effect on Credit Growth for Major European Banks

				Percentage Decrease July 2007–December 2007	Percentage Decrease July 2007–June 2008	Implied Annual Decrease in Real Credit Growth June–December 2007 (percentage point)	Implied Annual Decrease in Real Credit Growth June 2007–June 2008 (percentage point)	Average Credit Growth 1991–2006		Implied Annual Decrease in Real GDP Growth July 2007–July 2008 (percentage point)
Bank	29-06-2007	31-12-2007	30-06-2008					Real	Nominal	
BNP Paribas	6.3	3.3	1.4	−48	−77	−0.4	−1.0			−0.5
Credit Agricole	5.0	2.2	0.4	−55	−92	−0.5	−1.6			−0.9
Societe Generale	5.9	1.9	0.1	−68	−98	−0.7	−2.7			−1.5
France system	6.3	2.7	1.0	−57	−83	−0.5	−1.2			−0.7
France average ¹	5.7	2.5	0.6	−57	−89	−0.6	−1.8	2.9	4.6	−1.0
Swedbank	6.2	2.6	0.1	−59	−99	−0.6	−3.1			−1.7
Nordea Bank	5.7	4.8	1.7	−16	−70	−0.1	−0.8			−0.4
SEB AB	5.2	1.9	0.0	−64	−99	−0.7	−3.4			−1.9
Svenska Handels banken	5.7	4.8	2.6	−15	−54	−0.1	−0.5			−0.3
Sweden system	6.2	3.6	1.3	−42	−80	−0.4	−1.0			−0.6
Sweden average	5.7	3.5	1.1	−38	−81	−0.4	−1.9	3.2	5.6	−1.1
UBS ²	6.2	2.0	−0.8	−68	−113	−0.7	−4.5			−2.5
Credit Suisse	6.6	2.5	0.1	−63	−99	−0.6	−3.1			−1.8
Switzerland system ²	6.8	2.3	−0.4	−66	−105	−0.7	−4.5			−2.5
Switzerland average ^{1,2}	6.4	2.2	−0.4	−66	−106	−0.7	−3.8	1.9	3.2	−2.2
Banco Espanol DE	7.0	2.8	0.9	−60	−88	−0.6	−1.4			−0.8
Banco Popular ES	8.9	4.5	1.8	−49	−80	−0.4	−1.0			−0.6
Banco Santander	7.0	5.7	3.4	−18	−52	−0.1	−0.5			−0.3
Banco Bilbao Vizcaya Argentaria	6.6	4.7	2.2	−29	−67	−0.2	−0.7			−0.4
Spain system	7.5	5.5	3.0	−27	−60	−0.2	−0.6			−0.3
Spain average ¹	7.4	4.4	2.0	−39	−72	−0.3	−0.9	8.1	12.7	−0.5

Barclays	5.6	1.3	0.1	-77	-99	-1.0	-3.0		-1.7
HBOS ²	7.2	2.4	-0.8	-67	-111	-0.7	-4.5		-2.5
Hongkong and Shanghai Banking Corporation Holdings	7.6	5.4	3.3	-30	-56	-0.2	-0.5		-0.3
Lloyds TSB GP	7.6	3.2	0.5	-58	-93	-0.6	-1.7		-1.0
Royal Bank of Scotland ²	5.1	1.5	-0.1	-71	-101	-0.8	-4.5		-2.5
Standard Chartered	6.9	4.8	2.4	-30	-65	-0.2	-0.7		-0.4
UK system	7.8	3.2	1.3	-59	-83	-0.6	-1.1		-0.7
UK average¹	6.7	3.1	0.9	-55	-88	-0.6	-2.5	5.4	8.6
Bankgesellschaft	4.9	4.7	4.7	-4	-4	0.0	0.0		0.0
Bayer Hypo-Und- Vereinsbank	8.5	9.3	7.9	9	-7	0.1	0.0		0.0
Commerzbank	6.8	2.8	0.6	-59	-92	-0.6	-1.6		-0.9
Deutsche Bank	7.2	3.5	0.3	-51	-97	-0.5	-2.2		-1.2
Germany system	10.0	5.9	2.3	-41	-77	-0.3	-1.0		-0.5
Germany average¹	6.9	5.1	3.4	-26	-50	-0.3	-1.0	4.3	5.0
Unicredito Italiano	6.3	3.4	1.2	-45	-82	-0.4	-1.1		-0.6
San Paolo IMI	10.2	N/A	N/A	N/A	N/A	N/A	N/A		N/A
Capitalia	6.1	5.7	11.2	-6	82	0.0	0.4		0.2
Banca Intesa	7.6	6.3	3.6	-17	-53	-0.1	-0.5		-0.3
Banca Monte Dei Paschi	7.2	4.1	2.2	-44	-69	-0.4	-0.8		-0.4
Banca Nazionale del Lavoro	29.2	N/A	N/A	N/A	N/A	N/A	N/A		N/A
Italy system	10.1	6.4	3.4	-36	-66	-0.3	-0.7		-0.4
Italy average¹	11.1	4.9	4.5	-28	-30	-0.2	-0.5	5.3	8.9
Unweighted average	6.6	3.8	1.9	-43.0	-71.7	-0.4	-1.7	4.4	6.9
									-1.0

Sources: Authors' calculations; Datastream.

1. Simple average across banks.

2. Distance to default below zero set at 0.1 for calculation purposes.

crisis, which could exacerbate further the real effect of a global financial stress, especially if unleashing a general loss of confidence or herd behavior. Hence, using the estimation results to gauge what would happen in a crisis is surrounded by caveats that could possibly be reduced by including data from crisis periods. Nonetheless, the effects illustrated by the models are likely to underestimate the true effects in a crisis, as additional factors such as loss of confidence or animal spirits often play a role there. Redoing the estimations by including data from the current crisis therefore will likely yield stronger effects, which would come closer to actual losses in output and credit growth.

Our results present a concrete way to interpret stress testing outcomes. Often, the outcome of stress tests is presented in terms of (additional) losses for the banks, insurance companies, or pension funds, while implications for financial stability and hence the financial system as a whole are discussed qualitatively. The broader implications for credit and GDP growth, and hence economic welfare, normally are not analyzed. Such an approach could be characterized as somewhere halfway between the traditional microprudential approach of supervision and a macroprudential perspective. Modeling the feedback loop from financial sector stress to the real economy, and hence to economic welfare, enables a more complete macroprudential analysis. In addition, it enables estimates of costs in terms of foregone output. Such costs potentially could be weighted against the costs of fiscal and financial rescue packages to come up with a rough cost-benefit analysis for policy actions. Quantifying these fiscal costs presents a problem in itself and could be attempted by estimating a link between banks' capital and our risk variables, thus enabling estimates of the cost of reducing financial sector risk. Such analysis is left for future work.

B. Individual bank regressions

The bank-specific panel regressions paint a similar picture to the macro regressions. Effects of the financial fragility variables are found to be significant and substantial, with coefficient estimates in the same range.⁸ The significance of the different control variables differs somewhat, leading to the inclusion of different control variables in our preferred regression equation. Estimated effects on real bank credit from the increase in financial sector fragility are substantially larger than the effects found in the macroeconomic regressions. However, effects on GDP growth are of the same order of magnitude.

All three risk variables yield broadly similar significant coefficient estimates. The significance of both the one- and the five-year EDFs is robust across specifications, and the coefficient estimates always have the expected (negative) sign. The significance of the DD measure is also at the 1 percent level in our preferred specification, but in several other speci-

fications employing additional or different control variables, it has a lower significance, of 5 percent or lower.

We use the same control variables as in our macro regressions, with the addition of a competition variable. Lagged GDP growth is found universally to have a significant coefficient estimate and is included as a control in all regressions. In addition, in the EDF1 and EDF5, as well as in the DD regressions, both the DJ Stoxx index and the size of the financial sector are significant. The house price index is not significant in the EDF1 and EDF5 regressions but is significant in the DD regressions. We hence include the DJ Stoxx and the size variables in the EDF regressions. For the specification of the DD regression, we prefer to stick closely to the specification in the macroeconomic regressions and therefore include only the house price index as a control variable. The control variable indicating the financial health of a bank's direct competition (in the form of the EDF or DD of its domestic competitors) turns out to have some significance in the EDF1 regressions but none in the DD and EDF5 regressions.⁹ We hence elect to drop it.

Our preferred specification of the EDF1 and EDF5 regression equations therefore includes controls for the DJ Stoxx index, the financial sector size, and GDP but not the house price or competition variables. For the DD regressions, we prefer the specification including controls for the house prices and GDP but no other controls. The R^2 for these regressions is generally very good, in the range of 0.40–0.45 for the EDF regression. However, for the DD regressions, the fit is less well, with R^2 in the order of 0.02–0.10.

The bank-specific estimation results imply significant and pronounced effects on bank credit and real growth. Analogous to the results already reported, using the decline in DD between July 2007 and July 2008, our model predicts a decline in real bank credit growth by a little below 10 percentage points (Table 34.2) on average over the countries in our sample. In Switzerland, the country experiencing the largest decline in DD, the decline in real bank credit would be more than double that, at 21 percentage points. The implied decline in GDP growth, though, is very similar to that found in the macroeconomic regressions. The July 2007–08 decline in DD would result in a decline in GDP growth of 1.1 percentage points and up to 2.5 percentage points (for Switzerland).

The effect on bank credit found here is larger than the effect on private sector credit found in the macroeconomic regressions, which hints at some replacement of bank credit by other credit during a downturn. This is natural in an environment where the banks specifically are experiencing problems but the broader market is not, or at least less so. In a more universal market crisis, such as the recent Great Recession, such replacement is likely to be very modest, as market access is very limited for most borrowers. The ultimate effect on real GDP growth found in the bank-specific

⁸ See Appendix Tables 12–18 in Tieman and Maechler (2009) for detailed panel regression results.

⁹ See Appendix Tables 15–17 in Tieman and Maechler (2009) for detailed panel regression results.

TABLE 34.2

Bank-Specific Regressions: Distance-to-Default Indicators and Implied Effect on Credit Growth for Major European Banks

Bank	6/29/2007	12/31/2007	6/30/2008	Percentage Decrease July 2007–December 2007	Percentage Decrease July 2007–July 2008	Implied Annual Decrease in Real Credit Growth July 2007–July 2008 (percentage point)	Implied Annual Decrease in Real GDP Growth July 2007–July 2008 (percentage point)
BNP Paribas	6.3	3.3	1.4	–48	–77	–5.3	–0.6
Credit Agricole	5.0	2.2	0.4	–55	–92	–9.0	–1.1
Societe Generale	5.9	1.9	0.1	–68	–98	–14.8	–1.7
France system	6.3	2.7	1.0	–57	–83	–6.4	–0.8
France average¹	5.7	2.5	0.6	–57	–89	–9.7	–1.1
Swedbank	6.2	2.6	0.1	–59	–99	–16.9	–2.0
Nordea Bank	5.7	4.8	1.7	–16	–70	–4.3	–0.5
SEB A	5.2	1.9	0.0	–64	–99	–18.6	–2.2
Svenska Handelsbanken	5.7	4.8	2.6	–15	–54	–2.8	–0.3
Sweden system	6.2	3.6	1.3	–42	–80	–5.7	–0.7
Sweden average	5.7	3.5	1.1	–38	–81	–10.6	–1.3
UBS ²	6.2	2.0	–0.8	–68	–113	–24.7	–2.9
Credit Suisse	6.6	2.5	0.1	–63	–99	–17.2	–2.0
Switzerland system ²	6.8	2.3	–0.4	–66	–105	–24.7	–2.9
Switzerland average^{1,2}	6.4	2.2	–0.4	–66	–106	–21.0	–2.5
Banco Espanol DE	7.0	2.8	0.9	–60	–88	–7.5	–0.9
Banco Popular ES	8.9	4.5	1.8	–49	–80	–5.8	–0.7
Banco Santander	7.0	5.7	3.4	–18	–52	–2.6	–0.3
Banco Bilbao Vizcaya Argentaria	6.6	4.7	2.2	–29	–67	–4.0	–0.5
Spain system	7.5	5.5	3.0	–27	–60	–3.3	–0.4
Spain average¹	7.4	4.4	2.0	–39	–72	–5.0	–0.6
Barclays	5.6	1.3	0.1	–77	–99	–16.5	–1.9
HBOS ²	7.2	2.4	–0.8	–67	–111	–24.7	–2.9
Hongkong and Shanghai Banking Corporation Holdings	7.6	5.4	3.3	–30	–56	–2.9	–0.3
Lloyds TSB GP.	7.6	3.2	0.5	–58	–93	–9.6	–1.1
Royal Bank of Scotland ²	5.1	1.5	–0.1	–71	–101	–24.7	–2.9
Standard Chartered	6.9	4.8	2.4	–30	–65	–3.8	–0.4
UK system	7.8	3.2	1.3	–59	–83	–6.3	–0.7
UK average¹	6.7	3.1	0.9	–55	–88	–13.7	–1.6
Bankgesellschaft	4.9	4.7	4.7	–4	–4	–0.2	0.0
Bayer Hypo-Und-Vereinsbank	8.5	9.3	7.9	9	–7	–0.3	0.0
Commerzbank	6.8	2.8	0.6	–59	–92	–9.0	–1.1
Deutsche Bank	7.2	3.5	0.3	–51	–97	–12.0	–1.4
Germany system	10.0	5.9	2.3	–41	–77	–5.3	–0.6
Germany average¹	6.9	5.1	3.4	–26	–50	–5.4	–0.6
Unicredito Italiano	6.3	3.4	1.2	–45	–82	–6.1	–0.7
San Paolo IMI	10.2	—	—	—	—	—	—
Capitalia	6.1	5.7	11.2	–6	82	2.2	0.3
Banca Intesa	7.6	6.3	3.6	–17	–53	–2.7	–0.3
Banca Monte Dei Paschi	7.2	4.1	2.2	–44	–69	–4.2	–0.5
Banca Nazionale del Lavoro	29.2	—	—	—	—	—	—
Italy system	10.1	6.4	3.4	–36	–66	–3.8	–0.5
Italy average¹	11.1	4.9	4.5	–28	–30	–2.7	–0.3
Unweighted average	6.6	3.8	1.9	–43	–72	–9.6	–1.1

Sources: Authors' calculations; Datastream.

1. Simple average across banks.

2. Distance to default below zero set at 0.1 for calculation purposes.

regressions is similar to the effect found through the macro-economic model.

4. CONCLUSION

Our analysis finds that reductions in credit growth as a result of financial sector fragility are substantial. Between July and end-2007, the increased financial sector risk as perceived by the market would lead real credit growth to decrease by 0.4 percentage points in real terms in the countries in our sample. Set against an average real credit growth of 4.4 percent over the period 1991–2006, this implies a decrease of some 10 percent. When the turmoil in the first half of 2008 is taken into account—that is, looking out-of-sample at the impact over the period July 2007–July 2008 using the estimated coefficients—the total impact on real credit growth amounts to a decrease of 32 percent. Individual countries and banks in our sample experience even more severe impacts, in some cases leading to a significant credit contraction. Similar effects are found for GDP. We estimate that the increase in financial fragility over the period July 2007 to July 2008 could have a negative impact on GDP growth of over 1 percentage point on average, ranging up to 2.5 percentage points for specific countries.

Various other (unmodeled) channels and multiplier effects can result in effects even larger than found here. For example, our estimated effects are based on estimates calibrated to data from noncrisis periods. In other words, these effects are the noncrisis component of the total effects, which could be substantially larger during crises due to added factors such as loss of confidence or other multiplier effects. Our results hence suggest that credit could decrease substantially during a crisis.

Quantifying the real economic impact of financial sector fragility is important for several reasons. First, our results clearly illustrate that certain real economic variables depend on financial sector risk variables and that these risk variables play an important role in the credit channel. Most models of the credit channel to date do not take such risk variables into account and hence may be missing an important link. Second, estimates of the real economic impact of financial sector fragility can serve to inform policymakers about costs and benefit estimates of certain financial sector policies. Third, quantifying the real economic impact of financial sector stress can serve to improve macroprudential stress tests.

The policy implications of employing a model like ours are twofold:

- First, our results suggest that prudential supervision could be enhanced and broadened by taking the effects of financial instability on the real economy into account. A narrow “macroprudential” approach, which focuses on financial stability and contagious defaults, misses the essential element that when fragility in the system increases, there are substantial economic costs long before any banks come close to

default. Such costs should be taken into account in macroprudential analysis, by focusing more on co-movements in fragility in different financial institutions, even in the range where most institutions are nowhere near their respective default barriers. Such analysis might show, for instance, that countercyclical capital ratios (i.e., shoring up financial institutions’ capital base in good times, while relaxing capital adequacy ratios in bad times) might be preferable from a social welfare point of view compared with waiting until firms come close to their minimum regulatory capital ratio.

- Second, our results could be of interest to fiscal policymakers, as these results could serve as a better benchmark to estimate the real costs of financial sector stress. By assuming or estimating a link between banks’ capital and our risk variables, the benefit of fiscal measures that increase financial sector stability could be quantified. Of course, any such benefits should be weighted against possible costs, including a moral hazard that could emerge when stability is enhanced by distorting incentives.

Our results also can be used to inform stress tests for the banking system directly. Stress tests aim at quantifying the resilience of the financial system to extreme but plausible shocks. They are performed regularly by many banks’ risk management departments, as well as by supervisory and financial stability authorities. They also are a main ingredient of financial system stability assessments under the International Monetary Fund–World Bank Financial Sector Assessment Program (FSAP).¹⁰ In recent years, FSAP stress tests have focused on the impact of medium-term macroeconomic shock scenarios (often over a three-year horizon) on the health of individual financial institutions and the system as a whole. The macro scenarios are translated into financial risk variables, such as changes in banks’ capital adequacy ratios and probabilities of default for specific credit portfolios.

Such stress tests, however, normally assume that institutions do not react to the shocks. In reality, financial institutions, of course, do react to a radically different environment after an extreme shock. Most important, banks often choose to shrink their credit portfolios or at least grow them less rapidly. Our work has shown the magnitude of such a feedback effect in Europe. The feedback effect thus can be quantified and included in the macro stress scenarios, improving their realism and accuracy. Assuming adequate data availability, this approach can be applied to other countries and institutions.¹¹

¹⁰ For more detailed information on the FSAP, see <http://www.imf.org/external/np/fsap/fsap.asp>.

¹¹ The construction of reliable risk indicators from market data rests on data availability. An environment with illiquid or clearly inefficient markets hence might prohibit dependable and consistent estimation of the feedback effect. In addition, the quality of the macroeconomic statistics used should be adequate.

In addition, our results provide a useful metric for interpreting stress testing results. They serve to translate financial sector losses, the often used metric for stress testing results, to actual output losses. As output is the ultimately relevant variable in many policy decisions, such a translation brings the results a step closer to the actual policy debate. Even when stress test results are not directly related to any policy debate, knowledge about potential output losses can serve to put macroprudential issues on the policy agenda.

REFERENCES

- Altunbas, Yener, Otabek Fazylov, and Philip Molyneux, 2002, "Evidence on the Bank Lending Channel in Europe," *Journal of Banking and Finance*, Vol. 26, No. 11, pp. 2093–110.
- Angeloni, Ignazio, and Michael Ehrmann, 2003, "Monetary Policy Transmission in the Euro Area: Any Changes after EMU?" ECB Working Paper No. 240 (Frankfurt: European Central Bank). Available via the Internet: <http://www.ecb.europa.eu/pub/scientific/wps/date/html/wps2003.en.html>
- Bagehot, Walter, 1873, *Lombard Street: A Description of the Money Market* (London: Henry S. King).
- Bayoumi, Tamim, and Andrew Swiston, 2007, "Foreign Entanglements: Estimating the Source and Size of Spillovers across Industrial Countries," IMF Working Paper 07/183 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.cfm?sk=21192.0>
- Bernanke, Ben, 1983, "Nonmonetary Effects of the Financial Crisis in the Propagation of the Great Depression," *American Economic Review*, Vol. 73, No. 3, pp. 257–76.
- Bernanke, Ben, and Alan Blinder, 1988, "Credit, Money and Aggregate Demand," *American Economic Review*, Vol. 78, No. 2, pp. 435–39.
- Bernanke, Ben S., and Mark Gertler, 1995, "Inside the Black Box: The Credit Channel of Monetary Policy Transmission," *Journal of Economic Perspectives*, Vol. 9, No. 4, pp. 27–48.
- Black, Fisher, and Myron S. Scholes, 1973, "The Pricing of Options and Corporate Liabilities," *Journal of Political Economy*, Vol. 81, No. 3, pp. 637–54.
- Čihák, Martin, and Petya Koeva-Brooks, 2008, "From Subprime Loans to Subprime Growth? Evidence from the Euro Area," IMF Country Report 08/263, pp. 5–25 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=22225>
- De Nicoló, Gianni, Robert Corker, Alexander F. Tieman, and Jan-Willem van der Vossen, 2005, "European Financial Integration, Stability and Supervision," IMF Country Report 05/266, pp. 113–146 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=18458>
- Espoziza, Raphael, Fabio Fornari, and Marco Lombardi, 2009, "The Role of Financial Variables in Predicting Economic Activity," ECB Working Paper No. 1108 (Frankfurt: European Central Bank). Available via the Internet: <http://www.ecb.europa.eu/pub/scientific/wps/date/html/wps2009.en.html>
- Hunter, William C., George G. Kaufman, and Thomas H. Krueger, eds., 1999, *The Asian Financial Crisis: Origins, Implications and Solutions* (Boston: Kluwer Academic Publishers).
- Merton, Robert C., 1974, "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates," *Journal of Finance*, Vol. 29, No. 2, pp. 449–70.
- Noland, Markus, Li-Gang Liu, Sherman Robinson, and Zhi Wang, 1998, "Global Economic Effects of the Asian Currency Devaluations," Policy Analyses in International Economics No. 56 (Washington: Institute for International Economics).
- Tieman, Alexander F., and Andrea M. Maechler, 2009, "The Real Effects of Financial Sector Risk," IMF Working Paper 09/198 (Washington: International Monetary Fund). Available via the Internet: <http://www.imf.org/external/pubs/cat/longres.aspx?sk=23275>
- Vassalou, Maria, and Yuhang Xing, 2004, "Default Risk in Equity Returns," *Journal of Finance*, Vol. 59, No. 2, pp. 831–68.

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Appendix I

The Data

TABLE 34.3

Data Summary							
Data Series	France	Germany	Italy	Spain	Sweden	Switzerland	UK
DD							
Average	4.6	5.0	4.7	6.3	4.4	5.5	5.1
Standard deviation	1.5	2.2	3.3	2.1	2.3	3.5	1.8
Availability	1991–2007	1991–2007	1991–2007	1991–2007	1991–2007	1991–2007	1991–2007
Source	Datastream and IMF staff calculations						
EDF1							
Average	0.10	0.22	0.14	0.15	0.19	0.15	0.15
Standard deviation	0.08	0.22	0.09	0.14	0.20	0.13	0.09
Availability	1992–2007	1992–2007	1992–2007	1992–2007	1992–2007	1998–2007	1992–2007
Source	Moody's KMV						
EDF5							
Average	0.19	0.33	0.23	0.21	0.27	0.25	0.22
Standard deviation	0.10	0.23	0.12	0.16	0.21	0.14	0.10
Availability	1992–2007	1992–2007	1992–2007	1992–2007	1992–2007	1998–2007	1992–2007
Source	Moody's KMV						
PS Cred							
Average	1214210	2075328	786	629854	1465	652	1206
Standard deviation	269980	457760	339	431340	1035	86	534
Availability	1991–2007	1991–2007	1991–2007	1991–2007	1991–2007	1991–2007	1991–2007
Source	International Financial Statistics						
Dom Cred							
Average	1432659	2526224	1064	742080	1629	703	1233
Standard deviation	327493	586533	316	435755	1028	93	524
Availability	1991–2007	1991–2007	1991–2007	1991–2007	1991–2007	1991–2007	1991–2007
Source	International Financial Statistics						
GDP							
Average	344951	497567	277365	150434	527971	100954	226252
Standard deviation	63187	60924	63589	54920	121866	11653	61972
Availability	1990–2007	1990–2007	1990–2007	1990–2007	1990–2007	1990–2007	1990–2007
Source	International Financial Statistics						
Total loans							
Average	1990–2007	1990–2007	1990–2007	Bank specific	1990–2007	1990–2007	1990–2007
Standard deviation				Bank specific			
Availability				1990–2007			
Source				Datastream			
House prices							
Average	119.5	99.4	100.8	1018.3	264.5	360.2	322.6
Standard deviation	40.5	5.3	12.6	461.9	87.1	54.4	145.9
Availability	1995–2007	1990–2007	1991–2007	1990–2007	1990–2007	1990–2007	1990–2007
Source	Global Insight						
DJ Stoxx 600							
Average	1990–2007	1990–2007	1990–2007	222	1990–2007	1990–2007	1990–2007
Standard deviation				98.2			
Availability				1990–2007			
Source				Bloomberg			
V-Dax New							
Average	1992–2007	1992–2007	1992–2007	22.4	1992–2007	1992–2007	1992–2007
Standard deviation				10.3			
Availability				1992–2007			
Source				Bloomberg			
ITraXX Crossover							
Average	1992–2007	1992–2007	1992–2007	272.3	1992–2007	1992–2007	1992–2007
Standard deviation				37.6			
Availability				1992–2007			
Source				Bloomberg			

Sources: Authors; Datastream; Global Insight; International Financial Statistics; and Moody's KMV.

Note: DD = distance to default; Dom Cred = total domestic credit; EDF = expected default frequency; PS Cred = private sector credit.

Appendix II

Cointegrating Relationship

To study the dynamics of the relationship between credit and the financial risk variables, we start by estimating a vector error-correction model explaining the change in credit growth, that is,

$$\Delta Cred_t = \alpha (Cred_{t-1} + \beta_1 GDP_{t-1} + \beta_2) + \gamma_1 Risk_t + \gamma_2 Interest_t + \gamma_3 Controls_t + \gamma_4 + \varepsilon_t.$$

The relation between brackets is the long-term equilibrium relation between (the nonstationary variables) credit, *Cred* (both private sector and domestic credit), and *GDP*, with the speed of adjustment toward equilibrium determined by the coefficient α . The short-term variables determine the behavior of credit around the long-term equilibrium. Here, we use our risk measure (distance to default [DD] or expected default frequency [EDF]), as well as a price variable (an interest rate) to distinguish volume and price effects. We use GDP growth as a macroeconomic control.

The estimation results strongly support the existence of a cointegrating relationship between credit and GDP (Table 34.4). The significance is evidenced by *t*-values in the 4–6 range for the long-term equilibrium relationship between credit and GDP and in the 9–11 range for the adjustment coefficient. This holds for both the regressions explaining private sector credit and the regressions with total domestic credit as the dependent variable.

Results also indicate that the various financial sector risk indicators we employ show significant short- to medium-term effects. For both dependent variables, *t*-values for the different risk variables range from 4 to 7.5, while all estimated coefficients exhibit the expected sign (positive for DD-based measures and negative for EDF-based measures).

The significance of our price variable and controls is less strong. The coefficient estimates for interbank interest rates are often not statistically different from zero but generally do exhibit the expected (negative) sign. Lagged GDP growth exhibits the expected (positive) sign, but the coefficient estimate is only significantly different from zero at the 5 percent level for the domestic credit regressions. Adding other control variables does not change this picture. Adding dummies does illustrate the difference between countries, as both level dummies and dummies that interact with the risk variables show significance in some cases. However, adding six dummies to cover all seven countries in the sample individually often leads to overidentified estimations and hence does not yield reliable results. Adding a dummy indicating whether countries are in or outside of the euro area indicates that the risk variables play a stronger role in the euro area. However, this result may well have been unduly influenced by the data for Sweden in particular, as these series include the direct aftermath of the Swedish banking crisis.

TABLE 34.4

Vector Error Correction Model Regression Results (coefficient estimates, with *t*-values in parentheses)

Cointegrating relationship	Dependent Variable															
	Private Sector Credit Regression								Domestic Credit Regression							
	DD	AWDD	SYSDD	DDINDEX	EDF1	AWEDF1	EDF5	AWEDF5	DD	AWDD	SYSDD	DDINDEX	EDF1	AWEDF1	EDF5	AWEDF5
PSCred(−1)	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Dcred(−1)																
GDP(−1)	−1.917 (−4.8)	−1.661 (−4.1)	−2.480 (−5.8)	−2.377 (−5.3)	−2.477 (−5.7)	−2.282 (−5.1)	−2.279 (−5.5)	−2.124 (−5.0)	−1.972 (−4.2)	−1.593 (−3.5)	−2.630 (−5.20)	−2.485 (−4.8)	−2.697 (−5.4)	−2.371 (−4.8)	−2.496 (−5.2)	−2.224 (−4.7)
Adjustment coefficient	0.0122 (10.2)	0.0119 (10.1)	0.0117 (9.5)	0.0112 (9.1)	0.0119 (9.5)	0.0116 (9.2)	0.0124 (10.1)	0.0120 (9.7)	0.0108 (10.9)	0.0109 (11.2)	0.0104 (10.0)	0.0101 (9.7)	0.0108 (10.4)	0.0109 (10.5)	0.0113 (10.9)	0.0113 (11.0)
Short-term dynamics																
DD	2006 (5.4)								1873 (4.9)							
AWDD		1978 (5.4)								1895 (5.1)						
SYSDD			1689 (4.8)								1475 (4.1)					
DDINDEX				1401 (4.3)								1328 (4.0)				
EDF1					−37202 (−6.0)								−39434 (−6.2)			
AWEDF1						−35878 (−5.9)								−38627 (−6.2)		
EDF5							−41908 (−7.4)								−43159 (−7.5)	
AWEDF5								−39463 (−7.1)								−40943 (−7.2)
IBR	−179.6 (−0.6)	−154.1 (−0.5)	−545.9 (−1.7)	−584.0 (−1.9)	−323.4 (−0.9)	−326.7 (−0.9)	−132.5 (−0.4)	−164.0 (−0.5)	−12.69 (0.0)	35.35 (0.1)	−385.1 (−1.2)	−396.4 (−1.2)	23.70 (0.1)	47.28 (0.1)	207.2 (0.6)	198.0 (0.6)
DGDP(<i>t</i> −1)	0.0358 (0.7)	0.0194 (0.4)	0.0388 (0.7)	0.0392 (0.8)	0.0353 (0.7)	0.0200 (0.4)	0.0281 (0.1)	0.0150 (0.3)	0.1197 (2.4)	0.1045 (2.1)	0.1229 (2.3)	0.1232 (2.4)	0.1223 (2.3)	0.1088 (2.1)	0.1157 (2.2)	0.1042 (2.0)

Source: IMF Staff calculations.

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